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AATCC Manual of International Test Methods and Procedures

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Preface

THE standards in this volume of the *Manual of International Test Methods and Procedures* were current as of July 2022. New standards that have been added and other important changes made since the last volume are summarized on page 18.

AATCC standards are developed by research committees through extensive investigations and interlaboratory comparisons, often covering several years of work. Simplicity, reproducibility, applicability, cost of performing the test and the time required to perform the test are all important considerations in each development.

Before a standard is published in the *Manual of International Test Methods and Procedures*, it must be approved by the responsible research committee and the Technical Committee on Research (TCR).

During the first three years of publication, each new standard is reviewed annually, at which time, on recommendation of the research committee and approval by TCR, it may be reaffirmed, revised or withdrawn. After the first three years, each standard is reviewed at least once every five years by the research committee, and following approval by TCR, may be reaffirmed, revised or withdrawn. The record of these actions is published in an activity history at the beginning or end of each standard.

An important feature of all AATCC test methods is that test results are numerically quantified. Methods do not include pass/fail criteria for materials or processes. AATCC policy prohibits endorsement of such specifications, requirements, or acceptance criteria. These must be agreed upon by the interested parties.

Each standard is designated by AATCC, then the letters TM, LP, EP, or M as appropriate, a number (no space), a dash, and the four-digit year of the latest revision. Additional technical changes within the same year are signified by a small "t" following the year. Reaffirmations—with no changes—are signified by a year in parentheses following the year of latest revision. Editorial changes are signified by a small "e" following the year. The designation should be quoted in full in referring to a standard, as for example, TM90-2011(2016)e.

AATCC M12-2020, Style Guide for Writing Test Methods and Procedures (page 521) is the defining document used by research committees in writing these methods. M13-2017e, Rules of Procedure for AATCC Test Method and Technology Committees (page 528) is the defining document for the organization and functioning of these research committees.

Prior to 1969, the *Manual of International Test Methods and Procedures* also contained the Buyer's Guide and the AATCC Membership Directory, now both available online. The general content of the *Manual of International Test Methods and Procedures* has continued unchanged since 1969.

Major changes in layout of the *Manual of International Test Methods and Procedures* were made in 1985 as follows: (1) indexes were grouped at the front of the book; (2) test methods were arranged in numerical order; (3) A Glossary of AATCC Standard Terminology was added and (4) the AATCC Style Guide for Writing Test Methods and Procedures was added.

In 1989, evaluation procedures were listed immediately after the test methods.

In 2012, monographs were given number designations and were included as a separate item in the numerical and topical indexes for easy access.

In 2019, laboratory procedures were listed between test methods and evaluation procedures. The glossary and style guide were numbered as monographs.

In 2020, the official designations and titles for all standards were updated.

In 2021, new, revised, reaffirmed, editorially revised, and editorially revised and reaffirmed methods were aligned to the AATCC style guidelines. The official title was changed from *Technical Manual of the American Association of Textile Chemists and Colorists* to the *American Association of Textile Chemists and Colorists Manual of International Test Methods and Procedures*.

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Numerical List of Current AATCC Standards

Test Methods

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TM5-1962	Colorfastness to Dry and Wet Heat. Superseded by AATCC TM133.
TM7-1989	Colorfastness to Degumming.
TM9-1989	Colorfastness to Stoving.
TM10-1944	Colorfastness to Commercial Laundering and to Domestic Washing. Superseded by AATCC TM36.
TM11-1989	Colorfastness to Carbonizing.
TM12-1931	Colorfastness to Sea Water. Superseded by AATCC TM63.
TM13-1957	Colorfastness to Peroxide Bleaching with Peroxide: Silk. Superseded by AATCC TM101.
TM14-1953	Dimensional Changes in Cotton and Linen Textiles. Superseded by AATCC TM91.
TM16-2004	Colorfastness to Light. Superseded by AATCC TM16.1, AATCC TM16.2 and AATCC TM16.3.
TM16A-1988	Colorfastness to Light: Carbon-Arc Lamp, Continuous Light. Superseded by AATCC TM16.
TM16B-1977	Colorfastness to Light through Glass: Sunlight.
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TM16F-1988	Colorfastness to Light: Water-Cooled Xenon-Arc Lamp, Alternate Light and Darkness. Superseded by TM16.
TM16G-1985	Colorfastness to Light: Determination of Fastness Above L-7. Superseded by AATCC TM16.
TM18-1967	Water Resistance: Hydrostatic Pressure Test. Superseded by AATCC TM127.
TM19-1937	Mercerization of Cotton, Determination of; Degree of. Superseded by AATCC TM89.
TM21-1983	Water Repellency: Static Absorption Test.
TM24-2004	Insects, Resistance of Textiles to.
TM25-1957	Colorfastness to Drycleaning. Superseded by AATCC TM85.
TM28-2004	Insect Pest Deterrents on Textiles.
TM29-1957	Colorfastness to Bleaching with Peroxide: Cotton and Linen. Superseded by AATCC TM101.
TM31-1962	Colorfastness to Pleating. Superseded by AATCC TM131.
TM32-1952	Detection of Phototropism. Superseded by AATCC TM139.
TM33-1962	Flammability of Clothing Textiles.
TM34-1969	Fire Resistance of Textile Fabrics.
TM36-1972	Colorfastness to Washing: Characterization of Textile Colorants.
TM37-1952	Colorfastness to Commercial Laundering and Domestic Washing (Silk). Superseded by AATCC TM36.
TM38-1952	Colorfastness to Commercial Laundering and Domestic Washing (Wool). Superseded by AATCC TM36.
TM39-1980	Wettability, Evaluation of.
TM40-1957	Dimensional Changes in Textiles Other Than Wool. Superseded by AATCC TM91.
TM41-1952	Dimensional Changes in Wool Textiles: Accelerated Test. Superseded by AATCC TM99.

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Method	Title
TM44	Title unavailable. Discontinued. Superseded by AATCC TM54.
TM45	Title unavailable. Discontinued. Superseded by AATCC TM54.
TM46	Title unavailable. Discontinued. Superseded by AATCC TM63.
TM47-1950	Colorfastness to Water (Other than Silk and Wool). Superseded by AATCC TM63.
TM48-1952	Colorfastness to Water Spotting. Superseded by AATCC TM104.
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TM57-1952	Colorfastness to Storage (Acetate).
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TM59-1952	Colorfastness to Peroxide Bleaching (Wool). Superseded by AATCC TM13.
TM60-1952	Detergents on Wool: Detergency Comparator Method.
TM62-1989	Oils, Wool; Oxidation in Storage.
TM63-1961	Colorfastness to Water: Distilled or Demineralized; Sea, and Chlorinated Pool. Superseded by Methods 105, 106, 107.
TM64-1977	Scouring, Continuous Scouring of Raw Grease Wool.
TM65-1984	Snag Resistance of Women's Nylon Hosiery.
TM67-1957	Wrinkle Recovery of Fabrics: Roller Pressure Crease-Recovery Method.
TM68-1969	Colorfastness to Washing at 105F: Rapid Control Test.
TM69-1958	Damage Caused by Retained Chlorine. Superseded by AATCC TM92.
TM71-1956	Colorfastness to Perspiration: Rapid Control Test.
TM72-1969	Colorfastness to Washing and Shrinkage: Combined Rapid Control Test.
TM73-1953	Shrinkage of Wool Hose: Accelerated Test. Superseded by AATCC TM99.
TM74-1953	Relaxation and Felting Shrinkage of Wool Knit Fabrics (Except Hose): Accelerated Test. Superseded by AATCC TM99.
TM75-1971	Colorfastness to Oxides of Nitrogen in the Atmosphere: Rapid Control Test.
TM77-1977	Scourability of Spinning Lubricant.
TM78-1989	Ash Content of Bleached Cellulosic Textiles.
TM80-1954	Determining the Noncotton Content of Bleached Woven Cotton Cloth. Superseded by AATCC TM97.
TM83-1974	Colorfastness to Light and Washing: Alternate Exposure.
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TM87-1965	Colorfastness to Washing, Industrial Laundering: Accelerated.
TM88-1961	Appearance of Wash and Wear Fabrics after Home Laundering. Superseded by AATCC TM88A.
TM88A-1964	Appearance of Fabrics in Wash and Wear Items after Home Laundering. Superseded by AATCC TM124.
TM90-1982	Antibacterial Activity of Fabrics, Detection of: Agar Plate Method. Superseded by Reactivated/Revised AATCC TM90-2016.
TM91-1958	Dimensional Changes in Woven Textiles (Excluding Wool). Superseded by AATCC TM96.
TM95-1959	Dimensional Restorability of Woven Textiles after Laundering. Superseded by AATCC TM96.
TM96-2012	Dimensional Changes in Commercial Laundering of Woven and Knitted Fabrics Except Wool.
TM99-2004	Dimensional Changes of Woven or Knitted Wool Textiles: Relaxation, Consolidation and Felting.
TM105-1975	Colorfastness to Water: Chlorinated Pool. Superseded by AATCC TM162.
TM108-1963	Dimensional Changes in Drycleaning.
TM111A-1990	Weather Resistance: Sunshine Arc Lamp Exposure with Wetting. Superseded by AATCC TM111.
TM111B-1990	Weather Resistance: Exposure to Natural Light and Weather. Superseded by AATCC TM111.
TM111C-1990	Weather Resistance: Sunshine Arc Lamp Exposure without Wetting. Superseded by AATCC TM111.
TM111D-1990	Weather Resistance: Exposure to Natural Light and Weather through Glass. Superseded by AATCC TM111.
TM113-1978	Formaldehyde Odor in Resin Treated Fabric, Determination of: Steam Method.
TM123-2000	Carpet Soiling: Accelerated Soiling Method.
TM126-1991	Colorfastness to Water (High Humidity) and Light: Alternate Exposure.
TM136-2013	Bond Strength of Bonded and Laminated Fabrics
TM139-2005	Colorfastness to Light: Detection of Photochromism.

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Method	Title
TM142-2016	Appearance of Flocked Fabrics after Home Laundering and/or Coin-Op Drycleaning.
TM145-1985	Color Measurement of the Blue Wool Lightfastness Standards: Instrumental.
TM148-1989	Light Blocking Effect of Curtain Materials. Superseded by Reactivated/Revised AATCC TM148-2014 with new title.
TM151-2003	Soil Redeposition: Launder-Ometer Method.
TM152-1990	Soil Redeposition, Resistance to: Terg-O-Tometer Method.
TM153-1985	Color Measurement of Textiles: Instrumental. Superseded by AATCC EP6.
TM155-1991	Transfer of Disperse Dyes on Polyester.
TM156-1991	Transfer of Basic Dyes on Acrylics.
TM160-1992	Dimensional Restoration of Knitted and Woven Fabrics after Laundering.
TM166-1998	Dispersion Stability of Disperse Dyes at High Temperature.
TM173-2009	Test Method for Calculation of Small Color Differences for Acceptability
TM177-2000	Colorfastness to Light at Elevated Temperature and Humidity: Xenon Lamp Apparatus.
TM178-2004	Barré: Visual Assessment and Grading. Superseded by M10.
TM180-1997	Colorfastness to Light at High Temperatures: Daylight Temperature Controlled Apparatus.
TM181-2005	Colorfastness to Light at High Temperatures: Daylight Temperature and Humidity Controlled Apparatus.
TM190-2016	Colorfastness to Home Laundering with Activated Oxygen Bleach Detergent: Accelerated.

Evaluation Procedures

Procedure	Title
EP3-1996	AATCC 5-Step Chromatic Transference Scale.
EP4-2011	Standard Depth Scales for Depth Determination.

Monographs

Monograph	Title
M2-2005	2003 Standard AATCC Standard Reference Liquid Laundry Detergent. Superseded by M1.
M3-2021	High Efficiency Washers in North America.
M5-2007	Standardization of Hand Laundering for Fabrics and Apparel. Superseded by LP2.
M6-2016	Standardization of Home Laundry Test Conditions. Superseded by LP1.
M7-1991	Standard Laboratory Practice for Home Laundering Fabrics Prior to Flammability Testing to Differentiate Between Durable and Non-Durable Finishes
M8-1992	Nomenclature for Subjective Rating Processes.

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Changes in 2024 Manual of International Test Methods and Procedures

The following AATCC standards have been developed, revised, or reaffirmed since publication of the 2023 volume of the Manual of International Test Methods and Procedures. In accordance with AATCC M13, Rules of Procedure for AATCC Test Method and Technology Committees, all technical changes are unanimously approved by the responsible research committee and the Technical Committee on Research (TCR) before publication.

AATCC TM16.1-2023, Test Method for Colorfastness to Light: Outdoor. Revised in 2023 to align to AATCC style guide.

AATCC TM16.2-2023, Test Method for Colorfastness to Light: Carbon-Arc. Revised in 2023 to align to AATCC style guide.

AATCC TM20A-2021e, Test Method for Fiber Analysis: Quantitative. Editorially revised in 2023 to update numbering in section 17.

AATCC TM43-1952e6(2018)e3, Test Method for Wetting Agents for Mercerization. Editorially revised in 2023 to move the History section.

AATCC TM61-2013e(2020)e2, Test Method for Colorfastness to Laundering: Accelerated. Editorially revised in 2023 to update designation and History section.

AATCC TM79-2010e2(2018)e3, Test Method for Absorbency of Textiles. Editorially revised in 2023 to move History section.

AATCC TM88B-2018te, **Test Method for Seam Smoothness in Fabrics after Home Laundering.** Editorially revised in 2023 to add History section.

AATCC TM88C-2018te, Test Method for Crease Retention in Fabrics after Home Laundering. Editorially revised in 2023 to add History section.

AATCC TM96-2012e3, **Test Method for Dimensional Changes in Commercial Laundering of Woven and Knitted Fabrics Except Wool.** Withdrawn

AATCC TM107-2022e, Test Method for Colorfastness to Water. Editorially revised in 2023 to correct misspellings.

AATCC TM111-2023, Test Method for Weather Resistance of Textiles: Exposure to Daylight and Weather. Revised in 2023 to align with the AATCC style guide.

AATCC TM124-2018te, Test Method for Smoothness Appearance of Fabrics after Home Laundering. Editorially revised in 2023 to add History section.

AATCC TM142-2016e, Test Method for Appearance of Flocked Fabrics after Home Laundering and/or Coin-Op Drycleaning. Withdrawn

AATCC TM143-2018te, Test Method for Appearance of Apparel and Other Textile End Products after Home Laundering. Editorially revised in 2023 to change History section.

AATCC TM149-2023, Test Method for Chelation Value of Aminopolycarboxylic Acids and Their Salts: Calcium Oxalate. Revised in 2023 to clarify and align with prescribed AATCC style guidelines.

AATCC TM161-2023, Test Method for Color Change of Disperse Dyes Caused by Metals, with and without Chelating Agent. Revised in 2023 to clarify and align with prescribed AATCC style guidelines.

AATCC TM163-2013(2020)e4, Test Method for Colorfastness to Storage: Dye Transfer. Editorially revised in 2023 to update history.

AATCC TM168-2023, Test Method for Chelation Value of Polyaminopolycarboxylic Acids and Their Salts: Copper PAN. Revised in 2023 to clarify and align with prescribed AATCC style guidelines.

AATCC TM174-2022e, Test Method for Antimicrobial Activity of New Carpets. Editorially revised in 2023 to correct section formatting.

AATCC TM179-2023, Test Method for Skew Change in Fabrics After Home Laundering. Revised in 2023 to add that optionally, the direction of skew may be reported.

AATCC TM186-2023, Test Method for Weather Resistance: UV Light and Moisture Exposure. Revised in 2023 for clarity and to align with style guidelines.

AATCC TM190-2010e2(2016)e2, Test Method for Colorfastness to Home Laundering with Activated Oxygen Bleach Detergent: Accelerated. Withdrawn

AATCC TM192-2023, Test Method for Weather Resistance of Textiles: Sunshine-Arc Lamp Exposure with and Without Wetting. Revised in 2023 to align to the AATCC style guide.

AATCC TM193-2023, Test Method for Aqueous Liquid Repellency: Water/Alcohol Solution Resistance. Revised in 2023 to add History section and clarity.

AATCC TM194-2023, Test Method for Anti-House Dust Mite Properties of Textiles under Long-Term Test Conditions. Revised in 2023 to add dust mite food source and to align with the AATCC Style Guide.

AATCC TM214-2022, Test Method for Measuring Condensation on Textiles in a Humid Microclimate. New – Developed in 2022 by AATCC Committee RA63.

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AATCC TM215-2023, Test Method for Soil Release: Household Soils. New – Developed in 2023 by AATCC Committee RA56.

AACC EP7-2021e, Evaluation Procedure for Instrumental Assessment of the Change in Color of a Test Specimen. Editorially revised to update a reference in Section 2: Principle, and to place the history in correct order.

AATCC EP10-2018e2, Evaluation Procedure for Multifiber Adjacent Fabrics. Editorially revised 2023 to clarify the Notes section and to move the History section.

AATCC EP14-2021e2, AATCC Evaluation Procedure for Small Color Differences. Editorially revised in 2023 to correct a typo.

AATCC LP4-2023, Laboratory Procedure for Synthetic Soil. New – Developed in 2023 by AATCC Committee RA57.

AATCC LP5-2023, Laboratory Procedure for Home Laundering with Bleach. New – Developed in 2023 by AATCC Committee RA88.

AATCC M3-2008(2021)e2, Monograph of High Efficiency Washers in North America. Withdrawn. Superseded by AATCC LP5.

Committee RA31, Antimicrobial Activity – Committee Scope updated.

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Test Method for Colorfastness to Acids and Alkalis

1. Purpose and Scope

1.1 Test specimens are evaluated for change in color when spot tested with acid, or steeped in alkali fumes, in a laboratory setting. These test methods are applicable to textiles made from all fibers in the form of yarns or fabrics, whether dyed, printed or otherwise colored.

2. Principle

2.1 Test specimens are steeped in, or spotted with, the required solutions by means of simple laboratory equipment, allowed to dry at room temperature, then examined for change in color.

3. Referenced Documents

3.1 AATCC EP1, Evaluation Procedure for Gray Scale for Color Change (see 12.1).
3.2 AATCC EP7, Evaluation Procedure for Instrumental Assessment of the Change in color of a Test Specimen (see 12.1).
3.3 ASTM E1402, Standard Guide for Sampling Design.

4. Terminology

4.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials, or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

5. Safety Precautions

5.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.
5.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.
5.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

6. Apparatus, Materials and Reagents

6.1 Beaker, 250 mL.
6.2 Bell jar, 4 L, with a glass plate base.
6.3 Evaporating dish, 7.6 cm.
6.4 Hydrochloric acid (HCl), 35%.
6.5 Acetic acid (CH_3COOH), 56%.
6.6 Ammonium hydroxide (NH_4OH), anhydrous ammonia 28% (NH_3).
6.7 Sodium carbonate (Na_2CO_3), anhydrous, technical.
6.8 Calcium hydroxide [$Ca(OH)_2$], freshly prepared paste.
6.9 Equipment and environment for evaluating color change.
6.9.1 For visual evaluation, use Gray Scale for Color Change (see 12.1). Refer to AATCC EP1 for additional materials and use of gray scale.
6.9.2 For instrumental evaluation, use a spectrophotometer as specified in AATCC EP7.

7. Specimens

7.1 Specimens of any convenient size and shape may be used.

8. Procedure

8.1 Acid Tests.
8.1.1 Prepare hydrochloric solution by adding 100 mL of 35% acid to a volumetric flask half-filled with distilled water and then dilute to the 1-L mark. Spot a test specimen with the solution of hydrochloric acid at 21°C (70°F), and then dry the specimen at room temperature without rinsing.
8.1.2 Spot a test specimen with acetic acid (56%) and dry it at room temperature without rinsing.
8.1.3 An eyedropper may be used to apply the acid spots. If liquid sits on sample surface, use a stirring rod to gently rub in.
8.2 Alkali Tests.
8.2.1 Soak a test specimen in ammonium hydroxide (28% anhydrous ammonia) at 21°C (70°F) for 2 min and dry it at room temperature without rinsing.
8.2.2 Soak a test specimen in a 10% solution of sodium carbonate at 21°C (70°F) for 2 min and dry it at room temperature without rinsing.
8.2.3 Suspend a test specimen over a 7.6-cm (3-in.) evaporating dish containing 10 mL ammonium hydroxide (28% anhydrous ammonia) for 24 h in a 4-L bell jar placed on a glass plate.
8.2.4 Spot a test specimen with a freshly prepared paste of calcium hydroxide (prepared by mixing the hydroxide with a small amount of water), and dry the specimen. Then brush the specimen to remove the dry powder.

9. Evaluation

9.1 After conditioning for 1 h in an atmosphere of 65 ± 5% RH and a temperature of 21 ± 2°C (70 ± 4°F), evaluate the change in color in each test specimen by comparison with the Gray Scale for Color Change (AATCC EP1), or using AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen. Record the numerical rating that corresponds to the appropriate one on the Gray Scale.

10. Report

10.1 Describe or identify the sample tested.
10.2 Report that the sample was tested using AATCC TM6-2021.
10.3 Report the testing conditions:
10.3.1 Acid and alkali solutions used.
10.3.2 Method of application.
10.3.3 Evaluation procedure for color change (AATCC EP1 or AATCC EP7).
10.4 Report the test results:
10.4.1 Color change for each reagent and application method.
10.5 Describe any modification(s) of the published method.

11. Precision and Bias

11.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.
11.2 *Bias*. The colorfastness to acids and alkalis can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

12. Note

12.1 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

13. History

13.1 Revised in 2021 to clarify and align with prescribed AATCC style guidelines.
13.2 Editorially revised 2019. Revised 2016. Reaffirmed 2011. Editorially revised 2010. Reaffirmed 2006. Editorially revised 2004. Editorially revised and reaffirmed 2001. Editorially revised 1995. Editorially revised and reaffirmed 1994. Reaffirmed 1989. Editorially revised and reaffirmed 1986, 1981. Reaffirmed 1978, 1975, 1972. Revised 1957, 1952, 1945.
13.3 Developed in 1925 by AATCC Committee RR1. Maintained by RA99.

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Test Method for Colorfastness to Crocking: Crockmeter

1. Purpose and Scope

1.1 This test method determines the amount of color transferred from the surface of colored textile materials and other dyed materials (i.e., leather) to other surfaces by rubbing.

1.2 Test procedures employing white crocking cloth squares, both dry and wet with water, are given.

2. Principle

2.1 A colored test specimen is rubbed with a white crocking test cloth under controlled conditions.

2.2 Color transferred to the crocking cloth is assessed by a comparison with the Gray Scale for Staining (AATCC EP2), the AATCC 9-Step Chromatic Transference Scale (AATCC EP8), or Instrumental Assessment of Degree of Staining (AATCC EP12), and a grade is assigned.

3. Terminology

3.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials, or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

3.2 **crocking,** n.—a transfer of colorant from the surface of a colored yarn or fabric to another surface or to an adjacent area of the same fabric principally by rubbing.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5. Uses and Limitations

5.1 This method is not recommended for use for carpets or for prints where the singling out of areas may be too small using this method (see 14.1 and 14.2).

5.2 Since washing, drycleaning, shrinkage, ironing, finishing, etc. may affect the degree of color transfer from a material, the test may be made before, after, or before and after any such treatment.

5.3 This method is designed for use with an AATCC crocking cloth as described in 14.5. Alternatively, other adjacent textile substrates may be used as agreed between interested parties.

6. Apparatus and Materials (see 14.3)

6.1 Crockmeter (see 14.4 and Fig. 1).

6.2 AATCC Crocking Cloth, cut in 50 ± 1 mm squares (see 14.5).

6.3 AATCC 9-Step Chromatic Transference Scale (AATCC EP8) (see 14.6).

6.4 Gray Scale for Staining (AATCC EP2) (see 14.6).

6.5 White AATCC Textile Blotting Paper (see 14.6).

6.6 Specimen holder for crockmeter.

6.7 Spiral wire clip for securing crocking cloth to crockmeter finger.

6.8 Crockmeter verification fabric, or in-house fabric with known poor colorfastness to crocking properties, that provides repeatable and predictable crocking results.

6.9 Emery paper to provide friction on the base of the crockmeter (see manufacturer's recommendations).

7. Verification

7.1 Verification checks on the operation of the test and the apparatus should be made routinely and the results kept in a log. The following observations and corrective actions are extremely important to avoid incorrect test results where abnormal crock images can result and influence the rating process (see 14.7).



Fig. 1—Crockmeter.

7.2 Use the crockmeter verification fabric or in-house poor crocking fabric with known predictable and repeatable crocking behavior and conduct three dry and wet crock tests.

7.2.1 A poor circular image with uneven dye pick-up may indicate the crocking finger needs resurfacing (see 14.8).

7.2.2 A double, elongated image may indicate a loose clip (see 14.8).

7.2.3 A stretched and streaked crock image may be due to mounting the crock square diagonally.

7.2.4 Scuff marks to the sides of the specimen indicate the loops of the wire clip are positioned downwards and are not high enough to prevent rubbing the specimen surface.

7.2.5 A streak in the center of the crock cloth image and in the direction of rubbing may indicate damage to the metal base of the device. Contact the manufacturer for recommended solution.

7.2.6 If specimen holder is used, place the holder over the specimen on the tester base. Verify that the holder is not impeding the motion of the finger or coming in contact with it in any way, adjust the holder as necessary to ensure free movement. Without correction, this problem will cause a dark area on one side of the crocking image.

7.2.7 Confirm the wet pick-up techniques (see 10.2).

7.2.8 Replace the abrasive paper on the tester base if it is smooth to the touch in the crocking area compared to the adjacent area, if slippage of the specimen is noticed or if any ridges or bumps are noticed.

7.2.9 In routine testing, observe if multiple streaks are on the crocking image. Position specimen normally with the long dimension oblique to the warp and filling. If the direction of rubbing falls along a twill line or surface pattern, etc., then streaks may occur. If they do occur, slightly adjust the angle for testing.

8. Test Specimens

8.1 Two specimens are used, one each for the dry and the wet tests.

8.1.1 Additional specimens may be used to increase the precision of the aver-

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age (see 13.1).

8.2 Cut the specimens at least 50 × 130 mm and position for testing preferably with the long dimension oblique to warp and filling or wales and courses (see Fig. 2).

8.2.1 Larger or full width lab samples may be used without cutting individual specimens, when multiple tests are needed and when using for production testing. Wherever possible, large samples should be positioned obliquely. If this is not possible, make a note in the report. Avoid seams and other raised areas.

8.2.2 Samples should be prepared in an oblique direction to the warp and weft, neither parallel, right angled or at 45°; they should be slanted.

8.3 Yarns. Knit a piece of fabric at least 50 × 130 mm and position as per 8.2, or wind yarn tightly on a suitable sample card/plate at least 50 × 130 mm, with the yarn running in the long direction (see 14.9).

9. Conditioning

9.1 Prior to testing, precondition and condition the test specimens and the crocking cloth for crock testing as directed in ASTM D1776, Standard Practice for Conditioning and Testing Textiles. Condition each specimen for at least 4 h in an atmosphere of 21 ± 2°C (70 ± 4°F) and 65 ± 5% RH by laying each test specimen or crock square separately on a screen or perforated shelf of conditioning rack.

10. Procedures

10.1 Dry Crocking Test.

10.1.1 Place a test specimen on the base of the crockmeter, flat on the abrasive cloth, with its long dimension in the direction of rubbing.

10.1.2 Place specimen holder over specimen and apply sufficient tension to avoid slippage or wrinkling.

10.1.3 Mount a crocking cloth square, the warp or weft direction parallel to the direction of rubbing, over the end of the finger that projects downward from the weighted arm. Use the spiral wire clip to hold the square in place. Position the clip with loops upward. If the loops point downward, they can drag against the test specimen.

10.1.4 Lower the covered finger onto the test specimen. For manual crockmeters, beginning with the finger positioned at the front end, turn the handle 10 complete turns at the rate of one turn per second to slide the covered finger back 10 times and forth 10 times. For motorized crockmeters, set and run for 10 complete turns. Refer to individual specifications for any other required number of turns.

10.1.5 Remove the crocking cloth square, condition (see 9.1) and evaluate as directed in Section 11. In the case of napped, brushed or sanded material, when loose fiber might interfere with the rating, remove the extraneous fibrous material by pressing lightly on the crock circle with the sticky side of cellophane tape before evaluating.

10.2 Wet Crocking Test.

10.2.1 Weigh a conditioned dry crocking cloth square. Calculate the desired wet weight (65 ± 5% pickup) by multiplying the original dry weight by 1.65.

Apply water evenly over the crocking cloth square and weigh to verify 65 ± 5% pickup. If necessary, discard the square with insufficient or excessive weight and use a new crocking cloth square. Repeat this process each day (see 14.10).

An example of how to achieve this pickup is: Using a syringe tube, graduated pipette or automatic pipetter, draw up water in mL to 0.65 times weight of crocking square. Example: If the crocking square weight equals 0.24 g, the mL used would be 0.24 × 0.65 = 0.16 mL. Lay the crocking cloth square on white plastic mesh over a dish. If needed, adjust the amount of water used to wet the square and using a new crocking cloth square, repeat steps. When 65 ± 5% wet pickup is achieved, record the amount of water used. Draw up the recorded amount of water into the syringe tube, graduated pipette or automatic pipette for each wet crocking performed during the current day. Other methods are available and can be used.

10.2.2 Avoid evaporative reduction of the moisture content on the crocking cloth square below the specified level of 65 ± 5% before the actual crock test is run.

10.2.3 Continue as directed in 10.1.

10.2.4 Air dry the crocking cloth square, then condition (see 9.1) before evaluating. In the case of napped, brushed or sanded material when loose fiber might interfere with the rating, remove the extraneous fibrous material by pressing lightly on the crock circle with the sticky side of cellophane tape before evaluating.

11. Evaluation (see 14.11)

11.1 Back the crocked square with three layers of clean crocking cloth for evaluating.

11.2 Rate the amount of color transferred from the specimen to the crock square under examination by means of the AATCC Gray Scale for Staining (AATCC EP2), AATCC 9-Step Chromatic Transference Scale (AATCC EP8) or Instrumental Assessment of Degree of Staining (AATCC EP12) and record the numerical rating (see 14.12 and 14.13).

11.3 Average the individual results to the nearest 0.1 grade when multiple specimens are tested or when a panel of evaluators rate color transfer.

12. Report

12.1 State the grades of dry and wet crocking test, unless otherwise requested between parties.

12.2 Report the grade determined in 11.3.

12.3 State whether AATCC EP2, AATCC EP8 or AATCC EP12 was used for evaluating staining (see 14.12).

12.4 If any pretreatment or aftertreatment was given to any specimens (see 5.2), indicate method of treatment.

13. Precision and Bias (see 14.14)

13.1 *Precision*. An interlaboratory test was conducted in 1986 to establish the precision of the test method. Testing was conducted under the normal atmospheric conditions of each laboratory and not necessarily under ASTM D1776 standard conditions. Two operators at each of 12 laboratories evaluated 5 fabrics in 3 replications by both dry and wet test method.

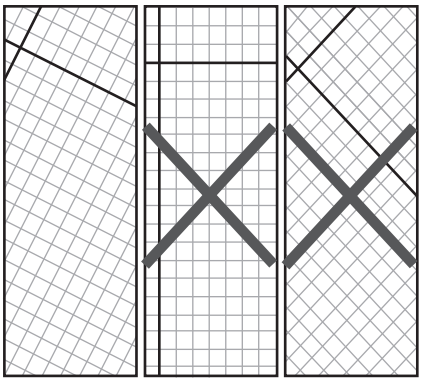


Fig. 2—Oblique alignment of specimen (left).

Table I—Components of Variance

	Dry		Wet	
	Chromatic	Gray	Chromatic	Gray
Single Operator/Rater	0.20	0.20	0.24	0.25
Within Laboratory	0.20	0.19	0.31	0.34
Between Laboratory	0.10	0.17	0.38	0.54

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Table II—Critical Differences

For the components of variance in Table I, two averages of observed values should be considered significantly different at the 95% probability level if the difference equals or exceeds the following critical differences.

	No. of Observations	Dry		Wet	
		Chromatic	Gray	Chromatic	Gray
Single Operator/Rater	1	0.55	0.54	0.68	0.70
	3	0.32	0.31	0.39	0.40
	5	0.24	0.24	0.30	0.31
Within Laboratory	1	0.77	0.75	1.08	1.17
	3	0.60	0.61	0.93	1.02
	5	0.60	0.57	0.90	1.00
Between Laboratory	1	0.82	0.89	1.53	1.90
	3	0.69	0.77	1.43	1.81
	5	0.66	0.74	1.41	1.79

The critical differences were calculated using $t = 1.96$ which is based on infinite degrees of freedom.

Table III—Sample Crock Test Results with Chromatic Scale

	Dry	Wet
Lab A	4.5	3.5
Lab B	4.0	1.5
Difference	0.5	2.0

Each of 3 raters independently rated the stained crocking cloth squares using both the Gray Scale for Staining and the AATCC 9-Step Chromatic Transference Scale. The original data is on file at the AATCC Technical Center.

13.1.1 The components of variance as standard deviations of the Gray Scale for Staining or AATCC 9-Step Chromatic Transference Scale rating units are given in Table I.

13.1.2 Critical differences are given in Table II.

13.1.3 Example for determining between laboratory differences using one observer and the chromatic scale are given in Table III.

Interpretation: For the dry crock test, since the difference between labs is less than the critical differences in Table II (0.82), the difference in results is not significant. For the wet crock test, since the difference between labs is greater than the critical difference (1.53), the difference in results is significant.

13.2 *Bias*. The true value of colorfastness to crocking can only be defined in terms of a test method. Within this limitation, this test method has no known bias.

14. Notes

14.1 For carpets, AATCC TM165, Test Method for Colorfastness to Crocking: Carpets—Crockmeter, under the jurisdiction of Committee RA57, Floor Covering Test Methods, should be used.

14.2 For prints where the singling out of areas too small to test with the standard crockmeter is necessary (see AATCC TM116, Test Method for Colorfastness to Cracking: Rotary Vertical Crockmeter). Specimens tested by both test methods may show dissimilar results. There is no known correlation between the two methods.

14.3 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

14.4 The crockmeter provides a reciprocating rubbing motion simulating the action of a human finger and forearm.

The crockmeter is so designed that the 16 ± 0.3 mm diameter finger moves back and forth, with each complete turn of the crank, in a straight line along a 104 ± 3 mm track on the specimen, with a downward force of 9 ± 0.9 N (2 ± 0.2 lb).

14.5 AATCC Crocking Cloth should meet the following specifications:

Fiber	100% 10.3-16.8 mm combed cotton staple with no optical brightener present
Yarn	15 tex (40/1 cotton count), 5.9 turns/cm "z"
Thread count	for greige fabric 32 ± 5 warp ends/cm: 33 ± 5 filling picks/cm
Weave	1/1 plain
Finished fabric	desized and bleached with no optical brightener or finishing material present
pH	7 ± 1
Mass/sq meter	100 ± 3 g finished
Whiteness	W = 78 ± 3 (AATCC TM110)

14.5.1 CAUTION: ISO crock test cloth results may not be equivalent to AATCC crocking cloth results/values based on crocking cloth study.

14.6 The AATCC 9-Step Chromatic Transference Scale, Gray Scale for Staining and the White AATCC Textile Blotting Paper are available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

14.7 For a discussion of crock testing, see the article by J. Patton, "Crock Test Problems can be Prevented," *Textile Chemist and Colorist*, Vol. 21, No. 3, March 1989, p13; and "Testing for Crocking: Some Problems and Pitfalls" by Allan E. Gore, *Textile Chemists and Colorists*, Vol. 21, No. 3, March 1989, p17.

14.8 Accidental damage to the rubbing finger, spiral clip or abrasive paper should be repaired as follows: neatly renew the abrasive paper; bend the clip further open or shut around a rod slightly smaller in diameter than the crock peg; resurface the finger by movement on an extra piece of fine emery cloth in a manner simulating regular use.

14.9 For more convenient crock testing of multiple strands of yarn or thread a dowel attachment is useful. This attachment was developed to avoid the tendency of the standard finger to dig into and push aside the yarns, or slide off them and possibly give erroneous results. This attachment is 25 mm in diameter by 51 mm long. Positioned on its side and held in place by the standard finger, it provides a wider test area, and holds the white test square by two spring loaded clips. For additional information on this development see the article by C. R. Trommer, "Modification of the AATCC Crockmeter for Yarn Testing," *American Dyestuff Reporter*, Vol. 45, No. 12, June 4, 1956, p357; also see articles by S. Korpanty and C. R. Trommer, "An Improved Crockmeter for Yarn Testing," *American Dyestuff Reporter*, Vol. 48, No. 6, March 23, 1959, p40.

14.10 Experienced operators do not have to repeat this weighing procedure during a test session once the technique is established.

14.11 CAUTION: It has been reported that the results for staining obtained by this method on fabrics dyed to dark shades (navy, black, etc.) that contain a combination of polyester and spandex, or their blends, may not show the full staining propensity of such fabrics in consumer use. It is, therefore, recommended that the staining results obtained by this test not be used for the acceptance testing of such fabrics.

14.12 It has been noted that different grades may result depending upon whether the Gray Scale for Staining or AATCC 9-Step Chromatic Transference Scale is used for the evaluation. It is, therefore, important to report which scale was used.

For very critical evaluations and in cases of arbitration, grades must be based on the Gray Scale for Staining.

14.13 An automated electronic grading system may be used as long as the system has been demonstrated to provide results that are equal to and provide equal or better repeatability and reproducibility than an experienced grader performing visual evaluation.

14.14 The precision of this test method is dependent on the combined variability of the material being tested, the test method itself, and the evaluation procedure utilized.

14.14.1 The precision statement in Section 13 was developed from results obtained by visual evaluation.

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14.14.2 It is expected that the use of instrumental evaluation procedures will result in greater precision than that obtained from visual evaluations.

15. History

15.1 Reaffirmed and editorially revised to add History section in 2022.

15.2 Editorially revised 2019; revised 2016, 2013; editorially revised 2011, 2010, 2009, 2008 (with title change); revised 2007, 2005, 2004; editorially revised 2002; editorially revised and reaffirmed 2001; revised 1996; editorially revised and reaffirmed 1995; reaffirmed 1989; revised 1988; editorially revised 1986; revised 1985; editorially revised and reaffirmed 1981, 1977, 1974; revised 1972, 1969; editorially revised and reaffirmed 1968; revised 1961,1957,1952; reaffirmed 1945; revised 1937.

15.3 Developed in 1936 by AATCC Committee RA38; Partly equivalent to ISO 105-X12.

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Test Method for Colorfastness to Perspiration

1. Purpose and Scope

1.1 This test method is used to determine the fastness of colored textiles to the effects of acid perspiration. It is applicable to dyed, printed or otherwise colored textile fibers, yarns and fabrics of all kinds and to the testing of dyestuffs as applied to textiles.

1.2 Work by Committee RA52 showed this test will correlate with limited field studies. Prior to this there were acid and alkaline tests; however, as a result of these studies the alkaline test was eliminated (see 13.1).

2. Principle

2.1 A specimen of colored textile in contact with other fiber materials (for color transfer) is wet out in simulated acid perspiration solution, subjected to a fixed mechanical pressure and allowed to dry slowly at a slightly elevated temperature. After conditioning, the specimen is evaluated for color change and the other fiber materials are evaluated for color transfer.

3. Terminology

3.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

3.2 **perspiration,** n.—a saline fluid secreted by the sweat glands.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Follow good laboratory practices. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 Observe wringer safety. Normal safe guards on pad should not be removed. Ensure adequate guard at the nip point. A foot operated kick off is recommended for a motorized wringer.

5. Apparatus, Materials and Reagents (see 13.2)

5.1 Perspiration tester (with acrylic plates) (see Figs. 1 and 2).

5.2 Drying oven—convection.

5.3 Balance with a weighing accuracy of ± 0.001 g.

5.4 Cold cut Multifiber test fabric (8 mm [0.33 in.] bands) containing acetate, cotton, nylon, silk, viscose rayon and wool shall be used for specimens containing silk. Multifiber test fabric (8 mm [0.33 in.] bands) containing acetate, cotton, nylon, polyester, acrylic and wool shall be used with specimens with no silk present (see 13.3).

5.5 pH meter accurate to ± 0.01.

5.6 AATCC 9-Step Chromatic Transference Scale (AATCC Evaluation Procedure 8) or Gray Scale for Staining (AATCC Evaluation Procedure 2) (see 13.4).

5.7 Gray Scale for Color Change (AATCC Evaluation Procedure 1 or 7) (see 13.4).

5.8 Wringer.



Fig. 1—Horizontal perspiration tester.

5.9 White AATCC Textile Blotting Paper (see 13.4).

5.10 Acid perspiration solution.

5.11 Petri dish with a depth greater than 1.5 cm and capable of containing a 6 × 6 ± 0.2 cm test specimen

5.12 Un-dyed adjacent fabric

6. Preparation of Reagent

6.1 Prepare the acid perspiration solution by filling a 1 L volumetric flask half full of distilled water. Add the following chemicals and mix to be sure that all chemicals are thoroughly dissolved:

10 ± 0.01 g sodium chloride (NaCl)

1 ± 0.01 g lactic acid, USP 85%

1 ± 0.01 g sodium phosphate, dibasic, anhydrous (Na_2HPO_4)

0.25 ± 0.001 g ℓ-histidine monohydrochloride ($C_6H_9N_3O_2{\cdot}HCl{\cdot}H_2O$)

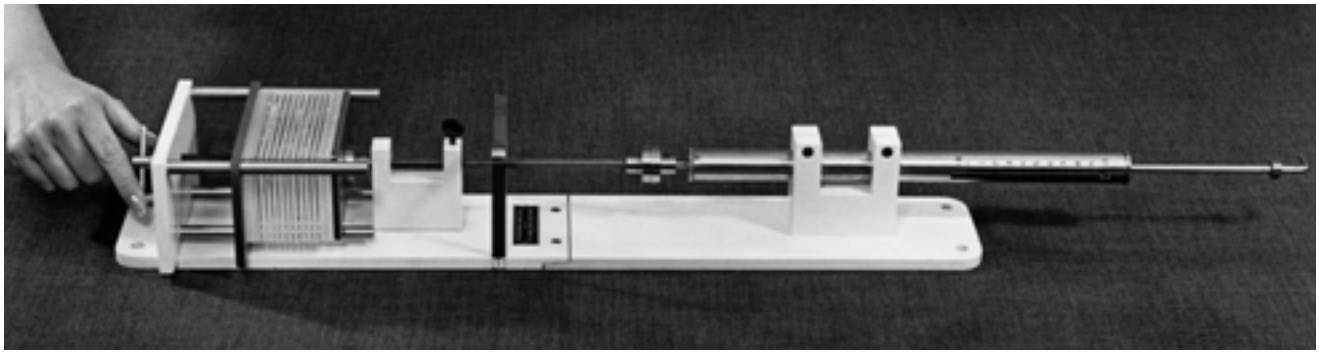


Fig. 2—Vertical perspiration tester.

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Fill the volumetric flask with distilled water to the 1 L mark.

6.2 Test the pH of the solution with a pH meter. If it is not 4.3 ± 0.2, discard it and prepare a new one, making sure all ingredients are weighed accurately. The use of pH test paper is not recommended for this purpose because of its lack of accuracy.

6.3 Do not use perspiration solution that is more than three days old (see 13.5).

7. Verification

7.1 Verification checks on the operation of the test and apparatus should be made routinely and the results kept in a log. The following observations and corrective actions are extremely important to avoid incorrect test results.

7.2 Use an in-house perspiration fabric with a mid-range visual grade on the most heavily stained stripe of the multifiber cloth as a calibration specimen and conduct a perspiration test using three specimens. Verification checks should be performed periodically as well as each time a new lot of multifiber or undyed adjacent fabric is used.

7.2.1 Non-uniform color transfer may be due to improper wet-out procedures or may be a result of uneven pressure on the specimens due to warped plates in the tester. Check the wet-out procedures to be sure that the balance is accurate and that the procedure is being carefully followed. Check all plates to be sure they are in good condition and not warped.

8. Test Specimens

8.1 Number and size of specimens.

8.1.1 For fabric testing, one specimen 6 × 6 ± 0.2 cm is needed. Attach a piece of multifiber adjacent fabric measuring 5 × 5 ± 0.2 cm to the face of the specimen by sewing a single seam stitch along one edge of the fabric.

8.1.2 For yarn or loose fiber testing, weigh a 5 × 5 ± 0.2 cm piece of multifiber fabric and a 6 × 6 ± 0.2 cm piece of the un-dyed adjacent fabric together. Then take a mass of the yarn or loose fiber approximately equal to one half of the combined mass of the adjacent fabrics. Place it between the 5 × 5 ± 0.2 cm piece of multifiber fabric and a 6 × 6 ± 0.2 cm piece of the un-dyed adjacent fabric, and sew along all four sides.

8.1.3 Do not use multifiber test fabric that has fused, sealed, or pre-sewn edges because it might have thickness variations at the edges which would cause uneven compression during testing.

9. Procedure

9.1 Weigh each test specimen (as prepared in 8.1) to the nearest 0.1g. Place each test specimen) in a petri dish. Add freshly prepared perspiration solution to a depth of 1.5 cm in the petri dish. Soak the test specimen in the solution for 30 ± 2 min with occasional agitation and squeezing to ensure complete wetting. For fabrics hard to wet out, alternately wet the specimen and pass it through the wringer until it is completely penetrated by the solution.

9.2 After 30 ± 2 min, pass each test specimen assembly through the wringer with the multifiber stripes perpendicular to the length of the wringer rolls (all stripes go through the wringer at the same time). Weigh each test specimen to be sure it weighs 2.25 ± 0.05 times its original weight. Because certain fabrics may not be able to retain this amount of solution when passing through a wringer, such fabrics may be tested after blotting to the required wet pickup with White AATCC Textile Blotting Paper (see 13.4). To obtain consistent results all specimens of a given construction in a test series should have identical pickup, as the degree of staining increases with the amount of retained solution.

9.3 Place each test specimen assembly on an acrylic plate with the multifiber stripes running perpendicular to the long dimension of the plate (see Fig. 3).

9.4 Depending upon equipment available, use the following alternates:

9.4.1 Horizontal Perspiration Tester (see Fig. 1): Place the plates in the perspiration tester with the specimen assemblies evenly distributed between the 21 plates. Place all 21 plates into the unit regardless of the number of specimens. After placing the final plate in position (on top) set the dual plates with compensating springs in position, place the 3.63 kg (8.0 lb) weight on top making a total of 4.54 kg (10.0 lb) under the pressure plate, and lock the pressure plate in position by turning the thumb screws. Remove the weight and place the unit lying on its side in the oven so that the sides of the perspiration tester are parallel to the oven walls (see Fig. 4).

9.4.2 Vertical Perspiration Tester (See Fig. 2): Assemble the plates in the perspiration tester with the specimens evenly distributed between the 21 plates. Place all 21 plates into the unit regardless of the number of specimens. The plates are held in a vertical position between an indicating scale with a fixed metal plate at one end and an adjustable metal plate at the other end. Use the adjusting screw to exert a 4.54 kg (10.0 lb) force against the plates. Lock the specimen unit containing the test specimens with a set screw. Remove the pressure gauge unit from the specimen unit and place the specimen unit in the oven such that the side of the perspiration tester is parallel to the oven walls. Another specimen unit may be added to the pressure gauge unit and the loading procedure repeated.

9.5 Heat the loaded specimen unit in an oven at 38 ± 1°C (100 ± 2°F) for 6 h ± 5 min. Check the oven temperature periodically to be sure it remains at the specified temperature throughout the test.

9.6 Remove the tester from the oven and for each test specimen assembly, separate the multifiber fabric and, if used, the undyed adjacent fabric from the test fabric. Place the multifiber fabric and test fabric specimens separately on a wire screen in a conditioned atmosphere (21 ± 2°C [70 ± 4°F]) and 65 ± 5% relative humidity overnight.

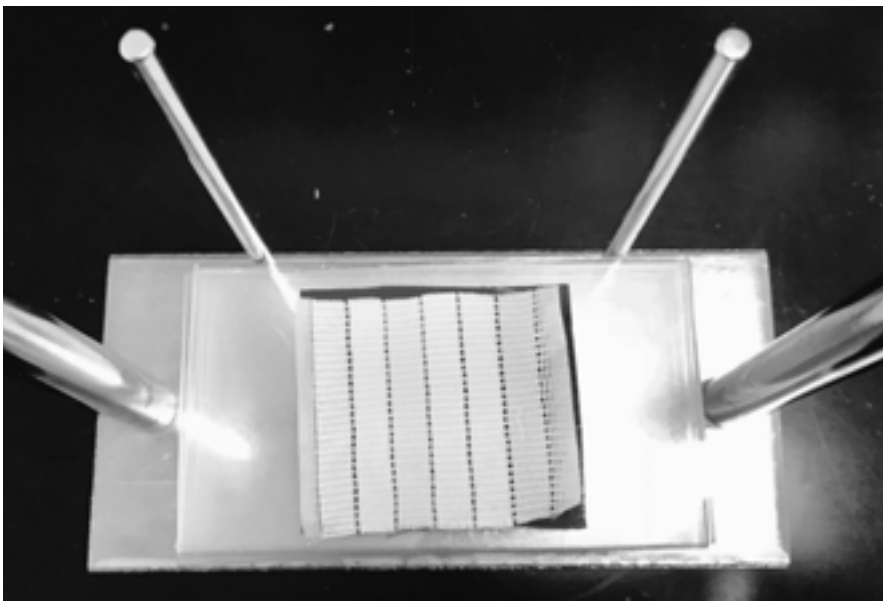


Fig. 3—Specimen in holder.

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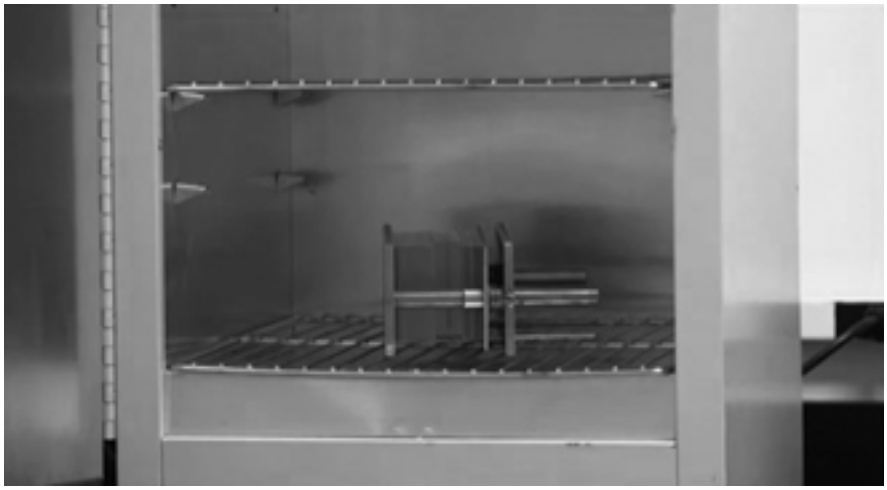


Fig. 4—Horizontal Perspiration tester placement in oven.

10. Evaluation (see 13.7)

10.1 General—Unsatisfactory perspiration fastness may be due to bleeding or migration of color or it may be due to change in color of the dyed material. It should be noted that objectionable change in color may be encountered with no apparent bleeding. On the other hand, there may be bleeding with no apparent change in color, or there may be both bleeding and change in color.

10.2 Rate the effect on the color of the test specimens by comparison with the Gray Scale for Color Change (AATCC Evaluation Procedure 1), or using AATCC Evaluation Procedure 7, Instrumental Assessment of the Change in Color of a Test Specimen, and record the numerical rating that corresponds to the appropriate one on the Gray Scale (see 13.4).

10.3 Rate the staining on each fiber type of the multifiber, and the undyed original fabric if used, by comparison with the Gray Scale for Staining (AATCC Evaluation Procedure 2), the AATCC 9-Step Chromatic Transference Scale (AATCC Evaluation Procedure 8) or Instrumental Assessment of Degree of Staining (AATCC Evaluation Procedure 12), and record the numerical rating that corresponds to the appropriate one on either of them. (see 13.4.)

11. Report

11.1 Report the color change grade and the staining grades for each fiber type in the multifiber test sample and state which scale (AATCC Evaluation Procedure 2, 8, or 12) was used in the staining evaluation (see 13.4).

12. Precision and Bias

12.1 *Precision*. Proficiency data generated by 188 different labs during five different time periods (June 2012, December 2012, June 2013, December 2013 and June 2014) was used to define the precision of stain rating and color change rating values. Each lab used the same fabric materials to obtain the stain rating and color change rating values. Each lab performed three replicate tests for each test material and used three different raters to assign ratings for staining and color change. Each lab also used multi-fiber fabric strips containing acetate, acrylic, cotton, nylon, polyester and wool fibers to assess the staining of individual fibers.

12.1.1 Tables 1-6 give the within lab and between lab precision values for the stain ratings of acetate, acrylic, cotton, nylon, polyester and wool fibers.

12.1.2 Table 7 gives the precision values for color change rating. As mentioned above, the multi-period data generated by 188 different labs was used to compute the precision values.

12.1.3 In addition to providing the precision values, the analysis of the multi-period data revealed the following statistical facts:

- The stain ratings assigned by **different labs** for **acetate** fiber **differed** significantly at 95% confidence level.
- The mean stain ratings of **acetate fiber** corresponding to five different **time periods differed** significantly at 95% confidence level.
- The stain ratings assigned by **different labs** for **acrylic** fiber **differed** significantly at 95% confidence level.
- The mean stain ratings of **acrylic fiber** corresponding to five different **time periods differed** significantly at 95% confidence level.
- The stain ratings assigned by **different labs** for **cotton** fiber **did not differ** significantly at 95% confidence level.
- The mean stain ratings of **cotton** fiber corresponding to five **different time** periods **differed** significantly at 95% confidence level.
- The stain ratings assigned by **different labs** for **nylon** fiber **did not** differ significantly at 95% confidence level.
- The mean stain ratings of **nylon** fiber corresponding to five different **time**

Table 1—Precision Table for Stain Rating of Acetate

No. of Specimens	Within Lab Precision	Between Lab Precision
1	0.01173	0.01519
2	0.0083	0.0108
3	0.0068	0.0088
4	0.0059	0.0076
5	0.0052	0.0068
6	0.0048	0.0062

Table 2—Precision Table for Stain Rating of Acrylic

No. of Specimens	Within Lab Precision	Between Lab Precision
1	0.02773	0.03646
2	0.0196	0.0258
3	0.0160	0.0210
4	0.0139	0.0182
5	0.0124	0.0163
6	0.0113	0.0149

Table 3—Precision Table for Stain Rating of Cotton

No. of Specimens	Within Lab Precision	Between Lab Precision
1	0.09994	0.1315
2	0.0708	0.0931
3	0.0577	0.0759
4	0.0500	0.0658
5	0.0447	0.0588
6	0.0408	0.0537

Table 4—Precision Table for Stain Rating of Nylon

No. of Specimens	Within Lab Precision	Between Lab Precision
1	0.040426	0.05312
2	0.0286	0.0376
3	0.0233	0.0307
4	0.0202	0.0266
5	0.0181	0.0238
6	0.0165	0.0217

Table 5—Precision Table for Stain Rating of Polyester

No. of Specimens	Within Lab Precision	Between Lab Precision
1	0.019723	0.026041
2	0.0140	0.0184
3	0.0114	0.0150
4	0.0099	0.0130
5	0.0088	0.0116
6	0.0081	0.0106

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Table 6—Precision Table for Stain Rating of Wool

No. of Specimens	Within Lab Precision	Between Lab Precision
1	0.01951	0.02552
2	0.0138	0.0181
3	0.0113	0.0147
4	0.0098	0.0128
5	0.0087	0.0114
6	0.0080	0.0104

Table 7—Precision Table for Color Change Ratings

No. of Specimens	Within Lab Precision	Between Lab Precision
1	0.02182	0.02552
2	0.0155	0.0181
3	0.0126	0.0147
4	0.0109	0.0128
5	0.0098	0.0114
6	0.0089	0.0104

periods differed significantly at 95% confidence level.

- The stain ratings assigned by **different labs** for **polyester** fiber **differed** significantly at 95% confidence level.
- The mean stain ratings of **polyester** fiber corresponding to five different **time periods differed** significantly at 95% confidence level.
- The stain ratings assigned by **different labs** for **wool** fiber **differed** significantly at 95% confidence level.
- The mean stain ratings of **wool** fiber corresponding to five different **time periods differed** significantly at 95% confidence level.
- The mean **color change ratings** of **different labs differed** significantly at 95% confidence level.
- The mean **color change ratings** of five different **time periods differed** significantly at 95% confidence level.

12.2 *Bias*. The colorfastness to perspiration can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

13. Notes

13.1 Background information on the committee's work and decision to eliminate the alkaline test was published in two articles in *Textile Chemist and Colorist*: "Colorfastness to Perspiration and Chemicals" (October 1974) and "Evaluating Colorfastness to Perspiration: Laboratory Test vs. Wear Test" (November 1974). Although the alkaline test has been eliminated from this method, there may be certain instances in foreign trade or special end-uses that require the alkaline test. In these instances the alkaline test should be run as in AATCC Method 15-1973. For convenient reference the composition of the alkaline solution is as follows: Alkaline Solution–10 g sodium chloride; 4 g ammonium carbonate, USP; 1 g sodium phosphate, dibasic, anhydrous (Na_2HPO_4); 0.25 g ℓ-histidine monohydrochloride. Make up to one liter with distilled water. This solution should give a pH of 8.0.

13.2 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at http://www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

13.3 The six fiber test fabrics without fused edges should be used in this method.

13.4 The AATCC 9-Step Chromatic Transference Scale, Gray Scale for Staining, Gray Scale for Color Change and White AATCC Textile Blotting Paper are available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919. 549.8933; e-mail: orders@aatcc.org; web site: www.aatcc.org.

13.5 Committee RR52 established that fungi begin to grow in the acid perspiration solution and that the pH gradually rises after three days of storage under ambient room temperatures, even when kept in a stoppered solution bottle.

13.6 For very critical visual evaluations and in the case of arbitration, ratings must be based on the Gray Scale for Staining as opposed to the 9 step chromatic transference scale.

13.7 CAUTION: It has been reported that the results for staining obtained by this method on fabrics dyed to dark shades (navy, black, etc.) that contain a combination of polyester and spandex, or their blends, may not show the full staining propensity of such fabrics in consumer use. It is, therefore, recommended that the staining results obtained by this test not be used for the acceptance testing of such fabrics.

14. History

14.1 Editorially revised in 2022 to fix a typo and update conditioning tolerances. Revised in 2021 for clarity and to add History section per the AATCC style guide. Revised 2013; editorially revised 2010; revised in 2009; editorially revised 2008, reaffirmed 2007; editorially revised 2005, 2004; editorially revised and reaffirmed 2002; revised 1997; editorially revised 1995; editorially revised and reaffirmed 1994; reaffirmed 1989; editorially revised 1986; reaffirmed 1985; editorially revised 1983, 1981; reaffirmed 1979; revised 1976, 1975; editorially revised 1974; revised 1973, 1972; reaffirmed 1967; editorially revised 1967; revised 1962; editorially revised 1961; revised 1960, 1957, 1952.

14.2 Developed in 1949 by AATCC Committee RR52; jurisdiction transferred to AATCC Committee RA23 in 2006; related to ISO 105-E04.

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Test Method for Colorfastness to Light: Outdoor

1. Purpose and Scope

1.1 This test method provides the general principles and procedures for determining the colorfastness to light of textiles outdoors under glass. The test options described are applicable to textile materials of all kinds and for colorants, finishes and treatments applied to textiles (see 15.1).

2. Principle

2.1 Specimens of the textile material to be tested and the agreed upon comparison standard(s) are exposed simultaneously to daylight behind window glass under specified conditions. The colorfastness to light of the specimen is evaluated by comparison of the color change of the exposed portion to the masked control portion of the test specimen or unexposed original material using the Gray Scale for Color Change (AATCC EP1), or by Instrumental Color Measurement (AATCC EP6). Lightfastness classification is accomplished by evaluation versus a simultaneously exposed series of AATCC Blue Wool Lightfastness Standards.

3. References

3.1 AATCC EP1, Evaluation Procedure for Gray Scale for Color Change (see 15.4).

3.2 AATCC EP6, Evaluation Procedure for Instrumental Color Measurement (see 15.4).

3.3 AATCC EP7, Evaluation Procedure for Instrumental Assessment of the Change in Color of a Test Specimen (see 15.4).

3.4 ASTM G24, Standard Practice for Conducting Exposures to Daylight Filtered through Glass (see 15.10).

3.5 ASTM G155, Operating Xenon Arc Lamp Apparatus for Exposure of Materials

4. Terminology

4.1 **AATCC Blue Wool Lightfastness Standard,** n.—one of a group of dyed wool fabrics distributed by AATCC for use in determining the amount of light exposure of specimens during lightfastness testing (see 15.2).

4.2 **AATCC Fading Unit (AFU),** n.—a specific amount of exposure made under the conditions specified in various test methods where one AFU is one-twentieth (1/20) of the light-on exposure required to produce a color change equal to Step 4 on the Gray Scale for Color Change or 1.7 ± 0.3 CIELAB units of color difference on AATCC Blue Wool Lightfastness Standard L4.

4.3 **broad bandpass radiometer,** n.—a relative term applied to radiometers that have a bandpass width of more than 20 nm and less than 70 nm at 50% of maximum transmittance.

4.4 **color change,** n.—*as used in colorfastness testing*, a change in color of any kind whether a change in lightness, hue or chroma or any combination of these, discernible by comparing the test specimen with a corresponding untested specimen.

4.5 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials, or both as a result of exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

4.6 **colorfastness to light,** n.—the resistance of a material to a change in its color characteristics as a result of exposure of the material to sunlight or an artificial light source.

4.7 **infrared radiation,** n.—radiant energy for which the wavelengths of the monochromatic components are greater than those for visible radiation and less than about 1 mm.

NOTE: The limits of the spectral range of infrared radiation are not well defined and may vary according to the user. Committee E-2.1.2 of the CIE distinguishes in the spectral range between 780 nm and 1 mm:

IR-A	780-1400 nm
IR-B	1.4-3.0 μm
IR-C	3 μm to 1 mm

4.8 **irradiance,** n.—radiant power per unit area incident on a receiver, typically reported in watts per square meter (W/m^2).

4.9 **"L" designation,** n.—the sequence number given each AATCC Blue Wool Lightfastness Standard according to the number of AATCC Fading Units required to produce a color change equal to Step 4 on the Gray Scale for Color Change.

NOTE: See Table I for the numerical relationship between "L" designations of the standards and their colorfastness to light in AFUs. The colorfastness to light of a fabric specimen can be determined by comparing its color change after light exposure with that of the most similar AATCC Blue Wool Lightfastness Standard as shown in Table II.

Table I—AATCC Fading Unit and Light Exposure Equivalents for AATCC Blue Wool Lightfastness Standards (see Section 14)[a]

AATCC Blue Wool Lightfastness Standard	AATCC Fading Units	Xenon Only kJ/(m^2nm) @ 420 nm	Xenon Only kJ/(m^2nm) 300-400 nm
L2	5	21	864
L3	10	43	1728
L4	20	85[b]	3456
L5	40	170	6912
L6	80	340[b]	13824
L7	160	680	27648
L8	320	1360	55296
L9	640	2720	110592

[a] For color change of Step 4 on the Gray Scale for Color Change.

4.10 **langley,** n.—a unit of total solar radiation equivalent to one gram calorie per square centimeter of irradiated surface.

NOTE: The internationally recommended units are: Joule (J) for quantity of radiant energy, watt (W) for quantity of radiant power, and meter squared (m^2) for area. The following factors are to be used: 1 langley = 1 cal/cm^2; 1 cal/cm^2 = 4.184 J/cm^2 or 41840 J/m^2.

4.11 **lightfastness,** n.—the property of a material, usually an assigned number, depicting a ranked change in its color characteristics as a result of exposure of the material to sunlight or an artificial light source.

4.12 **narrow bandpass radiometer,** n.—a relative term applied to radiometers that have a bandpass width of 20 nm or less at 50% of maximum transmittance and can be used to measure irradiance at wavelengths such as 340 or 420, ± 0.5 nm.

4.13 **photochromism,** n.—a qualitative designation for a reversible change in color of any kind (whether a change in hue or chroma) which is immediately noticeable upon termination of light exposures when the exposed area of a specimen is compared to the unexposed area.

NOTE: The reversal of the color change or instability of the hue or chroma upon standing in the dark distinguishes photochromism from fading.

4.14 **pyranometer,** n.—a radiometer used to measure the global solar irradiance or, if inclined, hemispherical solar irradiance.

4.15 **radiant power,** n.—energy per unit time emitted, transferred or received

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Table II—Classification by AATCC Blue Wool Lightfastness Standards[a]
Test Specimen Color Change

Less Than Standard	Equal To But Not Greater Than Standard	More Than Standard	Lightfastness Class	AATCC Fading Units (AFU)
—	—	L2	L1	
—	L2	L3	L2	5
L2	—	L3	L2-3	
—	L3	L4	L3	10
L3	—	L4	L3-4	
—	L4	L5	L4	20
L4	—	L5	L4-5	
—	L5	L6	L5	40
L5	—	L6	L5-6	
—	L6	L7	L6	80
L6	—	L7	L6-7	
—	L7	L8	L7	160
L7	—	L8	L7-8	
—	L8	L9	L8	320
L8	—	L9	L8-9	
—	L9	—	L9	640

[a] The following are examples for using Table II to assign lightfastness classifications:

The test specimen is exposed simultaneously with standards L4, L5, and L6. After exposure and conditioning, the color change exhibited by the test specimen is less than that exhibited by the standards L4 and L5 but greater than that exhibited by the standard L6. The test specimen would be assigned a Lightfastness Classification of L5-6, or use the following example.

The test specimen is examined after each exposure increment until it exhibits a color change equal to Step 4 on the Gray Scale for Color Change. If this occurs after 40 AFU and before 80 AFU exposure, the test specimen would be assigned a Lightfastness Classification of L5-6.

as radiation.

4.16 **radiant exposure,** n.- time integral of irradiance, typically reported in joules per square meter, J/m^2

4.17 **radiometer,** n.—an instrument used to measure radiant energy.

4.18 **total irradiance,** n.—radiant power integrated over all wavelengths at a point in time expressed in watts per square meter (W/m^2).

4.19 **ultraviolet radiation,** n.—radiant energy for which the wavelengths of the monochromatic components are smaller than those for visible radiation and more than about 100 nm.

NOTE: The limits of the spectral range of ultraviolet radiation are not well defined and may vary according to the user. Committee E.2.1.2 of the CIE distinguishes in the spectral range between 400 and 100 nm:

UV-A 315-400 nm
UV-B 280-315 nm
UV-C 100-280 nm

4.20 **visible radiation,** n.—any radiant energy capable of causing a visual sensation.

NOTE: The limits of the spectral range of visible radiation are not well defined and may vary according to the user. The lower limit is generally taken between 380 and 400 nm and the upper limit between 760 and 780 nm (1 nanometer, 1 nm = 10^{-9} m).

4.21 **wide bandpass radiometer**, n. – a relative term applied to radiometers that have a bandpass width of more than 70nm at 50% of maximum transmittance.

Note: Wide bandpass radiometers can be used to measure irradiance at wavelengths such as 300-800nm.

4.22 For definitions of other terms relative to lightfastness used in this test method, refer to AATCC M11.

5. Safety Precautions

5.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

5.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

5.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).
updated wording.

5.4 Daylight exposure of the skin and eyes for prolonged periods may be hazardous and therefore caution should be employed to protect these areas. Do not look directly at the sun under any circumstances.

5.5 Good laboratory practices should be followed.

6. Uses and Limitation

6.1 Not all materials are affected equally by the same light source and environment. Results obtained by the use of any one test option may not be representative of those of any other test option or any end-use application unless a mathematical correlation for a given material and/or a given application has been established. Enclosed Carbon-Arc, Xenon-Arc and natural Daylight have been extensively used in the trade for acceptance testing of textile materials. There may be a distinct difference in spectral irradiance, air temperature and humidity sensor locations, and test chamber size between test equipment supplied by different manufacturers that can result in differences in reported test results. Consequently, data obtained from equipment supplied by the different manufacturers, different test chamber size, or different light source and filter combinations cannot be used interchangeably, unless a mathematical correlation has been established. No correlations among differently constructed test apparatus are known to AATCC Committee RA50.

6.2 Since the spectral distribution of Xenon-Arc, Alternate Light and Dark, equipped with the specified window glass filter glass (according to ASTM G155 Table 2) is very close to that of average or typical daylight behind window glass, it is expected that similar photochemical degradation reactions in similar ratios appear and therefore the results should be in good agreement with the results obtained in natural Daylight or Daylight Behind Glass. The two Carbon-Arc options, Continuous and Alternating Light and Dark, under the conditions specified, will produce results which correlate with those obtained in the Daylight Behind Glass Method unless the material being tested is adversely affected by the differences in spectral characteristics of Enclosed Carbon-Arc and natural light.

6.3 When using this test method, the test method option selected should incorporate light, humidity, and heat effects based upon historical data and experience. The test method option selected should also reflect expected end-use conditions associated with the material to be tested.

6.4 When using this test method, use a standard of comparison which has a known change in lightfastness after a specific exposure for comparison to the material to be tested. AATCC Blue Wool Lightfastness Standards have been used extensively for this purpose.

7. Apparatus and Materials (see 15.3)

7.1 AATCC Blue Wool Lightfastness Standards L2 through L9 (see 15.2 and 15.4).

7.2 Gray Scale for Color Change (AATCC EP1) (see 15.4).

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7.3 Card stock: 163 g/m^2 (90 lb) one ply, White Bristol Index.

7.4 Test masks made of material approaching zero light transmittance, and suitable for multiple exposure levels, such as 10, 20, 40, etc. AFU.

7.5 Spectrophotometer or Colorimeter (see 3.2).

7.6 Daylight Exposure Cabinet (see 15.5, 15.8 and Appendix A).

8. Comparison Standards

8.1 AATCC Blue Wool Lightfastness Standards, as defined in AATCC TM16.1, are preferred for all options. However, the rate of fade of any AATCC Blue Wool Lightfastness Standard by one test option may not agree with that of other test options.

8.2 The reference standard can be any suitable textile material where a history of the rate of color change is known. Reference standards for comparison must be determined and agreed upon by the contractual parties. Standards must be exposed simultaneously with the test specimen. The use of the standard assists in determining time-to-time equipment and test procedure variations. If test results of the exposed standards differ by more than 10% from the known standard data, thoroughly review the test equipment operating conditions, and correct any malfunctions or defective parts. Then, repeat the test.

9. Test Specimen Preparation

9.1 Number of Specimens—For acceptance testing, use at least three replicate specimens of both the material to be tested and the standard for comparison to ensure accuracy unless otherwise agreed upon between the purchaser and the supplier.

NOTE: It is recognized that in practice one test and one control specimen are used for test purposes. While such a procedure cannot be accepted in cases of dispute, it may be sufficient in routine testing.

9.2 Specimen Cutting and Mounting—Identify each specimen using a label resistant to the environment encountered during the test. Mount in frames such that the surface of the test specimen and the reference specimen are the same distance from the light source. Use covers that avoid specimen surface compression, particularly when testing pile fabrics. The test specimen and the reference standards shall be of equal size and shape. Cut and prepare test specimens for exposure as follows:

9.2.1 Specimen Backing—For all options, mount the specimens and standards on white card stock. The card stock shall be white, non-reflecting cardboard. Consult the *AATCC Buyer's Guide* (www.aatcc.org/bg) for information. When mounted test specimens are masked, use test masks approaching zero light transmittance. Put the mounted, or mounted and masked, test specimens in frames with backing as directed in an applicable specification: such as open-backed, solid metal, or solid backing (white cardboard).

9.2.2 Fabric—Cut swatches of fabric with the long direction parallel to the machine (warp) direction, at least 70.0 mm x 120.0 mm (2.75 in x 4.7 in.) with the exposed area measuring not less than 30.0 mm x 30.0 mm (1.2 in x 1.2 in.). Secure the backed specimens in the frames supplied with the test apparatus. Ensure that front and back covers of the holders make good contact with the specimens and give a sharp line of demarcation between the exposed and unexposed areas without compressing the specimen unnecessarily (see 15.6). When required to prevent raveling, the specimens may be edged by sewing, pinking or fusing.

9.2.3 Yarns—Wind or fasten yarns on frames of white card stock to a length of approximately 150.0 mm (6.0 in.). Only that portion of the yarns directly facing the radiant energy is evaluated for color change. Wind the yarn on the frame closely packed to at least 25.0 mm (1.0 in.) width. The control specimen must contain the same number of strands as the specimen subjected to exposure. After the exposure has been completed, bind together those yarns facing the light source using 20.0 mm (0.75 in.) masking or other suitable tape to keep the yarns closely packed on the exposure frame for evaluation (see 15.7).

10. Test Procedures

10.1 General Conditions

10.1.1 Mount the AATCC Blue Wool Lightfastness Standard(s) and the test specimen(s) on cardboard with an opaque cover (mask) covering one-half of the standard.

10.1.2 Expose standards and test specimen(s) simultaneously to the same test conditions behind glass (see 15.9 and Appendix A). Ensure that the face of the exposed standard(s) and test specimen(s) are at least 75.0 mm (3.0 in.) below the inside surface of the plate glass cover and are positioned at least 150.0 mm (6.0 in.) in from the edges of the glass frame. The back of the exposure cabinet may be varied as follows to achieve the desired exposure conditions:

Backing	Exposure Condition
Open/Expanded Metal	Low Temperature
Solid	High Temperature

Standard(s) and specimen(s) remain exposed 24 h a day and are removed only for inspection.

10.1.3 Monitor temperature and relative humidity in the vicinity of the test cabinets (see 15.11).

10.2 Daylight Behind Glass Exposure to a Specified Amount of Radiant Energy

10.2.1 Use of AATCC Blue Wool Lightfastness Standards—Mount reference and test specimen(s) to be exposed as directed in 10.1.1 and expose simultaneously to the same test conditions behind glass as directed in 10.1.2. Monitor the effect of light by frequently removing the standard(s) from the test frame and evaluating the color change. Continue the exposure until the standard exhibits a difference in color between the exposed and masked portion as described in Section 14. When the test of the specimens is to be terminated after exposure to a specified number of AATCC Fading Units, choose the appropriate standard to achieve the end point. The standards may be used as a set, L2 through L9, or in replicate sets exposed consecutively to total a given end point; that is, singularly expose two L2 standards to reach 10 fading units, or expose one L3 standard to reach 10 fading units.

10.2.1.1 Remove the specimens from exposure when the desired AATCC Fading Units have been achieved and evaluate as specified in Evaluation of Results. For multiple step exposure, that is, 5 fading units and 20 fading units, a single specimen may be exposed, and portions covered (masked) at intervals measured by the standard. The result will be a specimen having an original masked, unexposed section and various sections which have been exposed and subsequently masked. Each section of the specimen, representing a stated exposure interval, can be evaluated versus masked control or an unexposed original portion of the specimen.

10.2.2 Use of Irradiation Monitors—Mount reference and test specimen(s) to be exposed as directed in 10.1.1 and expose them simultaneously to the same test conditions behind glass as directed in 10.1.2.

NOTE: The exposure of the AATCC Blue Wool Lightfastness Standards with their known performance can be helpful in determining whether any unusual conditions were present during the test duration (see 15.9).

10.2.2.1 Record any one, or a combination of global, broad bandpass (preferred from 295-385nm, or narrow bandpass irradiation with a radiometer, exposed under the same conditions as the specimens.

10.2.2.2 Remove the reference and test specimens from exposure when the desired radiant energy, as measured by the radiometer, has been achieved. For multiple step exposure, a single specimen may

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be exposed and portions covered (masked) at intervals of measured radiant exposure (see 10.2.1.1).

10.3 Daylight Exposure Using a Reference Specimen

10.3.1 Substitute reference specimen(s) for the AATCC Blue Wool Lightfastness Standards and proceed as directed in 10.2.1 and 10.2.2, as applicable.

10.4 Daylight Exposure for Lightfastness Classification

10.4.1 One-Step Method—Expose test specimen(s), as detailed in 10.1.1 and 10.1.2, simultaneously with a series of AATCC Blue Wool Lightfastness Standards or determine the number of AATCC Fading Units required to produce a color change in the test specimen equal to Step 4 on the Gray Scale for Color Change (see 15.12).

10.4.2 Two-Step Method—Proceed as directed in 10.4.1, except double the exposure area of the test specimens. After the specimen has been exposed to a color change equal to Step 4 on the Gray Scale for Color Change, remove the specimens from the test chamber and mask (cover) one-half of the exposed area and continue the exposure until the test specimen exhibits a color change equal to Step 3 on the Gray Scale for Color Change (see 15.12).

10.5 Conditioning After the Exposure

10.5.1 After the test exposure is completed, remove the test specimens and comparison standards from exposure. Condition in a dark room at standard conditions for testing textiles, as directed in ASTM D1776, Standard Practice for Conditioning and Testing Textiles, [65 ± 5% RH and 21 ± 2 ° C (70 ± 4 ° F)] for a minimum of 4 h before evaluation.

11. Evaluation and Interpretation

11.1 Assessment of Color Change

11.1.1 After conditioning according to 10.5.1 compare the exposed portion to the masked control or to an unexposed original portion of the specimen (preferred), as specified in a material specification or purchase order. Complete characterization of the lightfastness of a test specimen requires evaluation at more than one level of exposure. A difference in color between unexposed material and the masked portion of the exposed specimen indicates that the textile has been affected by some agent other than light, such as heat or a reactive gas in the atmosphere. Although the exact cause of this difference in color may not be known, it should be noted in the report when it occurs.

11.1.2 Quantify the color change using either AATCC EP1, or AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen, at the specified exposure level whether in AATCC Fading Units or compared to a reference standard (see Section 11.2 and 11.3)

11.1.3 Determine total color difference (DE_{CIELAB}) and the difference in lightness, chroma, and hue (DL^*, DC^*, DH^*). Use instruments that provide values based on the CIE 1976 equation using illuminant D_{65} and 10° observer data. For instruments with diffuse geometry, include the specular component of reflectance in the measurements (refer to AATCC EP6).

11.2 Classification Based on the AATCC Blue Wool Lightfastness Standards

11.2.1 One Step Exposure—Classify lightfastness of the material by:

(a) comparison of the color change of the test specimen to that of a simultaneously exposed series of AATCC Blue Wool Lightfastness Standards (see Table II), or

(b) determination of the number of AATCC Fading Units required to produce a color change in the test specimen equal to Step 4 of the Gray Scale for Color Change (see Table I).

11.2.2 Two Step Exposure—Classify lightfastness of the material by:

(a) determination of the number of AATCC Fading Units required to produce color changes in the test specimen equal to both a Step 4 and Step 3 on the Gray Scale for Color Change (see Table I).

11.2.3 Assign both classifications: the Step 3 level appears first, followed by the Step 4 level in parentheses. For example, a L5(4) classification would illustrate a L5 classification at Step 3 color change and a L4 classification at Step 4 color change. When only one classification number is assigned, it shall represent the number of AATCC Fading Units to produce a Step 4 color change.

11.3 Classification above L7 AATCC Blue Wool Lightfastness Standard

11.3.1 Using Table III, classify lightfastness above the L7 AATCC Blue Wool Lightfastness Standard according to the total number of consecutive L7 standards exposed to Step 4 on the Gray Scale for Color Change during the exposure cycle that is required to produce a Step 4 color change on the test specimen, and Table III.

12. Acceptance Based on Simultaneous Exposure of a Reference Specimen

12.1 Assess color change of the material (not blue wool) as directed in Section 14 in terms of the agreed upon reference specimen.

12.2 Assess the lightfastness of the material as follows:

12.2.1 Satisfactory—If the test specimen exhibits a color change equal to or less than the reference specimen at the exposure level when the reference specimen shows a color change equal to Step 4 on the Gray Scale for Color Change.

12.2.2 Unsatisfactory—If the test specimen exhibits a color change greater than the reference specimen at the exposure level when the reference specimen shows a color change equal to Step 4 on the Gray Scale for Color Change.

12.3 Alternatively, buyer and seller may make acceptance judgment based on color difference as described in 14.3.

13. Report (see Table IV)

13.1 Reporting, at the minimum, should include the following information:

13.1.1 Reference to AATCC TM16.1-2023.

Table III—Classification by AATCC Blue Wool Lightfastness Standards Above L7

Number of L7 Standards Exposed				
Less Than	Equal To But Not Greater Than	More Than	Lightfastness Class	Equivalent AATCC Fading Unit (AFU)
—	2	—	L8	320
3	—	2	L8-9	—
—	3	—	L8-9	480
4	—	3	L8-9	—
—	4	—	L9	640
5	—	4	L9-10	—
—	5	—	L9-10	800
6	—	5	L9-10	—
—	6	—	L9-10	960
7	—	6	L9-10	—
—	7	—	L9-10	1120
8	—	7	L9-10	—
—	8	—	L10	1280
etc.[a]	etc.[a]		etc.[a]	etc.[a]

[a] A classification increase of 1 represents the interval when the equivalent AATCC Fading Units are doubled from the previous whole number classification. Any test specimen for which the number of L7 Standards fall between two whole number classifications is assigned both the lower and higher classification defining that interval.

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Table IV—Reporting Form

Operator's Name _____ Date_____

Sample Identification _____

Material Exposed: Face _____ Back _____

Colorfastness to Light Rating_____ Lightfastness Classification _____

Acceptance Compared to Reference Sample (Yes/No)_____

Test Specimen Compared To: Masked Portion _____

Unmasked Portion_____ Unexposed Original _____

Colorfastness to Light Rating determined by:

Gray Scale for Color Change _____

Instrumentally, Name Type _____

Classification Method_____

Reference Standard_____

Temperature Controlled By: Ambient (Dry Bulb) _____ °C

Black Panel _____ °C Black Standard _____ °C

Exposure Controlled By: AATCC Blue Wool Lightfastness Standards _____

Radiant Energy _____ Other _____

Total Radiant Energy _____

Type of Test Apparatus _____ Model No. _____

Serial No._____ Manufacturer's Name_____

Specimen Rack: Inclined _____ 2-Tier _____ 3-Tier _____ Horizontal_____

Type of Water Supply_____

Option Employed _____ Elapsed Exposure Time _____

Mounting Procedure: Backed_____ Unbacked_____

Sample Rotation Schedule _____ % Relative Humidity_____

For Daylight Behind Glass, report the following:

Geographical Location_____

Exposure Dates: From _____ To _____

Exposure Latitude _____ Exposure Angle _____

Exposed Behind Window Glass: Yes/No _____ If Yes, Specify Type_____

Daily Ambient Temperature: Minimum _____°C Maximum _____°C Avg. _____°C

Daily Black Panel Temperature: Minimum _____°C Maximum _____°C Avg. _____°C

Test Environment Temperature: Minimum _____°C Maximum _____°C Avg. _____°C

Daily % Relative Humidity: Minimum _____ Maximum _____ Avg. _____

Hours of Wetness: Rain _____ Rain and Dew _____

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13.1.2 Operator's name and test date

13.1.3 Specimen identification

13.1.4 Colorfastness to light rating/ Light classification

13.1.5 Acceptance compared to reference specimen or masked area (Yes/No)

13.1.6 Colorfastness to light rating determined by AATCC Gray Scale or Instrumentally

13.1.7 Classification method

13.1.8 Reference standard (if any)

13.1.9 Exposure controlled by: AATCC Blue Wool, radiant energy or other

13.1.10 Total radiant energy

13.1.11 Elapsed exposures time

13.1.12 Mounting procedure (backed or unbacked)

13.1.13 Deviations from AATCC TM16.1 or the performance of the reference standard, if any

13.1.14 Geographical location

13.1.15 Exposure dates

13.1.16 Site latitude and exposure angle

13.1.17 Type of exposure

13.1.18 Daily ambient temperature (min., max. and avg. in °C), and relative humidity (min., max. and avg. in %)

14. Precision and Bias

14.1 Precision

14.1.1 Interlaboratory Test Summary—Committee RA50 has conducted extensive studies to evaluate the use of radiation monitoring devices to terminate exposures in lightfastness testing. Data has been collected in interlaboratory studies using controlled irradiance, xenon-arc equipment and in daylight exposures conducted during a two-year period in both Arizona and Southern Florida. In both studies, one laboratory conducted instrumental measurement of the color change for all exposed specimens.

14.1.2 The interlaboratory studies were under-taken, using eight different lightfastness standard fabrics, to determine the definition of 20 AATCC Fading Units in terms of measured radiant exposure. These studies showed that acceptable agreement between laboratories can be obtained for lightfastness testing providing the following variables are controlled: irradiance level, black-panel temperature, ambient temperature and relative humidity. Overall, there was less than 10% variability in the instrumentally determined color change of specimens exposed in different laboratories. For all specimens tested the standard deviation was equivalent to less than one-half step on the Gray Scale for Color Change. As a result of these tests, 20 AATCC Fading Units was established at 85 kJ/(m^2nm) when measured at 420 nm (approximately 21.5 continuous light-on operating hours) when tested at the conditions specified for Xenon-Arc Lamp, Continuous Light, Option 3.

14.1.3 For the daylight studies, 16 different fabrics, in addition to AATCC and ISO Blue Wool Lightfastness standard fabrics, were exposed. An exposure series was begun each quarter year at two locations over a two-year period. Exposures were terminated based on instrumental measurement of radiant energy dosage. A wide variation in climatic conditions was encountered during the test period. The data obtained clearly shows that the color change of individual specimens is affected differently by variations in temperature, humidity, atmospheric contaminants, etc.; however, the single most significant variable is radiation. The variation in color change resulting from exposure during different years, locations, and seasons, averaged ± 30%.

14.1.4 A more detailed summary of these test results was presented to the 14th meeting of ISO, Technical Committee 38, Subcommittee 1 as Document 38/ 1 N 993, USA Report on Monitoring of Radiation during Lightfastness Testing.

14.2. Bias

14.2.1 The colorfastness to natural and artificial light can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

15. Notes

15.1 This method is technically equivalent to ISO 105-B01.

15.2 Historically, AATCC Blue Wool Lightfastness Standards, except L2, were specially prepared by blending varying proportions of wool dyed with a very fugitive dyestuff, Erio Chrome Azurole B (C.I. 43830) and wool dyed with a fast dyestuff, Indigosol Blue AGG (C.I. 73801). Each resultant higher numbered standard is twice as colorfast as the preceding numbered standard. AATCC Blue Wool Lightfastness Standards and the ISO numbered Blue Wool Lightfastness Standards (as used in ISO 105-B01) produce different ratings and therefore cannot be used interchangeably. Lots 8 and 9 of the L2 AATCC Blue Wool Lightfastness Standard were batch dyed. L2 can be used to measure 5 or 20 AFU exposures, producing two distinct fading endpoints. L2 Standards of Fade for 5 AFU and 20 AFU are available (see 15.4).

15.3 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

15.4 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; web site: www.aatcc.org; e-mail: ordering@aatcc.org.

15.5 Refer to ASTM G24 for guidance on test cabinet selection.

15.6 Pile fabric, such as carpets, which have fibers that may shift position, or texture which may make evaluations in small areas difficult should be tested with an exposed area of not less than approximately 40.0 × 50.0 mm (1.6 × 2.0 in.). Expose sufficient size or multiple specimens to include all colors in the specimen.

15.7 Specimen frames must be made of stainless steel, aluminum, or suitably coated steel to avoid contaminating the specimens with metallic impurities that might catalyze or inhibit the degradation. When specimens are fastened with staples, they should be of the nonferrous type overcoated to avoid contamination of the specimen by corrosion products. Metal frames must have a dull finish and be designed to avoid reflectances that could influence the performance of the material. Frames shall conform to the curvature of the specimen rack. The size of the frame is determined by the type specimens required for individual property requirements.

The following references provide background information on radiation measurements by Light Control Systems:

15.7.1 *Handbook of Chemistry & Physics*, 61st Edition, 1980, edited by Robert C. Weast; The Chemical Rubber Co., Cleveland OH.

15.7.2 International Commission on Illumination (CIE) Publication No. 20, 1972.

15.7.3 *Atlas Sun Spots*, Vol. 4, No 9, Spring 1975, Atlas Material Testing Technology LLC, Chicago, IL.

15.8 Refer to ASTM G24 for guidance on window glass selection.

15.8.1 In order to reduce variability due to changes in UV transmission of glass, all new glass shall be exposed facing the equator, at the site latitude angle, or on an empty under glass exposure cabinet, for at least three months prior to installation in test cabinets.

15.8.2 After the three-month exposure period, it is recommended that the spectral transmittance of representative specimens from each lot of glass be measured. Typically, "single strength" glass will have a transmittance of 10-20% at 320 nm and at least 85% at wavelengths of 380 nm or higher after the three-month pre-aging procedure. If transmittance of the glass is measured, report the average for at least three pieces of the lot of glass being tested. Follow the instructions for measurement of transmittance of solid specimens recommended by the manufacturer of the UV-visible spectrophotometer used. If a spectrophotometer with an integrating sphere is used, the measurements shall be performed in accordance with ASTM E903, Test Method for Solar Absorptance, Reflectance, and Transmittance of Materials Using Integrating Spheres. Additional information on this subject is contained in the ASTM paper by W. D. Ketola and J. S. Robbins: "UV Transmission of Single Strength Window Glass," *Accelerated and Outdoor Durability Testing of Organic Materials*, ASTM STP1202, Warren D. Ketola and Douglas Grossman, Eds., American Society for Testing and Materials, Philadelphia, 1993.

15.9 In some cases high humidity, in combination with atmospheric contaminants, has been found to produce color changes as great as those produced by light. When requested, prepare a duplicate set of test specimens and standards mounted on cardboard, but not masked, and expose simultaneously in another

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cabinet of the same type used in the light exposures but with the glass covered with an opaque material so that the light is excluded. Since there is a combined effect of light, temperature, humidity and atmospheric contaminants, it cannot be assumed that a comparison between specimens exposed in the covered cabinet and in the uncovered cabinet under glass will permit separating the effects produced by light only. However, a comparison of the two sets of specimens with a piece of the original which has not been in an exposure cabinet will indicate whether a material is sensitive to moisture and atmospheric contaminants. This may also help to explain why different results may be obtained with the same amount of radiant energy in daylight exposures made at different times and in different locations.

15.10 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832.9500; fax: +1.610.832.9555; web site: www.astm.org.

15.11 For measuring temperature and relative humidity of the air under the same conditions that the specimens and reference materials are exposed and in the vicinity of test cabinets, any suitable indicating or recording instrument may be used. Continuous recording of temperature and relative humidity is preferred.

15.12 An automated electronic grading system may be used as long as the system has been demonstrated to provide results that are equal to and provide equal or better repeatability and reproducibility than an experienced grader performing visual evaluations.

16. History

16.1 Revised in 2023 to align to style guide.

16.2 Revised in 2022. Editorially revised 2021, 2019, 2016; revised 2014, 2012 (Supersedes AATCC TM16-2004); editorially revised 2009; revised 2004, 2003; reaffirmed 1998; editorially revised 1996, 1995; revised 1993, 1990 (Supersedes AATCC TM16-1987, TM16A-1988, TM16C-1988, TM16D-1988, TM16E-1987, TM16F-1988 and TM16G-1985); editorially revised 1986, 1984,1983; revised 1982, 1981, 1978; reaffirmed 1977; revised 1974, 1971.

16.3 Developed in 1964 by AATCC Committee RA50. Technically equivalent to ISO 105-B01.

Appendix A
Daylight Exposure Cabinet

A. Daylight Exposure Cabinet and Location

A1 The daylight exposure cabinet shall consist of a glass-covered enclosure of any convenient size constructed of metal, wood or other satisfactory material to protect the specimens from rain and weather, and be well ventilated to allow free flow of air over the specimens. The glass cover shall be a sheet 2.0-2.5 mm thick piece of good grade, clear and flat-drawn. It shall be single strength, free of bubbles or other imperfections.

A2 The enclosure or cabinet shall be equipped with a rack which supports the specimens in a plane parallel to that of the glass cover with the face of the specimen at a distance below it of not less than 75.0 mm (3.0 in.). The specimen mounting rack shall be constructed of a material that is compatible with the test specimens. It may be either the open type providing good ventilation on the back side of the specimen, or of a solid material as required. To minimize shadows from the top and the sides of the cabinet, the usable exposure area under glass shall be limited to that of the glass cover reduced by twice the distance from the cover to the specimens.

A3 The cabinet shall be located where it will receive direct daylight throughout the day and where shadows of objects in the vicinity will not fall upon it. When the cabinet is installed over soil, the distance between the bottom of the cabinet and the plane of the cleared area shall be great enough to prevent any undesirable effects of contact with grass or other plant growth during the period of exposure.

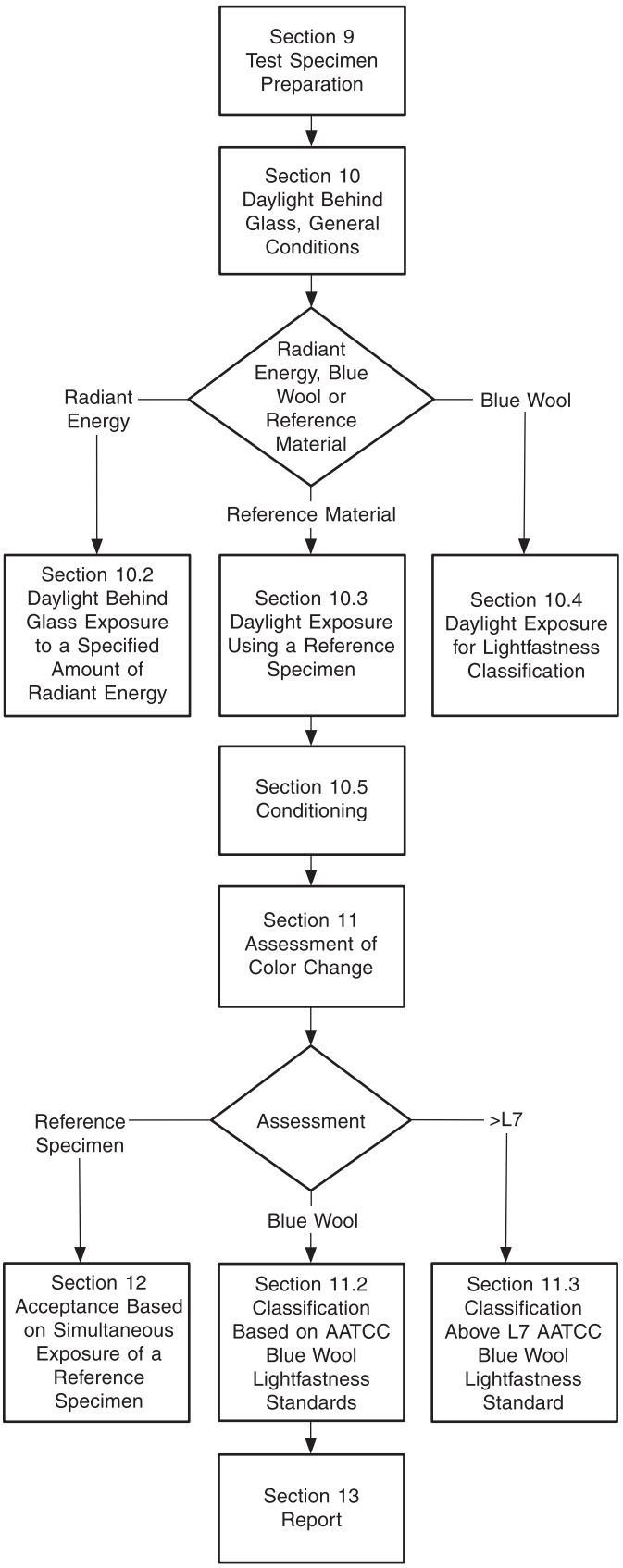
A4 The glass cover and the test specimen shall slope toward the equator at an angle from the horizontal equal in degrees to approximately the latitude of the location at which the tests are being made. Other angles of exposure, such as 45° may be used, but the angle must be reported in the results of test.

A5 Exposure cabinets shall be located in cleared areas, preferably at a suitable number of climatologically different sites representing the various conditions under which the material will be used. Major climatological variations include subtropical, desert, seashore (salt air), industrial atmosphere and areas exhibiting a wide range in percentage of available sunshine. The area beneath and in the vicinity of the cabinets should be characterized by low reflectance and be typical of the ground cover in that climatological area. In desert areas, it should be gravel whereas in most temperature and subtropical areas, it should be low cut grass. The type of ground cover should be indicated in the report.

A6 Instruments for determining climatological data during the exposure period shall be operated in the immediate area of the exposure cabinets. When requested, data obtained shall be reported as part of the results of the test. To characterize the conditions around the test frame, these instruments should be capable of recording: ambient temperature (daily minimum and maximum), relative humidity (daily minimum and maximum), hours of precipitation (rain), and total hours of wetness (rain and dew). To characterize the conditions within the test frame, these instruments should be capable of recording: ambient temperature under glass (daily minimum and maximum), black-panel temperature sensor under glass, total radiant exposure and ultraviolet radiant exposure (either broad or narrow bandpass) at the same angle of exposure as the test specimens, and relative humidity (daily minimum and maximum).

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Appendix B
Flow Chart for AATCC TM16.1-Outdoor



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Test Method for Colorfastness to Light: Carbon-Arc

1. Purpose and Scope

1.1 This test method provides the general principles and procedures for determining the colorfastness to light of textiles. The test options described are applicable to textile materials of all kinds and for colorants, finishes and treatments applied to textile materials. Test options included are:

1—Enclosed Carbon-Arc Lamp, Continuous Light

2—Enclosed Carbon-Arc Lamp, Alternate Light and Dark

1.2 The use of these test options does not imply, expressly or otherwise, an accelerated test for a specific application. The relationship between any lightfastness test and the actual exposure in use must be determined and agreed upon by the contractual parties.

2. Principle

2.1 Specimens of the textile material to be tested and the agreed upon comparison standard(s) are exposed simultaneously to a light source under specified conditions. The colorfastness to light of the specimen is evaluated by comparison of the color change of the exposed portion to the masked control portion of the test specimen or unexposed original material using the Gray Scale for Color Change, or by instrumental color measurement. Lightfastness classification is accomplished by evaluation versus a simultaneously exposed series of AATCC Blue Wool Lightfastness Standards.

3. References

3.1 AATCC EP1, Evaluation Procedure for Gray Scale for Color Change (see 30.5).

3.2 AATCC EP6, Evaluation Procedure for Instrumental Color Measurement (see 30.5).

3.3 AATCC EP7, Evaluation Procedure for Instrumental Assessment of the Change in Color of a Test Specimen (see 30.5).

3.4 ASTM D1776/D1776M, Standard Practice for Conditioning and Testing Textiles (see 30.10).

3.5 ASTM G151, Standard Practice for Exposing Nonmetallic Materials in Accelerated Test Devices that Use Laboratory Light Sources (see 30.10).

3.6 ASTM G153, Standard Practice for Operating Enclosed Carbon-Arc Light Apparatus for Exposure of Nonmetallic Materials (see 30.10).

4. Terminology

4.1 **AATCC Blue Wool Lightfastness Standard,** n.—one of a group of dyed wool fabrics distributed by AATCC for use in determining the amount of light exposure of specimens during lightfastness testing (see 30.1).

4.2 **AATCC Fading Unit (AFU),** n.—a specific amount of exposure made under the conditions specified in various test methods where one AFU is one-twentieth (1/20) of the light-on exposure required to produce a color change equal to Step 4 on the Gray Scale for Color Change or 1.7 ± 0.3 CIELAB units of color difference on AATCC Blue Wool Lightfastness Standard L4.

4.3 **black-panel thermometer,** n.—a temperature measuring device, the sensing unit of which is coated with black paint designed to absorb most of the radiant energy encountered in lightfastness testing (see 30.2).

4.3.1 This device provides an estimation of the maximum temperature a specimen may attain during exposure to artificial light. Any deviation from the geometry of this device described in 30.2 may have an influence on the measured temperature.

4.4 **black standard thermometer,** n.—a temperature measuring device, the sensing unit of which is coated with black material designed to absorb most of the radiant energy encountered in lightfastness testing and is thermally insulated by means of a plastic plate (see 30.2).

4.4.1 This device provides an estimation of the maximum temperature a specimen may attain during exposure to artificial light. Any deviation from the geometry of the device described in 30.2 may have an influence on the measured temperature. The temperature measured by the black standard thermometer will not be the same as that measured by the black-panel thermometer; therefore, they cannot be used interchangeably.

4.5 **broad bandpass radiometer,** n.—a relative term applied to radiometers that have a bandpass width of more than 20 nm and less than 70 nm at 50% of maximum transmittance.

4.6 **color change,** n.—*as used in colorfastness testing*, a change in color of any kind whether a change in lightness, hue or chroma or any combination of these, discernible by comparing the test specimen with a corresponding untested specimen.

4.7 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials, or both resulting from exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

4.8 **colorfastness to light,** n.—the resistance of a material to a change in its color characteristics resulting from exposure of the material to sunlight or an artificial light source.

4.9 **infrared radiation,** n.—radiant energy for which the wavelengths of the monochromatic components are greater than those for visible radiation and less than about 1 mm.

NOTE: The limits of the spectral range of infrared radiation are not well defined and may vary according to the user. Committee E-2.1.2 of the CIE distinguishes in the spectral range between 780 nm and 1 mm:

IR-A	780-1400 nm
IR-B	1.4-4.0 μm
IR-C	3 μm to 1 mm

4.10 **irradiance,** n.—radiant power per unit area incident on a receiver, typically reported in watts per square meter (W/m^2).

4.11 **"L" designation,** n.—the sequence number given each AATCC Blue Wool Lightfastness Standard according to the number of AATCC Fading Units required to produce a color change equal to Step 4 on the Gray Scale for Color Change.

NOTE: See Table I for the numerical relationship between "L" designations of the standards and their colorfastness to light in AFUs. The colorfastness to light of a fabric specimen can be determined by comparing its color change after light exposure with that of the most similar AATCC Blue Wool Lightfastness Standard as shown in Table II.

4.12 **langley,** n.—a unit of total solar radiation equivalent to one gram calorie per square centimeter of irradiated surface.

NOTE: The internationally recommended units are: Joule (J) for quantity of radiant energy, watt (W) for quantity of radiant power, and meter squared (m^2) for area. The following factors are to be used: 1 langley = 1 cal/cm^2; 1 cal/cm^2 = 4.184 J/cm^2 or 41840 J/m^2.

4.13 **lightfastness,** n.—the property of a material, usually an assigned number, depicting a ranked change in its color characteristics resulting from exposure of the material to sunlight or an artificial

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Table I—AATCC Fading Unit and Light Exposure Equivalents for AATCC Blue Wool Lightfastness Standards (see Section 31)[a]

AATCC Blue Wool Lightfastness Standard	AATCC Fading Units	Xenon Only kJ/(m²nm) @ 420 nm	Xenon Only kJ/(m²nm) 300-400 nm
L2	5	21	864
L3	10	43	1728
L4	20	85[b]	3456
L5	40	170	6912
L6	80	340[b]	13824
L7	160	680	27648
L8	320	1360	55296
L9	640	2720	110592

[a] For color change of Step 4 on the Gray Scale for Color Change.

[b] Verified by experiment using Daylight Behind Glass and Xenon-Arc, Continuous Light. All other values are calculated (see Section 31).

light source.

4.14 **narrow bandpass radiometer,** n.—a relative term applied to radiometers that have a bandpass width of 20 nm or less at 50% of maximum transmittance and can be used to measure irradiance at wavelengths such as 340 or 420, ± 0.5 nm.

4.15 **photochromism,** n.—a qualitative designation for a reversible change in color of any kind (whether a change in hue or chroma) which is immediately noticeable upon termination of light exposures when the exposed area of a specimen is compared to the unexposed area.

NOTE: The reversal of the color change or instability of the hue or chroma upon standing in the dark distinguishes photochromism from fading.

4.16 **pyranometer,** n.—a radiometer used to measure the global solar irradiance or, if inclined, hemispherical solar irradiance.

4.17 **radiant power,** n.—energy per unit time emitted, transferred, or received as radiation.

4.18 **radiant exposure,** n.-time integral of irradiance, typically reported in joules per square meter, J/m^2.

4.19 **radiometer,** n.—an instrument used to measure radiant energy.

4.20 **total irradiance,** n.—radiant power integrated over all wavelengths at a point in time expressed in watts per square meter (W/m^2).

4.21 **ultraviolet radiation,** n.—radiant energy for which the wavelengths of the monochromatic components are smaller than those for visible radiation and more than about 100 nm.

NOTE: The limits of the spectral range of ultraviolet radiation are not well defined and may vary according to the user. Committee E.2.1.2 of the CIE distinguishes in the spectral range between 400 and 100 nm:

UV-A 315-400 nm
UV-B 280-315 nm
UV-C 100-280 nm

4.22 **visible radiation,** n.—any radiant energy capable of causing a visual sensation.

NOTE The limits of the spectral range of visible radiation are not well defined and may vary according to the user. The lower limit is generally taken between 380 and 400 nm and the upper limit between 760 and 780 nm (1 nanometer, 1 nm = 10^{-9} m).

4.22 wide bandpass radiometer, n. – a relative term applied to radiometers that have a bandpass width of more than 70nm at 50% of maximum transmittance.

NOTE: Wide bandpass radiometers can be used to measure irradiance at wavelengths such as 300-800nm

4.23 For definitions of other terms relative to lightfastness used in this test method, refer to AATCC M11.

5. Safety Precautions

5.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

5.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

5.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

5.4 Do not operate the test equipment until the manufacturer's instructions have been read and understood. It is the operator's responsibility to conform to the manufacturer's directions for safe operation.

5.5 The test equipment contains high intensity light sources. Do not look directly at the light source. The door to the test chamber must be kept closed when the equipment is in operation.

5.6 Before servicing light sources, allow 30 min for cool-down after lamp operation is terminated.

5.7 When servicing the test equipment, shut off both the off switch and the main power disconnect switch. When equipped, ensure that the main power indicator light on the machine goes out.

5.8 Good laboratory practices should be followed.

6. Uses and Limitation

6.1 Not all materials are affected equally by the same light source and environment. Results obtained using any one test option may not be representative of those of any other test option or any end-use application unless a mathematical correlation for a given material and/or a given application has been established. Enclosed Carbon-Arc, Xenon-Arc and Natural Daylight have been extensively used in the trade for acceptance testing of textile materials. There may be a distinct difference in spectral irradiance, air temperature and humidity sensor locations, and test chamber size between test equipment supplied by different manufacturers that can result in differences in reported test results. Consequently, data obtained from equipment supplied by the different manufacturers, different test chamber size, or different light source and filter combinations cannot be used interchangeably, unless a mathematical correlation has been established. No correlations among differently constructed test apparatus are known to AATCC Committee RA50.

6.2 When using this test method, the test method option selected should incorporate light, humidity, and heat effects based upon historical data and experience. The test method option selected should also reflect expected end-use conditions associated with the material to be tested.

6.3 When using this test method, use a standard of comparison which has a known change in lightfastness after a specific exposure for comparison to the material to be tested. AATCC Blue Wool Lightfastness Standards have been used extensively for this purpose.

7. Apparatus and Materials (see 30.3)

7.1 AATCC Blue Wool Lightfastness Standards L2 through L9 (see 30.1, 30.4 and 30.5).

7.2 L4 AATCC Blue Wool Standard of Fade for 20 AATCC Fading Units (AFU) (see 30.5).

7.3 L2 AATCC Blue Wool (alternate) Standard of Fade for 20 AATCC Fading Units (AFU) (see 30.5).

7.4 Gray Scale for Color Change (AATCC EP1) (see 30.5).

7.5 Card stock: 163 g/m^2 (90 lb) one ply, White Bristol Index

7.6 Test masks made of material approaching zero light transmittance, and suitable for multiple exposure levels, such as 10, 20, 40, etc. AFU

7.7 Black-Panel Thermometer (see 4.3, 30.2).

NOTE: The Black-Panel Thermometer should not be confused with the Black Standard Thermometer which is used in AATCC TM16.3, Test Method for Colorfastness to Light: Xenon-Arc, Continuous Light, Option 2, and some European test procedures. Temperatures as measured by the two different devices generally will not agree at the same test condi-

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tion. The term *Black Thermometer*, as used in this method, refers to either the Black Panel or Black Standard Thermometers.

7.8 Spectrophotometer or Colorimeter (see 3.2).

7.9 Enclosed Carbon-Arc Lamp Fading Apparatus (see Appendix A and 30.6).

8. Comparison Standards

8.1 AATCC Blue Wool Lightfastness Standards, as defined in AATCC TM16.2, are preferred for all options. However, the rate of fade of any AATCC Blue Wool Lightfastness Standard by one test option may not agree with that of other test options.

8.2 The reference standard can be any suitable textile material where a history of the rate of color change is known. Reference standards for comparison must be determined and agreed upon by the contractual parties. Standards must be exposed simultaneously with the test specimen. The use of the standard assists in determining time-to-time equipment and test procedure variations. If test results of the exposed standards differ by more than 10% from the known standard data, thoroughly review the test equipment operating conditions, and correct any malfunctions or defective parts. Then, repeat the test.

Table II—Classification by AATCC Blue Wool Lightfastness Standards[a]
Test Specimen Color Change

Less Than Standard	Equal To But Not Greater Than Standard	More Than Standard	Lightfastness Class	AATCC Fading Units (AFU)
—	—	L2	L1	
—	L2	L3	L2	5
L2	—	L3	L2-3	
—	L3	L4	L3	10
L3	—	L4	L3-4	
—	L4	L5	L4	20
L4	—	L5	L4-5	
—	L5	L6	L5	40
L5	—	L6	L5-6	
—	L6	L7	L6	80
L6	—	L7	L6-7	
—	L7	L8	L7	160
L7	—	L8	L7-8	
—	L8	L9	L8	320
L8	—	L9	L8-9	
—	L9	—	L9	640

[a] The following are examples for using Table II to assign lightfastness classifications:

The test specimen is exposed simultaneously with standards L4, L5, and L6. After exposure and conditioning, the color change exhibited by the test specimen is less than that exhibited by the standards L4 and L5 but greater than that exhibited by the standard L6. The test specimen would be assigned a Lightfastness Classification of L5-6, or use the following example.

The test specimen is examined after each exposure increment until it exhibits a color change equal to Step 4 on the Gray Scale for Color Change. If this occurs after 40 AFU and before 80 AFU exposure, the test specimen would be assigned a Lightfastness Classification of L5-6.

Table III—Classification by AATCC Blue Wool Lightfastness Standards Above L7

Number of L7 Standards Exposed				
Less Than	Equal To But Not Greater Than	More Than	Lightfastness Class	Equivalent AATCC Fading Unit (AFU)
—	2	—	L8	320
3	—	2	L8-9	—
—	3	—	L8-9	480
4	—	3	L8-9	—
—	4	—	L9	640
5	—	4	L9-10	—
—	5	—	L9-10	800
6	—	5	L9-10	—
—	6	—	L9-10	960
7	—	6	L9-10	—
—	7	—	L9-10	1120
8	—	7	L9-10	—
—	8	—	L10	1280
etc.[a]	etc.[a]		etc.[a]	etc.[a]

[a] A classification increase of 1 represents the interval when the equivalent AATCC Fading Units are doubled from the previous whole number classification. Any test specimen for which the number of L7 Standards fall between two whole number classifications is assigned both the lower and higher classification defining that interval.

9. Test Specimen Preparation and Mounting

9.1 Number of Specimens—For acceptance testing, use at least three replicate specimens of both the material to be tested and the standard for comparison to ensure accuracy unless otherwise agreed upon between the purchaser and the supplier.

NOTE: It is recognized that in practice one test and one control specimen are used for test purposes. While such a procedure cannot be accepted in cases of dispute, it may be sufficient in routine testing.

9.2 Specimen Cutting and Mounting—Identify each specimen using a label resistant to the environment encountered during the test. Mount in frames such that the surface of the test specimen and the reference specimen are the same distance from the light source. Use covers that avoid specimen surface compression, particularly when testing pile fabrics. The test specimen and the reference standards shall be of equal size and shape. Cut and prepare test specimens for exposure as follows:

9.2.1 Specimen Backing—For all options, if the specimens do not have an integrated backing, mount the specimens and standards on white card stock. The card stock shall be white, non-reflecting cardboard. Consult the *AATCC Buyer's Guide* (www.aatcc.org/bg) for information. When mounted test specimens are masked, use test masks approaching zero light transmittance.

9.2.2 Fabric—Cut swatches of fabric with the long direction parallel to the machine (warp) direction, at least 70.0 × 120.0 mm (2.75 × 4.7 in.) with the exposed area measuring not less than 30.0 ×30.0 mm (1.2 × 1.2 in.). Secure the backed specimens in the frames supplied with the test apparatus. Ensure that front and back covers of the holders make good contact with the specimens and give a sharp line of demarcation between the exposed and unexposed areas without compressing the specimen unnecessarily (see 30.7 and 30.8). When required to prevent raveling, the specimens may be edged by sewing, pinking, or fusing.

9.2.3 Yarns—Wind or fasten yarns on frames of white card stock to a length of approximately 150.0 mm (6.0 in.). Only

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that portion of the yarns directly facing the radiant energy is evaluated for color change. Wind the yarn on the frame closely packed to at least 25.0 mm (1.0 in.) width. The control specimen must contain the same number of strands as the specimen subjected to exposure. After the exposure has been completed, bind together those yarns facing the light source using 20.0 mm (0.75 in.) masking or other suitable tape to keep the yarns closely packed on the exposure frame for evaluation (see 30.8).

Option 1
Enclosed Carbon-Arc Lamp, Continuous Light

Machine Operating Conditions

10. Preparation of Test Apparatus

10.1 Prior to running the test procedure, verify machine operation by using the following test protocol. To enhance the repeatability of test results, install test apparatus in a room where temperature and relative humidity are controlled in accordance with the manufacturer's recommendations.

10.2 Check to see that the machine has been calibrated and maintained within the manufacturer's recommended calibration schedule interval.

10.3 Turn off all rack and specimen spray units, if applicable.

10.4 Set machine operating conditions as follows:

Light source	Enclosed carbon Continuous light-on
Black panel temperature	63 ± 3°C (145 ± 6°F)
Chamber air temperature	43 ± 2°C (110 ± 4°F)
Relative Humidity %	30 ± 5
Filter type	Borosilicate
Irradiance	Not controlled
Water requirements (Input)	
Type	Demineralized, distilled or reverse osmosis
Solids ppm	Less than 17, preferably less than 8
pH	7 ± 1
Temperature	Ambient 16 ± 5°C (61 ± 9°F)

(See Appendix A.) Be sure that the temperature chosen is appropriate for the type of Black Thermometer to be used (see 30.2). Fill the specimen rack with framed white card stock and the required black thermometer unit. The white card stock is used to simulate air flow in the test chamber during the test exposure and should not include the actual test specimens. Support the black thermometer unit in the specimen drum or rack in the same manner as the test specimen frames. Operate and control the test apparatus as described above and further defined by the manufacturer. Operate the test apparatus in this mode and adjust the instrumentation to provide the required black panel temperature, chamber air temperature and relative humidity. When exterior indicators are not available, read the black thermometer unit through the window in the test chamber door.

10.5 Verify using AATCC Blue Wool Lightfastness Standards following the guidelines in 12.1-12.2.2. If the fade of the L2 or L4 standards do not meet these requirements follow the instrument manufacturer's instructions for calibration and repeat the 20 AFU exposure with fresh L2 or L4 standards. If the fade does meet the requirements described in Section 12, remove the white card stock from the specimen rack and proceed.

10.6 For additional information to prepare and operate the test apparatus refer to the manufacturer's instructions and the following:

10.6.1 For Carbon-Arc use Test Standard ASTM G151 and G153 (see 30.6).

11. Calibration, Verification and AATCC Fading Unit Measurement

11.1 Instrument Calibration—To ensure standardization and accuracy, the instruments associated with the exposure apparatus (that is, light monitor control system, Black Thermometers, chamber air sensor, humidity control system, UV sensors and radiometers) require periodic calibration. Whenever possible, calibration shall be traceable to national or international standards. Calibration schedule and procedure should be in accordance with manufacturer's instructions.

11.1.1 The accuracy of machine operation must be verified by exposure of an applicable AATCC Blue Wool Lightfastness Standard and assessment of the Standard after every 80-100 AATCC Fading Units. Always expose reference standards near the center position of the specimen rack adjacent to the black-panel temperature sensing unit.

12. Verification by AATCC Blue Wool Lightfastness Standards

12.1 Expose the L4 AATCC Blue Wool Lightfastness Standard at the specified temperature, humidity and operating conditions 20 ± 2 continuous light-on operating hours. After exposure, assess the exposed standard specimen, either visually or instrumentally. Increase or decrease the wattage of the lamps, the time of exposure, of both, and expose additional standard specimens until the change in color of the exposed standard meets one of the following criteria.

12.1.1 Visual Comparison—equals the change in color exhibited by the L4 Standard of Fade applicable to the Lot designation used.

12.1.2 Instrumental Color Measurement—Lot designations of AATCC Blue Wool Lightfastness Standard L4 equals the CIELAB units of color change specified on the calibration certificate supplied with the standard as determined by AATCC EP6, Instrumental Color Measurement.

12.2 Alternatively, the L2 AATCC Blue Wool Lightfastness Standard may be exposed at the specified temperature, humidity, and selected conditions for 20 ± 2 continuous light-on operating hours. After exposure, assess the exposed standard specimen, either instrumentally or compare to a L2 Standard of Fade for 20 AFU. When required, increase or decrease the wattage of the lamps, or the time of exposure, or both, and expose additional standard specimens until the change in color of the exposed standard meets one of the following criteria.

12.2.1 Visual Comparison—equals the change in color exhibited by the L2 Standard of Fade for 20 AFU applicable to the Lot designation used (see 30.5).

12.2.2 Instrumental Color Measurement—The performance of the Blue Wool lot designations of AATCC Blue Wool Lightfastness Standard L2 and L4 must equal the CIELAB unit of color change specified on the calibration certificate supplied with the standard as determined by AATCC EP6.

13. AATCC Fading Unit Measurement by AATCC Blue Wool Lightfastness Standards

13.1 The use of AATCC Blue Wool Lightfastness Standards and AATCC Fading Units provides a common exposure standard across the various exposure methods: daylight, carbon-arc lamp, and xenon-arc lamp. The terms *clock hours* and *machine hours* are not valid reporting methods.

13.2 Table I illustrates the number of AATCC Fading Units to produce a color change equal to Step 4 on the Gray Scale for Color Change (see 3.1) or instrumentally (see 3.3) on each of the AATCC Blue Wool Lightfastness Standards.

13.3 For instrumental color measurement, the colorimetric data are calculated using CIE 1964 10° observer data for Illuminant D_{65}. Express the color difference in CIELAB units as directed in AATCC EP6.

Machine Exposure Procedures

14. Machine Exposure, Test Procedures

14.1 Specimen Mounting. Mount the test material on the specimen rack. Make sure that all materials are adequately supported. Any displacement of the material

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toward or away from the source, even by a small distance, may lead to variation in fading between specimens (see 9.2). When alternate light and dark cycles are required, begin exposure at the start of the light cycle.

14.2 In the case of woven, knitted, and nonwoven fabrics, unless otherwise specified, ensure that the side normally used as the face is directly exposed to the radiant source.

14.3 Operate the test apparatus daily until the selected exposure has been completed. Avoid unnecessary delays when interrupting the exposure period to change filters, carbons, or lamps, as such delays may contribute to variations in results or lead to errors**.** When available, monitor exposure test chamber conditions with suitable recorders. If necessary, readjust the controls to maintain the specified test conditions.

15. Machine Exposure to a Specified Amount of Radiant Energy

15.1 One-Step Method—Expose the test specimens and applicable standards for 5, 10, 20 or multiples of 20 AATCC Fading Units until the specimen has been exposed to the desired amount of radiant energy defined in terms of AATCC Fading Units measured by simultaneous exposure of the appropriate Blue Wool Standard(s).

15.2 Two-Step Method—Proceed as directed in 15.1, except double the exposure area of the test specimens. After the specimen has been exposed to the first specified level of radiant energy, remove the specimens from the test chamber and mask (cover) one-half of the exposed area and continue the exposure for an additional 20 or multiples of 20 AATCC Fading Units until the specimen has been exposed to the higher desired amount of radiant energy.

NOTE: The two-step method is preferred for the complete characterization of the lightfastness of a test specimen.

16. Machine Exposure using a Reference Specimen

16.1 Expose the test specimen(s) and reference specimen(s) simultaneously to the required end point in terms of AATCC Fading Units or reference specimen performance (that is, the reference specimen shows a color change equal to Step 4 on the Gray Scale of Color Change).

17. Machine Exposure for Lightfastness Classification

17.1 One-Step Method—Expose test specimen(s) simultaneously with a series of AATCC Blue Wool Lightfastness Standards or determine the number of AATCC Fading Units required to produce a color change in the test specimen equal to Step 4 on the Gray Scale for Color Change (see 30.11).

17.2 Two-Step Method—Proceed as directed in 17.1, except double the exposure area of the test specimens. After the specimen has been exposed to a color change equal to Step 4 on the Gray Scale for Color Change, remove the specimens from the test chamber and mask (cover) one-half of the exposed area and continue the exposure until the test specimen exhibits a color change equal to Step 3 on the Gray Scale for Color Change (see 30.11).

Option 2
Enclosed Carbon-Arc, Alternate Light and Dark

Machine Operating Conditions

18. Preparation of Test Apparatus

18.1 Prior to running the test procedure, verify machine operation by using the following test protocol. To enhance the repeatability of test results, install test apparatus in a room where temperature and relative humidity are controlled in accordance with the manufacturer's recommendations.

18.2 Check to see that the machine has been calibrated and maintained within the manufacturer's recommended calibration schedule interval.

18.3 Turn off all rack and specimen spray units, if applicable.

18.4 Set machine operating conditions as follows:

Light source	Enclosed carbon Alternate light/dark (light on/off)
Black panel temperature	
Light cycle	63 ± 3°C (145 ± 6°F)
Chamber air temperature	
Light cycle	43 ± 2°C (110 ± 4°F)
Dark cycle	43 ± 2°C (110 ± 4°F)
Relative Humidity %	
Light cycle	35 ± 5
Dark cycle	90 ± 5
Light cycle hours	
Light ON	3.8
Light OFF	1.0
Filter type	Borosilicate
Irradiance	Not controlled
Water requirements (Input)	
Type	Demineralized, distilled or reverse osmosis
Solids ppm	Less than 17, preferably less than 8
pH	7 ± 1
Temperature	Ambient 16 ± 5°C (61 ± 9°F)

(See Appendix A.) Be sure that the temperature chosen is appropriate for the type of Black Thermometer to be used (see 30.2). Fill the specimen rack with framed white card stock and the required black thermometer unit. The white card stock is used to simulate air flow in the test chamber during the test exposure and should not include the actual test specimens. Support the black thermometer unit in the specimen drum or rack in the same manner as the test specimen frames. Operate and control the test apparatus as described above and further defined by the manufacturer. Operate the test apparatus in this mode and adjust the instrumentation to provide the required black panel or black standard temperature, chamber air temperature and relative humidity. When exterior indicators are not available, read the black thermometer unit through the window in the test chamber door.

18.5 Verify using AATCC Blue Wool Lightfastness Standards following the guidelines in 20.1-20.2.2. If the fade of the L2 or L4 standards do not meet these requirements follow the instrument manufacturer's instructions for calibration and repeat the 20 AFU exposure with fresh L2 or L4 standards. If the fade does meet the requirements described in Section 21, remove the white card stock from the specimen rack and proceed.

18.6 For additional information to prepare and operate the test apparatus refer to the manufacturer's instructions and the following:

18.6.1 For Carbon-Arc use Test Standard ASTM G151 and G153 (see 30.6).

19. Calibration, Verification and AATCC Fading Unit Measurement

19.1 Instrument Calibration—To ensure standardization and accuracy, the instruments associated with the exposure apparatus (that is, light monitor control system, Black Thermometers, chamber air sensor, humidity control system, UV sensors and radiometers) require periodic calibration. Whenever possible, calibration shall be traceable to national or international standards. Calibration schedule and procedure should be in accordance with manufacturer's instructions.

19.1.1 The accuracy of machine operation must be verified by exposure of an applicable AATCC Blue Wool Lightfastness Standard and assessment of the Standard after every 80-100 AATCC Fading Units. Always expose reference standards near the center position of the specimen rack adjacent to the black-panel temperature sensing unit.

20. Calibration by AATCC Blue Wool Lightfastness Standards

20.1 Expose the L4 AATCC Blue Wool Lightfastness Standard at the specified temperature, humidity, and selected option for 20 ± 2 continuous light-on operating hours. After exposure, assess the

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exposed standard specimen, either visually or instrumentally. Increase or decrease the wattage of the lamps, the time of exposure, of both, and expose additional standard specimens until the change in color of the exposed standard meets one of the following criteria.

20.1.1 Visual Comparison—equals the change in color exhibited by the L4 Standard of Fade applicable to the Lot designation used.

20.1.2 Instrumental Color Measurement—Lot designations of AATCC Blue Wool Lightfastness Standard L4 equals the CIELAB units of color change specified on the calibration certificate supplied with the standard as determined by AATCC EP6.

20.2 Alternatively, the L2 AATCC Blue Wool Lightfastness Standard may be exposed at the specified temperature, humidity, and selected conditions for 20 ± 2 continuous light-on operating hours. After exposure, assess the exposed standard specimen, either instrumentally or compare to a L2 Standard of Fade for 20 AFU. When required, increase, or decrease the wattage of the lamps, or the time of exposure, or both, and expose additional standard specimens until the change in color of the exposed standard meets one of the following criteria.

20.2.1 Visual Comparison—equals the change in color exhibited by the L2 Standard of Fade for 20 AFU applicable to the Lot designation used (see 30.5).

20.2.2 Instrumental Color Measurement—The performance of the Blue Wool lot designations of AATCC Blue Wool Lightfastness Standard L2 and L4 must equal the CIELAB unit of color change specified on the calibration certificate supplied with the standard as determined by AATCC EP6.

21. AATCC Fading Unit Measurement by AATCC Blue Wool Lightfastness Standards

21.1 The use of AATCC Blue Wool Lightfastness Standards and AATCC Fading Units provides a common exposure standard across the various exposure methods: daylight, carbon-arc lamp, and xenon-arc lamp. The terms *clock hours* and *machine hours* are not valid reporting methods.

21.2 Table I illustrates the number of AATCC Fading Units to produce a color change equal to Step 4 on the Gray Scale for Color Change (see 3.1) or instrumentally (see 3.3) on each of the AATCC Blue Wool Lightfastness Standards.

21.3 For instrumental color measurement, the colorimetric data are calculated using CIE 1964 10° observer data for Illuminant D_{65}. Express the color difference in CIELAB units as directed in AATCC EP6.

Machine Exposure Procedures

22. Machine Exposure, Test Procedures

22.1 Specimen Mounting. Mount the test material on the specimen rack. Make sure that all materials are adequately supported. Any displacement of the material toward or away from the source, even by a small distance, may lead to variation in fading between specimens (see 9.2). When alternate light and dark cycles are required, begin exposure at the start of the light cycle.

22.2 In the case of woven, knitted, and nonwoven fabrics, unless otherwise specified, ensure that the side normally used as the face is directly exposed to the radiant source.

22.3 Operate the test apparatus daily until the selected exposure has been completed. Avoid unnecessary delays when interrupting the exposure period to change filters, carbons, or lamps, as such delays may contribute to variations in results or lead to errors**.** When available, monitor exposure test chamber conditions with suitable recorders. If necessary, readjust the controls to maintain the specified test conditions.

23. Machine Exposure to a Specified Amount of Radiant Energy

23.1 One-Step Method—Expose the test specimens and applicable standards for 5, 10, 20 or multiples of 20 AATCC Fading Units until the specimen has been exposed to the desired amount of radiant energy defined in terms of AATCC Fading Units measured by simultaneous exposure of the appropriate Blue Wool Standard(s).

23.2 Two-Step Method—Proceed as directed in 23.1, except double the exposure area of the test specimens. After the specimen has been exposed to the first specified level of radiant energy, remove the specimens from the test chamber and mask (cover) one-half of the exposed area and continue the exposure for an additional 20 or multiples of 20 AATCC Fading Units until the specimen has been exposed to the higher desired amount of radiant energy.

NOTE: The two-step method is preferred for the complete characterization of the lightfastness of a test specimen.

24. Machine Exposure using a Reference Specimen

24.1 Expose the test specimen(s) and reference specimen(s) simultaneously to the required end point in terms of AATCC Fading Units or reference specimen performance (that is, the reference specimen shows a color change equal to Step 4 on the Gray Scale of Color Change).

25. Machine Exposure for Lightfastness Classification

25.1 One-Step Method—Expose test specimen(s) simultaneously with a series of AATCC Blue Wool Lightfastness Standards or determine the number of AATCC Fading Units required to produce a color change in the test specimen equal to Step 4 on the Gray Scale for Color Change (see 30.11).

25.2 Two-Step Method—Proceed as directed in 25.1, except double the exposure area of the test specimens. After the specimen has been exposed to a color change equal to Step 4 on the Gray Scale for Color Change, remove the specimens from the test chamber and mask (cover) one-half of the exposed area and continue the exposure until the test specimen exhibits a color change equal to Step 3 on the Gray Scale for Color Change (see 30.1).

26. Conditioning After the Exposure

26.1 After the test exposure is completed, remove the test specimens and comparison standards from exposure. Condition in a dark room at standard conditions for testing textiles, as directed in ASTM D1776/D1776M, Standard Practice for Conditioning and Testing Textiles, [65 ± 5% RH and 21 ± 2C (70 ± 4F)] for a minimum of 4 h before evaluation.

27. Evaluation and Interpretation

27.1 Assessment of Color Change

27.1.1 Compare the exposed portion to the masked control or to an unexposed original portion of the specimen (preferred), as specified in a material specification or purchase order. Complete characterization of the lightfastness of a test specimen requires evaluation at more than one level of exposure. A difference in color between unexposed material and the masked portion of the exposed specimen indicates that the textile has been affected by some agent other than light, such as heat or a reactive gas in the atmosphere. Although the exact cause of this difference in color may not be known, it should be noted in the report when it occurs.

27.1.2 Quantify the color change using either AATCC EP1, Gray Scale for Color Change, or AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen, at the specified exposure level whether in AATCC Fading Units or compared to a reference standard (see 30.11).

27.1.3 Determine total color difference (DE_{CIELAB}) and the difference in lightness, chroma, and hue (DL^*, DC^*, DH^*). Use instruments that provide values based on the CIE 1976 equation using illuminant

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D_{65} and 10° observer data. For instruments with diffuse geometry, include the specular component of reflectance in the measurements (refer to AATCC EP6).

27.2. Acceptance Based on Simultaneous Exposure of a Reference Specimen

27.2.1 Assess color change of the material (not blue wool) as directed in Section 27.1 in terms of the agreed upon reference specimen.

27.2.2 Assess the lightfastness of the material as follows:

27.2.2.1 Satisfactory—If the test specimen exhibits a color change equal to or less than the reference specimen at the exposure level when the reference specimen shows a color change equal to Step 4 on the Gray Scale for Color Change.

27.2.2.2 Unsatisfactory—If the test specimen exhibits a color change greater than the reference specimen at the exposure level when the reference specimen shows a color change equal to Step 4 on the Gray Scale for Color Change.

27.2.3 Alternatively, buyer and seller may make acceptance judgment based on color difference as described in 27.1.3.

27.3 Classification Based on the AATCC Blue Wool Lightfastness Standards (See Note 30.12)

27.4 Classification above L7 AATCC Blue Wool Lightfastness Standard (See Note 30.13)

28. Report (see Table IV)

28.1 Reporting at the minimum should include the following information:

28.1.1 Operator's name and test date

28.1.2 Specimen identification

28.1.3 Colorfastness to light rating/ Light classification

28.1.4 Acceptance compared to reference specimen or masked area (Yes/No)

28.1.5 Colorfastness to light rating determined by Gray Scale or Instrumentally

28.1.6 Classification method

28.1.7 Reference standard

28.1.8 Temperature ambient (dry bulb) °C, black panel °C or black standard °C

28.1.9 Exposure controlled by: AATCC Blue Wool, radiant energy or other

28.1.10 Total radiant energy

28.1.11 Type of apparatus, model no., serial no., manufacturer, specimen rack (inclined, 2-tier, 3-tier or horizontal) and type of water supply

28.1.12 Option employed

28.1.13 Elapsed exposures time

28.1.14 Mounting procedure (backed or unbacked)

28.1.15 Deviations from AATCC TM16.2 or the performance of the reference standard, if any

29. Precision and Bias

29.1 Precision

29.1.1 Test Summary—Committee RA50 has conducted extensive studies to evaluate the use of radiation monitoring devices to terminate exposures in lightfastness testing. Data has been collected in interlaboratory studies using controlled irradiance, xenon-arc equipment and in daylight exposures conducted during a two-year period in both Arizona and Southern Florida. In both studies, one laboratory conducted instrumental measurement of the color change for all exposed specimens.

29.1.2 The interlaboratory studies were under-taken, using eight different lightfastness standard fabrics, to determine the definition of 20 AATCC Fading Units in terms of measured radiation. These studies showed that acceptable agreement between laboratories can be obtained for lightfastness testing providing the following variables are controlled: irradiance level, black-panel temperature, ambient temperature, and relative humidity. Overall, there was less than 10% variability in the instrumentally determined color change of specimens exposed in different laboratories. For all specimens tested the standard deviation was equivalent to less than one-half step on the Gray Scale for Color Change. As a result of these tests, 20 AATCC Fading Units was established at 85 kJ/(m^2nm) when measured at 420 nm (approximately 21.5 continuous light–on operating hours) when tested at the conditions specified for Xenon-Arc Lamp, Continuous Light, Option 3.

29.1.3 For the daylight studies, 16 different fabrics, in addition to AATCC and ISO Blue Wool Lightfastness standard fabrics, were exposed. An exposure series was begun each quarter year at two locations over a two-year period. Exposures were terminated based on instrumental measurement of radiant energy dosage. A wide variation in climatic conditions was encountered during the test period. The data obtained clearly shows that the color change of individual specimens is affected differently by variations in temperature, humidity, atmospheric contaminants, etc.; however, the single most significant variable is radiation. The variation in color change resulting from exposure during different years, locations, and seasons, averaged ± 30%.

29.1.4 A more detailed summary of these test results was presented to the 14th meeting of ISO, Technical Committee 38, Subcommittee 1 as Document 38/ 1 N 993, USA Report on Monitoring of Radiation during Lightfastness Testing.

29.2 Bias

29.2.1 The colorfastness to natural and artificial light can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

30. Notes

30.1 Historically, AATCC Blue Wool Lightfastness Standards, except L2, were specially prepared by blending varying proportions of wool dyed with a very fugitive dyestuff, Erio Chrome Azurole B (C.I. 43830) and wool dyed with a fast dyestuff, Indigosol Blue AGG (C.I. 73801). Each resultant higher numbered standard is twice as colorfast as the preceding numbered standard. AATCC Blue Wool Lightfastness Standards and the ISO numbered Blue Wool Lightfastness Standards (as used in ISO 105-B01) produce different ratings and therefore cannot be used interchangeably. Lots 8 and 9 of the L2 AATCC Blue Wool Lightfastness Standard were batch dyed. L2 can be used to measure 5 or 20 AFU exposures, producing two distinct fading endpoints. L2 Standards of Fade for 5 AFU and 20 AFU are available (see 30.5).

30.2 Black Thermometers are used to control an artificial weathering device and to provide an estimate of the maximum temperature of specimens exposed to a radiant energy source. There are two types of Black Thermometers. One type is referred to as a "Black Panel Thermometer" which is uninsulated and is made of metal. The other type is referred to as a "Black Standard Thermometer" which is insulated and is made of metal with a plastic backing. As a point of information, some ISO specifications specify the use of a "Black Standard Thermometer." Typically, Black Standard Thermometers indicate higher temperatures than Black Panel Thermometers under the same exposure conditions.

The Black Thermometer units indicate the absorbed irradiance minus the heat dissipated by conduction and convection. Keep the black face of these thermometer units in good condition. Follow the manufacturer's recommendations for proper care and maintenance of Black Thermometers.

30.2.1 For Black Panel Thermometers: Testing temperature is measured and regulated by a Black Panel Thermometer unit mounted on the specimen rack to permit the face of it to receive the same exposure as the test specimen. Black Panel Thermometers shall consist of a metal panel at least 70 mm x 150 mm whose temperature is measured with a thermometer or thermocouple whose sensitive portion is in the center of and in good thermal contact with the panel. The side of the panel facing the light source shall be black with a reflectance of less than 10% throughout the spectrum of light source reaching the specimen; the side of the panel not facing the light source shall be open to the atmosphere within the exposure chamber.

30.2.2 For Black Standard Thermometers: Testing temperature is measured and regulated by a black standard thermometer unit mounted on the specimen rack to permit the face of it to receive the same exposure as the test specimen. The Black Standard Thermometer shall consist of a plane of stainless-steel plate measuring 70 mm x 40 mm with a thickness of about 0.5 mm, whose temperature is measured by a thermal resistor, with good heat-conducting properties, fitted to the reverse side. The metal plate is fixed to a plastic plate so that it is thermally insulated. The side of the panel facing the light source shall be black with a reflectance of less than 5% throughout the spectrum of light reaching the specimen.

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Table IV—Reporting Form

Operator's Name _____ Date_____

Sample Identification _____

Material Exposed: Face _____ Back _____

Colorfastness to Light Rating_____ Lightfastness Classification _____

Acceptance Compared to Reference Sample (Yes/No)_____

Test Specimen Compared To: Masked Portion _____

Unmasked Portion_____ Unexposed Original _____

Colorfastness to Light Rating determined by:

Gray Scale for Color Change _____

Instrumentally, Name Type _____

Classification Method_____

Reference Standard_____

Temperature Controlled By: Ambient (Dry Bulb) _____ °C

Black Panel _____ °C Black Standard _____ °C

Exposure Controlled By: AATCC Blue Wool Lightfastness Standards _____

Radiant Energy _____ Other _____

Total Radiant Energy _____

Type of Test Apparatus _____ Model No. _____

Serial No._____ Manufacturer's Name_____

Specimen Rack: Inclined _____ 2-Tier _____ 3-Tier _____ Horizontal_____

Type of Water Supply_____

Option Employed _____ Elapsed Exposure Time _____

Mounting Procedure: Backed_____ Unbacked_____

Sample Rotation Schedule _____ % Relative Humidity_____

For Daylight Behind Glass, report the following:

Geographical Location_____

Exposure Dates: From _____ To _____

Exposure Latitude _____ Exposure Angle _____

Exposed Behind Window Glass: Yes/No _____ If Yes, Specify Type_____

Daily Ambient Temperature: Minimum _____°C Maximum _____°C Avg. _____°C

Daily Black Panel Temperature: Minimum _____°C Maximum _____°C Avg. _____°C

Test Environment Temperature: Minimum _____°C Maximum _____°C Avg. _____°C

Daily % Relative Humidity: Minimum _____ Maximum _____ Avg. _____

Hours of Wetness: Rain _____ Rain and Dew _____

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30.3 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its corporate members, but AATCC does not qualify, or in any way approve, endorse, or certify that any of the listed equipment or materials meets the requirements in its test methods.

30.4 More uniform and reproducible fading of the AATCC Blue Wool Lightfastness Standards and test specimens is achieved when backed with white cardboard. The color difference values in the initial determination of the end point for the AATCC Blue Wool Lightfastness Standards were determined from exposures with such backing. Although tolerances are given for the AATCC Blue Wool Lightfastness Standards, every effort should be made to achieve the midpoint value given for these standards. For referee purposes, the Blue Wool Lightfastness Standards will be exposed in multiples of three and the average color change must equal the CIELAB unit of color change specified on the calibration certificate supplied with the applicable standard as determined by AATCC EP6 (see 30.5).

30.5 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; web site: www.aatcc.org; e-mail: ordering@aatcc.org.

30.6 Refer to ASTM G151 and G153 for guidance on design and performance requirements for instruments specified in this method. Further guidance is found in Appendix A. Consult the *AATCC Buyer's Guide* for potential equipment information.

30.7 Pile fabric, such as carpets, which have fibers that may shift position, or texture which may make evaluations in small areas difficult should be tested with an exposed area of not less than approximately 40.0 × 50.0 mm (1.6 × 2.0 in.). Expose sufficient size or multiple specimens to include all colors in the specimen.

30.8 Specimen frames must be made of stainless steel, aluminum, or suitably coated steel to avoid contaminating the specimens with metallic impurities that might catalyze or inhibit the degradation. When specimens are fastened with staples, they should be of the nonferrous type overcoated to avoid contamination of the specimen by corrosion products. Metal frames must have a dull finish and be designed to avoid reflectance that could influence the performance of the material. Frames shall conform to the curvature of the specimen rack. The size of the frame is determined by the type specimens required for individual property requirements.

30.9 In Table A1, the data are for a typical spectral irradiance for an enclosed carbon-arc with a borosilicate glass globe. The daylight data are for global irradiance on a horizontal surface with an air mass of 1.2, column ozone 0.294 atm cm, 30% relative humidity, altitude 2100 m (atmospheric pressure of 787.8 mb), and an aerosol represented by an optical thickness of 0.081 at 300 nm and 0.62 at 400 nm. Data from 701-800 nm is not shown.

The following references provide background information on radiation measurements by Light Control Systems:

30.9.1 *Handbook of Chemistry & Physics*, 61st Edition, 1980, edited by Robert C. Weast; The Chemical Rubber Co., Cleveland OH.

30.9.2 International Commission on Illumination (CIE) Publication No. 20, 1972.

30.9.3 *Atlas SunSpots*, Vol. 4, No 9, Spring 1975, Atlas Material Testing Technology LLC, Chicago, IL.

30.10 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832.9500; fax: +1.610.832.9555; web site: www.astm.org.

30.11 An automated electronic grading system may be used as long as the system has been demonstrated to provide results that are equal to and provide equal or better repeatability and reproducibility than an experienced grader performing visual evaluations.

30.12 **Classification Based on the AATCC Blue Wool Lightfastness Standards**

30.12.1 One Step Exposure—Classify lightfastness of the material by:

(a) comparison of the color change of the test specimen to that of a simultaneously exposed series of AATCC Blue Wool Lightfastness Standards (see Table II), or

(b) determination of the number of AATCC Fading Units required to produce a color change in the test specimen equal to Step 4 of the Gray Scale for Color Change (see Table I).

30.12.2 Two Step Exposure—Classify lightfastness of the material by:

(a) determination of the number of AATCC Fading Units required to produce color changes in the test specimen equal to both a Step 4 and Step 3 on the Gray Scale for Color Change (see Table I).

30.12.3 Assign both classifications: the Step 3 level appears first, followed by the Step 4 level in parentheses. For example, a L5(4) classification would illustrate a L5 classification at Step 3 color change and a L4 classification at Step 4 color change. When only one classification number is assigned, it shall represent the number of AATCC Fading Units to produce a Step 4 color change.

30.13 **Classification above L7 AATCC Blue Wool Lightfastness Standard**

30.13.1 Using Table III, classify lightfastness above the L7 AATCC Blue Wool Lightfastness Standard according to the total number of consecutive L7 standards exposed to Step 4 on the Gray Scale for Color Change during the exposure cycle that is required to produce a Step 4 color change on the test specimen, and Table III.

31. History

31.1 Revised in 2023 to align to style guide

31.2 Revised in 2022. Editorially revised in 2019, 2016; revised 2014 (Supersedes AATCC TM16-2004), 2012; editorially revised 2009; revised 2004, 2003; reaffirmed 1998; editorially revised 1996, 1995; revised 1993 (Supersedes AATCC TM16-1987, TM16A-1988, TM16C-1988, TM16D-1988, TM16E-1987, TM16F-1988 and TM16G-1985); revised 1990; editorially revised 1986, 1984, 1983; revised 1982, 1981, 1978; reaffirmed 1977; revised 1974, 1971.

31.3 Developed in 1964 by AATCC Committee RA50.

Appendix A
Enclosed Carbon-Arc Lamp Fading Apparatus

A. Enclosed Carbon-Arc Lamp Fading Apparatus

A1 Different types of carbon-arc test apparatus may be utilized. The design of the test chamber may vary, but it should be constructed from corrosion resistant material, and in addition to the radiant source, may provide for means of controlling temperature and relative humidity.

A1.1 Laboratory Light Source—Enclosed carbon-arc light sources typically use carbon rods which contain a mixture of metal salts. An electric current is passed between the carbon rods which burn and give off ultraviolet, visible, and infrared radiation. Use carbon rods recommended by the device manufacturer.

A1.2 Filter—The most used filters are borosilicate glass globes which fit around the carbon burners.

A1.3 The emission spectra of the enclosed carbon-arc shows strong emission in the long wavelength ultraviolet region. Emissions in the visible, infrared, and short wavelength ultraviolet below 350 nm are generally weaker than in daylight behind window glass (see Table A1). The enclosed carbon-arc does not provide a good match to natural daylight.

A1.3.1 Spectral Irradiance for Enclosed Carbon-Arc with Borosilicate Filters—Table A1 is representative of the spectral irradiance received by a test specimen mounted in the specimen plane.

A1.4 See 30.9 for additional information.

A1.5 Thermometer—A Black Panel or Black Standard Thermometer may be used and shall conform to the descriptions found in 30.2.1 and 30.2.2. The type of thermometer used, the method of mounting on specimen holder, and the exposure temperature shall be stated in the test report.

A1.6 Relative Humidity—The test chamber may be equipped with a means to measure and control the relative humidity. Such instruments shall be shielded from the light source.

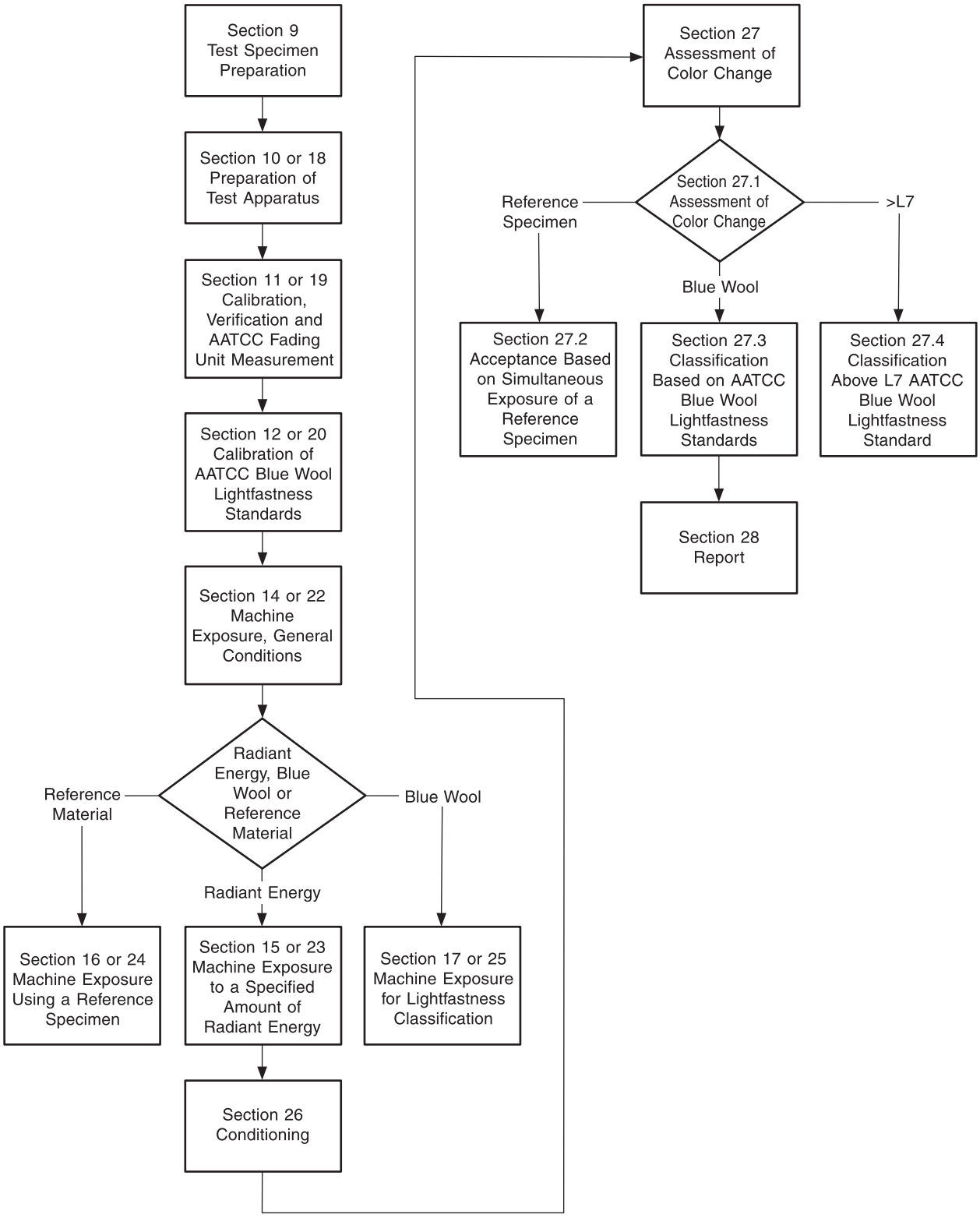
A1.7 Apparatus Maintenance—The test apparatus requires periodic maintenance to maintain uniform exposure conditions. Perform required maintenance in accordance with manufacturer's instructions.

Table A1—Typical Spectral Irradiance Distribution for Enclosed Carbon-Arc with Borosilicate Filters. Ultraviolet Wavelength Region Irradiance as a Percentage of Total Irradiance from 300-400 nm

Bandpass (nm)	Enclosed Carbon-Arc with Borosilicate Filters	Sunlight
290-320	0%	5.6%
320-360	20.5%	40.2%
360-400	79.5%	54.2%

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Appendix B
Flow Chart for AATCC TM16.2, Carbon-Arc



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Test Method for Colorfastness to Light: Xenon-Arc

1. Purpose and Scope

1.1 This test method provides the general principles and procedures for determining the colorfastness to light of textile materials. The test options described are applicable to textile materials of all kinds and for colorants, finishes and treatments applied to textile materials. Test options included are:

1—Xenon-Arc Lamp, Alternate Light and Dark

2—Xenon-Arc Lamp, Continuous Light, Black Standard Option

3—Xenon-Arc Lamp, Continuous Light, Black Panel Option

1.2 The use of these test options does not imply, expressly or otherwise, an accelerated test for a specific application. The relationship between any lightfastness test and the actual exposure in use must be determined and agreed upon by the contractual parties.

1.3 This test method contains the following sections that assist in the use and implementation of the various options for determining lightfastness of textile materials.

	Section
Principle	2
Terminology	3
Safety Precautions	4
Uses and Limitation	5
Apparatus and Materials	6
Comparison Standards	7
Test Specimen Preparation	8
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Report	40
Precision and Bias	41-42
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2. Principle

2.1 Specimens of the textile material to be tested and the agreed upon comparison standard(s) are exposed simultaneously to a light source under specified conditions. The colorfastness to light of the specimen is evaluated by comparison of the color change of the exposed portion to the masked control portion of the test specimen or unexposed original material using the Gray Scale for Color Change, or by instrumental color measurement. Lightfastness classification is accomplished by evaluation versus a simultaneously exposed series of AATCC Blue Wool Lightfastness Standards.

3. Terminology

3.1 **AATCC Blue Wool Lightfastness Standard,** n.—one of a group of dyed wool fabrics distributed by AATCC for use in determining the amount of light exposure of specimens during lightfastness testing (see 44.1).

3.2 **AATCC Fading Unit (AFU),** n.—a specific amount of exposure made under the conditions specified in various test methods where one AFU is one-twentieth (1/20) of the light-on exposure required to produce a color change equal to Step 4 on the Gray Scale for Color Change or 1.7 ± 0.3 CIELAB units of color difference on AATCC Blue Wool Lightfastness Standard L4.

3.3 **black-panel thermometer,** n.—a temperature measuring device, the sensing unit of which is coated with black paint designed to absorb most of the radiant energy encountered in lightfastness testing (see 44.2).

3.3.1 This device provides an estimation of the maximum temperature a specimen may attain during exposure to artificial light. Any deviation from the geometry of this device described in 44.2 may have an influence on the measured temperature.

3.4 **black standard thermometer,** n.—a temperature measuring device, the sensing unit of which is coated with black material designed to absorb most of the radiant energy encountered in lightfastness testing and is thermally insulated by means of a plastic plate (see 44.2).

3.4.1 This device provides an estimation of the maximum temperature a specimen may attain during exposure to artificial light. Any deviation from the geometry of the device described in 44.2 may have an influence on the measured temperature. The temperature measured by the black standard thermometer will not be the same as that measured by the black-panel thermometer; therefore, they cannot be used interchangeably.

3.5 **broad bandpass radiometer,** n.—a relative term applied to radiometers that have a bandpass width of more than 20 nm at 50% of maximum transmittance and can be used to measure irradiance at wavelengths such as 300-400 nm or 300-800 nm.

3.6 **color change,** n.—*as used in colorfastness testing*, a change in color of any kind whether a change in lightness, hue or chroma or any combination of these, discernible by comparing the test specimen with a corresponding untested specimen.

3.7 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials, or both as a result of exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

3.8 **colorfastness to light,** n.—the resistance of a material to a change in its color characteristics as a result of exposure of the material to sunlight or an artificial light source.

3.9 **infrared radiation,** n.—radiant energy for which the wavelengths of the monochromatic components are greater than those for visible radiation and less than about 1 mm.

NOTE: The limits of the spectral range of infrared radiation are not well defined and may vary according to the user. Committee E-2.1.2 of the CIE distinguishes in the spectral range between 780 nm and 1mm:

IR-A 780-1400 nm
IR-B 1.4-3.0 μm
IR-C 3 μm to 1 mm

3.10 **irradiance,** n.—radiant power per unit area incident on a receiver, typically reported in watts per square meter (W/m^2).

3.11 **"L" designation,** n.—the sequence number given each AATCC Blue Wool Lightfastness Standard according to the number of AATCC Fading Units required to produce a color change equal to Step 4 on the Gray Scale for Color Change.

NOTE: See Table I for the numerical relationship between "L" designations of the standards and their colorfastness to light in AFUs. The colorfastness to light of a fabric specimen can be determined by comparing its color change after light exposure with that of the most similar AATCC Blue Wool Lightfastness Standard as shown in Table II.

3.12 **langley,** n.—a unit of total solar radiation equivalent to one gram calorie per square centimeter of irradiated surface.

NOTE: The internationally recom-

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mended units are: Joule (J) for quantity of radiant energy, watt (W) for quantity of radiant power, and meter squared (m^2) for area. The following factors are to be used: 1 langley = 1 cal/cm^2; 1 cal/cm^2 = 4.184 J/cm^2 or 41840 J/m^2.

3.13 **lightfastness,** n.—the property of a material, usually an assigned number, depicting a ranked change in its color characteristics as a result of exposure of the material to sunlight or an artificial light source.

3.14 **narrow bandpass radiometer,** n.—a relative term applied to radiometers that have a bandpass width of 20 nm or less at 50% of maximum transmittance and can be used to measure irradiance at wavelengths such as 340 or 420, ± 0.5 nm.

3.15 **photochromism,** n.—a qualitative designation for a reversible change in color of any kind (whether a change in hue or chroma) which is immediately noticeable upon termination of light exposures when the exposed area of a specimen is compared to the unexposed area.

NOTE: The reversal of the color change or instability of the hue or chroma upon standing in the dark distinguishes photochromism from fading.

3.16 **pyranometer,** n.—a radiometer used to measure the global solar irradiance or, if inclined, hemispherical solar irradiance.

3.17 **radiant power,** n.—energy per unit time emitted, transferred or received as radiation.

3.18 **radiometer,** n.—an instrument used to measure radiant energy.

3.19 **total irradiance,** n.—radiant power integrated over all wavelengths at a point in time expressed in watts per square meter (W/m^2).

3.20 **ultraviolet radiation,** n.—radiant energy for which the wavelengths of the monochromatic components are smaller than those for visible radiation and more than about 100 nm.

NOTE: The limits of the spectral range of ultraviolet radiation are not well defined and may vary according to the user. Committee E.2.1.2 of the CIE distinguishes in the spectral range between 400 and 100 nm:

UV-A	315-400 nm
UV-B	280-315 nm
UV-C	100-280 nm

3.21 **visible radiation,** n.—any radiant energy capable of causing a visual sensation.

NOTE The limits of the spectral range of visible radiation are not well defined and may vary according to the user. The lower limit is generally taken between 380 and 400 nm and the upper limit between 760 and 780 nm (1 nanometer, 1 nm = 10^{-9} m).

3.22 For definitions of other terms relative to lightfastness used in this test method, refer to AATCC M11.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted on specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Do not operate the test equipment until the manufacturer's instructions have been read and understood. It is the operator's responsibility to conform to the manufacturer's directions for safe operation.

4.2 The test equipment contains high intensity light sources. Do not look directly at the light source. The door to the test chamber must be kept closed when the equipment is in operation.

4.3 Before servicing light sources, allow 30 min for cool-down after lamp operation is terminated.

4.4 When servicing the test equipment, shut off both the off switch and the main power disconnect switch. When equipped, ensure that the main power indicator light on the machine goes out.

4.5 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5. Uses and Limitation

5.1 Not all materials are affected equally by the same light source and environment. Results obtained by the use of any one test option may not be representative of those of any other test option or any end-use application unless a mathematical correlation for a given material and/or a given application has been established. Enclosed Carbon-Arc, Xenon-Arc and Daylight have been extensively used in the trade for acceptance testing of textile materials. There may be a distinct difference in spectral power distribution, air temperature and humidity sensor locations, and test chamber size between test equipment supplied by different manufacturers that can result in differences in reported test results. Consequently, data obtained from equipment supplied by the different manufacturers, different test chamber size, or different light source and filter combinations cannot be used interchangeably, unless a mathematical correlation has been established. No correlations among differently constructed test apparatus are known to AATCC Committee RA50.

5.2 Results from Xenon-Arc, for all materials should be in good agreement with the results obtained in Daylight Behind Glass (see Table I). Since the spectral distribution of Xenon-Arc, Alternate Light and Dark, equipped with the specified filter glass is very close to that of average or typical daylight behind window glass, it is expected that results should be in good agreement with the results obtained in Daylight, Daylight Behind Glass.

5.3 When using this test method, the test method option selected should incorporate light, humidity, and heat effects based upon historical data and experience. The test method option selected should also reflect expected end-use conditions associated with the material to be tested.

5.4 When using this test method, use a standard of comparison which has a known change in lightfastness after a specific exposure for comparison to the material to be tested. AATCC Blue Wool Lightfastness Standards have been used extensively for this purpose.

6. Apparatus and Materials (see 44.3)

6.1 AATCC Blue Wool Lightfastness Standards L2 through L9 (see 44.1, 44.4 and 44.5).

6.2 L4 AATCC Blue Wool Standard of Fade for 20 AATCC Fading Units (AFU) (see 44.5).

6.3 L2 AATCC Blue Wool (alternate) Standard of Fade for 20 AATCC Fading Units (AFU) (see 11.2, 44.5).

6.4 Gray Scale for Color Change (AATCC EP1) (see 44.5).

6.5 Card stock: 163 g/m^2 (90 lb) one ply, White Bristol Index.

6.6 Test masks made of material approaching zero light transmittance, and

Table I—AATCC Fading Unit and Light Exposure Equivalents for AATCC Blue Wool Lightfastness Standards (see Section 41)[a]

AATCC Blue Wool Lightfastness Standard	AATCC Fading Units	Xenon Only kJ/(m^2nm) @ 420 nm	Xenon Only kJ/(m^2nm) 300-400 nm
L2	5	21	864
L3	10	43	1728
L4	20	85[b]	3456
L5	40	170	6912
L6	80	340[b]	13824
L7	160	680	27648
L8	320	1360	55296
L9	640	2720	110592

[a] For color change of Step 4 on the Gray Scale for Color Change.

[b] Verified by experiment using Daylight Behind Glass and Xenon-Arc, Continuous Light. All other values are calculated (see Section 41).

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suitable for multiple exposure levels, such as 10, 20, 40, etc. AFU.

6.7 Black-Panel Thermometer (see 3.3, 44.2).

6.8 Black Standard Thermometer (see 3.4, 44.2).

NOTE: The Black-Panel Thermometer should not be confused with the Black Standard Thermometer. Temperatures as measured by the two different devices generally will not agree at the same test condition. The term *Black Thermometer*, as used in this method, refers to either the Black Panel or Black Standard Thermometers.

6.9 Spectrophotometer or Colorimeter (see 43.2).

6.10 Xenon-Arc Lamp Fading Apparatus optionally equipped with light monitors and control systems (see Appendix A and 44.6).

7. Comparison Standards

7.1 AATCC Blue Wool Lightfastness Standards, as defined in Method 16.3, are preferred for all options. However, the rate of fade of any AATCC Blue Wool Lightfastness Standard by one test option may not agree with that of other test options.

7.2 The reference standard can be any suitable textile material where a history of the rate of color change is known. Reference standards for comparison must be determined and agreed upon by the contractual parties. Standards must be exposed simultaneously with the test specimen. The use of the standard assists in determining time-to-time equipment and test procedure variations. If test results of the exposed standards differ by more than 10% from the known standard data, thoroughly review the test equipment operating conditions, and correct any malfunctions or defective parts. Then, repeat the test.

8. Test Specimen Preparation

8.1 Number of Specimens—For acceptance testing, use at least three replicate specimens of both the material to be tested and the standard for comparison to ensure accuracy unless otherwise agreed upon between the purchaser and the supplier.

8.2 Specimen Cutting and Mounting—Identify each specimen using a label resistant to the environment encountered during the test. Mount in frames such that the surface of the test specimen and the reference specimen are the same distance from the light source. Use covers that avoid specimen surface compression, particularly when testing pile fabrics. The test specimen and the reference standards shall be of equal size and shape. Cut and prepare test specimens for exposure as follows:

8.2.1 Specimen Backing—For all options, if the specimens do not have an integrated backing mount the specimens and standards on white card stock. The card stock shall be white, non-reflecting cardboard. Consult the *AATCC Buyer's Guide* (www.aatcc.org/bg) for information. When mounted test specimens are masked, use test masks approaching zero light transmittance

8.2.2 Fabric—Cut swatches of fabric with the long direction parallel to the machine (warp) direction, at least 70.0 × 120.0 mm (2.75 × 4.7 in.) with the exposed area measuring not less than 30.0 × 30.0 mm (1.2 × 1.2 in.). Secure the backed specimens in the frames supplied with the test apparatus. Ensure that front and back covers of the holders make good contact with the specimens and give a sharp line of demarcation between the exposed and unexposed areas without compressing the specimen unnecessarily (see 44.7 and 44.8). When required to prevent raveling, the specimens may be edged by sewing, pinking or fusing.

8.2.3 Yarns—Wind or fasten yarns on frames of white card stock to a length of approximately 150.0 mm (6.0 in.). Only that portion of the yarns directly facing the radiant energy is evaluated for color change. Wind the yarn on the frame closely packed to at least 25.0 mm (1.0 in.) width. The control specimen must contain the same number of strands as the specimen subjected to exposure. After the exposure has been completed, bind together those yarns facing the light source using 20.0 mm (0.75 in.) masking or other suitable tape to keep the yarns closely packed on the exposure frame for evaluation (see 44.8).

Option 1
Xenon-Arc Lamp, Alternate Light and Dark

Machine Operating Conditions

9. Preparation of Test Apparatus

9.1 Prior to running the test procedure, verify machine operation by using the following test protocol. To enhance the repeatability of test results, install test apparatus in a room where temperature and relative humidity are controlled in accordance with the manufacturer's recommendations.

9.2 Check to see that the machine has been calibrated and maintained within the manufacturer's recommended calibration schedule interval.

9.3 Turn off all rack and specimen spray units, if applicable.

9.4 Set machine operating conditions as follows:

Light source	Xenon Alternate light/dark (Light ON/OFF)
Black Standard temperature	70 ± 1ºC (158 ± 2ºF)
Chamber air temperature	
Light cycle	43 ± 2ºC (110 ± 4ºF)
Dark cycle	43 ± 2ºC (110 ± 4ºF)
Relative Humidity %	
Light cycle	35 ± 5
Dark cycle	90 ± 5
Light cycle hours	
Light – ON	3.8
Light – OFF	1.0
Filter type	See A3.3
Irradiance	
W/m²/nm (at 420 nm)	1.10 ± 0.03
W/m² at (300-400nm)	48 ± 1
Water requirements (Input)	
Type	Demineralized, distilled or reverse osmosis
Solids ppm	Less than 17, preferably less than 8
pH	7 ± 1
Temperature	Ambient 16 ± 5ºC (61 ± 9ºF)

(see Appendix A.) Be sure that the temperature chosen is appropriate for the type of Black Thermometer to be used (see 44.2). Fill the specimen rack with framed white card stock and the required black thermometer unit. The white card stock is used to simulate air flow in the test chamber during the test exposure and should not include the actual test specimens. Support the black thermometer unit in the specimen drum or rack in the same manner as the test specimen frames. Operate and control the test apparatus as described above and further defined by the manufacturer. Operate the test apparatus in this mode and adjust the instrumentation to provide the required black panel or black standard temperature, chamber air temperature and relative humidity. When exterior indicators are not available, read the black thermometer unit through the window in the test chamber door.

9.5 Verify using AATCC Blue Wool Lightfastness Standards following the guidelines in 11.1-11.2.2. If the fade of the L2 or L4 standards do not meet these requirements follow the instrument manufacturer's instructions for calibration and repeat the 20 AFU exposure with fresh L2 or L4 standards. If the fade does meet the requirements described in Section 11, remove the white card stock from the specimen rack and proceed.

9.6 For additional information to prepare and operate the test apparatus refer to the manufacturer's instructions and the following:

9.6.1 Test Standard ASTM G151 and G155 (see 43.4 and 43.5).

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10. Calibration, Verification and AATCC Fading Unit Measurement

10.1 Instrument Calibration—To ensure standardization and accuracy, the instruments associated with the exposure apparatus (that is, light monitor control system, Black Thermometers, chamber air sensor, humidity control system, UV sensors and radiometers) require periodic calibration. Whenever possible, calibration shall be traceable to national or international standards. Calibration schedule and procedure should be in accordance with manufacturer's instructions.

10.1.1 The accuracy of machine operation must be verified by exposure of an applicable AATCC Blue Wool Lightfastness Standard and assessment of the Standard after every 80-100 AATCC Fading Units. Always expose reference standards near the center position of the specimen rack adjacent to the black-panel temperature sensing unit.

11. Verification by AATCC Blue Wool Lightfastness Standards

11.1 Expose the L4 AATCC Blue Wool Lightfastness Standard at the specified temperature, humidity and operating conditions for 20 ± 2 continuous light-on operating hours. After exposure, assess the exposed standard specimen, either visually or instrumentally. Increase or decrease the wattage of the lamps, the time of exposure, of both, and expose additional standard specimens until the change in color of the exposed standard meets one of the following criteria.

11.1.1 Visual Comparison—equals the change in color exhibited by the L4 Standard of Fade applicable to the Lot designation used.

11.1.2 Instrumental Color Measurement—Lot designations of AATCC Blue Wool Lightfastness Standard L4 equals the CIELAB units of color change specified on the calibration certificate supplied with the standard as determined by AATCC EP6, Evaluation Procedure for Instrumental Color Measurement.

11.2 Alternatively, the L2 AATCC Blue Wool Lightfastness Standard may be exposed at the specified temperature, humidity and selected conditions for 20 ± 2 continuous light-on operating hours. After exposure, assess the exposed standard specimen, either instrumentally or compare to a L2 Standard of Fade for 20 AFU. When required, increase or decrease the wattage of the lamps, or the time of exposure, or both, and expose additional standard specimens until the change in color of the exposed standard meets one of the following criteria.

11.2.1 Visual Comparison—equals the change in color exhibited by the L2 Standard of Fade for 20 AFU applicable to the Lot designation used (see 44.5).

11.2.2 Instrumental Color Measurement—The performance of the Blue Wool lot designations of AATCC Blue Wool Lightfastness Standard L2 and L4 must equal the CIELAB unit of color change specified on the calibration certificate supplied with the standard as determined by AATCC EP6.

12. AATCC Fading Unit Measurement by AATCC Blue Wool Lightfastness Standards

12.1 The use of AATCC Blue Wool Lightfastness Standards and AATCC Fading Units provides a common exposure standard across the various exposure methods: daylight, carbon-arc lamp and xenon-arc lamp. The terms *clock hours* and *machine hours* are not valid reporting methods.

12.2 Table I illustrates the number of AATCC Fading Units to produce a color change equal to Step 4 on the Gray Scale for Color Change (see 43.1) or instrumentally (see 43.3) on each of the AATCC Blue Wool Lightfastness Standards.

12.3 For instrumental color measurement, the colorimetric data are calculated using CIE 1964 10° observer data for Illuminant D_{65}. Express the color difference in CIELAB units as directed in AATCC EP6.

NOTE: For Xenon-Arc, Alternating Light and Dark, Option 1, although calibration is conducted using continuous light-on operating hours, it may take more or less operating hours during the actual test cycle due to the inclusion of the dark periods.

13. AATCC Fading Unit Measurement based on Spectral Irradiation

13.1 The 20 AATCC Fading Units are produced by an exposure interval of 85 kJ/(m^2nm) measured at 420 nm when xenon-arc machines are operated at the conditions specified in this test method (see Table I).

Machine Exposure Procedures

14. Machine Exposure, General Conditions

14.1 Specimen Mounting. Mount the test material on the specimen rack. Make sure that all materials are adequately supported. Any displacement of the material toward or away from the source, even by a small distance, may lead to variation in fading between specimens (see 8.2). When alternate light and dark cycles are required, begin exposure at the start of the light cycle.

14.2 In the case of woven, knitted and nonwoven fabrics, unless otherwise specified, ensure that the side normally used as the face is directly exposed to the radiant source.

14.3 Operate the test apparatus on a daily basis until the selected exposure has been completed. Avoid unnecessary delays when interrupting the exposure period to change filters, carbons or lamps, as such delays may contribute to variations in results or lead to errors**.** When available, monitor exposure test chamber conditions with suitable recorders. If necessary, readjust the controls to maintain the specified test conditions.

15. Machine Exposure to a Specified Amount of Radiant Energy

15.1 One-Step Method—Expose the test specimens and applicable standards for 5, 10, 20 or multiples of 20 AATCC Fading Units until the specimen has been exposed to the desired amount of radiant energy defined in terms of AATCC Fading Units measured by simultaneous exposure of the appropriate Blue Wool Standard(s).

15.2 Two-Step Method—Proceed as directed in 15.1, except double the exposure area of the test specimens. After the specimen has been exposed to the first specified level of radiant energy, remove the specimens from the test chamber and mask (cover) one-half of the exposed area and continue the exposure for an additional 20 or multiples of 20 AATCC Fading Units until the specimen has been exposed to the higher desired amount of radiant energy.

15.3 In those machines equipped with irradiation monitors, the AATCC Fading Units of exposure can be determined and controlled by measuring kJ/(m^2nm) at 420 nm (see Table I).

NOTE: The two-step method is preferred for the complete characterization of the lightfastness of a test specimen.

16. Machine Exposure using a Reference Specimen

16.1 Expose the test specimen(s) and reference specimen(s) simultaneously to the required end point in terms of AATCC Fading Units, kilojoules per square meter of irradiance or reference specimen performance (that is, the reference specimen shows a color change equal to Step 4 on the Gray Scale of Color Change).

17. Machine Exposure for Lightfastness Classification

17.1 One-Step Method—Expose test specimen(s) simultaneously with a series of AATCC Blue Wool Lightfastness Standards or determine the number of AATCC Fading Units required to produce a color change in the test specimen equal to Step 4 on the Gray Scale for Color Change (see 44.10).

17.2 Two-Step Method—Proceed as di-

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rected in 17.1, except double the exposure area of the test specimens. After the specimen has been exposed to a color change equal to Step 4 on the Gray Scale for Color Change, remove the specimens from the test chamber and mask (cover) one-half of the exposed area and continue the exposure until the test specimen exhibits a color change equal to Step 3 on the Gray Scale for Color Change (see 44.10).

Option 2
Xenon-Arc Lamp, Continuous Light, Black Standard Option

Machine Operating Conditions

18. Preparation of Test Apparatus

18.1 Prior to running the test procedure, verify machine operation by using the following test protocol. To enhance the repeatability of test results, install test apparatus in a room where temperature and relative humidity are controlled in accordance with the manufacturer's recommendations.

18.2 Check to see that the machine has been calibrated and maintained within the manufacturer's recommended calibration schedule interval.

18.3 Turn off all rack and specimen spray units, if applicable.

18.4 Set machine operating conditions as follows:

Light source	Xenon
	Continuous Light-on
Black Standard temperature	60 ± 3ºC (140 ± 8ºF)
Chamber air temperature	32 ± 5ºC (90 ± 9ºF)
Relative Humidity %	30 ± 5
Filter type	See A3.3
Irradiance W/m²/nm (at 420nm)	1.25 ± 0.2
Irradiance W/m² (300–400nm)	65 ± 1
Water requirements (Input)	
Type	Demineralized, distilled or reverse osmosis
Solids ppm	Less than 17, preferably less than 8
pH	7 ± 1
Temperature	Ambient 16 ± 5ºC (61 ± 9ºF)

(see Appendix A.) Be sure that the temperature chosen is appropriate for the type of Black Thermometer to be used (see 44.2). Fill the specimen rack with framed white card stock and the required black thermometer unit. The white card stock is used to simulate air flow in the test chamber during the test exposure and should not include the actual test specimens. Support the black thermometer unit in the specimen drum or rack in the same manner as the test specimen frames. Operate and control the test apparatus as described above and further defined by the manufacturer. Operate the test apparatus in this mode and adjust the instrumentation to provide the required black panel or black standard temperature, chamber air temperature and relative humidity. When exterior indicators are not available, read the black thermometer unit through the window in the test chamber door.

18.5 Verify using AATCC Blue Wool Lightfastness Standards following the guidelines in 20.1-20.2.2. If the fade of the L2 or L4 standards do not meet these requirements follow the instrument manufacturer's instructions for calibration and repeat the 20 AFU exposure with fresh L2 or L4 standards. If the fade does meet the requirements described in Section 20 remove the white card stock from the specimen rack and proceed.

18.6 For additional information to prepare and operate the test apparatus refer to the manufacturer's instructions and the following:

18.6.1 Test Standard ASTM G151 and G155 (see 43.4 and 43.5).

19. Calibration, Verification and AATCC Fading Unit Measurement

19.1 Instrument Calibration—To ensure standardization and accuracy, the instruments associated with the exposure apparatus (that is, light monitor control system, Black Thermometers, chamber air sensor, humidity control system, UV sensors and radiometers) require periodic calibration. Whenever possible, calibration shall be traceable to national or international standards. Calibration schedule and procedure should be in accordance with manufacturer's instructions.

19.1.1 The accuracy of machine operation must be verified by exposure of an applicable AATCC Blue Wool Lightfastness Standard and assessment of the Standard after every 80-100 AATCC Fading Units. Always expose reference standards near the center position of the specimen rack adjacent to the black-panel temperature sensing unit.

20. Verification by AATCC Blue Wool Lightfastness Standards

20.1 Expose the L4 AATCC Blue Wool Lightfastness Standard at the specified temperature, humidity and operating conditions for 20 ± 2 continuous light-on operating hours After exposure, assess the exposed standard specimen, either visually or instrumentally. Increase or decrease the wattage of the lamps, the time of exposure, of both, and expose additional standard specimens until the change in color of the exposed standard meets one of the following criteria.

20.1.1 Visual Comparison—equals the change in color exhibited by the L4 Standard of Fade applicable to the Lot designation used.

20.1.2 Instrumental Color Measurement—Lot designations of AATCC Blue Wool Lightfastness Standard L4 equals the CIELAB units of color change specified on the calibration certificate supplied with the standard as determined by AATCC EP6.

20.2 Alternatively, the L2 AATCC Blue Wool Lightfastness Standard may be exposed at the specified temperature, humidity and selected conditions for 20 ± 2 continuous light-on operating hours. After exposure, assess the exposed standard specimen, either instrumentally or compare to a L2 Standard of Fade for 20 AFU. When required, increase or decrease the wattage of the lamps, or the time of exposure, or both, and expose additional standard specimens until the change in color of the exposed standard meets one of the following criteria.

20.2.1 Visual Comparison—equals the change in color exhibited by the L2 Standard of Fade for 20 AFU applicable to the Lot designation used (see 44.5).

20.2.2 Instrumental Color Measurement—The performance of the Blue Wool lot designations of AATCC Blue Wool Lightfastness Standard L2 and L4 must equal the CIELAB unit of color change specified on the calibration certificate supplied with the standard as determined by AATCC EP6.

21. AATCC Fading Unit Measurement by AATCC Blue Wool Lightfastness Standards

21.1 The use of AATCC Blue Wool Lightfastness Standards and AATCC Fading Units provides a common exposure standard across the various exposure methods: daylight, carbon-arc lamp and xenon-arc lamp. The terms *clock hours* and *machine hours* are not valid reporting methods.

21.2 Table I illustrates the number of AATCC Fading Units to produce a color change equal to Step 4 on the Gray Scale for Color Change (see 43.1) or instrumentally (see 43.3) on each of the AATCC Blue Wool Lightfastness Standards.

21.3 For instrumental color measurement, the colorimetric data are calculated using CIE 1964 10° observer data for Illuminant D_{65}. Express the color difference in CIELAB units as directed in AATCC EP6.

NOTE: For Xenon-Arc, Alternating Light and Dark, Option 1, although calibration is conducted using continuous light-on operating hours, it may take more or less operating hours during the actual test cycle due to the inclusion of the dark periods.

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Machine Exposure Procedures

22. Machine Exposure, General Conditions

22.1 Specimen Mounting. Mount the test material on the specimen rack. Make sure that all materials are adequately supported. Any displacement of the material toward or away from the source, even by a small distance, may lead to variation in fading between specimens (see 8.2). When alternate light and dark cycles are required, begin exposure at the start of the light cycle.

22.2 In the case of woven, knitted and nonwoven fabrics, unless otherwise specified, ensure that the side normally used as the face is directly exposed to the radiant source.

22.3 Operate the test apparatus on a daily basis until the selected exposure has been completed. Avoid unnecessary delays when interrupting the exposure period to change filters, carbons or lamps, as such delays may contribute to variations in results or lead to errors**.** When available, monitor exposure test chamber conditions with suitable recorders. If necessary, readjust the controls to maintain the specified test conditions.

23. Machine Exposure to a Specified Amount of Radiant Energy

23.1 One-Step Method—Expose the test specimens and applicable standards for 5, 10, 20 or multiples of 20 AATCC Fading Units until the specimen has been exposed to the desired amount of radiant energy defined in terms of AATCC Fading Units measured by simultaneous exposure of the appropriate Blue Wool Standard(s).

23.2 Two-Step Method—Proceed as directed in 23.1, except double the exposure area of the test specimens. After the specimen has been exposed to the first specified level of radiant energy, remove the specimens from the test chamber and mask (cover) one-half of the exposed area and continue the exposure for an additional 20 or multiples of 20 AATCC Fading Units until the specimen has been exposed to the higher desired amount of radiant energy.

23.3 In those machines equipped with irradiation monitors, the AATCC Fading Units of exposure can be determined and controlled by measuring kJ/(m^2nm) at 420 nm (see Table I).

NOTE: The two-step method is preferred for the complete characterization of the lightfastness of a test specimen.

24. Machine Exposure using a Reference Specimen

24.1 Expose the test specimen(s) and reference specimen(s) simultaneously to the required end point in terms of AATCC Fading Units, kilojoules per square meter of irradiance or reference specimen performance (that is, the reference specimen shows a color change equal to Step 4 on the Gray Scale of Color Change).

25. Machine Exposure for Lightfastness Classification

25.1 One-Step Method—Expose test specimen(s) simultaneously with a series of AATCC Blue Wool Lightfastness Standards or determine the number of AATCC Fading Units required to produce a color change in the test specimen equal to Step 4 on the Gray Scale for Color Change (see 44.10).

25.2 Two-Step Method—Proceed as directed in 25.1, except double the exposure area of the test specimens. After the specimen has been exposed to a color change equal to Step 4 on the Gray Scale for Color Change, remove the specimens from the test chamber and mask (cover) one-half of the exposed area and continue the exposure until the test specimen exhibits a color change equal to Step 3 on the Gray Scale for Color Change (see 44.10).

Option 3
Xenon-Arc Lamp, Continuous Light, Black Panel Option

Machine Operating Conditions

26. Preparation of Test Apparatus

26.1 Prior to running the test procedure, verify machine operation by using the following test protocol. To enhance the repeatability of test results, install test apparatus in a room where temperature and relative humidity are controlled in accordance with the manufacturer's recommendations.

26.2 Check to see that the machine has been calibrated and maintained within the manufacturer's recommended calibration schedule interval.

26.3 Turn off all rack and specimen spray units, if applicable.

26.4 Set machine operating conditions as follows:

Light source	Xenon Continuous light-on
Black Panel temperature	63 ± 1ºC (145 ± 2ºF)
Chamber air temperature	43 ± 2ºC (110 ± 4ºF)
Relative Humidity %	30 ± 5
Filter type	See A3.3
Irradiance	
W/m²/nm (at 420nm)	1.10 ± 0.03
W/m² (at 300-400nm)	48 ± 1
Water requirements (Input)	
Type	Demineralized, distilled or reverse osmosis
Solids ppm	Less than 17, preferably less than 8
pH	7 ± 1
Temperature	Ambient 16 ± 5ºC (61 ± 9ºF)

(see Appendix A.) Be sure that the temperature chosen is appropriate for the type of Black Thermometer to be used (see 44.2). Fill the specimen rack with framed white card stock and the required black thermometer unit. The white card stock is used to simulate air flow in the test chamber during the test exposure and should not include the actual test specimens. Support the black thermometer unit in the specimen drum or rack in the same manner as the test specimen frames. Operate and control the test apparatus as described and further defined by the manufacturer. Operate the test apparatus in this mode and adjust the instrumentation to provide the required black panel or black standard temperature, chamber air temperature and relative humidity. When exterior indicators are not available, read the black thermometer unit through the window in the test chamber door.

26.5 Verify using AATCC Blue Wool Lightfastness Standards following the guidelines in 28.1-28.2.2. If the fade of the L2 or L4 standards do not meet these requirements follow the instrument manufacturer's instructions for calibration and repeat the 20 AFU exposure with fresh L2 or L4 standards. If the fade does meet the requirements described in Section 28 remove the white card stock from the specimen rack and proceed.

26.6 For additional information to prepare and operate the test apparatus refer to the manufacturer's instructions and the following:

26.6.1 Test Standard ASTM G151 and G155 (see 43.4 and 43.5).

27. Calibration, Verification and AATCC Fading Unit Measurement

27.1 Instrument Calibration—To ensure standardization and accuracy, the instruments associated with the exposure apparatus (that is, light monitor control system, Black Thermometers, chamber air sensor, humidity control system, UV sensors and radiometers) require periodic calibration. Whenever possible, calibration shall be traceable to national or international standards. Calibration schedule and procedure should be in accordance with manufacturer's instructions.

27.1.1 The accuracy of machine operation must be verified by exposure of an applicable AATCC Blue Wool Lightfastness Standard and assessment of the Standard after every 80-100 AATCC Fading Units. Always expose reference

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standards near the center position of the specimen rack adjacent to the black-panel temperature sensing unit.

28. Verification by AATCC Blue Wool Lightfastness Standards

28.1 Expose the L4 AATCC Blue Wool Lightfastness Standard at the specified temperature, humidity and operating conditions for 20 ± 2 continuous light-on operating hours. After exposure, assess the exposed standard specimen, either visually or instrumentally. Increase or decrease the wattage of the lamps, the time of exposure, of both, and expose additional standard specimens until the change in color of the exposed standard meets one of the following criteria.

28.1.1 Visual Comparison—equals the change in color exhibited by the L4 Standard of Fade applicable to the Lot designation used.

28.1.2 Instrumental Color Measurement—Lot designations of AATCC Blue Wool Lightfastness Standard L4 equals the CIELAB units of color change specified on the calibration certificate supplied with the standard as determined by AATCC EP6.

28.2 Alternatively, the L2 AATCC Blue Wool Lightfastness Standard may be exposed at the specified temperature, humidity and selected conditions for 20 ± 2 continuous light-on operating hours. After exposure, assess the exposed standard specimen, either instrumentally or compare to a L2 Standard of Fade for 20 AFU. When required, increase or decrease the wattage of the lamps, or the time of exposure, or both, and expose additional standard specimens until the change in color of the exposed standard meets one of the following criteria.

28.2.1 Visual Comparison—equals the change in color exhibited by the L2 Standard of Fade for 20 AFU applicable to the Lot designation used (see 44.5).

28.2.2 Instrumental Color Measurement—The performance of the Blue Wool lot designations of AATCC Blue Wool Lightfastness Standard L2 and L4 must equal the CIELAB unit of color change specified on the calibration certificate supplied with the standard as determined by AATCC EP6.

29. AATCC Fading Unit Measurement by AATCC Blue Wool Lightfastness Standards

29.1 The use of AATCC Blue Wool Lightfastness Standards and AATCC Fading Units provides a common exposure standard across the various exposure methods: daylight, carbon-arc lamp and xenon-arc lamp. The terms *clock hours* and *machine hours* are not valid reporting methods.

29.2 Table I illustrates the number of AATCC Fading Units to produce a color change equal to Step 4 on the Gray Scale for Color Change (see 43.1) or instrumentally (see 43.3) on each of the AATCC Blue Wool Lightfastness Standards.

29.3 For instrumental color measurement, the colorimetric data are calculated using CIE 1964 10° observer data for Illuminant D_{65}. Express the color difference in CIELAB units as directed in AATCC EP6.

NOTE: For Xenon-Arc, Alternating Light and Dark, Option 1, although calibration is conducted using continuous light-on operating hours, it may take more or less operating hours during the actual test cycle due to the inclusion of the dark periods.

30. AATCC Fading Unit Measurement Based on Spectral Irradiation

30.1 The 20 AATCC Fading Units are produced by an exposure interval of 85 kJ(m^2nm) measured at 420 nm when xenon-arc machines are operated at the conditions specified in this test method (see Table I).

Machine Exposure Procedures

31. Machine Exposure, General Conditions

31.1 Specimen Mounting. Mount the test material on the specimen rack. Make sure that all materials are adequately supported. Any displacement of the material toward or away from the source, even by a small distance, may lead to variation in fading between specimens (see 8.2). When alternate light and dark cycles are required, begin exposure at the start of the light cycle.

31.2 In the case of woven, knitted and nonwoven fabrics, unless otherwise specified, ensure that the side normally used as the face is directly exposed to the radiant source.

31.3 Operate the test apparatus on a daily basis until the selected exposure has been completed. Avoid unnecessary delays when interrupting the exposure period to change filters, carbons or lamps, as such delays may contribute to variations in results or lead to errors**.** When available, monitor exposure test chamber conditions with suitable recorders. If necessary, readjust the controls to maintain the specified test conditions. Verify calibration of the test apparatus during the test cycle.

32. Machine Exposure to a Specified Amount of Radiant Energy

32.1 One-Step Method—Expose the test specimens and applicable standards for 5, 10, 20 or multiples of 20 AATCC Fading Units until the specimen has been exposed to the desired amount of radiant energy defined in terms of AATCC Fading Units measured by simultaneous exposure of the appropriate Blue Wool Standard(s).

32.2 Two-Step Method—Proceed as directed in 32.1, except double the exposure area of the test specimens. After the specimen has been exposed to the first specified level of radiant energy, remove the specimens from the test chamber and mask (cover) one-half of the exposed area and continue the exposure for an additional 20 or multiples of 20 AATCC Fading Units until the specimen has been exposed to the higher desired amount of radiant energy.

32.3 In those machines equipped with irradiation monitors, the AATCC Fading Units of exposure can be determined and controlled by measuring kJ/(m^2nm) at 420 nm (see 30.1 and Table I).

NOTE: The two-step method is preferred for the complete characterization of the lightfastness of a test specimen.

33. Machine Exposure using a Reference Specimen

33.1 Expose the test specimen(s) and reference specimen(s) simultaneously to the required end point in terms of AATCC Fading Units, kJ/m^2 of irradiance or reference specimen performance (that is, the reference specimen shows a color change equal to Step 4 on the Gray Scale of Color Change).

34. Machine Exposure for Lightfastness Classification

34.1 One-Step Method—Expose test specimen(s) simultaneously with a series of AATCC Blue Wool Lightfastness Standards or determine the number of AATCC Fading Units required to produce a color change in the test specimen equal to Step 4 on the Gray Scale for Color Change (see 44.10).

34.2 Two-Step Method—Proceed as directed in 34.1, except double the exposure area of the test specimens. After the specimen has been exposed to a color change equal to Step 4 on the Gray Scale for Color Change, remove the specimens from the test chamber and mask (cover) one-half of the exposed area and continue the exposure until the test specimen exhibits a color change equal to Step 3 on the Gray Scale for Color Change (see 44.10).

Evaluation of Results

35. Conditioning

35.1 After the test exposure is completed, remove the test specimens and comparison standards from exposure.

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Condition in a dark room at standard conditions for testing textiles, as directed in ASTM D1776, Standard Practice for Conditioning and Testing Textiles, [65 ± 5% RH and 21 ± 2°C (70 ± 4°F)] for a minimum of 4 h before evaluation.

36. Assessment of Color Change

36.1 Compare the exposed portion to the masked control or to an unexposed original portion of the specimen (preferred), as specified in a material specification or purchase order. Complete characterization of the lightfastness of a test specimen requires evaluation at more than one level of exposure. A difference in color between unexposed material and the masked portion of the exposed specimen indicates that the textile has been affected by some agent other than light, such as heat or a reactive gas in the atmosphere. Although the exact cause of this difference in color may not be known, it should be noted in the report when it occurs.

36.2 Quantify the color change using either AATCC EP1, Evaluation Procedure for Gray Scale for Color Change, or AATCC EP7, Evaluation Procedure for Instrumental Assessment of the Change in Color of a Test Specimen, at the specified exposure level whether in AATCC Fading Units, kilojoules of radiant energy, or compared to a reference standard (see 44.10).

36.3 Determine total color difference (ΔE_{CIELAB}) and the difference in lightness, chroma, and hue (ΔL^*, ΔC^*, ΔH^*). Use instruments that provide values based on the CIE 1976 equation using illuminant D_{65} and 10° observer data. For instruments with diffuse geometry, include the specular component of reflectance in the measurements (refer to AATCC EP6).

37. Acceptance Based on Simultaneous Exposure of a Reference Specimen

37.1 Assess color change of the material (not blue wool) as directed in Section 36 in terms of the agreed upon reference specimen.

37.2 Assess the lightfastness of the material as follows:

37.2.1 Satisfactory—If the test specimen exhibits a color change equal to or less than the reference specimen at the exposure level when the reference specimen shows a color change equal to Step 4 on the Gray Scale for Color Change.

37.2.2 Unsatisfactory—If the test specimen exhibits a color change greater than the reference specimen at the exposure level when the reference specimen shows a color change equal to Step 4 on the Gray Scale for Color Change.

37.3 Alternatively, buyer and seller may make acceptance judgment based on color difference as described in 36.3.

38. Classification Based on the AATCC Blue Wool Lightfastness Standards (See Note 44.11)

39. Classification above L7 AATCC Blue Wool Lightfastness Standard (See Note 44.12)

40. Report (See Table IV)

40.1 Reporting at the minimum should include the following information:

a) Operator's name and test date
b) Specimen identification
c) Colorfastness to light rating/Light classification
d) Acceptance compared to reference specimen or masked area (Yes/No)
e) Colorfastness to light rating determined by AATCC Gray Scale or Instrumentally
f) Classification method
g) Reference standard
h) Temperature ambient (dry bulb) ºC, black panel ºC or black standard ºC
i) Exposure controlled by: AATCC Blue Wool, radiant energy or other
j) Total radiant energy
k) Type of apparatus, model no., serial no., manufacturer, specimen rack (inclined, 2-tier, 3-tier or horizontal) and type of water supply
l) Option employed
m) Elapsed exposures time
n) Mounting procedure (backed or unbacked)
o) Deviations from AATCC TM16.3 or the performance of the reference standard, if any

Precision and Bias

41. Precision

41.1 Interlaboratory Test Summary—Committee RA50 has conducted extensive studies to evaluate the use of radiation monitoring devices to terminate exposures in lightfastness testing. Data has been collected in interlaboratory studies using controlled irradiance, xenon-arc equipment and in daylight exposures conducted during a two-year period in both Arizona and Southern Florida. In both studies, one laboratory conducted instrumental measurement of the color change for all exposed specimens.

The interlaboratory studies were undertaken, using eight different lightfastness standard fabrics, to determine the definition of 20 AATCC Fading Units in terms of measured radiation. These studies showed that acceptable agreement between laboratories can be obtained for lightfastness testing providing the following variables are controlled: irradiance level, black-panel temperature, ambient temperature and relative humidity. Overall, there was less than 10% variability in the instrumentally determined color change of specimens exposed in different laboratories. For all specimens tested the standard deviation was equivalent to less than one-half step on the Gray Scale for Color Change. As a result of these tests, 20 AATCC Fading Units was established at 85 kJ/(m^2nm) when measured at 420 nm (approximately 21.5 continuous light-on operating hr) when tested at the conditions specified for Xenon-Arc Lamp, Continuous Light, Option 3.

For the daylight studies, 16 different fabrics, in addition to AATCC and ISO Blue Wool Lightfastness standard fabrics, were exposed. An exposure series was begun each quarter year at two locations over a two-year period. Exposures were terminated based on instrumental measurement of radiant energy dosage. A wide variation in climatic conditions was encountered during the test period. The data obtained clearly shows that the color change of individual specimens is affected differently by variations in temperature, humidity, atmospheric contaminants, etc.; however, the single most significant variable is radiation. The variation in color change resulting from exposure during different years, locations, and seasons, averaged ± 30%.

A more detailed summary of these test results was presented to the 14th meeting of ISO, Technical Committee 38, Subcommittee 1 as Document 38/1 N 993, USA Report on Monitoring of Radiation during Lightfastness Testing.

42. Bias

42.1 The colorfastness to natural and artificial light can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

43. References

43.1 AATCC EP1, Evaluation Procedure for Gray Scale for Color Change (see 44.5).

43.2 AATCC EP6, Evaluation Procedure for Instrumental Color Measurement (see 44.5).

43.3 AATCC EP7, Evaluation Procedure for Instrumental Assessment of the Change in Color of a Test Specimen (see 44.5).

43.4 ASTM G151, Standard Practice for Exposing Nonmetallic Materials in Accelerated Test Devices that Use Laboratory Light Sources (see 44.9).

43.5 ASTM G155, Standard Practice for Operating Xenon-Arc Light Apparatus for Exposure of Nonmetallic Materials (see 44.9).

44. Notes

44.1 Historically, AATCC Blue Wool

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Lightfastness Standards, except L2, were specially prepared by blending varying proportions of wool dyed with a very fugitive dyestuff, Erio Chrome Azurole B (C.I. 43830) and wool dyed with a fast dyestuff, Indigosol Blue AGG (C.I. 73801). Each resultant higher numbered standard is twice as colorfast as the preceding numbered standard. AATCC Blue Wool Lightfastness Standards and the ISO numbered Blue Wool Lightfastness Standards (as used in ISO 105-B01) produce different ratings and therefore cannot be used interchangeably. Lots 8 and 9 of the L2 AATCC Blue Wool Lightfastness Standard were batch dyed. L2 can be used to measure 5 or 20 AFU exposures, producing two distinct fading endpoints. L2 Standards of Fade for 5 AFU and 20 AFU are available (see 44.5).

44.2 Black Thermometers are used to control an artificial weathering device and to provide an estimate of the maximum temperature of specimens exposed to a radiant energy source. There are two types of Black Thermometers. One type is referred to as a "Black Panel Thermometer" which is uninsulated and is made of metal. The other type is referred to as a "Black Standard Thermometer" which is insulated and is made of metal with a plastic backing. As a point of information, some ISO specifications specify the use of a "Black Standard Thermometer." Typically, Black Standard Thermometers indicate higher temperatures than Black Panel Thermometers under the same exposure conditions.

The Black Thermometer units indicate the absorbed irradiance minus the heat dissipated by conduction and convection. Keep the black face of these thermometer units in good condition. Follow the manufacturer's recommendations for proper care and maintenance of Black Thermometers.

44.2.1 For Black Panel Thermometers: Testing temperature is measured and regulated by a Black Panel Thermometer unit mounted on the specimen rack to permit the face of it to receive the same exposure as the test specimen. Black Panel Thermometers shall consist of a metal panel at least 70 × 150 mm and not less than 45 × 100 mm whose temperature is measured with a thermometer or thermocouple whose sensitive portion is located in the center of and in good contact with the panel. The side of the panel facing the light source shall be black with a reflectance of less than 5% throughout the spectrum of light reaching the specimen; the side of the panel not facing the light source shall be open to the atmosphere within the exposure chamber.

44.2.2 For Black Standard Thermometers: Testing temperature is measured and regulated by a black standard thermometer unit mounted on the specimen rack to permit the face of it to receive the same exposure as the test specimen. The Black Standard Thermometer shall consist of a plane of stainless steel plate measuring 70 × 40 mm with a thickness of about 0.5 mm, whose temperature is measured by a thermal resistor, with good heat-conducting properties, fitted to the reverse side. The metal plate is fixed to a plastic plate so that it is thermally insulated. The side of the panel facing the light source shall be black with a reflectance of less than 5% throughout the spectrum of light reaching the specimen.

44.3 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

44.4 More uniform and reproducible fading of the AATCC Blue Wool Lightfastness Standards and test specimens is achieved when backed with white cardboard. The color difference values in the initial determination of the end point for the AATCC Blue Wool Lightfastness Standards were determined from exposures with such backing. Although tolerances are given for both the AATCC Blue Wool Lightfastness Standards, every effort should be made to achieve the midpoint value given for these standards. For referee purposes, the AATCC Blue Wool Lightfastness Standards will be exposed in multiples of three and the average color change must equal the CIELAB unit of color change specified on the calibration certificate supplied with the applicable standard as determined by AATCC EP6 (see 44.5).

44.5 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; web site: www.aatcc.org; e-mail: ordering@aatcc.org.

44.6 Refer to ASTM G151 and G155 for guidance on design and performance requirements for instruments specified in this method. Further guidance is found in Appendix A. Consult the *AATCC Buyer's Guide* for potential equipment information.

44.7 Pile fabric, such as carpets, which have fibers that may shift position, or texture which may make evaluations in small areas difficult should be tested with an exposed area of not less than approximately 40.0 × 50.0 mm (1.6 × 2.0 in.). Expose sufficient size or multiple specimens to include all colors in the specimen.

44.8 Specimen frames must be made of stainless steel, aluminum, or suitably coated steel to avoid contaminating the specimens with metallic impurities that might catalyze or inhibit the degradation. When specimens are fastened with staples, they should be of the nonferrous type overcoated to avoid contamination of the specimen by corrosion products. Metal frames must have a dull finish and be designed to avoid reflectances that could influence the performance of the material. Frames shall conform to the curvature of the specimen rack. The size of the frame is determined by the type specimens required for individual property requirements.

44.9 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832.9500; fax: +1.610.832.9555; web site: www.astm.org.

44.10 An automated electronic grading system may be used as long as the system has been demonstrated to provide results that are equal to and provide equal or better repeatability and reproducibility than an experienced grader performing visual evaluations.

44.11 **Classification Based on the AATCC Blue Wool Lightfastness Standards**

44.11.1 One Step Exposure—Classify lightfastness of the material by:

(a) comparison of the color change of the test specimen to that of a simultaneously exposed series of AATCC Blue Wool Lightfastness Standards (see Table II), or

(b) determination of the number of AATCC Fading Units required to produce a color change in the test specimen equal to Step 4 of the Gray Scale for Color Change (see Table I).

44.11.2 Two Step Exposure—Classify lightfastness of the material by:

(a) determination of the number of AATCC Fading Units required to produce color changes in the test specimen equal to both a

Table II—Classification by AATCC Blue Wool Lightfastness Standards[a]
Test Specimen Color Change

Less Than Standard	Equal To But Not Greater Than Standard	More Than Standard	Lightfastness Class	AATCC Fading Units (AFU)
—	—	L2	L1	
—	L2	L3	L2	5
L2	—	L3	L2-3	
—	L3	L4	L3	10
L3	—	L4	L3-4	
—	L4	L5	L4	20
L4	—	L5	L4-5	
—	L5	L6	L5	40
L5	—	L6	L5-6	
—	L6	L7	L6	80
L6	—	L7	L6-7	
—	L7	L8	L7	160
L7	—	L8	L7-8	
—	L8	L9	L8	320
L8	—	L9	L8-9	
—	L9	—	L9	640

[a] The following are examples for using Table II to assign lightfastness classifications:

The test specimen is exposed simultaneously with standards L4, L5, and L6. After exposure and conditioning, the color change exhibited by the test specimen is less than that exhibited by the standards L4 and L5 but greater than that exhibited by the standard L6. The test specimen would be assigned a Lightfastness Classification of L5-6, or use the following example.

The test specimen is examined after each exposure increment until it exhibits a color change equal to Step 4 on the Gray Scale for Color Change. If this occurs after 40 AFU and before 80 AFU exposure, the test specimen would be assigned a Lightfastness Classification of L5-6.

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Table III—Classification by AATCC Blue Wool Lightfastness Standards Above L7

Number of L7 Standards Exposed				
Less Than	Equal To But Not Greater Than	More Than	Lightfastness Class	Equivalent AATCC Fading Unit (AFU)
—	2	—	L8	320
3	—	2	L8-9	—
—	3	—	L8-9	480
4	—	3	L8-9	—
—	4	—	L9	640
5	—	4	L9-10	—
—	5	—	L9-10	800
6	—	5	L9-10	—
—	6	—	L9-10	960
7	—	6	L9-10	—
—	7	—	L9-10	1120
8	—	7	L9-10	—
—	8	—	L10	1280
etc.[a]	etc.[a]		etc.[a]	etc.[a]

[a] A classification increase of 1 represents the interval when the equivalent AATCC Fading Units are doubled from the previous whole number classification. Any test specimen for which the number of L7 Standards fall between two whole number classifications is assigned both the lower and higher classification defining that interval.

Step 4 and Step 3 on the Gray Scale for Color Change (see Table I).

44.11.3 Assign both classifications: the Step 3 level appears first, followed by the Step 4 level in parentheses. For example, a L5(4) classification would illustrate a L5 classification at Step 3 color change and a L4 classification at Step 4 color change. When only one classification number is assigned, it shall represent the number of AATCC Fading Units to produce a Step 4 color change.

44.12 **Classification above L7 AATCC Blue Wool Lightfastness Standard**

44.12.1 Using Table III, classify lightfastness above the L7 AATCC Blue Wool Lightfastness Standard according to the total number of consecutive L7 standards exposed to Step 4 on the Gray Scale for Color Change during the exposure cycle that is required to produce a Step 4 color change on the test specimen, and Table III.

45 History

45.1 Revised in 2020 to update and added three flow charts in Appendix B to reflect each of the three options.

45.2 Revised 1971, 1974, 1978, 1981, 1982, 1990 (Supersedes AATCC TM16-1987, TM16A-1988, TM16C-1988, TM16D-1988, TM16E-1987, TM16F-1988 and TM16G-1985), 1993, 2003, 2004, 2012 (Supersedes AATCC TM16-2004), 2014; reaffirmed 1977, 1998; editorially revised 1983, 1984, 1986, 1995, 1996, 2009, 2016, 2019.

45.3 Developed in 1964 by AATCC Committee RA50; Related to Option 3-ISO 105-B02.

Appendix A
Xenon-Arc Lamp Fading Apparatus

A. Xenon-Arc Lamp Fading Apparatus

A1 Different types of xenon-arc test apparatus may be utilized provided that the test apparatus is capable of automatically controlling irradiance level, humidity level, chamber air temperature, and Black Panel or Black Standard Thermometer temperature.

A2 The design of the test chamber may vary, but it should be constructed from corrosion resistant material.

A3 Xenon-Arc Light Source. The xenon-arc test apparatus utilizes a long-arc quartz-jacketed xenon-arc lamp as the source of irradiance which emits radiation from below 270 nm in the ultraviolet through the visible spectrum and into the infrared.

While all of the xenon-arc lamps are of the same general type, different size lamps operated in different wattage ranges are employed in several sizes and types of apparatus. In each of the various models, the size and dimension of the specimen rack varies according to the lamp size and the wattage at which it is operated to provide an irradiance at the face of the specimen of 1.10 ± 0.03 W/(m^{2}nm) measured at 420 nm or equivalent when exposed in standard holders.

A3.1 Aging of the xenon burners or filters can result in changes in lamp spectrum. Changes in lamp spectrum may also be caused by accumulation of dirt or other residue in or on the burner envelope.

A3.2 Filter—In order for xenon-arcs to simulate terrestrial daylight, filters must be used to remove short wavelength UV radiation. In addition, filters to remove infrared radiation may be used to prevent unrealistic heating of test specimens that can cause thermal degradation not experienced during outdoor exposures. Filters to reduce irradiance at wavelengths shorter than 310 nm must be used to simulate daylight filtered through window glass.

The instrument manufacturers' recommendations should be used to provide the appropriate spectrum (see A3.4 below). Replace filters when chipped, cracked, or when discoloration or milkiness develops. Discard xenon lamp tubes and filters at the manufacturer's recommended time intervals or sooner, or when 20 AATCC Fading Units can no longer be attained in 20 ± 2 continuous light-on operating (clock) hours.

A3.3 Spectral Irradiance of Filtered Xenon-Arc—Fig. A1 shows the desired relative spectral power distribution for filtered xenon-arcs comply with these limits. The acceptable limits for variation of the relative spectral power distribution shown in Fig. A1 are on file at the AATCC Technical Center.

A3.4 Follow the device manufacturer's instructions for recommended maintenance.

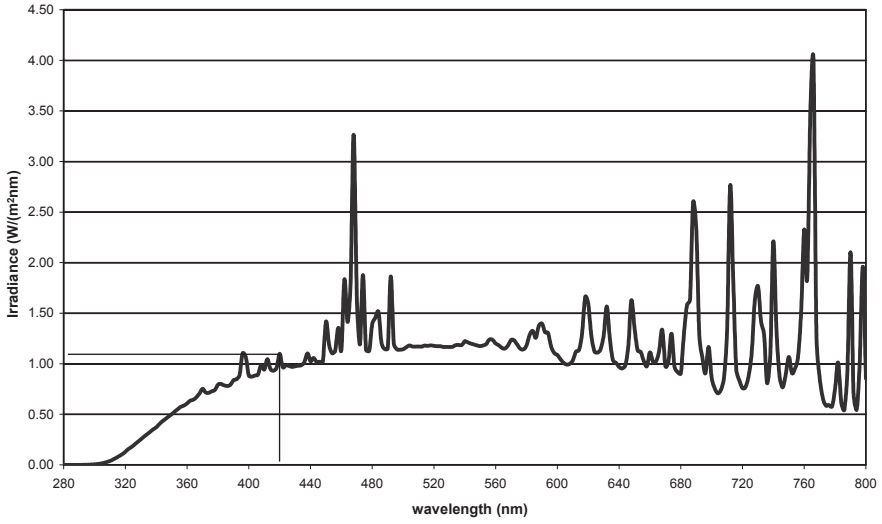


Fig. A1—Filtered Xenon Lamp Spectral Power Distribution Controlled at 1.1 W/(m^{2}nm) at 420 nm

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Table IV—Reporting Form

Operator's Name_____ Date _____

Sample Identification_____

Material Exposed: Face_____ Back_____

Colorfastness to Light Rating _____ Lightfastness Classification _____

Acceptance Compared to Reference Sample (Yes/No) _____

Test Specimen Compared To: Masked Portion _____

Unmasked Portion _____ Unexposed Original_____

Colorfastness to Light Rating determined by:

Gray Scale for Color Change _____

Instrumentally, Name Type _____

Classification Method _____

Reference Standard _____

Temperature Controlled By: Ambient (Dry Bulb)_____ °C

Black Panel_____°C Black Standard_____°C

Exposure Controlled By: AATCC Blue Wool Lightfastness Standards _____

Radiant Energy_____ Other _____

Total Radiant Energy_____

Type of Test Apparatus _____ Model No._____

Serial No. _____ Manufacturer's Name _____

Specimen Rack: Inclined _____ 2-Tier _____ 3-Tier _____ Horizontal _____

Type of Water Supply _____

Option Employed _____ Elapsed Exposure Time _____

Mounting Procedure: Backed _____ Unbacked _____

Sample Rotation Schedule _____ % Relative Humidity _____

For Daylight Behind Glass, report the following:

Geographical Location _____

Exposure Dates: From_____ To _____

Exposure Latitude_____ Exposure Angle _____

Exposed Behind Window Glass: Yes/No_____ If Yes, Specify Type _____

Daily Ambient Temperature: Minimum _____°C Maximum _____°C Avg. _____°C

Daily Black Panel Temperature: Minimum _____°C Maximum _____°C Avg. _____°C

Test Environment Temperature: Minimum _____°C Maximum _____°C Avg. _____°C

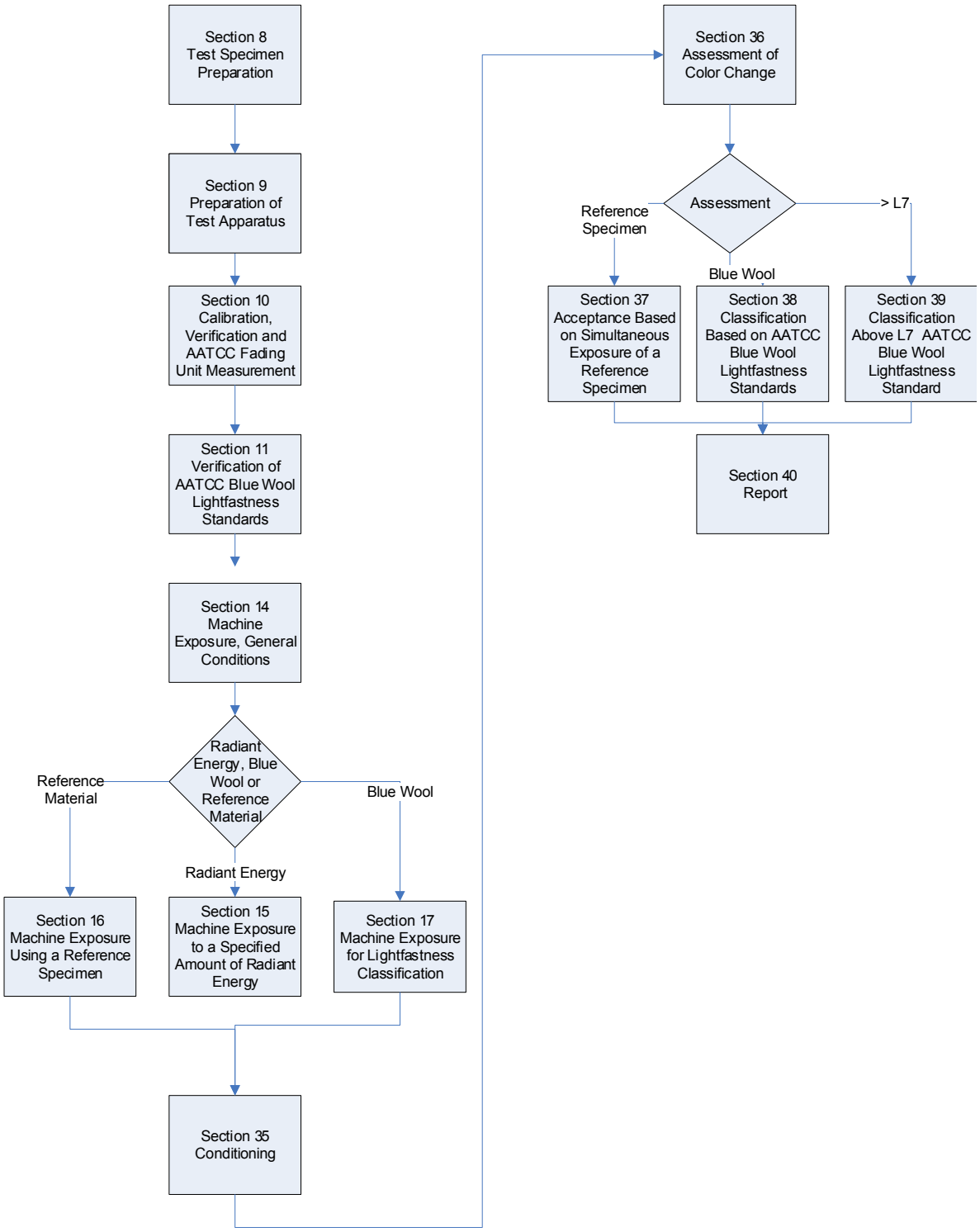
Daily % Relative Humidity: Minimum _____ Maximum _____ Avg. _____

Hours of Wetness: Rain_____ Rain and Dew_____

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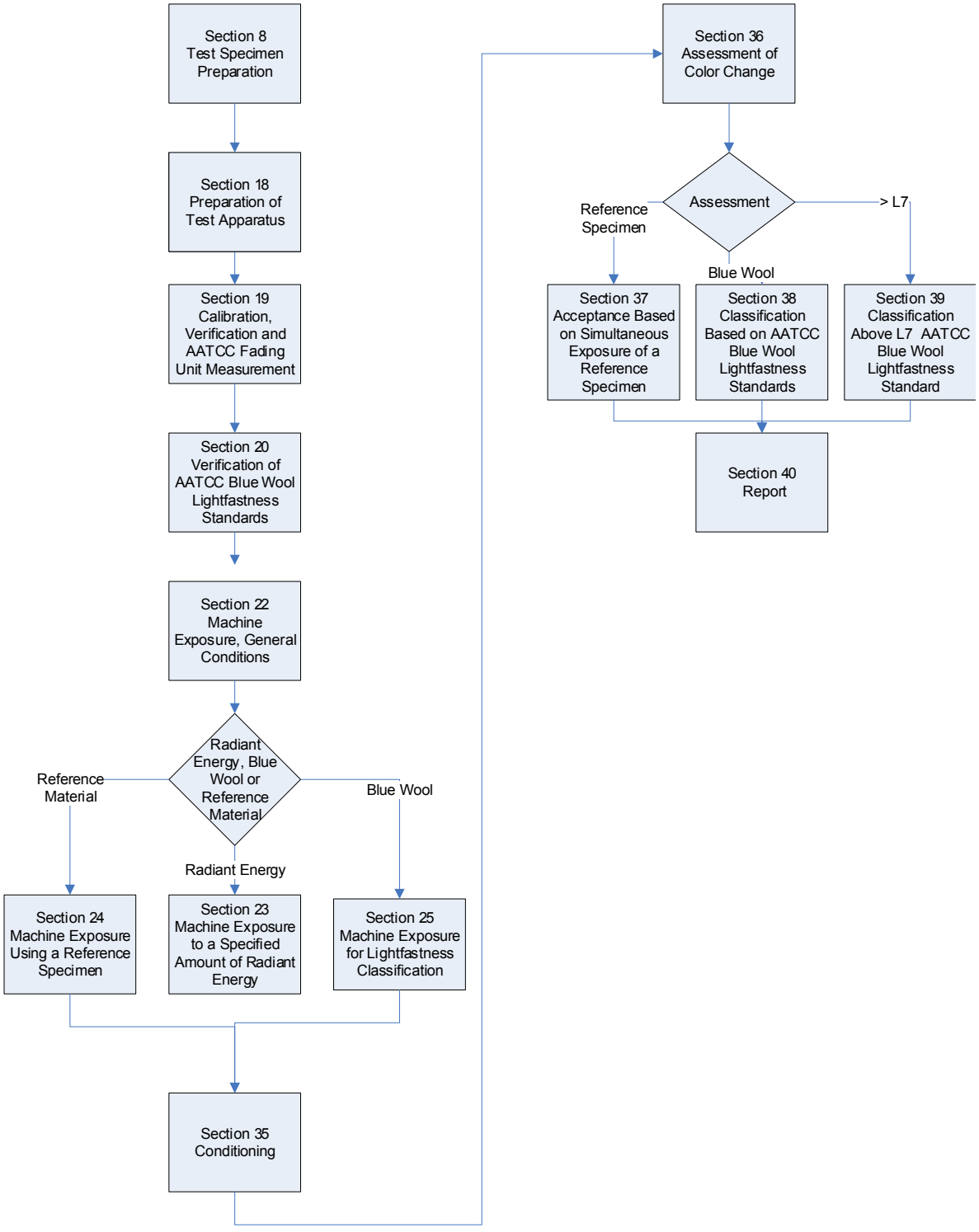
Appendix B
Flow Chart for AATCC TM16.3, Xenon-Arc

16.3 Flowchart 1 for Option 1 Xenon-Arc Lamp, Alternating Light and Dark



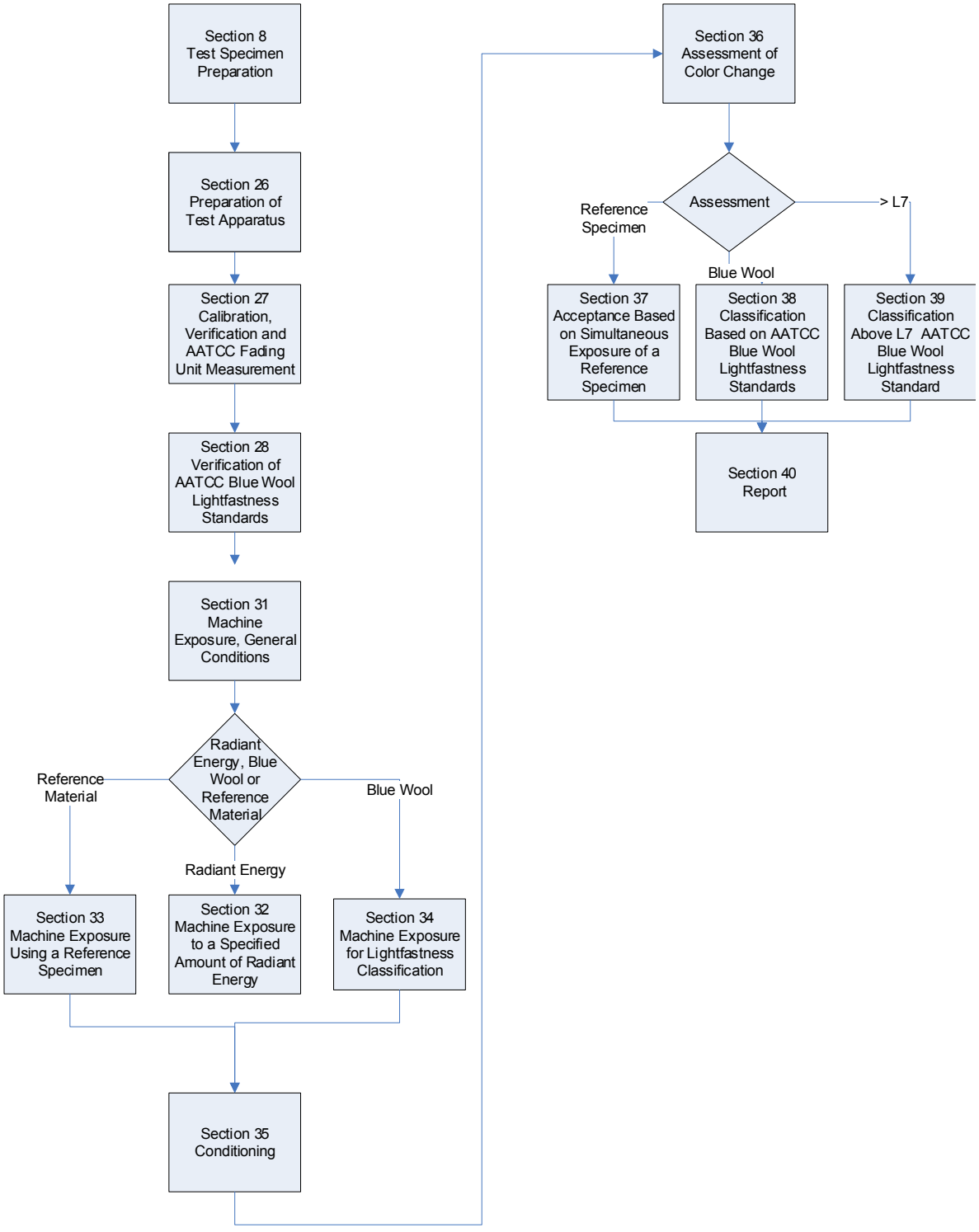
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16.3 Flowchart 2 for Option 2 Xenon-Arc Lamp, Continuous Light, Black Standard Option



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16.3 Flowchart 3 for Option 3 Xenon-Arc Lamp, Continuous Light, Black Panel Option



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Test Method for Wetting Agents, Evaluation of

Developed in 1932 by AATCC Committee RR8; jurisdiction transferred in 2003 to AATCC Committee RA63; jurisdiction transferred in 2022 to AATCC Committee RA114; reaffirmed 1943, 1971, 1977, 1980, 1989, 2005, 2010, 2014, 2018; revised 1952, 1999; editorially revised and reaffirmed 1974, 1985, 1994; editorially revised 1988, 1991, 2004, 2008, 2019, 2022.

1. Purpose and Scope

1.1 This test method determines the efficiency of ordinary commercial wetting agents.

2. Principle

2.1 A weighted cotton test skein is dropped into a tall cylinder containing a water solution of a wetting agent. The time required for a string stirrup connecting the weight and the skein to relax is recorded as the sinking time.

3. Terminology

3.1 **wetting agent,** n.—a chemical compound which when added to water lowers both the surface tension of the liquid and its interfacial tension against the solid material.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Wear chemical goggles, rubber gloves and apron in preparing wetting agent stock solution.

5. Apparatus and Materials (see 10.1)

5.1 Hooks of standard weight and attached anchors (see 10.2 and 10.3).

5.2 Volumetric flasks, 1000 mL.

5.3 Beaker, 1500 mL.

5.4 Graduated cylinders, 500 mL.

5.5 Bulb pipette (or aspirator), 100 mL.

5.6 Bulb transfer pipettes, assorted sizes.

5.7 Cotton yarn, greige unboiled, 2-ply, in 5-g skeins (see 10.4).

5.8 Water, distilled (see 10.5).

5.9 Graph paper, log-log.

6. Test Solutions

6.1 Stock solutions of the agents to be tested are normally prepared to contain 50.0 g of agent per liter unless the solubility is so poor that less must be employed. The wetting agent is first thoroughly dissolved in about a quarter of the necessary distilled water at a temperature above 80°C and is then diluted to the final volume with cold distilled water. Aliquot portions of 5, 7, 10, 15, 25, 35, 50, 75 and 100 mL of the 5% stock solution taken with delivery bulb pipettes and diluted with suitable water (see 10.4) to 1000 mL correspond, respectively, to concentrations of 0.25, 0.35, 0.50, 0.75, 1.25, 1.75, 2.50, 3.75 and 5.00 g of wetting agent per liter. This range of concentrations is sufficient for the study of any commercial product.

7. Procedure

7.1 The diluted solution for test is poured from the liter volumetric flask into a 1500 mL beaker to ensure mixing. The solution in the beaker is then divided equally between two 500 mL graduated cylinders. If the more dilute solutions are tested first, the mixing beaker and cylinders need not be rinsed out and dried each time. The operator must wait after the cylinders have been filled until all bubbles *below* the surface of the solution have risen to the top before the sinking tests are made. The operator may advantageously prepare solutions for 6 more cylinders while waiting for the bubbles to rise. Foam on the surface of the solution is removed either with a 100 mL bulb pipette or with an aspirator. Where there is little tendency for exhaustion of the wetting agent on the test skeins—practically always true for cotton—it is permissible to use the same diluted solution over again several times rather than to make a new diluted solution for each new skein. In this case only one 500 mL cylinder can be filled repeatedly from one liter of solution of a certain concentration.

7.2 Since temperature often markedly affects wetting, standard temperatures of 25°C, 50°C, 70°C and 90°C have been chosen for testing so as to include the complete commercially useful range. It is most convenient to attain a temperature of 25°C merely by using water which has been brought to the correct temperature in a large pail. For tests at higher temperatures, heat the diluted solution for test in the mixing beaker to a temperature somewhat above that required, pour the solution into the cylinder and then allow it to cool back to the testing temperature.

7.3 For a determination, a 5.00 g skein of yarn is folded enough times to form a loop 45.7 cm around. A 91.4 cm skein is most convenient and can be formed into a 45.7 cm loop with only 2 folds; a 137.2 cm skein requires 3 folds; a 182.9 cm skein, 4 folds and a 228.6 cm skein requires 5 folds. The hook with its anchor is fastened at one end of the folded skein and the skein is cut through with shears at the opposite end. The cut skein is drawn through the fingers when testing wetting agents in order to make it more compact. Any threads which have been tied around the skein to correct its weight are folded into the skein near the hook. The skein is held in one hand with the anchor dangling into the wetting solution contained in the 500 mL graduated cylinder. A stopwatch held in the other hand is started just as the skein is released into the solution and it is stopped when the buoyant skein definitely starts to sink to the bottom of the cylinder. The skein before sinking must be entirely covered with solution and yet it must possess enough buoyancy from the air within the yarn to keep the linen thread taut between the anchor and the hook (see Fig. 1). The average of at least four determinations of sinking time should be obtained for each concentration of wetting agent. An average deviation of 10-12% in sinking time may be expected (see 10.6).

8. Evaluation

8.1 The method of handling the data

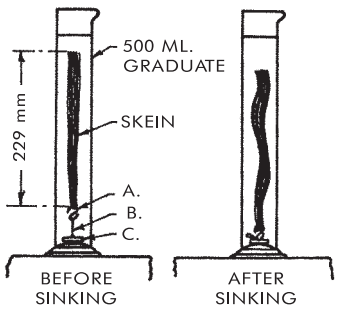


Fig. 1—Position of skein before and after sinking.

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obtained by the technique described above is very important. The most useful plot where a complete curve is desired is one made on log-log graph paper with logarithmic coordinates along both axes but reading directly in antilogarithms. Values of concentrations of wetting agent are shown on the horizontal scale, or X-axis, the value on the left being 0.1 g and the value on the right 10 g/L. In like manner, sinking times in seconds are shown on the vertical scale, or Y-axis, the value on the bottom being 1 or 10 s and the value at the top being 100 s. A smooth curve is drawn through the points. For most products this curve will be a straight line (see Fig. 2).

8.2 When sinking curves for two products have the same slopes for a 3.0 g hook, it has been found that they have the same slopes for any weight of hook by the method of testing described here, and they even have quite closely the same slopes for any other method of testing in which cotton is employed. Under such conditions it is then logical to assume that solutions that cause wetting in equal times for the same lot of cotton under similar conditions are equivalent. Comparisons of relative costs of wetting can then be made as illustrated in Table I, where the cost for 378.5 L of wetting agent is calculated from the following formula:

8.3 Cost for 378.5 L of solution at wetting conc. = 0.835 × (wetting conc. in g/L × (cost per g).

8.4 When two products have markedly different slopes for plots representing the relationship between sinking times and concentrations on log-log graph paper, great care must be exercised in interpreting the wetting data.

Table I—Comparison of Two Original Wetting Agents*

Standard		New Product
3.000 g	Wt. of hook	3.000 g
25°C	Temperature	25°C
20¢	Cost per lb	18¢
1.95	25 s wetting conc., g/L	2.44
100	Equivalent parts	125
32.5¢	Cost for 378.5 L of soln. at wetting conc.	36.7¢

*Medium: distilled water, neutral.

9. Precision and Bias

9.1 *Precision*. A *within-laboratory* test was conducted to establish the precision of the test method. Three testers, on three different days, performed four checks per level of surfactant. An average of the four checks was computed for each level of surfactant. Table II indicates the average, and the standard deviation, from the data of the three testers.

9.1.1 The coefficient of variation is being used to determine the bias of this method. The data used was generated from the *within-laboratory* results of three testers. Table III indicates the percentage per level of surfactant.

Table II—Sinking Time Average and Standard Deviation

Surfactant Level	Sinking Time	
	Avg. of 3 Testers	S.D. of 3 Testers
0.25 g/L	120.00 s	0 s
0.35 g/L	120.00 s	0 s
0.50 g/L	77.00 s	13.18 s
0.75 g/L	32.75 s	3.70 s
1.25 g/L	14.42 s	1.70 s
1.75 g/L	8.58 s	0.80 s
2.50 g/L	4.75 s	0.50 s
3.75 g/L	3.10 s	0.14 s
5.00 g/L	2.00 s	0 s

Table III—Coefficient of Variation at Different Surfactant Levels

Surfactant Level	CV %
0.25 g/L	0
0.35 g/L	0
0.50 g/L	17
0.75 g/L	11
1.25 g/L	12
1.75 g/L	9
2.50 g/L	11
3.75 g/L	5
5.00 g/L	0

9.2 *Bias*. Wetting agents can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

10. Notes

10.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

10.2 The hook of a standard weight and the attached anchor are prepared as follows: A piece of No. 10 B&S gauge copper wire about 6.51 cm long is bent into the form of a hook as illustrated by "A" in Fig. 3 and then the weight of the bent hook is adjusted to exactly 3.000 g. Nickel, silver and stainless steel wire are even more suitable than copper for this purpose because they are more corrosion resisting. The anchor, "C," is a flat, cylindrical, lead slug weighing over 40 g and having a diameter of 25 mm and a thickness of about 4.7 mm. In the center of the anchor is soldered a loop of wire to serve as a small ring, or eye, for attaching the anchor to the hook with a fine linen thread, "B," at a distance apart of 19 mm. If many products are to be tested, prepare at least two hooks and anchors.

10.3 In the comparison of wetting agents it has been found that a 3.0 g hook gives a concentration for 25 s wetting which is most often fairly close to the concentration employed in practice for original wetting in various mill processes. If now, however, the concentration of wetting agent found satisfactory for the particular work in the mill is much higher or much lower than the concentration obtained with the 3.0 g hook, then a hook of a different weight should be employed for making comparisons between products which are valid for the particular situation.

For comparisons at low concentrations a 6.0 g, or even a 9.0 g hook, is employed for a sinking time of 25 s. Only products with sinking curves of similar slopes show the same equivalent values for 0.5 g, 1.5 g, 3.0 g, 6.0 g,

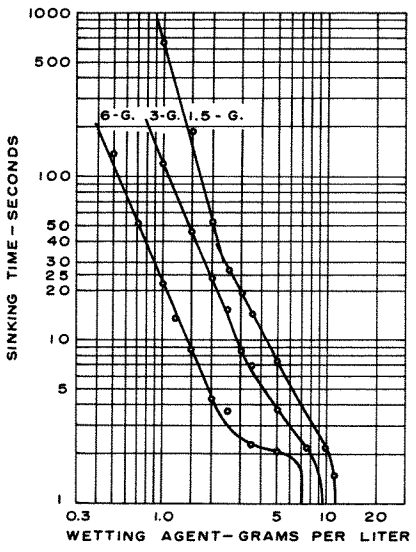


Fig. 2—Plot of results. (See text under Evaluation.)

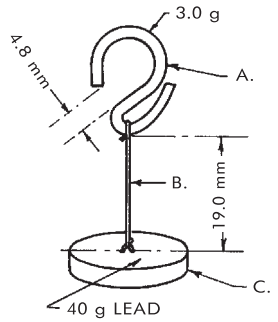


Fig. 3—Hook for use in test of wetting agents.

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and 9.0 g hooks at any standard sinking time.

For comparisons at higher concentrations than those corresponding to a sinking time of 25 s for a 3.0 g hook, a 0.5 g or a 1.5 g hook may be employed. For quicker and even more reliable results of higher concentrations it is advantageous to employ an electronic timing device and standard sinking times of 10 s and 4 s. The procedure otherwise is exactly the same as with the 3 g hook at 25 s.

10.4 40s/2 (30 tex) combed Peeler yarn with a lisle twist of 708.7-787.4 turns per meter and a balanced construction is suitable. All the tubes of greige cotton yarn used for making 5 g skeins for a given series of wetting tests must be from the same lot of cotton. To average out the slight differences still remaining between different tubes of the same lot and to increase the number of skeins which are closely alike in wetting properties for the series, it is urged that each skein be reeled simultaneously from 4-12 tubes of yarn. In the case of purchased skeins, the weights of the skein must be corrected individually to within 10 mg of 5 g.

10.5 The quality of the water used in the testing of wetting agents must be given careful consideration. The stock solution is best prepared with distilled water. When it is not known under what conditions the wetting agent is to be employed, distilled water may likewise be used for final solution. On the other hand, for the simulation of mill practice, the final solution for test—and even the first stock solutions—should be made up with water from the mill and with any ingredients which are necessary to duplicate exactly the chemical composition of the solutions as they will be used in practice. If this is done, the pH will take care of itself automatically, although the careful chemist will want to check the acidity or alkalinity of the final test solutions either colorimetrically or electrically.

For the purpose of uniformity, standard concentrations of acid and alkali are employed for routine tests made in other than neutral solution. It is recommended that tests be made at different temperatures in the presence of 5 or 10 g of conc. sulfuric acid (sp. gr. 1.84), 5 or 10 g of sodium carbonate and 5 or 10 g of caustic soda per liter of final solution, respectively.

10.6 The spread among readings for sinking time is considerably reduced by placing the cylinder for test on a vibrating surface. The bubbles are then more uniformly liberated, that is, occasional lingering is eliminated, and greater uniformity follows. It is to be observed that vibration produces lower average sinking concentrations for a given standard time and standard hook.

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Test Method for Fiber Analysis: Qualitative

1. Purpose and Scope

1.1 This test method describes physical, chemical and microscopical techniques for identifying textile fibers found in textile products and sold in the United States. Fibers may be examined in raw fiber form or taken from yarn or fabric.

1.2 These test methods may be used to identify generic fiber types as defined by the Textile Fibers Products Identification Act and subsequent rules and regulations of the Federal Trade Commission or ISO 2076, Textiles man-made fibres-generic names. Quantitative methods for determining percentages in blends of fibers are covered by AATCC TM20A, Test Method for Fiber Analysis: Quantitative.

1.3 The test methods apply to fibers which are shown below grouped by generic classifications:

Natural Fibers	Man-Made Fibers
Cellulose (Vegetable)	acetate
cotton	secondary
hemp	triacetate
jute	acrylic
linen	anidex
ramie	aramid
sisal (agave)	meta aramid
manila hemp (abaca)	para aramid
	azlon
	glass
Keratin (Animal)	metallic
alpaca	modacrylic
camel	novoloid
cashmere	nylon
horse	6
llama	6,6
mohair	11
rabbit	nytril
vicuna	olefin
wool	lastol
yak	polyethylene
	polypropylene
	polybenzimidazole
Fibroin (Animal)	polyester
silk	elastrelle
Bombyx (cultivated)	rayon
tussah (wild)	cuprammonium
	lyocell
	viscose
	rubber
	saran
	spandex
	triexta
Mineral	vinal
asbestos	vinyon

2. Use and Limitations

2.1 This test method describes a number of procedures—microscopical examination, solubility in solvents, melting point, refractive index, and micro-Fourier Transform Infrared Spectroscopy—which should be used in combination to identify a fiber type. For identifying certain fibers some procedures will be found to be more effective than others.

2.1.1 For example, microscopical examination is particularly useful in characterizing the natural fibers. It must be used with caution on man-made fibers since they are frequently produced in a number of modifications which alter the longitudinal or cross-sectional appearance. In addition, man-made fibers may contain some or no delusterant or other additive particles. Filaments of a given type may vary in size or cross-sectional shape. Individual filaments may have two or more component sections of the same or different generic types.

2.1.2 Even natural fibers show a fairly wide variation in typical cross-section. No specific specimen will look exactly like the pictures published. A sufficient number of fibers should be examined to cover the range of appearance in any specimen.

2.2 Successful identification of fibers depends upon experience and familiarity with the fibers. The identification of an unknown fiber is best made by comparison with properly identified fibers used as reference standards. For this reason it is desirable to have available at least one representative fiber sample from each generic class of fibers, which can be used for comparative identification.

2.3 This test method provides means for identifying the generic classification of the common fiber types. In special cases, as when dealing with fibers not described in this method or attempting to distinguish between products of different suppliers of the same generic types, one must consult standard texts on fiber identification or technical bulletins issued by suppliers of man-made fibers. See references Section 13.

3. Terminology

3.1 For definitions of technical terms, refer to the AATCC M11 in this *Manual of International Test Methods and Procedures*.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 In preparing, dispensing and handling the reagents outlined in Section 6, use chemical goggles or face shield, impervious gloves and an impervious apron. Concentrated acids should be handled only in an adequately ventilated laboratory hood. CAUTION: Always add acid to water.

4.4 All poisonous and flammable reagents should be mixed and handled only in an adequately ventilated laboratory hood. CAUTION: Acetone and ethyl alcohol are highly flammable and should be stored in the laboratory only in small containers away from heat, open flame and sparks.

4.5 An eyewash/safety shower should be located nearby and an organic vapor respirator should be readily available for emergency use.

4.6 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 12.1).

5. Apparatus (see 12.2)

5.1 Compound microscope with matched objectives and eye pieces to achieve magnifications of 100-500× and equipped with a polarizer and analyzer.

5.2 Glass slides and cover glasses.

5.3 Dissection needles.

5.4 Scissors (small) and tweezers (fine).

5.5 Cross-sectioning device of one type such as listed below.

5.5.1 Stainless steel plate, 2.54 × 7.62 × 0.0254 cm (1 × 3 × 0.01 in.) drilled with 0.09 cm (0.04 in.) diameter holes. Soft copper magnet wire AWG #34, 0.016 cm (0.0063 in.) diameter.

5.5.2 Microtome, hand.

5.6 Razor blades, sharp, thin, single edge or double edge with holder.

5.7 Density gradient tube, glass approximately 2.5 cm (1 in.) in diameter

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and 45 cm (18 in.) long with a sealed bottom and a 24/40 standard taper joint glass cap closure at top to avoid moisture pickup or evaporation of solvents. Tiny glass spheres of calibrated density may be used as density standards.

5.8 Melting point apparatus consisting of a heated block, temperature measuring device such as a thermometer and means for controlling the rate of heating and viewing the specimen at low magnification. The instrument should have a range of 100-300°C or more, and an accuracy of ± 1 degree over the entire range.

5.9 Micro-FTIR Instrument.

5.10 Differential Scanning Calorimeter.

6. Reagents and Materials (see 12.2)

6.1 Mounting Reagents.

6.1.1 Mineral oil, U.S.P. or other immersion fluid.

6.1.2 Collodion, solution of nitrocellulose (4 g/100 mL) in 1:3 alcohol/ethyl ether.

6.2 Bleaching Reagent.

6.2.1 Hydrosulfite-caustic solution. Dissolve 2 g sodium hydrosulfite and 2 g sodium hydroxide in 100 mL of water.

6.3 Staining Reagents.

6.3.1 Zinc chloro-iodide reagent. Dissolve 20 g of zinc chloride in 10 mL of water. Add 2.1 g of potassium iodide and 0.1 g of iodine dissolved in 5 mL of water. Add a leaf of iodine.

6.3.2 Acid phloroglucinol reagent. Dissolve 2 g of phloroglucinol in 100 mL of water. Use with equal volume of concentrated hydrochloric acid.

6.4 Refractive Index Immersion Liquids.

6.4.1 Hexadecane (cetane) C. P. Grade, RI = 1.434.

6.4.2 Alpha chloronaphthalene, RI = 1.633. Poisonous. Avoid inhaling vapors.

6.4.3 Blends of the above. Assume that the refractive index varies linearly by volume of ingredients. For example, to make a liquid of RI = 1.550 mix 42 parts by volume of hexadecane with 58 parts by volume of alpha chloronaphthalene.

6.5 Fiber Solvents.

6.5.1 Acetic acid, glacial. Corrosive. Do not get in eyes or on skin.

6.5.2 Acetone, reagent grade. (CAUTION: Highly flammable.)

6.5.3 Sodium hypochlorite solution, 5%. Home laundry bleach is satisfactory.

6.5.4 Hydrochloric acid, concentrated reagent, 20%. Dilute 50 mL of concentrated hydrochloric acid, 38%, to 95 mL with distilled water.

6.5.5 Formic acid, 85%. Corrosive. Do not get in eyes or on skin.

6.5.6 1,4-Dioxane.

6.5.7 *m*-Xylene.

6.5.8 Cyclohexanone.

6.5.9 Dimethylformamide. (CAUTION: If spilled on skin, wash off immediately.)

6.5.10 Sulfuric acid solution, 59.5 ± 0.25% by weight, density 1.4929 ± 0.0027 g per mL at 20°C. Weigh into a beaker 59.5 g of concentrated sulfuric acid (sp gr 1.84). Weigh into a 250 mL Pyrex Erlenmeyer flask 40.5 g of distilled water. Cautiously add *the acid to the water* swirling and cooling in ice water or under a tap. *Wear goggles!* The solution becomes very hot and may boil and spatter if not cooled during mixing. After the solution has cooled to 20°C (68°F) adjust the density to a value between 1.4902 and 1.4956 g per mL.

6.5.11 Sulfuric acid solution, (70 ± 1% by weight, density 1.6105 ± 0.0116 g per mL at 20°C). Weigh 70 g of concentrated sulfuric acid (sp gr 1.84) and 30 g of water and mix with the same precautions as in 6.5.10. After the solution has cooled to 20°C (68°F) adjust the density to a value between 1.5989 and 1.6221 g per mL.

6.5.12 *m*-Cresol, reagent grade. Poisonous. Use in ventilated hood.

6.5.13 Hydrofluoric acid, 49% reagent grade. Very dangerous. Use goggles and hood under suction. Do not inhale vapors or allow to contact skin.

7. Sampling

7.1 To obtain a representative sample for identification it is necessary to consider the following:

7.1.1 If the sample is loose fiber or yarn, it may contain one fiber only or it may be an intimate blend of two or more fibers.

7.1.2 A yarn sample may be a ply or wrapped core blend of two or more yarns. These yarns in turn may be the same or different and may themselves contain a blend of fibers.

7.1.3 Woven or knitted fabrics may be constructed of yarns from one fiber type or from yarns made with multiple fiber types. Further, fabrics may be made with yarns that have different individual fiber types or multiple fiber types in the length or width directions of the fabric construction, that may require separate analysis.

7.1.4 Different fiber types may be dyed to the same color. Contrarywise, the same fiber type may appear in different colors in the finished article by mixing stock dyed or yarn dyed materials or using fibers with modified dyeing characteristics.

7.1.5 It is essential that the specimens selected represent the entire sample of raw fiber, yarn or fabric under examination.

8. Specimen Preparation

8.1 In many cases the identity of an unknown fiber can be established without pretreatment.

8.2 When the presence of starch, wax, oil or other coating obscures the appearance of the fiber, try heating in warm to hot distilled water to remove the foreign matter. If this fails, try extracting with organic solvents, 0.5% hydrochloric acid or 0.5% sodium hydroxide. Some fibers such as nylon are damaged by acid and some such as azlon, silk and wool, are damaged by caustic treatment (see 9.7).

8.3 To separate vegetable fiber bundles, treat with 0.5% sodium hydroxide solution, rinse well with water and dry.

8.4 To strip dye from colored fibers, especially the cellulosic fibers, heat for 30 min at 50°C with hydrosulfite caustic solution (see 6.2.1).

9. Procedure

9.1 The identification of fibers is usually carried out by subjecting specimens to a variety of selected tests until enough information is obtained to make a satisfactory judgment as to the generic class or specific type. The selection of tests and order in which they are performed can change depending on the knowledge already available and results of the preliminary tests.

9.2 Visual and Microscopical Examination.

9.2.1 Examine the sample of material submitted for identification. Note form (loose fibers, yarn, fabric, etc.), color, fiber length and fineness, uniformity of appearance and probable end-use. If the sample is a fabric, separate out yarns by unraveling or cutting. If the fabric is woven, separate warp and filling yarns. If yarns differ in color, luster, size or other apparent ways, make a physical separation of these yarns for separate identification.

9.2.2 The fibers may be identified using either light or scanning electron microscopy. If using transmitted light microscopy, place a small quantity of the fibers on a glass slide, tease fibers apart, mount with a drop of mineral oil or other immersion fluid, cover with a cover slip and examine under the microscope.

9.2.3 Observe the fiber characteristics as seen in traverse section and classify into four general classes.

9.2.4 Fibers with surface scales. These are the animal hair fibers listed in Purpose and Scope (see 1.3). All of the natural protein fibers listed except silk are included in this group. Continue detailed microscopical examination including cross-section examination (see 9.3). Compare with the characteristics in Table I, photographs in the appendix of this method, and known specimens of hair fibers (see 12.3). Cashmere is the fine (dehaired) undercoat fibers produced by a Cashmere goat (Capria hircus laniger). The fiber is generally non-medullated and has a mean maximum diameter of 19 microns. The coefficient of variation

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around the mean shall not exceed 24%. There can be no more than 3% (by weight) of cashmere fibers over 30 microns. Additional confirmation of this class may be made by burning test (see 9.5); density (see 9.6); and solubility (see 9.7).

9.2.5 Fibers with Cross Markings. These are the vegetable fibers other than cotton listed in 1.3. Continue detailed microscopical examination including cross-section examination (see 9.3). Compare with characteristics as given in Table II, photographs and known specimens of vegetable fibers. To distinguish linen and ramie from hemp, observe the direction of rotation on drying (see 9.8). If the fibers are light in color, stain with zinc chloro-iodide reagent and with acid phloroglucinol as directed in 9.9. Additional confirmation of this class may be made by tests described in 9.4, 9.5, 9.6 and 9.7.

9.2.6 Convoluted Fibers. This class includes cotton and tussah silk. The two are easily distinguished by cross-section (see 9.2), burning test (see 9.5) and solubility (see 9.7) or micro-FTIR (see Appendix II). If fibers are light in color, they may be distinguished by staining with zinc chloro-iodide reagent (see 9.9).

9.2.7 Other Fibers. This class includes all of the man-made fibers, Bombyx silk and asbestos. The latter two can be identified by microscopical appearance including cross-section (see 9.3). Burning tests (see 9.5) and solubility tests (see 9.7) are especially significant for asbestos and are useful in confirming the presence of silk.

The man-made fibers are best identified by micro-FTIR, solubility, melting point, refractive indices and other optical characteristics and density: properties which relate to chemical nature rather than physical shape. Cross-sections may vary and staining tests can be misleading since modifications of a generic type may dye differently. Used with discrimination and in combination with other tests, cross-sections and dye tests may sometimes be helpful in narrowing down the number of possibilities. Metallic fibers have a distinctive shiny appearance, but a shiny residue may not indicate a metallic fiber. Using a transmitted light microscope at 5×, 10× and 20× magnification powers will not reveal diffraction, interference and gloss effects that can deceptively appear to be metallic in nature. Chemical (colorimetric) and/or instrumental (Atomic Absorption Spectrophotometric, Inductively Coupled Plasma Spectrometry and/or Energy Dispersive X-ray or X-ray Fluorescence) analyses should be performed to confirm their presence.

Examples of x-ray and elemental analysis are provided in the *AATCC Fiber Identification Supplement* (see 12.4).

9.3 Microscopical Cross-Section Examination.

Table I—Characteristics of Fibers with Scales on Surface

Note: The characteristic designated by a capital letter X in this table is that which is especially significant.

Microscopical Appearance	Alpaca	Camel	Cashmere[d,e]	Horse	Llama	Mohair	Vicuna	Wool[c]	Yak
Longitudinal:									
Epidermis:									
Pronounced	—	—	—	—	—	—	—	X	—
Faint	x	x	X	x	x	X	x	—	x
Coronal[a]	—	x	x	X	—	—	x	x	x
Imbricate[b]	x	x	—	—	x	x	—	x	x
Smooth edge	—	x	x	x	—	x	x	x	—
Serrated edge	X	—	—	x	X	—	—	—	x
Medulla:									
Occurrence:									
Usually present	x	—	—	x	X	—	—	—	—
Seldom present	—	x	X	—	—	x	x	x	—
Never present	—	—	—	—	—	—	—	—	x
Type:									
Fragmental	x	x	—	—	x	—	x	x	—
Interrupted	x	—	—	—	x	x	x	x	—
Continuous	x	—	—	x	x	x	—	—	—
Size (ratio to fiber diameter):									
Under ¼	—	—	—	—	—	x	x	x	—
¼ to ½	x	—	—	x	x	x	—	x	—
Over ½	—	—	—	X	—	—	—	—	—
Pigment:									
Diffuse	—	—	x	—	—	—	—	—	—
Streaky	x	X	x	—	x	—	x	—	x
Granular	—	x	—	x	—	—	—	—	x
None	—	—	—	—	—	X	—	X	—
Cross-Section:									
Contour:									
Round to oval	—	x	x	x	—	x	x	x	x
Oval to elongated	x	—	—	—	x	—	—	x	—
Kidney	x	—	—	—	x	—	—	—	—
Medulla contour:									
Round to oval	—	—	—	x	—	x	x	x	—
Oval to elongated	X	—	—	—	X	—	—	—	—
Kidney to dumbbell	X	—	—	—	X	—	—	—	—
Pigment distribution:									
Uniform	x	X	—	—	x	—	x	—	x
Centric	—	—	X	—	—	—	—	—	—
Eccentric	—	—	—	X	—	—	—	—	—
Fineness (in μm)									
Average	26 to 28	18	15 to 19.0	—	26 to 28	—	13 to 14	—	18 to 22
Range	10 to 50	9 to 40	5 to 30	—	10 to 40	10 to 90	6 to 25	10 to 70	8 to 50
Number of scales per 100 μm	—	—	6 to 7	—	—	Under 5.5	—	Over 5.5	Over 7

[a] Coronal means crownlike, and refers to scales in which the visible scale edge completely encircles the fiber.
[b] Imbricate means overlapping, and refers to scales in which the visible scale edges overlap like shingles on a roof and cover only a part of the fiber circumference.
[c] The term wool is used here to represent clothing wool and not carpet wool.
[d] The longitudinal/epidermal appearance of cashmere fiber while generally fainter than that of sheep's wool is not as faint as some specialty fibers such as camel and alpaca.
[e] The average diameter for cashmere fibers was broadened in 2000 based on a review of published research concerning Asian *capra hircus*.

Table II—Appearance of Fibers with Cross Markings or Swellings

Longitudinal Section:	Linen	Hemp	Ramie
Ratio, lumen of fiber diameter	under ⅓	usually over ⅓	over ⅓
Cell ends	pointed	blunt or forked	blunt
Cross-Section:			
Contour	sharp polygon	rounded polygon	elongated polygon
Lumen	round or oval	irregular	irregular

9.3.1 Obtain a parallel bundle of fibers or yarn. Thread a loop of copper wire through one hole in the stainless steel plate. Catch the bundle or yarn in the loop and pull it through the hole. Use sufficient fibers to pack the hole full. If necessary, use some extra, readily identifiable other fiber to fill up the hole.

9.3.2 With a sharp razor blade make a smooth cut on both sides of the plate.

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9.3.3 Examine the section in air, or covered by a mounting fluid like mineral oil under a cover slip. Use transmitted light and useful magnification of 200-500×. Compare with photographs in Appendix or cross-sections of known fibers.

9.3.4 If the Hardy type microtome is used, follow instructions with the instrument. Insert a bundle of fibers or yarn into the slot. Slide the tongue into the slot to compress the bundle. Adjust the quantity of fibers for a tight package. Cut off both sides. Apply a drop of collodion to one cut face, wait until it shows through on the other side and then apply a drop to the other side. Let dry thoroughly. Slice off excess collodion and fiber on both sides of the plate with a sharp razor blade.

9.3.5 Attach the auxiliary plunger, and with the screw advance the embedded tuft through the slot until it protrudes 20-40 μm above the plate. Add a drop of collodion and let dry about 5 min or until firm. Slice off collodion and embedded fiber section with a sharp razor blade held at about 45° angle with the slot. Cut with a single stroke.

9.3.6 Remove the section, transfer to a glass slide, mount with a drop of mineral oil or other immersion liquid, add a cover slip and examine. Continue to cut sections as described in 9.3.5 until good clear cross-sections are obtained. Examine and compare to known cross-sections.

9.4 Refractive Index.

9.4.1 Place a small sample of the fiber on a glass slide. Immerse in a drop of a chloronaphthalene and cetane mixture (or equivalent) of refractive index 1.55.

9.4.2 Insert the polarizer into the substage of the microscope to provide light polarized in the 6-12 o'clock direction. Align the long dimension of a fiber in the same direction. Close the diaphragm of the substage condenser so as to provide axial illumination.

9.4.3 Focus carefully on the outlines of the fiber. With the fine adjustment, raise the focus to just above the top of the fiber. If the fiber is roughly a cylinder, it will act like a lens. If the refractive index of the fiber is *higher* than the immersion liquid, the fiber will act like a positive lens. A bright line of light will move into the middle of the fiber as the focus is raised. If the refractive index of the fiber is *lower* than the immersion liquid, the light will flare out as the focus is raised and the middle of the fiber will become darker. Repeat until certain of the direction.

9.4.4 The test can be seen best with round fibers. On flat ribbons it may be easier to see movement of a bright line—the Becke line—at the outlines of the fiber. Movement is in the same direction, toward the medium of higher refractive index as the focus is raised.

9.4.5 Rotate the specimen 90° and repeat the test.

9.4.6 Rotate the specimen 45°. Insert the analyzer in the body tube or eyepiece to provide crossed polars. Observe if the fiber appears very bright (strong birefringence), dim (weak birefringence) or dark (no birefringence).

9.4.7 Refer to Table IV for refractive index of fibers in the long and cross direction and an estimate of birefringence. Consider that the retardation depends on the thickness of the fiber specimen as well as the extent of drawing of a given specimen. Compare with known fibers of similar diameter for birefringence.

9.4.8 Select other immersion liquids and repeat 9.4.1 through 9.4.5. As the refractive index of the liquid approaches that of the fiber, the outlines of the fiber become less distinct. Match the liquid to the fiber within 0.01 refractive index.

9.5 Burning Test.

9.5.1 Take a small tuft of fibers (held in tweezers) and place close to the side of a small flame. Note if the fibers melt or shrink from the flame.

9.5.2 Move the fibers into the flame. Note whether the fibers will burn when held in the flame. Remove from the flame very slowly and carefully. Note whether the fibers continue to burn outside of the flame. Make sure that the fiber was ignited before making this latter observation.

9.5.3 Blow out the flame, if still burning, and smell the smoke. Note the odor and examine the color and nature of any ash residue.

9.5.4 Compare the behavior observed with that listed in Table III and that of known reference fibers. Flame retardant modifications of some fibers—cotton, rayon, acetate and modacrylic, for example—are available. Burning is retarded; odor on burning and ash may be changed. Colored fibers—especially those colored by pigments—will retain color in the colored residue.

9.5.5 Odors of some burning fibers are quite characteristic. The animal fibers and man-made protein fibers (azlon) have the odor of burning hair or feathers. The vegetable fibers and regenerated cellulose (rayon) smell like burning paper. The smell of burning rubber is a familiar odor. Other man-made fibers—such as acrylic, nylon and spandex—have characteristic odors which can be recognized with experience.

9.6 Density.

9.6.1 Prepare a density gradient col-

Table III—Reaction to Flame

	Melts Near Flame	Shrinks From Flame	Burns in Flame	Continues to Burn	Appearance of Ash
Natural Fibers					
silk	yes	yes	yes	slowly	soft black bead
wool	yes	yes	yes	slowly	irregular black
cellulose	no	no	yes	yes	light grayish
asbestos	no	no	no	no	may blacken
Man-Made Fibers					
acrylic acetate azlon nytril	yes	yes	yes	yes	hard black irregular shaped bead
polybenzimidazole	no	no	no	no	may blacken
polyester	yes	yes	yes	yes	hard black round bead
nylon	yes	yes	yes	yes	hard gray round bead
olefin vinal	yes	yes	yes	yes	hard tan bead
modacrylic saran vinyon	yes	yes	yes	no	hard black irregular bead
metallic	yes	yes	no	no	metal bead
glass	yes	slowly	no	no	hard clear bead
rubber	yes	yes	yes	no	irregular mass
spandex	yes	no	yes	yes	fluffy black or gray
anidex	yes	no	yes	yes	brittle black irregular bead
rayon	no	no	yes	yes	none
aramid	no	yes	yes	no	hard black bead
novoloid	no	no	brief	no	carbon
triexta	yes	yes	yes	yes	hard black round bead

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Table IV—Physical Properties of Fibers

	Refractive Index				
	Long	Cross	Birefringence	Density g/cc	Melting Point Degree C
Natural Fibers					
asbestos (chrysotile)	1.50-1.55	1.49	strong	2.1-2.8	over 350
cellulose	1.58-1.60	1.52-1.53	strong	1.51	none
silk	1.59	1.54	strong	1.32-1.34	none
wool and other hair	1.55-1.56	1.55	weak	1.15-1.30	none
Man-Made Fibers					
acetate, secondary	1.47-1.48	1.47-1.48	weak	1.32	260
acetate, tri.	1.47-1.48	1.47-1.48	weak	1.30	288
acrylic	1.50-1.52	1.50-1.52	weak, negative	1.12-1.19	none
anidex	opaque		—	1.22	softens at 190
aramid	—	—	strong	1.38	chars at 400
azlon	1.53-1.54	1.53-1.54	none	1.30	none
glass	1.55	1.55	none	2.4-2.6	850
metallic	opaque		—	varies	over 300
modacrylic	1.54	1.53	weak	1.30 or 1.36	188* or 120
novoloid	1.5-1.7		none	1.25	none
nylon	1.57	1.51	strong	1.12-1.15	213-225
nylon 6,6	1.58	1.52	strong	1.12-1.15	256-265
nytril	1.48	1.48	nil	1.20	218
polybenzimidazole	>1.70	>1.70	moderate	1.4	none
polyester	1.71-1.73 or 1.63	1.53-1.54	intense	1.38 or 1.23	250-260 or 282
polyethylene	1.56	1.51	strong	0.90-0.92	135
polypropylene	1.56	1.51	strong	0.90-0.92	170
rayon	1.54-1.56	1.51-1.53	strong	1.51	none
rubber	opaque		—	0.96-1.06	none
saran	1.61	1.61	weak, negative	1.70	168
spandex	opaque		—	1.20-1.21	230
triexta	1.57	—	—	1.33	226-233
vinal	1.55	1.52	strong	1.26-1.30	—
vinyon	1.53-1.54	1.53	weak, negative	1.34-1.37	230 or 400

*Not sharp. Sticking point is 176°C.

umn as follows. Clamp the density gradient glass tube in a firm vertical stand. Pour into the tube 25 mL of tetrachloroethylene. Now prepare mixtures of xylene and tetrachloroethylene by volume in descending order of percent tetrachloroethylene: 90/10, 80/20, 70/30, 60/40, 50/50, 40/60, 30/70, 20/80 and 10/90. Pour 25 mL of each in order carefully down the side of the gradient tube. Finally, put 25 mL of xylene on top.

9.6.2 Take short fragments of dyed reference fibers and tie in a knot snipping off loose ends. Boil for about 2 min in xylene solvent to remove moisture and air. Place in the column. After about a half hour they should come to rest at the level representing their density. Calibrated glass spheres may be used to determine the actual density at various levels.

9.6.3 Prepare the unknown fiber in like manner, place in the gradient column and note at what level it floats. Densities of fibers are listed in Table IV.

9.7 Solubility Test.

9.7.1 For tests at room temperature (20°C) place a small sample of the fibers in a watch crystal, test tube or 50 mL beaker and cover with the test solvent (see Table V). Use about 1 mL of solvent per 10 mg of fiber.

9.7.2 If the test is conducted at the boiling point of the solvent, first bring the solvent to a boil in a beaker on an electric hot plate in a ventilated hood. Adjust the hot plate temperature to maintain slow boiling and keep watch so that the solvent does not boil dry. Drop fiber sample into the boiling solvent.

9.7.3 If the test is conducted at some intermediate temperature, heat a beaker of water on a hot plate and adjust the temperature with a thermometer. Place the fiber sample in the test solvent in a test tube and immerse in the heated water bath.

9.7.4 Note if the fiber dissolves completely, softens to a plastic mass or remains insoluble. Compare with data on fiber solubility in Table V.

9.7.5 Solubility can also be used as a test to determine the presence of a metal component in a fiber. On dissolving in meta-cresol a shiny residue is evidence of a metal component.

9.8 Drying Twist Test.

9.8.1 Separate out a few parallel fibers. Dip in water and squeeze off excess. Tap the end of the bundle to make separate fibers flare out. Hold over a hot plate in warm air to dry the fibers. Hold the fibers so that the free ends point toward the observer. Note the direction of twist as the fibers dry. Linen and ramie twist in a clockwise direction; hemp and jute twist in a counter clockwise direction.

9.9 Stain Tests.

9.9.1 Place a few fibers on a microscopical slide. Apply a drop of zinc chloro-iodide reagent and cover with a cover slip taking care to avoid bubbles. Examine the fibers for staining. Hemp, ramie and cotton stain violet; linen stains brownish violet; jute stains brown. Many other fibers stain shades of yellow brown including silk.

9.9.2 Warm a few fresh fibers on a microscopical slide with a drop of acid phloroglucinol reagent. Woody fibers such as unbleached jute stain a deep magenta because of the presence of lignin.

9.10 Melting Point.

9.10.1 Melting Point Apparatus.

9.10.2 Place a few fibers on the clean heating block. Cover with a cover slip. Turn the heat on and set the heating rate for fast heating. Observe the thermometer and the sample. When the temperature reaches 100°C, reduce the rate of heating. (If previous tests already indicate the fiber, the rate can be set at about 10°C per min until 10-20°C below the expected melting point.) Near the melting point the rate should be reduced to about 2°C per min.

9.10.3 Observe the fibers for evidence of softening. At the melting point, liquid begins to form which wets the cover slip. Fibers eventually merge and form a liquid mass. As the test is in progress it is sometimes helpful to press gently on the cover slip with tweezers to see if the fibers flatten under pressure. If this melting point is passed at a high heating rate, repeat the test with a fresh sample.

9.10.4 Compare the melting point found with those listed in Table IV.

9.10.5 Microscope Mounted Apparatus.

9.10.6 Place a few fibers on one of the small glass slides provided and cover with a cover slip. Place the slide on the stage of the microscope with the fibers over the central hole of the stage. Put the baffle and cover plate in place.

9.10.7 Insert the polarizing accessories in the optical path of the microscope to provide crossed polars. The fibers should be visible if aligned diagonally to the direction of polarization. If fibers are not visible under these lighting conditions, remove the polarizing accessories and observe in ordinary transmitted light.

9.10.8 Set a high heating rate with the voltage regulator until the temperature reaches 100°C. Reduce the rate as the expected melting point is approached as directed in 9.10.2.

9.10.9 Observe the fibers. As melting takes place there is a decrease in birefrin-

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Table V—Solubility of Fibers

	acetic acid	acetone	sodium hypochlorite	hydrochloric acid	formic acid	1,4 dioxane	*m*-xylene	cyclohexanone	dimethyl formamide	sulfuric acid	sulfuric acid	*m*-cresol	hydrofluoric acid
Concentration (%)	100	100	5	20	85	100	100	100	100	59.5	70	100	50
Temperature (°C)	20	20	20	20	20	101	139	156	90	20	38	139	50
Time (minutes)	5	5	20	10	5	5	5	5	10	20	20	5	20
acetate	S	S	I	I	S	S	I	S	S	S	S	S	
acrylic	I	I	I	I	I	I	I	I	S	I	I	P	I
anidex	I	I	I	I	I	I	I	I	I	I	I	I	
aramid	I	I	I	I	I	I	I	I	I	I	I	I	I
azlon	I	I	S										
cotton & linen	I	I	I	I	I	I	I	I	I	I	S	I	I
glass	I	I	I	I	I	I	I	I	I	I	I	I	S
modacrylic	I	SE	I	I	I	SP	I	S	SP*	I	I	P	
novoloid	I	I	I	I	I	I	I	I	I	I	I	I	I†
nylon	I	I	I	S	S	I	I	I	I	S	S	S	
nytril	I	I	I	I	I	I	I	S	S	I	I	SP	
olefin	I	I	I	I	I	I	S	S	I	I	I	I	
polybenzimidazole	I	I	I	I	I	I	I	I	I	I	I	I	I
polyester	I	I	I	I	I	I	I	I	I	I	I	S	I
rayon	I	I	I	I	I	I	I	I	I	S	S	I	I
saran	I	I	I	I	I	S	S	S	S	I	I	I	
silk	I	I	S	I	I	I	I	I	I	S	S	I	
spandex	I	I	I	I	I	I	I	S	S	SP	SP	SP	
teflon	I	I	I	I		I	I	I	I	I	I	I	I
triexta	I	I	I	I	I	I	I	I	I	I	I	S	I
vinal				S	S	I	I	I	I	S	S	I	
vinyon	I	S	I	I	I	S	S	S	S	I	I	S	
wool	I	I	S	I	I	I	I	I	I	I	I	I	

S = Soluble
I = Insoluble
P = Forms plastic mass
SP = Soluble or forms plastic mass
SE = Soluble except for one modacrylic fiber characterized by low flammability and liquid inclusions visible in cross-section.
* = Soluble at 20°C without plastic mass.
† = Novoloid turns red

gence; the fibers become dark. Read as melting point the temperature at which the fibers become completely dark. If crossed polars are not used, observe melting as described in 9.10.2.

9.10.10 Compare the melting point found with those listed in Table IV.

9.11 Micro-FTIR.

9.11.1 Compare FTIR spectra with FTIR spectra in Appendix II or other library sources (see Appendix II—Figs. 1-9).

10. Report

10.1 Report the fiber type and, if more than one type is present, where each is located. For example, the report could be "woven fabric with nylon 6,6 warp and cotton/rayon filling."

11. Precision and Bias

11.1 A precision and bias statement is not applicable because data are not generated by this method.

12. Notes

12.1 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1.513.742.2020; web site: www.acgih.org.

12.2 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

12.3 Reference samples, along with interlab data, of cashmere and wool, both separate and combined, are available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

12.4 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

13. References

13.1 The Textile Institute, *Identification of Textile Materials*, Sixth Edition, C. Tinling & Co., London, 1970.

13.2 Federal Trade Commission, "Rules and Regulations Under the Textile Fiber Products Identification Act," as amended 1969, Washington, DC 20580, www.ftc.gov.

13.3 Heyn, A. N. J., *Fiber Microscopy, A Text Book and Laboratory Manual*, Interscience, New York, 1954. Old, but good techniques and illustrations.

13.4 Wildman A. B., *The Microscopy of Animal Textile Fibers*, Wool Industries Research Association, Torridon, Leeds, England, 1954.

13.5 Appleyard, H. M., *Guide to the Identification of Animal Fibers*, same publisher as 13.4, 1960. Both contain excellent descriptions and photomicrographs.

13.6 Man-Made Fiber Producers Association, *Man-Made Fiber Fact Book*, New York, 1970. Revised annually. List of current fibers made in United States.

13.7 Man-Made Textiles, *Index to Man-Made Fibers of the World*, Third Edition, Harlequin Press, Manchester 2, England, 1967. Lists about 2,000 man-made fibers by trademark names and their suppliers.

13.8 Linton, G. E., *Natural and Man-Made Textile Fibers*, Duell, Sloan and Pearce. New York, 1966. History and technology, especially of natural fibers.

13.9 Potter, D. M. and Corbman, B. P., *Textiles: Fibers to Fabric*, Fourth Edition, McGraw-Hill, New York, 1967. A textbook with section on fiber identification.

13.10 Chamot, E. M., and Mason, C. W., *Chemical Microscopy, Vol. I on Physical Methods*, Third Edition, John Wiley & Sons, New York, 1950. A classical textbook with various references to fibers.

13.11 IWTO Draft Test Method 58-97, "Quantitative Analysis of Blends of Wool with Specialty Fibers by Scanning Electron Microscope." Describes method for distin-

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guishing between specialty animal fibers and wool and provides numerous references.

13.12 GSB 16-2262-2008, "Micrograph Collection for Cashmere Fiber Morphology" is available from the Inner Mongolia Erdos Cashmere Group Corporation; tel: +86. 4778543855; fax: +86.4778540114; e-mail: erdoscathy@yahoo.com; web site: www.chinaerdos.com.

14. History

14.1 Revised 2021 to update Table V.

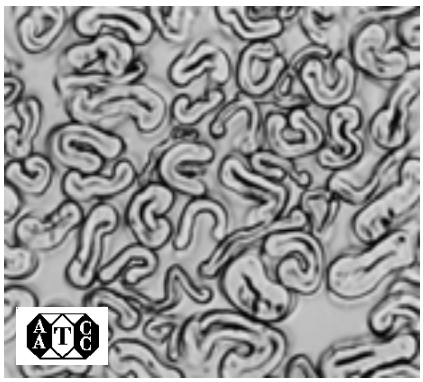
14.2 Editorially revised 2019; reaffirmed 2018; revised 2013, 2011, 2010; editorially revised 2009; editorially revised with technical correction 2008; revised 2007, 2005, 2004, 2002, 2001, 2000, 1999, 1998; editorially revised and reaffirmed 1995, 1990; editorially revised 1988; reaffirmed 1985; editorially revised 1984, 1983, 1982 (new title), 1977; revised 1976; editorially revised 1974; editorially revised and reaffirmed 1973; revised 1972, 1963, 1962, 1958; adopted as tentative 1955.

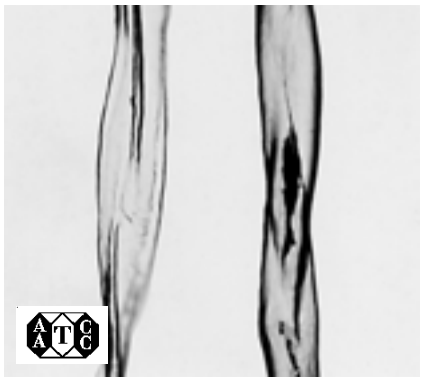
14.3 Developed by AATCC Committee RA24. Related to ISO 17751, ISO 1833, ISO 2017, and IWTO 58.

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Appendix I
Photomicrographs of Common Textile Fibers

Cross-Section 500X

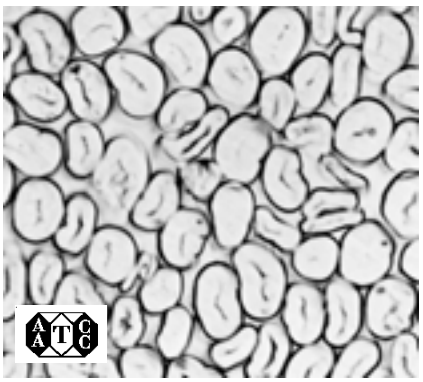


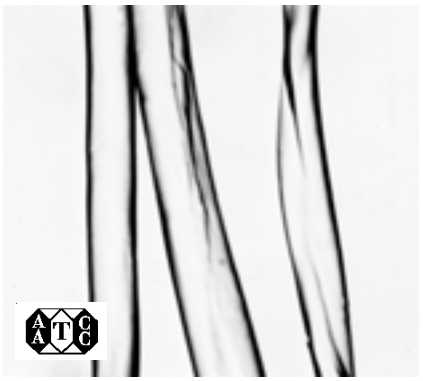


Longitudinal View 500X

Fig. 1—Cotton, not mercerized.

Cross-Section 500X

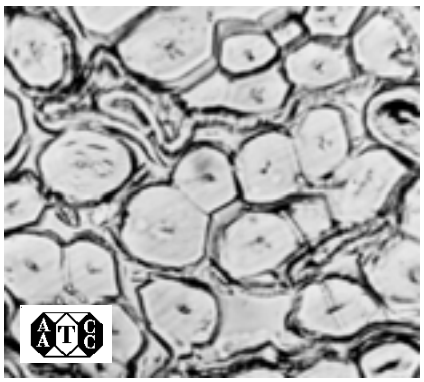




Longitudinal View 500X

Fig. 2—Cotton, mercerized.

Cross-Section 500X

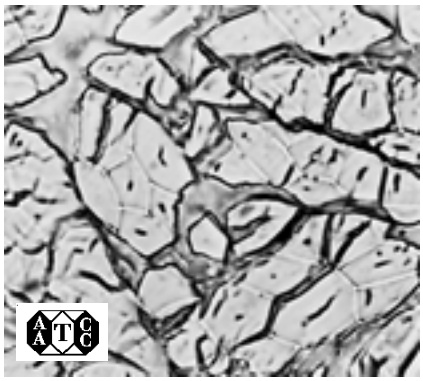


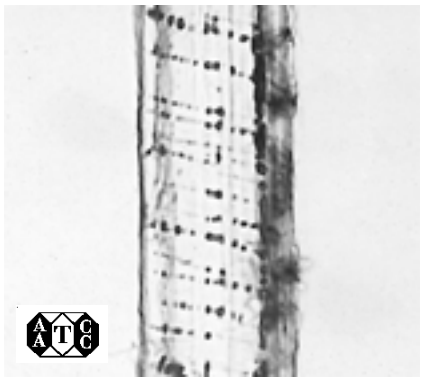


Longitudinal View 500X

Fig. 3—Linen.

Cross-Section 500X





Longitudinal View 500X

Fig. 4—Hemp.

Cross-Section 500X

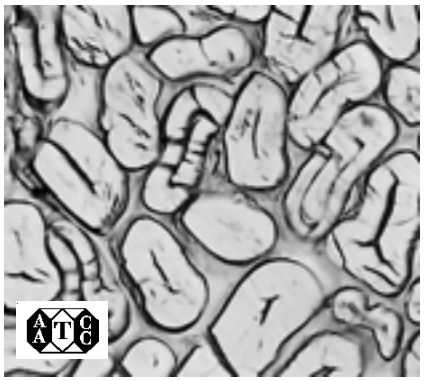


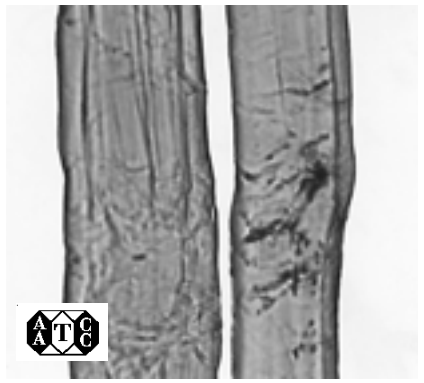


Longitudinal View 500X

Fig. 5—Jute.

Cross-Section 500X



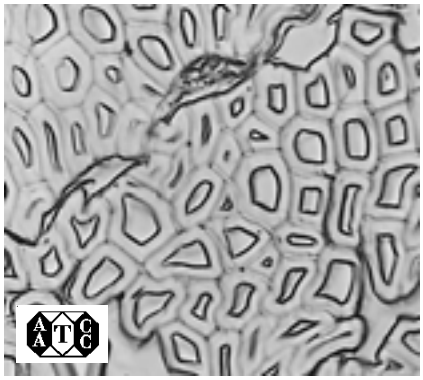


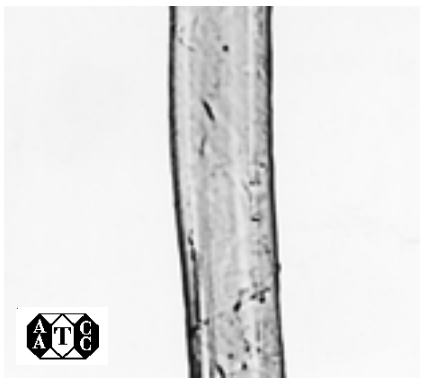
Longitudinal View 500X

Fig. 6—Ramie.

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Cross-Section 500X

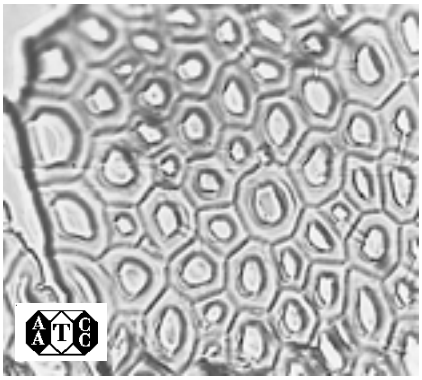


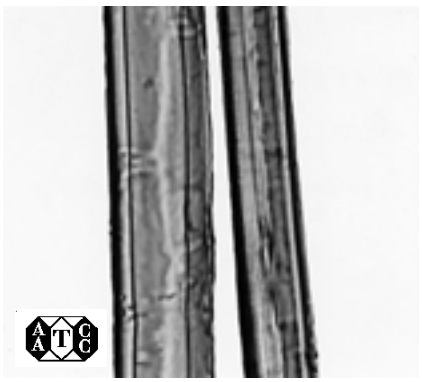


Longitudinal View 500X

Fig. 7—Sisal.

Cross-Section 500X

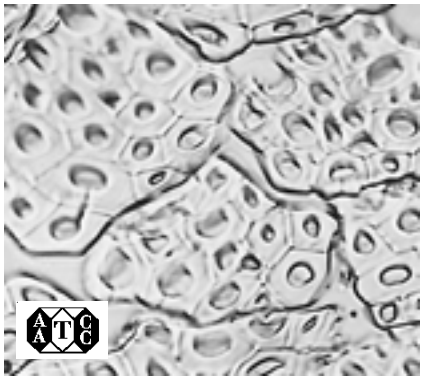




Longitudinal View 500X

Fig. 8—Abaca.

Cross-Section 500X

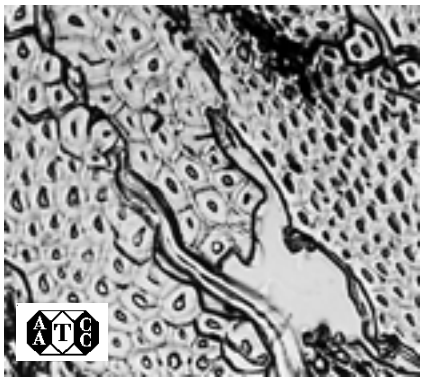




Longitudinal View 500X

Fig. 9—Kenaf.

Cross-Section 500X

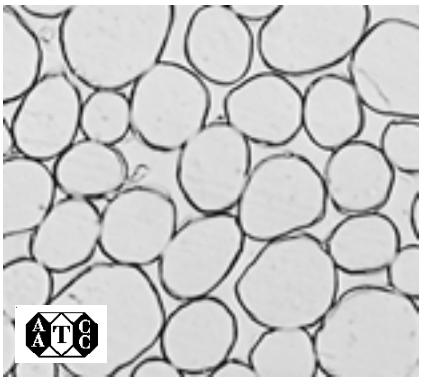




Longitudinal View 500X

Fig. 10—Phormium.

Cross-Section 500X

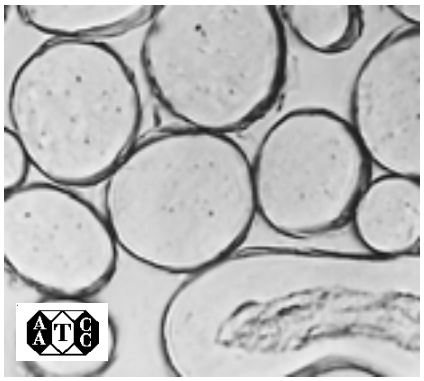




Longitudinal View 500X

Fig. 11—Wool.

Cross-Section 500X



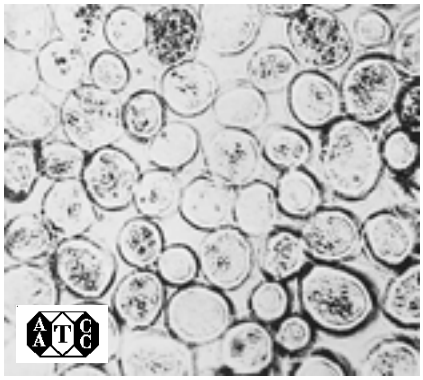


Longitudinal View 500X

Fig. 12—Mohair.

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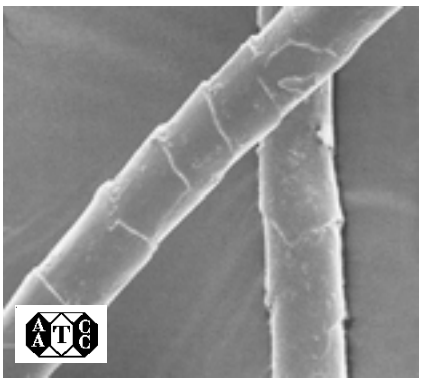
Cross-Section 500X





Longitudinal View 240X

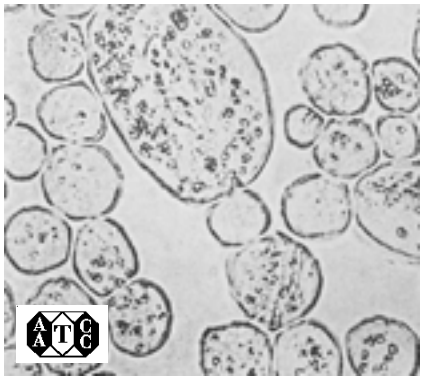
Fig. 13—Cashmere.

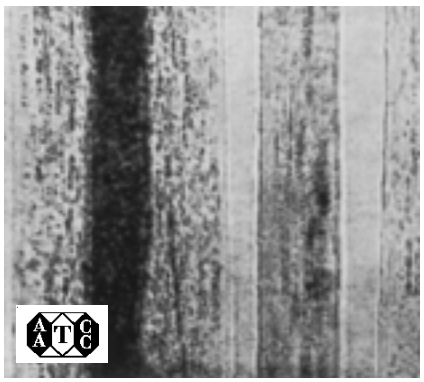


Longitudinal View 1500X

Fig. 13A—SEM (Cashmere).

Cross-Section 500X

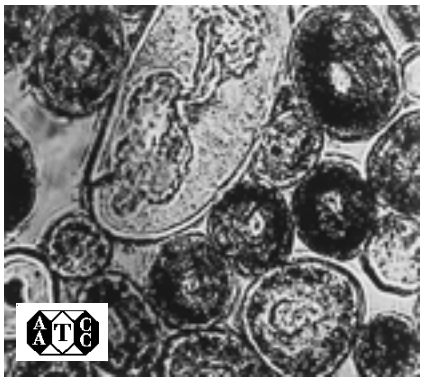


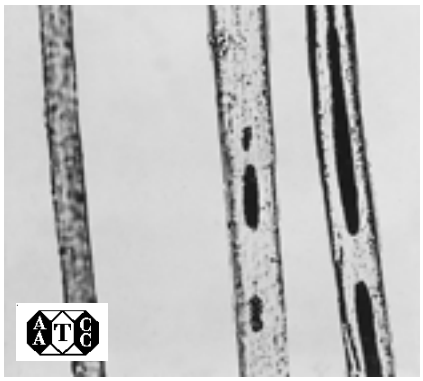


Longitudinal View 500X

Fig. 14—Camel hair.

Cross-Section 500X

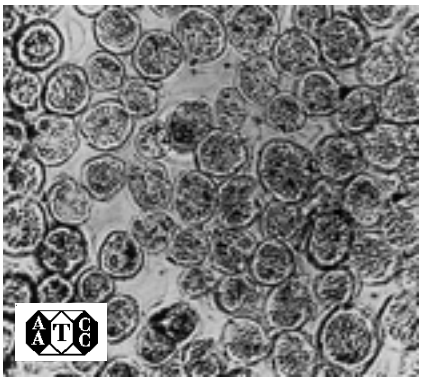




Longitudinal View 240X

Fig. 15—Alpaca.

Cross-Section 500X

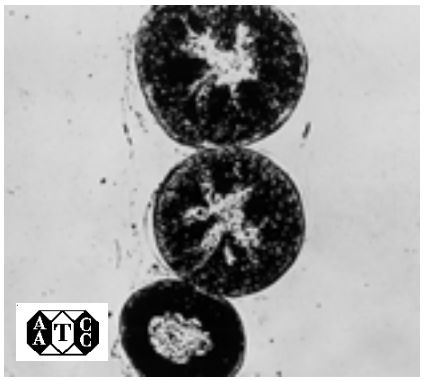


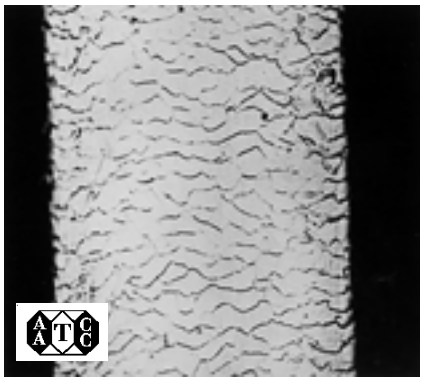


Longitudinal View 240X

Fig. 16—Vicuna.

Cross-Section 115X



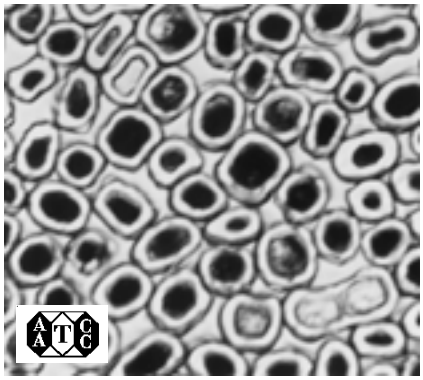


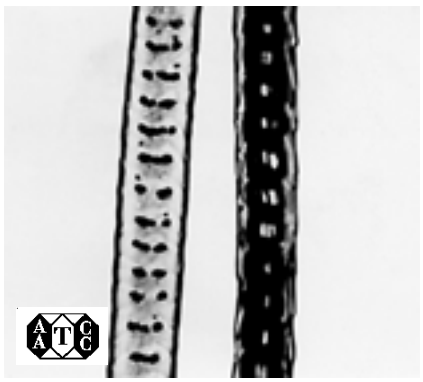
Longitudinal View 230X

Fig. 17—Horsehair.

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Cross-Section 500X

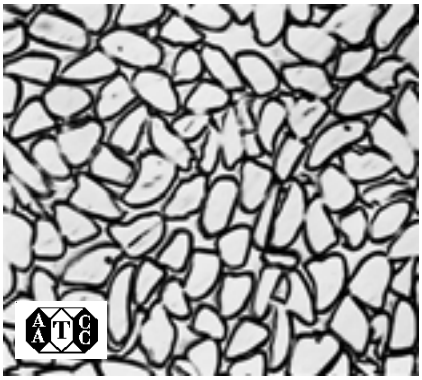




Longitudinal View 500X

Fig. 18—Rabbit fur.

Cross-Section 500X

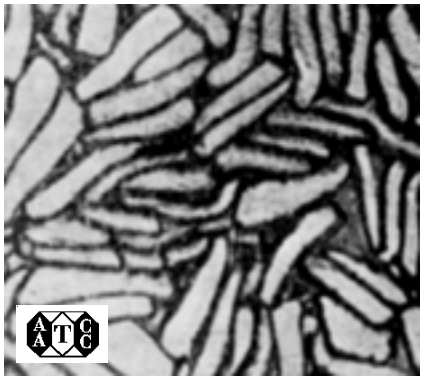


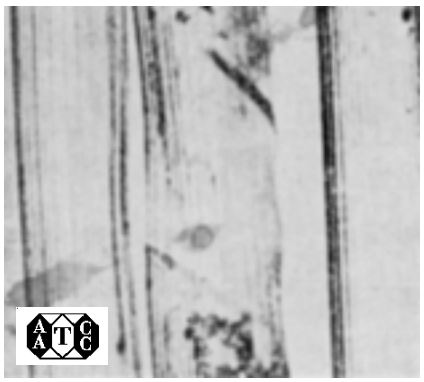


Longitudinal View 500X

Fig. 19—Silk.

Cross-Section 500X

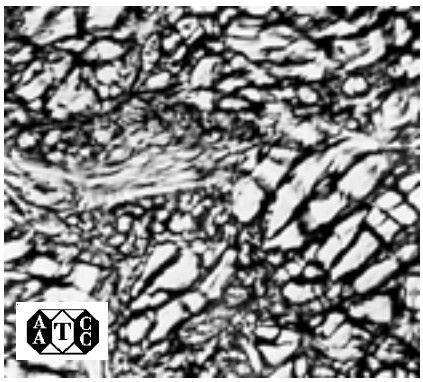


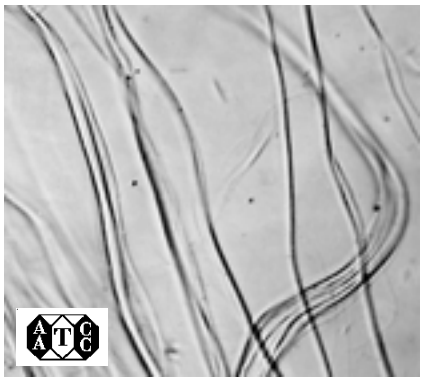


Longitudinal View 500X

Fig. 20—Silk, tussah.

Cross-Section 500X

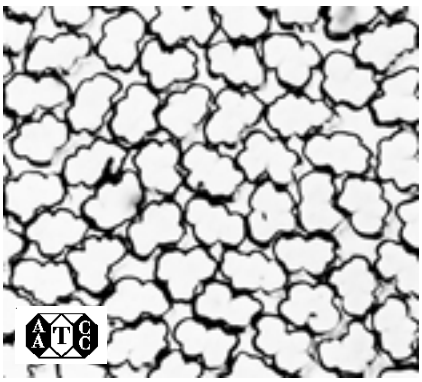


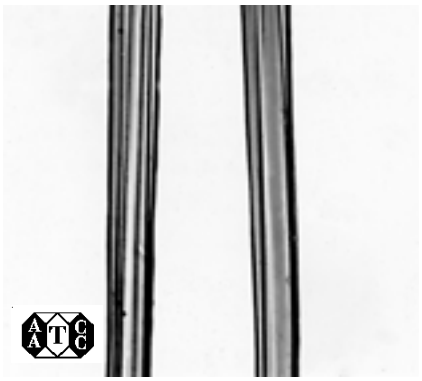


Longitudinal View 500X

Fig. 21—Asbestos.

Cross-Section 500X

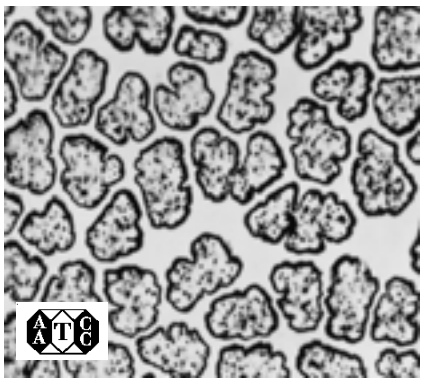


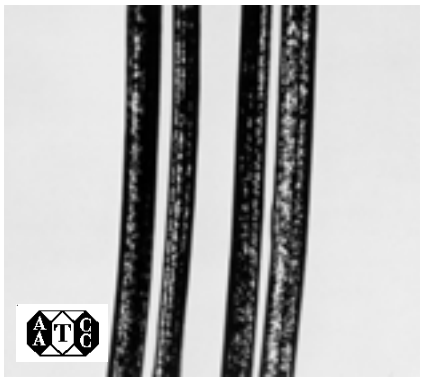


Longitudinal View 500X

Fig. 22—Acetate, secondary.

Cross-Section 500X



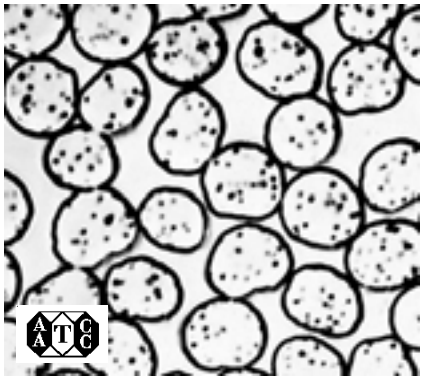


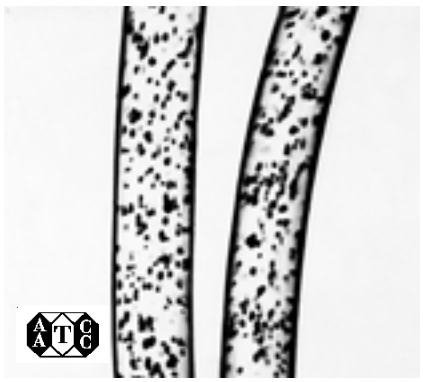
Longitudinal View 250X

Fig. 23—Triacetate, 2.5 denier (0.28 tex) per filament, dull luster.

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Cross-Section 500X

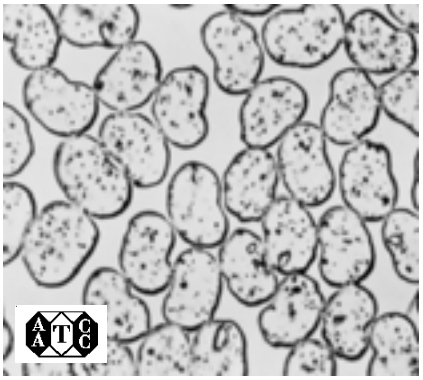


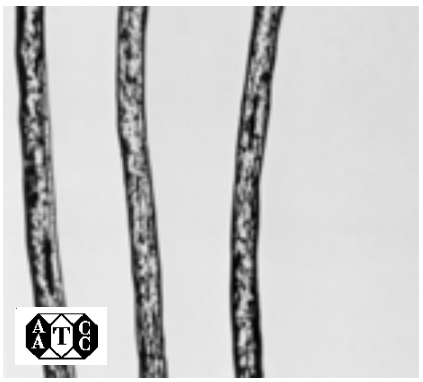


Longitudinal View 500X

Fig. 24—Acrylic, reg. wet spun, semi-dull.

Cross-Section 500X

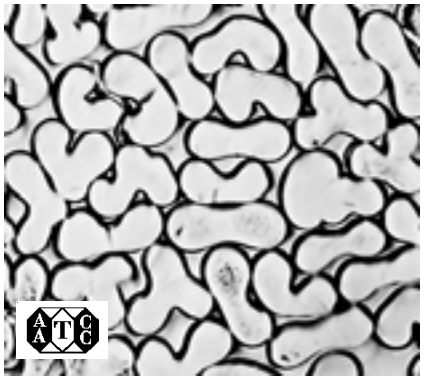


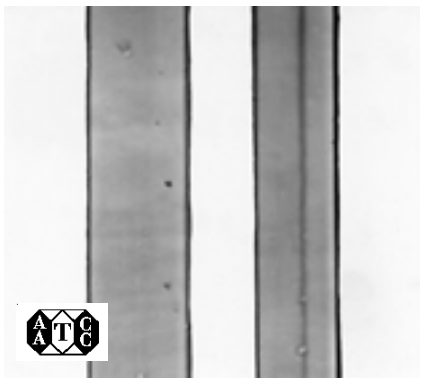


Longitudinal View 250X

Fig. 25—Acrylic, modified wet spun, 3.0 denier (0.33 tex) per filament, semi-dull luster.

Cross-Section 500X

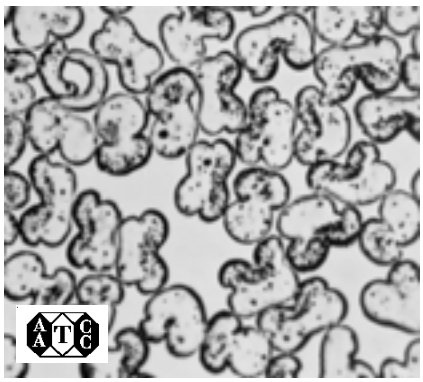


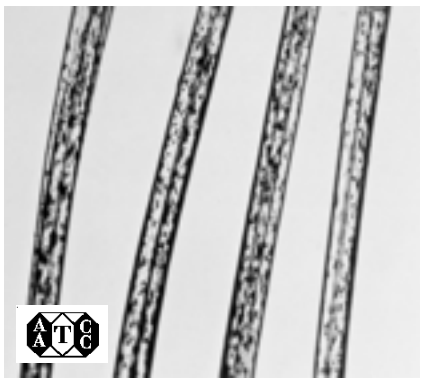


Longitudinal View 500X

Fig. 26—Acrylic, solvent spun.

Cross-Section 500X

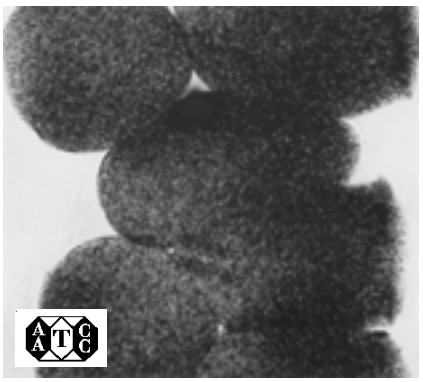


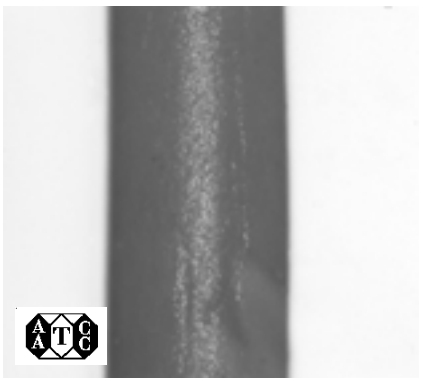


Longitudinal View 250X

Fig. 27—Acrylic, two-component, 3.0 denier (0.33 tex) per filament, semi-dull luster.

Cross-Section 100X

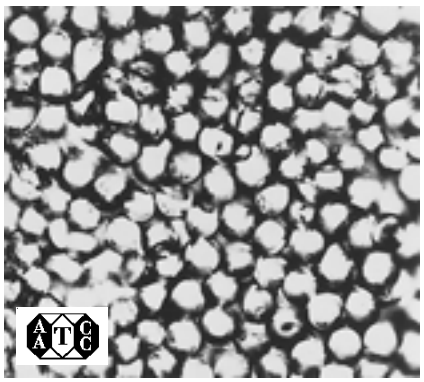


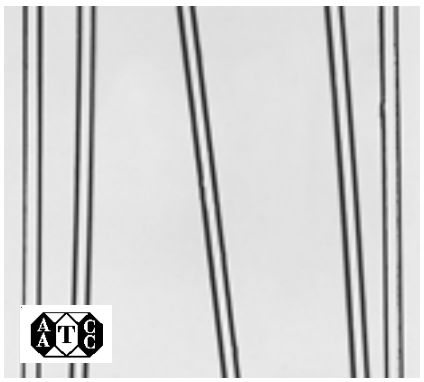


Longitudinal View 100X

Fig. 28—Anidex.

Cross-Section 500X



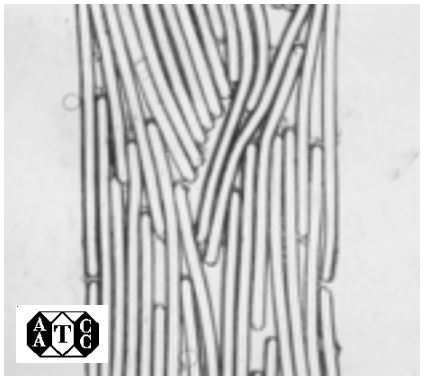


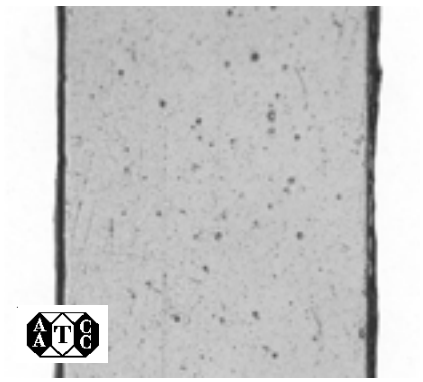
Longitudinal View 250X

Fig. 29—Glass.

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Cross-Section 100X

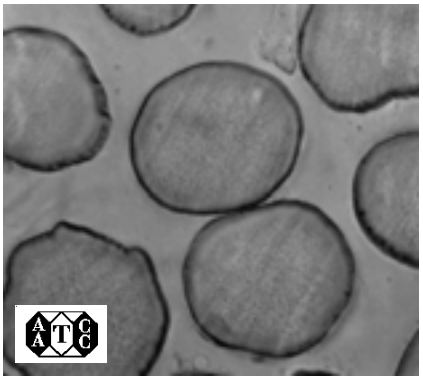


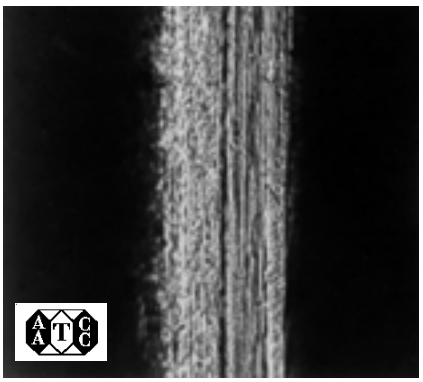


Longitudinal View 100X

Fig. 30—Metallic.

Cross-Section 500X

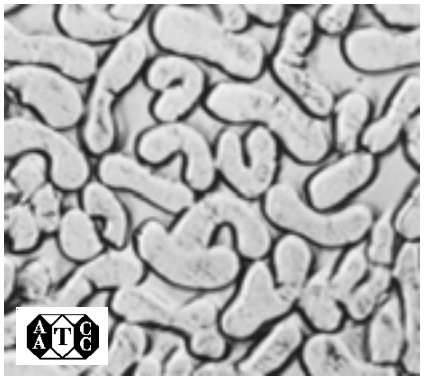


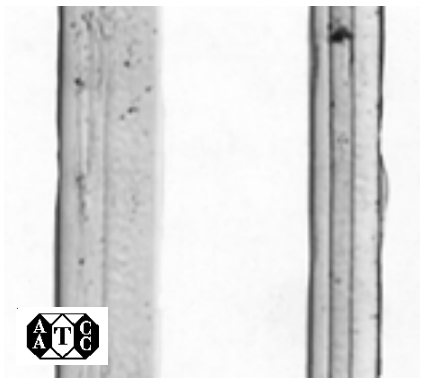


Longitudinal View 500X

Fig. 31—Modacrylic.

Cross-Section 500X

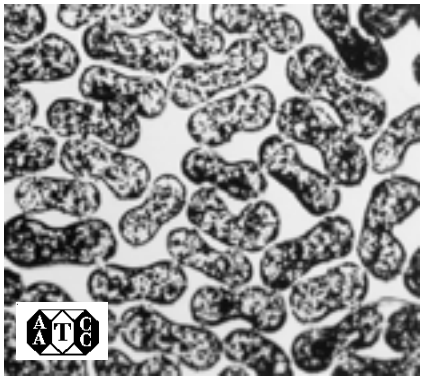


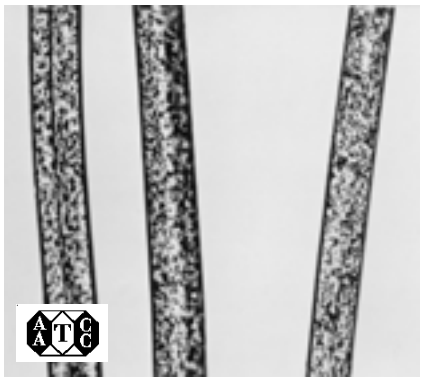


Longitudinal View 500X

Fig. 32—Modacrylic.

Cross-Section 500X

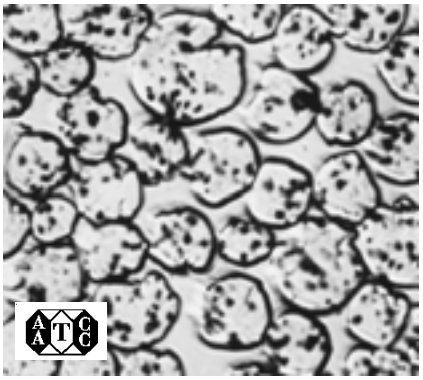


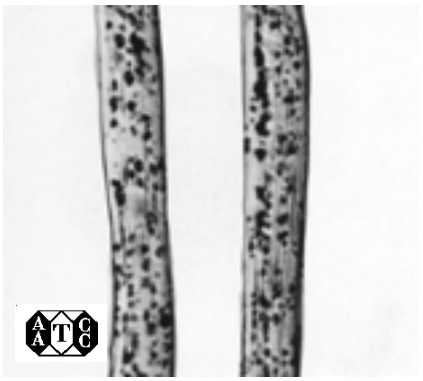


Longitudinal View 250X

Fig. 33—Modacrylic, 3.0 denier (0.33 tex) per filament, dull luster.

Cross-Section 500X

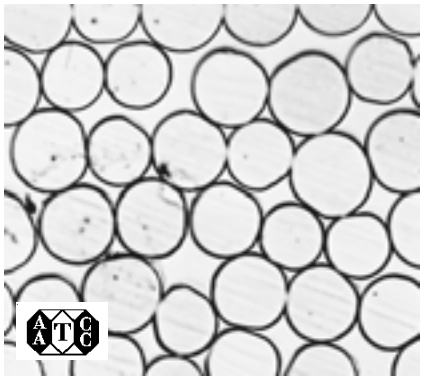


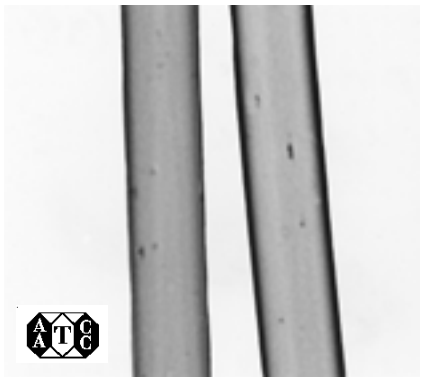


Longitudinal View 500X

Fig. 34—Modacrylic with liquid inclusions.

Cross-Section 500X



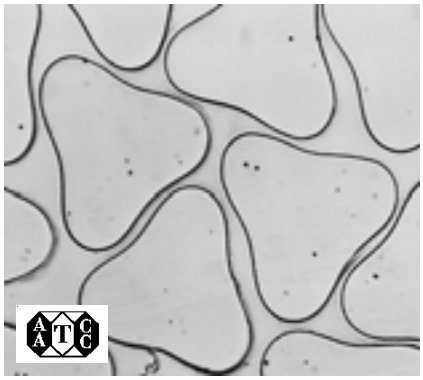


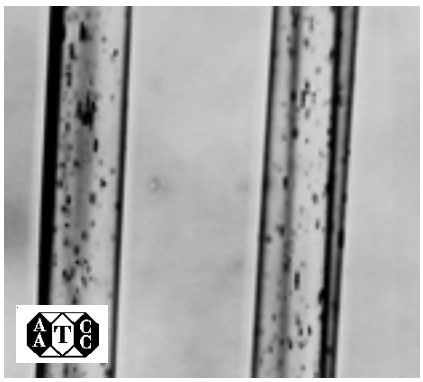
Longitudinal View 500X

Fig. 35—Nylon, bright.

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Cross-Section 500X

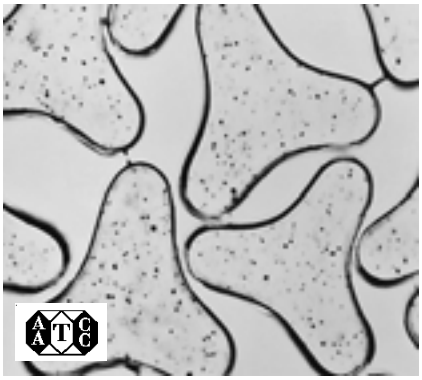


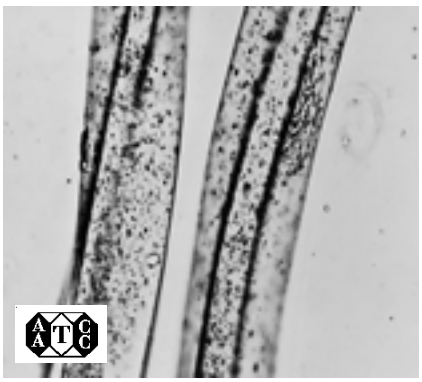


Longitudinal View 250X

Fig. 36—Nylon, low modification ratio trilobal, 15 denier (1.65 tex) per filament, bright luster.

Cross-Section 500X

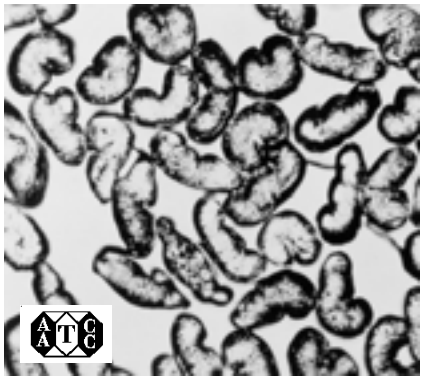


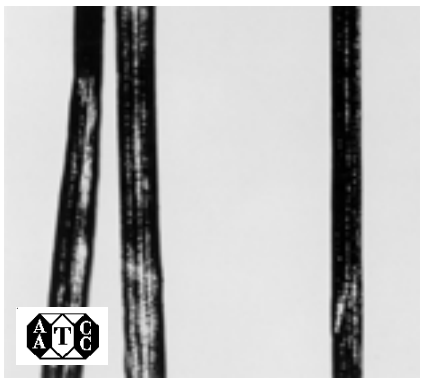


Longitudinal View 250X

Fig. 37—Nylon, high modification ratio trilobal, 18 denier (1.98 tex) per filament, semi-dull.

Cross-Section 500X

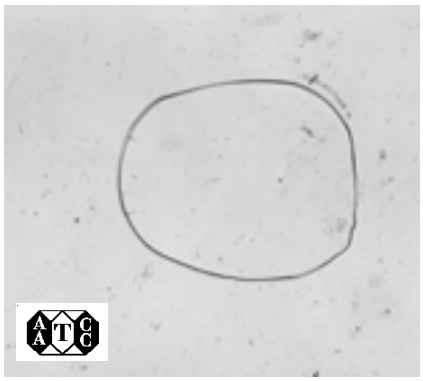


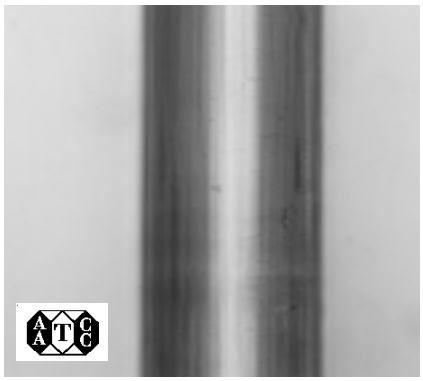


Longitudinal View 250X

Fig. 38—Nytril, 2.0 denier (0.22 tex) per filament, dull luster.

Cross-Section 500X

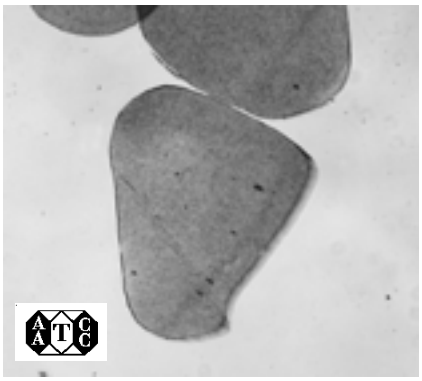


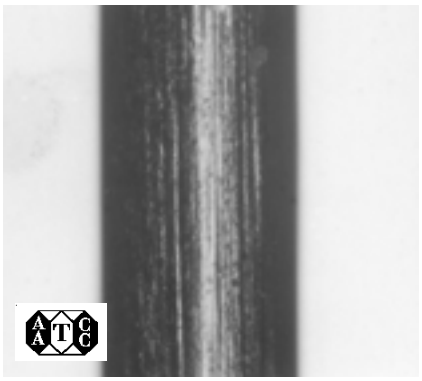


Longitudinal View 500X

Fig. 39—Polyethylene, low density.

Cross-Section 500X

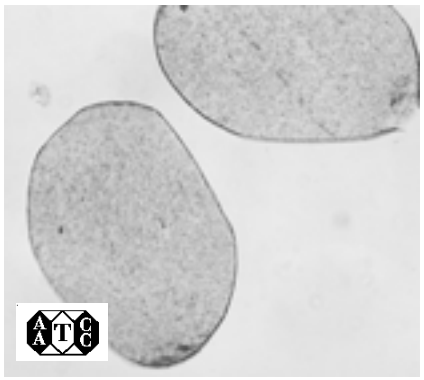


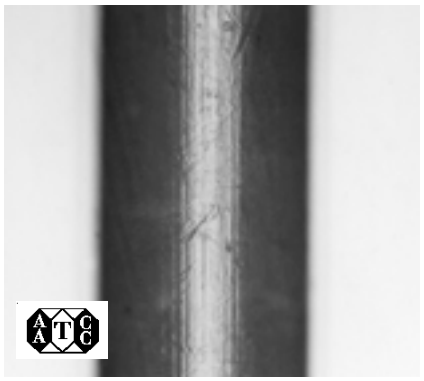


Longitudinal View 500X

Fig. 40—Polyethylene, medium density.

Cross-Section 500X



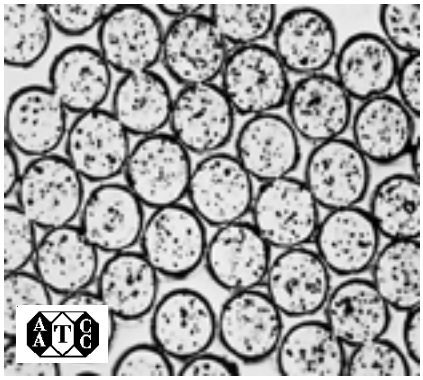


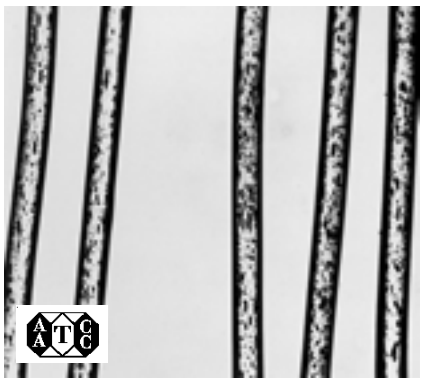
Longitudinal View 500X

Fig. 41—Polyethylene, high density.

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Cross-Section 500X

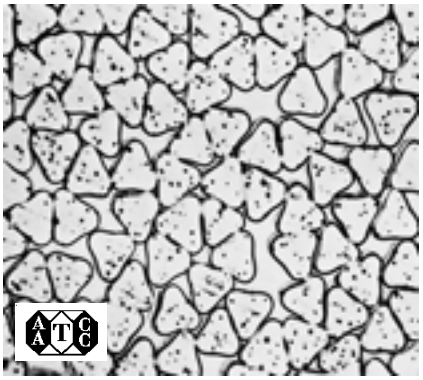


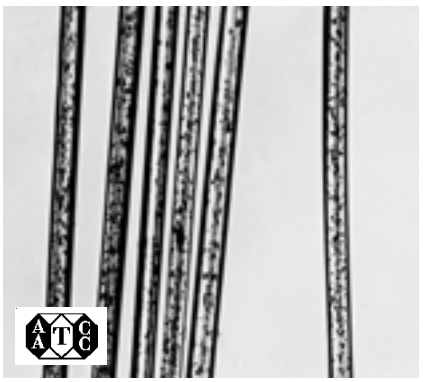


Longitudinal View 250X

Fig. 42—Polyester, regular melt spun, 3.0 denier (0.33 tex) per filament, semi-dull.

Cross-Section 500X

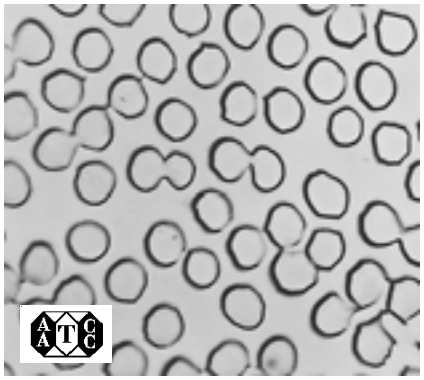


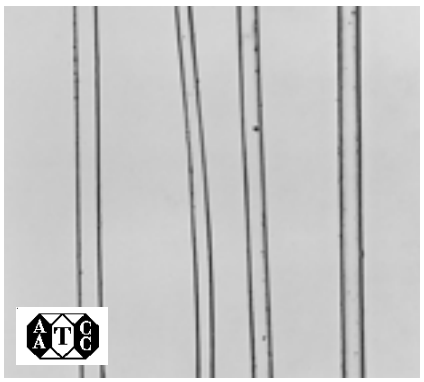


Longitudinal View 250X

Fig. 43—Polyester, low modification ratio trilobal, 1.4 denier (0.15 tex) per filament, semi-dull luster.

Cross-Section 500X

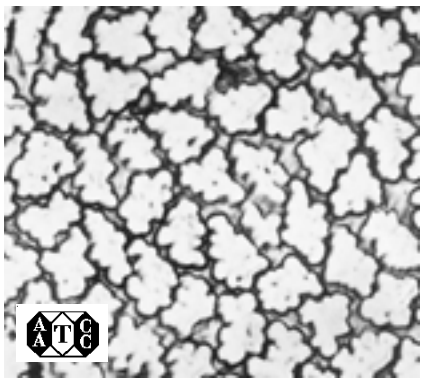


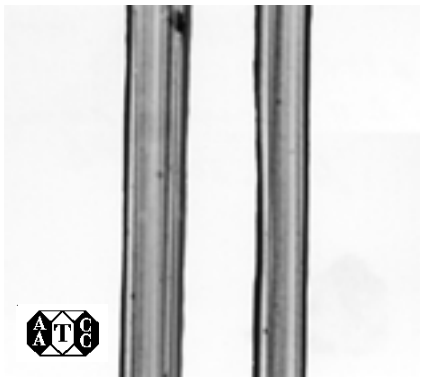


Longitudinal View 250X

Fig. 44—Rayon, cuprammonium, 1.3 denier (0.14 tex) per filament, bright luster.

Cross-Section 500X

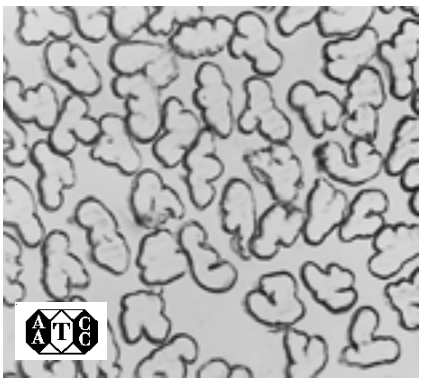


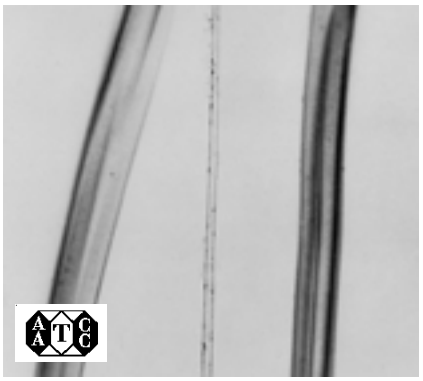


Longitudinal View 500X

Fig. 45—Rayon, viscose. Regular tenacity, brt.

Cross-Section 500X

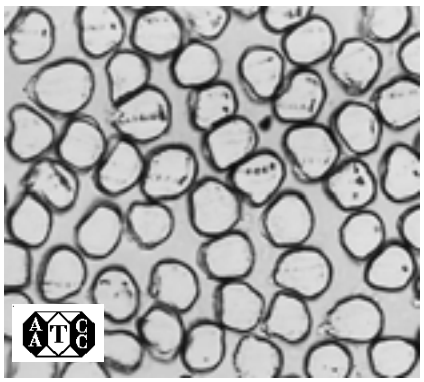


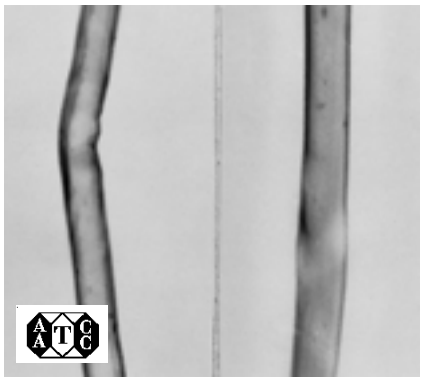


Longitudinal View 500X

Fig. 46—Rayon, viscose. High tenacity, high wet elongation.

Cross-Section 500X



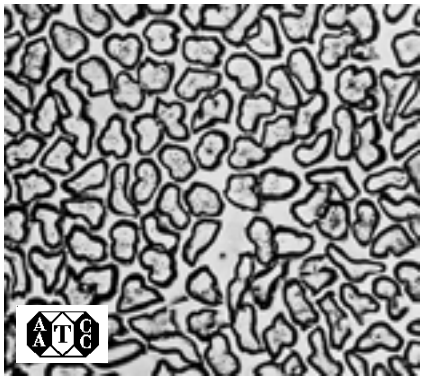


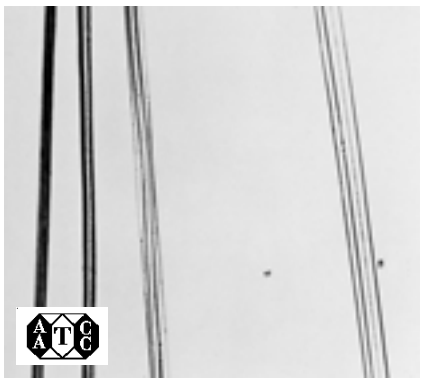
Longitudinal View 500X

Fig. 47—Rayon, viscose. High tenacity, low wet elongation.

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Cross-Section 500X

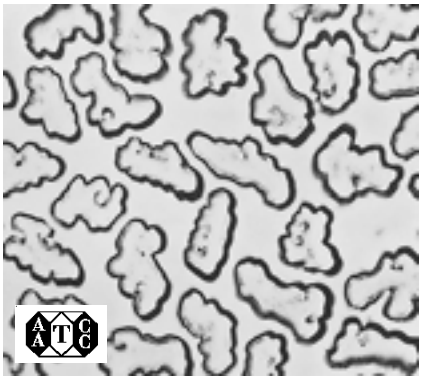


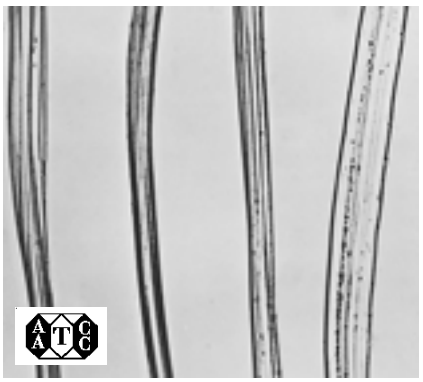


Longitudinal View 250X

Fig. 48—Rayon, saponified acetate. 0.8 denier (0.09 tex) per filament, bright luster.

Cross-Section 500X

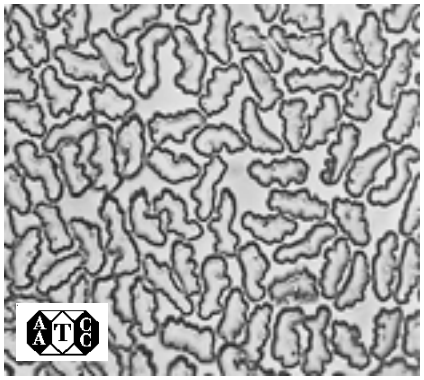


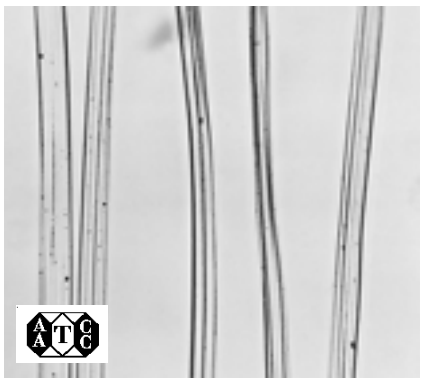


Longitudinal View 250X

Fig. 49—Rayon, viscose. Modified, 3.0 denier (0.33 tex) per filament, bright luster.

Cross-Section 500X

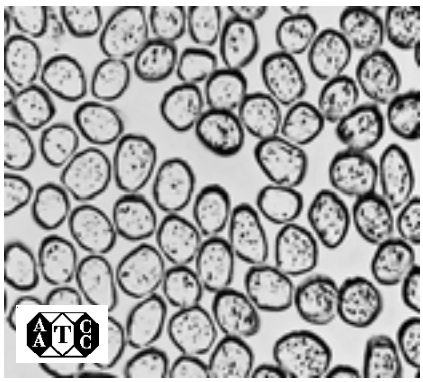


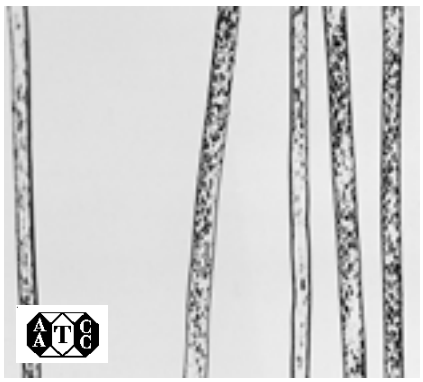


Longitudinal View 250X

Fig. 50—Rayon, viscose. Modified, 1.5 denier (0.17 tex) per filament, bright luster.

Cross-Section 500X

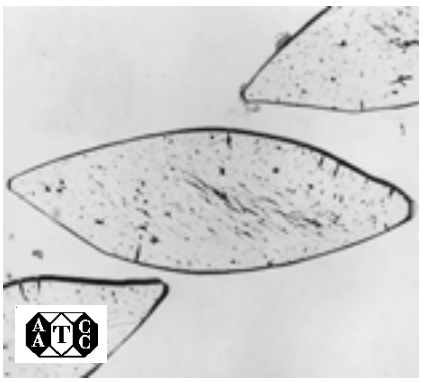


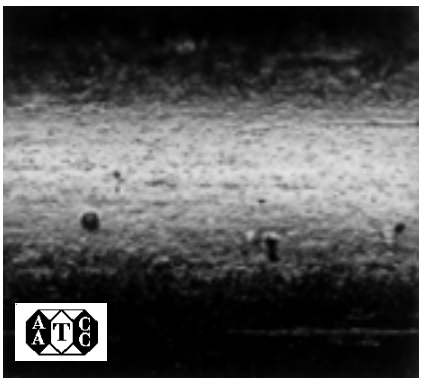


Longitudinal View 250X

Fig. 51—Rayon, viscose. Modified, 1.5 denier (0.17 tex) per filament, semi-dull luster.

Cross-Section 65X

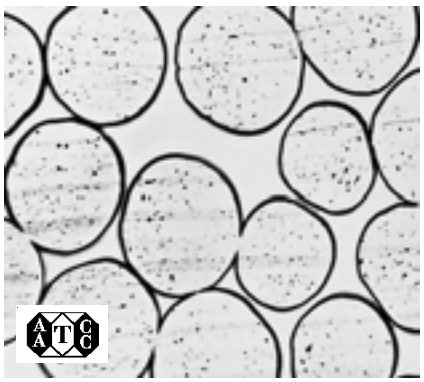


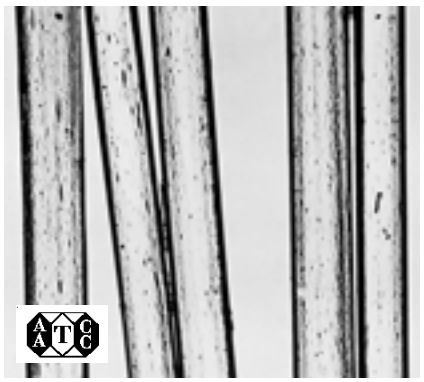


Longitudinal View 65X

Fig. 52—Saran.

Cross-Section 500X



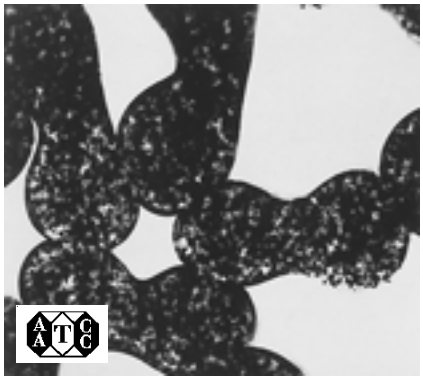


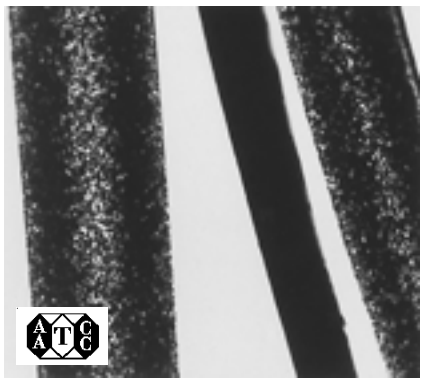
Longitudinal View 250X

Fig. 53—Saran, 16 denier (1.76 tex) per filament, bright luster.

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Cross-Section 500X

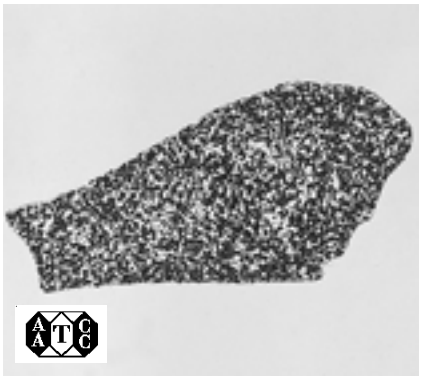


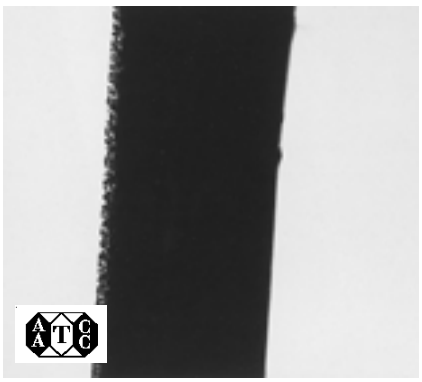


Longitudinal View 250X

Fig. 54—Spandex, adhering filaments, 12 denier (1.32 tex) per filament, dull luster.

Cross-Section 500X

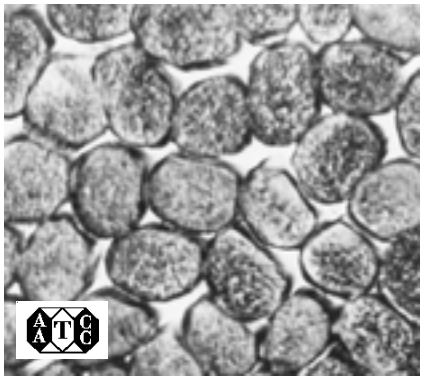


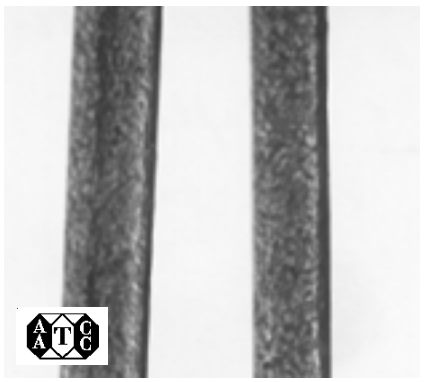


Longitudinal View 250X

Fig. 55—Spandex, coarse monofilaments, 250 denier (27.50 tex) per filament, dull luster.

Cross-Section 500X

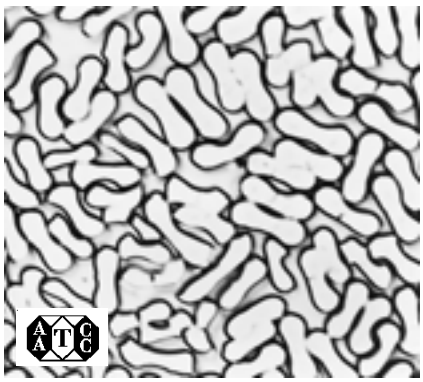


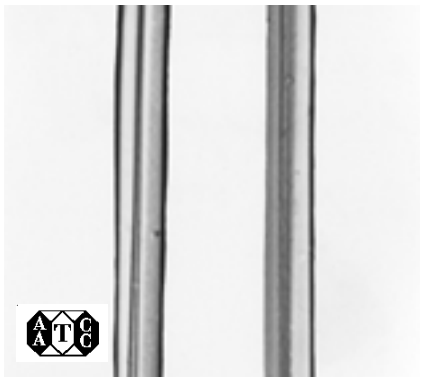


Longitudinal View 500X

Fig. 56—Fluorocarbon.

Cross-Section 500X

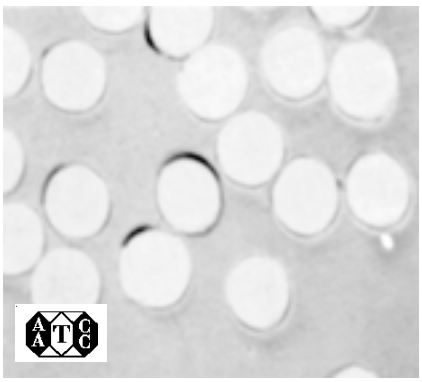


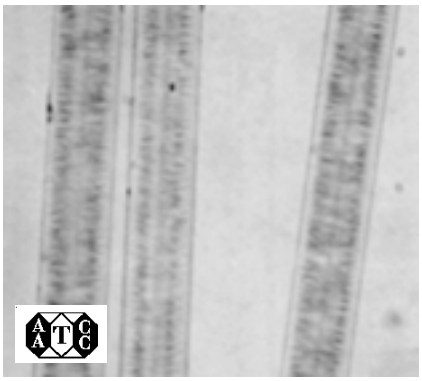


Longitudinal View 500X

Fig. 57—Vinyon.

Cross-Section 600X



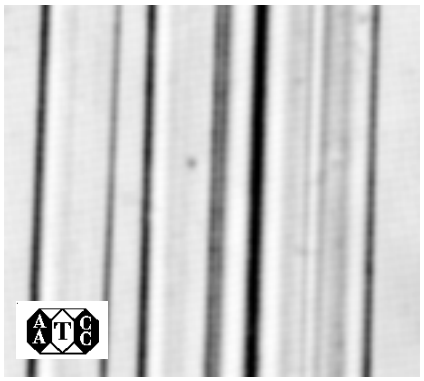


Longitudinal View 600X

Fig. 58—Aramid, round, high-tenacity filament.

Cross-Section 600X



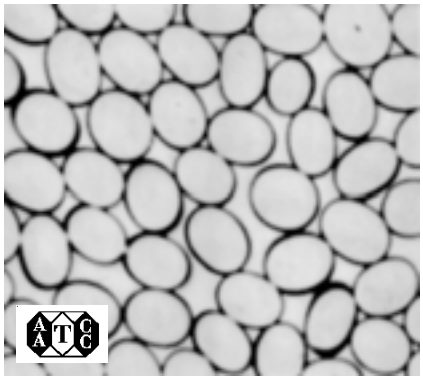


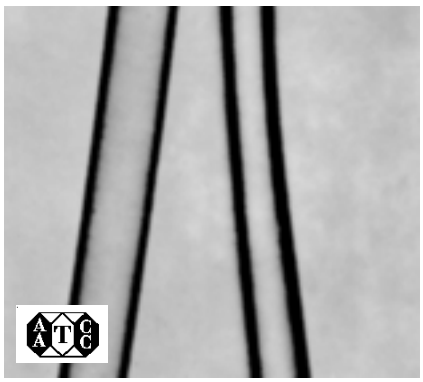
Longitudinal View 600X

Fig. 59—Aramid, FR staple fiber.

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Cross-Section 500X

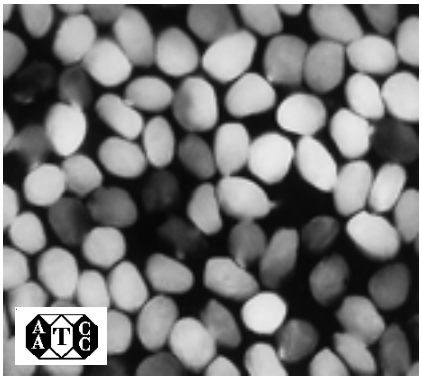


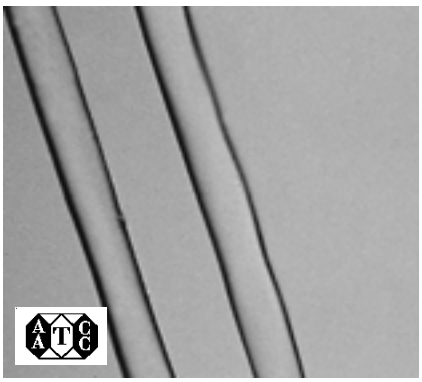


Longitudinal View 500X

Fig. 60—Novoloid.

Cross-Section 1500X

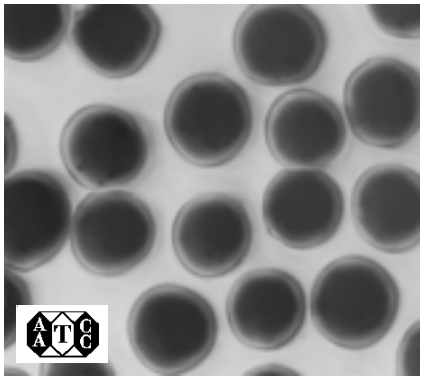


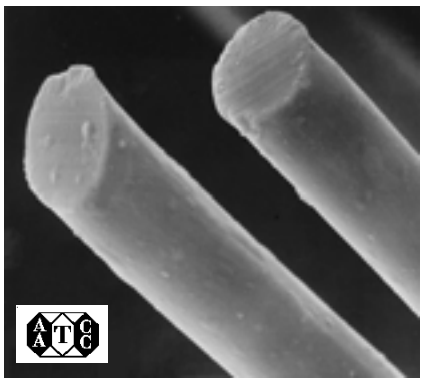


Longitudinal View 1500X

Fig. 61—Permanently crimped lyocell fiber.

Stained Transverse Cross-Section 1800X

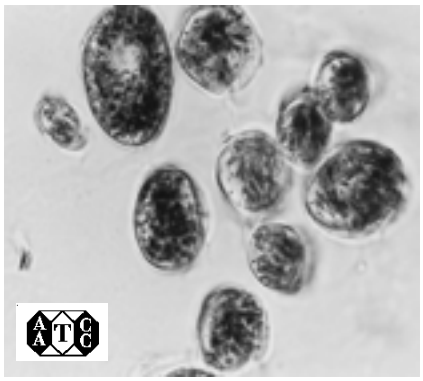


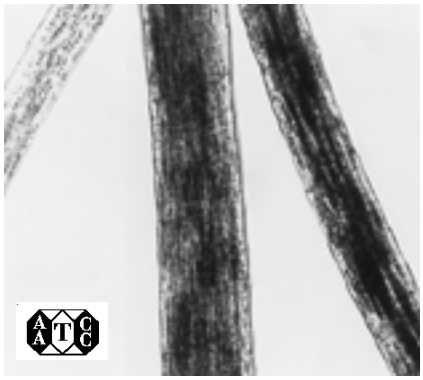


Scanning Electron Photomicrograph (10 µm)

Fig. 62—Uncrimped lyocell fiber.

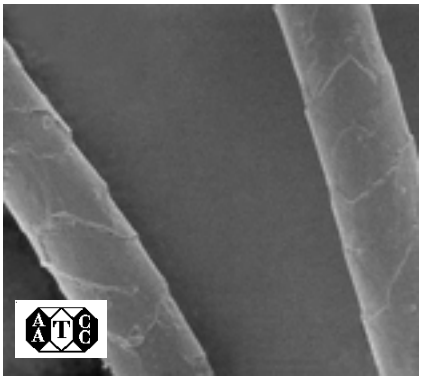
Cross-Section 500X





Longitudinal View 500X

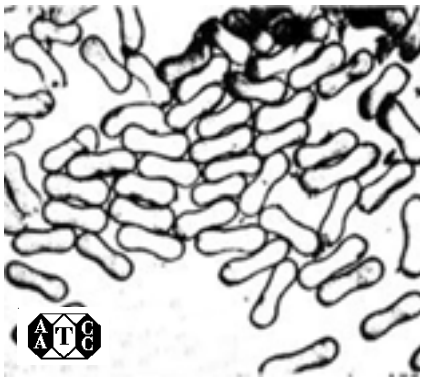
Fig. 63—Yak.

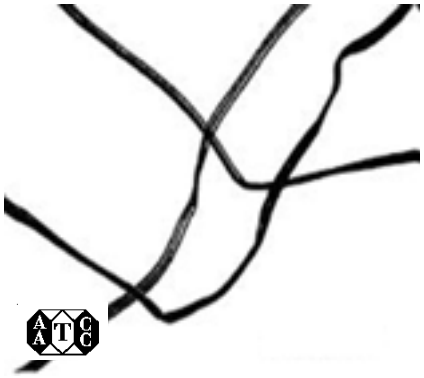


Longitudinal View 1500X

Fig. 63A—SEM of Yak.

Cross-Section 500X





Longitudinal View 100X

Fig. 64—Polybenzimidazole.

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Appendix II
Micro-Fourier Transform Infrared Spectra of Common Textile Fibers

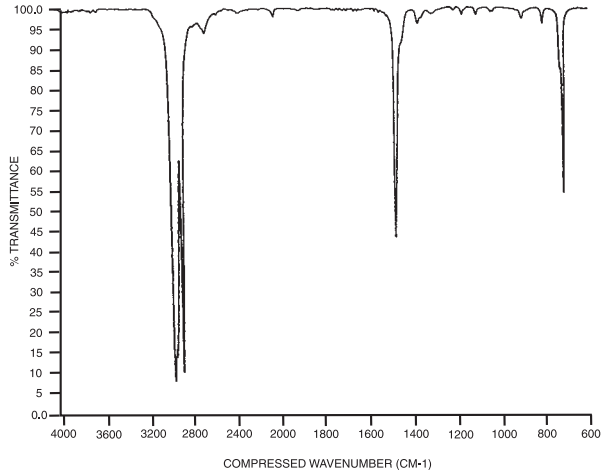


Fig. 1—Poly(ethylene) Ultra High Molecular Weight

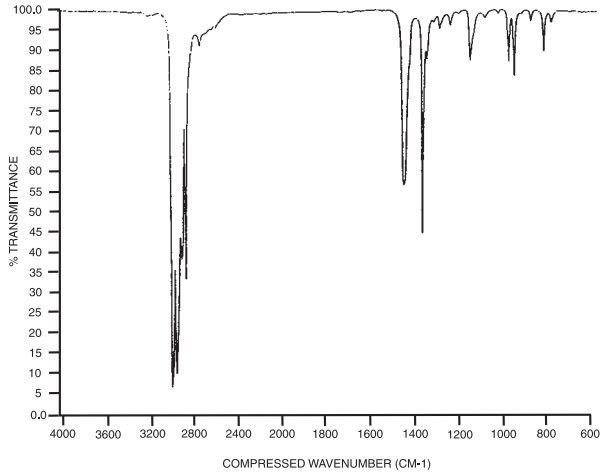


Fig. 2—Poly(propylene)

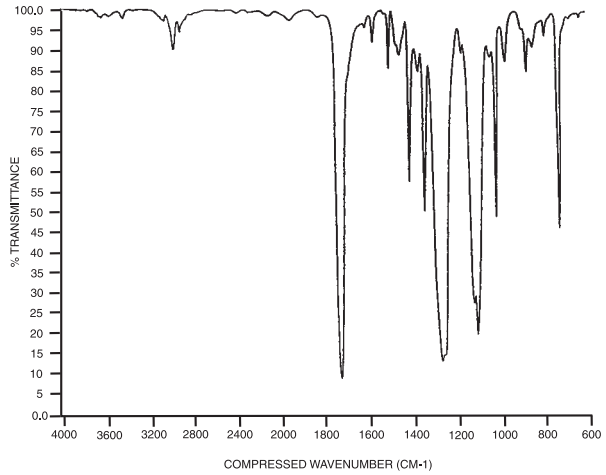


Fig. 3—Poly(ethyleneterephthalate)

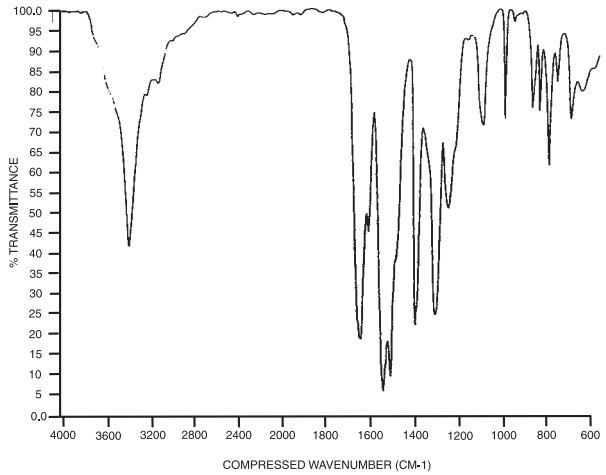


Fig. 4—Poly(phenyleneterephthalamide)

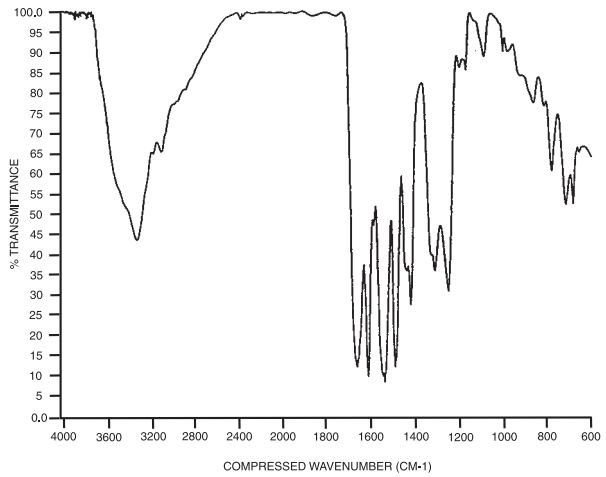


Fig. 5—Poly(m-phenyleneisophthalamide)

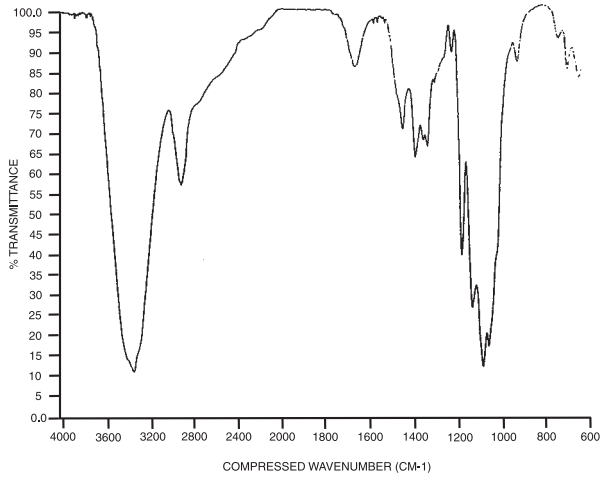


Fig. 6—Cotton

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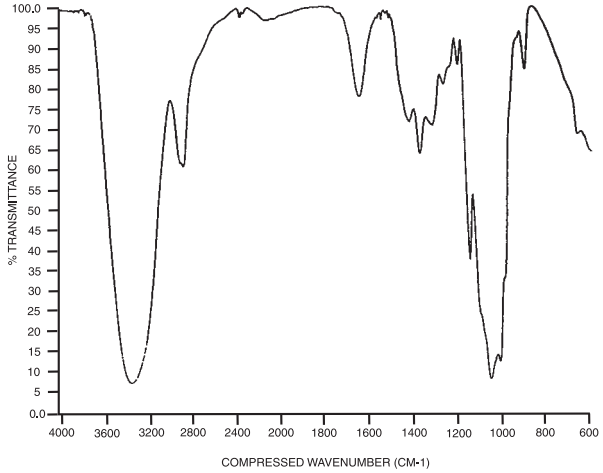


Fig. 7—Rayon

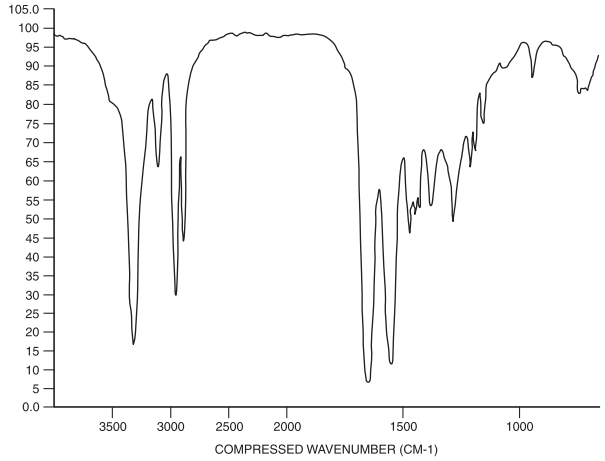


Fig. 8—Nylon 6,6 Fiber

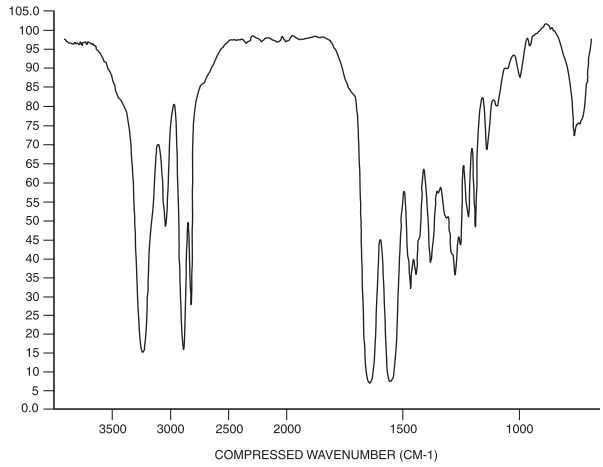


Fig. 9—Nylon 6 Fiber

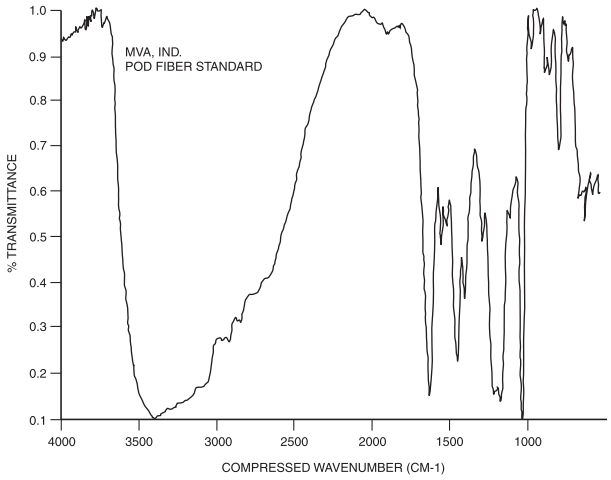


Fig. 10—Polybenzimidazole

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Test Method for Fiber Analysis: Quantitative

1. Purpose and Scope

1.1 This method presents individual procedures for the quantitative determination of moisture content, nonfibrous content and fiber composition of textiles.

1.2 The procedures for the determination of fiber composition include mechanical, chemical and microscopical methods. They are applicable to blends of the following generic classes:

Natural Fibers	Man-Made Fibers
Cotton	Acetate
Hair	Acrylic
Hemp	Aramid (see 17.17)
Linen	meta-aramid
Ramie	para-aramid
Silk	Modacrylic
Wool	Nylon (see 17.1)
	Olefin
	Polybenzimidazole (see 17.17)
	Polyester
	Rayon
	Spandex
	Triexta

2. Uses and Limitations

2.1 The procedure given for the removal of nonfibrous materials will remove most, but not all, of these components. Each treatment is applicable only to certain categories of these substances and no general scheme can be given that is all inclusive.

2.1.1 Some of the newer finishes may present special problems and the analyst will have to deal with these cases as they arise. Thermosetting resins and crosslinking latices are not only difficult to remove but in some cases cannot be wholly removed without destroying the fiber.

2.1.2 When it is necessary to modify a procedure, or use a new one, one should make sure that the fibrous portion of the specimen under test is not attacked.

2.2 Fiber composition is generally expressed in the laboratory either on the oven-dry weight of the textile as received or on the oven-dry weight of the clean fiber after nonfibrous materials are first removed from the textile before the fiber analysis is carried out, or if the treatments described in Section 9 are incapable of removing them, any such materials present will increase the percentage of the fiber constituent with which they are removed during the analysis.

When used in commerce for the representation of the nominal fiber content of end use items such as garments, moisture regain is typically added back to bone dry numbers generated. ASTM D1909, Standard Table of Commercial Moisture Regains for Textile Fibers, can be used for this purpose.

2.3 The procedure for determining fiber composition by mechanical separation is applicable to those textiles wherein the different fibers making up its composition are segregated in separate yarns, or plies, in the textile product.

2.4 The chemical procedures for fiber composition described herein are applicable to most of the current, commercial production fibers within each generic class listed. Known exceptions are noted in Table II. However, there may be instances in which a method may not be fully adequate for a newly developed fiber falling within one of the listed generic classes and for re-used and/or physically or chemically modified fibers. Caution should be exercised when applying these methods to such cases.

2.5 The microscopical procedures for fiber composition are applicable to all fibers and their accuracy depends to a considerable extent upon the ability of the analyst to identify the individual fibers present. However, owing to the tedious nature of this technique, its use is generally limited to those mixtures which cannot be separated mechanically or chemically; e.g., mixtures of hair and wool and mixtures of cotton, linen, hemp and/or ramie.

3. Terminology

3.1 **clean-fiber content,** n.—the amount of fiber after removal of nonfibrous content.

3.2 **fiber,** n.—*in textiles*, a generic term for any one of the various types of matter that form the basic elements of a textile and which are generally characterized by flexibility, fineness and high ratio of length to thickness.

3.3 **moisture content,** n.—that part of the total mass of a material that is absorbed or adsorbed water, compared to the total mass.

3.4 **nonfibrous content,** n.—products such as fiber finishes, yarn lubricants, slasher sizing, fabric softeners, starches, china-clay, soaps, waxes, oils and resins which are applied to fiber, yarn, fabric or apparel.

3.5 Additional terms used in this test method can be found in standard chemical dictionaries, in dictionaries of common terms or in AATCC M11.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 Perform the soxhlet extractions in Section 9, Nonfibrous Material—Clean Fiber Content, using hexane and ethyl alcohol inside an adequately ventilated laboratory hood. CAUTION: Hexane and ethyl alcohol are highly flammable.

4.4 Perform Chemical Analysis Procedure No. 1 (Table II, 100% acetone) inside a ventilated laboratory hood. CAUTION: Acetone is highly flammable.

4.5 Hexane, ethyl alcohol and acetone are flammable liquids and should be stored in the laboratory only in small quantities away from heat, open flame and sparks.

4.6 In preparing, dispensing, and handling hydrochloric acid (20%), sulfuric acids (59.5% and 70%), and formic acid (90%) used in Chemical Analysis Procedure Methods No. 2, 3, 4, and 6 (Table II), use chemical goggles or face shield, impervious gloves and an impervious apron. Concentrated acids should be handled only in an adequately ventilated laboratory hood. CAUTION: Always add acid to water.

4.7 In preparing ammonium hydroxide (8:92) for use in Chemical Analysis Procedure Method No. 4 (Table II, 70% sulfuric acid), use chemical goggles or face shield, impervious gloves and an impervious apron. Dispense, mix and handle ammonium hydroxide only in an adequately ventilated laboratory hood.

4.8 An eyewash/safety shower should be located nearby and a self-contained breathing apparatus should be readily available for emergency use.

4.9 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Govern-

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mental Industrial Hygienists (AC-GIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 17.2).

5. Apparatus

5.1 Analytical balance, capable of weighing to 0.1 mg.

5.2 Oven, maintained at 105-110°C.

5.3 Desiccator, containing anhydrous silica gel, calcium sulfate (such as Drierite) or its equivalent.

5.4 Soxhlet extractor, 200 mL capacity.

5.5 Constant temperature bath, adjustable, capable of controlling temperature to ± 1°C.

5.6 Weighing bottle, 100 mL capacity, glass, with ground glass cover. (Alternate: aluminum weighing can; same size, tight cover.)

5.7 Erlenmeyer flask, 250 mL capacity, ground glass stopper.

5.8 Beaker, borosilicate heat resistant glass, 250 mL capacity.

5.9 Filtering crucible, fritted glass, coarse porosity, 30 mL.

5.10 Suction flask, with adapter, to hold filtering crucible.

5.11 Weighing bottle, large enough to hold filtering crucible.

5.12 Microscope, equipped with a moveable stage and a cross-hair ocular, 200-250× magnification.

5.13 Projection microscope, capable of 500× magnification.

5.14 Fiber cutter: A device comprised of two razor blades, a threaded pin and an assemblage that will hold the blades rigidly in position. The device is operated by applying pressure vertically downward. It cuts fibers approximately 250 µm in length.

5.15 Wedge scale: Strips of heavy paper or Bristol board imprinted with a wedge for use at 500× magnification.

5.16 Flask cover (see 17.16).

5.17 Wet grinder/polisher equipped with 10 in platen for use with 10-in. abrasive discs.

5.18 Adhesive-backed abrasive discs, 10-in. (grit: 120, 240, 320, 400, 600, 800, 1200).

5.19 1-gallon vacuum chamber with pump capable of maintaining vacuum pressure of at least 25 in-Hg.

5.20 2-piece castable mounting cups, 1.5 in.

5.21 Rigid mounting card: non-absorbent yarn sample mounting card for use with epoxy resin mounting method. See Fig. 1 for dimensions suitable for 1.5-in. sample cups.

6. Reagents

6.1 Ethyl alcohol (95%), pure or denatured.

6.2 Hexane (C_6H_{14}).

6.3 Hydrochloric acid (HCl), 0.1*N*.

6.4 Enzyme solubilizing preparation.

6.5 Acetone (CH_3COCH_3), reagent grade.

6.6 Hydrochloric acid (HCl) (20%). Dilute HCl, sp gr 1.19, with water until the specific gravity of the solution is 1.10 at 20°C.

6.7 Sulfuric acid (H_2SO_4) (59.5%). Add H_2SO_4, sp gr 1.84, slowly to water. After the solution has cooled to 20°C, adjust the density to a value between 1.4902 and 1.4956 g/mL.

6.8 Sulfuric acid (H_2SO_4) (70%). Add H_2SO_4, sp gr 1.84, slowly to water. After the solution has cooled to 20 ± 1°C, adjust the density to a value between 1.5989 and 1.6221 g/mL.

6.9 Sulfuric acid (H_2SO_4) (1:19). Slowly stir 1 volume of H_2SO_4, sp gr 1.84, into 19 volumes of water.

6.10 Sodium hypochlorite (NaOCl). Prepare a solution of NaOCl, 5.25% available chlorine. Sodium hypochlorite based household bleach (nominally 5.25%) has been found to be acceptable.

6.11 Sodium bisulfite ($NaHSO_3$) (1%). Freshly prepared.

6.12 Formic acid (HCOOH) (90%), sp gr of 1.202 at 20°C.

6.13 Ammonium hydroxide (NH_4OH) (8:92). Mix 8 volumes of NH_4OH, sp gr 0.90, with 92 volumes of water.

6.14 Herzberg stain. Add the previously prepared solution A to solution B; allow to stand overnight; decant the clear liquid into a dark colored glass bottle and add a leaf of iodine.

Solution A		**Solution B**	
Zinc Chloride	50 g	Potassium Iodide	5.5 g
Water	25 mL	Iodine	0.25 g
		Water	12.5 mL

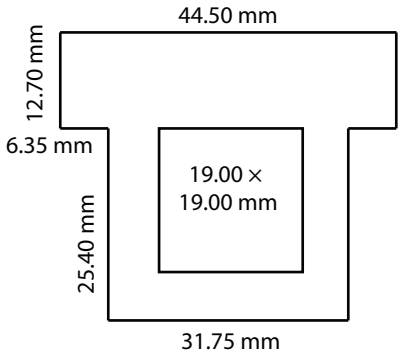


Fig. 1—Rigid Mounting Card (all tolerances are ±0.38 mm).

6.15 Isopropanol (C_3H_7)HO (70%)

6.16 N,N-Dimethylacetamide $CH_3C(O)N(CH_3)_2$

6.17 Methanol (CH_3OH), reagent grade.

6.18 Sodium Hydroxide (NaOH), pellet, 95% reagent grade.

6.19 Xylenes ($C_6H_4(CH_3)_2$), mixed, 95% reagent grade.

6.20 Lithium Chloride (LiCl), crystal reagent grade.

6.21 Epoxy release agent for use with castable mounting materials.

6.22 Epoxy resin suitable for creating castable mounts.

6.23 Epoxy hardener compatible with the selected epoxy resin.

6.24 Cyclohexanone $(CH_2)_5CO$.

7. Sampling

7.1 It is not possible to give specific instructions for taking a laboratory test sample from all types of textile materials to which these methods may be applicable; but a few general recommendations will be given.

7.1.1 The sample should be as representative as possible of the lot of material from which it was taken.

7.1.2 If a reasonably large lot is available, and if it is possible to do so, samplings should be taken from different, widely separated areas or parts of the lot.

7.1.3 In the case of fabrics where there is a definite repetition in the pattern, the sample should include all yarns in a complete pattern (see 17.4).

7.1.4 In the case of yarns, not less than a 2-meter length should be taken.

Test Methods

8. Moisture Content

8.1 Procedure. Place not less than 1 g of the textile to be tested in a previously tared weighing bottle and immediately replace the cover. Weigh to the nearest 0.1 mg using the analytical balance and record the weight. Place the uncovered weighing bottle containing the specimen in an oven maintained at 105-110°C for 1.5 h. At the end of the time period, remove the bottle from the oven, immediately replace the cover and put it in the desiccator. When the bottle and contents have cooled to room temperature, remove them from the desiccator and reweigh. Repeat the heating and reweighing process for periods of 30 min until the weight is constant to within ± 0.001 g and record the constant weight.

8.2 Calculations.

8.2.1 Calculate the moisture content of the specimen as follows:

$$M = \frac{A - B}{A - T} \times 100$$

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where:

M = moisture content, percent.
A = weight of sample before drying + bottle.
B = weight of sample after drying + bottle.
T = tare weight of weighing bottle.

9. Nonfibrous Material—Clean Fiber Content

9.1 Procedure. Take a specimen of not less than 5 g, dry it to constant weight in an oven at 105-110°C (see 8.1), record the oven-dry weight to the nearest 0.1 mg using an analytical balance and then subject it to one, or more, of the following treatments, as appropriate. When specific type of nonfibrous content is known, only that specific treatment, or treatments, need be performed; otherwise, all treatments must be applied.

9.1.1 Hexane Treatment (for removal of oils, fats, waxes, etc.). Extract the dried specimen with hexane in a soxhlet extractor, siphoning over a minimum of six times. Air dry, and then dry at 105-110°C to constant weight. For an alternative to soxhlet extractor, see 17.15.

9.1.2 Alcohol Treatment (for removal of soaps, cationic finishes, etc.). Extract the dried specimen with ethyl alcohol in a soxhlet extractor, siphoning over a minimum of six times. Air dry, and then dry at 105-110°C to constant weight. For an alternative to soxhlet extractor, see 17.15.

9.1.3 Aqueous Treatment (for removal of water soluble materials). Immerse the dried specimen for 30 min in water at 50°C using a 100:1 liquid to fabric ratio. Stir occasionally or use a mechanical shaker. Rinse 3 times in fresh portions of water and dry at 105-110°C to constant weight.

9.1.4 Enzyme Treatment (for removal of starch, etc.). Immerse the dried specimen in aqueous solution of the enzyme preparation following the manufacturer's recommendations as to concentration, liquid to fabric ratio and temperature and time of immersion. Rinse thoroughly with hot water and dry at 105-110°C to constant weight.

9.1.5 Acid Treatment (for removal of amino resins). Immerse the dried specimen in 100 times its weight of 0.1N HCl at 80°C for 25 min, stirring occasionally. Rinse thoroughly with hot water and dry at 105-110°C to constant weight.

9.2 Calculations.

9.2.1 Calculate the nonfibrous content of the specimen as follows:

$$N = \frac{C - D}{C} \times 100$$

where:

N = nonfibrous materials, percent.
C = dry weight, specimen, before treatment.
D = dry weight, specimen, after treatment.

9.2.2 Calculate the clean fiber content of the specimen as follows:

$$F = \frac{D}{C} \times 100$$

where:

F = clean fiber content, percent; other terms as in 9.2.1

9.2.3 Additional techniques for the extraction and analysis of textile finishes can be found in AATCC TM94, Test Method for Finishes in Textiles: Identification.

10. Mechanical Separation

10.1 Procedure. Remove the nonfibrous materials using the appropriate treatment (see 9.1). Separate the component yarns by mechanical dissection; combine those yarns, or plies, having the same fiber composition and determine the oven-dry weight of each generic type present.

10.2 Calculation. Calculate the content of each generic fiber as follows:

$$X_i = \frac{W_i}{E} \times 100$$

where:

X_i = content of fiber i, percent.
W_i = oven-dry weight of fiber i, after separation.
E = weight of clean, oven-dry specimen taken for analysis.

11. Chemical Analysis—General

11.1 Specimen Preparation. Before analyses are undertaken, the laboratory test sample should be disintegrated, homogenized and a portion of the homogenate taken for the chemical treatment(s). In the case of a fabric, one should unravel it into its individual yarns, cut the yarns into lengths not greater than 3 mm, thoroughly mix the cut pieces and then take a representative portion for the specific determination. An alternate procedure, suitable in many cases, is to grind the sample using a Wiley Mill, homogenize the ground fibers by slurrying them in a water suspension in a Waring Blender and taking the representative portion from the dried homogenate for the specific determination. Yarns are treated the same way but omitting the unnecessary steps.

11.2 Method Application. A tabulation of appropriate chemical treatments for binary fiber mixtures is given in Table I. To use this table one enters at the left side on the line listing one of the components of the binary mixture and moves to the box under the column listing the other component and the number therein is the method, or methods, that are applicable for that specific combination. The unbracketed methods are those that dissolve the fiber at the left side of the diagram while the bracketed ones dissolve the fiber at the top of the diagram. Mixtures of more than two components may be analyzed by proper application of a sequence of the individual methods. Table II presents the relative solubilities of the various fibers in all the reagents and, from this, one can select the proper methods and their sequence for the analysis of multifiber mixtures (see 17.5).

12. Chemical Analysis Procedures

12.1 Method No. 1, 100% Acetone: Weigh accurately a 0.5-1.5 g portion of the clean, dry, prepared specimen and record the weight to the nearest 0.1 mg. Transfer into a 250 mL Erlenmeyer flask. Add 100 times its weight of acetone and agitate vigorously for 15 min keeping the temperature at 40-50°C. Decant the liquid from the undissolved residue, add a fresh portion of acetone and agitate for a few more minutes. Repeat the decanting and agitation process one more time and then filter the undissolved residue by suction through a dried weighed, fritted-glass, filtering crucible. Dry the crucible and residue in air and then in an oven at 105-110°C to constant weight. Record the weight of the dried residue to the nearest 0.1 mg.

12.2 Method No. 2, 20% Hydrochloric acid: Weigh accurately a 0.5-1.5 g portion of the clean, dry, prepared specimen and record the weight to the nearest 0.1 mg. Transfer into a 250 mL Erlenmeyer flask. Add 50-150 mL of 20% hydrochloric acid (100 mL reagent/g of sample); shake vigorously and let stand for 5 min at 15-25°C. Shake again and let stand for 15 min. Shake for a third time (see 17.6) and filter the mixture through a dried weighed fritted-glass crucible. Wash into the crucible any residue left in the flask using a little more 20% hydrochloric acid. Apply suction to drain the excess liquor from the filter residue. Wash the residue in the crucible with about 40 mL of 20% hydrochloric acid and then with water until the filtrate is neutral to litmus. Disconnect the suction and add to the crucible about 25 mL of ammonium hydroxide (8:92) allowing the fiber residue to soak for 10 min before applying suction to drain it. Wash the residue with about 250 mL of water, allowing it to soak in the water for about 15 min. After the final washing, apply suction to remove rinse water, and dry the crucible and residue in an oven at 105-110°C to constant weight. Record the dry weight to the nearest 0.1 mg.

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Table I—Chemical Methods for Analysis of Fiber Mixtures

	Wool/ Hair***	Spandex	Silk	Rayon, Lyocell	Polyester, Triexta	Poly-amide-imide	PLA	Para-aramid	Olefin	Nylon	Mod-acrylic	Meta-aramid	Mela-mine	Cotton, Hemp, Linen, Ramie	Acrylic
Acetate	1 4 12 (5)	1	1 12 (5)	1 12	1 12 (9)	1 12	1 12	1 12	1 (10)	1 12 (2)	N/A	1 12	1 12	1 12	1 12
Acrylic	7 8 (5)	(12)	7 8 (3) (5)	7 8 (3)	7 8 (9)	7 8	7 8	7 8	7 8 (10) (12)	7 8 (2) (3) (6)	(1) (12)	7 8	7 8	7 8	
Cotton, Hemp, Linen, Ramie, Jute, Sisal	4 (5)	(7) (8) (12)	(5)	(3)	4 (9)	4	4	4	4 (10) (12)	(2) (6)	4 (1) (12)	4 (11)	4		
Melamine	(5)	(7) (8) (12)	(3) (4) (5)	(3) (4)	(9)	—	—	—	(10) (12)	(2) (6)	(1)(12)	(11)			
Meta-aramid	(5)	(7) (8) (12)	(3) (4) (5)	(3) (4)	(9)	11	—	11	(10) (12)	(2) (6)	(1)(12)				
Modacrylic	12 (5)	1*	1* 12 (3) (4) (5)	1* 12 (3) (4)	1* 12 (9)	1* 12	1* 12	1* 12	1* (10)	1* 12 (2) (6)					
Nylon	2** 3 6 (5)	2** 3 6 (7) (8) (12)	(5)	2** 6	2** 3 6 (9)	2** 3 6	2** 3 6	2** 3 6	2** 3 6 (10) (12)						
Olefin	10 12 (5)	10 (7) (8)	10 12 (3) (4) (5)	10 12 (4)	10 12 (9)	10 12	10 12	10 12							
Para-aramid	(5)	(7) (8) (12)	(3) (4) (5)	(3) (4)	(9)	—	—								
PLA	(5)	(7) (8) (12)	(3) (4) (5)	(3) (4)	(9)	—									
Polyamide-imide	(5)	(7) (8) (12)	(3) (4) (5)	(3) (4)	(9)										
Polyester, Triexta	(5)	9 (7) (8) (12)	(3) (4) (5)	9 (3) (4)											
Rayon, Lyocell	4 (5)	(7) (8) (12)	(5)												
Silk	3 4	(7) (8) (12)													
Spandex	7 8 12														

Notes:
The unbracketed methods are those that dissolve the fiber at the left side of the diagram while the bracketed ones dissolve the fiber at the top of the diagram.
NA = Chemical Methods not applicable to separate two different fiber types of the same generic class. Use the microscopy sections of AATCC TM20A.
— = Method Under Development.
Section 11.2 contains details of table use.

1	*100% acetone:	section 12.1
2	20% hydrochloric acid:	section 12.2
3	59.5% sulfuric acid:	section 12.3
4	70% sulfuric acid:	section 12.4
5	sodium hypochlorite:	section 12.5
6	90% formic acid:	section 12.6
7	dimethylformamide	section 12.7
8	N,N-dimethylacetamide	section 12.8
9	alkaline methanol	section 12.9
10	xylenes	section 12.10
11	4% lithium chloride in N,N-dimethylacetamide	section 12.11
12	Cyclohexanone	section 12.12

*Not suitable for all modacrylic fibers
**Not suitable for all nylon fibers
***Hair fibers referenced in AATCC TM20A are synonymous with fibers with scales on surface in Table I of AATCC TM20

12.3 Method No. 3, 59.5% Sulfuric acid: Weigh accurately a 0.5-1.5 g portion of the clean, dry, prepared specimen and record the weight to the nearest 0.1 mg. Transfer into a 250 mL Erlenmeyer flask. Add 50-150 mL of 59.5% sulfuric acid (100 mL reagent/g of sample) and shake vigorously for 1 min. Let stand for 15 min at a temperature of 15-25°C. Shake again and let stand for another 15 min, shake for a third time (see 17.6) and then filter the mixture through a dried weighed fritted-glass crucible. Wash into the crucible any residue left in the flask using three 10 mL aliquots of 59.5% sulfuric acid. Apply suction to drain the excess liquor from the fiber residue after the addition of each aliquot. Wash the residue in the crucible with 50 mL of sulfuric acid (1:19), then with water until the filtrate is neutral to litmus. Disconnect the suction and add to the crucible about 25 mL of ammonium hydroxide (8:92), allowing the fiber residue to soak for 10 min before applying suction to drain it. Wash the residue with about 150 mL of water, allowing it to soak in the water for about 15 min. After the final washing, apply suction to remove the rinse water and dry the crucible and fiber residue in an oven at 105-110°C to constant weight. Record the weight of the dried residue to the nearest 0.1 mg (see 17.7).

12.4 Method No. 4, 70% Sulfuric acid: Weigh accurately a 0.5-1.5 g portion of the clean, dry, prepared specimen and record the weight to the nearest 0.1 mg. Transfer into a 250 mL Erlenmeyer flask. Add 50-150 mL of 70% sulfuric acid (100 mL reagent/g of sample) and shake vigorously for 1 min. Let stand for 15 min at a temperature of 15-25°C. Shake again and let stand for another 15 min; shake for a third time (see 17.6) and then filter the mixture through a fritted-glass crucible which has been oven-dried, cooled in a desiccator and weighed to 0.1 mg. Wash into the crucible any residue left in the flask using three 10 mL aliquots of 70% sulfuric acid. Apply suction to drain the excess liquor from the fiber residue after the addition of each aliquot. Wash the residue in the crucible with 50 mL of sulfuric acid (1:19), then with water until the filtrate is neutral to litmus. Disconnect the suction and add to the crucible about 25 mL of ammonium hydroxide (8:92); allow the fiber residue to soak for 10 min before applying suction to drain it. Wash the residue with about 150 mL of water, allowing it to soak in the water for about 15 min. After the final washing, apply suction to remove excess water and dry the crucible and fiber residue in an oven at 105-110°C to constant weight. Record the weight of the dry residue to the nearest 0.1 mg.

12.5 Method No. 5, Sodium hypochlorite: Weigh accurately a 0.5-1.5 g portion of the clean, dry, prepared specimen and record the weight to the nearest 0.1 mg. Transfer into a 250 mL Erlenmeyer flask.

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Table II—Solubilities of Fibers in Reagents Used in the Chemical Methods

	Chemical Method											
	No. 1 100% CH_3COCH_3	No. 2 20% HCl	No. 3 59.5% H_2SO_4	No. 4 70% H_2SO_4	No. 5 NaOCl	No. 6 90% HCOOH	No. 7 $(CH_3)_2$ NC(O)H	No. 8 $CH_3C(O)N(CH_3)_2$	No. 9 NaOH + CH_3OH	No. 10 100% $C_6H_4(CH_3)_2$	No. 11 LiCl + $CH_3C(O)N(CH_3)_2$	No. 12 $(CH_2)_5$
Acetate	S	I	S	S	I	S	S	S	S	I	ND	S
Acrylic	I	I	I	I*	I	I	S	S	I	I	ND	I
Cotton	I	I	SS	S	I	I	I	I	I	I	ND	I
Hair	I	I	I	I	S	I	I	I	I	I	ND	I
Hemp	I	I	SS	S	I	I	I	I	I	I	ND	I
Linen	I	I	SS	S	I	I	I	I	I	I	ND	I
Modacrylic	S or I*	I	I	I	I	I	PS	PS	I	I	ND	S
Nylon	I	S	S	S	I	S	I	I	I	I	ND	I
Olefin	I	I	I	I	I	I	I	I	I	S	ND	S
Polyester**	I	I	I	I	I	I	I	I	S	I	ND	I
Ramie	I	I	SS	S	I	I	I	I	I	I	ND	I
Rayon	I	I	S	S	I	I	I	I	I	I	ND	I
Silk	I	PS	S	S	S	PS	I	I	I	I	ND	I
Spandex	I	I	PS	PS	I	I	S	S	PS	I	ND	S
Wool	I	I	I	I	S	I	I	I	I	I	ND	I
Meta-aramid	I	I	I	I	I	I	I	I	I	I	S	I
Para-aramid	I	I	I	I	I	I	I	I	I	I	I	I
Polyamideimide	I	I	I	I	I	I	I	I	I	I	I	I
Melamine	I	I	I	I	I	I	I	I	I	I	I	I
PLA	I	I	I	I	I	I	I	I	NA	I	ND	I
Lyocell	I	I	S	S	I	I	I	I	I	I	ND	I
Polybenzimidazole												

*Depending on type
**Compatible for triexta

KEY TO SYMBOLS: S = SOLUBLE
PS = PARTIALLY SOLUBE (Method not applicable)
SS = SLIGHTLY SOLUBLE (Usable but correction factor required)
I = INSOLUBLE
NA = Not Applicable ND = Not Determined

Section 11.2 contains details of table use.

Add 50-150 mL of sodium hypochlorite reagent (100 mL reagent/g of sample). Use a wrist shaker or stir the specimen vigorously in this solution for 20 min making sure the temperature is maintained at 25 ± 1°C (use constant temperature bath) (see 17.8) and then filter through a dried weighed, fritted-glass crucible. Wash thoroughly with sodium bisulfite (1%) followed by water and remove the excess water by suction. After the final washing, apply suction to remove excess water and dry in an oven at 105-110°C to constant weight. Record the weight of the dried residue to the nearest 0.1 mg.

12.6 Method No. 6, 90% Formic acid: Weigh accurately a 0.5-1.5 g portion of the clean, dry, prepared specimen and record the weight to the nearest 0.1 mg. Transfer into a 250 mL Erlenmeyer flask. Add 50-150 mL of 90% formic acid (100 mL reagent/g of sample) and shake frequently over a period of 15 min (see 17.9). Decant the supernatant liquid into a dried, weighed, fritted-glass crucible, add another equal portion of 90% formic acid to the flask and agitate for an additional 15 min. Filter the contents of the flask through the crucible, rinse with two 50 mL portions of 90% formic acid and drain with the aid of suction. Wash the residue with 50 mL of water and then allow it to soak in 25 mL of ammonium hydroxide (8:92) for about 10 min. Wash the residue thoroughly with water until the filtrate is neutral to litmus. Drain the residue with the aid of suction and dry in an oven at 105-110°C to constant weight. Record the weight of the dried residue to the nearest 0.1 mg.

12.7 Method No. 7, Dimethylformamide: Weigh accurately a 0.5-1.5 g portion of the clean, dried, prepared specimen and record the weight to the nearest 0.1 mg. Transfer to a 250 mL Erlenmeyer flask. Add 50-150 mL of dimethylformamide reagent (100 mL reagent/g of sample). Agitate for 20 min keeping the temperature at 98 ± 1°C. Decant the liquid from the undissolved residue, add a fresh portion of dimethylformamide and agitate for a few more minutes. Repeat the decanting and agitation process one more time rinsing well with 70% isopropanol, and then filter the undissolved residue by suction through a dried, weighed fritted glass filtering crucible. Dry the crucible and residue in air and then in an oven at 105-110°C to constant weight. Record the weight of the dried residue to the nearest 0.1 mg.

12.8 Method No. 8, N,N-Dimethylacetamide: Weigh accurately a 0.5-1.5 g portion of clean, dried, prepared specimen and record the weight to the nearest 0.1 mg. Transfer to a 250 mL Erlenmeyer flask. Add 50-150 mL of N,N-dimethylacetamide reagent. Agitate for 20 min keeping the temperature at 70 ± 1°C. Decant the liquid from the undissolved residue, add a fresh portion of the N,N-dimethylacetamide and agitate for a few more minutes. Repeat the decanting and agitation process one more time rinsing well with 70% isopropanol, and then filter the undissolved residue by suction through a dried, weighed fritted glass filtering crucible. Dry the crucible and residue in air then in an oven at 105-110°C to constant weight. Record the weight of the

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dried residue to the nearest 0.1 mg.

12.9 Method No. 9, Alkaline Methanol: Weigh accurately a 0.5-1.5 g portion of clean dried, prepared specimen and record the weight to the nearest 0.1 mg. Add 18 g of reagent grade sodium hydroxide in a pellet form to 200 mL of methanol (dissolution of the NaOH is mildly exothermic) in a 250 mL Erlenmeyer flask. Heat to 65°C mixing thoroughly. Add specimen and stir using a magnetic stirring bar and flask cover. After 5 min of immersion, decant the liquid from the undissolved residue, rinse using 70% isopropanol and then filter the undissolved residue by suction through a dried, weighed, fritted glass filtering crucible. Dry the crucible and residue in air and then in an oven at 105-110°C to constant weight. Record the weight of the dried residue to the nearest 0.1 mg.

12.10 Method No. 10, Xylene: Weigh accurately a 0.5-1.5 g of clean, dried, prepared specimen and record the weight to the nearest 0.1 mg. Transfer to a 250 mL Erlenmeyer flask. Add 50-150 mL of xylenes under a fume hood and cover with a flask cover. Heat to the boiling temperature and stir with a magnetic stirrer for 20 min. Decant the liquid from the undissolved residue. Rinse well with 70% isopropanol, and repeat the decanting and agitation process. Filter the undissolved residue by suction through a dried, weighed, fritted glass filtering crucible. Dry the crucible and residue in air and then in an oven at 105-110°C to constant weight. Record the weight of the dried residue to the nearest 0.1 mg.

12.11 Method No. 11, 4% Lithium Chloride in N,N-Dimethylacetamide: Weigh accurately a 0.5-1.5 g portion of the clean, dried, prepared specimen and record the weight to the nearest 0.1 mg. Transfer to a 250 mL Erlenmeyer flask. Add 50-150 mL of 4% lithium chloride in N,N-dimethylacetamide reagent (100 mL reagent/g of sample). Agitate for 180 min keeping the temperature at 65 ± 1°C. Decant the liquid from the undissolved residue, add a fresh portion of 4% lithium chloride and agitate for 5-10 min. Repeat the decanting and agitation process one more time rinsing well with 70% isopropanol, and then filter the undissolved residue by suction through a dried, weighed fritted glass filtering crucible. Dry the crucible and residue in air and then in an oven at 105-110°C to constant weight. Record the weight of the dried residue to the nearest 0.1 mg.

12.12 Method No. 12, Cyclohexanone: Weigh accurately a 0.5-1.5 g of clean, dried, prepared specimen and record the weight to the nearest 0.1 mg. Transfer to a 250 mL Erlenmeyer flask. Add 50-150 mL of cyclohexanone under a fume hood and cover with a flask cover. Heat to the boiling temperature and stir with a magnetic stirrer for 5-10 min. Decant the liquid from the undissolved residue. Rinse well with 70% isopropanol and repeat the decanting and agitation process. Filter the undissolved residue by suction through a dried, weighed, fritted glass filtering crucible. Dry the crucible and residue in air and then in an oven at 105-110°C to constant weight. Record the weight of the dried residue to the nearest 0.1 mg.

12.13 Calculations: Calculate the content of each generic fiber type as determined by any one of the above applicable chemical methods using one of the following equations:

12.13.1 If the fiber is *dissolved* by the test reagent:

$$X_i = \frac{G - H_i}{G} \times 100$$

12.13.2 If the fiber is insoluble in the test reagent:

$$X_i = \frac{H_i}{G} \times 100$$

where:

X_i = content of fiber *i*, percent.
G = weight of clean, dry, prepared specimen
H_i = weight of dried residue after treatment

13. Microscopical Analysis, General

13.1 The following procedure may be used for the quantitative analysis of textiles containing two or more fiber types which cannot be separated readily by mechanical or chemical methods. The procedures rely on the ability of a technician to identify and count, by means of a microscope, the relative number of fibers of each type in a prepared specimen. Such a count will result in a percent blend by *number* of fibers. In order to convert this result to a percent by weight, the size of the fibers being counted and their respective densities must be included in the calculation.

13.2 Microscope slides may be prepared to scan longitudinal or cross-section views of a fiber sample. The fiber images may be viewed either through a microscope or as projected onto a horizontal plane. While either viewing method may be used for identification and counting of fibers, the projection method is specifically used for measuring fiber diameters using a wedge scale (see 14.3.2).

13.3 Methods which may be used to identify fibers during the fiber counting procedures are discussed in AATCC TM20, Test Method for Fiber Analysis: Qualitative. They include the following:

	AATCC TM20
Herzberg stain (zinc chloro-iodide)	Section 9.9.1
Acid phloroglucinol reagent	Section 9.9.2
Longitudinal appearance	Tables I and II
Cross-section appearance	Tables I and II and Appendix I

It is recommended that reference tests be made on known fibers rather than placing total reliance on photographic reproductions and word descriptions of colors.

14. Microscopical Analysis Procedures

14.1 Preparation of Slides.

14.1.1 Longitudinal Sections of Vegetable Fibers (cotton, flax, ramie, etc.): A swatch of fabric measuring at least 5 × 5 cm should be available. Count the number of yarns in both the warp and filling and select from each direction at random a number of yarns that is proportional to the fabric count. The combined number of warp and filling yarns should total at least 20 (see 17.10). If the sample is in yarn form, take at least a two-meter length and, from it, cut not less than twenty 5-cm sections at random. Cut approximately 2.5 cm of each yarn, or yarn section, into lengths of 0.5-1 mm. The shorter the lengths the easier it is to prepare a homogeneous fiber suspension. Collect the cut fibers on a paper of contrasting color and transfer to a 125 mL Erlenmeyer flask. Add sufficient water so that after stoppering the flask and shaking the contents, a uniform and fairly dense fiber suspension is obtained. Quick boiling or the addition of a few glass pellets facilitates the separation of the fibers. Using a glass-marking pencil, draw two parallel lines about 1 in. apart across a glass slide. With a wide-mouth pipette, draw 0.5-1 mL of the well shaken suspension and place it between the two reference lines on the slide. The amount of liquid taken is dependent upon the density of the suspension. Just sufficient liquid should be placed on the slide so that—after evaporation—a thin, uniform film of fibers remains. After all of the moisture has evaporated from the slide, stain the fibers with Herzberg stain and cover with a cover glass.

14.1.2 Longitudinal Sections of Wool, Hair and other round Fibers: Select a representative swatch, or yarn sections, as in 14.1.1. With a fabric swatch, remove the outermost yarns in both directions so that the warp and filling yarns are protruding approximately 1 cm. Lay the sample flat on a table and, using a fiber cutter, force the blades vertically downward into the warp fringe. Repeat the operation with the filling fringe. Remove the device with the top plate up, release the tension on the cutting blades and remove them together by their ends between the thumb and

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forefinger. Carefully separate the blades so that the cut fiber sections will adhere to the edge of one or both blades. Place a few drops of mineral oil on a clean slide and, with a dissecting needle, scrape the fiber sections onto the oil. Thoroughly disperse the fibers in the oil with the dissecting needle and cover with a cover glass. For yarn samples, align the sections side-by-side and perform the above operation starting with the use of the fiber cutter.

14.1.3 Cross-Sections of Fibers: Select representative yarns, or yarn sections, from the sample and render them to staple by removing twist and drawing out fibers. Align the fibers parallel to each other so as to form a well blended tuft. Prepare a slide following the instructions of AATCC TM20, Section 9.3.

14.1.4 Cross-Sections of Cut-resistant Fibers or Yarns: For yarns containing highly cut resistant fibers that cannot be readily separated by mechanical or chemical means (e.g. blends containing aramids and/or polybenzimidazole), do not use the fiber cutting techniques described in AATCC TM20, Section 9.3. Mount representative yarns from the sample onto the rigid mounting card using double sided tape. Coat the inside surfaces of the sample cup with epoxy release agent. Place rigid mounting card, with sample affixed, into the sample cup taking care to keep the card centered and perpendicular. Mix epoxy resin and epoxy hardener according to manufacturer's instructions and fill the sample cup. Place sample cup containing samples and resin into vacuum chamber and evacuate to a vacuum pressure of 25 in-Hg for 10 minutes. Slowly release vacuum pressure and cure epoxy in standard atmospheric conditions according to manufacturer's instructions (typically 18-24 h). Remove cured epoxy mount from sample cup and cut away top tabs of rigid mounting card. Starting with 120 grit abrasive disc affixed to grinder/polisher platen, grind surface to a smooth finish, free of indentations. Gradually increase the grit rating (going to finer grit) of the abrasive disc, grinding/polishing to a smooth (scratch-free) clear finish on the epoxy mount. Mounted fibers are then counted according to 14.2.

14.2 Fiber Counting.

14.2.1 Fibers Viewed Through Microscope: Place the slide prepared as in 14.1 on the moveable stage on a microscope equipped with a crosshair ocular and having a magnification of 200-250×. Begin to count near either the upper or lower corner of the field and, as the slide is moved slowly across the field in the horizontal direction, identify and count all fibers passing through the center of the crosshairs. After each trip across the field, move the slide 1-2 mm vertically and identify and count the fibers as the field is again traversed. Repeat this procedure until the whole slide has been covered. The spacing between each traverse is dependent upon the number of fiber sections on the slide. If a fiber passes the crosshair more than once, record each passing (see 17.11). In a similar manner, count the fibers by moving the slide vertically. The combined horizontal and vertical counts should total at least 1000 fibers.

14.2.2 Projected Images of Fibers: Calibrate the microprojector so that it will give a magnification of 500× in the plane of the projected image. To do this, place a stage micrometer (having units of 0.01 mm) on the stage of the microprojector with its scale toward the objective and place a large sheet of white, nonglare paper in the projection plane. Lower or raise the microscope until an interval of 0.20 mm on the stage micrometer will measure 100 mm when sharply focused in the center of the paper. Place the slide prepared as in 14.1 on the stage of the microprojector with the cover glass toward the objective. Draw a 10-cm diameter circle in the center of the white paper in the projection plane. All measurements and counts are to be made within this circle. Begin to count near either the upper or lower corner of the field and proceed exactly as described in 14.2.1 (see 17.12).

14.2.3 Video Imaging of Fibers: Using a video monitor with a crosshair placed in the center of the screen coupled to a video camera attached to a correctly adjusted transmitted light microscope scan the slide as outlined in 14.2.1.

14.3 Fiber Measurement.

14.3.1 Fibers with Noncircular Cross-Sections:

14.3.1.1 Paper Tracing Method: Prepare a slide as described in 14.1.3. Place this slide on the stage of the microprojector and project the image onto a sheet of graph paper having one millimeter squares. Trace the image of the fibers on the graph paper using a sharpened pencil, taking care not to retrace fibers previously traced. If there are not sufficient fibers on the slide to provide 100 of each type, prepare another slide as described above using another tuft of fibers. Continue to trace and to count on the new slide until 100 fibers of each type have been tallied. By counting squares, and parts of squares, determine the cross-sectional area of each individual fiber of each type. Calculate the mean cross-sectional area for each fiber type present by summing the individual values recorded for that type and dividing the sum by the total number of fibers of that type measured. Final values should be in mm^2.

14.3.1.2 Digital Tracing Method: Prepare a slide as described in 14.1.3. Place this slide on the stage and adjust until a clear well formed image appears on the video monitor. Use of a 50× objective with C mount video camera coupling has been found acceptable for most fiber types. Using correctly calibrated image analysis software and a digital image capture board digitally capture the cross-sectional image. Using a mouse or stylus trace around the captured cross-sectional images. Use the image analysis software to store the resultant cross-sectional areas. If there are not sufficient fibers on the slide to provide 100 of each type, prepare another slide as described above using another tuft of fibers. Continue to trace and count on the new slide until 100 fibers of each type have been tallied. Calculate the mean cross-sectional area automatically using the statistical functions of the digital image analysis software. Final values should be in μm^2.

14.3.2 Fibers with Circular Cross-Sections: Prepare a slide as described in 14.1.2. Be sure to make measurements on the same day that the slide was prepared. Position the slide in the microprojector or microscope so that all areas of the slide can be reached.

14.3.2.1 Measurement with Wedge Scale and Microprojector: Bring each individual fiber being measured into sharp focus on the wedge scale. Adjust the position of the scale until the fiber image is projected on the wedge with a fine line and mark on the wedge the point that corresponds to the width of the fiber midway of its length. Traverse the slide and measure successive fibers of each type following a planned course. Measure fibers only when their midpoint falls within the 10-cm circle central located in the field. Exclude from measurement those fibers that cross another fiber at the point of measurement and those that are shorter than 150 µm. A minimum of 100 fibers of each type present should be measured. Calculate the mean cross-sectional area of each fiber type. Final values should be in μm^2 (see 17.13). Considerable variation can occur in the average diameter of hair and wool fibers. Thus, for accurate results on any specific sample, the fiber diameters must be measured. If, however, maximum accuracy is not required, the diameters given in Tables III and IV may be used.

14.3.2.2 Measurement with Digital Image Analysis Software: Using a correctly calibrated image analysis system measure the diameter of the individual fibers in question.

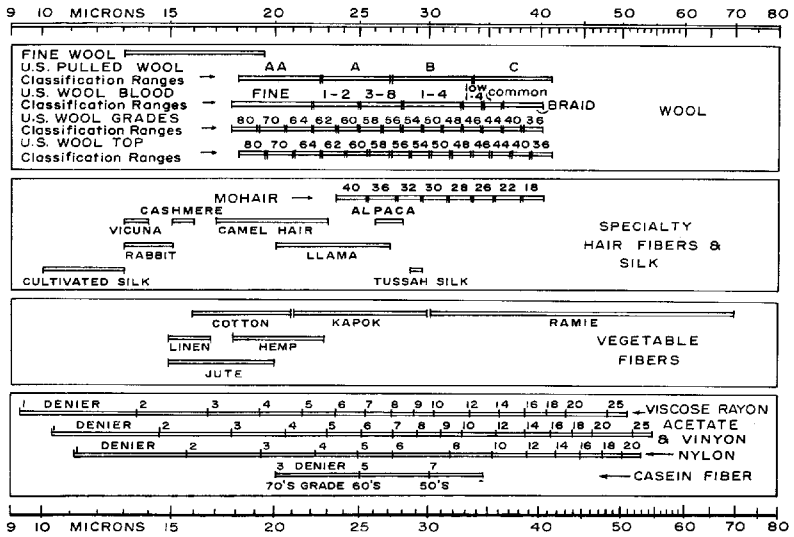
14.4 Calculations. Calculate the content of each fiber as percent by weight using equation 1 if 14.3.1 was used to determine average area of fiber cross-section images, and equation 2 if section 14.3.2 was used to determine fiber diameters (see 17.14).

$$X_i = \frac{N_i \times A_i \times S_i}{\Sigma(N \times A \times S)} \qquad \text{(Eq. 1)}$$

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Table III—Comparative Scale for Fineness of Various Textile Fibers in Micrometers (μm)

IWTO Super Fine Wools Grades			
'X' Value	**Average Fineness μ**	**'X' Value**	**Average Fineness μ**
Super 80's	19.5 ± 0.25	Super 150's	16.0 ± 0.25
Super 90's	19.0 ± 0.25	Super 160's	15.5 ± 0.25
Super 100's	18.5 ± 0.25	Super 170's	15.0 ± 0.25
Super 110's	18.0 ± 0.25	Super 180's	14.5 ± 0.25
Super 120's	17.5 ± 0.25	Super 190's	14.0 ± 0.25
Super 130's	17.0 ± 0.25	Super 200's	13.5 ± 0.25
Super 140's	16.5 ± 0.25	Super 210's	13.0 ± 0.25



$$X_i = \frac{N_i \times D_i^2\ \pi/4 \times S_i}{\Sigma(N \times D^2 \pi/4 \times S)} \qquad \text{(Eq. 2)}$$

where:

X_i is content of fiber i, percent (by weight)
N_i is relative number of fibers of type i
A_i is average area of fiber images of fiber i
D_i^2 is the mean of diameter squared of fibers of type i
D^2 $\pi/4$ is the mean cross-sectional area of round cross-section fibers of type i
S_i is specific gravity of type i fiber
$\Sigma(N_i \times A_i \times S_i)$ is sum of the respective N × A × S products for each fiber type in the blend
$\Sigma(N \times D_i^2\ \pi/4 \times S)$ is sum of the respective $N \times D_i^2\ \pi/4 \times S$ products for each fiber type in the blend

See Table V for specific gravity values.

15. Report

15.1 Report the percentage fiber content by weight of the sample analyzed. State if nonfibrous content has been removed or if results are based on other than oven-dry weights.

16. Precision and Bias

16.1 A chemical separation interlaboratory test was conducted using a PET/Wool intimate blend fabric with a nominal fiber content of 55% PET/45% Wool according to the fabric manufacturer, with results as noted in Tables VI and VII.

16.1.1 Between Laboratories

Standard Deviation = $\sqrt{0.6305}$ = 0.7940% polyester
Precision: ± t.975(6df)×S = 2.45 × 0.7940% = ± 1.9454% polyester

16.1.2 Between Operators within Laboratories.

Standard deviation = $\sqrt{0.0655}$ = 0.2559% polyester
Precision: ± t.975(6df)×S = 2.45 × 0.2559% = ± 0.627% polyester

16.2 *Interpretation*. The above statistics apply to the PET/Wool fabric tested which may represent a best case scenario for the determination of fiber content by chemical separation. Additional studies are underway in conjunction with Committee RA102, Statistics Advisory, involving different blend levels.

16.3 *Bias*. The quantitative fiber analysis can only be defined in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

17. Notes

17.1 Applicable only to Nylon 6 and Nylon 6,6.

17.2 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

17.3 Since sodium hypochlorite solutions lose strength on standing, it is recommended that they be standardized frequently. The following is a suitable method for determining the available chlorine content of such solutions: Dilute a 10 mL aliquot of the solution to be tested to 250 mL with water in a volumetric flask. Pipette 25 mL from the volumetric flask to an Erlenmeyer flask; add 3-5 mL of a 10% solution of potassium iodide (KI) and then add 2-3 mL acetic acid (CH_3COOH). Mix well and titrate with 0.1N sodium thiosulfate ($Na_2S_2O_3$) until the yellow color of the iodine is nearly destroyed. Add 5 mL of a starch indicator solution and titrate until the blue color entirely disappears. Calculate the percentage of available chlorine by weight as follows:

$$\text{Available Chlorine, \%} = 3.5A/B$$

where:

A = mL of 0.1N sodium thiosulfate used
B = g of 10 mL aliquot taken

17.4 Exception: when the pattern is smaller than 15 × 15 cm, a sufficient number of complete patterns should be taken to be equivalent to not less than 225 cm^2.

17.5 Whenever there is any doubt about the effectiveness of a given method in dissolving a specific fiber, or whenever there is an application of a method to a new type of fiber, one should always examine the residue in the filtering crucible after weighing. This precaution should always be taken when a fiber blend consists of (1) one predominate fiber with one (or more) minor components; or (2) one very minor fiber with one (or more) major components.

17.6 If a mechanical shaking machine is available, the flask may be shaken on it continuously for 30 min.

17.7 Cotton is not completely insoluble in H_2SO_4 (59.5%). Furthermore, a small amount of rayon remains undissolved in this solvent. In addition, some types of rayon, like lyocell, may gel instead of dissolve in 59.5% sulfuric acid (see Fig. 2). In the analysis of cotton/rayon blends, interlaboratory tests indicate that to allow for the above bias, the composition of the specimen should be calculated as follows:

$$\text{Corrected cotton, \%} = \frac{100\ aJ}{F} - 1.6$$

where:

for raw cotton, a is 1.062
for bleached cotton, a is 1.046
J is the oven-dry residue weight
F is the oven-dry weight of clean fiber before treatment.

Corrected Regenerated Cellulose Rayon,

% = 100 – corrected cotton percent.

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Table IV—Fineness Ranges and Fiber Diameters of Various Textile Fibers[a] in Micrometers (μm)

U.S. Wool Classification

Wool Grades			Wool Top Grades		Pulled Wool
Numerical System	Average Diameter	Blood System[c]	Numerical System	Average Diameter[d]	Grades
80s	17.7-19.1	Fine	80s	18.1-19.5	AA
70s	19.2-20.5	Fine	70s	19.6-21.0	AA
64s	20.6-22.0	Fine	64s	21.1-22.5	AA
62s	22.1-23.4	½	62s	22.6-24.0	A
60s	23.5-24.9	½	60s	24.1-25.5	A
58s	25.0-26.4	⅜	58s	25.6-27.0	A
56s	26.5-27.8	⅜	56s	27.1-28.5	B
54s	27.9-29.3	¼	54s	28.6-30.0	B
50s	29.4-30.9	¼	50s	30.1-31.7	B
48s	31.0-32.6	¼	48s	31.8-33.4	B
46s	32.7-34.3	Low ¼	46s	33.5-35.1	C
44s	34.4-36.1	Common	44s	35.2-37.0	C
40s	36.2-38.0	Braid	40s	37.1-38.9	C
36s	38.1-40.2	Braid	36s	39.0-41.2	C

Hair Fibers and Silk

Mohair (1)		Miscellaneous Hair Fibers (1)		Silk (1)	
Grade	Fineness Range	Fiber	Average Fineness	Fiber	Average Fineness
40s	23.55-25.54	Vicuna	13.0-14.0	Cultivated silk	10.0-13.0
36s	25.55-27.54	Cashmere	14.0-19.0	Tussah silk	28.5
32s	27.55-29.54	Camel hair	17.0-23.0		
30s	29.55-31.54	Alpaca	26.0-28.0		
28s	31.55-33.54	Llama	20.0-27.0		
26s	33.55-35.54				
22s	35.55-38.04				
18s	38.05-40.54				

Vegetable Fiber (1)		Glass Fiber (2)			
Fiber	Average Fineness	Filament Diameter Designation	Theoretical Diameter	Staple Fiber Diameter Designation	Average Diameter
Cotton	16.0-21.0	D	5.3	E	7.1
Flax (linen)	15.0-17.0	E	7.4	G	9.7
Jute	15.0-20.0	G	9.0	J	11.4
Hemp	18.0-23.0				
Kapok	21.0-30.0				
Ramie	25.0-30.0				

Theoretical Fiber Diameter[d]

Rayon (3), Acetate (3), Nylon (4), and Vinyon (3)				Casein Fiber (5)		
Filament Denier	Viscose Rayon	Acetate and Vinyon	Nylon	Grade	Denier	Fiber Diameter
1	9.6	10.3	11.1	70s	3	20
2	13.6	14.5	15.7	60s	5	25
3	16.7	17.8	19.3	50s	7	30
4	19.3	20.6	22.3			
5	21.6	23.0	24.9			
6	23.6	25.2	27.3			
7	25.5	27.3	29.5			
8	27.3	29.1	31.5			
9	28.9	30.9	33.4			
10	30.5	32.6	35.2			
12	33.4	35.7	38.5			
14	36.1	38.5	41.7			
16	38.6	41.2	44.5			
18	40.9	43.7	47.3			
20	43.1	46.1	49.9			

[a] Source of Data:
(1) Werner von Bergen and W. Krauss, *Textile Fiber Atlas*, Textile Book Publishers Inc., New York NY (1949).
(2) Owens-Corning Fiberglas Corp.
(3) American Viscose Corp.
(4) E. I. du Pont de Nemours and Co.
(5) Aralac Incorporated.

[c] Trade application.

[d] U.S. Standards, Federal Register, January 13, 1954; Specifications for Fineness of Wool Tops and Assignment of Grade (ASTM Designation: D3992).

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Table V

Fiber	Specific Gravity
Acetate	1.31
Acrylic	1.16-1.22
Cotton	1.55
Hair	1.32
Hemp	1.48
Linen	1.50
Modacrylic	1.28-1.38
Nylon	1.14
Olefin	0.93
Polyester	1.23-1.40
Polybenzimidazole	1.4
Ramie	1.51
Rayon	1.52
Silk	1.25
Spandex	1.0-1.2
Wool	1.31

In the case of a fiber with a range of values, knowledge of the specific type in the generic class may permit the selection of a precise specific gravity; or the density of a fiber may be determined using the procedure described in AATCC TM20, Section 9.6.

17.7.1 Linen is not completely insoluble in H_2SO_4 (59.5%). Furthermore, a small amount of rayon remains undissolved in this solvent. In addition, some types of rayon, like lyocell, may gel instead of dissolve in 59.5% sulfuric acid (see Fig. 2). In the analysis of linen/rayon blends, interlaboratory tests indicate that to allow for the above bias, the composition of the specimen should be calculated as follows:

$$\text{Corrected linen, \%} = \frac{100\ aJ}{F} - 1.6$$

where:

for undyed linen, a is 1.084
for dyed linen, a is 1.105
J is the oven-dry residue weight
F is the oven-dry weight of clean fiber before treatment.

Corrected Regenerated Cellulose Rayon,

% = 100 – corrected linen percent.

17.7.2 Some dyeing and finishing processes may increase the variability and decrease the accuracy of chemical separation using 59.5% H_2SO_4 to determine the percent fiber content by weight of cellulosic blends.

17.8 Extreme care must be taken to control both time of exposure to the reagent and the temperature of treatment. If either is insufficient, the desired fiber or fibers, will not be completely dissolved. If either becomes excessive, it will cause attack of other fibers.

17.9 A mechanical shaking machine may be used for this purpose.

17.10 For fancy woven fabrics, use all the yarns in one or more complete patterns or a representative fraction, if the pattern is large.

17.11 Linen fibers may be present in the fabric or yarn in the form of fiber bundles. Most bundles are reduced to single fibers during the preparation of the fiber suspension. If, however, some bundles do appear on the slide, an attempt should be made to count each of the individual fibers in the bundle.

17.12 A regular microscope may be used for the fiber counting and, if a suitable calibrating device is available, it may also be used for measuring fiber diameters.

17.13 For further information on marking cell numbers on a wedge scale, and for examples of calculations of how to determine the average fiber diameter using cell numbers, see ASTM D2130, Method of Test for Diameter of Wool and Other Animal Fibers by Microprojection.

17.14 One is cautioned not to mix units; e.g., if some fiber diameters have been determined in µm, then all diameters must be in µm. For ease of calculation, the terms D_i^2 and D^2 may be used in place of $D_i^2\ \pi/4$ and $D^2\ \pi/4$ respectively. If this is done one cannot use both cross-sectional areas and squared diameters in the same equation.

17.15 Any extractor capable of heating the sample in the solvent up to 150°C while pressurizing up to 2000 may be used. Solvent in use must have an auto ignition of higher than 200°C. The accelerated solvent extractor (ASE) has been found to be an acceptable alternative to Soxhlet extractor.

17.16 The flask covers (reflux caps) can be purchased from Fisher Scientific Company, Part #10-042.

17.17 Blends of aramids and polybenzimidazole are very difficult to cut cleanly using fiber cutting techniques (e.g. AATCC TM20, Section 9.3) without smearing the cross-section of the highly cut resistant fibers. Attempts at chemical separation of the fibers typically results in the either no dissolution or partial dissolution of both fiber types, making accurate quantitative analysis impossible. Hard epoxy mount cross-sections (see 14.1.4) provide very clean cross-sections that allow accurate microscopical quantitative blend analysis.

18. References

18.1 ASTM D276, Standard Test Methods for Identification of Fibers in Textiles.

18.2 ASTM D629, Standard Test Methods for Quantitative Analysis of Textiles.

18.3 ASTM D1776, Standard Practice for Conditioning and Testing Textiles.

18.4 ASTM D1909, Standard Table of Commercial Moisture Regains for Textile Fibers.

18.5 ASTM D2130, Method of Test for Diameter of Wool and Other Animal Fibers by Microprojection.

18.6 ASTM D4920, Standard Terminology Relating to Moisture in Textiles.

18.7 Available from ASTM International,

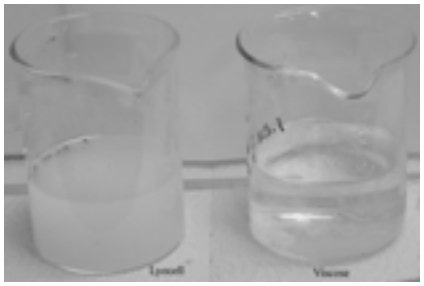


Fig. 2—Gelled Lyocell in left beaker; Dissolved Viscose in right beaker.

Table VI—Nested Factorial Design (Polyester %)

Laboratory	A		B		C		D		E	
Operator	1	2	3	4	5	6	7	8	9	10
	58.00	57.57	58.60	58.00	57.95	58.27	58.35	59.88	58.30	57.78
	58.09	57.65	58.00	57.70						
	58.04	57.60								
Totals	174.13	172.82	116.60	115.70	57.95	58.27	58.35	59.88	58.30	57.78

Table VII—ANOVA

	Degrees of Freedom (df)	Sum of Squares	Mean Square	F Ratio
Between Laboratories	4	2.5221	0.6305	2.161
Between Operators within Laboratories	5	0.3275	0.0655	0.224
Residual (or error)	6	1.7501	0.2917	—
Totals	15	4.5997		

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100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610. 832.9500; fax: +1.610.832. 9555; web site: www.astm.org.

18.8 IWTO Draft Test Method 58-97, "Quantitative Analysis of Blends of Wool with Specialty Fibers by Scanning Electron Microscope." Describes method for distinguishing between specialty animal fibers and wool and provides numerous references.

19. History

19.1 Editorially revised in 2023 to update numbering in section 17.

19.2 Revised in 2021 to updated Table I for correctness. Revised in 2020 for clarity and to update Tables I and II; editorially revised 2019; revised 2018, 2017; editorially 2015; revised 2014, 2013, 2012, 2011, 2010; editorially revised 2009; revised 2008, 2007, 2005, 2004; editorially revised 2002; revised 2000, 1995; reaffirmed 1989; editorially revised 1985, 1982 (new title); reaffirmed 1981; editorially revised 1980; reaffirmed 1978; revised 1975; reaffirmed 1971; revised 1959, 1958.

19.3 Developed in 1957 by AATCC Committee RA24. Related to ISO 17751 and 1833 and IWTO 58.

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Test Method for Water Repellency: Spray

Developed in 1941 by AATCC Committee RA63; revised 1952, 1996, 2001, 2005, 2017; reaffirmed 1943, 1961, 1964, 1967, 1971, 1974, 1977, 1980, 1985, 1989, 2010, 2014; editorially revised 1987, 2008, 2016, 2019. Technically equivalent to ISO 4920.

1. Purpose and Scope

1.1 This test method is applicable to any textile fabric, which may or may not have been given a water-repellent finish. It measures the resistance of fabrics to wetting by water. It is especially suitable for measuring the water-repellent efficacy of finishes applied to fabrics.

1.2 The results obtained with this test method depend on the resistance to wetting or water repellency of the fibers, yarns and finishes on the fabric, and upon the construction of the fabric.

2. Principle

2.1 Water sprayed against the taut surface of a test specimen under controlled conditions produces a wetted pattern whose size depends on the repellency of the fabric. Evaluation is accomplished by comparing the wetted pattern with pictures on a standard chart.

3. Terminology

3.1 **water repellency,** n.—*in textiles*, the characteristic of a fiber, yarn or fabric to resist wetting.

3.2 **face,** n.—*in textiles*, the side of a fabric that is intended to be the outer visible surface in an end product.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should he followed. Wear safety glasses in all laboratory areas.

5. Uses and Limitations

5.1 The portability and simplicity of the instrument, and the shortness and simplicity of the test procedure, make this method of test especially suitable for screening finishes. It is not intended, however, for use in predicting the probable rain penetration resistance of fabrics, since it does not measure penetration of water through the fabric. For testing rain penetration, AATCC TM35, Test Method for Water Resistance: Rain Test, should be used.

6. Apparatus and Materials (see 11.1)

6.1 AATCC Spray Tester (see 11.2, Figs. 1, 2, 2A and 3).

6.2 Graduated cylinder, 250 mL.

6.3 Water, distilled.

6.4 Stopwatch (see 8.3)

7. Test Specimens

7.1 Three test specimens 180.0 × 180.0 mm (7.0 × 7.0 in.) are needed and should be conditioned at 65 ± 5% relative humidity and 21 ± 2°C (70 ± 4°F) for a minimum of 4 h before testing.

7.2 Where possible, each specimen should contain different groups of lengthwise and widthwise yarns.

8. Procedure

8.1 Calibrate the apparatus by pouring 250 mL of distilled water at 27 ± 1°C (80 ± 2°F) into the funnel of the tester and measure the time required for the funnel to empty.

8.1.1 The spray time must be between 25-30 s, otherwise the nozzle should be checked to see if the holes are enlarged or blocked. Nominal hole size is indicated in Fig. 3, but spray time is the critical value for calibration.

8.2 Fasten the test specimen securely in the 152.4 mm (6.0 in.) diameter hoop

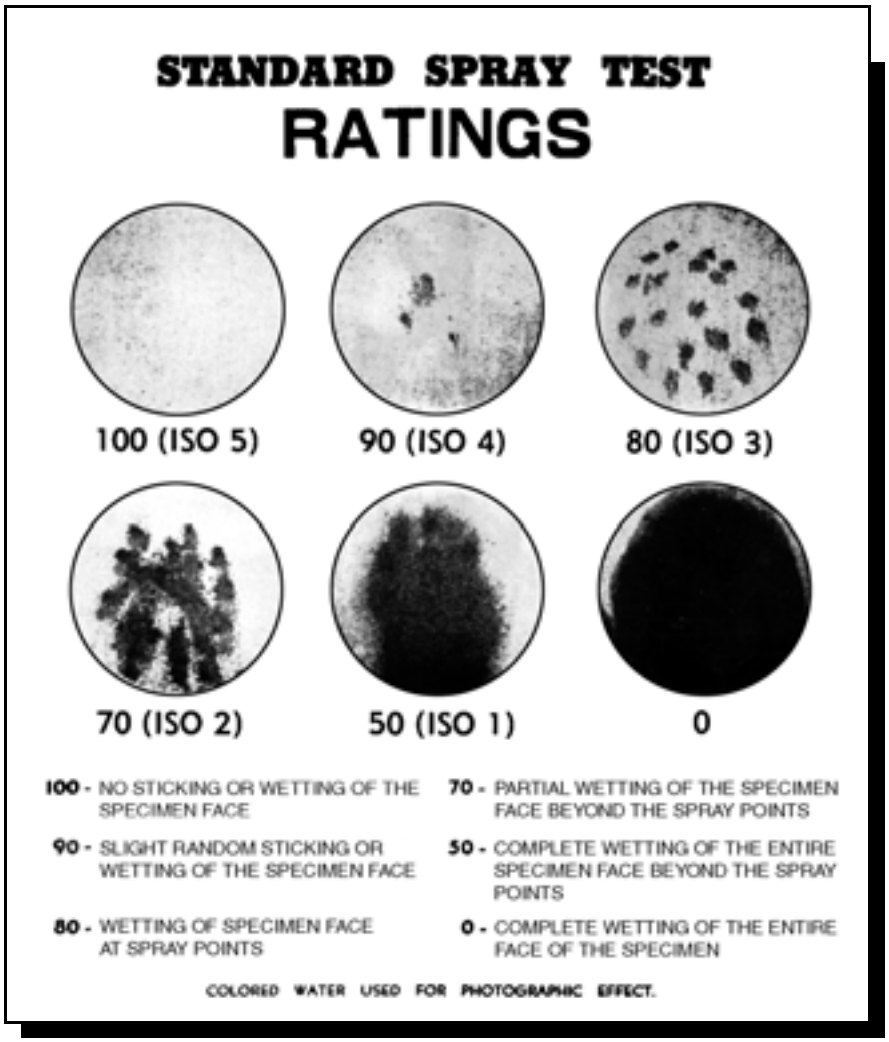


Fig. 1—Spray test rating chart.

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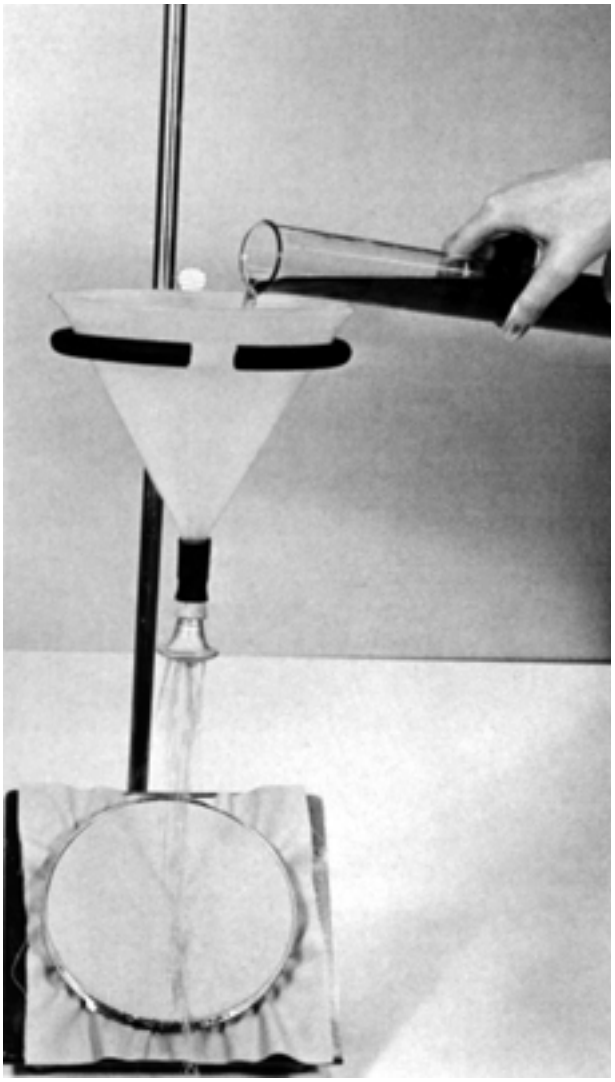


Fig. 2—AATCC spray tester.

so that the face of the fabric specimen will be exposed to the water spray. The surface of the specimen should be smooth and without wrinkles.

8.3 Place the hoop on the stand of the tester with the fabric uppermost in such a position that the center of the spray pattern coincides with the center of the hoop (see Fig. 2A).

8.3.1 In the case of twills, gabardines, piques or fabrics with similar ribbed construction, place the hoop on the stand in such a way that the fabric is oriented in the same direction as it will be used in the end product.

8.4 Pour 250 mL of distilled water at 27 ± 1°C (80 ± 2°F) into the funnel of the tester and allow it to spray onto the test specimen for 25-30 s.

8.4.1 Avoid touching the funnel with the graduated cylinder while pouring the distilled water. Movement of the funnel will alter the spray disposition on the specimen.

8.5 Take the hoop by the bottom edge and tap the opposite edge firmly once

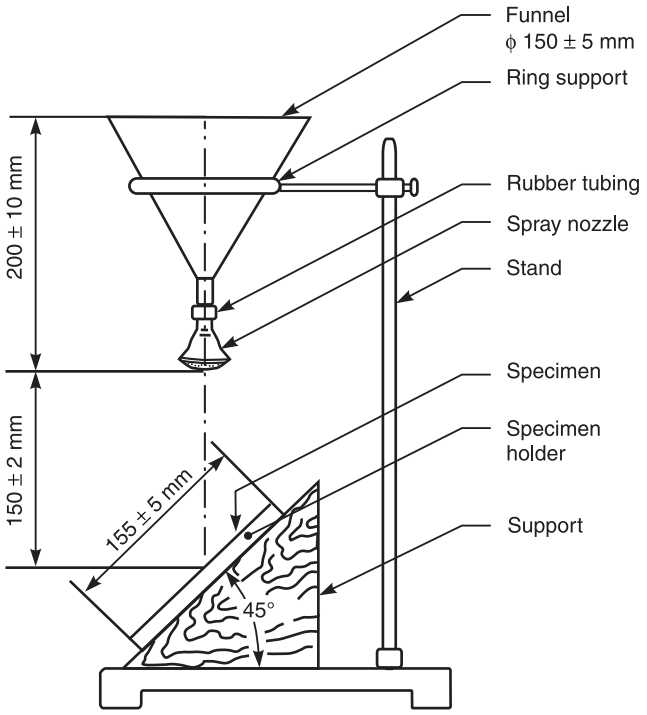
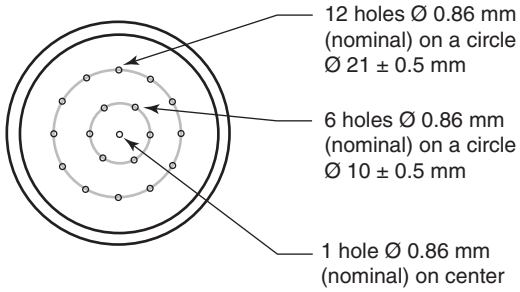


Fig. 2A—Details of AATCC spray tester.



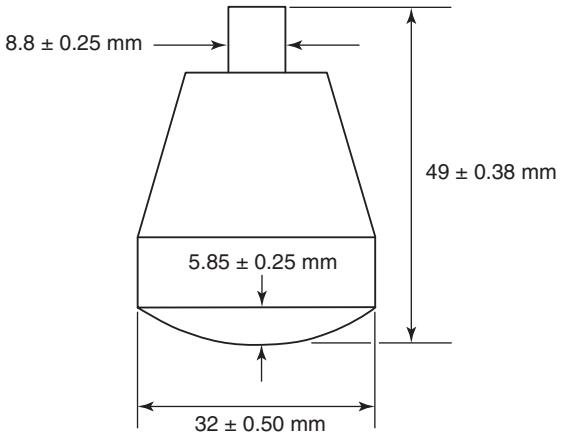


Fig. 3—Nozzle for spray tester.

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against a solid object with the fabric facing the object, then rotate the hoop 180° and tap once more on the point previously held.

8.6 Repeat steps 8.2 through 8.5 for all three specimens.

9. Evaluation and Report

9.1 Immediately after tapping, compare the wet or spotted pattern with the rating chart (see Fig. 1). Rate the face of the specimen. Each test specimen is assigned a rating corresponding to the nearest level on the rating chart.

9.1.1 Intermediate ratings can be used for ratings of 50 or higher (95, 85, 75, 60) (see flow chart in Appendix A).

9.1.2 In rating loosely woven or porous fabrics, such as voile, any passage of water through the openings of the fabric is disregarded.

9.2 Report the individual rating results for each test specimen. Do not average the results.

10. Precision and Bias

10.1 *Precision*. Precision of this test method is based on interlaboratory tests conducted in 1994, using six laboratories, three fabrics, two operators per laboratory and three determinations per fabric. Separate tests were run on two days. The data sets from each lab were combined for analysis, which required averaging the individual ratings. There were no significant differences found between days.

10.1.1 The Spray Rating Scale is discrete and discontinuous, but results are based on averages which tend to normality. Also, ratings are assigned against developed grade standards (photographs), rather than by arbitrary visual assignment. For these reasons, RA63 has made a judgment that analysis of variance be used in determining the precision of this method, without correction for discontinuity.

10.1.2 The three fabrics used in this study cover a range of 100 to 80 Spray Rating. Obviously, this is a limited study upon which to base a statement of precision for the method. It is the best and only estimate of precision currently available. Users of the test method should be guided accordingly. In making fabric comparisons for Spray Rating, laboratories are encouraged to establish their levels by running fabrics of known performance prior to any test comparisons, and to bring the practice of the test method under statistical control.

10.1.3 Plant experience consistently has shown variation in spray ratings near a grade of 100 to be quite small, with increasing variation as the grade goes lower. This has been verified again, in this limited study. For this reason, critical difference tables are shown for two ranges.

10.1.4 Single fabric precision parameters for two fabric levels are given in Tables I and II.

10.2 *Bias*. This test method has no known bias. There is no referee method by which to determine the true value of Spray Rating and to find any existing bias in this methodology.

Table I—Spray Rating Range—80
Components of Variance

V_{lab} = 17.2222
V_{op} = 9.2593
V_{err} = 9.3750

Single Fabric Precision Parameters

N	Single Operator	Within Laboratory	Between Laboratory
1	8.5	12.0	16.6
2	6.0	10.4	15.5
3	4.9	9.8	15.1
4	4.2	9.4	14.9
5	3.8	9.2	14.8

Table II—Spray Rating Range—100
Components of Variance

V_{lab} = 0
V_{op} = 0.6945
V_{err} = 4.4841

Single Fabric Precision Parameters

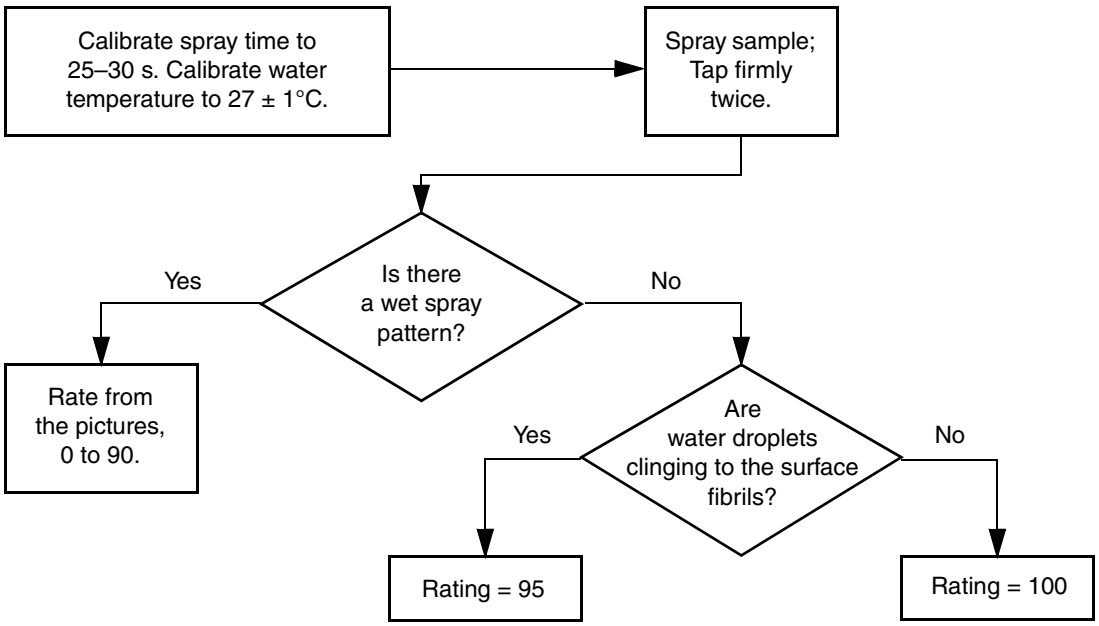
N	Single Operator	Within Laboratory	Between Laboratory
1	5.9	6.3	6.3
2	4.2	4.8	4.8
3	3.4	4.1	4.1
4	2.9	3.7	3.7
5	2.6	3.5	3.5

11. Notes

11.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

11.2 The AATCC Spray Test Unit consisting of hoop, nozzle, funnel, stand and Spray Test Rating Chart is available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

Appendix A
Flow Chart of Evaluation and Report



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Test Method for Colorfastness to Burnt Gas Fumes

1. Purpose and Scope

1.1 This test method is intended for assessing the resistance of the color of textiles of all kinds and in all forms when exposed to atmospheric oxides of nitrogen as derived from the combustion of natural gas. Exceptions are noted in 11.8.1.

1.2 This test method may be used for rating colorfastness of dyes by applying the dye to textiles by a specified procedure and at a specified depth of color and testing the dyed textiles.

2. Principle

2.1 A specimen of the textile and the test control fabric are exposed simultaneously to oxides of nitrogen from burnt gas fumes until the control shows a change in color corresponding to that of the standard of fading. The change in color of the specimen is assessed with the standard gray scale for assessing change in color. If no color change is observed in the specimen after one exposure period or cycle, exposure may be continued, for either a specified number of periods or for the number of periods required to produce a specified amount of color change in the specimen.

3. Terminology

3.1 **burnt gas fumes,** n.—atmospheric oxides of nitrogen as derived from the combustion of illuminating or heating gas.

3.2 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials, or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 Stoddard solvent is a combustible liquid and presents a moderate hazard. It should not be used near an open flame, and fabric saturated with this solvent should be dried in an adequately ventilated laboratory hood. Use chemical goggles or face shield, impervious gloves and an impervious apron when handling Stoddard solvent.

4.4 Perchloroethylene is toxic by inhalation, by repeated contact with skin and by ingestion; it MUST be used only in a well vented atmosphere. Toxicology studies with laboratory animals have shown evidence of cancer in rats and mice exposed to perchloroethylene vapors at 100-400 ppm concentrations for prolonged times. Fabric saturated with this solvent should be dried in an adequately ventilated laboratory hood. Use chemical goggles or face shield, impervious gloves and an impervious apron when handling perchloroethylene.

4.5 An eyewash/safety shower should be located nearby and an organic vapor respirator should be readily available for emergency use.

4.6 In previous versions of this test, recommendations were made for the use of a wire screen placed above the burner to decrease the test time. During recent studies it was found that the improper application of the steel screen may increase the generation of residual unburnt gas and, thereby, cause the potential for an explosion. The manufacturers of the devices are recommending against the use of this screen due to potential safety issues (see 11.6.1).

4.7 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 11.1).

5. Apparatus, Materials and Reagents

5.1 Exposure chamber (see 11.2).
5.2 Control Ribbon No. 1 (see 11.3).
5.3 Standard of Fading (Ribbon No. 1) (see 11.3).
5.4 Gray Scale for Color Change (AATCC EP1) (see 11.9).
5.5 Natural gas (see 11.6).
5.6 Urea solution (see 11.10).
5.7 Mineral spirits.
5.8 Perchloroethylene.
5.9 Trichloroethylene.
5.10 1993 AATCC Standard Reference Detergent WOB (see 11.9).

6. Test Specimens

6.1 Use test specimens that measure approximately 5.0 × 10.0 cm. If pressing is necessary to remove wrinkles, iron the specimens between two layers of closely woven cotton cloth or press them with a steam press (see 11.4).

6.1.1 For fastness to oxides of nitrogen in storage or use, test a specimen of the original fabric.

6.1.2 For fastness to oxides of nitrogen after drycleaning, immerse one specimen in cold mineral spirits for 10 min, then squeeze the solvent out and dry the specimen in air. One specimen is immersed in cold perchloroethylene for 10 min, squeezed and allowed to dry in the air. Trichloroethylene may be substituted for perchloroethylene. Prepare a drycleaned only specimen for comparison with the tested drycleaned specimen.

6.1.3 For fastness to oxides of nitrogen after washing (see 11.5), unless a specific wash test is specified, wash the test specimen in a detergent solution containing 5 g of 1993 AATCC Standard Reference Detergent WOB per liter of water of approximately zero hardness for 10 min at 41 ± 3°C, rinse it in warm water and dry it in air. Prepare a washed only specimen to compare with the tested washed specimen.

7. Procedure

7.1 Suspend the test specimens and a piece of the Control Ribbon freely in the test chamber so that they are not in contact with each other and do not come into direct contact with any hot metallic surface. Light the gas burner and adjust the flame and ventilating equipment so that the temperature in the chamber does not exceed 60°C (see 11.7 and 11.8). Leave the specimens to remain in the chamber until the control sample changes color to correspond to the color of the

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Standard of Fading when compared in daylight (ranging from average to slightly bluish north sky light) or equivalent artificial illumination.

7.2 Then remove the test specimens from the test chamber and immediately assess the change in color of each by use of the Gray Scale for Color Change.

7.3 Specimens exposed to oxides of nitrogen may continue to change color after removal from the test chamber. Optionally, another or more detailed visual or instrumental examination may be done. If this is to be done, without delay, plunge the test and control specimens and pieces of the original test and control fabric into buffered urea solution (see 11.10) for 5 min. Then squeeze out the specimens and pieces and thoroughly rinse them. Dry the specimens in air at a temperature not above 60°C.

7.4 After the first cycle, any specimens which do not show a color change and which are not treated with the buffered urea solution may be returned to the test chamber along with a fresh piece of the test control and the test continued until the second test control shows a change in color corresponding to that of the Standard of Fading.

7.5 Cycles may be repeated for either a specified number of cycles or until the specimens show a specified amount of color change.

8. Evaluation

8.1 At the end of each cycle immediately compare those specimens which are removed from the exposure chamber with the preserved originals from which they were drawn.

8.2 Rate the effect on color of test specimens after any specified number of cycles by reference to AATCC EP1, Gray Scale for Color Change, or using AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen, and record the numerical rating that corresponds to the appropriate one on the Gray Scale. Report the number of cycles run (see 11.11).

9. Report

9.1 Report the grade for each specimen and the number of cycles, to which it was exposed. Report the average temperature employed and, if humidity was increased, report the method used.

10. Precision and Bias

10.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

10.2 *Bias*. Burnt gas fumes can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

11. Notes

11.1 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

11.2 Exposure Chamber.

11.2.1 The exposure chamber may be of variable construction but must provide an enclosure in which test specimens can be exposed in an atmosphere of air which has passed over and contains the byproducts of combustion from a lighted gas burner. The apparatus should be equipped with a suitable means of supporting the test specimens so that there is a free circulation of the atmosphere around the specimens and so that only a minimum amount of the specimen at the point of its suspension is in direct contact with any hot metallic surface. Either a motor driven fan to move the air around in the test chamber or a motor driven rotating specimen rack should be used in order to insure as nearly as is possible that all specimens are exposed to the same conditions of gas concentration, temperature and humidity. Adjustable vents or dampers in the top of the unit together with the height of the flame of the gas burner are used to regulate the temperature of the exposure chamber, but temperature and humidity will vary with that of the room in which the exposure chamber is being operated.

11.2.2 Several forms of suitable apparatus are illustrated in the *American Dyestuff Reporter*, July 22, 1940, pp368-369. Blueprints of a suitable apparatus can be purchased from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.3 Test Control and Standard of Fading.

11.3.1 Historically, the control ribbon was a dyeing of 0.4% C.I. Disperse Blue 3 on cellulose acetate satin. Lot 21 uses 0.4% Disperse Violet 1 on cellulose acetate satin since its fading characteristics are well known and it yields more color change after the testing cycle than previous test control lots dyed with Disperse Blue 3.

11.3.2 Pieces of the original lot of control ribbon were hung in air in three separate places in southern New Jersey for six months. It was assumed that the atmosphere in these places contained an average content of oxides of nitrogen. At the end of the exposure period, samples from the three locations were assembled and compared with the original material. All had changed about equally, being decidedly duller and redder than the original. The faded shade was then matched with vat dyes on cellulose acetate satin and this became the original standard of fading for that lot of control fabric. This produced a standard of fading which was more permanent than the exposed pieces of the control fabric which would have continued to change with even very small amounts of oxides of nitrogen.

11.3.3 Since different lots and sources of both the dyestuff and undyed fabric will produce variations in both the original shade and its fading rate, it is therefore necessary to accurately establish a new standard of fading for each dyed lot of control ribbon so that comparable tests results can be obtained when using different lots of the control and their respective standards of fading. In conducting tests, only the standard of fading applicable to that lot of control ribbon may be used.

11.3.4 The standard of fading for the original lot of control ribbon was made with vat dyes on cellulose acetate satin. For subsequent lots of the control ribbon, it was found that the use of direct dyes on viscose satin for the standard of fading provided a better shade match.

11.3.5 Both the Control Ribbon and the Standard of Fading must be kept in suitable containers or enclosures to protect them from possible exposure and color change to oxides of nitrogen and other contaminants which might be present in the atmosphere during transportation and storage.

11.3.6 The Control Ribbon is also sensitive to other atmospheric contaminants such as ozone. Its fading rate will vary considerably at different humidities and temperatures and its use in natural or end-use testing as a measure of exposure to oxides of nitrogen is not recommended. The color change produced on the control will reflect the combined effects of the atmospheric contaminants present, and temperature-humidity variations; not just the effects of exposure to oxides of nitrogen.

11.3.7 A sealed unit of Test Control Fabric comprising 2.47 cm wide Control Ribbon No. 1 with a specific lot number indicated; and a specimen of the Standard of Fading for that lot of control ribbon is available as CR & SF No. 1 from Testfabrics Inc., 415 Delaware Ave., P.O. Box 3026, W. Pittston PA 18643; tel: +1.570.603.0432; fax: +1.570.603.0433; e-mail: info@testfabrics.com; web site: www.testfabrics.com.

11.4 Heat by Ironing.

11.4.1 When an acetate fabric is pressed with an iron which is hot enough to remove all wrinkles, there is danger of sealing the surface of the fabric which would increase the resistance to gas fading. This technique is not conducive to accurate test results and should be avoided on fabrics of this type.

11.5 Drycleaning and Washing Test Specimens.

11.5.1 All the inhibitors available at present are soluble to some extent in water and therefore tend to be removed by washing. These inhibitors are in general not soluble in the ordinary drycleaning solvents, and fabrics treated with suitable inhibitors should withstand several drycleanings without losing their resistance to gas fading (if the drycleaning operation does not include spotting or sponging with water). The inhibitors also tend to lose their efficiency if the fabrics come into frequent contact with perspiration.

11.6 Gas.

11.6.1 Illuminating gas furnished by the gas companies in Massachusetts, Connecticut, Rhode Island, New York, New Jersey, Pennsylvania and Delaware has been used in this work, including both natural and manufactured gas. The results were essentially the same in all cases. Any gas burner may be used, and either a luminous (yellow) or a blue-green flame is suitable, though the latter

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is to be preferred to avoid formation of soot. In cases of dispute or for referee testing compressed butane (c.p.) gas in cylinders shall be used.

11.7 Temperature for Testing.

11.7.1 Other conditions being equal, the fading of the specimens will vary according to the temperature in the exposure chamber, which in turn depends upon the amount of gas consumed in a given period. Exposure for 8-12 h at 60°C may cause as much color destruction as exposure for 96 h at 21-27°C. Aside from this, the temperature may vary somewhat in different parts of the exposure chamber from time to time.

11.8 Humidity for Testing.

11.8.1 The fading of dyes by oxides of nitrogen on acetate, triacetate and polyester will occur at relatively low humidities as would be obtained in the conventional chamber at temperatures approaching 60°C. For other fibers such as nylon, rayon or cotton it is necessary to use high humidities to produce results which correlate with service performance. A suggested procedure for raising the level of humidity in the chamber is to place containers of water on the floor surface of the chamber. If the humidity is raised by this or any other procedure, this should be noted in the report of the test results.

11.9 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.10 Urea Solution.

11.10.1 10 g of urea (NH_2-CO-NH_2) per liter buffered to pH 7 by addition of 0.4 g of monosodium orthophosphate ($NaH_2PO_4 \cdot 2H_2O$) and 2.5 g of sodium phosphate dibasic, dodecahydrate ($Na_2HPO_4 \cdot 12H_2O$) and 0.1 g or less of an anionic surfactant.

11.11 An automated electronic grading system may be used as long as the system has been demonstrated to provide results that are equal to and provide equal or better repeatability and reproducibility than an experienced grader performing visual evaluation.

12. History

12.1 Reaffirmed 2020.

12.2 Revised 2015, 2005, 1972, 1962, 1957, 1952, 1946. Reaffirmed 1989, 1983, 1975, 1971. Editorially Revised 2019, 2014, 2008, 1997, 1996, 1995, 1985, 1983, 1981. Editorially Revised and Reaffirmed 2010, 2004, 1999, 1994, 1988.

12.3 Developed in 1941 by AATCC Committee RA33. Technically equivalent to ISO 105-G02.

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Test Method for Ageing of Sulfur-Dyed Textiles: Accelerated

1. Purpose and Scope

1.1 This test method describes a procedure for determining whether sulfur-dyed textile materials will deteriorate under normal storage conditions and to establish the degree of such deterioration.

2. Principle

2.1 Specimens are steam-aged in a moist atmosphere under controlled conditions and then tested for loss in breaking strength to determine if storage deterioration is likely.

3. Referenced Documents

3.1 ASTM E145, Standard Specification for Gravity-Convection and Forced-Ventilation Ovens.

3.2 ASTM E1402, Standard Guide for Sampling Design.

4. Terminology

4.1 **accelerated ageing,** n.—*in textile processing and testing*, use of controlled environmental conditions to promote rapid physical and/or chemical change in a textile material.

4.2 **sulfur dye,** n.—a dye, containing sulfur both as an integral part of the chromophore and in attached polysulfide chains, normally applied in the alkali-soluble reduced (leuco) form from a sodium sulfide solution and subsequently oxidized to the insoluble form in the fiber.

5. Safety Precautions

5.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

5.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

5.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

6. Apparatus

6.1 Conventional laboratory drying oven (see 13.1), uniformly heated and controlled to ± 2°C. It should be provided with ventilating ports and some means for introducing moisture (see 13.2) or;

6.2 Steam ager (see 13.3) equipped with proper controls for uniform steam flow and temperature.

6.3 Tensile testing machine.

7. Sampling

7.1 Test results are valid only when the samples are statistically representative (see ASTM E1402).

7.2 Sampling must be random. Every unit of product must have a mathematically equal chance of becoming a sample; and every portion of each sample must be equally likely to become a test specimen.

7.3 All specimens must be alike within the variations due to pure chance. There must be no differences within samples assignable to known causes.

8. Test Specimens

8.1 Sufficient samples are taken for both control and test specimens. Tensile specimens are prepped from control and aged samples after conditioning (see 10.1).

9. Procedure

9.1 Oven Test, Option 1. Expose the specimens to 135 ± 2°C heat in the oven for 6 h continuously. At the beginning of the second, third, fourth, fifth and sixth hours, close the ports or vents and add 20 mL of water for each 0.03 m^3 (1.0 ft^3) of oven capacity. Keep vents closed for 5 min, then open and resume circulation for the rest of the period. At the end of 6 h of heating, remove the specimens from the oven and place in a conditioning room.

9.1.1 Oven Test, Option 2. Expose the specimens to 135 ± 2°C heat in the oven for 6 h. Remove the test specimens from the oven at the end of each hour and thoroughly humidify in an atmosphere of steam, then replace in the oven. At the end of 6 h of heating, remove the specimens from the oven and place in a conditioning room.

9.1.2 Oven Test, Option 3. Expose the specimens to 135 ± 2°C heat in the oven for 6 h continuously. At the beginning of the test, introduce 100 mL of water for each 0.03 m^3 (1.0 ft^3) of oven capacity using a vessel having a surface area of approximately 413 cm^2 (64 $in.^2$). Leave the vents open during the entire test. At the end of the 6 h of heating, remove the specimens from the oven and place in a conditioning room. Under the conditions above, it has been found that the water will require 1.75-2.0 h to evaporate.

9.2 Steam Ager Test. Suspend the test specimens while ageing, or tack on frames and place in the steam ager, pyramid style, with the bottom baffle functioning as a support. Age the specimens for 8 h with saturated steam at 103 kPa (15 psi) pressure, or for 16 h at 51.7 kPa (7.5 psi) pressure. After the ageing test is completed, remove the specimens from the ager and place in a conditioning room.

10. Evaluation

10.1 Condition test specimens and unaged controls for 16 h at standard conditions, 65 ± 5% RH and 21 ± 2°C (70 ± 4°F), before performing breaking strength tests. Perform testing under the same standard conditions.

10.2 Determine the breaking strength of the unaged control specimens. Since comparative, but not absolute, results are to be obtained, any suitable breaking strength machine may be used which is adaptable to breaking the specimens being tested. Perform a minimum of 10 significant breaks and average the results.

10.3 Determine the breaking strength of the aged specimens. Perform a minimum of 10 significant breaks and average the results.

10.4 Calculate percent loss (or gain) in strength based on the unaged control material.

11. Report

11.1 Describe or identify the sample tested.

11.2 Report that the sample was tested using AATCC TM26-2020.

11.3 Report the testing conditions:

11.3.1 Test procedure (oven test: Option 1, Option 2, Option 3, or steam ager test).

11.3.2 Test conditions.

11.4 Report test method used for breaking strength.

11.5 Report the test results:

11.5.1 Mean breaking strength and standard deviation of unaged control specimens.

11.5.2 Mean breaking strength and standard deviation of aged test specimens.

11.5.3 Percent loss (or gain) in strength based on the unaged control material.

11.6 Describe any modification(s) of the published method.

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12. Precision and Bias

12.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

12.2 *Bias*. The accelerated ageing of sulfur-dyed textiles can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

13. Notes

13.1 The oven used may be of any conventional design and constructed of conventional material as long as the heating elements or materials of construction do not give off gases when heated that would tend to distort the results. The quantity of materials tested must bear a fixed relationship to the capacity of the oven, 25 g of materials tested per 0.03 m^3 (1 ft^3) of oven capacity being considered correct.

13.2 The temperature control must be accurate within ± 2°C. Care must be exercised to ensure that the thermometer indicates the true and correct internal temperature of the area in which the samples are being tested.

13.2.1 The oven should be equipped with ports or vents of such size as to allow the air in the oven to be changed approximately every 2 min except when humidifying. This may be accomplished by natural convection or by a circulating system. During humidification, the ports or vents should be closed to confine the moisture within the oven for the specified time.

13.2.2 Controlled amounts of water (or steam) should be admitted into the oven at specified intervals. This may be through either the top or the side of the oven. The requisite amount of water should be added onto a heated mass of material to effect rapid vaporization. To effect this relatively rapid vaporization, the weight of material should be approximately 100 g for each gram of water admitted per hour. The amount of water admitted should be sufficient to cause the filling of the oven with steam at atmospheric pressure each hour (20 mL of water produces approximately 0.03 m^3 (1 ft^3) of vapor at atmospheric pressure and at 135°C). In addition to the desired rapid vaporization of the water admitted, the mass of metal or other materials acts as a temperature stabilizer and serves to prevent a sudden drop in temperature owing to the latent heat of vaporization of the water introduced. Ovens meeting the ASTM specifications for accelerated aging of rubber, equipped with automatic humidity control, are acceptable (see ASTM E145, Standard Specification for Gravity-Convection and Forced-Ventilation Ovens).

13.2.3 Provision must be made for holding specimens to be tested so that they will not contact the hot metal portion of the oven. A suitable means is to hang the specimens by strings from glass or wooden rods, avoiding contact with each other and with the sides of the oven. Materials tested must be held in that part of the oven where the temperature is constant and uniform and where all portions of tested specimens are exposed to the same temperature and the same conditions of humidity and atmospheric change.

13.3 A typical steam ager consists essentially of a closed cylinder, 58 cm (23 in.) long by 38 cm (15 in.) diameter, inside measurement, one end of which is a swinging door. A groove in the face of the cylinder holds a gasket against which the door is held tightly by a lever clamp. The apparatus is equipped with a pressure gauge, automatic temperature recorder, steam inlet with handcock flow control valve, petcock, trap and baffle of thin stainless steel sheet metal in the top half to prevent dripping from condensation. A wooden frame 56 × 36 cm (22 × 14 in.), covered with coarse cloth and supported by protruding bolts in the sides along the center, serves as a bottom baffle to prevent possible splashing. However, any steam ager equipped with the proper controls for uniform steam flow and pressure may be adapted to this test procedure.

14. History

14.1 Revised in 2020 to clarify and align with prescribed AATCC style guidelines.

14.2 Editorially revised 2016. Editorially revised and reaffirmed 2013, 2009, 2004. Reaffirmed 1999. Editorially revised and reaffirmed 1994. Editorially revised 1990. Reaffirmed 1989, 1988, 1983, 1978. Revised 1975. Reaffirmed 1972. Revised 1952. Reaffirmed 1944.

14.3 Developed in 1943 by AATCC Committee RR9. Maintained by RA99.

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Test Method for Wetting Agents: Evaluation of Rewetting Agents

Developed in 1944 by AATCC Committee RR8; jurisdiction transferred in 2003 to AATCC Committee RA63; jurisdiction transferred in 2022 to AATCC Committee RA114; revised 1952; reaffirmed 1971, 1974, 1977, 1980, 1989, 1999, 2009, 2013, 2018; editorially revised and reaffirmed 1985, 1994, 2004; editorially revised 1988, 1991, 1992, 2008, 2016, 2019, 2022.

1. Purpose and Scope

1.1 This test method provides a means of measuring the effectiveness of commercial rewetting agents on cotton fabrics.

2. Principle

2.1 A test specimen is padded in a water solution of a rewetting agent and dried. A drop of water is then carefully deposited on the taut surface of the specimen and the time required for the specular reflection of the water is recorded as rewetting time.

3. Terminology

3.1 **rewetting agent,** n.—*in textile preparation, dyeing and finishing*, a surfactant which, after application and drying onto textiles, promotes rapid wetting on subsequent exposure to an aqueous solution.

3.2 **wetting agent,** n.—a chemical compound which when added to water lowers both the surface tension of the liquid and its interfacial tension against the solid material.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Observe padder safety. Normal safeguards on pad should not be moved. Ensure adequate guard at the nip point. A foot-operated kickoff is recommended.

5. Apparatus and Materials (see 9.1)

5.1 Rewetting agents.

5.2 Cloth, fairly heavy cotton, in the greige; cloth must not have been boiled, bleached or desized. Cotton sateen is suggested for use in this test.

5.3 Small padder. A household wringer with improvised pad box is satisfactory in lieu of a small padder.

5.4 Embroidery hoop, 15.2 cm (6 in.).

5.5 Burette delivering 15-25 drops per mL.

5.6 Stopwatch.

5.7 Conditioned room in which standard temperature 21 ± 2°C (70 ± 4°F) and humidity (65 ± 5% RH) are maintained. Although satisfactory comparisons can be made in the ordinary laboratory, the results should be more reproducible under standard conditions.

6. Procedure

6.1 Application of rewetting agent—a solution of the rewetting agent is prepared by pouring 100 mL of hot water over the required amount of the product in a small beaker or casserole. This solution is heated above 97°C (200°F) for a moment and then is diluted with hot water to a final volume of 1 L. The temperature of the solution in the padder box for application should be 70 ± 3°C (158 ± 5°F).

6.2 The padder is adjusted to a uniformly firm squeeze from side to side. On a household wringer it has been found that a dozen half turns of the wing nuts from the place where they just begin to take hold is satisfactory.

6.3 A strip of the selected cotton goods is padded *three times* through the solution of the rewetting agent. Three times through ensures thorough and uniform penetration of the cloth. The pickup should be from 60-90%.

6.4 The padded cloth is dried in air at approximately 82°C (180°F) for 30 min.

6.5 Four strips of padded cotton are prepared for each rewetting agent representing, respectively, four concentrations for each product to be tested. Concentrations generally suitable are 2.50, 5.0, 10.0 and 20.0 g of rewetting agent per liter.

6.6 Rewetting—a square of the padded, dried and conditioned cloth is mounted in the embroidery hoop. The burette containing either distilled water or tap water, whichever is desired for the test, is adjusted so that it delivers one drop of water at 21 ± 3°C (70 ± 4°F) approximately every 5 s. The surface of the taut cloth is held about 1 cm (0.394 in.) below the tip of the burette and the stopwatch is started just as the drop falls on the cloth. The watch is stopped when the liquid on the surface of the cloth loses its specular reflective power. This point is determined by placing the hoop between the observer and a source of light—such as a window—at such an angle that the specular reflection of light from the surface of the flattened drop can plainly be seen. As the drop is gradually absorbed the area of this tiny mirror diminishes and finally vanishes entirely leaving only a dull wet spot behind. It is at this moment that the watch is stopped.

7. Evaluation

7.1 Readings of rewetting time are made so easily and quickly that ten are taken for each concentration. The average times of disappearance of a drop for 2.50, 5.0, 10.0 and 20.0 g of rewetting agent per liter of pad liquor are then plotted on log-log coordinate paper with concentration as the abscissa (X-axis). A straight line is drawn to approximate the four points as nearly as possible.

7.2 From this line relating disappearance time to concentration, one can find the concentration which would give a disappearance time of 10 s. This is called the rewetting concentration for the product. A corresponding value is then determined for the second rewetting agent. By using a simple proportion with these figures it can be calculated how many parts of the second product are equivalent to 100 parts of the first product (or Standard) for rewetting.

8. Precision and Bias

8.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

8.2 *Bias*. Rewetting agents can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

9. Notes

9.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

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Test Method for Antifungal Activity, Assessment on Textile Materials: Mildew and Rot Resistance of Textile Materials

Developed in 1946 by AATCC Committee RA31; revised 1952, 1957, 1971, 1981, 1987, 1988 (with title change), 1993, 1999, 2017; reaffirmed 1970, 1974, 1979, 1989, 1998, 2013; editorially revised and reaffirmed 1986, 2004; editorially revised 2010, 2019.

1. General Purpose and Scope

1.1 The two purposes of this test method are to determine the susceptibility of textile materials to mildew and rot and to evaluate the efficacy of fungicides on textile materials.

2. Principle

2.1 Tests I, II, III and IV can be used, singly or in combination, depending on the type of exposure to which the textile material will be subjected. For example, if the final product will come in contact with soil, Test I, which simulates this exposure, can be used; if the finished product will never come in contact with soil or tropical conditions, a much less severe test (e.g., II or III) can be used. Test II is specifically designed for cellulose-containing materials while Test III is for all others. For materials intended for outdoor and above ground use, Test IV can be used. The two important considerations when evaluating textile materials in relation to fungal growth are (1) the actual deterioration of the textile product (rot), and (2) growth not necessarily deteriorating the product, but making it unsightly (mildewy), often with an unpleasant and musty odor.

2.2 Certain pre-exposures of textile products may be needed when specific end-uses are critical (see Appendix A). When the end-use will be in high temperature environments and the fungicide may be volatile, a preliminary oven exposure may be desired. When the end-use will be in tropical exposures or outside with rainfall present, a leaching exposure should be performed before mildew evaluation is made. When at all possible, the textile material should be exposed to the expected conditions of use prior to performing this test.

3. Terminology

3.1 **fungicide**, n.—a compound used to kill fungi.

3.2 **fungistatic**, adj.—inhibiting the growth of fungi or fungal spores without destroying them.

3.3 **mildew resistance,** n.—*in textiles*, resistance to development of unsightly fungal growths and accompanying unpleasant, musty odors on textile materials exposed to conditions favoring such growths.

3.4 **rot resistance,** n.—*in textiles*, resistance to deterioration of a textile material as a result of fungal growth in or on it.

NOTE: Such deterioration is normally assessed by measuring loss in tensile strength.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 This test should be performed only by trained personnel. The U.S. Department of Health and Human Services publication *Biosafety in Microbiological and Biomedical Laboratories* should be consulted (see 44.1).

4.2 CAUTION: Some of the fungi used in these tests are allergenic and pathogenic; i.e., capable of infecting humans and producing disease. Therefore, every necessary and reasonable precaution must be taken to eliminate this risk to the laboratory personnel and to personnel in the associated environment. Wear protective clothing, respiratory protection, and impervious gloves when working with the organisms. NOTE: Choose respiratory protection that prevents penetration by the spores.

4.3 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.4 All chemicals should be handled with care.

4.5 An eyewash/safety shower should be located nearby for emergency use.

4.6 Sterilize all contaminated samples and test materials prior to disposal.

4.7 Exposure to chemicals used in this procedure must be controlled at or below levels set by government authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 44.2).

Test I
Soil Burial

5. Purpose and Scope

5.1 This procedure is used to evaluate the loss of strength of textile materials after exposure to soil and soil microorganisms. This procedure is generally considered to be the most severe test for textile products. Only those specimens that will come in direct contact with soil—such as sandbags, tarpaulins, tents—need to be tested by this procedure. It can also be used for testing experimental textile fungicides.

6. Principle

6.1 Test specimens are buried in a container of soil. After a defined incubation period, the test specimens are removed and evaluated for loss of strength.

7. Referenced Documents

7.1 ASTM D5035, Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Strip Method) (see 44.3).

8. Apparatus, Reagents, Materials

8.1 Untreated cotton cloth, 271 ± 20 g/m^2 (8 oz ± 0.6 oz/yd^2). For viability control.

8.2 Soil. See 44.4 for acceptable soil types.

8.3 Incubator. Incubating equipment to maintain a temperature of 28 ± 1°C (82 ± 2°F).

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8.4 Apparatus, reagents, and materials required to perform breaking strength by method ASTM D5035.

8.5 Autoclave. To sterilize test materials prior to disposal.

9. Specimens

9.1 Prepare the fabric specimens with dimensions 15.0 ± 1.0 × 4.0 ± 0.5 cm (6.0 ± 0.4 × 1.5 ± 0.2 in.) with the long dimension parallel to the warp and unraveling to 2.5 ± 0.1 cm wide (1.0 ± 0.04 in.), or, in the case of fabric with less than 20 threads per 2.5 cm (1.0 in.) to a predetermined number of threads to give a specimen 2.5 ± 1.0 cm wide (1.0 ± 0.4 in.). A sample cutter can also be used. The number of specimens will vary according to the number of variables. The suggested number of replicate specimens is 5 (i.e., 5 specimens for each treatment, 5 controls and 5 reference fabrics).

10. Procedure

10.1 Soil Bed: Place the air-dry test soil (see 44.4) in trays, boxes or suitable containers to a depth of 13.0 ± 1.0 cm (5.1 ± 0.4 in.) and bring to optimum moisture content by gradual addition of water accompanied by mixing to avoid puddling. After allowing it to stand for 24 h, sieve it through a 6.4-mm (0.25-in.) mesh screen. Maintain uniform moisture content by covering the soil container with a suitable lid. The moisture content of the soil during the test period shall be maintained at 25 ± 5% (based on dry weight). If the surrounding air is maintained at higher than 83 ± 3% relative humidity, the loss of moisture will be negligible.

10.2 Viability control: Expose untreated cotton cloth (see 8.1) in the soil bed for seven days during the test period to verify fungal activity. The soil bed shall be considered satisfactory if the viability control fabric loses 90% breaking strength after seven days exposure.

10.3 Incubation: After the soil bed has been determined to be satisfactory based on the viability control, place the specimens horizontally on 10.0 ± 1.0 cm (3.9 ± 0.4 in.) of soil, spaced at least 2.5 cm (1.0 in.) apart and then cover with 2.5 ± 0.5 cm (1.0 ± 0.2 in.) of test soil. Incubation periods can vary from 2 to 16 weeks, depending on severity of service requirements, and other factors of importance to the interested parties. Maintain the temperature at 28 ± 1°C (82 ± 2°F) during the test period.

10.4 Do not reuse soil as components from the material(s) tested may be present in the soil bed and may impact future tests. Fresh soil must be used for each new test.

11. Interpretation, Calculation, and Evaluation

11.1 Strength loss determination: Remove specimens, gently wash with water, dry at room temperature for 22 ± 4 h and then condition in an atmosphere of 64 ± 2% humidity and a temperature of 24 ± 3°C (76 ± 6°F) for 24 h. Determine breaking strength by the method outlined in ASTM D5035, using 25 × 75 mm (1 × 3 in.) jaw faces. The gage length is determined as 25% of the specimen length. Testing can be performed every two weeks or as specified by the end-user.

12. Report

12.1 The report shall include the following:

12.1.1 The test method and test part.

12.1.2 Composition of the soil bed.

12.1.3 Number of test specimen replicates used.

12.1,4 Incubation time period in the soil bed.

12.1.5 Percent retained breaking strength when compared to the unexposed textile.

12.1.6 Any pre-exposure of specimens before burying and percent retained breaking strength of untreated specimen and/or viability control.

13. Precision and Bias

13.1 The precision and bias of this test method have not been established. Refer to the precision and bias statement given in ASTM D5035.

Test II
Agar Plate, *Chaetomium globosum*

14. Purpose and Scope

14.1 This procedure is used for evaluating rot resistance of cellulose-containing textile materials that will not come in contact with soil. It may also be used for determining uniformity of fungicide treatment.

15. Principle

15.1 Test specimens are placed on an agar surface and inoculated with fungi. After a defined incubation period, the test specimens are evaluated for loss of strength and/or observed for fungal growth.

16. Referenced Documents

16.1 ASTM D5035, Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Strip Method) (see 44.3).

17. Uses and Limitations

17.1 If the test specimen is thick or non-porous, the fungal spores may not have access to minerals in the agar plate. This may cause a lack of fungal growth on the test specimen and cause misleading results.

18. Apparatus

18.1 Autoclave. To sterilize media and to sterilize test materials prior to disposal.

18.2 Incubator. Incubating equipment to maintain a temperature of 28 ± 1°C (82 ± 2°F).

18.3 Stereo microscope.

18.4 Counting chamber suitable for determining spore concentrations; e.g., hemocytometer.

18.5 Apparatus, reagents, and materials to perform breaking strength by method ASTM D5035 (optional).

19. Reagents and Materials

19.1 Organism: *Chaetomium globosum*. American Type Culture Collection No. 6205 (see 44.5).

19.2 Ammonium nitrate, NH_4NO_3.

19.3 Dipotassium hydrogen phosphate, K_2HPO_4.

19.4 Ferrous sulfate, $FeSO_4 \cdot 7H_2O$.

19.5 Magnesium sulfate, $MgSO_4 \cdot 7H_2O$.

19.6 Potassium dihydrogen phosphate, KH_2PO_4.

19.7 Agar.

19.8 Cellulose filter paper.

19.9 Non-ionic wetting agent (see 44.6).

19.10 Gauze or glass wool, sterile.

19.11 Sterile scraping instrument to harvest fungal spores. Platinum or nichrome wire, plastic loop or needle, glass rod, or swab.

20. Specimens

20.1 If strength loss is to be determined, prepare specimens as in Section 9. If only a visual examination of fungal growth is required, cut 3.8 ± 0.5 cm (1.5 ± 0.2 in.) diameter discs from both treated and untreated samples. If the fabric is different on both sides, face up and face down, shall be tested. Five replicate specimens should be used; however, any number can be tested depending on end-users request.

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21. Procedure

21.1 Culture medium. The mineral salts agar (see 44.7) should have the following composition:

Ammonium nitrate, NH_4NO_3	3.0 g
Potassium dihydrogen phosphate, KH_2PO_4	2.5 g
Dipotassium hydrogen phosphate, K_2HPO_4	2.0 g
Magnesium sulfate, $MgSO_4 \cdot 7H_2O$	0.2 g
Ferrous sulfate, $FeSO_4 \cdot 7H_2O$	0.1 g
Agar	20.0 g
Distilled water	to 1000 mL

Add components together and sterilize in an autoclave at 103 kPa (15 psi) and 121°C (250°F) for 15 min. Dispense media into sterile petri dish or other appropriate sterile container, and leave undisturbed until the agar hardens.

21.2 Inoculum: Under aseptic conditions, place a disc of filter paper (previously sterilized by dry heat in an oven at 71 ± 3°C (160 ± 5°F) for 1 h) on the surface of the mineral salts agar. Streak the filter paper with spores of *Chaetomium globosum* by use of a sterile instrument (see 19.11). Incubate at 28 ± 1°C (82 ± 2°F) for approximately 10-14 days to produce abundant growth. Remove the filter paper from the container and add it to a flask containing 50 ± 1 mL of sterile DI water and a few sterile glass beads. Shake vigorously to bring the spores into suspension. Filter the spore suspension using sterile gauze or glass wool. Utilizing a hemocytometer, or other cell counter, enumerate the number of spores present to a final population of approximately 8.0×10^5 to 1.2×10^6 CFU/mL. Use this suspension as the inoculum. Suspensions may be stored at 6 ± 4°C (43 ± 7°F) for up to 4 weeks.

21.3 Inoculation: Pre-wet the specimen (but do not rub, squeeze, or agitate) in water containing 0.05% of a non-ionic wetting agent. Absorbent specimens should be damp but not saturated. Place the specimen on the agar surface. Distribute 1.0 ± 0.1 mL of the inoculum evenly over the 15.0 ± 1.0 × 4.0 ± 0.5 cm (6.0 ± 0.4 × 1.5 ± 0.2 in.) specimens by means of a sterile pipette. Use 0.2 ± 0.01 mL of inoculum for the 3.8 ± 0.5 cm (1.5 ± 0.2 in.) discs. Set up a control specimen, cellulose filter paper or untreated control, in a similar way by using 1.0 ± 0.1 or 0.2 ± 0.01 mL of sterile water. Incubate all specimens at a temperature of 28 ± 1°C (82 ± 2°F) for 14 days.

22. Calculations, Interpretation, and Evaluation

22.1 Strength loss evaluation: Proceed as per 11.1 and report the change in breaking strength as compared to the sample before exposure or the control if available.

22.2 Visual assessment: Report the extent of fungal growth on the discs, using a stereo microscope (7-50×) where necessary, in accordance with the following scheme:

Observed Growth

No growth

Microscopic growth (visible only under the microscope)

Macroscopic growth (visible to the eye)

23. Report

23.1 The report shall include the following:

23.1.1 The test method and test part.

23.1.2 Size of test specimen used.

23.1.3 Number of test specimen replicates used.

23.1.4 Concentration of the fungal spore suspension.

23.1.5 For strength loss evaluation option, report the percent retained breaking strength when compared to the unexposed textile, any pre-exposure of specimens before burying and percent retained breaking strength of untreated specimen and/or viability control. For visual assessment option, report the observed growth according to 22.2.

24. Precision and Bias

24.1 The precision and bias of this test method have not been established. Refer to the precision and bias statement given in ASTM D5035.

Test III
Agar Plate, *Aspergillus niger*

25. Purpose and Scope

25.1 Certain fungi, including *Aspergillus niger*, can grow on textile products without causing measurable breaking strength loss within a laboratory experimental time frame. Nonetheless, their growth may produce undesirable and unsightly effects. This procedure is used to evaluate textile specimens where growth of these fungi is important. The procedure has the option of testing with or without 3% glucose in the agar. Although the evaluation of fungal growth is conducted on top of the sample, the incorporation of glucose in the agar simulates exposure of the sample to soiling conditions as might be encountered during use.

26. Principle

26.1 Test specimens are placed on an agar surface and inoculated with fungi. After a defined incubation period, the test specimens are observed for fungal growth.

27. Uses and Limitations

27.1 If the test specimen is thick or non-porous, the fungal spores may not have access to minerals or nutrients in the agar plate. This may cause a lack of fungal growth on the test specimen and cause misleading results.

28. Apparatus, Reagents, Materials

28.1 Organism: *Aspergillus niger*, American Type Culture Collection No. 6275 (see 44.5).

28.2 Autoclave. To sterilize media and to sterilize test materials prior to disposal.

28.3 Stereo microscope.

28.4 Incubator. Incubating equipment to maintain a temperature of 28 ± 1°C (82 ± 2°F).

28.5 Counting chamber suitable for determining spore concentrations; e.g., hemocytometer.

28.6 Culture media for preparation of inoculum. Mineral salts agar (see 21.1) containing 3.0 ± 0.1% glucose. Other suitable media are Czapek (Dox) Agar, Potato Dextrose Agar and Sabouraud Dextrose Agar.

28.7 Test media. Mineral salts agar (see 21.1) with or without 3.0 ± 0.1% glucose. Glucose may be added to the mineral salts agar prior to autoclave sterilization.

28.8 Non-ionic wetting agent (see 44.6).

28.9 Gauze or glass wool, sterile.

28.10 Sterile scraping instrument to harvest fungal spores. Platinum or nichrome wire, plastic loop or needle, glass rod, or swab.

29. Specimens

29.1 Cut duplicate 3.8 ± 0.5 cm (1.5 ± 0.2 in.) diameter discs from both treated and untreated samples. If the fabric is different on both sides, duplicate samples of each, face up and face down, shall be tested. Other shapes and sizes can be used, provided any anticipated size of the growth-free zone is taken into consideration.

30. Procedure

30.1 Inoculum: Add 10 mL of sterile DI water to a ripe (7-14 days) fruiting culture of *Aspergillus niger* grown on a

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medium described in 28.6. Scrape the surface of the culture gently with a sterile instrument (see 28.10) to liberate the spores. Agitate the liquid slightly to disperse the spores without detaching mycelial fragments, and gently decant the spore suspension into a flask containing 50 ± 1 mL of sterile DI water and a few glass beads. Shake the flask thoroughly to bring the spores into suspension. Filter the spore suspension using sterile gauze or glass wool. Utilizing a hemocytometer, or other cell counter, enumerate the number of spores present to a final population of approximately 8.0×10^5 to 1.2×10^6 CFU/mL. Use this suspension as the inoculum. Suspensions may be stored at 6 ± 4°C (43 ± 7°F) for up to 4 weeks.

30.2 Inoculation: Mineral salts agar with or without 3.0 ± 0.1% glucose may be used as the test media (see 28.7). Distribute evenly 0.5 ± 0.1 mL of the inoculum over the surface of the agar. Pre-wet the specimen (but do not rub, squeeze, or agitate) in water containing 0.05% of a non-ionic wetting agent. Absorbent specimens should be damp but not saturated. Place the specimen on the agar surface. Distribute evenly over each disc 0.2 ± 0.01 mL of the inoculum by means of a sterile pipette. A negative control of either filter paper or untreated cotton fabric (see 39.3) is required to ensure inoculum viability.

30.3 Incubate all specimens at a temperature of 28 ± 1°C (82 ± 2°F) for 14 days when mineral salts agar is used alone or for 7 days when 3% glucose is added.

31. Interpretation and Evaluation

31.1 If mineral salts agar containing 3.0 ± 0.1% glucose was used as the test media, report the size of the growth-free zone (in mm) if present.

31.2 For either medium, report the extent of fungal growth of *Aspergillus niger* on top of the specimens, using a stereo microscope (7-50×) where necessary, in accordance with the following scheme:

Observed Growth

No growth

Microscopic growth (visible only under the microscope)

Macroscopic growth (visible to the eye)

32. Report

32.1The report shall include the following:

32.1.1 The test method, and test part.

32.1.2 Size of test specimen used.

32.1.3 The number of test specimen replicates used.

32.1.4 The agar media used to harvest the fungi (see 28.6) and for the test media (see 28.7).

32.1.5 Concentration of the fungal spore suspension.

32.1.6 The incubation time.

32.1.7 The observed growth-free zone according to 31.1.

32.1.8 The observed fungal growth according to 31.2.

32.1.9 The magnification used to make observation of microscopic growth (if necessary).

33. Precision and Bias

33.1 The precision and bias of this test method have not been established.

Test IV
Humidity Jar, Mixed Spore Suspension

34. Purpose and Scope

34.1 This test method is designed to determine the fungistatic effectiveness of treatments intended to control mildew and non-pathogenic fungal growth on articles or surfaces composed of textile materials intended for outdoor and above ground use and which are usually waterproofed.

34.2 For this test method, visual assessment is used. Additionally, breaking strength may be determined.

35. Principle

35.1 Treated and untreated, nutrient-saturated strips of fabric are sprayed with a mixed spore suspension of mildew-causing organisms and incubated at 90 ± 2% relative humidity. Mildew growth on treated and untreated strips is rated at weekly intervals for up to 4 weeks.

36. Referenced Documents

36.1 ASTM D5035, Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Strip Method) (see 44.3).

37. Apparatus

37.1 Glassware: 500 mL French square jars or equivalent with fitting screw caps. Caps are modified by center drilling and inserting an appropriate size stainless steel or brass bolt to which a hook (formed from a 6.5 ± 0.5 cm length [2.6 ± 0.2 in.] of #22 nickel-chromium wire or other non-corrosive wire) is attached.

37.2 Plastic paper clips or nylon thread to suspend the specimens from the screw caps of the French jars.

37.3 Atomizer, DeVilbiss #152 (or equivalent) operated at 69 ± 7 kPa (10 ± 1 psi).

37.4 Counting chamber suitable for determining spore concentrations; e.g., hemocytometer.

37.5 Incubator. Incubating equipment to maintain a temperature of 28 ± 1°C (82 ± 2°F).

37.6 Autoclave. To sterilize media and to sterilize test materials prior to disposal.

37.7 Stereo microscope.

38. Reagents and Materials

38.1 Organisms (see 44.5):

38.1.1 *Aspergillus niger*, American Type Culture Collection No. 6275.

38.1.2 *Penicillium varians*, American Type Culture Collection No. 10509.

38.1.3 *Trichoderma viride*, American Type Culture Collection No. 28020.

38.2 Potato dextrose agar.

38.3 Malt agar.

38.4 Yeast extract.

38.5 Sodium chloride, NaCl.

38.6 Glycerol.

38.7 Ammonium nitrate, NH_4NO_3.

38.8 Dipotassium hydrogen phosphate, K_2HPO_4

38.9 Magnesium sulfate, $MgSO_4 \cdot 7H_2O$.

38.10 Non-ionic wetting agent (see 44.6).

38.11 Gauze or glass wool, sterile.

38.12 Sterile scraping instrument. To harvest fungal spores. Platinum or nichrome wire, plastic loop or needle, glass rod, or swab.

39. Specimens

39.1 If strength loss is to be determined, prepare specimens as in Section 9. If only a visual examination of fungal growth is required, cut 2.5 ± 0.5 cm × 7.5 ± 0.5 cm (1.0 ± 0.2 × 3.0 ± 0.2 in.) strips from sample weighing 170.0 ± 34.0 g/m² (5.0 ± 1.0 oz/yd²). For heavier fabrics use strips 2.0 ± 0.5 cm × 2.0 ± 0.5 cm (0.8 ± 0.2 × 0.8 ± 0.2 in.).

39.2 Use at least 4 specimens of each treated and untreated fabric.

39.3 Untreated fabric strips, identical in all other respects to the treated specimens under test, are required to establish the test validity. If untreated fabrics are not available, use a control cloth with the following requirements:

Cotton:	American type, good middling
Warp:	18.5 tex z 886 × 2S748 ± 2 tex
Weft:	30 tex z 630 × 2S748 ± 2 tex
Weave:	Plain 34 ends/cm and 17 picks/cm ± 5 yarn count

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Mass/unit area: 230.0 ± 20 g/m² (6.8 ± 0.6 oz/yd²)
Finish: Scoured only

40. Procedure

40.1 Culture medium:

40.1.1 Maintain fungal cultures individually on appropriate agar such as potato dextrose agar for *A. niger* and *P. varians*, and malt extract agar for *T. viride*.

40.1.2 Incubate cultures 7-10 days at 25 ± 1°C (77 ± 2°F), then store at 2- 10°C (36-50°F).

40.2 Preparation of conidial suspensions:

40.2.1 Conidial suspensions of fungal organisms are prepared by adding 10 mL of a sterile 0.5% saline solution containing 0.05% of a non-ionic wetting agent (see 44.6) to a 7-10 day culture.

40.2.2 Scrape the surface of the culture gently with a sterile instrument (38.13) to liberate the spores. Agitate the liquid slightly to disperse the spores without detaching mycelial fragments, and gently decant the mold suspension into a flask containing a few glass beads.

40.2.3 Shake the dispersion vigorously to break up any clumps of spores and then filter through a thin layer of sterile cotton or glass wool. Conidial suspensions may be stored at 6 ± 4°C (43 ± 7°F) for up to 4 weeks.

40.2.4 Inoculum for test should be adjusted using a hemocytometer or a Petroff-Hausser bacteria counter to contain $5 \times 10^6 \pm 2 \times 10^5$ conidia/mL on day of use by appropriate dilution of stock suspension with saline solution.

40.3 Preparation of test specimens:

40.3.1 To ensure luxuriant growth, both the test and control strips must be saturated with a sterilized glycerol nutrient solution of the following composition: 97.6% distilled water, 2.0% glycerol, 0.1% K_2HPO_4, 0.1% NH_4NO_3, 0.05% $MgSO_4 \cdot 7H_2O$, 0.1% yeast extract and 0.05% of a non-ionic wetting agent (see 44.6). Adjust the pH to 6.3 ± 0.1. Sufficient nutrient solution should be prepared to saturate all the specimens used in a single test.

40.3.2 Soak each strip in nutrient for three minutes or until saturated. Squeeze excess liquid and allow fabric strips to air dry before proceeding with application of test fungi.

40.4 Pre-mix equal volumes of well agitated conidial suspensions of *A. niger*, *T. viride* and *P. varians*. Evenly distribute 1.0 ± 0.1 mL of the above suspension onto both sides of each specimen either by spraying or by means of a pipette.

40.5 Suspend fabric strips using plastic paper clips or nylon thread from the caps of individual jars containing 90 ± 3 mL of water each. Hook position must be adjusted so that the bottom ends of attached strips are all at a uniform height above the water level. The caps are tightened, then backed off one-eighth turn to allow for some ventilation.

40.5 Incubate at 28 ± 1°C (82 ± 2°F) for 14 days (for non-coated cellulosic textiles) or 28 days (for non-cellulosic or coated cellulosic textiles).

41. Calculations, Interpretation, and Evaluation

41.1 A record of the percent of surface area covered with fungal growth for each strip is made at weekly intervals, or until heavy growth occurs on each sample replicate. Using a microscope (7-50×) where necessary, assess each specimen in accordance with the following scheme:

Observed Growth

No growth

Microscopic growth (visible only under the microscope)

Macroscopic growth (visible to the eye)

41.2 After 7 days each control strip must show macroscopic growth. If this is not the case, repeat the test since test conditions were not valid.

41.3 Any adverse effect of incubation on the fabric; e.g., color changes, flexibility, water repellency, should be qualitatively reported.

41.4 Strength loss determination can be carried out as per 11.1.

41.5 The results of this test method have to be correlated to claims and directions for use recommended for the mildew control product plus any other criteria agreed upon by the interested parties.

42. Report

42.1 The report shall include the following:

42.1.1 The test method and test part.

42.1.2 Size of test specimen used.

42.1.3 Number of test specimen replicates used.

42.1.4 Concentration of the fungal spore suspension.

42.1.5 Pressure of the atomizer.

42.1.6 The incubation time.

42.1.7 For strength loss evaluation, report the percent retained breaking strength when compared to the unexposed textile, any pre-exposure of specimens before burying and percent retained breaking strength of untreated specimen and/or viability control. For visual assessment, report the observed growth according to 41.1.

43. Precision and Bias

43.1 The precision and bias of this test method are being established. If the breaking strength loss is determined, then refer to the statement given in ASTM D5035.

44. Notes

44.1 Publication available from U.S. Department of Health & Human Services—CDC/NIH-HHS Publication No. (CDC) 84-8395; web site: www.hhs.gov.

44.2 Booklet available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1.513.742.2020; web site: www.acgih.org.

44.3 Available from ASTM International, 100 Barr Harbor, W. Conshohocken PA: tel: +1.610.832.9500; web site: www.astm.org.

44.4 Types of soil which have been found satisfactory for this purpose include garden and naturally fertile topsoils, composts and non-sterile greenhouse potting soils. An equal blend of good topsoil, well rotted and shredded manure, and coarse sand should be used. These usually possess the proper physical characteristics, along with an organic content sufficient to ensure a high degree of microbial activity and the presence of cellulose destroying organisms. The optimum moisture content of these is about 30% moisture above oven dry weight.

44.5 Available from the American Type Culture Collection, P.O. Box 1549, Manassas VA 20108; tel: +1.703.365.2700; fax: +1.703.365.2701; web site: www.atcc.org.

44.6 Triton X-100, dioctyl sodium sulfosuccinate and N-methyl-tauride derivatives have been found to be good wetting agents. Do not use Polysorbate 80 or Tween 80 as it is known to inactivate some antimicrobials.

44.7 Prepared culture medium having composition prescribed in 20.1 (Mineral Salts Agar) can be purchased from HiMedia Laboratories, A-516, Swastik Disha Business Park, via Vadhani Industrial Estate, L.B.S. Marg, Mumbai - 400 086, India. www.himedialabs.com (M232 AATCC Mineral Salts Agar)

44.8 If testing is being performed for Federal Standards, use AATCC TM30 (Test II). Other organisms can be used: *Myrothecium verrucaria* ATCC 9095, QM 460; *Trichoderma virens* ATCC 9645, QM 365; *Memnoniella echinata* ATCC 11973, QM 1225; *Aspergillus niger* ATCC 6275, QM 458; *Aspergillus claratus* ATCC 18214, QM 862.

Appendix A
Pre-Treatments

A1. Leaching

A1.1 The leaching should conform in principle to the following procedure: Water from a faucet is passed through a tubing into leaching vessels, care being taken that specimens having different amounts of the same treatment are in separate vessels. Flow is adjusted to ensure a complete change of water not less than three times in 24 h. The delivery tubes are inserted down through the center of wire mesh cylinders in the leaching ves-

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sels and held in the wire cylinders with rubber bands and leached for 24 h. The pH and temperature of the water is recorded and included in the report of the test results.

A2. Volatilization

A2.1 Standard specimens of the fabric to be tested are exposed continuously to dry heat at 100-105°C (212-221°F) for 24 h in a well ventilated oven.

A3. Weathering

A3.1 Portions of the material to be tested are exposed on a series of weathering racks at 45° to the horizontal facing South, between April 1 and October 1, in such a manner as to avoid sagging or flapping. It is recommended that such racks be set up in at least four locations within the United States; e.g., Washington DC; Miami FL; New Orleans LA; and suitable desert locations.

A3.2 Exposure of textile materials in an artificial weathering apparatus using controlled conditions may also be used. Refer to AATCC TM169, Test Method for Weather Resistance of Textiles: Xenon Lamp Exposure or ASTM D7869, Standard Practice for Xenon Arc Exposure Test with Enhanced Light and Water Exposure for Transportation Coatings. The use of an artificial weathering apparatus has limitations, which are listed in the methods mentioned. Results obtained by the use of any one artificial weathering test method may not be representative of any other artificial weathering test method or any outdoor weathering test.

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Test Method for Water Resistance: Rain

1. Purpose and Scope

1.1 This test method is applicable to any textile fabric, which may or may not have been given a water-resistant or water-repellent finish. It measures the resistance to the penetration of water by impact, and thus can be used to predict the probable rain penetration resistance of fabrics. It is especially suitable for measuring the penetration resistance of garment fabrics. With the instrument, tests may be made at different intensities (see 11.1) of water impact to give a complete overall picture of the penetration resistance of a single fabric or a combination of fabrics.

1.2 The results obtained with this test method depend on the water repellency of the fibers and yarns, and on the construction of the fabric.

2. Principle

2.1 A test specimen, backed by a weighed blotter, is sprayed with water for 5 min under controlled conditions. The blotter is then reweighed to determine the amount of water which has leaked through the specimen during the test.

3. Terminology

3.1 **water resistance,** n.—*of fabric*, the characteristic to resist wetting and penetration by water. (See also *water repellency.*)

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5. Apparatus and Materials

5.1 AATCC Rain Tester (see Figs. 1, 2 and 3 and 11.3).

5.2 White AATCC Textile Blotting Paper (see 11.4).

6. Test Specimens

6.1 A minimum of three specimens of 20 × 20 cm is cut from the test fabric. The fabric samples and the blotting paper should be conditioned in an atmosphere of 65 ± 5% RH and 21 ± 2ºC (70 ± 4°F) for at least 4 h before testing.

7. Procedure

7.1 The test specimen (see 11.5),



Fig. 1—Rain tester, complete assembly.

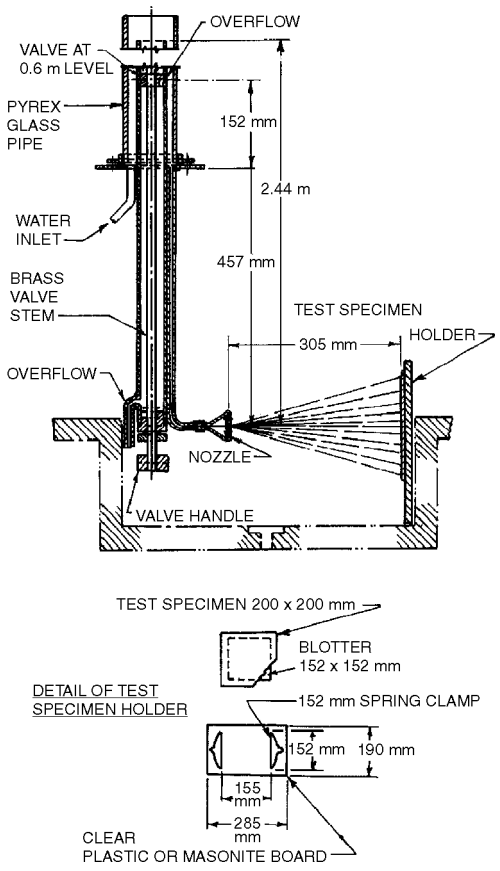


Fig. 2—Rain tester, structural details.

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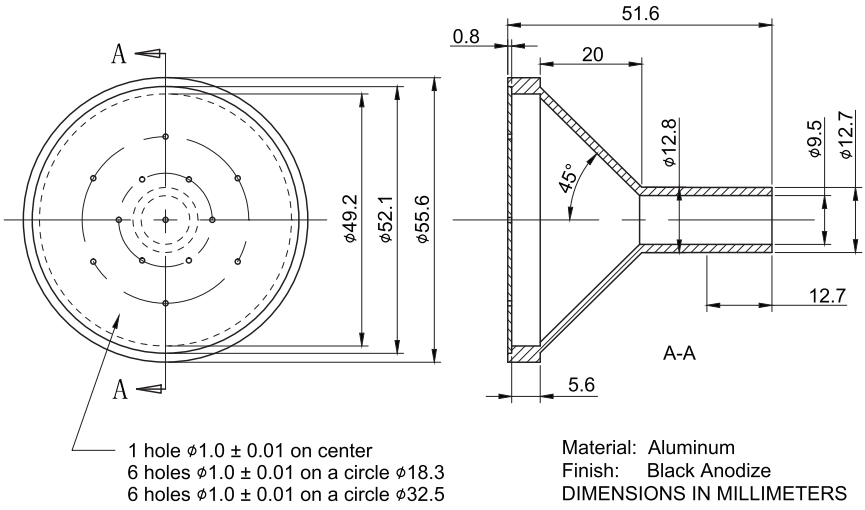


Fig. 3—Rain tester, nozzle.

backed by a 15.2 × 15.2 cm standard paper blotter weighed to the nearest 0.1 g is clamped in the specimen holder and the assembly is mounted in a vertical rigid support frame. The specimen assembly is positioned into the central portion of the spray at a distance of 30.5 cm from the face of the spray nozzle (see 11.6). A horizontal water spray at 27 ± 1°C (81 ± 2°F) (see 11.2) is directed against the specimen and is allowed to continue for a period of 5 min. At the end of the spray period the blotter is carefully removed and quickly reweighed to the nearest 0.1 g.

8. Evaluation

8.1 Water penetration as indicated by the increase in mass of the blotting paper during the 5 min test period is calculated, and the average for the three test specimens is reported. Individual determinations or average values of over 5.0 g may be simply reported to 5 + g or > 5 g.

8.2 In order to obtain a complete overall picture of the penetration resistance of a fabric or fabric combination the average penetration with different pressure heads on the nozzle should be obtained. The pressure head should be varied by 300 mm increments in order to determine (a) the maximum head at which no penetration occurs, (b) the change in penetration with increasing head and (c) the minimum head required to cause "breakdown" or the penetration of more than 5 g of water. At each pressure head a minimum of three specimens should be tested in order to obtain the average penetration for that head.

9. Report

9.1 Report the individual determinations. For values of over 5.0 g simply report as 5 + g or > 5 g.

10. Precision and Bias

10.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

10.2 *Bias*. The bias derived by this procedure can be defined only in terms of a test method. There is no independent, referee test method by which bias may be determined. This test method has no known bias.

11. Notes

11.1 The intensities are produced and controlled by means of a column of water which may be adjusted to 0.6, 0.9, 1.2, 1.5, 1.8, 2.1 and 2.4 m above the nozzle. This is done by means of a glass pressure column to which a nozzle is connected. The adjustment is made by a simple setting of a valve at the lower end of the drain or overflow pipe which extends up through the center of the glass column. A filtering device between the pressure gauge and the glass column may be used to prevent clogging of the nozzle openings. It may be eliminated in localities where the water supply is relatively free from iron rust or other suspended matter. A pressure gauge on the supply line also is an accessory which usually can be eliminated in the interest of economy.

11.2 The temperature of the supply water may be measured by means of a thermometer, but recent work has shown that it is more conveniently and accurately measured with a thermometer suspended in the glass pressure column or immersed in a beaker placed to catch water from the overflow.

11.3 The AATCC Rain Tester is available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org. Information relative to the principle of the instrument may be obtained by reference to the original article by Slowinske, G.A. and Pope, A.G., *American Dyestuff Reporter*, Vol. 36, 1947, p108.

11.4 Blotters suitable for this test can be obtained from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.5 The test specimen may comprise (a) a single layer of the test fabric, (b) two layers of the test fabric, or (c) a combination of two different fabrics such as the outer fabric of a raincoat and the lining fabric.

11.6 A loose-fitting cap may be placed over the end of the spray nozzle to shut off the spray during the mounting or the removal of the specimen holder in the support frame.

12. History

12.1 Editorially revised in 2021 to remove excess wording from Figure 2 and to add history section.

12.2 Reaffirmed 1952, 1963, 1964, 1967, 1969, 1971, 1974, 1977, 1980, 1985, 1989, 2006, 2013; editorially revised 1983, 1987, 1998, 2004, 2009, 2016, 2019; editorially revised and reaffirmed 1994; revised 2000, 2018. Technically equivalent to ISO 22958.

12.3 Developed in 1947 by AATCC Committee RA63.

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Test Method for Water Resistance: Impact Penetration

Developed in 1945 by AATCC Committee RA63; revised 1952, 2000, 2017; reaffirmed 1957, 1961, 1964, 1967, 1971, 1977, 1980, 1989, 2007, 2013; editorially revised and reaffirmed 1985, 1994; editorially revised 1986, 1987, 2009, 2014, 2016, 2019. Technically equivalent to ISO 18695.

1. Purpose and Scope

1.1 This test method is applicable to any textile fabric, which may or may not have been given a water-resistant or water-repellent finish. It measures the resistance of fabrics to the penetration of water by impact, and thus can be used to predict the probable resistance of fabrics to rain penetration. It is especially suitable for measuring the penetration resistance of garment fabrics.

1.2 The results obtained with this test method depend on the water repellency of the fibers and yarns and on the construction and finish of the fabric.

2. Principle

2.1 A volume of water is allowed to spray against a taut surface of a test specimen backed by a weighed blotter. The blotter is then reweighed to determine water penetration and the specimen is classified accordingly.

3. Terminology

3.1 **water resistance,** n.—*of fabric*, the characteristic to resist wetting and penetration by water. (See also *water repellency*.)

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5. Apparatus and Materials

5.1 Impact penetration testers

5.1.1 Type I tester (see 11.1, Figs. 1, 3 and 4)



Fig. 1—Impact penetration Type I tester.



Fig. 2—Impact penetration Type II tester.

5.1.2 Type II tester (see 11.1, Figs. 2, 3 and 4)

5.2 White AATCC Textile Blotting Paper (see 11.2)

5.3 Water, distilled, deionized or reverse osmosis

5.4 Balance accurate to 0.1 g

6. Test Specimens

6.1 A minimum of three specimens are taken, each 178 × 330 mm, with the warp (machine direction of the fabric) in the long direction. The specimens and the blotting paper should be conditioned in an atmosphere of 65 ± 5% RH and 21 ± 2°C (70 ± 4°F) for at least 4 h before testing.

7. Procedure

7.1 One end of the specimen is clamped under the 152 mm spring clamp at the top of the inclined stand. Another 152 ± 10 mm clamp, of 0.4536 kg ± 10% total mass, is clamped at the free end of the test specimen. A standard blotter paper 152 × 230 mm is weighed to the nearest 0.1 g and inserted beneath the test specimen.

7.2 A 500 ± 10 mL volume of distilled, deionized or reverse osmosis water at 27 ± 1°C is poured into a funnel of the tester and allowed to spray onto the test specimen. The water should be poured into the funnel without imparting any swirling motion of the water in the funnel. (A small

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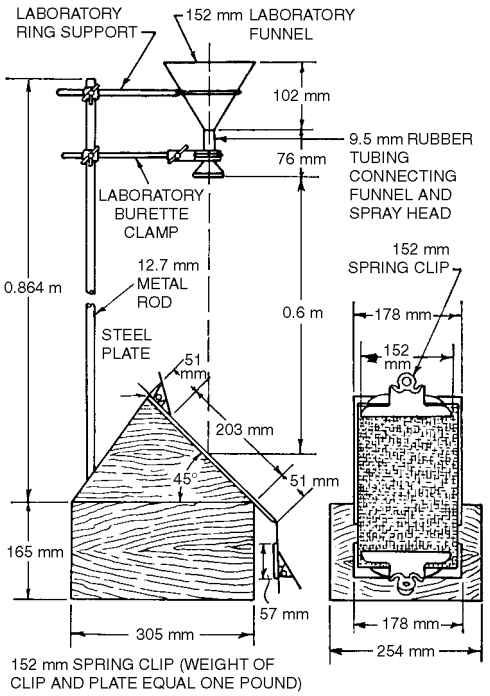


Fig. 3—Structural details of impact penetration tester.

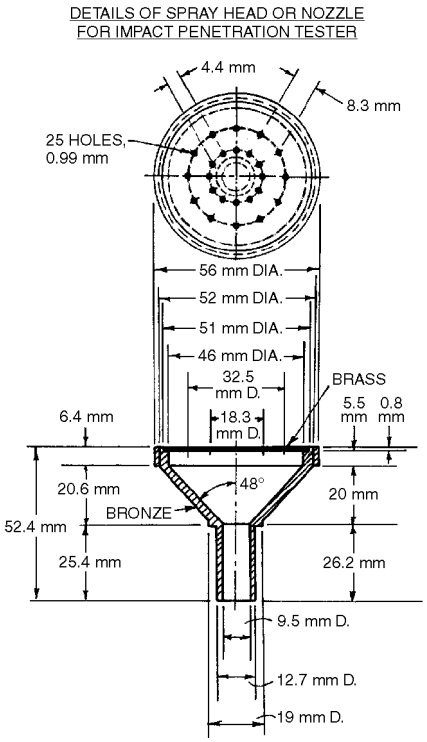


Fig. 4—Details of spray head.

blade fixed to the inside of the funnel and extending down its side, will prevent any swirling motion from being initiated.)

7.3 Upon completion of the spraying period, the test specimen is carefully lifted, the blotter beneath removed, and then quickly reweighed to the nearest 0.1 g.

8. Evaluation

8.1 The increase in mass of the blotter in grams is calculated and the average result of the three test specimens is reported. Individual determinations or average values of over 5.0 g may be simply reported as + 5.0 g or > 5.0 g.

9. Report

9.1 Report the individual determinations and the average. For values of over 5.0 g simply report as + 5.0 g or > 5.0 g.

9.2 Report the test method number and which piece of equipment was used.

10. Precision and Bias

10.1 *Precision*. In 1998, a limited intralaboratory study was completed. The same operator tested all samples.

10.1.2 Three sets of fabrics were analyzed using both pieces of equipment. Each fabric was evaluated 15 times and averages calculated for each group of three. This single laboratory data set was analyzed and used in writing a temporary precision statement, pending a full interlaboratory study. Until a full study is completed, users of the method are advised to exercise conventional statistical caution in making any comparisons of test results and apply these findings with due caution.

10.1.3 Variances ranged in value from 0.1-0.4 with an average value of 0.23 (standard deviation = 0.48) for the Type I tester, as determined by analysis of variance. Critical differences, based on these values and a 95% probability level, may be applied to determine significance (see Table I).

10.1.4 Variances ranged in value from 0.0-0.1 with an average value of 0.01 (standard deviation = 0.10) for the Type II tester, as determined by analysis of variance. Critical differences, based on these values and a 95% probability level, may be applied to determine significance (see Table II).

10.2 *Bias*. The values derived by this procedure can be defined only in terms of a test method. There is no independent, referee method for determining the true value. This test method has no known bias.

11. Notes

11.1 Impact Penetration Testers (see Figs. 1, 2, 3 and 4). These testers, Types I and II, are a combination of the spray test apparatus used in AATCC TM22, Test Method for Water Repellency: Spray Test, plus the impact penetration head and stand. The Type II tester is a more sturdy version of the Type I tester with the addition of a drip catcher. Both these testers are available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549. 8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.2 Blotters suitable for this test can be obtained from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

Table I—Within-Laboratory Critical Differences

Type I Tester—95% Probability Level

Det in Avg (*N*)	Standard Error	Critical Differences
1	0.48	1.11
3	0.28	0.64
5	0.21	0.50
7	0.18	0.41

N = Number of determinations per average
SE = Standard Error for *N* determinations
CD = 2.306 SE

Table II—Within-Laboratory Critical Differences

Type II Tester—95% Probability Level

Det in Avg (*N*)	Standard Error	Critical Differences
1	0.17	0.40
3	0.10	0.23
5	0.08	0.18
7	0.07	0.15

N = Number of determinations per average
SE = Standard Error for *N* determinations
CD = 2.306 SE

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Test Method for Wetting Agents for Mercerization

1. Purpose and Scope

1.1 This test method is applicable only to the evaluation of wetting agents in caustic soda of mercerizing concentration.

2. Principle

2.1 A 25 mm (1 in.) long bundle of cotton yarns composed of 120 ends is carefully dropped onto the surface of a test solution and the time for complete wetting of all ends is determined.

3. Terminology

3.1 **mercerization,** n.—a process for irreversibly altering the physical characteristics and appearance of natural cellulosic fibers by swelling in strong alkali.

3.2 **wetting agent,** n.—a chemical compound which when added to water lowers both the surface tension of the liquid and its interfacial tension against the solid material.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care. Use chemical goggles or face shield, impervious gloves and an impervious apron during dispensing and mixing of sodium hydroxide.

4.3 An eyewash/safety shower should be located nearby and a self-contained breathing apparatus should be readily available for emergency use.

4.4 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 9.1).

5. Apparatus and Materials (see 9.2)

5.1 Beakers, 250 mL.

5.2 Mohr measuring pipettes (0.1 graduation), 1 or 2 mL.

5.3 Volumetric transfer pipettes, 100 mL.

5.4 Stopwatch.

5.5 Scissors.

5.6 Ruler.

5.7 Cotton yarn, greige (unboiled), 40/2 combed *peeler*, preferably in form of chain warp or skein.

5.8 Standard mercerizing penetrant against which comparisons can be made.

5.9 Mercerizing caustic soda with a specific gravity of 48-52% Tw (271-299 g NaOH per liter of soln). Allow to stand several hours (until clear) after preparation.

5.10 Log-log coordinate paper.

6. Procedure

6.1 Into each of three 250 mL beakers measure out 100 mL of the mercerizing caustic soda solution at 26 ± 3°C (78 ± 5°F). Pipette into one beaker 0.75 mL, into the next 1.00 mL and into the third beaker 1.25 mL of the mercerizing penetrant using a suitable small pipette. Mercerizing wetting agents all seem to be liquids. Stir the contents of each beaker until the penetrant is completely dissolved in the caustic soda. These solutions are allowed to stand until all air bubbles have risen to the surface. It should be particularly noticed if the penetrant remains well dispersed or dissolved because the separation of an immiscible layer on the surface would entirely vitiate the results and give false values.

6.2 From the greige (unboiled), 40/2, combed peeler cotton yarn, cut a bundle of 120 parallel ends, 25 mm (1 in.) in length. This bundle of yarn ends is dropped carefully onto the surface of one of the caustic solutions of penetrant, and the time required for complete wetting of all the ends is noted with a stopwatch. The average of five determinations is taken as the sinking time for that concentration.

6.3 In like manner the average sinking time is determined at the other two concentrations for the penetrant under test. Then the average sinking times for the standard at the same three concentrations are determined by the same procedure.

7. Evaluation

7.1 The three average sinking times to the nearest tenth second for the product under test and for the standard are plotted as ordinates on log-log coordinate paper against concentrations of wetting agent in milliliters per 100 mL of caustic soda solution. Straight lines should be obtained when the points are joined.

7.2 The number of milliliters of product under test which are required to give the same wetting time as 1 mL of standard is read from the plot. If the product is either too efficient or too inefficient to obtain a comparison within the range of the volumes already employed, then either a smaller (0.5 mL) or a larger (2.0 or 2.5 mL) volume of the product must be tested. Making the assumption that the standard and the product are reasonably the same in specific gravity, the same wetting effect as 100 parts of standard is obtained from the following formula:

$$x = 100\ v$$

where:

x = parts of product being tested which are equivalent to 100 parts of standard and

v = volume of product in milliliter to give same wetting times as 1 mL of standard.

8. Precision and Bias

8.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

8.2 *Bias*. Wetting agents for mercerization can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

9. Notes

9.1 Available from Publications Office,

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ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

9.2 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

10. History

10.1 Editorially revised in 2023 to move the history section.

10.2 Editorially revised 2022 , 2019; reaffirmed 2018, 2013; editorially revised 2010; reaffirmed 2009; editorially revised 2008; editorially revised and reaffirmed 2004; reaffirmed 1999; editorially revised and reaffirmed 1994; editorially revised 1991; reaffirmed 1989; reaffirmed editorially revised 1986; reaffirmed 1985; reaffirmed 1980, 1977, 1974, 1971; revised 1952, 1945.

10.3 Developed in 1941 by AATCC Committee RR8; jurisdiction transferred in 2003 to AATCC Committee RA63; jurisdiction transferred in 2022 to AATCC Committee RA114. Related to ISO 6836.

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Test Method for Colorfastness to Laundering: Accelerated

1. Purpose and Scope

1.1 These accelerated laundering tests are to evaluate the colorfastness to laundering of textiles which are expected to withstand frequent laundering. The fabric color loss and surface changes resulting from detergent solution and abrasive action of five typical hand or home launderings, with or without chlorine, are roughly approximated by one 45 min test (see 9.2-9.6). However, the staining effect produced by five typical hand or home launderings cannot always be predicted by the 45 min test. Staining is a function of the ratio of colored to undyed fabrics, fiber content of fabrics in the wash load and other end-use conditions which are not always predictable.

1.2 When this test method was originally developed, various options of this method were intended to evaluate the color change and staining by five home or commercial launderings, on an accelerated basis. Throughout the years, commercial laundering procedures have changed and commercial cleaning today involves many different types of processes, dependent on the type of product being cleaned that cannot be duplicated by one accelerated laboratory procedure. In 2005, all references to commercial laundering were removed as it is not known if these procedures accurately replicate typical commercial laundering processes used today.

2. Principle

2.1 Specimens are tested under appropriate conditions of temperature, detergent solution, bleaching and abrasive action such that the color change is similar to that occurring in five hand or home launderings. The color change is obtained in a conveniently short time. The abrasive action is a result of the frictional effects of fabric against canister, the low liquor ratio and the impact of the steel balls on the fabric.

3. Terminology

3.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

3.2 **laundering,** n.—*of textile materials*, a process intended to remove soils and/or stains by treatment (washing) with an aqueous detergent solution and normally including subsequent rinsing, extracting and drying.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in the test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 The AATCC 1993 Standard Reference Detergents (with and without optical brighteners) may cause irritation. Care should be taken to prevent exposure to skin and eyes.

4.4 An eyewash/safety shower should be located nearby for emergency use.

4.5 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

5. Apparatus, Reagents and Materials (see 12.1)

5.1 Accelerated laundering machine.

5.1.1 A laundering machine for rotating closed canisters in a thermostatically controlled water bath at 40 ± 2 rpm.

5.1.2 Stainless steel lever lock canisters Type 1 500 mL (1 pt), 75 × 125 mm (3.0 × 5.0 in.) for Test No. 1A.

5.1.3 Stainless steel lever lock canisters Type 2 1200 mL, 90 × 200 mm (3.5 × 8.0 in.) for Tests No. 1B, 2A, 3A, 4A and 5A.

5.1.4 Adapter plates for holding canisters (see 5.1.3) on laundering machine shaft.

5.1.5 Stainless steel balls, 6 mm (0.25 in.) in diameter.

5.1.6 White Synthetic (SBR) Rubber Balls 9-10 mm (3/8 in.) dia. 70 Durometer hardness for Test 1B (see 12.1)

5.1.7 Teflon fluorocarbon gaskets (see 7.4.2 and 12.2).

5.1.8 Preheater/storage module (see 7.4, 12.1 and 12.3).

5.2 Scales for rating test results.

5.2.1 AATCC 9-Step Chromatic Transference Scale (AATCC EP8) (see 12.4).

5.2.2 Gray Scale for Color Change (AATCC EP1) (see 12.4).

5.2.3 Gray Scale for Staining (AATCC EP2) (see 12.4).

5.3 Reagents and materials.

5.3.1 Multifiber test fabrics (8 mm [0.33 in.] bands) containing acetate, cotton, nylon, silk, viscose rayon and wool. Multifiber test fabrics (8 mm [0.33 in.] bands) and (15 mm [0.6 in.] bands) containing acetate, cotton, nylon, polyester, acrylic and wool (see 12.5).

5.3.2 Bleached cotton test fabric, 32 × 32 ends × picks/cm (80 × 80 ends × picks/in.) construction, 100 ± 3 g/m^2 (3.0 ± 0.1 oz/yd^2), desized without optical brightener (see 12.5).

5.3.3 AATCC 1993 Standard Reference Detergent Without Brightener (WOB) (see 10.5 and 12.6).

5.3.4 AATCC 1993 Standard Reference Detergent With Brightener (see 10.5 and 12.6).

5.3.5 Water, distilled or deionized (see 12.7).

5.3.6 Sodium hypochlorite (NaOCl) bleach (see 12.8).

5.3.7 Sulfuric acid (H_2SO_4), 10% (see 12.8.1).

5.3.8 Potassium iodide (KI), 10% (see 12.8.1).

5.3.9 Sodium thiosulfate ($Na_2S_2O_3$), 0.1*N* (see 12.8.1).

5.3.10 Crockmeter test cloth cut in 50 mm (2 in.) squares (see 12.9).

5.3.11 White cards (specimen mounts) with *Y* tristimulus value at least 85%.

6. Test Specimens

6.1 The sizes of the specimens required for the various tests are as follows:

50 × 100 mm (2.0 × 4.0 in.) for Test No. 1A,

50 × 150 mm (2.0 × 6.0 in.) for Tests No. 1B, 2A, 3A, 4A and 5A.

6.2 Test only one specimen in each canister.

6.2.1 Test one specimen per laboratory sample. Replication may be advisable for improved precision.

6.3 To determine staining in Tests No. 1A and 2A, use multifiber test fabric. To determine staining in Test No. 3A, use either multifiber test fabric or bleached cotton test fabric. With respect to Test No. 3A, the use of multifiber test fabric is optional but the staining of acetate, nylon, polyester and acrylic is disregarded unless one of these fibers is present in the fabric being tested or known to be in the final garment. For Test 3A, multifiber test fabric with heat-sealed edges is recommended. Staining is not determined in Tests No. 4A and

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5A (see 12.10 and 12.11).

6.4 Specimen preparation.

6.4.1 Preparation with multifiber test fabric with individual component bands 8 mm (0.33 in.) wide or with bleached cotton test fabric. Prepare pieces with a 50 mm (2.0 in.) square of multifiber test fabric cloth or bleached cotton test fabric (as required) sewn, stapled or suitably attached along one 50 mm (2.0 in.) edge of the test specimen and in contact with the face of the material. When multifiber test fabric is used, attach it so that each of the six fiber bands is along the 50 mm (2.0 in.) edge of the specimen with the wool on the right. The fiber bands in the multifiber test fabric will be parallel to the lengthwise direction of the test specimen.

6.4.2 Preparation with multifiber test fabric with individual component bands 15 mm (0.6 in.) wide. Prepare pieces with a 50 × 100 mm (2.0 × 4.0 in.) rectangle of multifiber test fabric sewn, stapled or suitably attached centered along one 100 mm (4.0 in.) or 150 mm (6.0 in.) edge of the test specimen and in contact with the face of the material. Attach it so that each of the six fiber bands will be parallel to the widthwise direction of the specimen. Attach and secure the wool band at the top of the specimen to avoid fiber loss.

6.4.3 It is recommended that knitted fabrics be sewn or stapled at the four edges to equivalent size pieces of bleached cotton test fabric to avoid rolled edges and to assist in obtaining a uniform test result over the entire surface. Attach the multifiber test fabric to the face of the knitted fabric.

6.4.4 For pile fabric specimens with a pile lay direction, attach the multifiber test fabric at the top of the specimen with the pile lay direction pointing away from the top of the specimen.

6.5 When the textile to be tested is yarn, specimens may be tested using Option 1 or Option 2.

6.5.1 Option 1. Knit yarn on an appropriate sample knitting machine. Prepare specimens and multifiber test fabrics according to 6.1-6.4.3. Keep one knitted specimen of each sample as an unwashed original.

6.5.2 Option 2. Prepare two 110 m (120 yd) skeins of each yarn. Fold the skein so that there is a uniform amount of yarn across a 50 mm (2 in.) width with a length appropriate for the procedure to be used. Keep one skein of each sample as an unwashed original. Sew or staple Crockmeter test cloth squares (see 12.9) or squares of bleached cotton test fabric having approximately the same weight folded over each end of the layered yarn specimen. Attach a multifiber test fabric according to 6.4.1 or 6.4.2.

7. Procedure

7.1 Table I summarizes the conditions of the tests.

7.2 Adjust the laundering machine to maintain the designated bath temperature. Prepare the required volume of wash liquor. Preheat this solution to the prescribed temperature.

7.3 Run Test No. 1A in 75 × 125 mm (3.0 × 5.0 in.) lever lock stainless steel canisters. Run Tests No. 2A, 3A, 4A and 5A in 90 × 200 mm (3.5 × 8.0 in.) lever lock stainless steel canisters.

7.3.1 For Tests No. 1A, 1B, 2A and 3A, add to the canister the amount of detergent solution designated in Table I.

7.3.2 For Test No. 4A, prepare a 1500 ppm available chlorine solution. For 1 L, determine the amount of stock sodium hypochlorite bleach solution (see 12.8) to dilute as follows:

159.4/% NaOCl = g to add

Weigh the correct amount of bleach into a volumetric flask and dilute to 1 L. To each canister, add 5 mL of 1500 ppm available chlorine solution and 45 mL of detergent solution making a total volume of 50 mL.

7.3.3 For Test No. 5A, determine the amount of stock sodium hypochlorite bleach solution (see 12.8) to dilute as follows:

4.54/% NaOCl = g to add

Weigh the correct amount of bleach into a graduated cylinder and add detergent solution to make a total volume of 150 mL. Prepare this solution separately for each canister.

7.3.4 For all the tests, add the designated number of stainless steel balls or white rubber balls to each canister.

7.4 The two options for preheating the canisters to the test temperature are by use of the laundering machine or the preheater/storage unit. If the canisters are to be preheated in the laundering machine, proceed to 7.4.2.

7.4.1 Place the canisters in the preheater module at the prescribed test temperature. They are to remain in the module for at least 2 min. Enter a well crumpled test specimen into each canister.

7.4.2 Clamp the covers on the canisters. A Teflon fluorocarbon gasket (see 5.1.6) may be inserted between the neoprene gasket and the top of each canister to prevent contamination of the wash solution by the neoprene. Fasten the 75 × 125 mm (3.0 × 5.0 in.) lever lock canisters vertically and the 90 × 200 mm (3.5 × 8.0 in.) lever lock canisters horizontally in the adapters on the rotor of the laundering machine in such a manner that when the canisters rotate, the covers strike the water first. Place an equal number of canisters on each side of the shaft. For canisters preheated in the module, proceed to 7.7.

7.5 Start the rotor and run it for at least 2 min to preheat the canisters.

7.6 Stop the rotor and with a row of canisters in an upright position, unclamp the cover of one canister, enter a well crumpled test specimen into the solution and replace the cover, but do not clamp it. Repeat this operation until all the canisters in the row have been loaded. Then clamp the covers in the same order in which the canisters were loaded (delay clamping the covers to allow equalization of pressure). Repeat this operation until all rows of canisters have been loaded.

7.7 Start the laundering machine and run it at 40 ± 2 rpm for 45 min.

7.8 The rinsing, extracting and drying procedures are the same for all the tests. Stop the machine, remove the canisters and empty the contents into beakers, keeping each test specimen in a separate beaker. Rinse each test specimen three times, in beakers, in distilled or deionized water at 40 ± 3°C (105 ± 5°F) for 1 min periods with occasional stirring or hand squeezing. To remove excess water, centrifuge, blot or pass the test specimens through wringer rolls. Dry the specimens in an air circulating oven in which the temperature does not exceed 71°C (160°F), or tumble dry in a nylon mesh bag in an automatic tumble dryer at nor-

Table I—Test Conditions[a]

Test No.[b]	Temp °C (± °2)	Temp °F (± °4)	Total Liquor Volume (mL)	Percent Powder Detergent of Total Volume	Percent Liquid Detergent of Total Volume	Percent Available Chlorine of Total Volume	No. Steel Balls	No. of Rubber Balls	Time (Min)
1A	40	105	200	0.37	0.56	None	10	0	45
1B[c]	31	88	150	0.37	0.56	None	0	10	20
2A	49	120	150	0.15	0.23	None	50	0	45
3A	71	160	50	0.15	0.23	None	100	0	45
4A	71	160	50	0.15	0.23	0.015	100	0	45
5A	49	120	150	0.15	0.23	0.027	50	0	45

[a] Refer to Section 9 for objectives for each test method.
[b] All Tests include an alternate use for 2003 AATCC Standard Liquid Detergent which has been discontinued and is no longer available. (See 12.6.)
[c] Test 1B provides for the use of White Rubber Balls instead of Stainless Steel Balls.

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mal cycle, which has an exhaust temperature of 60-71°C (140-160°F), or air dry.

7.9 Allow specimens to condition at 65 ± 5% relative humidity and 21 ± 2°C (70 ± 4°F) for 1 h before evaluating.

7.10 Prepare tested fabric specimens and adjacent fabrics for evaluation by trimming off raveled yarns and lightly brushing off any loose fiber and yarn on the fabric surfaces. Brush pile fabric specimens in required direction to restore them as nearly as possible to the same pile angle as the untreated specimens. Specimens should be smoothed or flattened if they are wrinkled and messy due to washing and/or drying. Specimens may be mounted on cards to facilitate identification and handling in their evaluation. For consistency in backing material, use a white mounting card with *Y* tristimulus value of at least 85%. Mounting material must not be visible in the area to be viewed and must not interfere with rating as specified in 5.1 of both AATCC EP1 and AATCC EP2 for the Gray Scale for Color Change and Gray Scale for Staining or instrumental assessment of color according to AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen (see 12.4).

7.10.1 Yarn skein specimens should be combed and brushed for improved alignment of the yarns before comparison with the unwashed original. The original specimens may also need further combing and brushing for uniformity of appearance.

8. Evaluation (see 12.16)

8.1 Evaluation of color change.

8.1.1 Evaluate the color change of the test specimens by comparison with the Gray Scale for Color Change (AATCC EP1), or using AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen, and record the numerical rating that corresponds to the appropriate one on the Gray Scale. For improved precision and accuracy the specimens should be rated by more than one rater.

8.2 Evaluation of staining.

8.2.1 Evaluate staining (see 12.10) by comparison with the Gray Scale for Staining (AATCC EP2), the AATCC 9-Step Chromatic Transference Scale (AATCC EP8), or Instrumental Assessment of Degree of Staining (AATCC EP12), and record the numerical rating that corresponds to the appropriate one on either of them. The scale used should be indicated when reporting the test results.

8.2.2 The color transferred to the multifiber test fabric or bleached cotton test fabric square of 6.4.1 can be quantitatively determined by measuring the color difference between a piece of the original material and the stained material. Multifiber test fabrics (15 mm [0.6 in.]) have filling bands of sufficient width to be within the aperture diameter capability of many colorimeters and spectrophotometers (see AATCC EP6, Instrumental Color Measurement and 12.14).

9. Interpretation of Results

9.1 Results from these tests are intended to approximate the color change effects (see 1.1) of five typical home launderings. These are accelerated tests, and in obtaining the required degree of acceleration some of the conditions, such as temperature, were purposely exaggerated. The tests have remained largely the same over many years while laundry detergents, washers and dryers, laundry practices and fabrics have changed (see AATCC LP1.

9.2 Test No. 1A—This test is for evaluating the colorfastness of textiles that are expected to withstand repeated hand laundering at low temperature. Specimens subjected to this test should show color change similar to that produced by five typical careful hand launderings at a temperature of 40 ± 3°C (105 ± 5°F).

9.3 Test No. 1B—This test is for evaluating the colorfastness of textiles that are expected to withstand repeated hand laundering at cool temperatures. Specimens subjected to this test should show color change similar to that produced by five typical careful hand launderings at a temperature of 27 ± 3°C (80 ± 5°F).

9.4 Test No. 2A—This test is for evaluating the colorfastness to washing of textiles that are expected to withstand repeated low temperature machine laundering in the home. Specimens subjected to this test should show color change similar to that produced by five home machine launderings at medium or warm setting in the temperature range of 38 ± 3°C (100 ± 5°F).

9.5 Test No. 3A—This test is for evaluating colorfastness to washing of textiles considered washable under vigorous conditions. Specimens subjected to this test should show color change similar to that produced by five home machine launderings at 60 ± 3°C (140 ± 5°F), without chlorine.

9.6 Test No. 4A—This test is for evaluating the colorfastness to washing of textiles laundered in the presence of available chlorine. Specimens subjected to this test should show color change similar to that produced by five home machine launderings at 63 ± 3°C (145 ± 5°F) with 3.74 g per L (0.50 oz/gal) of 5% available chlorine per 3.6 kg (8.0 lb) load.

9.7 Test No. 5A—This test is for evaluating the colorfastness to washing of textiles that may be laundered in the presence of available chlorine. Specimens subjected to this test should show color change similar to that produced by five home machine launderings at 49 ± 3°C (120 ± 5°F) with 200 ± 1 ppm available chlorine.

10. Report

10.1 Report the test number.

10.2 Report the grade number determined for color change in 8.1 and the staining grade numbers for the evaluated fibers in the multifiber test fabric and/or bleached cotton test fabric as determined in 8.2.

10.3 State which scale (Gray Scale for Staining or AATCC 9-Step Chromatic Transference Scale) was used in evaluating staining (see 12.12).

10.4 Report the multifiber test fabric used and if bleached cotton test fabric was employed to avoid knit curling.

10.5 Report the detergent used with color change and staining results (see 12.6).

10.6 Report which laundering machine is used.

11. Precision and Bias (see 12.15)

11.1 Precision and bias statements have been developed for Tests No. 2A and 5A. Although correlation work has been done, no precision and bias statements have been developed for Tests No. 1A, 3A and 4A.

11.1.1 Because of changes in the detergents used in this method, these precision and bias statements may not apply to data or information obtained with the currently available detergents.

11.2 Test No. 2A.

11.2.1 *Summary.* An interlaboratory test was carried out in May 1985 to establish the precision of Test No. 2A. A part of the test was to determine if the wider 15 mm (0.6 in.) No. 10A multifiber test fabric could be substituted for the 8 mm (0.33 in.) wide No. 10. The complete test consisted of six laboratories evaluating 10 materials in duplicate by one operator using Test No. 2A.

11.2.2 *Color Change*. Three raters from six laboratories independently evaluated nine materials in duplicate using the Gray Scale for Color Change. The components of variance as standard deviations of the colorfastness grades (averages of variances for No. 10 and No. 10A multifiber test fabrics) were calculated as follows:

Single-Operator Component	0.29
Within-Laboratory Component	0.29
Between-Laboratory Component	0.29

11.2.3 *Critical Differences*. For the components of variance reported in 11.2.2, two averages of observed values should be considered significantly different at the 95% probability level if the dif-

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ference is equal to or exceeds the critical differences shown in Table II.

Table II—Critical Differences, Grades, for the Condition Noted[a]

No. of Observations	Single-Operator Precision	Within-Lab Precision	Between-Lab Precision
1	0.80	1.12	1.37
3	0.46	0.92	1.21
5	0.36	0.87	1.18

[a]The critical differences were calculated using t = 1.950 which is based on infinite degrees of freedom.

11.2.4 *Staining*. Three raters independently rated the six fibers of the multifiber test fabric (No. 10 and No. 10A) for 10 materials at six laboratories using the Gray Scale for Staining. Of the 60 possible fiber/fabric combinations, only 51 could be used. The components of variance were averaged for the No. 10 and No. 10A multifiber test fabrics and appear below as standard deviations of staining ratings:

Single Operator Component	0.27
Within-Laboratory Component	0.34
Between-Laboratory Component	0.25

11.2.5 *Critical Differences*. For the components of variance reported in 11.2.4, two averages of observed values should be considered significantly different at the 95% probability level if the difference equals or exceeds the critical differences shown in Table III.

Table III—Critical Differences, Grades, for the Condition Noted[a]

No. of Observations	Single-Operator Precision	Within-Lab Precision	Between-Lab Precision
1	0.75	1.20	1.39
3	0.43	1.03	1.25
5	0.33	1.00	1.22

[a]The critical differences were calculated using t = 1.950 which is based on infinite degrees of freedom.

11.2.6 *Bias*. Tests comparing five home launderings at 40°C (105°F) with one Launder-Ometer Test No. 2A indicate there is no bias between the two methods for the colorfastness and staining levels evaluated.

11.3 Test No. 5A, Chlorine Bleach.

11.3.1 *Summary.* An interlaboratory test was carried out in 1984 to establish the precision of Test No. 5A for determining the effect of chlorine bleach on the colorfastness of fabrics. All specimens were laundered in a Launder-Ometer by one operator. Color change in Test No. 5A was determined both visually and instrumentally. Details of the statistical analysis of the data can be found in the report, Third Interlaboratory Study of Proposed Launder-Ometer Test for Colorfastness of Fabrics to Chlorine and Non-Chlorine Bleaches, October 21, 1985, by J. W. Whitworth, Milliken Research Corp., Spartanburg, SC.

11.3.2 *Visual Assessment*. Four materials were tested at each of five laboratories. Three raters visually assessed the color change of four specimens. The components of variance as standard deviations of colorfastness grades were calculated as follows:

Single Operator Component	0.38
Within-Laboratory Component	0.28
Between-Laboratory Component	0.27

11.3.3 *Critical Differences*. For the components of variance in 11.3.2, two averages of observed values should be considered significantly different at the 95% probability level if the difference equals or exceeds the critical differences shown in Table IV.

Table IV—Critical Differences, Grades, for the Condition Noted[a]

No. of Observations	Single-Operator Precision	Within-Lab Precision	Between-Lab Precision
1	1.03	1.29	1.49
3	0.59	0.98	1.23
5	0.46	0.91	1.17

[a]The critical differences were calculated using t = 1.950 which is based on infinite degrees of freedom.

11.3.4 *Instrumental Assessment*. Color change as total color difference (CIELAB) was measured on a spectrophotometer or colorimeter using apertures ranging in size from 13-51 mm (0.5-2.0 in.) in diameter, illuminant D_{65}/10° observer or illuminant C/2° observer. Six materials were tested at each of six laboratories. One operator in each laboratory tested four specimens of each fabric. The components of variance for ΔE^* expressed as coefficients of variation were calculated to be:

Single-Operator Component	6.8%
Between-Laboratory Component	11.2%

11.3.5 *Critical Differences*. For the components of variance reported in 11.3.4, two averages of observed values should be considered significantly different at the 95% probability level if the difference equals or exceeds the critical differences shown in Table V.

Table V—Critical Differences, Percent of Grand Average for the Conditions Noted[a,b]

No. of Observations in Each Average	Single-Operator Precision	Between-Laboratory Precision
1	18.7	36.2
3	10.8	32.8
5	8.4	32.1

[a]The critical differences were calculated using t = 1.950 which is based on infinite degrees of freedom. [b]To convert the values of the critical differences to units of measure, multiply the critical differences by the average of the two specific sets of data being compared and then divide by 100.

11.3.6 *Bias*. Tests comparing five home launderings at 49°C (120°F) with one Launder-Ometer Test No. 5A indicate there is no bias between the two methods for the colorfastness levels evaluated (see 12.13).

12. Notes

12.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

12.2 Teflon is a registered trademark of the DuPont Co., Wilmington DE 19898.

12.3 The preheater/storage unit may be a side unit to the laundering machine or a separate module with its individual electric heater and thermostat to control water bath temperatures for heating containers and solutions prior to loading the laundering machine.

12.4 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

12.5 Bleached cotton test fabric in 32 × 32 ends × picks/cm (80 × 80 ends × picks/in.) construction, 100 ± 3 g/m^2 and without optical brightener should be used.

12.6 The AATCC 1993 Standard Reference Detergent Without Brightener (WOB), a compact formulation, is the primary detergent to be used in this test method. Where the effect of an optical brightener is to be evaluated, AATCC 1993 Standard Reference Detergent With Brightener should be used. The AATCC 2003 Standard Reference Liquid Detergents have been discontinued and are no longer available for purchase. Users who have a supply of AATCC 1993 Standard Reference Liquid Detergents may continue to use until depleted. All detergents are available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1. 919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

12.7 Use distilled water or deionized water

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of not more than 15 ppm hardness to dissolve the detergent and for the test solutions.

12.8 Use sodium hypochlorite bleach purchased within the last six months for a stock solution.

12.8.1 To confirm the stock solution's hypochlorite activity, weigh 2.00 g liquid sodium hypochlorite into an Erlenmeyer flask and dilute with 50 mL of deionized water. Add 10 mL of 10% sulfuric acid and 10 mL of 10% potassium iodide. Titrate with 0.1*N* sodium thiosulfate until colorless.

Calculation:

% sodium hypochlorite

$$= \frac{(\text{mL } Na_2S_2O_3)(0.1N)(0.03722)}{(2.00 \text{ g NaOCl})} \times 100$$

The factor 0.03722 is derived by multiplying the molecular weight of NaOCl (74.45 g/mol) by 0.001 (mL to L conversion) and dividing by 2 (mols of thiosulfate per hypochlorite).

12.8.2 Oxidizing power of sodium hypochlorite is typically expressed in terms of available chlorine, the equivalent amount of diatomic chlorine present. A 5.25% NaOCl solution contains 50,000 ppm available chlorine.

12.9 Crockmeter test cloth, 32 × 33 ends × picks/cm (80 × 84 ends × picks/cm) combed cotton, desized, bleached (no optical brightener or finishing material present) should be used.

12.10 If staining evaluations are needed for Tests No. 4A and 5A, they may be carried out using the corresponding Tests No. 2A or 3A, which use no bleach. Test No. 2A is the no-bleach alternate for Test No. 5A, and Test No. 3A is the no-bleach alternate for Test No. 4A.

12.11 If multifiber test fabric is used in conjunction with Tests 4A or 5A, the wool can absorb the chlorine leaving very little for bleaching action. The wool may be removed from the multifiber test fabric before testing to eliminate this effect.

12.12 For very critical evaluations and in cases of arbitration, grades must be based on the geometric Gray Scale for Staining.

12.13 For additional information pertaining to the bias between Test No. 5A and five home washes, refer to Fig. 1 in Interlaboratory Study of Proposed Launder-Ometer Test for Colorfastness of Fabrics to Chlorine and Non-Chlorine Bleaches, Report to AATCC Committee RA60, Colorfastness to Washing Test Methods, November 1984, New York NY by L. B. Farmer and J. W. Whitworth of Milliken Research Corp., Spartanburg SC, and J. G. Tew, AATCC Technical Center, Research Triangle Park NC.

12.14 AATCC EP7 gives a method for computing Gray Scale grades from color measurement data.

12.15 The precision of this test method is dependent on the combined variability of the material being tested, the test method itself, and the evaluation procedure utilized.

12.15.1 The precision statement in Section 11 was developed from results obtained by visual evaluation (AATCC EP1 and AATCC EP2).

12.15.2 It is expected that the use of instrumental evaluation procedures (AATCC EP7 and AATCC EP12) will result in greater precision than that obtained from visual evaluations.

12.16 CAUTION: It has been reported that the results for staining obtained by this method on fabrics dyed to dark shades (navy, black, etc.) that contain a combination of polyester and spandex, or their blends, may not show the full staining propensity of such fabrics in consumer use. It is, therefore, recommended that the staining results obtained by this test not be used for the acceptance testing of such fabrics.

13. History

13.1 Editorially revised in 2023 to update designation and history section.

13.2 Editorially revised 2022; reaffirmed 2020; reaffirmed 2019, 2016; revised 2013; reaffirmed 2012; revised 2010, 2009; reaffirmed 2008; revised 2007, 2006 (title change); reaffirmed 2004; revised 2003; reaffirmed 2002; editorially revised and reaffirmed 2001; reaffirmed 1998; revised 1996; reaffirmed 1995; revised 1994,1993; reaffirmed 1991; revised 1989,1986 (title change); reaffirmed 1984, 1983, 1981; reaffirmed 1976, 1975, 1974, 1973; revised 1972, 1970, 1961, 1960, 1957, 1954,1952.

13.3 Developed in 1950 by AATCC Committee RA60. Partly equivalent to ISO 105C06.

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Test Method for Wrinkle Recovery of Woven Fabrics: Recovery Angle

1. Purpose and Scope

1.1 This test method is used to determine the wrinkle recovery of woven fabrics. It is applicable to fabrics made from any fiber, or combination of fibers.

2. Principle

2.1 A test specimen is folded and compressed under controlled conditions of time and force to create a folded wrinkle. The test specimen is then suspended in a test instrument for a controlled recovery period, after which the recovery angle is recorded.

3. Terminology

3.1 **wrinkle recovery,** n.—that property of a fabric which enables it to recover from folding deformations.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5. Uses and Limitations

5.1 Two options for performing wrinkle recovery angle testing are included in this test method. The Option 1 procedure is applicable to equipment commercially available and similar to devices used in ISO 2313: Determination of the Recovery from Creasing of a Horizontally Folded Specimen by Measuring the Angle of Recovery (see 13.1). The Option 2 procedure is for those laboratories that still have the older recovery devices which are no longer available from their original source.

5.2 This method has been used as a research tool and for production quality control; however, it is not recommended to be used for acceptance testing (see 13.2).

5.3 Parameters to be controlled in the test are: relative humidity, temperature, applied pressure, time under pressure, and recovery time. This method specifies arbitrary selected values for the latter three factors, based on a compromise between conditions likely to be encountered in service, and the expeditious conduct of the test. Two conditions of temperature and relative humidity are specified as being representative of normal experience. For specialized purposes, it may be advisable to use other combinations of temperature and humidity.

5.4 Specimens from limp or heavy weight fabrics may twist or curl making angle readings difficult to determine (see 13.3).

6. Apparatus (see 13.4)

6.1 Wrinkle Recovery Tester and accessories, Option 1 (see Fig. 1).

6.1.1 Loading device with two flat plates (see Fig. 2).

6.1.2 Wrinkle Recovery Circular Scale divided into degrees over the range 10–180° (see Fig. 3).

6.1.3 Paper or aluminum foil not exceeding 0.04 mm thick.

6.2 Wrinkle Recovery Tester and accessories, Option 2 (see Fig. 4).

6.2.1 Disk and protractor with clamp mounted to the disk (see Fig. 4).

6.2.2 Specimen holder with two superimposed stainless steel leaves, 0.16 ± 0.01 mm thick, fastened together at one end. The top leaf is shorter than the bottom leaf.



Fig. 1—Option 1—Wrinkle Recovery Angle Tester and accessories.

6.2.3 Plastic press, consisting of two superimposed leaves 95 × 20 mm fastened together at one end. A 23 × 20 mm plastic section is attached to the outer surface of the free end of one leaf, flush with the outer edges that form a platform for a weight.

6.3 Tweezers with a jaw width of 25 mm and a line marked on both external faces 5 mm from the end and parallel to the jaw width (see Fig. 5). Another pair of tweezers, preferably plastic, is needed to handle specimens.

6.4 Clock or timer accurate to ± 1 s.

6.5 Template, 40 × 15 mm.

6.6 Weight, 500 ± 5 g.

6.7 Conditioning chamber suitable for the manipulation of the test specimen and the test apparatus that can produce atmospheres other than the standard atmosphere for testing textiles.

7. Test Specimens

7.1 Identify the face and back of the fabric sample. Avoid taking specimens from creased, wrinkled, or distorted sections of the sample.

7.2 Cut 12 specimens 40 × 15 mm, six with their long dimension parallel to the warp direction of the fabric and six with their long dimension parallel to the filling direction.

7.2.1 Cut warp specimens from sample locations with different warp yarns. Cut filling specimens from sample locations with different filling yarns (see Fig. 6). Unless otherwise specified, take specimens no nearer the selvage or edge of the fabric than one tenth the width of the fabric.

7.3 Mark the fabric face of each specimen.

7.4 Avoid handling or distorting the specimens. The use of a template or die for cutting the specimens and tweezers for handling them is recommended.

8. Conditioning

8.1 Lay specimens flat to condition at 65 ± 5% RH, 21 ± 2°C (70 ± 4°F) for at least 24 h prior to testing. Shorter conditioning periods which still allow attainment of equilibrium moisture content are acceptable.

8.2 If other atmospheres are used, they should be reported with the test results (see 11.3). For example, high humidity determinations may be made after condi-

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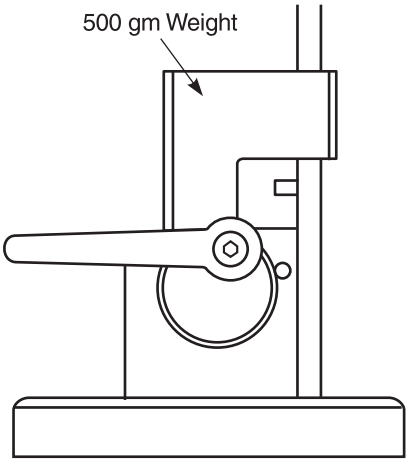


Fig. 2—Option 1—Loading device.

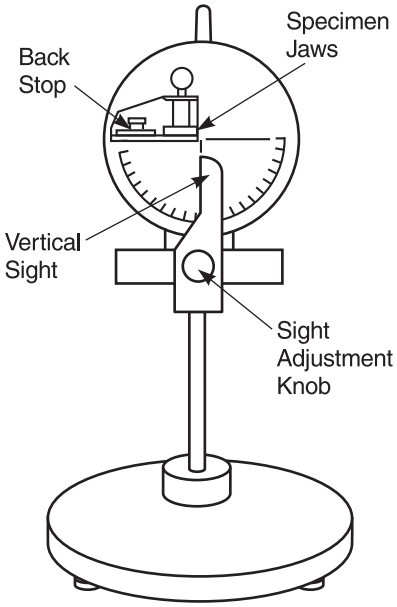


Fig. 3—Option 1—Wrinkle Recovery Angle Tester.



Fig. 4—Option 2—Wrinkle Recovery Tester and accessories.

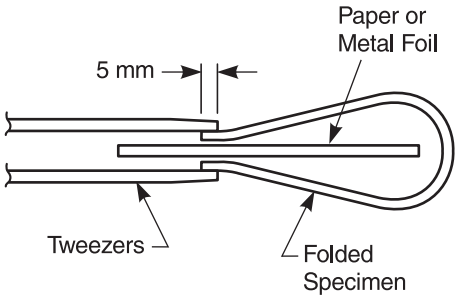


Fig. 5—Option 1—Folding of specimen.

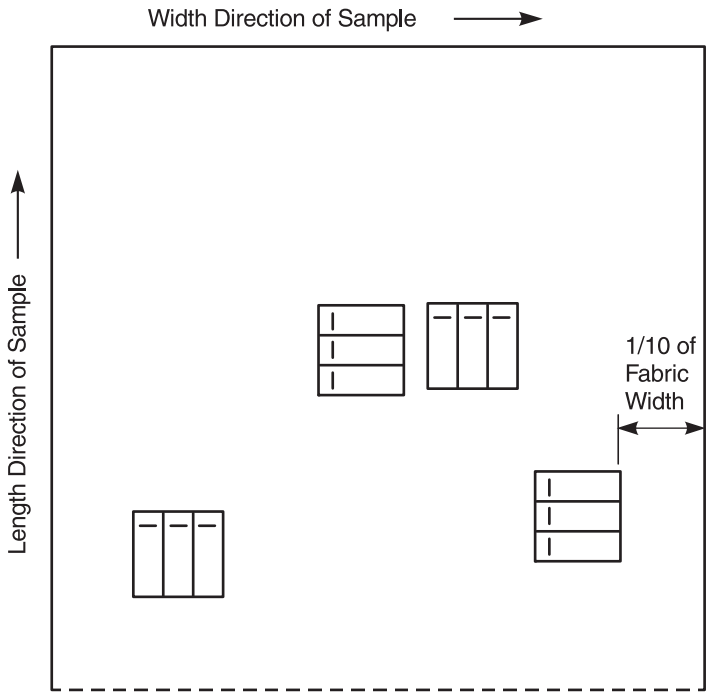


Fig. 6— Specimen sampling layout.

tioning samples at 35 ± 1°C (95 ± 2°F), 90 ± 2% RH for 24 h prior to testing.

9. Procedure

9.1 Option 1.

9.1.1 Three sets of loading devices, crease recovery scales and weights can be used to test three specimens sequentially, one after another every 60 s.

9.1.2 Fold a face-to-face specimen end-to-end and hold it in this position with tweezers, gripping no more than 5 mm from the ends. In instances where the surfaces of the specimens show a tendency to stick together, place a piece of paper or aluminum foil, 18 × 14 mm, between the ends of the specimen (see Fig. 5). Avoid touching the specimen with anything other than the tweezers.

9.1.3 Place the folded specimen between the two leaves of the loading device and immediately apply the weight. Start a timing device. Wait 60 ± 2 s to repeat steps 9.1.2 and 9.1.3 for the next specimen. Wait another 60 ± 2 s to start the third specimen's load timing.

9.1.4 After 5 min ± 5 s, quickly but smoothly, remove the weight from the first specimen so that the specimen press does not spring open.

9.1.5 Using tweezers, transfer the folded specimen to the instrument's circular scale specimen holder. If paper or foil was used, remove it at this time. Insert one end of the specimen between the clamp on the specimen holder, leaving the other end to hang freely. Avoid disturbing the folded formation by placing the specimen no farther into the clamp than the back stop. Start timing the recovery period. Wait 60 ± 2 s, sequentially, to load the second and third specimens in holders.

9.1.6 While the specimen is in the holder, adjust the instrument to keep the

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free hanging end of the specimen in alignment with the vertical mark. Frequent adjustment during the 5 min time period is necessary to avoid gravitational effects.

9.1.7 Read and record the recovery angle from the circular scale 5 min ± 5 s after inserting a specimen into a clamp. If the free end of the specimen twists, sight a vertical plane through its center and align it with the vertical mark on the circular scale. Read and record the recovery angle for each specimen after 5 min ± 5 s after inserting it into a clamp.

9.1.8 Repeat all steps for three specimens in the opposite direction and for specimens to be folded face-to-back in both fabric directions.

9.2 Option 2.

9.2.1 Using tweezers, place a specimen between the leaves of the metal holder with one end aligned under the 18 mm mark. With the tweezers, lift the free end of the specimen up and over to the 18 mm mark taking care to loop back rather than flatten the specimen. Hold the edge of the specimen firmly in place with a thumbnail.

9.2.2 Still holding the specimen end with a thumbnail, open the jaws of a plastic press with the other hand. Insert the holder with the specimen between the long and short jaws, releasing the thumbnail when bringing the end edge of the long jaw into contact with the specimen. The 18 mm mark on the metal holder, the unfolded end of the specimen, and the end edge of the plastic press are to be aligned before releasing the specimen. A fold should be formed 1.5 mm from the end of the short metal leaf. The plastic press should be in firm contact with the folded specimen but should not be squeezed.

9.2.3 Invert the press-holder combination on a flat surface with the small platform up. Gently apply a weight to the platform. Start a timing device. Repeat steps 9.2.1 through 9.2.3 for a second specimen after 60 ± 2 s has lapsed. After 60 ± 2 additional s, repeat all steps for the third specimen.

9.2.4 After 5 min ± 5 s, remove the weight. Pick up the press-holder combination by the plastic press and insert the exposed end of the specimen holder in the clip mount on the face of the recorder device. Open the jaws and remove the press rapidly taking care to avoid rolling the exposed end of the specimen or pulling it out of the holder.

9.2.4.1 Align the holder with the front edge of the clip mount shelf. The specimen fold should line up with the spot at the center of the recorder disk leaving the free hanging leg of the specimen aligned with the vertical guide line on the scale. Take special care to not touch, blow on or jam the specimen against the face of the recorder. Perform all operations as rapidly as possible. Repeat steps 9.2.4 and 9.2.4.1 for the second specimen, starting 60 ± 2 s after the weight is removed from the first specimen. Repeat all steps for the third specimen, 60 ± 2 s after the weight was removed from the second specimen.

9.2.4.2 To eliminate gravitational effects, keep the free hanging leg of the specimen aligned with the recorder's vertical guide line during the 5 min ± 5 s recovery period. Adjust every 15 ± 1 s for the first minute, and once a minute thereafter during the remaining recovery period. Make the final adjustment 15 ± 1 s before the 5 min ± 5 s recovery period ends. A procedure to run continuous specimen testing is given in 13.5.

9.2.5 Read and record the recovery angle from the scale 5 min ± 5 s after inserting a specimen into the clip mount on the recorder. If the free end of a specimen twists, sight a vertical plane through its center and align it with the vertical mark on the recorder scale. Read and record the recovery angle for each specimen after it has been inserted in the clip mount for 5 min ± 5 s.

9.2.6 Repeat all steps for three specimens in the opposite direction and for specimens folded face-to-back in both fabric directions.

10. Calculation

10.1 Calculate the average recovery in degrees for each group of three specimens; warp folded face-to-face; warp folded back-to-back; filling folded face-to-face; and filling folded back-to-back.

10.2 If the difference between face-to-face and back-to-back averages is not greater than 15°, average all warp readings and all filling readings separately. If the difference between face-to-face and back-to-back averages is greater than 15°, report the four averages separately.

11. Report

11.1 State that the tests were performed as directed in AATCC TM66, Option 1 or Option 2.

11.2 Report the average warp and filling (or when necessary warp-face, warp-back, filling-face and filling-back) recovery angles.

11.3 If other testing atmospheres are used, report the conditions under which the test was performed with the average values.

12. Precision and Bias

12.1 *Precision*.

12.1.1 In 2016, an inter-laboratory study was conducted comparing Option 1 to Option 2. Five laboratories participated in the study. Two of the participating laboratories performed Option 1 (Lab A: 3 technicians; Lab B: 3 technicians), two other laboratories performed Option 2 (Lab D: 1 technician; Lab E: 3 technicians), and the fifth laboratory (Lab C: 2 technicians) performed both Options 1 and 2. For the study, three different fabrics (100% Cotton Twill, Color Khaki; S/419 Cotton Broadcloth, Bleached/Merc; and S/7465 50/50 Poly/Cotton Print Cloth) were each treated with five finishes (Finish 1. No chemical; Finish 2. Softener only; Finish 3. 6% resin + softener; Finish 4. 12% resin + softener; and Finish 5. 18% resin + softener). Each test was performed using five face-to-face and five back-to-back determinations in warp and filling directions.

12.1.2 Results of t-test performed to determine the statistical significance of the mean values produced by Option 1 and Option 2 are shown in Table I. It can be seen that the mean and standard deviation values of Option 1 differ from the mean and standard deviation values of Option 2. A p-value of 0.000 for the two-sample t-test suggests that there is 0% probability for Option 1 results to be statistically similar to Option 2 results. The results, therefore are significantly different at 95% confidence level. Therefore, laboratories using Option 1 should not compare their results with that of laboratories using Option 2.

12.1.3 *Within-laboratory* and *between-laboratory* data for Option 1.

Within-laboratory and *between-laboratory* precision data for Option 1, with specimens ranging from 1 to 6 is given in Table II. The precision values capture the variability contributed by the participating laboratories and their respective operators.

12.1.4 *Within-laboratory* and *between-laboratory* data for Option 2.

Within-laboratory and *between-laboratory* precision data for Option 2, with specimens ranging from 1 to 6 is given in Table III. The precision values capture the variability contributed by the participating laboratories and their respective operators.

12.2 *Bias*.

12.2.1 Wrinkle recovery angle can be defined only in terms of a test method. There is no independent referee method for determining the true value of this property. This method has no known bias.

13. Notes

13.1 ISO 2313: Determination of the Recovery from Creasing of a Horizontally Folded Specimen by Measuring the Angle Recovery, may be obtained from ANSI, 11 West 42nd St., New York NY 10036; tel: +1.212.302.1286; fax: +1.212.398.0023; web site: www.ansi.org; or ISO web site: www.iso.org.

13.2 AATCC TM66 interlaboratory test results using the original crease recovery angle

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Table I—Two-sample t-test for Recovery Angle Versus Option

Option	*n*	Mean	StDev	SE Mean
1	1440	119.1	23.3	0.61
2	1080	209.3	35.7	1.1

Difference = mu (1) - mu (2)
Estimate for difference: 9.83
95% CI for difference: (7.39, 12.28)
t-test of difference = 0 (vs. not =); t-value = 7.89; p-value = 0.000; DF = 1743

Table II—Precision Table for Option 1

No. of Specimens	Within-Laboratory Precision	Between-Laboratory Precision
1	1.015	1.7565
2	0.7188	1.2440
3	0.5860	1.0141
4	0.5075	0.8783
5	0.4539	0.7855
6	0.4144	0.7171

Table III—Precision Table for Option 2

No. of Specimens	Within-Laboratory Precision	Between-Laboratory Precision
1	1.8511	3.1347
2	1.3110	2.2200
3	1.0687	1.8098
4	0.9256	1.5674
5	0.8278	1.4019
6	0.7557	1.2797

devices showed significant differences in *between-laboratory* precision, but good *within-laboratory* precision. The method, however, cannot be recommended for acceptance testing.

13.3 Most specimens will have a straight, free hanging leg. When the free hanging specimen leg is not straight, alignment problems may cause angle reading difficulty. If a specimen's free leg twists, align the center of the bottom edge of the specimen with the vertical guide line on the tester. If a specimen curls, align the section of the free hanging leg closest to the crease angle with the vertical guide line on the tester.

13.4 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

13.5 Practical Procedure for Option 2 Multiple Specimen Testing. This procedure requires six testers, six weights, 12 plastic presses, 18 specimen holders and one timer. The test run consists of running six specimens concurrently on cycles staggered at 7 min intervals.

13.5.1 After six specimens have been mounted in the press-holder combinations, the first cycle is started by placing weights on all six presses within 5 s. The weights are removed 5 min later in the same order, so that the folding time is the same for all specimens. The specimens are then placed in the clips on recorders as rapidly as possible. The specimens are to be adjusted and read in the same order so that the recovery time will be 5 min for each specimen. One minute after starting to remove the weights and inserting holders into recorders, place weights on the next set of holder press combinations. Start timing with the weight load application. This procedure requires a continuously running clock or timer.

13.5.1.1 Differential times among specimens occur between the folding and the recovery period when specimens are in the plastic presses without load. The differential was found not to have appreciable effect on test results, in that, the specimens held in holders without load are not free to recover.

13.5.2 Each set of six specimens follows the preceding one by 7 min, the creasing period being started (load application) at 0, 7, 14, etc. min. Continuous operation by this technique will yield 51 readings per h.

13.6 Referring to the inter-laboratory study performed in 2016, two additional laboratories participated by performing instrumental wrinkle recovery tests on the same fabrics and finishes that were provided to the other laboratories (see 12.1.1). However, since Options 1 and 2 were not statistically similar, no data analysis was performed to compare the instrumental methods to Option 1 or Option 2. If a laboratory wishes to use an instrumental option, the accuracy must be established by the user.

14. History

14.1 Editorially revised in 2022 to add History section.

14.2 Editorial revised 2019; 2017 revised, 2016 editorial revised, 2014 reaffirmed, 2008, 2006 editorially revised, 2003 reaffirmed, 1998 revised (with title change), 1996 reaffirmed and editorially revised, 1995 editorially revised, 1991 editorially revised, 1990 reaffirmed, 1986 editorial revised, 1984, 1978, 1975, 1972, 1968 reaffirmed, 1959, 1956, 1953, 1952 revised.

14.3 1951 Developed by AATCC Committee RR6; 1995 Jurisdiction transferred to AATCC Committee RA61.

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Test Method for Water Repellency: Tumble Jar Dynamic Absorption

1. Purpose and Scope

1.1 This test method is applicable to any textile fabric, which may or may not have been given a water-resistant or water-repellent finish. It measures the resistance of fabrics to wetting by water. It is particularly suitable for measuring the water-repellent efficacy of finishes applied to fabrics, because it subjects the treated fabrics to dynamic conditions similar to those often encountered during actual use. It is not intended for use in predicting the probable rain penetration resistance of fabrics, since it measures absorption of water into, but not through, the fabric.

1.2 The results obtained with this test method depend primarily on the resistance to wetting or water repellency of the fibers and yarns in the fabric, and not upon the construction of the fabric.

2. Principle

2.1 Preweighed specimens are tumbled in water for a fixed period of time and are reweighed after the excess water has been removed from them. The percentage increase in mass is taken as a measure of the absorption or resistance to internal wetting.

3. Terminology

3.1 **absorbency,** n.—the propensity of a material to take in and retain a liquid, usually water, in the pores and interstices of the material.

3.2 **water repellency,** n.—*in textiles*, the characteristic of a fiber, yarn or fabric to resist wetting.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Observe wringer safety especially at nip points. Normal safeguards should not be removed. A foot-operated kickoff is recommended.

4.3 The manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

5. Apparatus and Materials (see 11.1)

5.1 Dynamic Absorption Tester (see Fig. 1 and 11.2).

5.2 Wringer (motor-driven) (see 11.3).

5.3 Laboratory balance, accurate to 0.1 g.

5.4 White AATCC Textile Blotting Paper, 25 × 25 cm (see 11.4).

5.5 Water, distilled.

5.6 Plastic container, or gallon "zipper" type plastic bag.

6. Test Specimens

6.1 The fabric samples and the blotting paper should be conditioned at 65 ± 5% RH and 21 ± 2°C (70 ± 4°F) for a minimum of 4 h before testing.

6.2 Test two specimens from each sample. Each specimen consists of five 20 × 20 cm pieces cut on a 0.79 rad (45°) bias. Remove loose corner yarns and spread a drop of liquid latex or rubber cement on yarns at the corner to prevent raveling. Mark a corner of each piece of specimen for identification as a part of the specimen set.

7. Procedure

7.1 Rinse the tumble jar of the Dynamic Absorption Tester thoroughly to free it from all extraneous matter, particularly soaps, detergents and wetting agents.



Fig. 1—Dynamic absorption tester.

7.2 Roll together the five pieces of each specimen (constituting a specimen set) and weigh to the nearest 0.1 g.

7.3 Pour 2 L of distilled water at 27 ± 1°C (80 ± 2°F) in the tumble jar of the Dynamic Absorption Tester. Put both specimen sets (see 11.5) into the jar, one piece at a time, and rotate for 20 min.

7.4 Remove a piece of one specimen immediately passing it through the wringer at a rate of 2.5 cm/s with the edge of the piece parallel to the rolls and then sandwich the specimen between two pieces of unused blotter paper and pass it through the wringer again. Leave the piece sandwiched in the wet blotters. Repeat this process for the remaining four pieces of the same specimen set. Then remove the blotters and roll the five pieces together, put them in a tared plastic container or gallon-sized zippered plastic bag, close the container and weigh the wet specimen set to the nearest 0.1 g. The mass of the wet specimen should not be more than twice its dry mass.

7.5 Repeat 7.4 for the second specimen.

8. Calculations

8.1 Calculate the water absorbed for each specimen to the nearest 0.1% using the following equation:

$$WA = (W - C)/C \times 100$$

where:

WA = water absorbed, %
W = wet specimen weight, g
C = conditioned specimen weight, g

8.2 Calculate the dynamic absorption of the fabric sample by averaging together the water absorbed by each of the two specimens.

9. Report

9.1 Report the average of the two specimens as the percent of water absorbed by the fabric.

9.2 Report the test method number.

10. Precision and Bias

10.1 *Precision*. In 1998, a limited intralaboratory study was completed. All samples were tested by the same operator.

10.1.2 Four sets of fabrics were analyzed. There was no statistical difference found between the sets of data, thus precision parameters were calculated on the pooled data.

10.1.3 Users of the method are advised of the limited nature of this study and ad-

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vised to apply these findings with due caution.

10.1.4 Analysis of the combined data yielded a repeatability standard deviation of 1.8, and components of variance and critical differences as displayed in Table I. Differences between two averages of (*N*) determinations, for the appropriate precision parameter, should reach or exceed the table value to be statistically significant at the 95% confidence level.

Table I—Components of Variance and Critical Differences

Det in Avg (*N*)	95% Confidence Interval	Critical Differences
2	+ 16.2	32.4
3	+ 4.5	9.0
4	+ 2.9	5.8
5	+ 2.2	4.4

10.2 *Bias*. Dynamic absorption of fabrics can only be defined in terms of a test method. There is no independent, referee method for determining the true value. This test method has no known bias.

11. Notes

11.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

11.2 The Dynamic Absorption Tester (see Fig. 1) should consist of a motor driven, 6 L cylindrical or hexagonal-shaped jar approximately 15 cm in diameter and 30 cm in length, mounted to rotate end over end at 55 ± 2 rpm with a constant tangential velocity. The jar may be of glass, corrosion-resistant metal, or chemical stoneware.

11.2.1 The revolutions per min of the tumble jar should be determined by counting the number of revolutions in 1 min using a stopwatch calibrated in intervals of not more than 1 s.

11.3 The wringer, household laundry type, should be equipped with soft rubber squeeze rolls 5.1-6.4 cm in diameter and 28.0-30.5 cm in length, with a hardness of 70-80 when measured using the A Scale of a Shore Durometer. The wringer should be so constructed that the pressure on the top of the piece of fabric is maintained by a dead weight or lever system such that the total pressure (resulting from the total of the dead weight or lever system and the weight of the roller) is 27.2 ± 0.5 kg. It should be power-driven at a constant rate so that the piece of fabric passes through the rolls at a rate of 2.5 cm/s.

The diameters of the squeeze rolls should be determined with a pair of calipers or directly with a suitable micrometer. Measurements should be made at five different places along the length of each roll, and the average of these measurements taken as the diameter of the roll.

The load applied by the dead weight or lever system should be measured using a spring scale or balance, and suspending the top roll of the wringer from the scale by means of two tapes of equal length. The tapes should be placed between the rolls near their ends and should be provided with a means of holding them sufficiently far apart so that there is no contact between the tapes and the top structural member of the wringer and loading system. The spring scale or balance should be suspended from a suitable rigid support and provided with a turnbuckle or other device for adjusting the height of the scale. The usual precautions concerning the zero correction of spring scales should be observed. The turnbuckle or other device should then be adjusted to place the weight of the top roll and its weighing system on the spring scale or balance, and the system should be considered to be in equilibrium when the top roll of the wringer has been lifted just sufficiently from the bottom roll to permit vision between the bottoms of the tapes and the top of the bottom roll. At this point, the dead weight on the loading system should be adjusted until the spring scale or balance indicates a load of 27.2 ± 0.5 kg. The calibration of the spring scale or balance should be verified by the use of known certified dead weights of 24.95, 27.22 and 29.48 kg ± 0.23 kg total weight. The spring scale should be accurate to within ± 0.2268 kg at each of the three verification loads.

The linear speed of the rolls should be measured by feeding a thin steel tape through the rolls. The steel tape should be at least 150 cm long and should be accurate to within 3 mm per 150 cm. The time required for exactly 150 cm of this tape to pass through the nip of the rolls should be measured in s to the nearest s with a stopwatch calibrated in intervals of not more than 0.5 s. The speed of the rolls should be adjusted until the time required for 150 cm of tape to pass though the nip of the roll is 60 ± 2 s.

11.4 Blotters suitable for this test can be obtained from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549. 8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.5 Should it be necessary to run only one specimen, a specimen of similar material with respect to weight should be run as ballast with the specimen undergoing test. The cloth in the jar during any run should be the equivalent of two specimens (10 pieces).

12. History

12.1 Reaffirmed 2020.

12.2 Revised 2015, 2000, 1997, 1961. Reaffirmed 2014, 2010, 1989, 1988, 1983, 1978, 1975, 1972, 1967, 1964. Editorially revised and reaffirmed 2005, 1994. Editorially revised 2019, 2016, 2009, 2008, 1986, 1985.

12.3 Developed in 1952 by AATCC Committee RA63. Technically equivalent to ISO 18696.

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Test Method for Electrical Surface Resistivity of Fabrics

Developed in 1954 by AATCC Committee RA32; revised 1963, 1968, 1970, 1972, 1973, 1982, 1995, 2000 (with title change); reaffirmed 1969, 1975, 1978, 1989, 2011, 2018; editorially revised 1974, 1984, 1985, 1997, 2004, 2008, 2019; editorially revised and reaffirmed 1987, 2005.

1. Purpose and Scope

1.1 The purpose of this test method is to determine the electrical surface resistivity of fabrics. The surface electrical resistivity may influence the accumulation of electrostatic charge of a fabric (refer to AATCC TM84, Test Method for Electrical Resistance of Yarns).

2. Principle

2.1 Specimens at equilibrium with specified atmospheric conditions of relative humidity and temperature are measured for electrical resistance between parallel electrodes by means of an electrical resistance meter.

3. Terminology

3.1 **electrical resistivity,** n.—material property of a substance whose numerical value is equal to the ratio of the voltage gradient to the current density.

NOTE: For the purposes of this method, surface resistivity is calculated by using the measured electrical resistance between superficially positioned parallel plates or concentric rings and their spacing and is reported as ohms per square. This test method in effect measures the resistance of a material to the flow of current between two electrodes.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 The safety recommendations provided by the manufacturer of the radioactive bar should be followed.

4.3 The radioactive bar emits alpha radiation which is externally harmless to the human body. The radioactive isotope polonium 210 is toxic and precaution should be exercised to prevent ingestion or inhalation of the solid material. Do not take the radioactive bar apart or touch the radioactive strip under the grid. If the strip is touched or handled, wash hands thoroughly at once. Return the device to the manufacturer when it loses its effectiveness as a static eliminator or for disposal if use is to be discontinued. ***Do not discard as scrap.***

5. Apparatus and Materials

5.1 Electrical resistance meter (see 11.1).

5.2 Conditioning and test chamber (see 11.2).

5.3 Standard resistors.

5.4 Radioactive bar.

5.5 Two rectangular flat metal surfaces of suitable size to serve as electrodes.

5.5.1 As an alternative, two concentric ring electrodes of spacing suitable to the material being measured and the purpose of the results.

6. Test Specimens

6.1 Adjust the size of fabric test specimens to suit the electrodes of the particular equipment being used. When using parallel plate electrodes, the width of the specimen must not exceed the width of the electrodes. When using the concentric rings, any size specimen at least as large as the outer ring may be used. Avoid contaminating the area on which the measurement is to be made.

6.2 *Specimens for use with instruments providing parallel plate electrodes.* Prepare 2 sets of 3 test specimens each, 1 set each so that the direction of testing is parallel to the yarns in the length direction of the fabric and 1 set with the direction of testing parallel to the width direction of the fabric.

6.2.1 *Test specimens for use with instruments providing concentric ring electrodes.* Prepare 1 set (3 test specimens) since, with this type instrument, the resistance is measured in both fabric length and fabric width directions simultaneously.

6.2.2 Depending on the fabric construction or end-use, it may be advisable to differentiate between face and back measurements. Each test specimen should be taken from a different part of the fabric.

7. Procedure

7.1 Calibrate the electrical resistance meter according to the manufacturer's recommendations. This calibration should be repeated periodically.

7.2 Condition the test specimens in a suitable testing chamber or conditioning room at a predetermined relative humidity that will reflect the conditions at which information on the electrical resistivity of the fabric is needed.

7.2.1 For most fabrics which require antistatic treatments or whose static propensity is critical, measurements at 20% relative humidity will be most meaningful.

7.2.2 In less critical situations 40% relative humidity may be employed.

7.2.3 For special requirements other relative humidities may be used. For example, preconditioning at 50 ± 2% RH at 21 ± 2°C (70 ± 5°F) is required for antistatic sheeting, film and textiles for use in hospital operating rooms (see 11.2.1). Measurements may be made under other conditions or ranges of conditions that are applicable to the end use (such as 65% RH and 24°C [75°F]). It is preferable to keep the temperature or humidity consistent for all measurements.

7.2.4 If it is necessary to measure electrical resistivity under a broad range of conditions, additional tests may be run at 65% RH and at 24°C (75°F) or other settings applicable to the end use. It is preferable to keep the temperature or humidity consistent for all measurements (see 11.3).

7.3 Optionally remove static charges on the surface of the fabric by passing a radioactive bar over both sides of the fabric.

7.4 Place a test specimen in firm contact with the electrodes. The contact with the electrodes should be such that, when additional pressure is applied between the fabric and the electrodes, the test results are not affected.

7.4.1 In the case of equipment using parallel plate electrodes, place the test specimen in contact with the electrodes with the direction of the test perpendicular to the adjacent edges of the electrode. Measure the electrical resistance of the fabric in both the length and width directions. Since the charge will follow the path of least resistance, record only the lower reading by direction.

7.4.2 For concentric ring electrodes, the charge follows the path of least resistance automatically.

7.5 Measure the electrical resistance of the test specimen according to the operating instruction and procedures for the

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particular resistance meter being used. Allow the current to pass through the test specimen for 1 min or until a constant reading is obtained. The criterion for constant electrical resistivity is a change in the value of Log R of less than 0.1 units per minute. The time to reach a constant reading may vary with the applied voltage and with the resistance of the test sample. High voltages for prolonged periods of time may damage the fabric.

7.6 For referee purposes, 80-100 volts at 25 mm electrode separation for 1 min should be used for the parallel plate configuration, and a similar voltage gradient for the concentric ring case (see 11.4).

7.7 Avoid the use of any electrically conductive liquids on either the fabric test specimens or the apparatus.

8. Evaluation

8.1 Compute the resistivity to the nearest ohms per square as follows:

8.1.1 For the parallel electrode case:

$$R = O \times W/D$$

where:

R = resistivity in ohms per square
O = measured resistance in ohms
W = width of specimen
D = distance between electrodes

8.1.2 For the concentric ring case:

$$R = 2.73(O)/\log r_o/r_i$$

where:

R = resistivity in ohms per square
O = measured resistance in ohms
r_o = outer electrodes radius
r_i = inner electrodes radius

8.2 Calculate the average resistivity for each sample and for the lot.

8.3 Determine the logarithm (base 10) of the resistivity (Log R) for each sample and for the lot.

9. Report

9.1 Report the following information:

9.1.1 The Log R for each sample and for the lot.

9.1.2 the number of specimens tested.

9.1.3 The relative humidity and temperature used.

9.1.4 The direction of testing if parallel plate electrodes were used.

10. Precision and Bias

10.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages. Experience has shown that, with care, reproducibility should be ± ≤ 2% average Log R.

10.2 *Bias*. Electrical resistivity can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

11. Notes

11.1 The resistance meter in conjunction with the electrode system should be capable of measuring values in the range of 10^8-10^{15} for any critical uses. Equipment that is capable of measuring values in the range of 10^8-10^{13} ohms is suitable for work where the product performance is known to be acceptable if resistance is < 10^{13} ohms.

11.1.1 A description of an instrument that will not allow measurements > 10^{13} ohms is described by Hayek & Chromey in *American Dyestuff Reporter*, Vol. 40, 1951, pp164-168.

11.1.2 Other resistance testers also may be satisfactory for this test.

11.2 A conditioning and testing chamber is required capable of providing relative humidity control of ± 2% (preferably effective over a range of 20-65% RH) and temperature control of ± 1°C (± 2°F) with circulating air. Since equilibration of a test specimen from the dry versus the wet condition relative to the humidity of the test chamber may show hysteresis, it is recommended that specimens approach equilibrium from the dry side relative to the humidity in the test chamber whenever possible.

11.2.1 National Fire Protection Association, Standard NFPA Code #56 A-1973 Section 4663.

11.3 Accumulation of static electricity generally is greater the lower the relative humidity (and vice versa). Fabrics which show a low degree of static accumulation of 40% RH may exhibit severe static accumulation of 20-25% RH, while fabrics which exhibit static problems at 40% RH may exhibit a low degree of static accumulation of 65% RH. The relationship between the tendency for static accumulation and relative humidity varies with the specific antistatic agent, fiber, fabric construction, surface character, etc. Therefore, 40% RH may not provide significant information as to the probable antistatic performance properties of a fabric unless the tests are also run at 20-25% RH, an atmospheric condition not unusual in modern heated and air-conditioned buildings. Complete information may also require information of resistivity at the upper level of 65% RH.

11.4 For more detailed information regarding the measurement of resistances, see ASTM D257, Tests for D-C Resistance or Conductance of Insulating Materials (ASTM Committee D20).

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Test Method for Absorbency of Textiles

Foreword

The original test method for the measurement of absorbency of bleached textiles was developed to help textile dyeing and finishing mills determine the effect and efficiency of textile preparation and processing applications. Later, the test method was used as one of several measurements for judging water resistance and repellency (i.e., non-absorbency) of textile finish applications. After some time, the test method was also used to judge the durability of textile applications to home laundering.

Option B has been added to AATCC TM79 to recognize variations from the original procedure that have come into use by retailers and independent laboratories. Option B was first published as an unofficial method in the 2004 AATCC/ASTM International's *Concept 2 Consumer Technical Supplement: A Compilation of Procedures and Guidelines for Textile Products*, TS-018, and as MM-TS-01 Quick Water Absorbency procedure in the 2008 AATCC/ASTM International's *Moisture Management Technical Supplement: Applicable to Apparel, Linens and Soft Goods* (see 12.1).

Because the rate of water flow (drops per milliliter) and height above the specimen (mm) can be accurately and precisely determined for Option A, this option is more rigorously controlled. Therefore, in case of dispute, Option A is to be considered the reference procedure.

1. Purpose and Scope

1.1 The test method is for the determination of the water absorbency of yarns, fabrics and garments. It can be used on textiles of any fiber content or construction, including woven, knit and nonwoven.

2. Principle

2.1 A drop of water is allowed to fall from a fixed height onto the taut surface of a test specimen. The wetting time required for a drop of water to lose any light reflection and change to a dull, wet spot is recorded.

3. Terminology

3.1 **absorbency,** n.—the propensity of a material to take in and retain a liquid, usually water, in the pores and interstices of the material.

NOTE: Absorbency is sometimes referred to as wettability.

3.2 **wetting,** n.—the time required for a drop of water to lose its light reflection characteristic, i.e., when it changes to a dull, wet spot due to the absorbent propensity of a textile material.

3.3 **textile product,** n.—an article of fabric, or other flexible material, made to protect or decorate the body (garments), for use in the home (bedding, drapery, towels, table linens), or other uses (i.e., handkerchiefs, etc.).

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5. Uses and Limitations

5.1 Absorbency is one of several factors that influence textile processing such as fabric preparation, dyeing, and the application of finishes. Often interchanged with the term wettability, the absorbency characteristics of a fabric can influence the uniformity and completeness of bleaching and dyeing by the ability to take in water into the fiber, yarn, or fabric construction. The suitability of a fabric for a particular use, for example gauze or toweling, may also be dependent upon a fabric's ability and propensity to take in water.

5.2 Water absorbency may help judge or interpret "comfort." However, users of AATCC TM79 should be cautioned that its test results should not be the only criterion for the assessment of comfort (see the 2008 AATCC/ASTM International's *Moisture Management Technical Supplement: Applicable to Apparel, Linens and Soft Goods*, p19) (see 12.1).

5.3 Users of the AATCC TM79 should be cautioned about the interpretation of results that may be impacted by whether the face or the back of a textile fabric is tested. If the objective is to measure the absorbency of finishing, processing or durability to laundering, the face side of a sample would be exposed to water during the test. However, if the objective of performing the test is to evaluate the absorbency of a textile for further processing, then the side of the textile that would be worn next to the skin would be exposed to the water drops during testing. Further, if textiles are used in a product as a composite, the absorbency of the individual fabrics will be different than that of the end product.

5.4 If liquids other than distilled water are used to perform this test method, test results might not be comparable.

5.5 It is not known how test results from this test method would compare to other absorbency test methods.

5.6 No statement can be made about the comparison or correlation of the results between Options A and B in this method.

6. Apparatus (see 12.2)

6.1 Beaker, with opening of a size that allows for the support of the outer edge of the specified embroidery hoop.

6.2 Option A: Burette, 10 ± 0.05 mL with 0.5 mL graduations that allows a delivery rate of 15-25 drops per milliliter.

6.3 Option A: Burette stand.

6.4 Distilled or deionized water, 21 ± 2°C (70 ± 4°F).

6.5 Embroidery hoop, 152 ± 5 mm (6.0 ± 0.2 in.) in diameter (see 12.1).

6.6 Stopwatch or timer.

6.7 Option B: Medicine dropper, 76 mm glass, 2mL capacity, capable of dispensing 20 drops per milliliter.

7. Test Specimens

7.1 Take two 200 ± 5 × 200 ± 5 mm specimens from each sample on which a total of five water drop sites can be tested. If sufficient fabric is available, five specimens from different locations of a sample (i.e., different length × width locations; side-middle-side locations on a sample cut; or front-back-sleeve garment panels) can be prepared for each water drop test site. Otherwise, five water drop tests may be performed on one specimen if the test sites are at least 25 ± 2.5 mm (1 ± 0.12 in.) away from the hoop edge and no closer together than 25 ± 2.5 mm (1 ± 0.12 in.) from the previous water drop test site's outside edge. For performing the 5-drop testing on garments or full cuts of a sample, it may not be necessary to cut specimens from the actual sample but insert different areas in the embroidery hoop.

7.2 If performing tests on yarns, skeins

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should be prepared in such a way that there are no gaps between individual yarn wraps when inserted into the embroidery hoop.

7.3 Specimens should be brought to moisture equilibrium in a standard atmosphere according to ASTM D1776, Standard Practice for Conditioning and Testing Textiles) (see 12.3) having a relative humidity of 65 ± 5% at 21 ± 2°C (70 ± 4°F). If the absorbency of a fabric from a wet processing stage is required, such as after bleaching, specimens for testing should be air dried before conditioning prior to testing (see 12.4).

8. Procedure

8.1 Perform all tests in the standard atmosphere for testing.

8.2 Option A – Burette.

8.2.1 Select a location in the conditioned laboratory area that has overhead lighting to facilitate the judgment of the test end point; i.e., wetting (see 3.2).

8.2.2 Determine the burette's stopcock position that will deliver the specified number of water drops (see 6.2).

8.2.3 Mount a specimen in an embroidery hoop so that the side of the specimen to be tested is up and the surface specimen is taut and free of wrinkles but without stretching or distorting the structure of the fabric.

8.2.4 Place the embroidery hoop with the specimen surface 10 ± 1.0 mm (0.394 ± 0.04 in.) below the tip of the burette and allow one drop of distilled or deionized water to fall on the cloth. Start the stopwatch or timer immediately. Observe the water drop without moving the beaker with the specimen from under the burette to avoid disturbance of the water drop and its interface with the specimen surface.

8.2.5 Stop the timer or stopwatch when the drop of water loses its reflectivity (see 3.2). If the water drop does not immediately disappear, observe the water drop from other positions until it finally vanishes. The end point will be the time, less than 60 s, it takes for the water drop to no longer reflect light and appear only as a dull wet spot.

8.2.6 Record the elapsed time to the nearest second. If the water drop disappears immediately, record as "zero." If the wetting time exceeds 60 s, record the time as "60 + s."

8.2.7 Repeat steps 8.2.4-8.2.6 for the additional four test locations.

8.3 Option B – Medicine Dropper**.**

8.3.1 Follow steps 8.2.3-8.2.6 except using the specified medicine dropper that is held at an approximate 10 mm height from the test specimen surface is used. A 10 ± 1.0 mm height sighting device can be made to help assure consistent height placement of the medicine dropper. Do not move the specimen in the embroidery hoop until the water drop no longer reflects light and the site appears only as a dull wet spot. Record the time it takes for a water drop to disappear as stated in 8.2.6. Repeat the steps for all specimens.

9. Calculation and Interpretation

9.1 For Options A and B.

9.1.1 Calculate the average and the standard deviation of the five recorded times and report to the nearest second. In the case where the five tests include immediate absorption times ("zeroes"), the average should be calculated using all times including zero(s). In the case where some of the five tests include absorption times greater than 60 s (60 +), use all of the wetting times to calculate the average and report with a plus sign (+) and note the number of tests that exceeded 60 s.

9.1.2 Shorter times indicate better absorbency.

10. Report

10.1 Report the average absorbency time and standard deviation, if desired.

11. Precision and Bias

11.1 *Precision*.

11.1.1 *Interlaboratory Study*. Tests for absorbency of textiles using Options A and B were conducted in 2009, with one laboratory, three operators, and six samples. The samples were (a) bleached cotton nonwoven, (b) 100% cotton jersey knit, (c) 100% cotton interlock knit, (d) 100% cotton twill weave, (e) cotton/polyester blend woven, and (f) 100% polyester woven.

11.1.2 The analysis of variance technique was applied to the data set. The analysis of variance indicated that there was no significant variation due to the fabric, operator or day. Therefore, all data were combined to calculate the precision of the test method. All data and the analysis are being retained for reference at the AATCC Technical Center.

11.1.3 The data for Option A is shown below in Table I. The average for all 180 points was 11 s, the standard deviation was 16 s, and 95% confidence interval was ± 2 s.

11.1.4 The data for Option B is shown below in Table II. The average for all 180 points was 9 s, the standard deviation was 14 s, and 95% confidence interval was ± 2 s.

11.1.5 *Between-laboratory* precision has not been established for this test method. Until such precision information is available, users of the method should use standard statistical techniques in making any comparison of test results for *between-laboratory* averages.

11.2 *Bias*. The true value of absorbency of textile fabrics can be defined only in terms of a test method. There is no independent method for determining the true value. In estimating this property, the test method has no known bias.

12. Notes

12.1 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

12.2 These items may be obtained from any lab equipment supplier.

12.3 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832.9500; fax: +1.610.832.9555; web site: www.astm.org.

12.4 It has been observed that if specimens collected at drying cans are not brought to moisture equilibrium, the wetting times could falsely indicate poor absorbency for that fabric.

13. History

13.1 Editorially revised in 2023 to move history section.

13.2 Editorially revised 2022, 2019, reaffirmed 2018; editorially revised 2016, reaffirmed 2014; editorially revised 2012; revised 2010; revised 2007 (with title change); reaffirmed 2000; revised 1995; reaffirmed 1992; editorially revised and reaffirmed (with title change) 1986; reaffirmed 1979, 1975, 1972, 1968.

13.3 Developed in 1954 by AATCC Committee RA34; jurisdiction transferred in 2003 to AATCC Committee RA63; jurisdiction transferred in 2022 to AATCC Committee RA114.

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Table I—Option A—Data (seconds)

	Operator1 Day 1	Operator 1 Day 2	Operator 2 Day 1	Operator 2 Day 2	Operator 3 Day 1	Operator 3 Day 2
Nonwoven	2	4	1	1	1	2
	4	2	1	1	2	3
	4	3	1	1	1	2
	2	4	2	1	0	2
	2	4	1	1	0	1
Jersey	1	3	2	1	0	1
	1	3	2	1	0	1
	1	3	2	1	0	1
	1	2	4	1	1	2
	1	2	2	3	1	1
Interlock	9	17	9	16	12	25
	13	22	9	23	12	9
	15	28	4	60	11	9
	8	50	14	44	12	6
	8	20	14	35	11	6
Woven C	0	1	1	1	0	0
	0	1	1	1	0	0
	1	1	1	1	0	0
	1	1	1	1	0	0
	1	1	2	1	0	0
Woven Blend	2	2	3	1	0	1
	1	1	1	1	0	1
	2	1	2	1	0	0
	2	2	2	1	0	1
	1	1	2	1	0	0
Woven P	39	47	42	39	38	37
	41	48	45	34	36	36
	46	50	40	43	34	38
	42	48	46	43	35	34
	42	49	50	41	31	35

Table II—Option B—Data (seconds)

	Operator 1 Day 1	Operator 1 Day 2	Operator 2 Day 1	Operator 2 Day 2	Operator 3 Day 1	Operator 3 Day 2
Nonwoven	2	2	1	2	4	2
	2	3	1	2	0	2
	2	2	2	1	2	1
	2	3	2	2	1	2
	2	2	2	1	0	1
Jersey	2	2	1	1	1	1
	1	2	1	2	0	1
	1	2	1	2	1	1
	2	2	1	2	0	2
	2	3	1	1	0	2
Interlock	6	11	14	26	15	12
	5	16	60	21	12	8
	5	9	19	44	11	7
	8	10	5	29	14	5
	6	15	23	7	15	4
Woven C	1	1	1	1	0	0
	1	0	1	0	0	0
	0	1	0	1	0	0
	1	0	1	1	0	0
	1	0	1	1	0	0
Woven Blend	2	1	1	1	0	1
	2	2	1	1	0	1
	1	2	1	1	0	0
	1	1	1	1	0	1
	1	1	2	1	0	1
Woven P	34	33	30	24	30	37
	36	34	37	36	34	32
	33	42	33	28	33	34
	38	39	40	32	36	35
	39	47	36	42	38	38

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Test Method for pH of the Water-Extract from Wet Processed Textiles

1. Purpose and Scope

1.1 This test method determines the pH of wet processed textiles.

1.2 To make a quantitative determination, the chemicals which influence pH must be removed from the textile specimen, collected as a water extract and then accurately measured by a pH meter.

2. Principle

2.1 The specimen is boiled in distilled or deionized water. The water-extract is cooled to room temperature and the pH is determined.

3. Terminology

3.1 **bleaching,** n.—elimination of unwanted coloring matter from a textile substrate by oxidative or reductive chemical treatment.

3.2 **pH,** n.—the negative logarithm of the effective hydrogen ion concentration or hydrogen ion activity in gram equivalents per liter used in expressing both acidity and alkalinity on a scale whose values run from 0 to 14 with 7 representing neutrality, numbers less than 7 increasing acidity and numbers greater than 7 increasing alkalinity.

3.3 **wet processing,** n.—*in textile manufacturing*, a collective term for processes included in preparation, dyeing, printing and finishing in which the textile material is treated with a liquid, normally water, or with chemicals in solution or dispersion in a liquid.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

5. Limitations

5.1 pH can be used to determine the suitability of wet processed textiles for subsequent dyeing and/or finishing operations or to evaluate the washing and/or neutralizing efficiency following any wet processing operation.

5.2 This method should be used in conjunction with AATCC TM144, Test Method for Alkali in Wet Processed Textiles: Total, in order to quantitatively determine the amount of an alkali present. While pH gives an indication of relative alkali or acid content, the exact amount can be masked by the presence of strong buffering agents.

6. Apparatus and Materials

6.1 pH meter with 0.1 unit graduations.

6.2 Beakers, glass, 400 mL.

6.3 Buffer solutions, pH 4.0, 7.0, 10.0 or others as needed.

7. Calibration

7.1 Calibrate the pH meter in accordance with the manufacturer's instructions. Select buffer solutions for calibration which are in the estimated range of the specimens' pHs.

8. Specimens

8.1 Use a 10 ± 0.1 g specimen of the material to be tested. If the specimen is difficult to wet out, then it should be cut into small pieces.

9. Procedure

9.1 Boil 250 mL of distilled water at a moderate rate for 10 min. Immerse the specimen, cover the beaker with a watch glass and boil for an additional 10 min.

9.2 Allow the covered beaker and contents to cool to room temperature. Remove the specimen with tweezers, allowing the excess liquid to drip back into the extract.

9.3 Determine the pH of the extract using a pH meter operated according to manufacturer's instructions.

10. Evaluation

10.1 The pH of the water-extract depends on the chemical treatment previously given the textile, the pH of the wash water, and the efficiency of the washing operation.

10.2 Normally, the pH of the water-extract will be higher after caustic boiling than after bleaching. If the textile is scoured after bleaching, the pH may be lower.

10.3 Textiles with high pH values may exhibit yellowing tendencies, create shade changes, alter the exhaustion and fixation of dyes, and produce a decrease in the cure of resin finishes or exhaustion of softeners.

11. Report

11.1 Describe or identify the sample.

11.2 Report that sample was tested using AATCC TM81-2022.

11.3 Report the pH of the water extract.

12. Precision and Bias

12.1 *Precision*.

12.1.1 In late 1993, an interlaboratory study was completed, which included five laboratories, two operators in each, running three determinations per fabric, on four fabrics. No prior assessment was made of the relative level of performance of the participating laboratories.

12.1.2 Analysis of the data set (5×2×3×4=120 values) yielded components of variance as follows:

Laboratories	0.1203
Operators within laboratories	0.0150
Specimens within materials, laboratories, and operators	0.0188

12.1.3 Table I indicates the critical differences calculated using the values in 11.1.2.

Table I—Critical Differences for Two Averages—95% Probability Level

pH of Water-Extract from Bleached Textiles

N	Single Operator	Within Laboratory	Between Laboratory
1	0.38	0.51	1.09
2	0.27	0.43	1.05
4	0.19	0.39	1.04
8	0.13	0.37	1.03

12.1.4 Differences between two averages of *N* determinations, for the appropriate precision parameter, should reach or exceed the table value to be statistically significant at the 95% probability level.

12.2 *Bias*.

12.2.1 To the extent the pH meter used in this test method is capable of measuring values of pH that are consistent with true values of pH, the determination of the pH of the water-extract from a bleached textile, using this test method, has no known bias. During this study, no determination was made of the true value of this property by an independent, referee analytical method for the purpose of establishing presence or absence of bias.

13. History

13.1 Revised in 2022 to align to the AATCC Style Guide.

13.2 Editorially revised 2019. Reaffirmed 2016, 2012. Editorially revised 2008. Reaffirmed and editorially revised 2006. Reaffirmed 2001. Revised (with title change) 1996. Revised 1990. Reaffirmed 1989, 1988, 1983, 1980, 1977, 1974, 1969, 1968, Revised 1963.

13.3 Developed in 1954 by AATCC Committee RA34.

13.4 Related to ISO 3017.

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Test Method for Fluidity of Dispersions of Cellulose from Bleached Cotton Cloth

1. Purpose and Scope

1.1 The test procedures in this test method are applicable to bleached, unfinished cotton cloth. The fluidity of a dispersion of bleached cellulose fibers in cupriethylene diamine solvent is a sensitive measure of the degradation of the cellulose resulting from the action of acids, alkalis, oxidizing or reducing agents. It is therefore useful in determining the completeness of the bleaching treatment to which the cloth has been subjected, as well as the effect of chemical treatments on the tensile strength of the cloth. (Fluidity of dispersions in cuprammonium hydroxide has been dropped from this test method; see 11.1 for details.)

2. Principle

2.1 The length of chain of cellulose molecules is decreased when treated with acid, alkali, oxidizing or reducing agents, etc. The severity of the treatment determines the extent of the shortening of the cellulose chain and the fluidity, or ease of flow, of a dispersion of the cellulose in the solvents.

3. Terminology

3.1 **fluidity,** n.—o*f a cellulose solution*, a measure of the ease of flow or motion of a solution, and therefore, an indicator of the molecular weight of the cellulose.

3.2 **rhe,** n.—the unit of fluidity; the reciprocal of the unit of viscosity (the poise).

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturers recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed, Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care. Use chemical goggles or face shield, impervious gloves and an impervious apron during dispensing and mixing. CAUTION: Always add acid to water.

4.3 An eyewash/safety shower should be located nearby and a self-contained breathing apparatus should be readily available for emergency use.

4.4 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 11.2).

5. Materials and Reagents

5.1 Cupriethylene diamine, 1.0 M solution.

5.2 Nitrogen (N_2)

5.3 Ostwald Cannon-Fiske viscometers (see Fig. 1 and 11.3).

6. Preparation of Sample

6.1 Use a representative sample of bleached and unfinished cloth weighing not less than 5g.

6.2 Cut the sample by use of a cutting mill with a 20-mesh sieve and top delivery tube, so that short individual fibers about 1.5 mm (0.06 in.) long are obtained.

6.3 Run duplicate tests, using fibers from the one sample. Divide the cut sample into two portions of approximately equal weight and store in separate containers overnight in a desiccator containing conc. sulfuric acid. Samples conditioned in this manner will have a moisture content of 2%.

6.4 The purpose of dividing the sample into two portions is to reduce the time of exposure to the atmosphere during weighing. The weight of the specimen for the fluidity determination should be sufficient to give a 0.5% solution of the cotton.

7. Preparation of Solutions

7.1 Prepare 0.167 *M* cupriethylene diamine solution by diluting 161 mL of the 1.0 *M* solution with freshly boiled and cooled water to 1.0 L in a volumetric flask, observing all precautions to exclude air.

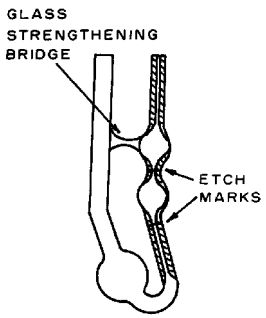


Fig. 1—Ostwald Cannon-Fenske cuen viscosimeter.

7.2 Store the reagents at all times under oxygen-free nitrogen to minimize decomposition.

7.3 Measure the 1.0 *M* and 0.161 *M* solutions from burettes of known accuracy. The burettes may be attached to the reagent storage bottles by means of a side arm sealed into the lower part of the burette and filled by application of nitrogen pressure on the storage bottle,

8. Procedure

8.1 Ostwald Cannon-Fenske-type viscometers (see Fig. 1) having the characteristics shown in Table I are suitable for determining fluidities in the range normally found for bleached cotton. It is recommended that the efflux time (see 8.9) exceed 100 s in order to ensure accuracy.

8.2 Weigh 0.4-0.45 g of the sample, prepared as previously described to the nearest 0.1 mg. Remove the samples from the desiccator one at a time, and make the weighings as quickly as possible to prevent additional moisture pickup from the air.

8.3 Transfer the fibers carefully from the weighing dish or watch glass to an approximately 120 mL (4 oz.) widemouth amber glass bottle.

8.3.1 To the specimen in the solution bottle, add *X* mL of 0.167 *M* cupriethylene diamine solution, where *X* is calculated as follows:

$$X = 120 \times \text{weight of specimen} \times 0.98$$

8.4 Sweep out the air in the bottle above the solution by applying a brisk stream of purified nitrogen over the sur-

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Table I—Viscosimeter Characteristics

Size No.	Approx. Range of Efflux Time, s	Capillary Diameter, cm	Fluidity Range Rhes
200	100 to 700	0.097 to 0.103	1.43 to 10.0
300	100 to 700	0.120 to 0.130	0.80 to 4.0
400	100 to 700	0.180 to 0.190	0.1175 to 0.83

face of the solution for 20 s, and quickly screw the cap of the bottle in place.

8.5 Shake the bottle by means of an automatic shaker at intermediate speed for 2 h.

8.6 Remove the bottle from the shaker and add to the solution *Y* mL of 1.0 M cuen, where *Y* is calculated as follows:

$$Y = 80 \times \text{weight of specimen} \times 0.98$$

8.7 Repeat the nitrogen treatment and bottle-shaking as described, with the exception that the shaking time should be 3 h.

8.8 Permit the bottle to stand for 30 min after removing from the shaker and before loading the viscometer. The viscometer should be flushed with nitrogen gas prior to being loaded with the dispersion. The viscometer is loaded by inverting it with the capillary side arm submerged in the cellulose dispersion and applying suction to the other arm of the instrument. Draw the dispersion into both the bulbs on the capillary arm to the etched mark. Rotate the pipette to its normal vertical position, and place the viscometer in a constant-temperature water bath at 25 ± 1°C (77 ± 2°F). The liquid will drain into the lower reservoir and will reach temperature equilibrium with the bath (approx. 5 min).

8.9 Determine the efflux time by drawing the liquid above the mark between the two bulbs and measuring the time for the meniscus to pass from the mark between the two bulbs to the mark below the lower bulb. Take the average of two or more observations.

9. Calculations

9.1 The fluidity in rhes, *F*, is calculated as follows:

$$F = 100/ctd$$

where:

c = instrument constant determined from a liquid of known viscosity.

t = time of discharge of the liquid within the ring interval in seconds.

d = density of cuen solution, 1.052.

10. Report

10.1 Describe or identify the bleached and unfinished cloth sample.

10.2 Report that sample was tested using AATCC TM 82 - 2022

10.3 Report the average efflux time.

11. Precision and Bias

11.1 Precision Estimate.

11.1.1 *Single Laboratory Study.* Estimates of Components of Variance for technicians (VO) and residual error (V), and their associated Critical Differences, CDm (multiple technicians in lab) and CDo (single technician in lab) were determined in a single laboratory in 2005. Three technicians performed duplicate tests on materials, according to 6.3 at three treatment levels of no bleach (NB), standard bleach (SB), and over bleached (OB), using the Cuen Method.

11.1.2 *95% Precision Estimates from Individual Technician and Treatment Values*. The results shown in Table II are the average of two determinations by a single technician. Based on these averages, the components of variance (Vo) and residual error (V) for each technician were computed. As can be seen even in this limited study, there was an apparent difference in the results reported by Technician 3, for each of the treatments. This is a very limited study, and caution is advised in the application of these estimates to specific testing scenarios (see Table II).

With three technicians involved in this limited study, it was possible to assess the effect of multiple technicians in test result comparisons by looking across technicians in treatments and calculating VO (technician) and V (residual) components of variance for each treatment. The above summary shows rather large values for CDm (multiple O in lab), due to the level differences among technicians. This is a very limited study, and caution is advised in application of these estimates to specific testing scenarios. All of the above considerations were based on a 95% confidence level.

11.1.3 95% *Precision Estimates from Analysis of the Whole Data Set*. Using Analysis of Variance, Components of Variance for technicians and residual were computed and are shown in Table III. As a result of the difference in results reported by Technician 3, the values generated for CDm (multiple operators in a single lab) were also high. These estimates pertain only to *within laboratory* precision. A much broader study, with multiple laboratory participation, must be made to yield reliable *between laboratory*

Table II—Data Summary for Technicians and Treatments

		TMT NB	TMT SB	TMT OB	Technician Summaries
Tech 01	Av(2)	0.564	0.811	4.033	V 0.0006
	V	0.0013	0.0004	0.0002	CDo 0.05
	CDso	0.07	0.04	0.03	
Tech 02	Av(2)	0.609	0.885	4.172	V 0.00014
	V	0.0003	0.0001	0.00004	CDo 0.02
	CDso	0.03	0.02	0.01	
Tech 03	Av(2)	0.397	0.745	3.625	V 0.0107
	V	0.0022	0.0238	0.0060	CDo 0.20
	CDso	0.09	0.31	0.15	
		Treatment Summaries			AV-Rhes Units DC-Rhes Units V-Rhes squared
		TMT1	TMT2	TMT3	
3 Tech (O=3)	VO	0.0118	0	0.0795	
	V	0.0012	0.0088	0.0021	
Single O	CDo	0.07	0.19	0.09	
Multiple O	CDm	0.31	0.19	0.79	

CD Equations: CDo=2.8 (V/2)^0.5 CDm=2.8 (V/2 + VO)^0.5

Table III—Components of Variance and 95% Precision Estimates

Components of Variance			
Source	**COV**	**Unit Var**	
O.L	VO	0.0307	
Res	V	0.0038	
95% Precision Estimates			
Single O in Lab	CDo	0.12	CDso=2.8 (V/2)^0.5
Multiple O in Lab	CDm	0.51	CDw=2.8 (V/2) + VO)^0.5

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estimates (see Table III).

11.2 *Bias*. The true value of fluidity of dispersions of cellulose from bleached cotton cloth can be defined only in terms of a test method. There is no independent method for determining the true value. In estimating this property, the test method has no known bias.

12. Notes

12.1 The cupriethylene diamine cuen method is predominant in the United States as well as some other parts of the world. The cuprammonium hydroxide cuam method has been predominantly used in Great Britain. AATCC Committee RA34 voted to drop the cuam method in 2005 and to refer to British Standard 2610:1978 "*Method of test for the determination of the cuprammonium fluidity of cotton and certain ce11uosic man-made fibres*." The Bleachers Handbook is also a good reference for a discussion of the cuam method (see 11.4).

12.2 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

12.3 Uncalibrated viscometers can be calibrated by means of standard oils or other reference liquids according to ASTM D445, Method 6B.

12.4 The examples of fluidities shown in the table in Fig. 2 are to be used as a guideline, Other tests for damage may be necessary. Consult "*Analytical Methods for a Textile Laboratory*," third edition, 1984, edited by J. William Weaver, published by AATCC, for further tests for damage. Although the source of the data and the graph in Fig. 2 are not known, it was maintained as a convenient reference for comparison of cuen to cuam data.

13.1 History

13.1 Revised in 2022 to align with the AATCC Style Guide.

13.2 Editorially revised 2019; reaffirmed 2016, 2012; editorially revised, 2010, 2008, revised 2007; editorially revised 2004; editorially revised and reaffirmed 2001;revised 1996; editorially revised, 1990; editorially revised and reaffirmed 1989; editorially revised 1988, 1985; reaffirmed 1984; editorially revised 1983; reaffirmed, 1979, 1975; editorially revised 1975, 1974; reaffirmed, 1972, 1968, 1961.

13.3 Developed in 1954 by AATCC RA34.

CLOTH	FLUIDITY– RHES CUEN	CUAM
GREIGE	0.2 - 0.4	1.0 - 2.0
MILDLY TREATED	0.3 - 0.7	1.5 - 2.5
AVERAGED BLEACHED	0.9 - 3.3	3.0 - 7.0
OVERBLEACHED	5.5 - 15.3	10.0 - 20.0

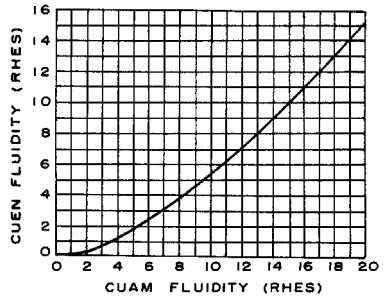


Fig. 2—Evaluation Cuam and Cuen fluidities

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Test Method for Electrical Resistance of Yarns

Developed in 1955 by AATCC Committee RA32; reaffirmed 1960, 1969, 1973, 1977, 1989, 2005, 2011, 2018; editorially revised 1974, 1984, 1985, 1997, 2008, 2019; revised 1982, 2000 (with title change); editorially revised and reaffirmed 1987, 1995.

1. Purpose and Scope

1.1 The purpose of this method is to determine the electrical resistance of any textile yarns containing natural or manmade fibers. The tendency of a textile yarn to accumulate electrical charges may depend upon the electrical resistance of the yarn. Due to the mechanism of conductance, this method is not applicable to yarns containing randomly situated stainless steel or other highly conductive fibers (refer to AATCC TM76, Test Method for Electrical Surface Resistivity of Fabrics).

2. Principle

2.1 Specimens of yarn are conditioned at specified relative humidity and temperature. The electrical resistance of the yarn held between two electrodes is measured by the use of an electrical resistance meter.

3. Terminology

3.1 **electrical resistance,** n.—the physical property of a material which is a measure of the ability of electrons to flow through it when a voltage is applied across two points on the material (resistance [in ohms] equal voltage [in volts] divided by current flow [in amperes]).

NOTE: This test method measures the electrical resistance of a material between two electrodes, and the results are expressed as resistance per unit of length per yarn strand, ohms per 10 mm.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standard and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 The safety recommendations provided by the manufacturer of the radioactive bar should be followed.

4.3 The radioactive bar emits alpha radiation which is externally harmless to the human body. The radioactive isotope polonium 210 is toxic and precaution should be exercised to prevent ingestion or inhalation of the solid material. Do not take the radioactive bar apart or touch the radioactive strip under the grid. If the strip is touched or handled, wash hands thoroughly at once. Return the device to the manufacturer when it loses its effectiveness as a static eliminator or for disposal if use is to be discontinued. ***Do not discard as scrap***.

5. Apparatus and Materials

5.1 Electrical resistance meter equipped with either fixed position parallel plate electrodes or with separate variable position parallel plate electrodes (see 11.1). The concentric ring electrode system recommended for fabrics in AATCC TM76 is not suitable for use in testing yarns.

5.2 Conditioning and test chamber (see 11.2).

5.3 Standard resistors.

5.4 Radioactive bar.

6. Test Specimens

6.1 The length of the test specimens depends on whether or not the electrodes used are in fixed or variable position. If the electrode system used provides for variable distances between the parallel plates, run a preliminary test to determine what distance between the plates provides the greatest sensitivity for resistance measurements for the meter being used.

6.2 To determine the uniformity of resistance along a single end of yarn, make measurements on at least 10 specimens of single strands.

6.3 For predicting the properties of woven or knitted fabrics to be made from yarn tested by this method, measurements should be made on multiple strands.

6.4 Prepare a minimum of three test specimens for each test, with parallel strands having the same tension, evenly spaced, with no overlap or touching along their length. When a limited supply of yarn is available, each specimen should contain 10 strands. If an adequate supply of yarn is available, prepare larger specimens by winding 50-100 turns on a skein winder, then fastening the strands with adhesive tape to prepare lengths suitable for the electrode system used. The number of strands in the specimen must be the same for each test specimen for reproducible results. The resistance may also be dependent on the cross-section of the yarn bundle; thus, similarly manufactured fibers may yield yarns with different resistance, due to different numbers of filaments in the yarn bundle or different numbers of plies.

7. Procedure

7.1 Calibrate the electrical resistance meter according to the manufacturer's recommendations.

7.2 Condition the test specimens in a suitable testing chamber or conditioning room at the pre-selected relative humidity and temperature at which the electrical resistance of the yarn is to be measured.

7.2.1 For yarns which require antistatic treatments or whose electrostatic propensity is critical, measurements at 20% RH will be most meaningful. For special requirements other relative humidities may be used. For example, preconditioning at 21 ± 2°C (70 ± 5°F) is required for antistatic sheeting, film and textiles for use in hospital operating rooms (see 11.2.1). If it is necessary to measure electrical resistance under a broad range of conditions, additional tests may be run at 65% RH and at 24°C (75°F) or other relevant conditions (see 11.3).

7.2.2 Condition the specimens at 24°C and at the predetermined relative humidity for a minimum of 4 h or until equilibrium is attained. A sufficient indication of equilibrium is no significant change of resistance on further conditioning. A significant change may be considered ± 5% of the logarithm of the resistance (Log *R*).

7.3 Remove static charges from the surface of the yarn by passing a radioactive bar over both sides of the specimen.

7.4 Place the yarn test specimen in firm contact with the electrodes with the direction of the yarn perpendicular to the adjacent edges of the electrodes. Apply sufficient tension to hold the strands straight.

7.5 Measure the electrical resistance of the specimen according to the operating instructions and procedures for the particular resistance meter being used.

7.6 Allow the current to pass through the sample for a minimum period of 1 min and until a constant reading is obtained. The time to reach a constant read-

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ing may vary with the applied voltage and with the resistance of the test specimen. High voltages for prolonged periods of time may damage the yarn.

7.7 For referee purposes, use 30-40 volts for a minimum period of 1 min or until a constant reading is obtained at an electrode separation of 10 mm (see 11.4).

7.8 Avoid the use of any electrically conductive liquids on either the yarn specimens or the apparatus.

8. Evaluation

8.1 Compute the resistance in ohms per 10 mm per strand of yarn.

$$R = \frac{S}{D} \times \frac{r_1 + r_2 + r_3 + \ldots + r_n}{n} \times 10$$

where:

R = resistance in ohms per mm per strand
S = number of strands per specimen
D = distance between electrodes in mm
r = resistance of individual specimens containing S strands per specimen
n = total number of specimens tested (multiplication times 10 converts to the same scale as previously used for this test method)

9. Report

9.1 Report the result as the logarithm of the resistance (Log R) in ohms per 10 mm per strand of yarn and the temperature and relative humidity used (see 10.1).

10. Precision and Bias

10.1 *Precision*. Experience has shown that the reproducibility is within ± 10% of the logarithm of the average resistance.

10.2 *Bias*. Electrical resistivity can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

11. Notes

11.1 The resistance meter in conjunction with the electrode system should be capable of measuring values in the range of 10^8-10^{15} ohms for any critical uses. Equipment that is capable of measuring values in the range of 10^8-10^{13} ohms is suitable for work where performance is known to be acceptable when the resistance is < 10^{13} ohms.

11.1.1 External fixed-position parallel plate electrodes may be used with this instrument and may be built according to specifications described by Hayek & Chromey in *American Dyestuff Reporter*, Vol. 40, 1951, pp164-168. For more versatility these electrodes may be designed so that the distance between them may be adjusted from 5-50 mm.

11.1.2 Other resistance testers may be satisfactory for this test.

11.2 A conditioning and testing chamber is required capable of providing relative humidity control of ± 2% (preferably effective over a range of 20-65% RH) and temperature control of ± 1°C with circulating air. Since equilibration of a test specimen from the dry versus the wet condition relative to the humidity of the test chamber may show hysteresis, it is recommended that specimens approach equilibrium from the dry side relative to the humidity in the test chamber whenever possible.

11.2.1 National Fire Protection Association, Standard NFPA Code #56 A-1973 Section 4663.

11.3 Accumulation of static electricity generally is greater the lower the relative humidity (and vice versa). Yarns which show a low degree of static accumulation at 40% RH may exhibit severe static accumulation at 20-25% RH, while yarns which exhibit static problems at 40% RH may exhibit a low degree of static accumulation at 65% RH. The relationship between the tendency for static accumulation and relative humidity varies with the specific antistatic agent, fiber, yarn construction, surface character, etc. Therefore, 40% RH may not provide significant performance properties of a yarn unless the tests are also run at 20 or 25% RH, an atmospheric condition not unusual in modern heated and air-conditioned buildings. Complete information may also require information on resistance at the upper level of 65% RH.

11.4 For more detailed information regarding the measurement of resistances, see ASTM D257, Tests for D-C Resistance or Conductance of Insulating Materials.

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Test Method for Drycleaning: Durability of Applied Designs and Finishes

Developed in 1957 by AATCC Committee RA43; revised 1963, 1968, 1970, 1973; reaffirmed 1976, 1979, 1989, 2000, 2005, 2016; editorially revised 1969, 1983, 1986, 1993, 1995, 2004, 2008, 2010, 2019; editorially revised and reaffirmed 1985, 1994, 2011.

1. Purpose and Scope

1.1 This test method indicates the effect of repeated (see 10.1) drycleanings on the durability of applied-design or finish on textiles and other materials. It is also applicable for evaluating the durability of applied design materials and finishing agents manufactured for use on fabrics and other products intended for apparel and household use which are renovated in consumer service by commercial drycleaning procedures. This test method is to be used for evaluating the resistance of colors to spot and stain removal procedures used by the drycleaner.

1.2 This test method is not intended for evaluating colorfastness to drycleaning. For colorfastness properties use AATCC TM132, Test Method for Colorfastness to Drycleaning.

2. Principle

2.1 A specimen is agitated in a solution of solvent and a drycleaning detergent with steel balls to simulate the kind of mechanical action that occurs in a drycleaning machine. A large specimen is provided to produce a correlation with commercial conditions.

3. Terminology

3.1 **drycleaning,** n.—the cleaning of fabrics with organic solvents such as petroleum solvent, perchloroethylene or fluorocarbon.

NOTE: The process also includes adding detergent and moisture to the solvent, up to 75% relative humidity, and hot tumble drying to 71°C (160°F).

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 Perchloroethylene is toxic by inhalation, by repeated contact with the skin and by ingestion. It should be used only in a well vented atmosphere. Toxicology studies with laboratory animals have shown evidence of cancer in rats and mice exposed to perchloroethylene vapors at 100-400 ppm concentrations for prolonged times. Fabric saturated with this solvent should be dried in an adequately ventilated hood. Use chemical goggles or face shield, impervious gloves and an impervious apron when handling perchloroethylene.

4.4 An eyewash/safety shower should be located nearby and an organic vapor respirator should be readily available for emergency use.

4.5 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 10.2).

5. Apparatus and Materials (see 10.3)

5.1 Accelerated laundering machine for rotating closed canisters in a thermostatically controlled water bath at 40 ± 2 rpm.

5.2 Stainless steel cylindrical containers, 203 × 89 mm (8.0 × 3.5 in.) in diameter equipped with solvent-resistant rings.

5.3 Teflon liners.

5.4 Metal adapters (to hold containers on shaft of accelerated laundering machine) (see 10.9).

5.5 Stainless steel balls, 6.3 mm (0.25 in.).

5.6 Gray Scale for Color Change (AATCC EP1) (see 10.4).

5.7 Perchloroethylene, drycleaning grade.

5.8 Drycleaning detergent (see 10.5).

5.9 Hand iron, steam press or flatbed press.

6. Test Specimens

6.1 One test specimen is required. The size of the specimen conforms to the weight range of the product tested as follows:

up to 135 g/m^2: 203 × 203 mm
(4 oz/yd^2: 8 × 8 in.)

139 to 203 g/m^2: 152 × 152 mm
(4.1 to 6.0 oz/yd^2: 6 × 6 in.)

207 to 305 g/m^2: 127 × 127 mm
(6.1 to 9.0 oz/yd^2: 5 × 5 in.)

over 305 g/m^2: 102 × 102 mm
(9 oz/yd^2: 4 × 4 in.)

6.2 For water-repellency evaluations, a minimum specimen size of 178 mm (7 in.) square is necessary (i.e., for AATCC TM22, Test Method for Water Repellency: Spray).

7. Procedure

7.1 Run Number 1: Place the specimen in a steel container with 150 mL of perchloroethylene, 1 mL drycleaning detergent (see 10.6) and 100 steel balls. Seal the container (see 10.7), clamp in the accelerated laundering machine and run at room temperature 27°C (81°F) for 10 min. Drain.

7.2 Run Number 2: Replace the solvent with an equal amount of perchloroethylene without detergent and run another 10 min. Drain.

7.3 Run Number 3: Repeat Run Number 2. Remove the specimen, blot thoroughly between paper towels and air dry.

7.4 Finish the specimen by one of the following methods:

7.4.1 Hand Pressing [for all except pile fabrics (see 10.8)]—Cover with a damp muslin cloth, weighing 135-153 g/m^2 (4.0-4.5 oz/yd^2), previously saturated with water and wrung out so as to retain moisture equal to approximately 75% of its dry weight and press with a hand iron having a temperature between 135-150°C (275-300°F) until dry.

7.4.2 Steam Pressing [for all except pile fabrics (see 10.8)]—A hotbed or polished metal-top type press *for flat fabrics*, or a cloth-top type press *for rough crepes*, operating at a steam pressure of between 448 and 482 kPa (65-70 psi). Lower the head of the machine and hold in contact with the fabric. During this period admit steam from the buck of the

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press for a period of 5-10 s.

8. Evaluation and Classification

8.1 Durability of Applied Designs—Rate the specimens containing flock, metallic or other applied designs for appearance as follows:

Grade A5—negligible or no change in appearance.

Grade A4—slightly changed in appearance.

Grade A3—noticeably changed in appearance.

Grade A2—considerably changed in appearance.

Grade A1—much changed in appearance.

8.2 Durability of Fabric Handle—Rate specimens for change in handle as follows:

Grade B5—negligible or no change in handle.

Grade B4—slightly changed in handle.

Grade B3—noticeably changed in handle.

Grade B2—considerably changed in handle.

Grade B1—much changed in handle.

8.3 Finish Performance—Evaluate finish performance characteristics in accordance with AATCC TM22.

9. Precision and Bias

9.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

9.2 *Bias*. The durability of applied designs and finishes to drycleaning can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

10. Notes

10.1 This test is based on an extensive series of interlaboratory tests which showed good correlation between laboratory tests and commercial drycleanings for as many as three repeated drycleanings. Since a major loss of finish material occurs in the first drycleaning, a single application of this test furnishes a good indication of the effect of repeated drycleanings.

10.2 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

10.3 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

10.4 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

10.5 Use any reputable supplier of drycleaning detergent.

10.6 Where residual soap affects the performance of the finish (e.g., where water-repellency evaluation is impaired by the presence of detergent) the detergent may be omitted from the test solution, but it is an artificial condition.

10.7 The jars need not be immersed in a water bath as required in washfastness tests.

10.8 Pile fabrics need only be air dried by any suitable means.

10.9 Necessary on older models.

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Test Method for Seam Smoothness in Fabrics after Home Laundering

Foreword

This test method and its accompanying photographic scales were developed for evaluation of woven fabrics with a durable-press finish. It is common industry practice to use the method and scales for evaluation of other textile materials although some specimens may have different appearance characteristics due to different fabric or seam construction.

Standard laundering procedures remain consistent to allow valid comparison of results. Standard procedures represent, but may not exactly replicate, current consumer practices, which vary over time and among households. Alternate laundering procedures and machine parameters may be found in AATCC LP1, Laboratory Procedure for Home Laundering: Machine Washing (see 12.3) and ISO 6330, Textiles — Domestic washing and drying procedures for textile testing (see 12.8).

1. Purpose and Scope

1.1 This test method is intended to determine the smoothness appearance of seams in fabrics after being subjected to home laundering procedures. Several washing and drying procedures provide standard parameters to represent common home care options.

1.2 Seams in any washable fabric (woven, knitted, or nonwoven) may be evaluated for smoothness using this method.

1.3 Techniques for seaming are not outlined, since the purpose is to evaluate seams as they will be supplied from manufacturing or as ready for use. Furthermore, seaming techniques would be controlled by fabric properties.

2. Principle

2.1 Seamed fabric specimens are subjected to standard home laundering practices. Appearance of specimens is evaluated in a standard viewing area by visually comparing the specimen seam smoothness after laundering with grades represented by the appropriate AATCC Photographic Seam Smoothness Scale. Results are reported as a seam smoothness (SS) grade of 1 to 5, with 1 representing the least smoothness (most puckering) and 5 representing most smoothness.

3. Terminology

3.1 **ballast,** n.—*in procedures for processing or testing of textiles*, material that is used to bring the total weight or volume of the textiles to an amount specified in the procedure.

3.2 **durable press,** adj.—having the ability to retain substantially the initial shape, flat seams, pressed-in creases and unwrinkled appearance during use and after laundering or drycleaning.

3.3 **grade,** n.—the number assigned to a test specimen resulting from comparison to a scale.

3.4 **laundering,** n.—*of textile materials*, a process intended to remove soils and/or stains by treatment (washing) with an aqueous detergent solution and normally including rinsing, extracting and drying.

3.5 **laundering creases,** n.—sharp folds or lines running in any direction in a washed or dried specimen.

NOTE: Laundering creases are an unintended result of restricted movement of specimens in the washer or the dryer.

3.6 **seam smoothness,** n.—*in fabrics*, the visual impression of planarity of a seamed specimen quantified by comparison with a set of reference standards.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Detergent may cause irritation. Care should be taken to prevent exposure to skin and eyes.

4.3 All chemicals should be handled with care.

4.4 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

5. Uses and Limitations

5.1 This test method is designed to be used only for evaluating the appearance of washable seamed fabrics after home laundering.

5.2 In general, it is preferable to conduct the test under relatively severe laundering conditions. It is recognized that special cycles or features are available on current washing machines and dryers to achieve improved performance on certain items; e.g., gentle cycles with reduced agitation to protect delicately constructed items, and durable press cycles, with cool-down or cold rinses and reduced spin speeds, to minimize wrinkling.

5.3 Prints and patterns may mask the mussiness present in seamed fabrics. The evaluation process is, however, based on the visual appearance of specimens including such effects.

5.4 The Seam Smoothness Scales were photographed from woven fabrics. It is understood that the photographs do not duplicate other fabric or seam possibilities. The scales are to be used as guides that represent various levels of seam smoothness.

5.5 Small specimen sizes will occasionally result in wrinkles or creases. These may not be characteristic of actual performance, in use.

5.6 The interlaboratory reproducibility of the results of this test method depends upon mutual agreement by users of the method on the washing and drying conditions as outlined in Tables I-IV.

6. Apparatus and Materials (see 12.1)

6.1 Steam or dry iron with appropriate fabric temperature settings.

6.2 Standard washing machine (see Table I, 12.2), for machine washing.

6.3 Wash tub, 9.5-L, for hand washing.

6.4 White towel, large enough to accommodate test specimen, any weight, for hand washing.

6.5 Standard tumble dryer (see Table IV, 12.2), or facilities for drip, screen or line drying.

6.6 Conditioning facilities and conditioning/drying racks with pull-out, perforated shelves (see 12.3).

6.7 Scale with at least 5-kg or 10-lb capacity.

6.8 AATCC 1993 Standard Reference Detergent (powder, with brightener, see 12.3).

6.9 Laundering ballast Type 1 or Type 3 (see Table V).

6.10 Evaluation area, as described in Appendix A.

6.11 AATCC Photographic Seam Smoothness Scales, for single- and double-needle seams (see Fig. 1, 12.3). The reproduction shown in Fig. 1 may not be used for rating.

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Table I—Standard Washing Machine Parameters (see 12.2, 12.7)

Cycle	(1) Normal		(2) Delicate		(3) Permanent Press	
Water Level, L (gal)	72 ± 4 (19 ± 1)		72 ± 4 (19 ± 1)		72 ± 4 (19 ± 1)	
Agitation Speed, strokes/min.	86 ± 2		27 ± 2		86 ± 2	
Washing Time, min.	16 ± 1		8.5 ± 1		12 ± 1	
Final Spin Speed, rpm	660 ± 15		500 ± 15		500 ± 15	
Final Spin Time, min.	5 ± 1		5 ± 1		5 ± 1	
Wash Temp, °C (°F)[1]	(II) Cold:	27 ± 3 (80 ± 5)	(II) Cold:	27 ± 3 (80 ± 5)	(II) Cold:	27 ± 3 (80 ± 5)
	(III) Warm:	41 ± 3 (105 ± 5)	(III) Warm:	41 ± 3 (105 ± 5)	(III) Warm:	41 ± 3 (105 ± 5)
	(IV) Hot:	49 ± 3 (120 ± 5)	(IV) Hot:	49 ± 3 (120 ± 5)	(IV) Hot:	49 ± 3 (120 ± 5)
	(V) Very Hot:	60 ± 3 (140 ± 5)	(V) Very Hot:	60 ± 3 (140 ± 5)	(V) Very Hot:	60 ± 3 (140 ± 5)

[1] The temperatures in this table are similar to those specified by the US Federal Trade Commission for care label verification (see Table VIII). Due to US Department of Energy requirements, many consumer washing machines use cooler water. An external control box may be used to override the machine set temperatures.

Table II—Standard Hand Washing and Rinsing Temperatures

Designation	Wash Temp, °C (°F)	Rinse Temp, °C (°F)
Very Cold	16 ± 3 (60 ± 5)	< 18 (< 65)
Cold	27 ± 3 (80 ± 5)	< 29 (< 85)
Warm	41 ± 3 (105 ± 5)	< 29 (< 85)
Hot[1]	49 ± 3 (120 ± 5)	< 29 (< 85)

[1] Hot water may not be a reasonable selection for hand washing or for the delicate items typically requiring hand washing.

Table III—Standard Drying Conditions

(A)	Tumble Dry
(Ai)	Tumble Dry Normal
(Aii	Tumble Dry Delicate
(Aiii)	Tumble Dry Permanent Press
(B)	Line/Hang Dry
(C)	Drip Dry
(D)	Screen Dry/Dry Flat

Table IV—Standard Tumble Dryer Parameters (see 12.2)

Cycle	(Ai) Normal	(Aii) Delicate	(Aiii) Permanent Press
Max. Exhaust Temp, °C (°F)	68 ± 6 (155 ± 10)	60 ± 6 (140 ± 10)	68 ± 6 (155 ± 10)
Cool Down Time, min.	≤ 10	≤ 10	≤ 10

Table V—Laundering Ballast Parameters

	Type 1	Type 3
Fiber Content	100% cotton	50% cotton/50% polyester ± 3%
Greige Fabric Yarns	16/1 ring spun	16/1 or 30/2 ring spun
Greige Fabric Construction	52 × 48 ± 5 yarns/in., plain weave	52 × 48 ± 5 yarns/in., plain weave
Finished Fabric Weight	155 ± 10 g/m^2	155 ± 10 g/m^2
Edges	All edges hemmed or over-edged	All edges hemmed or over-edged
Finished Piece Size	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)
Finished Piece Weight	130 ± 10 g	130 ± 10 g

7. Test Specimens

7.1 Cut three 380 x 380 mm (15 x 15 in) specimens, sides parallel to the length and width fabric directions such that existing seams run through the middle of each specimen. Where possible, each specimen should contain different groups of lengthwise and widthwise yarns. The specimens should be marked to indicate the lengthwise direction.

7.1.1 If fraying is expected in laundering, see 12.4.

7.1.2 If the fabric is wrinkled, it may be smoothed by appropriate ironing prior to laundering. See Table I, Safe Ironing Temperature Guide, in AATCC TM133, Test Method for Colorfastness to Heat: Hot Pressing (see 12.3). Take care to avoid altering quality of the seam itself.

8. Laundering Procedure

8.1 Machine Washing.

8.1.1 Select washing conditions for testing from Table I. Normal cycle is recommended. Set washing machine controls to generate the selected cycle parameters.

8.1.2 Load for laundering shall consist of all specimens, plus enough laundering ballast pieces to make a total load weight of 1.8 ± 0.1 kg (4.0 ± 0.2 lb). For very critical evaluation and in arbitration, limit the number of specimens per washer load to those from one sample.

8.1.3 Begin selected wash cycle. Allow machine to fill to specified water level.

8.1.4 Add 66 ± 1 g of AATCC 1993 Standard Reference Detergent to washing machine as directed by washing machine manufacturer's instructions. If detergent is added directly to wash water, agitate briefly to dissolve completely. Stop agitation before adding wash load.

8.1.5 Add wash load (test specimens and ballast), distributing evenly around center agitator. Restart wash cycle.

8.1.6 For specimens to be drip dried (drying procedure C), stop wash cycle before the water begins to drain from the final rinse cycle. Remove specimens soaking wet. For specimens to be tumble (A), line (B), or screen/flat (D) dried, allow washing to proceed through the final spin cycle.

8.1.7 After each washing cycle, separate tangled specimens and ballast pieces, taking care to minimize distortion.

8.1.8 Washer creases. Specimens may be in a folded or creased state after washing. Such folds or creases should be removed by hand prior to drying.

8.1.9 Proceed to appropriate drying procedure.

8.2 Hand Washing.

8.2.1 Select washing temperature from Table II. Add 7.6 ± 1.9 L (2.0 ± 0.5 gal) of water at this temperature to the wash tub.

8.2.2 Add 20 ± 1 g of AATCC 1993 Standard Reference Detergent to wash tub.

8.2.3 Agitate by hand to dissolve detergent.

8.2.4 Add specimen(s) to water and gently squeeze to distribute detergent solution. Do not twist or wring.

8.2.5 Let specimen(s) soak for 2 min.

8.2.6 Gently squeeze specimen(s) in detergent solution for 1 min. Do not twist or wring.

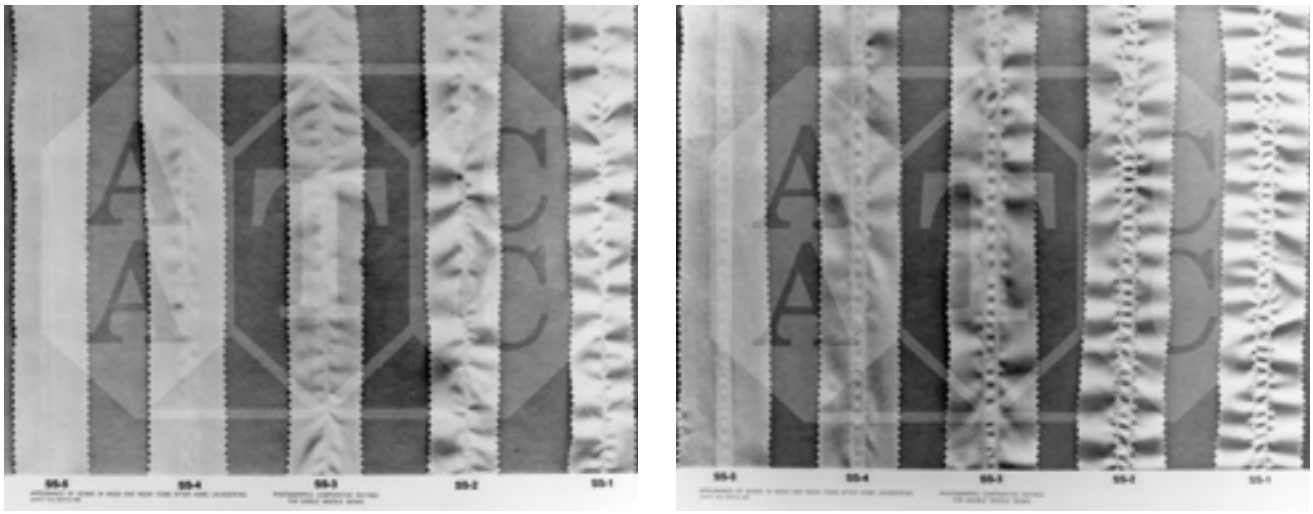
8.2.7 Repeat 2-min. soak and 1-min squeeze in detergent solution.

8.2.8 Remove specimen(s) from wash tub and gently squeeze to remove excess detergent solution. Do not twist or wring.

8.2.9 Place specimen(s) on clean white towel. Empty and rinse wash tub.

8.2.10 Add 7.6 ± 1.9 L (2.0 ± 0.5 gal) of clean water at the specified rinse tem-

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A. For single needle seams.

B. For double needle seams.

Fig. 1—AATCC photographic seam smoothness scales.

perature (see Table II) to the wash tub.

8.2.11 Place washed specimen(s) from towel in rinse water and gently squeeze to distribute. Do not twist or wring.

8.2.12 Let specimen(s) soak for 2 min.

8.2.13 Gently squeeze specimen(s) in rinse water for 1 min. Do not twist or wring.

8.2.14 Repeat 2-min. soak and 1-min squeeze in rinse water.

8.2.15 Remove specimen(s) from wash tub and gently squeeze to remove excess water. Do not twist or wring.

8.2.16 Using clean white towels, blot water from washed specimen(s). Do not wring or twist.

8.2.17 Proceed to appropriate drying procedure.

8.3 Drying.

8.3.1 Select drying conditions from Table III.

8.3.2 (A) Tumble Dry. Place the washed load (test specimens and ballast) in the tumble dryer, and set the temperature control to generate the selected cycle exhaust temperature (see Table IV). Allow the dryer to run until the load is fully dry. Remove specimens immediately.

8.3.3 (B) Line/Hang Dry. Hang each fabric specimen by two corners, with the fabric length in the vertical direction. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

8.3.4 (C) Drip Dry. Hang each dripping wet specimen by two corners, with the fabric length in the vertical direction. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

8.3.5 (D) Screen/Flat Dry. Spread each specimen on a horizontal screen or perforated surface, removing wrinkles without distorting or stretching it. Allow specimens to rest in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

8.3.6 For all drying methods. allow specimens to dry completely before washing again.

8.3.7 Laundering creases. If specimens are folded or creased after any drying cycle except the last, rewet and attempt to remove the creases by ironing with a hand iron at a temperature suitable for the fabric being tested (see 7.1.2), prior to additional laundering cycles. Do not attempt to remove wrinkles or creases by hand ironing after the final drying cycle.

8.3.8 Repeat the selected washing and drying procedures for a total of 5 complete cycles, or an agreed upon number of cycles.

8.4 Conditioning.

8.4.1 After the final drying cycle is complete, condition test specimens as directed in ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 12.5). (Use conditions indicated in Table 1 for Textiles, general. Estimate conditioning time as indicated in Table 2 for the appropriate fiber content.) Lay each test specimen flat, separately, on a screen or perforated shelf of a conditioning/drying rack.

9. Evaluation

9.1 Three trained observers should evaluate each test specimen independently.

9.2 Make all evaluations in the specified viewing area (see Appendix A). Illuminate the viewing area with the overhead fluorescent light only. Turn all other lights in the room off.

9.3 Each observer is to stand directly in front of the specimen, 1219 ± 25 mm (48 ± 1in.) away from the board. It has been found that normal variations in the height of the observer above and below the arbitrary 1524-mm (60-in.) eye level have no significant effect on the grade given.

9.4 Mount the test specimen on the viewing board with the seam in the vertical direction. Place the appropriate single- or double-needle AATCC Photographic Seam Smoothness Scale beside the specimen to facilitate comparative evaluation.

9.5 Confine observations to the area influenced by the seam and disregard the appearance of the surrounding fabric. The Seam Smoothness Scales were photographed from woven fabrics. It is understood that the photographs do not duplicate other fabric or seam possibilities. The scales are to be used as guides that represent various levels of seam smoothness.

9.6 The seam smoothness (SS) grade is that of the Photographic Scale seam that most nearly matches the appearance of the test specimen seam.

9.7 Average the nine observations made on each test fabric (three grades on each of three test specimens). Report the average to the nearest tenth of a grade. This average is the unit of measure of this test method.

10. Report

10.1 Report for each sample tested:

10.1.1 A description or identification of the sample.

10.1.2 Evaluation was performed using AATCC TM88B-2018t.

10.1.3 Number of specimens tested.

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10.1.4 Type of ballast used, i.e., type 1 or type 3.

10.1.5 Number of laundering cycles (default is 5 cycles).

10.1.6 Laundering conditions, including washing machine cycle, wash temperature, drying procedure and tumble drying temperature, if applicable. Alphanumeric designations may be used if they are clearly understood by all parties. For example, 1-IV-A(ii) indicates normal wash cycle at 49°C and Delicate tumble dry cycle.

10.1.7 Number of observers.

10.1.8 Seam smoothness scale used. i.e., single-needle or double-needle.

10.1.9 Any modifications of the test method.

10.1.10 Average seam smoothness (SS) grade.

11. Precision and Bias

11.1 *Interlaboratory Study.* Tests for seam smoothness were conducted in 1993 using Machine Wash Normal, 41°C, and Tumble Dry, Normal/Cotton Sturdy laundering settings. Six laboratories tested double-needle seams. Three observers in each laboratory independently evaluated the same three specimens from each of the fabrics, yielding nine grades to average for each test result, as required by the method.

11.2 *Precision*.

11.2.1 Double-Needle Seams. Components of variance are given in Table VI, and critical differences for single fabric and for multiple fabric comparisons are given in Tables VII and VIII, respectively.

11.2.2 If comparisons are made between laboratories on a single fabric, critical differences in Table VII are used.

11.2.3 If comparisons are made between laboratories on multiple fabrics, critical differences in Table VIII are used.

11.2.4 If two laboratories are shown to be in statistical control and operating at comparable levels, the magnitude of applicable critical differences may be less than those given in these tables, and may be determined by data from comparison trials between them.

11.3 *Bias*. The value of seam appearance can be defined only in terms of a test method. There is no independent method for determining the true value. This test method has no known bias.

Table VI—Components of Variance (as variance)

Component	Variance
Laboratory	0.113
FL Interaction	0.031
Specimen (FRL)	0.191

Table VII—Critical Differences for Single Fabric Comparisons (95% probability)

Within Laboratory	Between Laboratory
0.70	1.16

Table VIII—Critical Differences for Multiple Fabric Comparisons (95% probability)

Within Laboratory	Between Laboratory
0.70	1.26

12. Notes

12.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

12.2 For model number(s) of washer(s) and tumble dryers(s) reported to meet the standard parameters, visit www.aatcc.org/test/washers or contact AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org. An alternate load size was described in previous versions of this method (3.6-kg load, 83-L water level, 80 g AATCC 1993 Standard Reference Detergent), but no machine is reported to meet the standard parameters for this load size. In addition, dimensional change results obtained with the alternate load size may not be equal to those obtained with a standard load size.

12.3 Materials or drawings available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel:+1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

12.4 If excessive fraying occurs in laundering, specimen edges may be pinked, slashed or stitched. If edges of laundered specimens appear distorted, clip as necessary before evaluating.

12.5 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610. 832.9500; fax: +1.610.832. 9555; web site: www.astm.org.

12.6 The use of two 96-in. lamps for viewing laundered specimens is specified in this method. It is recognized, however, that physical space limitations in certain laboratories will prevent the use of 96-in. lamps. In those situations, two 48-in. lamps and a narrower viewing board may be used.

12.7 The laundering temperatures and other parameters listed in this test method are standard conditions for testing purposes. As with most lab procedures, they represent, but may not exactly replicate, current consumer practices. Consumer practices vary over time and among households; lab practices must be consistent to allow valid comparison of results. If laundering equipment or conditions other than those specifically listed in this test method are used, they must be described in detail and noted as a modification of the standard method. Alternate laundering conditions are outlined in AATCC LP1 and ISO 6330.

12.8 Available from ISO, www.iso.org.

13. History

13.1 Editorially revised 2023 to add History section.

13.2 Editorially revised and reaffirmed twice in 2018 (with title change); editorially revised 2018; editorially revised 2016; editorially revised and reaffirmed 2014; editorially revised 2012; editorially revised and reaffirmed 2011, 2010; editorially revised 2008; editorially revised and reaffirmed 2006; editorially revised 2005, 2004; editorially revised and reaffirmed 2003; editorially revised and reaffirmed 2001; editorially revised 1997; editorially revised and reaffirmed 1996; editorially revised and reaffirmed 1992; editorially revised 1991; editorially revised and reaffirmed 1989 (with title change); editorially revised 1986, 1985; editorially revised and reaffirmed 1984; editorially revised 1983; editorially revised and reaffirmed 1981; editorially revised and reaffirmed 1978; editorially revised and reaffirmed 1975; editorially revised 1974; reaffirmed 1973; editorially revised and reaffirmed 1970; reaffirmed 1969.

13.3 Developed in 1962 by AATCC Committee RA61. Technically equivalent to ISO 7770.

Appendix A
Evaluation Area

A1 Viewing Board.

A1.1 Plywood board, 1829 mm (72 in.) high × 6 mm (0.25 in.) thick × wide enough to accommodate scale and specimen side by side

A1.2 Painted to match the Grade 2 chip on the Gray Scale for Staining (see 12.3). Approximate CIELAB values for the chip are L* = 77, a* = 0, b* = 0. A tolerance of two units for each parameter is acceptable for the board color in this test method.

A1.3 Spring-loaded swatch mount or other means to secure specimen and scale such that center of specimen is 1524 mm (60 in.) above the floor. Mount may be fabricated using light sheet metal (22 ga.)

A2 Lighting.

A2.1 Overhead fluorescent tube light fixture (see 12.6).

A2.1.1 Two parallel F96 T12 cool white lamps (without baffle or glass),

A2.1.2 One white enamel reflector (without baffle or glass),

A2.1.3 Mount as shown in Fig. A1.

A2.2 Eliminate all light sources except the specified fluorescent lamps.

A3 Walls.

A3.1 It has been the experience of many observers that light reflected from the side walls near the viewing board can interfere with the rating results. It is recommended that the side walls be painted matte black (85° gloss less than 5 units) or that blackout curtains be mounted on both sides of the viewing board to eliminate the reflective interference.

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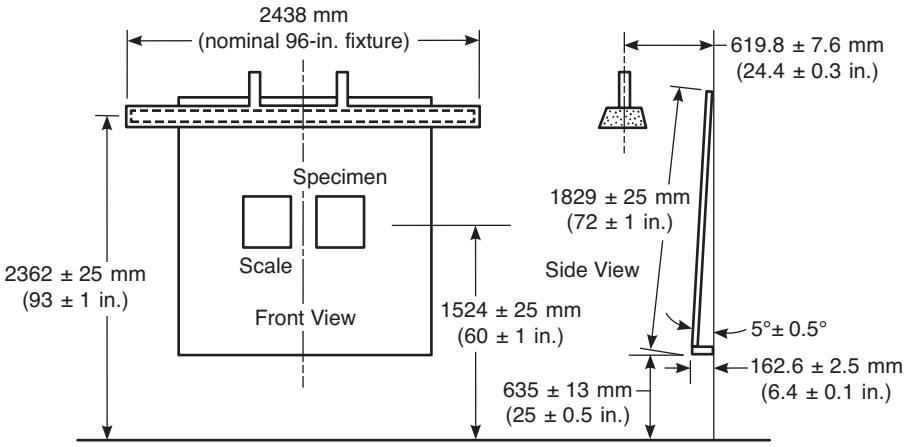


Fig. A1—Evaluation area for seam smoothness.

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Test Method for Crease Retention in Fabrics after Home Laundering

Foreword

This test method and its accompanying three-dimensional crease replica set were developed for evaluation of woven fabrics with a durable-press finish. It is common industry practice to use the method and replicas for evaluation of other textile materials although some specimens may have different appearance characteristics due to different fabric constructions.

Standard laundering procedures remain consistent to allow valid comparison of results. Standard procedures represent, but may not exactly replicate, current consumer practices, which vary over time and among households. Alternate laundering procedures and machine parameters may be found in AATCC LP1, Laboratory Procedure for Home Laundering: Machine Washing (see 12.3) and ISO 6330, Textiles — Domestic washing and drying procedures for textile testing (see 12.8).

1. Purpose and Scope

1.1 This test method is intended to determine the retention of pressed-in creases in fabrics after being subjected to home laundering procedures. Several washing and drying procedures provide standard parameters to represent common home care options.

1.2 Creases in any washable fabric (woven, knitted or nonwoven) may be evaluated for retention using this method.

1.3 Techniques for creasing are not outlined, since the purpose is to evaluate creases as they will be supplied from manufacturing or as ready for use. Furthermore, creasing techniques would be controlled by fabric properties.

2. Principle

2.1 Creased fabric specimens are subjected to standard home laundering practices. Appearance of specimens is evaluated in a standard viewing area by visually comparing the specimen crease after laundering with grades represented by AATCC Three-Dimensional Crease Replicas. Results are reported as a crease retention (CR) grade of 1 to 5, with 1 representing the least retention and 5 representing most retention (sharpest crease).

3. Terminology

3.1 **ballast,** n.—*in procedures for processing or testing of textiles*, material that is used to bring the total weight or volume of the textiles to an amount specified in the procedure.

3.2 **crease retention,** n.—*in fabrics*, the visual impression of an inserted crease quantified by comparison with a set of reference standards.

3.3 **durable press,** adj.—having the ability to retain substantially the initial shape, flat seams, pressed-in creases and unwrinkled appearance during use and after laundering or drycleaning.

3.4 **grade,** n.—the number assigned to a test specimen resulting from comparison to a scale.

3.5 **laundering,** n.—*of textile materials*, a process intended to remove soils and/or stains by treatment (washing) with an aqueous detergent solution and normally including rinsing, extracting and drying.

3.6 **laundering creases,** n.—sharp folds or lines running in any direction in a washed or dried specimen.

NOTE: Laundering creases are an unintended result of restricted movement of specimens in the washer or the dryer.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Detergent may cause irritation. Care should be taken to prevent exposure to skin and eyes.

4.3 All chemicals should be handled with care.

4.4 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

4.5 When evaluating crease retention, use of a light shield can aid in prevention of burns that could result from lamp heat.

5. Uses and Limitations

5.1 This test method is designed to be used only for evaluating the appearance of washable creased fabrics after repeated home laundering.

5.2 In general, it is preferable to conduct the test under relatively severe laundering conditions. It is recognized that special cycles or features are available on current washing machines and dryers to achieve improved performance on certain items; e.g., gentle cycles with reduced agitation to protect delicately constructed items, and durable press cycles, with cool-down or cold rinses and reduced spin speeds, to minimize wrinkling.

5.3 Prints and patterns may mask the appearance of a crease. The evaluation process is, however, based on the visual appearance of specimens including such effects.

5.4 The Crease Replicas were cast from woven fabrics. It is understood that the replicas do not duplicate other fabric possibilities (knits, nonwovens). The replicas are to be used as guides that represent various levels of fabric creasing.

5.5 Small specimen sizes will occasionally result in wrinkles or creases. These may not be characteristic of actual performance, in use.

5.6 The interlaboratory reproducibility of the results of this test method depends upon mutual agreement by users of the method on the washing and drying conditions as outlined in Tables I-IV.

6. Apparatus and Materials (see 12.1)

6.1 Steam or dry iron with appropriate fabric temperature settings.

6.2 Standard washing machine (see Table I, 12.2), for machine washing.

6.3 Wash tub, 9.5-L, for hand washing.

6.4 White towels, large enough to accommodate test specimen, any weight, for hand washing.

6.5 Standard tumble dryer (see Table IV, 12.2) or facilities for drip, screen or line drying.

6.6 Conditioning facilities and conditioning/drying racks with pull-out, perforated shelves (see 12.3).

6.7 Scale with at least 5-kg or 10-lb capacity.

6.8 AATCC 1993 Standard Reference Detergent (powder, with brightener, see 12.3).

6.9 Laundering ballast Type 1 or Type 3 (see Table V).

6.10 Evaluation area, as described in Appendix A.

6.11 AATCC Three-Dimensional Crease Replicas, set of five (see Fig. 1, 12.3).

7. Test Specimens

7.1 Cut three 380 × 380 mm (15 × 15 in) specimens, sides parallel to the length

Table I—Standard Washing Machine Parameters (see 12.2, 12.7)

Cycle	(1) Normal		(2) Delicate		(3) Permanent Press	
Water Level, L (gal)	72 ± 4 (19 ± 1)		72 ± 4 (19 ± 1)		72 ± 4 (19 ± 1)	
Agitation Speed, strokes/min.	86 ± 2		27 ± 2		86 ± 2	
Washing Time, min.	16 ± 1		8.5 ± 1		12 ± 1	
Final Spin Speed, rpm	660 ± 15		500 ± 15		500 ± 15	
Final Spin Time, min.	5 ± 1		5 ± 1		5 ± 1	
Wash Temp, °C (°F)[1]	(II) Cold:	27 ± 3 (80 ± 5)	(II) Cold:	27 ± 3 (80 ± 5)	(II) Cold:	27 ± 3 (80 ± 5)
	(III) Warm:	41 ± 3 (105 ± 5)	(III) Warm:	41 ± 3 (105 ± 5)	(III) Warm:	41 ± 3 (105 ± 5)
	(IV) Hot:	49 ± 3 (120 ± 5)	(IV) Hot:	49 ± 3 (120 ± 5)	(IV) Hot:	49 ± 3 (120 ± 5)
	(V) Very Hot:	60 ± 3 (140 ± 5)	(V) Very Hot:	60 ± 3 (140 ± 5)	(V) Very Hot:	60 ± 3 (140 ± 5)

[1] Due to US Department of Energy requirements, many washing machines use cooler water. An external control box may be used to override the machine set temperatures.

and width fabric directions. Where possible, each specimen should contain different groups of lengthwise and widthwise yarns. The specimens should be marked to indicate the lengthwise direction.

7.1.1 If fraying is expected in laundering, see 12.4.

7.1.2 If the fabric is wrinkled, it may be smoothed by appropriate ironing prior to laundering. See Table I, Safe Ironing Temperature Guide, in AATCC TM133, Test Method for Colorfastness to Heat: Hot Pressing (see 12.3).

7.1.3 Press a crease through the middle of each specimen in the fabric length direction.

8. Laundering Procedure

8.1 Machine Washing.

8.1.1 Select washing conditions for testing from Table I. Normal cycle is recommended. Set washing machine controls to generate the selected cycle parameters.

8.1.2 Load for laundering shall consist of all specimens, plus enough laundering ballast pieces to make a total load weight of 1.8 ± 0.1 kg (4.0 ± 0.2 lb). For very critical evaluation and in arbitration, limit the number of specimens per washer load to those from one sample.

8.1.3 Begin selected wash cycle. Allow machine to fill to specified water level.

8.3.3 Add 66 ± 1 g of AATCC 1993 Standard Reference Detergent to washing machine as directed by washing machine manufacturer's instructions. If detergent is added directly to wash water, agitate briefly to dissolve completely. Stop agitation before adding wash load.

8.1.4 Add wash load (test specimens and ballast), distributing evenly around center agitator. Restart wash cycle.

8.1.5 For specimens to be drip dried (drying procedure C), stop wash cycle before the water begins to drain from the final rinse cycle. Remove specimens soaking wet. For specimens to be tumble (A), line (B), or screen/flat (D) dried, allow washing to proceed through the final spin cycle.

8.1.6 After each washing cycle, separate tangled specimens and ballast pieces, taking care to minimize distortion.

8.1.7 Washer creases. Specimens may be in a folded or creased state after washing. Such folds or creases should be removed by hand prior to drying.

8.1.8 Proceed to appropriate drying procedure.

8.2 Hand Washing.

8.2.1 Select washing temperature from Table II. Add 7.6 ± 1.9 L (2.0 ± 0.5 gal) of water at this temperature to the wash tub.

8.2.2 Add 20 ± 1 g of AATCC 1993 Standard Reference Detergent to wash tub.

8.2.3 Agitate by hand to dissolve detergent.

8.2.4 Add specimen(s) to water and gently squeeze to distribute detergent solution. Do not twist or wring.

8.2.5 Let specimen(s) soak for 2 min.

8.2.6 Gently squeeze specimen(s) in detergent solution for 1 min. Do not twist or wring.

8.2.7 Repeat 2-min. soak and 1-min squeeze in detergent solution.

8.2.8 Remove specimen(s) from wash tub and gently squeeze to remove excess detergent solution. Do not twist or wring.

8.2.9 Place specimen(s) on clean white towel. Empty and rinse wash tub.

8.2.10 Add 7.6 ± 1.9 L (2.0 ± 0.5 gal) of clean water at the specified rinse temperature (see Table II) to the wash tub.

8.2.11 Place washed specimen(s) from towel in rinse water and gently squeeze to distribute. Do not twist or wring.

8.2.12 Let specimen(s) soak for 2 min.

8.2.13 Gently squeeze specimen(s) in rinse water for 1 min. Do not twist or wring.

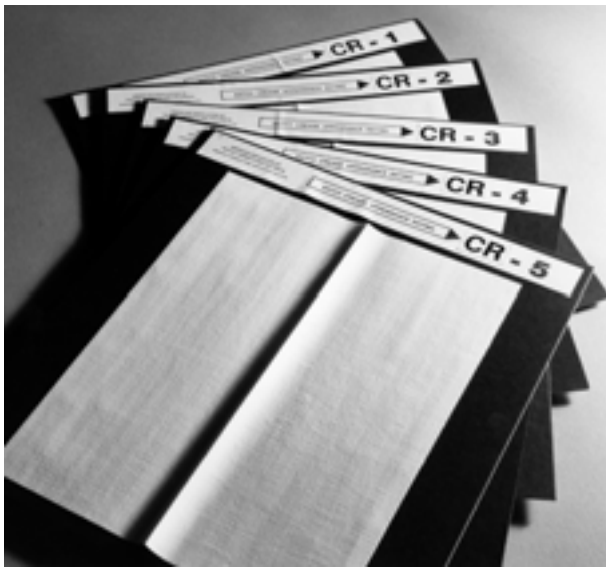


Fig. 1—AATCC three-dimensional crease replicas.

Table II—Standard Hand Washing and Rinsing Temperatures

Designation	Wash Temp, °C (°F)	Rinse Temp, °C (°F)
Very Cold	16 ± 3 (60 ± 5)	< 18 (< 65)
Cold	27 ± 3 (80 ± 5)	< 29 (< 85)
Warm	41 ± 3 (105 ± 5)	< 29 (< 85)
Hot[1]	49 ± 3 (120 ± 5)	< 29 (< 85)

[1] Hot water may not be a reasonable selection for hand washing or for the delicate items typically requiring hand washing.

8.2.14 Repeat 2-min. soak and 1-min squeeze in rinse water.

8.2.15 Remove specimen(s) from wash tub and gently squeeze to remove excess water. Do not twist or wring.

8.2.16 Using clean white towels, blot water from washed specimen(s). Do not wring or twist.

8.2.17 Proceed to appropriate drying procedure.

8.3 Drying.

8.3.1 Select drying conditions from Table III.

8.3.2 (A) Tumble Dry. Place the washed load (test specimens and ballast) in the tumble dryer, and set the temperature control to generate the selected cycle exhaust temperature (see Table IV). Allow the dryer to run until the load is fully dry. Remove specimens immediately.

8.3.3 (B) Line/Hang Dry. Hang each specimen by two corners, with the fabric length in the vertical direction. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

8.3.4 (C) Drip Dry. Hang each dripping wet specimen by two corners, with the fabric length in the vertical direction. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

8.3.5 (D) Screen/Flat Dry. Spread each specimen on a horizontal screen or perforated surface, removing wrinkles without distorting or stretching it. Allow specimens to rest in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

8.3.6 For all drying methods, allow specimens to dry completely before washing again.

8.3.7 Laundering creases. If specimens are folded or creased after any drying cycle except the last, rewet and attempt to remove the creases by ironing with a hand iron at a temperature suitable for the fabric being tested (see 7.1.2), prior to additional laundering cycles. Take care to avoid altering quality of the crease itself. Do not attempt to remove wrinkles or creases by hand ironing after the final drying cycle.

8.3.8 Repeat the selected washing and drying procedures for a total of 5 complete cycles, or an agreed upon number of cycles.

8.4 Conditioning.

8.4.1 After the final drying cycle is complete, condition test specimens as directed in ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 12.5). (Use conditions indicated in Table 1 for Textiles, general. Estimate conditioning time as indicated in Table 2 for the appropriate fiber content.) Lay each test specimen flat, separately, on a screen or perforated shelf of a conditioning/drying rack.

Table III—Standard Drying Conditions

(A)	Tumble Dry
(Ai)	Tumble Dry Normal
(Aii	Tumble Dry Delicate
(Aiii)	Tumble Dry Permanent Press
(B)	Line/Hang Dry
(C)	Drip Dry
(D)	Screen Dry/Dry Flat

9. Evaluation

9.1 Three trained observers should evaluate each test specimen independently.

9.2 Make all evaluations in the specified viewing area (see Appendix A). Illuminate the viewing area with the overhead fluorescent light and the flood lamp only. Turn all other lights in the room off.

9.3 Each observer is to stand directly in front of the specimen 1219 ± 25 mm (48 ± 1 in.) away from the board (see Fig. A2). It has been found that normal variations in the height of the observer above and below the arbitrary 1524-mm (60-in.) eye level have no significant effect on the grade given.

9.4 Mount the test specimen on the viewing board with the crease in the vertical direction. Place the most similar Three-Dimensional Crease Replicas on each side of the test specimen to facilitate comparative evaluation. Mount replicas 1, 3 and 5 on the left; mount replicas 2 and 4 on the right.

9.5 Confine observations to the crease itself and disregard the appearance of the fabric. The Crease Replicas were cast from woven fabrics. It is understood that the replicas do not duplicate other fabric possibilities (knits, nonwovens). The replicas are to be used as guides that represent various levels of fabric creasing.

9.6 The crease retention (CR) grade is that of the Crease Replica that most nearly matches the appearance of the test specimen crease.

9.7 Average the nine observations made on each test fabric (three grades on each of three test specimens). Report the average to the nearest tenth of a grade. This average is the unit of measure of this test method.

10. Report

10.1 Report for each sample tested.

10.1.1 A description or identification of the sample.

10.1.2 Evaluation was performed using AATCC TM88C-2018t.

10.1.3 Number of specimens tested.

10.1.4 Type of ballast used, i.e., Type 1 or Type 3.

10.1.5 Number of laundering cycles (default is 5 cycles).

10.1.6 Laundering conditions, including washing machine cycle, wash temperature, drying procedure and tumble drying temperature, if applicable. Alphanumeric designations may be used if they are clearly understood by all parties. For example, 1-IV-A(ii) indicates normal wash cycle at 49°C and Delicate tumble dry cycle.

10.1.7 Number of observers.

10.1.8 Any modifications of the test method.

10.1.9 Average crease retention (CR) grade.

11. Precision and Bias

11.1 *Interlaboratory Study.* Tests for crease retention were conducted in 1992 using Machine Wash Normal, 41°C, and Tumble Dry, Normal/Cotton Sturdy laundering settings. Six laboratories evaluated six fabrics for crease retention grades. Three observers in each laboratory independently evaluated the same three specimens from each of the fabrics, yielding nine grades to average for each test re-

Table IV—Standard Tumble Dryer Parameters (see 12.2)

Cycle	(Ai) Normal	(Aii) Delicate	(Aiii) Permanent Press
Max. Exhaust Temp, °C (°F)	68 ± 6 (155 ± 10)	60 ± 6 (140 ± 10)	68 ± 6 (155 ± 10)
Cool Down Time, min.	≤ 10	≤ 10	≤ 10

Table V—Laundering Ballast Parameters

	Type 1	Type 3
Fiber Content	100% cotton	50% cotton/50% polyester ± 3%
Greige Fabric Yarns	16/1 ring spun	16/1 or 30/2 ring spun
Greige Fabric Construction	52 × 48 ± 5 yarns/in., plain weave	52 × 48 ± 5 yarns/in., plain weave
Finished Fabric Weight	155 ± 10 g/m^2	155 ± 10 g/m^2
Edges	All edges hemmed or over-edged	All edges hemmed or over-edged
Finished Piece Size	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)
Finished Piece Weight	130 ± 10 g	130 ± 10 g

sult, as required by the method.

11.1.1 The participating laboratories were presumed to be performing the test method under statistical control, without verification.

11.1.2 Committee RA61 directed use of the analysis of variance technique. No correction was made for rating scale discontinuity.

11.1.3 Because variability was quite high in the study, especially residual variance, users of the method are alerted to check for possible sources of variability before initiating any test program.

11.1.4 The analysis has been retained for reference in the Committee RA61 files.

11.2 *Precision*.

11.2.1 Components of Variance are given in Table VI, and critical differences for single fabric and for multiple fabric comparisons are given in Tables VII, and VIII, respectively.

11.2.2 If comparisons are made between laboratories on a single fabric, critical differences in Table VII are used.

11.2.3 If comparisons are made between laboratories on multiple fabrics, critical differences in Table VIII are used.

11.2.4 If two laboratories are shown to be in statistical control and operating at comparable levels, the magnitude of applicable critical differences may be less than those given in these tables, and may be determined by data from comparison trials between them.

11.3 *Bias*. The value of crease retention can be defined only in terms of a test method. There is no independent method for determining the true value. This test method has no known bias.

Table VI—Components of Variance (as variance)

Component	Variance
Laboratory	0.0855
FL Interaction	0.2049
Specimen (FRL)	0.6304

Table VII—Critical Differences for Single Fabric Comparisons (95% probability)

Within Laboratory	Between Laboratory
1.37	1.59

Table VII—Critical Differences for Multiple Fabric Comparisons (95% probability)

Within Laboratory	Between Laboratory
1.37	2.03

12. Notes

12.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

12.2 For model number(s) of washer(s) and tumble dryers(s) reported to meet the standard parameters, visit www.aatcc.org/test/washers or contact AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org. An alternate load size was described in previous versions of this method (3.6-kg load, 83-L water level, 80 g AATCC 1993 Standard Reference Detergent), but no machine is reported to meet the standard parameters for this load size. In addition, dimensional change results obtained with the alternate load size may not be equal to those obtained with a standard load size.

12.3 Materials or drawings available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

12.4 If excessive fraying occurs in laundering, specimen edges should be pinked, slashed or stitched. If edges of laundered specimens appear distorted, clip as necessary before evaluating.

12.5 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610. 832.9500; fax: +1.610.832.9555; web site: www.astm.org.

12.6 The use of two 96-in. lamps for viewing laundered specimens is specified in this method. It is recognized, however, that physical space limitations in certain laboratories will prevent the use of 96-in. lamps. In those situations, two 48-in. lamps and a narrower viewing board may be used.

12.7 The laundering temperatures and other parameters listed in this test method are standard conditions for testing purposes. As with most lab procedures, they represent, but may not exactly replicate, current consumer practices. Consumer practices vary over time and among households; lab practices must be consistent to allow valid comparison of results. If laundering equipment or conditions other than those specifically listed in this test method are used, they must be described in detail and noted as a modification of the standard method. Alternate laundering conditions are outlined in AATCC LP1 and ISO 6330.

12.8 Available from ISO, www.iso.org.

13. History

13.1 Editorially revised 2023 to add History section.

13.2 Revised 2018 (with title change); editorially revised 2018; revised 2014; editorially revised 2012; revised 2011, 2010; editorially revised 2008; revised 2006; editorially revised 2005, 2004; revised 2003; editorial revised and reaffirmed 2001; editorially revised 1997; revised 1996; editorially revised 1991; revised 1992; revised 1989 (with title change); revised 1987; editorially revised 1986, 1985; editorial revised and reaffirmed 1984 (new title); revised 1979; revised 1975; reaffirmed 1973; editorially revised 1974; reaffirmed 1969.

13.3 Developed in 1963 by AATCC Committee RA61. Technically equivalent to ISO 7769.

Appendix A
Evaluation Area

A1 Viewing Board.

A1.1 Plywood board, 1829 mm (72 in.) high × 6 mm (0.25 in.) thick × wide enough to accommodate scale and specimen side by side

A1.2 Painted to match the Grade 2 chip on the Gray Scale for Staining (see 12.3). Approximate CIELAB values for the chip are L* = 77, a* = 0, b* = 0. A tolerance of two units for each parameter is acceptable for the board color in this test method.

A1.3 Spring-loaded swatch mount or other means to secure specimen and scale such that center of specimen is 1524 mm (60 in.) above the floor. Mount may be fabricated using light sheet metal (22 ga.)

A2 Lighting.

A2.1 Overhead fluorescent tube light fixture (see 12.6).

A2.1.1 Two parallel F96 T12 cool white lamps (without baffle or glass),

A2.1.2 One white enamel reflector (without baffle or glass),

A2.1.3 Mount as shown in Fig. A1.

A2.2 Incandescent flood lamp, 500 watts, 120 V, C9 filament, 90° flood spread.

A2.2.1 254-mm (10-in.) aluminum reflector, for use with flood lamp.

A2.2.2 Light shield.

A2.2.3 Position as shown in Fig. A2.

A2.3 Eliminate all light sources except the specified fluorescent lamps and flood lamp.

A3 Walls.

A3.1 It has been the experience of many observers that light reflected from the side walls near the viewing board can interfere with the rating results. It is recommended that the side walls be painted matte black (85° gloss less than 5 units) or that blackout curtains be mounted on both sides of the viewing board to eliminate the reflective interference.

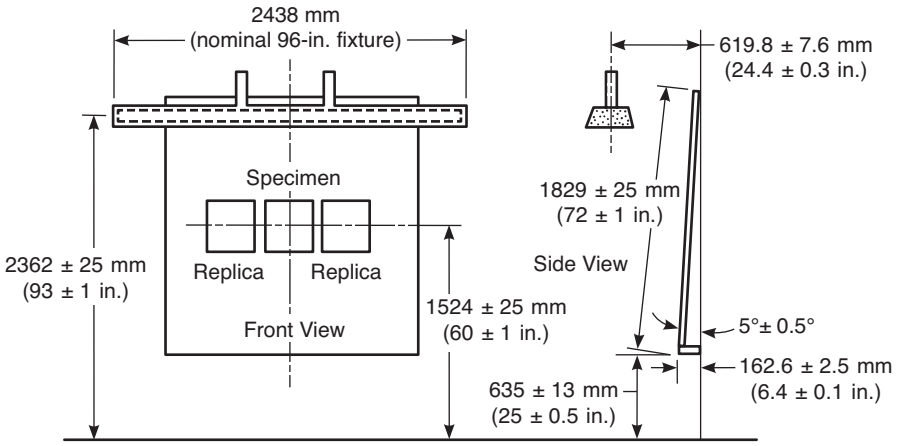


Fig. A1—Evaluation area for crease retention.

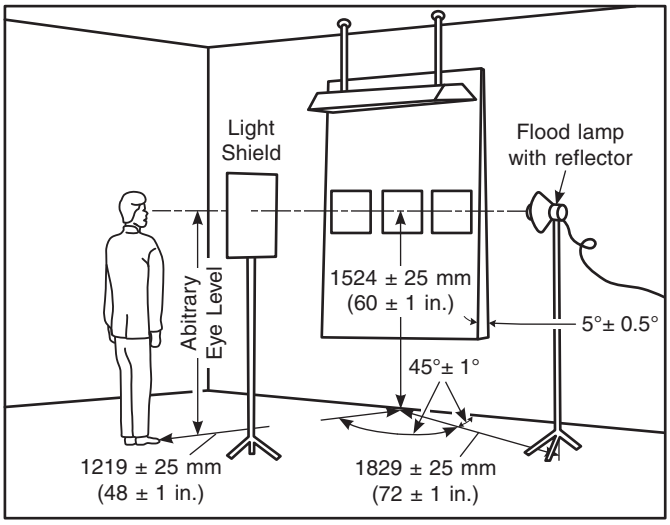


Fig. A2—Lighting and viewing arrangement for crease retention evaluation.

Test Method for Mercerization in Cotton

1. Purpose and Scope

1.1 This test method provides a method for the determination of the presence of mercerization in dyed and undyed cotton yarns and fabrics. In addition, the test will give an indication of the completeness of the reaction between the cotton and the mercerization bath.

2. Principle

2.1 Carefully scoured specimens of the cotton to be tested and unmercerized cotton are immersed in separate baths of barium hydroxide solution for a definite time period. Aliquot portions of each soaking bath and of fresh barium hydroxide solution are then titrated with hydrochloric acid.

2.2 The ratio of the amount of barium hydroxide absorbed by the mercerized specimen to that absorbed by the unmercerized specimen multiplied by 100 gives the barium activity number.

3. Terminology

3.1 **mercerization,** n.—a process for irreversibly altering the physical characteristics and appearance of natural cellulosic fibers by swelling in strong alkali.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 Use chemical goggles or face shield, impervious gloves and an impervious apron during dispensing and mixing of barium hydroxide, sodium carbonate and hydrochloric acid. Concentrated acids should be handled only in an adequately ventilated laboratory hood. CAUTION: Always add acid to water.

4.4 Petroleum solvent can be a combustible or flammable liquid, depending on which solvent is used, and presents a hazard. Ethanol and methanol are flammable liquids. Flammable liquids should be stored in the laboratory only in small containers away from heat, open flame and sparks. These chemicals should not be used near an open flame.

4.4.1 Carry out reflux procedure in a well ventilated hood with an electrical heating mantle or a water bath as the heat source.

4.4.2 Use chemical goggles or face shield, impervious gloves and an impervious apron when handling organic solvents.

4.5 An eyewash/safety shower should be located nearby and an organic vapor respirator and a self-contained breathing apparatus should be readily available for emergency use.

4.6 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities [e.g., Occupational Safety and Health Administration's (OSHA) permissible exposure limits (PEL) as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version]. In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 13.7).

5. Limitations

5.1 The test cannot be used satisfactorily if durable finishes or fibers other than cotton are present.

5.2 Near-infrared spectroscopy (NIR) prediction of barium number may be an alternate method to titration. NIR has the advantages of being fast (values in minutes versus days) and is non-destructive. However, an NIR library must first be established to accurately predict barium number. This library must contain multiple known samples encompassing the barium number range of interest. Typically, these known representatives will be from titrated fabrics.

6. Apparatus

6.1 Burette (preferably automatic) (see 13.4).

6.2 Flask, Erlenmeyer, with reflux tube.

6.3 Flasks, glass-stoppered, 250 mL.

6.4 Flasks, Erlenmeyer, 125 mL.

6.5 Bottles, storage, 250-500 mL.

6.6 Beaker, 1500 mL.

6.7 Pipette, 10 mL.

6.8 Drying oven.

7. Reagents and Materials

7.1 Hydrochloric acid (HCl) standardized approx. 0.1*N*.

7.2 Barium hydroxide reagent [$Ba(OH)_2$] approximately 0.25*N* (see 13.1).

7.3 Phenolphthalein.

7.4 Petroleum solvent [BP 30-60°C (86-140°F)].

7.5 Alcohol (95% ethanol or anhydrous methanol).

7.6 Enzyme, starch-solubilizing.

7.7 Soap, neutral, granular (see 13.2).

7.8 Water, distilled.

7.9 Cotton yarn, unmercerized, for reference (standard cotton) (see 13.3).

7.10 Cotton fabric, unmercerized, for reference (AATCC Crocking squares) (see 13.4).

8. Test Specimens

8.1 A minimum of 5 g of each sample and of the unmercerized standards are scoured as directed after which a 2 g specimen of each scoured sample is weighed and placed in clean, dry glass-stoppered flasks.

9. Procedure

9.1 Scouring. The purpose of the scouring operation is to remove all extraneous matter, leaving the cotton cellulose in as pure a form as possible and without changing it chemically.

9.1.1 The samples to be tested (at least 5 g each), together with the standard unmercerized cotton, are refluxed together successively for 1 h with petroleum solvent [boiling point 30-60°C (86-140°F)], 1 h with alcohol (95% USP ethanol, No. 30 specially denatured alcohol, 95% or anhydrous methanol may be used), and for 1 h with distilled water (see 13.4).

9.1.2 Following these three extractions, starches are removed as follows:

9.1.3 Cover the sample with distilled water containing 3% of a commercial starch-solubilizing malt enzyme solution and heat to 60 ± 5°C (140 ± 9°F). Maintain the solution at this temperature for a period of 1 h. Pour off the enzyme solution, rinse and then scour as follows.

9.1.4 Boil the samples together for 1 h in 1 L of water containing 10 g of a neutral soap and 2 g of soda ash. Wash repeatedly in warm water until free from soap and alkali, i.e., until neutral to phe-

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nolphthalein, squeeze and dry. The samples and the standard unmercerized cotton are dried in an oven at 100°C (212°F) until thoroughly dry. The samples are then allowed to come to room conditions. Each sample should then be cut into small pieces [approximately 3 mm (0.125 in.) square] for subsequent weighing.

9.2 Testing. Prepare and test duplicate specimens from each sample. Weigh 2 g of each scoured sample and of the scoured standard cotton into dry 250-mL flasks equipped with stoppers. (Ground glass stoppers are recommended.) Add 30 mL 0.25*N* barium hydroxide (see 13.5) to each flask containing a test specimen and to two empty flasks for blank determinations. Stopper each flask immediately upon addition of the barium hydroxide and store them for at least 2 h in a water bath at 20-25°C (68-77°F) (room temperature). Shake the flasks at frequent intervals. After 2 h, transfer 10 mL of solution (see 13.6) from each container, including the blanks and titrate with 0.1*N* hydrochloric acid, using phenolphthalein as an indicator.

10. Calculations

10.1 Using the titration figures, determine the ratio of barium hydroxide absorbed by a mercerized specimen to that absorbed by the unmercerized standard. Multiply this ratio by 100 to obtain the barium activity number.

Example: 10 mL of barium hydroxide solution (blank) required 24.30 mL of 0.1*N* HCl; 10 mL of barium hydroxide from an unknown sample of cotton required 19.58 mL of 0.1*N* HCl; 10 mL of barium hydroxide from unmercerized cotton (standard) required 21.20 mL of 0.1*N* HCl. Therefore the barium number of the unknown sample is:

$$\frac{24.30 - 19.58}{24.30 - 21.20} \times 100 = 152$$

10.2 Barium numbers should be run in duplicate and should be reported separately. Duplicate runs should not be off by more than four units. The titrations should be within 0.1 mL for check results. Skilled operators can estimate to within 0.05 mL. A difference of more than four units between duplicate runs indicates inaccuracy in running the test (see Table I).

11. Interpretation

11.1 A barium activity number in the range of 100-105 indicates no mercerization. A barium activity number above 150 indicates substantially complete reaction between the cotton and the mercerizing bath. Intermediate numbers indicate either incomplete reaction or use of a weak mercerizing bath.

Table I—Barium Numbers: Interlaboratory Tests on Mercerized Fabrics

Cloth	Lab A	Lab B	Lab C[1]	Lab C[2]	Lab C[3]
80 × 80—35° Tw	118	118	117	120	114
80 × 80—55° Tw	130	131	128	132	125
108 × 58—55° Tw	141	145	143	143	140
136 × 64—55° Tw	122	123	123	122	120
88 × 50—55° Tw	139	140	136	140	133

[1] Fabrics scoured by Lab C, unmercerized 80 × 80 as standard.
[2] Fabrics scoured by Lab A, unmercerized 80 × 80 as standard.
[3] Fabrics scoured by Lab C, 40/2 combed mercerized yarn as standard.

12. Precision and Bias

12.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

12.2 *Bias*. Mercerization in cotton can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

13. Notes

13.1 The barium hydroxide reagent is prepared by shaking distilled water with slightly more than the calculated quantity of barium hydroxide, allowing it to stand overnight in a stoppered bottle, and then siphoning the clear solution into a clean storage bottle.

13.2 Neutral Granular Soap is no longer available from AATCC. Interlaboratory testing determined that the soap prescribed in BS EN 20105 C10:2006 is a suitable substitute:

Free alkali (calculated as Na_2C0_3): 0.3% maximum.

Free alkali (calculated as NaOH): 0.1% maximum.

Total fatty content: 85.0% minimum.

Titer of mixed fatty acids prepared from the soap: 30°C maximum.

Iodine value: 50 maximum.

Moisture content: 5.0%.

Free from fluorescent brightening agents.

13.3 Unmercerized standard cotton skeins (40/2 ply) as used for the Draves Wetting Tests are particularly satisfactory.

13.4 AATCC Crocking cloth is useful as the unmercerized cotton reference fabric and should meet the specifications as outlined in the notes for AATCC TM8, Test Method for Colorfastness to Crocking: Crockmeter.

13.5 If it is known that all samples that test do not contain finish or starch, then the scouring procedure may begin with the soap-and-soda ash treatment. If one sample requires solvent extraction and the enzyme treatment, then all samples including the standard cotton must be treated together with the complete scouring procedure to insure the same final state for the whole set of samples.

13.6 An automatic burette is most convenient for the addition of the barium hydroxide solution to the specimens. The air outlet must be equipped with an absorption tube containing soda lime to remove carbon dioxide. The latter must not be allowed to enter any burette that might be used, because formation of barium carbonate not only affects the concentration of the reagent but forms a film which interferes with burette readings. A cork is fitted to the bottom of the burette in such a manner that the 250 mL Erlenmeyer flasks containing the specimens under test are locked into place without exposure to air during the addition of barium hydroxide. The barium hydroxide solution should cover the specimens, tilting the flasks if necessary to accomplish this end.

13.7 For the removal of 10 mL of barium hydroxide when equilibrium has been reached, a 10 mL pipette is used. The same pipette, burette, etc., should be used for the whole set of determinations and the same technique for emptying or filling pipettes and burettes should be used on each determination. The hydrochloric acid burette is also equipped with a cork to which the 125 mL flask can be attached during the titration of the 10 mL aliquot portions of barium hydroxide, thereby eliminating titration errors resulting from carbon dioxide absorption by the alkaline solution. In removing the 10 mL aliquot portion of barium hydroxide from flasks containing specimens, the operator should use the end of his pipette to push the cotton against the wall of the flask and to express the excess liquor. In this way, a larger amount of the solution will be available for drawing up into the pipette.

13.8 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

14. History

14.1 Revised in 2019 to add unmercerized reference fabric as an alternative to reference yarn.

14.2 Revised 2017. Editorially revised and reaffirmed 2012. Editorially revised 2010. Editorially revised 2009. Editorially revised and reaffirmed 2008. Reaffirmed 2003. Reaffirmed 1998. Editorially revised and reaffirmed 1994. Editorially revised 1992. Editorially revised 1990. Editorially revised 1988. Editorially revised 1986. Editorially revised and reaffirmed 1985. Editorially revised 1984. Reaffirmed 1980. Reaffirmed 1977. Editorially revised and reaffirmed 1974.

14.3 Developed in 1958 by AATCC Committee RR66, jurisdiction transferred to RA34 in 2009.

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Test Method for Antibacterial Activity of Textile Materials: Agar Plate

1. Foreword

1.1 The Agar Plate Method is a qualitative method to determine antibacterial activity of diffusible antimicrobial agents incorporated into treated articles. This method is an adaptation of a test method described by Ruehle and Brewer (see 16.1) and was originally adopted as an AATCC tentative test method in 1958. RA31 revised this method in 1962 in order to clearly define the objectives and limitations of this method. In 1989, the method was discontinued and removed from the *Manual of International Test Methods and Procedures*. The reason cited for removal was that pertinent aspects of the agar plate method were incorporated in the rewrite of AATCC TM147, Test Method for Antibacterial Activity Assessment of Textile Materials: Parallel Streak Method. In 2011, RA31 updated and reactivated the method in part because ASTM cites this method in their standard ASTM E1115, Standard Test Method for Evaluation of Surgical Hand Scrub Formulations, and because the agar plate method was found to be beneficial when testing odd-shaped or non-flat test samples, such as batting, fiber fill, and yarn, as well as non-textile materials such as foam or plastics. This method is used to demonstrate bacteriostatic activity by the diffusion of the antibacterial agent through agar. Molten agar is seeded with the test organisms and allowed to become semi-solid or solid depending on the test specimen to be tested. The test specimen is then placed in intimate contact with the agar surface. An advantage of the Agar Plate Method is that a non-embedded test specimen can be removed to determine inhibition in the contact area without fear of removing the test organism with the test sample. Both gram-positive and gram-negative bacteria are tested.

2. Purpose and Scope

2.1 The objective is to detect bacteriostatic activity on samples that are treated with antimicrobials and are capable of producing a zone of inhibition.

2.2 Some test samples do not have a flat surface and are not easily tested using AATCC TM147. This method provides the advantage of testing odd-shaped and non-flat samples such as batting, fiber fill, foam, plastic, yarn, etc. by partially embedding them into molten agar prior to solidification. This provides the intimate contact required between the test sample and the seeded agar.

3. Principle

3.1 Test samples, including corresponding untreated controls of the same material if available, are placed in intimate contact with agar which has been previously seeded with an inoculum of a test bacterium. After incubation, a clear area of interrupted growth underneath and/or adjacent to the test sample indicates antibacterial activity of the specimen. A standard strain of bacteria is used that is specific to the requirements of the sample under test. If no other bacterial species is specified, *Staphylococcus aureus* and *Klebsiella pneumoniae* may be used as representative gram-positive and gram-negative organisms. Other recommended strains can be used.

4. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

4.1 AATCC TM147, Test Method for Antibacterial Activity of Textile Materials: Parallel Streak (see 16.2)

4.2 ASTM E1115, Standard Test Method for Evaluation of Surgical Hand Scrub Formulations (see 16.3)

4.3 ASTM E1402, Standard Guide for Sampling Design (see 16.3)

4.4 Centers for Disease Control. Biosafety in Microbiological and Biomedical Laboratories (see 16.4).

5. Terminology

5.1 **activity,** n.—*of an antibacterial agent*, a measure of effectiveness of the agent.

5.2 **antibacterial agent,** n.—any chemical which kills bacteria (bactericide) or interferes with the multiplication, growth, or activity of bacteria (bacteriostat).

5.3 **zone of inhibition,** n.—clear area of no visible growth of a microorganism, cultured onto the surface of an agar growth medium, in proximity to the borders of a specimen placed in direct contact with this agar surface.

NOTE: A zone of inhibition occurs as a result of the diffusion of an antimicrobial agent from the specimen.

6. Safety Precautions

6.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

6.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

6.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g. OSHA standards and rules).

6.4 See 4.4, 9.1, 12.1 for additional safety precautions and personnel training related to handling bacteria.

7. Uses and Limitations

7.1 The method is not suitable for materials which tend to encapsulate and prevent the diffusion of the antibacterial agent. Not suitable for materials that contain substances that may neutralize the antibacterial agent. Not suitable for antimicrobial finishes that will react with the agar.

7.2 Being a qualitative method, the efficacy of the antimicrobial is not determined by zone size.

7.3 Some standard textile finishing chemicals, especially crease resistant and permanent-press reagents, will give antibacterial activity. To determine whether the antibacterial activity is due to the antibacterial agent, test a specimen of the same material treated in exactly the same way with whatever other finishing agents were used, but without the antibacterial agent.

8. Apparatus (see 16.5)

8.1 Incubator maintained at 37 ± 2°C (99 ± 4°F).

8.2 Water bath maintained at 45-50°C (113-122°F).

8.3 Bunsen Burner or equivalent.

8.4 Stereomicroscope, minimum 40x magnification.

8.5 Autoclave.

8.6 Vortex or equivalent.

9. Materials (see 16.5)

9.1 Bacteria species suitable for the intended end-use of the test sample. Some

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bacteria are pathogenic (i.e., capable of infecting humans and producing disease). Take every necessary and reasonable precaution to eliminate risk to laboratory personnel and others in the associated environment. Sterilize all contaminated samples and test materials prior to disposal. Any or all the following bacteria may be used:

9.1.1 Gram-positive: *Staphylococcus aureus*, ATCC 6538, CIP 4.83, DSM 799, NBRC 13276, NCIMB 9518 or equivalent strain (see 16.6).

9.1.2 Gram-negative: *Klebsiella pneumoniae*, ATCC 4352, CIP 104216, DSM 789, NBRC 13277, NCIMB 10341 or equivalent strain (see 16.6).

9.1.3 Other species agreed upon by the interested parties.

9.2 Broths/agars; Nutrient broth/agar, Trypticase Soy broth/agar, Brain-Heart Infusion broth/agar, Müller-Hinton broth/agar or other capable of growing and maintaining the chosen bacterial test organism.

9.3 Distilled or deionized water.

9.4 Inoculating loop.

9.5 Pipets, 1 mL, sterile.

9.6 Culture Tubes with caps; minimum 10 mL capacity, sterile

9.7 Petri dishes, 100 mm diam. × 15 mm deep, sterile.

9.8 Forceps, sterile.

9.9 Ruler.

10. Sampling

10.1 Test results are valid only when the samples are statistically representative (see ASTM E1402).

10.2 Sampling must be random. Every unit of product must have a mathematically equal chance of becoming a sample; and every portion of each sample must be equally likely to become a test specimen.

10.3 All specimens must be alike within the variations due to pure chance. There must be no differences within samples assignable to known causes.

11. Specimens

11.1 Test specimens are cut by hand or with a die. Any convenient size can be used if the material is odd-shaped. Standard samples are circular with a diameter of 2.5 cm (1 in.).

11.2 One replicate is needed. Additional test specimens may be used to increase the precision of the average.

12. Procedure

12.1 This test should be performed only by trained personnel. Consult the U.S. Department of Health and Human Services publication *Biosafety in Microbiological and Biomedical Laboratories* (see 4.4).

12.2 Prepare the test organism by taking a sterilized inoculation loop and inoculating separate culture tubes of the appropriate broth with the organism to be used. Incubate the cultures at 37 ± 2°C (99 ± 4°F) for 18-24 h.

12.3 Mix the culture by vortex or equivalent method. Pipette 1.0 mL of each challenge organism that has been incubated for 18-24 h into separate 150 mL portions of sterile, molten agar maintained in a 45-50°C (113-122°F) water bath. An equivalent ratio would be 0.1 mL of test organism to 15 mL of molten agar. Gently swirl manually (vortexing will cause unnecessary bubbles) to achieve even distribution of the bacteria within the molten agar. For each organism, aseptically transfer 15 mL of the inoculated agar into separate petri dishes. Allow the agar to cool to a semi-gelatinous state for irregular test specimens that need to be embedded in the agar for intimate contact and to a solid state for flat surface regular samples.

12.4 Using flame-sterilized or autoclave-sterilized forceps that have been cooled, gently press the test specimen into the agar for embedded test specimens and on top of the agar for non-embedded test specimens. For embedded test samples, aseptically pour additional seeded agar containing the same organism into the plate to achieve a depth of approximately 6 mm.

12.5 Cover the plate with the Petri dish lid and allow the agar to solidify. Incubate at 37 ± 2°C (99 ± 4°F) for 18-24 h.

NOTE: Care must be taken to use sufficient agar to contact the sample, but not to deeply embed the test specimen and obscure the observation of any zone of inhibition.

13. Evaluation

13.1 Examine the incubated plates for a growth-free zone around the test specimen and for clearing directly beneath the test sample. Avoid parallax by looking directly down on the plate bottom. If a zone is not observed with the naked eye, then the test samples should be placed under a stereomicroscope at a minimum magnification of 40X to detect any small zones or colonies.

13.2 Calculation of zone of inhibition size is optional. The average width of a zone of inhibition on either side of the test specimen may be calculated using Eq. 1:

$$W = (T - D)/2$$

where:

W = width of clear zone of inhibition in mm;

T = total diameter of test specimen and clear zone in mm; and

D = diameter of the test specimen in mm.

13.3 The size of the zone cannot be construed as a quantitative evaluation of antibacterial activity. Treated samples should be compared to a corresponding untreated sample and a sample specimen with known bacteriostatic activity. The criterion for passing the test must be agreed upon by the interested parties.

14. Report

14.1 Describe or identify the sample tested.

14.2 Report that the sample was tested using AATCC TM90-2022.

14.3 Report the testing conditions:

14.3.1 Size of test specimen used.

14.3.2 Number of test replicates used.

14.3.3 If the test specimen was sterilized prior to testing and the method of sterilization.

14.3.4 Agar media used.

14.3.5 Incubation time.

14.3.6 Organism used.

14.4 Report the test results:

14.4.1 Any observed zone of inhibition.

14.5 Describe any modification(s) of the published standard.

15. Precision and Bias

15.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

15.2 *Bias*. The antimicrobial activity can be defined on in terms of a test method. There is no independent method for determining the true value. As a means of evaluating this property, the method has no known bias.

16. Notes

16.1 Ruehle, G. G. A. and Brewer, C. M., United States Food and Drug Administration Methods of Testing Antiseptics and Disinfectants. *U.S. Dept. Agric. Circular, 198, 20pp* (1931).

16.2 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org

16.3 Available from ASTM, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA, 19428-2959, USA; Tel: +1.610.832.9500, Fax: +1.610.832.9555, service@astm.org.

16.4 Available from CDC. Website: https://www.cdc.gov/labs/BMBL.html.

16.5 For potential apparatus, reagents or materials sources, visit the AATCC Buyer's Guide at www.aatcc.org/bg. AATCC provides its Corporate members the option to list their items and services. AATCC does not qualify, or in any way approve, endorse or certify that any of the listings meet the specifications in its standards.

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16.6 ATCC is the American Type Culture Collection (USA), P.O. Box 1549, Manassas VA 20108; tel: +1.703.365.2700; fax: +1.703.365.2701; web site: www.atcc.org; CIP is the Pasteur Institute Collection (France); DSM is the German Collection of Microorganisms and Cell Cultures (Germany); NBRC is the NITE Biological Resource Center (Japan); and NCIMB is the National Collection of Industrial Bacteria (UK). Equivalent bacteria strains obtained from agencies of the World Federation of Culture Collection (WFCC) may be used by agreement between the interested parties. The strains used in the test shall be documented with their supply source.

16.7 Consistent and accurate testing requires maintenance of a pure, uncontaminated, non-mutant test culture. Avoid contamination by using good sterile technique in plating and transferring. Avoid mutation by strict adherence to monthly stock transfers. Check culture purity by making streak plates periodically and observing for a single species-characteristic type of colonies.

17. History

17.1 Revised in 2022 to clarify and align with prescribed AATCC style guidelines.

17.2 Editorially revised (with title change) 2019. Reaffirmed 2016. Reactivated/revised 2011. Revised 2010. Removed 1989. Editorially revised 1986, 1985, 1982. Reaffirmed 1977. Editorially revised 1974, 1972, 1971. Reaffirmed 1970. Revised 1965, 1962.

17.3 Developed in 1958 by AATCC Committee RA31.

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Test Method for Chlorine, Retained, Tensile Loss: Single Sample

1. Purpose and Scope

1.1 This is an accelerated test method to determine the potential damage caused by retained chlorine.

1.2 It has been designed for use on cotton and rayon fabrics, but may be used on any fabric or finish that is not damaged by heat alone (see 11.1). Fabrics must be appropriate for tensile strength testing (typically woven fabrics).

2. Principle

2.1 The fabrics are treated in sodium hypochlorite solution, rinsed, dried and pressed between hot metal plates. The damaging action of the retained chlorine is calculated from the difference in tensile strength before and after pressing.

3. Terminology

3.1 **retained chlorine,** n.—*in textiles bleached with chlorine-type bleaches*, available chlorine that remains in the material after washing and drying.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Users MUST consult manufacturers for specific details such as safety data sheets and other recommendations. Consult and follow all applicable OSHA standards and rules.

4.1 Follow good laboratory practices. Wear safety glasses in all laboratory areas.

4.2 Follow manufacturer's safety recommendations when operating laboratory testing equipment.

4.3 Handle all chemicals with care. Use chemical goggles or face shield, impervious gloves and an impervious apron when handling sodium carbonate and sodium hypochlorite bleach solutions; these are corrosive chemicals.

4.4 Ensure an eyewash/safety shower is located nearby and a self-contained breathing apparatus is readily available for emergency use.

4.5 Control exposure to chemicals used in this procedure at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 11.2).

5. Apparatus and Reagents (see 11.3)

5.1 Beakers, 800 mL.

5.2 pH meter (see 11.4).

5.3 Constant temperature bath (or suitable alternative means of controlling temperatures).

5.4 Wringer, laboratory or household type.

5.5 Heating device, providing even heat transfer to the specimen from the top by close contact at a controlled temperature and giving a pressure on the specimen of 8.8 g/cm^2 (see Fig. 1 and 11.5).

5.6 Tensile-strength tester.

5.7 Sodium hypochlorite stock solution having an available chlorine content of approximately 5.0%.

5.8 Water, distilled.

6. Preparation of Specimens

6.1 Normally, only warp-wise specimens are tested. Cut a swatch approximately 35.6 cm (14 in.) in the warp-wise direction and 20.3 cm (8 in.) in the filling-wise direction (if testing filling specimens, reverse these dimensions). Specimens will be cut from this swatch after chlorination.

7. Preparation of Reagents

7.1 Prewetting and Rinse Solutions. Determine the pH of the distilled water; if outside the range of 6-7, report the actual pH (see 11.6).



Fig. 1—Heating device.

7.2 Chlorination Solution. Prepare a chlorination solution containing 0.25% available chlorine at a pH of 9.5 ± 0.1 as follows:

7.2.1 Determine the percent available chlorine (Cl) of the sodium hypochlorite stock solution (see 11.7).

7.2.2 Calculate the number of grams (g) of stock solution required to make one liter of a 0.25% available chlorine solution using Eq. 1:

$$g = (1000 \times 0.25)/Cl \qquad \text{Eq. 1}$$

where:

g = grams of stock solution required.
Cl = % available chlorine determined.

7.2.3 To 900 mL of distilled water add the calculated grams (g) of stock solution required. Adjust the pH to 9.5 ± 0.1 by the addition of either sodium bicarbonate to lower the pH or sodium carbonate to raise it. Finally, add distilled water to make 1 L and recheck the pH.

8. Procedure

8.1 Because many factors, such as pH, concentration and time, significantly influence the amount of damage caused by retained chlorine, it is *important* to follow closely the conditions of the test. Should any deviations be necessary, include details in the test. As a means of checking the test procedure, it is good practice to run, at the same time, a cotton fabric whose chlorine-retentive properties have been determined previously.

8.2 Chlorination Step.

8.2.1 Prewetting Bath. Measure into an 800-mL beaker a volume of distilled water equal to 50 times the dry weight of the fabric sample. Immerse the sample for 3 min at 71 ± 3°C (160 ± 5°F), stirring frequently. Remove sample from bath, allow to drain and cool at room temperature.

8.2.2 Chlorination Bath. Place the swatch in an 800-mL beaker containing the chlorination solution (equal to 50 times the dry weight of the fabric) that is maintained at a temperature of 25 ± 1°C (77 ± 2°F). Agitate the swatch frequently and gently with a glass stirring rod during the 15-min chlorination period. At the end of 15 min, remove swatch, allow to drain momentarily, and pass through a wringer to remove as much solution as possible using care to keep the swatch flat and free from wrinkles. Rinse equipment free of chlorination solution with distilled water to avoid contamination of

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specimens during the subsequent rinsing procedure.

8.2.3 Rinsing. Immerse the swatch in an 800-mL beaker containing distilled water (equal to 50 times the weight of the swatch) at a temperature of 21-32°C (70-90°F) for 2 min, agitating the swatch gently and continually. Remove the swatch from the bath, allow to drain momentarily, and then pass through a wringer, as before, taking all precautions mentioned previously.

8.2.4 Repeat this rinsing procedure five (5) more times, making a *total of six* (6) rinses. For greater precision, rinse *all swatches separately*, to avoid danger of contamination.

8.2.5 Drying. *Air-dry* swatches. Hang on a line or dry flat on a noncorrosive rack, away from heat, until just dry (do *not* press!) Transfer at once to a conditioned atmosphere at 65 ± 5% RH and 21 ± 2°C (70 ± 4°F) and allow to remain through the scorching step and tensile strength testing.

8.3 Scorching Step.

8.3.1 Prepare warp specimens by carefully cutting five (5) strips approximately 35.6 × 3.2 cm (14.0 × 1.25 in.). Ravel the strips to exactly 2.54 cm (1.0 in.) wide following ASTM D5035, Test for Breaking Force and Elongation of Textile Fabrics (Strip Test). Then trim the 35.6 cm (14.0 in.) length to 30.5 cm (12.0 in.). From this set of five 2.54 × 30.5 cm (1.0 × 12.0 in.) strips, make two sets (from the same set of warp yarns) of 2.54 × 15.2 cm (1.0 × 6.0 in.) raveled strip specimens by cutting the strips in two. These sets should be kept separate; one set is to be scorched, the other used as a control. The total conditioning time should be not less than 4 nor more than 24 h.

8.3.2 Prepare heating device in advance so that the heating plates of the instrument maintain a temperature of 185 ± 1°C (365 ± 2°F). If necessary, to avoid air currents, enclose instrument in a cabinet. Make sure the two heating plates are clean and in good adjustment (uniformly in contact at all points). Scorch each strip of one set separately (one at a time) by placing the 2.54 × 15.2 cm (1.0 × 6.0 in.) specimen in the heating device with the long direction of the raveled strip perpendicular to the long direction of the heating plates so that it is scorched across the center of the raveled strip (see Figs. 1 and 2). Scorch the strips for 30 s. Check reading of the thermometers frequently during use. Condition the specimens again; this time conditioning at least 16 h before making tensile strength measurements.

8.4 Tensile Strength.

8.4.1 Make tensile strength tests on the unscorched and scorched specimens and record the individual values. From those, calculate the average tensile strength for each set.

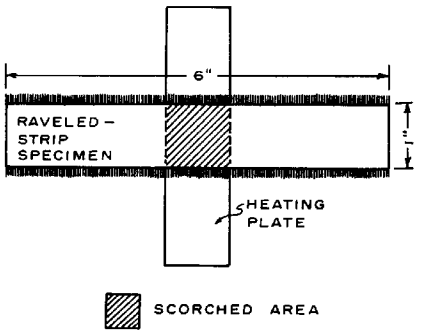


Fig. 2—Specimen on heating plate.

9. Calculation of Damage Caused by Retained Chlorine

9.1 Use the formula as follows:

$$(Tc - Tcs)/Tc \times 100 =$$

% loss in tensile strength due to damage caused by retained chlorine.

where:

Tc = average tensile strength of chlorinated specimens, unscorched, and

Tcs = average tensile strength of chlorinated specimens, scorched.

10. Precision and Bias

10.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

10.2 *Bias*. The retained chlorine as determined by tensile loss-single sample method can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

11. Notes

11.1 Where it is suspected that either fabric or finish is susceptible to damage on heating, this susceptibility can be determined by running a distilled water control. That is, the fabric is run through the entire procedure, but using distilled water in place of the hypochlorite solution. The loss in strength due to heat alone is calculated by the equation:

$$(Tw - Tws)/Tw \times 100 =$$

% tensile strength loss due to scorching alone.

where:

Tw = average tensile strength of the water-treated specimens, unscorched, and

Tws = average tensile strength of the water-treated specimens, scorched.

If this loss is appreciable, there is some question as to whether the chlorine damage test is applicable.

It is usually not necessary to determine the effect of wet chlorination on the fabric, as this factor is canceled out in the calculation for chlorine damage. However, where it is desirable to determine the effect of the wet chlorination step, this may be calculated as follows:

$$(Tw - Twc)/Tw \times 100 =$$

% tensile strength loss due to wet chlorination.

where:

Tw = average tensile strength of the water-treated specimens, unscorched (handled as above), and

Tc = average tensile strength of the original fabric chlorinated and unscorched.

11.2 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240, USA; +1. 513.742.2020; www.acgih.org.

11.3 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

11.4 Any standard laboratory pH meter having suitable provision for high pH readings may be used. Colorimetric methods cannot be used with NaOCl.

11.5 A pair of electrically heated hot plates whose temperature is accurately controlled and which can be adjusted to provide a pressure on the specimen of 8.8 g/cm^2. To obtain the specified pressure of 8.8 g/cm^2 the four (4) spring-loaded pins must contact the upper housing so that they counterbalance the proper proportion of the weight of the upper housing to provide the specified pressure on the test specimen. Other similar devices providing identical test conditions and results may be used.

11.6 There is reason to believe that the pH of the prewetting and rinsing solutions may influence the results obtained in this test method. However, in view of the widespread variation of pH in commercial practice, AATCC has not fixed a specific value for the test procedure.

11.7 For determining the percent available chlorine, pipette a 1.00-mL portion of sodium hypochlorite solution into an Erlenmeyer flask and dilute to 100 mL with distilled water. Add 20 mL of 6*N* H_2SO_4 and 6 mL of 12% KI. Titrate with 0.1*N* sodium thiosulfate solution.

Calculation:

% available Chlorine =

$$\frac{\text{mL Na thiosulfate} \times 0.1N \times 0.0355}{1\text{ mL} \times \text{sp gr NaOCl soln}} \times 100$$

12. History

12.1 Revised in 2019 to align format with AATCC style.

12.2 Reaffirmed 1962, 1967, 1971, 1977, 1980, 1989, 1999, 2009, 2013; editorially revised 1974, 1988, 1992, 2008, 2010, 2016; editorially revised and reaffirmed 1985, 1994, 2004.

12.3 Developed in 1958 by AATCC Committee RR35. Maintained by RA99.

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Test Method for Abrasion Resistance of Fabrics: Accelerotor

1. Purpose and Scope

1.1 This test method is intended for evaluating the resistance of fabrics and other flexible materials to abrasion (see 14.1).

2. Principle

2.1 An unfettered fabric specimen is driven by an impeller (rotor) along a zigzag course in a generally circular orbit within a cylindrical chamber, so that it repeatedly impinges on the walls and abradant liner of the chamber while at the same time being continually subjected to extremely rapid, high velocity impacts. The specimen is subjected to flexing, rubbing, shock, compression, stretching and other mechanical forces during the test. Abrasion is produced throughout the specimen by rubbing of yarn against yarn, fiber against fiber, surface against surface and surface against abradant.

2.2 Evaluation is made on the basis of weight loss of the specimen or grab strength loss of the (woven) specimen when broken at an abraded fold line. Generally, flat woven fabrics may be evaluated by either method. Tufted and other fabrics with raised surfaces, and knit fabrics, are evaluated by the weight loss method.

2.3 Changes in other fabric characteristics may also be useful for evaluating abrasion resistance with the Accelerotor (see 14.1).

3. Terminology

3.1 **abrasion,** n.—the wearing away of any part of a material by rubbing against another surface.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 The door of the Accelerotor must be kept closed whenever the motor is running. Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

4.3 A face mask is recommended for operators who work regularly with the Accelerotor to guard against inhalation of fiber dust.

5. Uses and Limitations

5.1 The results obtained by this method are affected by duration; the size, shape and angular velocity of the rotor; and by the type of liner used. These effects are interrelated, and testing conditions may be varied to produce different degrees of abrasion in a test specimen. For example, it may require only 2 or 3 min at 209.44 rad/s (2000 rpm) with the standard offset rotor to produce a reasonable degree of abrasion in a delicate or fragile fabric, while a heavier or more durable fabric may require 6 min at 314.16 rad/s (3000 rpm).

5.2 When fabrics curl or, for other reasons, do not move about freely in the test chamber, testing should be discontinued.

5.3 The results of this test should not be equated with service life.

6. Apparatus and Materials

6.1 Accelerotor (Fig. 1; see 14.3) equipped with the following parts:

6.1.1 Rotor, offset (elongated S-shape), 114 mm (4.5 in.) (Fig. 2) (see 14.4 for descriptions of alternate rotors).

6.1.2 Collar insert, plastic, lined with 3.2 mm (0.125 in.) polyurethane foam.

6.2 Liner, fine abrasive, 500J grit aluminum oxide cloth (see 14.3; an alternate abrasive liner is described in 14.5).

6.3 Neon lamp or other stroboscopic device.

6.4 Timer, automatic, accurate to ± 1 s.

6.5 White glue.



Fig. 1—Accelerotor fitted with abrasive liner over foam rubber cushion and with 114 mm (4.5 in.) S-shape rotor.



Fig. 2—Elongated S-shape rotor.

6.6 Pinking shears and marking templates, or cutting dies (see 14.8).

6.7 Brush, nylon, for cleaning Accelerotor chamber or small portable vacuum cleaner for chamber and specimens.

6.8 Thread, size E, Type I, Class 1 or 2, Fed. Spec. V-T-295.

6.9 Cotton fabric, 78 ends per in. × 76 picks per in. print cloth (carded yarns) 1.2 m (46 in.) wide, about 8 m/kg (4 yd/lb), bleached and desized, free from blueing, optical bleach or finishing materials.

6.10 Analytical balance accurate to ± 0.001 g.

7. Test Specimens

7.1 In the absence of applicable specifications, take a minimum of three specimens from each sample to be tested not nearer the selvage than one-tenth the width of the fabric or 64 mm (2.5 in.). Select specimens that are representative of the whole sample to be tested.

7.2 Size of Specimens.

7.2.1 Method A (Evaluation by weight loss)—Cut specimens of heavier or bulkier fabrics smaller than those of lighter fabrics to minimize variations in impact produced when the specimens hit the chamber wall or abradant liner. Table I is a guide to the relationship between fabric weight, in grams per square meter (ounces per square yard), and specimen size. Specimens may be taken with the edges parallel to the warp and filling (courses and wales) or on the bias.

7.2.2 Method B (Evaluation by strength loss)—The specimen size is 100 × 150 mm (4 × 6 in.) with the greater dimension in the direction of the yarns to be broken. Take specimens with their edges parallel to the warp and filling yarns.

7.3 Specimen Preparation.

7.3.1 Method A—Cut a specimen with pinking shears or die (see 14.8). If woven fabric is die-cut parallel to yarns, ravel 3.2 mm (0.125 in.) along each edge. Place the specimen on plastic sheet (to protect bench top) and apply a 3.2 mm (0.125 in.) strip of adhesive to each cut or raveled edge (Fig. 3) (see 14.4 and 14.9). Allow the adhesive to dry at room

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Table I—Selection of Specimen Size

Weight Range of Fabrics[a]	Size of Specimens[b]
300-400 (9-12)	95 (3.75)
200-300 (6-9)	115 (4.5)
100-200 (3-6)	135 (5.25)
less than 100 (3)	150 (6)

[a] Grams per square meter (ounces per square yard)
[b] Millimeters (inches) square

temperature.

7.3.2 Method B—Cut specimens 100 × 300 mm (4 × 12 in.) (twice the length required for the grab breaking strength test). Number each specimen at both ends, and then cut in half. Use one half for a control to determine the original grab breaking strength, and the other to determine grab breaking strength after abrading. Apply adhesive to the edges of the specimens to be abraded as in Method A. Then fold each specimen perpendicular to the long dimension and 50 mm (2 in.) from one end, making it into a 100 × 100 mm (4 × 4 in.) square. Attach the 50 × 100 mm (2 × 4 in.) flap with a seam 6 mm (0.25 in.) from its edge to the body of the specimen (Fig. 4). Use 4 stitches to the centimeter (11 stitches per inch).

8. Accelerotor

8.1 Adjustment of Tachometer.

8.1.1 Rotor—Select and install appropriate rotor. The 114 mm (4.5 in.) offset rotor is standard) (see 14.4).

8.1.2 Neon Lamp—To check the accuracy of the tachometer, the neon lamp is used as a simple stroboscope to view the spinning rotor. With the test chamber door closed and the neon bulb held close to the window of the door, the rotor gives distinct patterns at several useful speeds. With some practice, the following patterns will be recognized: If the neon lamp is being operated on 60 Hz AC, at 188.50 rad/s (1800 rpm) the rotor appears as a stationary distinct two-bladed figure. At 377 rad/s (3600 rpm) the hub of the rotor appears as a stationary blur, with two slight lobes apparent on the sides of the hub. If the neon lamp is operated on 50 Hz AC, as in many European countries, the above described test patterns will appear at 157.08 and 314.16 rad/s (1500 and 3000 rpm). If the tachometer does not read the appropriate speed, turn the small screw on the dial face to correct it.

8.1.3 Stroboscope—Set the stroboscope dial at 314.16 rad/s (3000 rpm). Close the Accelerotor door, turn the unit on, and bring the speed of the rotor to a point at which it appears as a stationary, two-bladed figure. If the tachometer does not read 314.16 rad/s (3000 rpm), turn the small screw on its dial face to correct it.

8.2 Liners (see 14.5).

8.2.1 Installation of Liners—Place liner into collar insert and, with the fingers, work liner around the insert wall until it fits snugly and smoothly without any crease.

8.2.2 Break-in of Fresh Liner—Insert the prepared collar insert into the Accelerotor equipped with the selected rotor. Break in the abrasive liner by running two successive 114 mm (4.5 in.) square specimens of 78 × 76 finish-free cotton print cloth (see 6.9), edged with adhesive.

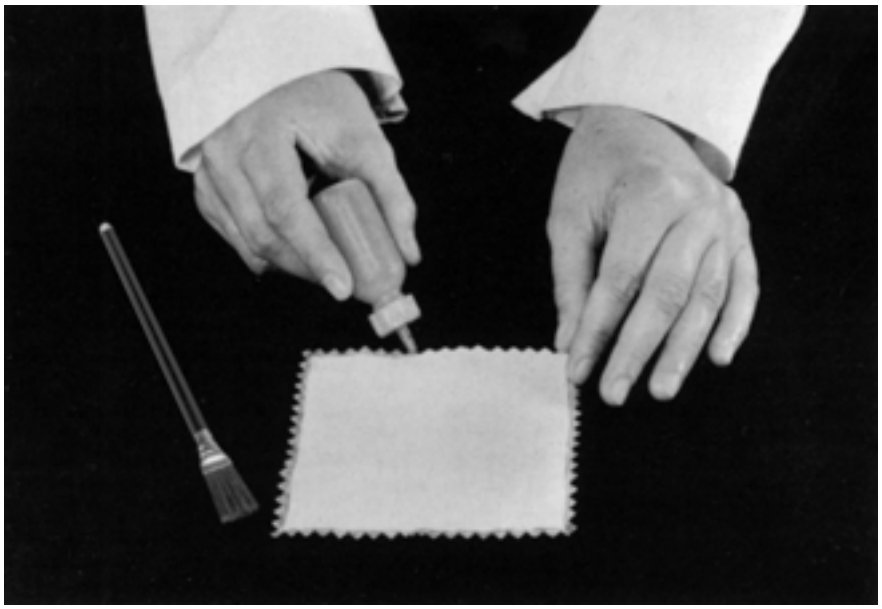


Fig. 3—Application of adhesive (see 6.5) to pinked edges of specimen from plastic squeeze bottle.

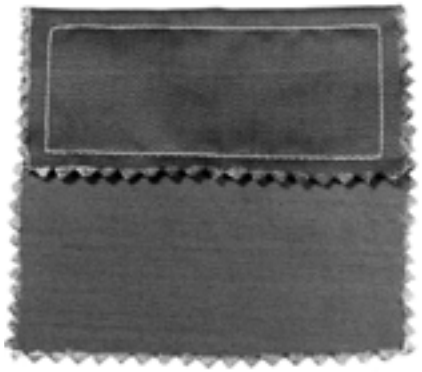


Fig. 4—Specimen prepared for testing by Method B.

Close the door, start the Accelerotor, and maintain rotor speed at 314.16 rad/s (3000 rpm) for 6 min. Replace with the second specimen and continue until total running time is 12 min. Stop the Accelerotor and remove the specimen. Brush or vacuum the abrasive liner to remove detritus.

8.2.3 Reversal of Line—For greater reproducibility, it is recommended that after six test specimens have been run, the collar assembly be removed from the Accelerotor and reversed so that the rim which was next to the door is placed at the back of the chamber.

8.2.4 Change of Liner—It is recommended that an abrasive liner be discarded after 12 specimens have been run. If the liner has not been subjected to severe duty, it may be used for more than 12 tests. To check the condition of a liner, determine the weight loss of finish-free 78 × 76 cotton print cloth (see 6.9) at the beginning of a series of tests and after intervals of six test specimen runs. The life of some abrasive liners can be extended to a considerable degree by laying them on a flat surface and scrubbing them with the nylon brush and soapy water to remove finishes or other materials deposited on them by some fabrics. After washing, the liners should be thoroughly rinsed with water and dried before their effectiveness is rechecked.

9. Conditioning

9.1 Let prepared specimens reach approximate moisture equilibrium in the standard atmosphere for preconditioning and then condition them in the standard atmosphere for testing textiles, as specified in ASTM D1776, Standard Practice for Conditioning and Testing Textiles.

10. Testing Procedures

10.1 Method A—Weight Loss

10.1.1 Weigh the conditioned specimen (see 9) on an analytical balance (see 6.10).

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10.1.2 Set up Accelerotor with the elongated, S-shaped, offset, 114 mm (4.5 in.) long rotor, and a fine abrasion, 500J grit aluminum oxide cloth (see 14.4, 14.5 and 14.7).

10.1.3 Crumple the test specimen and place it in the chamber.

10.1.4 Close the door (see 4), start the Accelerotor and timer, and maintain accurately, ± 10.48 rad/s (100 rpm), at the selected speed for the desired time. The operating speed usually is maintained at 314.16 ± 10.48 rad/s (3000 ± 100 rpm) for a period of time sufficient to abrade the specimen substantially without tearing it; e.g., 2-6 min (see 14.7).

10.1.5 Stop the Accelerotor at the end of the time (± 2 s) and remove the test specimen.

10.1.6 Brush or vacuum the liner to remove detritus.

10.1.7 Shake or vacuum the specimen free of detritus.

10.1.8 Condition the tested specimen (see 9.1).

10.1.9 Reweigh the specimen on an analytical balance to ± 0.001 g.

10.2 Method B—Strength Loss

10.2.1 Perform Steps 10.1.2 through 10.1.7.

10.2.2 Remove stitching thread to restore specimen to 100 × 150 mm (4 × 6 in.) dimensions.

10.2.3 Condition specimen as directed in 9.1.

10.2.4 Determine breaking strength by the Grab Method given in ASTM D5034, Test for Breaking Force and Elongation of Textile Fabrics (Grab Test), placing the abraded fold of the test specimen parallel to and equidistant from the jaws of the tensile tester (see 14.6). The specimen must break along the fold for a valid test.

10.2.5 Determine breaking strength on conditioned (see 9.1) original (unabraded) (see 7.3.2) specimen.

11. Calculations and Evaluation

11.1 Method A—Weight Loss. Calculate the percent loss in weight for each specimen to ± 0.1%.

11.2 Method B—Strength Loss. Calculate the percent strength loss for each pair (see 7.3.2) of specimens.

11.3 Calculate the average for each method.

12. Report

12.1 Method A—Report the average percent weight loss for the three test specimens.

12.2 Method B—Report the average percent strength loss for the three test specimens.

12.3 The report must state the exact conditions employed; i.e., rotor speed, time, size and type of rotor and type of liner. For Method A also state the specimen size used.

13. Precision and Bias

13.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

13.2 *Bias*. The abrasion resistance of fabrics (Accelerotor method) can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

14. Notes

14.1 Although they are not included as part of the standard procedure, changes in such characteristics as air permeability, light transmission, visual appearance, hand, etc., could be used for evaluation, depending on the type of fabric and its intended end-use.

14.2 For additional information see: T. F. Cooke, Abrasion Testing with the Accelerotor: Reproducibility in Interlaboratory Tests, *American Dyestuff Reporter*, Vol. 47, No. 20, 1958, pp679-683; H. W. Stiegler, H. E. Glidden, G. J. Mandikos and G. R. Thompson, "The Accelerotor for Abrasion Testing and Other Purposes," *American Dyestuff Reporter*, Vol. 45, No. 19, 1956, pp685-700.

14.3 Available from SDL Atlas L.L.C., 3934 Airway Dr., Rock Hill, SC 29732-9200, tel: +1.803.329.2110, fax: +1.803.329.2133, e-mail: info@sdlatlas.com; web site: www.sdlatlas.com.

14.4 For special applications, pitched-blade rotors 108 mm (4.25 in.), 114 mm (4.50 in.) and 121 mm (4.75 in.) long and a 108 mm (4.25 in.) offset S-shaped rotor are available (see 14.8).

14.5 For special applications, a medium fine abrasive, 240J grit aluminum oxide cloth liner is available (see 14.8). The liner is installed the same way as the fine abrasive, 500J grit aluminum oxide cloth (see 8.2).

14.6 Tensile strength tester as described in ASTM D76, Standard Specifications for Tensile Testing Machines for Textiles.

14.7 By mutual agreement, the rate and type of abrasion produced in the Accelerotor may be varied by the alternate use of the medium fine abrasive, 240J grit aluminum oxide cloth liner, the use of the shorter S-shaped rotor or the use of one of the 0.26 rad (15 deg.) pitched-blade rotors, and variations of the rotor speed from 157.08 ± 10.48 to 418.88 ± 10.48 rad/s (1500 ± 100 to 4000 ± 100 rpm), but details of these must be made a part of the report of test results.

14.8 It is convenient to mark specimens for cutting by means of square templates made of metal, plastic or cardboard. If available, cutting dies of appropriate dimensions may also be used; however, die-cut specimens should be raveled prior to the edge sealing (see 14.9).

14.9 To prevent loss of specimen weight by fraying, white glue is applied to the pinked or raveled edges of all specimens from a plastic squeeze bottle (see Fig. 3). The cement strip should not exceed 3.2 mm (0.125 in.) in width. For fabrics which have been die-cut, or cannot be pinked, specimens should be raveled 3.2 mm (0.125 in.) along each edge and adhesive applied as above.

15. History

15.1 Revised in 2019 to clarify construction for cotton fabric.

15.2 Reaffirmed 2016. Editorially revised and reaffirmed 2011. Editorially revised 2008. Revised 2005. Editorially revised and reaffirmed 2004. Editorially revised and reaffirmed 1999. Editorially revised 1995. Editorially revised and reaffirmed 1994. Reaffirmed 1989. Editorially revised 1986. Editorially revised 1985. Revised 1984. Editorially revised 1978. Reaffirmed 1977. Reaffirmed 1974. Revised 1966.

15.3 Developed in 1959 by AATCC Committee RR29. Maintained by RA99.

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Test Method for Finishes in Textiles: Identification

1. Purpose and Scope

1.1 This test method gives basic procedures for qualitative identification of various finish components present on textile materials.

1.2 The method is applicable to textile fabrics, yarns or fibers.

2. Principle

2.1 The identification scheme may involve any or all of the following approaches:

2.1.1 Sequential solvent extractions followed by identification of extracts by infrared spectroscopy (IR), gas chromatography (GC), high performance liquid chromatography (HPLC), thin layer chromatography (TLC), mass spectrometry (MS) or other instrumental or wet-chemical methods.

2.1.2 Direct measurement of elemental or chemical species on fabric by X-ray fluorescence (XRF) spectroscopy, infrared reflectance spectroscopy, furnace atomic absorption (AA) spectroscopy, inductively coupled plasma (ICP) spectroscopy and other instrumental or wet-chemical analysis methods.

2.1.3 Identification of specific finishing components by chemical spot tests on the textile or extracts from textiles.

3. Referenced Documents

3.1 AATCC TM20, Test Method for Fiber Analysis: Qualitative (see 16.1).

3.2 *Analytical Methods for a Textile Laboratory*, Fourth Edition, pending, American Association of Textile Chemists and Colorists, Research Triangle Park, NC (see 16.1).

3.3 *Infrared Spectra for the Identification of Chemical Finishing Agents from Textiles*, Part I, ARDS-S-47, October 1974, Agricultural Research Service, U.S. Dept. of Agriculture.

3.4 *Handbook of Chemical Specialties*, John E. Nettles, Chapter on Textile Fiber Processing, Preparation and Bleaching, John Wiley & Sons, New York.

3.5 *Textile Chemist and Colorist*, Vol. 5, No. 12, December 1973, p279 (see 16.1).

3.6 Walker, J. F. *Formaldehyde*, Reinhold, NY, 1964.

3.7 Feigl, F., *Spot Tests in Organic Analysis*, 7th Edition, Elsevier, Amsterdam, 1966.

3.8 VanLoo, W. J., Jr., et al., *American Dyestuff Reporter*, Vol. 45, 1956, p397.

3.9 Feigl, F., *Spot Tests in Inorganic Analysis*, 7th Edition, Elsevier, Amsterdam, 1966.

3.10 *Textile Chemist and Colorist*. Vol. 21, No. 26, June 1989, p13 (see 16.1).

4. Terminology

4.1 **chemical finish,** n.—chemical material other than colorants and residual processing chemicals added to textiles to impart desired functional or aesthetic properties to the textile product.

5. Safety Precautions

5.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

5.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

5.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

5.4 Refer to each procedure for additional safety precautions regarding handling acids, hexane and methanol flammability, and reflux precautions.

6. Limitations

6.1 This test method is a flexible set of guidelines for identification of textile finishes. Appropriate adjustments and modifications will be needed as finishing chemicals change and as new analytical test methods are developed.

6.2 Any chemical which is on the textile prior to finishing may be present on the finished fabric and could be removed and/or detected as finish. Warp sizes, [e.g., starches, acrylics, waxes, polyesters, poly(vinyl alcohol)], dyes, dye fixatives, binders, spin finishes, natural gums, sugars, optical brighteners, and processing chemicals (e.g., surfactants, bleach stabilizers, dye carriers, and oils) may be present by design or unintentionally. A working knowledge of these chemicals is needed to properly detect and identify them. Where possible, supplier should be consulted for further information.

7. Sample Preparation

7.1 Dry textile samples at 105 ± 1°C (221 ± 2°F) and desiccate to constant weight prior to weighing for subsequent extractions.

7.2 Additional preparation of samples, apparatus, and reagents is described in subsequent sections. Read all sections to ensure proper preparation and safety precautions before starting.

8. Solvent Extraction Procedures

8.1 Extraction with various solvents is intended to remove as many finishing chemicals as possible in a sequential order. The extraction scheme presented here (see Table I) was chosen on the basis of solvent polarity, solubility, volatility, economy and safety.

8.2 Use chemical goggles or face shield, impervious gloves and an impervious apron during dispensing and mixing of alkalis, acids and organic solvents. Handle concentrated acids only in an adequately ventilated laboratory hood. CAUTION: Always add acid to water.

8.2 Changes in chemical finishes, safety considerations or personal preferences may dictate the use of other solvents. The user is responsible for determining which finish types may be removed by each solvent.

8.3 Some finishes may not be soluble in any solvent. An alternate method for insoluble finishes such as fully-crosslinked polymers [e.g., some acrylics, polyurethanes, silicones and poly(vinyl acetates)] is to dissolve all of the fibrous material with a suitable solvent, leaving the residue of finish for further analysis. Refer to AATCC TM20, Table V (see 3.1).

8.4 Procedure. Extract a weighed textile sample (1-10 g depending on the size of the extractor) in sequence by the solvents in Table I, using Soxhlet reflux, beaker or pressurized solvent extractions depending on equipment availability. Carry out reflux procedure in a well-ventilated hood with an electrical heater or a water bath as the heating source. Use chemical goggles or face shield, impervious gloves and an impervious apron when handling organic solvents.

Extraction times may range from 30 min to several hours depending on the boiling point, exchange rate, apparatus and knowledge of the history of finish removal for the textile sample in question. Dry samples completely before beginning the next solvent in the sequence.

After the extractions are complete, filter each solvent and evaporate or distill, leaving the residual finish extract. The extract may be transferred to a weighing

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Table I

Group	Solvent	Polarity Index*	Typical Finishing Chemicals Removed
1	Cold Hexanes	0.1	Oils, silicone oils
2	Hot Hexanes	1.0	Oils, waxes, silicones, some softeners, ester lubricants
3	Methanol	5.1	Unfixed cellulose reactants, organic salts, sulfonated organics
4	Water	10.2	Unfixed cellulose reactants, urea, linear starches, poly(vinyl alcohol)
5	0.1 *N* Hydrochloric acid	10.2	Fixed cellulose reactants, branched starches, inorganic salts

*Burdick and Jackson

NOTE: Some types of fibers may be partially dissolved by the solvents. Pre-check the fiber types to avoid errors. Refer to AATCC TM20 (see 3.1).

CAUTION: Hexanes and methanol are flammable liquids and should be stored in the laboratory only in small containers away from heat, open flames and sparks. These chemicals should not be used near an open flame.

pan prior to complete dryness; drying can then be completed in an oven at 105 ±1°C.

It may be useful to neutralize the acid extract with dropwise addition of dilute sodium hydroxide to avoid further degradation of the finish during evaporation. The quantitative results will reflect the presence of sodium chloride in the dried residue. In addition, contamination of the solution by sodium carbonate from the reaction of sodium hydroxide with atmospheric carbon dioxide may interfere with interpretation of subsequent infrared spectra.

Calculate the percent extract using Eq. 1:

$$\text{Extract} = 100 \times \frac{\text{Weight of Residue}}{\text{Weight of Original Fabric}} \quad \text{(Eq. 1)}$$

The percent extract, coupled with subsequent identification, will be useful in distinguishing true finish components from residual non-finishing chemicals that may have been left on the fabric; e.g., natural waxes from softeners.

Identify the extracted finish by one of the analytical techniques described below. Additional information may be found in *Handbook of Chemical Specialties*, Textile Fiber Processing, Preparation and Bleaching chapter (see 3.4).

9. Infrared Spectroscopic Procedures

9.1 Infrared spectroscopy offers a convenient way to identify extracted finish from textiles. The basis for infrared spectroscopy is the structural information about a compound that can be found from its infrared spectrum. Infrared absorption bands in the 4000 to 250 wavenumbers region result from specific functional groups in a molecule. These can be compared with spectra of finishes in published or personal libraries. Visual inspection by a trained analyst will quickly identify the general type of finish. Computerized search-and-match capabilities on many infrared systems will be helpful. For more complete information, including typical wavenumbers for comparison of spectra of textile materials, descriptions of newer techniques such as Fourier transform infrared spectroscopy (FTIR), attenuated total reflectance spectroscopy (ATR) and primary references, refer to *Analytical Methods for a Textile Laboratory* (see 3.2). For representative spectra of chemical finishes, refer to *Infrared Spectra for the Identification of Chemical Finishing Agents from Textiles* (see 3.3).

9.2 FTIR spectrometers are capable of producing high quality spectra for identification and reference purposes. Other spectroscopic techniques such as laser-Raman spectroscopy, mass spectrometry and UV-Visible spectroscopy may also be utilized, if available, to further identify the extracts. Figs. 1-6 represent infrared spectra run on an FTIR spectrometer at a resolution of 4 wavenumbers of a fatty ester softener, a silicone softener, a dimethyldihydroxyimidazolidinone (DMDHI, aka DMUG) cellulose reactant, a glycolated dimethyloldihydroxyethylene-urea (DMDHEU) cellulose reactant, a tetrakis-(hydroxymethyl)phosphonium chloride (or sulfate) flame retardant and a C-6 fluorochemical oil/water repellent, respectively. Fig. 7 represents a laser-Raman spectrum of an aqueous solution of a DMDHEU cellulose reactant using an argon ion laser source.

9.3 Infrared Sample Preparation.

9.3.1 Techniques. Prepare finish samples for infrared analysis by dissolving the extract in suitable solvent for analysis as a liquid or as a cast film on an ATR crystal.

9.4 Infrared Analysis.

9.4.1 Obtain a spectrum of the sample or extract prepared by one of the above techniques. Compare the sample spectrum with infrared spectra of known finishes. For laboratories that possess an FTIR instrument, it is strongly recommended that an in-house spectral library of appropriate finishing agents be developed. For more accurate identification, the spectrum of the known compound should be obtained with the same application and extraction history as the unknown finish.

9.4.2 Softeners. Quaternary ammonium compounds, polyethylenes, polyoxyethylene glycols, polyurethanes, polyacrylates and silicones are soluble or partially soluble in hot hexanes. Concentrate the extract and transfer to a clean ATR crystal.

9.4.3 Durable Press Reactants. Since most durable press reactants for cellulose can be extracted with either water or hydrochloric acid, this provides an alternate method for removing finishes for infrared spectroscopy.

Use chemical goggles or face shield, impervious gloves and an impervious apron during dispensing and mixing of alkalis, acids and organic solvents. Handle concentrated acids only in an adequately ventilated laboratory hood. CAUTION: Always add acid to water.

Reflux approximately 0.2 g fabric, cut into small squares, for 5 min in 0.1 *N* HCl. Carry out reflux procedure in a well-ventilated hood with an electrical heater or a water bath as the heating source. Use chemical goggles or face shield, impervious gloves and an impervious apron when handling organic solvents.

Decant the solution into a 50-mL round-bottom flask with 350 mg spectral grade KBr.

Evaporate to dryness on a rotary evaporator while the flask is immersed in an ice-salt bath to prevent further hydrolysis.

Complete final drying over phosphorus pentoxide.

Mix the residue with more KBr (in a 50 mg:300 mg ratio) and press into a disk.

Obtain a spectrum and compare to that of known compounds, with spectra of hydrolysates of known compounds preferable for accurate comparison.

Another method of sample preparation for identification of durable press cellulose reactants by infrared spectroscopy is to grind finished and unfinished fabrics and pelletize the two samples with spectral grade KBr. The unfinished pellet is run as the background and its spectrum subtracted from the spectrum of the finished sample pellet. It is important to use the same weights of both fabric samples to compensate for the cellulose bands. The resulting spectrum from the subtraction process represents the spectrum of the finish.

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9.4.4 A quick alternative for acquiring infrared spectra of finishes was described in the 1972 Intersectional Technical Paper Competition presented by AATCC Rhode Island Section (see 3.5). The method relies on the transfer of finish by solvent from a small rectangle of fabric placed on the surface of an ATR crystal. The solvent is evaporated leaving a thin film of finish residue on the crystal.

10. Direct Spectroscopic Analysis of Fabric

10.1 With certain types of accessories and spectroscopic techniques, it is possible to acquire direct spectra of both the fabric substrate and the finishing components on it. These techniques may include infrared reflectance with ATR and diffuse reflectance accessories, laser-Raman spectra, electron spectroscopy for chemical analysis (ESCA) and near infrared reflectance spectroscopy (NIR). In some cases, it may be possible to use computerized spectral subtraction techniques in identifying small amounts of finish on fabric.

11. Chromatographic Procedures

11.1 Chromatography is a useful method for the separation and tentative identification of some of the finishes on textile fabrics. After separation is accomplished, identification is made by chromatographic comparison to known compounds either in pure form or after being subjected to the same application and extraction history as the unknown; identification requires comparison under at least two different sets of chromatographic conditions. Alternatively, the chromatographic peak of interest may be identified spectroscopically online or after isolation. For most analyses, extracted residues or their derivatives may be analyzed by GC, HPLC or TLC. Applications included in this section are simple techniques for use with basic GC or HPLC. For information on more complex and diverse instruments and methods refer to *Analytical Methods for a Textile Laboratory* (see 3.2).

11.2 Gas Chromatography. Gas Chromatography involves the separation of a mixture of volatile substances as they are carried through a coated capillary column, where individual components are retained for different times, depending on their retention in the stationary phase. That differential retention is a function of volatility (boiling point) and polarity.

11.2.1 Softeners, Lubricants, and Emulsifiers. Solvents, such as hexanes, are used to extract softeners, lubricants and emulsifiers from fabrics. Hexanes are flammable liquids and should be stored in the laboratory only in small containers away from heat, open flames and sparks. These chemicals should not be used near an open flame.

These 3 groups of products are commonly made from fatty acids and fatty acid esters or their derivatives. Some of these compounds can be chromatographed directly and some must be made into derivatives to be better analyzed by GC.

Other long chain fatty compounds are more easily analyzed after derivatization. Monocarboxylic acids are esterified, usually with methanol, and these methyl esters, also called fatty acid methyl esters or FAMEs) are analyzed as follows: Column: 60 m × 0.25 mm (50%-Cyanopropyl)-methylpolysiloxane, 0.25 µm capillary column, using an appropriate temperature program. Methanol is a flammable liquid and should be stored in the laboratory only in small containers away from heat, open flames and sparks. Methanol should not be used near an open flame.

Silylation, another method of derivatization of fatty acids and fatty alcohols, forms non-polar, volatile, thermally stable ethers. These ethers form easily, and the reactions are performed in screw-capped vials. The column used for analysis is chosen according to the type of compound that was silylated. The silyl derivatives of glycerol and mono- and diglycerides for example, are chromatographed on a 100% Dimethylpolysiloxane column, 5 m × 0.25 mm, 0.25 µm, with an appropriate temperature program.

Paraffin waxes, often used as softeners, lubricants, water repellents and winding lubricants are analyzed directly by GC. Instrument conditions for paraffin wax or alpha-olefin C-30+ type hydrocarbons are:

Column: 100% Dimethylpolysiloxane Column 15 m × 0.53 mm × 0.15 µm
Oven: 100°C to 360°C at 10°C/min for 20 min.
Carrier: Hydrogen, 6.7 mL/min
Injector: Splitless at 350°C
Detection: Flame Ionization Detector (FID) (430°C)

11.2.2 Polyols. Water extracts of fabrics may be analyzed by GC to detect ethylene and diethylene glycols. Glycols are frequently used with or in durable press cellulose reactants. Instrument conditions are:

Column: Polyethylene glycol, 30 m × 0.53 mm, 1 µm
Oven: 80°C to 230°C at 15°C/min for 10 min
Carrier: Helium at 6.6 mL/min
Detector: Flame Ionization Detector (FID) at 325°C
Injector: Splitless at 225°C

11.3 Thin Layer Chromatography. TLC is the separation of components of a substance by differential travel in a solvent phase over an absorbent plate and can be an analytical tool for a variety of textile extracts such as surfactants, dye carriers, cellulose reactants, polyester oligomers and, more commonly, dyes. Reasons for utilizing TLC include small sample size, preliminary work for HPLC method development or two-dimensional separation.

11.4 High Performance Liquid Chromatography. Several fabric finishes and extracts can be detected by HPLC, which relies on separation of components carried by a liquid phase through a column and detection by ultraviolet (UV), refractive index (RI) or mass spectrometry. Formaldehyde, resins and other auxiliaries as well as residual chemicals from prior processing can be detected.

11.4.1 Formaldehyde. Formaldehyde (CH_2O), extracted from the fabric in water, precipitated as formaldehyde 2,4-dinitrophenylhydrazone, and analyzed by liquid chromatography, is detectable at very low levels. For preparation of 2,4-DNP standards and samples, refer to the article Optimization of the AATCC Sealed Jar and HPLC Methods for Measurement of Low Levels of Formaldehyde (see 3.10). Fig. 8 is a chromatogram of the 2,4-DNP derivative of formaldehyde using the procedure in 14.10. Instrument conditions are:

Column: C18 (reverse phase)
Flow rate: 1.0 mL/min
Eluent: 40% water/60% acetonitrile
Detector: UV at 360 nm

11.4.2 Durable Press Finishes. Some cellulose reactants for durable press finishes can be extracted with water. HPLC is ideal for critical comparison of durable press finishes. See Figs. 9 and 10 for chromatograms of representative DMDHEU and DMDHI finishes, respectively.

The following conditions were used:
Column: C-18 (reverse phase)
Eluent: 100% water
Detector: Refractive index
Flow rate: 1.0 mL/min

Under these conditions, other finishes including uron, triazones and glycols in the cellulose reactant can also be identified.

11.4.3 Dyes. For identification of various dye classes, refer to *Analytical Methods for a Textile Laboratory*, Chapter 6 (see 3.2).

12. Elemental Analysis

12.1 Determination of the elemental content of finished textiles can be used to indicate the presence of various finishing agents. Increased nitrogen content on treatment of fabric, as determined by traditional Kjeldahl analysis or by instrumental nitrogen analyzers, can suggest the presence of durable press reactants. Nonmetallic elements such as phosphorus, chlorine and bromine from flame re-

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tardant finishes, as well as fluorine from soil-release finishes, can be analyzed by chemical digestion or oxygen flask combustion followed by titration, specific ion analysis or other instrumental methods.

Metals on fabric, such as antimony or titanium from flame retardants, magnesium, aluminum or zinc from catalysts for durable press reactants and sodium or potassium salts of other finishes, can be determined by acid extractions or dry ashing followed by atomic absorption (AA) analysis or inductively coupled plasma (ICP) analysis. X-ray fluorescence (XRF) can be used to simultaneously determine the content of many elements directly on the fabric.

13. Chemical Spot Tests

13.1 Qualitative spot tests are detection methods based on chemical reactions employing a drop of test solution or reagent (Table II). The chemical reaction usually results in a characteristic color or odor. Spot test methods should be sensitive, reliable and capable of being conducted satisfactorily with small sample amounts and simple equipment. Identification limit is the minimum quantity of a substance dissolved in the test solution that can be detected. Spot tests are not designed to furnish a complete analysis of the kind or amounts of all constituents in a given material, but rather to determine whether certain elements, compounds or classes of compounds are present or absent. Always run a blank and a known sample containing the substance in question, along with the unknown sample. Further, repeat tests for confirmation of reliability.

13.2 Several procedures are available for spot tests: (1) a drop of test solution and a drop of reagent can be brought together on a nonporous surface; (2) a drop of test solution can be placed onto a medium impregnated with a reagent; (3) a drop of reagent can be placed on a solid specimen; (4) a drop of reagent or a strip of reagent paper can be exposed to the action of gases evolved from the test solution or solid specimen (see Table II); and (5) a drop of solution can be placed on a textile and color development observed on white filter paper held beneath the specimen. For analysis of textile finishing agents, the sample may be prepared for one of the above treatments in several ways. First, the test may be carried out directly on a fabric sample with no preliminary preparation. Second, the fabric may be extracted with a selected solvent and the test carried out with the extract as the test solution. Third, the fabric (or textile sample) may be destroyed by dry or wet ashing methods, and the residue tested in solid or dissolved form. Finally, a fusion method may be used for substances that form insoluble oxides on ignition. This method involves heating the sample with metallic sodium or potassium.

Table II—Chemical Spot Tests

Substance	Reagents Employed	Reaction	Reference
Formaldehyde	Chromotropic acid and sulfuric acid	Purple color	3.6, 3.7
	Phenylhydrazine hydrochloride, hydrochloric acid, and ferric chloride	Cherry red color	3.8
Starch	Iodine and potassium iodide in water	Purple color	3.2
Urea	*p*-Dimethylaminobenzaldehyde in methanol and hydrochloric acid	Yellow color	3.2
Zinc	Dithiazone (diphenylthiocarbozone) in acetone	Cherry red color	3.2, 3.9
Zirconium	Alizarin in alcohol	Red to red-violet precipitate	3.9
Magnesium	Quinalizarin in methanol, and sodium hydroxide	Blue precipitate	3.2, 3.9
Aluminum	Aluminum and ammonium acetate	Red color	3.2, 3.9
Iron	Hydrochloric acid and potassium ferrocyanide	Deep blue color	3.2, 3.9
	Hydrochloric acid and potassium thiocyanate	Deep red color	3.2, 3.9
	Hydrochloric acid and heat	Yellow color	
Ammonium ion	Sodium hydroxide and heat	Odor of ammonia	3.2
Chloride ion	Silver nitrate and nitric acid	White to gray precipitate	3.2, 3.9
Nitrate ion	Ferrous sulfate and sulfuric acid	Brown ring forms	3.2, 3.9
Phosphate ion	Ammonium molybdate, benzidine hydrochloride and ammonium acetate	Blue color	3.2, 3.9
Sulfate ion	Barium chloride and hydrochloric acid	White precipitate	3.2
*Phosphorus	Nitric acid and ammonium molybdate	Yellow color	3.2, 3.9
*Nitrogen	Ferrous sulfate and ferric chloride	Deep blue color	3.2, 3.9
*Sulfur	Sodium nitroferricyanide	Red violet color	3.2, 3.9
*Chlorine	Silver nitrate in water, and nitric acid	White to gray precipitate	3.2
*Bromine	Silver nitrate and nitric acid	Yellow precipitate	3.2
Polyvinyl alcohol	Chromic acid and sodium hydroxide	Brown stain	3.2
	Iodine and potassium iodide and boric acid	Purple color	3.2
Polyvinyl acetate	Iodine and potassium iodide	Red-brown color	3.2
Sodium carboxymethyl-cellulose	Uranyl nitrate	Yellow precipitate	3.2
Proteins	Potassium hydroxide, heat, copper sulfate	Violet color	3.2
Methylol melamine	Phenylhydrazine hydrochloride, hydrochloric acid, and ferric chloride	Light pink color	3.8
Methylol urea	Phenylhydrazine hydrochloride, hydrochloric acid, and ferric chloride	Pale brown color	3.8
Methylol ethylene	Phenylhydrazine hydrochloride, hydrochloric acid, and ferric chloride	Deep red brown color	3.8

*Sodium Fusion Method

14. Report

14.1 Describe or identify the sample tested.

14.2 Report that the sample was tested using AATCC TM94-2020.

14.3 Report the testing conditions:

14.3.1 Extraction solvents.

14.3.2 Method of extract analysis.

14.3.3 Instrument conditions for extract analysis.

14.3.4 Other relevant testing conditions.

14.4 Report the test results:

14.4.1 Finish or finish components detected.

14.5 Describe any modification(s) of the published standard.

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15. Precision and Bias

15.1 Precision and bias statements are not applicable because data are not generated by this method.

16. Notes

16.1 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

17. History

17.1 Revised in 2020 to update the spectra, revise the text in existing sections and to add new sections for clarity.

17.2 Reaffirmed 2017, 2012. Editorially revised 2010. Reaffirmed 2007, 2002. Editorially revised and reaffirmed 1997, 1992. Revised 1987. Reaffirmed 1985, 1977. Editorially revised 1974. Reaffirmed 1973, 1969. Revised 1965, 1962, 1961.

17.3 Developed in 1959 by AATCC Committee RA45.

Appendix 1—Spectra

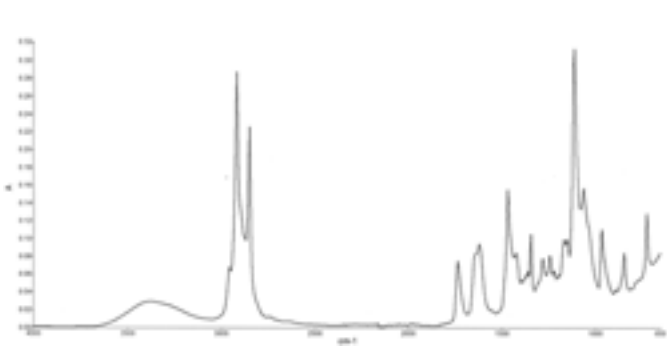


Fig. 1—Infrared spectrum of a fatty ester softener.

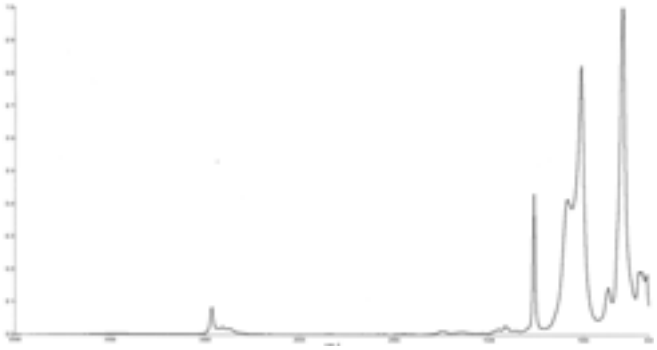


Fig. 2—Infrared spectrum of a silicone softener.

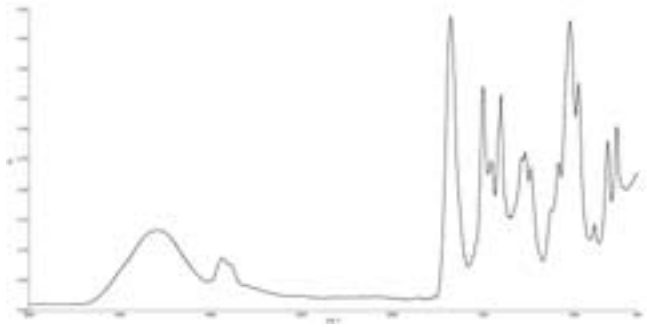


Fig. 3—Infrared spectrum of a DMDHI non-formaldehyde durable press reactant.

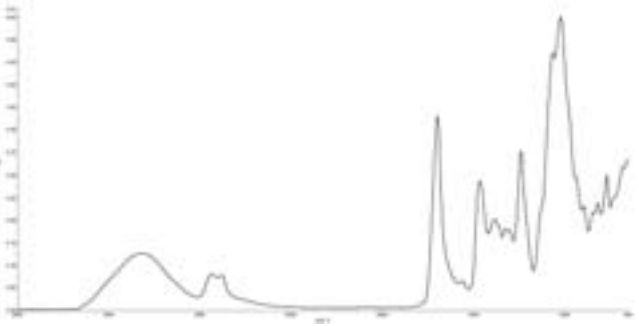


Fig. 4—Infrared spectrum of glycolated DMDHEU durable press reactant.

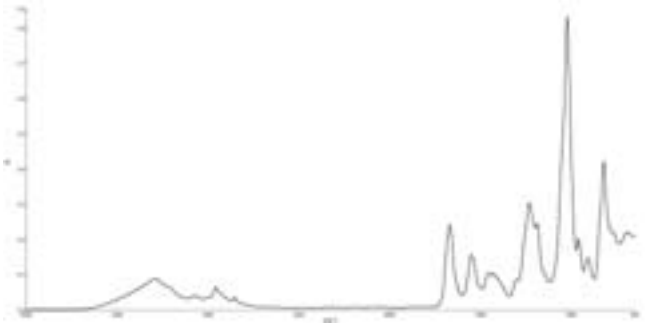


Fig. 5—Infrared spectrum of an organophosphorous flame retardant.

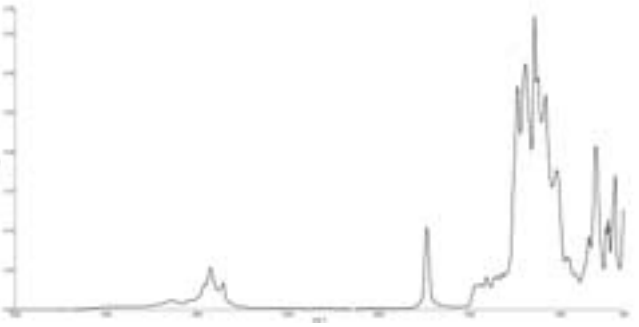


Fig. 6—Infrared spectrum of a C-6 fluorocarbon oil/water repellent.

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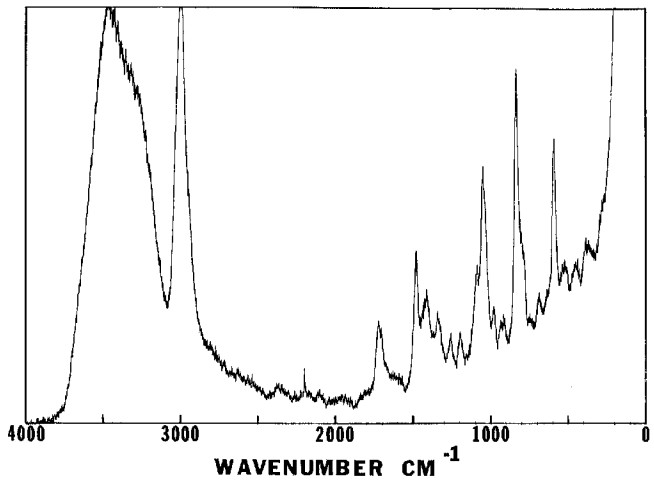


Fig. 7—Laser-Raman spectrum of DMDHEU durable press finish.

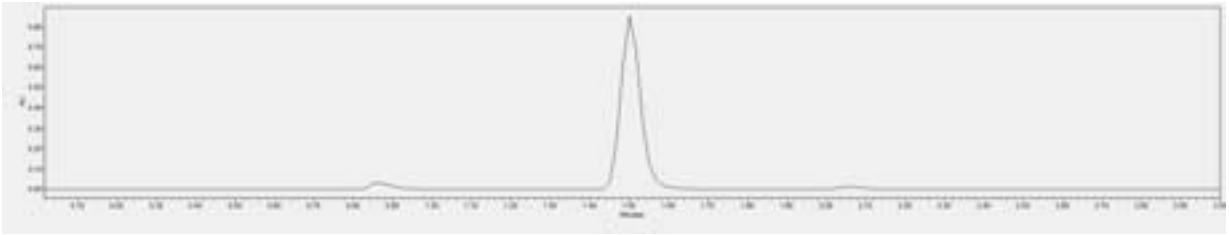


Fig. 8—HPLC chromatogram of 2,4-dinitrophenylhydrazone of formaldehyde.



Fig. 9—HPLC chromatogram of DMDHI durable press reactant.

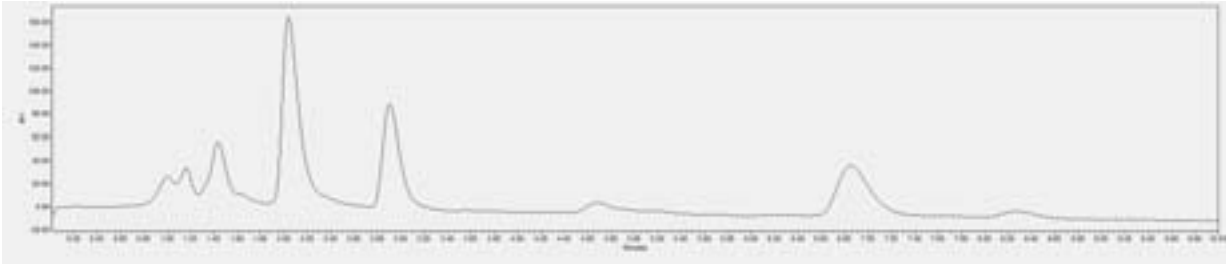


Fig. 10—High performance liquid chromatogram of a glycolated DMDHEU durable press reactant.

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Test Method for Extractable Content of Textiles

1. Purpose and Scope

1.1 This test method is for determining the amount of water, enzyme and organic-solvent extractable matter of fibers, yarns or fabrics containing cellulosic fibers and blends of cellulose and other fiber types in their greige and/or prepared state of processing.

2. Principle

2.1 Water and enzyme soluble non-fibrous matter is removed from a test specimen by specified exposure to hot water, followed by an amylase solution. Oils, fats and waxes are removed by hexanes solvent extraction (see 5.11 and 11.1).

3. Terminology

3.1 **extractable matter,** n.—nonfibrous material in or on a textile substrate, not including water, which is removable by water, enzyme or a specified solvent or solvents as directed in a specified procedure.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Users MUST consult manufacturers for specific details such as safety data sheets and other recommendations. Consult and follow all applicable OSHA standards and rules.

4.1 Follow good laboratory practices. Wear safety glasses in all laboratory areas.

4.2 Handle all chemicals with care.

4.3 Store hexanes in the laboratory only in small containers away from heat, open flames and sparks. Do not use hexanes near an open flame.

4.4 Ensure an eyewash/safety shower is located nearby and an organic vapor respirator should be available for emergency use.

4.5 Control exposure to chemicals used in this procedure at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 11.2).

5. Apparatus (see 11.3)

5.1 Analytical balance with a sensitivity of 0.1 mg.

5.2 Circulating air oven capable of maintaining 105-110°C (221-230°F).

5.3 Soxhlet extraction assembly (Option 1).

5.4 Accelerated Solvent Extractor, 22 mL cell and glass beads (Option 2) (see 11.3). (Liquid Nitrogen with gaseous withdrawal has to be supplied to the instrument).

5.5 Glass weighing bottles with covers.

5.6 Extraction thimbles, cellulose.

5.7 Beakers, 300 mL, tall-form.

5.8 Watch glasses, 90 mm (sized to cover 300 mL tall form beakers).

5.9 Sieve, stainless steel, 100-mesh.

5.10 Calcium chloride ($CaCl_2$) or equivalent desiccant.

5.11 Hexanes, extraction grade (Hexanes are a blend of hexane isomers. More refined grades, such as HPLC, are expensive and may be unnecessary for this test method).

5.12 Bacterial amylase, *bacillus subtilis* origin, 1600-1800 B.A.U. [see AATCC TM103, Test Method for Bacterial Alpha-Amylase Enzymes used in Desizing].

5.13 Laboratory fume hood.

6. Sampling

6.1 Take one 10 g specimen from each sample. If the sample is a woven fabric, cut on the bias to minimize loose fibers or yarns.

6.2 For improved accuracy, test duplicate specimens from each sample.

7. Procedure

7.1 Specimen Preparation. Place the specimen in a tared weighing bottle being careful to fold the edges of the specimen to prevent the loss of any fiber or loose yarn. Dry to a constant weight (see AATCC TM20A, Test Method for Fiber Analysis: Quantitative) in a circulating air oven at 105-110°C (221-230°F). Cool the weighing bottle in a desiccator. Weigh the bottle and specimen to the nearest 0.1 mg (see 11.4).

7.2 Water Extraction. Remove the dried specimen from the bottle and place it in a beaker containing 200 mL of distilled or deionized water at 82 ± 3°C (180 ± 5°F). Cover the beaker with a watch glass. Maintain the temperature of the water at 82 ± 3°C (180 ± 5°F) for 2 h. Pour the solution and specimen into the sieve and rinse the specimen, washing with two successive 100 mL portions of distilled water at 82 ± 3°C (180 ± 5°F). Recover any loose fibers on the sieve and fold them into the specimen. Return the specimen to its bottle and dry it to a constant weight as specified in 7.1. Determine the amount of water extractable matter as the difference in the specimen weights obtained in 7.1 and 7.2 using the equation in 8.1.

7.3 Enzyme Extraction. Remove the dried specimen from the water extraction procedure (see 7.2) from the bottle and place in a beaker containing 200 mL of a 2% bacterial amylase solution at 74 ± 3°C (165 ± 5°F) for 1 h. Pour the solution and specimen into a sieve and rinse the specimen by washing with 10 successive 100 mL portions of distilled water at 82 ± 3°C (180 ± 5°F). Recover any loose fibers on the sieve and fold them into the specimen. Return the specimen to its bottle and dry to a constant weight as specified in 7.1. Determine the amount of enzyme extractable matter as the difference in the weights obtained in 7.2 and 7.3 using the equation in 8.1.

7.3.1 If an extraction thimble has not been exposed to hexanes, it is recommended that it should be subjected to a Soxhlet extraction before use with a test specimen to remove possible contaminants.

7.4 Oils, Fats and Waxes (select one of the following options).

7.4.1 Option 1 (Soxhlet procedure). Place the dried specimen from the water and subsequent enzyme extractions (see 7.2 and 7.3) in a Soxhlet extractor. If the specimen contains loose, fibrous material, then use an extraction thimble. Extract for 12-16 times with hexanes (see 11.1). Remove the specimen from the extractor and evaporate the remaining hexanes on the specimen in a laboratory fume hood. Return the specimen to the weighing bottle. If a thimble is used, remove the specimen from the thimble prior to placing in the weighing bottle. Return the specimen to its bottle and dry to a constant weight as specified in 7.1.

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Determine the amount of solvent extracted matter as the difference in the specimen weights obtained in 7.3 and 7.4 using the equation in 8.1.

7.4.2 Option 2 (Accelerated solvent extractor procedure). Extract the dried specimen using the following conditions with an accelerated extractor.

Cell size — 22 mL (see 11.3)
Heat — 5 min
Static — 15 min
Flush Volume — 90%
Purge — 90 s
Cellulose filter
Temperature — 100°C
Pressure — 1500
Solvent — Hexanes
Cycles — 3
Glass Beads

Air dry the specimen, then return it to the weighing bottle and dry to a constant weight as specified in 7.1. Determine the amount of solvent extracted matter as the difference in the specimen weights obtained in 7.3 and 7.4 using Eq. 1.

8. Calculations

8.1 Calculate to two (2) decimal places the matter extracted from each specimen by water, enzyme or solvent using Eq. 1:

$$E = [(B-A/X](100) \qquad \text{(Eq. 1)}$$

where:

E = the material extracted by water, enzyme solution or organic solvent, %,
B = the mass of the specimen before the particular extraction, g,
A = the mass of the specimen after the particular extraction, g, and
X = the mass of the oven-dried specimen before the first extraction, g.

8.1.1 If the extracted matter for any of the three extractions is less than 0.02%, record the value for that extraction as "less than 0.02%."

8.2 Determine the total extractable content for each specimen as the sum of the water, enzyme and solvent extraction procedures. Those values reported as "less than 0.02% are arbitrarily added in as 0.01%.

9. Report

9.1 Report:

9.1.1 Data reported is from the use of AATCC TM97-2019.

9.1.2 Report any deviations from the procedure(s) used.

9.1.3 Report the brand and model of the extraction device used.

9.2 The percent of matter from water extraction procedure.

9.2.1 The percent of matter from enzyme extraction procedure.

9.2.3 Report total extracted matter.

10. Precision and Bias

10.1 Interlaboratory Study. A limited study was conducted in 2005 to estimate the precision of the Soxhlet and accelerated solvent extraction procedures and their relative bias. The study consisted of three laboratories, three fabrics and two operators per laboratory. The fabrics used included a 100% cotton bleached fabric, a 50/50 cotton/polyester greige fabric and a 100% cotton greige fabric. An accelerated solvent extractor made by Dionex Corporation, Sunnyvale, California was used in the study. Two determinations were conducted by each operator for each combination of test procedure and fabric. Averages and standard deviations for total extractable content are shown in Table I. Based on the limited data set on which this precision and bias statements have been made, users are cautioned when making critical decisions.

10.1.1 Analysis. The analyses used in this precision and bias statement are based on an analysis of all fabrics within each extraction method. The decision to conduct the all-fabric analysis was made based on the small study size and the results of the single-material computations. Variance components reported here were estimated using ASTM D2904-97, Annex A1. Details on the data and analysis can be obtained by contacting the AATCC Technical Center. Precision is reported in terms of critical differences. Calculations outlined in ASTM D2906-97 (2002) were used to compute the one-sided critical differences between two means for sample sizes.

10.1.2 Variance Components, Soxhlet Procedure. The variance components calculated for the Soxhlet procedure are reported in Table II. All metrics reported in this section are in units of% of total extracted content.

10.1.3 Critical Differences—Soxhlet Method. The 95% critical differences calculated for the Soxhlet method are shown in Table III.

10.1.4 The variance components for the accelerated solvent extraction procedure are reported in Table IV.

10.1.5 The subsequent 95% critical differences estimated for the accelerated solvent procedure are shown in Table V.

10.2 Bias. The amount of extractable matter on a textile material is dependent on the extraction procedure used. Therefore, it is not possible to determine abso-

Table I—Average Values of Total Content (Reported in Percent)

Method	Material	Mean	Std. Dev.	n
Accelerated	Bleached Cotton	0.412	0.113	12
Soxhlet	Bleached Cotton	0.429	0.060	12
Accelerated	50/50 Cotton/Poly	2.615	0.190	12
Soxhlet	50/50 Cotton/Poly	2.669	0.057	12
Accelerated	100% Cotton	8.127	0.267	12
Soxhlet	100% Cotton	8.066	0.362	12

Table II—Estimated Components of Standard Deviation for the Soxhlet Procedure

Variance Component	Estimated Standard Deviation
L	0.000
M-L Interaction	0.000
O within L	0.000
M-O Interaction within L	0.242
S within Combination of M, O and L	0.081

where:
L = Laboratories
M = Materials (fabrics)
O = Operators
S = Test Specimens

Table III—Estimated 95% Critical Differences for the Soxhlet Procedure

Samples per Mean	Within Operator Critical Differences	Within Lab Critical Difference	Between Lab Critical Difference
1	0.225	0.706	0.706
2	0.159	0.688	0.688
3	0.130	0.682	0.682
4	0.112	0.679	0.679

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Table IV—Estimated Components of Standard Deviation for the Accelerated Solvent Extraction Procedure

Variance Component	Estimated Standard Deviation
L	0.121
M-L Interaction	0.000
O within L	0.080
M-O Interaction within L	0.018
S within Combinations of M, O and L	0.134

Table V—Estimated 95% Critical Differences for the Accelerated Solvent Extraction Procedure

Samples per Mean	Within Operator Critical Differences	Within Lab Critical Difference	Between Lab Critical Difference
1	0.371	0.488	0.591
2	0.263	0.411	0.530
3	0.214	0.382	0.508
4	0.186	0.367	0.496

lute bias of these test procedures against an absolute value.

10.2.1 Relative Bias. The data generated in this study was used to check for relative bias between the two extraction methods. There was no indication of a consistent bias between the methods. However, there were specific instances where bias was observed between methods within a laboratory for a specific fabric. Laboratories should verify themselves if bias exists between procedures for their specific applications and testing situation.

11. Notes

11.1 Alternative or additional solvents may be used in lieu of hexanes for removal of oils, fats and waxes; however, other such solvents have not been evaluated by the RA34 committee. Therefore, no statement on the reliability or reproducibility regarding the use of other solvents can be made at this time. If a different solvent other than hexanes (or an additional solvent) is used, note the solvent on the test report.

11.2 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; +1.513.742.2020; www.acgih. org.

11.3 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods. If the volume of fabric is too great for the 22 mL cell, a larger cell size is recommended.

11.4 Synthetic fibers sensitive to drying conditions in 7.1 will not yield accurate data by this test method. Since this test method is based on differential weighing, such fiber decomposition would lead to erroneous results.

12. History

12.1 Revised in 2020 to clarify the Safety requirements.

12.2 Revised in 2019 to align with AATCC style.

12.3 Reaffirmed 2013. Editorially revised 2010. Revised (with title change) 2009. Revised 1999. Revised 1995. Editorially revised 1993. Reaffirmed 1989. Reaffirmed 1988. Editorially revised 1987. Revised (with title change) 1982. Revised 1979. Reaffirmed 1975. Reaffirmed 1972. Reaffirmed 1968.

12.4 Developed in 1960 by AATCC Committee RA34.

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Test Method for Alkali in Bleach Baths Containing Hydrogen Peroxide

1. Purpose and Scope

1.1 This test method determines the total alkali content of a bleach bath containing hydrogen peroxide and alkali from any source. The total alkali content is expressed as % sodium hydroxide.

1.2 Alkali in bleach baths may be supplied by sodium hydroxide, sodium silicate, sodium carbonate or by various other alkaline materials, including potassium compounds, ammonia, lime or by peroxygen solids which contain alkaline salts.

1.3 The alkali content of a bleach bath is a critical parameter in determining the rate and extent of bleaching and may also determine the extent to which the textile being bleached is affected by the bleaching process.

1.4 This test method is used for laboratory determinations and for process control.

2. Principle

2.1 A weighed specimen of the bleach bath is titrated with a standardized solution of sulfuric acid to a Phenol Red endpoint, or to the pH range 6.8-8.4 on a pH meter. The total alkali, expressed as % NaOH, is calculated based on the weight of the bath.

3. Terminology

3.1 **bleaching,** n.—elimination of unwanted coloring matter from a substrate by oxidative or reductive chemical treatment.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The safety precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care. Sulfuric acid is corrosive. When preparing the dilute solutions from pure or concentrated reagents, use chemical goggles or face shield, impervious gloves, and an impervious apron. CAUTION: Always add acid to water.

4.3 An eyewash/safety shower should be located nearby and a high efficiency particulate respirator with organic vapor cartridge and full facepiece should be readily available for emergency use.

4.4 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 12.1).

5. Apparatus and Materials

5.1 Potassium acid phthalate ($C_8H_5O_4K$), ACS Reagent.

5.2 Phenol Red (phenol-sulfonphthalein), or 0.06% solution thereof, if a pH meter is not used as an alternative.

5.3 Methanol (CH_3OH) (not needed if 0.06% Phenol Red solution is available).

5.4 Sulfuric acid (0.1*N*) (H_2SO_4), ACS Reagent, 95-98%, or 0.1*N* (see 7.1 and 12.2).

5.5 Sodium hydroxide (NaOH), ACS Reagent.

6. Specimens

6.1 Obtain a 10 ± 0.01 g specimen from the bleach bath. (see 8.2 and 12.3)

7. Preparation of Reagents

7.1 Preparation and standardization of 0.1*N* H_2SO_4 (see 12.2).

7.1.1 For each liter of 0.1*N* H_2SO_4 solution to be prepared, weigh 5.5 ± 0.001 g C.P. sulfuric acid and add it to 500 ± 100 mL deionized water with stirring. CAUTION: *Dilution of sulfuric acid is exothermic. Use appropriate safety precautions*. Cover the solution and allow it to cool to 20 ± 1°C (68 ± 2°F). Add the solution to a 1 L volumetric flask, and dilute to mark with deionized water.

7.1.2 Weigh 4.0 ± 0.01 g sodium hydroxide and dissolve it in about 100 mL deionized water in a 250 mL beaker. Transfer the solution to a 1 L volumetric flask, and rinse the beaker 5 times with deionized water, adding the rinses to the volumetric flask. Then dilute to mark with deionized water.

7.1.3 Weigh 20.4080 ± 0.0002 g potassium acid phthalate (National Bureau of Standards sample No. 84) and dissolve it in approximately 100 mL deionized water in a 250 mL beaker. Transfer the solution to a 1 L volumetric flask, and rinse the beaker 5 times with deionized water, adding the rinses to the volumetric flask. Dilute to mark with deionized water.

7.1.4 Using a volumetric pipette, transfer 25 mL of the potassium acid phthalate solution into a 250 mL Erlenmeyer flask. Add 5 drops of 0.06% Phenol Red indicator solution and titrate with the sodium hydroxide solution to a greenish-yellow end-point, or to pH 7.6 ± 0.8 if using a pH meter to detect the end-point.

7.1.5 Calculate the normality of the sodium hydroxide solution (N_h) to the nearest 0.001 by the following equation:

$$N_h = \frac{(25)(0.1000)}{\text{mL}}$$

where:

mL = the number of mL of sodium hydroxide solution required.

7.1.6 Using a volumetric pipette, transfer 25 mL of the sulfuric acid solution (see 9.1) into a 250 mL Erlenmeyer flask. Add 5 drops of 0.06% Phenol Red indicator solution and titrate with the sodium hydroxide solution to a greenish-yellow end-point, or to pH 7.6 ± 0.8 if using a pH meter to detect the end-point.

7.1.7 Calculate the normality of the sulfuric acid solution to the nearest 0.001 by the following equation:

$$N_s = \frac{(\text{mL})(N_h)}{(25)}$$

where:

N_s = normality of the sulfuric acid solution

mL = number of mL of sodium hydroxide solution

N_h = normality of the sodium hydroxide solution

7.1.8 Adjustment of normality of sulfuric acid solution.

7.1.8.1 If the normality of the sulfuric acid solution is less than 0.0990, discard it and make up a new solution.

7.1.8.2 If the normality of the sulfuric acid solution is greater than 0.1010, adjust it by adding an amount of water calculated using the following equation:

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$$\text{mL } H_2O = \frac{(N)(950)}{0.1000} - 950$$

where:

N = the normality of the sulfuric acid solution

950 = mL sulfuric acid solution after removal of two samples for titration.

7.1.8.3 Re-standardize the adjusted sulfuric acid solution using 7.1.6 and 7.1.7.

7.2 If the indicator solution is prepared in-lab, preparation of 0.06% Phenol Red indicator solution (see 12.2).

7.2.1 Weigh 1.0 ± 0.1 g Phenol Red and dissolve it in 833 mL methanol.

7.2.2 Add 833 mL deionized water.

7.2.3 Cover the solution and stir with a magnetic stirrer until the solution is completely clear.

8. Procedure

8.1 Measure 20 ± 1 mL deionized water into a 250 mL beaker and add 2-3 drops of Phenol Red indicator solution. If a pH meter is used to detect the end-point, do not add Phenol Red indicator solution.

8.2 Weigh a 10 ± 0.01 g specimen from the bleach bath. Add the specimen to the beaker and mix (see 12.3).

8.3 Titrate the contents of the beaker with 0.1*N* sulfuric acid to a greenish-yellow end-point, or to pH 7.6 ± 0.8 if using a pH meter to detect the end-point.

8.4 Record the number of the 0.1*N* sulfuric acid used to the nearest 0.1 mL.

9. Calculations

9.1 Calculate the total alkali as NaOH to the nearest 0.1% using the following equation:

% Total Alkali, as NaOH

$$= \frac{(\text{mL})(N)(0.040)(100)}{W}$$

where:

mL = the number of mL of the sulfuric acid solution required, as recorded in 7.1.4.

N = the normality of the sulfuric acid solution, as determined in 7.1.7.

0.040 = the milliequivalent weight of sodium hydroxide.

W = the mass of the specimen, as determined in 8.2.

9.2 For alternative calculations see 12.4.

10. Report

10.1 Describe or identify the sample bleach bath.

10.2 Report that sample was tested using AATCC TM98 2022.

10.3 Report the % Total Alkali as NaOH.

11. Precision and Bias

11.1 *Precision*.

11.1.1 In 1995 an interlaboratory study included five laboratories with two operators each, evaluating three concentrations of alkali in aqueous solutions containing 1.15% hydrogen peroxide (H_2O_2). Each operator made three determinations on each concentration. Data were analyzed using the ASTM Tex-pac program (see 12.5).

11.1.2 Analysis showed the residual variances of the three concentrations could be pooled as shown in Table I.

11.2 *Bias*.

11.2.1 The results obtained in the interlaboratory study indicate 97.6% average recovery as compared with the calculated values of alkali concentration based on chemical additions. However, the concentration of alkali can be defined only in terms of a test method, and within this limitation this method has no known bias in the determination of alkali concentration.

Table I—Critical Differences for Two Averages for the Conditions Noted, 95% Probability Level, % Alkali Concentration

Number of Test Results in Each Average	Single-Material Comparisons		
	Single-Operator Precision	Within-Laboratory Precision	Between-Laboratory Precision
1	0.013	0.027	0.069
2	0.009	0.025	0.069
3	0.008	0.025	0.068
4	0.007	0.025	0.068

12. Notes

12.1 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

12.2 Standardized solutions of sulfuric acid, sodium hydroxide, and Phenol Red indicator are available from various laboratory supply companies in either prepared solution or as concentrates. Standardization of purchased reagent solutions should be verified by the procedures in 7.1.4-7.1.7.

12.3 For process control purposes, specimen volume is often substituted for weight. In this case the% alkali may be in error to the extent of the bleach bath density. If results are to be communicated between installations, specimen mass should be used.

12.4 Simplifications and conversions.

12.4.1 If 0.1*N* H_2SO_4 solution is used, and the specimen mass is 10.0 g, the equation in 9.1 is simplified to:

$$\% \text{ alkali as NaOH} = (0.04)\,(\text{mL})$$

12.4.2 Alkalinity may be defined in terms other than "% as NaOH." Some convenient alternatives, using 0.1*N* sulfuric acid solution, a specimen mass of 10.0 g, and the term "mL" as defined in 9.1, are:

(mL) (2.5) = lb sodium silicate per 100 gal, if the sodium silicate is 10.5% Na_2O

(mL) (2.9) = lb sodium silicate per 100 gal, if the sodium silicate is 8.9% Na_2O

(mL) (0.332) = lb NaOH per 100 gal

12.5 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832.9500; fax: +1.610.832.9555; web site: www.astm.org.

13. History

13.1.1 Revised in 2022 to align to the AATCC Style Guide.

13.1.2 Editorially revised 2019. Reaffirmed 2016, 2012. Editorially revised 2010. Reaffirmed 2007, 2002. Revised 1997. Reaffirmed 1989, 1988. Editorially revised 1987, 1985, 1984. Revised 1982 (with title change). Reaffirmed 1979, 1975, 1972, 1968.

13.2 Developed in 1960 by AATCC Committee RA34.

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Test Method for Antibacterial Finishes on Textile Materials: Assessment of

1. Purpose and Scope

1.1 This test method provides a quantitative procedure for the evaluation of the degree of antibacterial activity. Assessment of antibacterial finishes on textile materials is determined by the degree of antibacterial activity intended in the use of such materials. If only bacteriostatic activity (inhibition of multiplication) is intended, a qualitative procedure which clearly demonstrates antibacterial activity as contrasted with lack of such activity by an untreated specimen may be acceptable. However, if bactericidal activity is intended or implied, quantitative evaluation is necessary. Quantitative evaluation also provides a clearer picture for possible uses of such treated textile materials.

2. Principle

2.1 This test method provides a quantitative procedure for the comparison and evaluation of the degree of antibacterial activity after a 24 h exposure to the test bacteria on the test fabric compared directly against an untreated control. After incubation, the bacterial challenge is eluted from the swatches and enumerated and a percent reduction by the fabric specimen is calculated.

3. Terminology

3.1 **activity,** n.—*of an antibacterial agent*, a measure of effectiveness of the agent.

3.2 **antibacterial agent,** n.—any chemical which kills bacteria (bactericide) or interferes with the multiplication, growth or activity of bacteria (bacteriostat).

4. Safety Precautions

4.1 The safety precautions specified in the method are ancillary to the testing procedures and are not intended to be all inclusive.

4.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

4.3 Users MUST be consult manufacturers for specific details such as safety data sheets, equipment operating instructions, and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

4.4 This test should be performed only by appropriately trained personnel. The U.S. Department of Health and Human Services publication, *Biosafety in Microbiological and Biomedical Laboratories*, should be consulted (see 13.1).

4.5 CAUTION: Some of the microorganisms used in these tests are allergenic and pathogenic; i.e., capable of infecting humans and producing disease. Therefore, every necessary and reasonable precaution must be taken to eliminate this risk to the laboratory personnel and to personnel in the associated environment. Wear protective clothing, respiratory protection, and impervious gloves when working with the organisms.

4.6 Sterilize all contaminated samples and test materials prior to disposal.

5. Test Organisms

5.1 Test bacteria:

5.1.1 *Staphylococcus aureus*, ATCC No. 6538, Gram positive organism, CIP 4.83, DSM 799, NBRC 13276, NCIMB 9518 or equivalent strain (see 13.2 and 13.3).

5.1.2 *Klebsiella pneumoniae*, ATCC No. 4352, Gram negative organism, CIP 104216, DSM 789, NBRC 13277, NCIMB 10341 or equivalent strain (see 13.2 and 13.3).

5.1.3 Other suitable species can also be used depending on the intended end-use of the test sample.

6. Materials, Media and Reagents

6.1 Media and reagents. Suitable broth/agar media are:

6.1.1 Nutrient broth/agar.

6.1.2 Trypticase Soy broth/agar.

6.1.3 Sterile distilled or deionized water.

6.1.4 Triton X-100.

6.1.5 Neutralizing broth: Letheen broth, Dey Engley broth or other appropriate for the antimicrobial compound being tested.

6.1.6 Suspension Medium, 1:20 (5%) Nutrient broth or Trypticase Soy broth + 0.05% Triton X-100. Dilute the nutrient broth (TSB) 1:20 with distilled or deionized water containing 0.05% Triton X-100. Sterilize by autoclaving at 121 ± 2°C. If not used immediately after preparation, store at 5-10°C.

6.2 Materials.

6.2.1 Incubator maintained at 37 ± 2°C (99 ± 4°F).

6.2.2 Inoculating loop.

6.2.3 Bunsen Burner or equivalent.

6.2.4 Water bath maintained at 45-50°C (113-122°F).

6.2.5 Pipettes and tips: 10-1000 µL pipettes and appropriate tips.

6.2.6 Culture tubes with non-screw caps; minimum 10 mL capacity.

6.2.7 Petri dishes, 100 mm diam. × 15 mm deep, sterile.

6.2.8 Forceps, sterile.

6.2.9 Stereomicroscope, minimum 40 × magnification.

6.2.10 Ruler.

6.2.11 Analytical balance, capable of measuring 10.0 ± 0.1 g.

6.2.12 Sterile specimen container with screw cap, 150-mL capacity or similar.

6.2.13 Exam gloves, nitrile or other.

6.2.14 Vortex mixer.

6.2.15 Steam sterilizer or autoclave.

6.2.16 Viability control fabric (see 13.4).

7. Preparation of Bacterial Inocula

7.1 Grow a fresh 18-h shake culture of each test specimen in sterile Tryptic Soy (or Nutrient) broth at 37 ± 2°C and 150-250 rpm prior to performing the test. These cultures should originate from a single colony selected from stock culture plates or growth on agar slants. Fresh stock plates should be generated weekly.

7.1.1 Each 18-h shake culture is then diluted with suspension medium (6.1.6), as appropriate for the estimated bacterial concentration, to obtain a bacterial concentration that is between 1.0×10^5 and 3.0×10^5 CFU/mL. These suspensions are used as the test inoculum. Prepared test inoculum is used within 2 h of preparation and bacterial concentration should be verified by appropriate enumeration method.

8. Test Specimens

8.1 Preparation. The following description will be in terms of fabric swatches. Textile materials not in fabric form can likewise be tested with the appropriate modification.

8.1.1 Size and shape of test swatches: Cut circular swatches 4.8 ± 0.1 cm (1.9 ± 0.03 in.) in diameter or square swatches 3.8 × 3.8 ± 0.1 cm, from the test fabric. Use the number of swatches needed to equal 1.0 ± 0.1g. Stack the swatches in a sterile specimen container with screw cap or other appropriate closed container. The number of swatches to be used is dependent on the fiber type and fabric construction. The number of swatches used per jar should be reported.

8.1.2 Untreated Controls. Swatches of the same fiber type and fabric construc-

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tion as test sample but containing no antibacterial finish will be required if calculating percent reduction using formula outlined in 10.2.

8.1.3 Viability Controls. Viability control fabric is required and should be known to demonstrate > 1 log bacterial growth as defined in 10.4.

8.1.4 Test specimens should not be sterilized prior to testing. If sterilization is performed, method and reason for sterilization must be noted on the test report.

9. Procedure

9.1 Size of inoculum per sample. Apply 1.0 ± 0.1 mL of test inoculum (7.1.1) so that recovery from (1) viability control fabric swatches or (2) test fabric swatches at "0" contact time (plated as soon as possible after inoculation) will show counts of $1\text{-}3 \times 10^5$ organisms.

9.1.1 As soon as possible after inoculation ("0" contact time), add 100 ± 1 mL of neutralizing solution to each of the jars containing the inoculated untreated control swatches, the inoculated test swatches and the viability control fabric swatches.

9.1.2 The neutralizing solution should include ingredients to neutralize the specific antibacterial fabric treatment and to take care of any pH requirements of the fabrics (from finishes, antibacterial agents, etc.). The neutralizing solution employed should be reported.

9.1.3 Shake the jars vigorously for one minute. Make serial dilutions with sterile distilled water and plate (in duplicate) on nutrient agar. Dilutions of 10^0, 10^1, 10^2 are usually suitable.

9.1.4 Incubation over contact periods. Incubate additional jars containing inoculated viability control swatches and jars containing inoculated test swatches at 37 ± 2°C (99 ± 3°F) for 24 h. Similar jars may be incubated over other periods (e.g., 1 or 6 h) to provide information about the bactericidal activity of the treatment over such periods.

9.1.5 Sampling of viability control and test swatches. After incubation, add 100 ± 1 mL of neutralizing solution to jars containing viability control swatches and to jars containing test swatches. Shake the jars vigorously for 1 minute. Make serial dilutions and plate (in duplicate) on nutrient agar. Dilutions of 10^0, 10^1, 10^2 are usually suitable for treated test fabrics. Dilutions of 10^3 or 10^4 may be required for viability control and untreated control swatches depending on the incubation period.

9.1.6 Incubate all plates for 24-48 h at 37 ± 2°C (99 ± 3°F).

10. Calculation

10.1 Report bacterial counts as the number of bacteria per sample (swatches in jar) not as the number of bacteria per mL of neutralizing solution. Report "0" counts at 10^0 dilution as "less than 100."

10.2 Calculate percent reduction of bacteria by the specimen treatments using Eq. 1.

$$100(B - A)/B = R \qquad \text{(Eq. 1)}$$

where:

R = % reduction

A = the number of bacteria recovered from the inoculated treated test specimen swatches in the jar incubated over the 24-h contact period

B = the number of bacteria recovered from the inoculated untreated test specimen swatches in the jar incubated over the 24-h contact period.

10.3 If an untreated control is not available, use Eq. 2.

$$100(C - A)/C = R \qquad \text{(Eq. 2)}$$

where:

C = the number of bacteria recovered from the inoculated, test specimen swatches in the jar immediately after inoculation (at "0" contact time)

10.4 For a valid test there should be: (1) "0" colonies of test organism recovered from the uninoculated test specimen swatches and (2) a significant (≥ 1 log) increase in the numbers of bacteria recovered from the inoculated viability control specimen swatches incubated for the specified contact time over the numbers of bacteria recovered from the inoculated viability specimen swatches at "0" contact time (immediately after inoculation).

11. Report

11.1 Report percent reduction of bacteria by the specimen treatment against each test organism. Report should include the calculation method used.

11.2 The criterion for passing the test must be determined by the interested parties.

11.3 All variables and materials used in the test including sample size, sterilization and method, media used, neutralizer used, and the dilution medium used.

12. Precision and Bias

12.1 Precision for this test method has not been established. Until a precision statement is generated for this test method, caution should be used when testing materials with this method. In most cases the use of standard statistical techniques in making any comparisons of test results for either within-laboratory or between-laboratory averages have been found to be generally accepted.

13. Notes and References

13.1 Publication available from U.S. Department of Health and Human Services, CDC/ NIH-HHS Publication No. (CDC) 84-8395; web site: www.hhs.gov.

13.2 ATCC is the American Type Culture Collection (USA), P.O. Box 1549, Manassas VA 20108; tel: +1.703.365.2700; fax: +1.703.365.2701; web site: www.atcc.org. CIP is the Pasteur Institute Collection (France), DSM is the German Collection of Microorganisms and Cell Cultures (Germany), NBRC is the NITE Biological Resource Center (Japan), NRRL is the Northern Regional Research Lab (USA), NCIMB is the National Collection of Industrial Bacteria (UK), and CUG is the Culture Collection University of Göteborg (Sweden). Equivalent bacteria strains obtained from agencies of the World Federation of Culture Collection (WFCC) may be used by agreement between the interested parties. The strains used in the test shall be documented with their supply source.

13.3 Consistent and accurate testing requires maintenance of a pure, uncontaminated, nonmutant test culture. Avoid contamination by use of good sterile technique in plating and transferring. Avoid mutation by strict adherence to monthly stock transfers. Check culture purity by making streak plates periodically and observing for single species-characteristic type of colonies.

13.4 A suitable viability control fabric known to demonstrate > 1 log growth under the standard conditions in this method has been identified and can be purchased through the International Antimicrobial Council (www.amcouncil.org).

14. History

14.1 Last revised 2019 to remove several issues that led to confusion among users. Ambiguity was reduced by defining the parameters for use of untreated viability controls, concentration of nutrient in inoculum medium, inoculum preparation, fabric swatch preparation, sample sterilization, and final test report contents.

14.2 Revised 2012. Editorially revised 2010, 2009. Reaffirmed 2008. Editorially revised and reaffirmed 2004. Revised 1999. Reaffirmed 1998. Revised 1993. Reaffirmed 1989. Revised (with title change) 1988. Editorially revised and reaffirmed 1986. Editorially revised 1985. Revised 1981. Reaffirmed 1977. Editorially revised 1974, 1971, 1969. Revised 1965.

14.3 Developed in 1961 by AATCC Committee RA31.

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Test Method for Colorfastness to Bleaching with Hydrogen Peroxide

1. Purpose and Scope

1.1 This test method is intended for evaluating the resistance of the color of textiles of all kinds except polyamide, in all forms, to the action of bleach baths containing hydrogen peroxide at concentrations commonly employed in textile processing.

2. Principle

2.1 A specimen of the textile in contact with specified white cloths is immersed in the bleaching solution, rinsed and dried. The change in color of the specimen and the staining of the test cloths are evaluated.

3. Terminology

3.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials, or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 In preparing bleaching baths, use appropriate personal protective equipment while handling hydrogen peroxide (35%) and sodium hydroxide concentrates. Chemical goggles or face shield, rubber gloves and rubber apron should be worn during preparation steps involving these materials.

4.3 An eyewash/safety shower should be located nearby for emergency use.

4.4 Use proper protective equipment, such as gloves and metal tongs, during handling of hot sample tubes.

4.5 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 12.1).

5. Apparatus and Materials (see 12.2)

5.1 Test tube or beaker, diameter and length such that the test specimen roll will have a reasonably snug fit and be covered by the bleach solution.

5.2 Bleach bath of the applicable composition given in Table I.

5.3 Two white test cloths, each 10.2 × 3.8 cm (4.0 × 1.5 in.), one piece made of the same kind of fiber as that of the textile to be tested, the second piece made of the fiber indicated in Table II. A white multifiber test fabric (see 12.3) may be used instead of the second white cloth.

5.4 Gray Scale for Color Change (AATCC EP1) (see 12.7).

5.5 Gray Scale for Staining (AATCC EP2) (see 12.7).

5.6 AATCC 9-Step Chromatic Transference Scale (AATCC EP8) (see 12.7).

6. Test Specimens

6.1 If the textile to be tested is cloth, place a 10.2 × 3.8 cm (4.0 × 1.5 in.) specimen of it between the two pieces of white cloth, and sew along all four sides to form a composite specimen.

6.2 If the textile to be tested is yarn, knit it into cloth and sew the knitted cloth as above, or form a layer of parallel lengths of yarn between the two white cloths, and sew around the four sides to hold the yarns in place.

6.3 If the textile to be tested is fiber, comb and compress some of it into a sheet 10.2 × 3.8 cm (4.0 × 1.5 in.) in size, place the sheet between the two white cloths, and sew around the four sides to form a composite specimen.

7. Procedure (see 12.4)

7.1 For Tests I, II, III, loosely roll the composite specimen in the direction of the long edge, place the roll in the test tube. Immerse cloth in the appropriate bleach solution (Table I), and keep it submerged for the indicated time and temperature (Table I).

7.2 For Test IV, saturate the composite specimen with 100% of its weight of the appropriate bleach solution (Table I), roll in the direction of the long edge and place in saturated steam (see 12.5) at 99-101°C (210-214°F) for 1 h.

7.3 Remove the composite specimen, rinse for 10 min in flowing cold tap water and squeeze. Remove the stitching on two long sides and one short side, open the specimen, and dry at a temperature not exceeding 60°C (140°F) with the three parts in contact only at the one line of stitching.

8. Evaluation Method for Alteration in Color (Shade and Strength)

8.1 Evaluate the test specimen for change in color by comparison with the Gray Scale for Color Change (AATCC

Table I—Composition of Bleaching Bath and Conditions of Use

	Test I Wool	Test II Silk	Test III Cotton	Test IV Cotton
	(Quantities per liter of distilled water)			
Hydrogen peroxide, 35%[a]	15.4 mL (17.5 g)	8.8 mL (10.0 g)	8.8 mL (10.0 g)	8.8 mL (10.0 g)
Sodium silicate, 42° Bé[b]		5.1 mL (7.2 g)	4.2 mL (6.0 g)	7.0 mL (10.0 g)
Sodium pyrophosphate[c]	5.0 g			
Sodium hydroxide[d]			0.5 g	0.5 g
Wetting agent[e]			2.0 mL	
pH (initial)[f]	9.0-9.5	10.5	10.5	10.5
Time	2 h	1 h	2 h	1 h
Temperature	49°C (120°F)	82°C (180°F)	88°C (190°F)	100°C (212°F)
Liquor to cloth ratio	30:1	30:1	30:1	1:1

[a] By weight (H_2O_2) w/w.
[b] 42° Bé, SiO_2:Na_2O ratio = 2.5:1, 10.6% Na_2O, 26.0% SiO_2.
[c] $Na_4P_2O_7 \cdot 10H_2O$.
[d] C.P. grade.
[e] Doubly sulfonated castor oil.
[f] Adjust with NaOH solution if necessary.

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Table II—Choice of Test Cloth

If first piece is	Second piece to be
Wool, silk, linen, rayon	Cotton or multifiber test fabric
Cotton, acetate	Viscose rayon or multifiber test fabric (see 12.3)

EP1), or using AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen, and record the numerical rating that corresponds to the appropriate one on the Gray Scale. For improved precision and accuracy, the specimens should be rated by more than one rater.

9. Evaluation Method for Staining

9.1 Evaluate the staining of the fabric used in the test by comparison with the Gray Scale for Staining (AATCC EP2), the AATCC 9-Step Chromatic Transference Scale (AATCC EP8), or Instrumental Assessment of Degree of Staining (AATCC EP12), and record the numerical grade that corresponds to the appropriate one on either of them (see 12.6). The scale used should be indicated when reporting the test results.

10. Report

10.1 Report the following information for each type of white fiber used in the test:

The bleach solution used (test number in Table I).

The color change grade.

The staining grade.

The evaluation procedure used (AATCC EP2, AATCC EP8 or AATCC EP12).

11. Precision and Bias

11.1 *Precision*. In 2000 a single laboratory study was performed using a single operator.

11.1.1 Samples tested consisted of six fabrics with three replicates each. Refer to Table 1 for the bleach formulas and conditions used. Color change and staining were evaluated instrumentally for each sample.

11.1.2 *Within-laboratory* standard errors and sample variance are shown in Table III. Data is on file at the AATCC Technical Center.

11.2 *Bias*. Colorfastness to bleaching with hydrogen peroxide can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

12. Notes

12.1 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; +1. 513.742.2020; www.acgih.org.

12.2 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

12.3 The white cloths should be plain weave, *medium* weight, free from finishes, residual chemicals and chemically damaged fibers. Cotton and linen should be bleached; other materials should be cleaned to their usual degree of whiteness. A white multifiber test fabric may be used instead of the second white cloth.

12.4 Select from Table I the test most suitable for the conditions under which the fiber is to be used. For example, if a colored silk is to be used as an effect thread in a woolen or worsted fabric, check its colorfastness by the wool method. If a colored silk is to be used as an effect thread in a silk fabric, check its colorfastness by the silk method.

12.5 Saturated steam may be obtained by placing approximately 20 mL of water in the bottom of a test tube fitted with a flared glass rod long enough to keep the specimens above the water level. Heat to active boil. Use a reflux condenser to maintain volume of liquor. Invert a small watch glass over the specimen to prevent direct impingement on the sample of water drops from the condenser.

12.6 For very critical evaluations and in cases of arbitration, ratings must be based on the geometric gray scale for evaluating staining.

12.7 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

13. History

13.1 Revised in 2019 to align format with AATCC style.

13.2 Reaffirmed 2013. Editorially revised 2010. Editorially revised and reaffirmed 2009. Editorially revised 2008. Revised 2004. Editorially revised 2002. Editorially revised 2001. Reaffirmed 1999. Editorially revised 1995. Editorially revised and reaffirmed (with title change) 1994. Editorially revised and reaffirmed 1989. Editorially revised 1987. Editorially revised 1985. Reaffirmed 1984. Reaffirmed 1979. Reaffirmed 1975. Revised 1972. Revised 1968. Revised 1963.

13.3 Developed in 1961 by AATCC Committee RA34.

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Table III—Within-Laboratory Standard Errors and Sample Variance

Sample Identification	Test Parameter	Mean Value	Standard Deviation	Standard Error	Sample Variance	Critical Difference
Brown Wool (Test 1)	Color Change	3.00	0.00	0.00	0.00	0.00
	Color Transfer	4.17	0.29	0.17	0.08	0.50
	Multifiber Stain					
	Acetate	4.17	0.58	0.33	0.33	1.01
	Cotton	4.67	0.29	0.17	0.08	0.50
	Nylon	3.33	0.58	0.33	0.33	1.01
	Polyester	4.67	0.29	0.17	0.08	0.50
	Acrylic	5.00	0.00	0.00	0.00	0.00
	Wool	3.67	0.29	0.17	0.08	0.50
Green Wool (Test 1)	Color Change	2.50	0.50	0.29	0.25	0.87
	Color Transfer	1.67	0.29	0.17	0.08	0.50
	Multifiber Stain					
	Acetate	3.33	0.29	0.17	0.08	0.50
	Cotton	3.33	0.29	0.17	0.08	0.50
	Nylon	2.00	0.00	0.00	0.00	0.00
	Polyester	4.50	0.50	0.29	0.25	0.87
	Acrylic	4.17	0.29	0.17	0.08	0.50
	Wool	3.50	0.87	0.50	0.75	1.51
Red Silk (Test 2)	Color Change	3.17	0.29	0.17	0.08	0.50
	Color Transfer	2.00	0.00	0.00	0.00	0.00
	Multifiber Stain					
	Acetate	1.67	0.29	0.17	0.08	0.50
	Cotton	3.33	0.58	0.33	0.33	1.01
	Nylon	1.67	0.29	0.17	0.08	0.50
	Polyester	2.83	0.29	0.17	0.08	0.50
	Acrylic	4.33	0.29	0.17	0.08	0.50
	Wool	2.50	0.00	0.00	0.00	0.00
Blue Silk (Test 3)	Color Change	2.00	0.00	0.00	0.00	0.00
	Color Transfer	4.67	0.29	0.17	0.08	0.50
	Multifiber Stain					
	Acetate	5.00	0.00	0.00	0.00	0.00
	Cotton	4.50	0.87	0.50	0.75	1.51
	Nylon	5.00	0.00	0.00	0.00	0.00
	Polyester	5.00	0.00	0.00	0.00	0.00
	Acrylic	5.00	0.00	0.00	0.00	0.00
	Wool	4.33	0.29	0.17	0.08	0.50
Purple Cotton (Test 3)	Color Change	3.83	0.29	0.17	0.08	0.50
	Color Transfer	4.67	0.29	0.17	0.08	0.50
	Multifiber Stain					
	Acetate	4.33	0.29	0.17	0.08	0.50
	Cotton	4.67	0.29	0.17	0.08	0.50
	Nylon	4.50	0.00	0.00	0.00	0.00
	Polyester	4.50	0.00	0.00	0.00	0.00
	Acrylic	4.50	0.00	0.00	0.00	0.00
	Wool	3.50	0.00	0.00	0.00	0.00
Gray Cotton (Test 3)	Color Change	3.67	0.29	0.17	0.08	0.50
	Color Transfer	5.00	0.00	0.00	0.00	0.00
	Multifiber Stain					
	Acetate	4.50	0.00	0.00	0.00	0.00
	Cotton	4.83	0.29	0.17	0.08	0.50
	Nylon	4.67	0.29	0.17	0.08	0.50
	Polyester	4.50	0.00	0.00	0.00	0.00
	Acrylic	4.50	0.00	0.00	0.00	0.00
	Wool	3.50	0.00	0.00	0.00	0.00

*Note: Because the interlaboratory test included less than five laboratories, estimates of standard error and sample variance may be either underestimated or overestimated to a considerable extent and should be used with special caution. The values should be viewed as minimal data with regards to precision. Confidence levels are not well established.

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Test Method for Determination of Hydrogen Peroxide by Potassium Permanganate Titration

1. Purpose and Scope

1.1 This test method determines the concentration of hydrogen peroxide (H_2O_2) in aqueous solutions, particularly those used in textile bleaching.

2. Principle

2.1 A specimen is acidified with sulfuric acid and titrated with standardized potassium permanganate solution. The concentration of hydrogen peroxide is calculated using the volume and normality of the permanganate solution used.

3. Terminology

3.1 **bleaching,** n.—elimination of unwanted coloring matter from a textile substrate by oxidative or reductive chemical treatment.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The safety precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care. Sodium oxalate, potassium permanganate and sulfuric acid are corrosive. When preparing the dilute solutions from pure or concentrated reagents, use chemical goggles or face shield, impervious gloves, and an impervious apron. CAUTION: Always add acid to water.

4.3 An eyewash/safety shower should be located nearby and a high efficiency particulate respirator with organic vapor cartridge and full facepiece should be readily available for emergency use.

4.4 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 13.1).

5. Apparatus

5.1 Filter funnel, fritted glass, fine porosity, 250 mL.

5.2 Filtration flask, 2000 mL.

6. Reagents

6.1 Sodium oxalate ($Na_2C_2O_4$), crystals, CP.

6.2 Sulfuric acid (H_2SO_4), 95-98%.

6.3 Potassium permanganate ($KMnO_4$), crystals.

7. Specimens

7.1 Obtain 10 ± 0.1 g of the aqueous hydrogen peroxide specimen into a 250 mL flask (see 13.3).

8. Preparation of Reagents

8.1 Standard 0.100*N* sodium oxalate solution.

8.1.1 Dry at least 7 g sodium oxalate in an oven at 105 ± 1°C (221 ± 2°F) for 4 h and cool in a desiccator.

8.1.2 Weigh 6.7000 ± 0.0002 g sodium oxalate (see 8.1.1) and dissolve in 250-300 mL distilled or deionized water at 70 ± 10°C (158 ± 18°F).

8.1.3 Cool and quantitatively transfer to a 1 L volumetric flask and cover. Allow to stand at least 12 h, then dilute to the mark with distilled or deionized water.

8.2 Sulfuric acid, approximately 20% by volume (see Section 4 for safety precautions).

8.2.1 Slowly add 200 mL 95-98% sulfuric acid to 800 mL water with stirring. Cool to 20 ± 2ûC (68 ± 4ûF).

8.2.2 Add water to 1 L.

8.3 Potassium permanganate solution 0.588*N* (see 13.2).

8.3.1 Weigh 18.6 ± 0.1 g potassium permanganate and add it to 900 mL water.

8.3.2 Boil the solution for 15 min and cool.

8.3.3 Filter the solution through a filter funnel (see 5.1) into a 1 L volumetric flask. Dilute to the mark with water.

8.3.4 Store solutions of potassium permanganate in brown bottles or otherwise protect the solutions from light.

8.4 Standardization of 0.588*N* potassium permanganate solution. Make three determinations as follows.

8.4.1 Pipette 100 mL 0.100*N* sodium oxalate solution into a 250 mL Erlenmeyer flask and add 10 ± 1 mL 20% sulfuric acid.

8.4.2 Heat to boiling. Remove from heat and immediately titrate with the potassium permanganate solution being standardized, adding the titrant drop by drop at the beginning and at the end of the titration. The end-point is a faint but permanent pink color.

8.4.3 Calculate the normality of the potassium permanganate solution to the nearest 0.001 using the following equation:

$$N_k = (V_o)(N_o)/(V_k)$$

where:

V_o = volume of the sodium oxalate solution used, in mL
N_o = normality of the sodium oxalate solution
V_k = volume of the potassium permanganate solution used, in mL

8.4.4 Average the three normality values and use that average for all calculations.

9. Procedure

9.1 Weigh 10 ± 0.1 g of the aqueous hydrogen peroxide specimen into a 250 mL flask (see 7.1 and 13.3).

9.2 Add 20 ± 1 mL 20% sulfuric acid to the flask (see 8.2). Gently mix by swirling or stirring.

9.3 Titrate the solution with standardized potassium permanganate solution (see 8.3 and 8.4) to a faint pink color which lasts at least 30 s. Record the volume of titrant in milliliters as V_t.

9.3.1 If less than 2 mL of titrant are used, repeat the procedure using either a standardized potassium permanganate solution of known lower normality or a larger specimen.

9.4 Titrate a blank, and record the mL titrant required as V_b.

10. Calculations

10.1 Calculate the % H_2O_2 to the near-

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est 0.01% using the following equation:

% H_2O_2 as 100%

$$= (V_t - V_b)(N_t)(0.017)(100)/(W_s)$$

where:

V_t = volume of titrant in mL
V_b = volume of titrant in mL for blank sample
N_t = normality of the titrant solution
W_s = specimen mass, in g

10.2 If 0.588*N* $KMnO_4$ solution is used as the titrant and the specimen mass is 10 g, the equation reduces to:

$$\% H_2O_2 \text{ as } 100\% = (0.1)(V_t - V_b)$$

10.3 Using the conditions of 9.2, for reporting % H_2O_2 on other bases, such as 35% or 50%, V_t may be multiplied by the factor 10/*B*, where *B* is the basis expressed as% H_2O_2. For example, if the basis is 35%,

% H_2O_2 as 35%

$$= (V_t - V_b)(10)/(35) = (V_t - V_b)(0.286)$$

11. Report

11.1 Describe or identify the aqueous hydrogen peroxide specimen

11.2 Report that specimen was tested using AATCC TM 102-2022.

11.3 Report the percentage of Hydrogen Peroxide (%H_2O_2) and the basis / concentration of the Hydrogen Peroxide (i.e. 35%)

12. Precision and Bias

12.1 *Precision*.

12.1.1 In 1993 an interlaboratory study included five laboratories, with two operators each, evaluating two concentrations of hydrogen peroxide in aqueous solution. Two batches of each concentration were prepared. Each operator made duplicate determinations on four separate occasions for each of the two batches, a total of 16 determinations per operator per concentration ($2 \times 4 \times 2$). The data set was analyzed using the ASTM Tex-Pac program (see 13.4).

12.1.2 Analysis showed the residual variances of the two concentrations were different. Accordingly, critical difference tables, with the number of determinations per average shown as *n*, were prepared for each concentration

12.2 *Bias*.

12.2.1 Concentration of hydrogen peroxide in aqueous solution can be defined only in terms of a test method. Within this limitation, this method has no known bias in the determination of percent hydrogen peroxide.

Table I—Critical Differences for Two Averages, 95% Probability Level

Hydrogen Peroxide (Grand Average = 0.65%)

n	Single Operator	Within-Laboratory	Between-Laboratories
1	0.03	0.04	0.05
2	0.02	0.03	0.05
4	0.01	0.03	0.04
8	0.01	0.02	0.04

Table II—Critical Differences for Two Averages, 95% Probability Level

Hydrogen Peroxide (Grand Average = 2.88%)

n	Single Operator	Within-Laboratory	Between-Laboratories
1	0.05	0.06	0.11
2	0.04	0.04	0.11
4	0.03	0.04	0.10
8	0.02	0.03	0.10

13. Notes

13.1 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

13.2 The most widely used standard solution of potassium permanganate is 0.588*N*. This concentration and 0.1*N* solution are available from laboratory supply companies.

13.3 If a volumetric aliquot of the aqueous hydrogen peroxide solution is used as the specimen, the specimen mass may be determined from the relationship:

Specimen mass

= volume of aliquot (mL) × density (g/mL)

The aliquot must be taken at the temperature at which the density is determined. For mill control purposes the density is often disregarded if the bleaching formula remains constant from day to day.

13.4 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832.9500; fax: +1.610.832.9555; web site: www.astm.org.

14. History

14.1 Revised in 2022 to align to AATCC Style Guide.

14.2 Editorially revised (with title change) 2019. Reaffirmed 2016, 2012. Editorially revised (with title change) 2010. Reaffirmed 2007, 2002. Revised 1997. Editorially revised and reaffirmed 1992. Revised 1987 (with title change). Editorially revised and reaffirmed 1985. Editorially revised 1983. Reaffirmed 1979, 1975, 1972, 1968, 1962.

14.3 Developed in 1957 by AATCC Committee RA34.

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Test Method for Bacterial Alpha-Amylase Enzymes used in Desizing

1. Purpose and Scope

1.1 This test method is intended for the assay of the bacterial amylases employed commercially for textile desizing. It is not applicable to products that contain *beta*-amylase in addition to *alpha*-amylase.

2. Principle

2.1 Dextrogenic amylase activity is measured in terms of the digestion time required to produce a color change denoting a definite stage of dextrinization of the starch substrate. The amylase content of the sample, expressed in Bacterial Amylase Units (BAU), is readily calculated from the dextrinizing time (see 13.1).

3. Terminology

3.1 **Bacterial Amylase Unit (BAU),** n.—a measure of starch degradation as shown by the quantity of an enzyme that will dextrinize one milligram of starch per minute under the specified experimental conditions.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Users MUST consult manufactures for specific details such as safety data sheets and other recommendations. Consult and follow all applicable OSHA standards and rules.

4.1 Follow good laboratory practices. Wear safety glasses in all laboratory areas.

4.2 Handle all chemicals with care. Use chemical goggles or face shield, impervious gloves and an impervious apron during dispensing and mixing of iodine; this is a corrosive chemical.

4.3 Ensure an eyewash/safety shower is located nearby and a self-contained breathing apparatus is readily available for emergency use.

4.4 Control exposure to chemicals used in this procedure at or below levels set by governmental authorities [e.g., Occupational Safety and Health Administration's (OSHA) permissible exposure limits (PEL) as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version]. In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 13.2).

5. Uses and Limitations

5.1 The calibration of the method as presented here further restricts its use to amylases of bacterial origin.

5.2 Some surface-active agents interfere with the color development which is an essential part of this method. The method is therefore not suitable for assessing the effect of surfactants on enzymes.

6. Apparatus (see 13.3)

6.1 Comparator (see 13.4).

6.2 Water bath, controlled at constant temperature 30 ± 1°C (86 ± 2°F).

6.3 Electric timer.

6.4 Glassware (see 13.5).

6.4.1 Borosilicate test tubes, 13 × 100 mm—at least six dozen should be on hand.

6.4.2 Standard volumetric pipettes to deliver 2 mL, 5 mL, 10 mL, 20 mL, fast 1 mL blow-out pipette for sampling hydrolyzing mixture, plus assorted standard pipettes to accomplish necessary sample preparation dilutions.

7. Reagents

7.1 Iodine (crystalline); reagent grade; I_2

7.2 Potassium iodide; KI

7.3 Potassium dihydrogen phosphate; KH_2PO_4 (monobasic)

7.4 Sodium phosphate; Na_2HPO_4 (dibasic)

7.5 Merck's Lintner Starch

8. Preparations

8.1 Test Sample Preparation.

8.1.1 Prepare the sample solution by dilution with water so that 10 mL of the final solution (see Table I) will give a dextrinizing time of 15-35 min. With dry samples, insoluble materials may be present but it is generally unnecessary to filter the solution.

8.1.2 Table I shows sample weight to be used with samples of varying alpha-amylase content. If the sample can be placed in either of two ranges, it is preferable to use the sample weight corresponding to the higher range since this will give a longer dextrinization time, making for easier and more accurate measurements.

Table I—Guide to Selection of Sample Weight

Sample alpha-amylase content (BAU per g)	Sample weight to be used mg sample per 10 mL final dilution
70-250	200
125-500	100
300-900	50
600-1800	25
1000-4000	10
3000-9000	5
6000-18000	2.5
12500-50000	1

8.1.3 Liquid products should be weighed as the specific gravity is usually greater than 1.0.

8.1.4 In all cases, a sufficient amount of sample should be weighed to minimize weighing errors. If necessary, the amount of sample to be applied to the test may be obtained by secondary volume dilution.

8.2 Stock Iodine Solution.

8.2.1 Weigh in glass, 5.5 g reagent grade crystalline iodine; to dissolve, add 11 g reagent grade potassium iodine dissolved in minimal (10-12 mL) water. When iodine has completely dissolved dilute to 250 mL.

8.2.2 Preserve this solution in an amber glass-stoppered bottle and store under refrigeration. Solution may be used for three (3) months.

8.3. Dilute Iodine Solution.

8.3.1 Dissolve 2.0 mL of stock iodine solution and 20 g reagent grade potassium iodide in water, and dilute to 500 mL.

8.3.2 The dilute iodine solution may be stored under refrigeration and used for one (1) week. However, at the time of its use (see 9.1), it should be at 30 ± 1°C (86 ± 2°F).

8.4 Buffer Solution, pH 6.6.

8.4.1 Solution A. Dissolve 9.078 g of KH_2PO_4 in water and dilute to 1 L.

8.4.2 Solution B. Dissolve 9.472 g of Na_2HPO_4 in water and dilute to 1 L.

8.4.3 Mix 600 mL of Solution A and 400 mL of Solution B to obtain pH 6.6 buffer.

8.5 Buffered Starch Substrate.

8.5.1 Determine the dry weight of Merck's Lintner Starch (Special for Diastatic Power Determinations) by drying 20.00 g of starch at 103-104°C (217-

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219°F) for 3 h. Weigh after cooling in desiccator and continue drying until weight is constant. Discard after dry weight has been determined.

8.5.2 Calculate the amount of starch equivalent to 10.00 g dry basis for 500 mL starch substrate.

8.5.3 Starch should be retained in tightly closed jar and not exposed to environment where the moisture content would be subject to change.

8.5.4 Quantitatively transfer a slurry of 10.00 g (dry weight basis) of Merck's Lintner Starch into about 300 mL of vigorously stirred boiling water contained in a 1-L Pyrex beaker. Leave the stirring rod in the solution after boiling resumes and boil for exactly 3 min.

8.5.5 Cool to room temperature in a cold-water bath with continuous stirring to avoid skin formation (surface dehydration).

8.5.6 Quantitatively transfer the starch into a 500-mL volumetric flask using a small quantity of water to complete the transfer.

8.5.7 Add 10 mL of pH 6.6 buffer and dilute to the mark.

8.5.8 Check the pH of the starch substrate with a standardized pH meter.

8.5.9 Starch substrate should be free of lumps or flakes and prepared fresh daily. Contamination of the starch substrate with even minute traces of enzyme will render the substrate unsuitable for use.

9. Procedure

9.1 Dispense 5.0 mL dilute iodine solution in each of a series of test tubes and allow to attemperate in the 30 ± 1°C (86 ± 2°F) bath.

9.2 Transfer 20.0 mL of buffered starch substrate into a 50-mL Erlenmeyer (or equivalent) (lead rings are convenient to weight the flask), stopper, place in 30 ± 1°C (86 ± 2°F) bath allowing about 15 min to attain uniform temperature within the flask.

9.3 Likewise, attemperate an appropriate amount of freshly prepared sample solution. Rapidly add 10.0 mL of the sample solution, using a blow-out type pipette, and start the timer. After pipette has drained, replace rubber stopper and swirl flask vigorously to ensure proper mixing.

9.4 At appropriate time intervals, add 1 mL (blow-out pipette with cotton plug) of the hydrolyzing mixture to 5 mL of dilute iodine solution at 30 ± 1°C (86 ± 2°F), shake to mix thoroughly, pour into the 13 mm precision square tube, and compare with standard alpha-amylase color disc in the comparator. After completing the comparison, empty the square tube, giving it a quick shake so that very little liquid remains. The tube may now be used for another test.

9.5 During the initial stages of reaction, it is not necessary that the 1-mL sample be measured precisely before adding to the dilute iodine solution. As the end point is approached, the addition must be made accurately. The contents of the pipette are blown into the iodine solution in order that the time may be more accurately measured.

9.6 Around the time of the end point, samples should be taken every 0.5 min. In case two (2) samples 0.5 min apart show that one is darker than the standard color and the other one is lighter, record the end-point time as the quarter minute between these two times.

9.7 Care must be exercised to avoid contact of the 1-mL hydrolyzing pipette with the dilute iodine solution. A carry-back of iodine to the hydrolyzing mixture will interfere with enzyme action.

10. Calculation

10.1 Calculate the alpha-amylase content of the sample by using the following equation:

$$\text{BAU per g} = 40(F)/T$$

10.2 Explanation of equation (see 13.1):

BAU = Bacterial Amylase Units per gram (g)
F = Dilution Factor (total dilution volume/sample weight in grams)
T = Dextrinizing time in minutes

10.3 The preceding equation follows from:

(a) the definition of the BAU as the quantity of enzyme that dextrinized one mg of starch per minute, and

(b) the assay practice of dextrinizing 400 mg of starch (20 mL of 2% solution) with a 10 mL aliquot of enzyme solution. Thus:

$$\text{BAU per gram} = 400/T \times F/10 = 40F/T$$

10.4 Example of Calculations:

10.4.1 If we assume that for a particular sample being tested the anticipated BAU/g is 800, reference to Table I indicates that 25 mg of sample per 10 mL of final dilution should be employed. Therefore, 2.5 g of the sample would be weighed out, and diluted to 1000 mL. It now becomes possible to calculate the Dilution Factor, F:

F = Total Dilution Volume/Sample Weight in grams = 1000/2.5 = 400

10.4.2 If the dextrinizing time is 20 min, the BAU/g can be calculated:

$$\text{BAU/g} = 40\,F/T = 40 \times 400/20 = 16000/20 = 800$$

10.4.3 If BAU does not fall within expected limits, a retest should include the practice of making up an entirely new sample preparation. This constitutes a check on the preparation of the test solution where errors may have occurred resulting in erroneous BAU.

11. Report

11.1 Report the amylase content of the sample as Bacterial Amylase Units (BAU) per gram.

12. Precision and Bias

12.1 *Precision*. The mean of duplicate tests should check within ± 6.5% of the true mean at the 95% confidence level. This information is based on an interlaboratory study.

12.2 *Bias*. The values, Bacterial Assay Units (BAU), can only be defined in terms of a test method. There is no independent method for determining the true value. Based on the information available, the method has no known bias.

13. Notes

13.1 If approximate or anticipated BAU of material to be tested is known, determine dilution factor (F) by multiplying selected test time (T) by anticipated BAU and dividing by 40, thus: $T \times \text{BAU}/40 = F$.

13.2 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240, USA; +1. 513.742.2020; www.acgih.org.

13.3 For potential apparatus, reagents or materials sources, visit the AATCC Buyer's Guide at www.aatcc.org/bg. AATCC provides its Corporate members the option to their items and services. AATCC does not qualify, or in any way approve, endorse or certify that any of the listings meet the specifications in its standards.

13.4 Light Source. Should be either daylight or daylight fluorescent lamps. Incandescent lamps should not be used since they give slightly lower results.

13.5 All glassware must be clean. This applies particularly to pipettes where a hang-up of droplets interferes with correct volume delivery. Sulfuric acid-potassium dichromate cleaning solution is an efficient cleanser but must be thoroughly removed by repeated rinsing.

14. History

14.1 Revised (with title change) in 2019 to align format with AATCC style.

14.2 Reaffirmed 2013. Editorially revised 2010. Reaffirmed 2009. Editorially revised 2008. Reaffirmed 2004. Revised 1999. Editorially revised and reaffirmed 1994. Editorially revised 1991. Editorially revised and reaffirmed 1989. Editorially revised 1986. Editorially revised 1985. Reaffirmed 1984. Reaffirmed 1979. Reaffirmed 1976. Reaffirmed 1973. Reaffirmed 1970. Reaffirmed 1965.

14.3 Developed in 1962 by AATCC Committee RR41. Jurisdiction transferred to RA34 in 1987, returned to RR41 in 1993. Currently maintained by RA99.

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Test Method for Colorfastness to Water Spotting

Developed in 1962 by AATCC Committee RA23; reaffirmed 1966, 1969, 1972, 1975, 1978, 1988, 1989, 1999; editorially revised and reaffirmed 1981, 1983, 1994, 2004, 2014; revised 2010; editorially revised 2019. Technically equivalent to ISO 105-E07.

1. Purpose and Scope

1.1 This test method is designed to evaluate the resistance to water spotting of dyed, printed or otherwise colored textile fabrics. Whites can also exhibit color change, such as yellowing.

1.2 The test method does not determine whether the discoloration is removable.

2. Principle

2.1 The test specimen is spotted with distilled or deionized water and evaluation is made of change of color, after wetting and after drying.

3. Terminology

3.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5. Apparatus and Materials

5.1 Glass rod.

5.2 Gray Scale for Color Change (AATCC EP1) (see 11.1).

5.3 Pipette, graduated, 1 mL.

5.4 Distilled or deionized water.

6. Test Specimens

6.1 Colored cloth specimen, approximately 15.2 × 15.2 ± 0.4 cm (6 × 6 ± 0.16 in.).

7. Procedure

7.1 With tip of pipette in contact with cloth, run onto the specimen 0.15 mL water at room temperature. If necessary, work water in with a rounded glass rod to assist penetration.

7.2 Evaluate the change in color in the periphery of the spot after 2 min wetting and after drying at room temperature by use of the Gray Scale for Color Change (AATCC EP1), or using AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen.

8. Evaluation

8.1 Rate the effect on the color of the test specimens by the Gray Scale for Color Change (AATCC EP1), or using AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen, and record the numerical rating that corresponds to the appropriate one on the Gray Scale (see 11.1).

9. Report

9.1 Report type of water and pH of water used for the test.

9.2 Report the grade for change in color after 2 min and after drying at room temperature.

10. Precision and Bias

10.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

10.2 *Bias*. Colorfastness to water spotting can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

11. Note

11.1 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

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Test Method for Colorfastness to Water: Sea

Developed in 1962 by AATCC Committee RA23; revised 1967, 1968, 1972, 1981, 2009; reaffirmed 1975, 1978, 1989, 2007; editorially revised 1985, 1994, 2001, 2005, 2008, 2010, 2016, 2019; editorially revised and reaffirmed 1986, 1991, 1997, 2002, 2013. Partly equivalent to ISO 105-E02.

1. Purpose and Scope

1.1 This test method is designed to measure the resistance to sea water of dyed, printed, or otherwise colored textile yarns and fabrics of all kinds.

1.2 Artificial sea water is used in this test because natural sea water is variable in composition, and is often difficult to obtain.

2. Principle

2.1 The specimen, backed by multifiber test fabric, is immersed in artificial sea water under specified conditions of temperature and time, and then placed between glass or plastic plates under specified conditions of pressure, temperature and time. The change in color of the specimen and the staining of the attached multifiber test fabric are observed.

3. Terminology

3.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

4.4 Observe padder safety. Ensure adequate guard at the nip point. Normal safeguards on pad should not be removed.

5. Apparatus and Materials (see 12.1)

5.1 Perspiration tester (plastic or glass plates are available with the equipment) (see 12.2).

5.2 Drying oven—convection.

5.3 Multifiber test fabrics (8 mm [0.33 in.] bands) containing acetate, cotton, nylon, silk, viscose rayon and wool shall be used for specimens containing silk, or (8 mm [0.33 in.] bands) containing acetate, cotton, nylon, polyester, acrylic and wool shall be used for specimens with no silk present.

5.4 AATCC 9-Step Chromatic Transference Scale (AATCC EP8) (see 12.3).

5.5 Gray Scale for Color Change (AATCC EP1) and Gray Scale for Staining (AATCC EP2) (see 12.3).

5.6 Wringer.

6. Test Solution (Artificial Sea Water)

6.1 Per liter:

Sodium chloride (NaCl), tech. 30 g

Magnesium chloride ($MgCl_2$), anhyd. 5 g

Distilled water to make 1000 mL solution

7. Test Specimens

7.1 If the specimen to be tested is a fabric, attach a piece of multifiber adjacent fabric also measuring 5 × 5 ± 0.2 cm to the specimen measuring 6 × 6 ± 0.2 cm by sewing along one of the shorter sides, with the multifiber fabric next to the face of the specimen.

7.2 If the specimen to be tested is a yarn or loose fiber, take a mass of the yarn or loose fiber approximately equal to one half of the combined mass of the adjacent fabrics. Place it between a 5 × 5 ± 0.2 cm piece of multifiber fabric and a 6 × 6 ± 0.2 cm piece of the non-dyeable fabric, and sew along all four sides.

8. Procedure

8.1 Immerse the test specimen in the test solution at room temperature with occasional agitation to ensure thorough wetting out (approximately 15 min generally required for average fabrics) (see 12.4).

8.2 Remove the test specimen from the test solution and only pass between squeeze rolls (wringer) to remove excess liquor when the wet weight of the test specimen is more than 3 times its dry weight. Whenever possible, the wet weight should be 2.5-3.0 times the dry weight.

8.3 Place the test specimen between glass or plastic plates and insert in the specimen unit of the perspiration tester. Adjust the perspiration tester to produce a pressure of 4.5 kg (10.0 lb) on the test specimen (see 12.2).

8.4 Heat the loaded specimen unit in an oven at 38 ± 1°C (100 ± 2°F) for 18 h.

8.5 Remove the tester from the oven and for each test specimen assembly, separate the multifiber fabric and, if used, the adjacent fabric from the test fabric. Place the multifiber fabric and test fabric specimens separately on a wire screen in a conditioned atmosphere 21 ± 2°C (70 ± 4°F) and 65 ± 5% relative humidity overnight.

9. Evaluation Method for Color Change

9.1 Evaluate the test specimen for change in color by comparison with the Gray Scale for Color Change (AATCC EP1) or using AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen, and record the numerical rating that corresponds to the appropriate one on the Gray Scale.

10. Evaluation Method for Staining

10.1 Evaluate the staining of the multifiber fabric used (see 12.5) by comparison with the Gray Scale for Staining (AATCC EP2), the AATCC 9-Step Chromatic Transference Scale (AATCC EP8) or Instrumental Assessment of Degree of Staining (AATCC EP12), and record the numerical rating that corresponds to the appropriate one on either of them. Report which scale is used (see 12.6).

11. Precision and Bias

11.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

11.2 *Bias*. Colorfastness to sea water can be defined only in terms of a test method. There is no independent method

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for determining the true value. As a means of estimating this property, the method has no known bias.

12. Notes

12.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

12.2 Horizontal Perspiration Tester: Put all 21 glass or plastic plates into the unit regardless of the number of specimens. After the final glass or plastic plate is put in position on top, set the dual plates with compensating springs in position. Place a 3.6 kg (8.0 lb) weight on top making a total of 4.5 kg (10.0 lb) under the pressure plate. Lock the pressure plate in position by turning the thumb-screws. Remove the weight. Place the unit in the oven on its side, so that the plates and the specimens are vertical.

Vertical Perspiration Tester: The plates are held in a vertical position between an indicating scale with a fixed metal plate at one end and an adjustable metal plate at the other end. By means of adjusting screws, the movable plate may be made to exert increasing pressure against the test specimens. When the desired pressure of 4.5 kg (10.0 lb) is indicated on the scale, lock the specimen unit by a set screw. The specimen unit can now be removed from the section applying the pressure. Another specimen unit may be added to the pressure section and the loading procedure repeated.

12.3 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

12.4 Or immerse the test specimen in the test solution at room temperature, pass through squeeze rolls (wringer) and reimmerse. Repeat, if necessary, to attain thorough wetting out.

12.5 Classify according to the fiber showing the greatest stain.

12.6 For very critical evaluations and in cases of arbitration, ratings must be based on the Gray Scale for Staining.

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Test Method for Colorfastness to Water

1. Purpose and Scope

1.1 This test method is designed to measure the resistance to water of dyed, printed, or otherwise colored textile yarns and fabrics.

1.2 Distilled water or deionized water is used in this test method because natural (tap) water is variable in composition.

2. Principle

2.1 The specimen, backed by multifiber test fabric, is immersed in water and then subjected to specified conditions of pressure, temperature and time. After conditioning, the change in color of the specimen and the staining of the attached multifiber test fabric are observed.

2.2 Results are reported as a Gray Scale for Change and Staining of grade 5 to 1. A grade of 5 represents no color change and a grade of 1 represents the most color change.

3. Referenced Documents

3.1 ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 12.1).

4. Terminology

4.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

5. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

5.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5.2 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

5.3 See 9.1 for safety precautions related to using a wringer

6. Apparatus and Materials (see 12.2)

6.1 Perspiration tester (with acrylic plates) (see Figs. 1 and 2).

6.2 Drying oven—convection.

6.3 Cold cut multifiber test fabrics (8 mm [0.33 in.] bands) containing acetate, cotton, nylon, silk, viscose rayon and wool shall be used for specimens containing silk, or (8mm [0.33 in.] bands) containing acetate, cotton, nylon, polyester, acrylic and wool shall be used for specimens with no silk present (see 12.3).

6.4 Gray Scale for Staining (AATCC EP2 or EP12) or AATCC 9-Step Chromatic Transference Scale (AATCC EP8) (see 11.4).

6.5 Gray Scale for Color Change (AATCC EP1 or EP7) (see 12.4).

6.6 Wringer.

6.7 Petri dish with a depth greater than 1.5cm and capable of containing a 6 x 6 ± 0.2 cm test specimen

6.8 Un-dyed adjacent fabric

6.9 White AATCC Textile Blotting Paper (see 12.4).

6.10 Balance with a weighing accuracy of ± 0.001 g.

7. Test Solution

7.1 Distilled water or deionized water from an ion-exchange device.

8. Test Specimens

8.1 Number and size of specimens

8.1.1 For fabric testing, one specimen, 6 x 6 ± 0.2 cm, is needed. Attach a 5 x 5 ± 0.2 cm multifiber test fabric swatch to the face of the fabric by sewing a single seam stitch along one edge of the fabric.

8.1.2 For yarn or loose fiber testing, weigh a 5 × 5 ± 0.2 cm piece of multifiber fabric and a 6 × 6 ± 0.2 cm piece of the un-dyed adjacent fabric together. Then take a mass of the yarn or loose fiber approximately equal to one half of the combined mass of the adjacent fabrics. Place it between a 5 × 5 ± 0.2 cm piece of multifiber fabric and a 6 × 6 ± 0.2 cm piece of the un-dyed adjacent fabric, and sew a single seam stitch along all four sides.

9. Procedure

9.1 Weigh each test specimen (as prepared in 8.1) to the nearest 0.1g. Immerse the test specimen in a petri dish in the test solution which is at room temperature with occasional agitation to ensure thorough wetting out (approximately 15 min generally required for average fabrics). For fabrics hard to wet out, immerse the test specimen in the test solution at room temperature, pass through squeeze rolls (wringer) and re-immerse. Repeat, if necessary, to attain thorough wetting out. When using the wringer, ensure adequate guard at the nip point. Normal safeguards on pad should not be removed.

9.2 Remove the test specimen from the test solution and pass each test specimen assembly through the wringer with the multifiber stripes perpendicular to the length of the wringer rolls (all stripes go through the wringer at the same time). Weigh each test specimen to be sure it weighs greater than or equal to 2.5 times its original weight. Whenever possible, the wet weight should be 2.5-3.0 times the original weight. Because certain fabrics may not be able to retain this amount of solution when passing through a wringer, such fabrics may be tested after blotting to the required wet pickup with White AATCC Textile Blotting Paper (see 12.4).

9.3 Place the test specimen assembly between acrylic plates and insert in the specimen unit of the perspiration tester with the multifiber stripes running perpendicular to the long dimension of the plate (see Fig. 3).

9.4 Depending upon equipment available, use the following alternates:

9.4.1 Horizontal Perspiration Tester (see Fig. 1): Place the plates in the perspiration tester with the specimen assem-



Fig. 1—Horizontal perspiration tester.

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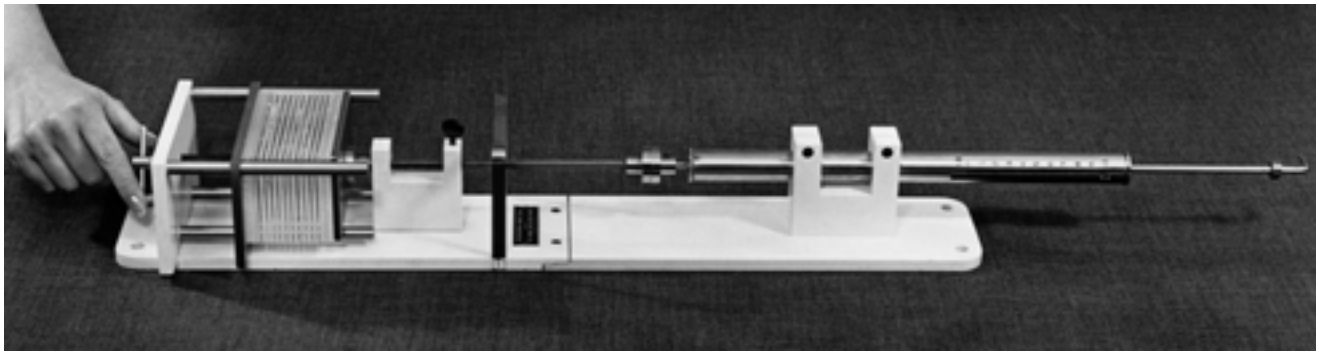


Fig. 2—Vertical perspiration tester.

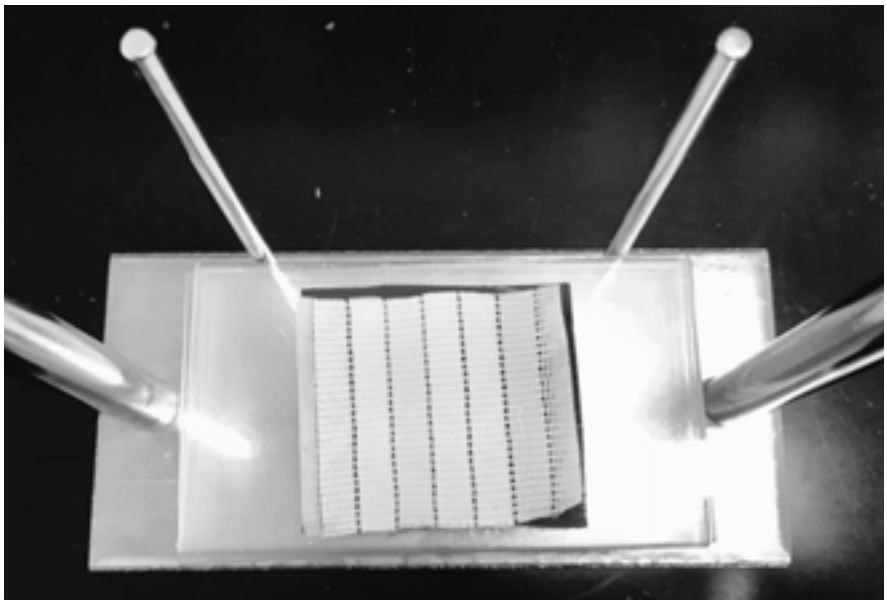


Fig. 3—Specimen in holder.

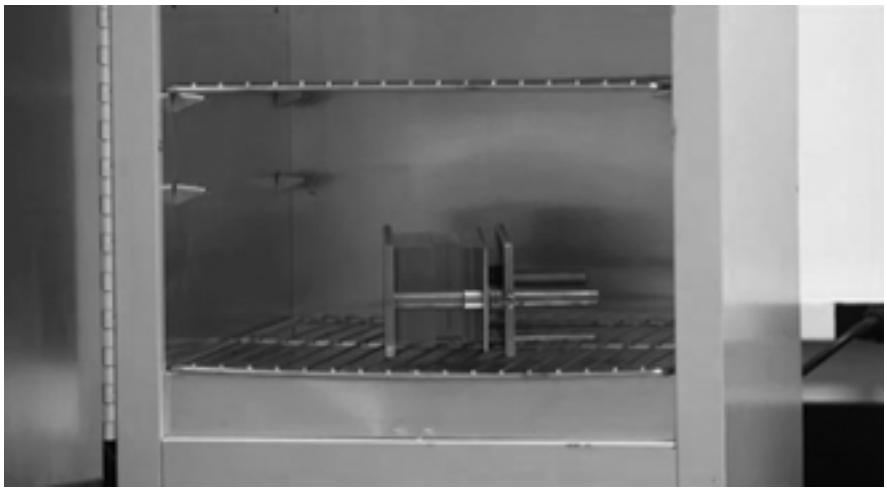


Fig. 4—Horizontal perspiration tester placement in oven.

blies evenly distributed between the 21 plates. Place all 21 plates into the unit regardless of the number of specimens. After placing the final plate in position (on top) set the dual plates with compensating springs in position, place the 3.63 kg (8.0 lb) weight on top making a total of 4.54 kg (10.0 lb) under the pressure plate, and lock the pressure plate in position by turning the thumb screws. Remove the weight and place the unit lying on its side in the oven so that the sides of the perspiration tester are parallel to the oven walls (see Fig. 4).

9.4.2 Vertical Perspiration Tester (see Fig. 2): Assemble the plates in the perspiration tester with the specimens evenly distributed between the 21 plates. Place all 21 plates into the unit regardless of the number of specimens. The plates are held in a vertical position between an indicating scale with a fixed metal plate at one end and an adjustable metal plate at the other end. Use the adjusting screw to exert a 4.54 kg (10.0 lb) force against the plates. Lock the specimen unit containing the test specimens with a set screw. Remove the pressure gauge unit from the specimen unit and place the specimen unit in the oven such that the side of the perspiration tester is to the oven walls. Another specimen unit may be added to the pressure gauge unit and the loading procedure repeated.

9.5 Heat the loaded specimen unit in an oven at 38 ± 1°C (100 ± 2°F) for 18 h ± 5 min.

9.6 Remove the tester from the oven and for each test specimen assembly, separate the multifiber fabric and, if used, the adjacent fabric from the test fabric. Place the multifiber fabric and test fabric specimens separately on a wire screen in accordance with ASTM D1776 overnight.

10. Evaluation

10.1 Color Change.

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10.1.1 Evaluate the test specimen for change in color by comparison with the Gray Scale for Color Change (AATCC EP1), or by using Instrumental Assessment of the Change in Color of a Test Specimen (AATCC EP7), and record the numerical rating that corresponds to the appropriate one on the Gray Scale.

10.2 Staining (see 12.7).

10.2.1 Evaluate the staining of the multifiber fabric used (see 12.5) by comparison with the Gray Scale for Staining (AATCC EP2), the AATCC 9-Step Chromatic Transference Scale (AATCC EP8), or Instrumental Assessment of Degree of Staining (AATCC EP12), and record the numerical rating that corresponds to the appropriate one on either of them. Report which scale was used (see 12.6).

11. Precision and Bias

11.1 *Precision*. Proficiency data generated by 91 different labs during five different time periods (June 2012, December 2012, June 2013, December 2013 and June 2014) was used to define the precision of stain rating and color change rating values. Each lab used the same fabric materials to obtain the stain rating and color change rating values. Each lab performed three replicate tests for each test material and used three different raters to assign ratings for staining and color change. Each lab also used multi-fiber fabric strips containing acetate, acrylic, cotton, nylon, polyester and wool fibers to assess the staining of individual fibers.

11.1.1 Tables I-VI give the within lab and between lab precision values for the stain ratings of acetate, acrylic, cotton, nylon, polyester and wool fibers.

Table I—Precision Table for Stain Rating of Acetate

No. of Specimens	Within Lab Precision	Between Lab Precision
1	0.02879	0.03715
2	0.0204	0.0263
3	0.0166	0.0214
4	0.0144	0.0186
5	0.0129	0.0166
6	0.0118	0.0152

Table II—Precision Table for Stain Rating of Acrylic

No. of Specimens	Within Lab Precision	Between Lab Precision
1	0.02983	0.03849
2	0.0211	0.0273
3	0.0172	0.0222
4	0.0149	0.0192
5	0.0133	0.0172
6	0.0122	0.0157

Table III—Precision Table for Stain Rating of Cotton

No. of Specimens	Within Lab Precision	Between Lab Precision
1	0.11251	0.14512
2	0.0797	0.1028
3	0.0650	0.0838
4	0.0563	0.0726
5	0.0503	0.0649
6	0.0459	0.0592

Table IV—Precision Table for Stain Rating of Polyester

Specimens	Within Lab Precision	Between Lab Precision
1	0.06906	0.08911
2	0.0489	0.0631
3	0.0399	0.0514
4	0.0345	0.0446
5	0.0309	0.0399
6	0.0282	0.0364

Table V—Precision Table for Stain Rating of Polyester

No. of Specimens	Within Lab Precision	Between Lab Precision
1	0.03017	0.03894
2	0.0214	0.0276
3	0.0174	0.0225
4	0.0151	0.0195
5	0.0135	0.0174
6	0.0123	0.0159

Table VI—Precision Table for Stain Rating of Wool

No. of Specimens	Within Lab Precision	Between Lab Precision
1	0.03743	0.04831
2	0.0265	0.0342
3	0.0216	0.0279
4	0.0187	0.0242
5	0.0167	0.0216
6	0.0153	0.0197

11.1.2 Table VII gives the precision values for color change rating. As mentioned above, the multi-period data generated by 91 different labs was used to compute the precision values.

11.1.3 In addition to providing the precision values, the analysis of the multi-period data revealed the following statistical facts:

- The stain ratings assigned by different labs for **acetate** fiber differed significantly at 95% confidence level.
- The mean stain ratings of **acetate** fiber corresponding to five different time periods differed significantly at 95% confidence level.
- The stain ratings assigned by different labs for **acrylic** fiber differed significantly at 95% confidence level.
- The mean stain ratings of **acrylic** fiber corresponding to five different time periods differed significantly at 95% confidence level.
- The stain ratings assigned by different labs for **cotton** fiber **did not** differ significantly at 95% confidence level.
- The mean stain ratings of **cotton** fiber corresponding to five different time periods differed significantly at 95% confidence level.
- The stain ratings assigned by different labs for **nylon** fiber **did not** differ significantly at 95% confidence level.
- The mean stain ratings of **nylon** fiber corresponding to five different time periods differed significantly at 95% confidence level.
- The stain ratings assigned by different labs for **polyester** fiber differed significantly at 95% confidence level.
- The mean stain ratings of **polyester** fiber corresponding to five different time periods differed significantly at 95% confidence level.
- The stain ratings assigned by different labs for **wool** fiber **did not** differ significantly at 95% confidence level.
- The mean stain ratings of **wool** fiber corresponding to five different time periods differed significantly at 95% confidence level.
- The mean **color change** ratings of different labs **did not** differ significantly at 95% confidence level. The mean **color change** ratings of five different time periods differed significantly at 95% confidence level.
- In general the between period variation was more than the between lab variation.

Table VII—Precision Table for Color Change Rating

No. of Specimens	Within Lab Precision	Between Lab Precision
1	0.04181	0.05376
2	0.0296	0.0381
3	0.0241	0.0310
4	0.0209	0.0269
5	0.0187	0.0240
6	0.0171	0.0219

11.2 *Bias*. Colorfastness to water can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

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12. Notes

12.1 Available from ASTM International, www.astm.org.

12.2 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at http://www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

12.3 The six fiber test fabrics without fused edges should be used in this method.

12.4 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: orders@aatcc.org; web site: www.aatcc.org.

12.5 Classify according to the fiber showing the greatest stain.

12.6 For very critical evaluations and in cases of arbitration, ratings must be based on the Gray Scale for Staining.

12.7 CAUTION: It has been reported that the results for staining obtained by this method on fabrics dyed to dark shades (navy, black, etc.) that contain a combination of polyester and spandex, or their blends, may not show the full staining propensity of such fabrics in consumer use. It is, therefore, recommended that the staining results obtained by this test not be used for the acceptance testing of such fabrics.

13. History

13.1 Editorially revised in 2023 to correct misspelling.

13.2 Revised 2022, 2013, 2012; editorially revised 2010; revised 2009; editorially revised 2008; reaffirmed 2007; editorially revised 2005; editorially revised and reaffirmed 2002; editorially revised 2001; editorially revised and reaffirmed 1997; editorially revised 1994; editorially revised and reaffirmed 1991; reaffirmed 1989; editorially revised and reaffirmed 1986; editorially revised 1985, 1983; revised 1981; reaffirmed 1978, 1975; revised 1972, 1968, 1967.

13.3 Developed in 1962 by AATCC Committee RA23. Technically equivalent to ISO 105-E01.

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Test Method for Colorfastness to Ozone in the Atmosphere under Low Humidities

Developed in 1963 by AATCC Committee RA33; revised 1972, 1986, 1987, 2011; reaffirmed 1971, 1975, 1983, 1992, 2002, 2005, 2016; editorially revised 1981, 1982, 1983, 1985, 1989, 1995, 2008, 2010, 2019; editorially revised and reaffirmed 1997. Partly equivalent to ISO 105-G03.

1. Purpose and Scope

1.1 This test method is intended for determining the resistance of the color of textiles to the action of ozone in the atmosphere at room temperatures with relative humidities not exceeding 67%.

2. Principle

2.1 A test specimen is exposed to ozone in an atmosphere under ambient temperatures (64-82°F) and relative humidities not exceeding 67%. The cycle is repeated (1) until the specimen shows a definite color change or (2) for a prescribed number of cycles.

3. Terminology

3.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The safety precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment. To protect the eyes further, do not look at ozone generator while it is operating.

4.3 Ozone is a sensory irritant. The test cabinet should be vented to the outside atmosphere in accordance with the manufacturer's specifications. Ozone in even moderate concentrations is injurious to health.

4.4 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 12.1).

5. Uses and Limitations

5.1 Although this test works well for some substrates, it is not suitable for nylon carpets. For nylon carpets, refer to AATCC TM129, Test Method for Colorfastness to Ozone in the Atmosphere Under High Humidities.

6. Apparatus and Materials (see 12.2)

6.1 Ozone exposure chamber for ambient temperatures and relative humidities not to exceed 67% (see 12.3).

6.2 Gray Scale for Color Change (AATCC EP1) (see 12.4).

7. Test Specimens

7.1 Cut each test specimen at least 10.0 × 6.0 cm (4.0 × 2.375 in.). For subsequent color comparisons, keep the unexposed sample in an airtight container away from light to avoid shade change.

7.2 When a laundered or drycleaned specimen is used, the effects of ozone are based on comparisons with the color of the unexposed laundered or drycleaned specimen. To prepare specimens for testing after laundering or drycleaning, use AATCC TM61, Test Method for Colorfastness to Laundering: Accelerated, and AATCC TM132, Test Method for Colorfastness to Drycleaning, respectively.

8. Procedure

8.1 Suspend the test specimens in the exposure chamber (see 12.3). The test apparatus should be located in a room in which the ambient air temperature is 18-28°C (64-82°F) and the relative humidity does not exceed 67%. For reference and interlab testing, conduct tests in a room or chamber with the standard atmospheric conditions of 21 ± 2°C (70 ± 4°F) and 65 ± 5% RH. Ozone should be present in concentrations of 4.5 ± 1 ppm for 4.5 ± 1 h. This constitutes one cycle (see 12.5).

8.2 Remove those specimens which exhibit a color change at the end of one cycle. One cycle will generally produce a measurable color change in specimens which are ozone sensitive.

8.3 Run additional similar cycles as necessary.

9. Evaluation

9.1 At the end of each cycle immediately compare those specimens which are removed from the exposure chamber with the preserved originals from which they were drawn.

9.2 Rate the effect on color of test specimens after any specified number of cycles by reference to the Gray Scale for Color Change (AATCC (EP1) or using AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen, and record the numerical rating that corresponds to the appropriate one on the Gray Scale. Report the number of cycles run (see 12.6).

10. Report

10.1 Report the numerical grade for the change in color of each test specimen, the number of cycles run and the temperature and relative humidity at which the test was performed.

11. Precision and Bias

11.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

11.2 *Bias*. Colorfastness to ozone in the atmosphere under low humidities can be defined only in terms of a test method. There is no independent method for determining the true values. As a means of estimating these properties, the method has no known bias.

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12. Notes

12.1 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

12.2 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

12.3 The ozone exposure chamber for room temperatures and relative humidities not exceeding 67% consists of an ozone generator, fan, baffle system, specimen rack and housing. Any form of an ozone generator may be used to produce the required concentration; however, ultraviolet light from mercury bulb generators or from spark-gap generators should be prevented from reaching the specimen on the rack by a suitable shield.

12.4 The Gray Scale for Color Change is available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1. 919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

12.5 For information regarding the measurement of ozone concentrations, see the following references:

Schulze, Fernand, "Versatile Combination Ozone and Sulfur Dioxide Analyzer," Analytical Chemistry 38, May 1966, pp748-752.

"Selected Methods of the Measurement of Air Pollutants," Public Health Service Publication No. 999-AP-11, May 1965. Office of Technical Information and Publications (OTIP), Springfield VA. PB 167-677.

12.6 An automated electronic grading system may be used as long as the system has been demonstrated to provide results that are equal to and provide equal or better repeatability and reproducibility than an experienced grader performing visual evaluation.

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Test Method for Whiteness of Textiles

1. Purpose and Scope

1.1 This test method provides procedures for measuring the whiteness and tint of textiles using formulas recommended by the CIE (see 12.1). The method describes the procedures to be used and also the limitations and restrictions that apply.

2. Principle

2.1 The CIE tristimulus values are measured using a reflectance spectrophotometer or colorimeter, and the whiteness and tint calculated from formulas based on the CIE chromaticity coordinates.

2.2 Many impurities in textiles absorb short wavelength light, resulting in a yellowish appearance to observers. Therefore, a measurement of whiteness may be an indication of the degree to which a textile is free from impurities.

2.3 The presence of bluing components or fluorescent whitening agents (FWAs) in textiles may also be determined using the whiteness measurement.

3. Terminology

3.1 **CIE chromaticity coordinates,** n.—the ratio of each of the tristimulus values of a psychophysical color to the sum of the tristimulus values (see 12.1) [ASTM E284].

3.2 **CIE tristimulus values,** n.—amounts of three non-real reference color stimuli required to give a color match with the color stimulus considered and defined by the CIE for the CIE 1931 standard colorimetric observer and the CIE 1964 standard colorimetric observer and for a particular illumination condition (see 12.1).

3.3 **fluorescent whitening agent (FWA),** n.—colorant that absorbs near ultraviolet (UV) radiation and re-emits visible longer wavelength radiation. This enhances the whiteness appearance of yellowish materials to which it has been applied. [ASTM E284].

3.4 **perfect reflecting diffuser,** n.—ideal reflecting surface that neither absorbs nor transmits light, but reflects diffusely, with the radiance of the reflecting surface being the same for all reflecting angles, regardless of the angular distribution of the incident light. [ASTM E284].

NOTE: The perfect reflecting diffuser is the basis of calibration of reflectance measuring instruments. The equations for whiteness and tint are formulated so that the CIE concept of the perfect reflecting diffuser has a whiteness index of 100.0 and a tint value of 0.0.

3.5 **tint,** n.—*in whiteness measurement*, the hue of a white material as influenced by the wavelength of peak emission or reflectance. [CIE 15.2].

NOTE: Tint, if other than zero, is an indication of a reddish or greenish hue having shifted away from a bluish hue with dominant wavelength of 466 nm.

3.6 **whiteness,** n.—whiteness is the attribute by which an object color is judged to approach a preferred white. [ASTM E284].

NOTE: Whiteness as measured by this test method is an indication of how white the textile appears to an average viewer.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The safety precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 CAUTION: Protect the eyes from UV light. The safety recommendations provided by the UV light manufacturer should be followed.

4.3 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

5. Uses and Limitations

5.1 Because reflectance is affected by the nature of the surface of the textile, comparisons can be made only between samples of the same type of textile.

5.2 The application of the formulas is restricted to specimens that are called "white" commercially, and that do not differ much in color and fluorescence.

5.3 When employing measuring instruments that are not equipped with capabilities for calibrating the UV content of the specimen illumination, the formulas provide for relative evaluations of whiteness that are restricted to specimens measured on the same instrument at nearly the same time. When employing measuring instruments that are equipped with capabilities for calibrating the UV content of the specimen illumination, the formulas provide evaluations of whiteness that can be directly compared to results from similarly calibrated instruments and that are adequate for commercial use.

6. Apparatus and Materials

6.1 Color Measuring Instrument. A reflectance spectrophotometer or colorimeter capable of measuring or calculating CIE tristimulus values with at least one of the CIE specified geometries (45°:0°, 0°:45°, d:8°, 8°:d). When integrating spheres are used for measuring fluorescent specimens, the spectral power distribution of the illuminating system is altered by the reflected and emitted power from the specimen. The use of the 45°:0°, or 0°:45° condition is therefore preferable (see 12.1). To be able to make direct comparisons between results from different instruments, identical measuring geometries should be used and all instruments should have the capability of adjusting the amount of ultraviolet energy in the illumination source.

6.2 Reference Standard. The primary standard is the perfect reflecting diffuser (see 3.4). Secondary Reference Standards are standards that are calibrated in terms of the perfect reflecting diffuser and are used in the standardization of the instrument.

6.3 UV Lamp. Used for visual determination of presence of FWA on textile specimens.

7. Test Specimen

7.1 Condition each specimen for several hours in an atmosphere of 21 ± 2°C (70 ± 4°F) and 65 ± 5% RH by layering each test specimen separately on a screen or perforated shelf of a conditioning rack (see ASTM D1776, Standard Practice for Conditioning and Testing Textiles, and 12.5). Keep the samples free of dirt and stains. The exact size necessary will depend on the aperture of the reflectance measuring instrument used and on the translucency of the textile material.

8. Procedure

8.1 Before proceeding with color measurements, determine whether or not a fabric contains FWA by viewing a specimen in a dark room under ultraviolet light. A fabric that contains FWA will fluoresce under the UV light.

8.1.1 If fluorescent whitening agents (FWA) are present on the textile material, the measurement must be made with an

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instrument that illuminates the specimen with polychromatic light (full spectrum) and has a relative spectral power distribution approximating CIE illuminant D_{65} from 360-700 nm. Consult instrument manufacturers for suitable equipment. The user should also verify the suitability of the instrument if flash illumination is being used.

8.1.2 For the determination of an approximate relative efficiency of an FWA, an instrument which permits the insertion of an ultraviolet cut-off filter into the incident light beam may be used. The difference between the measurements taken before and after the insertion of the ultraviolet cut-off filter can provide an indication of the enhancement of the apparent whiteness due to the addition of an FWA. Because of possible variations in light sources and/or ultraviolet cut-off filters, the user is cautioned to use this procedure only for "relative in-house" determination.

8.2 Operate the color measuring instrument according to the manufacturer's instructions for standardization and measurement, in accordance with AATCC EP6, Instrumental Color Measurement.

8.2.1 If the spectrophotometer is equipped with adjustable ultraviolet energy, standardize the amount of ultraviolet energy in the spectrophotometer light source (see 12.2).

9. Calculations, Interpretation and Limitations

9.1 Average the measurement values taken for each test sample.

9.2 For each averaged measurement, determine the CIE tristimulus values X_{10}, Y_{10} and Z_{10} for CIE standard illuminant CIE D_{65} and 1964 10° observer (see ASTM E308 for details on calculation of tristimulus values from reflectance data). Determine the chromaticity coordinates Y_{10}, x_{10}, y_{10}.

9.3 The whiteness index (W_{10}) for any sample shall be calculated by the equation in 9.4 and the tint ($T_{W,10}$) by the equation in 9.5. Direct comparisons of whiteness and tint values should only be made for similar samples. Whiteness values are suitable for direct comparison to results from similarly calibrated instruments only when the ultraviolet energy in the spectrophotometer light source has been standardized and measuring geometries are identical. Tint values are not suitable for direct comparison between instruments. The degree of difference that will be accepted or rejected is the sole responsibility of the user, as the requirements are extremely dependent on the particular use and material being measured. The higher the value of W_{10}, the greater the whiteness. Equal differences in W_{10} value do not necessarily indicate equal differences in perceived whiteness nor equal concentration differences of FWA. Similarly, equal differences in $T_{W,10}$ value do not always represent equal perceptual differences in greenness or redness of whites.

9.4 Whiteness (see 12.2 and 12.3). (For illuminant D_{65} and 1964 10° observer):

$$W_{10} = Y_{10} + 800\,(0.3138 - x_{10}) + 1700\,(0.3310 - y_{10})$$

where W_{10} is the whiteness value or index; Y_{10}, x_{10}, y_{10} are the chromaticity coordinates of the specimen, and 0.3138 and 0.3310 are, respectively, the x_{10}, y_{10} chromaticity coordinates for the perfect diffuser.

Limited to: $40 < W_{10} < 5Y_{10} - 280$.

9.5 Tint (for illuminant D_{65} and 1964 10° observer):

$$T_{W,10} = 900\,(0.3138 - x_{10}) - 650\,(0.3310 - y_{10})$$

where $T_{W,10}$ is the tint value; x_{10}, y_{10} are the chromaticity coordinates of the sample, and 0.3138 and 0.3310 are, respectively, the x_{10}, y_{10} chromaticity coordinates for the perfect diffuser.

Limited to: $-4 < T_{W,10} < +2$.

Values of $T_{W,10}$, when positive, indicate a greenish hue; when negative, indicate a reddish hue; and when zero, indicate a bluish hue with a dominant wavelength of 466 nm.

10. Report

10.1 Report the numerical whiteness value, the tint value if required, the illuminant and the CIE standard colorimetric observer used in the calculations, the instrument and its measuring geometry used, and whether or not the ultraviolet energy of the spectrophotometer light source has been standardized using EP11, Spectrophotometer UV Energy Calibration Procedure for Optically Brightened Textiles (see 12.2).

11. Precision and Bias

11.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

11.2 *Bias*. The whiteness and tint of textiles can be defined only in terms of a test method. There is no independent method for determining the true values. As a means of estimating these properties, the method has no known bias.

12. Notes and References

12.1 For a description of the CIE colorimetric system, instrument geometry and a complete description of the whiteness and tint formulas used above, see CIE Publication 15:2018, *Colorimetry*, *4th Edition*. Available through US National Committee of the CIE or via the CIE webstore (www.techstreet.com/cie/).

12.2 See EP11, Spectrometer UV Energy Calibration Procedure for Optically Brightened Textiles, as included in the AATCC *Manual of International Test Methods and Procedures*, for details regarding ultraviolet energy standardization of spectrophotometer light sources. The Procedure can also be purchased separately from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: 919/549-8141; fax: 919/549-8933; e-mail: ordering@aatcc.org; web site: www. aatcc.org.

12.3 The equation used in prior versions of this test method was:

$$W = 4B - 3G \text{ (AATCC TM 110-1979)}$$

where W = Whiteness, B = Blue reflectance and G = Green reflectance for CIE illuminant C and CIE 1931 2° standard observer.

12.4 Users who wish to explore the prediction of data under true D_{65} illuminant for samples containing FWAs can refer to the following: F. W. Billmeyer Jr., Metrology, Documentary Standards, and Color Specifications for Fluorescent Materials, *Color Research and Application*, Vol. 19, 1994, pp413-425, and Publication ISO 23603 (CIE S 012/E) "Standard method of assessing the spectral quality of daylight simulators for visual appraisal and measurement of colour."

12.5 ASTM Standard Test Methods and Practices are available from ASTM, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832.9500; fax: +1.610.832.9555; web site: www.astm.org.

12.6 For a more complete description of the proper procedures in color measurement, refer to AATCC EP6, Evaluation Procedure for Instrumental Color Measurement.

13. History

13.1 Revised in 2021 for clarity and to add history section to align with AATCC Style Guide.

13.2 Editorially revised 2019 and 2016; revised 2015; reaffirmed 2011, 2005 and 2000; revised 1995; editorially revised and reaffirmed 1994; revised 1989 (title change); editorially revised 1980 and 1979; reaffirmed 1979, 1975, and 1972; reaffirmed 1968.

13.3 Developed in 1964 by AATCC Committee RA34; jurisdiction transferred in 1983 to AATCC Committee RA36. Adopted as ISO 105-J02 in 1987.

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Test Method for Weather Resistance of Textiles: Exposure to Daylight and Weather

1. Purpose and Scope

1.1 This test method provides a means for determining the weather resistance of textile materials.

1.2 The test options described below are applicable to fibers, yarns and fabrics and products made therefrom, including coated fabrics, whether natural, colored, finished or unfinished. Test options included are:

Option A—Direct Exposure to Natural Light and Weather

Option B—Behind Glass Exposure to Natural Light and Weather without Direct Wetting

1.3 Outdoor testing as described under this method must be performed to verify accelerated laboratory testing. It is up to interested parties to determine frequency and scope of outdoor testing. Refer to AATCC TM169, Test Method for Weather Resistance of Textiles: Xenon Lamp Exposures, and AATCC TM186, Test Method for Weather Resistance: UV Light and Moisture Exposure, for information on the limitations of accelerated laboratory testing.

1.4 This test method contains the following sections that assist in the use and implementation of the various options for determining the weather resistance of textiles.

2. Principle

2.1 Test specimens and the agreed upon comparison standard(s) are exposed simultaneously to natural weathering, either Directly or Behind Window Glass, either to a specified degree of degradation such as color change, strength loss, etc., or alternately to a specific amount of radiant energy. Exposures can be timed by calendar days, months, or years as well. However, this method can result in more variation when comparing test exposures for equal duration at different time periods. The weather resistance of the material is determined by the exposed portion of the test specimens to an unexposed counterpart, using one or more of the procedures recommended in this test method.

3. References

3.1 **ASTM *Standards*** (see 15.9).

3.1.1 D5034, Test Method for Breaking Force and Elongation of Textile Fabrics (Grab Test).

3.1.2 D5035, Test Method for Breaking Force and Elongation of Textile Fabrics (Strip Test).

3.1.3 D2256, Test Method for Tensile Properties of Yarns by the Single-Strand Method.

3.1.4 D3787, Test Method for Bursting Strength of Knitted Goods-Constant-Rate-of-Traverse (CRT) Ball Burst Test.

3.1.5 D3786, Test Method for Hydraulic or Pneumatic Bursting Strength of Textile Fabrics-Diaphragm Bursting Strength Tester Method.

3.1.6 D1424, Test Method for Tearing Strength of Fabrics by Falling Pendulum (Elmendorf) Apparatus.

3.1.7 D5587, Test Method for Tearing Strength of Fabrics by Trapezoid Procedure.

3.1.8 E903-1996, Test Method for Solar Absorptance, Reflectance, and Transmittance of Materials Using Integrating Spheres.

3.1.9 E824, Standard Test Method for Transfer of Calibration from Reference to Field Pyranometers.

3.1.10 G24, Standard Practice for Conducting Exposures to Daylight Filtered through Glass.

3.1.11 D2905, Standard Practice for Statements on Number of Specimens for Textiles.

3.1.12 D1776, Standard Practice for Conditioning and Testing Textiles.

3.1.13 G7, Standard Practice for Atmospheric Environmental Exposure Testing of Nonmetallic Materials.

3.1.14 G183, Standard Practice for Field use of Pyranometers, Pyrheliometers and UV Radiometers.

3.2 **AATCC *Test Methods*** (see 15.10).

3.2.1 AATCC TM16.1, Test Method for Colorfastness to Light: Outdoor.

3.2.2 AATCC EP1, Evaluation Procedure for Gray Scale for Color Change.

3.2.3 AATCC EP7, Evaluation Procedure for Instrumental Assessment of the Change in Color of a Test Specimen.

3.3 **ISO *Standards*** (see 4.4, 4.12 and 4.30).

3.3.1 ISO 9370 Plastics—Instrumental Determination of Radiant Exposure in Weathering Tests—General Guidance.

4. Terminology

4.1 **AATCC Blue Wool Lightfastness Standard,** n.—one of a group of dyed wool fabrics distributed by AATCC for use in determining the amount of light exposure of specimens during lightfastness testing.

4.2 **black panel thermometer,** n.—a temperature measuring device, the sensing unit of which is coated with black designed to absorb most of the radiant energy encountered in lightfastness testing.

4.3 **breaking strength,** n.—the maximum force applied to a specimen in a tensile test carried to rupture.

4.4 **broad bandpass radiometer,** n.—a relative term applied to radiometers that have a bandpass width of more than 20 nm and less than 70 nm at 50% of maximum transmittance.

4.5 **bursting strength,** n.—the force or pressure required to rupture a textile by distending it with a force applied at right angles to the plane of the fabric, under specified conditions.

4.6 **center wavelength,** n.—the specified wavelength midway between the half power points, for example, 340 nm ± 2 nm.

4.7 **color change,** n.—*as used in colorfastness testing*, a change in color of any kind whether in lightness, hue or chroma or any combination of these, discernible by comparing the test specimen with a corresponding untested specimen.

4.8 **half-power bandpass,** n.—the interval between wavelengths at which transmittance is 50% of peak transmittance in a bandpass filter.

4.9 **irradiance,** n.—radiant power per unit area as a function of wavelength expressed as watts per square meter, W/m^2.

4.10 **irradiation,** n.—the time integral of irradiance expressed in joules per square meter (J/m^2).

4.11 **laboratory sample,** n.—a portion of material taken to represent the lot sample, or the original material, and used in the laboratory as a source of test specimens.

4.12 **narrow bandpass radiometer,** n.—a relative term applied to radiometers that have a bandpass width of 20 nm or less at 50% of maximum transmittance and can be used to measure irradiance at wavelengths such as 340 or 420, ± 0.5 nm.

4.13 **pyranometer,** n.—a radiometer used to measure the global solar irradiance or if inclined, hemispherical solar irradiance.

4.14 **radiant power,** n.—power traveling through space in the form of photons or electromagnetic waves of various lengths.

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4.15 **radiant exposure,** n.—time integral of irradiance.

4.16 **radiant flux density,** n.—rate of flow of radiant energy past the specimen.

4.17 **radiant power,** n. energy per unit time, emitted, transferred or received as radiation.

4.18 **radiometer,** n.—an instrument used to measure radiant energy.

4.19 **reference fabric,** n.—one or more blue wool lightfastness standards selected for exposure as a check on a test apparatus and operating conditions.

4.20 **specimen,** n.—a specific portion of a material or a laboratory sample upon which a test is performed or which is selected for that purpose.

4.21 **spectral irradiance,** n.—the variation of energy due to the source over the wavelength span of the emitted radiation.

4.22 **spectral transmittance,** n.—the percent of incident radiant energy passing through a given material and not absorbed in the process, as a function of wavelength.

4.23 **standard atmosphere for testing textiles,** n.—air maintained at 21 ± 2 °C (70 ± 4 °F) and 65 ± 5% relative humidity.

4.24 **standard atmosphere for storing textiles,** n.—laboratory conditions maintained as required in ASTM D1776.

4.25 **tearing strength,** n.—the average force required to continue a tear previously started in a fabric.

4.26 **ultraviolet radiation,** n.—radiant energy for which the wavelengths of the monochromatic components are smaller than those for visible radiation and more than about 100 nm.

NOTE: The limits of the spectral range of ultraviolet radiation are not well defined and may vary according to the user. Committee E-2.1.2 of the CIE distinguishes in the spectral range between 400 and 100 nm:

UV-A	315-400 nm
UV-B	280-315 nm
UV-C	100-280 nm

4.27 **visible radiation,** n.—any radiant energy capable of causing a visual sensation.

NOTE: The limits of the spectral range visible radiation are not well defined and may vary according to the user. The lower limit is generally taken between 380 and 400 nm and the upper limit between 760 and 780 nm (1 nanometer, 1 nm = 10^{-9} m).

4.28 **weather,** n.—climatic conditions at a given geographic location, including such factors as sunlight, rain, humidity and temperature.

4.29 **weather resistance,** n.—ability of a material to resist degradation of its properties when exposed to climatic conditions.

4.30 **wide bandpass radiometer**, n.—a relative term applied to radiometers that have a bandpass width of more than 70 nm at 50% of maximum transmittance.
Note: Wide bandpass radiometers can be used to measure irradiance at wavelengths such as 300-800 nm.

5. Safety Precautions

5.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

5.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

5.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).
updated wording.

5.4 Daylight exposure of the skin and eyes for prolonged periods may be hazardous; therefore, caution should be employed to protect these areas. Do not look directly at the sun under any circumstances.

5.5 Good laboratory practices should be followed.

6. Uses and Limitations

6.1 The use of these options does not imply, expressly or otherwise, an accelerated test for a specific application. The degree of correlation between any weather resistance test and the actual exposure in use must be determined mathematically and agreed upon by the contractual parties.

A. Uses

6.2 Option A is intended for use in evaluating the resistance to degradation of a textile and related material, including coated fabrics, when subjected to natural weathering including wetting, and radiant energy factors, and are further intended to check the level of durability represented in a material specification. Option B is intended for use in evaluating the resistance to degradation due to weathering and sunlight exposure under the milder conditions of a protected atmosphere, such as found in indoor environments, where wetting is not a factor and also milder due to reduction in short wavelength UV radiation by the glass.

6.3 The test option selected should reflect expected end-use conditions associated with the material to be tested.

6.4 When using this test method, use a standard of comparison. This comparison material should have a known history in terms of a change in property value after a specific exposure.

B. Limitations

6.5 Not all materials are affected equally by the same light source and environment. Results obtained by the use of any one test option may not be representative of those of any other test option or any end-use application unless a mathematical correlation for a given material or a given application has been established between contractual parties.

6.6 In interpreting the results of these test options, consider that resistance to degradation under simulated actual sunlight exposure is dependent upon such factors as:

6.6.1 The inherent properties of the material, its physical state, mass and compactness.

6.6.2 Spectral energy distribution and density of the radiant flux (from the sun).

6.6.3 The temperature and relative humidity of the air around the textile specimen during exposure.

6.6.4 The effect of leaching or degradation of additive products, including fiber stabilizers, by rain.

6.6.5 Atmospheric contaminants.

6.7.6 The action of additive finishes and colorants including spectral absorption characteristics.

6.6.7 Where applicable, the action of residual laundry or drycleaning chemicals.

6.7 The relative rates of degradation of different textile materials do not necessarily change to the same degree as those factors themselves change. Consequently, the relative durability of textile and related materials under the varying conditions of use cannot be predicted with certainty with any one test. Therefore, it is common practice to investigate the durability of materials by exposure to the variety of conditions under which they can be used to arrive at a full understanding of the performance that can be expected from them.

6.8 Variations in results may be expected when operating conditions are varied within the accepted limits of this method, or tests are conducted in different geographical locations. Therefore, no reference shall be made to results from the use of this method unless accompanied by a report detailing the specific operating conditions and locations in conformance with Section 12.

7. Apparatus and Materials (see 15.1)

7.1 Daylight Exposure Rack that allows wetting by rainfall (see 15.2, 15.4 and ASTM G7).

7.2 Behind Glass Daylight Exposure Cabinet that is glass covered and does not allow wetting by natural rainfall (see 15.3, Appendix A and ASTM G24, Type A).

7.3 Black Panel Thermometer.

7.4 Pyranometer.

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7.5 UV Radiometer.

7.6 RH Sensor.

7.7 Blue Wool Standards (see AATCC TM16.1).

8. Maintenance

8.1 Maintain in accordance with the manufacturer's instructions. For the Behind Glass Daylight Exposure Cabinet, wash both the exterior surface and the interior surface of the glass cover monthly (or more frequently, if required) to remove dust particles and other undesirable material.

9. Test Specimen Preparation

9.1 Prepare specimens for the specified test procedures of at least the size given in Table I. When applicable, cut specimens with the longer direction parallel to the warp or filling direction as required in a material specification or contract order (see 9.1.1.3).

9.1.1 Prepare specimens as required to meet the needs of the applicable test method given in Table I.

9.1.1.1 Unless otherwise agreed upon, as when specified in an applicable material specification, take a number of specimens such that the user may expect at the 95% probability level that the test result is not more than 5.0% of the average above the true average of the lot. Determine the number of specimens using standard deviation with one-sided limits as directed in ASTM D2905.

9.1.1.2 Pile fabrics, such as carpets, which have fibers that may shift position, or texture which may make evaluations in small areas difficult should be tested with an exposed area of not less than approximately 40.0 mm (1.6 in.) by 50.0 mm (2.0 in.). Expose sufficient size or multiple specimens to include all colors in the sample.

9.1.1.3 In general, the warp direction is used, but the filling direction may be used in conjunction or in place of the warp direction when specified. Warp yarns sometimes are protected from the radiation due to fabric structure. When the filling direction is used, it must be reported.

9.1.2 These dimensions are given as a general guide and in most cases will be adequate to perform the required evaluation. Certain materials may exhibit a dimensional change with respect to the exposure. The test equipment manufacturer, physical test apparatus and number of specimens required will affect the needed specimen size. In any event, the test procedures given in Table I should be reviewed to ensure sufficient specimens are exposed for individual needs.

9.1.3 Unless otherwise specified, test specimens shall be a maximum thickness of 25 mm (1 in.). For specimens over 25 mm (1 in.) thick, formed or configured specimens, and large components, specific instructions must be agreed upon between the purchaser and the supplier.

9.1.4 When required to prevent raveling, the specimens may be edged using a flexible epoxy resin or similar material, or by sewing, pinking or fusing.

9.1.5 Labeling each specimen is preferred to provide a record of the changes encountered at different radiant exposure levels. Retain an unexposed specimen for comparison to the exposed specimen.

9.2 Mounting

9.2.1 For ***Options A and B***. To more closely simulate specific end-use conditions, the cabinet backing materials shown below have been found to be acceptable:

Cabinet Backing Materials

End-Use	Cabinet Backing	Approximate Temperatures
Automotive	Solid (Plywood)	82°C (180°F)
Household (Drapery, etc.)	Expanded Metal	63°C (145°F)
Clothing and Sensitive Materials	None (Open)	43°C (110°F)
Fabric Roof Structures	None (Open)	43°C (110°F)

9.2.2 **Fabrics.** Secure the specimens to the frames to hold them smoothly without curling of the edges. Fabrics may be stitched to gauze backing. For colorfastness determinations, mount the specimens in frames as directed in AATCC TM16.1.

9.2.3 **Yarns.** Wind yarns on frames. Only that portion of the yarns directly facing the radiant energy is tested for breaking strength. Single strand or multiple strand testing is performed. When multiple strand testing may be performed, the yarns are wound on the frame closely packed to 2.5 cm (1.0 in.) width. The control specimens must contain the same number of strands as the specimen subjected to exposure. After the exposure has been completed and before the yarns are removed from the frame for testing, those yarns facing the light source are bound together using 2.0 cm (0.75 in.) wide masking or other suitable tape to maintain the closely packed arrangements on the exposure frame.

10 Conditioning

10.1 When the exposure cycle is completed, remove the test and control specimens from the exposure rack and transfer them to a standard atmosphere [air maintained at 21 ± 2 °C (70 ± 4 °F) and 65 ± 5% relative humidity] for testing textiles for conditioning.

10.2 If the specimens are wet upon removal from the racks, dry them at ambient laboratory conditions or at a temperature not exceeding 71 °C (160 °F) before placing them in the conditioning atmosphere. The unexposed reference standard (comparison standard) and the retained unexposed original should be treated under exactly that same conditions of drying and conditioning as the test specimens.

10.3 Bring all specimens, control and test, to moisture equilibrium in the standard atmosphere for testing textiles. Equilibrium is considered to have been reached when the increase in mass of the specimen in successive weightings made at intervals of not less than 2 h does not exceed 0.2% of the mass of the specimen. In general practice, the industry approaches equilibrium from the "as received" side.

10.3.1 It is recognized that, in practice, textile materials are frequently not weighed to determine when moisture equilibrium has been reached. While such a procedure cannot be accepted in cases of dispute, it may be sufficient in routine testing to expose the material to the standard atmosphere for testing for a reasonable period of time before the specimens are tested. Twenty-four hours has been found acceptable in most cases; however, certain fibers may exhibit slow

Table I—Specimen Size and Preparation

Property	Test Method	Specimen Size
Breaking Strength:		
Strip Test	ASTM D5035	5 × 20 cm (2 × 8 in.)
Grab Test	ASTM D5034	13 × 28 cm (5 × 7 in.)
Single Strand	ASTM D2256	15 cm (6 in.)
Bursting Strength:		
Woven Fabric	ASTM D3786	15 × 15 cm (6 × 6 in.)
Nonwoven Fabric	ASTM D3786	15 × 15 cm (6 × 6 in.)
Knit Fabric	ASTM D3787	15 × 15 cm (6 × 6 in.)
Tear Strength:		
Elmendorf	ASTM D1424	10 × 13 cm (4 × 5 in.)
Trapezoid	ASTM D5587	10 × 18 cm (4 × 7 in.)
Colorfastness to Light: Outdoor	AATCC TM15.1	3 × 6 cm (1.25 × 2.4 in.)

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moisture equalization rates from the "as received" "wet side." When this is known, a preconditioning cycle, as described in ASTM D1776 (see 3.1) may be agreed upon between contractual parties.

10.4 For each test to be made on the test material and control, exposed and unexposed, prepare test specimens by marking and raveling or cutting the central portion of each exposed specimen to the dimensions specified in the respective test methods given in Table I. The marking and raveling or cutting of the test specimens is preferred after the exposure but may be done before exposure. Control specimens not exposed are similarly prepared. For samples exposed with wetting, control specimens are wet-out and allowed to dry without tension before testing. All specimens, control and test, are conditioned simultaneously to the standard atmosphere for testing textiles for a minimum of 24 h and preferably longer, depending upon the material, and tested at the same time.

11. Procedure

11.1 Mount on the exposure rack the appropriate number of replicates (see Section 8) of the specimens as required to average out variability and to ensure accuracy of results and the necessary number of replicates of the standard, when such are available. The specimens shall be so positioned as to avoid shadowing of the specimens at the sides of the cabinet.

11.2 Expose the specimens to sunlight and the general elements of weather for a prescribed period of time during which period the radiant energy may be recorded, or when a prescribed amount of radiant exposure has been attained as determined by an applicable specification (see 15.5, 15.7 and 15.8) using a pyranometer and UV radiometer.

11.2.1 ***For Option A***, Direct Exposure to Daylight and Weathering, use Direct Exposure Rack described in Section A2 of Appendix A.

11.2.2 ***For Option B***, Behind Glass Exposure to Natural Light without Wetting, use cabinet described in Section A3 of Appendix A. Ensure that the face of the exposed standard(s) and test specimen(s) are at least 7.5 cm (3 in.) below the inside surface of the glass cover and are positioned at least 15 cm (6 in.) in from the edges of the glass frame. The back of the exposure cabinet may be varied as follows to achieve the required exposure conditions:

Backing	Exposure Condition
Open	Low Temperature
Expanded Metal	Medium Temperature
Solid	High Temperature

11.3 Expose standard(s) and specimen(s) 24 h a day and remove for inspection or for physical properties testing.

11.4 Monitor temperature and relative humidity in the vicinity of the test cabinets or racks.

11.5 After exposure and conditioning (according to section 9) conduct physical test using the test methods listed in Section 14, as applicable.

11.5.1 Average the data for the various replicates, or handle statistically as appropriate, and record the value of breaking, tearing, or bursting strength retention and/or colorfastness after exposure in relation to original strength or color, as applicable. It is appropriate to record the percent elongation characteristics of the unexposed controls and exposed specimens at rupture or at some prescribed point in the force-elongation curve as this is frequently significant supplementary information.

12. Evaluation

12.1 Classify the durability or resistance to degradation of the material with respect to the standard of comparison by one of the following:

12.1.1 *Percent Strength Retained or Percent Strength Loss*. Record percent strength loss or percent strength retained (breaking, tearing, or bursting as appropriate) of the material after the prescribed exposure time period.

12.1.2 *Residual Strength*. Record the initial and final strength values along with all other pertinent data as above.

12.1.3 *Colorfastness*. Classify the colorfastness of the material as directed in AATCC TM16.1

12.1.4 *In Terms of an Agreed upon Comparison Sample or Standard*. Durability of the sample in terms of the standard is determined by the following**:** As *durable* or *more durable* than the comparison sample at the amount of radiant exposures and/or exposure time period prescribed in the material standard; or *less durable* than the comparison sample at the amount of radiant exposure and/or exposure time period prescribed in the material standard.

12.1.5 For the purpose of defining the relative durability of a test material to an agreed upon comparison standard an index, S_nX, defined as the ratio of the percent residual strength in the tested specimen to that of the untested material, may be used. With the index a specimen is considered to be as resistant to degradation as the standard when the S_nX value is 1; more durable when the S_nX index exceeds 1; less durable when the S_nX index is less than 1.

NOTE: An index of this nature has particular value when recording the relative durability of a series of materials to a common standard and can be actually more useful in research than in the routine assessment of the durability of materials for purpose of commerce.

13. Report

13.1 Reference to AATCC TM16.1-2022.

13.2 Use the guide below and report all applicable information:

13.3 Report any deviation from

Exposure Option (A or B)_____

Geographical Location_____

Exposure Date: From _____ to_____

Radiant Energy:_____

Exposure Latitude_____ Exposure Angle_____

Exposed behind window glass: Yes _____ No_____ If Yes, Type_____

Daily Ambient Temperature: Minimum _____°C Maximum _____°C Avg._____°C

Daily Black Panel Temperature: Minimum _____°C Maximum _____°C Avg._____°C

Daily Relative Humidity %: Minimum _____ Maximum _____ Avg. _____

Moisture: Rain Amount (mm)_____

Time of Wetness (hrs): Rain plus Condensation_____

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AATCC TM111 or the performance of the comparison standard.

13.4 Report all information in 13.2 for the same conditions that the samples and reference materials are exposed.

13.5 Report all applicable properties evaluated in Section 11 (also see 15.8).

13.6 Report the direction of the test specimens if not in warp direction.

13.7 Under certain conditions, by agreement between principals to a transaction, tearing strength of the material can be used in lieu of, or to complement, the breaking or bursting strength. The material specification may further define the use of wet breaking, tearing, or bursting strength tests in lieu of or to complement those tests done under standard textile testing conditions. Such conditions of test shall be reported along with the data.

14. Precision and Bias

14.1 This test method may be used for acceptance testing of commercial shipments. However, caution is advised since information about *between-laboratory* precision shows a high variability between results. Interlaboratory testing using Option A (Daylight) showed a high variability between results for samples exposed at different times of the year. Efforts to minimize the seasonal effect by time versus irradiance test duration did not compensate for this phenomenon for all fabrics. For some fabrics, exposure to a specific level of irradiance may result in a lower variation of results, while for other fabrics little difference may be seen between time and irradiance exposure. Further, the nature of the fabric, its finish or coating and the climate may affect the results. It is strongly recommended to begin exposures at the same time of the year for comparative purposes. This may minimize the seasonal effect.

14.1.1 In case of a dispute arising from differences in reported test results when using this test method for acceptance testing of commercial shipments, the purchaser and the supplier should conduct comparative tests to determine if there is a statistical bias between their laboratories. Statistical analysis is recommended for the investigation of bias. As a minimum, the two parties should take a group of test specimens that are as homogeneous as possible and that are from a lot of material of the type in question. Test specimens should then be randomly assigned in equal numbers to each laboratory for testing. The average results from the two laboratories should be compared using the appropriate Student's *t*-test and an acceptable probability level chosen by the two parties before testing is begun. If a bias is found either its cause must be found and corrected or the purchaser and the supplier must agree to interpret future test results for that material with consideration for the known bias.

14.1.2 *Interlaboratory Test Data, Breaking Strength Determinations*. An interlaboratory test was conducted in 1990 and 1991 in which randomly drawn samples of six materials were tested at three exposure sites in each, southern Florida and Arizona, in accordance with AATCC TM111B. NOTE: Previous versions of AATCC TM111 contain actual tabular data for the precision and bias.

14.2 *Interlaboratory Test Data, Summary.* Interlaboratory testing using Option B (Daylight) showed a high variability between results for samples exposed at different times of the year. Efforts to minimize the seasonal effect by time versus irradiance test duration did not compensate for this phenomenon for all fabrics. For some fabrics, exposure to a specific level of irradiance may result in a lower variation of results, while for some fabrics little difference may be seen between time and irradiance exposure. Further, for some, the nature of the fabric, its finish or coating and the climate may effect the results. Differences between laboratories were small as shown by the zero, or near-zero components of variance reported in Tables II and III. The higher values for *between-laboratory* critical differences reported in Tables IV and V were driven by the high values for between seasonal quarter exposures. Consequently, it is strongly recommended to begin exposures for Options A and B at the same time of the year for comparative purposes. This may minimize the seasonal effect and components of variance would more likely be represented by the single operator precision.

14.4 *Precision*. For the components of variance reported in Tables II and III, two averages of observed values should be considered significantly different at the 95% probability level if the difference equals or exceeds the critical differences listed in Tables IV and V.

NOTE 1: The square roots of the components of variance are being reported to express the variability as a percent for Table II and as units of measure for Table III, rather than the square of those values. Critical differences were calculated using z =1,960.

NOTE 2: Since the interlaboratory test included only three laboratories for each geographical location, estimates of *between-laboratory* precision should be used with special caution. The tabulated values of the critical differences should be considered to be a general statement, particularly with respect to *between-laboratory* precision. Before a meaningful statement can be made about two specific laboratories, the amount of statistical bias, if any, between them must be established with each comparison being based on recent data obtained on specimens taken from a lot of material of the type being evaluated so as to be as nearly homogeneous as possible and then randomly assigned in equal numbers to each of the laboratories.

14.5. *Bias*. The values of weather resistance can only be defined in terms of a specific test method. Within this limitation, the procedure in AATCC TM111 for measuring weather resistance of textiles using breaking strength criteria has no known bias.

15. Notes

15.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

15.2 Refer to ASTM G7 and G24 for guidance on test rack or test cabinet selection (see 3.1).

15.3 Refer to ASTM G24 for guidance on window glass selection (see 3.1).

15.4 Specimen frames must be made from suitably inert materials that will not contaminate the specimens. When specimens are fastened with staples, they should be of the nonferrous type overcoated to avoid contamination of the specimen by corrosion products. Metal frames must have a dull finish and be designed to avoid reflectances that could influence the performance of the material. Frames shall conform to the curvature of the specimen rack. The size of the frame is determined by the type specimens required for individual property requirements.

15.5 Refer to ASTM G183 for guidance on pyranometers used for measuring total sunlight radiant energy and on UV radiometers used for measuring total UV radiant energy from 295-385 nm of sunlight (see 3.1).

15.6 Internationally recommended units for measuring and reporting radiant energy, factors to be used when converting from one unit to another and pyranometer/radiometer descriptions and classifications are taken from *Guide to Meteorological Instrument and Observing Practices*, World Meteorological Organization, WMO-No. 8 TP.3.

15.7 Refer to ASTM G183 for guidance on pyranometers used for measuring global (total sun and sky) radiation (see 3.1).

15.8 A difference in color change properties between original material and the covered portion of the exposed specimen indicates that the textile has been affected by some agent other than light, such as heat or a reactive gas in the atmosphere. Although the exact cause of this difference in color may not be known, it should be noted in the report when it occurs.

15.9 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832.9500; fax: +1.610.832.9555; web site: www.astm.org.

15.10 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

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Table II—Breaking Strength after Weathering ASTM D5035, Strip Procedure

Fabric Exposure Location	Components of Variance, Coefficient of Variation, %							
	Grand Average Pounds		Single Operator Component		Within-Laboratory Component		Between-Laboratory Component	
	3 MO.	75 kJ	3 MO.	75 kJ	3 MO.	75 kJ	3 MO.	75 kJ
MIL-C-44103								
ARIZONA	204	203	4.2	5.3	9.2	6.7	0	0
SOUTHERN FLORIDA	201	201	4.0	3.7	3.2	4.7	0	0
MIL-C-7219								
ARIZONA	150	162	4.2	5.4	46.6	31.6	0	0
SOUTHERN FLORIDA	183	182	6.0	6.7	31.4	19.2	0	0
MIL-C43285-B								
ARIZONA	236	244	10.3	6.5	26.6	19.8	0	0
SOUTHERN FLORIDA	246	248	8.5	6.6	18.0	12.2	0	0
MIL-C-4362-7A								
ARIZONA	64	69	4.1	3.9	13.8	14.4	4.8	0
SOUTHERN FLORIDA	79	77	3.6	4.2	4.5	7.4	0.1	0
ALLIED A-609-029-D								
ARIZONA	248	265	10.8	12.7	81.9	55.1	0	0
SOUTHERN FLORIDA	286	284	9.9	17.0	46.2	33.6	0	0
MIL-C-44103								
ARIZONA	210	211	4.6	3.9	7.2	6.8	0	0
SOUTHERN FLORIDA	208	208	4.6	6.0	3.2	4.5	0	0

NOTE: *Within-laboratory* component represents variation between quarterly exposure start times.

Table III—Instrumental Color Measurement, ΔE, After Weathering (AATCC EP6)

Fabric Exposure Location	Components of Variance, Standard Deviation, Units of Measure, Single Material Comparison							
	Grand Average Pounds		Single Operator Component		Within-Laboratory Component		Between-Laboratory Component	
	3 MO.	75 kJ	3 MO.	75 kJ	3 MO.	75 kJ	3 MO.	75 kJ
MIL-C-44103								
ARIZONA	2.00	1.88	0.19	0.14	0.61	0.49	0.15	0.10
SOUTHERN FLORIDA	2.14	2.07	0.09	0.19	0.47	0.43	0.11	0.23
MIL-C-7219								
ARIZONA	8.99	8.40	1.60	1.13	2.56	1.24	0	0.90
SOUTHERN FLORIDA	7.89	8.00	0.78	1.41	1.34	0	0	0
MIL-C43285-B								
ARIZONA	1.45	0.94	0.27	0.47	0.19	0.31	0.13	0
SOUTHERN FLORIDA	2.30	2.25	0.27	0.52	0.39	0	0.08	0
MIL-C-4362-7A								
ARIZONA	5.77	5.77	1.88	1.55	0.94	0.61	0	1.58
SOUTHERN FLORIDA	0.78	0.88	0.17	0.22	0	0	0	0
ALLIED A-609-029-D								
ARIZONA	14.2	13.2	0.78	0.90	5.23	2.28	0.42	0.92
SOUTHERN FLORIDA	11.9	11.8	0.72	0.98	1.94	1.37	0.51	0.80
MIL-C-44103								
ARIZONA	2.99	2.88	0.51	0.83	0.65	0	0	0
SOUTHERN FLORIDA	5.37	5.41	0.72	0.92	1.68	1.48	0	0

NOTE: *Within-laboratory* component represents variation between quarterly exposure start times.

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Table IV—Breaking Strength After Weathering, ASTM D5035, Strip Procedure

Fabric Exposure Location	Number of Observations in Each Average	Critical Difference for Conditions Noted, % of Average					
		Single-Operator Component		Within Quarterly Exposure Component		Between-Laboratory Component	
		3 MO.	75 kJ	3 MO.	75 kJ	3 MO.	75 kJ
MIL-C-44103							
ARIZONA	1	5.7	7.2	13.7	11.6	13.7	11.6
	2	4.1	5.1	13.1	10.4	13.1	10.4
	5	2.7	3.2	12.7	9.7	12.7	9.7
SOUTHERN FLORIDA	1	5.5	5.0	10.5	8.3	10.5	8.3
	2	3.9	3.6	9.8	7.5	9.8	7.5
	5	2.5	2.3	9.3	6.9	9.3	6.9
MIL-C-7219							
ARIZONA	1	7.9	9.2	86.5	54.9	86.5	54.9
	2	5.6	6.5	86.3	54.5	86.3	54.5
	5	3.5	4.1	86.2	54.3	86.2	54.3
SOUTHERN FLORIDA	1	9.0	10.2	48.5	31.0	48.5	31.0
	2	6.4	7.2	48.1	30.1	48.1	30.1
	5	4.0	4.6	47.8	29.6	47.8	29.6
MIL-C43285-B							
ARIZONA	1	12.1	7.4	33.5	23.7	33.5	23.7
	2	8.5	5.2	32.3	23.1	32.4	23.1
	5	5.4	3.3	31.7	22.8	31.7	22.8
SOUTHERN FLORIDA	1	9.6	7.4	22.4	15.5	22.4	15.5
	2	6.8	5.2	21.4	14.6	21.4	14.6
	5	4.3	3.3	20.7	14.1	20.7	14.1
MIL-C-4362-7A							
ARIZONA	1	17.9	15.8	62.7	59.9	66.2	59.9
	2	12.6	11.1	61.4	58.9	65.0	58.8
	5	8.0	7.0	60.7	58.2	64.2	58.2
SOUTHERN FLORIDA	1	12.7	15.2	20.4	30.8	20.4	30.8
	2	9.0	10.8	18.2	28.8	18.3	28.8
	5	5.7	6.8	16.9	27.6	16.9	27.6
ALLIED A-609-029-D							
ARIZONA	1	12.1	13.3	92.3	59.1	92.3	59.1
	2	8.5	9.4	91.9	58.4	91.9	58.4
	5	5.4	6.0	91.7	57.9	91.7	57.9
SOUTHERN FLORIDA	1	9.6	16.6	45.8	36.7	45.8	36.7
	2	6.8	11.7	45.3	34.8	45.3	34.8
	5	4.3	7.4	45.0	33.6	45.0	33.6
MIL-C-44103							
ARIZONA	1	6.0	5.1	11.3	10.3	11.3	10.3
	2	4.3	3.6	10.4	9.6	10.4	9.6
	5	2.7	2.3	9.9	9.2	9.9	9.2
SOUTHERN FLORIDA	1	6.1	8.0	7.4	10.9	7.4	10.0
	2	4.3	5.6	6.0	8.3	6.0	8.3
	5	2.7	3.6	5.0	7.0	5.0	7.0

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Table V—Instrumental Color Measurement, ΔE, After Weathering (AATCC EP6)

Fabric Exposure Location	Number of Observations in Each Average	Critical Difference for Conditions Noted, Standard Deviation, Units of Measure					
		Single-Operator Component		Within Quarterly Exposure Component		Between-Laboratory Component	
		3 MO.	75 kJ	3 MO.	75 kJ	3 MO.	75 kJ
MIL-C-44103							
ARIZONA	1	0.52	0.40	1.76	1.42	1.81	1.44
	2	0.37	0.28	1.72	1.39	1.77	1.41
	5	0.23	0.18	1.70	1.37	1.75	1.40
SOUTHERN FLORIDA	1	0.25	0.53	1.33	1.30	1.36	1.45
	2	0.18	0.37	1.32	1.25	1.35	1.40
	5	0.13	0.24	1.31	1.21	1.34	1.37
MIL-C-7219							
ARIZONA	1	4.43	3.14	8.37	4.66	8.37	5.28
	2	3.13	2.22	7.76	4.10	7.76	4.79
	5	1.98	1.40	7.37	3.72	7.37	4.47
SOUTHERN FLORIDA	1	2.17	3.90	4.29	3.90	4.29	3.90
	2	1.53	2.75	4.01	2.75	4.01	2.75
	5	0.97	1.74	3.83	1.74	3.83	1.74
MIL-C43285-B							
ARIZONA	1	0.76	1.32	0.92	1.58	0.99	1.58
	2	0.54	0.93	0.75	1.27	0.83	1.27
	5	0.34	0.59	0.63	1.05	0.72	1.05
SOUTHERN FLORIDA	1	0.74	1.44	1.30	1.44	1.32	1.44
	2	0.52	1.02	1.19	1.02	1.21	1.02
	5	0.33	0.64	1.12	0.64	1.14	0.64
MIL-C-4362-7A							
ARIZONA	1	5.20	4.28	5.82	4.60	5.82	6.36
	2	3.68	3.02	4.51	3.46	4.51	5.59
	5	2.32	1.92	3.50	2.55	3.50	5.08
SOUTHERN FLORIDA	1	0.47	0.60	0.47	0.60	0.47	0.60
	2	0.34	0.42	0.34	0.42	0.34	0.42
	5	0.21	0.27	0.21	0.27	0.21	0.27
ALLIED A-609-029-D							
ARIZONA	1	2.16	2.49	14.7	6.79	14.7	7.25
	2	1.52	1.76	14.6	6.56	14.6	7.03
	5	0.96	1.12	14.5	6.41	14.6	6.90
SOUTHERN FLORIDA	1	1.98	2.72	5.72	4.67	5.89	5.17
	2	1.40	1.93	5.55	4.26	5.72	4.79
	5	0.89	1.22	5.44	3.99	5.62	4.56
MIL-C-44103							
ARIZONA	1	1.41	2.30	2.29	2.30	2.29	2.30
	2	1.00	1.63	2.06	1.63	2.06	1.63
	5	0.71	1.03	1.93	1.03	1.93	1.03
SOUTHERN FLORIDA	1	2.00	2.56	5.06	4.83	5.06	4.83
	2	1.41	1.81	4.86	4.48	4.86	4.48
	5	0.89	1.14	4.74	4.26	4.74	4.26

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16. History

16.1 Revised in 2023 to align with the style guide.

16.2 Revised 2022; editorially revised 2019, 2016; revised 2015; reaffirmed and editorially revised 2009; editorially revised 2008, 2007; revised 2003 (with title change).

16.3 Developed in 1996 by AATCC Committee RA64 to supersede 111A-1990, 111B-1990, 111C-1990 and 111D-1990 originally developed in 1964 and last revised/reaffirmed in 1990; jurisdiction transferred in 2007 to Committee RA50.

Appendix A
Apparatus and Material-Daylight Exposure

A1. General Conditions, Daylight Exposure, Options A and B

A1.1 The cabinet or rack shall be located where it will receive direct sunlight throughout the day and where shadows of objects in the vicinity will not fall upon it. When the cabinet or rack is installed over soil, the distance between the bottom of the cabinet or rack and the plane of the cleared area shall be great enough to prevent damage which could possibly occur during maintenance (grass cutting, regraveling, weed control, etc.).

A1.2 Exposure cabinets or racks shall be located in clear areas, preferably at a suitable number of climatologically different sites representing the various conditions under which the material will be used. Major climatological variations include subtropical, desert, seashore (salt air), industrial atmosphere and areas exhibiting a wide range in percentage of available sunshine. The area beneath and in the vicinity of the cabinets or racks should be characterized by low reflectance and be typical of the ground cover in that climatological area. In desert areas, it should be gravel whereas in most temperate and subtropical areas, it should be low cut grass.

A1.3 Instruments for determining climatological data during the exposure period shall be operated in the immediate area of the exposure cabinets or racks. To characterize the conditions around the test frame, these instruments should be capable of recording: ambient temperature (daily minimum and maximum), Black Panel temperature, relative humidity (daily minimum and maximum), hours of precipitation (rain), and total hours of wetness (rain and dew).

A1.4 Other methods of radiation measurement may be used when mutually agreed between principals to a transaction. Total ultraviolet radiation (typically from 295-385 nm) reported in MJ/m^2 has been used in the past, however, the relative degradative effect of a given amount of radiant energy obtained at different locales in different seasons may be found to be different due to temperature and moisture effects and to variations in ratios of UV, visible and infrared components.

A2. Option A-Daylight Exposure with Wetting

A2.1 Exposure racks. A suitable exposure rack consists of a structure elevated above ground facing the equator. The angle of the rack from the horizontal must be at the same angle as the location at which the tests are made or any angle, such as 45°, when agreed upon. When the specimens are mounted on the rack without backing there must be free air access to the back of the specimens at all times.

A2.2 Under certain conditions, to simulate certain conditions of end-use when specified, expose specimens with a backing. Report that a backing was used, the type and its specific nature. Exposures are continuous 24 h a day until the required exposure has been attained.

A3. Option B-Behind Glass Daylight Exposure Cabinet and Location, Without Wetting

A3.1 The daylight exposure cabinet shall consist of a glass-covered enclosure of any convenient size constructed of metal, wood or other satisfactory material to protect the specimens from rain and weather, and be well ventilated to allow free flow of air over the specimens. The glass cover shall be a sheet 2.0-3.2 mm thick piece of good grade, clear and flat-drawn. In order to reduce variability due to changes in UV transmission of glass, all new glass shall be exposed facing the equator, at the site latitude angle, according to ASTM G24, or an empty under glass exposure cabinet, for at least three months prior to installation in test cabinets. It shall be single strength, free of bubbles or other imperfections. Typically, "single strength" or "double strength" glass will have a transmittance of 10-20% at 320 nm and at least 85% at wavelengths of 380 nm or higher after the three month pre-aging procedure. If transmittance of the glass is measured, report the average for at least three pieces of the lot of glass being tested. Follow the instructions for measurement of transmittance of solid samples recommended by the manufacturer of the UV-visible spectrophotometer used. If a spectrophotometer with an integrating sphere is used, the measurements shall be performed in accordance with ASTM E903 (1996) (see 3.1).

A3.2 The enclosure or cabinet shall be equipped with a rack which supports the specimens in a plane parallel to that of the glass cover with the face of the specimen at a distance below it of not less than 75.0 mm (3.0 in.). The specimen mounting rack shall be constructed of a material that is compatible with the test specimens. It may be either the open type providing good ventilation on the back side of the specimen, or of a solid material as required. To minimize shadows from the top and the sides of the cabinet, the usable exposure area under glass shall be limited to that of the glass cover reduced by twice the distance from the cover to the specimens.

A3.3 The glass cover and the test specimen shall slope toward the equator at an angle from the horizontal equal in degrees to approximately the latitude of the location at which the tests are being made. Other angles of exposure, such as 45°, may be used, but the angle must be reported in the results of test.

A3.4 If required to characterize the conditions within the test frame, instruments should be capable of recording: ambient temperature under glass (daily minimum and maximum), Black Panel Temperature under glass, total radiant energy and ultraviolet radiant exposure (either broad or narrow bandpass) at the same angle of exposure as the test specimens, and relative humidity (daily minimum and maximum).

A3.5 If required, an ultraviolet radiometer shall be used to measure irradiance, and connected to an integrator for computing ultraviolet radiant exposure. The ultraviolet radiometer shall be mounted behind glass of the same type used in the exposure cabinet, or equivalent, in a ventilated enclosure to avoid overheating the instrument. The enclosure containing the radiometer shall maintain the same orientation as the test cabinet. Two different types of radiometers may be used:

A3.5.1 A broad-band ultraviolet (UV) radiometer (see 15.5, 15.7, and 15.8), an instrument which measures irradiance at wavelengths 295-385 nm.

A3.5.2 A narrow-band ultraviolet radiometer (see 15.5, 15.7 and 15.8), instrument which measures irradiance centered at a wavelength of 340 ± 2 nm.

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Test Method for Formaldehyde Release from Fabric: Sealed Jar

1. Purpose and Scope

1.1 This test method is applicable to textile fabrics that may release formaldehyde, particularly fabrics finished with chemicals containing formaldehyde. It provides accelerated storage conditions and an analytical means for determining the amount of formaldehyde released under the conditions of accelerated storage (see Section 5 and 10.1). This test method determines free and hydrolyzed formaldehyde and any-additional formaldehyde which may be generated under accelerated storage conditions.

1.2 An optional accelerated incubation procedure is available (see 13.6).

1.3 An optional submersion method which approximates the amount of hydrolyzed formaldehyde while closely simulating normal user conditions is AATCC TM 206.

2. Principle

2.1 A weighed fabric specimen is suspended over water in a sealed jar. The jar is placed in an oven at a controlled temperature for a specified length of time (see 13.5). The amount of formaldehyde absorbed by the water is then determined colorimetrically.

3. Terminology

3.1 **formaldehyde release,** n.—That formaldehyde exuded from textiles under the accelerated storage conditions of this test, including that which is free (unbound or occluded) from unreacted chemicals, or from finish degradation as a result of this test.

3.2 **free formaldehyde,** n.—Formaldehyde that is not bonded to a finish or a fabric is considered to be free formaldehyde. In this form, formaldehyde can be readily extracted from the fabric by immersion in water.

3.3 **hydrolyzed formaldehyde,** n—Although originally part of a larger chemical structure, such as a crosslinking reactant, during the course of this test the formaldehyde is cleaved from the larger molecule and dissolved into water.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practice should be followed. Wear safety glasses in all laboratory areas.

4.2 When handling glacial acetic acid to prepare Nash reagent, use chemical goggles or face shield, impervious gloves and an impervious apron during preparation. Concentrated acids should be handled only in an adequately ventilated laboratory hood. CAUTION: Always add acid to water.

4.3 Formaldehyde is a sensory irritant and potential sensitizer. Its chronic toxicity has not been fully established. Use in an adequately ventilated laboratory hood. Avoid inhalation or skin contact. Use chemical goggles or face shield, impervious gloves and an impervious apron when working with formaldehyde (see 8.1).

4.4 An eyewash/safety shower should be located nearby and a self-contained breathing apparatus should be readily available for emergency use.

4.5 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 13.7).

5. Uses and Limitations

5.1 The procedure is intended for use in the range of releasable formaldehyde on hydrophobic and non-hydrophobic fabrics up to about 3500 µg/g. The upper limits are 500 µg/g if a 1:1 ratio of Nash reagent to sample solution is used in the analytical portion of the test and 3500 µg/g if a 10:1 ratio is used. The procedure promotes formaldehyde release from odor-free, fully cured durable press fabrics that have been afterwashed (Vail, S. L. and B. A. K. Andrews, *Textile Chemist and Colorist*, Vol. 11, No. 1, January 1979, p48). For these reasons, the procedure should not be used to estimate µg/g formaldehyde in air for compliance with any mandated or voluntary standards. The procedure was originally developed to measure the "propensity of a resin-treated fabric to liberate an excessive amount of formaldehyde under hot, humid conditions." (Nuessle, A. C., *American Dyestuff Reporter*, Vol. 55, No. 17, 1966, pp48-50; also Reid, J. D., R. L. Arcenaux, R. M. Reinhardt and J. A. Harris, *American Dyestuff Reporter*, Vol. 49, No. 14, 1960, pp29-34.)

5.2 Alternatively, Test Method 206, Free and Hydrolyzed Formaldehyde, Determination of: Water Extraction Method determines free and hydrolyzed formaldehyde on the fabric from 20 µg/g up to about 3500 µg/g. This method approximates hydrolyzed formaldehyde and closely simulates normal user conditions. The appropriate test should be based on that distinction. However, since TM 206 is an aqueous immersion method, fabrics with a propensity for dye bleeding could be of concern; if the dye has an absorbance in the range of 412 nm, it could interfere with the spectrophotometer measurement. TM 112 does not use aqueous immersion; therefore there is less chance for dye contamination interfering with the spectrophotometer measurement.

6. Apparatus and Materials

6.1 Clear glass jars, Mason or equivalent canning jars, 0.95 L (1 qt) and gas sealing caps. These are clear, square glass jars with approximate dimensions of 9 cm wide × 17 cm tall, contain approximately 1L and have a screw on lid.

6.2 Small wire mesh baskets (or other suitable means for suspending fabric above the water level inside the jars; see 13.2). As an alternative to the wire mesh baskets, a double strand of sewing thread may be used to make a loop in the fabric that has been folded in half twice, suspended above the water level. The two double-thread ends are draped over the top of the jar and held securely by the jar cap.

6.3 Weighing balance, with minimum

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sensitivity of 0.01 g.

6.4 Thermostatically controlled oven, 49 ± 1°C (120 ± 2°F) (see 13.6).

6.5 Nash reagent prepared from ammonium acetate, acetic acid and acetylacetone and water (see 7.1).

6.6 Formaldehyde solution, approximately 37%.

6.7 Volumetric flasks: 50, 250, 500 and 1000-mL, Class A

6.8 Volumetric pipettes: 5, 10, 15, 20, 25, 30 and 50-mL volumetric pipettes, all meeting Class A volume accuracy and flow time requirements (see 13.3). Or a digital, variable volume pipette capable of meeting Class A requirements.

6.9 A 1-L clean, brown glass bottle with lid.

6.10 Spectrophotometer, capable of reading absorbance to minimum of three decimal places at wavelength of 412 nm (see 10.6).

6.11 Test tubes (see 13.3).

6.11 Water bath; general purpose and temperature controlled (see 10.5).

6.12 Distilled or deionized water. Hereafter, referred to as deionized water.

7. Preparation of Nash Reagent

7.1 In a 1000-mL volumetric flask, dissolve 150.0 ± 0.1 g of ammonium acetate in about 800-mL of deionized water; add 3.0 ± 0.1 mL of glacial acetic acid and 2.0 ± 0.1 mL of acetylacetone. Make up to the mark with deionized water and mix. Store in a capped brown bottle.

7.2 The reagent darkens in color slightly on standing over the first 12 h. For this reason the reagent should be held 12 h before use. Otherwise, the reagent is usable over a considerable period of time, at least 6-8 weeks. However, since the sensitivity may change slightly over a long period of time, it is good practice to run a calibration curve weekly to correct for slight changes in the standard curve.

8. Preparation of Standard Solution and Calibration (Caution)

8.1 Prepare an approximately 1500 µg/mL stock solution of formaldehyde by diluting 3.8 ± 0.1 mL of reagent grade formaldehyde solution (approximately 37%) to 1.0 liter with deionized water. Equilibrate the stock solution for at least 24 h before standardization. Determine the concentration of formaldehyde in the stock solution by a standard method (see 13.5 or any other suitable procedure such as sodium sulfite titration using 0.1 *N* HCl. Reference: J. Frederick Walker, *Formaldehyde*, 3rd Ed. Reinhold Publ. Co., New York, 1964, p486). An alternate reference is the *Analytical Methods for a Textile Laboratory*, Third Edition. Record the actual concentration of this standardized stock solution. This stock solution will keep for at least four weeks and is used to prepare standard dilutions. A 1:10 dilution of the standardized formaldehyde stock solution is prepared by pipetting 25.0 ± 0.03 mL of the standardized stock solution into a 250-mL volumetric flask and diluting to the mark with deionized water. When the stock solution is titrated, the concentration is determined in µg/mL, prepare the calibration curves by:

8.1.1 Pipette 5.0, 10.0, 15.0, 20.0 and 30.0-mL of the 1:10 dilution into a 500-mL volumetric flask and dilute to mark with deionized water. (If, for example, the standardized stock solution were found to be 1470 µg/mL by titration, calculate new values for the calibration curve abscissa, i.e., 1.47, 2.94, 4.41, 5.88, 8.82 µg/mL using linear regression analysis.)

8.2 When 5.0, 10.0, 15.0, 20.0 and 30.0-mL aliquots of the 1:10 dilution of the standardized stock solution from 8.1 are diluted with deionized water in 500-mL volumetric flasks, formaldehyde solutions containing approximately 1.5, 3.0, 4.5, 6.0 and 9.0 µg/mL formaldehyde, respectively will be obtained. Record accurately the concentration of solutions. The equivalent concentrations of the formaldehyde in the test fabric based on the weight of 1.0 g of the test fabric and 50.0-mL of water in the test jars will be 50 times the accurate concentrations of these standard solutions.

8.3 Use 5.0-mL aliquots of each of the standard solutions and the procedure described in 10.4-10.7 to prepare a calibration chart in which µg/mL formaldehyde are plotted against absorbance.

9. Test Specimens

9.1 Samples should immediately be placed into separate zipper-type closure plastic bags when sampled or cut from rolls. As an extra precaution for maintaining their original state the samples may be wrapped in aluminum foil before being placed in the plastic bags.

9.2 Cut approximately 1-g specimens; weigh each one to ± 0.01 g. Perform in duplicate for each sample. If there is a delay in testing the 1-g specimens once they are cut, they should be put back into the zipper-type closure plastic bags (and rewrapped in foil if previously done) until testing commences.

10. Procedures

10.1 Pipette 50.0-mL of deionized water in the bottom of each jar. Suspend one fabric specimen above the water in each jar, using a wire mesh basket or other means (see Fig. 1). Seal the jars and place them in the oven at 49 ± 1°C (120 ± 2°F) for 20 h ± 10 min (see 13.6). A control of an empty basket (no fabric) in a jar with 50-mL of deionized water may also be used.

10.2 Remove and cool the jars for at least 30 min.

10.3 Remove the fabric and baskets, or other support, from the jars. Recap the jars and shake them to mix any condensation formed on the jar sides.

10.4 Pipette 5.0-mL of Nash reagent into a suitable number of test tubes then add 5.0-mL aliquots from each of the sample incubation jars to the tubes also

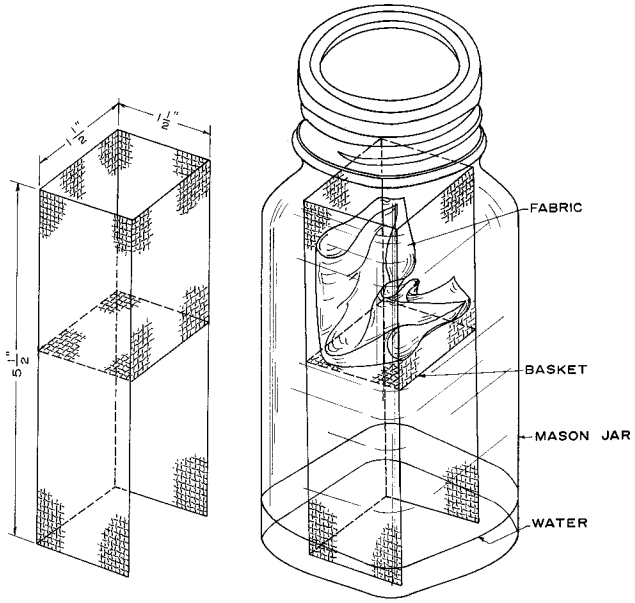


Fig. 1—The wire mesh basket detailed on the left is suspended in a sealed jar with one fabric specimen as shown on the right.

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and 5.0-mL of deionized water to a tube and used as a reagent blank. Perform duplicate solutions of each jar.

10.5 Mix and place the tubes in a 58 ± 1°C (136 ± 2°F) water bath for 6 min ± 15 sec. Remove and cool.

10.6 Read the absorbance in the spectrophotometer against the reagent blank using a wavelength of 412 nm. CAUTION: Exposure of the developed yellow color to direct sunlight for a period of time will cause some fading. If there is appreciable delay in reading the tubes after color development and strong sunlight is present, care should be exercised to protect the tubes from light. Otherwise the color is stable for considerable time (at least overnight) and reading may be delayed.

10.7 Determine the µg/mL formaldehyde (HCHO) in the sample solutions using the prepared calibration curve (see 8.3 and 13.3).

11. Calculation

11.1 Calculate the amount of formaldehyde released for each specimen to the nearest µg/g using the following equation:

$$F = (C)\,(50)/W$$

where:

F = concentration of formaldehyde (µg/g),
C = concentration of formaldehyde in solution as read from the calibration curve, and
W = weight of the test specimen, g.

12. Precision and Bias

12.1 *Precision*.

12.1.1 *Interlaboratory tests*. Two interlaboratory studies (ILS) of AATCC TM112 were conducted in 1990 and 1991, with a 20 h incubation at 49°C and a 5/5 sample to Nash solution ratio. Single operators in each participating laboratory ran triplicate determinations on each fabric. In the first ILS, results from nine laboratories testing one fabric each at three low formaldehyde levels in the range of 100-400 µg/g were analyzed by analysis of variance (ANOVA). In the second ILS, results from eight laboratories testing ten fabrics of nominal 0 µg/g were analyzed by ANOVA. The analyses have been deposited for reference in the RA68 committee files.

12.1.2 Critical differences were calculated for zero-formaldehyde fabrics, shown in Table I, and for low level-formaldehyde fabrics shown in Table II.

12.1.3 When two or more laboratories wish to compare test results, it is recommended that laboratory level be established between them prior to beginning test comparisons.

12.1.4 If comparisons are made between laboratories on a single fabric level of formaldehyde release, the critical differences in the column, single level, in Table II should be used.

12.1.5 If comparisons are made between laboratories on a series of fabrics of a range of formaldehyde levels, the critical differences in the column, multiple levels, in Table II should be used.

12.1.6 The number of determinations per laboratory average (det/avg) also determines the critical difference.

12.2 *Bias*.

12.2.1 The formaldehyde release of a fabric can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating formaldehyde released from a fabric under the conditions of accelerated storage in AATCC TM112, the method has no known bias.

12.2.2 AATCC TM112 generally is accepted by the textile and apparel industries as a referee method.

Table I—Critical Differences for Zero Formaldehyde

	Critical Differences for Averages 95% Probability, µg/g		
Det/Avg	Within Lab	Single Fabric Between Labs	Multiple Fabric Between Labs
1	7.7	12.0	13.8
2	5.5	10.6	12.7
3	4.5	10.2	12.3

Table II—Critical Differences for Low Level Formaldehyde

	Critical Differences for Averages 95% Probability, µg/g		
Det/Avg	Within Lab	Single Fabric Between Labs	Multiple Fabric Between Labs
1	21.6	80.3	116.0
2	15.2	78.9	115.0
3	12.4	78.4	114.7

13. Notes

13.1 Formaldehyde is a component of, or precursor to, many fabric finishes. As part of the application the finish is cured, resulting in self-crosslinking of the finish and or crosslinking of the finish with the fabric. Incomplete crosslinking, or hydrolysis in later storage, can result in the presence of hydrolyzed formaldehyde on the fabric. Warm moist conditions can (further) hydrolyze cross-linked finishes and release more formaldehyde. Test Method 112 determines free and hydrolyzed formaldehyde and additional formaldehyde collectively referred to as "released formaldehyde" as generated under more severe conditions. Test Method 206 determines the amount of free and hydrolyzed formaldehyde. The appropriate test should be chosen based on that distinction.

13.2 A simple cloth support for insertion in the mason jars can be constructed as follows: A piece of aluminum wire screening 15.2 × 14.0 cm (6.0 × 5.5 in.) is bent around a length of wood 3.8 cm (1.5 in.) square and fastened together to form a rectangular, open-ended cage. One side is cut at the corners about halfway up the side and the cut section folded inward and fastened. This folded piece forms the bottom of the wire basket while the other three sides form the support legs. Fastening can be accomplished by twisting short lengths of wire through or around the appropriate part. The wire screening should not be a source of formaldehyde. Test before using.

13.3 The ratio of reagent to sample solutions can be adjusted, within limits, to suit the individual absorbance range and optical path length of the sampling tubes or cuvettes of the particular photometric instrument being used. For example, although 5 mL reagent to 5 mL sample solution has been found convenient for several types of instruments, other 1:1 ratios, such as a ratio of 2 mL reagent to 2 mL sample may be more suitable for others. The same ratio should be used with the standards as with the sample. The use of spectrophotometer tubes directly for color developments avoids the transfer step, test tubes to spectrophotometer cells, and may save considerable time when many determinations are to be made. Repipettes or similar devices can be used for reagent dispensing, and disposable tip automatic pipettes can be used for the sample solutions.

13.4 The procedure in Section 10 has been set up to cover the range from about 0 µg/g on the weight of the fabric to about 500 µg/g. In fabrics containing releasable formaldehyde in the range from 500 µg/g on the weight of the fabric to about 3500 µg/g, it is recommended that a ratio of 10-mL Nash reagent to 1 mL sample be used. If this is done, it is necessary to prepare an additional calibration chart with 10:1 ratios of standard solutions to Nash reagent by diluting 5.0, 10.0, 15.0 and 20.0 µg/mL, respectively, of the approximately 1500 µg/mL standardized stock solution of formaldehyde to the mark with deionized water in 500-mL volumetric flasks. Formaldehyde solutions containing approximately 15, 30, 45 and 60 µg/ mL respectively will be obtained (see 8.3).

13.5 The chromotropic acid colorimetric method can be used as an alternate to the Nash reagent for the determination of the formaldehyde content of the sample jars after oven incubation. It should be noted that the Precision and Bias Statement was not developed using the chromotropic acid method. A suitable procedure is given in J. Frederick Walker, Formaldehyde, 3rd Edition, Reinhold Publishing Co., NY, 1964, p470. When using this method, a change may be necessary in the size of both the aliquots taken from the sample jars (see 10.2) and the standard formaldehyde solutions used in preparing the calibration curve (see 8.3). CAUTION: Since concentrated sulfuric acid is used with the chromotropic acid method, adequate care should be exercised to protect operating personnel and spectrophotometric equipment.

13.6 Incubation conditions of 65 ± 1°C (149 ± 4°F) for 4 h can be used as an alternate to the incubation conditions of 49 ± 1°C (120

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± 2°F) for 20 h (see 5.3 and 10.1). The incubation conditions of time and temperature must be reported. Upon completion of the 4-h incubation period, remove and cool the jars for at least 30 min and remove the fabric from the jars. Recap the jars and shake them to mix any condensation formed on the jar sides. Procedures for sample preparation and color development following the incubation period are performed as outlined in 10.4-10.7.

13.7 If extracted solutions are colored in flask refer to Dilip Pasad's publication "Optimization of the AATCC Sealed Jar and HPLC Methods for Measurement of Low Levels of Formaldehyde." Textile Chemist and Colorist, June 1989, Volume 21, No 6.

13.8 If the extracted solutions contain particulate matter, the solutions should be centrifuged or filtered to remove contaminant.

13.9 Standardization of Formaldehyde Stock Solution. General: The stock solution containing approximately 1500 µg/mL of formaldehyde must be accurately standardized in order to make precise calculations from the calibration curve used in colorimetric analysis.

An aliquot of the stock solution is reacted with an excess of sodium sulfite followed by a back-titration with standard acid solution in the presence of thymolphthalein as indicator.

Apparatus: 10-mL volumetric pipette, 50-mL volumetric pipette, 50-mL burette, 150-mL Erlenmeyer flask.

Reagents: 1 M sodium sulfite (126 g Anhydrous Na_2SO_3/L), 0.1% Thymolphthalein Indicator in ethanol, 0.02 *N* sulfuric acid (can be purchased in standardized form from chemical supply companies or must be standardized from standard NaOH solution). Do not use commercial standardized sulfuric acid that has been stabilized with formaldehyde. If there is a doubt, check with the chemical supplier.

Procedure:

A. Pipette 50.0-mL of 1 M Na_2SO_3 into the Erlenmeyer flask.

B. Add 2 drops of thymolphthalein indicator.

C. Add a few drops of standard acid until blue color disappears (if necessary).

D. Pipette 10.0-mL of the stock formaldehyde solution to the flask. (Blue color will reappear.)

E. Titrate the solution with the standard 0.02 *N* H_2SO_4 until the blue color is discharged. Record the volume of acid used. (The volume of acid should be in the range of 25 mL for 0.02 *N* acid.) An alternative is to titrate to an endpoint of 9.5 using no thymolphthalein nor additional drops of standard acid to make blue color disappear.

Calculations:

$$C = (30{,}030)\ (A)\ (N)/10$$

where:

C = Wt/Vol concentration of formaldehyde (µg/mL)
A = Vol of acid used (mL)
N = Normality of acid

Perform the standardization in duplicates. Average the results and use the accurate concentration in preparing the calibration curve for the colorimetric analysis.

13.7 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

14. History

14.1 Revised in 2020 to update multiple sections for clarity.

14.2 Reaffirmed 1968, 1972, 1989, 1998, 2003; revised 1975, 1978, 1982, 1984, 1993; technical correction 1983; editorially revised 1985, 1986, 2010, 2011; editorially revised and reaffirmed (with new title) 1990, 2008.

14.3 Developed in 1965 by AATCC Committee RR68; jurisdiction transferred to AATCC Committee RA45 2012.

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Test Method for Retained Chlorine: Tensile Loss

1. Foreword

1.1 This test method is based on AATCC TM92. AATCC TM114 uses an automatic home washing machine to chlorinate multiple specimens while AATCC TM92 uses a beaker for accelerated chlorination of a single specimen.

2. Purpose and Scope

2.1 This test method is a procedure to determine the potential damage that may be caused by chlorine bleaching.

2.2 The method is applicable to woven textile fabrics. The tensile strength test is not recommended for knitted fabrics because of their high stretch.

3. Principle

3.1 Samples are chlorine bleached in a domestic-type laundry machine, rinsed, dried and pressed between hot metal plates. Multiple samples may be bleached together.

3.2 The damaging action of the retained chlorine is calculated from the difference in tensile strength before and after pressing (see 14.1). Results are reported as a percentage of the original tensile strength. A higher percent loss indicates more damage due to retained chlorine.

4. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

4.1 AATCC LP1 Laboratory Procedure for Home Laundering: Machine Washing (see 14.2).

4.2 AATCC TM92 Test Method for Chlorine, Retained, Tensile Loss: Single Sample (see 14.2).

4.3 ASTM D5035, Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Strip Method) (see 14.3).

5. Terminology

5.1 **retained chlorine,** n.—*in textiles bleached with chlorine-type bleaches*, available chlorine that remains in the material after washing and drying.

6. Safety Precautions

6.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

6.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

6.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

7. Apparatus and Reagents (see 14.4)

7.1 Standard washing machine (see Table I, 14.5).

7.2 Laundering ballast, Type 1 (see Table II).

7 3 Standard tumble dryer (see Table III, 14.5).

7.4 Heating device, providing even heat transfer to the specimen from the top by close contact at a controlled temperature and giving a pressure on the specimen of 9 g/cm^2.

7.5 Apparatus and conditioning facilities as needed for ASTM D5035. (Refer to standard for details.)

7.6 Reagents

7.6.1 Sodium hypochlorite solution, approx. 5% available chlorine (see 9.2).

7.7.1 Buffer solution (see 9.1).

7.8.1 Sodium tetraphosphate ($Na_6P_4O_{13}$).

7.9.1 Sodium carbonate.

7.10.1 Sodium bicarbonate

8. Specimens

8.1 Specimens.

8.1.1 Normally, only the warp strength is determined. Cut two pieces for each sample approximately 35.6 cm (14 in.) in the warp direction and 20.3 cm (8 in.) in the filling direction. (If testing filling specimens, reverse these figures.)

9. Preparation of Reagents

9.1 Buffer Solution: Prepare a solution containing 290 g of sodium tetraphosphate and 93 g of monosodium phosphate ($NaH_2PO_4 \cdot H_2O$) in sufficient water to make 1000 mL of solution.

Table I—Standard Washing Machine Parameters (see 14.5, 14.7)

Cycle	Normal	
Water Level, L (gal)[1]	45 ± 4 (12 ± 1)	
Agitation Speed (strokes/min.)	86 ± 2	
Stroke Length	Up to 220°	
Washing Time, min.[2]	12 ± 1	
Rinse	1	
Final Spin Speed, rpm	660 ± 15	
Final Spin Time, min.	5 ± 1	
Wash Temp, °C (°F)[1]	Hot:	60 ± 3 (140 ± 5)
	Warm:	41 ± 3 (105 ± 5)
	Cold:	27 ± 3 (80 ± 5)

[1] Water Level and Wash Temperature indicated in the table may not correspond with standard machine settings. Verify and adjust as needed to achieve the indicated parameters.

Table II—Laundering Ballast Parameters

	Type 1
Fiber Content	100% Cotton
Greige Fabric Yarns	16/1 ring spun
Greige Fabric Construction	52 × 48 ± 5 yarns/in., plain weave
Finished Fabric Weight	155 ± 10 g/m^2
Edges	All edges hemmed or over-edged
Finished Piece Size	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)
Finished Piece Weight	130 ± 10 g

Table III—Standard Tumble Drying Parameters

Cycle	Normal	Delicate	Permanent Press
Max. Exhaust Temp, °C (°F)	68 ± 6 (155 ± 10)	60 ± 6 (140 ± 10)	68 ± 6 (155 ± 10)
Cool Down Time, min.	≤ 10	≤ 10	≤ 10

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9.2 Available Chlorine. Verify the percent available chlorine. Pipette a 1.00-mL portion of nominal 5% sodium hypochlorite solution into an Erlenmeyer flask and dilute to 100 mL with distilled water. Add 6 mL of 12% KI followed by 20 mL of 6N H_2SO_4. Titrate with 0.1*N* sodium thiosulfate solution and calculate using Eq. 1. Use 1.08 as the value for specific gravity.

Percent available chlorine =

$$\frac{\text{mL Na thiosulfate} \times 0.1N \times 0.0355}{1\text{ mL} \times \text{sp. grav. Na hypochlorite}} \times 100 \qquad \text{(Eq. 1)}$$

9.2.1 A 5% available chlorine solution, purchased from a laboratory chemical supplier, may also be used. COA's may be used to ascertain strength. Chlorine does degrade. Once opened, store chemical in refrigerator and use by manufacturer's expiration date.

10. Procedure

10.1 Preparation of Washing Machine. If the washing machine has been used for anything other than chlorine bleaching, run a full wash cycle with warm water, adding 0.15% sodium tetraphosphate at the beginning of the cycle. Determine water volume at low-water level and use in calculating liquor-to-bath ratio.

After the machine is free of impurities, set washing machine controls to obtain the parameters listed in Table I. Select water temperature based on care instructions.

10.2 Chlorination.

10.2.1 Begin selected wash cycle. Allow machine to fill to specified water level (50:1 liquor ratio).

10.2.2 Add sufficient diluted sodium hypochlorite to wash water to obtain a bath liquor concentration of 0.10% by weight of available chlorine.

10.2.3 Stir well. Adjust the pH to 9.5 with sodium carbonate to raise the pH or sodium bicarbonate to lower it.

10.2.4 Add all the test samples and sufficient untreated ballast to make a 50:1 liquor-to-cloth ratio. Run machine through wash portion of cycle.

10.3 Rinsing. After the wash water has drained and the machine has refilled, add 10 mL of the buffer solution for each gallon of water and continue operation through rinse cycle and the final spin cycle.

10.4 Drying and Conditioning. Tumble dry the washed samples with ballast until just dry. Set tumble dryer to obtain the parameters listed in Table III. Select drying temperature based on care instructions. Do not reuse ballast for this test method due to retained chlorine. Transfer at once to a conditioned atmosphere at 65 ± 5% RH and 21 ± 2°C (70 ± 4°F) and allow to remain through the scorching step and tensile strength testing. The samples should be conditioned not less than 4 nor more than 24 h before scorching, and not less than 16 h before tensile testing.

10.5 Preparation of Specimens for Scorching. Prepare warp specimens by carefully cutting five strips approx. 35.6 × 3.2 cm (14.0 × 1.25 in.). Ravel the 3.2-cm (1.25-in.) strips to exactly 2.54 cm (1 in.) following ASTM D5035. Then trim the 35.6 cm (14 in.) length of each strip to 30.5 cm (12 in.) From this set of five 2.54 × 30.5 cm (1 × 12 in.) strips, make two sets (from the same set of warp threads) of 2.54 × 15.2cm (1 × 6 in.) raveled strip specimens by cutting the strips in two. Keep these sets separate; one set to be scorched, the other to be used as a control.

10.6 Preparation of Heating Device. Prepare heating device in advance so that the heating plates of the instrument maintain a temperature of 185 ± 1°C (365 ± 2°F). If necessary, to avoid air currents, enclose instrument in a cabinet. Make sure the two heating plates are clean and in good alignment (uniformly in contact at all points) (see 14.6). Scorch each strip of one set separately (one at a time) by placing the 2.54 × 15.2 cm (1 × 6 in.) specimen in the heating device with the long direction of the raveled strip perpendicular to the long direction of the heating plates so that it is scorched across the center of the raveled strip (see Fig. 1). Scorch the strips for 30 s. Check reading of the thermometers frequently during use. Condition the specimens again, this time conditioning at least 16 h before performing tensile strength tests.

10.7 Tensile Strength. Perform tensile strength tests on the unscorched and scorched specimens and record the individual values. From these, calculate the average (mean) tensile strength for each set.

11. Calculations

11.1 Use Eq. 2 (see 14.1):

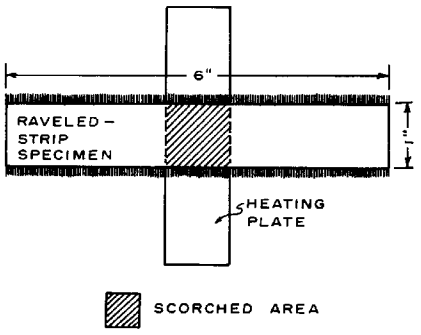


Fig. 2—Specimen on heating plate.

$$(T_c - T_{cs})/T_c \times 100 =$$

% loss in tensile strength due to damage caused by retained chlorine (Eq. 2)

where:

T_c = average tensile strength of chlorinated specimens, unscorched, and

T_{cs} = average tensile strength of chlorinated specimens, scorched.

12. Report

12.1 Describe or identify the samples tested.

12.2 Report that the sample was tested using AATCC TM114-2021.

12.3 Report the testing conditions:

12.3.1 Number of specimens tested.

12.3.2 Laundering conditions.

12.4 Report the test results:

12.4.1 Average percent loss in tensile strength for *warp* due to damage caused by retained chlorine.

12.4.2 Standard deviation of average

12.4.3 If filling strength loss is determined, report average and standard deviation of percent loss in tensile strength for *filling*.

12.5 Describe any modification(s) of the published standard.

13. Precision and Bias

13.1 *Precision*. The standard deviation obtained in developing this method was s = 5.98.

The confidence interval based on 5 specimens is C.I. ± 5.2.

13.2 *Bias*. The tensile strength loss due to retained chlorine can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

14. Notes

14.1 Where it is suspected that either fabric or finish is susceptible to damage on heating, this susceptibility can be determined by running a distilled-water control, that is, the fabric is run through the entire procedure, but using distilled water in place of the hypochlorite solution. The loss in strength due to heat alone is calculated by Eq 3:

$$(T_w - T_{ws})/T_w \times 100 =$$

Percent tensile strength loss due to scorching alone (Eq. 3)

where:

T_w = average tensile strength of the water-treated specimens, unscorched (handled as above) and

T_c = average tensile strength of the original fabric chlorinated and unscorched.

If this loss is appreciable, there is some question as to whether the chlorine damage test is applicable. It is usually not necessary to

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determine the effect of wet chlorination on the fabric, as this factor is cancelled out in the calculation for chlorine damage. However, where it is desirable to determine the effect of the wet chlorination step, this may be calculated using Eq. 4:

$$(T_w - T_c)/T_w \times 100 =$$

Percent tensile strength loss due to wet chlorination (Eq. 4)

where:

T_w = average tensile strength of the water-treated specimens, unscorched (handled as above) and

T_c = average tensile strength of the original fabric chlorinated and unscorched.

14.2 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

14.3 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428, USA; +1.610.832.9500; www.astm.org.

14.4 For potential apparatus, reagents or materials sources, visit the AATCC Buyer's Guide at www.aatcc.org/bg. AATCC provides its Corporate members the option to list their items and services. AATCC does not qualify, or in any way approve, endorse or certify that any of the listings meet the specifications in its standards.

14.5 For model number(s) of washer(s) and tumble dryer(s) reported to meet the standard parameters, visit www.aatcc.org/testing/laundering or contact AATCC, P.O. Box 12215, Research Triangle Park NC 27709; +1.919.549.8141; ordering@aatcc.org.

14.6 The test block should exert a force of 255 g (9 oz), or 9 g/cm^2 (2 oz/in^2). It can be calibrated by attaching a spring balance to the upper plate with a small thread sling and determine the force required to just separate the plates. If readjustment is necessary, follow instructions supplied by the manufacturer.

14.7 Prior to 2021, water temperature and agitation time were the only washing parameters defined for this test method. The additional parameters listed in Table I are standard conditions for testing purposes, selected to align with machines specified in other AATCC laundering methods. Note that this test method must be performed in a traditional top-loading washing machine.

15. History

15.1 Revised in 2021 to clarify and align with prescribed AATCC style guidelines.

15.2 Editorially revised 2019. Revised 2016. Editorially revised 2012. Editorially revised and reaffirmed 2011. Technical correction 2010. Editorially revised 2008. Editorially revised and reaffirmed 2005. Editorially revised 2004. Reaffirmed 1999. Editorially revised and reaffirmed 1994. Editorially revised 1992. Reaffirmed 1989. Editorially revised and reaffirmed 1985. Reaffirmed 1980, 1977, 1974, 1971, 1967.

15.3 Developed in 1965 by AATCC Committee RR35. Maintained by RA99.

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Test Method for Electrostatic Clinging of Fabrics: Fabric-to-Metal

Developed in 1965 by AATCC Committee RA32; revised 1969, 1973, 2000; editorially revised 1974, 1976, 1977, 1978, 1991, 1999, 2008, 2019; reaffirmed 1977, 1980, 1989, 2005, 2011; editorially revised and reaffirmed 1986, 1995.

1. Purpose and Scope

1.1 This test method evaluates the relative clinging tendency of certain fabrics due to electrostatic charge generation. The test integrates the effect of fabric weight, stiffness, construction, surface character, finish application and other fabric parameters which affect the tendency of fabrics to cling.

2. Principle

2.1 Clinging of positively or negatively charged fabrics to the human body occurs due to instantaneous induction of an equal and opposite charge on the surface of the body when the charged fabric is brought close to it. A basic law of physics states that oppositely charged materials attract each other. A metal plate exhibits a similar phenomenon of instantaneous charge induction as would the human body when placed in the field of a charged material. Therefore, a metal plate can be used to simulate the problems of clinging observed between charged garments and the human body. Some individuals are more prone to electrostatic clinging than others, and a given individual may be more prone at one time than at another time. Therefore, fabric-to-metal cling times cannot be related directly to fabric-to-body clinging for different individuals.

2.2 In this test method, the time (t_d) is measured for the charge on a fabric specimen to decay to a level where the electrical attractive forces between the specimen and the metal plate are overbalanced by the gravitational forces of the specimen acting to pull the specimen away from the plate (see 12.1).

3. Terminology

3.1 **electrostatic clinging,** n.—adherence of one substance to another caused by an electrical charge on one or both surfaces.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 The radioactive bar emits alpha radiation which is externally harmless to the human body. The radioactive isotope polonium 210 is toxic and precaution should be exercised to prevent ingestion or inhalation of the solid material. Do not take the radioactive bar apart or touch the radioactive strip under the grid. If the strip is touched or handled, wash hands thoroughly at once. Return the device to the manufacturer when it loses its effectiveness as a static eliminator for reactivation when tested in accordance with 12.3.1 or for disposal if use is to be discontinued. ***Do not discard as scrap***.

5. Uses and Limitations

5.1 This test method is not intended for use in determining the suitability of fabric for use in hazardous areas where the electrostatic generation of sparks might result in fires or explosions.

5.1.1 Certain fabrics, particularly those of heavy construction, which do not exhibit a clinging propensity by this method might cling under some conditions of use.

5.1.2 This test method is primarily useful for determining clinging properties of lightweight apparel fabrics, such as those intended for use in lingerie.

6. Apparatus and Materials (see 12.2)

6.1 Test plates.

6.1.1 Standard Test Plate—A test plate (see 12.3) is made by bending 100 × 450 mm strips of 18 gauge Type 304 stainless steel across the width 150 mm from one end so that an included angle of 1.22 ± 0.04 RAD is formed between the 100 × 150 mm base and the 100 × 300 upright plate. The grain of the finish should be such that the 450 direction of the test plate corresponds to the length or grain direction of the stainless steel sheets from which they are made. The face of the 100 × 300 plate must have a No. 4 finish (see 12.3.1) on the included angle side, and should be kept clean and smooth at all times. A fine line is indented across the test plate surface 230 mm from the top of the plate and this is used to position the lower edge of the specimen (see 12.3).

6.1.2 Variable Angle Experimental Test Plate—The sensitivity of this test depends on the included angle of the metal plate and can be diminished by decreasing the included angle or vice versa. A more versatile version of the test plate, suitable for research work, which allows facile change of the sensitivity, uses a 25 × 100 × 100 mm aluminum base with several slots machined into the base at different angles (e.g., 0.017, 0.087, 0.175, 0.35, 0.52, 0.70, 0.87, 1.05 RAD) from the vertical axis into which a 100 × 360 mm stainless steel plate fits.

6.2 Grounding Plate—This is a 200 × 360 mm, flat, 18 gauge Type 304 stainless steel plate, attached to a ground, such as a water pipe, with wire (e.g., No. 18 stranded, plastic-covered electronic wire). One such grounding plate is required for each test plate.

6.3 Radioactive bar (see 12.4).

6.4 Rubbing blocks, white pine, approx. 20 × 50 × 150 mm adjusted to weigh 65 g and with strips of 20 mm wide, double-faced tape on the ends of each block to attach ends of the rubbing fabric.

6.5 Polyurethane foam, 25 × 100 × 300 mm non-rigid, having a density of 21 kg/m^3 and an ILD (Identification Load Deflection) firmness of 6.8 kg when tested (see 12.3) as directed in ASTM D3574.

6.6 Clamps, metal (e.g., No. 3 Bulldog Clamp or Di-acro Toggle Clamp, see 12.4), with 70 mm edges covered with 20 mm electrical tape to prevent scratching of the surface of the test plates.

6.7 Stopwatch, with 0.01 min or one of equivalent precision marked in other units.

6.8 Forceps or tweezers, non-conducting, ivory-tipped, forceps used with analytical balances.

6.9 Oven, forced-draft type, capable of maintaining a temperature of 105 ± 2°C (221 ± 5°F).

6.10 Conditioning chamber, capable of providing relative humidity control of 40 ± 2% and temperature at 24 ± 1°C (75 ± 2°F) with circulating air.

6.10.1 If tests are to be conducted at relative humidities or temperatures other than 40 ± 2% RH and 24 ± 1°C (75 ± 2°F), the conditioning test chamber should

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be capable of providing the necessary range of test conditions (such as 20-65 ± 2% RH and from 10-30 ± 1°C [50-86 ± 2°F).

6.11 Rubbing fabrics.

6.11.1 Nylon rubbing fabric—100% spun nylon 6,6.

6.11.2 Polyester rubbing fabric— 100% spun polyester.

6.12 Fabric iron, domestic hand type, with appropriate settings (see Table I).

6.13 Cleaning solvent: halogenated hydrocarbon metal-cleaning solvent hereinafter called HH Solvent.

Table I—Safe Ironing Temperature Guide

Class 0 Below 121° C	Class I 121-135° C	Class II 149-163° C	Class III 177-191° C	Class IV 204° C and Above
Modacrylic 93-121°C	Acetate		Nylon 6,6	Cotton
Olefin (Polyethylene) 79-121°C	Olefin (Polypropylene)	Acrylic	Polyester	Fluorocarbon
		Azlon		
		Nylon 6		Glass
				Hemp, Jute, Ramie
Rubber 82-93°C	Silk	Spandex		Linen
		Wool		Rayon, Viscose
Saran 66-93°C				Triacetate (heatset)
Vinyon 54°C				

7. Test and Rubbing Specimens

7.1 Test fabrics—Twelve 75 × 230 mm test specimens are required. Using die or scissors, cut six specimens with the longer dimension parallel to the warp, or wale, direction and six with the longer dimension parallel to the fill, or course, direction. (Do not cut fabric specimens with a hot soldering iron or heated wire since the edges can cause localized static problems due to heat degradation of the fibers and antistatic agents.)

7.2 Rubbing fabrics—Both nylon and polyester rubbing fabrics are required. Using die or scissors, cut six nylon and six polyester rubbing fabrics, each 75 × 230 mm, with the longer dimension parallel to the warp, or wale, direction of the fabric.

7.3 Do not contaminate the test specimens or the rubbing fabric by unnecessary handling or by allowing them to come in contact with transferable materials.

7.4 If either the test or rubbing fabric specimens do not lay completely flat, press them free of creases or wrinkles with a dry, clean iron at the appropriate setting as directed in Table I. Do not use test specimens with curled edges.

8. Conditioning

8.1 Since equilibration of a test specimen from the dry versus the wet condition relative to the humidity of the test chamber may show hysteresis, precondition the specimens from the dry side relative to the humidity in the test chamber. To accomplish this, place the test and rubbing specimens in the forced-draft oven at 105 ± 2°C (221 ± 4°F) for 30 min. Transfer them immediately to the controlled humidity chamber allowing no more than 15 s for the transfer.

8.2 Condition and test the test and rubbing specimens at 40 ± 2% RH and 24 ± 1°C (75 ± 2°F) in the controlled humidity chamber for at least 16 h. If the test is conducted at lower humidities (such as 30 ± 2% or at 20 ± 2%), or at higher humidities (such as 65 ± 2%), keep the temperature at 24 ± 1°C (75 ± 2°F) for standard testing. If other temperature conditions are used for nonstandard testing, state the temperatures used in the report. For any humidity or temperature test conditions that are to be used, the test and rubbing specimens should be conditioned for at least 16 h (see 12.8).

9. Procedure

9.1 Clean the metal test plate before testing every different lot of fabric. To clean, swab with a facial tissue saturated with HH Solvent (wear protective gloves). Allow the plate to dry at least 5 min in the conditioning test chamber before using. Make sure that HH Solvent does not accumulate in the conditioning test chamber as a result of cleaning the metal test plate. If the No. 4 finish on the surface of the test plate is changed noticeably due to use, replace the test plate with a new one.

9.2 Handle the fabric test specimens and rubbing fabrics at only one marked corner to minimize transfer of impurities to the remainder of the fabric.

9.3 Adhere double-faced adhesive tape around all four 20 mm thick sides of the rubbing block. Completely discharge the rubbing fabric on both sides with the radioactive bar (*Caution*, see 4.2). Place a 50 × 150 mm side of wood block on the rubbing fabric in such a way that the 150 mm dimension of the block parallels the 230 mm dimension of the rubbing fabric and the block is centered on the fabric. Bring up the protruding edges of the fabric so that they adhere tightly to the double-faced tape on all four sides of the rubbing block. Take care that the double-faced tape does not extend to the area between the rubbing surface of fabric and the block.

9.4 Grasp a test specimen at one corner and allow it to hang free. Discharge both sides of the test specimen using the radioactive bar, holding the bar not over 25 mm away from the fabric as it moves up or down (*Caution*, see Section 4 and Fig. 1, Step A). Fasten the test specimen at the top of the long arm of the metal plate on the 1.22 RAD included angle side with a metal bulldog or toggle clamp with the face of the test specimen out. Position the lower edge of the test specimen even with the indented line on the metal plate.

9.5 Attach the metal grounding plate with the wire to a suitable ground and place it horizontally in the controlled humidity chamber and lay the polyurethane foam on top.

9.6 Place the metal test plate with attached test specimen in a horizontal position on the polyurethane foam in such a way that the long arm of the plate completely rests on the polyurethane foam with the short arm of the test plate closest to the operator and pointing upward. The clamp should extend over the edge of the supporting foam so that the back of the plate is completely supported.

9.7 Place the rubbing block with attached rubbing fabric at the top of the plate so that the long axis of the rubbing fabric and the long axis of the test specimen are at right angles (see Fig. 1, Step B). The side of the rubbing block that does not expose the cut edge of the rubbing fabric should be down. Keep the test plate and rubbing block perfectly flat on their facing surfaces to ensure complete contact between test and rubbing fabrics during charge generation.

9.8 Place the middle fingers on the back edge of the rubbing block and pull the rubbing assembly the full length of the test specimen (see Fig. 1, Step C). Grasp the lock between the thumbs and middle fingers at the ends of the block, lift the assembly from the test specimen and return the rubbing assembly to its original position at the top of the test specimen. Repeat this rubbing sequence 12 times at a cycle of approx. 1 rubbing cycle per second. The total 12 rubbing cycles should be completed in no more than 15 s. For constant rubbing pressure, take care that the only downward pressure on the test specimen comes from the weight of the rubbing block and fabric.

9.9 Quickly place the assembly in an

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upright position on a grounded plate. Grasp the lower right corner of the test specimen with the nonconducting tweezers (see Fig. 1, Step D), pull the unclamped part of the test specimen completely away from the metal plate to a point where the test specimen is positioned vertically for ± 0.5 s (see Fig. 1, Step E), and then release the test specimen from the tweezers. Start the stopwatch immediately (see Fig. 1, Step F).

9.10 After every 30 ± 2 s, grasp the lower corner of the test specimen with nonconducting tweezers, pull the test specimen completely away from the metal plate to a point where the test specimen is positioned vertically for 1 ± 0.5 s, and then release the test specimen from the tweezers.

9.11 Record the time, to the nearest 0.1 min, for the fabric to decling of its own accord (see Fig. 1, Step G). Disregard any tendency for the test specimen to continue clinging within 25 mm of the bottom of the holding clamp.

9.11.1 If the time is greater than 10 min, discontinue the test and report t_d as >10 min (greater than 10 min). In some cases, a different "maximum cling time" period may be chosen. In this case, report the actual maximum cling time used for the testing.

9.11.2 If the test specimen declings *almost* completely but persists in clinging in a small localized area in addition to the area within 25 mm of the bottom of the holding clamp, discard and test an additional specimen. If this condition is observed in all test specimens, note and report the condition without assigning cling time (see 12.9).

9.12 Test three specimens in both warp (or wale) and filling (or course) directions against both nylon and polyester rubbing fabrics, using fresh test specimens and fresh rubbing fabric surfaces for each determination (see 12.10).

10. Report

10.1 Report the time (t_d) to the nearest 0.1 min required for a test specimen to decling from the metal plate as specified in 9.9 and 9.11.2. Report the three individual values of tests in both warp (or wale) and filling (or course) directions as well as an average of each of the three determinations, reporting separately the times (t_d) obtained with both nylon and polyester rubbing fabrics.

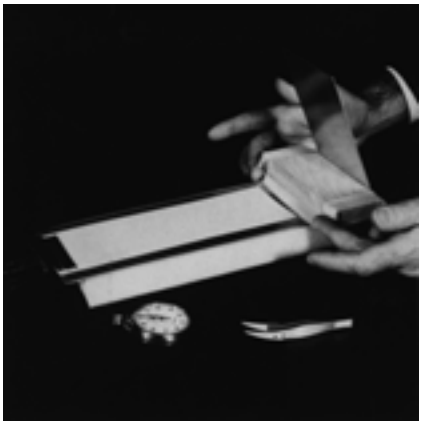
10.1.1 If the t_d value of two of a set of three test specimens is less than the selected maximum cling time selected from 9.11.1 but the t_d value of the third test specimen is greater than the selected maximum cling time, report the average of the two determinations which are less



A



B



C



D



E



F



G

Fig. 1—Principal steps in conducting test.

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than the selected maximum cling time and disregard the t_d value of the third test specimen. Report that the third specimen exceeded the maximum cling time.

10.1.2 If the t_d value of two of a set of three test specimens is greater than the selected maximum cling time, report the t_d value as greater than the selected maximum cling time and disregard the t_d value of the third test specimen.

10.2 Report the percent relative humidity and the temperature at which the specimens were tested.

11. Precision and Bias

11.1 *Precision*. Extensive interlaboratory evaluations run in 1955 on woven fabrics indicated that, for the fabrics tested, the 95% confidence limits of this test are ± 0.5 min (see 12.1).

11.2 An interlaboratory test on tricot fabrics was run in 1975 which involved the following variables:

5 laboratories
4 fabrics
2 levels of laundering
2 fabric directions
2 relative humidities
2 types of nylon and 2 types of polyester rubbing fabrics

Based on the results of this study, the 95% confidence limits for the average decling time of a set of three observations, none of which exceeds the maximum cling time of 10 min, is ± 2.0 min for tests at 20% RH and ± 1.7 min for tests at 40% RH.

Some of the fabrics tested did not cling at all (0 min) while others clung more than the maximum 10 min, at which point testing is discontinued. In this interlaboratory test, 576 different test conditions were listed which involved 1,728 individual decling measurements. Of the 576 test conditions, 469 contain at least one 0 or 10 value. Only 109 contain no 0 or 10. Any set of these replicates containing a 0 or 10 cannot give a valid estimate of the experimental error (see 12.7).

11.3 *Bias*. Electrostatic clinging can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

12. Notes and References

12.1 For details, see *American Dyestuff Reporter*, Vol. 56, 1967, pp345-350.

12.2 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

12.3 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

12.3.1 The specified No. 4 finish on the standard test plate shall be as specified in the *ASTM Metals Handbook*, Vol. 2, 8th Edition, 1964, p599, which states: "Following initial grinding with coarser abrasives, sheets are finally finished with lubricated 120-150 mesh abrasive belts." The resultant No. 4 finish surface roughness expressed in micro-inches is 45 maximum.

12.4 Since the polonium in this radioactive element has a limited half life (approx. 6 months), the radioactive bar must be checked periodically to be sure that it is sufficiently active for complete removal of charge. The radioactive bar may be checked by testing a specimen known to exhibit a cling time of at least 10 min when tested as directed in Section 6. A jersey stitch nonantistatic heatset 40 denier nylon tricot fabric free of finish or scoured and in the range of 50-100 g/m^2 is satisfactory for this purpose. The fabric should be charged as directed and then deionized with the radioactive bar. If the test specimen shows zero cling time, the radioactive bar is sufficiently active for complete removal of charge.

12.5 The clinging behavior by this method of test of clean, finish-free 100% cotton fabrics can be used as a reference point for antistatic behavior of test fabrics of similar construction and weights. Garments made from such clean, finish-free 100% cotton fabrics are generally considered to be free of clinging problems under use conditions.

12.6 Gayler, J., R. E. Wiggins, and J. B. Arthur, Department of Textile Technology, School of Textiles, North Carolina State University, "Static Electricity Generation, Measurement, and Its Effect on Textiles," May 1965, pp11-12.

12.7 The statistical analyses and related data are on file at AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

12.8 At a given temperature, accumulation of static electricity generally is greater the lower the relative humidity (and vice versa). For example, fabrics which show a low degree of static accumulation at 40% RH and 24°C (75°F) may exhibit severe static accumulation at 25% RH and 24°C (75°F) while fabrics which exhibit static problems at 40% RH and 24°C (75°F) may exhibit a low degree of static accumulation at 65% RH and 24°C (75°F). The relationship between the tendency for static accumulation and relative humidity varies with the specific antistatic agent, fiber, fabric construction, surface charger, etc. Therefore, although 40% RH and 24°C (75°F) are the standard conditions for the test, significant information as to the probable antistatic performance properties of a fabric may require that tests be run also at a lower relative humidity, such as 20%, and 24°C (75°F), an atmospheric condition not unusual in modern heated and air conditioned buildings. Complete evaluation may also require tests at the upper level of 65% RH and 24°C (75°F).

12.9 Clinging in a small localized area can be indicative of the presence of foreign material or some irregularity and may not be characteristic of the clinging properties of the fabric lot from which the test specimens were taken.

12.10 Since several determinations sometimes can be run concurrently, time can be saved by having several test plates and grounding plates available.

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Test Method for Colorfastness to Crocking: Rotary Vertical Crockmeter

1. Purpose and Scope

1.1 This test method is used to determine the amount of color transferred from the surface of colored textile materials to other surfaces by rubbing. It is applicable to textiles made from all fibers in the form of yarn or fabric, whether dyed, printed or otherwise colored and especially where the singling out of colored areas smaller than possible to test with the standard crockmeter (AATCC TM8, Test Method for Colorfastness to Crocking: Crockmeter) is required (see 14.1).

1.2 Test procedures employing test squares either dry or wet with water or other liquids are within the scope of this method.

2. Principle

2.1 A test specimen held at the base of the rotary vertical crockmeter is rubbed with standard test squares under controlled conditions (see 14.1).

2.2 Color transferred to the test squares is assessed by a comparison with the Gray Scale for Staining, AATCC 9-Step Chromatic Transference Scale or Instrumental Assessment of Degree of Staining (see 14.2 and 14.3).

3. Terminology

3.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

3.2 **crocking,** n.—a transfer of colorant from the surface of a colored yarn or fabric to another surface or to an adjacent area of the same fabric principally by rubbing.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5. Uses and Limitations

5.1 Since manufacturing processes such as dyeing, finishing, etc., and washing, drycleaning, ironing, etc. may affect the degree of color transfer from a material, the test may be made before or after, any such treatment, depending upon the information desired.

5.2 This method is designed for use with an AATCC crocking cloth. Alternatively, other adjacent textile substrates may be used as agreed between interested parties.

6. Apparatus and Materials (see 14.4)

6.1 Rotary vertical crockmeter with 16 ± 0.1 mm diameter finger (see Fig. 1).

6.2 AATCC Crocking Cloth, cut in 50 ±1 mm squares (see 14.5).

6.3 AATCC 9-Step Chromatic Transference Scale (AATCC EP8) (see 14.2).

6.4 Gray Scale for Staining (AATCC EP2) (see 14.2).

6.5 White AATCC Textile Blotting Paper (see 14.2).

6.7 Spiral wire clip for securing crocking cloth to rotary vertical crockmeter finger.

6.8 Crockmeter verification fabric, or in-house fabric with known poor colorfastness to crocking properties, that provides repeatable and predictable crocking results.



Fig. 1—Rotary vertical crockmeter.

7. Verification

7.1 Verification checks on the operation of the test and the apparatus should be made routinely and the results kept in a log. The following observations and corrective actions are extremely important to avoid incorrect test results where abnormal crock images can result and influence the rating process.

7.2 Use the crockmeter verification fabric or in-house poor crocking fabric with known predictable and repeatable crocking behavior.

7.2.1 A poor circular image with uneven dye pick-up may indicate the crocking finger needs resurfacing.

7.2.2 A double image may indicate a loose clip.

7.2.3 An uneven stained ring around the circumference may indicate that the peg is uneven.

7.2.4 Scuff marks to the sides of the specimen indicate the loops to the wire clips are positioned downwards and are not high enough to prevent rubbing the specimen surface.

7.2.5 Confirm the wet pick-up techniques (see 10.2).

7.2.6 Replace the abrasive paper on the tester base if it is smooth to the touch in the crocking area compared to the adjacent area, or if slippage of the specimen is noticed.

8. Test Specimens

8.1 Two specimens are used, one each for the dry and the wet tests.

8.1.1 Additional specimens may be used to increase the precision of the average (see 13.1).

8.2 Specimens must exceed the size of the crocking finger such that it is in full contact with the specimen during the complete rotation of the finger. Avoid seams and other raised areas.

8.3 For yarns, knit a piece of fabric, or wind yarn tightly on a suitable sample card/plate.

9. Conditioning

9.1 Prior to testing, precondition and condition the test specimens and the crocking cloth for crock testing as directed in ASTM D1776, Standard Practice for Conditioning and Testing Textiles. Condition each specimen for at least 4 h in an atmosphere of 21 ± 2°C (70 ± 4°F) and 65 ± 5% RH by laying each test specimen or

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crock square separately on a screen or perforated shelf of conditioning rack.

10. Procedure

10.1 Dry Crocking Test.

10.1.1 Tilt the upper half of the machine away to allow access to the base of the machine. Place a test square on the end of the vertical crocking finger and fasten it with spring clip.

10.1.2 Hold the test specimen on the machine base at the point the crocking finger contacts the base. Tilt the upper half of the machine back to the operating position with the test square at the end of the finger in contact with the test specimen. Place the weight supplied with the Crockmeter on the vertical shaft to give the rubbing finger a downward force of 11.1 N ± 10% (40 oz ± 10%).

10.1.3 Hold the test specimen in position on the base. Rotate the crank 20 times at approximately 1 rotation per second. This produces approximately 40 reciprocal turns of the vertical shaft.

10.1.4 Tilt the upper part of the machine back and remove the test specimen and the test square, condition as in 9.1 and evaluate as in Section 11.

10.2 Wet Crocking Test.

10.2.1 Weigh the conditioned dry crock square. Using a syringe tube, graduated pipette or automatic pipette, draw water in mL to 0.65 times the weight of crocking square. Example: If the crocking square weight equals 0.24 g, the mL used would be 0.24 × 0.65 = 0.16 mL. Lay the crocking square on white plastic mesh over a dish. Apply water evenly over the crocking square to achieve a 65 ± 5% wet pickup when weighed. Calculate wet pickup by multiplying the original dry conditioned weight by 1.65. If needed, adjust the amount of water used to wet the square and using a new crocking square, repeat steps. When 65 ± 5% wet pickup is achieved, record the amount of water used. Draw up the recorded amount of water into the syringe tube, graduated pipette or automatic pipette for each wet crocking performed during the current day. Repeat this process each day (see 14.6).

10.2.2 Avoid evaporative reduction of the moisture content on the crock square below the specified level of 65 ± 5% before the actual crock test is run.

10.2.3 Continue as directed in 10.1.

10.2.4 Air dry the crock test square, then condition (see 9.1) before evaluating. When loose fiber might interfere with the rating, remove the extraneous fibrous material by pressing lightly on the crock circle with the sticky side of cellophane tape before evaluating.

11. Evaluation (see 14.7)

11.1 Back the crocked test square with three layers of clean AATCC Crocking Cloth for evaluating.

11.2 Rate by means of the Gray Scale for Staining (AATCC EP2), the AATCC 9-Step Chromatic Transference Scale (AATCC EP8) or Instrumental Assessment of Degree of Staining (AATCC EP12), and record the numerical rating (see 14.3 and 14.8).

11.3 It will be noted that the color transfer is usually greater near the edge of the tested circle than the center.

11.4 Rate the color transfer at the edge of the tested circle.

11.5 Average individual results to the nearest 0.1 grade when multiple specimens are tested or when a panel of evaluators rate color transfer.

12. Report

12.1 Report the grade determined in 11.5.

12.2 Indicate whether dry or wet test was performed.

12.3 Indicate whether Gray Scale for Staining, AATCC 9-Step Chromatic Transference Scale or Instrumental Assessment of Staining was used.

13. Precision and Bias

13.1 *Precision*. In 2017, 5 laboratories participated in an interlaboratory study to rate the colorfastness of fabrics after dry crocking and wet crocking. All 5 labs conducted colorfastness testing of 3 fabrics, following this test method. Fabrics tested included 2 vat-dyed denim fabrics, and 1 basic-dyed cotton fabric. Each lab used 2 operators, and each operator performed 10 replications by both dry and wet crocking. Each of the stained crock squares were rated instrumentally using a spectrophotometer to determine the Gray Scale for Staining grade. The basic-dyed cotton fabric (Fabric 1) exhibited a grade of 3.5, as a result of dry crocking, and 1.5 as a result of wet crocking. The denim fabrics (Fabric 1 and 2) exhibited a grade of 4.5 and 5, as a result of dry crocking, respectively; and 1 and 1.5 as a result of wet crocking, respectively. Table I gives the *within-lab* and *between-lab* precision values for this study.

13.2 *Bias*. The true value of colorfastness to crocking can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

14. Notes

14.1 The rotary vertical crockmeter provides a reciprocating rotary motion to the test finger and a selected pressure on the test finger. Specimens tested by AATCC TM8 versus AATCC TM116 may show dissimilar results. There is no known correlation between the two methods.

14.2 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@ aatcc.org; www.aatcc.org.

14.3 It has been noted that different grades may result depending upon whether the Gray Scale for Staining or AATCC 9-Step Chromatic Transference Scale is used for the evaluation. It is, therefore, important to report which scale was used. For very critical evaluations and in cases of arbitration, ratings must be based on the Gray Scale for Staining.

14.4 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

14.5 AATCC Crocking Cloth should meet the following specifications:

Fiber	100% 10.3-16.8 mm combed cotton staple with no optical brightener present
Yarn	15 tex (40/1 cotton count), 5.9 turns/cm "z"
Thread count	for greige fabric 32 ± 5 warp ends/cm: 33 ± 5 filling picks/cm
Weave	1/1 plain

Table I—Precision Values

	Dry		Wet	
n	Within-Laboratory	Between-Laboratory	Within-Laboratory	Between-Laboratory
1	0.116	0.4006	0.307	1.067
2	0.082	0.2837	0.218	0.7557
3	0.067	0.2313	0.178	0.6160
4	0.058	0.2003	0.154	0.5335
5	0.052	0.1792	0.138	0.4772
6	0.047	0.1635	0.126	0.4356
7	0.044	0.1515	0.116	0.4034
8	0.041	0.1417	0.109	0.3773
9	0.039	0.1335	0.103	0.3557
10	0.037	0.1267	0.097	0.3374

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Finished fabric	desized and bleached with no optical brightener or finishing material present
pH	7 ± 1
Mass/sq meter	100 ± 3 g finished
Whiteness	W = 78 ± 3 (AATCC TM110)

CAUTION: ISO crock cloth results may not be equivalent to AATCC crock cloth results/values based on crock cloth study.

14.6 Experienced operators do not have to repeat the weighing procedure during a test session once the technique has been established.

14.7 CAUTION: It has been reported that the results for staining obtained by this method on fabrics dyed to dark shades (navy, black, etc.) that contain a combination of polyester and spandex, or their blends, may not show the full staining propensity of such fabrics in consumer use. It is, therefore, recommended that the staining results obtained by this test not be used for the acceptance testing of such fabrics.

14.8 An automated electronic grading system may be used as long as the system has been demonstrated to provide results that are equal to and provide equal or better repeatability and reproducibility than an experienced grader performing visual evaluation.

15. History

15.1 Reaffirmed and editorially revised in 2022 to add the History section.

15.2 Editorially revised 2019; revised 2018, 2013; editorially revised 2011; editorially revised and reaffirmed 2010; editorially revised 2008; revised 2005; editorially revised 2004, 2002; editorially revised and reaffirmed 2001; revised 1996, 1994; reaffirmed 1989, 1988; editorially revised 1986, 1985; reaffirmed 1983; editorially revised 1981; reaffirmed 1977,1974; revised 1972; revised 1969.

15.3 Developed in 1966 by AATCC Committee RA38. Technically equivalent to ISO 105-X16.

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Test Method for Colorfastness to Heat: Dry (Excluding Pressing)

Foreword

Several tests differing in temperature are provided; one or more of them may be used, depending on the requirements and the stability of the fibers. This test excludes hot pressing, which is addressed in AATCC TM133; however, the heating device does provide some pressure on the specimen and adjacent undyed cloth.

1. Purpose and Scope

1.1 This test method is intended for assessing the degree of color change and staining caused by the action of dry heat, excluding pressing.
1.2 It is applicable to textiles of all kinds and in all forms.

2. Principle

2.1 A specimen in contact with undyed fabrics is exposed to dry heat by close contact under controlled conditions. Color change of the test specimen is reported as a grade from 5 to 1, with 5 representing no color change and 1 representing the most color change. Staining on each undyed fabric or fiber is reported as a grade from 5 to 1, with 5 representing no staining and 1 representing the most staining.

3. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.
3.1 AATCC EP1, Evaluation Procedure for Gray Scale for Color Change (see 13.1).
3.2 AATCC EP2, Evaluation Procedure for Gray Scale for Staining (see 13.1).
3.3 AATCC EP7, Evaluation Procedure for Instrumental Assessment of the Change in Color of a Test Specimen (see 13.1).
3.4 AATCC EP8, Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale (see 13.1).
3.5 AATCC EP12, Evaluation Procedure for Instrumental Assessment of Degree of Staining (see 13.1).
3.6 AATCC TM133, Test Method for Colorfastness to Heat: Hot Pressing (see 13.1).
3.7 ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 13.2).
3.8 "The Determination of Colour Fastness to Dry-heat Pleating and Setting." *Journal of the Society of Dyers and Colourists*. March 1960. 76. 158-168.

4. Terminology

4.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

5. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Users MUST consult manufacturers for specific details such as safety data sheets and other manufacturer's recommendations. Consult and follow all applicable OSHA standards and rules.

5.1 Follow good laboratory practices. Wear safety glasses in all laboratory areas.

6. Uses and Limitations

6.1 When this test method is used for assessing color changes and staining in dyeing, printing and finishing processes, other chemicals and physical factors may influence the results.

6.2 A color change of the undyed fabric may be caused by factors other than staining from the test specimen. To determine this, pieces of the undyed fabric may be tested alone.

7. Apparatus and Materials (see 13.3)

7.1 Heating device, providing even heat transfer at controlled temperatures by close contact on both sides of the specimen (see 13.4).

7.2 Undyed fabrics, two pieces, with dimensions appropriate for the heating device used.

7.2.1 One piece shall be of the same kind of fiber as that of the textile to be tested or that predominating in the case of blends.

7.2.2 The fiber content of the second piece shall be the second component of the blend; a multifiber fabric, 15 mm (0.6 in.) bands, containing acetate, cotton, nylon, polyester, acrylic and wool; or any other fabric of particular interest as specified in the test request.

7.3 Conditioning facilities and racks with pull-out, perforated shelves for laying specimens flat (see 13.1).

7.4 Equipment and environment for evaluating color change.

7.4.1 For visual evaluation, use Gray Scale for Color Change (see 13.1). Refer to AATCC EP1 for additional materials and use of gray scale.

7.4.2 For instrumental evaluation, use a spectrophotometer as specified in AATCC EP7.

7.5 Equipment and environment for evaluating staining.

7.5.1 For visual evaluation, use Gray Scale for Staining or AATCC 9-Step Chromatic Transference Scale (see 13.1). Refer to AATCC EP1 or AATCC EP8 for additional materials and use of scales. (see 13.5)

7.5.2 For instrumental evaluation, use a spectrophotometer as specified in AATCC EP12.

8. Specimens

8.1 If the textile to be tested is fabric, place a test specimen with dimensions appropriate for the heating device between the two pieces of undyed fabric (see 7.2), and sew along one of the shorter sides to form a composite test specimen.

8.2 If the textile to be tested is yarn, knit it into fabric and treat it as in 8.1 or form a uniform layer of parallel yarn lengths between two pieces of undyed fabric (see 7.2), the amount of yarn taken being approximately equal to half the combined mass of the undyed fabrics. Sew along one side to hold the yarn in place and to form a composite test specimen.

8.3 If the textile to be tested is loose fiber, comb and compress an amount approximately equal to half the combined mass of the undyed fabrics (see 7.2) into a uniform sheet of the required size. Place the sheet between the two undyed fabrics and sew along all four sides to form a composite test specimen.

9. Procedure

9.1 Set the heating device to one of the following temperatures, based on expected end use and stability of the specimen.

I	150 ± 2°C
II	180 ± 2°C
III	210 ± 2°C

9.2 Set the device pressure to 4 ± 1 kPa.

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9.3 Place the composite test specimen in the heating device and leave it there for 30 s.

9.4 Remove the composite specimen and condition as directed in ASTM D1776. (Use conditions indicated in Table 1 for Textiles, general. Estimate conditioning time as indicated in Table 2 for the appropriate fiber content.) Lay each swatch or product, separately, on a screen or perforated shelf of a conditioning/drying rack.

10. Evaluation

10.1 Evaluate color change of the test specimen.

10.1.1 For visual evaluation, evaluate color change of the specimen as directed in AATCC EP1.

10.1.2 For instrumental evaluation, evaluate color change of the specimen as directed in AATCC EP7.

10.2 Evaluate staining on each undyed fabric and each strip of the multifiber fabric, as appropriate.

10.2.1 For visual evaluation, evaluate staining on each undyed fabric and/or multifiber fabric strip as directed in AATCC EP2 or AATCC EP8 (see 13.5).

10.2.2 For instrumental evaluation, evaluate staining on each undyed fabric and/or multifiber fabric strip as directed in AATCC EP12.

11. Report

11.1 Describe or identify the sample tested.

11.2 Report that the sample was tested using AATCC TM117-2019.

11.3 Report the testing conditions:

11.3.1 Temperature of test.

11.3.2 Heating device used.

11.3.3 Nominal fiber content of test specimen and undyed fabrics.

11.3.4 Evaluation procedure for color change (AATCC EP1 or AATCC EP7).

11.3.5 Evaluation procedure for staining (AATCC EP2, AATCC EP8, or AATCC EP12).

11.4 Report the test results:

11.4.1 Color change grade for tested specimen.

11.4.2 Staining grade for each undyed fabric and/or multifiber fabric strip.

11.5 Describe any modification(s) of the published method.

12. Precision and Bias

12.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

12.2 *Bias*. The colorfastness to dry heat (excluding pressing) can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

13. Notes

13.1 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

13.2 Available from ASTM International, 100 Barr Harbor Dr, W Conshohocken PA 19428, USA; +1.610.832. 9500; www.astm.org.

13.3 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

13.4 One of the following heating devices may be used:

(a) A pair of electrically-heated hot plates whose temperature is accurately controllable at temperatures over the specified range and which can be adjusted to provide a pressure on the specimen of 4 ± 1 kPa. To obtain the specified pressure, the total area of the composite specimen should be at a suitable relationship to that of the plates so that the four spring-loaded pins acting against the housing of the upper plate is utilized. This is accomplished by using a small-diameter rod—e.g., a small Allen wrench—to push the upper protruding end of each pin down through the plastic plate and then moving it slightly to one side so that the pin remains engaged against the underside of the plate. These spring-loaded pins, when in their normal up position, provide a counter balance to the weight of the upper plate and its housing so that the pressure between the plates is 8.8 g/cm^2. Other similar devices providing identical test conditions and results may be used.

(b) Molten metal bath, in which a holder containing the composite specimen is immersed, as described in "The Determination of Colour Fastness to Dry-heat Pleating and Setting.").

13.5 For very critical evaluations and in the cases of arbitration, ratings must be based on the geometric Gray Scale for Staining.

14. History

14.1 Revised in 2019 for clarification and to align format with AATCC style.

14.2 Revised (with title change) 2018. Editorially revised 2016. Editorially revised and reaffirmed 2013. Editorially revised 2010. Editorially revised and reaffirmed 2009. Editorially revised 2008. Editorially revised and reaffirmed 2004. Editorially revised 2002. Editorially revised 2001. Editorially revised and reaffirmed 1999. Editorially revised and reaffirmed 1994. Reaffirmed 1989. Editorially revised 1988. Editorially revised 1985. Reaffirmed 1984. Editorially revised 1983. Editorially revised 1981. Reaffirmed 1979. Reaffirmed 1976. Revised 1973. Revised 1971. Revised 1967.

14.3 Developed in 1966 by AATCC Committee RR54. Maintained by RA99.

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Test Method for Oil Repellency: Hydrocarbon Resistance

Foreword

This test method was developed to detect the presence of a fluorochemical finish by evaluating the fabric's resistance to wetting by liquid hydrocarbons of different surface tensions.

1. Purpose and Scope

1.1 This test method is used to determine the existence of finishing compounds capable of imparting a low energy surface by evaluating the fabric's resistance to wetting.

1.2 This test method is applicable to all types of fabrics.

2. Principle

2.1 Drops of standard test liquids, consisting of a selected series of hydrocarbons (oils) with varying surface tensions, are placed on the fabric surface and observed for wetting, wicking and contact angle. The oil repellency grade is reported, which is the numerical value of the highest numbered test liquid that does not wet the fabric surface. Results are reported as a grade from 8 to 0, with 8 representing the most oil repellency and 0 representing the least oil repellency. Half-grades may be assigned. A grade of zero (0) can be assigned when the fabric fails the mineral oil test liquid.

3. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

3.1 AATCC TM193, Aqueous Liquid Repellency: Water/Alcohol Solution Resistance (see 13.1)

3.2 ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 13.2).

3.3 29 CFR 1910.1000, Air Contaminants (see 13.3).

4. Terminology

4.1 **grade,** n.— the number assigned to a test specimen resulting from comparison to a scale.

4.2 **oil repellency,** n.—*in textiles*, the characteristic of a fiber, yarn or fabric whereby it resists wetting by oily liquids.

4.3 **wetting,** n.—the point at which a drop of liquid loses its light reflection characteristic, i.e., when it changes to a dull, wet spot due to the absorbent propensity of textile materials.

4.4 **wicking,** n.—*in textiles*, the movement of a liquid, by capillary action, along or through a material.

5. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

5.1 Follow good laboratory practices. Wear safety glasses and impervious gloves when handling test liquids in all laboratory areas.

5.2 The hydrocarbons specified in this method are flammable. Keep away from heat, sparks and open flame. Use with adequate ventilation. Avoid prolonged breathing of vapor or contact with skin. Do not take internally.

5.3 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities [e.g., Occupational Safety and Health Administration's (OSHA) permissible exposure limits (PEL) as found in 29 CFR 1910.1000; see www.osha.gov for latest version]. In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met.

6. Uses and Limitations

6.1 This test method is not intended to give an absolute measure of the resistance of the fabric to staining by all oily materials. Other factors, such as composition and viscosity of the oily substances, fabric construction, fiber type, dyes, other finishing agents, etc., also influence stain resistance. This test can, however, provide a rough index of oil stain resistance, in that generally the higher the oil repellency grade, the better resistance to staining by oily materials, especially liquid oil substances. This is particularly true when comparing various finishes for a given fabric.

7. Apparatus and Materials (see 13.4)

7.1 Dropping bottles (see 13.5).

7.2 Test liquids prepared and numbered according to Table I.

7.3 AATCC blotting paper (see 13.1).

7.4 Laboratory gloves (general purpose is sufficient).

8. Specimens

8.1 Test two (2) specimens of the same size from each sample (see 13.6).

8.2 Specimen size should be sufficient to allow for the complete range of test liquids to be evaluated, but shall be no smaller than 20 × 20 cm (8 × 8 in.) and no larger than 20 × 40 cm (8 × 16 in.)

9. Conditioning

9.1 Prior to testing, condition test specimens as directed in ASTM D1776. (Use conditions indicated in Table 1 for Textiles, general. Estimate conditioning time as indicated in Table 2 for the appropriate fiber content.) Lay each swatch or product, separately, on a screen or perforated shelf of a conditioning/drying rack.

10. Procedure and Evaluation

10.1 Use silicone-free equipment, benches and gloves. Use of silicone-containing products could adversely affect the oil repellency grade.

10.2 Place two pieces of blotting paper on a smooth, horizontal surface.

10.3 Place a test specimen flat over each piece of blotting paper.

10.3.1 When evaluating open weave fabrics, conduct the test on at least two layers of the fabric; otherwise, the test

Table I—Standard Test Liquids (see 13.7)

AATCC Oil Repellency Grade	Composition	Melting or Boiling Point Range	N*
0	None (does not repel mineral oil)		
1	Mineral oil	174-177°C	31.5
2	65:35 mineral oil: n-hexadecane (by volume)	N/A	N/A
3	n-hexadecane	17-18°C	27.3
4	n-tetradecane	4-6°C	26.4
5	n-dodecane	-10.5–-9.0°C	24.7
6	n-decane	173-175°C	23.5
7	n-octane	124-126°C	21.4
8	n-heptane	98-99°C	19.8

*N = dynes/cm at 25°C.

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liquid may wet the underlying surface, not the actual test fabric, and thereby cause confusion in the reading of the results.

10.4 Wearing clean laboratory gloves, brush the pile of napped or pile fabrics with your hand in the direction causing the pile to lay flat prior to placing the drops of the test liquid.

10.5 Beginning with the lowest-numbered test liquid (AATCC Oil Test Grade Liquid No. 1), *carefully* place small drops [approximately 5 mm (0.187 in.) in diameter or 0.05 mL volume] on each test specimen in five locations along the filling direction. Place drops approximately 4.0 cm (1.5 in.) apart (see Fig. 1). Hold the dropper tip approximately 0.6 cm (0.25 in.) from the fabric surface while placing drops. DO NOT TOUCH THE FABRIC WITH THE DROPPER TIP.

10.6 Observe the drops for 30 ± 2 s, from approximately a 45° angle.

10.7 If no penetration or wetting of the fabric at the liquid-fabric interface and no wicking around ~~the~~ 3 of the 5 drops occur, place drops of the next higher-numbered test liquid at an adjacent site on the fabric and again observe for 30 ± 2 s.

10.8 Continue this procedure until one of the test liquids shows obvious wetting or wicking of the fabric around 3 of the 5 drops under or around the drop within 30 ± 2 s. If wetting occurs, the test is stopped, and user should proceed to grading.

10.9 Compare drops to the examples in Fig. 2 and descriptions below. Assign the corresponding letter grade (A-D) to each drop.

Description of Fig. 2 [Grade D]—When 3 or more of the 5 drops of a test liquid applied show complete wetting.

Description of Fig. 2 [Grade C]—When 3 or more of the 5 drops of a test liquid applied show wicking with loss of contact angle.

Description of Fig. 2 [Grade B]—When 3 or more of the 5 drops of a test liquid applied show a rounded drop with partial darkening of the test specimen.

Description of Fig. 2 [Grade A]—When 3 or more of the 5 drops of a test liquid applied show a clear well rounded appearance with high contact angle.

10.10 The Oil Repellency Grade is then determined by subtracting 0.5 from the highest numbered borderline pass (3 out of 5 Grade B or higher) test liquid number. For example:

Test Liquid 5: A, A, A, B, B
Test Liquid 6: A, B, B, B, C
Test Liquid 7: B, B, B, C, C
Test Liquid 8: B, B, C, C, C

The Oil Repellency Grade for the above example would be calculated by test liquid 7 minus 0.5 equals grade 6.5.

10.10.1 Different types of wetting may be observed depending on the finish, fiber, construction, etc.; evaluation can be difficult on certain fabrics.

10.10.2 Wetting of the fabric is normally evidenced by a darkening of the fabric at the liquid-fabric interface or wicking and/or loss of contact angle of the drop.

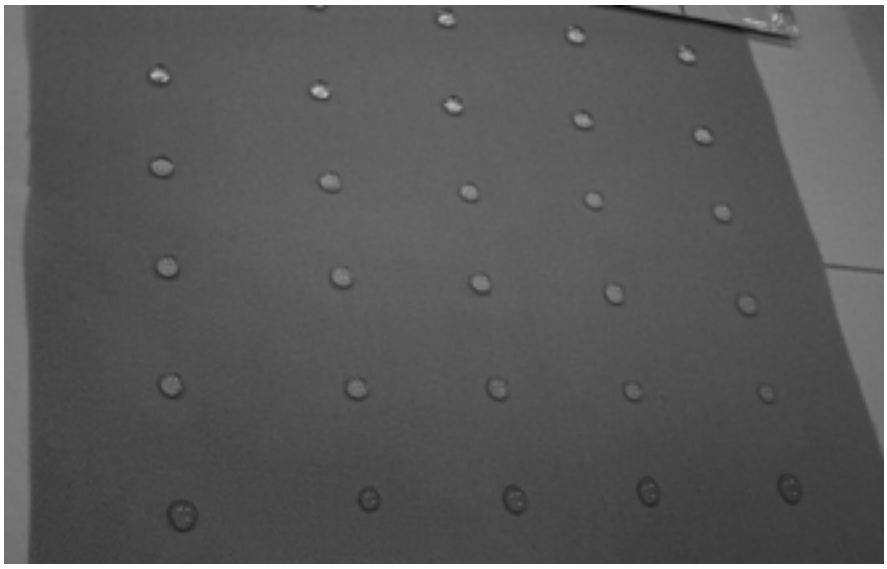
10.10.3 Many fabrics will show complete resistance to wetting by a given test liquid (Fig. 2 [A]), followed by immediate penetration by the next higher-numbered test liquid making the end point and oil repellency grade obvious. However, some fabrics will show progressive wetting under several test liquids as evidenced by a partial darkening of the fabric at the liquid-fabric interface (Fig 2 [B, C and D]. For these fabrics, the end point and oil repellency grade is the test liquid number which exhibits complete darkening of the interface or any wicking within 30 ± 2 s.

10.10.4 Wicking is the key. If the sample does not exhibit wicking but spreads (as in glass) then it is grade A. If sample exhibits wicking but still has a contact angle, then it is a grade C.

10.10.5 On black or dark fabrics, wetting can be detected by loss of "sparkle" within the drop; the aid of a small desk lamp or flashlight may be used for this detection.

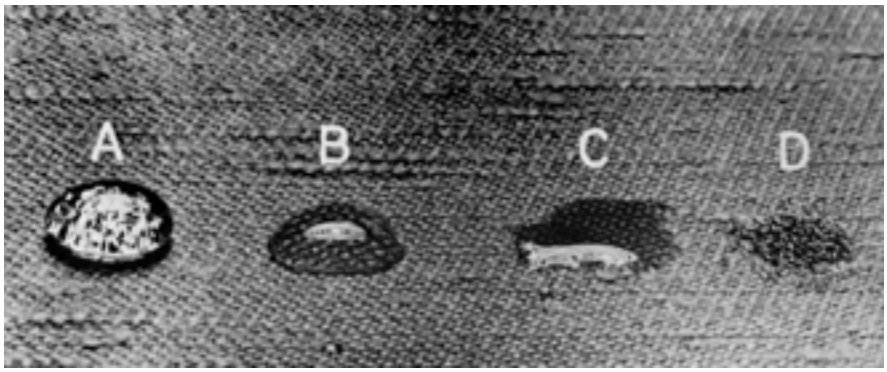
10.11 When the AATCC Oil Repellency Grade for both specimens of a fabric are the same, this is the grade for the sample.

10.12 When the AATCC Oil Repellency Grade for 2 specimens of a fabric



Bottom Row shows Test Liquid 6; Top Row shows Test Liquid 0

Fig. 1—Test liquid placement example.



A = Passes; clear well-rounded drop
B = Borderline pass; rounding drop with partial darkening
C = Fails; wicking apparent and/or complete wetting
D = Fails; complete wetting

Fig. 2—Grading example.

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are NOT the same, test a third specimen.

10.12.1 If any 2 of the 3 specimen grades are the same, this is the grade for the sample.

10.12.2 If all 3 specimen grades are different, the median is the grade for the sample. For example, if the first two specimen grades are 3 and 4, and the third specimen grade is 4.5, the sample grade is 4 (median of 3, 4, 4.5). Do not calculate the mean of specimen grades.

11. Report

11.1 Describe or identify the fabric tested.

11.2 Report that the sample was tested using AATCC Test Method 118-2019.

11.3 Report the testing conditions:

11.3.1 Report the specimen size.

11.3.2 Report the number of specimens tested.

11.4 Report the test results:

11.4.1 Report the sample AATCC Oil Repellency Grade determined.

11.5 Describe any modification(s) of the published method.

12. Precision and Bias

12.1 *Summary.* Interlaboratory tests were conducted in September 1990 and April 1991 to establish the precision of this test method. The September interlab involved 2 operators at each of 9 laboratories rating 2 specimens of each of 4 fabrics each day for 3 days. The grades of this interlab were concentrated into the 1-2 and 4-5 regions of the scale. The April interlab was conducted with fabrics responding in the 2-3 and 5-7 portions of the scale. This interlab involved 2 operators at each of 7 laboratories rating 2 specimens of each of 2 fabrics each day for 2 days. (Day interaction was shown not to be a significant factor in the analysis of the September interlab.) Results from both interlabs were combined for calculation of critical values. All materials necessary for the interlabs were provided to each laboratory by AATCC including the standard test liquids. A video of the grading procedure prepared at the AATCC Technical Center by the subcommittee and visual examples were included in the protocol. The fabrics were limited to polyester/cotton materials. The unit of measure was the median of the grades of the 2 (or 3) specimens rated each day.

12.2 The components of variance as standard deviations of the AATCC Oil Repellency Grade were calculated to be as follows:

	AATCC Oil Repellency Test
Single operator	0.27
Between operators/ within laboratories	0.30
Between laboratories	0.39

12.3 *Critical differences*. For the components of variance in 12.2, 2 observations should be considered significantly different at the 95% probability level if the difference equals or exceeds the critical differences shown in Table II.

12.4 *Bias*. The true value of the oil repellency grade can only be defined in terms of this test method. Within this limitation, this test method has no known bias.

13. Notes

13.1 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc. org.

13.2 Available from ASTM International, 100 Barr Harbor Dr, W Conshohocken PA 19428, USA; +1.610.832.9500; www.astm.org.

13.3 Available from U.S. Government Publishing Office; www.govinfo.gov.

Table II—Critical Differences[a]

No. of Observations[b]	Single Operator	Within Laboratory	Between Laboratory
1	0.75	1.12	1.55
2	0.53	0.99	1.45
3	0.43	0.94	1.42

[a] The critical differences were calculated using t - 1.950, which is based on infinite degrees of freedom.

[b] An observation is a unit of measure obtained from the median of the grades for 2 (or 3) specimens.

13.4 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

13.5 For convenience, transfer the test liquids from stock solutions to dropping bottles, each marked with the appropriate AATCC Oil Repellency Grade number. A typical system found useful consists of 60-mL dropping bottles with ground-in pipettes and Neoprene bulbs. Prior to use, the bulbs should be soaked in heptane for several hours and then rinsed in fresh heptane to remove soluble substances. It has been found helpful to place the test liquids in sequential order in a wooden platform on the evaluation table.

13.6 If AATCC TM193 and AATCC TM118 are done concurrently, use the same specimen sizes for each test.

13.7 Purity of test liquids does affect surface tension of the liquid. Use only analytical grades of test liquids. Mineral oil, for the purpose of this test, is defined as the commercial product commonly known as White Mineral Oil, USP, Food Grade or better, CAS #8042-47-5, with a Saybolt viscosity at 38°C (100°F) of 340-360, boiling point of 174-177°C (346-350°F), specific gravity of 0.840-0.890, and flash point greater than 188°C (370°F); Kaydol and Crystal Plus are two well-known trade names in the United States; other brands may be suitable. All test liquids are available from local chemical suppliers.

14. History

14.1 Editorially revised 2021 to update from CAS# 8012-95-1 (which is paraffin oil) to CAS #8042-47-5 (which is white mineral oil) to correct the reference.

14.2 Revised 2020, 2019; editorially revised 2016; revised 2013; reaffirmed 2012; editorially revised 2010, 2008; revised 2007; editorially revised 2004; reaffirmed 2002; editorially revised and reaffirmed 1997; editorially revised 1995; revised 1992; editorially revised 1990; reaffirmed 1989; editorially revised 1986, 1985; reaffirmed 1978, 1975, 1972.

14.3 Developed in 1966 by AATCC Committee RA56. Technically equivalent to ISO 14419.

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Test Method for Colorfastness to Flat Abrasion (Frosting): Screen Wire

Foreword

This test method was developed by an industry committee, and was known as the Industry Committee Test. It was first published as "A Proposed Test Method to Evaluate Frosting Potential Caused by Abrasive Wear" in 1965. This method approximates localized color change similar to that produced on some garments over relatively short periods of actual wear in which the garment is exposed to relatively mild abrasive action. To evaluate the effect of relatively severe abrasion action, refer to AATCC TM120.

1. Purpose and Scope

1.1 This test method is intended for evaluating the resistance of colored fabrics to change in shade caused by flat abrasion.

1.2 It can be used for all colored fabrics, but is especially sensitive to the color change of durable-press, cross-dyed blend fabrics, in which one fiber is abraded away faster than another.

2. Principle

2.1 The fabric specimen is mounted over a foam rubber cushion and rubbed multi-directionally against a wire screen mounted on a weighted head. After rinsing and conditioning, specimens are evaluated for color change. Color change is reported as a grade from 5 to 1, with 5 representing no color change and 1 representing the most color change.

3. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

3.1 AATCC EP1, Evaluation Procedure for Gray Scale for Color Change (see 14.1).

3.2 AATCC EP7, Evaluation Procedure for Instrumental Assessment of the Change in Color of a Test Specimen (see 14.1).

3.3 AATCC LP1, Laboratory Procedure for Home Laundering: Machine Washing (see 14.1).

3.4 AATCC TM120, Test Method for Colorfastness to Flat Abrasion (Frosting): Emery (see 14.1).

3.5 ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 14.2).

3.6 "A Proposed Test Method to Evaluate Frosting Potential Caused by Abrasive Wear," Gobeil, N.B. and D'Alessandro, P.L. *American Dyestuff Reporte*r, November 22, 1965. 54(24), 42-49 (see 14.1).

4. Terminology

4.1 **abrasion,** n.—the wearing away of any part of a material by rubbing against another surface.

4.2 **frosting,** n.—*in textiles*, a change of fabric color caused by localized abrasive wear. (Syn: *differential wear*, *fibrillation*).

NOTE: Frosting may be the result of differential wear, as in multicomponent blends in which the fibers do not match in shade, or of the abrasion of single-fiber constructions in which there is a variation in or incomplete penetration of dye.

5. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Users MUST consult manufacturers for specific details such as safety data sheets and other manufacturer's recommendations. Consult and follow all applicable OSHA standards and rules.

5.1 Follow good laboratory practices. Wear safety glasses in all laboratory areas.

5.2 Follow manufacturer's safety recommendations when operating laboratory testing equipment.

6. Apparatus and Materials (see 14.3)

6.1 Conditioning facilities and racks with pull-out, perforated shelves for laying specimens flat (see 14.1).

6.2 Stoll—Universal Wear Tester (see 9.2, 9.3 and Fig. 1).

6.3 Frosting unit (see Fig. 2).

6.3.1 Circular frosting specimen holder (see 9.4).

6.3.2 Rubber O ring.

6.3.3 Conical mounting piece.

6.3.4 Side clamps.

6.4 Abradant, stainless steel wire screening, 16 mesh, 0.23 mm wire (see 9.5).

6.5 Towels, for blotting specimens after rinsing.



Fig. 1—Universal Wear Tester prepared for frosting test.

6.6 Cloth, white cotton, for ironing.

6.7 Hand iron.

6.8 Equipment and environment for evaluating color change.

6.8.1 For visual evaluation, use Gray Scale for Color Change (see 14.1). Refer to AATCC EP1 for additional materials and use of gray scale.

6.8.2 For instrumental evaluation, use a spectrophotometer as specified in AATCC EP7.

7. Specimens

7.1 Cut a minimum of 2 specimens 127 × 127 mm from each sample (see 14.4). Take no specimen from within one-tenth of the width of the fabric from either selvage. Cut specimens with sides parallel to fabric length and width.

7.2 Mark length direction on back side of specimen.

8. Conditioning

8.1 Prior to testing, condition test specimens as directed in ASTM D1776. (Use conditions indicated in Table 1 for Textiles, general. Estimate conditioning time as indicated in Table 2 for the appropriate fiber content.) Lay each swatch or prod-

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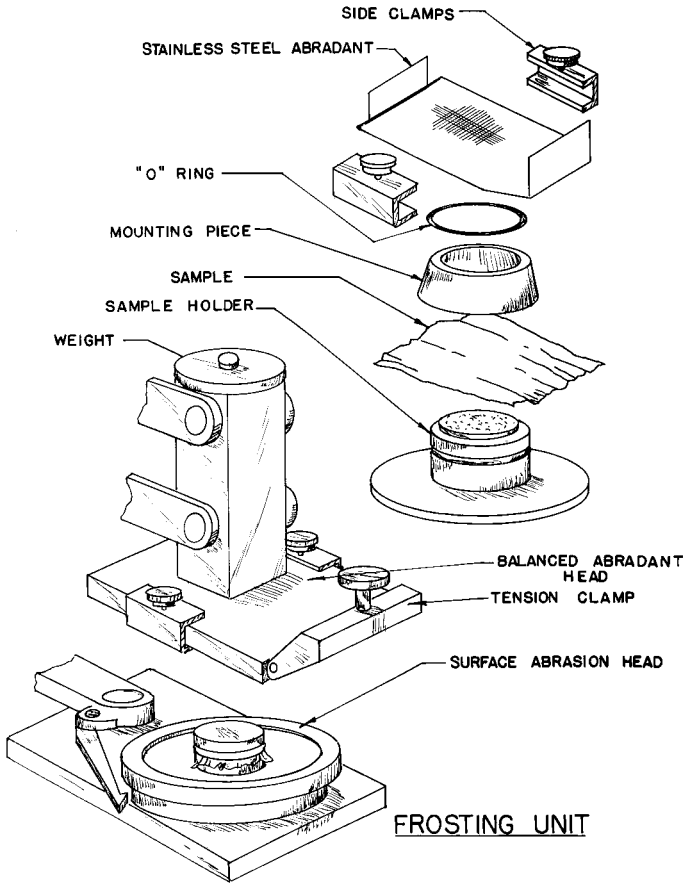


Fig. 2—Schematic drawing of frosting unit.

uct, separately, on a screen or perforated shelf of a conditioning/drying rack.

9. Preparation of Specimens and Apparatus

9.1 To obtain information pertinent to the behavior of the fabric in actual use, it may sometimes be desirable to pretreat test specimens, as by laundering, before subjecting them to the frosting test. See AATCC LP1 for standard home laundering procedures.

9.2 At regular intervals, check the operation of the frosting tester with one or more standard fabrics of known frosting properties.

9.3 Verify that instrument head will neither fall forward nor tilt backward when not loaded with weights. If the head is not balanced, the abrasion pressure between screen and fabric will be altered.

9.4 When the specimen holder with the foam rubber cushion is first received, or after it has been used extensively, the plane of the surface of the rubber head may not be in perfect alignment with the abradant plate. This may lead to poor or uneven wear pattern. Even by abrading the rubber surface (without specimen) on the tester with "0" emery cloth under a 1134-g head weight.

9.5 It is imperative that the warp of the abradant screen be parallel to the reciprocating motion of the upper head. Wash new screens to remove oil. After each test, clean the screen of detritus with a compressed air gun with a safety nozzle. Wash the screen periodically in a mild detergent to remove possible build-up of matter which cannot be blown off with an air gun. Replace worn or defective screens promptly.

10. Procedure

10.1 Conduct the test in the standard atmosphere for testing textiles: 65 ± 5% relative humidity and 21 ± 2ºC (70 ± 4ºF).

10.2 Remove the circular surface abrasion head from the reciprocating table of the abrasion tester by removing the circular locking stud in the center. Use the special two-pronged wrench supplied with the frosting unit.

10.3 Insert the circular frosting specimen holder in the surface abrasion head, place the metal O ring over specimen holder, and lock in place with screw collar.

10.4 Place the first test specimen face up and centrally over the specimen holder, and then place the conical mounting piece over the specimen. (This mounting piece holds the specimen over the foam rubber cushion and facilitates the insertion of the rubber O ring over the test specimen into the groove. The rubber O ring holds the sample firmly in place without distortion during the test.) Insert the rubber O ring over the specimen and into the groove. Remove the mounting piece.

10.5 Return the surface abrasion head (with the test specimen) to the reciprocating table.

10.6 Attach the abradant screen to the abradant head. Apply necessary tension by means of the front tension clamp. Secure sides with side clamps.

10.7 Apply head weights totaling 1134 g to the balanced abradant head.

10.8 Adjust rotary motor pawl to rotate the surface abrasion head approximately one revolution per 100 cycles.

10.9 Gently lower the head until it contacts the test specimen and push the "start" button.

10.10 Abrade for 1200 cycles and stop the machine.

10.11 Remove abrasion head with the abraded specimen and remove the test specimen.

10.12 Hand rinse the test specimen in clear lukewarm water at 38°C (100ºF) to remove detritus; blot between towels to remove excess water.

10.13 Place specimen face down between two pieces of clean white cotton cloth and press with a hand iron set at approximately 149°C (300°F) until dry.

10.14 Repeat the procedure for all remaining specimens.

10.15 Recondition specimens at as described in Section 8 before evaluating.

11. Evaluation

11.1 For visual evaluation, evaluate color change of each abraded specimen as directed in AATCC EP1, with modified viewing angles.

11.1.1 Place the specimen flat on the viewing table with the illumination overhead. The light should be sufficiently diffused so that shadows are not caused by viewing the specimen from overhead. Align the specimen with machine (warp or wale) direction in the line of sight.

11.1.2 Keep the specimen stationary while the viewer moves to adjust viewing angle. Scan the specimen at 0.2-1.57 rad (15°-90°) overhead.

11.1.3 Turn the specimen to align the cross (filling or course) direction with the line of sight. Scan at 0.2-1.57 rad (15°-90°) overhead.

11.2 For instrumental evaluation, evaluate color change of each abraded specimen as directed in AATCC EP7.

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12. Report

12.1 Describe or identify the sample tested.

12.2 Report that the sample was tested using AATCC TM119-2019.

12.3 Report the testing conditions:

12.3.1 Number of specimens.

12.3.2 Evaluation procedure for color change (AATCC EP1 or AATCC EP7).

12.4 Report the test results:

12.4.1 Individual and average color change grade for tested specimens.

12.5 Describe any modification(s) of the published method.

13. Precision and Bias

13.1 *Precision.* Interlaboratory tests were conducted in 1966 and 1967 in which 5 laboratories each made 2 observations on each of 8 fabrics. All evaluations were performed with AATCC EP1.

13.1.1 The precision of this test method is dependent on the combined variability of the material being tested, the test method itself, and the evaluation procedure utilized.

13.1.2 The components of variance, expressed as standard deviations, were calculated to be: *Within-laboratory* component, 0.3 Gray Scale units; *Between-laboratory* component, 0.5 Gray Scale units.

13.1.3 Two averages of observed values should be considered significantly different at the 95% probability level if the difference equals or exceeds the critical differences listed in Table I.

13.2 *Bias.* The colorfastness to flat abrasion can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

Table I—Critical Difference, Gray Scale Units, Using AATCC EP1, For the Conditions Noted*

n	Within-Laboratory Precision	Between-Laboratory Precision
2	0.4	1.4
4	0.3	1.4
8	0.2	1.4

*The critical differences were calculated using t = 1.960, which is based on infinite degrees of freedom.

Note: The tabulated values of the critical differences shown are based on results obtained using EP1. The critical differences for results obtained using EP7 may not be the same. Moreover, the values shown in Table I should be considered a general statement only, particularly with respect to between-laboratory precision. Before a meaningful statement can be made about two specific laboratories, the statistical bias between them, if any, must be established, with each comparison being based on data obtained on randomized specimens from one sample of the material to be tested.

14. Notes

14.1 Available from AATCC, PO Box 12215, Research Triangle Park, NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

14.2 Available from ASTM International, 100 Barr Harbor Dr, W Conshohocken PA 19428, USA; +1.610.832. 9500; www.astm.org.

14.3 For potential equipment information pertaining to this test method, please visit the online AATCC Buyer's Guide at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

14.4 For the optimum number of specimens, unless otherwise agreed upon (for instance, provided for by an applicable materials specification), test the number of test specimens required to secure a precision of the mean of the test result of ± 0.5 units on the Gray Scale at a probability level of 95%, calculated as follows:

$$n = 15.4\ \sigma^2$$

where:

n = number of test specimens and

σ = standard deviation of results for individual specimens determined from extensive past records on similar material.

If σ is not known, test 4 specimens for within-laboratory determinations; test 9 specimens for between-laboratory determinations.

The number of tests prescribed above is based on σ = 0.5 Gray Scale units for single-operator, single-laboratory results, and σ = 0.75 Gray Scale units for *between-laboratory* results, somewhat higher values of σ than will usually be found in practice. Knowledge of the correct σ is, therefore, likely to permit testing fewer than the number of specimens specified above.

15. History

15.1 Revised (with title change) in 2019 for clarification and to align format with AATCC style.

15.2 Editorially revised 2016. Editorially revised and reaffirmed 2013. Editorially revised 2010. Reaffirmed 2009. Reaffirmed 2004. Reaffirmed 1999. Editorially revised 1997. Editorially revised and reaffirmed 1994. Reaffirmed 1989. Editorially revised 1986. Editorially revised 1985. Editorially revised and reaffirmed 1984. Editorially revised 1980. Editorially revised and reaffirmed 1979. Reaffirmed 1977. Reaffirmed 1974. Reaffirmed 1970.

15.3 Developed in 1967 by AATCC Committee RR29. Maintained by RA99.

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Test Method for Colorfastness to Flat Abrasion (Frosting): Emery

Foreword

This test method was developed from the Celanese Corp's test procedure APD; EL 9C, and was known as the Celanese Test. This method approximates localized color change similar to that produced on some garments over relatively long periods of actual wear in which the garment is exposed to relatively severe abrasive action. To evaluate the effect of relatively mild abrasion action, refer to AATCC TM119.

1. Purpose and Scope

1.1 This test method is intended for evaluating the resistance of colored fabrics to changes in shade caused by flat abrasion.

1.2 It can be used for all colored fabrics, but is especially sensitive to poor dye penetration in all-cotton fabrics and to abrasion-caused changes in color of union-dyed blends.

2. Principle

2.1 The fabric specimen is mounted over a diaphragm inflated under controlled air pressure and rubbed multi-directionally against an emery abradant surface mounted on a weighted head. After rinsing and conditioning, specimens are evaluated for color change. Color change is reported as a grade from 5 to 1, with 5 representing no color change and 1 representing the most color change.

3. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

3.1 AATCC EP1, Evaluation Procedure for Gray Scale for Color Change (see 14.1).

3.2 AATCC EP7, Evaluation Procedure for Instrumental Assessment of the Change in Color of a Test Specimen (see 14.1).

3.3 AATCC LP1, Laboratory Procedure for Home Laundering: Machine Washing (see 14.1).

3.4 AATCC TM119, Test Method for Colorfastness to Flat Abrasion (Frosting): Screen Wire (see 14.1).

3.5 ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 14.2).

4. Terminology

4.1 **abrasion,** n.—the wearing away of any part of a material by rubbing against another surface.

4.2 **frosting,** n.—*in textiles*, a change of fabric color caused by localized abrasive wear. (Syn: *differential wear*, *fibrillation*).

NOTE: Frosting may be the result of differential wear, as in multicomponent blends in which the fibers do not match in shade, or of the abrasion of single-fiber constructions in which there is a variation in, or incomplete penetration of dye.

5. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Users MUST consult manufacturers for specific details such as safety data sheets and other manufacturer's recommendations. Consult and follow all applicable OSHA standards and rules.

5.1 Follow good laboratory practices. Wear safety glasses in all laboratory areas.

5.2 Follow manufacturer's safety recommendations when operating laboratory testing equipment.

6. Apparatus and Materials (see 14.3)

6.1 Conditioning facilities and racks with pull-out, perforated shelves for laying specimens flat (see 14.1).

6.2 Stoll-Universal Wear Tester

6.3 Rubber diaphragm, without electrical contact point.

6.4 Emery polishing paper, 600 grit, 3.8 × 22.8 cm (see 14.4).

6.5 Vacuum cleaner.

6.6 Towels, for blotting specimens after rinsing.

6.7 Cloth, white cotton, for ironing.

6.8 Hand iron.

6.9 Equipment and environment for evaluating color change.

6.9.1 For visual evaluation, use Gray Scale for Color Change (see 14.1). Refer to AATCC EP1 for additional materials and use of gray scale.

6.9.2 For instrumental evaluation, use a spectrophotometer as specified in AATCC EP7.

7. Specimens

7.1 Cut 2 circular specimens approximately 10.8 cm in diameter. Take no specimen within one-tenth of the width of the fabric from either selvage.

7.2 Mark length direction on back side of specimen.

8. Conditioning

8.1 Prior to testing, condition test specimens as directed in ASTM D1776. (Use conditions indicated in Table 1 for Textiles, general. Estimate conditioning time as indicated in Table 2 for the appropriate fiber content.) Lay each swatch or product, separately, on a screen or perforated shelf of a conditioning/drying rack.

9. Preparation of Specimens and Apparatus

9.1 To obtain information pertinent to the behavior of the fabric in actual use, it may sometimes be desirable to pretreat test specimens, as by laundering, before subjecting them to the frosting test. See AATCC LP1 for standard home laundering procedures.

9.2 At regular intervals, check the operation of the frosting tester with one or more standard fabrics of known frosting properties.

9.3 Verify that instrument head will neither fall forward nor tilt backward when not loaded with weights. If the head is not balanced, the abrasion pressure between screen and fabric will be altered.

10. Procedure

10.1 Conduct the test in the standard atmosphere for testing textiles: 65 ± 5% relative humidity and 21 ± 2ºC (70 ± 4ºF).

10.2 Place an unused strip of emery paper around the abradant plate and clamp at each end. Apply the minimum amount of tension on the paper that is necessary to make it lie flat against the lower surface of the abradant plate.

10.3 Place the first test specimen (face up) over the rubber diaphragm (without electrical contact) so that the fabric is free from wrinkles. Clamp the test specimen against the diaphragm with the clamping ring, taking care not to distort the fabric.

10.4 Set the air pressure on the diaphragm to 20.68 kPa and place a 1361-g load on the balanced head.

10.5 Engage the rotation mechanism of the specimen clamp to cause multidirectional abrasion on the specimen.

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10.6 Gently lower the abradant plate with the emery paper until it contacts the specimen, and push the "start" button. (The lower carriage should oscillate at a speed of approximately 120 cycles/min).

10.7 Allow the carriage to oscillate for 100 continuous cycles, and then stop the machine.

10.8 Remove the test specimen from the machine and vacuum it to remove fiber and abradant residue.

10.9 Hand rinse the test specimen in clear lukewarm water at 38ºC (100ºF) to remove remaining detritus; blot between towels to remove excess water.

10.10 Place specimen face down between two pieces of clean white cotton cloth and press with a hand iron set at approximately 149ºC (300ºF) until dry.

10.11 Repeat the procedure for the remaining specimen.

10.12 If a specimen slips in the clamp, the air pressure does not remain constant or an abnormal wear pattern is obtained, discard the specimen and repeat the test on an additional specimen.

10.13 Recondition specimens as described in Section 8 before evaluating.

11. Evaluation

11.1 For visual evaluation, evaluate color change of each abraded specimen as directed in AATCC EP1, with modified viewing angles.

11.1.1 Place the specimen flat on the viewing table with the illumination overhead. The light should be sufficiently diffused so that shadows are not caused by viewing the specimen from overhead. Align the specimen with machine (warp or wale) direction in the line of sight.

11.1.2 Keep the specimen stationary while the viewer moves to adjust viewing angle. Scan the specimen at 0.27-1.57 rad (15°-90°) overhead.

11.1.3 Turn the specimen to align the cross (filling or course) direction with the line of sight. Scan at 0.2-1.57 rad (15°-90°) overhead.

11.2 For instrumental evaluation, evaluate color change of each abraded specimen as directed in AATCC EP7.

12. Report

12.1 Describe or identify the sample tested.

12.2 Report that the sample was tested using AATCC TM120-2019.

12.3 Report the testing conditions:

12.3.1 Emery paper used.

12.3.2 Evaluation procedure for color change (AATCC EP1 or AATCC EP7).

12.4 Report the test results:

12.4.1 Individual and average color change grade for tested specimens.

12.5 Describe any modification(s) of the published method.

13. Precision and Bias

13.1 *Precision.* Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory or between-laboratory* averages.

13.2 *Bias.* The color change due to flat abrasion, frosting, emery method, can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

14. Notes

14.1 Available from AATCC, PO Box 12215, Research Triangle Park, NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

14.2 Available from ASTM International, 100 Barr Harbor Dr, W Conshohocken PA 19428, USA; +1.610.832. 9500; www.astm.org.

14.3 For potential equipment information pertaining to this test method, please visit the online AATCC Buyer's Guide at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

14.4 The method originally called for 0 grade emery paper. The current specification of 600 grit provides the same abrasive action, but is a more precise measure and should be consistent among manufacturers.

15. History

15.1 Revised (with title change) in 2019 for clarification and alignment of format with AATCC style.

15.2 Editorially revised 2016. Editorially revised and reaffirmed 2013. Editorially revised 2010. Reaffirmed 2009. Editorially revised and reaffirmed 2004. Reaffirmed 1999. Editorially revised 1997. Editorially revised and reaffirmed 1994. Reaffirmed 1989. Editorially revised 1986. Editorially revised and reaffirmed 1984. Editorially revised and reaffirmed 1980. Reaffirmed 1977. Reaffirmed 1974. Reaffirmed 1970.

15.3 Developed in 1967 by AATCC Committee RR29. Maintained by RA99.

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Test Method for Carpet Soiling: Visual Rating

Developed in 1967 by AATCC Committee RA57; reaffirmed 1970, 1973, 1976, 1979, 1982, 1989, 2000, 2005, 2010; editorially revised 1986, 1991, 2008, 2019; editorially revised and reaffirmed 1987, 2014; revised 1995.

1. Purpose and Scope

1.1 This test method covers the measurement of the degree of cleanness of pile floor coverings, in the range from clean to a middle degree of soiling. It can be used to measure the accumulation of soil, or the removal of soil by a cleaning procedure. It can be used with pile floor coverings of any color, pattern, structure or fiber content.

1.2 This test method does not evaluate structural appearance changes (see 14.1).

2. Principle

2.1 The degree of difference in cleanness between an original or clean area, and the area under examination is determined by visual matching with a stepwise series of differences in gray chips selected to form a geometrical scale of differences on the light-dark axis.

3. Terminology

3.1 **carpet,** n.—all textile floor coverings not designated as rugs.

3.2 **cleanness,** n.—*in carpet soiling tests*, the absence of change in appearance due to soil, specifically the degree to which the specimen approaches the original clean, unsoiled condition.

NOTE: Cleanness is independent of changes of physical structure which may be present because of exposure to traffic or action of cleaning procedures.

3.3 **soil,** n.—dirt, oil or other substances not normally intended to be present on a substrate such as a textile material.

3.4 **soiling,** n.—*in textiles*, a process by which a textile substrate becomes more or less uniformly covered with/or impregnated with soil.

3.5 **textile floor covering,** n.—an article having a use-surface composed of textile material and generally used for covering floors.

3.6 **use-surface,** n.—*of textile floor covering*, the part of a textile floor covering directly exposed to foot traffic.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5. Uses and Limitations

5.1 The complex surface structure of pile floor coverings, the depth of the pile and its compressibility make it difficult to specify a method of measurement of reflectance appropriate for all classes of pile floor covering or for comparison between classes. The interpretation of differences in reflectance, particularly for materials of different color or original reflectance, is also uncertain. The present method avoids both sets of difficulties by obtaining comparison of one difference, that between clean and soiled, with a standardized scale of difference in reflectance, by visual recognition of equivalent magnitudes of difference.

6. Apparatus

6.1 Gray Scale for Color Change (AATCC EP1) (see 14.2).

7. Verification of Gray Scales

7.1 Visual verification. Inspection of the Gray Scales should show a recognizable, approximately-equal, apparent difference between each successive pair. If such visual stepwise progression is apparent, the scale is satisfactory. The Gray Scale must be clean for proper evaluation.

7.2 Reflectance measurement. If there is doubt about the visual test, it may be checked by reflectance measurement. Use any reflectance measuring equipment appropriate to chip size (10 × 38 mm). The measured reflectance for each of the chips of Grade 5 and each of the reference chips of all the other pairs should differ from their average by less than one-half the difference between the pair in Step 4 for a nine-step scale which includes half steps. The difference between reflectances of the chips in a pair, when plotted on semilog paper against the grade step or half step, should form a straight line. The difference between chips at the 4-5 step may be in line with the others since this difference is on the order of the variation in the Grade 5 chips. If the other steps form a regular series, the 4-5 step is less critical.

8. Preparation of Specimens

8.1 Clean reference specimen. Use a specimen of original or clean carpet large enough to show any pattern of structure of color in a representative way. If there are light and dark colors, be sure to locate the light colors at an edge. Cut a sharp edge on the reference specimen.

8.2 Leave the soiled floor covering, or the floor covering after cleaning, intact on the floor, or cut out a specimen convenient for handling. If a specimen is cut, use a portion which is representative of the exposure to soil or to the cleaning process, as well as representative of the pattern, and which includes the lighter colors.

9. Conditioning

9.1 Use the prevailing room conditions. Be sure that the floor covering is dry, within normal regain, if on-location cleaning methods have been used.

10. Procedure

10.1 Place the clean reference specimen on the specimen to be examined, or beside it, with no gap between the specimens. Orient the two specimens in the same way with respect to the structure and pattern. Use a standard lighting system which may include a daylight and artificial light source. Any other lighting should be standardized as much as possible for visual testing.

10.2 Compare the pairs of chips in the Gray Scale with the pair of specimens, until the nearest corresponding pair of chips has been found. Use the dark shield to expose only one pair of chips at a time.

10.3 Record the number of the step or half step which most nearly corresponds in difference to the difference in cleanness between the specimens. Keep in mind that it is degree of difference which is being matched; do not attempt to match gray shades in the less clean specimen on the Gray Scale.

10.4 Repeat the ratings with at least

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four observers. For greater precision, use five or six observers. Arrange that no observer knows the ratings given by other observers. If observers who do not have previous experience with this method are used, give practice by going through a set of three or more comparisons, including the full range of differences which are to be compared, without recording the ratings, before doing the series of ratings which is recorded.

11. Calculation of Results

11.1 Average the ratings given by the four or more observers, to the nearest 0.1 scale unit.

12. Report

12.1 Report the average grade of cleanness, and the number of observers, on the scale from 5, no difference from clean standard, to 1, the largest difference in the reference scale.

13. Precision and Bias

13.1 The variance (the square of the standard deviation) of ratings given the same specimen by different observers ranges from 0.2 for experienced observers to 0.5 for learners. Values of 0.3-0.4 are likely. From this, four to six observers are required to ensure that a difference of 0.5 scale units will be reported by another similar group of observers at the 95 or 90% confidence level.

13.2 Information on differences between laboratories is limited but variations from 0.2-0.5 scale units have been encountered.

13.3 The bias of the test method can be considered in terms of its definition as a means of measuring magnitude of subjective appreciation of a difference in cleanness by means of a geometric scale of physical differences. The results obtained are by definition subjectively accepted as valid for different structures, patterns and colors, and show general correlation with reflectance measurements on any one type of floor covering, if the original reflectance is high. For darker colors, the visual rating is independent of change of reflectance.

14. Notes

14.1 Procedures for the determination of change of the structural aspects of appearance caused by traffic are covered in ASTM D7330, Standard Test Method for Assessment of Surface Appearance Change in Pile Floor Coverings using Standard Reference Scales, available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832. 9500; fax: +1.610.832.9555; web site: www. astm.org.

14.2 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

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Test Method for Carpet Soiling: Service Soiling

Developed in 1967 by AATCC Committee RA57; reaffirmed 1970, 1973, 1976, 1979, 1982, 1989, 2000, 2013; editorially revised 1985, 1990; editorially revised and reaffirmed 1987, 1995; revised 2009, 2019.

1. Purpose and Scope

1.1 This test method evaluates the service soiling of textile floor coverings.

2. Principle

2.1 Specimens of carpets and selected control samples are exposed to normal foot traffic in a controlled test area. The test specimens and controls are removed and rated at predetermined intervals corresponding to different degrees of soiling or exposure to soiling.

3. Terminology

3.1 **carpet,** n.—all textile floor coverings not designated as rugs.

3.2 **soil,** n.—dirt, oil or other substances not normally intended to be present on a substrate such as a textile material.

3.3 **soiling,** n.—*in textiles*, a process by which a textile substrate becomes more or less uniformly covered with/or impregnated with soil.

3.4 **textile floor covering,** n.—an article having a use-surface composed of textile material and generally used for covering floors.

3.5 **use-surface,** n.—*of textile floor covering*, the part of a textile floor covering directly exposed to foot traffic.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5. Soiling Location

5.1 The soiling location shall be sufficiently removed from any street or other outside area so that the specimen will not be subjected to wet traffic. A distance of at least 15 m (50 ft) is suggested.

5.2 It shall be sufficiently isolated from greasy, oily or other nontypical soiling which might be present in factory areas and the like.

5.3 The traffic pattern in the soiling area shall be such that all specimens are subjected to all of the traffic as in the case of a narrow aisle or passageway with access only at either end or, alternately, if the traffic cannot be so confined or directed, it shall be randomized to such a degree as to produce the effect of uniform soiling.

5.4 In the case of the long narrow arrangement with traffic only in two opposite directions, a 2 m (6.5 ft) length of buffer carpet shall be installed at both ends of the test area. For other arrangements, or where there is a random traffic pattern, the buffer carpet shall be installed around the perimeter of the test specimens and may extend for a proportionately shorter distance but in no case for less than 2 m (6.5 ft) in any direction. The buffer carpet may be of any available fiber or construction but should be light in color in order to avoid errors due to the transfer of fiber or dyestuff to the test specimens.

6. Carpet Specimens

6.1 Specimen size. The minimum size of the test specimens shall be 30×30 cm (12×12 in.).

6.2 Specimen mounting. The carpet specimens shall be mounted or installed in such a way that the specimens can be removed or transported easily. Examples include the use of double sided tape and the mounting of the specimens on 12 mm (0.5 in.) plywood with staples.

7. Arrangement and Rotation of Specimens

7.1 The requirements are related to the dimensions and traffic pattern of the particular soiling area. Therefore, it is not possible to propose a procedure which will be universally applicable. Specimen rotation is necessary to achieve uniform soiling.

7.2 To find a suitable rotation for each particular soiling area, select a material with a known level of soiling (the control sample for example). Install the correct number of specimens to the soiling area such that all spaces are filled. Soil the known material to the level desired, rotating samples such that consistent soiling is obtained on all specimens. Once the rotation process has been defined in this manner, document the rotation sequence for future use.

8. Soiling Levels

8.1 Control Carpet Soiling Level.

8.1.1 The standard levels of soiling must be predetermined prior to conducting the service soiling test. For general use of this test the soiling intervals may correspond to those required to obtain soiling levels of an arbitrarily selected carpet (control sample) exposed to give a light, medium and high degree of soil (see 13.1). Each laboratory should select and prepare control soiled samples consistent with the needs of the laboratory. If two or more laboratories are involved, identical control samples should be available for each laboratory.

8.1.2 Test and control specimens are removed from the test pattern at specified intervals corresponding to different degrees of soiling or exposure to soiling.

8.1.3 Remove all the specimens from the first group whenever the control test specimen matches the soiled control sample (first soiling level). This is done regardless of the appearance of the other specimens at that time.

8.1.4 Remove the other specimens and control samples in a similar manner at the second and third soiling levels, respectively.

8.1.5 As each soiling level is reached, replace spaces left vacant by the removal of test and control specimens. Use portions of the buffer carpet for this purpose.

8.2 Soiling Level Based on Traffic.

8.2.1 An alternative to the above method is to use a specific amount of traffic to all the test specimens for every test. The amount of traffic the test specimens would be exposed to would be established prior to testing by the laboratory. After that specific level of traffic was attained, the samples would be removed from the area.

9. Evaluation

9.1 After the samples are removed, compare the control sample with the con-

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trol soiled sample daily as soon as possible after vacuuming. Once the decision is made to remove a set of specimens, vacuum the set immediately before they are removed from the test area. Soiling continues with the specimens being advanced one position each day.

9.2 Records of the traffic count and the duration of the test may be maintained.

10. Maintenance

10.1 Clean the test specimens daily using an upright motor-driven brush-type of vacuum cleaner.

10.1.1 A proper cleaning shall consist of four passes of the cleaner over a particular area of the carpet. The first pass when starting shall be away from the operator, the second back toward the operator. The last stroke shall be back toward the operator. On the next adjacent section of the specimen to be cleaned, the first stroke shall be away from the operator. Overlaps due to the width of the specimen not being an even multiple of the cleaner shall be ignored.

11. Evaluation

11.1 Visual evaluation can be used to determine differences in degrees of soiling (see AATCC TM121, Test Method for Carpet Soiling: Visual Rating).

12. Precision and Bias

12.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

12.2 *Bias*. Carpet soiling (service soiling method) can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

13. Notes

13.1 In the work of Committee RA57, the soiling levels were defined by nylon cut-pile carpet samples soiled to three different levels (light, medium, heavy) (see Appendix A for Synthetic Soil Preparation).

Appendix A
Synthetic Soil Preparation

A1. Synthetic Soil Formulation:

Ingredient	*% by Weight*
Peat moss (dark)	38
Portland cement	17
Kaolin clay	17
Silica (200 mesh)	17
Carbon black (Lamp or furnace black)	1.75
Black iron oxide	0.50
Mineral oil (medicinal grade)	8.75

A1.1 Peat moss should be dry and free of lumps.

A1.2 Portland cement must be dry. If it contains lumps, discard and use fresh supply.

A1.3 Black iron oxide was incorrectly listed as “red iron oxide” in previous versions of this test method.

A1.4 Mix all dry ingredients together thoroughly before adding mineral oil. Dry mix at 50°C (122°F) for 6-8 h.

A1.5 Place dry mixture in ball mill with ceramic balls. Turn mill on and run for approximately 24 h.

A1.6 Store mixture in vapor tight containers with a desiccant.

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Test Method for Smoothness Appearance of Fabrics after Home Laundering

Foreword

This test method and its accompanying three-dimensional smoothness appearance replica set were developed for evaluation of woven fabrics with a durable-press finish. It is common industry practice to use the method and scales for evaluation of other textile materials although some specimens may have different appearance characteristics due to different fabric constructions.

Standard laundering procedures remain consistent to allow valid comparison of results. Standard procedures represent, but may not exactly replicate, current consumer practices, which vary over time and among households. Alternate laundering procedures and machine parameters may be found in AATCC LP1, Laboratory Procedure for Home Laundering: Machine Washing, (see 12.3) and ISO 6330, Textiles — Domestic washing and drying procedures for textile testing (see 12.8).

1. Purpose and Scope

1.1 This test method is intended to determine the smoothness appearance of fabrics after being subjected to home laundering procedures. Several washing and drying procedures provide standard parameters to represent common home care options.

1.2 Any washable fabric (woven, knitted or nonwoven) may be evaluated for smoothness appearance using this method.

2. Principle

2.1 Fabric specimens are subjected to standard home laundering practices. Appearance of specimens is evaluated in a standard viewing area by visually comparing the specimen smoothness after laundering with grades represented by AATCC Three-Dimensional Smoothness Replicas. Results are reported as a smoothness appearance (SA) grade of 1 to 5, with 1 representing the least smoothness and 5 representing the most smoothness (no wrinkles).

3. Terminology

3.1 **ballast,** n.—*in procedures for processing or testing of textiles*, material that is used to bring the total weight or volume of the textiles to an amount specified in the procedure.

3.2 **durable press,** adj.—having the ability to retain substantially the initial shape, flat seams, pressed-in creases and unwrinkled appearance during use and after laundering or drycleaning.

3.3 **grade**, n.—the number assigned to a test specimen resulting from comparison to a scale.

3.4 **laundering,** n.—*of textile materials*, a process intended to remove soils and/or stains by treatment (washing) with an aqueous detergent solution and normally including rinsing, extracting and drying.

3.5 **laundering creases,** n.—sharp folds or lines running in any direction in a washed or dried specimen.

NOTE: Laundering creases are an unintended result of restricted movement of specimens in a washer or the dryer.

3.6 **smoothness appearance,** n.—*in fabrics*, the visual impression of planarity of a specimen quantified by comparison with a set of reference standards.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Detergent may cause irritation. Care should be taken to prevent exposure to skin and eyes.

4.3 All chemicals should be handled with care.

4.4 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

5. Uses and Limitations

5.1 This test method is designed to be used only for evaluating the appearance of washable fabrics after home laundering.

5.2 In general, it is preferable to conduct the test under relatively severe laundering conditions. It is recognized that special cycles or features are available on current washing machines and dryers to achieve improved performance on certain items; e.g., gentle cycles with reduced agitation to protect delicately constructed items, and durable press cycles, with cool-down or cold rinses and reduced spin speeds, to minimize wrinkling.

5.3 Prints and patterns may mask the mussiness present in fabrics. The evaluation process is, however, based on the visual appearance of specimens including such effects.

5.4 The Smoothness Appearance Replicas were cast from woven fabrics. It is understood that the replicas do not duplicate other fabric possibilities (knits, nonwovens). The replicas are to be used as guides that represent various levels of fabric smoothness.

5.5 Small specimen sizes will occasionally result in wrinkles or creases. These may not be characteristic of actual performance, in use.5.6 The interlaboratory reproducibility of the results of this test method depends upon mutual agreement by users of the method on the washing and drying conditions as outlined in Tables I-IV.

6. Apparatus and Materials (12.1)

6.1 Steam or dry iron with appropriate fabric temperature settings.

6.2 Standard washing machine (see Table I, 12.2), for machine washing.

6.3 Wash tub, 9.5-L, for hand washing.

6.4 White towels, large enough to accommodate test specimen, any weight, for hand washing.

6.5 Standard tumble dryer (see Table IV, 12.2) or facilities for drip, screen or line drying.

6.6 Conditioning facilities and conditioning/drying racks with pull-out, perforated shelves (see 12.3).

6.7 Scale with at least 5-kg or 10-lb capacity.

6.8 AATCC 1993 Standard Reference Detergent (powder with brightener, see 12.3).

6.9 Laundering ballast, Type 1 or Type 3 (see Table V).

6.10 Evaluation area, as described in Appendix A.

6.11 AATCC Three-Dimensional Smoothness Appearance Replicas, set of six (see Fig. 1 and 12.3).

6.12 Digital imaging system, for specimen evaluation (optional).

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Fig. 1—AATCC three-dimensional smoothness appearance replicas.

7. Test Specimens

7.1 Cut three 380 × 380 mm (15 × 15 in) specimens, sides parallel to the length and width fabric directions. Where possible, each specimen should contain different groups of lengthwise and widthwise yarns. The specimens should be marked to indicate the lengthwise direction.

7.1.1 If fraying is expected in laundering, see 12.4.

7.1.2 If the fabric is wrinkled, it may be smoothed by appropriate ironing prior to laundering. See Table I, Safe Ironing Temperature Guide, in AATCC TM133, Test Method for Colorfastness to Heat: Hot Pressing (see 12.3).

8. Laundering Procedure

8.1 Machine Washing.

8.1.1 Select washing conditions for testing from Table I. Normal cycle is recommended. Set washing machine controls to generate the selected cycle parameters.

8.1.2 Load for laundering shall consist of all specimens, plus enough laundering ballast pieces to make a total load weight of 1.8 ± 0.1 kg (4.0 ± 0.2 lb). For very critical evaluation and in arbitration, limit the number of specimens per washer load to those from one sample.

8.1.3 Begin selected wash cycle. Allow machine to fill to specified water level.

8.1.4 Add 66 ± 1 g of AATCC 1993 Standard Reference Detergent to washing machine as directed by washing machine manufacturer's instructions. If detergent is added directly to wash water, agitate briefly to dissolve completely. Stop agitation before adding wash load.

8.1.5 Add wash load (test specimens and ballast), distributing evenly around center agitator. Restart wash cycle.

8.1.6 For specimens to be drip dried (drying procedure C), stop wash cycle before the water begins to drain from the final rinse cycle. Remove specimens soaking wet. For specimens to be tumble (A), line (B), or screen/flat (D) dried, allow washing to proceed through the final spin cycle.

8.1.7 After each washing cycle, separate tangled specimens and ballast pieces, taking care to minimize distortion.

8.1.8 Washer creases. Specimens may be in a folded or creased state after washing. Such folds or creases should be removed by hand prior to drying.

8.1.9 Proceed to appropriate drying procedure.

8.2 Hand Washing.

8.2.1 Select washing temperature from Table II. Add 7.6 ± 1.9 L (2.0 ± 0.5 gal) of water at this temperature to the wash tub.

8.2.2 Add 20 ± 1 g of AATCC 1993 Standard Reference Detergent to wash tub.

8.2.3 Agitate by hand to dissolve detergent.

8.2.4 Add specimen(s) to water and gently squeeze to distribute detergent solution. Do not twist or wring.

8.2.5 Let specimen(s) soak for 2 min.

8.2.6 Gently squeeze specimen(s) in detergent solution for 1 min. Do not twist or wring.

8.2.7 Repeat 2-min. soak and 1-min squeeze in detergent solution.

8.2.8 Remove specimen(s) from wash tub and gently squeeze to remove excess detergent solution. Do not twist or wring.

8.2.9 Place specimen(s) on clean white towel. Empty and rinse wash tub.

8.2.10 Add 7.6 ± 1.9 L (2.0 ± 0.5 gal) of clean water at the specified rinse temperature (see Table II) to the wash tub.

8.2.11 Place washed specimen(s) from towel in rinse water and gently squeeze to distribute. Do not twist or wring.

8.2.12 Let specimen(s) soak for 2 min.

8.2.13 Gently squeeze specimen(s) in rinse water for 1 min. Do not twist or wring.

8.2.14 Repeat 2-min. soak and 1-min squeeze in rinse water.

8.2.15 Remove specimen(s) from wash tub and gently squeeze to remove excess water. Do not twist or wring.

8.2.16 Using clean white towels, blot water from washed specimen(s). Do not wring or twist.

8.2.17 Proceed to appropriate drying procedure.

8.3 Drying.

8.3.1 Select drying conditions from Table III.

8.3.2 (A) Tumble Dry. Place the washed load (test specimens and ballast) in the tumble dryer, and set the temperature control to generate the selected cycle exhaust temperature (see Table IV). Allow the dryer to run until the load is fully dry. Remove specimens immediately.

8.3.3 (B) Line/Hang Dry. Hang each specimen by two corners, with the fabric

Table II—Standard Hand Washing and Rinsing Temperatures

Designation	Wash Temp, °C (°F)	Rinse Temp, °C (°F)
Very Cold	16 ± 3 (60 ± 5)	< 18 (< 65)
Cold	27 ± 3 (80 ± 5)	< 29 (< 85)
Warm	41 ± 3 (105 ± 5)	< 29 (< 85)
Hot[1]	49 ± 3 (120 ± 5)	< 29 (< 85)

[1] Hot water may not be a reasonable selection for hand washing or for the delicate items typically requiring hand washing.

Table III—Standard Drying Conditions

(A)	Tumble Dry
(Ai)	Tumble Dry Normal
(Aii	Tumble Dry Delicate
(Aiii)	Tumble Dry Permanent Press
(B)	Line/Hang Dry
(C)	Drip Dry
(D)	Screen Dry/Dry Flat

Table I—Standard Washing Machine Parameters (see 12.2, 12.7)

Cycle	(1) Normal		(2) Delicate		(3) Permanent Press	
Water Level, L (gal)	72 ± 4 (19 ± 1)		72 ± 4 (19 ± 1)		72 ± 4 (19 ± 1)	
Agitation Speed, strokes/min.	86 ± 2		27 ± 2		86 ± 2	
Washing Time, min.	16 ± 1		8.5 ± 1		12 ± 1	
Final Spin Speed, rpm	660 ± 15		500 ± 15		500 ± 15	
Final Spin Time, min.	5 ± 1		5 ± 1		5 ± 1	
Wash Temp, °C (°F)[1]	(II) Cold:	27 ± 3 (80 ± 5)	(II) Cold:	27 ± 3 (80 ± 5)	(II) Cold:	27 ± 3 (80 ± 5)
	(III) Warm:	41 ± 3 (105 ± 5)	(III) Warm:	41 ± 3 (105 ± 5)	(III) Warm:	41 ± 3 (105 ± 5)
	(IV) Hot:	49 ± 3 (120 ± 5)	(IV) Hot:	49 ± 3 (120 ± 5)	(IV) Hot:	49 ± 3 (120 ± 5)
	(V) Very Hot:	60 ± 3 (140 ± 5)	(V) Very Hot:	60 ± 3 (140 ± 5)	(V) Very Hot:	60 ± 3 (140 ± 5)

[1] Due to US Department of Energy requirements, many washing machines use cooler water. An external control box may be used to override the machine set temperatures.

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Table IV—Standard Tumble Dryer Parameters (see 12.2)

Cycle	(Ai) Normal	(Aii) Delicate	(Aiii) Permanent Press
Max. Exhaust Temp, °C (°F)	68 ± 6 (155 ± 10)	60 ± 6 (140 ± 10)	68 ± 6 (155 ± 10)
Cool Down Time, min.	≤ 10	≤ 10	≤ 10

Table V—Laundering Ballast Parameters

	Type 1	Type 3
Fiber Content	100% cotton	50% cotton/50% polyester ± 3%
Greige Fabric Yarns	16/1 ring spun	16/1 or 30/2 ring spun
Greige Fabric Construction	52 × 48 ± 5 yarns/in., plain weave	52 × 48 ± 5 yarns/in., plain weave
Finished Fabric Weight	155 ± 10 g/m^2	155 ± 10 g/m^2
Edges	All edges hemmed or over-edged	All edges hemmed or over-edged
Finished Piece Size	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)
Finished Piece Weight	130 ± 10 g	130 ± 10 g

length in the vertical direction. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

8.3.4 (C) Drip Dry. Hang each dripping wet specimen by two corners, with the fabric length in the vertical direction. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

8.3.5 (D) Screen/Flat Dry. Spread each specimen on a horizontal screen or perforated surface, removing wrinkles without distorting or stretching it. Allow specimens to rest in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

8.3.6 For all drying methods, allow specimens to dry completely before washing again.

8.3.7 Laundering creases. If specimens are folded or creased after any drying cycle except the last, rewet and attempt to remove the creases by ironing with a hand iron at a temperature suitable for the fabric being tested (see 7.1.2), prior to additional laundering cycles. Do not attempt to remove wrinkles or creases by hand ironing after the final drying cycle.

8.3.8 Repeat the selected washing and drying procedures for a total of 5 complete cycles, or an agreed upon number of cycles.

8.4 Conditioning.

8.4.1 After the final drying cycle is complete, condition test specimens as directed in ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 12.5). (Use conditions indicated in Table 1 for Textiles, general. Estimate conditioning time as indicated in Table 2 for the appropriate fiber content.) Lay each test specimen flat, separately, on a screen or perforated shelf of a conditioning/drying rack.

9. Evaluation

9.1 Three trained observers should evaluate each test specimen independently.

9.2 Make all evaluations in the specified viewing area (see Appendix A). Illuminate the viewing area with the overhead fluorescent light only. Turn all other lights in the room off.

9.3 Each observer is to stand directly in front of the specimen 1219 ± 25 mm (48 ± 1 in.) away from the board. It has been found that normal variations in the height of the observer above and below the arbitrary 1524-mm (60-in,) eye level have no significant effect on the grade given.

9.4 Mount the test specimen on the viewing board with the fabric length in the vertical direction. Place the most similar Three-Dimensional Smoothness Appearance Replicas on each side of the test specimen to facilitate comparative evaluation.

9.5 The Smoothness Appearance Replicas were cast from woven fabrics. It is understood that the replicas do not duplicate all possible fabric surfaces. The replicas are to be used as guides that represent various levels of fabric smoothness or freedom from wrinkles. The observer should mentally integrate degree and frequency of wrinkles in the specimen to determine a level of smoothness that can be identified with the SA replica number which most nearly represents that smoothness appearance level (see Table VI).

9.6 The smoothness appearance (SA) grade is that of the Smoothness Appearance Replica that most nearly matches the smoothness appearance of the test specimen. Observers may assign a grade midway between those whole-number replicas that have no half-number replica separating them (SA-1.5, SA-2.5, SA-4.5) if the appearance of the test specimen warrants it.

9.6.1 If laundering creases are present on any specimens to be evaluated, take care in rating the specimens. Some laundering creases can be disregarded (commonly called "reading out"). When the grade of a laundering-creased specimen differs from the other specimens by more than one grade, the test should be repeated with new specimens, taking all precautions to avoid the occurrence of laundering creases.

9.7 A digital imaging system may be used in place of the visual replicas if it has been established by the user that its accuracy is equivalent.

9.8 Average the nine observations made on each test fabric (three grades on each of three test specimens). Report the average to the nearest tenth of a grade. This average is the unit of measure of this test method.

10. Report

10.1 Report for each sample tested:

10.1.1 A description or identification of the sample.

10.1.2 Evaluation was performed using AATCC TM124-2018t.

10.1.3 Number of specimens tested.

10.1.4 Type of ballast used, i.e., Type 1 or Type 3).

10.1.5 Number of laundering cycles (default is 5 cycles).

10.1.6 Laundering conditions, including washing machine cycle, wash temperature, drying procedure and tumble drying temperature, if applicable. Alphanumeric designations may be used if they are clearly understood by all parties. For example, 1-IV-A(ii) indicates normal wash cycle at 49°C and Delicate tumble dry cycle.

10.1.7 Number of observers.

10.18 Any modification of the test method.

10.1.9 Average smoothness appearance (SA) grade.

11. Precision and Bias

11.1 *Interlaboratory tests*. Tests were conducted in 1980 using Machine Wash Normal, 41°C, and Tumble Dry, Normal/

Table VI—Fabric Smoothness Grades by SA Replica Equivalents

Grade	Description
SA-5	Equivalent to the SA-5 Replica. Very smooth, pressed, finished appearance.
SA-4	Equivalent to the SA-4 Replica. Smooth, finished appearance.
SA-3.5	Equivalent to the SA-3.5 Replica. Fairly smooth but nonpressed appearance.
SA-3	Equivalent to the SA-3 Replica. Mussed, nonpressed appearance.
SA-2	Equivalent to the SA-2 Replica. Rumpled, obviously wrinkled appearance.
SA-1	Equivalent to the SA-1 Replica. Crumpled, creased and severely wrinkled appearance.

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Cotton Sturdy laundering settings. Eight laboratories evaluated four fabrics for smoothness appearance grades. The analysis of variance technique was judged not to be applicable to this data set because its distribution was not normal, and because of the limited and discontinuous scale of replica grades. The data were analyzed by calculating expected laboratory test results from the distribution of individual specimen grades. This analysis has been deposited for reference in the RA61 committee files.

11.2 *Observer repeatability.* From the data it was determined that single observers rated three specimens on the following frequency:

3 specimens to same replica 0.55
2 specimens to same replica and one different 0.40
3 specimens different 0.05

Only rarely did the separation in specimen grades exceed the next replica step. This is indicative of the high degree of repeatability in observer rating of smoothness appearance.

11.3 *Laboratory test result distribution (within-laboratory repeatability)*. From the observed grade distribution, a distribution of laboratory test results was calculated for each replica level with half grades included. Precision over the whole SA replica range was improved.

11.4 *Precision*. From the frequency distribution of laboratory test results, a calculation was made of the critical difference, D, between two laboratory test results. With laboratories at the same level:

Critical Difference	Confidence Level
$D > 0.17$	$P \geq 0.95$
$D \geq 0.25$	$P \geq 0.99$

When two or more laboratories wish to compare test results, it is recommended that laboratory level be established between them prior to commencing test comparisons. Fabrics of known history and performance may be used for this purpose.

Differences between laboratory test results (on the same fabric, under the same washing and drying conditions) equal to or greater than a quarter replica unit are statistically significant at $P \geq 0.99$. A difference of this magnitude or greater suggests a difference in laboratory levels and indicates the need for laboratory level comparisons.

11.5 *Bias*. The true value of smoothness appearance in durable press fabrics after repeated home launderings can be defined only in terms of a test method. There is no independent method for determining the true value. As an estimate of this property, this test method has no known bias.

12. Notes

12.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

12.2 For model number(s) of washer(s) and tumble dryers(s) reported to meet the standard parameters, visit www.aatcc.org/test/washers or contact AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org. An alternate load size was described in previous versions of this method (3.6-kg load, 83-L water level, 80 g AATCC 1993 Standard Reference Detergent), but no machine is reported to meet the standard parameters for this load size. In addition, dimensional change results obtained with the alternate load size may not be equal to those obtained with a standard load size.

12.3 Materials or drawings available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

12.4 If excessive fraying occurs in laundering, specimen edges should be pinked, slashed or stitched. If edges of laundered specimens appear distorted, clip as necessary before evaluating.

12.5 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610. 832.9500; fax: +1.610.832.9555; web site: www.astm.org.

12.6 The use of two 96-in. lamps for viewing laundered specimens is specified in this method. It is recognized, however, that physical space limitations in certain laboratories will prevent the use of 96-in. lamps. In those situations, two 48-in. lamps and a narrower viewing board may be used, but replicas identified as SA-4, SA-3 and SA-1 should always be placed on the left side of the viewing board as the board is viewed from the front. Replicas identified as SA-5, SA-3.5 and SA-2 should always be placed on the viewing board to the right side as the board is viewed from the front.

12.7 The laundering temperatures and other parameters listed in this test method are standard conditions for testing purposes. As with most lab procedures, they represent, but may not exactly replicate, current consumer practices. Consumer practices vary over time and among households; lab practices must be consistent to allow valid comparison of results. If laundering equipment or conditions other than those specifically listed in this test method are used, they must be described in detail and noted as a modification of the standard method. Alternate laundering conditions are outlined in AATCC LP1, and ISO 6330.

12.8 Available from ISO, www.iso.org.

Appendix A
Evaluation Area

A1 Viewing Board.

A1.1 Plywood board, 1829 mm (72 in.) high × 6 mm (0.25 in.) thick × wide enough to accommodate scale and specimen side by side

A1.2 Painted to match the Grade 2 chip on the Gray Scale for Staining (see 12.3). Approximate CIELAB values for the chip are L* = 77, a* = 0, b* = 0. A tolerance of two units for each parameter is acceptable for the board color in this test method.

A1.3 Spring-loaded swatch mount or other means to secure specimen and scale such that center of specimen is 1524 mm (60 in.) above the floor. Mount may be fabricated using light sheet metal (22 ga.)

A2 Lighting.

A2.1 Overhead fluorescent tube light fixture (see 12.6).

A2.1.1 Two parallel F96 T12 cool white lamps (without baffle or glass),

A2.1.2 One white enamel reflector (without baffle or glass),

A2.1.3 Mount as shown in Fig. A1.

A2.1.4 Eliminate all light sources except the specified fluorescent lamps.

A3 Walls.

A3.1 It has been the experience of many observers that light reflected from the side walls near the viewing board can interfere with the rating results. It is recommended that the side walls be painted matte black (85° gloss less than 5 units) or that blackout curtains be mounted on both sides of the viewing board to eliminate the reflective interference.

13. History

13.1 Editorially revised 2023 to add History section.

13.2 Editorially revised 2018; revised 2018 (with title change); revised 2014; editorially revised 2012; revised 2011; revised 2010, 2009 (with title change); editorially revised 2008; revised 2006, 2005, 2004; editorially revised and reaffirmed 2001; editorially revised 1997, 1996, 1992; editorially revised 1991; revised 1989 (with title change); editorially revised 1988, 1985; editorially revised and reaffirmed 1984; editorially revised 1983; revised 1982; editorially revised and reaffirmed 1978; revised 1975; editorially revised 1974; reaffirmed 1973; revised 1969.

13.3 Developed in 1967 by AATCC Committee RA61. Technically equivalent to ISO 7768.

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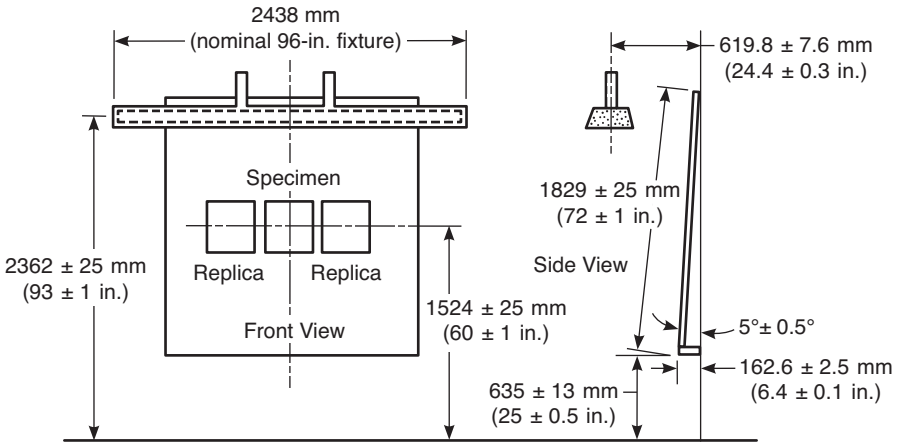


Fig. A1—Evaluation area for smoothness appearance evaluation.

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Test Method for Colorfastness to Perspiration and Light

1. Purpose and Scope

1.1 The purpose of this method is to determine the effect of the combination of perspiration solution and light exposure on the colorfastness of a colored textile specimen. Therefore, only perspiration solutions will be used in this procedure.

2. Principle

2.1 A colored test specimen is immersed in a perspiration test solution for a specified period of time and immediately exposed to light in a fading apparatus for a specified period of time.

2.1.1 Fading apparatus will be the xenon-arc lamp test apparatus as described in AATCC TM16.3, Test Method for Colorfastness to Light: Xenon-Arc.

3. Terminology

3.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

3.2 **lightfastness**, n.—the property of a material, usually an assigned number, depicting a ranked change in its color characteristics as a result of exposure of the material to sunlight or an artificial light source.

3.3 **perspiration,** n.—a saline fluid secreted by the sweat glands (in this test an artificial perspiration solution will be used).

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5. Apparatus, Materials and Test Solutions

5.1 Xenon-arc lamp fading apparatus (see AATCC TM16.3, Option 3)

5.2 Balance with a weighing accuracy of 0.0001 g.

5.3 pH meter accurate to 0.01.

5.4 Cardboard: 41 kg (91 lb) White Bristol Index (no backing of exposed area of colored test specimen).

5.5 Acid perspiration solution (see 12.4).

5.6 Gray Scale for Color Change (AATCC EP1) (see 12.1).

5.7 White AATCC Textile Blotting Paper (see 12.1).

5.8 Wringer (see 12.5).

6. Preparation of Reagent

6.1 Acid Perspiration Solution.

6.1.1 Prepare the acid perspiration solution by filling a 1 L volumetric flask half full of distilled water. Add the following chemicals and mix to be sure that all chemicals are thoroughly dissolved:

10 ± 0.01 g sodium chloride (NaCl)

1 ± 0.01 g lactic acid, USP 85%

1 ± 0.01 g disodium hydrogen phosphate, anhydrous (Na_2HPO_4) (see 12.2)

0.25 ± 0.001 g ℓ-histidine monohydrochloride ($C_6H_9N_3O_2{\cdot}HCl{\cdot}H_2O$)

Fill the volumetric flask with distilled water to the 1 L mark.

6.1.2 Test the pH of the solution with a pH meter. If it is not 4.3 ± 0.2 discard it and prepare a new one, making sure all ingredients are weighed accurately. The use of pH test paper is not recommended for this purpose because of its lack of accuracy.

6.1.3 Do not use perspiration solution that is more than three days old (see 12.3).

7. Test Specimens

7.1 Cut a specimen of colored fabric 5.1 × 7.0 cm (2.0 × 2.75 in.).

8. Procedure

8.1 Weigh the specimen to ± 0.01g.

8.2 Place each test specimen (as prepared in 7.1) in a 9 cm diameter, 2 cm deep petri dish. Add freshly prepared perspiration solution to a depth of 1.5 cm in the petri dish. Soak the test specimen in the solution for 30 ± 2 min with occasional agitation and squeezing to ensure complete wetting. For fabrics hard to wet out, alternately wet the specimen and pass through a laboratory wringer until it is completely penetrated by the solution.

8.3 After 30 ± 2 min, pass each test specimen through the wringer with the multifiber stripes perpendicular to the length of the wringer rolls (all stripes go through the wringer at the same time) or press between White AATCC Textile Blotting Paper such that each test specimen weighs 200 ± 5% its original weight (e.g., if the original specimen weighed 10 g, it should weigh between 19.5 and 20.5 g after soaking and wringing).

8.4 Mount wet, unbacked specimen in exposure frame or mount specimen on white cardboard stock that has been wrapped with a thin plastic film (see 12.6) to prevent wetting of the backing.

8.5 Expose specimens in fading apparatus in accordance with AATCC TM16.3, Option 3 for 20 AFUs.

8.6 Remove the specimens.

9. Evaluation

9.1 Evaluation of color change.

9.2 Evaluate the change in color in the test specimen by comparison with the Gray Scale for Color Change (AATCC EP1), or using AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen, and record the numerical rating that corresponds to the appropriate one on the Gray Scale (see 12.1).

9.3 For improved precision and accuracy more than one rater should rate the specimens.

10. Report

10.1 Report the color change grade.

10.2 Report the Fading apparatus used.

10.3 Report the reagent used.

11. Precision and Bias

11.1 *Precision*. In 2002 a single laboratory study was performed using a single operator. This study was intended to be a temporary table of variances to give some indication of test variability. A complete interlaboratory study is to be conducted in the near future for the purposes of precision and bias. Table values do not reflect different types of material tested to this standard. *Between-Laboratory* variability is not indicated either. Special care and consideration of the variances reported must be used when examining test variability problems.

11.1.1 Samples tested consisted of four fabrics, with three replicates each. Lightfastness exposure conditions were those found in AATCC TM16-1998, Option E.

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Each sample was evaluated instrumentally three times and averages were calculated. The data is found in Table I.

11.1.2 *Within-laboratory* standard errors and Sample Variance are shown in Table II. Data is on file at the AATCC Technical Center.

11.2 *Bias*. The colorfastness to natural and artificial light can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

12. Notes

12.1 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

12.2 Also sold as sodium phosphate, dibasic, anhydrous.

12.3 AATCC Committee RR52, Colorfastness to Perspiration, established that fungi begin to grow in the acid perspiration solution and that the pH rises after three days of storage under ambient room temperatures even when kept in a stoppered solution bottle.

12.4 There may be instances where alkaline perspiration may be used in this test method. In these instances the alkaline test solution should be run as described in AATCC TM15, Test Method for Colorfastness to Perspiration. For convenient reference the composition of the alkaline solution is as follows: Alkaline Solution - 10 g sodium chloride; 4 g ammonium carbonate, USP; 1 g disodium hydrogen phosphate, anhydrous (Na_2HPO_4); 0.25 g ℓ-histidine monohydrochloride. Make up to 1 L with distilled water. This solution should give a pH of 8.0 (see 12.2).

12.5 No specific wringer model is suggested. The purpose of the wringer is to achieve the total wet pick up after soaking in the solution. In lieu of the wringer, other suitable methods, like pressing between White AATCC Textile Blotting Paper, may also be used.

12.6 The purpose of the film used to wrap the card is to prevent the card from becoming wet. It shall be a transparent film such as food grade wrapping film.

Table I—ΔE

	Brown #1	Brown #2	Green	Blue
Specimen 1	1.26	4.37	6.25	7.83
Specimen 2	0.95	4.89	8.18	6.42
Specimen 3	1.17	5.78	5.23	4.87
Average	1.127	5.013	6.553	6.373

Table II—Within-Laboratory Standard Errors and Sample Variance

Sample Identification	Standard Dev.	Standard Error	Sample Variance	95% Confidence
Brown #1	0.159	0.092	0.025	0.396
Brown #2	0.713	0.412	0.508	1.771
Green	1.498	0.865	2.245	3.722
Blue	1.481	0.855	2.192	3.678

*Note: Because the interlaboratory test included less than five laboratories, estimates of standard error and sample variance may be either underestimated or overestimated to a considerable extent and should be used with special caution. The values should be viewed as minimal data with regards to precision. Confidence intervals are not well established.

13. History

13.1 Reaffirmed 2020.

13.2 Revised 2013, 2004 (with title change), 1982. Reaffirmed 1991, 1989, 1978, 1974, 1971. Editorially revised 2019, 2014, 2010, 2005, 1996, 1993, 1990. Editorially revised and reaffirmed 2009, 1986.

13.3 Developed in 1967 by AATCC Committee RA23; jurisdiction transferred in 1996 to AATCC Committee RA50. Partially equivalent to ISO 105-B07.

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Test Method for Water Resistance: Hydrostatic Pressure

Developed in 1968 by AATCC Committee RA63; reaffirmed 1971, 1974, 1977, 1980, 1989, 2003, 2013, 2018; editorially revised 1982, 1986, 2006, 2016, 2019; editorially revised and reaffirmed 1985, 2008; revised 1995, 1998, 2014, 2017. Related to ISO 811.

1. Purpose and Scope

1.1 This test method measures the resistance of a fabric to the penetration of water under hydrostatic pressure. It is applicable to all types of fabrics, including those treated with a water resistant or water repellent finish.

1.2 Water resistance depends on the repellency of the fibers and yarns, as well as the fabric construction.

1.3 The results obtained by this method may not be the same as the results obtained by the AATCC methods for resistance to rain or water spray.

1.4 This test method is performed without a restraint. If a restraint is required, use AATCC TM208, Test Method for Water Resistance: Hydrostatic Pressure Using a Restraint.

2. Principle

2.1 One surface of the test specimen is subjected to a hydrostatic pressure, increasing at a constant rate, until three points of leakage appear on its other surface. The water may be applied from above or below the test specimen.

2.2 This test is performed without a restraint.

3. Terminology

3.1 **hydrostatic pressure,** n.—the force distributed over an area exerted by water.

3.2 **water resistance,** n.—*of fabric*, the characteristic to resist wetting and penetration by water.

3.3 **water repellency,** n.—*in textiles*, the characteristic of fiber, yarn, or fabric to resist wetting.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details, such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

5. Apparatus and Materials (see 11.1)

5.1 Hydrostatic Tester.

5.1.1 For Option 1, Hydrostatic Pressure Tester (see 11.2).

5.1.2 For Option 2, Hydrostatic Head Tester (see 11.3).

5.2 Water, distilled or de-ionized.

6. Test Specimens

6.1 A minimum of three fabric specimens should be taken diagonally across the width of the fabric to be representative of the material. Cut specimens at least 200 × 200 mm to allow proper clamping.

6.2 Handle the specimens as little as possible and avoid folding or contaminating the area to be tested.

6.3 Condition the test specimens at 21 ± 2°C (70 ± 4°F) air at 65 ± 5% RH for at least 4 h before testing.

6.4 The surface of the fabric to be exposed to water must be specified because different results may be obtained on the face and the back. Identify that surface on a corner of each specimen.

7. Procedure

7.1 Verify the water in contact with the test specimen is regulated at 21 ± 2°C (70 ± 4°F) (see 11.4).

7.2 Dry the clamping surface.

7.3 Clamp the specimen with the surface to be tested facing the water (see 11.5).

7.4 Operation.

7.4.1 Option 1—Hydrostatic Pressure Tester. Turn on the motor, press the lever to raise the overflow device at the rate of 10 mm/s, and close the air vent as soon as water flows from it.

7.4.2 Option 2—Hydrostatic Head Tester. Select the gradient of 60 mbar/min (1 mbar/s), press the start button (see 11.6).

7.5 Disregarding water droplets that appear within approximately 3 mm of the specimen clamping ring, record the hydrostatic pressure at the moment water droplets penetrate the fabric in three different places (see 11.7).

8. Calculation

8.1 Calculate the average hydrostatic pressure for each sample.

9. Report

9.1 Results for each specimen and the average for each sample.

9.2 The material and the side tested.

9.3 Water temperature and type.

9.4 Gradient (rate of increasing water pressure).

9.5 Tester option used.

9.6 Any modification to the method.

10. Precision and Bias

10.1 *Precision*. The test results are tester dependent. Precision statements for each tester are given in 10.2 and 10.3.

10.2 Suter Hydrostatic Pressure Tester (Option 1).

10.2.1 In 1993, a limited interlaboratory study was completed, which included six laboratories, two operators in each, running determinations on three specimens of two fabrics. No prior assessment was made of the relative level of the participating laboratories on performance of the test method.

10.2.2 The two fabrics were at different levels (Fabric 1 approximately 810 mm and Fabric 2 approximately 340 mm), and residual variances of the two fabrics were found to be different. Accordingly, precision is reported separately for each fabric.

10.2.3 Users of the method are advised of the limited nature of this study and advised to apply these findings with due caution.

10.2.4 Analysis of the data sets for each fabric yielded components of variance and critical differences as displayed in Tables I, II and III. Differences between two averages of (N) determinations, for the appropriate precision parameter, should reach or exceed the table value to be statistically significant at the 95% confidence level.

10.3 Hydrostatic Head Tester (Option 2).

10.3.1 In a single-laboratory study, six different laboratory technicians ran determinations on three specimens of five fabrics.

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10.3.2 The five fabrics were at different levels of approximately: A=103, B=33, C=37, D=12, and E=77. Data obtained in this study is recorded in millibars (SI standard). The residual variance of the five fabrics were found to be different, therefore, precision is reported separately for each.

10.3.3 Analysis of the data sets for each fabric yielded critical differences as shown in Tables IV, V, VI, VII and VIII. Differences between two averages of (*N*) determinations, for the appropriate precision parameter, should reach or exceed the table value to be statistically significant at the 95% confidence level.

10.3.4 Between laboratory precision has not been established for this option. Until such precision information is available, users of this method should use standard statistical techniques in making any comparison of test results for between laboratory averages.

10.4 *Bias*.

10.4.1 Water resistance of fabrics can only be defined in terms of a test method. There is no independent, referee method for determining the true value. This test method has no known bias.

11. Notes

11.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

11.2 Hydrostatic Pressure Tester (Suter).

11.2.1 The apparatus consists essentially of an inverted conical well equipped with a coaxial ring clamp to fasten the cloth specimen under the well bottom. The apparatus introduces water from above the specimen over an area 114 mm in diameter and at a rate of 10.0 ± 0.5 mm of hydrostatic head per second. A mirror is affixed below the specimen to enable the operator to ascertain penetration of the specimen by drops of water. A valve is provided for venting the air in the well.

11.2.2 Hydrostatic testing apparatus of the type described is no longer available for sale.

11.3 Hydrostatic Head Tester.

11.3.1 The apparatus applies hydrostatic pressure at 60 mbar/min (selectable) to the bottom side of the fabric. A reservoir with a circular test area of 100 ± 5 cm^2 (15.5 ± 0.8 $in.^2$) contains distilled or deionized water which is applied to the fabric surface. The fabric specimen is secured with a clamp.

11.4 Some laboratories use water at ambient temperature. If water is other than 21 ± 2°C (70 ± 4°F), so state.

11.5 Lateral water leakage can be minimized by sealing the fabric with paraffin at the clamping area.

11.6 1 mbar = 10.2 mm H_2O.

11.7 An automated droplet detection system may be used in place of visual observation of water droplets if it is established that its accuracy is equivalent to visual observation.

Table I—Components of Variance for Two Fabrics (Option 1 Tester)

Component	Variance Fabric 1	Variance Fabric 2
Laboratory	13.450	7.323
Operator	3.127	2.145
Specimen	30.253	5.382

Table II—Fabric 1—Critical Differences—95% Confidence (Option 1 Tester)

Det in Avg (*N*)	Single Operator	Within Laboratory	Between Laboratory
1	15.25	16.02	18.97
2	10.78	11.84	15.61
3	8.80	10.08	14.31
4	7.62	9.06	13.62
5	6.82	8.04	13.19

Table III—Fabric 2—Critical Differences—95% Confidence (Option 1 Tester)

Det in Avg (*N*)	Single Operator	Within Laboratory	Between Laboratory
1	6.43	7.61	10.68
2	4.55	6.10	9.67
3	3.71	5.50	9.30
4	3.22	5.18	9.12
5	2.88	4.98	9.00

Table IV—Material A—Critical Differences—95% Confidence (Option 2 Tester)

Det in Avg (*N*)	Single Operator	Within Laboratory
1	72.49	72.49
2	51.26	51.26
3	41.85	41.85
4	36.25	36.25
5	32.42	32.42

Table V—Material B—Critical Differences—95% Confidence (Option 2 Tester)

Det in Avg (*N*)	Single Operator	Within Laboratory
1	10.08	12.85
2	7.13	9.09
3	5.82	7.42
4	5.04	6.43
5	4.51	5.75

Table VI—Material C—Critical Differences—95% Confidence (Option 2 Tester)

Det in Avg (*N*)	Single Operator	Within Laboratory
1	16.13	16.13
2	11.40	11.40
3	9.31	9.31
4	8.06	8.06
5	7.21	7.21

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Table VII—Material D—Critical Differences—95% Confidence (Option 2 Tester)

Det in Avg (*N*)	Single Operator	Within Laboratory
1	2.88	3.50
2	2.04	2.47
3	1.66	2.02
4	1.44	1.75
5	1.29	1.57

Table VIII—Material E—Critical Differences—95% Confidence (Option 2 Tester)

Det in Avg (*N*)	Single Operator	Within Laboratory
1	15.04	16.55
2	10.63	11.70
3	8.68	9.55
4	7.52	8.27
5	6.72	7.40

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Test Method for Wrinkle Recovery of Fabrics: Appearance

1. Purpose and Scope

1.1 This test method is for determining the appearance of textile fabrics after induced wrinkling. It is applicable to fabrics made from any fiber or combination of fibers.

1.2 The method can be used to evaluate fabrics at their original, unwashed state or after home laundering.

2. Principle

2.1 A test specimen is wrinkled under standard atmospheric conditions in a standard wrinkling device under a predetermined load for a prescribed period of time. The specimen is then reconditioned in the standard atmosphere for textile testing and evaluated for appearance by comparison with 3-dimensional reference standards.

3. Terminology

3.1 **smoothness appearance,** n.—*in fabrics*, the visual impression of planarity of a specimen quantified by comparison with a set of reference standards.

3.2 **wrinkle recovery,** n.—that property of a fabric which enables it to recover from folding deformations.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5. Apparatus and Materials

5.1 AATCC Wrinkle Tester (see Fig. 1, and 11.1 and 11.2).

5.2 AATCC 3-Dimensional Wrinkle Recovery Replicas (see Fig. 2 and 11.3).

5.3 Standard conditions room at 21 ± 2°C (70 ± 4°F) and 65 ± 5% RH.

5.4 Clothes hangers with clips.

5.5 A lighting and evaluation area in a darkened room using overhead lighting arrangement shown in Fig. 3.

5.6 Digital Imaging System (see 11.8).



Fig. 1—AATCC Wrinkle Tester.

6. Test Specimens

6.1 Take three test specimens from the fabric to be tested by cutting each 15 × 28 cm (6 × 11 in.) with the long dimension running in the direction of the warp of woven fabric or the wale of knit fabrics (see 11.4). Identify each specimen along one edge of the face side.

6.2 Condition specimens for a minimum of 8 h at 21 ± 2°C (70 ± 4°F) and 65 ± 5% RH.

7. Procedure

7.1 Perform all further steps at 21 ± 2°C (70 ± 4°F) and 65 ± 5% RH.

7.2 Raise top flange of wrinkle tester and hold in top position with locking pin.

7.3 Wrap one long edge (28 cm) (11 in.) of the specimen around the top flange of the AATCC Wrinkle Tester with the face side of the specimen on the outside and clamp it in position using the steel spring and clamp provided. Arrange the ends of the specimen so that they are opposite the opening in the spring clamp.

7.4 Wrap the opposite long edge of the specimen around the bottom flange and clamp as described above.

7.5 Adjust the specimen by pulling on its bottom edge, so that it lies smooth without sagging between the top and the bottom flanges.

7.6 Withdraw the locking pin and lower the top flange gently with one hand until it comes to rest.

7.7 Immediately place a total of 3500 grams weight on the top flange and record the exact time.

7.8 After 20 min, remove the weights, remove the springs and clamps, raise the top flange and gently remove the specimen from the tester so as not to distort any induced wrinkles (see 11.5).

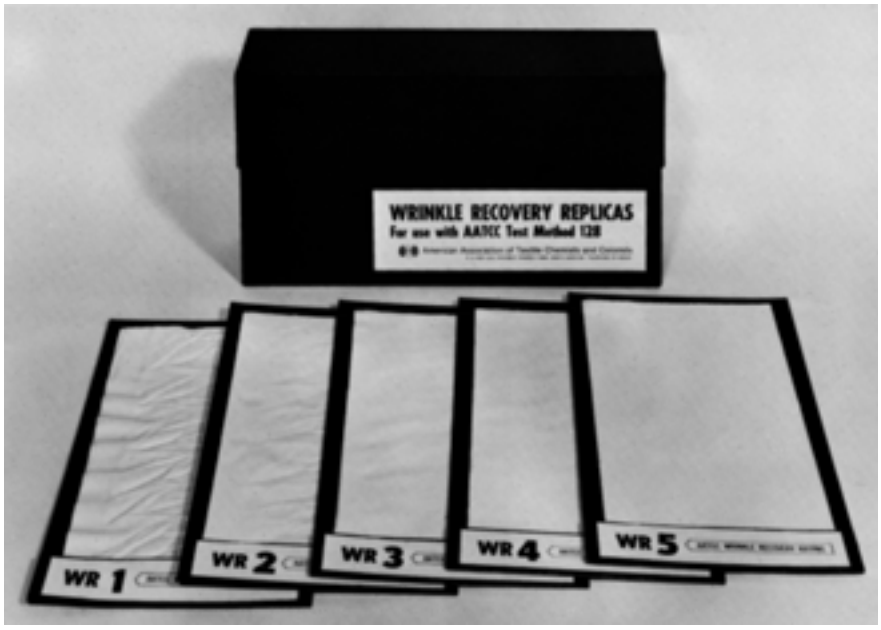


Fig. 2—AATCC Wrinkle Recovery Replicas.

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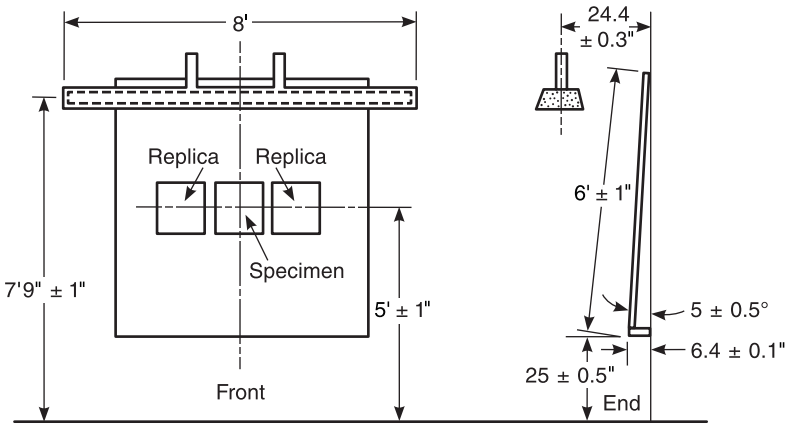


Fig. 3—Lighting equipment for viewing test specimens.

Materials list: (a) Two 8-ft Type F96 CW (Cool White) preheat Rapid Star fluorescent lamps (without baffle or glass). (b) One white enamel reflector (without baffle or glass). (c) One general type swatch mount, spring loaded. Fabricate using light sheet metal (22 ga.). (d) One ¼ in. plywood mounting board painted to match No. 2 gray chip on Gray Scale for Staining (see 11.9).

7.9 With a minimum of handling, place the shorter edge (15 cm) (6 in.) under the clips on the clothes hanger and allow the specimen to hang vertically in the long direction.

7.10 After 24 h in the standard atmosphere gently remove the hanger with the specimen to the evaluation site (see 11.6).

8. Evaluation

8.1 Three trained observers should rate each test specimen independently (see 11.7).

8.2 Mount the test specimen on the viewing board as illustrated in Fig. 3, with the warp in vertical direction. Place three-dimensional plastic replicas on each side of the test specimen to facilitate comparative rating.

8.2.1 The overhead fluorescent light should be the only light source for the viewing board, and all other lights in the room should be turned off.

8.2.2 It has been the experience of many observers that light reflected from the side walls near the viewing board can interfere with the rating results. It is recommended that the side walls be painted black or that blackout curtains be mounted on either side of the viewing board to eliminate the reflective interference.

8.3 The observer is to stand directly in front of the specimen 4 ft away from the board. It has been found that normal variations in the height of the observer above and below the arbitrary 5-ft eye level have no significant effect on the rating given.

8.4 Assign the number of the replica which most nearly matches the appearance of the test specimens.

8.4.1 A No. 5 rating is equivalent to the WR-5 Replica and represents the smoothest appearance and best retention of original appearance, while a No. 1 rating is equivalent to the WR-1 Replica and represents the poorest appearance and poorest retention of original appearance (see Table I).

8.5 Similarly, the observer independently rates each of the other two test specimens. The other two observers proceed in the same manner assigning ratings independently.

8.6 A digital imaging system may be used (see 11.8).

9. Calculating and Reporting Results

9.1 Average the nine observations made on each test fabric (three judgments on each of three specimens).

9.2 Report the average to the nearest tenth of a rating.

10. Precision and Bias

10.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

10.2 *Bias*. The wrinkle recovery of fabrics (appearance method), can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

Table I—Fabric Smoothness Ratings

5—an appearance equivalent to the WR-5 Replica
4—an appearance equivalent to the WR-4 Replica
3—an appearance equivalent to the WR-3 Replica
2—an appearance equivalent to the WR-2 Replica
1—an appearance equivalent to or worse than the WR-1 Replica

11. Notes

11.1 The AATCC Wrinkle Tester is provided for introducing wrinkles in the test specimen under controlled conditions. This apparatus is based on a development of ENKA/AKU Research. The apparatus is obtainable from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1. 919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.2 All three weights (500, 1000 and 2000 g) provided with this tester are used in the test. The top flange assembly weighs 500 g which is 200 g heavier than that of the AKU Wrinkle Tester. The AKU Wrinkle Tester may be used for this test by adding 200 g to the top flange.

11.3 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.4 Cut specimens from a wrinkle-free area of the fabrics. If any wrinkles are unavoidably present in the test specimens, press each lightly with a steam iron before conditioning.

11.5 If the test has been run properly, diagonal wrinkles should run through the center portion of the specimen.

11.6 Evaluation area should also be maintained at 21 ± 2°C (70 ± 4°F) and 65 ± 5% RH.

11.7 Since previous tests have proven that specimens do change in appearance during the first few hours, it is important that accurate times be observed and that a minimum time elapses while the three observers are evaluating the samples. Because of these changing conditions, the duration of recovery before rating in this method has been standardized at 24 h.

11.8 A digital imaging system may be used in place of the visual replicas if it has been established by the user that its accuracy is equivalent.

11.9 Approximate CIELAB values for the Step 2 chip on the Gray Scale for Staining are L* = 77, a* = 0, b* = 0. A tolerance of two units for each parameter is acceptable for the board color in this test method.

12. History

12.1 Editorially revised 2022 to add History section.

12.2 Editorially revised 2019; revised 2017; editorially revised 2016; reaffirmed 2013; editorially revised 2010; reaffirmed and editorially revised 2009, 2004; reaffirmed 1999; editorially revised 1995; reaffirmed and editorially revised 1994; editorially revised 1995 1992,1990; reaffirmed 1989; editorially revised 1988; reaffirmed and editorially revised 1985; reaffirmed 1980, 1977; revised 1974, 1970; reaffirmed and editorially revised 1969.

12.3 1968 Developed by AATCC Committee RR6; 1995 Jurisdiction transferred to AATCC Committee RA61. Technically equivalent to ISO 9867.

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Test Method for Colorfastness to Ozone in the Atmosphere under High Humidities

Developed in 1962 by AATCC Committee RA33; editorially revised 1973, 1974, 1981, 1989, 1997, 2008, 2019; revised 1972, 1975, 1985, 2005, 2011; reaffirmed 1990, 2001, 2016; editorially revised and reaffirmed 1996, 2010. Partly equivalent to ISO 105-G03.

1. Purpose and Scope

1.1 This test method is used for determining the resistance of the color of textiles to the action of ozone in the atmosphere at elevated temperatures with relative humidities above 85%.

2. Principle

2.1 A test specimen and a swatch of control sample are simultaneously exposed to ozone in an atmosphere which is maintained at 87.5 ± 2.5% RH and a temperature of 40 ± 1°C (104 ± 2°F) until the control sample shows a color change corresponding to that of a standard of fading. The cycles are repeated until the specimen shows a definite color change or for a prescribed number of cycles.

2.2 On certain fibers, dye fading does not readily take place at humidities below 85%. Testing at high humidities is required to produce color change that predicts service fading under warm, humid conditions (see 11.1).

3. Terminology

3.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OHSA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Ozone is a sensory irritant. The test cabinet should be ventilated to the outside atmosphere in accordance with the manufacturer's instructions.

4.3 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 11.2).

5. Apparatus and Materials (see 11.3)

5.1 Ozone Chamber for elevated temperatures and relative humidities above 85% (see 11.4).

5.2 Control Sample No. 129 for high humidities (see 11.5, 11.7 and 11.8).

5.3 Standard of Fading No. 129 for high humidities (see 11.6, 11.7 and 11.8).

5.4 Gray Scale for Color Change (AATCC EP1) (see 11.7).

6. Test Specimens

6.1 Cut each test specimen at least 100 × 60 mm (4.25 × 2.375 in.). For subsequent color comparison, the unexposed sample should be kept in an airtight container away from light to avoid further shade change.

6.2 When a laundered or drycleaned specimen is used, base the effects of ozone exposure on comparisons with the color of the laundered or drycleaned swatch. In the preparation of specimens for testing after laundering or drycleaning, use AATCC TM61, Test Method for Colorfastness to Laundering: Accelerated, and AATCC TM132, Test Method for Colorfastness to Drycleaning.

7. Procedure

7.1 Suspend test and control specimens (see 11.5 and 11.7) in the exposure chamber (see 11.4) which is maintained at 87.5 ± 2.5% RH and a temperature of 40 ± 1°C (104 ± 2°F). Ozone concentrations must range from 10-35 pphm which should produce one cycle of fading in 3-28 h of test.

7.2 Examine the control specimen periodically until the resulting color corresponds to that of the standard of fading (see 11.6 and 11.7) when compared in daylight ranging from average to slightly bluish north sky light, or equivalent artificial light. This constitutes one cycle (see 11.9).

7.3 Remove those specimens which exhibit a color change at the end of one cycle. One cycle will generally produce a measurable color change in specimens which are ozone sensitive.

7.4 Suspend a fresh control specimen and continue exposure of any remaining specimens for a second cycle.

7.5 Run additional similar cycles as necessary.

8. Evaluation

8.1 At the end of each cycle immediately compare those specimens which are removed from the exposure chamber with the preserved original from which they were drawn.

8.2 Rate the effect on color of test specimens after any specified number of cycles by reference to the Gray Scale for Color Change (AATCC EP1) or using AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen, and record the numerical rating that corresponds to the appropriate one on the Gray Scale. Report the number of cycles run (see 11.7, 11.10 and 11.11).

9. Report

9.1 Report the numerical rating for the change in color of the test specimen, the number of cycles run and the temperature and relative humidity at which the test was performed.

10. Precision and Bias

10.1 *Precision*.

10.1.1 Interlaboratory Study. Four laboratories participated in this study. Twenty-one carpet materials were evaluated for colorfastness under conditions of the test method. Four specimens were assigned to each laboratory for each material. Three specimens were exposed in an

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ozone-humidity chamber, and one was retained in the original condition to evaluate color retention using EP1. The ILS exposure plan contained these factors: laboratories 4, materials 21, specimens 3, with 1 determination per specimen. Each laboratory assigned grades to specimens using EP1.

10.1.2 Components of Variance. Variance components were derived for carpet materials (VF), laboratories (VL), interaction (VFL), and residual (V). Components VL and V were needed to compute precision estimates for tests done on a single carpet material, as shown in Table I and using equations:

$$95\ \text{CD Within Lab} = \frac{2.83(V/3)}{0.5}$$

and

$$95\ \text{CD Between Lab} = \frac{2.83(V/3 + VL)}{0.5}$$

stated in spreadsheet notation. These equations give expected critical differences not to be exceeded (1) in two test results of a single carpet materials within one laboratory, or (2) in single test results of two laboratories from a single carpet material, when either difference is not significant.

10.1.3 Precision Estimates. Table II values are computed at the 95% confidence level. Values are Critical Differences. This study used three specimens (noted in bold). Other specimen values show the effect of the number of specimens in an average on the magnitude of a critical difference.

Table I—Components of Variance

COV	Unit Var
VL	0.038
V	0.026

Table II—Critical Differences

Specimens in Average	95 CD Within Lab	95 CD Between Lab
1	0.46	0.70
3	**0.26**	**0.61**
5	0.20	0.40

10.2 *Bias*.

10.2.1 This test method has no known bias. There is no known referee test method, nor any known standard materials by which test method bias may be established.

11. Notes

11.1 At high humidities, the fading of dyes by ozone on small fibers (such as nylon) is altered greatly by relatively small variations in relative humidity. Therefore, to achieve reproducibility and good interlaboratory correlation in test results, temperature and relative humidity must be closely controlled.

11.2 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

11.3 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

11.4 The ozone exposure chamber for elevated temperatures and relative humidities above 85% can be of variable construction as long as it meets the conditions of 87.5 ± 2.5% RH, 40 ± 1°C (104 ± 2°F) temperature and contains an ozone concentration in the range of 10-35 pphm.

11.4.1 Blueprints of a suitable apparatus are available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1. 919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.4.2 For information regarding the measurement of ozone concentration, see the following references:

Schulze, Fernand, "Versatile Combination Ozone and Sulfur Dioxide Analyzer," *Analytical Chemistry*, Vol. 38, May 1966, pp748-752.

"Selected Methods of the Measurement of Air Pollutants," Public Health Service Publication No. 999-AP-11, May 1965. Office of Technical Information and Publication (OTIP), Springfield VA, PB167-677.

11.5 The control sample for tests made at elevated temperatures and relative humidities above 85% is a tertiary shade, avocado, prepared by dyeing a Y-cross-section nylon 6 unautoclaved knitted sleeve with the following formula: 0.15% C.I. Disperse Red 4, 0.63% C.I. Disperse Yellow 3 and 0.25% C.I. Disperse Blue 3 based on the weight of the fabric.

11.6 The standard of fading for tests made at elevated temperatures and relative humidities above 85% is a dyeing of the same Y-cross-section nylon 6 unautoclaved knitted sleeve with the following formula: 0.557% C.I. Acid Yellow 79, 0.081% C.I. Acid Red 361 and 0.102% C.I. Acid Blue 277.

11.7 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.8 CAUTION: Both the control sample and the standard of fading must be kept in airtight containers to prevent color change encountered in normal atmospheres. The control fabric is also sensitive to other atmospheric contaminants such as oxides of nitrogen. Its fading rate will vary considerably at different humidities and temperatures and its use in natural or end-use testing as a measure of exposure to ozone is not recommended. The color change produced on the control will reflect the combined effects of the atmospheric contaminants present, and temperature, humidity variations and not just the effects of exposure to ozone.

11.9 An alternative method of determining one cycle of fade is to terminate the exposure cycle when the Control Sleeve 129 exhibits a color change of 22.5 ± 1.4 CIELAB units for Lot No. 10 (see 11.10).

11.10 Make colorimetric or spectrophotometric measurement of Control Sleeve 129 with the specular component included. Calculate colorimetric data using the CIE 1964 10° observer data for Illuminant D_{65}. The color difference shall be expressed in CIELAB units.

11.11 An automated electronic grading system may be used as long as the system has been demonstrated to provide results that are equal to and provide equal or better repeatability and reproducibility than an experienced grader performing visual evaluation.

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Test Method for Soil Release: Oily Stain Release

Developed in 1969 by AATCC Committee RA56; reaffirmed 1970, 1974, 1977; editorially revised 1978, 1983, 1986, 1997, 2004, 2005, 2008, 2011, 2012, 2016; revised 1981, 1990, 1995, 2010, 2018 (twice); editorially revised and reaffirmed 2000, 2015.

Foreword

This test method measures the ability of fabrics to release an oily soil during home laundering. Standard laundering procedures represent, but may not exactly replicate, current consumer practices, which vary by sample, over time and among households. Alternate stains or care procedures may be addressed by other methods or modifications of this method.

In this method, corn oil is forced into the fabric before laundering. To evaluate a fabric's ability to resist or repel oily stains, use AATCC TM118, Oil Repellency: Hydrocarbon Resistance Test.

1. Purpose and Scope

1.1 This test method is for the evaluation of fabrics to release an oily soil after exposure to home laundering conditions.

1.2 This test may be performed on garments or other end products suitable for home laundering.

2. Principle

2.1 Corn oil is applied to a test specimen. The oil is forced into the fabric by using a specified weight. The stained fabric is then laundered in a prescribed manner and the residual stain evaluated by comparison with a stain release scale showing a graduated series of stains. Results are reported as a grade from 5 to 1, with 5 representing no residual stain and 1 representing the most residual stain.

3. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

3.1 AATCC LP1, Home Laundering: Machine Washing (see 15.1).

3.2 AATCC TM118, Oil Repellency: Hydrocarbon Resistance Test (see 15.1).

3.3 AATCC TM133, Colorfastness to Heat: Hot Pressing (see 15.1).

3.4 ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 15.2).

4. Terminology

4.1 **soil,** n.—dirt, oil or other substances not normally intended to be present on a substrate such as a textile material.

4.2 **soil release,** n.—the degree to which a soiled substrate approaches its original, unsoiled appearance as a result of a care procedure.

4.3 **stain,** n.—a local deposit of soil or discoloration on a substrate that exhibits some degree of resistance to removal, as by laundering or drycleaning.

5. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

5.1 Follow good laboratory practices. Wear safety glasses in all laboratory areas.

5.2 Detergent may cause irritation. Take care to prevent contact with skin and eyes.

5.3 Follow manufacturer's safety recommendations when operating laboratory testing equipment.

6. Uses and Limitations

6.1 In this method, soils are purposely forced into the fabric before laundering. To evaluate a fabric's ability to resist or repel oily soils, use AATCC TM118.

6.2 Standard laundering conditions are not intended to replicate consumer care for all samples. By laundering all samples under the same conditions, this method provides a relative measure of stain release properties. The method may be modified to use recommended care for a particular sample, but results should not be compared with those for samples laundered under different conditions.

6.3 Specimens prepared for this method should not be used for dimensional change or appearance evaluation. Specimens for this method should be laundered separately from specimens for dimensional change or appearance. Although specimen dimensions are the same for these methods, oily stains may interfere with evaluation of other properties.

7. Apparatus and Materials (see 15.3)

7.1 Conditioning facilities and racks with pull-out, perforated shelves for laying specimens flat (see 15.1).

7.2 AATCC Blotting Paper (see 15.1).

7.3 Corn oil (see 15.4).

7.4 Glassine paper (see 15.1).

7.5 Timer.

7.6 Cylindrical weight, 6.4 cm (2.5 in.) diam., 2.268 ± 0.045 kg (5.0 ± 0.1 lb) (stainless steel is preferable, see 15.1). Multiple weights are recommended to efficiently process multiple samples.

7.7 Amber bottle (see 15.5) with medicine dropper, 76 mm glass, 2-mL capacity, capable of dispensing
20 drops/mL.

7.8 Automatic washing machine (see Table I, 15.6).

7.9 Tumble dryer, normal cycle: maximum exhaust temperature 68°C ± 6°C (155°F ± 10°F), cool down time ≤ 10 min. (see 15.6).

7.10 AATCC 1993 Standard Reference Detergent (powder, without brightener [WOB]) (see 15.1).

7.11 Balance with at least 5.0-kg (10.0-lb) capacity.

7.12 Laundering ballast Type 1 (see Table II, 15.7).

7.13 Water hardness test kit, residential use.

7.14 Evaluation area, as described in Appendix A.

7.15 AATCC Stain Release Scale (see 15.1)

8. Specimens

8.1 Use two test specimens 38 × 38 ± 1 cm (15.0 × 15.0 ± 0.4 in.) for each determination.

9. Conditioning

9.1 Prior to the application of corn oil, condition test specimens as directed in ASTM D1776. (Use conditions indicated in Table 1 for Textiles, general. Estimate conditioning time as indicated in Table 2 for the appropriate fiber content.) Lay each swatch or product, separately, on a screen or perforated shelf of a conditioning/drying rack.

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Table I—Standard Washing Machine Parameters (see 15.9)

Cycle	(1) Normal	
Water Level, L (gal)	72 ± 4 (19 ± 1)	
Agitation Speed, strokes/min.	86 ± 2	
Washing Time, min.	16 ± 1	
Final Spin Speed, rpm	660 ± 15	
Final Spin Time, min.	5 ± 1	
Wash Temp, °C (°F)[1]	(III) Warm:	41 ± 3 (105 ± 5)

[1] Due to US Department of Energy requirements, many washing machines use cooler water. An external control box may be used to override the machine set temperatures.

Table II—Type 1 Laundering Ballast Parameters

	Type 1
Fiber Content	100% cotton
Greige Fabric Yarns	16/1 ring spun
Greige Fabric Construction	52 × 48 ± 5 yarns/in., plain weave
Finished Fabric Weight	155 ± 10 g/m^2
Edges	All edges hemmed or over-edged
Finished Piece Size	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)
Finished Piece Weight	130 ± 10 g

10. Soil Application Procedure

10.1 Place a single thickness of blotting paper on a smooth, horizontal surface.

10.2 Place the specimen face up, flat, on the blotting paper.

10.3 Using the medicine dropper, place 5 drops (approx. 0.2 mL) of corn oil in the approximate center of the test specimen. Apply all drops in the same location.

10.4 Place a 7.6 × 7.6 cm (3.0 × 3.0 in.) square of glassine paper over the soil application area.

10.5 Place the weight on the glassine paper, directly over the soil application area.

10.6 Allow weight to sit undisturbed for 60 ± 5 s. Then remove the weight and discard the glassine paper.

10.7 Repeat 10.2-10.6 for the second specimen. If testing multiple samples to be laundered together, the use of multiple weights allows greater testing efficiency. Do not use blotting paper sites that have previously been used for soil application.

10.8 Wash specimens 20 ± 5 minutes after corn oil has been applied to allow stains to set in. Do not allow test specimens to contact each other in a manner that would transfer corn oil. The maximum number of test specimens per washer load shall be 30. If testing multiple samples and to comply with the 20 ± 5 minutes dwell time requirement, it is recommended that the prescribed washing machine temperature, detergent and ballast pieces to be used are ready for all specimens to be added at the same time.

11. Laundering Procedure

11.1 Washing.

11.1.1 Set washing machine controls to generate the parameters indicated in Table I.

11.1.2 Load for laundering shall consist of all specimens in the sample, plus enough laundering ballast pieces to make a total load weight of 1.8 ± 0.1 kg (4.0 ± 0.2 lb).

11.1.3 Begin selected wash cycle. Allow machine to fill to specified water level.

11.1.4 Measure and record washing water hardness (see 15.8).

11.1.5 Add 66 ± 1 g of AATCC 1993 Standard Reference Detergent (without brightener) to washing machine as directed by washing machine manufacturer's instructions. If detergent is added directly to wash water, agitate briefly to dissolve completely. Stop agitation before adding wash load.

11.1.6 Add wash load (test specimens and ballast), distributing evenly around center agitator. Restart wash cycle.

11.1.7 Allow washing to proceed through the final spin cycle.

11.1.8 Separate tangled specimens and ballast pieces, taking care to minimize distortion. Proceed to drying procedure.

11.2 Tumble Drying.

11.2.1 Place the washed load (test specimens and ballast) in the tumble dryer, and set the temperature control to generate a maximum exhaust temperature of 68°C ± 6°C (155°F ± 10°F).

11.2.2 Allow the dryer to run until the load is fully dry.

11.2.3 Remove specimens from dryer immediately on completion of the cycle and lay flat to prevent formation of wrinkles or creases that can affect the stain release rating.

11.3 Ironing (optional)

11.3.1 If the specimens are extremely wrinkled and the consumer would ALWAYS expect to iron the end product, hand iron swatches or products prior to evaluation. Use caution due to the extremely high variability in hand ironing procedures performed by individual operators (no standard test method exists for hand ironing procedures).

11.3.2 Use safe ironing temperature for the sample. See AATCC TM133, Table I, Safe Ironing Temperature Guide for guidance regarding textile fibers. Exert the minimum pressure necessary to smooth specimen.

11.4 Evaluate the grade of stain release of each specimen within 4 h after drying. Reconditioning is not required.

12. Evaluation

12.1 Mount the Stain Release Scale on the viewing board, with the center of the scale 114 ± 3 cm (45 ± 1 in.) from the floor (see Appendix A).

12.2 Place the test specimen flat with face up in the center of the table. Rotate the fabric to the angle that produces the lowest grade.

12.3 Stand directly in front of the specimen for evaluation. Stand 76 ± 3 cm (30 ± 1 in.) from the viewing board, with eyes 157 ± 15 cm (62 ± 6 in.) from the floor. Varying the viewing angle either horizontally or vertically can affect grades obtained on some fabrics.

12.4 Two observers shall independently assign a grade to each test specimen. The grade is that of the Stain Release Scale that corresponds most closely to the residual stain on the test specimen. Half-step grades (i.e., 1.5, 2.5, 3.5, 4.5) may be assigned to residual stains that are between grades on the Stain Release Scale.

12.5 Calculate the mean of 4 grades for the sample (2 observers, 2 specimens) to nearest 0.1. This is the unit of measure for this test method.

13. Report

13.1 Describe or identify the fabric tested.

13.2 Report that the sample was tested using AATCC TM130-2018t.

13.3 Report the testing conditions:

13.3.1 Water hardness of the washing procedure in terms of parts per million (ppm).

13.4 Report the test results:

13.4.1 Mean stain release grade calculated in 12.5.

13.4.2 Individual stain release grades for each specimen and observer, or standard deviation of the mean.

13.5 Describe any modification(s) of the published method, including alternate staining or laundering procedures.

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14. Precision and Bias

14.1 *Summary.* An interlaboratory test was carried out in 1987 to establish the precision of this test method. One operator at each of 5 laboratories tested 2 specimens of each of 5 fabrics each day for 3 days using 1 washer and 1 dryer. A detergent, widely available in the U.S., which contained 9.8% phosphate, was used (see 15.10). The reference vegetable oil used was Mazola brand corn oil. Two observers independently evaluated the specimens using the AATCC Stain Release Scale. The unit of measure was the average of the 2 grades for the 2 specimens tested each day. The fabrics were limited to polyester and polyester/cotton materials, most having soil release finishes.

14.2 The components of variance as standard deviations of stain release grades were calculated to be as follows:

	AATCC Stain Release Scale
Single operator/washer	0.30
Between laboratories	0.23

14.3 *Critical Differences.* For the components of variance in 14.2, two averages of observed values should be considered significantly different at the 95% probability level if the difference equals or exceeds the critical differences shown in Table III.

14.4 *Bias*. The true value of the oily stain release grade can only be defined in terms of this test method. Within this limitation, this test method has no known bias.

15. Notes

15.1 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1. 919.549.8141; ordering@aatcc.org; www.aatcc.org.

15.1.1 The AATCC Stain Release Scale should be replaced every 12 months and stored in the dark to prevent fading.

15.2 Available from ASTM International, 100 Barr Harbor Dr, W Conshohocken PA 19428, USA; +1.610.832.9500; www.astm.org.

15.3 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

15.4 Mazola, a trademark of ACH Food Companies Inc., is a pure corn oil and was used for the interlaboratory test of this method because it was widely available and consistent in color and quality. Each bottle has a purchase by date printed on the label. It should not be used after that date.

15.5 The amber bottle is specified to protect the corn oil from degradation.

15.6 For model number(s) of washer(s) and tumble dryers(s) reported to meet the standard parameters, visit www.aatcc.org/test/washers or contact AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org.

15.7 There is a possibility of finish and/or oil redeposition on ballast leading to subsequent transfer to test specimens. Fresh ballast should be used whenever it is deemed that buildup has occurred to the point where stain release results would be affected significantly or in cases where results are critical in adopting new finishes. The ballast should be replaced when it becomes obviously worn or frayed.

15.8 Water hardness may be measured with a kit designed for residential use, following manufacturer's instructions. Hardness can make detergent less effective and result in lower stain release grades. Hardness range is not specified for this method, but use caution in comparing test results obtained with vastly different water hardness values.

15.9 The laundering temperatures and other parameters listed in this test method are standard conditions for testing purposes. As with most lab procedures, they represent, but may not exactly replicate, current consumer practices. Consumer practices vary over time and among households; lab practices must be consistent to allow valid comparison of results. If laundering equipment or conditions other than those specifically listed in this test method are used, they must be described in detail and noted as a modification of the standard method. Alternate laundering conditions are outlined in AATCC LP1.

15.10 The 1995 revision of this method reflected the change in formulation of the reference detergent to eliminate phosphates. The revision also provided an additional choice of ballast (Type 3). Preliminary comparative studies with the new reference detergent and new ballast Type 3 were conducted by the AATCC Laboratory. These early results indicated a statistical difference between the 1993 AATCC Standard Reference Detergent and the previous AATCC Standard Reference Detergent 124, the new detergent tending to be less efficient toward oily stain release removal. No statistical difference at the 95% probability level was observed for the new ballast Type 3.

Revisions to the test method in 2010 aligned wash fill level and detergent amount with that of other AATCC standards at that time. A 2007/2008 interlab test study among 5 laboratories showed: a) although there was some differences observed between fill levels, there is no statistical indication that any one fill level is more consistent than the other; b) although there was some differences observed between detergent use levels, there is no statistical indication that any one detergent use level is more consistent than another; and c) raters within a lab were consistent with themselves, replicate pieces were consistent to each other, rater-to-rater measurements were consistent with each lab, and Analysis of Variance for these factors showed values in line with the published historical values in the test method.

Laundering conditions in other AATCC standards and available laundering equipment have continued to change. In 2018, "standard" laundering conditions were defined in AATCC LP1. These vary only slightly from the previously defined AATCC TM130 laundering conditions for Machine Wash Warm, Normal Cycle, Tumble Dry Normal. Previous AATCC TM130 washing conditions prescribed a 12-min. wash rather than the 16-min. wash prescribed in Table I. Warm wash temperature is the same. Previous AATCC TM130 drying conditions prescribed a 65°C maximum exhaust temperature and 5-min. cool down time; the method now prescribes a 68°C maximum exhaust temperature and ≤ 10-min. cool down time to align with AATCC LP1 and related methods.

15.11 The use of two 96-in. lamps for viewing laundered specimens is specified in this method. It is recognized, however, that physical space limitations in certain laboratories will prevent the use of 96-in. lamps. In those situations, two 48-in. lamps and a narrower viewing board may be used.

Table III—Critical Differences, Grades, for the Conditions Noted[a]

(Observation = Unit of Measure = Average of 2 grades for 2 specimens)

No. of Observations	Single Operator/Washer	Between Laboratories
	AATCC Stain Release Scale	AATCC Stain Release Scale
1	0.82	1.04
2	0.58	0.86
3	0.47	0.79

[a] The critical differences were calculated using $t = 1.950$, which is based on infinite degrees of freedom.

Appendix A
Evaluation Area

A1 Viewing Board.

A1.1 Plywood board, 1829 mm (72 in.) high × 6 mm (0.25 in.) thick × wide enough to accommodate scale.

A1.2 Painted to match the Grade 2 chip on the Gray Scale for Staining (see 15.1). Approximate CIELAB values for the chip are L* = 77, a* = 0, b* = 0. A tolerance of two units for each parameter is acceptable for the board color in this test method.

A2 Lighting.

A2.1 Overhead fluorescent tube light fixture (see 15.11).

A2.1.1 Two parallel F96 T12 cool white lamps (without baffle or glass),

A2.1.2 One white enamel reflector (without baffle or glass),

A2.1.3 Mount as shown in Fig. A1.

A2.1.4 Eliminate all light sources ex-

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cept the specified fluorescent lamps.

A3. Walls.

A3.1 It has been the experience of many observers that light reflected from the side walls near the viewing board can interfere with the evaluation results. It is recommended that the side walls be painted matte black (85° gloss less than 5 units) or that blackout curtains be mounted on both sides of the viewing board to eliminate the reflective interference.

A4. Specimen Table.

A4.1 Table with non-glare black top 610 × 920 mm (24 × 36 in.); 890 ± 30 mm (35 ± 1 in.) high.

A4.2 One long edge of table centered and touching the viewing board (see Fig. A1).

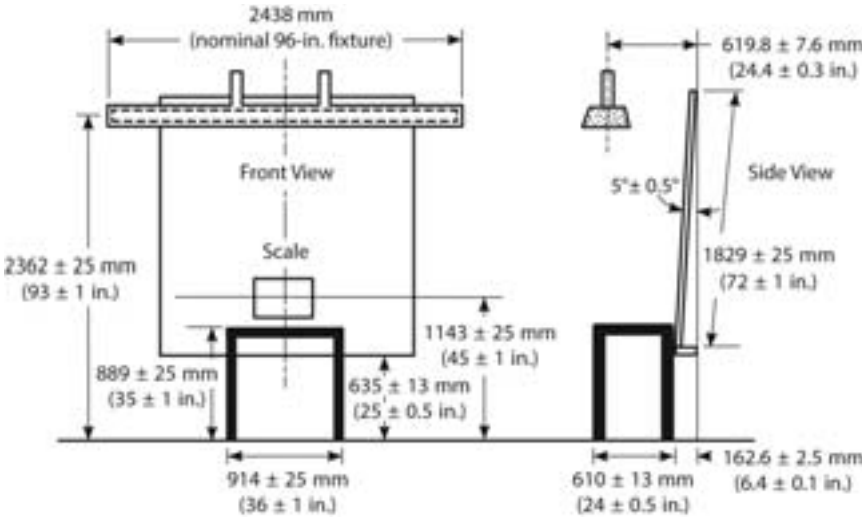


Fig. A1—Evaluation area for stain release evaluation.

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Test Method for Colorfastness to Steam Pleating

1. Purpose and Scope

1.1 This test method is intended for assessing the colorfastness of textiles of all kinds and in all forms to the action of steam pleating processes. The materials are not pleated during the test, and it is emphasized that the test is not intended for assessing the quality of the pleating process (see 12.1).

1.2 Three (3) tests, differing in severity, are provided; one (1) or more of them may be used.

1.3 The severe test is intended primarily for wholly manufactured textiles such as those made from polyamide and polyester fibers. It should not be used for textiles containing wool.

2. Principle

2.1 A specimen of the textile in contact with specified undyed cloths is steamed under specific conditions of pressure and time and then dried. The color change of the specimen and staining on the undyed cloths are assessed with standard gray scales. Color change and staining results are reported as a grade from 5 to 1, with 5 representing no color change or staining and 1 representing the most color change or staining.

3. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

3.1 AATCC EP1, Evaluation Procedure for Gray Scale for Color Change (see 12.2).

3.2 AATCC EP2, Evaluation Procedure for Gray Scale for Staining (see 12.2).

4. Terminology

4.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

4.2 **pleating,** n.—the process of making one or more desirable folds in a cloth by doubling the material over on itself.

5. Safety Precautions

NOTE: These safety statements are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Users MUST consult manufacturers for specific details such as safety data sheets and other recommendations. Consult and follow all applicable OSHA standards and rules.

5.1 Follow good laboratory practices. Wear safety glasses in all laboratory areas.

5.2 Follow manufacturer's safety recommendations when operating laboratory testing equipment.

6. Apparatus and Materials (see 12.3)

6.1 Specimen holder (see Fig. 1 and 12.4).

6.2 Jacketed steamer (see 12.5) or pressure cooker (see 12.6).

6.3 Two (2) undyed scoured cloths, each measuring 5 × 4 cm (2.0 × 1.6 in.), made of the same kind of fiber as that of the textile to be tested or as otherwise specified. In the case of blends, two different undyed scoured cloths are required, corresponding to the two predominating fibers (see 12.7).

6.4 Gray Scale for Color Change (see 12.2). Refer to AATCC EP1 for additional materials, environment and use of gray scale.

6.5 Gray Scale for Staining (see 12.2). Refer to AATCC EP2 for additional materials, environment and use of the gray scale.

7. Specimens

7.1 If the textile to be tested is fabric, cut a specimen 5 × 4 cm (2.0 × 1.6 in.)

7.1.1 Place fabric specimen between two (2) pieces of undyed cloth and sew along one side to form a composite specimen.

7.2 If the textile to be tested is yarn, knit it into fabric and treat it as in 7.1, or form a layer of parallel lengths of yarn between the two (2) pieces of undyed cloth, the amount of yarn taken being approximately equal to half the combined mass of the undyed cloths. Sew along one side to hold the yarn in place and to form a composite specimen.

7.3 If the textile to be tested is loose fiber, comb and compress an amount approximately equal to half the combined mass of the undyed cloths into a sheet 5 ×

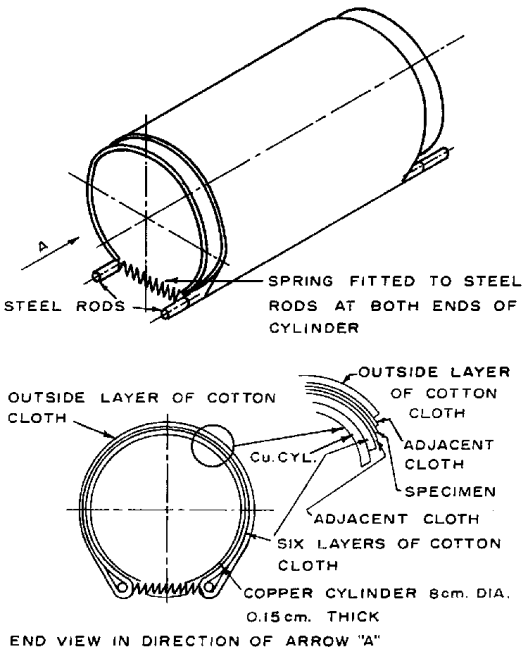


Fig. 1—Specimen holder.

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Table I—Steaming Conditions

Test	Duration	Pressure (Gauge)	Temperature
I. Mild	5 min	0.35 kg/cm^2 (5 lb/in.^{2})	108°C (226°F)
II. Intermediate	10 min	0.7 kg/cm^2 (10 lb/in.^{2})	115°C (239°F)
III. Severe	20 min	1.76 kg/cm^2 (25 lb/in.^{2})	130°C (266°F)

4 cm (2.0 × 1.6 in.) Place the sheet between the two (2) undyed cloths, and sew along all sides to form a composite specimen.

8. Procedure

8.1 Mount the composite specimen in the holder between the two undyed cloths (see 6.3) as shown in Fig. 1.

8.2 Place the holder containing the composite specimen in a jacketed steamer or a pressure cooker.

8.3 Steam under one of the sets of conditions listed in Table I.

8.4 When steaming is complete, release the pressure over a period not exceeding 2 min.

8.5 Separate layers of the composite specimen layers so they are in contact only at one line of stitching. Dry in air at a temperature not exceeding 60°C (140°F). Condition at 20 ± 2°C (68 ± 3°F) and 65 ± 2% RH for 4 h.

8.6 Test specimens that liberate formaldehyde under steam pleating conditions separately.

9. Evaluation

9.1 Evaluate the color change of the test specimen as directed in AATCC EP1.

9.2 Evaluate the staining on each undyed adjacent cloth as directed in AATCC EP2.

10. Report

10.1 Describe or identify the sample tested.

10.2 Report that the sample was tested using AATCC TM131-2019.

10.3 Report the testing conditions:

10.3.1 Test number (I, II or III).

10.3.2 Nominal fiber content of test sample and undyed adjacent cloth(s).

10.4 Report the test results:

10.4.1 Color change grade for tested specimen.

10.4.2 Staining grades for the undyed adjacent cloth(s). If the two cloths are identical and show different degrees of staining, only the heavier should be reported.

10.5 Describe any modification(s) of the published method.

11. Precision and Bias

11.1 *Precision.* Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

11.2 *Bias.* Colorfastness to steam pleating can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

12. Notes

12.1 Papers used in commercial pleating occasionally contain reducing agents, that with certain coloring matters, can produce a much greater change in color than occurs under the test conditions.

12.2 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

12.3 For potential equipment information pertaining to this test method, please visit the online AATCC Buyer's Guide at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

12.4 The specimen holder consists of a copper tube 8 cm (3 in.) in external diameter. The thickness of the copper is 0.15 cm (0.06 in.) The copper tube is wrapped with six layers of bleached cotton cloth weighing 125 g/m^2 (3.7 oz/yd^2). The outside layer is made from bleached cotton cloth weighing 186 g/m^2 (5.5 oz/yd^2). The edge of the outside layer is folded back and sewn to provide a place for insertion of the rods. The rods at each end are made from 0.6-cm (0.25-in.) diameter soft steel. The strength of the springs is not critical, but it should be sufficient to hold the layer tightly against the tube. The springs are fastened to one of the steel rods and should hook easily onto the other steel rod (see Fig. 1). A 15.2-cm (6-in.) long specimen holder is preferred.

12.5 A jacketed steamer may be used, provided that the pressure can be accurately determined and that no water splashes on the specimen during the test.

12.6 A domestic pressure cooker may be used as an alternative to a jacketed steamer. Its dimensions should be sufficiently large to avoid water splashing onto the specimen during the test; it is suggested that the minimum size should be 23 cm (9 in.) in diameter and 26 cm (10 in.) high. It should be fitted with an accurate pressure gauge. If a domestic pressure cooker is used, the specimen holder should be loosely wrapped in one layer of polyester film, which projects 1 cm (0.5 in.) over each end of the tube and is not closed at the ends. The specimen holder (see Fig. 2) should then be placed in a rectangular metal container containing 10 0.1 cm (0.06 in.) holes equally spaced along the center of the bottom. The container should be sufficiently deep to reach to 1 cm (0.5 in.) from the top of the specimen

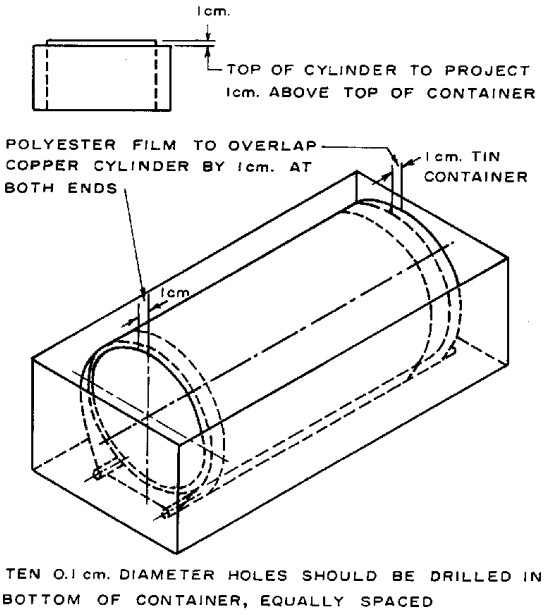


Fig. 2—Specimen holder in the container (for testing in pressure cooker).

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holder. The bottom of the container should be slightly concave to ensure that condensed water rapidly drains away. The container is then placed on a stand that holds the container 5 cm (2 in.) from the surface of the water. The quantity of water in the cooker is not critical, but water to a depth of 3 cm (1 in.) is suggested. Expel air from the pressure cooker for 2 min before raising the pressure.

12.7 If wool is used as one of the adjacent cloths (see 5.3) it may have an adverse effect on the dye in the specimen, particularly under alkaline conditions.

13. History

12.1 Revised (with title change) in 2019 to align format with AATCC style.

12.2 Editorially revised 2013. Reaffirmed 2011. Editorially revised 2008. Reaffirmed 2005. Editorially revised 2004. Editorially revised and reaffirmed 2000. Editorially revised and reaffirmed 1995. Reaffirmed 1990. Editorially revised 1988. Reaffirmed 1985. Editorially revised 1981. Reaffirmed 1980. Reaffirmed 1977. Reaffirmed 1974. Reaffirmed 1971. Reaffirmed 1970.

12.3 Developed in 1969 by AATCC Committee RR53. Maintained by RA99.

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Test Method for Colorfastness to Drycleaning

Developed in 1969 by AATCC Committee RA43. Supersedes AATCC TM85-1968; reaffirmed 1973, 1976, 1979, 1989, 1998, 2003; editorially revised 1981, 1986, 1990, 1995, 2001, 2002, 2008, 2010, 2016, 2019; editorially revised and reaffirmed 1985, 2009, 2013; revised 1993, 2004. Technically equivalent to ISO 105-D01.

1. Purpose and Scope

1.1 This test method is intended for determining the colorfastness of textiles to all kinds to drycleaning.

1.2 The test method is neither suitable for the evaluation of the durability of textile finishes, nor is it intended for use in evaluating the resistance of colors to spot and stain removal procedures used by the drycleaner (see 11.1).

1.3 This test method gives an indication of results to be obtained from three commercial drycleanings.

2. Principle

2.1 A specimen of the textile in contact with cotton fabric, multifiber swatch and noncorrodible steel discs is agitated in perchloroethylene (see 11.2) and then dried in air. Any change in color of the specimen is then assessed with the standard Gray Scale for Color Change (AATCC EP1).

2.2 Any staining is assessed by using the Gray Scale for Staining (AATCC EP2) or the AATCC 9-Step Chromatic Transference Scale (AATCC EP8).

3. Terminology

3.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

3.2 **drycleaning,** n.—the cleaning of fabrics with organic solvents such as petroleum solvent, perchloroethylene or fluorocarbon.

NOTE: The process also includes adding detergent and moisture to the solvent, up to 75% RH, and hot tumble drying to 71°C (160°F).

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 Perchloroethylene is toxic by inhalation, by repeated contact with the skin and by ingestion; it should be used only in a well vented atmosphere. Toxicology studies with laboratory animals have shown evidence of cancer in rats and mice exposed to perchloroethylene vapors at 100-400 ppm concentrations for prolonged times. Fabric saturated with this solvent should be dried in an adequately ventilated hood. Use chemical goggles or face shield, impervious gloves and an impervious apron when handling perchloroethylene.

4.4 An eyewash/safety shower should be located nearby and an organic vapor respirator should be readily available for emergency use.

4.5 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 11.3).

4.6 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

4.7 Any perchloroethylene residue must be sent to a permitted waste management facility. Any disposal must be in compliance with federal, state or local regulations.

5. Apparatus and Materials (see 11.4)

5.1 An accelerated laundering machine for rotating closed canisters in a thermostatically controlled water bath at 40 ± 2 rpm.

5.2 Standard stainless-steel containers, 7.5 cm diam × 12.5 cm (3 in. diam × 5 in.), of approximately 500 mL (1 pt) capacity, which can be closed using solvent-resistant gaskets.

5.3 Noncorrodible steel discs, 30.0 ± 2.0 mm (1.18 ± 0.08 in.) diam × 3.0 ± 0.5 mm (0.12 ± 0.02 in.) smooth and free from rough edges, weight 20.0 ± 2.0 g (0.7 ± 0.07 oz).

5.4 Undyed cotton twill cloth weighing 270 ± 70 g/m^2 (8 ± 2 oz/yd^2), free from finishes and cut into swatches 12.0 × 12.0 cm (4.75 × 4.75 in.).

5.5 Multifiber test fabrics (8 mm [0.33 in.] bands) containing acetate, cotton, nylon, silk, viscose rayon and wool. Multifiber test fabrics (8 mm [0.33 in.] bands) and (15 mm [0.6 in.] bands) containing acetate, cotton, nylon, polyester, acrylic and wool (see 11.5).

5.6 Perchloroethylene, commercial drycleaning grade.

5.7 Gray Scale for Color Change (AATCC EP1) (see 11.6).

5.8 Gray Scale for Staining (AATCC EP2) or AATCC 9-Step Chromatic Transference Scale (AATCC EP8) (see 11.6).

5.9 Colorimeter or spectrophotometer.

5.10 Crockmeter Test Cloth.

5.11 Detergent, Perk-Sheen (see 11.7).

6. Specimens

6.1 If the textile to be tested is fabric, use three specimens, each 10 × 5 cm (4 × 2 in.) with the longer dimension parallel to the warp or wale direction of the fabric.

6.2 If the textile to be tested is a yarn, knit it into a fabric and use three specimens 10 × 5 cm (4 × 2 in.) with the longer dimension parallel to the wale direction of the fabric.

6.3 Sample preparation.

6.3.1 White cotton fabric, multifiber 8 mm (0.33 in.) bands, of appropriate type. Prepare pieces with a 5 cm (2 in.) square of multifiber cloth or white cotton fabric (as required) sewn, stapled or suitably attached along one 5 cm (2 in.) edge of the test specimen and in contact with the face of the material. When multifiber test cloth is used, attach it so that each of the six fiber stripes is along the 5 cm (2 in.) edge of the specimen with wool on the right. The fiber stripes in the multifiber fabric will be parallel to the lengthwise direction of the test specimen.

6.3.2 Multifiber 15 mm (0.6 in.) bands. Prepare pieces with a 5 × 10 cm (2 × 4 in.) rectangle of multifiber cloth sewn, stapled or suitably attached centered

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along a 10 cm (4 in.) edge of the test specimen and in contact with the face of the material. Attach it so that each of the six fiber stripes will be parallel to the widthwise direction of the specimen. Attach and secure the wool stripe at the top of the specimen to avoid fiber loss.

6.3.3 It is recommended that knitted fabrics be sewn or stapled at the four edges to equivalent size pieces of 80 × 80 cm bleached cotton fabric to avoid rolled edges and to assist in obtaining a uniform test result over the entire surface.

6.3.4 For pile fabric specimens with pile lay direction, attach the multifiber fabric at the top of the specimen with the pile lay direction pointing away from the top of the specimen.

6.3.5 When the textile to be tested is yarn, specimens may be tested in one of two ways.

6.3.5.1 Option 1. Knit yarn on an appropriate sample knitting machine. Prepare specimens and multifiber test fabrics according to 6.1-6.3. Keep one knitted specimen of each sample as an original.

6.3.5.2 Option 2. Prepare four 120-yd skeins of each yarn. Fold the skeins so that there is a uniform amount of yarn across a 2 in. width with a length appropriate for the procedure to be used. Keep one skein of each sample as an original. Sew or staple crock squares folded over each end of the layered yarn specimen. Attach a multifiber test fabric to one end.

7. Procedure

7.1 Prepare a bag for each specimen with inside dimensions of 10 × 10 cm (4 × 4 in.) using the undyed cotton twill cloth (see 5.4) by sewing together two squares of this cloth around three sides. Place the specimen and 12 steel discs inside the bag (see 5.3). Close the bag by any convenient means, such as sewing.

7.2 In a hood, prepare the perchloroethylene/detergent solution. To a 1000 mL volumetric flask filled partially with perchloroethylene, add 10 mL of detergent, (Perk Sheen 324). Shake or stir. Add perchloroethylene until the total volume equals 1000 mL. Add 0.6 mL water. Shake or stir until the solution is no longer cloudy. This mixture produces a 1% charge volume/volume at 75% RH.

7.3 In a hood, place the bag containing the specimen and the steel discs in the standard 500 mL (1 pt) stainless steel container and add 200 mL perchloroethylene detergent solution at 30 ± 2°C (86 ± 4°F). Close the container and agitate the specimen for 30 min at 30 ± 2°C (86 ± 4°F) in the accelerated laundering machine.

7.4 Place the container in an adequately ventilated hood. Remove the bag from the container, withdraw the specimen and place the specimen between layers of absorbent paper or cloth to remove surplus solvent. Air dry the specimen in a hood. Air at a temperature not exceeding 65°C (149°F).

7.5 Allow specimens to condition at 65 ± 5% RH and 21 ± 2°C (70 ± 4°F) for 1 h before evaluating.

7.6 Prepare the tested specimens and staining fabric for evaluation by trimming off raveled yarns and lightly brushing off any loose fiber and yarn on the fabric surfaces. Brush pile fabric specimens in required direction to restore them as nearly as possible to the same pile angle as the untreated specimens. Specimens should be smoothed or flattened if there are wrinkles. Specimens may be mounted on cards to facilitate identification and handling in their evaluation. Mounting must not interfere with rating as specified in AATCC EP1, AATCC EP2 and AATCC EP8. For consistency in backing material, use a white mounting card with Y tristimulus value of at least 85%.

7.6.1 Yarn skein specimens should be combed and brushed for improved alignment of yarns before comparison with the unwashed original. The original specimens may also need further combing and brushing for uniformity of appearance.

8. Evaluation and Calculation

8.1 Evaluation of Color Change.

8.1.1 Evaluate each test specimen for change in color by comparison with the Gray Scale for Color Change (AATCC EP1), or using AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen, and record the numerical rating that corresponds to the appropriate one on the Gray Scale. For improved precision and accuracy, the specimens should be rated by more than one rater.

8.1.2 The color change can be quantitatively determined between the original sample and a test specimen using a suitable colorimeter or spectrophotometer (see AATCC EP7).

8.2 Evaluation of Staining.

8.2.1 Evaluate the staining of the multifiber fabric used in the test by comparison with AATCC EP2, AATCC EP8 or AATCC EP12, Instrumental Assessment of Degree of Staining, and record the numerical rating that corresponds to the appropriate one on either of them. The scale used should be indicated when reporting.

8.2.2 The color transferred to the multifiber fabric or white cotton square of 6.3.1 can be quantitatively determined by measuring the color difference between a piece of the original material and the stained material. Using a suitable colorimeter or spectrophotometer, the 15 mm (0.6 in.) bands multifiber material has sufficient width to be within the aperture diameter capability of many instruments.

8.3 Calculation.

8.3.1 Average the observations made on each sample (means of judgement of 3 specimens).

9. Calculating and Reporting Results

9.1 Average the number of observations made on each sample (means of judgments on each of 3 specimens). Report the average to the nearest tenth of a grade point.

9.2 Report the test method number.

9.3 Report the grade number determined for color change in 9.1 and the staining grade number for the evaluated fibers in the multifiber test fabric and/or staining fabric as determined in 9.2.

9.4 State which scale (Gray Scale for Staining or AATCC 9-Step Chromatic Transference Scale) was used in evaluating staining (see 11.8).

9.5 State which multifiber was used, and if the cotton print cloth was employed to avoid knit curling.

10. Precision and Bias (see 11.9)

10.1 *Precision*. On one fabric, three specimens were tested by each of two operators in three trials. The specimens were rated by two evaluators. The average grade for the two raters and three specimens was determined. This was considered the unit of measure of this test.

10.2 The components of variance as standard deviations of the color change grades were calculated to be as follows:

Single operator	0.03
Within laboratory	0.11

10.3 *Bias*. The true value of the color change can only be defined in terms of the test method. Within this limitation, the test method has no known bias.

11. Notes

11.1 This test method covers colorfastness to drycleaning only; it does not cover the effects of water spotting, solvent spotting and steam pressing which are normally involved in commercial drycleaning practice. For testing the durability of applied designs and finishes see AATCC TM86, Test Method for Durability of Applied Designs and Finishes to Drycleaning.

11.2 Perchloroethylene is used in this test because (a) it is the most used solvent in commercial drycleaning in the USA, and (b) it is slightly more severe in solvent action than petroleum. A color which is not affected by perchloroethylene will not be affected by petroleum solvents, whereas the converse is not always true.

11.3 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

11.4 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.

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aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

11.5 Bleached cotton test fabric in 32 × 32 ends × picks/cm (80 × 80 ends × picks/in.) construction, 136 ± 10 g/m^2 (4.0 ± 0.3 oz/yd^2) and without optical brightener.

11.6 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.7 Available from Adco Inc., 900 W. Main St., P.O. Box 999, Sedalia MO 65301; tel: +1.660.826.3300 or +1.800.821.7556; fax: +1.660.826.1361; e-mail: sales@adco-inc.com; web site: www.adco-inc.com.

11.8 For very critical evaluations and in cases of arbitration, grades must be based on the Gray Scale for Staining.

11.9 The precision of this test method is dependent on the combined variability of the material being tested, the test method itself, and the evaluation procedure utilized.

11.9.1 The precision statement shown in Section 10 was developed from results obtained by visual evaluations (AATCC EP1 for color change and AATCC EP2 and AATCC EP8 for staining).

11.9.2 It is expected that the use of instrumental evaluations (AATCC EP7 and AATCC EP12) will result in greater precision than that obtained from visual evaluations.

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Test Method for Colorfastness to Heat: Hot Pressing

1. Purpose and Scope

1.1 This test method is intended for determining the resistance of the color of textiles of all kinds and in all forms to color change, and color transfer when subjected to hot pressing. Tests are given for hot pressing when the fabric is dry, damp and wet. The textile end use usually determines which tests should be made.

2. Principle

2.1 Dry Pressing. The dry specimen is pressed with a heating device at a specified temperature and pressure for a specified time.

2.2 Damp Pressing. The dry specimen is covered with a wet, undyed cotton cloth and pressed with a heating device at a specified temperature and pressure for a specified time.

2.3 Wet Pressing. The wet specimen is covered with a wet, undyed cotton cloth and pressed with a heating device at a specified temperature and pressure for a specified time.

2.4 Color change of specimen and staining of cotton cloth are each reported as a grade from 5 to 1, with 5 representing no color change or staining and 1 representing the most color change or staining.

3. Referenced Documents

3.1 AATCC EP1, Evaluation Procedure for Gray Scale for Color Change (see 14.1).

3.2 AATCC EP2, Evaluation Procedure for Gray Scale for Staining (see 14.1).

3.3 AATCC EP7, Evaluation Procedure for Instrumental Assessment of the Change in Color of a Test Specimen (see 14.1).

3.4 AATCC EP8, Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale (see 14.1).

3.5 AATCC EP12, Evaluation Procedure for Instrumental Assessment of Degree of Staining (see 14.1).

3.6 ASTM E1402, Standard Guide for Sampling Design.

4. Terminology

4.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

4.2 **hot pressing,** n.—a process for smoothing and possibly shaping textile products by applying mechanical pressure with heat, either dry or in the presence of moisture.

5. Safety Precautions

5.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

5.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

5.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

6. Apparatus and Materials (see 14.2)

6.1 Heating device, providing even heat transfer to the specimen from the top by close contact at a controlled temperature (110-200°C, see 10.1) and giving a pressure on the specimen of 40 ± 10 g/cm^2 (see 14.3, 14.4).

6.2 Smooth, heat-resistant sheet (see 14.5).

6.3 Wool flannel, approximately 260 g/m^2 (see 14.3). Two layers of this material are used to make a pad approximately 3 mm thick. Other smooth wool fabrics or felt may also be used to achieve this thickness.

6.4 Cotton cloth, undyed, bleached, unmercerized, with a smooth surface, 100-130 g/m^2.

6.5 Equipment and environment for evaluating color change.

6.5.1 For visual evaluation, use Gray Scale for Color Change (see 14.1). Refer to AATCC EP1 for additional materials and use of gray scale.

6.5.2 For instrumental evaluation, use a spectrophotometer as specified in AATCC EP7.

6.6 Equipment and environment for evaluating staining.

6.6.1 For visual evaluation, use Gray Scale for Staining or AATCC 9-Step Chromatic Transference Scale (see 14.1). Refer to AATCC EP2 for additional materials and use of the gray scale. Refer to AATCC EP8 for additional materials and use of the chromatic scale.

6.6.2 For instrumental evaluation, use a spectrophotometer as specified in AATCC EP12.

7. Sampling

7.1 Test results are valid only when the samples are statistically representative (see ASTM E1402).

7.2 Sampling must be random. Every unit of product must have a mathematically equal chance of becoming a sample; and every portion of each sample must be equally likely to become a test specimen.

7.3 All specimens must be alike within the variations due to pure chance. There must be no differences within samples assignable to known causes.

8. Specimens

8.1 If the textile to be tested is fabric, a 12 × 4 cm specimen is used.

8.2 If the textile to be tested is yarn or thread, knit it into a fabric specimen 13 × 4 cm or wind it around a piece of thin, inert material 12 × 4 cm.

9. Conditioning

9.1 Condition specimens, wool flannel, and undyed cotton cloth for at least 4 h in an atmosphere of 21 ± 2°C (70 ± 4°F) and 65% ± 5% RH by laying each piece separately on a screen or perforated shelf of a conditioning rack.

10. Procedure

10.1 Use one of the following temperatures (see Table I, 14.6):

110 ± 2°C
150 ± 2°C
200 ± 2°C

When necessary, other temperatures may be used, provided they are specifically noted in the report. Preheat the top plate of heating device to test temperature for at least 15 min. During the preheating period and between test periods, both plates of the heating device should be in contact to ensure uniform heat distribution.

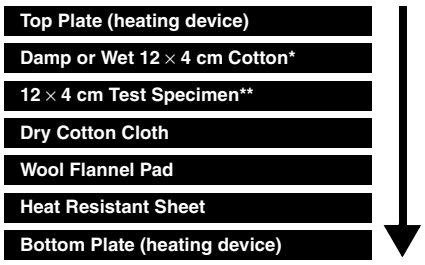
10.2 Cover the bottom plate of the heating device with the heat resistant sheet. Place wool flannel over the sheet. Place dry, undyed cotton cloth over wool flannel (see Fig. 1). Allow the sheet to cool and fabric to dry between tests.

10.3 Dry Pressing. Place dry test specimen on top of the cotton cloth (see Fig. 1).

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Table I—Safe Pressing Temperature Guide (see 14.6)

Class 0 < 121°C	Class I 121-135°C	Class II 149-163°C	Class III 177-191°C	Class IV ≥ 204°C
Modacrylic 93-121°C	Acetate	Acrylic	Nylon 6,6	Cotton
Olefin (Polyethylene) 79-121°C	Olefin (Polypropylene)	Azlon	Polyester	Fluorocarbon
		Nylon 6		Glass
				Hemp, Jute, Ramie
Rubber 82-93°C	Silk	Spandex		Linen
		Wool		Rayon, Viscose
Saran 66-93°C				Triacetate (heat set)
Vinyon 54°C				



*Damp or Wet Pressing only. Evaluate staining after 4 h conditioning.
**Evaluate color change immediately and after 4 h conditioning.

Fig. 1—Testing set-up for AATCC TM133.

Lower top plate of heating device and leave specimen for 15 s at the specified pressing temperature.

10.4 Damp Pressing. Place dry specimen on top of the cotton cloth. Soak a second piece of undyed cotton cloth (12 × 4 cm) in distilled water, and squeeze or extract it to contain its own weight of water (100% wet pickup). Place the wet cloth on top of the dry specimen (see Fig. 1). Lower the top plate of the heating device and leave specimen for 15 s at the specified pressing temperature.

10.5 Wet Pressing. Soak the test specimen and a piece of undyed cotton cloth (12 × 4 cm) in distilled water and squeeze or extract them to contain their own weight of water (100% wet pickup). Place the wet specimen on top of the dry cotton cloth. Place the wet cotton cloth over the specimen. Lower the top plate of the heating device and leave specimen for 15 s at the specified pressing temperature.

11. Evaluation

11.1 Immediately after testing, evaluate the color change of the test specimen as directed in AATCC EP1 or AATCC EP7.

11.2 After conditioning for 4 h in an atmosphere of 65 ± 5% RH and a temperature of 21 ± 2°C (70 ± 4°F), evaluate the color change of the test specimen as directed in AATCC EP1 or AATCC EP7.

11.3 For Damp or Wet Pressing, condition the top cotton cloth (placed damp or wet over the specimen for testing) for 4 h in an atmosphere of 65 ± 5% RH and a temperature of 21 ± 2°C (70 ± 4°F). Evaluate the staining on the conditioned cloth as directed in AATCC EP2, AATCC EP8, or AATCC EP12 (see 14.7, 14,8).

12. Report

12.1 Describe or identify the sample tested.

12.2 Report that the sample was tested using AATCC TM133-2020e.

12.3 Report the testing conditions:

12.3.1 Test procedure (dry, damp or wet).

12.3.2 Temperature of the heating device.

12.3.3 Wet pickup (%) of cotton cloth and tested sample, if applicable.

12.3.4 Evaluation procedure for color change (AATCC EP1 or AATCC EP7).

12.3.5 Evaluation procedure for staining (AATCC EP2, AATCC EP8, or AATCC EP12).

12.4 Report the test results:

12.4.1 Color change grade for tested specimen immediately after pressing.

12.4.2 Color change grade for tested specimen after pressing and conditioning.

12.4.3 for Damp or Wet Pressing, staining grade for cotton cloth after pressing and conditioning.

12.5 Describe any modification(s) of the published method.

13. Precision and Bias

13.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

13.2 *Bias*. Colorfastness to heat, hot pressing, can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

14. Notes

14.1 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

14.2 For potential apparatus, reagents or materials sources, visit the *AATCC Buyer's Guide* at www. aatcc.org/bg. AATCC provides its Corporate members the option to list their items and services. AATCC does not qualify, or in any way approve, endorse or certify that any of the listings meet the specifications in its standards.

14.3 To obtain the pressure per unit area 40 ± 10 g/cm^2 the total area of the wool flannel padding should bear a suitable relationship to the mass of the plate pressing down on the padding. If the fabric to be tested has an appreciable thickness, it is necessary either to increase the area of the test specimen or to augment the pressure-bearing surface using a suitable template made from the same material as the test specimen. If the plates of the heating device are smaller than the specimen size, the pressure depends on the design of the apparatus (ratio of mass and area of top plate).

14.4 For less critical testing, a household iron may be used; but its temperature should be measured with a surface pyrometer or with temperature sensitive papers. The iron should be weighted so that its area and total weight are in the approximate ratio to exert a pressure of 40 ± 10 g/cm^2. However, due to temperature fluctuation during on-off differences over the iron surface, the accuracy and reproducibility are limited. When a hand iron is used, it must be stated in the report.

14.5 The heat resistant sheet is used for insulation and should be smooth and not warped. It is best to complete specimen assembly on the heat resistant sheet before placing it on the heating device. This insulating material must be used, whether the bottom plate is heated or not, to prevent heat to or from the bottom of the test assembly.

14.6 The pressing temperature selection depends to a large extent on the type of fiber and on the construction of the fabric or garment. In the case of blends, use the temperature appropriate to the fiber with the lowest heat resistance.

14.7 For very critical evaluations and in the case of litigation, ratings must be based on the geometric Gray Scale for Staining or the instrumental equivalent.

14.8 CAUTION: It has been reported that the results for staining obtained by this method on fabrics dyed to dark shades (navy, black, etc.) that contain a combination of polyester and spandex, or their blends, may not show the full staining propensity of such fabrics in consumer use. It is, therefore, recommended that the staining results obtained by this test not be used for the acceptance testing of such fabrics.

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15. History

15.1 Editorially revised in 2021 to update test method reference to match current title.

15.2 Revised in 2020 to align with prescribed AATCC style guidelines. Editorially revised 2016. Revised 2013. Editorially revised 2010. Editorially revised and reaffirmed 2009. Editorially revised 2008. Editorially revised and reaffirmed 2004. Editorially revised 2002, 2001. Editorially revised and reaffirmed 1999, 1994. Editorially revised 1991. Reaffirmed 1989. Editorially revised 1986, 1985. Reaffirmed 1984. Editorially revised 1981. Reaffirmed 1979, 1976. Revised 1973.

15.3 Developed in 1969 by AATCC Committee RR54 (supersedes AATCC TM5-1962). Maintained by RA99.

15.4 Adopted as ISO 105-X11 in 1987.

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Test Method for Electrostatic Propensity of Carpets

Developed in 1969 by AATCC Committee RR32; jurisdiction transferred in 2007 to AATCC Committee RA57; revised 1975, 1979, 1991, 2001, 2011, 2016, 2019; reaffirmed 1986, 1996, reaffirmed and editorially revised 2006; editorially revised 2007, 2008, 2013. Related to ISO 6356.

1. Purpose and Scope

1.1 This test method assesses the propensity of carpets to generate static when a person walks across them. This method uses controlled laboratory simulation of the conditions, that may be encountered in use. The conditions used in this test include those known from experience to be contributors to excessive accumulation of static charges.

1.2 This test method does not include standards of performance. The specification requirements pertinent to any given application can vary considerably, depending on the specific demands of the application. It is recommended that specifiers evaluate results with caution as large variations were noted during inter- and intralab testing using this test method.

2. Principle

2.1 A carpet brought to moisture equilibrium at controlled atmospheric conditions is walked on by a test operator in a specified manner with specified shoe soles and heels. The static charge, which builds up on the operator, is monitored continuously by a voltage indicator with a recorder.

2.2 The maximum voltage, generated on the person by the charge accumulated, and measured during the test period is considered to be the static-generating propensity of the carpet under the conditions of this test.

3. Terminology

3.1 **electrostatic propensity,** n.—*of carpets*, the ability to produce and accumulate an electrostatic charge.

NOTE: For the purposes of this test, it is the *resultant voltage* on a person walking across the surface of a textile floor covering under specified conditions, which has been caused by the accumulation of an electrostatic charge on the body.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 Isopropyl alcohol is a flammable liquid and should be stored in the laboratory only in small containers away from heat, open flames and sparks.

4.4 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

4.5 Ground all electrical equipment.

4.6 High-voltage sources should have an internal impedance of not less than 1×10^8 ohm (or the maximum output current limited to 1 milliampere) to avoid shock hazard in calibration of the detection system.

5. Uses and Limitations

5.1 Build-up of a static charge on a person walking across a carpet under conditions of low atmospheric humidity has been recognized as a classic example of the triboelectric effect, whereby a separation of electrical charges is produced when two dissimilar surfaces in contact are separated. The magnitude of the charge separation and the resultant voltage on the person vary under the influence of many factors. The most important factors for the purpose of this test are:

(a) the chemical and physical characteristics of the two materials brought into contact, rubbed and separated; i.e., the shoe soles and carpet;

(b) the surface contamination on one or both;

(c) the nature of the rubbing and/or separation, i.e., the method of walk, including the height of the shoe above the carpet; and

(d) the ambient conditions (especially the relative humidity).

6. Apparatus and Materials

6.1 A room that can be maintained at a temperature of 21 ± 2°C (70 ± 4°F) and at 20 ± 5% RH, and suitable means for monitoring these conditions. Special requirements of end-use may dictate the use of other testing conditions (e.g., for aircraft or for controlled-humidity environments) and the test facility should be capable of achieving and maintaining those conditions.

6.1.1 The room should be of such dimensions and so arranged that the test operator comes no closer than approximately 600 mm to extraneous grounded or charged surfaces (such as walls or workbenches) during testing and should be equipped with open-wire shelves, racks or horizontal rods suitable for hanging carpet samples for conditioning. Air must pass freely around all surfaces of the samples to facilitate moisture equilibration.

6.2 Test sandals should be used only for testing carpet. New test sandals should be cleaned before use as prescribed in 9.7.1. Soles must be attached to the sandals by gluing and/or stitching. Tacks or rivets must not be used.

6.2.1 One pair of AATCC TM134 sandals (see 13.1) manufactured with XS 664 P-HK Neolite soles. A second pair of Neolite test shoes or sandals with optional self-adhesive (i.e., replaceable) suede leather attached to the shoe sole. The second pair of Neolite shoes or sandals must be dedicated to this use only.

6.2.2 Reagent grade isopropyl alcohol and cheesecloth.

6.3 Electrostatic detection and recording means for voltages up to at least 20 kV (see 13.2 and 13.3).

6.3.1 The system should have a high input impedance and be suitably damped such that a steady recorder or meter indication of 3000 V will, when the input is grounded, decay to 1500 V in not less than 1 s or more than 3 s (this corresponds to a time constant of 1.4-4.3 s). To achieve reasonable accuracy in use, the input capacitance, including lead-in wire, should not exceed 30 pF. (See Appendix A for additional information on measurement of damping and Appendix B for information on damping techniques.)

6.4 A grounded metal plate approximately 1200 × 1200 mm.

6.5 A standard 1350 g/m^2 rubberized jute/hair cushion underlay (see 13.1) at least 1200 × 1200 mm. The vertical resistance through the cushion shall not be less than 10^{12} ohms (see 13.6).

6.6 A set of AATCC TM134 Static Control Carpets (see 13.1) consisting of static unprotected and static protected versions.

6.7 An optional metronome.

6.8 An optional handheld blowing-fan balanced ionizer (see 13.4).

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7. Specimens

7.1 Cut each carpet test specimen to a size of approximately 900 × 900 mm or 1000 × 1000 mm. If test specimens are less than this size, multiple specimens may be pieced together to reach the required test size.

8. Conditioning

8.1 Condition specimens under the required test conditions, 21 ± 2°C (70 ± 4°F) and 20 ± 5% RH, for a minimum of 48 h prior to testing. This will equilibrate the moisture content of the specimens to the test conditions. The temperature and relative humidity should be automatically recorded during the conditioning period with a chart recorder or other electronic recording device.

8.1.1 Hang or lay specimens in such manner that free circulation of air over both surfaces is possible.

9. Procedure

9.1 Test AATCC TM134 Static Unprotected Control Carpet *without* an underlay (to avoid charging the underlay) and the Static Protected Control Carpet *with* an underlay at the beginning of each testing session. If the results are outside of the test laboratory's Control Chart Control Limits, then seek the cause of the deviation and correct the problem. Check relative humidity and temperature (current values and recorded values during the specimen conditioning period) (see 6.1). Test backup Control Carpets if available (see 6.6), reclean soles (see 9.7) and check instrument calibration. Use a blowing-fan, handheld, balanced high-voltage ionizer to neutralize any charge on the underlay pad between tests and prior to testing any specimen, moving it slowly over the entire surface at a height of about 100-200 mm above it for at least 1 min.

9.2 Test the specimen once a day until two reproducible voltages are obtained. Voltages are considered reproducible when consecutive tests agree within 10% or 0.5 kV, whichever is greater. Leaving the specimen in the test chamber overnight should allow any static charges to be dissipated. This process may be accelerated by using a blowing-fan, balanced high-voltage ionizer to neutralize any charges. If an ionizer is used to thoroughly neutralize the specimen (held about 100-200 mm above the specimen and moved slowly over its entire surface, requiring approximately 1-2 min), the same specimen may be retested the same day.

9.3 Test the specimen over the standard underlay (see 6.5), which is placed on the grounded metal plate.

9.3.1 Residual static charges on the specimen and underlay have been found to be a major source of error. Residual charges should decay naturally over a period of several hours. Specimens should be hung undisturbed for a minimum of 4 h, or thoroughly neutralized with a blowing-fan, balanced high-voltage ionizer, prior to each testing, as above.

9.3.2 Lay the underlay on the grounded metal plate, rubberized side up. Avoid dragging the underlay across the surface or rubbing different parts of it together.

9.3.3 If a blowing-fan, balanced high-voltage ionizer is available, thoroughly and slowly cover the entire surface of the underlay with the ionizer output, holding the ionizer about 100-200 mm above the underlay. This will neutralize any residual charges from handling.

9.3.4 Lay the specimen on the underlay carefully to avoid excessive friction against the mat or rubbing different parts of it together.

9.3.5 If a blowing-fan, balanced high-voltage ionizer is available, thoroughly and slowly cover the entire surface of the sample with the ionizer output, holding the ionizer about 100-200 mm above the sample. This will neutralize any residual charges from handling.

9.3.6 Zero the test apparatus. Place clean test sandals (see 9.7) on the specimen, grasp the test probe and ground the test operator while standing off the specimen in stocking feet. Ensure that the recorder reads "0." Step into the clean test sandals specified for the test while standing on the specimen and holding the test probe, being careful not to move the sandals around on the specimen.

9.4 After following the procedure in 9.3, there should be very little voltage indication. Presence of significant voltage (more than 200 V) indicates excessive initial charge somewhere in the system and the procedures in 9.2 and 9.3 must be repeated to avoid possible significant errors. Since the location of the spurious charge is important in affecting the final test results, it is not sufficient to simply reground the test subject, although this will obviously create a new zero value.

9.5 If the carpet is not intended for installation over an underlay, then the specimen shall be tested without an underlay. In such cases, test the specimen on the grounded metal plate and record the fact that the test was performed on the specimen mounted directly on a grounded surface. Otherwise, follow the procedure outlined in 9.3.

9.6 Proceed with the tests described below in 9.6.1 with the neolite soles, and if required, the suede leather soles. NOTE: Be sure to keep SEPARATE shoes or sandals for the Neolite and suede leather tests; do not remove the leather and test with the underlying Neolite, since it will be contaminated by the adhesive. Note that the samples and underlay must be discharged between repeat tests or between types of tests or between tests with different soles, in order to avoid errors. Discharge may be achieved by allowing natural discharge over 4 h or overnight, or neutralization with a blowing-fan, handheld balanced high-voltage ionizer, as above.

9.6.1 Tests.

Test I—Step Test/Neolite Soles

Test II—Scuff Test/Neolite Soles (Optional)

Test III—Step Test/Self-adhesive suede leather soles (Optional)

Test IV—Scuff Test/Self-adhesive suede leather soles (Optional)

9.6.2 **Step Test Procedure.** While holding the hand probe, walk on the specimen using a simple box step, lifting the sandals as close as possible to 80 mm above the specimen at the rate of 120 ± 10 steps per min (a metronome can be used). Always keep the plane of the sandal soles parallel to the plane of the specimen. Do not scuff or rub the sandals over the specimen. Cover the entire surface of the specimen during stepping, for a total test time of 1 min or until a consistent peak voltage is observed for each step. Touch the ground with the hand probe to bring the body voltage to "0." Neutralize the carpet and underlay prior to any further testing.

9.6.3 **Scuff Test Procedure.** While holding the probe, scuff (or wipe) as though wiping chewing gum from the bottom of both sandals. The wiping action is always in a backward motion. As above, the bottom of the sandal should reach a height of as close as possible to 80 mm above the specimen, parallel to the sample. Repeat scuffing at a rate of 60 ± 5 steps per min and continue the motion for about 1 min while covering the entire specimen. Touch the ground with the hand probe to bring the body voltage to "0." Neutralize the carpet and underlay prior to any further testing.

9.7 Remove and clean the sandal. Hang or lay the specimen on the conditioning rack. The procedures in 9.7.1 and 9.7.2 will usually suffice to clean the sandals. Sandals must be cleaned with extreme care after testing specimens, which have had surface-type (spray-on, topical) antistatic treatments (see 13.7). Failure to do so may transfer material from one specimen to another.

9.7.1 Clean Neolite sandal soles by swabbing with cheesecloth or paper towel moistened with isopropyl alcohol, using a fresh cheesecloth or paper towel. In case of extensive contamination, repeat this procedure, sand the sole with fine sandpaper to expose fresh material and clean again.

9.7.2 The suede leather soles are difficult to clean, once contaminated. Sanding

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the sole may remove the contamination. Other cleaning methods may contaminate the leather or change its electrical properties (such as water absorption from isopropanol/water cleaning solution). Replace these soles if sanding does not readily remove the contamination.

9.7.3 Store the test sandals in the controlled relative humidity of the test area.

9.8 Record the test parameters, including specimen identification, specimen tested "as received" or "cleaned," date, temperature, relative humidity, sandal sole and walking procedures (Step or Scuff).

9.9 In case of any uncertainty about the consistency of the test conditions, retest the AATCC Static Control Carpets at the end of and throughout the testing session to assure that testing conditions have not changed. If the Control Carpet measurements are significantly different (outside of the test laboratory's established Control Limits for the control specimens), then the specimen test results are unreliable.

10. Analysis of Results

10.1 The chart trace serves as a permanent record of the test and characterizes the carpet for electrostatic propensity. The maximum voltage (at the highest point of each step), the sign of the voltage and the rate-of-rise of voltage are characteristics available from the chart trace and have been found to be relevant to the performance of a carpet under use conditions similar to those of the test.

10.2 Maximum voltage, Step and Scuff. The maximum voltage is defined as the maximum achieved for several successive steps. An example is given in Fig. 1.

10.2.1 Rate of rise of voltage may be reported, Step and Scuff. The average rate of increase of voltage (in kV/s) is taken as the reported maximum voltage divided by the time required to reach that voltage from the start of the test walk.

10.3 Effect of contaminants. Carpets that have been contaminated by spills, soiling, contact with other carpets or materials that have removable chemical components, etc., will yield results that are different from clean carpets. Since the triboelectric effect depends upon the difference between the flooring and the shoes, these contaminants may increase or decrease the measured electrostatic propensity. Similarly, non-permanent topical treatments will affect the results; the variation of such treatments over time and after cleaning should be considered by the test requester. To remove contaminants, a hot-water extraction procedure is recommended, such as AATCC TM171, Test Method for Carpets: Cleaning of; Hot Water Extraction.

10.4 Effect of sole materials. Very large differences in laboratory and field results will often occur as a result of use of different shoe sole materials. The triboelectric, surface roughness and conductivity characteristics of the materials are important. Higher conductivity soles will tend to suppress the accumulation of charge on a person particularly when carpets having static-control filaments are tested.

10.4.1 Neolite XS 664 P-HK has been chosen as the primary reference material because its static performance is much like that of many common shoe soles; it is easy to keep clean and its chemical and physical properties are believed to be quite uniform. Its triboelectric properties are distinct from those of nylon, the dominant carpet fiber polymer. Other types of carpet should also be tested with the secondary reference material for a more complete characterization of the static propensity of the carpet.

10.4.2 Suede leather has been chosen as the secondary reference material because typically it is representative of a certain class of leathers whose triboelectric performance differs significantly from that of Neolite soles, in that they tend to give high values on acrylic, polyester and polypropylene carpets.

10.4.3 For some purposes it may be desirable or necessary to characterize carpet performance with special shoes, such as ESD (electrostatic discharge) control footwear. Equivalent test procedures can be conducted with any relevant sole materials or shoes, but such tests should be considered supplementary information only.

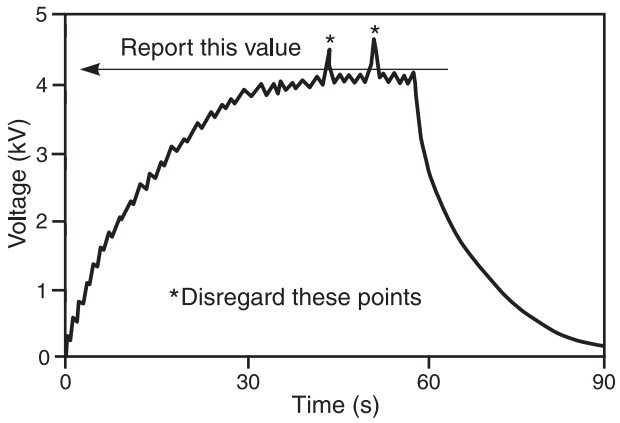


Fig. 1—Typical chart trace indicating maximum voltage.

11. Report

11.1 Two reproducible voltages and their average should be reported (see 9.2) for the Step test method with neolite soles with the highest voltage or for each optional test method as requested, with the polarity indicated as positive or negative. (The polarity does not affect the impact of static buildup on people or equipment, but does serve a diagnostic purpose. Only the magnitude of the results should be considered when comparing the test results with floor covering static-control specifications.) Optional scuff test measurements should also be reported, if required, for comparison; Scuff test measurements are more variable *within-lab* and *between-lab* than step test results. If Scuff test measurements differ from Step test results by more than 2.5 kV, the report should explain that this may be an indication that a topical treatment or contaminant may be present on the specimen. When a carpet is evaluated for the full triboelectric effect the step and scuff methods should be performed with neolite and suede leather soles.

11.2 Report the AATCC Static Control Carpet test results for the same day(s) as the specimen test(s), the laboratory's Upper Control Limit and Lower Control Limit, and the laboratory's established measurement standard deviation for each control, for each type of test (see 13.5).

11.3 The test report should include the testing conditions (relative humidity and temperature), test method and version, and any observations or apparent anomalies.

11.4 The report should state whether an ionizer was used to neutralize charges on the samples prior to each test. If specimens were tested without an underlay, the presence or absence of the underlay should be noted.

11.5 The report should state whether the carpet was tested "as received" or what preparation technique (e.g., "cleaned using AATCC TM171") was used. A caveat on the use of AATCC TM134 should be included which states, "The results of this test relate to the sample of carpet tested. Its static performance may be altered in service as a result of wear, soil-

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ing, cleaning, temperature, relative humidity, etc."

12. Precision and Bias

12.1 Users of data from AATCC TM134 should be aware that large variations in results have been noted during interlab and intralab testing. Preliminary evaluation of the round robin test results, in the range of 0-6 kV, shows that differences of less than approximately 0.5 kV are not significant, and this must be considered when comparing test results to specification requirements. An initial study on TM134 Static Protected Control Carpet with shoes fitted with XS 664 P-HK Neolite soles (not AATCC TM134 sandals) produced a Step procedure average voltage of 2.7 kV with a standard deviation of 0.3 kV, based on data from seven test sites.

13. Notes

13.1 AATCC TM134 sandals, both large and small sizes, suede leather, rubberized jute/hair cushion carpet underlay and AATCC TM134 Static Control Carpets (Protected and Unprotected) are available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@ aatcc.org; web site: www.aatcc.org.

13.2 A suitable electrometer, (digital) or (analog), a 1000:1 voltage divider probe. This apparatus may require slight modification to conform to requirements in 6.3.1.

13.3 Two methods for system damping testing are outlined in Appendix A.

13.4 Handheld blowing-fan, balanced high-voltage ionizers are available from a number of distributors under various names including ionized-air blowers. These are typically 4" fans with integral electronics and high-voltage ion-emission points.

13.5 The test laboratory must maintain a record of AATCC Static Control Carpet results, and generate suitable Control Charts. The Upper Control Limit and Lower Control Limit Values established through this analysis are to be used to define when all of the experimental conditions and equipment parameters are operating within the expected range. Non-random long-term drift in control carpet readings should be investigated to determine the nature of the changes that have taken place and corrective action taken.

13.6 If required, use National Fire Protection Association Test Method 99 or ESD/EOS STM 7.1 to measure resistance. For products intended to be installed in electrostatic-discharge-sensitive areas (i.e., electronics component manufacturing or assembly areas), several test methods developed by the Electrostatic Discharge Association (ESD Association) may be appropriate (ESD/EOS Standard Test Methods).

13.7 If the shoe soles or heels become permanently contaminated by testing carpets with topical antistats or yarn overfinishes, they should then be resoled or discarded immediately. The standard reference carpets recommended in the method can be very helpful in determining when soles or heels are contaminated, but these are also subject to contamination by contaminated shoes. Spare reference carpets should be stocked in case of contamination carry-over when testing the shoe condition.

Appendix A
Damping Testing

A.1 Two methods to test the damping are as follows:

A.1.1 With a high-voltage power supply capable of producing at least 3000 volts:

(a) Set up the electrometer (and recorder, if available) for at least 3 kV maximum range (10 kV preferred).

(b) Connect the power supply high-voltage output to the electrometer input to the divider probe.

(c) Check the power line grounds, and if in doubt, run a line from both the electrometer and power supply ground terminals to a common ground.

(d) After allowing the electrometer and power supply to warm up per the manufacturer's specifications, adjust the power supply output for a steady 3 kV reading on the electrometer.

(e) Switch the power supply off for a moment and observe the electrometer response. Most commercial high-voltage units have a built-in grounding of the output when turned off for safety reasons. If the unit in use does not show grounding when switched off, a properly shielded single pole, double throw switch must be connected to the electrometer input so it can be quickly switched from high-voltage to ground.

(f) With a steady 3 kV reading on the electrometer (and recorder), switch the electrometer input to ground (see Section 5) and measure the time it takes the voltage reading to drop to 1.5 kV. This should be between 1-3 s.

A.1.2 Electrometer without a high-voltage power supply:

(a) Set up as for a standard carpet test using a sample known to generate reading of 5 kV or over (AATCC TM134 Static Unprotected Control Carpet).

(b) Have the operator proceed to walk (or scuff) test normally and observe the electrometer reading.

(c) When the reading reaches 5 kV or more, the operator should stop moving and watch the reading without touching anything.

(d) When the voltage has leaked off enough to drop the reading to 3 kV, the operator will ground the electrometer input (touch probe tip to ground, use a jumper wire, or a wire and switch combination). Record the time it takes the reading to drop to 1.5 kV.

(e) This procedure should be repeated five times or more, and the tests averaged to smooth out reading inaccuracies.

A.1.3 The decay to 50% of the original signal has been selected as a convenience to simplify the readout of the electrometer and recorder. The 1-3 s delay corresponds to a time constant (t = 1/e) 1.4-4.3 s.

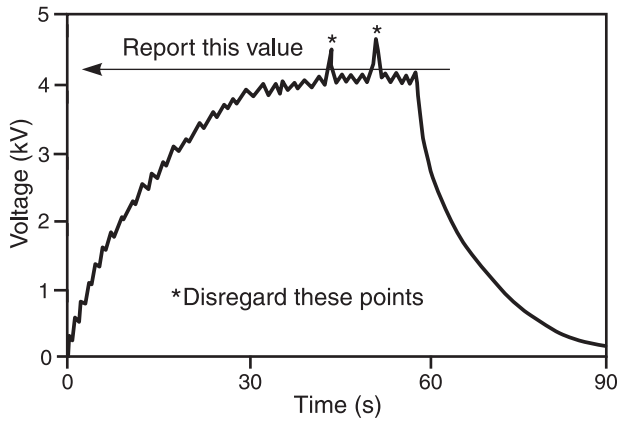


Fig. 1—Typical chart trace indicating maximum voltage.

Appendix B
Damping Techniques

B.1 For electrometers with recorders, the preferred method of adjustment of response time is a filter network between electrometer and recorder. Details of how to select a network can be found in electronic handbooks under integrating or low pass filter headings.

B.2 An electrometer with an analog display often has a built-in response in the desired range. Modification of a non-complying unit should be referred to the manufacturer or an electronics engineer.

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Test Method for Dimensional Changes of Fabrics after Home Laundering

Developed in 1970 by AATCC Committee RA42; reaffirmed 1973, 2000; revised 1978, 1987, 1995, 2001, 2003 (with title change), 2004, 2010, 2012, 2014, 2015, January 2018, November 2018; editorially revised 1982, 1985, 1989, 1990, 1991, 1996, 1997, 2006, 2008, 2016, June 2018; editorially revised and reaffirmed 1992. Related to ISO 3759.

Foreword

Standard laundering procedures remain consistent to allow valid comparison of results. Standard procedures represent, but may not exactly replicate, current consumer practices, which vary over time and among households. Alternate laundering procedures and machine parameters may be found in AATCC LP1, Laboratory Procedure for Home Laundering: Machine Washing (see 12.3), AATCC LP2, Laboratory Procedure for Home Laundering: Hand Washing (see 12.3) and ISO 6330, Textiles — Domestic washing and drying procedures for textile testing (see 12.10).

1. Purpose and Scope

1.1 This test method is intended for the determination of dimensional (length and width) changes of fabrics when subjected to home laundering procedures. Four washing temperatures, three agitation cycles, and four drying procedures provide standard parameters to represent common home care options.

1.2 This test is applicable to all fabrics suitable for home laundering.

2. Principle

2.1 Pairs of benchmarks are applied to the fabric before laundering. The distance between each pair is measured before laundering and remeasured after a specified number of standard laundering cycles. Percent change is calculated. A negative value indicates shrinkage; a positive value indicates growth. A value close to zero indicates little or no change.

3. Terminology

3.1 **dimensional change,** n.—a generic term for changes in length or width of a fabric specimen subjected to specified conditions. The change is usually expressed as a percentage of the initial dimension of the specimen.

3.2 **growth,** n.—a dimensional change resulting in an increase of length or width of a specimen.

3.3 **laundering,** n.—*of textile materials*, a process intended to remove soils and/or stains by treatment (washing) with an aqueous detergent solution and normally including rinsing, extraction and drying.

3.4 **shrinkage,** n.—a dimensional change resulting in a decrease in the length or width of a specimen.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Detergent may cause irritation. Care should be taken to prevent exposure to skin and eyes.

4.3 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

5. Apparatus and Materials (see 12.1)

5.1 Standard washing machine (see Table I, 12.2).

5.2 Standard tumble dryer (see Table III, 12.2) or facilities for drip/line drying.

5.3 Conditioning facilities and conditioning/drying racks with pull-out, perforated shelves (see 12.3).

5.4 Balance with at least 5.0-kg or 10.0-lb capacity.

5.5 AATCC 1993 Standard Reference Detergent (powder, with brightener, see 12.3).

5.6 Laundering ballast, Type 1 or Type 3 (see Table IV).

5.7 Indelible ink marking pen (see 12.3) for use with suitable rule, tape, marking template or other marking device (see 12.3). Sewing thread may be used for making benchmarks.

5.8 Measuring devices.

5.8.1 Tape or rule marked in millimeters, eighths or tenths of an inch.

5.8.2 Tape or ruled template marked directly in percent dimensional change to 0.5% or smaller increment (see 12.3).

5.8.3 Digital Imaging System (see 12.4).

6. Test Specimens

6.1 Sampling and Preparation.

6.1.1 Samples from which dimensional change specimens are to be taken should be representative of the fabric processing stage, finishing treatment, research lab trial, pallet, lot or end-product stage.

6.1.2 Fabrics that are distorted in their unlaundered state may give deceptive dimensional change results when laundered by any procedure. In such cases, it is recommended that specimens not be taken from any distorted area of a fabric sample.

6.1.3 Tubular knitted samples should be slit and handled flat in a single layer. Only circular knitted fabrics produced on body-width machines are to be used as specimens in their tubular form. Circular knitted fabrics made on body-width machines are ones to be used in garments with no side seams. Body-width tubular circular knitted garments and seamless garments (knit-to-wear) should be tested according to AATCC TM150, Dimensional Changes of Garments after Home Laundering (see 12.3).

6.1.4 Specimens tested in accordance with AATCC TM135 may also be used for testing for skewness change in accordance with Options 1 or 2 in AATCC TM179, Skew Change in Fabrics After Home Laundering (see 12.3).

6.1.5 If fraying of specimens is expected in laundering, refer to 12.6.

6.1.6 Prior to marking, condition test specimens as directed in ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 12.5). (Use conditions indicated in Table 1 for Textiles, general. Estimate conditioning time as indicated in Table 2 for the appropriate fiber content.) Lay each test specimen flat, separately, on a screen or perforated shelf of a conditioning rack.

6.1.7 Lay the sample on a flat surface. Do not allow any section of the sample to hang over the edge of the work table. Using a template for the selected test size, mark specimens parallel to the selvage or fabric length direction. Avoid use of the sample area within ten percent of the sample width. Specimens should be taken

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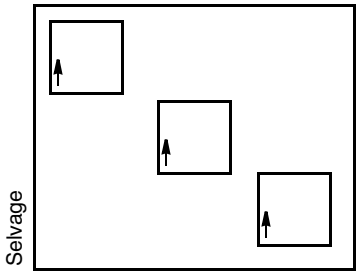


Fig. 1—Diagram for cutting fabric specimens.

from areas with different lengthwise and widthwise yarns (see Fig. 1). Identify the length direction of the specimens before cutting them out of the sample. When possible, three specimens from each fabric should be used. One or two specimens may be used when insufficient fabric sample is available.

6.2 Marking. Mark specimens without tension on a flat, smooth horizontal surface.

6.2.1 Option 1: *250-mm (10.0-in.) benchmarks*. Mark each 380 × 380 mm (15 × 15 in.) test specimen with three 250-mm (10-in.) pairs of benchmarks parallel to the test specimen length and three 250 mm (10 in.) pairs of benchmarks parallel to the test specimen width. Each benchmark must be at least 65 mm (2.5 in.) from all test specimen edges. Pairs of benchmarks in the same direction must be spaced approximately 125 mm (5 in.) apart (see Fig. 2 and 12.7).

6.2.2 Option 2: *460-mm (18.0-in.) benchmarks*. Mark each 610 × 610 mm (24.0 × 24.0 in.) test specimen with three 460-mm (18.0-in.) pairs of benchmarks parallel to the test specimen length and three 460-mm (18.0-in.) pairs of benchmarks parallel to the test specimen width. Each benchmark must be at least 65 mm (2.5 in.) from all test specimen edges. Pairs of benchmarks in the same directions must be spaced approximately 230 mm (9 in.) apart.

6.2.3 Narrow Fabrics.

6.2.3.1 For test specimens greater than 125 mm (5 in.) and less than 380 mm (15 in.) wide, take full width of test fabrics and cut each specimen 380 mm (15 in.) long. Mark the length as in 6.2.1. Measurement of the width is optional.

6.2.3.2 For test specimens 25-125 mm (1-5 in.) wide, take full width of test fabrics and cut each specimen 380 mm (15 in.) long. Use only two pairs of 250-mm (10-in.) benchmarks parallel to the length. Measurement of the width is optional.

6.2.3.3 For test specimens less than 25 mm (1 in.) in width, take full width of test fabrics and cut each specimen 380 mm (15 in.) long. Use only one pair of 250-mm (10-in.) benchmarks parallel to the length. Measurement of the width is optional.

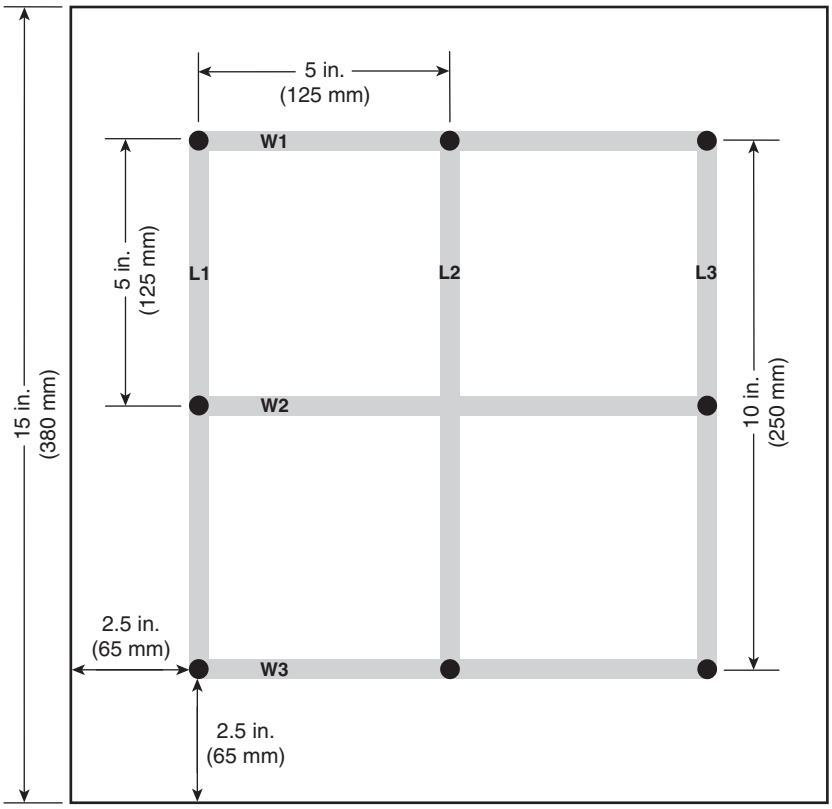


Fig. 2—Diagram for placing benchmarks on fabric specimens.

6.3 Original Measurements and Specimen Size. Make all measurements without tension on the same surface used for marking.

6.3.1 Specimen size and benchmarks distances used must be indicated in the report.

6.3.2 Dimensional change results may not be comparable when different specimen sizes, different benchmark lengths, different number of specimens, or different number of benchmarks are used.

6.3.3 To improve the accuracy and precision of the dimensional change calculations based on the benchmarks applied to the fabrics as instructed in 6.2, measure and record the distance between each pair of benchmarks with suitable tape or rule to nearest millimeter, eight or tenth of an inch. This is measurement A. In case of narrow fabrics less than 380 mm (15 in.) wide, measure and record width if width measurement will be used. If using a calibrated template for marking and measuring in percent dimensional change directly, an initial measurement is not needed.

7. Laundering Procedure

7.1 Washing.

7.1.1 Select washing conditions for testing from Table I. Set washing machine controls to generate the selected cycle parameters.

7.1.2 Load for laundering shall consist of all specimens in the sample, plus enough laundering ballast pieces to make a total load weight of 1.8 ± 0.1 kg (4.0 ± 0.2 lb).

7.1.3 Begin selected wash cycle. Allow machine to fill to specified water level.

7.1.4 Add 66 ± 1 g of AATCC 1993 Standard Reference Detergent to washing machine as directed by washing machine manufacturer's instructions. If detergent is added directly to wash water, agitate briefly to dissolve completely. Stop agitation before adding wash load.

7.1.5 Add wash load (test specimens and ballast), distributing evenly around center agitator. Restart wash cycle.

7.1.6 For specimens to be drip dried (drying procedure C), stop wash cycle before the water begins to drain from the final rinse cycle. Remove specimens soaking wet. For specimens to be tumble (A), line (B), or screen/flat (D) dried, allow washing to proceed through the final spin cycle.

7.1.7 After each washing cycle, separate tangled specimens and ballast pieces, taking care to minimize distortion. Proceed to appropriate drying procedure.

7.2 Drying.

7.2.1 Select drying conditions from Table II.

7.2.2 (A) Tumble Dry. Place the washed load (test specimens and ballast) in the tumble dryer, and set the tempera-

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Table I—Standard Washing Machine Parameters (see 12.2, 12.9)

Cycle	(1) Normal	(2) Delicate	(3) Permanent Press
Water Level, L (gal)	72 ± 4 (19 ± 1)	72 ± 4 (19 ± 1)	72 ± 4 (19 ± 1)
Agitation Speed, strokes/min.	86 ± 2	27 ± 2	86 ± 2
Washing Time, min.	16 ± 1	8.5 ± 1	12 ± 1
Final Spin Speed, rpm	660 ± 15	500 ± 15	500 ± 15
Final Spin Time, min.	5 ± 1	5 ± 1	5 ± 1
Wash Temp, °C (°F)[1]	(II) Cold: 27 ± 3 (80 ± 5) (III) Warm: 41 ± 3 (105 ± 5) (IV) Hot: 49 ± 3 (120 ± 5) (V) Very Hot: 60 ± 3 (140 ± 5)	(II) Cold: 27 ± 3 (80 ± 5) (III) Warm: 41 ± 3 (105 ± 5) (IV) Hot: 49 ± 3 (120 ± 5) (V) Very Hot: 60 ± 3 (140 ± 5)	(II) Cold: 27 ± 3 (80 ± 5) (III) Warm: 41 ± 3 (105 ± 5) (IV) Hot: 49 ± 3 (120 ± 5) (V) Very Hot: 60 ± 3 (140 ± 5)

[1]Due to US Department of Energy requirements, many washing machines use cooler water. An external control box may be used to override the machine set temperatures.

Table II—Standard Drying Conditions

(A)	Tumble Dry
(Ai)	Tumble Dry Normal
(Aii	Tumble Dry Delicate
(Aiii)	Tumble Dry Permanent Press
(B)	Line/Hang Dry
(C)	Drip Dry
(D)	Screen Dry/Dry Flat

Table III—Standard Tumble Dryer Parameters (see 12.2)

Cycle	(Ai) Normal	(Aii) Delicate	(Aiii) Permanent Press
Max. Exhaust Temp, °C (°F)	68 ± 6 (155 ± 10)	60 ± 6 (140 ± 10)	68 ± 6 (155 ± 10)
Cool Down Time, min.	≤ 10	≤ 10	≤ 10

Table IV—Laundering Ballast Parameters

	Type 1	Type 3
Fiber Content	100% cotton	50% cotton/50% polyester ± 3%
Greige Fabric Yarns	16/1 ring spun	16/1 or 30/2 ring spun
Greige Fabric Construction	52 × 48 ± 5 yarns/in., plain weave	52 × 48 ± 5 yarns/in., plain weave
Finished Fabric Weight	155 ± 10 g/m^2	155 ± 10 g/m^2
Edges	All edges hemmed or over-edged	All edges hemmed or over-edged
Finished Piece Size	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)
Finished Piece Weight	130 ± 10 g	130 ± 10 g

ture control to generate the selected cycle exhaust temperature (see Table III). Allow the dryer to run until the load is fully dry. Remove specimens immediately.

7.2.3 (B) Line/Hang Dry. Hang each specimen by two corners, with the fabric length in the vertical direction. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

7.2.4 (C) Drip Dry. Hang each dripping wet specimen by two corners, with the fabric length in the vertical direction. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

7.2.5 (D) Screen/Flat Dry. Spread each specimen on a horizontal screen or perforated surface, removing wrinkles without distorting or stretching it. Allow specimens to rest in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

7.2.6 For all drying methods, allow specimens to dry completely before washing again.

7.2.7 Repeat the selected washing and drying procedures for a total of 3 complete cycles, or an agreed upon number of cycles.

7.3 Conditioning.

7.3.1 After the final drying cycle is complete, condition test specimens as described in 6.1.6.

7.4 Ironing

7.4.1 If the specimens are extremely wrinkled and the consumer would ALWAYS expect to iron a garment made from the fabric, test specimens may be hand ironed prior to re-measurement of benchmarks.

7.4.2 Use safe ironing temperatures appropriate to the fibers in the fabric being ironed. See AATCC TM133, Test Method for Colorfastness to Heat: Hot Pressing, Table I, Safe Ironing Temperature Guide (see 12.3). Exert the minimum pressure necessary to remove wrinkles.

7.4.3 Due to the extremely high variability of hand ironing procedures performed by individual operators (no standard test method exists for hand ironing procedures), the reproducibility of dimensional change results after hand ironing has been found to be extremely poor. Consequently, caution is advised when comparing dimensional change results after laundering and hand ironing, reported by different operators.

7.4.4 After ironing, condition test specimens as described in 6.1.6.

8. Measurement

8.1 After conditioning, lay each test specimen without tension on the same surface used for marking and measuring in 6.2 and 63.

8.2 Measure and record the distance between each pair of benchmarks to the nearest millimeter, eighth or tenth of an inch. This is measurement B. If using a scale calibrated in percent dimensional change, measure each benchmark to nearest 0.5% or smallest increment on the scale and record the percent dimensional change directly. If using a digital imaging system, follow manufacturer's instructions.

8.3 The wrinkles in most fabrics flatten sufficiently under pressure of a measuring instrument at the time of measurement not to cause measurement bias.

9. Calculation and Interpretation

9.1 Calculation.

9.1.1 If measurements were made directly in percent dimensional change, average the measurements in each direction made on the specimens after the first, third, or other specified number of washing and drying cycles. Calculate length and width averages separately to the nearest 0.1%.

9.1.2 If measurements were made to the nearest millimeter or eight or tenth of an inch, calculate the dimensional change after the first and third or other specified washing and drying cycle using equation 1:

$$\text{Average\% DC} = 100\,(B - A)/A \quad \text{(Eq. 1)}$$

where:

DC = Average dimensional change
A = Average original dimension
B = Average dimension after laundering

Both the average original and average

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final dimensions are the averages of the measurements in each direction made on all test specimens. Calculate length and width averages separately to the nearest 0.1% (see 12.8).

9.1.3 A final measurement smaller than the original measurement results in a negative dimensional change, which is shrinkage. A final measurement larger than the original measurement results in a positive dimensional change, which is growth.

9.2 Interpretation.

9.2.1 If the dimensional change after one full laundering cycle as calculated in 9.1, is within a specification previously agreed on, continue test procedures as directed in 7.1-7.4 until an agreed upon number of cycles has been completed.

9.2.2 If the dimensional change after one washing, drying, and, if used, hand ironing cycle as calculated in 9.1 exceeds a specification previously agreed on, terminate the test.

10. Report

10.1 Report for each sample tested:

10.1.1 A description or identification of the sample.

10.1.2 Evaluation was performed using AATCC TM135-2018t.

10.1.3 Type of ballast used, i.e., Type 1 or Type 3.

10.1.4 Number of laundering cycles (default is 3 cycles, see 9.2).

10.1.5 Laundering conditions, including washing machine cycle, wash temperature, drying procedure and tumble drying temperature, if applicable. Alphanumeric designations may be used if they are clearly understood by all parties. For example, 1-IV-A(ii) indicates normal wash cycle at 49°C and Delicate tumble dry cycle.

10.1.6 If fabrics were distorted or wrinkled in their original state.

10.1.7 If fabrics were hand ironed.

10.1.8 Any modifications of the test method. If laundering equipment or conditions other than those specifically listed in the test method were used, they must be described in detail and noted as a modification of the standard method (see 12.9).

10.2 Report for each direction:

10.2.1 Direction measured, i.e., length or width.

10.2.2 Average dimensional change, to the nearest 0.1%, with a minus sign (-) for shrinkage or a plus (+) sign for growth.

10.2.3 Number of measurements (default is 9: 3 measurements on each of 3 specimens).

10.2.4 Distance between original benchmarks, i.e., 250 mm (Option 1) or 460 mm (Option 2).

11. Precision and Bias

11.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

11.2 *Bias*. Dimensional changes in automatic home laundering of fabrics can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

12. Notes

12.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

12.2 For model number(s) of washer(s) and tumble dryers(s) reported to meet the standard parameters, visit www.aatcc.org/test/washers or contact AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org. An alternate load size was described in previous versions of this method (3.6-kg load, 83-L water level, 80 g AATCC 1993 Standard Reference Detergent), but no machine is reported to meet the standard parameters for this load size. In addition, dimensional change results obtained with the alternate load size may not be equal to those obtained with a standard load size.

12.3 Materials or drawings are available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

12.4 A digital imaging system may be used as a measuring device in place of the prescribed manual measurement devices if it is established that its accuracy is equivalent to the manual devices.

12.5 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610. 832.9500; fax: +1.610.832.9555; web site: www.astm.org.

12.6 If excessive fraying occurs in laundering, specimen edges may be pinked or slashed. Sewing or over-edging a specimen is not recommended as it may influence actual dimensional change results. However, in the case where AATCC TM124 (Test Method for Smoothness Appearance of Fabrics after Repeated Home Laundering) and AATCC TM135 are performed on the same specimens, some woven fabric constructions may require the specimen edges to be sewn or over-edged to prevent severe raveling that could cause entanglement in washing or drying, and therefore influence the assessment of both dimensional change and smoothness.

12.7 When using the same specimens for AATCC TM135 and AATCC TM179, extend corner markings to form right angles.

12.8 If information on the dimensional change variability within or between specimens is desired, calculate dimensional change based on the individual pairs of benchmarks for within-specimen data or based on the average of the three pair of benchmarks for between specimen data.

12.9 The laundering temperatures and other parameters listed in this test method are standard conditions for testing purposes. As with most lab procedures, they represent, but may not exactly replicate, current consumer practices. Consumer practices vary over time and among households; lab practices must be consistent to allow valid comparison of results. If laundering equipment or conditions other than those specifically listed in this test method are used, they must be described in detail and noted as a modification of the standard method. Alternate laundering conditions are outlined in AATCC LP1, AATCC LP2 and ISO 6330.

12.10 Available from ISO, www.iso.org.

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Test Method for Rug Back Staining on Vinyl Tile

Developed in 1972 by AATCC Committee RA57; reaffirmed 1973, 1989, 2000, 2007, 2012; editorially revised 1974, 1986, 1991, 2001, 2010, 2013, 2019; revised 1983, 2002; editorially revised and reaffirmed 1995.

1. Purpose and Scope

1.1 This test method determines the degree of color transferred from the back or face surface of colored rugs to vinyl tile.

1.2 The wet testing in this method is a means of acceleration and has been found to correlate with both dry and wet in-service use.

2. Principle

2.1 A rug specimen is wet out and placed between two pieces of tile and subjected to a weighted load for 24 h at room temperature.

2.2 Color transferred to the tile is assessed by a comparison with the AATCC 9-Step Chromatic Transference Scale (AATCC EP8) or Gray Scale for Staining (AATCC EP2).

3. Terminology

3.1 **colorant staining,** n.—the unintended pickup of colorant by a substrate due to (1) exposure to a colored or contaminated liquid medium, or (2) direct contact with dyed or pigmented material, from which colorant transfers by sublimation or mechanical action (as in crocking).

3.2 **rug,** n.—a textile floor covering of limited area which is complete in itself and is intended for use as a partial covering of a floor or another floor covering.

3.3 **rug back,** n.—(1) that part of a rug normally in contact with the floor; (2) the underside of a rug as opposed to the use surface.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Observe wringer safety, especially at the nip point. The safety recommendations provided by the wringer manufacturer should be followed.

5. Apparatus and Materials

5.1 Two 76 × 76 mm (3 × 3 in.) 100% vinyl tiles, solid white and smooth surfaced (see 11.1).

5.2 Laboratory wringer.

5.3 Weight, with mass of 0.91 kg (2.0 lb) (see 11.2).

5.4 Rating Scales.

5.4.1 AATCC 9-Step Chromatic Transference Scale (AATCC EP 8) (see 11.3).

5.4.2 Gray Scale for Staining (AATCC EP2) (see 11.3).

5.5 Distilled water.

5.6 AATCC Detergent #171 (see 11.5).

6. Test Specimens

6.1 A 51 × 51 mm (2 × 2 in.) specimen is cut from a representative sample of the rug, binding tape or fringe. It is recommended that rugs having multiple colors or fibers be tested in areas representative of each.

7. Procedure

7.1 Prewash tile with AATCC Detergent #171 prepared according to the directions supplied (see 11.5.1). Dry with toweling.

7.2 Completely wet out the test specimen with distilled water.

7.3 Pass the test specimen between squeeze rolls (wringer) such that a pickup of 40-60% is obtained.

7.4 Place the rug specimen back side down on the face side of one of the tiles.

7.5 Cover the test specimen with the other tile, face side down on rug face pile.

7.6 Apply a 0.91 kg (2.0 lb) mass to sandwiched test assembly (see 11.2).

7.7 Allow test assembly to remain fixed for 24 h at room temperature.

7.8 Do not permit movement of rug specimen or tile after applying weight as it can affect test results.

7.9 Remove weight and rug specimen and let tiles condition at room temperature face up for 30 min.

7.10 Wash tile with AATCC Detergent #171 solution. Dry with toweling.

8. Evaluation

8.1 Evaluate the degree of color transferred from the rug backing or face fiber on to the face of the vinyl tile by comparison with AATCC EP2, AATCC EP8 or Instrumental Assessment of Degree of Staining (AATCC EP12), and record the numerical rating that corresponds to the appropriate one on either of them (see 11.3).

8.1.1 Where fringe or binding tape is a factor, color transference to the vinyl tile can also be evaluated from the tile which was used as the cover in the test assembly.

8.2 Evaluate the color transfer at the most severe area on the tile.

9. Report

9.1 Report the grade determined in 8.1.

9.2 Report the color transference determined for rug back and face. As an option, the color transference may be reported for fringe or binding tape.

9.3 Report whether Gray Scale for Staining or AATCC 9-Step Chromatic Transference Scale was used for evaluating staining (see 11.4).

10. Precision and Bias (see 11.6)

10.1 *Precision*. A preliminary intralaboratory study was run in 1999 with a single laboratory with four materials, with a total of eight test results for each material. Refer to Table I for *within-laboratory* precision.

10.2 *Bias*. Rug back staining, on vinyl

Table I—Summary of Precision Parameters

Material*	Observations Each Average	Overall Average	Repeatability Standard Deviation (*within-laboratory*)	95% Repeatability Limit (*within-laboratory*)
A	8	3.4	0.42	0.35
B	8	4.1	0.35	0.30
C	8	3.4	0.32	0.27
D	8	2.9	0.32	0.27

*Material A-Tan Cotton Rug; Material B-Pink Cotton Rug; Material C-Blue Nylon Rug; Material D-Red Nylon Rug

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tile, can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

11. Notes

11.1 AATCC Standard Vinyl Tile 137-2 available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1. 919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.2 A laboratory beaker filled with sufficient water is satisfactory.

11.3 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.4 It has been noted that different ratings may result depending upon whether the Gray Scale for Staining or AATCC 9-Step Chromatic Transference Scale is used for the evaluation. It is, therefore, important to report which scale was used.

11.5 AATCC Detergent #171 available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc org; web site: www.aatcc.org.

11.5.1 Dilution: 3.0-4.5 g/L.

11.5.2 Contents of detergent:

(a) Sodium metasilicate pentahydrate

(b) Sodium stearate

(c) Dodecylbenzene sulfonate

(d) Sodium bicarbonate

(e) Sodium tripolyphosphate

11.6 The precision of this test method is dependent on the combined variability of the material being tested, the test method itself, and the evaluation procedure utilized.

11.6.1 The precision statement shown in Section 10 was developed from results obtained by visual evaluation (AATCC EP2 or AATCC EP8).

11.6.2 It is expected that the use of instrumental evaluation (AATCC EP12) will result in greater precision than that obtained from visual evaluation.

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Test Method for Cleaning: Washing of Textile Floor Coverings

Developed in 1972 by AATCC Committee RA57 (as shampooing procedure); revised 1978, 1982, 1995 (to cleaning procedure with title change), 2000; editorially revised 1986, 1998, 2008, 2019; editorially revised and reaffirmed 1987; reaffirmed 2005, 2010, 2014.

1. Purpose and Scope

1.1 This test method is a laboratory procedure designed to simulate changes that occur in washing of a textile floor covering during cleaning.

1.2 This method is recommended only for small specimens with dimensions not exceeding the capacity of commercially-available laboratory equipment.

1.3 This test method may be used on soiled, as well as unsoiled, textile floor coverings.

1.4 This wet cleaning procedure can be used for the evaluation of:

- wetfastness durability and permanency of antimicrobial properties
- colorfastness behavior
- permanency of finishes and other topical treatments to the pile of textile floor coverings before, during or after manufacturing
- cleanability
- dimensional stability

1.5 This procedure may also be used for the removal of soil or extraneous matter from specimens of textile floor coverings to prepare them for testing.

2. Principle

2.1 A specimen is washed by hand scrubbing using a surfactant cleaning agent solution, rinsed and dried.

3. Terminology

3.1 **carpet,** n.—all textile floor coverings not designated as rugs.

3.2 **textile floor covering,** n.—an article having a use-surface composed of textile material and generally used for covering floors.

3.3 **use-surface,** n.—*of textile floor covering*, the part of a textile floor covering directly exposed to foot traffic.

3.4 **washing,** n.—*in testing textile floor coverings*, a specific wet cleaning process involving the use of detergent and scrub brush to remove soil and/or extraneous matter residing in the pile fibers.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 Eye protection should be used when mixing, handling or applying the detergent and detergent solutions.

4.4 Gloves or protective hand cream are recommended for handling detergents and solutions.

4.5 Observe operating procedures and precautions by the manufacturer of the extraction machine. Liquid under pressure and high vacuum, can cause excessive splashing of liquids and/or hose "kick back."

4.6 Special care should be used at the nip point of the wringer. Follow the safety guidelines of the manufacturer.

5. Apparatus and Materials (see 10.1)

5.1 Cleaning Agent: Sodium lauryl sulfate type of surfactant. AATCC Standard Detergent (see AATCC TM171, Test Method for Carpets: Cleaning of; Hot Water Extraction) can be used. A 1.0% solution of this surfactant should have a pH of 8.0 ± 0.5 (see 10.2).

5.2 Scrub Brush: A brush with stiff nylon or polypropylene bristles approximately 25 mm long. The width of the brush should be approximately 50 mm; the length preferably no less than one dimensional length of the test specimen, 200 mm.

5.3 Extraction Unit: A laboratory wringer, padder or hydro-extractor.

5.4 Drying Unit: Circulating air-type oven recommended.

6. Test Specimens

6.1 Textile floor covering samples, finished with back coating, approximately 200 × 200 mm square.

7. Procedure

7.1 Thoroughly wet out each specimen by soaking for 1 min in clean, tap water at 50 ± 3°C. Extract excess water by passing through a laboratory wringer or hydro-extractor.

7.2 Secure the specimen to the floor by a suitable means (double-stick tape) or use the scrub block (see Fig. 1) to secure during the hand scrubbing process.

7.3 For every 645 mm^2 of pile surface, apply 0.30 mL of the 1.0% solution of sodium lauryl sulfate preheated to 50 ± 3°C to the face fibers. For example, a 200 × 200 mm specimen would require approximately 20 mL of this cleaning solution. Hand scrub *the entire surface of the pile* in specimen with the scrub brush, using 5 full strokes in each direction. A full stroke implies motion forward and backward in this case. Scrub 5 strokes along one dimension and repeat with 5 strokes across the specimen.

7.4 Thoroughly rinse each specimen with clean tap water at 50 ± 3°C. Remove excess water by squeezing through a lab wringer or hydro-extracting.

7.5 Place each specimen in the oven (see 5.4) set at 105 ± 5°C. Do not remove specimen until dry.

7.6 Cool down the specimen after drying before test work is allowed to proceed. Conditioning at standard laboratory temperature and relative humidity is recommended. Also, a Scrub Block can be used to secure the specimens during the hand scrubbing process (see 10.3 and Fig. 1).

8. Evaluation

8.1 This is a washing procedure which is not evaluated.

8.1.1 Specimens cleaned according to this procedure are suitable substrates for evaluating the effect of washing on colorfastness, cleanability, dimensional change and the durability of finishes, topicals and antimicrobial treatments. State that the test specimens were cleaned according to AATCC TM138. Report the number of wash cycles (see 7.1, 7.2, 7.3 and 7.4) performed.

9. Precision and Bias

9.1 Precision and Bias statements are not applicable because data are not generated by this test method.

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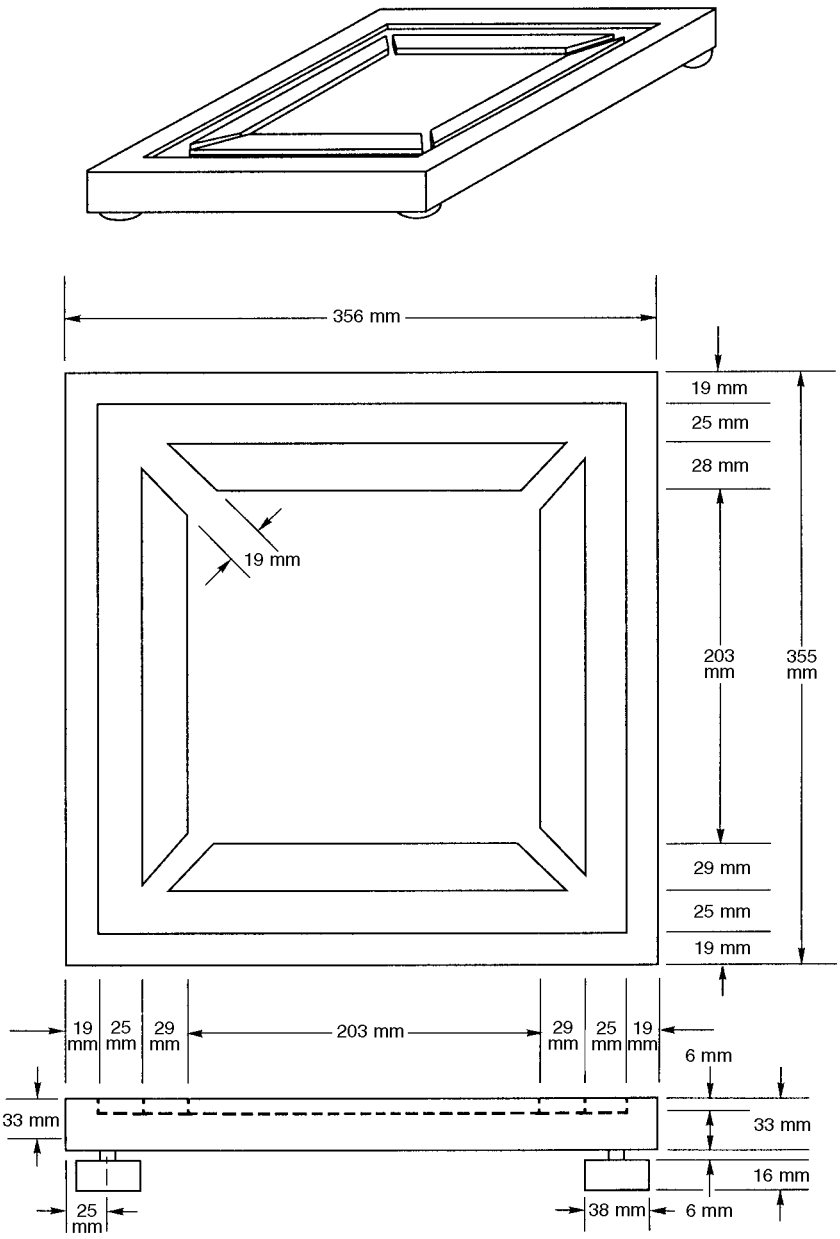


Fig. 1—Scrub block.

10. Notes

10.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

10.2 AATCC Standard Detergent 171 can be used. Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

10.3 Some laboratories rely on special jigs to hold the 200 × 200 mm square specimen in place during the scrubbing process. A schematic of a Scrub Block designed to hold carpet specimens during the scrubbing process is included as Fig. 1.

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Test Method for Dye and Pigment Migration in a Pad-Dry Process

Developed in 1974 by AATCC Committee RA87; reaffirmed 1976, 1977, 1980, 2006, 2011; editorially revised and reaffirmed 1985, 1990, 2001; editorially revised 1987, 1991, 1998, 2010, 2019 (with title change); revised 1992 (with title change), 1996 (with title change), 2018. Technically equivalent to ISO 105-Z06.

1. Purpose and Scope

1.1 This test method provides a means of assessing the migration propensity of a pad liquor system containing dyes or pigments, subsequently referred to as colorants, and which may also contain different types and amounts of migration inhibitors.

1.2 When drying conditions are not constant and/or uniform, uneven migration may occur, causing shade variations during a run, or shade differences between the face and back, or between side and center of the fabric.

2. Principle

2.1 Fabric impregnated with colorant alone, or with colorant and auxiliaries, is dried while partially covered with a watch glass permitting differential drying and, therefore, migration to occur. The degree of migration is evaluated by visual examination, or by reflectance measurements of the covered and uncovered areas.

3. Terminology

3.1 **colorant,** n.—a material which is applied to a substrate for the express purpose of changing the transmittance or reflectance of visible light.

NOTE: Dyes, pigments, tints and optical brighteners are examples of colorants; soils are not colorants.

3.2 **dye,** n.—a colorant applied to or formed in a substrate, via the molecularly dispersed state, which exhibits some degree of permanence.

3.3 **migration,** n.—*in textile processing, testing, storage and use*, movement of a chemical, dye or pigment between fibers within a substrate or between substrates due to capillary forces (see also *transfer*).

3.4 **pigment,** n.—a colorant in particulate form which is insoluble in a substrate, but which can be dispersed in the substrate to modify its color.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Follow good laboratory practices. Wear safety glasses in all laboratory areas.

4.2 Observe padder safety. Normal safeguards on padders should not be removed. Ensure that there is an adequate guard at the nip point. A foot-operated kickoff is recommended.

4.3 Handle all chemicals with care. Use chemical goggles or face shield, impervious gloves and an impervious apron when preparing buffered mock dyebaths with concentrated acids and sodium hydroxide.

4.4 An eyewash/safety shower should be located nearby and a self-contained breathing apparatus should be readily available for emergency use.

4.5 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 13.1).

5. Uses and Limitations

5.1 This method offers two alternative procedures:

5.1.1 Procedure A. The fabric/glass assembly (see 8.2) is allowed to dry at room temperature. The procedure is very simple, but time consuming (overnight).

5.1.2 Procedure B. The fabric/glass assembly (see 8.3) is dried in a laboratory dryer or oven, either with or without air circulation. This procedure is faster, but somewhat more complicated.

5.2 The test method may be used to compare the migration propensity of dyes and the effect on migration of different types of migration inhibitors, thickeners and electrolyte.

5.3 The test method may be used to evaluate a pad liquor with which migration has been found on a continuous dye range. The pad liquor composition may be modified by varying the amount or type of migration inhibitors and tested in the laboratory prior to the application on the range. The colorant concentrations, substrate and wet pickup should be the same in the laboratory as used on the range. It is then possible to correlate the test results with the improvement experienced in practice.

6. Apparatus and Materials

6.1 Laboratory padder.

6.2 Glass plate, 600 × 350 mm (Procedure A).

6.3 Watch glasses (90 mm), 22 mm arch punch.

6.4 Aluminum rings: 110 mm outer diameter, 80 mm inner diameter and 1 mm thickness (Procedure B).

6.5 Clips (Procedure B).

6.6 Laboratory dryer or oven (Procedure B).

6.7 Gray Scale for Color Change (AATCC EP1) (for visual evaluation) (see 13.2).

6.8 Spectrophotometer (AATCC EP7) (see 13.2).

6.9 Fabric swatches (see 13.3 and 8.2.1 or 8.3.1).

7. Test Specimen

7.1 Colorant(s) in required concentration(s).

7.2 Migration inhibitors, thickening agents and other auxiliaries (e.g., electrolyte for reactive dyes) may be used as appropriate.

8. Procedure

8.1 Pad Bath Preparation.

8.1.1 Prepare a pad bath appropriate to

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the colorant(s) with or without a migration inhibitor.

8.2 Procedure A: Drying in Air at Room Temperature.

8.2.1 Pad a 150 × 300 mm swatch of fabric at 20 ± 2°C (68 ± 4°F). Other padding temperatures may be used, but must then be stated in the report. Generally, use a wet pickup of 60%, but adjust as necessary to simulate wet pickup of a particular fabric to be run on a particular range (see 13.4).

8.2.2 Immediately after padding place the fabric on a flat glass plate. Place the watch glass on the fabric as shown in Fig. 1 and leave the fabric to dry at room temperature. Record the room temperature and relative humidity during drying.

8.2.3 Remove the watch glass.

8.3 Procedure B: Pin Frame Oven Drying.

8.3.1 Pad a 110 × 220 mm swatch of fabric at 20 ± 2°C (68 ± 4°F). Other padding temperatures may be used, but must then be stated in the report. Generally, use a wet pickup of 60%, but adjust as necessary to simulate wet pickup of a particular fabric to be run on a particular range (see 13.4).

8.3.2 Immediately after padding mount the fabric taut on a pin frame, and sandwich it between two watch glasses, one on the fabric face and one directly underneath on the back of the fabric. Hold the watch glasses in place with two aluminum rings and clamps as shown in Fig. 2.

8.3.3 Dry the fabric horizontally at 100 ± 2°C (212 ± 5°F) for approximately 7 min (or until dry) in a laboratory dryer or oven, with or without air circulation.

8.3.4 Remove the watch glasses.

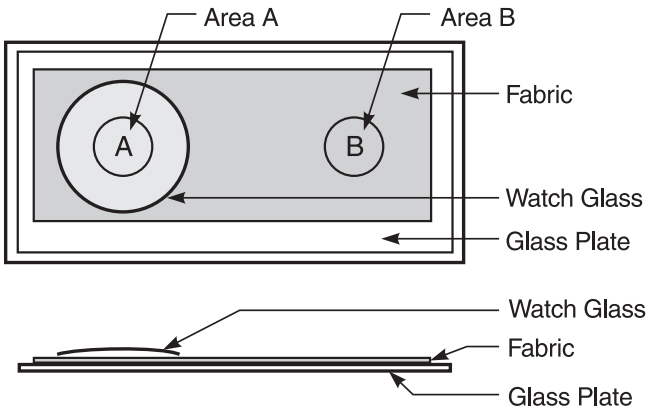


Fig. 1—Layout of apparatus for inducing colorant migration using procedure A.

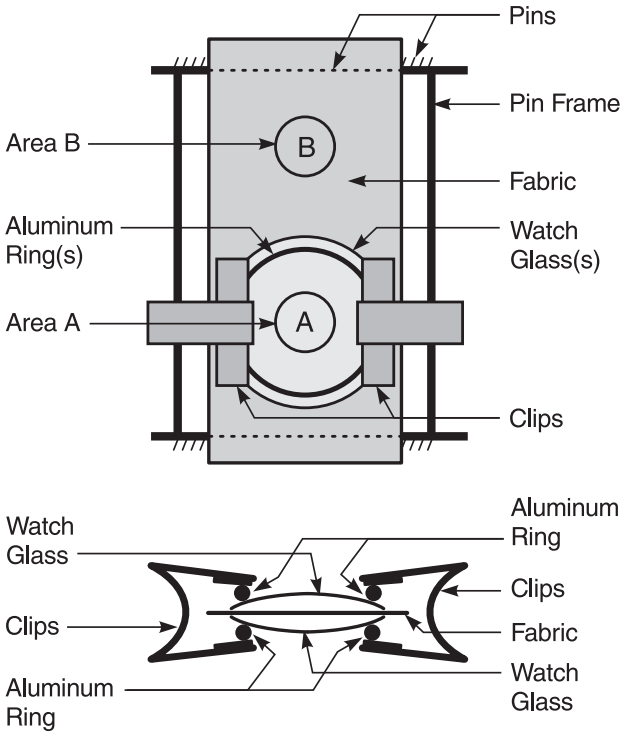


Fig. 2—Layout of apparatus for inducing colorant migration using procedure B.

9. Gray Scale Evaluation

9.1 Reference the following evaluation procedures.

9.1.1 AATCC EP1, Evaluation Procedure for Gray Scale for Color Change.

9.2 AATCC EP7, Evaluation Procedure for Instrumental Assessment of the Change in Color of a Test Specimen.

10. Report

10.1 Report substrate, pad liquor composition, including dyes, chemicals, auxiliaries, etc. and wet pickup.

10.2 For Procedure A, report room temperature and relative humidity during drying.

10.3 For Procedure B, report whether dried with or without air circulation.

10.4 For visual examination, report assessment on a scale of 1-5 by reference to the Gray Scale for Color Change.

10.5 For reflectance measurements, report percent migration using the outer dyed area as reference.

11. Precision and Bias

11.1 Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

12. References

12.1 Etters, J. N., *Textile Chemist and Colorist*, Vol. 4, No. 6, 1972, p160.

12.2 Etters, J. N., *Modern Knitting Management*, Vol. 51, No. 2, 1973, p24.

12.3 Gerber, H., *Melliand Texblberichte*, Vol. 53, No. 3, 1972, p335.

12.4 Lehmann, H., and Somm, F., *Textile Praxis International*, Vol. 28, No. 1, 1973, p52.

12.5 Northern Piedmont Section, AATCC, *Textile Chemist and Colorist*, Vol. 7, No. 11, 1975, p192.

12.6 Urbanik, A., and Etters, J. N., *Textile Research Journal*, Vol. 43, 1973, p657.

13. Notes

13.1 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

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13.2 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

13.3 For disperse dyes, vat dyes and pigments, generally the preferred fabric is a 65/35 polyester/cotton gabardine or heavy twill, heat-set, bleached and mercerized. For soluble dyes with affinity to cellulose (e.g., reactive dyes) a bleached and mercerized 100% cotton gabardine or heavy twill should be used. However, any other fabric intended for use on a continuous dye range may be chosen.

13.4 The desired wet pickup is obtained by adjusting the pressure at the nip of the pad rolls. Wet pickup is the weight gained by the substrate at ambient conditions after padding:

$$\% \text{ wet pickup} = 100 \times (A/B - 1)$$

where:

A = weight after padding
B = weight before padding

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Test Method for Compatibility of Basic Dyes for Acrylic Fibers

Developed in 1974 by AATCC Committee RA87; reaffirmed 1976, 1977, 1980, 1989, 2009, 2013; revised 1984, 2019; editorially revised 1985, 1988, 2008, 2010; editorially revised and reaffirmed 1987, 1994, 1999, 2004. Technically equivalent to ISO 105-Z03.

Foreword

This test method is based on one first proposed by the Society of Dyers and Colourists (see 12.1). The standard dyes were chosen because: (a) they form two series encompassing the compatibility behavior of nearly all basic dyes recommended for acrylic fibers; (b) the dyes in each series are spaced to produce approximately equal visual effects; and (c) the corresponding standard dyes from both scales are compatible.

1. Purpose and Scope

1.1 In the dyeing of acrylic fibers with basic dyes, the classic parameters—e.g., the time of half-dyeing—of individual dyes do not give a true indication of their dyeing behavior in admixture with other basic dyes.

1.1.1 Since basic dyes show virtually no migration in acrylic fibers under normal dyeing conditions, compatibility is of major importance in selecting dye combinations with optimum level dyeing behavior.

1.2 This test method determines the compatibility of a basic dye when applied to acrylics.

2. Principle

2.1 Compatibility of a basic dye is assessed by determining its dyeing behavior in combination with each of the standard dyes in a yellow or blue scale—whichever shows the greatest difference in hue from that of the dye under test.

2.2 The dye is assigned a grade from 1 through 5 corresponding to the standard dye with which it gives on-tone dyeings throughout the sequence (half-steps, < 1, or > 5 may also be assigned).

3. Terminology

3.1 **basic dye,** n.—a dye that dissociates in an aqueous medium to give a positively-charged ion (cation) with affinity for fibers containing acidic groups.

3.2 **compatibility,** n.—*in textile dyeing*, propensity of individual dye components in a combination shade to exhaust at similar rates, resulting in a buildup of shade that is constant, or nearly constant, in hue throughout the dyeing process.

3.3 **grade**, n.—the number assigned to a test specimen resulting from comparison to a scale.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas and a single- use dust respirator while handling powder dyes.

4.2 All chemicals should be handled with care.

4.3 Use chemical goggles or face shield, impervious gloves and an impervious apron during handling of concentrated acetic acid.

4.4 An eyewash/safety shower should be located nearby and an organic vapor respiratory with full facepiece should be readily available for emergency use.

4.5 The basic dyes listed in this method belong to the following classes:

C.I. Basic Orange 42—Azo-methineazo
C.I. Basic Yellow 29—Methine
C.I. Basic Yellow 28—Monoazo
C.I. Basic Yellow 15—Monoazo
C.I. Basic Orange 48—Azo
C.I. Basic Blue 69—Methine
C.I. Basic Blue 45—Anthraquinone
C.I. Basic Blue 47—Anthraquinone
C.I. Basic Blue 22—Anthraquinone
C.I. Basic Blue 77—Triarylmethane

4.6 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 12.2).

5. Uses and Limitations

5.1 The test is valid only under the conditions defined. Compatibility may be affected by the presence of anionic substances—e.g., anionic surfactants or anionic dyes—in the dyebath.

5.2 The yellow and blue standard dyes of compatibility grade 2.0 are not fully compatible when applied in combination. The differences observed are exaggerated if wet-spun acrylic fibers are used.

5.3 The test is one for compatibility and not a measure of the extent to which incompatibility may lead to unlevelness when combinations of basic dyes are applied to acrylic fibers.

5.4 Although some of the specified dyes are not in current commercial production, the set of 10 dyes (50 g each), is available for purchase by laboratories interested in performing this method (see 12.3).

6. Apparatus, Reagents and Materials

6.1 Acrylic fiber, 30 samples (pieces of skeins) of equal weight.

6.2 Dyeing apparatus, for dyeing at a 40:1 liquor:goods ratio at the required constant temperature.

6.3 Standard dyes, 5, yellow or blue (see Table I and 12.3).

6.3.1 Use the dye scale (yellow or blue) that shows the greatest difference

Table I—Standard Dye Scales

C.I. Basic Dye	% owf	Compatibility Grade
Yellow		
Orange 42	0.45	1.0
Yellow 29 (200%)	0.25	2.0
Yellow 28 (200%)	0.15	3.0
Yellow 15	0.75	4.0
Orange 48	0.65	5.0
Blue		
Blue 69	0.55	1.0
Blue 45	2.7	2.0
Blue 47 (200%)	0.6	3.0
Blue 77	0.6	4.0
Blue 22 (200%)	1.2	5.0

Note: Dye strength is 100% unless otherwise noted (see 12.6).
The percentages given produce dyeings approximately one half of 1/1 standard depth on many acrylic fibers (see 12.4).

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in hue from that of the dye under test (e.g., to evaluate a blue dye, use the yellow standard dye scale).

6.4 Acetic acid (glacial).

6.5 Sodium acetate crystals ($CH_3COONa \cdot 3H_2O$).

6.6 Deionized or distilled water, which does not change significantly in pH on boiling.

7. Specimens

7.1 A specimen of the dye under test is required for each of 5 dyebaths. The % owf is selected to give dyeings approximately one half of 1/1 standard depth on the pretreated acrylic fiber (see 12.4).

8. Procedure

8.1 Pretreat the acrylic fiber for 10 min at approximately 95°C (203°F) in a solution containing 1% (owf) acetic acid (glacial) and 1% (owf) $CH_3COONa \cdot 3H_2O$ (to give pH 4.5 ± 0.2) at a liquor:goods ratio of 40:1.

8.2 Lightly squeeze the pretreated fiber and keep it wet, ready for use.

8.3 Set 5 dyebaths, one for each of the standard dyes in the selected scale, at 95°C (203°F), a liquor:goods ratio of 40:1 and a pH value of 4.5 ± 0.2 with:

X% Standard dye (see Table I)

Y% Dye under test (see 7.1)

1% Acetic acid (glacial)

1% $CH_3COONa \cdot 3H_2O$.

8.3.1 These percentages and the liquor:goods ratio refer to the weight of an individual fiber sample, not to the combined weight of the 6 samples required for each series of dyeings.

8.4 Use 6 fiber samples for each dyebath. Enter the first fiber sample into the dyebath at the recommended temperature, for the recommended time (see 12.5). Remove the first sample and replace it with the second. Repeat this dyeing procedure for a total of 5 samples.

8.4.1 Dye the last sample to complete exhaustion in the same dyebath as the first 5.

8.4.2 Rinse and dry each fiber sample immediately after it is removed from the dyebath. Mount the samples in the sequence in which they were dyed.

8.5 Repeat the dyeing procedure for the remaining dyebaths (6 fiber samples in each of 5 baths; 30 fiber samples in total).

Table II—Sample Test Results for a Blue Dye

Compatibility Grade (Yellow Scale)	Appearance of Dyed Samples
1.0	Progressively much bluer
2.0	Progressively bluer
3.0	On-tone
4.0	Progressively yellower
5.0	Progressively much yellower

The compatibility grade of the blue dye is 3.0.

9. Evaluation

9.1 Visually assess each dyed fiber sample. For each dyebath, report whether the 6 dyeings are progressively bluer, progressively yellower, or consistently on-tone.

9.1 Assign a compatibility grade to the dye under test.

9.2.1 If one of the dyebaths produced consistently on-tone dyeings, the grade is the number associated with that bath (see Table I). A typical example of the results obtained in assessing a blue dye is shown in Table II.

9.2.2 It is possible for the compatibility of a dye to lie between two adjacent standards; e.g., to be 1.5, 2.5, 3.5 or 4.5. In these instances, a series of on-tone dyeings will not be obtained with any of the standard dyes.

9.2.3 If a dye lies outside the scale as defined, assign a grade of < 1, or > 5, whichever is appropriate.

10. Report

10.1 Describe or identify the tested dye.

10.2 Report that sample was tested according to AATCC TM141-2019.

10.3 Report the testing conditions:

10.3.1 Scale used (yellow or blue).

10.3.2 Acrylic fiber description.

10.3.3 Dyeing apparatus, temperature and time.

10.3.4 Test dye and standard dyes % owf.

10.4 Report the test results:

10.4.1 Compatibility grade of the tested dye.

10.4.2 Relevant observations regarding the dyeings.

10.5 Describe any modification(s) of the published method.

11. Precision and Bias

11.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

11.2 *Bias*. The compatibility of basic dyes for acrylic fibers can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

12. Notes

12.1 Compatibility Test for Combinations of Basic Dyes in the Dyeing of Acrylic Fibres, *Journal of the Society of Dyers and Colourists*, Vol. 88, June 1972, pp220-222.

12.2 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240, USA; +1. 513.742.2020; www.acgih.org.

12.3 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

12.4 For ease in visual assessment, use the dyes under test and the standard dyes in a concentration that produces approximately one half of 1/1 standard depth when dyed on most acrylic fibers. If standard depth of the dye being tested is not known, a dyeing concentration of similar depth to the Standard Dye Scales in Table 1 can be used. Additionally, standard depth information may be available from the dye supplier. It should be understood, however, that it is not critical that the compatibility is assessed at standard depth, since identical grades are obtained at other depths and in other ratios of standard and test dye.

12.5 Acrylic fibers can vary widely in their dyeing characteristics. The test is to be conducted in such a manner that all available dye is exhausted in approximately equal portions onto the number of fiber samples used. To accomplish this, it may be necessary to adjust the dyeing temperature within a range of 90-100°C and/or the dyeing time for each sample in the range of 5-10 min. Suggested times and temperatures for some typical fibers are:

Acrilan 16—93°C, 5 min intervals

Acrilan SEF—96°C, 5 min intervals (except for blue C-1 and yellow C-2 standard, use 93°C) Orlon 75—96°C, 10 min intervals (except for C-1 standard, use 5 min)

Creslan—88°C, 5 min intervals

12.6 The strengths of some standard dyes were changed in August 1984. The % owf was adjusted accordingly. For example, C. I. Basic Yellow 29 (200%) at 0.25% replaced C. I. Basic Yellow 29 (100%) at 0.50%.

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Test Method for Appearance of Apparel and Other Textile End Products after Home Laundering

Foreword

This test method and its accompanying evaluation tools were developed for evaluation of woven fabrics with a durable-press finish. It is common industry practice to use the method and scales for evaluation of other textile materials although some specimens may have different appearance characteristics due to different fabric constructions.

Standard laundering procedures remain consistent to allow valid comparison of results. Standard procedures represent, but may not exactly replicate, current consumer practices, which vary over time and among households. Alternate laundering procedures and machine parameters may be found in AATCC LP1, Laboratory Procedure for Home Laundering: Machine Washing (see 12.3) and ISO 6330, Textiles — Domestic washing and drying procedures for textile testing (see 12.7).

1. Purpose and Scope

1.1 This test method is intended to determine the smoothness appearance of fabric, seams, and pressed-in creases in garments and other textile products after being subjected to home laundering procedures.

1.2 Any washable textile end product may be evaluated for smoothness appearance, seam smoothness and crease retention using this method.

1.3 Textile end products containing fabrics of any construction, such as woven, knit and nonwoven, may be evaluated according to this method.

1.4 Techniques for seaming and creasing are not outlined since the purpose is to evaluate textile end products as they will be supplied from manufacturing or as ready for use.

2. Principle

2.1 Textile end product items are subjected to standard home laundering practices. Appearance of specimens is evaluated in a standard viewing area by visually comparing the appearance of specimens with appropriate reference standards. Results may be reported as an overall percentile value from 0% to 100%, with 100% representing the most smoothness and crease retention. Results may also be reported as separate smoothness appearance (SA), seam smoothness (SS) and crease retention (CR) grades. All grades may range from 1 to 5, with 1 representing the least smoothness or crease retention and 5 representing the most smoothness or crease retention.

3. Terminology

3.1 **appearance of textile end products,** n.—the overall visual impression of a textile end product quantified by comparison of individual components with appropriate reference standards.

3.2 **ballast,** n.—*in procedures for processing or testing of textiles*, material that is used to bring the total weight or volume of the textiles to an amount specified in the procedure.

3.3 **crease retention,** n.—*in fabrics*, the visual impression of an inserted crease quantified by comparison with a set of reference standards.

3.4 **durable press,** adj.—having the ability to retain substantially the initial shape, flat seams, pressed-in creases and unwrinkled appearance during use and after laundering or drycleaning.

3.5 **grade**, n.—the number assigned to a test specimen resulting from comparison to a scale.

3.6 **laundering,** n.—*of textile materials*, a process intended to remove soils and/or stains by treatment (washing) with an aqueous detergent solution and normally including rinsing, extracting and drying.

3.7 l**aundering creases,** n.—sharp folds or lines running in any direction in a washed or dried specimen. NOTE: Laundering creases are an unintended result of restricted movement of specimens in a washer or the dryer.

3.8 **seam smoothness,** n.—*in fabrics*, the visual impression of planarity of a seamed specimen quantified by comparison with a set of reference standards.

3.9 **smoothness appearance,** n.—*in fabrics*, the visual impression of planarity of a specimen quantified by comparison with a set of reference standards.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Detergent may cause irritation. Care should be taken to prevent exposure to skin and eyes.

4.3 All chemicals should be handled with care.

4.4 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

4.5 When evaluating crease retention, use of a light shield can aid in prevention of burns that could result from lamp heat.

5. Uses and Limitations

5.1 This test method is designed to be used for evaluating the appearance of apparel or other textile end products produced with washable fabric after home laundering. Although this method applies to laundering of garments, it is acknowledged that common industry practice is to also rate the product in its original state.

5.2 In general, it is preferable to conduct the test under relatively severe laundering conditions. It is recognized that special cycles or features are available on current washing machines and dryers to achieve improved performance on certain items; e.g., gentle cycles with reduced agitation to protect delicately constructed items, and durable press cycles, with cool-down or cold rinses and reduced spin speeds, to minimize wrinkling.

5.3 Prints and patterns may mask the mussiness present in textile end products. The evaluation process is, however, based on the visual appearance of specimens including such effects.

5.4 The evaluation replicas and scales were created from woven fabrics. It is understood that they do not duplicate other fabric possibilities (knits, nonwovens). The replicas and scales are to be used as guides that represent various levels of smoothness or crease retention.

5.5 The interlaboratory reproducibility of the results of this test method depends upon mutual agreement by users of the method on the washing and drying conditions as outlined in Tables I-IV and the weighting factors to use for individual components of the items as described in 9.7.3.

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Fig. 1—AATCC three-dimensional smoothness appearance replicas.

6. Apparatus and Materials (see 12.1)

6.1 Steam or dry iron with appropriate fabric temperature settings.

6.2 Standard washing machine (see Table I, 12.2), for machine washing.

6.3 Wash tub, 9.5-L, for hand washing.

6.4 White towels, large enough to accommodate test specimen, any weight, for hand washing.

6.5 Standard tumble dryer (see Table IV, 12.2) or facilities for drip, screen or line drying.

6.6 Conditioning facilities and conditioning/drying racks with pull-out, perforated shelves (see 12.3)

6.7 Scale with at least 5 kg or 10 lb capacity.

6.8 AATCC 1993 Standard Reference Detergent (powder with brightener, see 12.3).

6.9 Laundering ballast, Type 1 or Type 3 (see Table V).

6.10 Evaluation area, as described in Appendix A.

6.11 AATCC Three-Dimensional Smoothness Appearance Replicas, set of six (see Fig. 1 and 12.3).

6.12 AATCC Photographic Seam Smoothness Scales, for single-and double-needle seams (see Fig. 2 and 12.3). The reproductions shown in Fig. 2 may not be used for rating.

6.13 AATCC Three-Dimensional Crease Replicas, set of five (see Fig. 3 and 12.3).

7. Test Specimens

7.1 For textile end product tests, select three items for testing.

7.1.1 If the fabric is wrinkled, it may be smoothed by appropriate ironing prior to laundering. See Table I, Safe Ironing Temperature Guide, in AATCC TM133, Test Method for Colorfastness to Heat: Hot Pressing (see 12.3).

8. Laundering Procedure

8.1 Machine Washing.

8.1.1 Select washing conditions for testing from Table I. Normal cycle is recommended. Set washing machine controls to generate the selected cycle parameters.

8.1.2 Load for laundering shall consist of all specimens, plus enough laundering ballast pieces to make a total load weight of 1.8 ± 0.1 kg (4.0 ± 0.2 lb). For very critical evaluation and in arbitration, limit the number of specimens per washer load to those from one sample.

8.1.3 Begin selected wash cycle. Allow machine to fill to specified water level.

8.1.4 Add 66 ± 1 g of AATCC 1993 Standard Reference Detergent to washing machine as directed by washing machine manufacturer's instructions. If detergent is added directly to wash water, agitate briefly to dissolve completely. Stop agitation before adding wash load.

8.1.5 Add wash load (test specimens and ballast), distributing evenly around center agitator. Restart wash cycle.

8.1.6 For specimens to be drip dried (drying procedure C), stop wash cycle before the water begins to drain from the final rinse cycle. Remove specimens soaking wet. For specimens to be tumble (A), line (B), or screen/flat (D) dried, allow washing to proceed through the final spin cycle.

8.1.7 After each washing cycle, separate tangled specimens and ballast pieces, taking care to minimize distortion.

8.1.8 Washer creases. Specimens may be in a folded or creased state after washing. Such folds or creases should be removed by hand prior to drying.

8.1.9 Proceed to appropriate drying procedure.

8.2 Hand Washing.

8.2.1 Select washing temperature from Table II. Add 7.6 ± 1.9 L (2.0 ± 0.5 gal) of water at this temperature to the wash tub.

8.2.2 Add 20 ± 1 g of AATCC 1993 Standard Reference Detergent to wash tub.

8.2.3 Agitate by hand to dissolve detergent.

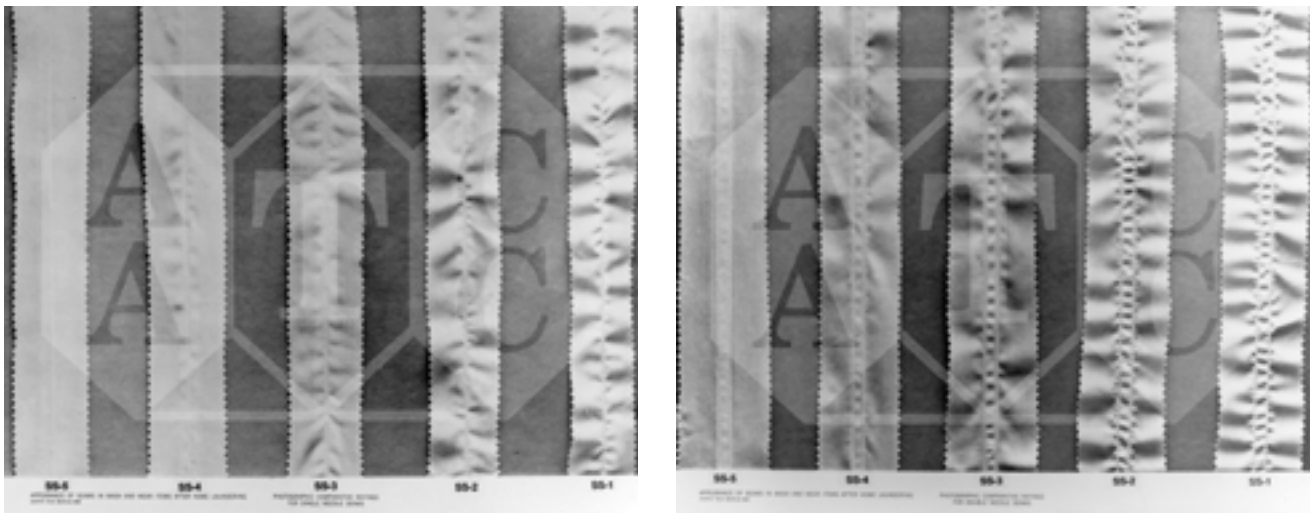
8.2.4 Add specimen(s) to water and gently squeeze to distribute detergent solution. Do not twist or wring.

8.2.5 Let specimen(s) soak for 2 min.

8.2.6 Gently squeeze specimen(s) in detergent solution for 1 min. Do not twist or wring.

8.2.7 Repeat 2-min. soak and 1-min squeeze in detergent solution.

8.2.8 Remove specimen(s) from wash



A. For single needle seams.

B. For double needle seams.

Fig. 2—AATCC photographic seam smoothness replicas.

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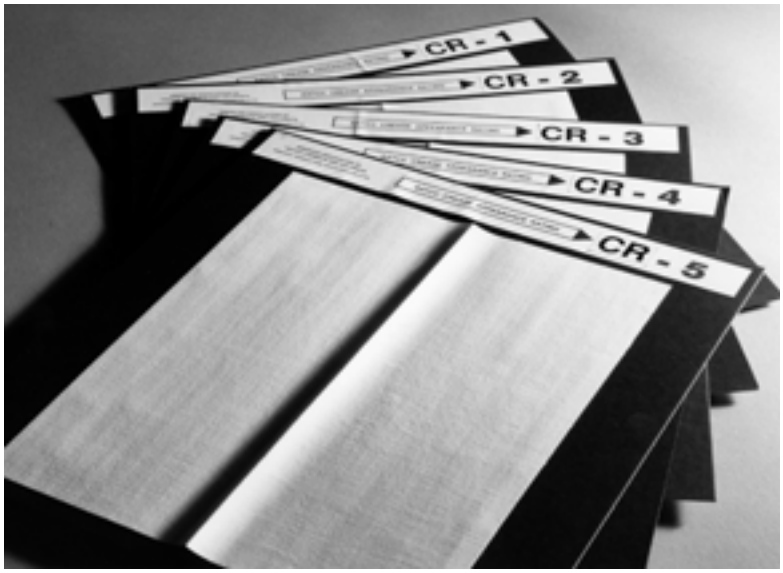


Fig. 3—AATCC three-dimensional crease replicas.

Table II—Standard Hand Washing and Rinsing Temperatures

Designation	Wash Temp, °C (°F)	Rinse Temp, °C (°F)
Very Cold	16 ± 3 (60 ± 5)	< 18 (< 65)
Cold	27 ± 3 (80 ± 5)	< 29 (< 85)
Warm	41 ± 3 (105 ± 5)	< 29 (< 85)
Hot[1]	49 ± 3 (120 ± 5)	< 29 (< 85)

[1] Hot water may not be a reasonable selection for hand washing or for the delicate items typically requiring hand washing.

Table III—Standard Drying Conditions

(A)	Tumble Dry
(Ai)	Tumble Dry Normal
(Aii	Tumble Dry Delicate
(Aiii)	Tumble Dry Permanent Press
(B)	Line/Hang Dry
(C)	Drip Dry
(D)	Screen Dry/Dry Flat

tub and gently squeeze to remove excess detergent solution. Do not twist or wring.

8.2.9 Place specimen(s) on clean white towel. Empty and rinse wash tub.

8.2.10 Add 7.6 ± 1.9 L (2.0 ± 0.5 gal) of clean water at the specified rinse temperature (see Table II) to the wash tub.

8.2.11 Place washed specimen(s) from towel in rinse water and gently squeeze to distribute. Do not twist or wring.

8.2.12 Let specimen(s) soak for 2 min.

8.2.13 Gently squeeze specimen(s) in rinse water for 1 min. Do not twist or wring.

8.2.14 Repeat 2-min. soak and 1-min squeeze in rinse water.

8.2.15 Remove specimen(s) from wash tub and gently squeeze to remove excess water. Do not twist or wring.

8.2.16 Using clean white towels, blot water from washed specimen(s). Do not wring or twist.

8.2.17 Proceed to appropriate drying procedure.

8.3 Drying.

8.3.1 Select drying conditions from Table III.

8.3.2 (A) Tumble Dry. Place the washed load (test specimens and ballast) in the tumble dryer, and set the temperature control to generate the selected cycle exhaust temperature (see Table IV). Allow the dryer to run until the load is fully dry. Remove specimens immediately.

8.3.3 (B) Line/Hang Dry. Hang each specimen on an appropriate hanger, straightening and smoothing facings, seams, etc. Specimens should generally be oriented in the direction in which they are worn. Do not fold or stretch specimens over the hanger. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

8.3.4 (C) Drip Dry. Hang dripping wet specimen on an appropriate hanger, straightening and smoothing facings, seams, etc. Specimens should generally be oriented in the direction in which they are worn. Do not fold or stretch specimens over the hanger. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

8.3.5 (D) Screen/Flat Dry. Spread each specimen on a horizontal screen or perforated surface, removing wrinkles without distorting or stretching it. Allow specimens to rest in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

8.3.6 For all drying methods, allow specimens to dry completely before washing again.

8.3.7 Laundering creases. If specimens are folded or creased after any drying cycle except the last, rewet and attempt to remove the creases by ironing with a hand iron at a temperature suitable for the fabric being tested (see 7.1.1), prior to additional laundering cycles. Do not attempt to remove wrinkles or creases by hand ironing after the final drying cycle.

8.3.8 Repeat the selected washing and drying procedures for a total of 5 complete cycles, or an agreed upon number of cycles.

8.4 Conditioning.

8.4.1 After the final drying cycle is complete, condition test specimens as directed in ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 12.4). (Use conditions indicated in Table 1 for Textiles, general. Estimate conditioning time as indicated in Table 2 for the appropriate fiber content.) Lay

Table I—Standard Washing Machine Parameters (see 12.2, 12.6)

Cycle	(1) Normal		(2) Delicate		(3) Permanent Press	
Water Level, L (gal)	72 ± 4 (19 ± 1)		72 ± 4 (19 ± 1)		72 ± 4 (19 ± 1)	
Agitation Speed, strokes/min.	86 ± 2		27 ± 2		86 ± 2	
Washing Time, min.	16 ± 1		8.5 ± 1		12 ± 1	
Final Spin Speed, rpm	660 ± 15		500 ± 15		500 ± 15	
Final Spin Time, min.	5 ± 1		5 ± 1		5 ± 1	
Wash Temp, °C (°F)[1]	(II) Cold:	27 ± 3 (80 ± 5)	(II) Cold:	27 ± 3 (80 ± 5)	(II) Cold:	27 ± 3 (80 ± 5)
	(III) Warm:	41 ± 3 (105 ± 5)	(III) Warm:	41 ± 3 (105 ± 5)	(III) Warm:	41 ± 3 (105 ± 5)
	(IV) Hot:	49 ± 3 (120 ± 5)	(IV) Hot:	49 ± 3 (120 ± 5)	(IV) Hot:	49 ± 3 (120 ± 5)
	(V) Very Hot:	60 ± 3 (140 ± 5)	(V) Very Hot:	60 ± 3 (140 ± 5)	(V) Very Hot:	60 ± 3 (140 ± 5)

[1] Due to US Department of Energy requirements, many washing machines use cooler water. An external control box may be used to override the machine set temperatures.

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Table IV—Standard Tumble Dryer Parameters (see 12.2)

Cycle	(Ai) Normal	(Aii) Delicate	(Aiii) Permanent Press
Max. Exhaust Temp, °C (°F)	68 ± 6 (155 ± 10)	60 ± 6 (140 ± 10)	68 ± 6 (155 ± 10)
Cool Down Time, min.	≤ 10	≤ 10	≤ 10

Table V—Laundering Ballast Parameters

	Type 1	Type 3
Fiber Content	100% cotton	50% cotton/50% polyester ± 3%
Greige Fabric Yarns	16/1 ring spun	16/1 or 30/2 ring spun
Greige Fabric Construction	52 × 48 ± 5 yarns/in., plain weave	52 × 48 ± 5 yarns/in., plain weave
Finished Fabric Weight	155 ± 10 g/m^2	155 ± 10 g/m^2
Edges	All edges hemmed or over-edged	All edges hemmed or over-edged
Finished Piece Size	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)
Finished Piece Weight	130 ± 10 g	130 ± 10 g

each test specimen flat, separately, on a screen or perforated shelf of a conditioning/drying rack.

9. Evaluation

9.1 Three trained observers should evaluate each test specimen independently.

9.2 Make all evaluations in the specified viewing area (see Appendix A). Illuminate the viewing area with the overhead fluorescent light only. Turn all other lights in the room off except when rating crease appearance. In that case, the floor lamp with reflector and light shield, positioned as shown in Fig. A2, is also required.

9.3 Each observer is to stand directly in front of the specimen 1219 ± 25 mm (48 ± 1in.) away from the board. It has been found that normal variations in the height of the observer above and below the arbitrary 1524-mm (60-in.) eye level have no significant effect on the grade given.

9.4 Smoothness Appearance.

9.4.1 Mount the test specimen on the viewing board with the fabric length in the vertical direction. Place the most similar Three-Dimensional Smoothness Appearance Replicas on each side of the test specimen to facilitate comparative evaluation.

9.4.2 The Smoothness Appearance Replicas were cast from woven fabrics. It is understood that the replicas do not duplicate all possible fabric surfaces. The replicas are to be used as guides that represent various levels of fabric smoothness or freedom from wrinkles. The observer should mentally integrate degree and frequency of wrinkles in the specimen to determine a level of smoothness that can be identified with the SA replica number which most nearly represents that smoothness appearance level (see Table VI).

9.4.3 The smoothness appearance (SA) grade is that of the Smoothness Appearance Replica that most nearly matches the smoothness appearance of the test specimen. Observers may assign a grade midway between those whole-number replicas that have no half-number replica separating them (SA-1.5, SA-2.5, SA-4.5) if the appearance of the test specimen warrants it.

9.4.4 If laundering creases are present on any specimens to be evaluated, take care in rating the specimens. Some laundering creases can be disregarded (commonly called "reading out"). When the grade of a laundering-creased specimen differs from the other specimens by more than one grade, the test should be repeated with new specimens, taking all precautions to avoid the occurrence of laundering creases.

9.5 Seam Smoothness.

9.5.1 Mount the test specimen on the viewing board with the seam in the vertical direction. Place the appropriate single-or double-needle AATCC Photographic Seam Smoothness (SS) Scale beside the specimen to facilitate comparative evaluation.

9.5.2 Confine observations to the area influenced by the seam and disregard the appearance of the surrounding fabric. The Seam Smoothness Scales were photographed from woven fabrics. It is understood that the photographs do not duplicate other fabric or seam possibilities. The scales are to be used as guides that represent various levels of seam smoothness.

9.5.3 The seam smoothness (SS) grade is that of the Photographic Scale seam that most nearly matches the appearance of the test specimen seam.

9.6 Crease Retention.

9.6.1 Mount the test specimen on the viewing board with the crease in the vertical direction. Place the most similar Three-Dimensional Crease Replicas on each side of the test specimen to facilitate comparative evaluation. Mount replicas 1, 3 and 5 on the left; mount replicas 2 and 4 on the right.

9.6.2 Position the flood lamp with reflector and light shield in the viewing area, as shown in Fig. A2, and used during the evaluation process.

9.6.3 Confine observations to the crease itself and disregard the appearance of the fabric. The Crease Replicas were cast from woven fabrics. It is understood that the replicas do not duplicate other fabric possibilities (knits, nonwovens). The replicas are to be used as guides that represent various levels of fabric creasing.

9.6.4 The crease retention (CR) grade is that of the Crease Replica that most nearly matches the appearance of the test specimen crease.

9.7 Appearance of Textile End Products.

9.7.1 The individual components to be evaluated in each test item shall be determined and entered on a Rating Chart (Fig. 4).

9.7.2 If it is desired to designate certain components as more or less important to the overall appearance of the item, weighting factors should be added to the Rating Chart.

9.7.3 The weighting factors to be assigned to each component are:

3—Very important to overall item appearance.

2—Moderately important to overall item appearance.

1—Slightly important to overall item appearance.

9.7.4 Mount the item on the viewing board so that the center of the area or component to be rated is approximately 1524 mm (60 in.) from the floor as illustrated in Fig. A1. Place the appropriate replicas or photographs in proper position to facilitate comparative evaluation (refer to 9.4, 9.5 or 9.6).

9.7.5 If the item is exceptionally large, such as a sheet, comforter, bedspread, curtain or drapery, fold the item lengthwise to produce a panel half the original width. Place this half-panel over a rod so that the fabric length is vertical and the

Table VI—Fabric Smoothness Grades by SA Replica Equivalents

Grade	Description
SA-5	Equivalent to the SA-5 Replica. Very smooth, pressed, finished appearance.
SA-4	Equivalent to the SA-4 Replica. Smooth, finished appearance.
SA-3.5	Equivalent to the SA-3.5 Replica. Fairly smooth but nonpressed appearance.
SA-3	Equivalent to the SA-3 Replica. Mussed, nonpressed appearance.
SA-2	Equivalent to the SA-2 Replica. Rumpled, obviously wrinkled appearance.
SA-1	Equivalent to the SA-1 Replica. Crumpled, creased and severely wrinkled appearance.

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Identification:

Component (Characteristic, Attribute)	Weighting Factor (10.1)	Average Grade (10.2)	Point Value
	X		=
	X		=
	X		=
	X		=
	X		=
	X		=
	X		=
	X		=

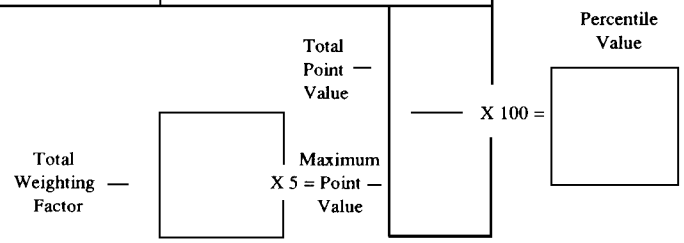


Fig. 4—Rating chart.

folded item is in equal quarters. The rod should be sufficiently long to accommodate the half-width item. Attach the rod with the large item to the rating board at approximately 1829 mm (72 in.) from the floor. Position the standard replicas or photographs to facilitate comparative evaluation. Evaluate the area across the full width of the quartered panel at the same eye level as the replicas. Evaluate all four quarters in the same manner and report average grade for each component evaluated in the test item.

10. Report

10.1 Option 1—Using Weighting Factors—Total the weighting factors assigned to each component in the rating chart (Fig. 4) and multiply by 5. This gives the maximum point value achievable by the item. Multiply the average grade recorded for each component by its assigned weighting factor. Total these values to obtain the actual point value achieved by the item. Report, as the percentile value of the item, the number obtained by dividing the actual point value by the maximum point value achievable and multiplying by 100%. This value is the unit of measure of this test method.

10.2 Option 2—Report average grades on each individual component of each test item, using the Average Grade column of the Rating Chart.

10.3 Report for each sample tested:

10.3.1 A description or identification of the sample.

10.3.2 Evaluation was using AATCC TM143-2018t.

10.3.3 Number of specimens used.

10.3.4 Type of ballast used, i.e., Type 1 or Type 3).

10.3.5 Number of laundering cycles (default is 5 cycles).

10.3.6 Laundering conditions, including washing machine cycle, wash temperature, drying procedure and tumble drying temperature, if applicable. Alphanumeric designations may be used if they are clearly understood by all parties. For example, 1-IV-A(ii) indicates normal wash cycle at 49°C and Delicate tumble dry cycle.

10.3.7 Number of observers.

10.3.8 Seam smoothness scale used. i.e., single-needle or double-needle.

10.3.9 Any modifications of the test method.

10.3.10 For Option 1, the overall percentile value for the item.

10.3.11 For Option 2, the average SA grade, average SS grade, and average CR grade, each to the nearest tenth of a grade.

10.4 If fraying occurs in laundering at seams or elsewhere within the product, the location and amount should be noted.

11. Precision and Bias

11.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

11.2 *Bias*. The appearance of apparel and other textile end products after home laundering can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

12. Notes

12.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at http://www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

12.2 For model number(s) of washer(s) and tumble dryers(s) reported to meet the standard parameters, visit www.aatcc.org/test/washers or contact AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org. An alternate load size was described in previous versions of this method (3.6-kg load, 83-L water level, 80 g AATCC 1993 Standard Reference Detergent), but no machine is reported to meet the standard parameters for this load size. In addition, dimensional change results obtained with the alternate load size may not be equal to those obtained with a standard load size.

12.3 Materials or drawings available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

12.4 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610. 832.9500; fax: +1.610.832.9555; web site: www.astm.org.

12.5 The use of two 96-in. lamps for viewing laundered specimens is specified in this method. It is recognized, however, that physi-

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cal space limitations in certain laboratories will prevent the use of 96-in. lamps. In those situations, two 48-in. lamps and a narrower viewing board may be used, but replicas identified as SA-4, SA-3 and SA-1 should always be placed on the left side of the viewing board as the board is viewed from the front. Replicas identified as SA-5, SA-3.5 and SA-2 should always be placed on the viewing board to the right side as the board is viewed from the front.

12.6 The laundering temperatures and other parameters listed in this test method are standard conditions for testing purposes. As with most lab procedures, they represent, but may not exactly replicate, current consumer practices. Consumer practices vary over time and among households; lab practices must be consistent to allow valid comparison of results. If laundering equipment or conditions other than those specifically listed in this test method are used, they must be described in detail and noted as a modification of the standard method. Alternate laundering conditions are outlined in AATCC LP1 and ISO 6330.

12.7 Available from ISO, www.iso.org.

Appendix A
Evaluation Area

A1 Viewing Board.

A1.1 Plywood board, 1829 mm (72 in.) high × 6 mm (0.25 in.) thick × wide enough to accommodate scale and specimen side by side

A1.2 Painted to match the Grade 2 chip on the Gray Scale for Staining (see 12.3). Approximate CIELAB values for the chip are L* = 77, a* = 0, b* = 0. A tolerance of two units for each parameter is acceptable for the board color in this test method.

A1.3 Spring-loaded swatch mount or other means to secure specimen and scale such that center of specimen is 1524 mm (60 in.) above the floor. Mount may be fabricated using light sheet metal (22 ga.)

A2 Lighting.

A2.1 Overhead fluorescent tube light fixture (see 12.5).

A2.1.1 Two parallel F96 T12 cool white lamps (without baffle or glass),

A2.1.2 One white enamel reflector (without baffle or glass),

A2.1.3 Mount as shown in Fig. A1.

A2.2 Incandescent flood lamp, 500 watts, 120 V, C9 filament, 90° flood spread, for crease evaluation.

A2.2.1 254-mm (10-in.) aluminum reflector, for use with flood lamp.

A2.2.2 Light shield.

A2.2.3 Position as shown in Fig. A2.

A2.3 Eliminate all light sources except the specified fluorescent lamps. Use flood lamp for crease evaluation only.

A3 Walls.

A3.1 It has been the experience of many observers that light reflected from the side walls near the viewing board can interfere with the rating results. It is recommended that the side walls be painted matte black (85° gloss less than 5 units) or that blackout curtains be mounted on both sides of the viewing board to eliminate the reflective interference.

13. History

13.1 Editorially revised 2023 to add History section.

13.2 Editorially revised 2018; revised 2018 (with title change); editorially revised 2012; revised 2011; 2010; editorially revised 2008; revised 2006; editorially revised 2005, 2004; editorially revised and reaffirmed 2001; editorially revised 1997; revised 1996, 1992; editorially revised 1991; revised 1989; editorially revised 1986; editorially revised and reaffirmed 1984; revised 1982.

13.3 Developed in 1975 by AATCC committee RA61. Technically equivalent to ISO 15487.

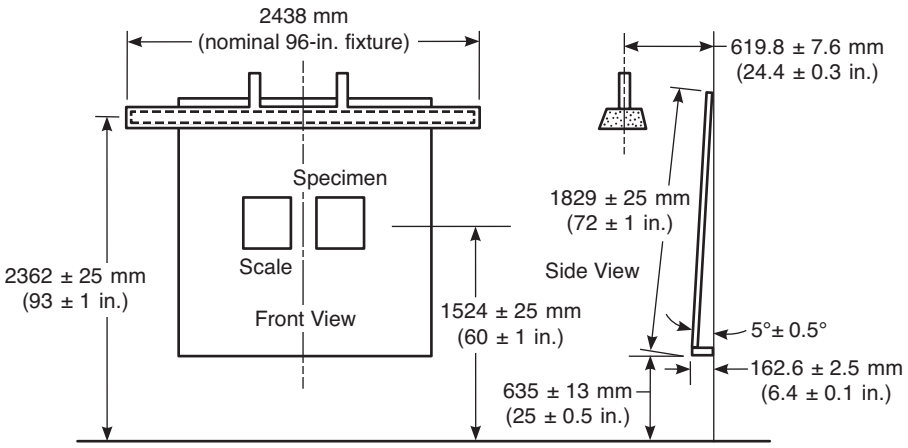


Fig. A1—Evaluation area.

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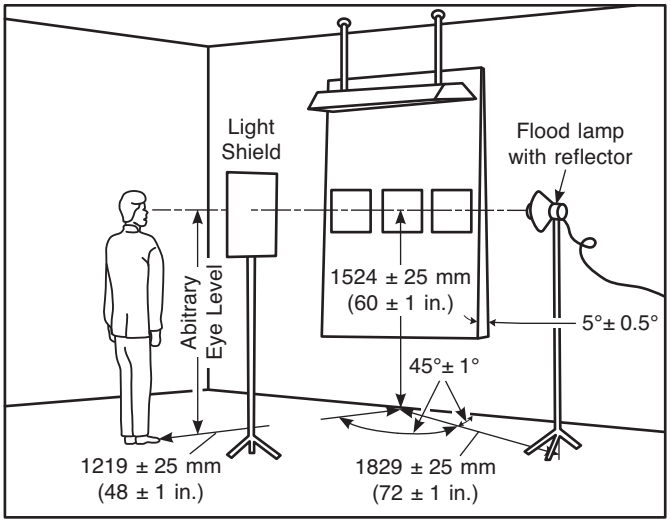


Fig. A2—Lighting and viewing arrangement for crease-retention evaluation.

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Test Method for Alkali in Wet Processed Textiles: Total

1. Purpose and Scope

1.1 This test method is used to determine the total alkali content of wet processed textiles. Total alkali can be used to determine washing and/or neutralizing efficiency after certain wet processing steps, particularly bleaching, and can be used as a measure of the suitability of prepared fabric for subsequent dyeing and finishing operations.

2. Principle

2.1 Each specimen is immersed in distilled or deionized water, and the solution is then titrated with a standard acid to a predetermined end point. The percentage of alkali can then be calculated from the amount of acid used and the weight of the specimen.

3. Terminology

3.1 **bleaching,** n.—elimination of unwanted coloring matter from a substrate by oxidative or reductive chemical treatment.

3.2 **pH,** n.—the negative logarithm of the hydrogen ion activity expressed in gram equivalents per liter used in expressing both acidity and alkalinity on a scale whose values run from 0 to 14 with 7 representing neutrality, numbers less than 7 representing increasing acidity, and numbers greater than 7 representing increasing alkalinity.

3.3 **total alkali,** n.—*in textile wet processing*, residual alkaline materials in a wet processed textile expressed as percent sodium hydroxide on dry weight of the textile.

3.4 **wet processing,** n.—*in textile manufacturing*, a collective term for processes included in preparation, dyeing, printing, and finishing in which the textile material is treated with a liquid, normally water, or with chemicals in solution or dispersion in a liquid.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 If concentrated sulfuric acid is diluted to prepare the 0.10*N* sulfuric acid (see 6.5), use chemical goggles or face shield, impervious gloves, and an impervious apron. Concentrated acids should be handled only in an adequately ventilated laboratory hood. CAUTION: Always add acid to water.

4.4 An eyewash/safety shower should be located nearby and a high efficiency particulate respirator with a full facepiece should be readily available for emergency use.

4.5 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVS) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL), and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 14.1).

5. Limitations

5.1 In this test method all alkalis detected, such as sodium hydroxide, sodium carbonate, sodium bicarbonate, and other alkaline salts, are calculated as sodium hydroxide. The results are reported as total alkali, expressed as NaOH.

5.2 Since distilled or deionized water can contain dissolved carbon dioxide, care should be taken to remove it by boiling water before use.

6. Apparatus and Reagents

6.1 pH meter with 0.1 unit graduations.

6.2 Beakers, glass, 600 mL.

6.3 Burette, glass, 10 mL with 0.10 mL graduations.

6.4 Buffer solution, pH 4.0 and 7.0.

6.5 Sulfuric acid (H_2SO_4), 0.10*N* (see 14.2).

7. Calibration

7.1 Calibrate the pH meter with pH 4.0 buffer solution in accordance with the manufacturer's instructions. Dual calibration instruments should be calibrated to pH 4.0 and pH 7.0.

8. Specimens

8.1 Take samples in the dry or wet state.

8.2 Dry cloth. Select two or more representative dried specimens. Place in oven at 100°C (212°F) for 1 h. Cool in a desiccator and weigh to the nearest 0.1 g. Specimens should weigh 5-10 g.

8.3 Wet cloth. Select two or more representative specimens weighing approximately 10-20 g in the wet state.

9. Procedure

9.1 Place each specimen in a 600 mL beaker containing 450-500 mL distilled or deionized water at room temperature and stir vigorously for 1 min. Cover the beaker and keep the specimen immersed for 15 min. Finally, stir the specimen again and insert the pH meter electrode(s), being careful to avoid contact with the specimen.

9.2 Titrate the water and specimen with 0.10*N* H_2SO_4 to a steady (10 s) pH 3.9 endpoint. Add titrant dropwise followed by gentle agitation of the specimen without touching the electrode(s). Read burette to the nearest graduation (see 14.3).

9.3 If the specimen was taken from a wet sample, rinse and dry the titrated specimen to a constant weight (to the nearest 0.1 g) as in 8.2, being careful to collect any large threads that may have unraveled during the titration procedure.

9.4 Titrate a water blank by repeating 9.2 without a specimen present.

10. Calculations

10.1 Compute the percentage total alkali from the titer according to the following formula:

$$X = [(A - B)(0.04)(N)\ 100]/W$$

where:

X = % total alkali as sodium hydroxide (NaOH),
A = volume of acid used for specimen, mL,
B = volume of acid used for blank, mL,
N = normality of sulfuric acid (0.10*N*),
W = weight of specimen.

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10.2 Compute the sample average by using the formula:

$$x = (x_1 + x_2 + \ldots + x_n)/n$$

11. Evaluation

11.1 Total alkalinity is important because of its effect on the subsequent processing operations to which the textile is subjected. Interpretation of test results is usually based on a combination of total alkalinity and pH of the water extracted from the fabric as determined by AATCC TM81, Test Method for pH of the Water Extract from Wet Processed Textiles.

11.2 Total alkalinity quantifies the amount of residual alkali in a specimen, per unit of weight, providing the specimen contains alkali.

12. Report

12.1 Describe or identify the sample.

12.2 Report that sample was testing using AATCC TM144-2022.

12.3 Report the average total alkalinity.

13. Precision and Bias

13.1 *Precision*.

13.1.1 In 1993, an interlaboratory study was completed, which included five laboratories, two operators in each, running three determinations per fabric, on four fabrics. No prior assessment was made of the relative level of the participating laboratories in performance of the test method.

13.1.2 Analysis of the data set ($5 \times 2 \times 3 \times 4 = 120$ values) yielded components of variance as follows.

Components of Variance	Variance
Laboratories	0.000245
Operators within laboratories	0.000061
Specimens within materials, laboratories, and operators	0.000047

13.1.3 Table I (critical differences) was calculated using the values in 13.1.2.

13.1.4 Differences between two averages of N determinations, for the appropriate precision parameter, should reach or exceed the table value to be statistically significant at the 95% probability level.

13.2 *Bias*.

13.2.1 The values of the total alkali in wet processed textiles can only be defined in terms of a test method. There is no known bias in the determination of this property by this test method. During this study, no determination of the true value was made by an independent, referee analytical method for the purpose of establishing presence or absence of bias.

Table I—Critical Differences for Two Averages—95% Probability Level

Alkali in Wet Processed Textiles: Total (% weight total alkali as sodium hydroxide)

N	Single Operator	Within Laboratory	Between Laboratory
1	0.0190	0.0288	0.0509
2	0.0134	0.0230	0.0491
4	0.0095	0.0209	0.0481
8	0.0067	0.0198	0.0477

14. Notes

141 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

14.2 For method of preparation of standard acid refer to W. W. Scott, *Standard Methods of Chemical Analysis*, 6th Ed., Van Nostrand, New York, 1962, p1343.

143 If preferred, titration may be made with a suitable colorimetric indicator such as methyl orange (pH 3.1-4.4). The precision may be less with this method than with a good pH meter.

15. History

15.1 Revised in 2022 to align to the AATCC style guide.

15.2 Editorially revised 2019. Reaffirmed: 2016, 2012. Editorially revised 2010. Reaffirmed 2007, 2002. Revised (with title change) 1997. Reaffirmed 1992. Editorially revised 1990. Revised (with title change) 1987. Reaffirmed 1986. Editorially revised 1985. Reaffirmed 1980, 1977.

15.3 Developed in 1975 by AATCC Committee RA34.

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Test Method for Dispersibility of Disperse Dyes: Filter

1. Purpose and Scope

1.1 This test determines the dispersibility as evaluated by filtering time and filter residue of disperse dyes under standard conditions in aqueous media. Additional information such as variables which affect accuracy and repeatability of the test are specified in Section 10.

1.2 This test method is to be used for determining the degree of dispersion at the specified conditions in aqueous media only.

2. Principle

2.1 A quantity of disperse dye is diluted, heated and passed through filter papers of specified micron size. The time needed for the dye dispersion to pass through the filter is then recorded.

2.2 Three tests are outlined according to intended dye application (see 10.5 and Table I).

3. Terminology

3.1 **disperse dye,** n.—an essentially water insoluble dye having affinity, when properly dispersed, for polyester, polyamide and some other manufactured polymeric fibers.

3.2 **dispersibility,** n.—the degree to which particles can be broken down to some minimum size such that they will pass through the interstices of a standard filter paper.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas and a single use dust respirator while handling powder dyes.

4.2 All chemicals should be handled with care. The compounds used in this test method at the concentrations specified are considered to present no significant hazard when used in accordance with good laboratory practices.

4.3 If concentrated acetic acid is diluted to prepare the 10% acetic acid, use chemical goggles or face shield, impervious gloves and an impervious apron during preparation. Concentrated acids should be handled only in an adequately ventilated laboratory hood. CAUTION: Always add acid to water.

4.4 An eyewash/safety shower should be located nearby and an organic vapor respirator with full facepiece should be readily available for emergency use.

4.5 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administrator's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 10.15).

Table I. Selection of Applicable Test

Test	Whatman Filter Paper Combination	Application of Dyes	pH of Dispersion
I	#2 over #4	Package dyeing of polyester	4.5-5.0
II	#4 over #4	Beck dyeing of polyester	4.5-5.0
III	#4 over #4	Dyeing of nylon carpet and apparel	9.0-10.0

5. Apparatus and Materials

5.1 Buchner funnel: Coors glazed one-piece porcelain, 110 mm diameter (see 10.2).

5.2 Filter papers: Whatman #2, 110 mm diameter: Whatman #4, 110 mm diameter or exact equivalent filter paper from other sources (see 10.4).

5.3 Stainless steel ring with approximate dimensions 103 mm (4.05 in.) ID, 111 mm (4.33 in.) OD and 8 mm (0.32 in.) height to hold filter paper in place (see 10.6 and 10.12).

5.4 Filtering flask with side tube, capacity 1000 mL.

5.5 Manometer to measure vacuum (pressure drop).

5.6 Vacuum pump or water aspirator to obtain vacuum of 560 ± 100 mm Hg (22 ± 4 in. Hg).

5.7 Vacuum control valve.

5.8 Rubber tubing, vacuum.

5.9 Stopwatch.

5.10 Beakers, 400 mL or larger.

5.11 Balance, analytical.

5.12 Tetrasodium salt of ethylenediaminetetraacetic acid (EDTA) as 25% solution, such as Cheelox BF-12 or equivalent.

5.13 Tetrasodium pyrophoshate, anhydrous (TSPP) ($Na_4P_2O_7$), 10% solution.

5.14 Acetic acid (CH_3COOH), 10% solution.

5.15 Filter Residue Scale (see Fig. 1 and 10.12).

5.16 Distilled or deionized water, which does not change significantly in pH on heating. This water is modified with 0.25 g/L of chelating agent (EDTA) as specified in 5.12.

5.17 Dye samples (see 6.2).

5.18 Stirrer, laboratory, magnetic.

5.19 Graduated cylinder, 250-mL.

5.20 pH meter.

6. Procedure

6.1 Water Preparation.

6.1.1 Preheat 400 mL of water to 71°C (160°F). This will be used to preheat the funnel before conducting test (see 10.7 and 6.6).

6.1.2 Prepare special water (see 5.16) for use in dye dispersion as follows: Add 0.25 g/L EDTA (see 5.12) to distilled or deionized water.

6.2 Weighing of dyes.

6.2.1 Weigh out 2.0 ± 0.1 g of dye

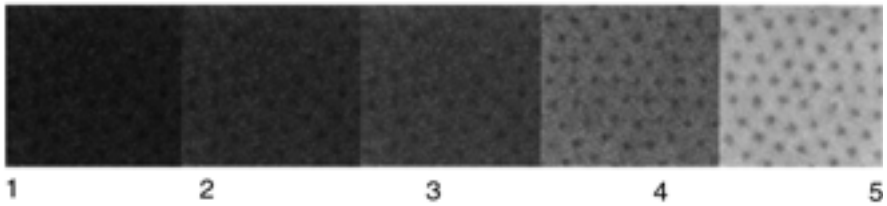


Fig. 1—Filter residue scale.

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powder or 4.0 ± 0.2 g of 50% liquid of the dye sample to be tested. Also weigh out an equal quantity of the standard dye (see 10.3).

6.2.2 If the dyes are of high tinctorial strength, the weighed amount can be reduced to 1 g/200 mL but should be reported as only 1g used (see 10.8).

6.2.3 When evaluating two different dyes of approximately equal shade but of different tinctorial values, one dye is always selected as the standard. Then the other product is run on an equal concentration (grams/200 mL) basis. When evaluating two samples of the same dye, the standard is the one furnished as standard material by the respective dye manufacturer. When evaluating the same dye from two different manufacturers, one should determine if the two products are of equal strength before running the test. If a strength difference is determined, the two products should be evaluated as outlined in the first two sentences of this paragraph.

6.3 Add the weighed dye (see 6.2.1) slowly and completely to 200 mL of the vigorously agitated specially treated water (see 5.16) at 43-49°C (110-120°F) in a 400 mL beaker (see 10.9) using a magnetic stirrer. Adjust the pH of the dye dispersion as follows:

6.3.1 Tests I and II: adjust to pH 4.5-5.0 with a 10% solution of acetic acid (see 5.13 and 10.14).

6.3.2 Test III: adjust to pH 9-10 with TSPP (see 5.13 and 10.14).

6.4 Heat the adjusted solution 5-10 min to 71°C (160°F). Stir by means of a magnetic stirrer to prevent localized heating (see 10.1 and 10.10).

6.5 Pour 200-300 mL of 71°C (160°F) water prepared in 6.1.1 through a 110 mm funnel without filter paper to preheat the funnel. Wait 25 ± 10 s. Turn on the vacuum until the water has passed through the funnel. Quickly wipe funnel dry (see 10.6 and 10.7).

6.6 Cut vacuum off and immediately place Whatman filter papers in dry Buchner funnel (see 10.4, 10.5, 10.6 and Table I):

Test I:	Whatman #2 on top
	Whatman #4 on bottom
Test II:	Whatman #4 on top
	Whatman #4 on bottom
Test III:	Whatman #4 on top
	Whatman #4 on bottom

6.7 Insert stainless steel ring in funnel over filter papers and turn on vacuum. The vacuum should be 560 ± 100 mm (22 ± 4 in.). This range is obtained by using either a standard vacuum pump or house vacuum. A water aspirator may not give this vacuum until a liquid is introduced into the system. In this event, the test should proceed after pouring the dye dispersion into the funnel and waiting 10-13 s for full vacuum.

6.7.1 With the vacuum on, immediately pour dye dispersion which has been heated to 71°C (160°F) (see 6.4) into funnel and begin timing. The end point is reached when the filter paper changes from a wet to a dry visual appearance. Record time up to 120 s to the nearest s.

6.7.2 In order to achieve more reproducible results, especially with dyes in Class 3 (intermediate values are interpolated) or lower, it may be advisable to rinse the filter paper while it is in the funnel with 10-15 mL of the specially prepared water (see 6.1.2) before drying (see 10.13).

6.8 Allow filter paper to dry and evaluate as outlined in Section 7).

7. Evaluation

7.1 Record time for filtration and rate as follows:

Class A—0-24 s
Class B—25-49 s
Class C—50-74 s
Class D—75-120 s
Class E—> 120 s

7.2 Examine residue on filter paper versus the Filter Residue Scale (standard photographic replicas) (see 10.13, 10.14). Also examine the paper for any visible coarse or granular particles (see 10.13). If present the test is automatically rated as Class 1. If none is present, rate as follows:

Class 5—Excellent
Class 4
Class 3—Intermediate values are interpolated
Class 2
Class 1—Poor

8. Report

8.1 Classify and report the following:

8.1.1 Test Number (see 10.6 and Table I).

8.1.2 Class according to filtration rate (see 7.1).

8.1.3 Class according to amount of residue (see 7.2).

Example: I-A-3, where I is the Test Number (Whatman #2 over Whatman #4); A indicates the time of filtration (0-24 s); and 3 represents amount of residue on filter paper as compared with a standard photographic chart.

9. Precision and Bias

9.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

9.2 *Bias*. The dispersibility of disperse dyes can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

10. Notes

10.1 The results of this test method can vary widely unless all details are run under the conditions outlined. Any variation in the conditions can cause the test to be invalid. Results have been shown to be repeatable in several laboratories when the specified conditions were met.

10.2 Variations in results can be caused by differences in the diameter of the filter funnel due to differences in surface area.

10.3 The dye standard is always included so that the influence of any operator induced variable on the results is minimized.

10.4 Whatman paper is specified because it was used by all participants in developing this test method. Any other filter paper which gives equivalent results may be used. Two filter papers are used to provide a more uniform vacuum across the filtering surface. The filter paper is used as received in the box with the slick side up to avoid variations due to texture differences on the rough side. The coarser paper should always be on the bottom. Other papers may be used if quality and microretention are exactly the same as the Whatman paper.

Whatman #2—Filters particles above 8 µm.
Whatman #4—Filters particles above 25 µm.

10.5 Tests I, II and III are defined as follows:

Test I—Whatman #2 filter paper over Whatman #4 filter paper. This test is designed to evaluate the most critical dispersion requirements such as for package dyeing of polyester.

Test II—Whatman #4 filter paper over Whatman #4 filter paper. This test is designed to evaluate dispersion requirements for dyeing at longer liquor ratios (such as beck dyeing).

Tests I and II are carried out at an acid pH of 4.5-5.0 to reflect common dyeing practices for disperse dyes on polyester fiber.

Test III—Whatman #4 filter paper over Whatman #4 filter paper. This test is designed to evaluate the dispersion requirements of disperse dyes for application on nylon (carpet and apparel dyeing). The pH of 9-10 specified for this test reflects common industrial practice.

10.6 Since the Whatman filter paper is cellulosic in nature, it can swell quite readily when wet out. Therefore, if the paper is prewet, the time of filtration will go up as a result of the wet out temperature and/or time lag. Thus it is *imperative* that the paper not be wet prior to the test, since the micron retention of the paper is reduced. This is the reason for the stainless steel ring, which holds the filter paper in place without prewetting. On dyes which have a very small particle size, the speed of filtration will not be changed to a sizable degree by a vacuum alteration. However, a reduced vacuum can result in faster filtration times on dyes of larger particle size, since the suction is not as great. On dyes of larger particle size, an increased vacuum will block the paper faster, slowing up the filtration rate, and thus swelling the paper more. This results in higher filtration times and more residual dye on the paper.

10.7 The funnel should always be preheated with 200-300 mL of water heated to 71°C (160°F) and then wiped dry so that the dispersion will not be reduced in temperature by a cold funnel. The preheating cycle should always come just prior to running the tests with-

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out the paper in place.

10.8 A concentration of 2 g/200 mL was chosen as representing the upper limit of normal dye application. On dyes of high tinctorial strength, this can be reduced to 1 g/200 mL, but it should always be reported that 1 g was used.

10.9 It is very important to use 43-49°C (110-120°F) water. Wide variations can result with certain dyes if water at 71°C (160°F) is used at the beginning. The specified procedure allows the dye to wet out under conditions similar to those used in plant practice.

10.10 Excessive time lags and holding temperatures may affect the outcome of the test. Test should be completed within 15 min.

10.11 The Filter Residue Scale consists of photographs representing five classes of retention of filter residue and is used as part of the evaluation procedure (see 7.2).

10.12 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

10.13 Rinsing the filter paper before drying with a small amount (10-15 mL) of the specially prepared water will aid in the removal of any colloidal dye that actually has a micron size smaller than the paper being employed.

10.14 It is necessary to adjust the pH at this point since various dyes have wide pH ranges themselves. For example, the pH of dispersions of paste dyes usually is more alkaline than of dispersions of the same dyes in powder form.

10.15 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

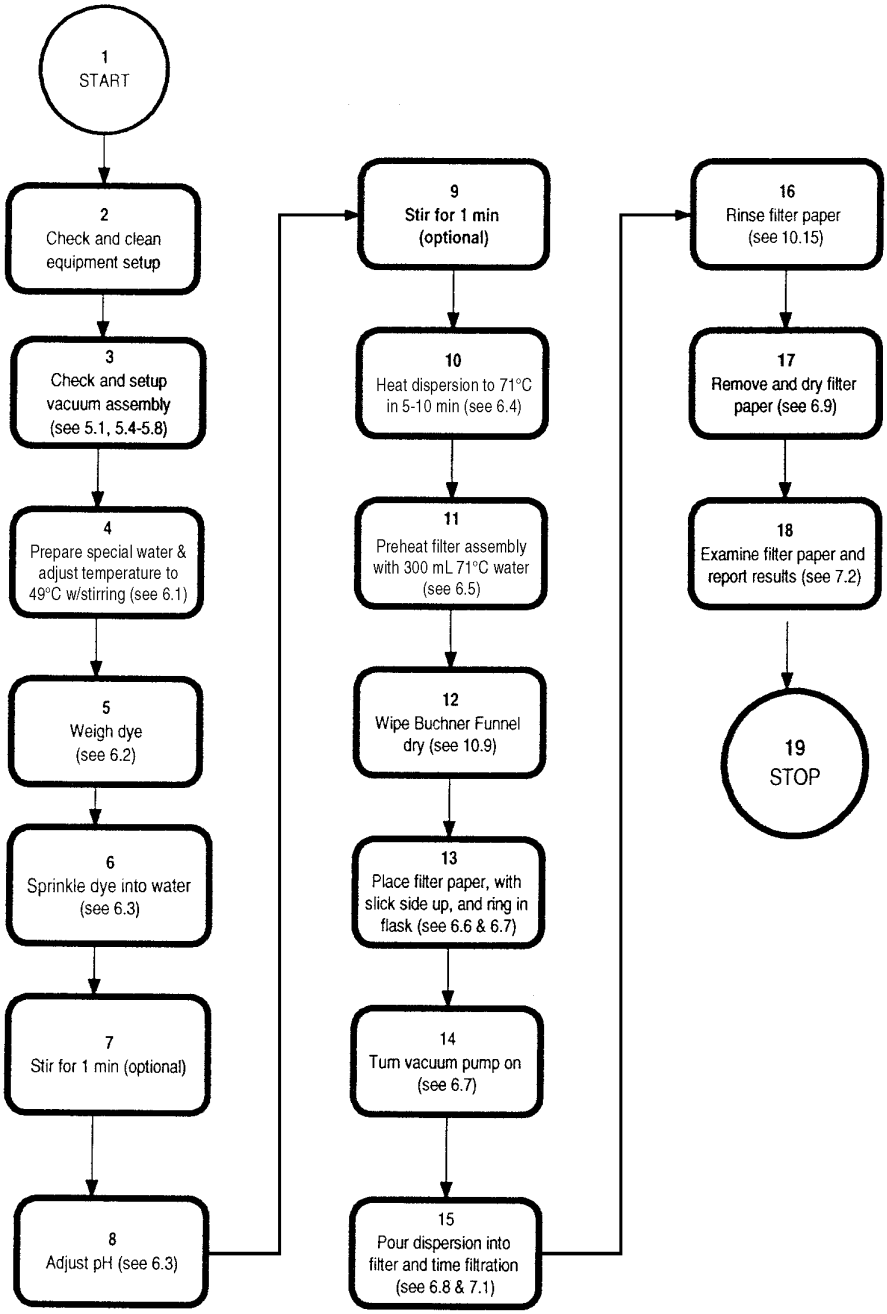
11. History

11.1 Reaffirmed and editorially revised to add the History section.

11.2 Editorially revised 2019; reaffirmed 2015;revised 2011; editorially revised 2010, 2008; reaffirmed 2006; editorially revised 2004, 2001: revised 1996;editorially revised 1995; editorially revised and reaffirmed 1994; reaffirmed 1989;editorially revised 1987; editorially revised 1985; editorially revised and reaffirmed 1984; editorially revised 1983, 1980; revised 1979; reaffirmed 1977, 1976

11.3 Developed in 1975 by AATCC Committee RA87. Related to ISO 105-Z04.

Appendix A. Flow Chart for AATCC TM146



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Test Method for Antibacterial Activity of Textile Materials: Parallel Streak

Developed in 1976 by AATCC Committee RA31; reaffirmed 1977, 1982, 1998, 2016; editorially revised 1980, 1982, 1983, 1986, 2010, 2019 (with title change); revised 1987, 1988 (with title change), 1993, 2011; editorially revised and reaffirmed 2004.

Foreword

The Parallel Streak Method has filled a need for a relatively quick and easily executed qualitative method to determine antibacterial activity of diffusible antimicrobial agents on treated textile materials.

AATCC TM100, Test Method for Antibacterial Finishes on Textile Materials, Assessment of, is a quantitative procedure which is adequately sensitive but is cumbersome and time consuming for routine quality control and screening tests. Therefore, when the intent is to demonstrate bacteriostatic activity by the diffusion of the antibacterial agent through agar, AATCC TM147 fulfills this need. In the Parallel Streak Method, the agar surface is inoculated making it easier to distinguish between the test organism and contaminant organisms which may be present on the unsterilized specimen. The Parallel Streak Method has proven effective over a number of years of use in providing evidence of antibacterial activity against both Gram positive and Gram negative bacteria.

1. Purpose and Scope

1.1 The objective is to detect bacteriostatic activity on textile materials. The results of using this procedure have been demonstrated by Committee RA31 to be reproducible by various laboratories working with materials containing residual amounts of antibacterial agents (as determined by chemical assay) after multiple standard washings. The method is useful for obtaining a rough estimate of activity in that the growth of the inoculum organism decreases from one end of each streak to the other and from one streak to the next resulting in increasing degrees of sensitivity. The size of the zone of inhibition and the narrowing of the streaks caused by the presence of the antibacterial agent permit an estimate of the residual antibacterial activity after multiple washings.

2. Principle

2.1 Specimens of the test material, including corresponding untreated controls of the same material, are placed in intimate contact with the agar surface which has been previously streaked with an inoculum of a test bacterium. After incubation, a clear area of interrupted growth underneath and along the sides of the test material indicates antibacterial activity of the specimen. A standard strain of bacteria is used which is specific to the requirements of the material under test. If no other bacterial species is specified, *Staphylococcus aureus* may be used as a representative Gram positive organism. Other recommended strains are listed below in Section 6.

3. Terminology

3.1 **activity,** n.—*of an antibacterial agent*, a measure of effectiveness of the agent.

3.2 **antibacterial agent,** n.—any chemical which kills bacteria (bactericide) or interferes with the multiplication, growth or activity of bacteria (bacteriostat).

3.3 **zone of inhibition,** n.—clear area of no growth of a microorganism, cultured onto the surface of an agar growth medium, in proximity to the borders of a specimen placed in direct contact with this agar surface.

NOTE: A zone of inhibition occurs as a result of the diffusion of an antimicrobial agent from the specimen.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 This test should be performed only by trained personnel. The U.S. Department of Health and Human services publication *Biosafety in Microbiological and Biomedical Laboratories* should be consulted (see 12.1).

4.2 CAUTION: Some of the bacteria used in this test are pathogenic; i.e., capable of infecting humans and producing disease. Therefore, every necessary and reasonable precaution must be taken to eliminate this risk to the laboratory personnel and to personnel in the associated environment. Wear protective clothing and respiratory protection that prevents penetration by the bacteria.

4.3 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.4 All chemicals should be handled with care.

4.5 An eyewash/safety shower should be located nearby for emergency use.

4.6 Sterilize all contaminated samples and test materials prior to disposal.

4.7 Exposure to chemicals used in this procedure must be controlled at or below levels set by government authorities (e.g., Occupational Safety and Health Administrations [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 12.2).

5. Uses and Limitations

5.1 The method is not suitable for materials which tend to encapsulate and prevent the diffusion of the antibacterial agent or contain antibacterial-neutralizing substances.

6. Test Organisms

6.1 Test bacteria:

6.1.1 *Staphylococcus aureus*, ATCC 6538, CIP 4.83, DSM 799, NBRC 13276, NCIMB 9518 or equivalent strain (see 12.3).

6.1.2 *Klebsiella pneumoniae*, ATCC 4352, CIP 104216, DSM 789, NBRC 13277, NCIMB 10341 or equivalent strain (see 12.3).

6.1.3 Other suitable species can also be used depending on the intended end-use of the test sample.

6.2 Maintain test organisms according to good laboratory practice standard (see 12.4).

6.3 Whenever possible, test the activity

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of the culture to be used against a standard control specimen (a positive control) with known antibacterial activity.

6.4 To determine whether the antibacterial activity is due to the antibacterial agent, test a specimen of the same material treated in exactly the same way with whatever other finishing agents were used, but without the antibacterial agent. Many standard textile finishing chemicals, especially crease resistant and permanent press reagents, will often give strong antibacterial activity even after many washes.

7. Materials, Media and Reagents

7.1 Media and Reagents. Suitable broth/agar media are:

7.1.1 Nutrient broth/agar.

7.1.2 Trypticase Soy broth/agar.

7.1.3 Brain-Heart Infusion broth/agar.

7.1.4 Müller-Hinton broth/agar.

7.1.5 Other appropriate broths/agars can be used depending on test organisms selected.

7.2 Materials.

7.2.1 Incubator maintained at 37 ± 2°C (99 ± 4°F).

7.2.2 Inoculating loop.

7.2.3 Bunsen Burner or equivalent.

7.2.4 Water bath maintained at 45-50°C (113-122°F).

7.2.5 Pipets, 1 mL, sterile.

7.2.6 Culture Tubes with caps; minimum 10 mL capacity.

7.2.7 Petri dishes, 100 mm diam. × 15 mm deep, sterile.

7.2.8 Forceps, sterile.

7.2.9 Stereomicroscope, minimum 40× magnification.

7.2.10 Ruler.

8. Test Specimens

8.1 Test specimens (non-sterile) are cut by hand or with a die. They may be any convenient size. Rectangular specimens cut 25 × 50 mm are recommended. A 50 mm length permits the specimens to lie across five parallel inoculum streaks each of diminishing width from about 8 to 4 mm wide.

9. Procedure

9.1 Dispense appropriate sterilized agar [cooled to 47 ± 2°C (117 ± 4°F)] by pouring 15 ± 2 mL into each standard (15 × 100 mm) flat bottomed petri dish. Allow agar to gel firmly before inoculating.

9.2 Prepare inoculum by transferring 1.0 ± 0.1 mL of a 24 h broth culture into 9.0 ± 0.1 mL of sterile distilled water contained in a test tube or small flask. Mix well using appropriate agitation.

9.3 Using a 4 mm inoculating loop, load one loopful of the diluted inoculum and transfer to the surface of the sterile agar plate by making five streaks approximately 60 mm in length, spaced 10 mm apart covering the central area of a standard petri dish (see 10.1) without refilling the loop. Take care not to break the surface of the agar while making the streaks.

9.4 Gently press the test specimen transversely across the five inoculum streaks to ensure intimate contact with the agar surface. This may be accomplished more easily by pressing the specimen to the agar surface with a biological section lifter or with a spatula which has been sterilized by flaming and then air cooled immediately before use.

9.5 If the specimen curls, preventing intimate contact with the inoculated surface, place sterile glass slides on the ends of the specimen to hold it in place.

9.6 Incubate at 37 ± 2°C (99 ± 4°F) for 18-24 h.

10. Evaluation

10.1 Examine the incubated plates for interruption of growth along the streaks of inoculum beneath the specimen and for a clear zone of inhibition beyond its edge. The average width of a zone of inhibition along a streak on either side of the test specimen may be calculated using the following equation:

$$W = (T - D)/2$$

where:

W = width of clear zone of inhibition in mm

T = total diameter of test specimen and clear zone in mm

D = diameter of the test specimen in mm

10.2 The size of the zone cannot be construed as a quantitative evaluation of antibacterial activity. Treated materials should be compared to an untreated corresponding material and a material specimen with known bacteriostatic activity. Report of results will include an observation of zones of inhibition and growth under the specimen if present. The criterion for passing the test must be agreed upon by the interested parties. To constitute acceptable antibacterial activity, there must be no bacterial colonies directly under the sample in the contact area.

11. Precision and Bias

11.1 Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

12 Notes and References

12.1 Publication available from U.S. Department of Health and Human Services, CDC/NIH-HHS Publication No. (CDC) 84-8395; web site: www.hhs.gov.

12.2 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1.513.742.2020; web site: www.acgih.org.

12.3 ATCC is the American Type Culture Collection (USA), P.O. Box 1549, Manassas VA 20108; tel: +1.703.365.2700; fax: +1.703.365.2701; CIP is the Pasteur Institute Collection (France); DSM is the German Collection of Microorganisms and Cell Cultures (Germany); NBRC is the NITE Biological Resource Center (Japan); and NCIMB is the National Collection of Industrial Bacteria (UK). Equivalent bacteria strains obtained from agencies of the World Federation of Culture Collection (WFCC) may be used by agreement between the interested parties. The strains used in the test shall be documented with their supply source.

12.4 Consistent and accurate testing requires maintenance of a pure, uncontaminated, non-mutant test culture. Avoid contamination by using good sterile technique in plating and transferring. Avoid mutation by strict adherence to monthly stock transfers. Check culture purity by making streak plates periodically and observing for a single species-characteristic type of colonies.

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Test Method for Light Blocking Effect of Textiles and Related Materials: Photodetector

Foreword

AATCC TM148-1989, Light Blocking Effect of Curtain Materials, was discontinued in 1994. During the time of its discontinuance, the need for a method to measure this property continued. The method has now been rewritten and retitled for reactivation, in parallel with an alternative method using a standard laboratory sphere spectrophotometer used in transmission mode (see AATCC TM203, Test Method for Light Blocking Effect of Textiles: Spectrophotometric). The two methods are not equivalent (see 14.1).

1. Purpose and Scope

1.1 The test method measures the light blocking properties of textile and related materials using a photometer.

1.2 This method is applicable to all types of fabrics, including coated textiles and other textile-based constructions.

2. Principle

2.1 By measuring the amount of light from a test lamp, and the amount of light from the same lamp transmitted through a test material, the percent of light blocked by the material is determined. This represents the material's light blocking effect.

3. Terminology

3.1 **diffuse transmission** n.—process by which incident light, while being transmitted through an object, is redirected or scattered over a range of angles.

3.2 **light** n.—electromagnetic radiation weighted by the luminosity function of the human eye.

3.3 **light blocking,** n.—the ability of a textile to prevent the transmission of light.

3.4 **non-diffuse transmission,** n.—process by which incident light is transmitted through an object in a rectilinear, straight-through manner, without diffusion.

3.5 **photometer,** n.—an instrument used for measuring light

3.6 **total transmission,** n.—the amount of light transmitted comprising diffuse plus non-diffuse transmission.

3.7 **transmission,** n.—process by which incident light is transmitted through an object.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as materials safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 The manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

5. Uses and Limitations

5.1 The results may depend on the specific light source used in the instrument. This is quoted as part of the report.

5.2 For materials of high light blocking effect, the method is limited by the ability of the system to measure very small amounts of light, in turn depending on the sensitivity of the instrument, together with the output of the light source used (see 14.3).

5.3 The test measures non-diffuse transmission through the material under test. It is designed to exclude diffuse transmission (scattered light) (see 14.1).

6. Calibration

6.1 The measurement is of an SI-based quantity (lux), and the instrument should be calibrated to traceable standards at intervals recommended by the manufacturer, typically on an annual basis.

7. Apparatus

7.1 The apparatus (see Fig. 1) consists of a rectilinear box constructed of light-impermeable materials with internal dimensions of 360 ± 20 mm high, 280 ± 20 mm wide, and 200 ± 20 mm deep with matte black internal surfaces. On one side is a circular opening of dimensions to fit the sensor of a photometer with a clamp to hold the sensor in place at the internal surface of the box. Directly opposite is a circular opening 50 ± 3 mm in diameter. A sliding shutter of opaque material allows this opening to be completely blocked. Magnetic bars provide a convenient means of clamping a fabric specimen in close contact with the outside wall of the opening, but other means of fixing a specimen against the opening may be used. A lamp housed within a metal reflector is mounted so that it is aligned with the two holes in the box. The distance from the lamp to the surface of the test fabric is 300 mm. To accommodate different lamps, it is convenient to have the housing mounted on a track so that the distance to the specimen can remain constant.

8. Test Specimens

8.1 For a (purportedly) evenly-blocking material, cut or mark three representative test specimens from the sample. Do not take specimens from within 10% of the width from the selvedge or machine edge of the fabric.

8.2 If testing a complete curtain or roll of fabric, test three locations across the width.

8.3 If testing reveals that the transmission characteristics vary excessively between specimens, even if not obvious to the eye, increase the number of specimens (see section 11 and 14.2).

8.4 Materials with small-scale (less than the 50 mm size of the specimen exposed) variation in transmission characteristics (based on fabric structure, or printed decoration, for example) may require additional specimens to be cut.

8.5 Materials with large (greater than the 50 mm size of the specimens exposed) repeats of varying transmission characteristics may have the light blocking effect of each area measured separately.

8.6 Include sampling information in the test report.

9. Conditioning

9.1 Prior to testing, condition test specimens as directed in ASTM D1776, Standard Practice for Conditioning and Testing Textiles. Condition the specimens for at least 4 h in an atmosphere of 21 ± 2°C (70 ± 4°F), 65 ± 5% RH, by laying each specimen separately on a screen or perforated shelf of a conditioning rack.

9.2 Perform all tests in the standard atmosphere for testing. Alternatively, conditions of testing may be adjusted to correspond to conditions of use. Report the use of alternative test conditions.

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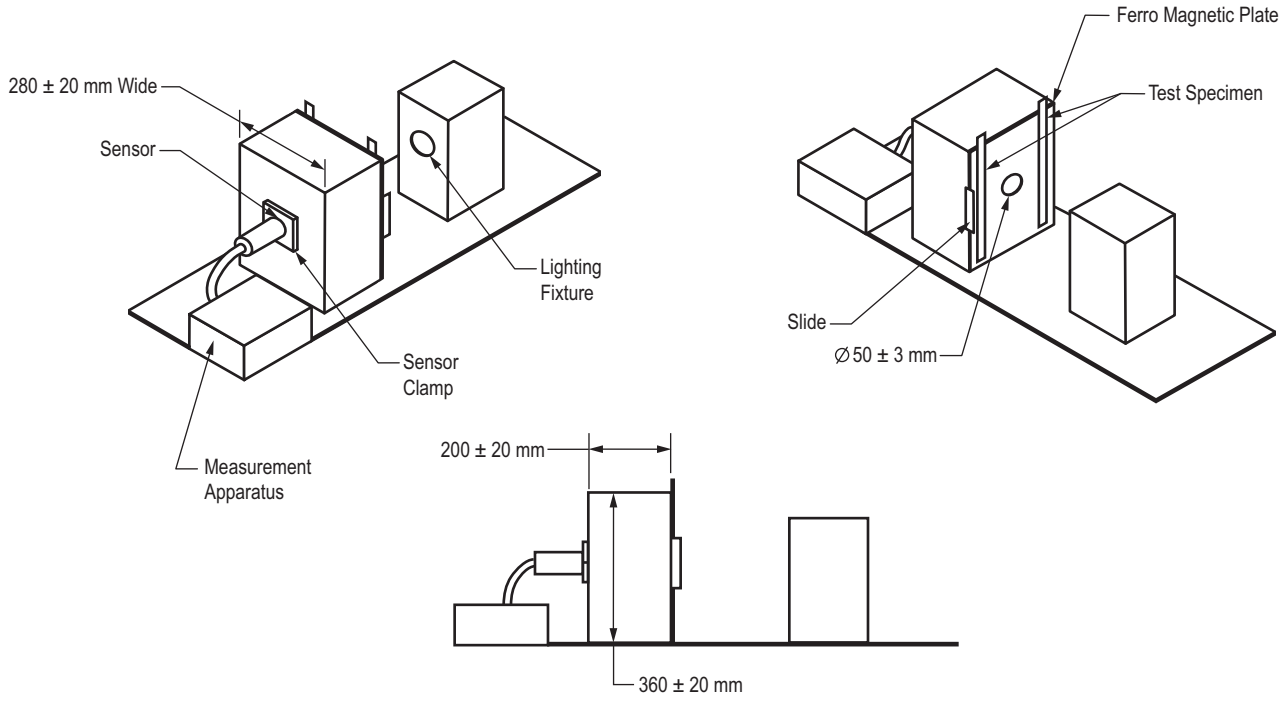


Fig. 1—Photometer

10. Procedure

10.1 Mount light source, box and photometer in a room with consistent and low ambient light (see 14.1). Close the slider over the specimen port.

10.2 Turn on the light source. Allow light to warm up to constant output.

10.3 Zero the instrument. This will take account of any ambient light, which should be low, and held constant while measurements are taken.

10.4 Adjust the instrument sensitivity (if applicable) to low sensitivity for measurement of high light levels.

10.5 Open the sliding aperture.

10.6 Measure and record the amount of light reaching the photometer (in units of lux, or foot candles) with no material in place ("A").

10.7 Close the sliding aperture.

10.8 Place the test specimen at the specimen aperture and clamp it in place.

10.9. Open the sliding aperture.

10.10 Measure and record the light transmitted by the material ("B") that reaches the photometer. If necessary and applicable, increase the instrument sensitivity to a level appropriate for the amount of light being transmitted.

10.11 Close the aperture, remove the specimen, mount the next specimen, open the aperture and read the transmitted light ("B"). Repeat until all specimens have been measured.

10.12 Repeat the measurement of amount of light reaching the photometer from the test lamp "A" with no material in place as in 9.6.

11. Calculation

11.1 Calculate the average of the two measurements of the amount of light ("A") entering the box from the test lamp (see 10.6 and 10.12)

11.2 Calculate the light blocking by each specimen of the test material by one of the two methods as follows (see 13.6).

11.2.1 Calculate the percentage of light blocked thus:

$$\% \text{ Light Blocked} = 100 \times [1 - (B/A)]$$

where:

B = amount of light transmitted through the specimen reaching the photometer.

A = average amount of light reaching the photometer from the test lamp with no material in place.

or:

11.2.2 Calculate the Absorbance of the material thus:

$$\text{Absorbance} = -\log 10\ (B/A)$$

11.3 Calculate the average % light blocked, or average Absorbance of the three (or more) specimens.

11.4 Calculate the standard deviation of measurements averaged in 10.3.

12. Report

12.1 Cite this test method as having been used.

12.2 Cite the light source: type, wattage, nominal output and measured output ("A" as calculated in 11.1) used in the apparatus.

12.3 Report the average light blocking capacity for all measurements of the specimens tested, either as % light blocked or average Absorbance (see 11.3).

12.4 Report the number of samples/specimens tested.

12.5 Report the standard deviation of the data.

12.6 Report the temperature and humidity conditions of the test.

13. Precision and Bias

13.1 *Precision.* The light blocking effect of several coated fabrics was measured. Three readings at varying positions on the fabrics as described above were recorded by the same operator three times, and by different operators. The precision (95% confidence limits) were analyzed using probability functions outlined in *Handbook of Mathematical Tables*, Richard Burington, McGraw Hill, Library of Congress 63-23531, p358–359 (Table 22).

For single operator consistency:

On 3 different fabrics, Operator #1 had 95% confidence intervals of 99.64300 to 99.66925%, 99.99996 to 99.99999%, and 99.99801% blockages.

For operator to operator consistency:

Two operators were compared on 3 fabrics giving similar results with single operator comparison. Sample B2 gave lower and upper limits of 99.65611 to

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99.65615% (95% confidence interval). Sample A5 produced lower and upper limits of 99.99738 to 99.99769%. Sample A2 yielded 99.99997 to 99.99998%, as shown in Table I.

13.2 *Bias.* Light blocking capacity of fabrics can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

14. Notes

14.1 The black box between the test material and the detector shields the detector from stray light in the test area. Foam or other material around the sensor clamp may augment the shielding. While ambient light is included in the zeroing of the instrument, this exclusion of stray light minimizes any effects of variation in ambient light. The black box also excludes the detection of light scattered by the substrate being measured. AATCC TM203 uses a reflectance spectrophotometer and has an integrating sphere between material and detector that will *include* such scattered light. This distinction means that agreement between the methods is limited. The extent to which the scattered light is relevant to real life situations will, in part, guide the choice of method. Light scattered as it passes through curtain materials into a room with light colored walls will reflect from those surfaces and contribute to the light seen by an observer. Light scattered as it passes through curtain materials into a room with dark colored walls will contribute less to the light seen by an observer in that room. Light scattered as it passes from within a lit structure through a textile material into a dark night will not contribute to the light seen by an observer. This method is thus most directly relevant to the last two cases.

14.2 Light blocking materials are often based on coated textile fabrics. Variation in amounts of coating applied will cause corresponding variations in light blocking. In this way, the method may have utility as a measure of coating evenness. In any case, the measurement of light may be carried out to a high degree of precision and any variation of results from specimen to specimen within a sample is likely to derive more from variation within the material than variability in measurement. Multiple determinations of light blocking of the same specimen will provide an estimate of measurement repeatability.

14.3 The test deals with "light," electromagnetic radiation weighted according to the luminosity function of the human eye (i.e. the standard observer). The weighting can be achieved by software calculation from a spectral analysis of light using (typically) an array of diodes to collect light of different wavelengths, or from a single measurement of total energy passing through a filter with the appropriate characteristics. These two means of weighting the light transmissions to correspond to human eye sensitivity will affect relative instrument sensitivities, which may in turn be a factor in the choice of method (see 14.1).

14.4 The method as described is not appropriate for analysis of non-visual e/m radiation, but might be modified to do so with other filters or software that correspond to the sensitivity of the detection device being simulated.

14.5 At very low light levels the human observer uses scotopic ("night") vision, and thus at low levels of light, the measurement might most appropriately be measured through a (physical or mathematical) scotopic filter and reported scotopically. However, since the original lamp measurement will be photopically weighted, consistency requires that the transmitted light be weighted the same way.

14.6 The method provides for two means of expressing the results obtained, % light blocking and Absorbance. The two are based on the same data, but use a different calculation. Absorbance provides more convenient numbers for materials that block very high percentages of light.

15. History

15.1 Reaffirmed and editorially revised in 2021 to add history per AATCC style guide.

15.2 Editorially revised in 2019, and 2016; reaffirmed and editorially revised 2015. Reactivated and revised 2014 (with new title); discontinued 1994; reaffirmed 1989, editorially revised in 1986, 1985 and 1984; reaffirmed 1984; editorially revised in 1983 and reaffirmed 1979.

15.3 Developed in 1976 by AATCC Committee RA36.

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Table I—Measurements Used to Calculate Precision Data

							95% Confidence Interval	95% Confidence Lower Limit	95% Confidence Upper Limit
Single Operator Consistency									
Operator 1	B2	99.622%	99.653%	99.659%					
Operator 1	B2	99.628%	99.667%	99.653%					
Operator 1	B2	99.666%	99.677%	99.680%	99.65614%	0.02011%	0.03942%	99.64300%	99.66928%
Operator 1	A2	100.000%	100.000%	100.000%					
Operator 1	A2	100.000%	100.000%	100.000%					
Operator 1	A2	100.000%	100.000%	100.000%	99.99997%	0.00003%	0.00005%	99.99996%	99.99999%
Operator 1	A5	99.998%	99.998%	99.996%					
Operator 1	A5	99.998%	99.998%	99.998%					
Operator 1	A5	99.998%	99.999%	99.998%	99.99761%	0.00061%	0.00120%	99.99721%	99.99801%
Operator-Operator Comparison									
Operator 1	A5	99.998%	99.998%	99.996%					
Operator 1	A5	99.998%	99.998%	99.998%					
Operator 1	A5	99.998%	99.999%	99.998%	99.99761%	0.00061%			
Operator 2	A5	99.997%	99.996%	99.997%					
Operator 2	A5	99.998%	99.998%	99.998%					
Operator 2	A5	99.998%	99.998%	99.998%	99.99746%	0.00063%			
					99.99754%	0.00011%	0.00022%	99.99738%	99.99769%
Operator 1	B2	99.622%	99.653%	99.659%					
Operator 1	B2	99.628%	99.667%	99.653%					
Operator 1	B2	99.666%	99.677%	99.680%	99.65614%	0.02011%			
Operator 2	B2	99.624%	99.624%	99.627%					
Operator 2	B2	99.669%	99.634%	99.656%					
Operator 2	B2	99.707%	99.671%	99.692%	99.65612%	0.03100%			
					99.65613%	0.00001%	0.00002%	99.65611%	99.65615%
Operator 1	A2	100.000%	100.000%	100.000%					
Operator 1	A2	100.000%	100.000%	100.000%					
Operator 1	A2	100.000%	100.000%	100.000%	99.99997%	0.00003%			
Operator 2	A2	100.000%	100.000%	100.000%					
Operator 2	A2	100.000%	100.000%	100.000%					
Operator 2	A2	100.000%	100.000%	100.000%	99.99997%	0.00001%			
					99.99997%	0.00000%	0.00000%	99.99997%	99.99998%

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Test Method for Chelation Value of Aminopolycarboxylic Acids and Their Salts: Calcium Oxalate

1. Foreword

1.1 The titration in this method yields a value that includes partially-substituted EDTA, HEDTA, DTPA, nitrilotriacetic acid (NTA), iminodiacetic acid (IDA), glycolate and other weak chelating agents found in some commercial products. An alternative procedure, excluding the effect of partially-substituted products, is provided in AATCC TM168, Test Method for Chelation Value of Polyaminopolycarboxylic Acids and Their Salts: Copper PAN.

2. Purpose and Scope

2.1 The active content of ethylenediaminetetraacetic acid (EDTA), N-hydroxyethylethylenediaminetriacetic acid (HEDTA) and diethylenetriaminepentaacetic acid (DTPA) and their salts is usually expressed by the calcium chelation value (CaCV). This value indicates the amount of calcium (as calcium carbonate) which will be chelated by a known weight of chelating agent.

3. Principle

3.1 The chelation value is determined experimentally by titrating a weighed sample of chelating agent with a calcium ion solution of known concentration. A precipitating anion (oxalate) is present during the titration. As the calcium solution is added, the ions are chelated by the chelating agent and no permanent precipitate is formed as long as there is free chelating agent present. Beyond this point, any excess calcium ion that is added reacts with oxalate anion to form a cloudy precipitate which is the endpoint.

3.2 The CaCV indicates the amount of calcium (as calcium carbonate) that will be sequestered by a known weight of chelating agent and is expressed as milligrams of calcium carbonate ($CaCO_3$) per gram of chelating agent (mg calcium carbonate/g chelating agent).

4. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

4.1 AATCC TM168, Test Method for Chelation Value of Polyaminopolycarboxylic Acids and Their Salts: Copper PAN.

5. Terminology

5.1 **chelating agent,** n.—*in textile chemistry*, a chemical capable of deactivating metal ions by formation of a water-soluble complex. SYN.—*sequestering agent*.

6. Safety Precautions

6.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

6.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

7. Uses and Limitations

7.1 Analysis by this method will yield a CaCV that includes partially-substituted EDTA, HEDTA and DTPA as well as nitrilotriacetic acid (NTA), iminodiacetic acid (IDA), glycolate and other weak chelating agents. These compounds may be present in some commercial products.

7.2 AATCC TM168 is an alternative procedure that yields a CaCV that does not include the partially-substituted products and weaker chelating agents.

8. Reagents

8.1 Calcium carbonate ($CaCO_3$).

8.2 Sodium oxalate ($Na_2C_2O_4$).

8.3 Diethylamine hydrochloride [$(C_2H_5)_2NH \cdot HCl$]

8.4 Sodium hydroxide (NaOH) solution, 50% W/W.

8.5 Hydrochloric acid (HCl) conc.

9. Sampling

9.1 Conduct the test in triplicate; i.e., analyze 3 test specimens from a sample.

10. Specimens

10.1 Do not use aluminum or metallic weighing pans.

10.2 If solid chelating agent is being evaluated, dry a 2-g sample at the appropriate temperature for at least 2 h and cool in desiccator before weighing specimens.

10.2.1 Dry free acid forms of chelating agents at 120°C.

10.2.2 Dry salt forms of chelating agents at 80°C.

10.3 For analysis of solid EDTA, HEDTA and DTPA (either free acid or salt forms) weigh a 0.49-0.51 g specimen of the dried chelating agent to an accuracy of 0.01 g.

10.4 For analysis of commercial solutions of salts of the chelating agents (see 10.2) weigh a 1.00-1.20 g specimen of the solution to an accuracy of 0.01 g.

11. Preparation of Reagents

11.1 Calcium Titrant Solution (0.250 M): Weigh 25.0 g of calcium carbonate to an accuracy of 0.1 g and add it to 300 mL of distilled water in a 600-mL beaker with stirring on a magnetic stirrer. Dissolve by adding 43 mL of concentrated hydrochloric acid. Add the hydrochloric acid slowly and carefully to prevent bubbling over and excessive splattering. After the calcium carbonate has dissolved (the solution becomes clear), heat the solution to boiling and allow it to boil for at least 5 min to expel carbon dioxide. Cool the solution to room temperature, transfer it to a 1-L volumetric flask and dilute it to 1 L. Prepare a fresh solution monthly.

11.2 Sodium Oxalate Indicator: Dissolve 5 g of sodium oxalate in 250 mL of distilled water. Prepare a fresh solution monthly.

11.3 Sodium Hydroxide (50%): 50% solution (W/W) of sodium hydroxide is available from many laboratory supply vendors. It can be prepared by dissolving 100 g of sodium hydroxide pellets in 100 mL of distilled water in a polyethylene bottle. Let stand for a week to allow the insoluble sodium carbonate to settle.

11.4 pH 12.0 Buffer Solution: Dissolve 41.0 g of diethylamine hydrochloride in 400 mL of distilled water. Add 40 mL of 50% sodium hydroxide and dilute to 500 mL. Store in a polyethylene bottle. Prepare a fresh solution monthly.

12. Procedure

12.1 Transfer the specimen into a 250-mL Erlenmeyer flask and add 85 mL of distilled water. Wash any residual chelating agent from the weighing paper or dish into the flask.

12.2 Add 5 drops of 50% sodium hydroxide (10 drops if the free acid form of a chelating agent is being analyzed) and swirl to dissolve and/or mix the chelating agent.

12.3 Add 1 mL of pH 12.0 buffer solution and swirl to mix.

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12.4 Add 10 mL of sodium oxalate solution and swirl to mix.

12.5 Using a 10-mL burette, titrate with calcium titrant solution until a drop produces a slight turbidity that persists for at least 30 s.

12.5.1 Turbidity is easier to observe when a piece of black paper is placed beneath and behind the flask, and the flask is illuminated from the side by a white fluorescent light. Add the calcium titrant dropwise once the turbidity begins to persist for more than 2-3 s. The endpoint is the first sign of turbidity that persists for at least 30 s. If there is uncertainty as to whether a slight turbidity has been obtained, compare the appearance of the flask to that of one containing only water, which has a distinct clear appearance.

13. Calculation

13.1 Calculate the CaCV for each specimen to 3 significant digits using Equation 1:

$$\text{CaCV} = \frac{(100.1)(0.250)(V)}{W} \quad \text{(Eq. 1)}$$

where:

CaCV= Calcium chelation value, expressed as mg $CaCO_3$/g chelating agent

100.1 = Molecular weight of $CaCO_3$ expressed in mg/mmole

0.250 = Molarity of calcium titrant solution expressed in mmoles/mL

V = Volume of calcium titrant solution in mL

W = Weight of specimen in grams

13.2 Average the CaCV values for the 3 specimens tested.

14. Report

14.1 Report for each sample tested:

14.1.1 A description or identification of the sample.

14.1.2 Report that the sample was tested using AATCC TM 149-2022.

14.1.3 Average CaCV, to 3 significant digits.

14.1.4 Describe any modifications of the test method.

15. Precision and Bias

15.1 *Summary.* An interlaboratory study was conducted in March 1990 to establish the precision of the test method. Four chelating agents (EDTA, tetrasodium salt; EDTA, 40% solution of tetrasodium salt; DTPA, free acid; and DTPA, 40% solution) were evaluated.

15.2 *Precision*. The components of variance as standard deviations of CaCV were calculated as follows:

Single Operator Component	1.46
Interlaboratory Component	0.00

15.3 *Critical Differences*. For the components of variances reported in 13.2, two averages of observed values should be considered significantly different at the 95% probability level if the difference is equal to or exceeds the following critical differences:

Single Operator	4.04
Interlaboratory	4.04

15.4 *Bias*. Based on the theoretical calcium chelation values of the four materials used in the interlaboratory study, the average bias of the five laboratories was 2.24%. The individual biases for the four materials and five laboratories ranged from 2.50-4.25%.

16. History

16.1 Revised in 2023 to clarify and align with prescribed AATCC style guidelines.

16.2 Editorially revised 2019; revised 2018; reaffirmed 2012; editorially revised 2010; reaffirmed 2007; reaffirmed 2002; editorially revised and reaffirmed 1997; revised 1992; editorially revised 1988, 1986; editorially revised and reaffirmed 1985; editorially revised 1984; reaffirmed 1980; editorially revised and reaffirmed 1977.

16.3 Developed in 1976 by AATCC Committee RR90. Maintained by RA99.

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Test Method for Dimensional Changes of Garments after Home Laundering

Developed in 1977 by AATCC Committee RA42; editorially revised and reaffirmed 1979, 1984, 1992; editorially revised 1983, 1985, 1989, 1990, 1991,1996, 1997, 2004, 2005, 2006, 2008, 2016; revised 1987, 1995, 2001, 2003 (with title change), 2010, 2012, January 2018, November 2018; reaffirmed 2000. Related to ISO 3759.

Foreword

Although this method calls for measurement of dimensional changes in unseamed areas of garment specimens, results may be affected by garment construction, tensions, sewing threads or trims in addition to fabric dimensional change. An option is also provided for measuring dimensional change between or along garment seams.

Standard laundering procedures remain consistent to allow valid comparison of results. Standard procedures represent, but may not exactly replicate, current consumer practices, which vary over time and among households. Alternate laundering procedures and machine parameters may be found in AATCC LP1, Laboratory Procedure for Home Laundering: Machine Washing, AATCC LP2, Laboratory Procedure for Home Laundering: Hand Washing (see 12.3) and ISO 6330, Textiles — Domestic washing and drying procedures for textile testing (see 12.7).

1. Purpose and Scope

1.1 This test method is for the determination of dimensional (length and width) changes of garments when subjected to home laundering procedures. Four washing temperatures, three agitation cycles, and four drying procedures provide standard parameters to represent common home care options.

1.2 This method may not be applicable to garments made of certain stretch fabrics.

2. Principle

2.1 Pairs of benchmarks are applied to the garment before laundering. The distance between each pair is measured before laundering and remeasured after a specified number of standard laundering cycles. Percent change is calculated. A negative value indicates shrinkage; a positive value indicates growth. A value close to zero indicates little or no change.

3. Terminology

3.1 **dimensional change,** n.—a generic term for changes in length or width of a fabric specimen subjected to specified conditions. The change is usually expressed as a percentage of the initial dimension of a specimen.

3.2 **garment,** n.—a shaped article of textile fabric, or other flexible material, intended to cover portions of the human body.

3.3 **growth,** n.—a dimensional change resulting in an increase of length or width of a specimen.

3.4 **laundering,** n.—*of textile materials*, a process intended to remove soils and/or stains by treatment (washing) with an aqueous detergent solution and normally including rinsing, extraction and drying.

3.5 **shrinkage,** n.—a dimensional change resulting in a decrease in the length or width of a specimen.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Detergent may cause irritation. Care should be taken to prevent exposure to skin and eyes.

4.3 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

5. Apparatus and Materials (see 12.1)

5.1 Standard washing machine (see Table I and 12.2).

5.2 Standard tumble dryer (see Table III and 12.2) or facilities for drip/line drying.

5.3 Conditioning facilities and conditioning/drying racks with pull-out, perforated shelves (see 12.3).

5.4 Balance with at least 5.0-kg or 10.0-lb capacity.

5.5 AATCC 1993 Standard Reference Detergent (powder, with brightener, see 12.3).

5.6 Laundering ballast, Type 1 or Type 3 (see Table IV).

5.7 Indelible ink marking pen (see 12.3) for use with suitable rule, tape, marking template or mechanical marking device (see 12.3). Sewing thread may be used for making benchmarks.

5.8 Measuring devices.

5.8.1 Tape or rule marked in millimeters, eighths or tenths of an inch.

5.8.2 Tape or ruled template marked directly in percent dimensional change to 0.5% or smaller increment (see 12.3).

6. Test Specimens

6.1 Sampling and Preparation**.**

6.1.1 Each garment will be considered a specimen. Garments used as specimens should be representative of the production lot. When possible, three specimens should be used. One or two specimens may be used when an insufficient number of garments is available.

6.1.2 Garments that are distorted in their unlaundered state due to faulty fabric finishing, garment assembly, or packing may give deceptive dimensional change results when laundered by any procedure. In such cases, it is recommended that distorted garments not be tested.

6.1.3 Prior to marking, condition test specimens as directed in ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 12.4). (Use conditions indicated in Table 1 for Textiles, general. Estimate conditioning time as indicated in Table 2 for the appropriate fiber content.) Hang each specimen on an appropriate hanger. If the garment would not normally be hung on a hanger; e.g., T-shirts, boxer shorts, etc., lay each test specimen flat, separately, on a screen or perforated shelves of a conditioning rack.

6.2 Marking. Mark specimens without tension on a flat, smooth horizontal surface.

6.2.1 Using Table V as a guide for the selection of garment specimen areas to use as measurement sites, place benchmarks on the garment at the selected locations. Use at least three length and three width locations for marking benchmarks on each garment specimen. Location of benchmarks should be as agreed upon by the buyer and seller. If garment

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Table I—Standard Washing Machine Parameters (see 12.2, 12.6)

Cycle	(1) Normal		(2) Delicate		(3) Permanent Press	
Water Level, L (gal)	72 ± 4 (19 ± 1)		72 ± 4 (19 ± 1)		72 ± 4 (19 ± 1)	
Agitation Speed, strokes/min.	86 ± 2		27 ± 2		86 ± 2	
Washing Time, min.	16 ± 1		8.5 ± 1		12 ± 1	
Final Spin Speed, rpm	660 ± 15		500 ± 15		500 ± 15	
Final Spin Time, min.	5 ± 1		5 ± 1		5 ± 1	
Wash Temp, °C (°F)[1]	(II) Cold:	27 ± 3 (80 ± 5)	(II) Cold:	27 ± 3 (80 ± 5)	(II) Cold:	27 ± 3 (80 ± 5)
	(III) Warm:	41 ± 3 (105 ± 5)	(III) Warm:	41 ± 3 (105 ± 5)	(III) Warm:	41 ± 3 (105 ± 5)
	(IV) Hot:	49 ± 3 (120 ± 5)	(IV) Hot:	49 ± 3 (120 ± 5)	(IV) Hot:	49 ± 3 (120 ± 5)
	(V) Very Hot:	60 ± 3 (140 ± 5)	(V) Very Hot:	60 ± 3 (140 ± 5)	(V) Very Hot:	60 ± 3 (140 ± 5)

[1] Due to US Department of Energy requirements, many washing machines use cooler water. An external control box may be used to override the machine set temperatures.

areas are sufficient, place 460-mm (18.0-in.) benchmarks on the garment. Benchmark distances used will depend on the size of the garment. For some garments, especially children's clothing, sufficient area may require shorter benchmarks such as 250-mm (10.0-in) or less. Benchmarks must be at least 25 mm (1 in.) from all edges or seams, except as described in 6.2.2.

6.2.2 Benchmarks across garment panels (between or along garment seams) are sometimes used such as, from side seam-to-side seam, the full length or width of a garment and in other selected garment locations. For these types of benchmarks, the measurement points should be clearly marked.

6.2.3 Dimensional change results from different garment sizes, styles and/or benchmark lengths may not be comparable.

6.2.4 Benchmarks distances used must be indicated in the report.

6.3 Original Measurements. Make all measurements without tension, on the same surface used for marking.

6.3.1 To improve the accuracy and precision of the dimensional change calculations based on the benchmarks applied to the garment in 6.2, measure and record the distance between each pair of benchmarks with suitable tape or rule to nearest millimeter or eighth or tenth of an inch. This is measurement A. If using a calibrated template for marking and measuring in percent dimensional change directly, an initial measurement is not needed.

7. Laundering Procedure

7.1 Washing.

7.1.1 Select washing conditions for testing from Table I. Set washing machine controls to generate the selected cycle parameters.

7.1.2 Load for laundering shall consist of all specimens in the sample, plus enough laundering ballast pieces to make a total load weight of 1.8 ± 0.1 kg (4.0 ± 0.2 lb).

7.1.3 Begin selected wash cycle. Allow machine to fill to specified water level.

7.1.4 Add 66 ± 1 g of AATCC 1993 Standard Reference Detergent to washing machine as directed by washing machine manufacturer's instructions. If detergent is added directly to wash water, agitate briefly to dissolve completely. Stop agitation before adding wash load.

7.1.5 Add wash load (test specimens and ballast), distributing evenly around center agitator. Restart wash cycle.

7.1.6 For specimens to be drip dried (drying procedure C), stop wash cycle before the water begins to drain from the final rinse cycle. Remove specimens soaking wet. For specimens to be tumble (A), line (B), or screen/flat (D) dried, allow washing to proceed through the final spin cycle.

7.1.7 After each washing cycle, separate tangled specimens and ballast pieces, taking care to minimize distortion. Proceed to appropriate drying procedure.

7.2 Drying.

7.2.1 Select drying conditions from Table II.

Table II—Standard Drying Conditions

(A)	Tumble Dry
(Ai)	Tumble Dry Normal
(Aii	Tumble Dry Delicate
(Aiii)	Tumble Dry Permanent Press
(B)	Line/Hang Dry
(C)	Drip Dry
(D)	Screen Dry/Dry Flat

7.2.2 (A) Tumble Dry. Place the washed load (test specimens and ballast) in the tumble dryer, and set the temperature control to generate the selected cycle exhaust temperature (see Table III). Allow the dryer to run until the load is fully dry. Remove specimens immediately.

7.2.3 (B) Line/Hang Dry. Hang each specimen by two corners, with the fabric length in the vertical direction. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

7.2.4 (C) Drip Dry. Hang each dripping wet specimen by two corners, with the fabric length in the vertical direction. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

7.2.5 (D) Screen/Flat Dry. Spread each specimen on a horizontal screen or perforated surface, removing wrinkles without distorting or stretching it. Allow specimens to rest in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

7.2.6 For all drying methods, allow specimens to dry completely before washing again.

7.2.7 Repeat the selected washing and drying procedures for a total of 3 complete cycles, or an agreed upon number of

Table III—Standard Tumble Dryer Parameters (see 12.2)

Cycle	(Ai) Normal	(Aii) Delicate	(Aiii) Permanent Press
Max. Exhaust Temp, °C (°F)	68 ± 6 (155 ± 10)	60 ± 6 (140 ± 10)	68 ± 6 (155 ± 10)
Cool Down Time, min.	≤ 10	≤ 10	≤ 10

Table IV—Laundering Ballast Parameters

	Type 1	Type 3
Fiber Content	100% cotton	50% cotton/50% polyester ± 3%
Greige Fabric Yarns	16/1 ring spun	16/1 or 30/2 ring spun
Greige Fabric Construction	52 × 48 ± 5 yarns/in., plain weave	52 × 48 ± 5 yarns/in., plain weave
Finished Fabric Weight	155 ± 10 g/m^2	155 ± 10 g/m^2
Edges	All edges hemmed or over-edged	All edges hemmed or over-edged
Finished Piece Size	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)
Finished Piece Weight	130 ± 10 g	130 ± 10 g

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Table V—Examples of Benchmark Locations

Type of Garment	Benchmark Locations
Shirt	Collar Collar Band Body Lengths Sleeve Lengths Width at Chest Cuffs
Trousers	Front Rise Back Rise Inseams Outseams Waist Seat
Coveralls	Body Lengths Front Rise Back Rise Inseams Underarm Lengths Sleeve Lengths Shoulder Waist Chest Seat Width
Boxer Shorts	Overall Lengths Front Rise Back Rise Waist
Pajama Top	Lengths Sleeves Hem Chest
Pajama Bottom	Inseams Lengths Hip Waist
Shorts	Lengths Front Rise Back Rise Leg Width Inseam Width Seat Width Waist
Sweater	Lengths Sleeve Lengths Chest Waist Shoulders
Slip	Lengths Hem Waist Hip
Blouse	Lengths Sleeve Lengths Shoulders Chest Waist
Skirt	Lengths Hem Hip Waist
Uniform/Dress	Bodice Lengths Skirt Lengths Sleeve Lengths Shoulders Chest Waist Hip Hem
Overalls	Lengths Outseams Front Rise Back Rise Inseams Waist Seat Width

cycles.

7.3 Conditioning.

7.3.1 After the final drying cycle is complete, condition test specimens as described in 6.1.3.

7.4 Ironing.

7.4.1 If the specimens are extremely wrinkled and the consumer would ALWAYS expect to iron a garment made from the fabric, test specimens may be hand ironed prior to re-measurement of benchmarks.

7.4.2 Use safe ironing temperatures appropriate to the fibers in the fabric being ironed. See AATCC TM133, Test Method for Colorfastness to Heat: Hot Pressing, Table I, Safe Ironing Temperature Guide (see 12.3). Exert the minimum pressure necessary to remove wrinkles.

7.4.3 Due to the extremely high variability of hand ironing procedures performed by individual operators (no standard test method exists for hand ironing procedures), the reproducibility of dimensional change results after hand ironing has been found to be extremely poor. Consequently, caution is advised when comparing dimensional change results after laundering and hand ironing, reported by different operators.

7.4.4 After ironing, condition test specimens as described in 6.1.3.

8. Measurement

8.1 After conditioning, lay each garment without tension, on the same surface used for marking and measuring in 6.2 and 6.3.

8.2 Measure and record the distance between benchmarks or between marks made across garment panels to the nearest millimeter, or eighth or tenth of an inch. This is measurement B. If using a scale calibrated in percent dimensional change, measure to nearest 0.5% or smallest increment on the scale and record the percent dimensional change directly.

9. Calculation and Interpretation

9.1 Calculation.

9.1.1 If measurements were made directly in percent dimensional change, average the measurements in each location made on the test garments after the first, the third, or other specified number of washing and drying cycles.

9.1.2 If measurements were made to the nearest millimeter, or eighth or tenth of an inch (measurement B), calculate the dimensional change after the first and, the third or other specified washing and drying cycle using equation 1:

$$\% DC = 100\ (B - A)/A \qquad \text{(Eq. 1)}$$

where:

DC = Dimensional change

A = Original dimension

B = Dimension after laundering

Average the dimensional change for each measurement location made on all test garments. If desired, calculate length and width averages separately to nearest 0.1% (see 12.5).

9.1.3 A final measurement smaller than the original measurement results in a negative dimensional change, which is shrinkage. A final measurement larger than the original measurement results in a positive dimensional change, which is growth.

9.2 Interpretation.

9.2.1 If the dimensional change after one full laundering cycle as calculated in 9.1, is within a specification previously agreed on, continue test procedure as directed in 7.1-7.4 until an agreed upon number of cycles has been completed.

9.2.2 If the dimensional change after one washing, drying, and, if used, hand ironing cycle as calculated in 9.1 exceeds a specification previously agreed on, terminate the test.

10. Report

10.1 Report for each sample test:

10.1.1 A description or identification of the sample.

10.1.2 Evaluation was performed using AATCC TM150-2018t.

10.1.3 Type of ballast used, i.e., Type 1 or Type 3.

10.1.4 Number of laundering cycles (default is 3 cycles, see 9.2).

10.1.5 Laundering conditions, including washing machine cycle, wash temperature, drying procedure and tumble drying temperature, if applicable. Alphanumeric designations may be used if they are clearly understood by all parties. For example, 1-IV-A(ii) indicates normal wash cycle at 49°C and Delicate tumble dry cycle.

10.1.6 If garments were distorted or wrinkled in their original state.

10.1.7 If garments were hand ironed.

10.1.8 Any modifications of the test method. If laundering equipment or conditions other than those specifically listed in the test method were used, they must be described in detail and noted as a modification of the standard method (see 12.6).

10.2 Report for each garment location:

10.2.1 Direction measured, i.e., length or width.

10.2.2 Dimensional change, to the nearest 0.1%, with a minus sign (-) for shrinkage or a plus (+) sign for growth (see 12.5).

10.2.3 Distance between original benchmarks, i.e., 250 mm or 460 mm.

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11. Precision and Bias

11.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

11.2 *Bias*. Dimensional changes in automatic home laundering of garments can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

12. Notes

12.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

12.2 For model number(s) of washer(s) and tumble dryers(s) reported to meet the standard parameters, visit www.aatcc.org/test/washers or contact AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc. org. An alternate load size was described in previous versions of this method (3.6-kg load, 83-L water level, 80 g AATCC 1993 Standard Reference Detergent), but no machine is reported to meet the standard parameters for this load size. In addition, dimensional change results obtained with the alternate load size may not be equal to those obtained with a standard load size.

12.3 Materials or drawings are available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549. 8933; e-mail: ordering@aatcc.org; web site: www. aatcc.org.

12.4 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610. 832.9500; fax: +1.610.832. 9555; web site: www.astm.org.

12.5 If information on the dimensional change variability between specimens is needed, calculate dimensional change based on the individual pairs of benchmarks.

12.6 The laundering temperatures and other parameters listed in this test method are standard conditions for testing purposes. As with most lab procedures, they represent, but may not exactly replicate, current consumer practices. Consumer practices vary over time and among households; lab practices must be consistent to allow valid comparison of results. If laundering equipment or conditions other than those specifically listed in this test method are used, they must be described in detail and noted as a modification of the standard method. Alternate laundering conditions are outlined in AATCC LP1, AATCC LP2 and ISO 6330.

12.7 Available from ISO, www.iso.org.

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Test Method for Thermal Fixation Properties of Disperse Dyes

Developed in 1978 by AATCC Committee RA87; reaffirmed 1981, 2006, 2011; editorially revised and reaffirmed 1986, 1991, 1996, 2001; editorially revised 2008, 2010, 2019; revised 2017.

1. Purpose and Scope

1.1 This test method is used to determine the fixation properties of disperse dyes on polyester/cellulose blends as a function of fixation conditions. The variable described herein is temperature; however, variations in time and/or concentration of dyes and/or auxiliaries may be studied by this method.

2. Principle

2.1 The dye is padded onto the blend fabric at a specified concentration; the fabric is dried and the dye is fixed by exposure to controlled conditions of time and temperature. The cotton portion of the blend is dissolved in concentrated sulfuric acid followed by neutralization and thorough rinsing.

2.2 The reflectance values of the dyed fabrics, exposed to the several fixation conditions, are determined by spectrophotometry and the concentrations of dye relative to the heaviest dyeing (as 100%) are evaluated by applying the Kubelka-Munk function (K/S). These results express the relative dye fixation for a particular set of dyeing conditions. Alternatively, the dye on the fabric can be extracted with a suitable solvent and the concentrations of dye determined by transmittance spectrophotometry. When the dye concentrations on the fixed samples are compared to the corresponding concentration on a padded but *unfixed* sample, a measure of the true dye fixation, as a function of applied dye, may be obtained.

3. Terminology

3.1 **disperse dye,** n.—an essentially water insoluble dye having affinity, when properly dispersed, for polyester, polyamide and some other manufactured polymeric fibers.

3.2 **thermal fixation,** n.—the use of dry heat to achieve a degree of permanence when applying colorants to textile materials.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 Perform leaching procedure (see 7.3) using sulfuric acid, 70%, in a hood. In preparing, dispensing and handling sulfuric acid, use chemical goggles or face shield, impervious gloves and an impervious apron. Concentrated acids should be handled only in an adequately ventilated hood. CAUTION: Always add acid to water.

4.4 In preparing, dispensing and handling ammonium hydroxide, use chemical goggles or face shield, impervious gloves and an impervious apron. Dispense, mix and handle ammonium hydroxide only in an adequately ventilated hood.

4.5 Perform the extraction procedure (see 8.2) using chlorobenzene inside an adequately ventilated hood. CAUTION: Chlorobenzene vapors are toxic and flammable.

4.6 Chlorobenzene is a flammable liquid and should be stored in the laboratory only in small containers away from heat, open flame and sparks.

4.7 An eyewash/safety shower should be located nearby and a self-contained breathing apparatus should be readily available for emergency use.

4.8 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 11.1).

5. Uses and Limitations

5.1 Evaluations by reflectance measurements are limited to relatively low dye concentrations because of the nonlinearity of the K/S function at high dye concentrations (see 11.2).

5.2 Fixation values obtained by reflectance measurements can differ from those determined by extraction techniques because of differences in dye distribution within the fiber (see 11.2).

5.3 Every effort should be made during padding, drying and fixation to avoid two-sided dyeings. Should this occur to any significant extent, the preferred method of measuring fixation is extraction.

6. Apparatus and Materials (see 11.3)

6.1 Laboratory padder.

6.2 Forced air thermofixation unit (see 11.4).

6.3 Analytical balance.

6.4 Spectrophotometer (see 11.5).

6.5 Volumetric pipettes and flasks.

6.6 65/35 polyester/cotton fabric; adequately pre-scoured and ready for dyeing.

6.7 Sulfuric acid, 70%.

6.8 Chlorobenzene.

6.9 Ammonium hydroxide, 5%.

6.10 Acetic acid, 56%.

7. Procedure

7.1 Pad polyester/cotton blend cloth at 50-60% wet pickup with:

10 g/L—Disperse dye

20 g/L—Antimigrant (see 11.6)

Acetic acid (56%) to pH 5.5-6.0

Use the same pad liquor for all the dyeings in the set.

7.2 Dry each sample thoroughly under minimum migration conditions (see 11.5), then thermofix for 90 s at 196°C (385°F), 205°C (400°F), 213°C (415°F) and 221°C (430°F), respectively, in a properly calibrated oven (see 11.4).

7.3 Leach out the cotton portion of the blend with 70% sulfuric acid for 3-4 min at 55°C (130°F). Rinse the sample in cold running water and neutralize with 5% ammonium hydroxide solution for 1 min followed by another cold water rinse.

7.4 In cases where a change in shade of a disperse dye is suspected due to leaching, the following procedure may be employed: Attach a 100% polyester rider to the blended cloth to be dyed by the method in 7.1. Leach a portion of the rider separately by the method described in 7.3 and evaluate for any changes that

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may occur in the leaching process.

8. Evaluation

8.1 Make reflectance measurements in the visible portion of the spectrum (see 11.2) to determine relative strengths on the leached polyester using the deepest dyeing fixed at 221°C (430°F) as 100%. *K*/*S* values of the minimum reflectance value are obtained using the Kubelka-Munk equation:

$$K/S = (1 - R)^2/2R$$

where:

K/S = absorption function
R = minimum reflectance value (see 11.2)

Alternatively, K/S for a summation of the visible wavelengths can be used for the strength calculation. In the example above, the K/S value used is for the wavelength of minimum reflectance. Either K/S calculation method can be used, but the chosen method should be noted by the user for replication by other users.

Calculate the relative fixation values in comparison to the deepest dyeing using the following equation:

$$C_i = 100 \times \frac{(K/S)_i}{(K/S)_{max}} \quad \text{(see 11.2)}$$

where:

C_i = percent of dye fixed on sample i
$(K/S)_i$ = absorption function of dyeing i
$(K/S)_{max}$ = absorption function of the deepest dyeing

8.2 As an alternate method of evaluation, an accurately weighed sample (250 mg) may be individually extracted in several small portions of chlorobenzene at the boil (132°C or 270°F). When the dye is completely removed from the substrate, transfer the extract quantitatively to a volumetric flask and dilute with additional chlorobenzene (see 11.7). Transmittance measurements made at the point of minimum transmittance or maximum absorption in the visible area of the spectrum of the extracts of fixed samples are compared to the corresponding measurements of extracts of the padded, unfixed sample (as 100%). Transmittance values are converted to absorbance values from tables (see 11.8) or as follows:

$$A = \log_{10}(1/T)$$

where:

A = the absorbance value
T = the transmittance value

Fixation values are calculated by the following equation:

$$C_i = 100 \times \frac{A_i}{A_u} \quad \text{(see 11.9)}$$

where:

C_i = percent of dye fixed on sample i
A_i = absorbance of the extract of dyeing i
A_u = absorbance of the extract of the unfixed dyeing

9. Report

9.1 Fixation data obtained by the above methods may be presented in graphic form or in tabular form expressed as the percent of fixation versus temperature. The effects of the time and/or dye concentration may be shown as a family of curves depending upon the objectives of the particular study under consideration.

10. Precision and Bias

10.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

10.2 *Bias*. The thermal fixation property of a disperse dye can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

11. Notes

11.1 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

11.2 Refer to AATCC EP6, Evaluation Procedure for Instrumental Color Measurement, paragraph 4.3, Color Strength Value Using Reflectance Measurement, and 4.5, Relative Strength, available from AATCC Technical Center, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.3 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

11.4 The equipment used should be carefully calibrated with respect to temperature, residence time and uniformity of air flow, and should be named in the written report.

11.5 A continuous or abridged reflectance spectrophotometer is suitable for measurements at the wavelength of maximum absorption (minimum reflectance).

11.6 A number of suitable gums for this purpose are available; e.g., natural gums, alginates and synthetic acrylic acid polymers. The use of electrolytes in the pad liquor is to be avoided since the resulting agglomeration is very difficult to control.

11.7 Some disperse dyes may be partially decomposed during the extraction and are not suitable for this procedure.

11.8 Refer to AATCC EP6, Evaluation Procedure for Instrumental Color Measurement, paragraph 4.4, Color Strength Value Using Transmittance Measurement, and 4.5, Relative Strength, available from AATCC Technical Center, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919. 549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.9 As a general guide for obtaining transmittance measurements, refer to A General Procedure for the Determination of Relative Dye Strength by Spectrophotometric Transmittance Measurements, *Textile Chemist and Colorist*, Vol., 4, No. 5, May 1972, p43.

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Test Method for Colorfastness to Solvent Spotting

1. Purpose and Scope

1.1 This test method is designed to measure the degree of color migration that occurs when a fabric is spotted with a drycleaning solvent.

1.2 Previous versions of this test method were based on perchloroethylene (often referred to as tetrachloroethene). However, due to environmental, health and safety issues related to perchloroethylene, drycleaner associations and test method organizations have been reviewing, evaluating, and approving alternatives.

1.3 The present recognized alternatives for cleaning of textile fabrics and garments by organizations include hydrocarbon solvents, dibutoxymethane, decamethylpentacyclosiloxane, and simulated wetcleaning. These solvent alternatives will be referred to as "solvent of choice" (see 11.2).

1.4 For users of this test method with perchloroethylene to evaluate the degree of color migration of disperse dyes, these proposed alternatives (see 1.3 and 5.5) are noticeably less effective. At this time, only acetone was found to be comparable to perchloroethylene for disperse dye migration.

1.5 During this period of transition away from perchloroethylene, brands, wet processors of dyed and finished fabrics, and testing laboratories should mutually agree on the "Solvent of Choice" to use when evaluating for solvent spotting of textile products using this test method.

2. Principle

2.1 A specimen, attached to a piece of white blotting paper, is placed face side up on a glass plate. The solvent of choice is placed on the center of the specimen. The test is carried out at room temperature. The staining of the blotting paper is evaluated.

3. Terminology

3.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

3.2 **migration,** n.—the nonuniform movement and distribution of dyes, pigments, finishes or other materials from one part of a material to another.

3.3 **spotting,** n.—*in cleaning*, application of solvent or solution to a material prior to or after commercial scouring, laundering, or drycleaning to remove or enhance removal of soil or stains in a local area.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 Test should be run in a well ventilated area such as under a hood.

4.4 An eyewash/safety shower should be located nearby, and an organic vapor respirator should be readily available for emergency use.

4.5 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met.

5. Apparatus, Materials and Reagents

5.1 Pipette, graduated to deliver 1.5 mL.

5.2 Plate, glass.

5.3 AATCC 9-Step Chromatic Transference Scale (AATCC EP8) (see 11.1).

5.4 White AATCC Textile Blotting Paper (see 11.1).

5.5 The present solvent of choice is represented in 11.2.

6. Test Specimen

6.1 Piece of dyed fabric, 25 × 25 mm (1 × 1 in.).

7. Procedure

7.1 Attach test specimen to the center of a 150 × 150 mm (6 × 6 in.) piece of white blotting paper. Make sure the sample is flat against the blotting paper. If necessary, staple the edges of the fabric down to keep the sample flat on the blotting paper.

7.2 Place blotting paper on the glass.

7.3 Apply 1.5 mL of solvent of choice (by pipette) on the center of the test specimen.

7.4 Let stand in air until dry; then evaluate the staining on the blotting paper.

8. Evaluation Method for Staining

8.1 Rate the degree of stain on the blotter using the AATCC 9-Step Chromatic Transference Scale (AATCC EP8) or Instrumental Assessment of Degree of Staining (AATCC EP12) and record the numerical rating that corresponds to the appropriate one on it.

9. Report

9.1 Cite the test method used and any modifications.

9.2 Report the staining grade assigned (see 8.1).

10. Precision and Bias

10.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

10.2 *Bias*. Colorfastness to solvent spotting can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

11. Notes

11.1 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.2 The solvent of choice may include hydrocarbon solvents (CAS 64742-48-9), dibutoxymethane (CAS 2568-90-3), decamethylpentacyclosiloxane (CAS 541-02-6), and simulated washing (to be determined by users).

12. History

12.1 Revised in 2022 to provide replacement chemistries for perchloroethylene, and title change.

12.2 Editorially revised 2020 to add History section and change Jurisdiction.

12.3 Editorially revised 2019; reaffirmed 2017; revised 2016; editorially revised and reaffirmed 2010; reaffirmed 2005; editorially revised 2002, and 2001; reaffirmed 2000; editorially revised and reaffirmed 1995; reaffirmed 1990. Editorially revised, 1986. Editorially revised and reaffirmed 1985 and 1981.

12.4 Developed in 1985 by Committee RA92. Jurisdiction transferred to Committee RA87 in 2020.

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Test Method for Dimensional Changes on Drycleaning in Perchloroethylene: Machine

Developed in 1978 by AATCC Committee RA43; editorially revised and reaffirmed 1979, 1985, 1995, 2011; editorially revised 1986, 1991, 1993, 2004, 2008, 2010, 2019; reaffirmed 1990, 2000, 2005, 2016. Partly equivalent to ISO 3175.

Foreword

Drycleaning, a process for cleaning textiles in an organic solvent, dissolves oils and fats and disperses particulate dirt substantially without the swelling and creasing associated with washing or wet cleaning. Small quantities of water and surfactant may be incorporated in the solvent to obtain better soil and stain removal. Some moisture-sensitive articles are preferably drycleaned without the addition of water to the solvent but a surfactant can be used in order to assist soil removal and prevent graying. Different solvents can be used for drycleaning, of which perchloroethylene (tetrachloroethylene) is the most common in many countries. For this reason, the present method prescribes the use of perchloroethylene. Drycleaning is normally followed by an appropriate restorative finishing procedure. In most cases, this comprises some form of steam treatment and/or hot pressing. Dimensional change on drycleaning and steaming and/or pressing is progressive, and in some cases a single treatment may give little indication of the extent of dimensional change that may arise after repeated treatments. Generally, most of the potential dimensional change is released after 3-5 drycleaning and finishing treatments.

1. Purpose and Scope

1.1 This test method specifies a drycleaning procedure, using a commercial drycleaning machine, for the determination of dimensional change of fabrics and garments after drycleaning in perchloroethylene. It comprises processes for normal materials and sensitive materials (see 13.1).

1.2 Very sensitive materials, which can be cleaned only when special precautions are taken, are excluded from the scope of this method (see 13.1).

1.3 The test method is intended only for the assessment of dimensional changes undergone by a specimen subjected to a single drycleaning and finishing operation. When it is desired to determine the amount of progressive dimensional change, the method may be repeated a specified number of times, normally not exceeding five cycles.

2. Principle

2.1 Conditioned fabrics or garments are marked and measured, then subjected to a drycleaning procedure, followed by an appropriate finishing procedure. They are afterwards conditioned and measured. The dimensional change is expressed as a percentage of the original dimensions.

3. Terminology

3.1 **dimensional change,** n.—a generic term for variation in length or width of a garment or fabric specimen subjected to specified conditions.

3.2 **drycleaning,** n.—the cleaning of fabrics with organic solvents such as petroleum solvent, perchloroethylene or fluorocarbon.

NOTE: The process also includes adding detergent and moisture to the solvent, up to 75% RH, and hot tumble drying to 71°C (160°F).

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 Perchloroethylene is toxic by inhalation, by repeated contact with the skin and by ingestion; it should be used only in a well vented atmosphere. Toxicology studies with laboratory animals have shown evidence of cancer in rats and mice exposed to perchloroethylene vapors at 100-400 ppm concentrations for prolonged times. Fabric saturated with this solvent should be dried in an adequately ventilated hood. Use chemical goggles or face shield, impervious gloves and an impervious apron when handling perchloroethylene.

4.4 An eyewash/safety shower should be located nearby and an organic vapor respirator should be readily available for emergency use.

4.5 The drycleaning apparatus should be ventilated to the outside atmosphere in accordance with the manufacturer's instructions.

4.6 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 13.2).

5. Reagents

5.1 Perchloroethylene (commercial), $CCl_2 = CCl_2$, drycleaning grade.

5.2 Sorbitan mono-oleate (a typical chemical is Span 80).

6. Apparatus and Materials

6.1 Drycleaning machine. This shall consist of a commercial rotating cage type, totally enclosed machine for use with perchloroethylene. The diameter of the rotating cage shall be not less than 600 mm and not more than 1080 mm. Its depth shall be not less than 300 mm. It shall be fitted with three or four lifters. The speed shall be such as to give a g-factor (see 13.3) between 0.5 and 0.8 for cleaning and between 35 and 120 for extraction. The machine shall be equipped with a thermometer for the measurement of the solvent temperature. The machine shall have suitable facilities so that the emulsion can be introduced gradually into the solvent between the inner and outer cages below the level of the solvent. The machine may be of the single-unit washer/dryer type or a separate dryer may be used. Both types of dryers shall be equipped with temperature control of either the incoming or the outgoing air during the drying cycle. (Information on

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solvent recovery is given in 13.4.) The dimensions of the rotating drum of the separate dryer shall comply with those given for the drycleaning machine.

6.2 Apparatus for applying the appropriate finishing treatment to the test specimen.

6.3 Means of producing the standard atmosphere for testing textiles.

6.4 Ballast consisting of clean textile pieces or garments. These shall be white or a light color and consist of about 80% wool and 20% cotton or rayon.

6.5 Pen and ink (indelible) or other suitable marking device.

6.6 Stable measuring scale of dimensions suitable for the article being tested, graduated in millimeters.

6.7 Flat table of dimensions such that the article being tested can be laid flat for measurement.

7. Test Specimens

7.1 Test garments as such.

7.2 Cut fabrics into test specimens preferably not smaller than 500 × 500 mm and stitched on all sides with polyester thread to prevent ravelling.

7.3 Carefully open elastic circular knits by cutting in the direction of the ribs without causing distortion. After marking and measuring according to Section 8, restore the circular state by sewing together the cut edges. After the test, the seam shall be cut again and distances between markings determined in the open state.

8. Preparation of Specimens

8.1 Condition the test specimens and dummy load from the dry side at 21 ± 2°C (70 ± 4°F) and 65 ± 5% RH by laying each test specimen separately on a flat, smooth, horizontal surface for at least 24 h.

8.2 When marking fabric specimens, lay out the specimen without tension on a flat, smooth surface, taking care to see that it is free from wrinkles and creases. Making three pairs of marks, each at least 250 mm apart in the length direction, and three similar pairs of marks in the width direction of the fabric. If the specimen is a garment, mark and measure different parts of the outer fabrics and linings separately.

9. Test Procedures

9.1 Procedure for normal materials.

9.1.1 The mass of the complete load shall be 50 ± 2 kg for each cubic meter of the volume of the inner cage. Ensure that the mass of the textile part of the test specimen(s) is not more than 10% of the total load, the remainder consisting of the ballast unless the mass of the test piece of garment as such exceeds 10% of the total load.

When loaded into the machine, the specimen(s) and the ballast shall be in equilibrium with the standard atmosphere for testing textiles. Equilibrium is deemed to be attained after exposure for 24 h to this atmosphere.

9.1.2 Place the conditioned load in the machine and introduce perchloroethylene containing l g/L of sorbitan mono-oleate so that the liquor ratio, calculated on the volume of solvent in the outer and inner cages, is 6.5 ± 0.5 L for each kg of load (this corresponds to a solvent level of about 30% of the inner cage diameter). Maintain the solvent at 30 ± 3°C (86 ± 5°F) throughout the cleaning operation.

9.1.3 Prepare an emulsion by mixing 1 part (by volume) of the sorbitan mono-oleate with 3 parts of perchloroethylene and then adding 2 parts of water (with stirring). Start the machine with the filter circuit shut off and slowly (over a period of not less than 2 min and not more than 12 min) add an amount of emulsion corresponding to 2% of water calculated on the mass of the load, to the machine between the inner and outer cages below the level of the solvent.

9.1.4 Keep the machine running for 15 min after switching it on. The filter circuit test is not used during any part of this test except the initial solvent fill in 9.1.2.

9.1.5 Drain the solvent and centrifugally extract the solvent from the load for 2 min (at least 1 min being at full extraction speed).

9.1.6 Introduce pure dry solvent at the same liquor ratio (see 9.1.2) and rinse for 5 min. Drain and extract again for 3 min (at least 2 min being at full extraction speed).

9.1.7 Dry the load in the machine or in a separate drying tumbler, by tumbling in circulating warm air for an appropriate time, preferably using an automatic solvent-dryness control. Either the outlet air temperature shall not exceed 60°C (140°F), or the inlet temperature shall not exceed 80°C (176°F) (see 13.5). After drying, blow air at ambient temperature through the rotating load for 3-5 min.

9.1.8 Remove the specimens from machine immediately. Place garments individually on hangers and place fabric specimens on a flat surface for not less than 30 min before finishing.

NOTE: If additional information on stability to drycleaning only is required, condition and remeasure the specimen at this stage before completing the procedure. Note details of this procedure in the test report.

9.1.9 Carry out a finishing treatment by the method appropriate for the type of garment or fabric under test. In most cases, this will involve pressing on a garment (steam) press supplied with steam at a pressure of 370-490 kPa[2] (3.8-5 kg/cm[2]) (oven-pressure) or steaming on a steam/air garment former for 5-20 s followed by drying with warm air for 5-20 s.

9.1.10 Condition the specimen according to 8.1 and measure, according to 8.2, the distances previously marked and measured.

9.2 Procedure for sensitive materials.

9.2.1 Proceed as in 9.1.1 but use a total load of mass 33 ± 2 kg for each cubic meter of the volume of the inner cage.

9.2.2 Proceed as in 9.1.2 but increase the amount of solvent to 10 ± 1 L/kg of load (this gives about the same solvent level as in 9.1.2).

9.2.3 Proceed as in 9.1.3 but operating at solvent relative humidity of 63 ± 2%. This means that no water emulsion addition is necessary.

9.2.4 Proceed as in 9.1.4 but reduce the running time to 10 min.

9.2.5 Proceed as in 9.1.5-9.1.10 but reduce the period of extraction at full extraction speed to 1 min.

9.2.6 Lay the specimen out according to 8.2 and measure the distance between marks to the nearest millimeter. Measure overall dimensions of garments to within ± 2 mm. Make all measurements in the standard atmosphere for conditioning and testing textiles as given in 8.1.

10. Calculation and Expression of Results

10.1 Calculate the dimensional change in the length and width directions of each fabric specimen or in the principal dimensions of a garment. Express as average percentage dimensional change, rounded to the nearest 0.2%, using a minus sign to indicate shrinkage and a plus sign to indicate an increase in dimensions.

11. Test Report

11.1 State that the tests were made in accordance with this method and give the following information:

(a) Details of any optional or additional requirements that have been met.

(b) Whether the procedure in 9.1 or 9.2 was used.

(c) The results obtained according to Section 10.

(d) The percentage by mass of specimen(s) in the load, and the type of material comprising the ballast.

(e) The maximum inlet or outlet air temperature during drying.

(f) Details of the finishing treatment used.

(g) The number of treatments given.

(h) Details of the dimensions of the garment or fabric specimen as required by the method.

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12. Precision and Bias

12.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

12.2 *Bias*. Dimensional change on drycleaning in perchloroethylene by machine can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

13. Notes

13.1 Examples of sensitive materials are woolen knitwear; fur fabrics; crepes; open set, loose spun wool fabrics; loom finished fabrics; boucles; Bedford cords; cavalry twills; raschel fabrics; chenille; cloque fabrics; pure silk fabrics; etc.

13.2 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

13.3 The *g* factor is calculated according to the following formula:

$$g = \frac{5.6n^2d}{10{,}000{,}000}$$

where:

n = the number of revolutions per min
d = the diameter, in mm, of the rotating cage

13.4 When using commercial drycleaning equipment, official regulations and normal safety precautions should be observed.

13.5 Textiles made of, or containing, heat sensitive fibers—for example, modacrylic fibers—shall be dried at an outlet air temperature not exceeding 40°C (104°F) [inlet air temperature up to 60°C (140°F)]. Very heat sensitive textiles, such as articles containing polyvinyl chloride fibers, shall be removed from the machine after extraction and dried at room temperature.

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Test Method for Transfer of Acid and Premetallized Acid Dyes on Nylon

Developed in 1979 by AATCC Committee RA87; editorially revised and reaffirmed 1984, 1989 (title change), 1994, 1999; editorially revised 1985, 1987, 1997, 2004, 2010, 2019; reaffirmed 2006; revised 2011, 2017.

1. Purpose and Scope

1.1 This test method evaluates the transfer of an acid (anionic) or premetallized dye from a dyed to an undyed nylon substrate under mock dyebath conditions. A general method of dyeing acid dyes on nylon is included for uniformity of dyeing procedure.

2. Principle

2.1 A test specimen of nylon fabric is dyed to 1/1 standard depth with the dye to be evaluated. Part of the dyed fabric together with an equal weight of undyed fabric is subjected to transfer in a mock dyebath.

2.2 Transfer is made at pH 4.5, 6.0 and 7.5, and at a temperature of 95°C.

3. Terminology

3.1 **acid dye,** n.—an anionic dye having substantivity for fibers which contain cationic groups usually in acidic or neutral aqueous dyebaths.

3.2 **premetallized acid dye,** n.—an acid dye manufactured by reacting an equivalent of a suitable metal ion with one equivalent of a dye (1:1 premetallized acid dye), or with two equivalents of the same or different dyes (1:2 premetallized acid dyes), capable of chelating the metal.

3.3 **transfer,** n.—*in textile processing, testing, storage and use*, movement of a chemical, dye or pigment between fibers within a substrate or between substrates (see also *migration*, *leveling*).

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas and a single-use dust respirator while handling powder dyes.

4.2 All chemicals should be handled with care. Use chemical goggles or face shield, impervious gloves and an impervious apron when dispensing concentrated acids and sodium hydroxide.

4.3 An eyewash/safety shower should be located nearby and an organic vapor respirator with full facepiece should be readily available for emergency use.

4.4 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 9.1).

5. Apparatus and Materials

5.1 Dyed nylon.

5.2 Undyed nylon (from same piece as dyed sample).

5.3 Nonionic surfactant.

5.4 Sodium carbonate (Na_2CO_3).

5.5 Phosphoric acid (usually 85%) (H_3PO_4).

5.6 Acetic acid, glacial (CH_3COOH).

5.7 Boric acid (H_3BO_3).

5.8 Sodium hydroxide (NaOH).

5.9 Acetic acid, 56% (CH_3COOH).

5.10 EDTA (ethylenediaminetetraacetic acid, tetrasodium salt, 100% powder).

5.11 Means of evaluating color change of specimen.

5.11.1 Gray Scale for Color Change (AATCC EP1) (see 9.2), or

5.11.2 Spectrophotometer meeting any one of the geometric definitions described in CIE publication 15.2, subclause 1.4 (see 9.3).

5.12 Means of evaluating color staining on specimen

5.12.1 Gray Scale for Staining (AATCC EP2) (see 9.2), or

5.12.2 AATCC 9-Step Chromatic Transference Scale (AATCC EP8) (see 9.2), or

5.12.3 Spectrophotometer meeting any one of the geometric definitions described in CIE publication 15.2, subclause 1.4 (see 9.3).

6. Procedure

6.1 Fabric Preparation.

6.1.1 Scour undyed fabric in a bath containing 1.5 g/L nonionic surfactant (5.3) and 1.5 g/L sodium carbonate for 30 min at 50-60°C.

6.1.2 Rinse thoroughly in water.

6.1.3 Neutralize with acetic acid, 1 mL/L 56% acetic acid.

6.1.4 Final rinse in water.

6.1.5 Air dry.

6.2 Dyeing Procedure.

6.2.1 Set 30:1 bath at 38°C (100°F); X.XX% dye; 0.25% EDTA sequestrant (omit for neutral premetallized dyes; and 2.00% acetic acid [56%]).

6.2.2 Enter nylon substrate (cloth, yarn, stock) at 38-48°C.

6.2.3 Raise temperature to boil at 1.5°C/min.

6.2.4 Dye at or near boil for 45 min.

6.2.5 If necessary, add an additional 1.0% acetic acid (56%) to exhaust dye.

6.2.6 Run an additional 15 min.

6.2.7 Rinse in hot 80-90°C water, then in 38°C water.

6.2.8 Air dry.

6.3 Transfer Bath Preparation (see Appendix I).

6.3.1 Prepare 200 mL mock dyebaths (one for each dye to be tested) for pH 4.5, 6.0 and 7.5 as described in Appendix I.

6.3.2 Add one 5 g piece of dyed fabric and one 5 g piece of undyed fabric to each mock dyebath; B/F = 20/1.

6.3.3 It has been shown that the degree of agitation is a contributing factor affecting the rate of diffusion of dyes into and out of a substrate. Agitation should be the same for all comparative experiments.

6.3.4 Raise temperature at 2°C/min to 95°C/min (5°F/min to 203°F/min) and run with agitation for 50 min. Save the bath (see 6.3.6).

6.3.5 Remove both pieces of fabric. Rinse with water and air dry.

6.3.6 Measure pHs of part of the transfer baths at room temperature and record in the table attached.

6.3.7 Add a piece of undyed nylon to the transfer bath and exhaust using the conditions outlined in 6.3.4. Adjust pH to 4.5 to accomplish. Record pH in table.

7. Evaluation and Reporting

7.1 Comparison of Dyes at pH 4.5.

7.1.1 For best observations, samples should be butted together on white board as shown in Fig. 1.

7.1.2 Mount pH 6.0 and 7.5 samples similarly.

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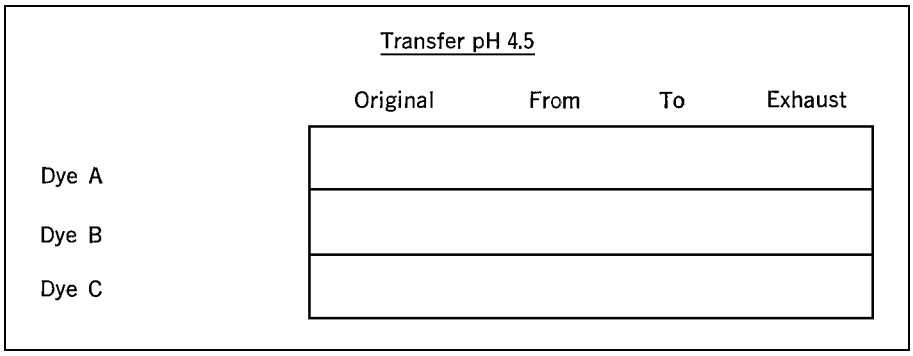


Fig. 1—Mounting for comparison of dyes at pH 4.5.

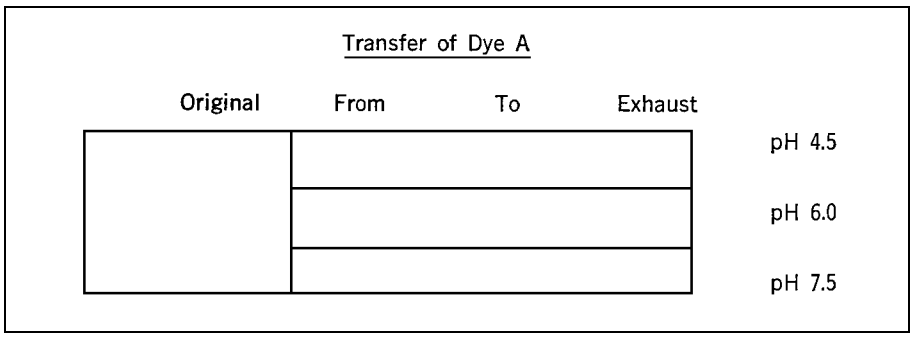


Fig. 2—Mounting for comparison of Dye A at pH 4.5, 6.0, and 7.5.

7.2 Comparison of Dye A at pH 4.5, 6.0 and 7.5.

7.2.1 Mount samples on white board as shown in Fig. 2.

7.2.2 Mount Dye B and Dye C samples similarly.

7.3 Comparison of Shade.

7.3.1 Rate the degree of color difference between "to" and "from" samples using the Gray Scale for Color Change (AATCC EP1) or AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen (see 9.2). Record the appropriate rating.

7.4 Exhaust Example Evaluation.

7.4.1 Rate the degree of staining on the "Exhaust" sample using the Gray Scale for Staining (AATCC EP2) or AATCC 9-Step Chromatic Transference Scale (AATCC EP8) or AATCC EP12, Instrumental Assessment of Degree of Staining (see 9.2). Record the appropriate numerical rating.

7.5 Record type of nylon and manufacturer on all test boards.

8. Precision and Bias

8.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

8.2 *Bias*. The transfer of acid and premetalized acid dyes on nylon can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

9. Notes

9.1 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

9.2 Available from AATCC Technical Center, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919. 549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

9.3 CIE Publication 15:2004 Colorimetry, 3rd Edition. Available through the US National Committee of the CIE or via the CIE webshop (www.techstreet.com/cie/).

Appendix I
Preparation of Buffered Mock Dyebaths

A1. Prepare two solutions as follows using distilled water (tap water can be used if distilled water is not available):

SOLUTION A—5 g phosphoric acid (100%); 2.4 g acetic acid (100%); and 1.76 g boric acid (HBO_2 100%). Make to 1L of solution.

SOLUTION B—8 g sodium hydroxide. Make to 1L of solution.

A1.1 Prepare mock dyebaths as shown in Table I using a pH meter. The amount of Solution B required to obtain the desired pH may vary slightly. Add Solution B to Solution A, checking end point with pH meter. Dilute to 600 mL with deionized water. This is sufficient for three mock dyebaths at each pH value.

Table I—Preparation of Mock Dyebaths

	Solution A	Solution B	Solution C
pH 4.5	100 mL	ca. 35 mL	600 mL
pH 6.0	100 mL	ca. 50 mL	600 mL
pH 7.5	100 mL	ca. 10 mL	600 mL

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Test Method for Color Change of Disperse Dyes Caused by Metals, with and without Chelating Agent

1. Purpose and Scope

1.1 This test method provides a laboratory procedure for:

1.1.1 Determining the effectiveness of chelating agents in disperse dyebaths for inactivating heavy metals that can cause color change.

1.1.2 Evaluating the sensitivity of disperse dyes to metal-induced color change during dyeing.

2. Principle

2.1 The effectiveness of chelating agents in sequestering metals in a dyebath, thereby preventing metal-induced color change, is determined by the protocol in Fig. 1.

2.2 Metal sensitivity of disperse dyes is determined by comparing control dyes and test dyes by the protocol shown in Fig. 2.

3. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

3.1 AATCC EP1, Evaluation Procedure for Gray Scale for Color Change.

3.2 AATCC EP7, Evaluation Procedure for Instrumental Assessment of the Change in Color of a Test Specimen.

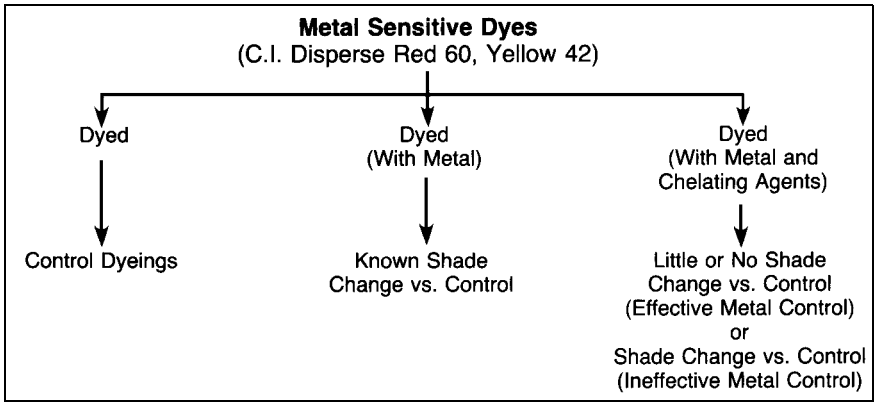


Fig. 1

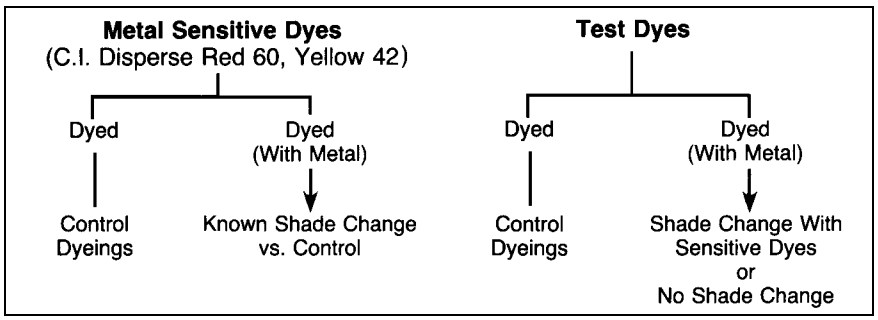


Fig. 2

4. Terminology

4.1 **chelating agent,** n.—*in textile chemistry*, a chemical capable of deactivating metal ions by formation of a water-soluble complex. SYN: *sequestering agent*.

4.2 **color change,** n.—*as used in colorfastness testing*, a change in color of any kind whether in lightness, hue or chroma, or any combination of these, discernible by comparing the test specimen with a corresponding untested specimen.

4.3 **grade,** n.—the number assigned to a test specimen resulting from comparison to a scale.

4.4 **Gray Scale,** n.—a scale consisting of pairs of standard gray chips, the pairs representing progressive differences in color or contrast corresponding to numerical colorfastness grades.

NOTE: Gray Scale grades are expressed as half-steps in decimal form (e.g., 1 or 1.5). Dashed form is also acceptable (e.g., 1 or 1-2).

4.5 **metal sensitivity,** n.—the propensity of a dye to produce an abnormal color on a textile material when in the presence of certain metal ions.

5. Safety Precautions

5.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

5.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

5.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

6. Uses and Limitations

6.1 Accurate results and expected shade changes depend on the metal dye and carrier used, as well as close control of all dyeing parameters.

6.2 Extraneous metal contamination can give misleading results. Use clean fabrics, distilled water, and rinse the dye containers with a 10% phosphoric acid solution followed by a distilled water rinse.

6.3 To avoid erroneous conclusions, test conditions must be established with dyes of known metal sensitivity to verify that the expected shade change can be obtained. Some samples of control dyes or carriers, or certain dyeing conditions, may not produce the expected shade change.

6.4 This test method is not intended for use with disperse dyes in combination with other dye classes that may contain metal complexes potentially damaged by chelating agents.

7. Apparatus and Materials (see 15.1)

7.1 Test fabric, 100% polyester fabric (filament or staple), desized (as needed) and prescoured (see 15.2).

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7.2 Control dyes, C. I. Disperse Red 60 and C. I. Disperse Yellow 42 (see 15.3).

7.3 Test dyes.

7.4 Cupric sulfate ($CuSO_4$), MW 159.606.

7.5 Ferrous ammonium sulfate ($Fe(NH_4)_2 (SO_4)_2 \cdot 6H_2O$), MW 392.158.

7.6 Carrier (see 15.4).

7.7 Chelating agent (see 15.5).

7.8 Distilled water.

7.9 Acetic acid (CH_3COOH), 99.7%.

7.10 Phosphoric acid (H_3PO_4), 85%.

7.11 Nonionic detergent

7.12 Sodium carbonate

7.13 Laboratory dyeing apparatus, capable of providing good agitation and uniform temperature control.

7.14 Analytical balance

7.15 pH meter

7.16 AATCC Gray Scale for Color Change (AATCC EP1) (see 15.6) or spectrophotometer as specified in AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen.

8. Sampling

8.1 Cut one specimen of polyester test fabric for each dyeing required (see Figs. 1 and 2).

8.2 Use the same dye, metal, carrier, and chelating agent solutions for all dyeings, as needed.

9. Specimens

9.1 Cut equal-weight specimens of polyester test fabric. Recommended weight for each specimen is 10 g.

9.2 Create dye and auxiliary solutions as described in Section 10.

10. Preparation of Reagents

10.1 Dye solutions (for *each* control and test dye).

10.1.1 Weigh accurately 2.0 g of dye. If test dyes are to be evaluated, use test dye amounts to give a comparable depth of shade to that for concentrations of C.I. Disperse Red 60 and C.I. Disperse Yellow 42 dyes specified in Tables I and II.

10.1.2 Measure 100 mL of distilled water and heat to 49-60°C (120-140°F). Add a small quantity of this water to the dye.

10.1.3 Paste well with a clean stirring rod until the dye is dispersed. Add several aliquots of the warm water with stirring.

10.1.4 Dilute to 1000 mL with distilled water to give a 2.0 g/L dye dispersion.

10.2 Cupric sulfate solution.

10.2.1 Weigh accurately 2.51 g of cupric sulfate.

10.2.2 Prepare a solution in 1000 mL of distilled water. This will result in a solution of 1.0 g/L metal ions.

10.2.3 Check the metal solutions before use. Discard if there is any flocculation or sedimentation.

10.2.4 The ferrous salt solution is unstable and will form a precipitate on standing. Make up fresh daily.

10.3 Ferrous ammonium sulfate solution.

10.3.1 Weigh accurately 7.02 g of ferrous ammonium sulfate.

10.3.2 Prepare a solution in 1000 mL of distilled water. This will result in a solution of 1.0 g/L metal ions (see 10.2.3 and 10.2.4).

10.4 Carrier solution.

10.4.1 Weigh accurately the carrier.

10.4.2 Prepare a 50.0 g/L carrier solution using distilled water.

10.4.3 With C.I. Disperse Red 60 and ferrous ions, some carriers may cause shade change due to dye dispersion breakage. An uneven, bluish-red color which crocks is characteristic of this broken dispersion.

10.5 Chelating agent solution.

10.5.1 Weigh accurately the chelating agent.

10.5.2 Prepare a 10.0 g/L chelating agent solution using distilled water.

10.5.3 Introduction of a sufficient amount of an EDTA chelating agent to dyebaths of C.I. Disperse Red 60 or Disperse Yellow 42 with cupric or ferrous ions present has been found to give essentially the expected dyed shade without metals.

11. Procedure

11.1 To test the *effectiveness of chelating agents* in sequestering metals in a dyebath to prevent metal-induced color change, perform dyeings with C.I. Disperse Red 60 and C.I. Disperse Yellow 42 under "control," "with metal," and "with metal and chelating agent" conditions (see Fig. 1 and Tables I and II.)

11.2 To test *metal sensitivity of disperse dyes*, perform dyeings with C.I. Disperse Red 60, C.I. Disperse Yellow 42, and test dyes under "control" and "with metal" conditions (see Fig. 2 and Tables I and II.)

11.3 Prior to dyeing, rinse all dyeing containers with a 10% solution of phosphoric acid followed by distilled water rinse.

11.4 Prepare dyebaths according to Table I and/or Table II.

11.5 Wet out specimens in distilled water. Squeeze to remove excess water. Add to the dyebath with stirring.

11.6 Use a pH meter to adjust dyebath, including specimen, to pH 5.0 with acetic acid.

11.6.1 Calibrate the pH meter before use with standard pH 4.0 and pH 7.0 buffer solutions.

11.7 Load pressure-dyeing containers into appropriate equipment. Load atmospheric boil dyeings into suitable dyeing apparatus.

11.8 Raise to dyeing temperature at 2.2°C/min (4°F/min.)

11.8.1 Run atmospheric dyeings at the boil for 60 min.

11.8.2 Run pressure dyeings for 30 min at 130°C (265°F).

11.9 Cool down and remove dye containers.

11.10 Afterscour specimens 10 min at 71°C (160°F) with:

1.0% owf nonionic detergent
2.0% owf sodium carbonate

11.12 Rinse.

11.13 Neutralize to pH 7 ± 0.5 with acetic acid.

11.14 Rinse (distilled water optional for after scour).

11.15 Dry.

11.16 Prior to evaluation, condition test specimens for 1 h in an atmosphere of 65 ± 5% RH and a temperature of 21 ± 2°C (70 ± 4°F),

12. Evaluation

12.1 Make all comparisons as directed in AATCC EP1 or AATCC EP7. Use "control" dyeings in place of original specimens. Use dyeings "with metal" and "with metal and chelating agent" as test specimens.

12.2 Compare C.I. Disperse Red 60 and C.I. Disperse Yellow 42 dyeings "with metal" to the relevant "control" dyeings (see 15.3). If there is no color difference, repeat the test, with dye from another manufacturer, if necessary.

12.3 To evaluate the *effectiveness of chelating agents*:

12.3.1 Compare C.I. Disperse Red 60 and C.I. Disperse Yellow 42 dyeings "with metal and chelating agent" to relevant "control" dyeings. Little or no color change indicates effective sequestering of metals.

12.4 To evaluate *metal sensitivity of test dyes*:

12.4.1 Compare test-dye dyeings "with metal and chelating agent" to relevant "control" dyeings.

13. Report

13.1 Test report should include, at a minimum:

13.1.1 A description or identification of each dye and chelating agent tested.

13.1.2 Evaluation was performed using AATCC TM161-2022.

13.1.3 Dyeing conditions (i.e. atmospheric, under pressure).

13.1.4 Evaluation procedure used (EP1 or EP7).

13.1.5 Gray Scale for Color Change grade for each dyeing. If dyeings are performed in replicate, report average grade for each dyebath composition.

13.1.6 Any modifications of the test method.

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Table I—Atmospheric Dyebath Composition (See 15.8)

Parameters	Control C.I. Disperse Red 60	
Fabric	10 g	
Water	259 mL	
Carrier Solution (see 1 0.4)[a]	8.0% (16 mL)	
Dye Solution (see 10.1)[b]	0.5% (25 mL)	
Liquor Ratio[c]	30:1	
	with Metal (No Chelating Agent)	
	Cupric C.I. Disperse Red 60	**Ferrous C.I. Disperse Red 60**
Fabric	10 g	10 g
Water	256 mL	181 mL
Metal Solution (see 10.2 and 10.3)	10 ppm (3 mL)	10 ppm (3 mL)
Carrier Solution (see 10.4)	8.0% (16 mL)	8.0% (16 mL)
Dye Solution	0.5% (25 mL)	2.0% (100 mL)
Liquor Ratio	30:1	30:1
	with Metal & Chelating Agent	
	Cupric C.I. Disperse Red 60	**Ferrous C.I. Disperse Red 60**
Fabric	10 g	10 g
Water	251 mL	176 mL
Chelating Agent Solution (see 10.5)	0.5% (5 mL)	0.5% (5 mL)
Metal Solution	10 ppm (3 mL)	10 ppm (3 mL)
Carrier Solution	8.0% (16 mL)	8.0% (16 mL)
Dye Solution	0.5% (25 mL)	2.0% (100 mL)
Liquor Ratio	30:1	30:1

NOTE: Perform all atmospheric dyeings at the boil, 100°C (212°F), for 60 min.

[a] Use an appropriate carrier and amount with adjusted amounts of distilled water as necessary.
[b] Dye concentrations were selected to yield maximum shade change.
[c] A 30:1 liquor ratio was selected for laboratory convenience. Prior evaluations indicated no appreciable difference among 10:1, 20:1 and 30:1 liquor ratios.

Table II—Pressure Dyebath Composition (See 15.8)

Parameters	Control C.I. Disperse Red 60	C.I. Disperse Yellow 42	
Fabric	10 g	10 g	
Water	275 mL	275 mL	
Dye Solution (see 10.1)[a]	0.5% (25 mL)	0.5% (25 mL)	
Liquor Ratio[b]	30:1	30:1	
	with Metal (No Chelating Agent)		
	Cupric C.I. Disperse Red 60	**Cupric C.I. Disperse Yellow 42**	**Ferrous C.I. Disperse Red 60**
Fabric	10 g	10 g	10 g
Water	272 mL	272 mL	189 mL
Metal Solution (see 10.2 and 10.3)	10 ppm (3 mL)	10 ppm (3 mL)	10 ppm (3 mL)
Carrier Solution (see 10.4)[c]	No Carrier	No Carrier	4.0% (8 mL)
Dye Solution	0.5% (25 mL)	0.5% (25 mL)	2.0% (100 mL)
Liquor Ratio	30:1	30:1	30:1
	with Metal & Chelating Agent		
Fabric	**Cupric C.I. Disperse Red 60**	**Cupric C.I. Disperse Yellow 42**	**Ferrous C.I. Disperse Red 60**
Water	10 g	10 g	10 g
Chelating Agent Solution (see 10.5)	267 mL	267 mL	184 mL
Metal Solution	10 ppm (3 mL)	10 ppm (3 mL)	10 ppm (3 mL)
Carrier Solution	No carrier	No Carrier	4.0% (8 mL)
Dye Solution	0.5% (25 mL)	0.5% (25 mL)	0.5% (25 mL)
Liquor Ratio	30:1	30:1	30:1

NOTE: Perform all dyeings under pressure at 130°C (265°F), for 30 min.

[a] Dye concentrations were selected to yield maximum shade change.
[b] A 30:1 liquor ratio was selected for laboratory convenience. Prior evaluations indicated no appreciable difference among 10:1, 20:1 and 30:1 liquor ratios.
[c] Use an appropriate carrier and amount with adjusted amounts of distilled water as necessary.

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14. Precision and Bias

14.1 *Precision*. Precision of this method has not been established. Until a precision statement is generated for this method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

14.2 *Bias*. The true value of disperse dye color change cause by metals can only be defined in terms of a test method. Within this limitation, this method has no known bias.

15. Notes

15.1 Use ACS reagent grade chemicals to minimize metal contamination.

15.2 Fabric from mill production is suggested.

15.3 C.I. Disperse Red 60 or Disperse Yellow 42 dyes from different manufacturers may yield different results. Control dyeings are necessary to verify that shade change is possible under the test dyeing conditions. If shade change does not occur, repeat the test.

15.4 Tests were developed without carrier where possible. If essential to the test, carrier is indicated.

15.5 EDTA was used in the establishment of this test method. Other commercially available chelating agents may give effective results. Their effectiveness can be evaluated with this test method.

15.6 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; +1.919.549.8141; ordering@ aatcc.org; www.aatcc.org.

15.7 No appreciable shade change was found to occur with C.I. Disperse Red 60 or Disperse Yellow 42 at dyeing conditions other than those listed in Tables I and II. For example, C.I. Disperse Yellow 42 did not show sufficient shade change with iron or copper at the boil, or with iron under pressure to give a positive test.

16. History

16.1 Revised in 2023 to clarify and align with prescribed AATCC style guidelines.

16.2 Editorially revised 2019; revised 2018; editorially revised and reaffirmed 2012; editorially revised 2010, 2008; reaffirmed 2007; editorially revised 2004; editorially revised and reaffirmed 2002, 1992; editorially revised 1988 (with title change); editorially revised and reaffirmed 1987,1986; editorially revised 1985.

16.3 Developed in 1976 by AATCC Committee RR90. Maintained by RA99.

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Test Method for Colorfastness to Water: Chlorinated Pool

Developed in 1984 by Committee RA23. Supersedes AATCC TM105-1975; editorially revised 1985, 1995, 2008, 2010, 2012, 2019; editorially revised and reaffirmed 1986, 1997; reaffirmed 1991, 2002, 2009; revised 2011. Related to ISO 105-E03.

1. Purpose and Scope

1.1 This test method is used for evaluating the resistance to chlorinated pool water of dyed, printed or otherwise colored textile yarns and fabrics of all kinds.

2. Principle

2.1 Yarn or fabric specimens are agitated at a fixed rate in diluted chlorine solution under specified conditions of temperature, time, pH and hardness. The dried specimens are evaluated for change in color.

3. Terminology

3.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

3.2 **pool water,** n.—water to which various chemicals are added to maintain purity and clarity, normally used for swimming.

4. Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 When using 6*N* sulfuric acid and acetic acid, use chemical goggles or face shield, impervious gloves and an impervious apron. Concentrated acids should be handled only in an adequately ventilated laboratory hood. CAUTION: Always add acid to water.

4.4 An eyewash/safety shower should be located nearby and a high efficiency particulate respirator with a full facepiece should be readily available for emergency use.

4.5 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

4.6 Observe padder safety. Ensure adequate guard at the nip point. Normal safeguards on pad should not be removed. A foot-operated kickoff is recommended.

4.7 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 11.1).

5. Apparatus and Materials (see 11.2)

5.1 Equipment:

5.1.1 Option 1-Accelerated laundering machine (see 7.1 and 11.3).

5.1.2 Stainless steel lever lock canisters Type 2, 1200 mL, 90 × 200 mL (3.5 × 8.0 in.).

5.2 Option 2-Drycleaning Cylinder (see 7.2 and 11.4).

5.3 Wringer.

5.4 Gray Scale for Color Change (AATCC EP1) (see 11.6).

5.5 Test Control Fabric 162 (see 11.6).

5.6 Reagents:

5.6.1 Household sodium hypochlorite solution (NaOCl), approximately 5% available chlorine (see 11.5).

5.6.2 Calcium chloride, anhydrous ($CaCl_2$).

5.6.3 Magnesium chloride, hexahydrate ($MgCl_2 \cdot 6H_2O$).

5.6.4 Sulfuric acid (H_2SO_4), 6*N*.

5.6.5 Potassium iodide, KI, 12%.

5.6.6 Starch solution, 1%.

5.6.7 Sodium thiosulfate ($Na_2S_2O_3$), 0.01*N*, (see 11.7).

5.6.8 Distilled or deionized water.

5.6.9 Sodium carbonate (Na_2CO_3).

5.6.10 Acetic acid (CH_3COOH).

6. Preparation of Reagents

6.1 Deionized or distilled water to which 5 ppm chlorine and 100 ppm salts (hardness concentrate) are added; adjusted to 21°C (70°F) and pH 7.0 with sodium carbonate or acetic acid.

7. Test Specimens

7.1 Option 1-Accelerated Laundering Machine.

7.1.1 Specimen size 5 × 5 cm; control fabric size 5 × 5 cm. Total weight of specimen and control fabric 1.0 ± 0.05 g. Multiple specimens of the same sample can be used to achieve a 1.0 g load.

7.2 Option 2-Drycleaning Cylinder Equipment.

7.2.1 Colored specimens, approximately 6 × 6 cm. Total weight of test specimens should be 5.0 ± 0.25 g. If specimens to be tested do not weigh 5.0 g, add multiple specimens to make 5.0 g. Include the Test Control Fabric in the 5.0 g weight. Different colored specimens can be mixed when making up the 5.0 g load.

8. Procedure

8.1 Solution Preparation.

8.1.1 Add 800 mL deionized or distilled water to a 1 L volumetric flask and add 8.24 g calcium chloride and 5.07 g magnesium chloride while stirring to dissolve. Bring volume up to 1 L with water. This is a "hardness concentrate" which can be retained and used for 30 days.

8.1.2 Dilute 51 mL hardness concentrate to 5100 mL with deionized or distilled water. Add 0.5 mL household sodium hypochlorite solution which is no more than 60 days old. Determine actual ppm Cl by titration and adjust to 5 ppm (see 11.7).

8.1.3 0.01*N* Sodium thiosulfate can be purchased in this form or can be diluted 10 to 1 volumetrically from 0.1*N* solution.

8.1.4 Adjust solution to pH 7.0 with sodium carbonate or acetic acid as necessary.

8.2 Option 1-Accelerated Laundering Tester:

8.2.1 Place 1000 mL of solution (from 8.1) in the canister of the testing machine. Adjust temperature to 21°C (70°F).

8.2.2 Place specimen and control fabric in canister with solution. Specimen size 5 × 5 cm; control fabric size 5 × 5 cm. Total weight of specimen and control fabric 1.0 ± 0.05 g. Multiple specimens from the same sample can be used to achieve a

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1.0 g load. Place lid with gasket on top of canister and lock in place. Invert canister and check for leakage.

8.2.3 The closed canister is placed in the testing machine with the test specimens and the control fabric at 21°C. Run each fabric and control fabric separately for this test. Tumble the specimen with solution for 60 min. Note: Steel balls are not added to this solution.

8.3 Option 2-Drycleaning Cylinder Tester:

8.3.1 Purge the drycleaning cylinder. Add approximately 5000 mL deionized water and 0.5 mL household sodium hypochlorite solution (see 11.5). Close cylinder and run for 10 min (purging is necessary only if cylinder has been used for testing other than chlorinated pool water, or has not been used for a period of two weeks). Discard this solution.

8.3.2 Place the 5000 mL solution (from 8.1) in the cylinder of the testing machine. Adjust temperature to 21°C (70°F).

8.3.3 Place colored test specimens including test control, close and tumble for 60 min.

8.4 For both options, remove the test specimens and pass through wringer to remove excess solutions. Rinse thoroughly with deionized or distilled water. Wring again and allow to dry on absorbent, bleached paper toweling at room temperature.

9. Evaluation Method for Color Change

9.1 Classify the change in color of the Test Control Fabric 162 with the Gray Scale for Color Change (AATCC EP1) or AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen. If the classification is not equivalent to a 2-3 or 3, the test is considered to be invalid. If classification is equivalent to a 2-3 or 3, proceed with 9.2.

9.2 Rate the change in color of test specimens with the Gray Scale for Color Change (AATCC EP1), or using AATCC EP7, and record the numerical rating that corresponds to the appropriate one on the Gray Scale (see 11.6 and 11.8).

10. Precision and Bias (see 11.8)

10.1 An interlaboratory study involving five different laboratories, eight different colored fabric samples, eight control fabrics, two test methods (drycleaning cylinder and accelerated laundering machine) and two colorfastness rating procedures (manual and photometer) was conducted to define the various statistical parameters for this test method. All five laboratories ran the colorfastness test on the group of eight fabrics using both test methods and both rating procedures. The main observations of the study are stated in Table I.

10.1.1 The drycleaning cylinder method showed a mean fastness rating of 3.483 and the accelerated laundering method showed a mean fastness rating of 3.3213. The difference between the mean values was not statistically significant at 95% confidence level.

10.1.2 Manual rating procedure gave a mean fastness rating of 3.24 while spectrophotometer rating gave a mean fastness rating of 3.23. The difference between the mean values was not statistically significant at 95% confidence level.

10.1.3 Among the five laboratories, three laboratories showed very close mean rating values of 3.249, 3.229 and 3.261. The two outliers showed mean rating values of 3.415 and 3.055

10.2 *Bias*. The true value of Colorfastness to Chlorinated Pool Water can only be defined in terms of a test method. Within this limitation, this test method has no known bias.

11. Notes

11.1 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

Table I—Computed Values of Precision for Colorfastness Rating

No. of Specimens	Within Lab	Between Lab
3	0.207	0.397
2	0.256	0.487
1	0.357	0.689

11.2 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

11.3 Option 1-Accelerated laundering machine for rotating closed canisters in a thermostatically controlled water bath at 40 ± 2 rpm.

11.4 Option 2-Drycleaning Cylinder; stainless steel cylinder of approximately 33 cm (13 in.) high and about 22 cm (8.75 in.) in diameter. The cylinder is mounted in a vertical position on an axis inclined 50° and is rotated at a speed of 45-50 rpm.

11.5 Household sodium hypochlorite solution. When purchased, the strength will be 5.25% or slightly higher. All sodium hypochlorite solutions lose strength in time. Strength loss is accelerated by exposure to light and heat. Do not use after 60 days from purchase.

11.6 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.7 Use a 100 mL sample of test solution. Add 20 mL of 6*N* sulfuric acid, 6 mL of 12% potassium iodide and 3 drops of 1% starch solution. Mix thoroughly; the solution should be brownish blue color. Back titrate with 0.01*N* sodium thiosulfate until the first disappearance of the blue starch indicator color. Calculate the available chlorine as follows:

$$\%\ \text{available chlorine} = \frac{\text{mL Na thiosulfate} \times 0.01N \times 0.0355}{100\ \text{mL} \times \text{sp. gr. Na hypochlorite}} \times 100$$

(Specific gravity of household sodium hypochlorite solution is 1.08. Using the above equation, the use of 1.6 mL of sodium thiosulfate in back titration would yield a result of 0.0005% or 5.0 ppm.)

11.8 The precision of this test method is dependent on the combined variability of the material being tested, the test method itself and the evaluation procedure utilized.

11.8.1 The precision statement shown in Section 10 was developed from results obtained by visual and instrumental evaluations (AATCC EP1 and AATCC EP7).

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Test Method for Colorfastness to Storage: Dye Transfer

1. Purpose and Scope

1.1 The storage of garments made with component parts of more than one shade sometimes results in the transfer of dye from one area to another, normally from the darker to the lighter shade. This phenomenon is distinct from sublimation because it occurs below the sublimation temperature of dyes as well as with non-subliming dyes.

1.2 Dye transfer occurs when garments are folded with different shades in intimate contact with each other. Normally the amount of dye transferred is intensified when moisture is present, and thus the problem is more severe during warm, humid weather or when garments are steamed immediately before storage. Storage in plastic bags tends to maintain the initial relative humidity of the garment environment and can aggravate or reduce transfer depending on the condition of the fabric when it enters the bag.

1.3 This test method is intended to estimate if color transfer might be expected to occur during prolonged periods of storage. In general, no problems will be expected under normal conditions of storage or time-delay, if color transfer does not occur under the conditions specified in Section 7 (Procedure).

1.4 This test method may be used also to assess potential color transfer problems that relate to finishing. Some dyes are inherently more prone to transfer than others, and some chemical finishes and finishing conditions accelerate transfer.

2. Principle

2.1 Dyed and finished test specimens are sandwiched between pre-wet multifiber fabric and a selected second fabric, placed in a Perspiration Tester for 48 h at room temperature, dried and rated.

3. Terminology

3.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

3.2 **dye,** n.—a colorant applied to or formed in a substrate, via the molecularly dispersed state, which exhibits some degree of permanence.

3.3 **transfer,** n.—*in textile processing, testing, storage and use*, movement of a chemical, dye or pigment between fibers within a substrate or between substrates.

4. Safety Precautions

NOTE: These safety precautions are for information purposes and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials and equipment in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

4.3 Observe padder safety. Ensure adequate guard at the nip point. Normal safeguards on pad should not be removed. A foot-operated kickoff is recommended.

5. Apparatus and Materials (see 11.1)

5.1 Perspiration tester.

5.2 Plastic bag, polyethylene, large enough to contain perspiration tester.

5.3 Multifiber Test Fabric (8 mm [0.33 in.] bands) containing acetate, cotton, nylon, silk, viscose rayon, and wool shall be used for specimens containing silk. Multifiber Test Fabric containing acetate, cotton, nylon, polyester, acrylic, and wool shall be used with specimens with no silk present.

5.4 White fabric of same fiber composition as test specimens. If not available, cotton cloth as defined in AATCC TM8, Test Method for Colorfastness to Crocking: Crockmeter.

5.5 Distilled or deionized water.

5.6 Evaporating dish or glass beaker having the capacity to hold 50 mL of water. Wet sponge may also be used.

5.7 AATCC 9-Step Chromatic Transference Scale (AATCC EP8) (see 11.2)

5.8 Gray Scale for Staining (AATCC EP2) (see 11.2)

5.9 Gray Scale for Color Change (AATCC EP1) (see 11.2).

6. Test Specimens

6.1 Prepare 5.7 × 5.7 cm (2.25 × 2.25 in.) test specimens from each sample of the dyed fabric to be tested.

6.2 Prepare 5.7 × 5.7 cm (2.25 × 2.25 in.) multifiber and 5.7 × 5.7 cm (2.25 × 2.25 in.) white fabric swatches.

7. Procedure

7.1 Option I.

7.1.1 Immerse in distilled or deionized water at 24 ± 3°C (75 ± 5°F) both the multifiber fabric and the fabric selected for the opposite side of the sandwich (see 5.4). The wet pickup of these fabrics should be 100-110% (see 11.3). Do not wet out the specimen to be tested to prevent removal of dye or finish before testing begins.

7.1.2 Prepare a sandwich with the dyed test specimen between pre-wet multifiber cloth and the pre-wet white swatch (see 5.4).

7.1.3 Clamp the sandwich (prepared as directed in 7.1.2) between two clean plates of the Perspiration Tester. Set the Perspiration Tester as instructed in 9.4.1 of AATCC TM15, Colorfastness to Perspiration, except that the unit is not placed in an oven.

7.1.4 Enclose the Perspiration Tester in a polyethylene bag along with a dish (see 5.6) containing at least 50 mL of distilled or deionized water to maintain high relative humidity in the polyethylene bag. Seal the opening in the bag with a twist tie or rubberband if zipping bag is not used (see 5.2). Allow to remain 48 h at room temperature (24 ± 3°C [75 ± 5°F]). Take care that the Perspiration Tester does not fall into the water dish (see 11.4).

7.1.5 Remove the cover from the Perspiration Tester, remove the sandwiched specimens from the tester and separate the swatches. Allow the swatches to dry at room temperature.

7.2 Option II.

7.2.1 An accelerated test procedure has been used showing approximately the same results on some dyed or finished fabrics by placing the test apparatus (see 7.1.4) in a force draft oven at 38 ± 1°C (100 ± 2°F) for 4 h (see 11.4).

8. Evaluation (see 11.8)

8.1 After the swatches are dried, examine the multifiber and the white cloth for color transfer. Rate each of the fiber types in the multifiber fabric and the swatch on the opposite side of the sandwich separately, by comparison with the Gray Scale for Staining (AATCC EP2), the AATCC 9-Step Chromatic Transference Scale (AATCC EP8 or AATCC EP12), and record the numerical rating that corresponds to the appropriate one on either of them. Record which scale was used.

8.2 If a change in color is noted in any of the test specimens, rate such change with AATCC EP1 or AATCC EP7, and record the numerical rating that corre-

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sponds to the appropriate one on the Gray Scale.

9. Report

9.1 Identify the white fabric used in the sandwich (see 5.4 and 11.6).

9.2 Report the moisture pickup if other than 100-110% (see 11.3).

9.3 Report the grade for staining of each of the fibers in the multifiber fabric, as well as on the white swatch used in the test, and the rating scale used.

9.4 Report the grade for color change in any of the test specimens, and the evaluation procedure used (see 11.7.2).

10. Precision and Bias (see 11.7)

10.1 *Interlaboratory Data*. An interlaboratory comparison was run in 1982 in which samples from two different textile materials were tested in each of four laboratories, each material having five different finishes. Both the AATCC 9-Step Chromatic Transference Scale and the Gray Scale for Staining were used and a correlation within one unit was established between laboratories.

10.2 *Precision*. Based on the observations described in 10.1 and on general practice in the trade, a lot or consignment is generally considered as having a rating that is significantly worse than a specified value when a specimen from the lot or consignment has a rating of dye transfer that is more than one unit below the specified rating on the Gray Scale for Staining.

10.3 *Bias*. The true value of dye transfer can only be defined in terms of a test method. Within this limitation, this method has no known bias.

11. Notes

11.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

11.2 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.3 With some fabrics and constructions, 100% pickup causes a dripping wet condition. For those materials, use the highest pickup that is possible without water dripping from the specimen when it is suspended.

11.4 If any temperature and/or time other than those specified in 7.2 are used, report the change and reason for same.

11.5 For very critical evaluation and in case of arbitration, rating must be based on the geometric gray scale for evaluating staining.

11.6 All materials are not affected equally. Results obtained by the use of any one test procedure should not be representative of those of any other test procedure unless the degree of quantitative correlation for a given material has been established. In any event, the test procedure should be described in a material specification.

11.7 The precision of this test method is dependent on the combined variability of the material being tested, the test method itself, and the evaluation procedure utilized.

11.7.1 The precision statement in Section 10 was developed from results obtained by visual evaluations (AATCC EP2 and AATCC EP8).

11.7.2 It is expected that the use of instrumental evaluation (AATCC EP7 or AATCC EP12) will result in greater precision than that obtained from visual evaluations.

11.8 CAUTION: It has been reported that the results for staining obtained by this method on fabrics dyed to dark shades (navy, black, etc.) that contain a combination of polyester and spandex, or their blends, may not show the full staining propensity of such fabrics in consumer use. It is, therefore, recommended that the staining results obtained by this test not be used for the acceptance testing of such fabrics.

12. History

12.1 Editorially revised in 2023 to update history.

12.2 Editorially revised 2022, 2021; reaffirmed and editorially revised 2020; editorially revised 2019 (with a title change); revised 2013; editorially revised and reaffirmed 2012; editorially revised 2011, 2010, 2008; reaffirmed 2007; editorially revised and reaffirmed 2002; editorially revised 2001; editorially revised and reaffirmed 1997; editorially revised 1995; editorially revised and reaffirmed 1992, 1986; reaffirmed 1987.

12.3 Developed in 1985 by Committee RA92. Jurisdiction transferred to Committee RA87 in 2020.

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Test Method for Colorfastness to Oxides of Nitrogen in the Atmosphere under High Humidities

1. Purpose and Scope

1.1 This test method is used to determine the resistance of the color of textiles to the action of oxides of nitrogen in the atmosphere at elevated temperatures with relative humidities above 85%.

1.2 On certain fibers, dyes do not readily fade at humidities below 85%. Testing at higher humidities is required to produce color changes that predict service fading under warm, humid conditions (see 11.1).

2. Principle

2.1 A test specimen and a swatch of control ribbon are simultaneously exposed to oxides of nitrogen in an atmosphere which is maintained at 87.5 ± 2.5% RH and a temperature of 40 ± 1°C (104 ± 2°F) until the control ribbon shows a color change corresponding to that of a standard of fading. The cycles are repeated until the specimen shows a definite color change or for a prescribed number of cycles.

3. Terminology

3.1 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. They are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Oxides of nitrogen in high concentrations are injurious to health and must be exhausted to the atmosphere or trapped in water and neutralized with a 10% solution of sodium hydroxide or sodium bicarbonate. The maximum concentration in a work area must not exceed 500 pphm.

4.2 Chain or clamp the gas cylinder to a wall or other suitable support so that it cannot fall or be knocked down.

4.3 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 11.2).

5. Apparatus and Materials

5.1 Exposure chamber suitable for containing oxides of nitrogen (NO_2) and maintaining a constant elevated temperature and relative humidity (see 11.3).

5.2 Control Ribbon No. 1 (see 11.4).

5.3 Standard of Fading (Ribbon No. 1) (see 11.4).

5.4 Gray Scale for Color Change (AATCC EP1) (see 11.5).

5.5 Oxides of nitrogen supply tank (see 11.6).

6. Test Specimens

6.1 Cut each specimen at least 60 × 60 mm. For subsequent color comparison, the unexposed sample should be kept in an airtight container away from light to avoid further shade changes.

6.2 If the test involves laundered or drycleaned material, launder or dryclean material for both the control and test specimens. For the preparation of specimens for testing after laundering or drycleaning, follow the procedures prescribed in AATCC TM61, Test Method for Colorfastness to Laundering: Accelerated, and/or AATCC TM132, Test Method for Colorfastness to Drycleaning.

7. Procedure

7.1 Suspend the test specimens and a swatch of Control Ribbon No. 1 in the exposure chamber which is maintained at 87.5 ± 2.5% RH and a temperature of 40 ± 1°C (104 ± 2°F). NO_2 concentrations must range from 400-600 pphm which should produce a cycle of fade within 5-15 h of exposure.

7.2 Examine the Control Ribbon No. 1 periodically until its color corresponds to that of the Standard of Fading when compared in daylight ranging from average to slightly bluish north skylight or equivalent artificial light. This constitutes one cycle.

7.3 Remove those specimens which exhibit a noticeable color change at the end of one cycle. One cycle will generally produce a measurable color change in specimens which are sensitive to oxides of nitrogen.

7.4 Suspend an unexposed swatch of Control Ribbon No. 1 for each additional cycle of fade until the required number of cycles have been completed.

7.5 Optional Aftertreatment

7.5.1 Specimens exposed to oxides of nitrogen might continue to change color after removal from the test chamber. The color may be stabilized by plunging the specimens into a buffered urea solution (see 11.7) for 5 min. They should then be squeezed, thoroughly rinsed in clean water and dried in air at a temperature not above 60°C (140°F). CAUTION: Do not treat any specimen with the urea solution that must be returned to the test chamber for additional exposure.

8. Evaluation

8.1 At the end of each cycle, immediately compare those specimens which are removed from the exposure chamber with the preserved originals from which they are drawn.

8.2 Rate the effect on color of test specimens after any specified number of cycles by reference to the Gray Scale for Color Change (AATCC EP1), or using AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen, and record the numerical rating that corresponds to the appropriate one on the Gray Scale. Report the number of cycles run.

9. Report

9.1 Report the numerical grade for the change in color of the test specimen, the number of cycles run, the temperature and relative humidity at which the test was performed.

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10. Precision and Bias

10.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

10.2 *Bias*. Colorfastness to oxides of nitrogen in the atmosphere under high humidities can be defined only in terms of a test method. There is no independent method for determining the true values. As a means of estimating these properties, the method has no known bias.

11. Notes

11.1 **Humidity for Testing.** The fading of dyes by oxides of nitrogen on some fibers such as nylon and acetate is altered greatly by relatively small variations in relative humidity at high humidities. Therefore closely control temperature and relative humidity.

11.2 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

11.3 **Exposure Chamber.** The oxides of nitrogen chamber for elevated temperatures and relative humidities above 85% should be made of stainless steel which has been coated on the inside with a resistant coating. It must maintain the conditions of 87.5 ± 2.5% RH, 40 ± 1°C (104 ± 2°F) temperature and contain an NO_2 concentration between 400-600 pphm.

11.3.1 Blueprints of a suitable apparatus are available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1. 919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.4 **Test Control and Standard of Fading.**

11.4.1 Historically, the test control fabric, Control Ribbon No. 1, was a dyeing of 0.4% C.I. Disperse Blue 3 (Celliton Blue FFRN) on cellulose acetate satin. Lot 21 uses 0.4% Disperse Violet 1 on cellulose acetate stain since its fading characteristics are well known and it yields more color change after the testing cycle than previous test control lots dyed with Disperse Blue 3.

11.4.2 Since different lots and sources of both the dyestuff and undyed fabric will produce variations in both the original shade and its fading rate, it is therefore necessary to accurately establish a new standard of fading for each dyed lot of control fabric ribbon so that comparable tests results can be obtained when using different lots of the control and their respective standards of fading. In conducting tests, only the standard of fading applicable to that lot of control fabric ribbon may be used.

11.4.3 The standard of fading for the original lot of control fabric ribbon was made with vat dyes on cellulose acetate satin. For subsequent lots of the fabric ribbon, it was found for the standard of fading that the use of direct dyes on viscose satin for the standard of fading provided a better shade match.

11.4.4 Both the Control Ribbon No. 1 and the Standard of Fading must be kept in suitable containers or enclosures to protect them from possible exposure and color change to oxides of nitrogen and other contaminants which might be present in the atmosphere during transportation and storage.

11.4.5 The Control Ribbon is also sensitive to other atmospheric contaminants such as ozone. Its fading rate will vary considerably at different humidities and temperatures and its use in natural or end-use testing as a measure of exposures to oxides of nitrogen is not recommended. The color change produced on the control will reflect the combined effects of the atmospheric contaminants present and temperature-humidity variations, not just the effects of exposure to oxides of nitrogen.

11.4.6 A sealed unit of Test Control Fabric comprising 2.47 cm wide Control Ribbon No. 1 with a specific lot number indicated; and a specimen of the Standard of Fading for that lot of control ribbon is available as CR & SF No. 1 from Testfabrics Inc., 415 Delaware Ave., P.O. Box 3026, W. Pittston PA 18643; tel: +1.570.603.0432; fax: +1.570.603.0433; e-mail: info@testfabrics.com; web site: www.testfabrics. com.

11.5 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.6 Use bottled gas which contains approximately 1% NO_2 in nitrogen in cylinders equipped with the proper reducing valves.

11.7 **Urea Aftertreatment.**

11.7.1 The use of this treatment is completely optional. Experience has shown that color change after removal of specimens from the exposure chamber is negligible. The urea treatment itself will often cause a color change in specimens. Therefore, if this procedure is used, it is essential that both the exposed and unexposed control specimens be treated in an identical manner.

11.7.2 Urea Solution: 10 g of urea ($NH_2 \cdot CO \cdot NH_2$) per liter of water buffered to pH7 by addition of 0.4 g of sodium dihydrogen orthophosphate ($NaH_2PO_4 \cdot 2H_2O$) and 2.5 g of disodium hydrogen orthophosphate ($Na_2HPO_4 \cdot 12H_2O$) and 0.1 g or less of a rapid wetting surface active agent; for example, sodium dioctyl sulfosuccinate.

12. History

12.1 Reaffirmed 2020.

12.2 Revised 2015. Reaffirmed 2013, 2006, 2001, 1992, 1987. Editorially Revised 2019, 2010, 2008, 2004, 1995, 1989. Editorially Revised and Reaffirmed 1997, 1986.

12.3 Developed in 1985 by AATCC Committee RA33. Technically equivalent to ISO 105-G04.

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Test Method for Colorfastness to Crocking: Textile Floor Coverings—Crockmeter

1. Purpose and Scope

1.1 This test method is designed to determine the degree of color transfer from the surface of textile floor coverings to other surfaces by rubbing. The intent is to reproduce as near as possible true-to-life situations in all constructions whether dyed, printed or otherwise colored.

1.2 Test procedures employing white test cloths, both dry and wet with water, are given.

1.3 As the use surface of a textile floor covering may be exposed to various conditions in true-to-life situations, such as soil, stains, cleaning, shampooing and application of chemical treatments such as antisoils, antistats, antimicrobials, etc., the test may be made before, after or before and after such situations.

2. Principle

2.1 A colored test specimen held on the base of a crockmeter is rubbed with a white test cloth under controlled conditions.

2.2 Color transferred to the white test cloth is rated by comparison with the AATCC 9-Step Chromatic Transference Scale or Gray Scale for Staining and a grade is assigned.

3. Terminology

3.1 **carpet, n.**—all textile floor coverings not designated as rugs.

3.2 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorants(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

3.3 **crocking,** n.—a transfer of colorant from the surface of a colored yarn or fabric to another surface or to an adjacent area of the same fabric principally by rubbing.

3.4 **rug,** n.—a textile floor covering of limited area which is complete in itself and is intended for use as a partial covering of a floor or another floor covering.

3.5 **textile floor covering,** n.—an article having a use-surface composed of textile material and generally used for covering floors.

3.6 **use-surface,** n.—*of textile floor covering*, the part of a textile floor covering directly exposed to foot traffic.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. These safety precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5. Apparatus and Materials (see 13.1)

5.1 Crockmeter (see 13.2 and 13.3).

5.2 Crock Block (see 13.3).

5.3 Crockmeter Test Cloth (see 13.4).

5.4 AATCC 9-Step Chromatic Transference Scale (AATCC EP8) (see 13.5).

5.5 Gray Scale for Staining (AATCC EP 2) (see 13.5).

5.6 White AATCC Textile Blotting Paper (see 13.5).

6. Test Specimens

6.1 Two specimens are used, one each for the dry and the wet tests.

6.1.1 Additional specimens may be used to increase the precision of the average.

6.2 Floor covering material: pieces at least 50 × 150 mm. If the pile lay is distinguishable, cut the specimen with the pile lay pointing in the long direction.

6.2.1 Larger or full width lab samples may be used without cutting individual specimens, when multiple tests are needed and when using for production testing.

7. Verification

7.1 Verification checks on the operation of the test and the apparatus should be made routinely and the results kept in a log. The following observations and corrective actions are extremely important to avoid incorrect test results where abnormal crock images can result and influence the rating process.

7.2 Use an in-house poor crocking carpet or rug as a calibration specimen and conduct three dry crock tests.

7.2.1 A double, elongated image may indicate a loose sleeve.

7.2.2 A stretched and streaked crock image may be due to mounting the crock test cloth diagonally.

7.2.3 Arc shaped indentions in the crocked area of the test cloth may indicate the crock block is mounted improperly, most likely perpendicular to the specified placement (see 13.3).

7.3 Replace the abrasive paper on the tester base if it is smooth to the touch in the crocking area compared to the adjacent area, or if slippage of the specimen is noticed.

8. Conditioning

8.1 Prior to testing, condition the test specimens and the crock test cloth for dry crock testing as directed in ASTM D1776, Standard Practice for Conditioning and Testing Textiles. Condition each specimen for at least 4 h in an atmosphere of 21 ± 2°C (70 ± 4°C) and 65 ± 5% RH by laying each test specimen or crock test cloth separately on a screen or perforated shelf of conditioning rack.

9. Procedures

9.1 Dry Crocking Test.

9.1.1 Place a test specimen on the base of the crockmeter so that it rests flat on the abrasive cloth with its long dimension in the direction of rubbing (see 13.5). The pile lay should point to the back of the crockmeter if direction of pile is distinguishable.

9.1.2 Mount a 25 × 100 mm swatch of white cotton test cloth over the rubbing surface of the crock block. The test cloth should be mounted so that the long dimension will be moving in the long direction during the test. This operation can be carried out with the crock block attached to the weighted arm of the crockmeter using an Allen socket cap screw (see 13.6) or with the crock block detached from the arm and inverted in a suitable crock block holder (see 13.3 and 13.7). Fasten the test cloth to the crock block by means of a rectangular sleeve that slips over the cotton fabric on the crock block. When ready for testing, the test cloth on the crock block projects downward from the reciprocating arm of the crockmeter. CAUTION: *Do not drop the crock block*. Nicks or dents on the rubbing surfaces might be irreparable. ***Handle With Care***.

9.1.3 Lower the covered crock block

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gently onto the test specimen. While holding the specimen firmly with extended thumb and forefinger of the left hand, execute ten (10) complete turns of the crank at the rate of one (1) turn per s.

9.1.4 Remove the white test cloth, condition (see 8.1) and evaluate as directed in Section 10. In the case of napped, brushed or sheared material when loose fiber might interfere with the ratings, remove the extraneous fibrous material by pressing lightly on the crock area with the sticky side of clear self-adhesive tape before evaluating.

9.2 Wet Crocking Test.

9.2.1 Thoroughly wet the white test cloth in distilled water.

9.2.2 By any convenient means, such as squeezing between filter papers through a hand wringer, bring the wet pickup to 65 ± 5% based on the conditioned weight of the dry test cloth under standard conditions (21 ± 2°C [70° ± 4°F] and 65 ± 5% RH) before testing (see 13.8).

9.2.3 Take care to prevent evaporation which would reduce the moisture content below the specified level before the actual crock test is run.

9.2.4 Continue testing as directed in 9.1.

9.2.5 Air-dry the test cloth before evaluating. In the case of napped, brushed or sheared material when loose fiber might interfere with the rating, remove the extraneous fibrous material by pressing lightly on the test cloth with the sticky side of clear self-adhesive tape before evaluating.

10. Evaluation

10.1 Evaluate the amount of color transferred from the specimen to the white test cloth under examination by comparison with the Gray Scale for Staining (AATCC EP2), the AATCC 9-Step Chromatic Transference Scale (AATCC EP8) or Instrumental Assessment of Degree of Staining (AATCC EP12), and record the numerical rating that corresponds to the appropriate one on either of them (see 13.5 and 13.9).

10.2 Place the test cloth on three layers of clean white cloth for the evaluation.

10.3 Classify dry and wet fastness to crocking. Different ratings may result depending upon whether the Gray Scale for Staining or AATCC 9-Step Chromatic Transference Scale is used for the evaluation. It is, therefore, important to report which scale was used. For very critical evaluations and in cases of arbitration, ratings must be based on the Gray Scale for Staining:

11. Report

11.1 Identify the sample tested.

11.2 Report the grade determined in 10.3.

11.3 Indicate whether dry or wet crocking was performed.

11.4 Indicate whether AATCC EP2 or AATCC EP8 was used for evaluating crocking (see 10.3).

11.5 If the carpet or rug was subjected to the conditions in 1.3 state this in the report.

12. Precision and Bias (see 13.9)

12.1 *Precision*. An interlaboratory test was conducted in 1997 to establish the precision of the test method. Testing was conducted under the normal atmospheric conditions of each laboratory and not necessarily under ASTM standard conditions. Two operators at each of four laboratories tested and rated six fabrics on two consecutive days, both wet and dry crock. Rating was performed using the Gray Scale for Staining and the AATCC 9-Step Chromatic Transference Scale.

12.1.1 The results for one operator chosen from each laboratory were evaluated in an ANOVA two factor with replication analysis. The components of variance are given in Table I.

12.1.2 Critical differences are given in Table II.

12.1.3 Example for determining between-laboratory differences using one observer and the Chromatic Scale are given in Table III.

Table III—Crock Test Results

	Dry	Wet
Laboratory A	4.0	4.0
Laboratory B	3.5	3.0
Difference	0.5	1.0

Interpretation: For the dry crock test, because the difference between labs is less than the critical difference in 12.1.2 (0.63), the difference in results is not significant. For the wet crock test, since the difference is greater than the critical difference (0.86), the difference in results is significant.

12.2 *Bias*. The true value of colorfastness to crocking can only be defined in terms of a test method. Within this limitation, this test method has no known bias.

13. Notes

13.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.

Table I—Components of Variance

	Dry		Wet	
	Gray	Chromatic	Gray	Chromatic
Within-Laboratory	0.0312	0.0417	0.125	0.0938
Interaction	0.0135	0.0403	-0.0201	-0.0031
Between-Laboratory	0.0264	0.0101	0.0028	0.0031

Table II–Critical Differences

For the components of variance in Table I, two averages of observed values should be considered significantly different at the 95% probability level if the difference equals or exceeds the following critical differences.

Single Fabric

	No. of Observation	Dry		Wet	
		Gray	Chromatic	Gray	Chromatic
Within-Laboratory	1	0.49	0.56	0.98	0.85
	3	0.28	0.33	0.68	0.49
	5	0.22	0.25	0.44	0.38
Between-Laboratory	1	0.66	0.63	0.99	0.86
	3	0.53	0.43	0.58	0.51
	5	0.50	0.38	0.46	0.41

Multiple Fabric

	No. of Observation	Dry		Wet	
		Gray	Chromatic	Gray	Chromatic
Within-Laboratory	1	0.59	0.79	0.98	0.85
	3	0.43	0.64	0.68	0.49
	5	0.39	0.61	0.44	0.38
Between-Laboratory	1	0.74	0.84	0.99	0.86
	3	0.62	0.70	0.58	0.51
	5	0.60	0.67	0.46	0.41

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13.2 The Crockmeter provides a reciprocating rubbing motion simulating the action of a human finger and forearm. The counting device is useful when prolonged rubbings are required and is available at extra cost.

13.3 The Crockmeter is so designed that the crock block with rectangular rubbing surface 19.0 × 25.4 mm moves back and forth, with each complete turn of the crank along a straight path approximately 100 mm in length on a textile floor covering specimen with a downward force in the order of 9 Newtons. The crock block must be mounted in one direction only; a cutout in the side of the stem receives the screw that holds it in the reciprocating arm. The flat head screw can be replaced by an Allen socket cap screw (see 13.6) or similar device that can be tightened and released by hand. A crock block holder can also be used to assist in the mounting of the cotton test cloth if so desired. Note that, with the rectangular slip-on sleeve provided with the crock block, the test cloth can be mounted with ease and speed while the crock block is still attached to the weighted arm of the crockmeter.

13.4 Crockmeter Test Cloth should meet the following specifications:

Fiber	100% 10.3-16.8 mm combed cotton staple, desized, bleached, with no optical brightener or finishing material present
Yarn	15 tex (40/1 cotton count), 5.9 turns/cm "z"
Thread count	32 ± 5 warp ends/cm: 33 ± 5 filling picks/cm
Weave	1/1 plain
pH	7 ± 0.5
Mass/sq meter	100 ± 3 g finished
Whiteness	W = 78 ± 3 [AATCC TM110)

13.4.1 CAUTION: ISO crock cloth results may not be equivalent to AATCC crock cloth results/values based on crock cloth study.

13.5 AATCC 9-Step Chromatic Transference Scale, Gray Scale for Staining and White AATCC Textile Blotting Paper are available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1. 919.549. 8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

13.6 Allen socket head cap screw, 10-32 threads, 20 mm long.

13.7 A crock block holder is simply a weighted block with a hole bored in the top to accommodate the 16 mm diameter stem of the crock block when the latter has been inverted for mounting of the test cloth.

13.8 Experienced operators do not have to repeat this weighing procedure during a test session once the technique is established.

13.9 The precision of this test method is dependent on the combined variability of the material being tested, the test method itself, and the evaluation procedure utilized.

13.9.1 The precision statement shown in Section 12 was developed from results obtained by visual evaluations (AATCC EP2 and AATCC EP8).

13.9.2 It is expected that the use of instrumental evaluation (EP12) will result in greater precision than that obtained from visual evaluations.

14. History

14.1 Reaffirmed and editorially revised to include history section per the AATCC style guide.

14.2 Editorially revised 2019, 2016; editorially revised and reaffirmed 2013; editorially revised 2011, 2010, 2009; editorially revised and reaffirmed 2008 (with title change); editorially revised 2004, 2002, 2001; revised 1999 (with title change); editorially revised 1996; editorially revised and reaffirmed 1993; reaffirmed 1988, 1987.

14.3 Developed in 1986 by AATCC Committee RA57. Partly equivalent to ISO 105-X12.

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Test Method for Foaming Propensity of Disperse Dyes

Developed in 1986 by AATCC Committee RA87; reaffirmed 1987, 1988, 1993, 1998, 2003, 2013, 2018; editorially revised 1989, 1991, 1997, 2010, 2019; reaffirmed and editorially revised 2008.

1. Purpose and Scope

1.1 Foam generation must be controlled in dyeing equipment such as jet machines which operate with high circulation and high agitation of the dye liquor at low bath ratios.

1.2 This test method provides a standard for measuring the foaming properties of disperse dye under controlled conditions and determines the relative contribution of a single disperse dye to generate dyebath foam.

2. Principle

2.1 A quantity of disperse dye is prediluted, preheated and placed in a regular kitchen blender (see 12.1). Using a predetermined blender speed, the dye dispersion is blended for a specific time, transferred to a glass graduated cylinder and the foam and liquid levels are measured.

3. Terminology

3.1 **foam,** n.—a dispersion of a gas in a liquid or solid (see 12.2).

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The safety precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers must be consulted for specific details such as material data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Handle all chemicals carefully. Use chemical goggles or face shield, impervious gloves and an impervious apron when handling the chemicals used in this procedure. An eyewash/safety shower should be located nearby for emergency use.

4.2 Wear safety glasses and a dust mask when handling powdered dyestuff.

4.3 If concentrated acetic acid is diluted to prepare the acetic acid for pH adjustment (7.1.4), use chemical goggles or face shield, impervious gloves and an impervious apron during preparation. Handle concentrated acids only in an adequately ventilated laboratory hood. An eyewash/safety shower should be located nearby for emergency use. CAUTION: Always add acid to water.

4.4 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities [e.g., Occupational Safety and Health Administration's (OSHA) permissible exposure limits (PEL) as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version]. In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of Time Weighted Averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 12.3).

5. Uses and Limitations

5.1 The test method provides a means of comparing lot-to-lot variation of foam generation in dye shipments versus a control or standard reference sample of the same dye.

5.2 The test procedure is not complex, uses readily available equipment and provides results that correlate with results from other more elaborate lab dye testing equipment.

5.3 The test method offers a specific procedure and set of conditions that are useful for comparative studies by an individual laboratory. Variability between laboratories using different equipment and operators can produce slightly different measurements but will distinguish among low, moderate and high foaming disperse dyes.

5.4 Variables which affect accuracy and reproducibility of the test are specified in the notes (see 12.5, 12.6 and 12.7).

6. Apparatus and Materials

6.1 Equipment:
6.1.1 500 mL glass graduated cylinder.
6.1.2 Kitchen blender (see 12.1).
6.1.3 Glass blender container.
6.1.4 Stopwatch.
6.1.5 pH meter
6.2 Materials:
6.2.1 Distilled water.
6.2.2 Acetic acid, 10% solution.
6.2.3 1993 AATCC Standard Reference Detergent WOB (see 12.4).

7. Preparation

7.1 Prepare dye dispersion.

7.1.1 Weigh out 5 g of 100% strength dye. (For other dye strengths adjust accordingly. Example: 200% strength dye–weigh 2.5 g.)

7.1.2 Paste dye in 25 mL distilled water in a 400 mL beaker.

7.1.3 Further dilute the dispersion with 175mL of distilled water and heat to 50°C (122°F) while stirring.

7.1.4 Dilute dye dispersion to 1 L with distilled water. Using a pH meter, adjust to pH 5.5 ± 0.2 with acetic acid (see 12.5). Final temperature of dispersion should be 30°C (86°F) for the test.

7.2 Prepare detergent solution.

7.2.1 Prepare a 0.5 g/L solution of 1993 AATCC Standard Reference Detergent WOB in tap water at 30°C (86°F) to check for any blender contamination that would prevent foaming.

7.3 Check the blender for contamination.

7.3.1 Pour 200 mL of detergent solution in clean 1.4 L (1.5 qt) capacity container of the blender.

7.3.2 Operate the blender at its highest speed for 30 s and then stop it. Allow the dispersion to stand for 30 s. A minimum of 2.5 cm of foam should be present on top of the solution. If less foam is present, clean the container and repeat the procedure until enough foam is generated (see 12.6).

7.3.3 Thoroughly rinse the blender container and add 200 mL of distilled water.

7.3.4 Operate the blender for 30 s at highest speed and then stop it. No foam should be present after 30 s. If foam is present, rinse the blender container thoroughly and repeat the procedure until no foam is present.

8. Foam Test

8.1 Pour 200 mL of dye dispersion into the blender container. Place a lid on the container.

8.2 Select blender speed which corresponds to the blade speed 14,000-15,000 rpm, usually highest speed of blender (see 12.1).

8.3 Start the stopwatch simultaneously with starting the blender. Hold the container, but not the lid, to prevent any spills.

8.4 After 30 s, stop the blender but allow the stopwatch to continue running.

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8.5 Immediately pour the contents of the blender into a dry 500 mL graduated cylinder. When pouring, tilt the container only once, and allow the dye dispersion and foam to freely flow into the cylinder. Keep the container tilted over the cylinder until 60 s have elapsed on the stopwatch. Then remove the container.

8.6 After 150 s, read the bottom of the meniscus of both the foam and liquid levels in the cylinder. The reading will be erroneous if the top of the foam which clings to the sides of the cylinder is read as the foam level (see 12.8).

8.7 Record the volume of foam and liquid volume and subtract the liquid volume to determine the mL of foam produced.

9. Evaluation

9.1 Rate dyes according to following classifications:

Class A: 0-30 mL—Very Low Foam
Class B: 31-60 mL—Low Foam
Class C: 61-90 mL—Moderate Foam
Class D: 91-120 mL—High Foam
Class E: Greater than 120 mL—Very High Foam

10. Report

10.1 Report the class (see 9.1) for each dye tested.

10.2 A standard, control or reference sample of the same product should be tested and reported under the same conditions if the test is used for lot-to-lot comparisons of the dye.

11. Precision and Bias

11.1 *Precision Between-Laboratories*. Interlaboratory testing of 14 dyes by five laboratories in 1982 was conducted with a single operator in each laboratory making two tests with each dye. A standard deviation of 7.8 was calculated for the study, but some laboratories had data that were significantly different from others. Blender speed was the major contributor to variation in test results. Blender speeds can vary with both model of blender and line voltage; there is even 10-15% variation in speed between blenders from the same manufacturer. Therefore, the test is not suggested for comparison between laboratories because variations were larger than expected.

11.2 *Precision Within-Laboratory.* The standard deviation of low, moderate and high foam dyes as evaluated by a single operator was determined by conducting five tests with each of five dispersions of each dye. Using a Duncan's multiple range test for variable value, less than 0.1% of the total variation is due to dispersion preparation. About 1.2% of the total variation is due to the test. Of the total variation, 98.7% is due to foam level. The test will distinguish with 95% confidence level or higher between low, moderate and high foam generating disperse dyes. Multiple dispersions of the same dye are not necessary. Unless test results are significantly different from those expected, multiple tests of the same dye dispersion are not necessary. Listed below are the standard deviations of low, moderate and high foam generating disperse dyes:

Dye Type	Standard Deviation (mL foam)
Low Foam	2.04
Moderate Foam	4.85
High Foam	9.16

11.3 *Bias*. No statement on bias can be made since the true value cannot be established by an independent method.

12. Notes

12.1 A standard kitchen blender having an agitator blade speed selection of 14,000-15,000 rpm can be used. Blade speed selection can be determined by consulting the manufacturer, but is usually near the highest speed selection.

12.2 The mass of bubbles formed by agitation of a dye dispersion in a regular kitchen blender is very similar to that which dyers commonly observe on the surface of highly agitated dyebaths.

12.3 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

12.4 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

12.5 Measure pH with a pH meter rather than pH paper or liquid indicators. Excess acetic acid producing pH values in the range of 3.5-4.5 can significantly increase the foam generated by some disperse dyes.

12.6 Excess silicone lubricants in the bearing of the agitator blade of the blender in new propeller assemblies can cause defoaming action. In such cases, rinse the assembly in an appropriate solvent to remove the lubricants.

12.7 Reproducible readings for foam generation are best taken at a 120-180 s interval after blending rather than the 60 s or less. A 60-s reading is too variable because some dyes require longer foam-liquid separation time than others.

12.8 A detailed description of the development of this procedure can be found in the 1982 AATCC Intersectional Technical Paper Competition from the Piedmont Section entitled, A Method for Measuring the Foam Propensity of Disperse Dyes, *Textile Chemist and Colorist*, Vol. 15, No. 1, January 1983, p21.

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Test Method for Chelation Value of Polyaminopolycarboxylic Acids and Their Salts: Copper PAN

1. Foreword

This test method is an alternate to AATCC TM149, Test Method for Chelation Value of Aminopolycarboxylic Acids and Their Salts: Calcium Oxalate. Titration with a calcium salt using either oxalate or a dye indicator, as in AATCC TM149, yields a value that includes partially-substituted EDTA, HEDTA, DTPA, nitrilotriacetic acid (NTA), iminodiacetic acid (IDA), glycolate and other weak chelating agents found in some commercial products. This alternative procedure excludes the effect of the partially-substituted products, providing a more stringent value.

2. Purpose and Scope

2.1 The active ingredient content of ethylenediaminetetraacetic acid (EDTA), N-hydroxyethylethylenediaminetriacetic acid (HEDTA) and diethylenetriaminepentaacetic acid (DTPA) and their salts is usually expressed by the calcium chelation value (CaCV). This value indicates the amount of calcium (as calcium carbonate) which will be chelated by a known weight of chelating agent. This method excludes the effect of partially-substituted products.

3. Principle

3.1 Calcium chelation value is determined experimentally by titrating a weighed specimen of chelating agent with a copper nitrate solution of known concentration in the presence of PAN (1-(2-pyridylazo)-2-naphthol). Initially, the color of the solution is yellow-green due to free indicator. When all of the chelating agent has reacted with copper nitrate, the color of the solution changes to a permanent purple which is the end point.

3.2 The CaCV indicates the amount of calcium (as calcium carbonate) that will be sequestered by a known weight of chelating agent and is expressed as milligrams of calcium carbonate ($CaCO_3$) per gram of chelating agent (mg calcium carbonate/g chelating agent).

4. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

4.1 AATCC 149 Test Method for Chelation Value of Aminopolycarboxylic Acids and Their Salts: Calcium Oxalate

5. Terminology

5.1 **chelating agent,** n.—*in textile chemistry*, a chemical capable of deactivating metal ions by formation of a water-soluble complex. SYN: *sequestering agent*.

6. Safety Precautions

6.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

6.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

7. Reagents

7.1 Acetic acid, 1.0*N* (CH_3COOH)

7.2 Cupric nitrate 2.5 hydrate ($Cu(NO_3)_2 \cdot 2.5H_2O$)

7.3 Ethylene diaminetetraacetic acid (free acid) (EDTA) ($C_{10}H_{16}O_8N_2$)

7.4 Methanol

7.5 1-(2-Pyridylazo)-2-naphthol, (PAN) ($C_{15}H_{11}ON_3$)

7.6 Sodium acetate trihydrate ($CH_3COONa \cdot 3H_2O$)

7.7 Sodium hydroxide, 1.0*N* (NaOH)

8. Sampling

8.1 Conduct the test in triplicate; i.e., analyze 3 test specimens from a sample.

9. Specimens

9.1 Do not use aluminum or metallic weighing pans.

9.2 If solid chelating agent is being evaluated, dry a 2-g sample at the appropriate temperature for at least 2 h and cool in a desiccator before weighing specimens.

9.2.1 Dry free acid forms of chelating agents at 120°C.

9.2.2 Dry salt forms of chelating agents at 80°C.

9.3 For analysis of solid EDTA, HEDTA, and DTPA (either free acid or salt forms) weigh a 0.24-0.26 g specimen of the dried chelating agent to an accuracy of 0.01 g.

9.4 For analysis of commercial solution, of salts of the chelating agents, weigh a 0.49-0.51 g specimen of the solution to an accuracy of 0.01 g.

10. Preparation of Reagents

10.1 Cupric Nitrate Solution: Dissolve 23.30 g of cupric nitrate 2.5 hydrate and dilute to 1.000 L with distilled water in a volumetric flask. Calculate the molarity of the cupric nitrate solution (see 10.4).

10.2 PAN Indicator: Dissolve 0.025 g of 1-(2-pyridylazo)-2-naphthol in 50 mL of methanol. Store in a refrigerator in a stoppered bottled. Prepare fresh weekly.

10.3 Acetate Buffer Solution: (pH = 4.65). Dissolve 34.0 g of sodium acetate trihydrate in 500 mL of distilled water and add 15 mL of glacial acetic acid. Mix well and store in a closed container.

10.4 Standardization of Copper-Nitrate Solution: Weigh to the nearest 0.0001 g approximately 0.5 g of pure reagent grade EDTA acid powder into each of the three 250-mL Erlenmeyer flasks. Add 150 mL of distilled water and 6.5 mL of 1.0*N* sodium hydroxide (or five drops of 50% sodium hydroxide) and stir the mixture until the EDTA is dissolved. Add 25 mL of sodium acetate buffer. Adjust to pH 4.5-5.5 with 1.0*N* acetic acid or 1.0*N* sodium hydroxide, if necessary. Add 1 mL PAN indicator and titrate immediately with copper nitrate to a permanent purple end point. Calculate the molarity of copper using Equation 1:

$$M = \frac{(W)(P)(K)}{V} \qquad \text{(Eq. 1)}$$

where:

M = Molarity of copper nitrate, expressed in moles/L
W = Weight of EDTA acid in grams
P = Percent purity of EDTA acid
V = Volume of copper nitrate titrant in mL
K = 0.042994

$$= \frac{(1000)\ \text{mL/L}}{(100)(232.59\ \text{g/mole})}$$

$$= 0.042994\ \frac{\text{mole} - \text{mL}}{\text{g} - \text{L}}$$

11. Procedure

11.1 Transfer the specimen to a 500-mL Erlenmeyer flask and add 150 mL of distilled water (see 9.2, 9.3 and 9.4). Wash any residual chelating agent from the weighing paper or dish into the flask, and swirl to mix (dissolve).

11.2 Add 25 mL of pH 4.65 acetate

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buffer solution to the flask and swirl to mix. Check the pH and adjust to 4.5-5.5 with 1.0*N* acetic acid or 1.0*N* sodium hydroxide, if necessary.

11.3 Add 1 mL of PAN indicator to the flask and swirl to mix.

11.4 Titrate with 0.1000 M cupric nitrate solution to a purple end point that persists for at least 1 min.

12. Calculation

12.1 Calculate the CaCV for each specimen to 3 significant digits for each titration using the Equation 2:

$$\text{CaCV} = \frac{(100.1)(V)(M)}{W} \qquad \text{(Eq. 2)}$$

where:

CaCV = Calcium chelation value, expressed as mg $CaCO_3$/g chelating agent

V = Volume of cupric nitrate titrant in mL

M = Molarity of cupric nitrate solution, expressed in mmoles/mL

100.1 = Molecular weight of calcium carbonate, expressed in mg/mmole

12.2 Average the CaCV values for the 3 specimens tested.

13. Report

13.1 Report for each sample tested:

13.1.1 A description or identification of the sample.

13.1.2 Evaluation was performed using AATCC TM168-2022.

13.1.3 Average CaCV, to 3 significant digits.

13.1.4 Any modifications of the test method.

14. Precision and Bias

14.1 *Precision*. The interlaboratory, single operator, single apparatus, same day precision of the test method is ± 2 (1.60) standard deviations. The interlaboratory precision of the test method (over five participating laboratories) is ± 1 (0.54) standard deviations.

14.2 *Bias*. Chelation value can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

15. History

15.1 Revised in 2023 to clarify and align with prescribed AATCC style guidelines.

15.2 Editorially revised 2019; revised 2018; reaffirmed 2012; editorially revised 2010; editorially revised and reaffirmed 2007; reaffirmed 2002; editorially revised and reaffirmed 1997; revised 1992; editorially revised and reaffirmed 1989; reaffirmed 1988.

15.3 Developed in 1987 by AATCC Committee RR90. Maintained by RA99.

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Test Method for Weather Resistance of Textiles: Xenon Lamp Exposure

1. Purpose and Scope

1.1 This test method provides a procedure for the exposure of textile materials of all kinds, including coated fabrics and products made thereof, in an artificial weathering apparatus using controlled conditions of test. This test method includes procedures for both controlled wetting and no wetting of the specimen.

1.2 Resistance to degradation is measured either as percent strength loss or percent residual strength (breaking, tearing, or bursting) and/or colorfastness of the material when evaluated under standard textile testing conditions.

2. Principle

2.1 Samples of the textile material to be tested and the agreed upon comparison standard are exposed simultaneously to a xenon lamp source under specified conditions. Resistance of the test material to degradation is compared to that of a comparison standard.

3. Referenced Documents

3.1 AATCC EP1, Evaluation Procedure for Gray scale for Color Change.

3.2 AATCC EP7, Evaluation Procedure for Instrumental Assessment of the Change in Color of a Test Specimen.

3.3 AATCC TM16.3, Test Method for Colorfastness to Light: Xenon-Arc.

3.4 ASTM D123, Standard Terminology Relating to Textiles.

3.5 ASTM D1424, Tearing Strength of Fabrics by Falling Pendulum Type (Elmendorf) Apparatus.

3.6 ASTM D1776, Standard Practice for Conditioning and Testing Textiles.

3.7 ASTM D2256, Breaking Load (Strength) and Elongation of Yarn by the Single-Strand Method.

3.8 ASTM D2261, Tearing Strength of Fabrics by the Tongue (Single Rip) Procedure (Constant-Rate-Of-Extension Tensile Testing Machine).

3.9 ASTM D3512, Pilling Resistance and Other Related Surface Changes of Textile Fabrics: Random Tumble Tester Method.

3.10 ASTM D3786, Hydraulic Bursting Strength of Knitted Goods and Nonwoven Fabrics-Diaphragm Bursting Strength Tester Method.

3.11 ASTM D3787, Bursting Strength of Knitted Goods-Constant-Rate-of Traverse (CRT) Ball Burst Test.

3.12 ASTM D3884, Abrasion resistance of Textile Fabrics (Rotary Platform, Double-Head Method

3.13 ASTM D5035, Breaking Force and Elongation of Textile Fabrics (Strip Test).

3.14 ASTM D5587, Test Method for Tearing Strength of Fabrics by Trapezoid Procedure.

3.15 ASTM G151, Standard Practice for Exposing Non-Metallic Materials in Accelerated Test Devices that use Laboratory Light Sources.

3.16 ASTM G155, Standard Practice for Operating Xenon-Arc Light Apparatus for Exposure of Non-Metallic Materials.

4. Terminology

4.1 **breaking strength,** n.— the ability or capacity of a specific material to withstand the ultimate tensile load or force required for rupture. (ASTM D123).

4.2 **bursting strength,** n.—the force or pressure required to rupture a textile by distending it with a force, applied at right angles to the plane of the fabric, under specified conditions. (ASTM D3786, D3787).

4.3 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials, or both, as the result of exposure of the material to any environment that might be encountered during processing, storage, use or testing of the material.

4.4 **irradiance,** n.—radiant power per unit area as a function of wavelength expressed as watts per square meter, W/m^2.

4.5 **irradiation**, n.—the time integral of irradiance expressed in joules per square meter (J/m^2).

4.6 **radiant energy,** n.—energy traveling through space in the form of photons or electromagnetic waves of various lengths.

4.7 **radiant flux density,** n.—rate of flow of the radiant energy past the specimen.

4.8 **radiant power**, n.—energy per unit time emitted, transferred, or received as radiation.

4.9 **spectral energy distribution,** n.—the variation of energy due to the source over the wavelength span of the emitted radiation.

4.10 **spectral transmittance,** n.—the percent of incident radiant energy passing through a given material and not absorbed in the process, as a function of wavelength.

4.11 **standard atmosphere for testing**, n.—*in textiles*, an atmosphere for testing in which the air is maintained at a relative humidity of 65 ± 2% and at a temperature of 21 ± 1°C (70 ± 2°F). (ASTM D123)

4.12 **tearing strength,** n.—*in fabrics*, the capacity of a material to withstand the ultimate tearing force required to propagate a tear after its initiation. (ASTM D1424, D2261, D5587, D5034)

4.13 **total irradiance,** n.—radiant power integrated over all wavelengths at a point in time expressed in watts per square meter (W/m^2).

4.14 **weather,** n.—climatic conditions at a given geographical location, including such factors as sunlight, rain, humidity, and temperature.

4.15 **weather resistance,** n.—ability of a material to resist degradation of its properties when exposed to climatic conditions.

5. Safety Precautions

5.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

5.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

5.3 Users MUST consult manufacturers for specific details such as safety data sheets, equipment operating instructions, and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules.

5.4 The test equipment contains high intensity lamps. The door of the test machine must be kept closed whenever it is running.

5.5 Before servicing xenon lamps, allow time for cool down after test machine is shut down.

5.6 When servicing the test machine, shut off both the off switch and, if applicable, unplug the machine from the wall electrical socket. Insure that the main power indicator light on the machine front panel goes out.

6. Uses and Limitations

6.1 Results obtained by this test method should not be considered equivalent to those results obtained by weathering in an outdoor environment unless a mathematical correlation for a given material has been established and agreed upon by the contractual parties. Outdoor environments vary with respect to season, geography and topography and, as a consequence, the effects of outdoor ex-

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posure will vary accordingly. Not all materials are affected equally by the same environment. The instruments for determining weather resistance described in this test method are considered satisfactory, since they have been used extensively in the trade for acceptance testing of textile materials. The decision as to which machine type to use should be agreed upon by the purchaser and supplier based on their historical data and experience. There may be distinct difference in spectral distribution, water spray application, air and humidity sensor location, and test chamber size between weathering test machines supplied by different manufacturers that can result in differences in reported test results. Consequently, data obtained from machines supplied by different manufacturers and different size test chambers and xenon lamps cannot be used interchangeably unless a mathematical correlation has been established and agreed upon. No comparison between differently manufactured test apparatus is known to AATCC Committee RA50 using this test method.

6.2 When using this test method, the contracting parties must agree upon a reasonable test program cycle incorporating light, temperature, humidity, and wetting effects. The cycle selected must reflect expected environmental conditions relative to season, geography, and topography associated with the use of the material (see Option 1, 8.2.1).

6.3 When using this test method, use a standard of comparison (such as a weathering reference material or other control specimen), which has a known change in property value after a specific exposure.

6.4 Other procedures for measuring resistance to degradation can be used as agreed upon between the contracting parties.

7. Apparatus (see 16.1)

7.1 Different types of xenon-arc test apparatus may be utilized provided that the test apparatus is constructed of corrosion resistant material and shall provide for means of automatically controlling irradiance level, humidity level, chamber air temperature, and Black Panel or Black Standard Thermometer temperature. Xenon-arc test apparatus may be water or air cooled (see 17.2).

7.2 Xenon-Arc Light Source. The xenon-arc test apparatus utilizes a long-arc quartz-jacketed xenon-arc lamp as the source of irradiance, which emits radiation from below 270 nm in the ultraviolet through the visible spectrum and into the infrared. While all of the xenon-arc lamps are of the same general type, different size lamps operated in different wattage ranges are employed in several sizes and types of apparatus. In each of the various models, the specimen exposure area varies according to the lamp size and the wattage at which it is operated to provide an irradiance at the face of the specimen of 0.35 W/m^2 measured at 340 nm or equivalent. Operate the xenon-arc test apparatus in accordance to one of the selected test cycle Options 1-4 (see 8.2.1-8.2.4).

7.2.1 Optical Filter—In order for xenon-arcs to simulate terrestrial daylight, optical filters must be used to remove short wavelength UV radiation. In addition, optical filters to remove infrared radiation may be used to prevent unrealistic heating of test specimens that can cause thermal degradation not experienced during outdoor exposures.

Xenon-Arc test apparatus shall be equipped with a Daylight Filter to provide the appropriate spectrum. The Daylight Filter shall comply with the relative spectral power distribution requirements specified in Appendix A.

7.2.2 Follow the xenon-arc test apparatus manufacturer's instructions for recommended maintenance. Unless otherwise specified, control the level of irradiance to 0.35 ± 0.02 W/m^2/nm bandpass at 340 nm, or 40 ± 2.0 W/m^2 bandpass at 300-400 nm. Operate the xenon-arc test apparatus in accordance to one of the selected test cycle Options 1-4 (see 8.2.1-8.2.4).

8. Test Cycle Determination

8.1 The test cycle is determined by the influencing factors of the end-use, in particular, the climatic conditions. Not all materials are affected equally by the same environment. Results obtained by the use of any one test cycle may not be representative of those of any other test cycle or any outdoor weathering test. Acceleration factors derived for one geographic location do not necessarily apply to any other geographic location. However, certain test cycles have been used to group similar climates with respect to the test cycle.

8.2 The nature of the test material contributes to the selection of the appropriate test cycle with respect to UV exposure, wetting, wet time and temperature. The apparatus may be equipped to monitor specific portions of a continuous spectrum, control the level of irradiance to 0.35 ± 0.02 W/m^2/nm bandpass at 340 nm, or 40 ± 2.0 W/m^2 bandpass at 300-400 nm unless otherwise specified. The following test cycle options have been used for textile materials.

8.2.1 Option 1—This cycle has been used to approximate a subtropical climate such as found in South Florida: 120 min cycle, 90 min light only, 70 ± 5% RH, black panel temperature 77 ± 3°C (170 ± 5°F) alternating with 30 min light and water spray. Black panel temperature and relative humidity control is not necessary during periods of water spray.

8.2.2 Option 2—This cycle has been used to approximate a subtropical climate such as found in South Florida when appropriate machine water supply is restricted: 120 min cycle, 60 min light only, 70 ± 5% RH, alternating with 60 min dark, black panel temperature 77 ± 3°C (170 ± 5°F), no water spray.

8.2.3 Option 3—This cycle has been used to approximate a desert climate such as found in Phoenix, Arizona: Light on cycle, continuous light only, no water spray, black panel temperature 77 ± 3°C (170 ± 5°F), 27 ± 3% RH.

8.2.4 Option 4—This cycle has been used to approximate a temperate climate such as found in Columbus, Ohio: 120 min cycle, 102 min light only, 50 ± 5% RH, black panel temperature 63 ± 3°C (145 ± 5°F) alternating with 18 min of light and water spray. Black panel temperature and relative humidity control is not necessary during periods of water spray.

8.3 The use of these cycles does not imply, expressly or otherwise, an accelerated weathering test. This test method is not restricted to the use of these cycles.

9. Standards for Comparison

9.1 Standards for comparison must be determined and agreed upon by the contractual parties. The standards can be any suitable textile material where a history of the rates of strength degradation or color change is known. Standards must be exposed simultaneously with the test specimen. The use of the standard is to determine time-to-time equipment and test procedure variations. If test results of the exposed standards differ by more than 10% from the known standard data, thoroughly review the test machine operating conditions, and correct any malfunctions or defective parts. Then, repeat the test. If the data still differ by more than 10% from the known data and there is no evidence of machine malfunction, then the standard should be questioned and re-evaluated.

9.2 When colorfastness is the sole evaluation criterion, Blue Wool Standards, as defined in AATCC TM16.3, are acceptable for options using no water spray. However, use caution since the rate of any fade of the Blue Wool Standards by any one test method may not agree with that of other test methods.

9.3 Visual Comparison—For AATCC Blue Wool Lightfastness Standards L4, equals the Step 4 color change of the Gray Scale for Color Change, or equals the L4 Standard of Fade for 20 AATCC Fading Units (AFU) applicable to the Lot designation used as determined by AATCC EP 1. For AATCC Blue Wool Lightfastness Standards L2, equals the Step 2-3 color change of the Gray Scale for Color Change, or equals the L2 Standard of Fade for 20 AATCC Fading Units (AFU) applicable to the Lot designation used as determined by AATCC EP1.

9.4 Instrumental Color Measure-

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Table I—AATCC Fading Unit and Light Exposure Equivalents for AATCC Blue Wool Lightfastness Standards

AATCC Blue Wool Lightfastness Standard	AATCC Fading Units	Xenon Only kJ/m²/nm @ 420 nm	Xenon Only kJ/m² 300-400 nm
L2	5	21	864
L3	10	43	1728
L4[a]	20	85[b]	3456
L5	40	170	6912
L6	80	340[b]	13824
L7	160	680	27648
L8	320	1360	55296
L9	640	2720	110592

[a] For color change of 1.7 ± 0.3 CIELAB units or Step 4 on the Gray Scale for Color Change.

[b] Verified by experiment using Daylight Behind Glass and Xenon-Arc, Continuous Light. All other values are calculated (see AATCC TM16.3).

ment—for Lot 5, AATCC Blue Wool Lightfastness Standard L4, equals 1.7 ± 0.3 units of color change as determined by AATCC EP7. Other L designations or lots of AATCC Blue Wool Lightfastness Standards equal the units of color change specified on the calibration certificate supplied with the standard as determined by AATCC EP7.

10. Test Specimens

10.1 Number of Specimens—It is strongly recommended to use replicate specimens (see 17.3) of both the material to be tested and the standard for comparison as required to ensure accuracy.

10.2 Fabric Specimen Size, Initial—These dimensions are given as a general guide and in most cases will be adequate to perform the required evaluation. Certain materials may undergo dimensional change during exposure. The test equipment manufacturer, physical test apparatus, and number of replicate specimens required will affect the needed sample size. Review the test procedures given in 14.3 to ensure that a sufficient number of specimens are exposed for the individual tests of degradation (see 17.4). Minimize the effect of fabric variation by randomly selecting pairs of test specimens from a fabric sample. One specimen from each pair is used for the exposure, the second, is used for the control (unexposed) test. Each pair of specimens shall contain the same warp yarns for warp-wise tests and the same fill yarns for filling-wise tests. No two pairs of test specimens cut parallel to the warp should contain the same set of warp ends, and no two pairs of test specimens cut parallel to the filling should contain the same set of filling picks. Cut samples for the specified test procedures as follows:

10.2.1 Breaking Strength—When strip breaking strength is required, cut strips of fabric at least 5 × 20 cm (2 × 8 in.) with the longer dimension parallel to the warp or machine direction unless otherwise specified. When grab breaking strength is specified, cut swatches of fabric at least 13 × 18 cm (5 × 7 in.) (see 17.5). Reference ASTM D5035 or ASTM D2256.

10.2.2 Bursting Strength—Cut swatches of fabric at least 15 × 15 cm (6 × 6 in.) Reference ASTM D3787 or ASTM D3786.

10.2.3 Tongue Tear Strength—Cut swatches of fabric at least 10 × 23 cm (4 × 9 in.) with the shorter direction parallel to the warp or machine direction unless otherwise specified (see 17.5 and 17.6). Reference ASTM D2261.

10.2.4 Elmendorf Tear Strength—Cut swatches of fabric at least 10 × 13 cm (4 × 5 in.) with the longer dimension parallel to the warp or machine direction unless otherwise specified (see 17.4 and 17.5). Reference ASTM D1424.

10.2.5 Trapezoid Tear Strength—Cut swatches of fabric at least 10 × 18 cm (4 × 7 in.) with the longer dimension parallel to the warp or machine direction unless otherwise specified (see 17.5 and 17.6). Reference ASTM D5587.

10.2.6 Colorfastness—Cut swatches of fabric at least 3 × 6 cm (1.2 × 2.4 in.) with the exposed area measuring not less than 3 × 3 cm (1.2 × 1.2 in.) adjacent to an unexposed area having the same dimension.

10.2.7 When required to prevent raveling, the samples may be edged using a flexible epoxy or similar resin.

10.2.8 Identify each sample using a label resistant to the environment encountered during the test.

10.3 Mounting.

10.3.1 Mount the samples on open-back frames, which are installed in the cabinet without backing unless otherwise specified (see 17.7).

10.3.2 Fabrics—Secure the samples to the frames to hold them smooth without curling of the edges. Fabrics may be stitched to a gauze backing.

10.3.3 Yarns—Wind or fasten yarns on frames to a length of 15 cm (6 in.) minimum. Only that portion of the yarns directly facing the radiant energy is tested for breaking (tensile) strength. Either single strand or multiple strand tests may be used. In multiple strand tests, wind the yarns on the frame closely packed to 2.54 cm (1.0 in.) width. The control sample must contain the same number of strands as the sample subjected to exposure. After the exposure has been completed and before the yarns are unwound for testing, bind together those yarns facing the light source using 2 cm (0.75 in.) masking or other suitable tape to keep the yarns closely packed on the exposure frame.

11. Conditioning

11.1 Condition samples after exposure by bringing all specimens, control and test, to moisture equilibrium in the atmosphere for testing textiles as directed in ASTM D1776. Equilibrium is considered to have been reached when the increase in mass of the sample in successive weighings made at intervals of not less than 2 h does not exceed 0.2% of the mass of the sample. Generally, the industry approaches equilibrium from the "As Received" side.

11.2 It is recognized that in practice textile materials are frequently not weighed to determine when moisture equilibrium has been reached. While such a procedure cannot be accepted in cases of dispute, it may be sufficient in routine testing to expose the material to the standard atmosphere for testing for a reasonable period of time before the specimens are tested. Twenty-four hours has been found acceptable in most cases; however, certain fibers or treatments may exhibit slow moisture equalization rates from the "As Received" side. When this is known, a preconditioning cycle, as described in ASTM D1776 may be agreed upon between contractual parties.

12. Preparation, Maintenance and Calibration of Test Apparatus

12.1 Install the test apparatus in a room where temperature and relative humidity are controlled to minimize the effects of air supply variations.

12.2 Prior to the start of each test, the operator shall verify that the test apparatus has been calibrated within the limits and time intervals recommended by the manufacturer, but not less than annually. The instruments associated with the exposure apparatus (that is, light monitor control system, Black Thermometers, chamber air sensor, humidity control system, UV sensors and radiometers) require periodic calibration. Whenever possible, calibration should be traceable to national or international standards. Calibration schedule and procedure should be in accordance with manufacturer's instructions.

12.3 Aging of the xenon lamps or optical filters can result in changes in lamp spectrum. Changes in lamp spectrum may also be caused by accumulation of dirt or other residue in or on the burner envelope.

12.4 Replace xenon lamp(s) and optical filters as per manufacturer's recommendations or when damaged.

12.5 Since it is not practical to monitor the surface temperature of individual test specimens, a specified black-panel sensor is used to indirectly measure and control test specimen temperatures. This temperature is influenced by the absorbed irradiance minus the heat dissipated by con-

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duction and convection. Keep the face of the black panel in good condition. Although it is coated with a high quality finish, when exposed in the weathering apparatus it is subject to deterioration. Maintain a control black panel unit to periodically check the operating black panel unit for conformance. When an operating black panel unit falls outside the limits established for the test procedure, compared to the control unit, refinish or replace the unit.

12.6 Spray water must have a maximum of 1 ppm solids and a maximum of 0.2 ppm silica, and leave no observable stains or deposits on the specimens. Very low levels of silica in spray water can cause significant deposits on the surface of test specimens. In addition to distillation, a combination of deionization and reverse osmosis can effectively produce water of the required quality. The pH of the water used should be reported.

12.7 The power and water supply are to be monitored during the test as specified in the detailed description of the apparatus furnished by the manufacturer. Ensure the designated black panel temperature sensor and relative humidity are functioning properly.

12.8 Control the test environment as specified by the chosen test option.

12.9 Program or adjust the instrument to provide a continuous light test. Fill the specimen holders with metal blanks. These are used to ensure uniform air flow in the test chamber during the test exposure and should not include the actual test samples. Support the black panel thermometer in the specimen exposure area in the same manner as the material. Operate and control the test apparatus as defined by the manufacturer, adjusting the black panel temperatures, chamber temperatures, and humidity controls to the parameters as defined by sections 8.2.1 through 8.2.4 If performing Option 1 or 4, program the instrument to provide light plus water spray. Ensure that all specimens receive spray during a spray cycle.

12.10 Shut down the test apparatus after operating at the controlled conditions for 60 min. Remove the material from the specimen area.

13. Procedure

13.1 Program the xenon-arc test apparatus to run test conditions specified in 8.2.

13.2 Mount the framed test specimens on the specimen area as recommended by the test apparatus manufacturer. Make sure that all materials are adequately supported and in proper alignment. Any displacement of the material toward or away from the source, even by a small distance, may lead to variation in degradation between specimens. The specimen area must be completely filled so that air flow

Table II—CIE Delta E Color Change Testing CIE Delta E*

sample ID	409.5kJ		819kJ		1638 kJ	
2-1	4.02		8.15		20.83	
2-2	2.40		9.18		20.43	
2-3	3.26		8.49		18.75	
2-4	3.42		5.08		18.23	
2-5	1.72		5.69		17.72	
Avg 2	**2.96**	0.91	**7.32**	1.82	**19.19**	1.37
3-1	9.62		6.51		7.80	
3-2	9.26		6.04		6.79	
3-3	9.65		6.55		6.88	
3-4	9.61		6.68		6.85	
3-5	9.47		6.54		7.41	
Avg 3	**9.52**	0.16	**6.46**	0.25	**7.15**	0.44
4-1	1.14		0.60		1.07	
4-2	1.39		0.52		0.95	
4-3	1.23		0.62		0.97	
4-4	1.41		0.64		0.92	
4-5	1.52		1.05		0.88	
Avg 4	**1.34**	0.15	**0.69**	0.21	**0.96**	0.07
5-1	6.65		11.50		25.34	
5-2	6.64		11.86		26.56	
5-3	6.55		12.80		26.30	
5-4	6.64		12.23		26.23	
5-5	7.32		13.13		27.09	
Avg 5	**6.76**	0.32	**12.30**	0.67	**26.30**	0.64

Table III—Tensile Strength Testing

409.5 kJ/m²	Breaking Load, lbs							%
Sample No.	Individual Values					Average	Std Dev	Strength Loss
2	620	588	550	650	643	610	41.45	
	396	424	398	428	426	414	15.96	32.1%
3	459	515	508	430	425	467	42.37	
	498	430	435	473	487	465	30.66	0.6%
4	229	221	220	213	222	221	5.70	
	162	169	152	154	156	159	6.91	28.2%
5	728	823	832	842	871	819	54.08	
	527	571	562	563	584	561	21.15	31.5%
819 kJ/m²	**Breaking Load, lbs**							**%**
Sample No.	**Individual Values**					**Average**	**Std Dev**	**Strength Loss**
2	620	588	550	650	643	610	41.45	
	324	328	330	297	322	320	13.35	47.5%
3	459	515	508	430	425	467	42.37	
	333	349	338	344	322	337	10.43	27.9%
4	229	221	220	213	222	221	5.70	
	105	98	101	96	98	100	3.51	54.9%
5	728	823	832	842	871	819	54.08	
	432	447	391	443	427	428	22.20	47.8%
1228 kJ/m²	**Breaking Load, lbs**							**%**
Sample No.	**Individual Values**					**Average**	**Std Dev**	**Strength Loss**
2	620	588	550	650	643	610	41.45	
	214	211	216	221	207	214	5.26	65.0%
3	459	515	508	430	425	467	42.37	
	174	180	175	174	171	175	3.27	62.6%
4	229	221	220	213	222	221	5.70	
	55	49	56	52	49	52	3.27	76.4%
5	728	823	832	842	871	819	54.08	
	200	201	212	206	209	206	5.13	74.9%

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is not able to pass through the specimen plane. Using a solid specimen tray or filling unused spaces in the specimen area fulfills this requirement.

13.2.1 For single strand tests when material length exceeds 23 cm (9 in.) center the test specimens in the exposure area.

13.3 In the case of woven, knitted, and nonwoven fabrics, ensure that the side directly exposed to the radiant source is that normally used as the face. If for any reason, the face of the material is not exposed, report it with the data.

13.4 For test program cycles with water spray, ensure that the water spray strikes the test samples as a fine spray equally distributed over the surface exposed to the radiation.

13.5 Monitor exposure test chamber conditions with suitable recorders or data acquisition system (optional).

13.6 Operate the test apparatus continually until the selected exposure has been completed. Avoid unnecessary delays when filters or lamps are changed and the exposure period continued, as such delays may contribute to variations in results or lead to errors.

13.7 If irradiance varies by more than 10% within the exposure area, reposition specimens to ensure uniform total irradiation over the specimen surface, reposition specimens following manufacturer's instructions, which will provide each specimen equivalent exposure periods in each location to ensure uniform total irradiation over the specimen surface. For short duration tests (less than one week), reposition specimens at least once per day. For longer duration tests, reposition specimens at least five times during the duration of the test. Other test methods of achieving uniform total irradiation may be employed if mutually agreed upon by the contractual parties.

13.8 Replace filters according to manufacturer's instructions, or when pronounced discoloration or milkiness develops, whichever occurs first.

13.9 When the exposure cycle is completed, remove the replicate standard and test specimens from the exposure area. Transfer them to a standard atmosphere for testing textiles for conditioning to perform agreed upon physical tests.

13.9.1 If upon removal from the specimen area, the specimens are wet, dry them without tension at ambient laboratory conditions or at a temperature not exceeding 71°C (160°F) before transferring them to the conditioning atmosphere. Wet-out the unexposed reference standard (comparison standard) and the retained unexposed original. Treat them under the same conditions of drying and conditioning as the exposed test specimens before physical testing or colorfastness evaluation.

14. Evaluation

14.1 Rate either the durability or resistance to degradation of the material with respect to the standard of comparison by one or more of the following:

14.1.1 Percent strength retained or percent strength loss—Record percent strength loss or percent strength retained (breaking, tearing, or bursting as appropriate) of the material after the prescribed exposure time period.

14.1.2 Residual Strength—Record the initial and final strength values along with all other pertinent data.

14.1.3 In Terms of the Agreed Upon Comparison Sample or Standard:

a) Satisfactory—As durable or more durable than the comparison standard at the amount of exposure time period prescribed in the material standard.

b) Unsatisfactory—Less durable than the comparison sample at the amount of exposure time period prescribed in the material standard.

14.2 Test specimen preparation after exposure is preferred but may be done before exposure (see 18.8). For each test to be made on the material and control, exposed and unexposed, prepare test specimens by marking and raveling or cutting the central portion of each exposed sample to the dimensions specified in the respective test procedures (see 14.3.)

14.3 Physical Properties. Determine the change in properties of fabrics as directed in one or more of the following test methods:

14.3.1 Determine the breaking strength of fabrics as directed in ASTM D5035, using the appropriate procedure as specified.

14.3.2 Determine the breaking strength of yarns as directed in ASTM D2256.

14.3.3 Determine the ball bursting strength of fabrics as directed in ASTM D3787.

14.3.4 Determine the diaphragm bursting strength of fabrics as directed in ASTM D3786. Use the Model C type tester for fabrics having a bursting strength of 200 psig or less; and the Model A type tester for fabrics having a bursting strength greater than 200 psig.

14.3.5 Determine the Tongue tear strength as directed in ASTM D2261.

14.3.6 Determine the Elmendorf tear strength as directed in ASTM D1424.

14.3.7 Determine the trapezoid tear strength as directed in ASTM D5587.

14.4 Measure the color change of AATCC lightfastness standards as directed in AATCC EP7.

14.5 Average the data for the various replicates, or handle statistically as appropriate, and record a significant value of breaking, tearing, or bursting strength retention and/or colorfastness after exposure in relation to original strength or color, as applicable. It is appropriate to record the percent elongation characteristics of the unexposed controls and exposed specimens at rupture or at some prescribed point in the force-elongation curve as this is oftentimes significant supplementary information (see 16.9).

15. Report

15.1 Include in the report the type of test apparatus, including model, serial number, and manufacturer's name, exposure program, elapsed exposure time, sample rotation schedule, irradiance level, black panel temperature, chamber air temperature, relative humidity, if black panel temperature was controlled by ambient (chamber air) or black panel and type of water supply.

15.2 Report the type of equipment used in measuring the strength characteristics of the fabric, the type of fiber(s) of which the material is composed, which side of the fabric was exposed (in the event that the fibers differ on the face and back of the fabric), the fabric weight in g/m^2 (oz/ yd^2) and the nature of the fabric finish, if known.

15.3 Report any deviation from this method or the performance of the standard of comparison.

15.4 Report the rating and relative data for each property evaluated.

15.5 Report the operating schedule of the test apparatus.

16. Precision and Bias

16.1 *Precision*.

16.1.1 CIE Delta E Color Change Testing. Four samples of coated fabrics (Geo Textile Samples), 5 replicates of each, were tested according to AATCC TM169-1991 in a Ci65 Weather-Ometer. The test cycle consisted of 90 min of light only followed by 30 min of light and spray. Black Panel temperature 70 ± 3°C (158 ± 5°F), relative humidity 55 ± 5%, specimen rotation—none.

16.1.2 *Tensile Testing*. Tensile tests were made on samples before and after exposure intervals of 409.5, 819 and 1228 kJ/m^2 @ 340 nm (325, 650 and 975h) Test Method: ASTM Two-inch Cut Strip Test

Test Device:	Instron Model 1000 Weight Beam Capacity: 1,000 lb
Full Scale Load:	100, 200, 500, 1000 lbs Cross head Speed: 12 in. per min
Gage Length:	3 in. Number of specimens per direction per sample: 5
Test Direction:	machine

16.1.3 Note Tables A3 (Color Change) and A4 (Tensile Strength) in Appendix A for data on precision.

16.2 *Bias*. There is no referee method

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for determining the true value of weather resistance of textile materials. The bias of a xenon lamp exposure for acceptance testing cannot be determined; therefore, this method has no known bias.

17. Notes

17.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

17.2 Refer to ASTM G151 and ASTM G155 for guidance on the design and performance requirements for instruments specified in this method.

17.3 The required number of specimens to be tested in order to have a certain minimum level of confidence in the value of the measured property is a function of the standard deviation, *s*, of the property being measured and the acceptable percentage error, *E*, in the measurement. If the standard deviation, *s*, is high, more specimens are required. Similarly, if the acceptable level of error percent, *E*, is low, more specimens are required. The generic formula for computing sample size, *k*, is:

$$k = (ts/E)^2$$

where:

- k = the required number of specimens
- s = the standard deviation of the property being measured
- E = the allowable percent error (E ranges between 0% and 5%. Actual E value depends on the property itself and the requirements of the situation)
- t = the degree of confidence in the estimated value of s. It also reflects the level of confidence we have in the computed value of k. The value of t should be obtained from standard t-tables for a one-sided test at 95% confidence, for the actual number of tests used to estimate the standard deviation, s

17.4 The material specification may further define the use of wet breaking, tearing, or bursting strength tests in lieu of or to complement those tests done under standard textile testing conditions. Report conditions of test along with the data.

17.5 In general, the warp direction is used, but the filling direction may be used to complement or in place of the warp direction when specified. Warp yarns sometimes are protected from the irradiance due to fabric structure. When the filling is used, it must be reported.

17.6 Under certain conditions, by agreement between principals to a transaction, tearing strength of the material can be used in lieu of, or to complement, the breaking or bursting strength.

17.7 Specimen frames must be made from suitably inert materials that will not contaminate the specimens. When samples are fastened with staples, they should be of the nonferrous type over-coated to avoid contamination of the specimen by corrosion products. Metal frames must have a dull finish and be designed to avoid reflectance that could influence the performance of the material. Frames should conform to the curvature of the specimen rack. The size of the frame is determined by the type specimens required for individual property requirements.

17.8 When preparing specimens from the exposed material, consideration must be given to the usable part of the sample and the type test apparatus employed.

17.9 In the case of some fabric blends, the cellulosic component may be seriously damaged, but the high strength of the synthetic component may mask any significant strength loss. In those cases, it may be advisable to evaluate the degradation effects by the weight loss procedure in test methods such as: ASTM D3884 or ASTM D3512.

18. History

18.1 Revised in 2020 to update referenced standards, add Appendix B, and make various editorial changes.

18.2 Revised 2017. Editorially revised 2016. Editorially revised and reaffirmed 2009. Editorially revised 2008, 2007. Revised 2003. Editorially revised and reaffirmed 1995. Revised 1990. Reaffirmed 1989, 1988.

18.3 Developed in 1987 by Committee RR64; jurisdiction transferred in 2007 to Committee RA50.

Appendix A
Xenon-Arc Light Spectrum

A1 Xenon-Arc test apparatus shall be equipped with a Daylight Filter to provide the appropriate spectrum. The Daylight Filter shall comply with the relative spectral power distribution requirements specified in Table AI and in Fig. A1.

Table AI—Relative Spectral Power Distribution Specification for Xenon-Arc with Daylight Filter

Wavelength bandpass (nm)	Relative irradiance in percent (%)[1)] minimum	maximum	CIE No.85, Table 4
< 290		0.14	
290-320	1.0	7.8	5.4
320-360	27.6	42.8	38.2
360-400	47.3	71.7	56.4

[1)] expressed as a percentage of the total irradiance < 400 nm.

The UV irradiance (290-400 nm) is 11% and the VIS irradiance (400-800 nm) is 89% expressed as a percentage of the total irradiance from 290-800 nm (CIE No.85, Table 4). These values measured on specimen plane in instruments may vary when performing a test due to the reflectance properties and number of specimens.

Appendix B
Correlation with Natural Exposures

B1 Options 1-4 listed in Sections 8.2.1 through 8.2.4 are noted as approximations of specific climatic regions. However, this does not confirm that the amount of property change seen after a certain amount of exposure in these climates will match the amount of change in the xenon-arc instrument after an equivalent radiant dosage, as materials exposed may be more or less sensitive to specific wavelengths of radiation or other weathering factors. It is recommended that the user validate the results with outdoor exposures to determine acceleration rates and correlation between natural and xenon-arc exposures listed within this test method.

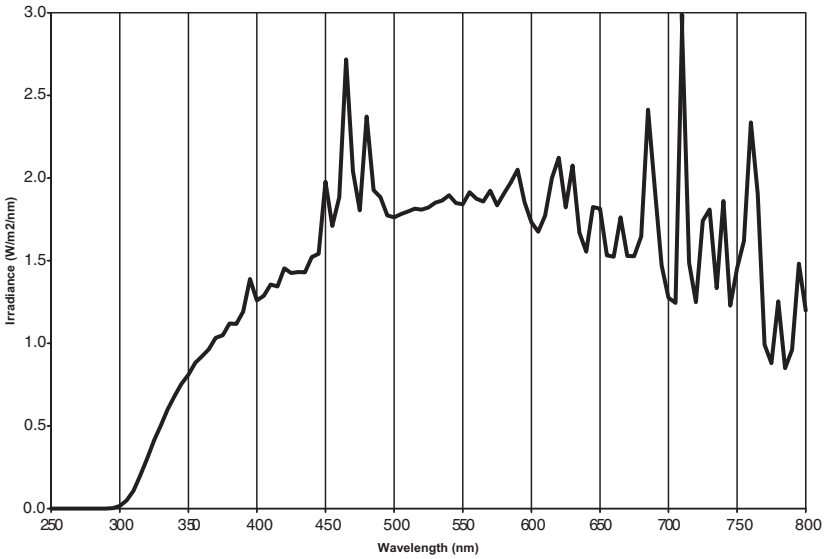


Fig. A1—Xenon Arc Spectral Power Distribution with Daylight Filters

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Test Method for Dusting Propensity of Powder Dyes

Developed in 1987 by Committee RA87; editorially revised and reaffirmed 1988, 1989 (title change), 1996; editorially revised 1992, 2004, 2008, 2019; reaffirmed 2001, 2006, 2011, 2016, 2017.

1. Purpose and Scope

1.1 This test method provides a standard method for the evaluation of dusting properties of powder dyes.

1.2 The method of assessment permits the assignment of a numerical rating describing the degree of dusting; or, conversely, the degree of non-dusting of powder dyes.

1.3 This test method is not intended for a quantitative determination of dusting. Water soluble dyes may give an inferior rating than an equal amount of dusting caused by disperse dyes. Further studies indicated that not all factors; i.e., humidity, are fully controlled. Also the method does not distinguish between slow and fast settling dust.

2. Principle

2.1 Three, approximately equal, weighed portions of a 10 g powder dye sample are dropped consecutively and rapidly into a funnel which has a wet filter paper collar on the stem and is fitted into a cylinder. Any dust which forms rises and settles on the pre-wet filter paper collar.

2.2 The resulting coloration on the filter paper is then compared to a series of five standards and a numerical rating is assigned.

3. Terminology

3.1 **dusting,** n.—the tendency of particles of sufficiently low mass to become airborne when a powder material is handled or agitated.

3.2 **dye,** n.—a colorant applied to or formed in a substrate, via the moleculary dispersed state, which exhibits some degree of permanence.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas and a single-use dust respirator while handling powder dyes.

5. Apparatus and Materials

5.1 Funnel, 1.5 mm thick stainless steel, 110 mm diameter, 230 mm stem length with 15 mm stem diameter (see Fig. 1).

5.2 Cylinder, stainless steel, 50 mm diameter, 355 mm tall (approx. capacity 500 mL) with capability of grounding to prevent interference from static electricity (see 11.1).

5.3 Filter paper, Whatman No. 2, 39 mm diameter circle with 16 mm diameter center hole (see 11.1 and 11.2).

5.4 Pipettes, Pasteur.

5.5 Beakers, Griffin, 50 mL.

5.6 Photographic Scale for Rating (see 11.3, Fig. 2).

6. Preparation and Assembly

6.1 Slip a filter paper collar on the funnel stem and position it 100 mm from the bottom of the stem.

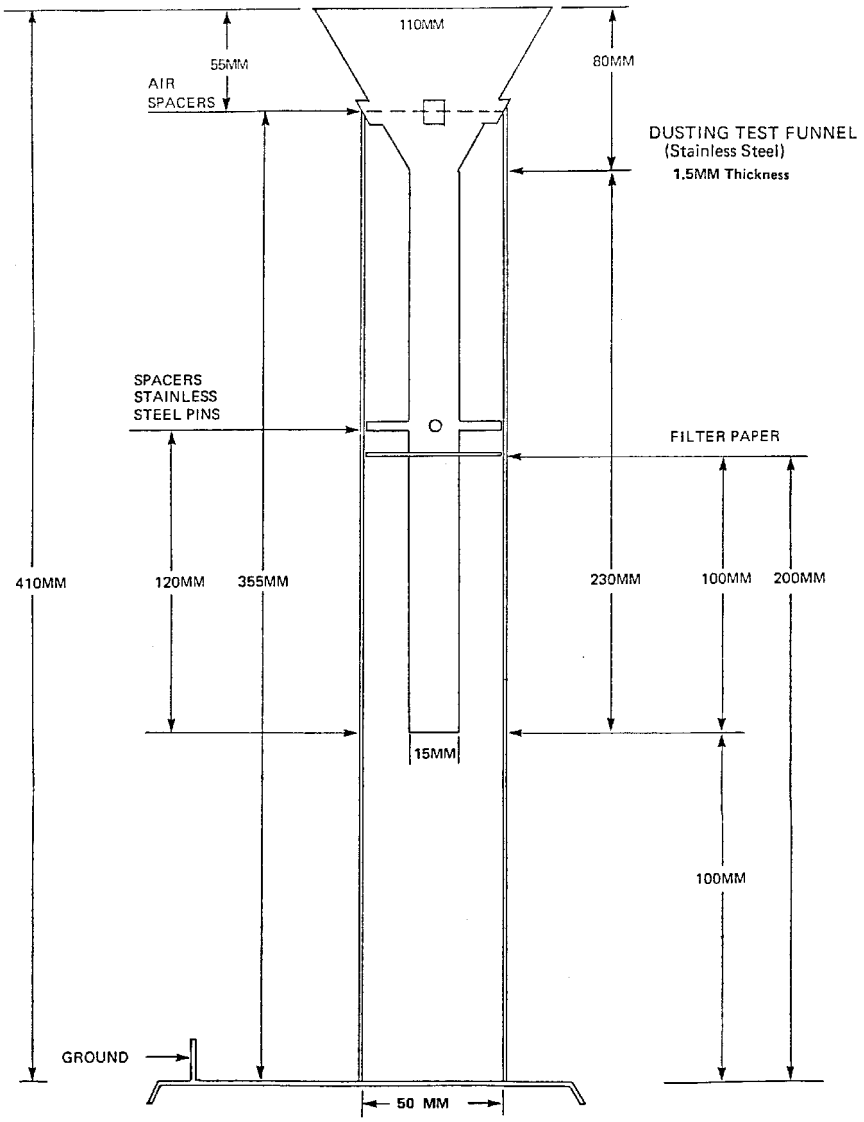


Fig. 1—Dusting test funnel.

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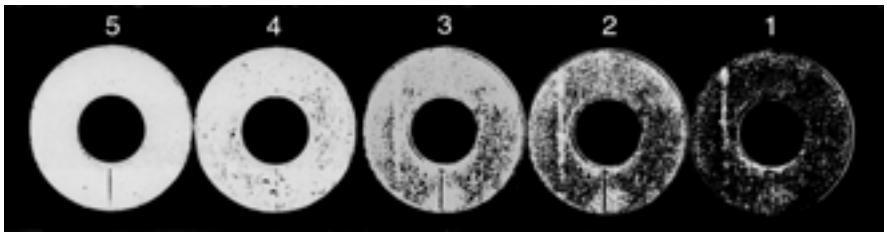


Fig. 2—Dusting test scale.

6.2 Moisten the paper collar with three drops of distilled water at points approximately equidistant from one another around the stem of the funnel. Use care to avoid water running down the funnel stem.

6.3 Place the funnel, with wet collar, in the steel cylinder. Ground the cylinder (see 11.4). The paper collar should be 200 mm from the bottom of the steel cylinder.

7. Procedure

7.1 Use a 10 g sample of powder dye, weighing it out as 3.3 g portions, into three 50 mL beakers.

7.2 Pour the three portions of dye onto the side of the funnel at intervals of approximately 2-3 s.

7.3 Allow the apparatus to remain undisturbed for 3 min to avoid erroneous results due to any external effects on the air in the cylinder.

7.4 Remove the funnel and, with scissors, snip off the filter paper collar from its periphery to the stem for easy removal (see 11.5). Air dry the collar by placing it on a sheet of filter paper away from possible sources of atmospheric "fly" or contaminants.

8. Evaluation

8.1 Compare the specimen with the scale (see 11.3) and assign the appropriate numerical rating. Intermediate ratings; e.g., 4-5 may be estimated.

8.2 The matching standard establishes a numerical rating for degree of dusting. A rating of 1 describes a very dusty material and a rating of 5 describes a nondusting or minimally dusting dye.

8.3 Duplicate or triplicate tests may be conducted and the average noted.

9. Report

9.1 Report the numerical rating for degree of dusting.

10. Precision and Bias

10.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

10.2 *Bias*. Dusting propensity can only be defined in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

11. Notes

11.1 Stainless steel funnels, cylinders, and filter paper collars are available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.2 Filter paper collars required for this test may be prepared in a number of different ways.

11.2.1 Larger circles of paper may be die cut or hand cut to size with scissors and the center hole punched out with a light blow on a #6 cork borer.

11.3 The illustration of the dusting scale in the text of this method is inadequate for the evaluation. A standard scale (photographic replicas) must be used and is available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.4 The steel cylinder is grounded by connecting one end of a copper conductor (No. 14 solid wire) to the grounding screw at the base of the cylinder and the other end clamped to a metallic (not plastic) water pipe or any metallic electrical conduit.

11.5 Pulling the paper collar off the stem of the funnel without using the scissors causes a "squeegee" effect which could lead to erroneous results.

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Test Method for Carpets: Cleaning of; Hot Water Extraction

Developed in 1987 by AATCC Committee RA57; reaffirmed 1988, 1989, 2000, 2005, 2014; editorially revised 1991, 1997, 2008; revised 1995 (with title change), 2019; editorially revised and reaffirmed 2010.

Foreword

Over the years, Committee RA57, Floor Covering Test Methods, has evaluated the various techniques for on-site cleaning of carpets as they have emerged. Information exchanged with professional carpet cleaning organizations led to the selection of the system described in this test method as typical of a majority of actual cleaning operations.

1. Purpose and Scope

1.1 This test method provides a laboratory procedure to clean textile floor coverings by a technique simulating cleaning of installed floor coverings by hot water extraction, sometimes erroneously called "steam cleaning."

1.2 This test method may be used to clean a carpet in a manner that duplicates the most popular (70%) means of actual on-floor cleaning of carpets.

1.3 This test method prepares specimens that can be tested for many properties; e.g., colorfastness, dimensional stability, permanency of finishes, ease of cleaning, etc.

2. Principle

2.1 The test is performed by securing a specimen of carpet, face up, on a floor or a mounting board. An extraction cleaning head is passed over each section of the specimen. The pile is erected with a brush or pile rake and the specimen is dried at room temperature.

3. Terminology

3.1 **carpet,** n.—all textile floor coverings not designated as rugs.

3.2 **carpet pile brush,** n.—a hand operated brush having long, semi-rigid bristles intended specifically for erecting the pile of small areas of carpet (see also *pile lifter*).

3.3 **cleaning head,** n.—a vacuum head modified with spray nozzles for cleaning solution application. Some types include a powered brush unit to facilitate wetting and soil release.

3.4 **hot water extraction,** n.—a method of cleaning carpet by injecting a heated cleaning solution into the pile and quickly removing the solution and soil by vacuum (see also *steam cleaning*).

NOTE: Hot water extraction is often erroneously called "steam cleaning." Hot water is used at a temperature of 60 ± 3°C (140 ± 5°F). These temperatures are far below the temperature of steam of 100°C (212°F).

3.5 **pilate,** n.—see *pile lifting*.

3.6 **pile lifter,** n.—a vacuum cleaning device with motorized rotating brushes designed to erect and agitate carpet pile to facilitate soil removal (see also *carpet pile brush*).

3.7 **pile lifting,** n.—the act of erecting the pile tufts of a carpet with a *carpet pile brush*, *pile lifter*, or *pile rake* to allow imbedded soil to be more easily removed and erecting the pile after cleaning to restore a uniform overall appearance (also called *pilate*).

3.8 **pile rake,** n.—a hand tool with smooth circular plastic tines used for pile lifting.

3.9 **rug,** n.—a textile floor covering of limited area which is complete in itself and is intended for use as a partial covering of a floor or another floor covering.

3.10 **steam cleaning,** n.—see the correct term, *hot water extraction*.

3.11 **textile floor covering,** n.—an article having a use-surface composed of textile material and generally used for covering floors.

3.12 **use-surface,** n.—*of textile floor covering*, the part of a textile floor covering directly exposed to foot traffic.

3.13 **wand,** n.—a tool used to deliver cleaning solution to carpets and apply vacuum to remove the solution.

NOTE: A *wand* usually consists of an extension handle and a *cleaning head*. Various types are light weight wand, drag wood (heavy duty) and power wand which has a motorized rotating or vibrating part to aid soil release.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe, and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 Eye protection should be used when mixing, handling or applying the detergent and detergent solutions.

4.4 Gloves or protective hand cream are recommended for handling detergents and solutions.

4.5 Observe operating procedures and precautions by the manufacturer of the extraction machine. Liquid under pressure and high vacuum can cause excessive splashing of liquids and/or hose "kick back."

5. Apparatus and Materials

5.1 AATCC Detergent #171, for all synthetic fiber carpets (see 11.1).

5.2 Hot water extraction unit (see 11.2).

5.3 Carpet brush or pile rake.

5.4 Mounting boards (see 11.3).

5.5 Extraction wand with a vacuum slot that is free of obstructions.

6. Test Specimens

6.1 The preferred minimum specimen size is 30 × 70 cm (12 × 27 in.) with the pile lay in the 70 cm (27 in.) direction (see 11.4).

6.2 If the entire specimen requires uniform cleaning, a border carpet of similar thickness is necessary for uniform extraction of the cleaning solution.

7. Procedure

7.1 Secure the specimen to the floor or mounting board with staples, tacks or other suitable means (see 11.4).

7.2 Prepare the cleaning solution according to the directions supplied (see 11.1.1).

7.3 Purge the system of cold cleaning solution before beginning each cleaning cycle.

7.3.1 Visually inspect and remove any obstructions from the vacuum slot (see 8.2).

7.4 Pour 8 L (2 gal) or more of preheated solution, 60°C (140°F), into the empty solution tank of the hot water extraction unit.

7.5 With water-feed spray and vacuum on, and the cleaning head in contact with the carpet surface, draw the cleaning head across the surface of the carpet in a direction against the lay of the pile. Without changing the position of the carpet,

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place the cleaning head in the same starting position of the carpet surface as before and with only the vacuum on (water-feed spray off) draw the cleaning head across the carpet as in the first pass. These two passes of the cleaning head constitute one cleaning cycle. In most cases, two cleaning cycles will be necessary to obtain measurable results. However, the number of cycles may be increased depending on the degree of soiling and the purpose of the test.

7.6 The wand should be moved at about 46.0 cm/s (1.5 ft/s) to deliver between 0.4-0.6 L/m^2 (0.10-0.14 gal/yd^2) of cleaning solution. The speed of the second (vacuum) stroke should be approximately the same as above, but may be decreased to achieve maximum liquid retrieval. CAUTION: Some jet size - jet height combinations may cause pile distortion from excessive liquid impact (see 11.5).

7.6.1 An effective professional unit should remove between 90-95% of the liquid, depending on fiber absorbency. A properly operating vacuum system should leave no more than 40.0 g/m^2 (1.2 oz/yd^2) of liquid (that is in excess of natural regain) in a carpet specimen.

7.7 Should the cleaning of a larger area of carpet than covered by a single cleaning cycle be required, reposition the cleaning head on the carpet so that in successive cleaning cycles the cleaning head will overlap the previously cleaned area by 5 cm (2 in.).

7.8 Erect the pile with a carpet brush or pile rake in the direction opposite to the lay of the pile.

7.9 Remove the specimen from the mounting board or floor and dry it at ambient conditions in a horizontal position.

8. Evaluation

8.1 Specimens which are subjected to the cleaning procedure of this test method may be tested for many properties; e.g., colorfastness, dimensional change, permanency of finishes, ease of cleaning, etc.

8.2 If an obstruction exists, a banding or striped look may appear after removing specimen for evaluation. This may be caused by blockages in the extraction wand's vacuum slot. Banding is not normal; therefore, the specimen should be retested after ensuring the vacuum slot is clear of obstruction.

9. Report

9.1 State that the specimens were cleaned according to this procedure.

9.2 Report the number of cleaning cycles used.

9.3 State the pH of the diluted cleaning solution after it has been prepared according to directions provided by the supplier.

9.4 Report evaluation data and cite the test method for each property evaluated.

10. Precision and Bias

10.1 Precision and bias statements are not applicable because data are not generated by this test method.

11. Notes

11.1 AATCC Detergent 171 available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

11.1.1 Dilution:

U.S. 11-17 g/gal of tap water
U.K. 13-18 g/gal (Imperial)
Metric: 3.0-4.5 g/L

11.2 A commercial hot water extraction device of professional quality, either portable, dolly, or truck mount. Small units and "add on" devices are generally unsuitable because of insufficient liquid capacity, delivery pressure, and/or vacuum for liquid retrieval. The device should meet or exceed the following specifications:

The tank shall have a means of maintaining the liquid at 60 ± 3°C (140 ± 5°F) and a thermometer to measure this.

A liquid delivery pump delivering at least 3 L/min (0.75 gal/min) at a pressure of 207 kPa (30 psi) or more.

The liquid recovery vacuum pump shall have a minimum lift of 250 mm (100 in.) sealed and a flow of 43 L/s (90 cfm) open orifice.

A standard floor wand of 25 cm (10 in.) width.

11.3 The mounting boards should be constructed from l.27 cm (0.5 in.) CD grade exterior plywood or better. Each board should measure 78.7 × 96.5 cm (31.0 × 38.0 in.).

11.4 The specimen and mounting board size may be adjusted to conform with other tests to be run on the sample. Examples: AATCC TM122, Test Method for Carpet Soiling: Service Soiling, or AATCC TM134, Test Method for Electrostatic Propensity of Carpets.

11.5 Spray jet(s) to deliver 3.0-4.0 L/min (0.75-1.0 gal/min) with a line pressure of 207 kPa (30 psi). The spray pattern should give even liquid delivery over the entire wand width. There should be no heavy or light streaks. The liquid impact on the face of the carpet should not cause matting or other pile distortion. Jet(s) orifice size and/or distance from the carpet face should be adjusted to avoid pile distortion and give even liquid distribution. EXAMPLE: If spray pattern is too narrow and pile matts, increase height. If pattern is wide enough and pile matts, increase orifice size.

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Test Method for Colorfastness to Powdered Non-Chlorine Bleach in Home Laundering

Developed in 1988 by AATCC Committee RA60; reaffirmed 1989, 1995, 2002; revised 1990, 1996, 1997, 2003, 2007 (with title change), 2010; editorially revised 1994, 2004, 2005, 2008, 2012, 2019; reaffirmed and editorially revised 2016.

1. Purpose and Scope

1.1 This test method is designed to evaluate the colorfastness to powdered non-chlorine bleach in home laundering of textiles which are expected to withstand frequent laundering. The fabric color changes resulting from the powdered non-chlorine bleach, detergent solution and abrasive action of five home launderings are evaluated.

1.2 First establish the level of performance with powdered non-chlorine bleach and if an effect is found, then retest with detergent alone. It may be necessary to use a water only wash to separate the effects of components like hardness, pH or chlorine content.

1.3 Since the powdered non-chlorine bleaches currently available contain ingredients other than the powdered non-chlorine bleach; such as fluorescent whitening agents, bluing, etc., it is the total effect of these chemicals on the color change that is being evaluated.

2. Principle

2.1 Specimens are laundered under appropriate conditions of temperature, detergent solution, powdered non-chlorine bleach solution and abrasive action for five home laundering cycles. The laundered specimens are evaluated for color change.

3. Terminology

3.1 **bleach,** n.—*in home laundering*, a product that will clean, whiten, brighten and aid in the removal of soils and stains from textile materials by oxidation that is inclusive of chlorine and non-chlorine products.

3.2 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the materials.

3.3 **laundering,** n.—*of textile materials*, a process intended to remove soil and/or stains by treatment (washing) with an aqueous detergent solution and normally including subsequent rinsing, extracting and drying.

3.4 **non-chlorine bleach,** n.—a product that does not release the hypochlorite ion in solution; i.e., sodium perborate, sodium percarbonate.

3.4.1 **liquid non-chlorine bleach,** n.—a product that contains hydrogen peroxide as its active ingredient.

3.4.2 **powdered non-chlorine bleach,** n.—a product that contains either sodium perborate or sodium percarbonate as its active ingredient.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 The 1993 AATCC Standard Reference Detergent WOB and the 2003 AATCC Standard Reference Liquid Detergent WOB may cause irritation. Care should be taken to prevent exposure to skin and eyes.

4.3 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

4.4 All chemicals should be handled with care.

4.5 The procedure for determining the percent active oxygen content outlined in 13.7 should be handled only in an adequately ventilated laboratory hood. In preparing, dispensing and handling the reagents, use chemical goggles or face shield, impervious gloves and an impervious apron.

4.6 An eyewash/safety shower should be located nearby and an organic vapor respirator should be available for emergency use.

4.7 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 13.1).

5. Uses and Limitations

5.1 Non-chlorine bleaches are available in two different forms—powdered or liquid. This method only evaluates the colorfastness of textiles to powdered non-chlorine bleach in home laundering. Due to differences in active ingredients, the results obtained by this method may not be representative of results obtained by the use of liquid non-chlorine bleach on textiles during laundering.

6. Apparatus and Materials (see 13.2)

6.1 Automatic washing machine (see 13.3).

6.2 Automatic tumble dryer (see 13.4).

6.3 Conditioning/drying racks with pull-out screens or perforated shelves (see 13.5).

6.4 Ballast of $92 \times 92 \pm 3$ cm ($36 \times 36 \pm 1$ in.) hemmed pieces of bleached cotton sheeting (Wash load ballast type 1) or 50/50 polyester/cotton bleached of plain weave construction (Wash load ballast type 3).

6.5 1993 AATCC Standard Reference Detergent WOB or 2003 AATCC Standard Reference Liquid Detergent WOB (see 13.6).

6.6 Facilities for drip and line drying.

6.7 Balance with at least 5 kg (20 lbs) capacity.

6.8 Powdered non-chlorine bleach (see 13.7 and 13.8).

7. Test Specimens

7.1 The weight of the test specimen can influence the results of this test. Only one test specimen is to be laundered in a load. For ease in weighing fabrics, the test specimen weight is to be 110.0 ± 10.0 g (0.25 ± 0.02 lb). For garment testing, one garment is to be tested. If the gar-

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Table I—Washing and Drying Conditions (see 13.10)

Machine Cycle	Washing Temperatures	Drying Procedures
(1) Normal/Cotton Sturdy	(II) 27 ± 3°C (80 ± 5°F)	(A) Tumble:
(2) Delicate	(III) 41 ± 3°C (105 ± 5°F)	i. Cotton Sturdy
(3) Permanent Press	(IV) 49 ± 3°C (120 ± 5°F)	ii. Delicate
	(V) 60 ± 3°C (140 ± 5°F)	iii. Permanent Press
		(B) Line
		(C) Drip
		(D) Screen

ment weight exceeds the specified load weight of 1.8 kg or 4 lb, the total weight is to be included in the report. See 8.2.2 for exact weight specifications on load.

8. Procedure

8.1 Table I summarizes washing and drying conditions. Information on the machine and laundering conditions may be found in the AATCC LP1 (see 13.10).

8.2 Washing.

8.2.1 Fill the washer with the specified water volume at the selected water temperature using the normal cycle setting. Measure and record the water hardness in any test.

8.2.2 Add 66 ± 1 g of 1993 AATCC Standard Reference Detergent WOB or 100 ± 1 g of 2003 AATCC Standard Reference Liquid Detergent WOB. Add the amount of the powdered non-chlorine bleach recommended by the manufacturer. Add test specimen and enough ballast to make a 1.8 ± 0.1 kg (4.00 ± 0.25 lb) load. Set the washer on the cycle selected for washing and set the wash timer (see Table I and 8.1). Start the agitation and independently time the wash for reproducible results. Advance the dial to the end of the wash cycle and rinse at a temperature below 29°C (85°F). If this rinse temperature is not attainable, record the actual temperature of the rinse water used.

8.2.3 For specimens to be dried by procedures A, B or D, allow washing to proceed automatically through the final spin cycle. Remove test specimens immediately after the final spin cycle, separate any tangled pieces—taking care to minimize distortion—and dry by procedures A, B or D (see Table I and 8.1).

8.2.4 For specimens to be dried by procedure C, Drip Dry, stop the washer at the end of the final rinse cycle just before the water begins to drain. Remove the specimens soaking wet.

8.3 Drying.

8.3.1 (A) Tumble Dry. Place the washed load (test specimens and ballast) in the tumble dryer, set on the cycle which generates the correct exhaust temperatures as specified in AATCC LP1 (see 8.1). For fibers that are heat sensitive, lower temperatures consistent with producer's recommendations are required and must be reported. Operate the dryer until the total load is dry. Remove the load immediately after machine stops.

8.3.2 (B) Line Dry. Hang each specimen by two corners with the fabric length in the vertical direction. Allow the test specimens to hang in still air at room temperature until dry.

8.3.3 (C) Drip Dry. Hang each dripping wet specimen by two corners, with the fabric length in the vertical direction. Allow specimens to hang in still air at room temperature until dry.

8.3.4 (D) Screen Dry. Spread each specimen on a horizontal screen or perforated surface, removing wrinkles but not distorting or stretching, and allow to dry in still air at room temperature.

8.4 Repeat 8.2 and 8.3 for a total of five launderings.

8.5 After drying, condition the specimens for at least 4 h by laying each specimen separately on the screen or perforated shelf of a conditioning rack in an atmosphere of 21 ± 2°C (70 ± 4°F) and 65 ± 5% RH before evaluating color change.

9. Evaluation

9.1 Evaluate the change in color in the test specimen by comparison with the Gray Scale for Color Change (AATCC EP1), or using AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen, and record the numerical rating that corresponds to the appropriate one on the Gray Scale.

10. Interpretation of Results

10.1 This test is a satisfactory end use test for demonstrating the effect of the 1993 AATCC Standard Reference Detergent WOB or the 2003 AATCC Standard Reference Liquid Detergent WOB and powdered non-chlorine bleach on textile materials in home laundering.

10.2 If a significant color change is observed, an untreated duplicate sample could be retested using either this method with detergent only, or the 2A Test in AATCC TM61, Test Method for Colorfastness to Laundering: Accelerated, without the addition of powdered non-chlorine bleach. This will determine whether the color change is due to dye instability or a bleach effect.

11. Report

11.1 Report the numerical average color change grade for each specimen tested.

11.2 Report washing procedure (Arabic and Roman Numerals) and drying procedure (capital letter from Table I; [i.e., (1) IIIA (iii)] means NORMAL cycle, wash water temperature of 41 ± 3°C (105 ± 5°F), tumble dry (permanent press).

11.3 Report the type and amount of detergent used.

11.4 Record the brand and quantity of the powdered non-chlorine bleach used.

11.5 Report the percent available oxygen.

11.6 Record the weight of fabric specimen or garment tested.

12. Precision and Bias (see 13.9)

12.1 *Interlaboratory Test Data*. An interlaboratory study was carried out in 1986 to establish the precision of this test method for determining the effect of powdered non-chlorine bleach in home laundering on the colorfastness of textile materials.

Four materials were tested separately for five cycles of home laundering in duplicate at each of the five laboratories using one brand of powdered non-chlorine bleach. After completing the laundering, three raters independently assessed the color change on each specimen of each fabric using the Gray Scale for Color Change. The washing and drying conditions were (1)IVa (see Table I for interpretation of code). Each of the 24 × 24 in. specimens that were washed separately ranged in weight from 72-97 g (0.16-0.21 lb). The fabrics were 100% cotton and polyester/cotton. Dyes were direct, pigment and disperse. The color change average grades for the three raters at the five laboratories were 3.33-3.93.

12.2 *Precision*. The components of variance as standard deviations of colorfastness grades were calculated to be as follows:

Single-operator component	0.22
Within-laboratory component	0.21
Between-laboratory component	0.00

For the components of variance reported above, the average of two color change grades should be considered significantly different at the 95% probability level if their difference equals or exceeds the critical differences listed in Table II.

12.3 *Bias*. The true value of colorfastness to non-chlorine bleach in home laundering can only be defined in terms of a test method. Within this limitation, this method has no known bias.

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Table II—Critical Differences, Rating Units for Conditions Noted[a]

Number of Observations in each Average	Single-Operator Precision	Within-Laboratory Precision	Between-Laboratory Precision
1	0.61	0.85	0.85
3	0.35	0.69	0.69
5	0.27	0.65	0.65

[a] The critical differences were calculated using $t = 1.950$, which is based on infinite degrees of freedom.

13. Notes

13.1 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

13.2 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

13.3 Contact AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1. 919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org, for model number(s) and source(s) of current recommended washer(s). Any other washer which is known to give comparable results can be used. Washing machine conditions given in AATCC LP1 represent the actual speeds and times available on current specified model(s). Other washers may vary in one or more of these settings.

13.4 Contact AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1. 919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org, for model number(s) and source(s) of current recommended dryer(s). Any other dryer which is known to give comparable results can be used. Dryer machine conditions given in AATCC LP1 represent the actual speeds and times available on current specified model(s). Other dryers may vary in one or more of these settings.

13.5 Screen or perforated conditioning/drying racks available from Somers Sheet Metal Inc., 5590 N. Church St., Greensboro NC 27405; tel: +1.336.643.3477; fax: +1.336.643. 7443. Rack drawings are available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

13.6 All detergents are available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1. 919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org. AATCC Standard Reference Detergent WOB has been changed to a 0% phosphate compact formula named 1993 AATCC Standard Reference Detergent WOB. Critical evaluations made using AATCC Standard Reference Detergent WOB and used for standard comparisons should be repeated using the 1993 AATCC Standard Reference Detergent WOB.

13.7 Powdered non-chlorine bleaches are available under a large number of trade names in supermarkets. Different brands will contain different ingredients and the same ingredients in differing quantities. Follow manufacturer's instruction for recommended usage. In conducting experiments with these materials, since they are blends of materials of different granule sizes and densities, it is strongly recommended that sufficient product to conduct a group of experiments in the same time frame be thoroughly mixed in a composite before and during use. Keep in air tight containers. The product used to develop this method was Clorox 2.

13.8 To determine the percent available oxygen in powdered non-chlorine bleaches or synthetic detergent products, grind the granules in a Waring blender prior to weighing. Using an analytical balance which is accurate to 0.002 g, weigh 3.000 ± 0.001 g of sample into a 250-300 mL Erlenmeyer flask. Place the sample on a magnetic stirrer protecting it from the light with a cardboard box. Put a hole in the box top that allows a funnel to be used to add chemicals and support it with a ring and stand. Use a Teflon coated stir bar. Add 100 mL of 1.5*N* sulfuric acid (H_2SO_4), 15.0 mL of 15.0% potassium iodide (150 g KI/L) and 10.0 mL of methylene chloride (CH_2Cl_2). Stir, *in the dark* provided by the cardboard box, for at least 15 min but do not exceed 30 min. Provide indirect light as you titrate slowly but steadily with 0.1*N* sodium thiosulfate ($Na_2S_2O_3$) with constant stirring, to a pale yellow. Add about 2.0 mL of starch indicator (5.0% solution) and continue the titration a drop at a time until the disappearance of the blue starch-iodine color.

Calculation:

Percent Available Oxygen =

$$\frac{\text{mL } Na_2S_2O_3 \times N \times 0.008 \times 100}{\text{Sample Size}}$$

(where N = Normality of $Na_2S_2O_3$ solution) Variation of ± 3% on the percent of available oxygen is acceptable within product test.

13.9 The precision of this test method is dependent on the combined variability of the material being tested, the test method itself, and the evaluation procedure utilized.

13.9.1 The precision statement shown in Section 12 was developed from results obtained by visual evaluations (EP1).

13.9.2 It is expected that the use of instrumental evaluations (EP7) will result in greater precision than that obtained from visual evaluations.

13.10 Laundering temperatures and other parameters listed in M6 may not be the same as those listed in various test methods. Parameters in M6 are updated periodically to reflect consumer experience, taking into account current technology and regulations regarding home water heater temperatures, particularly in the United States.

Typically, test method committees pursue a deliberate effort to maintain the same test method conditions over a long period of time in order to produce comparable results over time. Additionally, significant changes may invalidate precision data developed using the original conditions.

Temperatures listed in AATCC test methods were established at the time the methods were developed or revised, and closely mirror the temperature ranges specified in 16 CFR Part 423 for care labels. It should also be noted that AATCC test methods were developed using traditional top loading washing machines. Specified water level, detergent, and other details are not applicable to high efficiency top loading or front loading machines. New, commercially available home laundering machines may have to be modified by the user to meet the specifications in various test methods. All test reports should indicate the exact testing conditions, and any modification of the standard method, including use of M6 parameters. Results of tests performed with different conditions may not be comparable.

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Test Method for Antimicrobial Activity of New Carpets

1. Purpose and Scope

1.1 This test method is designed to determine the antimicrobial activity of new carpet materials and consists of three procedures: a qualitative antibacterial assessment, a quantitative antibacterial assessment and qualitative antifungal assessment. It may also be used to evaluate the effect of a cleaning process (agreed upon by the interested parties) on the antimicrobial resistance of carpets.

2. Principle

2.1 This test method consists of three procedures, Test I, II and III, which can be used singly or in combination.

2.1.1 Test I is a qualitative assessment of antibacterial activity. Specimens of the test material, including corresponding untreated controls of the same material (if available but not required), are placed in intimate contact with the agar surface which has been previously streaked with a bacterial culture. After incubation, a clear area of interrupted growth underneath and along the sides of the test material indicates antibacterial activity of the specimen. Standard strains of bacteria are used with *Staphylococcus aureus* (gram-positive) and *Klebsiella pneumoniae* (gram-negative), the representative organisms.

2.1.2 Test II provides a quantitative procedure for the comparison and evaluation of the degree of antibacterial activity after a 24 h exposure to the test bacteria on the carpet swatches. Test carpets are inoculated with the test organisms. After incubation, the bacteria are eluted from the swatches by shaking in known amounts of neutralizing liquid. The number of bacteria present in this liquid is determined, and the percent reduction by the specimen is calculated.

2.1.3 Test III is a qualitative assessment of the antifungal activity of carpet and resistance to a common fungal organism on a sabouraud dextrose agar medium.

3. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

3.1 AATCC TM138, Test Method for Cleaning: Washing of Textile Floor Coverings (see 31.1).

3.2 AATCC TM171, Test Method for Carpets: Cleaning of; Hot Water Extraction (see 31.1).

3.3 ASTM E2471, Standard Test Method for Using Seeded-Agar for the Screening Assessment of Antimicrobial Activity In Carpets (see 31.2).

3.4 ASTM E1402, Standard Guide for Sampling Design (see 31.2).

3.5 Centers for Disease Control. Biosafety in Microbiological and Biomedical Laboratories (see 31.3).

4. Terminology

4.1 **activity,** n.—*of an antimicrobial agent*, a measure of effectiveness of the agent.

4.2 **antibacterial agent,** n.—any chemical which kills bacteria (bactericide) or interferes with the multiplication, growth or activity of bacteria (bacteriostat).

4.3 **antifungal agent,** n.—any chemical which kills or inhibits the growth of fungi.

4.4 **antimicrobial agent,** n.—any chemical which kills or inhibits the growth of microorganisms.

4.5 **bacterial resistance,** n.—*in textiles*, resistance to the development of visible bacterial growth and accompanying odors, resulting from bacterial degradation of fibers or soil on them, as distinguished from musty fungal odors.

4.6 **mildew resistance,** n.—*in textiles*, resistance to the development of unsightly fungal growths and accompanying unpleasant, musty odors on carpet materials exposed to conditions favoring such growths.

4.7 **rot resistance,** n.—*in textiles*, resistance to deterioration of a carpet material as a result of fungal growth in or on it.

4.8 **zone of inhibition,** n.—clear area of no visible growth of a microorganism, cultured onto the surface of agar growth medium, in proximity to the borders of a specimen placed in direct contact with this agar surface.

NOTE: A zone of inhibition occurs as result of the diffusion of an antimicrobial agent from the specimen.

5. Safety Precautions

5.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

5.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

5.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g. OSHA standards and rules).

5.4 See 3.5, 8.1, 11.1 for additional safety precautions and personnel training related to handling bacteria and fungi.

6. Uses and Limitations

6.1 This test method is designed for use only for new carpets and must not be used for carpets that have been laid down and used.

I. Qualitative Assessment of Antibacterial Activity on Carpets: Single Streak Method

7. Apparatus (see 31.4)

7.1 Incubator maintained at 37 ± 2°C (99 ± 4°F).

7.2 Water bath maintained at 45-50°C (113-122°F).

7.3 Bunsen Burner or equivalent.

7.4 Stereomicroscope, minimum 40× magnification.

7.5 Autoclave.

7.6 Vortex or equivalent.

8. Materials (see 31.4)

8.1 Bacteria species suitable for the intended end-use of the test sample. Some bacteria are pathogenic (i.e., capable of infecting humans and producing disease). Take every necessary and reasonable precaution to eliminate risk to laboratory personnel and others in the associated environment. Sterilize all contaminated samples and test materials prior to disposal. Any or all the following bacteria may be used:

8.1.1 Gram-positive: *Staphylococcus aureus*, ATCC 6538, CIP 4.83, DSM 799, NBRC 13276, NCIMB 9518 or equivalent strain (see 31.5 and 31.6).

8.1.2 Gram-negative: *Klebsiella pneumoniae*, ATCC 4352, CIP 104216, DSM 789, NBRC 13277, NCIMB 10341 or equivalent strain (see 31.5 and 31.6)

8.1.3 Other species agreed upon by the interested parties.

8.2 Broths/agars; Nutrient broth/agar, Trypticase Soy broth/agar, Brain-Heart Infusion broth/agar, Müller-Hinton broth/agar or other capable of growing and maintaining the chosen bacterial test organism.

8.3 Distilled or deionized water, sterile.

8.4 Inoculating loop, 4-mm.

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8.5 Pipets, 1 mL, sterile or equivalent.
8.6 Culture Tubes with caps; minimum 10 mL capacity, sterile
8.7 Petri dishes, 100 mm diam. × 15 mm deep, flat-bottom, sterile.
8.8 Forceps, sterile.
8.9 Ruler, marked in millimeters.

9. Sampling

9.1 Test results are valid only when the samples are statistically representative (see ASTM E1402).
9.2 Sampling must be random. Every unit of product must have a mathematically equal chance of becoming a sample; and every portion of each sample must be equally likely to become a test specimen.
9.3 All specimens must be alike within the variations due to pure chance. There must be no differences within samples assignable to known causes.

10. Specimens

10.1 Test specimens (non-sterile) are cut by hand or with a die. They may be any convenient size although oblong specimens cut 25 × 50 mm are recommended.
10.2 If possible, test a specimen of the same material treated in exactly the same way with whatever other finishing agents were used, but without the antibacterial agent. However, this is not essential for the validity of the test. Many standard finishing chemicals will give strong antibacterial activity even after many cleanings.
10.3 Test 1 test specimen. Additional test specimens may be used to increase the precision of the average.

11. Procedure

11.1 This test should be performed only by trained personnel. Consult the U.S. Department of Health and Human Services publication *Biosafety in Microbiological and Biomedical Laboratories* (see 3.5).
11.2 If durability data is desired, carpet specimens can be tested before and after being cleaned by a test method agreed upon by the interested parties i.e. AATCC TM138 (see 3.1) or AATCC TM 171 (see 3.2)
11.3 Dispense sterilized (appropriate medium) agar cooled to 45 ± 2°C (117 ± 4°F) by pouring 15.0 ± 2.0 mL into each standard (15 × 100 mm) flat bottomed Petri dish. Allow the agar to gel firmly before inoculating.
11.4 Prepare the test organism by taking a sterilized inoculation loop and inoculating separate culture tubes of the appropriate broth with the organism to be used. Incubate the cultures at 37 ± 2°C (99 ± 4°F) for 18-24 h.
11.5 Prepare inoculum by transferring 1.0 ± 0.1 mL of a 18-24 h broth culture into 9.0 ± 0.1 mL of sterile distilled water contained in a test tube or small flask. Mix well using appropriate agitation or by using a vortex.
11.6 Using a 4-mm inoculating loop, load one loop full of the diluted inoculum and transfer it to the surface of the sterile agar plate by making one long streak of approximately 75 mm in length across the center of the plate. Do not break the surface of the agar while making the streak.
11.7 Gently press the test specimen transversely across the inoculum streak to ensure intimate contact with the agar surface. An easy technique is to press the specimen to the agar surface with a biological section lifter or with a spatula which has been sterilized in a flame and then air-cooled immediately before use. Test both the pile (face fibers) and the backing of the carpet on separate agar plates.
11.8 Incubate the plates at 37 ± 2°C (99 ± 3°F) for 18-24 h.

12. Evaluation

12.1 Examine the incubated plates for interruption of growth along the streak of inoculum beneath the specimen and for a clear zone of inhibition beyond the specimen edge.
12.2 Calculation of zone of inhibition size is optional. The width of the zone of inhibition around the test specimen may be calculated using Eq 1:

$$W = (T - D)/2 \qquad \text{(Eq.1)}$$

where:
W = width of clear zone of inhibition in mm
T = total width of test specimen and clear zone in mm
D = width of the test specimen in mm

12.3 The criterion for passing the test must be agreed upon by the interested parties.
12.4 The size of the zone cannot be construed as a quantitative evaluation of antibacterial activity. The report of results will include an observation of zones of inhibition and growth under the specimen if present.

13. Report

13.1 Describe or identify the sample tested.
13.2 Report that the sample was tested using AATCC TM174-2022 Test I.
13.3 Report the testing conditions:
13.3.1 Size of test specimen used.
13.3.2 Number of test replicates used.
13.3.3 Number of carpet cleanings (if any) and the method used.
13.3.4 If the test specimen was sterilized prior to testing and the method of sterilization.
13.3.5 Agar media used.
13.3.6 Organism used.
13.3.7 incubation time.
13.4 Report the test results:
13.4.1 Any observed growth underneath the sample.
13.4.2 Any observed zone of inhibition.
13.5 Describe any modification(s) of the published standard.

14. Precision and Bias

14.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.
14.2. *Bias*. The antimicrobial activity can be defined on in terms of a test method. There is no independent method for determining the true value. As a means of evaluating this property, the method has no known bias.

II. Quantitative Assessment of Antibacterial Activity on Carpets

15. Apparatus (see 31.4)

15.1 Incubator maintained at 37 ± 2°C (99 ± 4°F).
15.2 Water bath maintained at 45-50°C (113-122°F).
15.3 Bunsen Burner or equivalent.
15.4 Autoclave.
15.5 Vortex or equivalent.

16. Materials (see 31.4)

16.1 Bacteria species suitable for the intended end-use of the test sample. Some bacteria are pathogenic (i.e., capable of infecting humans and producing disease). Take every necessary and reasonable precaution to eliminate risk to laboratory personnel and others in the associated environment. Sterilize all contaminated samples and test materials prior to disposal. Any or all the following bacteria may be used:
16.1.1 Gram-positive: *Staphylococcus aureus*, ATCC 6538, CIP 4.83, DSM 799, NBRC 13276, NCIMB 9518 or equivalent strain (see 31.5 and 31.6).
16.1.2 Gram-negative: *Klebsiella pneumoniae*, ATCC 4352, CIP 104216, DSM 789, NBRC 13277, NCIMB 10341 or equivalent strain (see 31.5 and 31.6).
16.1.3 Other species agreed upon by the interested parties.
16.2 Broths/agars; Nutrient broth/agar, Trypticase Soy broth/agar, Brain-Heart Infusion broth/agar, Müller-Hinton broth/agar or other capable of growing and maintaining the chosen bacterial test organism.

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16.3 Distilled or deionized water.

16.4 Inoculating loop.

16.5 Pipettes, 1 mL, sterile or equivalent.

16.6 Culture Tubes with caps; minimum 10 mL capacity, sterile

16.7 Petri dishes, 100 mm diam. × 15 mm deep, flat-bottom, sterile.

16.8 Forceps, sterile.

16.9 Ruler, marked in millimeters.

16.10 0.85% Saline solution, sterile.

16.11 Wetting agent (see 31.7).

16.12 Neutralizing broth; Letheen, D/E broth or other appropriate for the antimicrobial compound being tested (see 31.8).

16.13 Container with lid. Minimum 120 mL capacity, to fit carpet specimens. Sterile.

17. Sampling

17.1 Test results are valid only when the samples are statistically representative (see ASTM E1402).

17.2 Sampling must be random. Every unit of product must have a mathematically equal chance of becoming a sample; and every portion of each sample must be equally likely to become a test specimen.

17.3 All specimens must be alike within the variations due to pure chance. There must be no differences within samples assignable to known causes.

18. Specimens

18.1 Cut a circular disc of approximately 48 mm in diameter from the test carpet (preferably with a steel die). Place the disc in a sterile container. The carpet disc should lie flat at the bottom of the container.

18.2 An uninoculated treated carpet may be used to determine the level of background organisms present on the carpet.

18.3 If the sample is sterilized, note the method of sterilization in report.

18.4 Test 3 replicates. Additional test specimens may be used to increase the precision of the average.

19. Procedure

19.1 This test should be performed only by trained personnel. Consult the U.S. Department of Health and Human Services publication *Biosafety in Microbiological and Biomedical Laboratories* (see 3.5).

19.2 If durability data is required, test carpet specimens before and after being cleaned in accordance with a method agreed upon by the interested parties i.e. AATCC TM138 (see 3.1) or AATCC TM171 (see 3.2).

19.3 The carpet disc may be prewetted by dipping it in sterile deionized water or in water containing 0.05% of a non-bacteriocidal wetting agent (see 31.7) and then briefly blotting it on sterile filter paper.

19.4 Prepare the test organism by taking a sterilized inoculation loop and inoculating separate culture tubes of the appropriate broth with the organism to be used. Incubate the cultures at 37 ± 2°C (99 ± 4°F) for 18-24 h.

19.5 Apply 0.1-0.5 mL of an 18-24 h broth bacterial inoculum adjusted to 1-2 $\times 10^5$ CFU/mL on the prewetted carpet fibers. The dilution of the test organism may be prepared in sterile 0.85% saline solution or suitable buffer if a steady-state culture is needed during the contact period. If, however, this test is to be performed under in-use conditions, use nutrient broth as the dilution medium.

19.6 Inoculate the carpet fibers evenly using a sterile pipette and place the specimen in a sterile container. Close or tighten the lid tightly to prevent evaporation.

19.7 Immediately after inoculation (0 contact time), add 100 ± 0.1 mL of neutralizer solution to the sterile container

19.8 Shake the container, either mechanically or by hand, vigorously for 1 min. Make serial dilutions and plate (in duplicate) on nutrient (or appropriate) agar to obtain colonies in a countable range. Incubate all plates for 24 h at 37 ± 2°C (99 ± 3°F). Record the number of bacteria recovered.

19.9 Incubation over contact periods. Incubate additional containers containing inoculated carpet discs at 37 ± 2°C (99 ± 3°F) for 6-24 h. Similar containers may be incubated over other periods (i.e., 1 or 6 h) to provide information about the bacteriocidal activity of the treatment over such periods.

19.10 Sampling of inoculated and incubated swatches. After incubation, add 100 ± 0.1 mL of neutralizer solution to containers containing treated carpet discs. Shake the Containers vigorously for 1 min. Make serial dilutions and plate (in duplicate) on nutrient (or appropriate) agar to obtain colonies in a countable range. Several different dilutions may be required for untreated control carpets depending on the incubation period.

19.11 Incubate all plates for 24 h at 37 ± 2°C (99 ± 3°F). Record the number of bacteria recovered.

20. Calculations, Interpretation, Evaluation

20.1 Report bacterial counts as the number of bacteria per specimen, not as the number of bacteria per mL of neutralizer solution. If there are no bacterial colonies on the dilution plates, then report as "less than 100."

20.2 Calculate percent reduction of bacteria by the specimen treatments by one of the following formulas:

1) $100(B - A)/B = R$ (Eq.2)
2) $100(C - A)/C = R$ (Eq.3)
3) $100(D - A/D) = R$ (Eq.4)

where:

A = the number of bacteria recovered from the inoculated treated test carpet in the container incubated over the desired contact period.

B = the number of bacteria recovered from the inoculated treated test carpet in the container immediately after inoculation (at 0 contact time).

C = the number of bacteria recovered from the inoculated untreated control carpet in the container immediately after inoculation (at 0 contact time). If B and C are not similar, the larger number should be used. If B and C are not significantly different, $(B + C)/2$ should be used.

D = $(B + C)/2$.

R = % reduction.

20.3 If an untreated control for carpet is not available, use the following calculation which allows for any background organisms that might interfere with the test:

$$Bg = 100\ \{(B-E)-(A-F)/B-E\} \quad \text{(Eq.5)}$$

where:

A = the number of bacteria recovered from the inoculated treated test carpet in the container incubated over the desired contact period.

B = the number of bacteria recovered from the inoculated treated test carpet in the container immediately after inoculation (at 0 contact time).

E = the number of bacteria initially (0 contact time) recovered from the uninoculated, treated test carpet (existing background organisms).

F = the number of bacteria recovered from the uninoculated, prewet treated test carpet after incubation in the container over the desired contact period (existing background organisms after contact period).

Bg = background organisms.

20.4 The criterion for passing the test must be determined by the interested parties.

21. Report

21.1 Describe or identify the sample tested.

21.2 Report that the sample was tested using AATCC TM174-2022 Test II.

21.3 Report the testing conditions:

21.3.1 Size of test specimen used.

21.3.2 Number of test replicates used.

21.3.3 Number of carpet cleanings (if

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any) and the method used.

21.3.4 If the test specimen was sterilized prior to testing and the method of sterilization.

21.3.5 Wetting agent, neutralizing broth, dilution liquid, and agar media used.

21.3.6 Organism used.

21.3.7 Incubation time.

21.4 Report the test results:

21.4.1 Assessment according to section 20.

21.5 Describe any modification(s) of the published standard.

22. Precision and Bias

22.1 Studies (see 31.9) indicate the following *within-laboratory* precision of the Standard Plate Count (SPC) Test: (a) *among-analyst* variation of 18%, and (b) *within-analyst* variation of 8%.

III. Antifungal Activity Assessment of Carpet Materials: Mildew and Rot Resistance of Carpet Materials

23. Apparatus (see 31.4)

23.1 Incubator maintained at 28 ± 1°C (82 ±2°F).

23.2 Water bath maintained at 45-50°C (113-122°F).

23.3 Bunsen Burner or equivalent.

23.4 Autoclave.

23.5 Vortex or equivalent.

23.6 Stereo-Microscope.

23.7 Counting chamber or equivalent; e.g., hemocytometer.

24. Materials (see 31.4)

24.1 *Aspergillus niger* ATCC 6275, DSM1957, NRRL 334, CBS 769.97 or 131.52, CCUG 26806 (see 31.5 and 31.6).

24.2 Sabouraud Dextrose Agar.

24.3 Distilled or deionized water.

24.4 0.85% Saline solution

24.5 Wetting agent (see 31.7).

24.6 Gauze or glass wool, sterile.

24.7 Sterile scraping instrument; Platinum or nichrome wire, plastic loop or needle, glass rod, or swab.

24.8 Glass beads, sterile.

24.9 Pipette, sterile.

24.10 Petri dishes, 100 mm diam. × 15 mm deep or 100 mm diam. × 20 mm deep, flat-bottom, sterile.

24.11 ruler, marked in millimeters.

25. Sampling

25.1 Test results are valid only when the samples are statistically representative (see ASTM E1402).

25.2 Sampling must be random. Every unit of product must have a mathematically equal chance of becoming a sample; and every portion of each sample must be equally likely to become a test specimen.

25.3 All specimens must be alike within the variations due to pure chance. There must be no differences within samples assignable to known causes.

26. Specimens

26.1 Cut 38.0 ± 1.0 mm (1.5 ± 0.04 in.) diameter discs from the sample. Other shapes and sizes can be used provided the anticipated sizes of growth-free zones are taken into consideration.

26.2 For primary backing assessment, an additional set of carpet discs may be included; as per ASTM E2471, which have the face fibers removed via electric clippers to a height of 3 ± 1 mm. This permits the fungal spore inoculum to be placed and assessed directly at the base of the fiber/primary backing and must be noted in the report.

26.3 Test 1 test specimen. Additional test specimens may be used to increase the precision of the average.

27. Procedure

27.1 This test should be performed only by trained personnel. Consult the U.S. Department of Health and Human Services publication *Biosafety in Microbiological and Biomedical Laboratories* (see 3.5).

27.2 If cleaning durability data are required, new carpet specimens can be tested before and after being cleaned in accordance with a method agreed upon by the interested parties i.e. AATCC TM138 (see 3.1) or AATCC TM171 (see 3.2).

27.3 Inoculum: The spore suspension of *Aspergillus niger* is prepared by adding 10 mL of a sterile 0.9% saline solution containing 0.05% of a nonionic wetting agent (see31.7) to a 7-10 day agar culture and scrapes the surface of the culture to liberate spores. Transfer the scrapings to a sterile flask or other suitable container containing 50 ± 1 mL of sterile 0.9% saline solution with 0.05% of a nonionic wetting agent and sterile glass beads. Shake the dispersion vigorously to break up any clumps of spores and then filter through a thin layer of sterile cotton or glass wool. The spore suspension may be stored at 6 ± 4°C (43 ± 7°F) for up to 4 weeks. Inoculum for test should be adjusted to produce a final population of 8.0×10^5 to 1.2×10^6 per mL on day of use by appropriate dilution of stock suspension with sterile 0.9% saline solution. Use this spore suspension for inoculation.

27.4 Inoculation: Mix the spore suspension using appropriate agitation or by using a vortex. Distribute 1.0 ± 0.1 mL of the inoculum over the surface of the agar. Prewet the carpet fibers by dipping the carpet disc in sterile deionized water or in water containing 0.05% of a nonionic wetting agent, and then briefly blotting on filter paper. Place the carpet disc on the surface of the agar. Distribute evenly over each disc 0.2 mL of fungal spore inoculum by means of a sterile pipette. Inoculate the carpet specimens with the face fibers up and also with the fibers down in separate Petri dishes. If testing carpet samples with fibers shaven, the samples should be placed both in the shaven fibers up and also with fibers shaven down orientation in separate Petri dishes. Incubate the inoculated plates at 28 ± 1°C (82 ± 2°F) for 7 days. Longer periods of incubation may be used to provide information about antifungal activity.

28. Evaluation

28.1 Assess the activity of the carpet as follows:

28.1.1 From the plate with the specimen whose pile is down and whose backing is up, observe and measure the size of any growth-free zone (in mm) produced by the pile fibers and also do the same for specimens that have face fibers shaven. Also from the same plate, record fungal growth on the backing according to the scheme below.

28.1.2 From the other plate with the specimen whose backing is down and whose pile or the shaven pile layer is uppermost, observe and measure the size of any growth-free zone (in mm) from the backing and score the fungal growth on the pile side or the shaven pile side according to the scheme given below.

28.1.3 Scoring Scheme:

Observed Growth

0 = No growth (if present, report the size of the growth-free zone in mm)

1 = Microscopic growth (visible only under the microscope)

2 = Macroscopic growth (visible to the eye)

If macroscopic growth is observed report growth according to the following coverage:

Traces of Growth (<10%), Light Growth (10-30%), Medium Growth (30-60%), Heavy Growth (60% to totally covered).

29. Report

29.1 Describe or identify the sample tested.

29.2 Report that the sample was tested using AATCC TM174-2022 Test III.

29.3 Report the testing conditions:

29.3.1 Size of test specimen used.

29.3.2 Number of test replicates used.

29.3.3 Number of carpet cleanings (if any) and the method used.

29.3.4 If the test specimen was sterilized prior to testing and the method of sterilization.

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29.3.5 Organism used and concentration of the fungal spore suspension.

29.3.6 Wetting agent used.

29.3.7 Incubation time.

29.4 Report the test results:

29.4.1 The assessment according to section 28.

29.5 Describe any modification(s) of the published standard.

30. Precision and Bias

30.1 Precision. Precision for this test method has not been established. Until a precision statement is generated for this method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

30.2 Bias. The antimicrobial activity can be defined on in terms of a test method. There is no independent method for determining the true value. As a means of evaluating this property, the method has no known bias.

31. Notes

31.1 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

31.2 Available from ASTM, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA, 19428-2959, USA; Tel: +1.610.832.9500, Fax: +1.610.832.9555, service@astm.org.

31.3 Available from CDC. Website: https://www.cdc.gov/labs/BMBL.html.

31.4 For potential apparatus, reagents or materials sources, visit the AATCC Buyer's Guide at www.aatcc.org/bg. AATCC provides its Corporate members the option to list their items and services. AATCC does not qualify, or in any way approve, endorse or certify that any of the listings meet the specifications in its standards.

31.5 ATCC is the American Type Culture Collection (USA), CIP is the Pasteur Institute Collection (France), DSM is the German Collection of Microorganisms and Cell Cultures (Germany); NBRC is the NITE Biological Resource Center (Japan), NRRL Northern Regional Research Lab (USA), NCIMB is the National Collection of Industrial Bacteria (UK), CUG is the Culture Collection University of Göteborg (Sweden). Equivalent bacteria strains obtained from agencies of the World Federation of Culture Collection (WFCC) may be used by agreement between the interested parties. The strains used in the test shall be documented with their supply source.

31.6 Consistent and accurate testing requires maintenance of a pure, uncontaminated, non-mutant test culture. Avoid contamination by using good sterile technique in plating and transferring. Avoid mutation by strict adherence to monthly stock transfers. Check culture purity by making streak plates periodically and observing for single species-characteristic type of colonies.

31.7 Suitable wetting agents include dioctyl sodium sulfosuccinate, N-methyl-tauride derivatives, Polysorbate 80, Triton X-100.

31.8 The following are examples of ingredients and their concentrations that may be added to the culture medium to neutralize inhibitory substances present in the sample: soy lecithin, 0.5%; and Polysorbate™ 20 and 80, 4.0%

31.9 Peeler, J. T., J. W. Leslie and J. W. Messer, Replicate counting errors by analysts and bacterial colony counters. *Journal of Food Protection*, Vol. 45, 1982, pp238-240.

32. History

32.1 Editorially revised in 2023 to fix section formatting.

32.2 Revised in 2022; editorially revised 2019; reaffirmed 2016; revised 2011, 2010, 2004; reaffirmed 1998; editorially revised and reaffirmed 1993; revised 1992.

32.3 Developed in 1991 by AATCC Committee RA31.

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Test Method for Stain Resistance: Pile Floor Coverings

Developed in 1991 by AATCC Committee RA57; reaffirmed 1992; reaffirmed and editorially revised 1993, 1998, 2003, 2019; editorially revised 2006, 2016; reaffirmed and editorially revised with technical correction 2008; revised 2013.

1. Purpose and Scope

1.1 This test method is used to determine the resistance to staining by acid food colors on pile floor coverings.

2. Principle

2.1 A specimen of pile floor covering is stained with a small volume of a diluted aqueous solution of Food Drug & Cosmetic (FD&C) Red 40 adjusted to an acid pH. After allowing the stained specimen to remain at controlled conditions for 24 ± 4 h, it is rinsed in water to remove all unused FD&C Red 40 dye. Any residual stain is assessed after drying.

3. Terminology

3.1 **stain,** n.—*for pile floor covering*, a discoloration due to a color adding material, such as food or liquid, that exhibits resistance to removal by standard cleaning methods.

3.2 **stainblocker,** n.—a chemical substance which, when applied to a textile substrate, imparts partial or total resistance to staining.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in the handling of materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

5. Apparatus and Materials

5.1 AATCC stain cup and 50 mm (2.0 in.) diam staining ring (see 12.1 and Fig. 1)

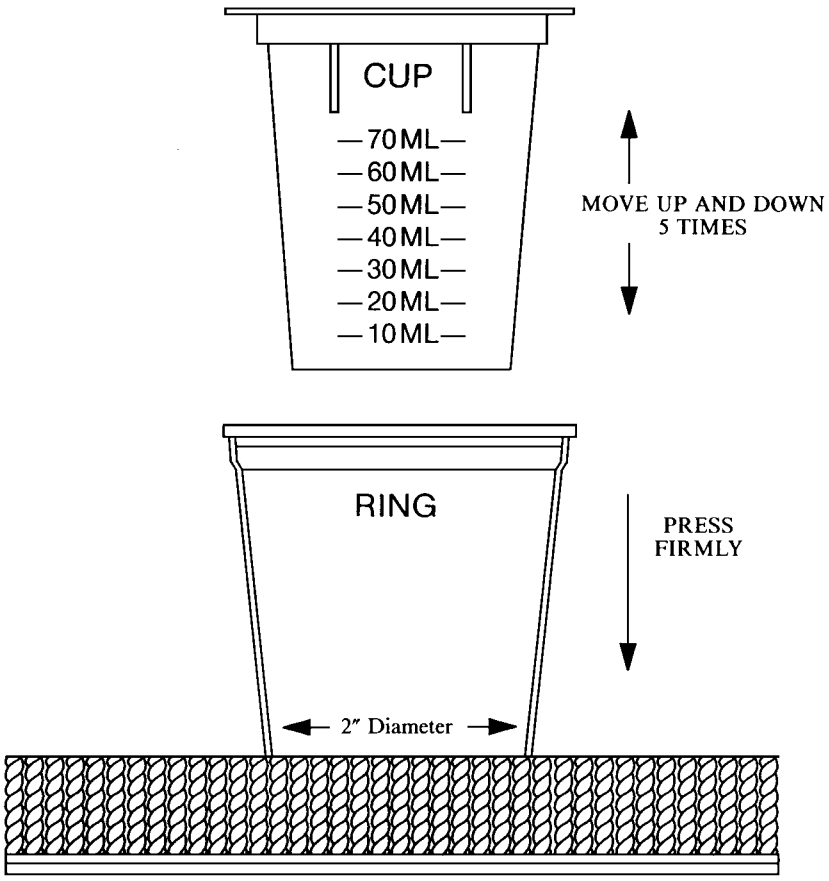


Fig. 1—AATCC stain cup and ring.

5.2 AATCC Red 40 Stain Scale (see 12.1)

5.3 FD&C Red 40 (C.I. Food Red 17) (see 12.1)

5.4 Citric acid (technical grade or better)

5.5 Deionized or distilled water

5.6 pH meter

5.7 Buffer solutions, pH 2.0 and 4.0

6. Staining Solution

6.1 Weigh 100 ± 1 mg of FD&C Red 40 and dissolve in 1 L ± 10 mL of distilled or deionized water at 24 ± 3°C (75 ± 5°F).

6.2 Adjust the pH of this solution to 2.8 ± 0.1 with citric acid (approximately 3.2 g). Check the pH of the solution with a meter that has been calibrated with pH 2.0 and 4.0 buffer solutions. Due to their lack of precision, the use of pH strips is not advised. If the pH of the prepared staining solution is below 2.7 discard this solution and prepare a new one.

7. Test Specimen

7.1 Cut one specimen at least 150 mm (6 in.) square for each test to be run.

7.2 Remove any extraneous surface materials by brushing and or vacuuming.

8. Procedure

8.1 Condition all test specimens, pile side up, on a nonabsorbent surface for 24 h at standard conditions of 21 ± 2°C (70 ± 4°F) and 65 ± 5% RH. Avoid contamination by contact with extraneous materials (see 12.2).

8.2 Place the 50 mm (2 in.) diameter staining ring in the center of the test specimen. While pressing down on the ring pour 20 mL of the staining solution into the center of the ring. Press the bottom of the stain cup through the ring into the pile, moving it up and down 5 times to facilitate wetting of the pile fiber from top to bottom. Do not rotate or twist the cup inside the ring since this may result in

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surface abrasion which may remove the stainblocker from the fiber surface. Carefully remove the staining ring and stain cup.

8.3 Recondition the stained specimen for 24 ± 4 h in a flat position, pile side up, at standard conditions of 21 ± 2°C (70 ± 4°F) and 65 ± 5% RH (see 12.2). Avoid any air draft which could result in an accelerated drying of the stained surface.

8.4 Rinse the stained specimen with running water at 21 ± 6°C (70 ± 10°F) until the rinse water is clear which indicates removal of all unfixed red dye. Make sure to rinse the backing thoroughly for removal of red dye. Removal of unfixed surface red dye is enhanced by squeezing the specimen in the rinse water.

8.5 Extract (centrifuge or vacuum) the wet specimen to remove excess water.

8.6 Oven dry the test specimen in a flat position, pile side up, at 100 ± 5°C (212 ± 9°F) for 90 min maximum or allow to air dry. NOTE: Excessive time in the oven can cause discoloration of the specimen.

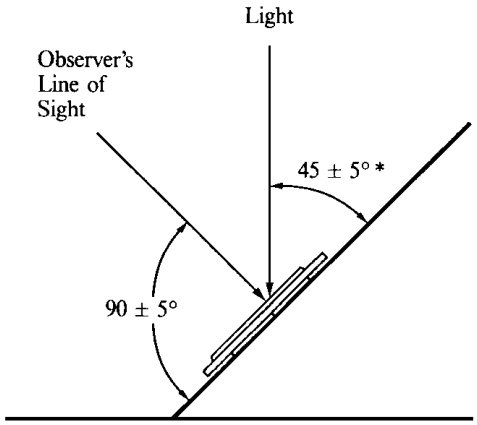
8.7 If red dye wicks to the surface during drying, repeat 8.4, 8.5 and 8.6.

9. Evaluation of Results

9.1 Rate the stain resistance of the test specimen by use of the AATCC Red 40 Stain Scale, where a grade of 10 is no staining and 1 is severely stained (see 12.3).

9.2 Rotate the specimen and/or gently brush the pile to produce the maximum display of visual red stain. Place the Red 40 Stain Scale over the test specimen with the stained area of the test specimen between the two reference columns and the non-stained portion (original) of the test specimen directly under the numbered colored film square.

9.3 Illuminate the surfaces with north sky light or equivalent source with illumination of 538 lx (50 lumens per square foot) or more. The light should be incident upon the surfaces at 45 ± 5° and the direction of viewing 90 ± 5° to the plane of the surfaces (see Fig. 2). Those angles of incident light and viewing should minimize any gloss from the scale. If necessary, alter the angle of illumination to eliminate direct reflection of the incident light from the scale surface (see 12.4).



*Adjust this angle to reduce light reflection (gloss) from the staining scale

Fig. 2—Illumination and viewing angles for rating samples.

9.4 Compare the color of the stained portion of the test specimen to that of the closest numbered colored square of the Stain Scale. Record the Stain Scale number that most closely matches the specimen's stained area (see 12.3). Interpolation of Stain Scale numbers is not recommended.

10. Report

10.1 Report the numerical grade for stain resistance as determined in Section 9.

10.2 Report the presence of staining at the base of the pile floor covering.

10.3 Report any deviation from the test procedure such as lighting conditions, angle of viewing, temperature, RH, etc.

11. Precision and Bias

11.1 Three interlaboratory tests were conducted in 1989 and 1990. The results of these tests indicate that the within-lab method precision standard deviation is 0.5 stain units. The within-lab 95% confidence interval is ± 1. There were significant level differences between labs related to the application of the staining agent, so the total method precision sigma (including both the between and within-lab components) is 1. Therefore the between-lab 95% confidence interval is ± 2.

11.2 No statement of bias can be made because the true stain value cannot be established by an independent method.

12. Notes

12.1 AATCC Red 40 Stain Scale is available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1. 919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

12.2 The presence of vapors and fumes from a strong base; i.e., ammonia, etc., in the test environment can increase the severity of staining.

12.3 Any test specimen which has a change in color or contrast decidedly greater than a grade of 1 should be given a grade a 0.

12.4 Reflection of the incident light from the scale surface will make it difficult to properly evaluate the test specimen. The problem generally occurs when either the light source is too close to the scale or is too bright. These conditions should be avoided.

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Test Method for Speckiness of Colorant Dispersions

Developed in 1992 by AATCC Committee RA87; reaffirmed 1993, 2006, 2017; editorially revised 1994, 2004, 2008, 2009, 2010, 2013, 2019; revised 1995, 1996 (with title change); editorially revised and reaffirmed 2001, 2011. Technically equivalent to ISO 105-Z11.

1. Purpose and Scope

1.1 Agglomerates in colorant dispersions may become apparent as specks on a continuously dyed (padded) or on a printed fabric, especially when pale and light shades are produced.

1.2 This test method is used to determine speckiness, primarily of disperse dye, vat dye and pigment dispersions.

2. Principle

2.1 A dispersion of a dye is filtered through a polyester/cotton blend fabric.

2.2 The degree of speckiness is assessed visually.

3. Terminology

3.1 **disperse dye,** n.—an essentially water insoluble dye having affinity, when properly dispersed, for polyester, polyamide and some other manufactured polymeric fibers.

3.2 **dispersion,** n.—*in textile wet processing*, a suspension of very fine particles in a liquid phase.

3.3 **pigment,** n.—a colorant in particulate form which is insoluble in the substrate, but which can be dispersed in the substrate to modify its color.

3.4 **speck,** n.—a small particle, such as an agglomerate in a liquid dispersion, or a very small spot of dark color on a dyed substrate.

3.5 **speckiness,** n.—*in textile dyeing and printing*, the quality or state of containing specks.

3.6 **vat dye,** n.—a water-insoluble dye, usually containing keto groups, which is normally applied to the fiber from an alkaline aqueous solution of the reduced enol (leuco) form which is subsequently oxidized in the fiber to the insoluble form.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 Use heat resistant gloves when working at hot air ovens and thermofixation devices.

4.4 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 12.1).

5. Apparatus and Materials

5.1 Filter cloth: 65/35 polyester/cotton, bleached woven broadcloth (shirting weight), approx. 240 × 240 mm square. A fabric of similar construction and of varying blend ratios may be used, but it must then be stated in the report.

5.2 Funnel, Buchner, polypropylene, 110 mm diameter. Cut out the perforated bottom smoothly and even with the rim using a knife or other suitable implement. Both parts are used in the test.

5.3 Filtering flask, heavy wall with side tabulation, 2 L.

5.4 Rubber stopper, 1 hole, to fit the filtering flask.

5.5 Stirrer, small propeller type, diameter approx. 20 mm, speed up to 2,200 rpm.

5.6 Dishes, evaporating, glass or porcelain, 1 L (approx. 150 mm diameter), three, for testing vat dyes.

5.7 Ovens.

5.7.1 Drying, without air circulation.

5.7.2 Thermofixation.

6. Reagents (only for testing vat dyes)

6.1 Sodium hydroxide (NaOH) (30%).

6.2 Sodium hydrosulfite conc. powder ($Na_2S_2O_4$).

6.3 Hydrogen peroxide (H_2O_2) (30%).

6.4 Acetic acid (CH_3COOH) (80%).

7. Preparation of Liquid Dispersions for Sampling

7.1 On prolonged storage, liquid dispersions tend to settle and may develop a more or less tacky sediment. It is imperative to ensure complete homogeneity of the dispersion prior to testing. Thoroughly mix dispersions in drums using a mechanical mixing device, such as a propeller stirrer or homogenizer. The liquid must be smooth and free of sediment and lumps before removing a sample for testing. Thoroughly shake laboratory samples to ensure that all dried material adhering to the cover and sides of the container is re-incorporated into the liquid. Then remove the cover and stir the sample either mechanically or by hand until all sediment and/or lumps are completely redispersed. Replace the cover and again shake the sample to ensure complete homogeneity. Test the sample immediately after homogenization. When storing the sample for future use, thoroughly clean the lid and the container lip prior to closure.

8. Procedure

8.1 Identify the filter cloth with the experiment or sample number in one of the corners and ensure that it is free of extraneous specks.

8.2 Clean and dry the funnel. Prepare the funnel assembly by placing the top part of the funnel upside down on a clean surface and drape the fabric over it as flat as possible with the identification mark facing down towards the surface. The side of the fabric with the identification will then be on top during filtration and is used for subsequent evaluation (see 9.1). Snap and press the bottom half of the funnel to the top half, causing the fabric to become a tight, smooth filter in the funnel.

8.3 Place the assembled funnel straight on the filtering flask using the rubber stopper to ensure that the funnel remains upright (and the filter fabric horizontal) during filtration and rinsing.

8.4 Weigh 7.5 ± 0.075 g of 100% powder or 15.0 ± 0.15 g of 50% liquid (or an equivalent weight of other concentrations) in a weighing cup and transfer to a 400 mL graduated beaker containing approx. 200 mL deionized or distilled water at room temperature. Rinse the weighing cup with water from a spray bottle.

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8.5 With powder and granular forms: stir 3 min with the propeller of the stirrer in the center and just above the container bottom. Adjust the stirrer and speed to create a vortex terminating at the top of the propeller.

8.6 With liquid forms: stir as in 8.5 except for 30 s.

8.7 After stirring, transfer the dispersion to the 1 L beaker. Rinse the 400 mL beaker with 200 mL deionized or distilled water at room temperature and pour the rinse water into the 1 L beaker. Dilute the dispersion further to 800 mL.

8.8 Pre-wet the fabric in the filter with 200 mL water.

8.9 Stir the dispersion in the beaker for approx. 30 s and pour it into the funnel.

8.10 Rinse the beaker with 200 mL water and pour it into the funnel.

8.11 Rinse the funnel with an additional 200 mL water and let stand for about 1 min until dripping has completely stopped.

8.12 Carefully remove the filter cloth from the funnel and place it on a blotting paper to remove excess water.

8.13 For disperse dyes:

8.13.1 Dry the fabric in an oven without air circulation at 80 ± 5°C (176 ± 9°F).

8.13.2 Thermosol (heat) for 60 s at 210-220°C (410-428°F).

8.14 For vat dyes proceed as follows:

8.14.1 Prepare 400 mL of a fresh reducing solution in an evaporating dish (see 12.2) at 60-70°C (140-158°F) containing 15.7 mL/L sodium hydroxide, 50%, and 20 g/L sodium hydrosulfite conc. powder (88-92%).

8.14.2 Fully immerse the filter cloth in the reducing solution in the evaporating dish for 5 min. Do not move the cloth during this time.

8.14.3 Immerse the filter cloth for 1 min without agitation in an evaporating dish containing deionized or distilled water at 15-25°C (59-77°F).

8.14.4 Prepare 100 mL of an oxidizing solution in an evaporating dish at 40-50°C (104-122°F) containing 10 mL/L hydrogen peroxide, 30%.

8.14.5 Fully immerse the filter cloth in the oxidizing solution in the evaporating dish for 2 min. Do not move the cloth during this time.

8.14.6 Neutralize the filter cloth for 2 min at 15-25°C (59-77°F) with 200 mL/L acetic acid, 80%, in a 400 mL beaker.

8.14.7 Rinse for approx. 30 s in cold running water.

8.14.8 Dry the fabric in an oven at 80 ± 5°C (176 ± 9°F).

8.15 For pigments:

8.15.1 Dry the fabric in an oven without air circulation at 80 ± 5°C (176 ± 9°F).

9. Evaluation

9.1 Visually examine the fabric for specks on the side with the identification mark. Count individual specks. Base the acceptability on user requirements; for example, acceptable, marginal, unacceptable.

10. Report

10.1 Report the dyestuff tested and weight used, the number of specks and the speckiness rating.

11. Precision and Bias

11.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this method, use standard statistical techniques in making any comparisons of test results for either comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

11.2 *Bias*. Speckiness of colorant dispersions can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

12. Notes

12.1 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

12.2 The reason for using evaporating dishes for the reduction and oxidation of vat dyes is to keep the fabric flat and without agitation. This prevents any specks from dissolving and leveling.

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Test Method for Skew Change in Fabrics After Home Laundering

Foreword

AATCC TM179 is used to evaluate the *change* in fabric skew *after* home laundering. AATCC TM207 is for evaluation of seam twist *before and after* home laundering. ASTM D3882 is only used to evaluate fabric skew in its original (*unlaundered*) state. The relationship among results of these methods is unknown; results of one method should not be compared to those of another method.

1. Purpose

1.1 This test method is used to determine change in skew of fabrics as a result of home laundering.

1.2 This test is applicable to all woven and knitted fabrics suitable for home laundering.

2. Principle

2.1 Marks are applied to fabric or garment specimens before laundering. The alignment of marks is measured after 3 standard laundering cycles. Percent skew is calculated and direction of skew recorded. A value close to zero indicates little or no skew; a higher value indicates greater skew; the calculations in this method do not produce negative skew values.

3. Referenced Documents

3.1 AATCC LP1, Home Laundering: Machine Washing (see 15.1).

3.2 AATCC LP2, Home Laundering: Hand Washing (see 15.1).

3.3 AATCC TM124, Smoothness Appearance of Fabrics after Home Laundering (see 15.1).

3.4 AATCC TM133, Colorfastness to Heat: Hot Pressing (see 15.1).

3.5 AATCC TM135, Dimensional Changes of Fabrics after Home Laundering (see 15.1).

3.6 AATCC TM143, Appearance of Apparel and Other Textile End Products after Home Laundering (see 15.1).

3.7 AATCC TM150, Dimensional Changes of Garments after Home Laundering (see 15.1).

3.8 AATCC TM207, Seam Twist in Garments Before and After Home Laundering (see 15.1).

3.9 ASTM D123, Standard Terminology Relating to Textiles (see 15.2).

3.10 ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 15.2).

3.11 ASTM D3882, Standard Test Method for Bow and Skew of Woven and Knitted Fabrics (see 15.2).

3.12 ISO 6330, Textiles—Domestic washing and drying procedures for textile testing (see 15.3).

4. Terminology

4.1 **laundering,** n.—*of textile materials*, a process intended to remove soils and/or stains by treatment (washing) with an aqueous detergent solution and normally including rinsing, extraction and drying.

4.2 **skew,** n.—a fabric condition resulting when filling yarns or knitted courses are angularly displaced from a line perpendicular to the edge or side of the fabric (see ASTM D123).

5. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

5.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5.2 Detergent may cause irritation. Care should be taken to prevent exposure to skin and eyes.

5.3 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

6. Apparatus and Materials (see 15.4)

6.1 Conditioning facilities and conditioning/drying racks with pull-out, perforated shelves (see 15.1).

6.2 Indelible ink marking pen (see 15.1).

6.3 Right-angle marking device (triangle, L-square or template).

6.4 Tape or rule marked in millimeters, tenths of an inch or smaller increments.

6.5 Standard washing machine (see Table I and 15.5).

6.6 AATCC 1993 Standard Reference Detergent (powder, with brightener) (see 15.1).

6.7 Balance with at least 5.0 kg (10.0 lb) capacity.

6.8 Laundering ballast, Type 1 or Type 3 (see Table IV).

6.9 Standard tumble dryer (see Table III and 15.5), or facilities for drip/line drying.

6.10 Digital Imaging System (optional, see 15.6).

7. Test Specimens

7.1 Test Specimens from Fabric Samples.

7.1.1 Consider each roll of fabric to be a sample. Randomly select rolls from each production lot as agreed.

7.1.2 Test 3 specimens from each sample. Cut specimens for Marking Method 1 at least 380 × 380 mm (15 × 15 in.) Cut specimens for Marking Method 2 at least 380 × 660 mm (15 × 26 in.)

7.1.3 Where possible, cut specimens from different areas of the fabric so that each specimen will contain different groups of lengthwise and widthwise yarns. Identify fabric face; mark lengthwise direction on each specimen.

7.1.4 If it is of interest to determine whether the magnitude of the skew change is different from one side of the fabric to the other, prepare 3 specimens from each area of interest. Identify area from which each set of specimens was taken; i.e., left-side, center or right-side.

7.1.5 Test specimens prepared for AATCC TM124 and/or AATCC TM135 may also be tested by this method, using Marking Method 1 (see 15.7), or Marking Method 2, provided specimens are cut large enough to accommodate the required markings.

7.2 Test Specimens from Garment Samples.

7.2.1 Consider each garment to be a sample. Randomly select garments from each production lot as agreed.

7.2.2 Test 3 garments or a total of 3 panels from 2 garments. Use the largest panels in the garments. Consider each tested panel to be a specimen.

7.2.3 Test specimens prepared for AATCC TM143, AATCC TM150, and/or AATCC TM207 may also be tested by this method, using Marking Method 1 or 2.

8. Conditioning

8.1 Prior to marking, condition test specimens as directed in ASTM D1776. Use conditions indicated in Table 1 for Textiles, general and conditioning time as indicated in Table 2 for the appropriate fiber content.

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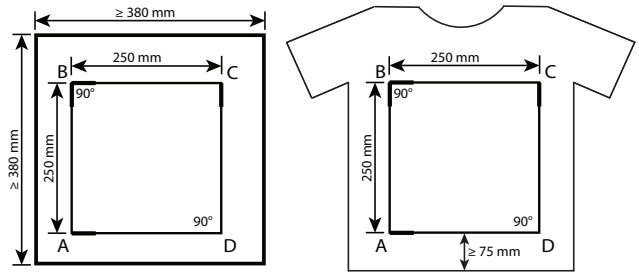


Fig. 1—Marking Method 1 for fabric (left) and garment (right).

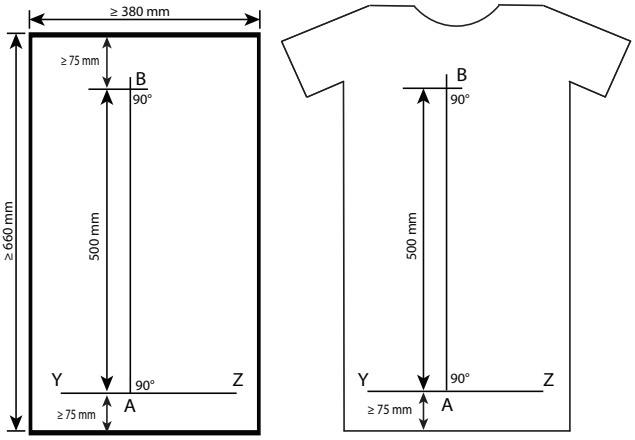


Fig. 2—Marking Method 2 for fabric (left) and garment (right).

8.1.1 Lay specimens flat, separately, on a screen or perforated shelf of a conditioning rack. For garments normally stored hung, hang separately on hangers to condition. Allow space for air movement on all sides of the garment.

8.2 Specimens will be conditioned again after laundering.

9. Marking of Specimens

9.1 Marking Method 1 (see 15.8 and Fig. 1).

9.1.1 Mark specimens without tension on a flat, smooth, horizontal surface.

9.1.2 On each specimen, mark 250-mm (10-in.) pairs of benchmarks parallel to the length of the specimen and 250-mm (10-in.) pairs of benchmarks perpendicular to the length of the specimen. Indicate any alternate benchmark distances in the test report (see 15.9).

9.1.3 Extend and connect benchmarks so that a square is formed. Label corners as *A*, *B*, *C* and *D* in clockwise direction starting with lower left corner.

9.1.4 For circular knitted fabrics to be used in tubular form (e.g., underwear, sweatshirts, etc.), test in tubular form. For others, slit tube and test in open-width form.

9.2 Marking Method 2 (see 15.8 and Fig. 2).

9.2.1 Mark specimens without tension on a flat, smooth, horizontal surface.

9.2.2 On each specimen, mark reference line *YZ* perpendicular to the length of the specimen, at least 75 mm (3 in.) above the bottom edge.

9.2.3 Mark *A* midway on *YZ*.

9.2.4 Place one side of a right-angle marking device along *YZ* so that the perpendicular side extends upward from point *A*. Mark a line along this side of the right-angle marking device, upward from point *A*, perpendicular to *YZ*, at least 500 mm (20-in.) long.

9.2.5 Mark point *B* along the second line, 500 mm (20-in) above point *A*. If the specimen size is insufficient to mark a 500-mm (20-in.) length, mark *B* at the farthest available point along the line that is at least 75 mm (3 in.) below the upper edge of the test specimen. Indicate any alternate distance in the test report.

9.2.6 or circular knitted fabrics to be used in tubular form (e.g., underwear, sweatshirts, etc.), test in tubular form. For others, slit tube and test in open-width form.

10. Laundering Procedure

10.1 Washing.

10.1.1 Select washing conditions for testing from Table I. Set washing machine controls to generate the selected cycle parameters.

10.1.2 Load for laundering shall consist of all specimens in the sample, plus enough laundering ballast pieces to make a total load weight of 1.8 ± 0.1 kg (4.0 ± 0.2 lb).

10.1.3 Begin selected wash cycle. Allow machine to fill to specified water level.

10.1.4 Add 66 ± 1 g of AATCC 1993 Standard Reference Detergent to washing machine as directed by washing machine manufacturer's instructions. If detergent is added directly to wash water, agitate briefly to dissolve completely. Stop agitation before adding wash load.

10.1.5 Add wash load (test specimens and ballast), distributing evenly around center agitator. Restart wash cycle.

10.1.6 For specimens to be drip dried (drying procedure C), stop wash cycle before the water begins to drain from the final rinse cycle. Remove specimens soaking wet. For specimens to be tumble (A), line (B), or screen/flat (D) dried, allow washing to proceed through the final spin cycle.

10.1.7 After each washing cycle, sepa-

Table I—Standard Washing Machine Parameters (see 15.5, 15.10)

Cycle	(1) Normal		(2) Delicate		(3) Permanent Press	
Water Level, L (gal)	72 ± 4 (19 ± 1)		72 ± 4 (19 ± 1)		72 ± 4 (19 ± 1)	
Agitation Speed, strokes/min.	86 ± 2		27 ± 2		86 ± 2	
Washing Time, min.	16 ± 1		8.5 ± 1		12 ± 1	
Final Spin Speed, rpm	660 ± 15		500 ± 15		500 ± 15	
Final Spin Time, min.	5 ± 1		5 ± 1		5 ± 1	
Wash Temp, °C (°F)[1]	(II) Cold:	27 ± 3 (80 ± 5)	(II) Cold:	27 ± 3 (80 ± 5)	(II) Cold:	27 ± 3 (80 ± 5)
	(III) Warm:	41 ± 3 (105 ± 5)	(III) Warm:	41 ± 3 (105 ± 5)	(III) Warm:	41 ± 3 (105 ± 5)
	(IV) Hot:	49 ± 3 (120 ± 5)	(IV) Hot:	49 ± 3 (120 ± 5)	(IV) Hot:	49 ± 3 (120 ± 5)
	(V) Very Hot:	60 ± 3 (140 ± 5)	(V) Very Hot:	60 ± 3 (140 ± 5)	(V) Very Hot:	60 ± 3 (140 ± 5)

[1] Due to US Department of Energy requirements, many washing machines use cooler water. An external control box may be used to override the machine set temperatures.

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Table II—Standard Drying Conditions

(A)	Tumble Dry
(Ai)	Tumble Dry Normal
(Aii	Tumble Dry Delicate
(Aiii)	Tumble Dry Permanent Press
(B)	Line/Hang Dry
(C)	Drip Dry
(D)	Screen Dry/Dry Flat

Table III—Standard Tumble Dryer Parameters (see 15.5)

Cycle	(Ai) Normal	(Aii) Delicate	(Aiii) Permanent Press
Max. Exhaust Temp, °C (°F)	68 ± 6 (155 ± 10)	60 ± 6 (140 ± 10)	68 ± 6 (155 ± 10)
Cool Down Time, min.	≤ 10	≤ 10	≤ 10

Table IV—Laundering Ballast Parameters

	Type 1	Type 3
Fiber Content	100% cotton	50% cotton/50% polyester ± 3%
Greige Fabric Yarns	16/1 ring spun	16/1 or 30/2 ring spun
Greige Fabric Construction	52 × 48 ± 5 yarns/in., plain weave	52 × 48 ± 5 yarns/in., plain weave
Finished Fabric Weight	155 ± 10 g/m^2	155 ± 10 g/m^2
Edges	All edges hemmed or over-edged	All edges hemmed or over-edged
Finished Piece Size	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)
Finished Piece Weight	130 ± 10 g	130 ± 10 g

rate tangled specimens and ballast pieces, taking care to minimize distortion. Proceed to appropriate drying procedure.

10.2 Drying.

10.2.1 Select drying conditions from Table II.

10.2.2 (A) Tumble dry. Place the washed load (test specimens and ballast) in the tumble dryer, and set the temperature control to generate the selected exhaust temperature (see Table III). Allow the dryer to run until the load is fully dry. Remove specimens immediately.

10.2.3 (B) Line/hang dry. Hang each specimen on an appropriate hanger, straightening and smoothing facings, seams, etc. Specimens should generally be oriented in the direction in which they are worn. Do not fold or stretch specimens over the hanger. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

10.2.4 (C) Drip dry. Hang dripping wet specimen on an appropriate hanger, straightening and smoothing facings, seams, etc. Specimens should generally be oriented in the direction in which they are worn. Do not fold or stretch specimens over the hanger. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

10.2.5 (D) Screen/flat dry. Spread each specimen on a horizontal screen or perforated surface, removing wrinkles without distorting or stretching it. Allow specimens to rest in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

10.2.6 For all drying methods, allow specimens to dry completely before washing again.

10.2.7 Repeat the selected washing and drying procedure for a total of 3 complete cycles, or an agreed upon number of cycles.

10.3 Conditioning.

10.3.1 After final drying cycle is complete, condition test specimens as described in section 8.

10.4 Ironing (optional).

10.4.1 The wrinkles in most fabrics flatten sufficiently under pressure of a measuring instrument at the time of measurement, so as not to cause measurement error. If the specimens are extremely wrinkled and the consumer would ALWAYS expect to iron a garment made from the fabric, test specimens may be hand ironed prior to measurement.

10.4.2 Use safe ironing temperatures appropriate to the fibers in the fabric being ironed. See TM133, Table I, Safe Ironing Temperature Guide (see 15.1). Exert only the minimum pressure necessary to remove wrinkles.

10.4.3 Due to the extremely high variability of hand ironing procedures performed by individual operators (no standard test method exists for hand ironing procedures), the reproducibility of dimensional change results after hand ironing has been found to be extremely poor. Consequently, caution is advised when comparing dimensional change results after laundering and hand ironing, reported by different operators.

10.4.4 After ironing, condition test specimens as described in 8.1.

11. Measurement

11.1 After conditioning, lay each test specimen without tension on the same surface used for marking in section 9. Use the same measuring device as for initial marking and measurement.

11.1.1 Calculation Option 1 or 2 can be used with Marking Method 1. Calculation Option 3 should be used with Marking Method 2.

11.2 Marking Method 1, Calculation Option 1 (see Fig. 3a).

11.2.1 Measure and record the diagonal length of *AC* and *BD* after laundering, to the nearest millimeter, or eighth or tenth of an inch (see Fig. 3a).

11.3 Marking Method 1, Calculation Option 2 (see Fig. 3b).

11.3.1 Extend line *AD* in each direction across the width of the specimen.

11.3.2 Place one side of a right-angle marking device along line *AD* so that the perpendicular side extends to point *B*. Mark point *A* where this side of the right-angle marking device meets line *AD*.

11.3.3 Next place one side of the right-angle marking device along line *AD* so that the perpendicular side extends to point *C*. Mark *D* where this side of the right-angle marking device meets line *AD*

11.3.4 Measure and record the length of *AA*, *DD*, *AB*, and *CD* after laundering, to the nearest millimeter, or eighth or tenth of an inch.

11.4 Marking Method 2, Calculation Option 3 (see Fig. 3c).

11.4.1 Place one side of a right-angle marking device along line *YZ* so that the perpendicular side extends to point *B*. Mark point *A* where this side of the right-angle marking device meets line *YZ*.

11.4.2 Measure and record the length of *AA* and *AB* after laundering, to the nearest millimeter, or eighth or tenth of an inch.

11.5 Point *A*, for Calculation Options 2 and 3, plus *D*, for Calculation Option 2, are marked after 3 (or an agreed upon number of) washings. Points after subsequent washings should be indicated by numbers or other symbols to differentiate measurements (e.g., A).

12. Calculation

12.1 Skew Change Calculation Option 1 (see Fig. 3a).

12.1.1 Calculate percent change in skew to the nearest 0.1% using equation 1,

$$X = \left|100 \times [2\ (AC - BD)/(AC + BD)]\right| \quad \text{(Eq. 1)}$$

where:

X = % change in skew

12.2 Skew Change Calculation Option 2 (see Fig. 3b).

12.2.1 Calculate percent change in skew to the nearest 0.1% using equation 2,

$$X = 100 \times [(AA' + DD')/(AB + CD)] \quad \text{(Eq. 2)}$$

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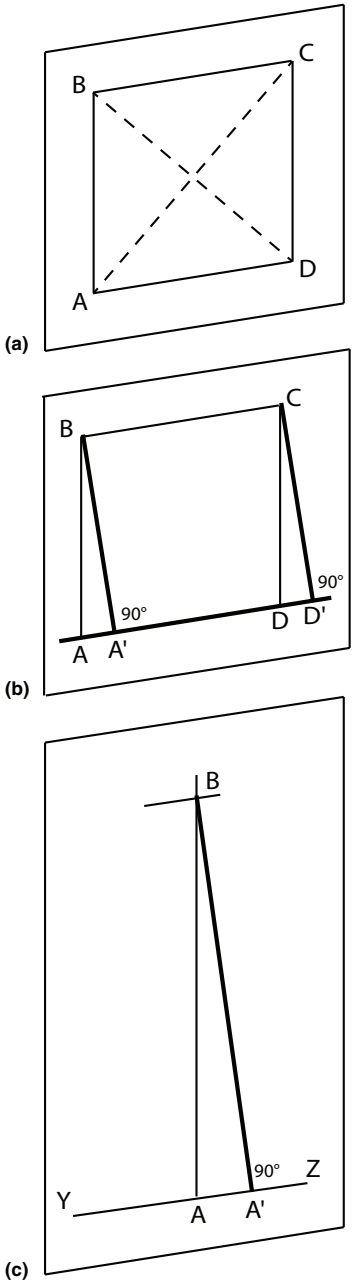


Fig. 3—Final (after laundering) markings for measurement and skew Calculation Option 1 (a), Calculation Option 2 (b), and Calculation Option 3 (c).

where:

X = % change in skew

12.3 Skew Change Calculation Option 3 (see Fig. 3c).

12.3.1 Calculate percent change in skew to nearest 0.1% using equation 3,

$$X = 100 \times (AA'/AB) \quad \text{(Eq. 3)}$$

where:

X = % change in skew

12.4 The calculation of Option 1 is related on a mathematical basis to the calculations of Options 2 and 3; however, when comparing data between laboratories, the same option should be used for calculations.

12.5 Average the results obtained from equation 1, 2 or 3, for all specimens in the sample. Note that results of equation 1 are reported as absolute values; there should be no negative values included in the average.

12.5.1 If the test is used to establish the condition of the fabric across the width (see 7.1.4), calculate averages for each area separately.

12.6 Interpretation of Skew Change.

12.6.1 If the skew change for fabric or garment specimens after 1 full laundering cycle is within a specification previously agreed on, continue until an agreed upon number of cycles have been completed.

12.6.2 If the change in skew exceeds a specification previously agreed on, terminate the test.

13. Report

13.1 Report for each sample tested:

13.1.1 A description or identification of the sample.

13.1.2 Evaluation was performed using AATCC TM179-2019.

13.1.3 Type of ballast used, i.e., Type 1 or Type 3.

13.1.4 Number of laundering cycles (default is 3 cycles, see 10.2.7).

13.1.5 Laundering conditions, including washing machine cycle, wash temperature, drying procedure and tumble-drying temperature, if applicable. Alphanumeric designations may be used if they are clearly understood by all parties. For example, 1-IV-A(ii) indicates normal wash cycle at 49°C and Delicate tumble dry cycle.

13.1.6 If specimens were distorted or wrinkled in their original state.

13.1.7 If specimens were hand ironed.

13.1.8 Marking method used (Method 1 or 2)

13.1.9 Calculation option used (Option 1, 2 or 3)

13.1.10 Number of specimens tested (default is 3).

13.1.11 Average skew change, to the nearest 0.1%, for the sample.

13.1.12 Average skew change, to the nearest 0.1%, for each area evaluated (e.g., left-side, center, right-side), if applicable.

13.1.13 Any modifications of the test method, including alternate benchmark distances.

13.1.14 Optionally, the direction of skew (left or right) may be reported.

14. Precision and Bias

14.1 *Precision*. Four different laboratories participated in an interlaboratory study to measure fabric skew change after home laundering. All 4 labs measured the skew change of 3 different garments (jersey leggings, jersey T-shirt, denim jeans), following this test method. Each lab used only 1 operator and repeated the test 2 times, on separate days. Laundering was performed in traditional top-loading washing machines with 1993 AATCC Standard Reference Detergent (powder), and tumble dryers. Precision of tests performed with other laundering conditions has not been determined.

14.1.1 Table V gives the within-lab and between-lab precision values for this study, using Marking Method 1 and Calculation Option 1 on jersey T-shirt samples.

14.1.2 Table VI gives the within-lab and between-lab precision values for this study, using Marking Method 1 and Calculation Option 2 on jersey T-shirt samples.

14.1.3 Table VII gives the within-lab and between-lab precision values for this study, using Marking Method 2 and Calculation Option 3 on jersey leggings and denim jeans samples.

14.2 *Bias*. Skew resulting from home

Table V—Precision Values for Marking Method 1, Calculation Option 1

n	Within Laboratory	Between Laboratory
1	0.8900	2.1820
2	0.6303	1.5453
3	0.5138	1.2597

Note: Mean change in skew after laundering for all interlab measurements of jersey T-shirt was 3.7%.

Table VI—Precision Values for Marking Method 1, Calculation Option 2

n	Within Laboratory	Between Laboratory
1	0.9531	2.4617
2	0.6750	1.7434
3	0.5503	1.4212

Note: Mean change in skew after laundering for all interlab measurements of jersey T-shirt was 3.2%.

Table VII—Precision Values for Marking Method 1, Calculation Option 3

n	Within Laboratory	Between Laboratory
1	0.3693	1.4140
2	0.2615	1.0014
3	0.2132	0.8164

Note: Grand mean of change in skew after laundering for all interlab measurements was 1.4%. Mean change in skew for jersey leggings was 1.3%. Mean change in skew for denim jeans was 1.6%.

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laundering can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

15. Notes

15.1 Available from AATCC, PO Box 12215, Research Triangle Park, NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

15.2 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428, USA; +1.610.832.9500; www.astm.org.

15.3 Available from ISO, www.iso.org.

15.4 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

15.5 For model number(s) of washer(s) and tumble dryers(s) reported to meet the standard parameters, visit www.aatcc.org/test/washers or contact AATCC, PO Box 12215, Research Triangle Park, NC 27709, USA; +1.919.549.8141; www.aatcc.org. An alternate load size was described in previous versions of this method (3.6-kg load, 83-L water level, 80 g AATCC 1993 Standard Reference Detergent), but no machine is reported to meet the standard parameters for this load size. In addition, skew change results obtained with the alternate load size may not be equal to those obtained with a standard load size.

15.6 A digital imaging system may be used as a measuring device in place of the prescribed manual measurement devices if it is established that its accuracy is equivalent to the manual devices.

15.7 When using specimens for AATCC TM135, extend corner markings to form right angles.

15.8 Select only one marking method and one calculation option for each test. Marking Method 2 may be more suitable for long, narrow specimens (e.g., pant leg panels).

15.9 Marks of 460 mm (18 in.) are commonly used on fabric specimens to give better measurement precision. If alternate size specimens or benchmarks are used, skew change results may not be equal to those obtained with 250-mm (10-in.) marks.

15.10 The laundering temperatures and other parameters listed in this test method are standard conditions for testing purposes. As with most lab procedures, they represent, but may not exactly replicate, current consumer practices. Consumer practices vary over time and among households; lab practices must be consistent to allow valid comparison of results. If laundering equipment or conditions other than those specifically listed in this test method are used, they must be described in detail and noted as a modification of the standard method. Alternate laundering conditions are outlined in AATCC LP1, AATCC LP2, and ISO 6330.

15.11 Skew direction is based on the apparent shift of the bottom edge. Optionally, the direction of skew may be reported.

16. History

16.1 Revised in 2023 to add that optionally, the direction of skew may be reported.

16.2 Revised 2019; revised 2017 (with title change); editorially revised 2014; revised 2012, 2010; editorially revised 2009, 2008, 2005; revised 2004; editorially revised and reaffirmed 2001; reaffirmed 1996; revised 1995.

16.3 Developed in 1994 by AATCC Committee RA42.

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Test Method for Relative Color Strength of Dyes in Solution

1. Purpose and Scope

1.1 This is a test method to determine the color strength of a dye sample relative to a reference dye by means of spectrophotometric transmittance measurements on solutions of each. It is a comparison of the color strength of the dyes in a chosen solvent which may or may not correspond to strengths when applied to a textile substrate.

1.2 To be valid as a measurement of the relative color strength of a dye for a particular textile end-use, additional tests must be performed to demonstrate that the relative strength in solution is equal to the relative strength when applied to the textile substrate (see 5.4).

1.3 The test is often used by dye manufacturers and users because of its speed, repeatability and reproducibility.

2. Principle

2.1 Solutions of the test dye and the reference dye are prepared in known concentrations and their transmittance values are determined using a spectrophotometer. The relative color strength for the test dye is then calculated using absorbance and concentration values.

3. Terminology

3.1 **absorbance,** n.—the logarithm to the base 10 of the reciprocal of transmittance.

NOTE: Absorbance is proportional to the mass of absorbing material in the path length of a spectrophotometer cell (syn: *absorbance value*).

3.2 **color strength,** n.—a measure of the ability of a dye to impart color to other materials.

NOTE: Color strength is evaluated by light absorption in the visible region of the spectrum (see 13.1).

3.3 **relative color strength,** n.—*in spectrophotometric testing of dyes*, the percentage of the color strength of a sample relative to that of a reference dye assigned a color strength of 100% (see 10.2).

3.4 **transmittance,** n.—*of light*, that fraction of the incident light of a given wavelength which is not reflected or absorbed, but passes through a substance.

NOTE: In this test, the transmittance of the material is measured on a spectrophotometer, and corrected for that of pure solvent for the same path length.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in the test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 Use chemical goggles or face shield, impervious gloves, and an impervious apron during dispersing and mixing of acids, alkalis and organic solvents. Concentrated acids, alkalis and organic solvents should be handled in a well ventilated hood. CAUTION: Always add acid to water.

4.4 Acetone, methyl alcohol and ethyl alcohol are highly volatile liquids and should be stored in the laboratory only in small containers away from heat, open flames and sparks. These chemicals should not be used near an open flame.

4.5 Cellosolve (2-ethoxyethanol) is combustible and may be harmful or fatal if ingested or inhaled. It is also a reproductive hazard and should only be used when absolutely necessary.

4.6 An eyewash/safety shower should be located nearby and an organic vapor respirator and a self contained breathing apparatus should be readily available for emergency use.

4.7 Exposure to chemicals used in this procedure must be controlled at or below levels set by government authorities [e.g., Occupational Safety and Health Administration's (OSHA) permissible exposure limits (PEL) as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version]. In addition, the American Conference of Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 14.1).

5. Uses and Limitations

5.1 This method is not applicable to all dyestuffs; e.g., pigments, due to solubility and/or other limitations. Dyes that are commonly measured include acid, basic, direct and disperse. Many reactive dyes are not easily measured using this procedure (see 5.4).

5.2 Basic requirements for this method are that the dye solutions do not scatter light and do obey the Lambert-Beer or Beer-Bouguer Laws as well as having identical or similar absorption curves of the samples and the reference dyes in the visible region of the spectrum (see 13.9).

5.3 Identical or similar absorption curves are usually obtained if the tests are carried out for the purpose of dye production control of batches/deliveries of the same dye. This method is not applicable for the evaluation of dyes with distinctly different hues or strengths greater than 20% or chemical compositions.

5.4 The method may be useful in the prediction of relative dye strength in process applications; e.g., dyeing. It is generally accepted that some reasonable correlation can be obtained between application and solution measurements. Exceptions occur where there is a significant difference in the relative amount of colored but non-exhausting or non-fixing impurities between the specimen under test and the reference dye; e.g., reactive dyes. Exceptions may also occur for some reactive dyes which exhibit shade differences between their hydrolyzed and unreacted forms (see additional caution for reactive dyes in 8.3.3).

5.5 As the relative color strength is determined by a comparison with a reference dye, the reference dye is assumed to remain constant. It is essential, therefore, to ensure careful storage and control of the reference dye. Many dyes are hygroscopic and sensitive to oxidation. Protect the reference dye from exposure to light in firmly sealed containers that are impervious to moisture.

6. Apparatus and Materials

6.1 Flasks, volumetric Class A, TC.

6.2 Pipettes, volumetric Class A, TD.

6.3 Analytical balance with accuracy of weighing to 0.0005 g.

6.4 Cells for spectrophotometers, 5 mm or 10 mm path length as needed, analytical grade or optical quality. May be cuvette or flow-through cell. (Note: the use of a 5-mm path length requires that the solution concentration be doubled to achieve the equivalent absorbance.)

6.5 Spectrophotometer.

7. Reagents

7.1 Sulfuric acid—H_2SO_4.

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7.2 Acetic acid—CH_3COOH.
7.3 Sodium hydroxide—NaOH.
7.4 Sodium carbonate—Na_2CO_3.
7.5 Buffering agents.
7.6 Solvents (see 13.3).
7.6.1 For water-soluble dyes:
7.6.1.1 Water, demineralized.
7.6.2 For water-insoluble dyes:
7.6.2.1 Methyl alcohol (CH_3OH, methanol), anhydrous.
7.6.2.2 Acetone (CH_3COCH_3).
7.6.2.3 N-methyl-2-pyrrolidone.
7.6.2.4 Ethylene glycol monoethylether (2-ethoxyethanol, Cellosolve).
7.6.2.5 Mixtures of the above solvents as well as other solvents appropriate for the test dye.

8. Preparation of Dye Solutions

8.1 Storage of dyes.
8.1.1 Store the samples in tightly sealed containers to avoid erroneous results arising from changes in moisture levels. Condition powder samples in a controlled atmosphere for 4 h as described in ASTM D49, Standard Test Methods of Chemical Analysis of Red Lead.
8.2 Stock solutions.
8.2.1 Weigh an amount of not less than 0.5 g to avoid errors attributable to microinhomogeneities to the nearest 0.0005 g (see 13.4 and 13.8).
8.2.2 Transfer the dye weighed quantitatively to a volumetric flask containing approximately one-third of the total volume of solvent, then dissolve/disperse it. In general, pre-dissolve/disperse the dye in about 20 mL of the selected solvent. A mixture of solvents or an additive may be needed.
8.2.2.1 Water-soluble dyes may need to be heated to effect dissolution. If heated, cool the dissolved/dispersed mixture to room temperature (see 13.5). (Do not heat in volumetric glassware; transfer to a beaker before heating, then re-transfer to the volumetric flask.)
8.2.3 Make up the volumetric flasks to the mark and homogenize the solution/dispersion by stirring or by turning the flask end over end.
8.3 Test solutions.
8.3.1 Dilute the concentrated stock solution as directed (see 13.6) to obtain maximum absorption (minimum transmittance) within the range of 10-60% transmittance (see 13.5). (The same solvents used to make the concentrated stock solution may not be the appropriate choice for dilution. For example, many disperse dyes are dispersed into water for the stock solution, and then pipetted into a solvent/water mixture for dilution.)
8.3.2 Some additives may be recommended to improve the stability and reproducibility.
8.3.3 To adjust the pH of water soluble dyes use an acid such as acetic or sulfuric, a base such as sodium carbonate or sodium hydroxide, or a buffering agent. (However, sodium carbonate or sodium hydroxide should not be used with reactive dyes.)
8.3.4 To eliminate the influence of unwanted metal ions use a sequestering agent.
8.3.5 To prevent agglomeration in water use a surfactant; e.g., ethylene oxide adducts to alkanols.
8.3.6 Use a dispersing agent or an antioxidant, if appropriate.
8.3.7 Volumetric equipment and solvents should be at ambient temperature.
8.4 Gravimetric transfer(s) and dilution(s) may be substituted for the volumetric ones mentioned in 8.1 through 8.3 under the premise that 1 mL @STP = 1 g. The premise applies to water dilution only, as solvent dilution requires the additional calculation: *Dilution (g) = (Required) × (Specific Gravity of Solvent)*. This eliminates the cost of Class A glassware and removes the restriction of temperature dependency (although the reference and sample must be at ambient temperature prior to measurement in a cuvette or flow-cell. It also facilitates the accurate transfer of odd quantities to account for differences in strength. Use a four-place balance (0.0001 g) for dye-weights and transfers and a three-place balance for dilutions (0.001 g).

9. Procedure

9.1 When a dye specimen is measured for the first time, it is advisable to confirm the validity of the relationship (Beer's Law) under the conditions selected (concentration, solvent) within a range of concentrations (10 mg/L to 100 mg/L similar to a primary set to determine dyeing characteristics).
9.2 Determine the wavelength (λ) of maximum absorbance (minimum transmittance.
9.2.1 Quantitative measurements (analogous to HPLC, MS, etc.) of the test and reference specimens should have the same wavelength of maximum absorbance (λ_{MAX}).
9.2.2 Qualitative measures (analogous to visual strength/appearance) should be determined by using weighted sum color strength determination (see 13.1 and 13.2).
9.3 Carry out the measurement as soon as possible after preparation of the solution to exclude the influence of changes in solution.
9.4 Adopt appropriate techniques with light sensitive solutions; e.g., using low actinic flasks, or working in dimly lit surroundings (see 13.7).
9.5 Measure the transmittance of the solution in cells (cuvette or flow-through cell) of identical path length (usually 5 mm or 10 mm).
9.5.1 The difference in absorbance between the test dye and reference dye shall not exceed 20%, to minimize the influence of variations in the spectrophotometric response over a relatively wide range of concentrations.
9.5.2 The transmittance, *T*, of the dye solution is obtained differently with a double-beam spectrophotometer compared to a single-beam spectrophotometer, but yields the same result. With a double-beam instrument, simultaneous measurements may be made with a cell of pure solvent placed in the reference beam, and the dye solution in the sample beam. Single-beam instruments require that the instrument first be standardized with pure solvent. Then, the dye specimens are measured successively in the same cell.
9.5.3 Some narrow (0.5-2.0 nm) bandpass, analytical spectrophotometers used for dye strength determination measure both transmittance and absorbance, directly. In this case, it is not necessary for the system or the user to calculate the absorbance separately.

10. Calculations

10.1 Base the evaluation on Beer's law, using the following equation:

$$A = \log_{10}(1/T_\lambda) \qquad \text{(Eq. 1)}$$

10.1.1 A is the absorbance at wavelength of maximum absorbance (see 13.2).
10.1.2 T_λ is the transmittance (in decimal fraction form, where 100% = 1.0) at wavelength of maximum absorbance.
10.2 Calculate the relative color strength to the nearest 0.1% using the following equation:

$$F_s = 100\ (A_2C_1/A_1C_2) \qquad \text{(Eq. 2)}$$

where:
F_s is relative color strength in percent
C is the dye concentration
subscript 1 refers to the reference dye
subscript 2 refers to the dye under test

10.2.1 Assuming $C_1 = C_2$, the following simplified equation is obtained:

$$F_s = 100\ (A_2/A_1) \qquad \text{(Eq. 3)}$$

11. Report

11.1 Identification of test dye.
11.2 Dye mass.
11.3 Solvent used (if solvent system is light-sensitive, note illumination conditions).
11.4 Dissolving conditions (e.g., temperature, also include names and concentrations of any additives used to improve stability and reproducibility, per 8.4).
11.5 Dilution factor.

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11.6 Concentration and temperature of measured solution.

11.7 Measuring instrument (e.g., type and band width).

11.8 Cell path length.

11.9 Wavelength of analysis, also note bandpass and reporting interval of spectrophotometer.

11.10 Relative color strength.

11.11 Conditions under which dye was stored.

11.12 Identify test method and any modifications.

12. Precision and Bias

12.1 *Precision*.

12.1.1 An interlaboratory study by ten laboratories was conducted in 1987. Absorbance was measured using aqueous visible transmittance spectroscopy. A round-robin test battery yielded the results of the test. Several different spectrophotometers were used to determine variance between instruments as well as laboratories. Sample preparation varied as described by this test method for readily soluble dyes and solvent/aqueous solution soluble dyes (disperse dyes). These data are on file at the AATCC Technical Center.

12.1.2 Intra-Lab Variation: The maximum percent coefficient of variation (%C.V.) calculated as:

%C.V. = s/mean × 100 for any given lab was 5.20% for Disperse Blue 56.

12.1.3 Typical %C.V. for water-soluble dyes are below 1%, provided they are sufficiently dissolved. Acid dyes used in the test had typical %C.V. values of 2.0%. Many of the labs in the study indicated that their internal standards for day-to-day and intra-lab variation is ± 0.5%.

12.1.4 Inter-Lab Variation: The maximum %C.V. for between-lab analysis was 6.55% for Disperse Blue 56. Typical %C.V. for water-soluble dyes are below 5%, provided they are sufficiently dissolved.

12.1.5 Spectroscopic measurement of disperse dyes in aqueous/solvent mixtures often yield worse precision than shown for water-soluble dyes, due largely to variances in the solubilities of the disperse dyes. As solubility decreases, light scattering increases, which will lead to higher error levels. For this reason, a given dye is best characterized when it is best dissolved.

12.1.6 One major source of deviation was the use of different spectrophotometers. In several studies, no one single wavelength of maximum absorbance (transmittance minimum) was used for any of the dyes. Other possible sources of error often include differences in time after initial dissolution; a dye's absorbance can increase over time after initial dilution until it reaches a maximum. Solution temperature and cis-trans (and azo-hydrazone) isomerism can lead to variability. It is possible and recommended that laboratories develop correction factors (correlation models) between instruments and laboratories to reduce these errors.

12.1.7 Table I shows *inter-* and *intra-laboratory variability* for the disperse dyes (worst case) studied. Some data points from the study failed the 1.5 IQR fit test and were not included in the data analysis.

12.2 *Bias*.

12.2.1 There is no independent referee analytical method for determining a true value of this property. This test method has no known bias.

13. Notes

13.1 Color strength is traditionally a coloristic concept, being based essentially on visual assessment. Therefore, color strength determined instrumentally should not contradict visual assessment. Such a contradiction would not be expected if the samples to be compared exhibit concentration-dependent differences in absorption only; i.e., their absorption curves in the visible region are identical or show only small differences when the strength of the absorption curves is made equal.

13.2 If the absorption curves differ more markedly (hue or chroma differences), use weighted sum color strength determination to improve the correlation to visually assessed strength.

13.3 When selecting solvents the following aspects are important:

13.3.1 Solubility of dye.

13.3.2 Stability of solutions.

13.3.3 Reproducibility of test.

13.3.4 Applicability of result to other media or practical applications.

13.4 It is essential to ensure that uptake of moisture by hygroscopic dye samples do not lead to errors during weighing.

13.5 It is essential to ensure that the solubility limit is not exceeded during subsequent cooling. Solubility specifications are available from the dye suppliers.

13.6 To avoid errors in dilution, use a minimum transfer of 5 mL (Class A, TD) and a minimum dilution of 100 mL (Class A, TC).

13.7 In certain instances, the solution to be measured may be influenced adversely by the light energy of the source in the measuring instrument (e.g., heat-sensitive or photo-sensitive products). If so, monochromatic illumination, shutter or flash light source should be used.

13.8 When measuring liquid dye samples, it is essential that the sample be thoroughly mixed before an aliquot is taken. If the sample is retained for future use, it must be sealed in a container protected from exposure to light and impervious to moisture.

13.9 A simple method to determine qualitatively the presence of solid dye/dispersing agents in what may appear to be clear solutions is to filter the dye solution through a submicrometer filter using a simple syringe filter as is often used in HPLC and determine the absorbance before and after filtration.

Table I—Inter- and Intra-Laboratory Variability

Disperse Dye	Yellow 54	Red 60	Blue 56
Intra-Lab %C.V. Maximum	0.75%	1.27%	5.20%
Inter-Lab %C.V. Maximum	1.07%	1.42%	6.55%

14. References

14.1 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati, OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

14.2 General procedure for the determination of relative dye strength by spectrophotometric transmittance measurement (report of the ISCC): R. G. Kuehni, *Textile Chemist and Colorist*, Vol. 4, 1972, p133.

14.3 Difficulties in preparing dye solutions for accurate strength measurement: T. R. Commerford, *Textile Chemist and Colorist*, Vol. 6, 1974, p14.

14.4 Reproducibility of dye strength evaluation by spectrophotometric transmission measurement (report of the ISCC): C. D. Sweeny, *Textile Chemist and Colorist*, Vol. 8, 1976, p31.

14.5 ISO proposed test method 105-Z10, 94/341270.

14.6 AATCC (Committee RR98) training video tape "Techniques for Solution Measurement Used in Dye Strength Determination," (1995) AATCC Committee RR98, Assessment of Dye Strength and Shade Test Methods.

14.7 Possible Test Method for Determining Strength of Acid Dyes: B. L. McConnell, *Textile Chemist and Colorist*, Vol 24, No. 2, February 1992, p23.

14.8 Can A Standard Test Be Developed For Measuring the Strength of Disperse Dyes?, M. D. Hurwitz, *Textile Chemist and Colorist*, Vol 25, No. 9, September 1993, p71.

15. History

15.1 Revised 2021 to clarify the wavelength requirements, update terms and add History section to align with style guide.

15.2 Editorially revised in 2019; revised in 2017; editorially revised and reaffirmed in 2011; editorially revised 2010, reaffirmed in 2005; editorially revised and reaffirmed in 2000; reaffirmed in 1999.

15.3 Established by AATCC Committee RR98, Test Method for Relative Color Strength of Dyes in Solution in 1998; under jurisdiction of AATCC Committee RA36. Technically equivalent to ISO 105-Z10.

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Test Method for Transmittance or Blocking of Erythemally Weighted Ultraviolet Radiation through Fabrics

Foreword

AATCC TM183 may be used in conjunction with two other standards developed by ASTM D13 Textiles to meet voluntary requirements for labeling UV protective clothing or products in the United States. Appendix B of this standard is a summary of the methodology for textile UV testing and labeling.

1. Purpose and Scope

1.1 This standard test method is used to determine the ultraviolet radiation transmitted or blocked by textile fabrics intended to be used for UV protection.

1.2 This method provides procedures for measuring this fabric property with specimens in either the dry or wet states.

1.3 Preparation and labeling of products with a UV protection (UPF) value are described in ASTM D6544 and ASTM D6603. Refer to Appendix B for a summary of methodology.

2. Principle

2.1 The transmission of ultraviolet radiation (UVR) through a specimen is measured on a spectrophotometer or spectroradiometer at known wavelength intervals.

2.1.1 The ultraviolet protection factor (UPF) is computed as the ratio of the erythemally weighted ultraviolet radiation (UVR) irradiance at the detector with no specimen to the erythemally weighted UVR irradiance at the detector with a specimen present. A higher UPF indicates more UV protection.

2.1.2 The erythemally weighted UVR irradiance at the detector with no specimen present is equal to the summation between wavelength intervals of the measured spectral irradiance times the relative spectral effectiveness for the relevant erythemal action spectra times the UVR weighting function of the appropriate solar radiation spectrum times the appropriate wavelength interval.

2.1.3 The erythemally weighted UVR irradiance at the detector with a specimen present is equal to the summation between wavelength intervals of the measured spectral irradiance times the relative spectral effectiveness for the relevant erythemal action spectrum times the spectral transmittance for the specimen times the wavelength interval.

2.1.4 The percent blocking of UVA and UVB radiation is also calculated.

3. Referenced Documents

3.1 ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 15.2).

3.2 ASTM D6544 Standard Practice for Preparation of Textiles Prior to Ultraviolet (UV) Transmission Testing (see 15.2).

3.3 ASTM D6603, Standard Specification for Labeling of UV-Protective Textiles (see 15.2).

3.4 ASTM E275, Practice for Describing and Measuring Performance of Ultraviolet, Visible and Near-Infrared Spectrophotometers (see 15.2).

3.5 CIE 106/4-1993, A reference action spectrum for ultraviolet induced erythema in human skin. (see 15.3).

3.6 *Information and Standards for UV Protective Textiles*; AATCC & ASTM International, 2017. (see 15.1).

3.7 Sayre, R. M., et al., "Spectral Comparison of Solar Simulators and Sunlight," *Photodermatol Photoimmunol. Photomed.*, 7, 159-165 (1990).

4. Terminology

4.1 **erythema,** n.—abnormal redness of the skin (sunburn) due to capillary congestion (as in inflammation).

4.2 **percent UV blocking,** n.—100 minus the UV transmittance.

4.3 **percent UV transmittance,** n.—that fraction of the incident ultraviolet radiation that is not reflected or absorbed, but passes through a substance.

4.4 **ultraviolet protection factor (UPF),** n.—the ratio of the average effective ultraviolet radiation (UVR) irradiance transmitted and calculated through air to the average effective UVR irradiance transmitted and calculated through fabric.

4.5 **ultraviolet radiation,** n.—radiant energy for which the wavelengths of the monochromatic components are smaller than those for visible radiation and more than about 100 nm.

NOTE: The limits of the spectral range of ultraviolet radiation are not well defined and may vary according to the user. Committee E-2.1.2 of the International Commission on Illumination (CIE) distinguishes in the spectral range between 400 and 100 nm:

UVA	315-400 nm
UVB	280-315 nm
UVR	280-400 nm

5. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Users MUST consult manufacturers on specific details such as safety data sheets and other recommendations. Consult and follow all applicable OSHA standards and rules.

5.1 Under any circumstances, do not look directly at the equipment and materials that may increase the light source intensity, such as mirrors.

5.2 Follow good laboratory practices. Wear prescribed safety glasses in all laboratory areas.

6. Uses and Limitations

6.1 UV-protective textiles may be worn in such a manner that stretching of the fabric over the shoulders, back, and arms of users will occur. This method can be used to determine the UPF of fabrics in a stretched state; however, there is no standardized technique known to replicate for stretching of specimens during testing.

7. Apparatus and Materials (see 15.4)

7.1 A spectrophotometer or spectroradiometer equipped with an integrating sphere (see Appendix A).

7.2 Filter, Schott Glass UG11 (see 15.5).

7.3 Clear plastic food wrap for use with wet samples (polyvinylidene chloride or polyvinylchloride film).

7.4 AATCC Blotting Paper (see 15.1).

7.5 AATCC Light Transmission Verification Fabric (see 15.1) or in-house fabric with known predictable and repeatable light transmission behavior and a UPF between 15 and 50.

8. Instrument Verification and Calibration

8.1 Calibration. Calibrate the spectrophotometer or spectroradiometer according to manufacturer instructions. The use of physical standards is recommended for validating the measurement of spectral transmittance.

8.1.1 When running a wet sample place the plastic wrap over the port and repeat calibration.

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8.2 Wavelength scale. Calibrate the wavelength scale of the spectrophotometer or spectroradiometer using the spectral emission lines of an electrical discharge in mercury vapor. The wavelength calibration of a spectrophotometer can be performed using the absorption spectra of a holmium oxide glass filter. Reference wavelengths for both mercury arc emission and holmium oxide absorption are provided in ASTM E275.

8.2.1 Transmittance scale. Set the transmittance scale to a 100% value by operating the instrument without a sample in the optical path; therefore, referenced to air. The 0% value can be subsequently verified by fully obstructing the sample beam path with an opaque material. Validate the linearity of the transmittance scale by means of either calibrated neutral density filters or calibrated perforated screens supplied by the instrument manufacturer or standardizing laboratories.

8.3 Verification checks on the operation of the test and the apparatus should be made and kept in a log. The use of a verification fabric is recommended at least once a month unless no testing is to be performed that month.

9. Specimens

9.1 Test a minimum of two (2) specimens from each sample, one each for the wet and the dry testing. Cut each specimen at least 50 × 50 mm (2.0 × 2.0 in.) or 50 mm (2.0 in.) in diameter. Avoid distorting the specimen during preparation and handling.

9.1.1 Should the fabric have areas of different colors or structure, test each color or structure if its size is sufficient to cover the measuring port.

9.2 See Appendix A5 for specimens exhibiting fluorescence.

10. Conditioning

10.1 For dry specimens, bring to moisture equilibrium in a standard atmosphere according to ASTM D1776. Use conditions listed for Textiles, General in Table I of the standard.

11. Procedure

11.1 Dry Evaluation.

11.1.1 Place the specimen flush against the sample transmission port opening in the sphere.

11.1.2 Make one UV transmission measurement with the specimen oriented in one direction, a second measurement at 0.79 rad (45°) to the first and a third at 0.79 rad (45°) to the second. Record the individual measurements.

11.1.3 On multicolored specimens, determine the area of highest UV transmission and make the three measurements at 0.79-rad (45°) rotation intervals in that area.

11.2 Wet Evaluation

11.2.1 Weigh the test specimen. Thoroughly wet out the specimen in distilled water by placing it flat in the bottom of a beaker and then pour distilled water into the beaker until the specimen is covered. Allow the specimen to remain submerged for 30 minutes. Press and move the specimen from time to time to ensure a good and uniform penetration. Prepare only one specimen at a time.

11.2.2 Bring the wet pick-up to 140 ± 5% by squeezing the wet specimen between blotting paper through a hand wringer or similar convenient means (such as squeezing between two glass rods). If the fabric has low moisture absorption, repeat the soaking and wringing steps. Note, some samples may not be capable of achieving the specified wet pick-up such as tightly woven synthetic fabrics.

Other wet pickup values may be used as agreed upon between parties.

11.2.3 Use plastic wrap over viewing port to protect instrument from water.

11.2.4 Continue as directed for dry evaluation. Avoid evaporative reduction of the moisture content below the specified level before the actual UV transmission measurements are made.

12. Calculations

12.1 Calculate the Ultraviolet Protection Factor (UPF) for each measurement using Equation 1:

$$\mathrm{UPF} = \frac{\sum_{280\ \mathrm{nm}}^{400\ \mathrm{nm}} E_\lambda \times S_\lambda \times \Delta\lambda}{\sum_{280\ \mathrm{nm}}^{400\ \mathrm{nm}} E_\lambda \times S_\lambda \times T_\lambda \times \Delta\lambda} \quad (1)$$

where:

E_λ = relative erythemal spectral effectiveness (see Table I)

S_λ = solar spectral irradiance (see Table II)

T_λ = spectral transmittance (measured)

$\Delta\lambda$ = measured wavelength interval (nm)

NOTE: Although integration is indicated from 280 nm to the stated wavelengths, little or no contribution will occur in the 280-290 nm region.

12.2 Calculate the mean UPF for all three (3) rotations at each measurement location. If more than one location was measured on a specimen, or if multiple specimens were measured in the same condition (dry, wet, prepared according to ASTM D6544, etc.), calculate the mean UPF for all comparable measurements (see 15.6).

12.3 Calculate the A-range ultraviolet (UVA) transmittance using Equation 2:

Table I—Relative Erythemal Effectiveness Function (E_λ)[a]

nm	response	nm	response	nm	response
280	1.00e+00	320	8.55e-03	360	4.84e-04
282	1.00e+00	322	5.55e-03	362	4.52e-04
284	1.00e+00	324	3.60e-03	364	4.22e-04
286	1.00e+00	326	2.33e-03	366	3.94e-04
288	1.00e+00	328	1.51e-03	368	3.67e-04
290	1.00e+00	330	1.36e-03	370	3.43e-04
292	1.00e+00	332	1.27e-03	372	3.20e-04
294	1.00e+00	334	1.19e-03	374	2.99e-04
296	1.00e+00	336	1.11e-03	376	2.79e-04
298	1.00e+00	338	1.04e-03	378	2.60e-04
300	6.49e-01	340	9.66e-04	380	2.43e-04
302	4.21e-01	342	9.02e-04	382	2.26e-04
304	2.73e-01	344	8.41e-04	384	2.11e-04
306	1.77e-01	346	7.85e-04	386	1.97e-04
308	1.15e-01	348	7.33e-04	388	1.84e-04
310	7.45e-02	350	6.84e-04	390	1.72e-04
312	4.83e-02	352	6.38e-04	392	1.60e-04
314	3.13e-02	354	5.96e-04	394	1.50e-04
316	2.03e-02	356	5.56e-04	396	1.40e-04
318	1.32e-02	358	5.19e-04	398	1.30e-04
				400	1.22e-04

Note: The intervals in Table I are 2 nm. For 5 nm UV transmission data, use the interpolated data between those ending in a "4" and a "6."

[a] Data from CIE Publication 106/4.

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Table II—Solar Spectral Irradiance of Noonday, July 3, Sunlight, Albuquerque, NM (S_λ)[a]

nm	W/cm²/nm	nm	W/cm²/nm	nm	W/cm²/nm
280	4.12e-11	320	3.14e-05	360	5.64e-05
282	2.37e-11	322	3.32e-05	362	6.00e-05
284	3.14e-11	324	3.61e-05	364	6.48e-05
286	4.06e-11	326	4.45e-05	366	7.18e-05
288	6.47e-11	328	5.01e-05	368	7.62e-05
290	3.09e-10	330	5.32e-05	370	7.66e-05
292	2.85e-09	332	5.33e-05	372	7.50e-05
294	2.92e-08	334	5.23e-05	374	6.61e-05
296	1.28e-07	336	5.04e-05	376	6.66e-05
298	3.37e-07	338	4.99e-05	378	7.46e-05
300	8.64e-07	340	5.39e-05	380	7.54e-05
302	2.36e-06	342	5.59e-05	382	6.42e-05
304	4.35e-06	344	5.35e-05	384	5.85e-05
306	7.19e-06	346	5.34e-05	386	6.26e-05
308	9.68e-06	348	5.37e-05	388	6.72e-05
310	1.34e-05	350	5.59e-05	390	7.57e-05
312	1.75e-05	352	5.89e-05	392	7.16e-05
314	2.13e-05	354	6.13e-05	394	6.55e-05
316	2.43e-05	356	6.06e-05	396	6.81e-05
318	2.79e-05	358	5.38e-05	398	8.01e-05
				400	1.01e-04

Note: The intervals in Table II are 2 nm. For 5 nm UV transmission data, use the interpolated data between those ending in a "4" and a "6."

[a] Data from Sayre, R. M., et al., "Spectral Comparison of Solar Simulators and Sunlight."

$$T(\text{UV-A})_{AV} = \frac{\sum_{315\text{ nm}}^{400\text{ nm}} T_\lambda \times \Delta\lambda}{\sum_{315\text{ nm}}^{400\text{ nm}} \Delta\lambda} \quad (2)$$

12.3.1 Calculate the mean T(UVA) for all comparable measurements (see 15.6).

12.4 Calculate the B-range ultraviolet (UV–B) transmittance using Equation 3:

$$T(\text{UV-B})_{AV} = \frac{\sum_{280\text{ nm}}^{315\text{ nm}} T_\lambda \times \Delta\lambda}{\sum_{280\text{ nm}}^{315\text{ nm}} \Delta\lambda} \quad (3)$$

12.4.1 Calculate the mean T(UVB) for all comparable measurements (see 15.6).

12.5 Calculate the percent blocking for UVA and for UVB using Equation 4 and Equation 5:

$$= 100\%\text{-}T(\text{UVA}) \quad (4)$$

$$= 100\%\text{-}T(\text{UVB}) \quad (5)$$

where:

T(UVA) or T(UVB) is the mean of comparable measurements, expressed as a percentage.

13. Report

13.1 Describe or identify the sample tested.

13.2 Report that the sample was tested using AATCC TM183-2020.

13.3 Report the testing conditions:

13.3.1 Illumination and viewing geometry (0/T or T/0)

13.3.2 Illumination (monochromatic or polychromatic).

13.3.3 Whether specimens were tested before or after exposure described in ASTM D6544.

13.3.4 Whether specimens were tested wet or dry.

13.3.5 For wet specimens, the actual percent wet pick-up if other than 140 ± 5%.

13.4 Report the test results:

13.4.1 UPF for each measurement (see 15.6).

13.4.1 The mean UPF for each condition, color and/or structure tested.

13.4.2 The mean UVA transmittance, T(UVA).

13.4.3 The mean UVB transmittance, T(UVB).

13.4.4 The percent blocking UVA.

13.4.5 The percent blocking UVB.

14. Precision and Bias

14.1 *Precision*. In 2001, a precision and bias study was conducted with six different materials and one laboratory (see Table III).

14.2 *Bias*. Transmittance or blocking of erythemally weighted ultraviolet radiation through fabrics can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

15. Notes

15.1 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

15.2 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; +1.610.832.9500; www.astm.org.

15.3 Available from International Commission on Illumination, CIE Central Bureau, Babenbergerstraße 9/9A, 1010 Vienna, AUSTRIA; +43 1 714 31 87; ciecb@cie.co; www.cie.co.at.

15.4 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

15.5 Available from Schott; www.schott.com.

15.6 Some software programs calculate the average transmittance of measurements before calculating UPF or UVA/UVB transmittance. The mean UPF, T(UVA), or T(UVB) result is the same, but UPF of individual measurements are required to determine a label value according to ASTM D6603.

Table III—Precision and Bias Study Data

	Average UPF		Standard Deviation		Sample Variance		95% Confidence	
Fabric	Dry	140% WPU	Dry	140% WPU	Dry	140% WPU	Dry	140% WPU
D	41.655	28.105	0.700	0.884	0.490	0.781	0.580	0.707
G	40.615	81.718	9.246	17.959	85.492	322.533	7.398	14.370
J	23.733	42.603	1.394	5.978	1.943	35.733	1.115	4.783
N	7.797	3.723	0.367	0.238	0.134	0.056	0.294	0.190
O	57.947	15.779	4.281	3.021	18.325	9.125	3.425	2.471
R	22.32	10.686	2.492	0.547	6.209	0.299	1.994	0.438

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16. History

16.1 Editorially revised in 2022 to remove the UPF compilation as it has been discontinued.

16.2 Revised in 2020 to add references to verification fabric and UPF compilation, and to align format with AATCC style guide. Editorially revised 2016. Reaffirmed 2014. Revised 2010. Revised 2004. Reaffirmed 2000. Reaffirmed 1999.

16.3 Developed in 1998 by AATCC Committee RA106.

Appendix A

Spectrophotometer or Spectroradiometer Specifications

A1. The integrating sphere surface is internally coated or constructed using a material that is both diffuse and highly reflecting in the ultraviolet region. The total surface area consumed by all port openings required shall not exceed 3% of the total surface area of the integrating sphere.

A2. Illumination and viewing geometries.

A2.1 Directional illumination/hemispherical collection (0/T). In this geometry, the specimen is illuminated with a unidirectional beam whose axis is not greater than 0.14 rad (8°) from the surface normal of the sample. Any ray of this beam shall not exceed 0.09 rad (5°) from the beam axis. The cross-sectional area of the illuminating beam shall be at least 10 times the dimension of the largest hole in the test material. The total flux transmitted by the specimen is collected by the integrating sphere.

A2.2 Hemispherical illumination/directional viewing (T/0). In this geometry the specimen is illuminated by an internally illuminated integrating sphere. The specimen is viewed unidirectionally with an axis not greater than 0.14 rad (8°) from the surface normal of the sample. Any ray of this beam shall not exceed 0.09 rad (5°) from the beam axis.

The cross-sectional area of the viewing beam shall be at least 10 times the dimension of the largest hole in the test material.

A2.3 Sample substitution errors. Integrating spheres can experience "sample substitution" error due to contributions of the reflectance of the specimen on the internal Illumination of the sphere. The error can be eliminated in either geometry by use of a separate reference beam that traverses its own port opening in the sphere. The reference beam impinges on either a portion of the sphere wall or a reference material mounted at a diametrically opposed port opening.

A3. Spectral requirements. The spectrophotometer or spectroradiometer shall have a spectral bandpass of 5 nm or less over the spectral range of 280 nm (or less) to 400 nm (or more). The measured wavelength interval over this spectral range should not be greater than 5 nm.

A4. Stray radiation. The contribution of stray radiation within the instrument, including that due to sample fluorescence, shall produce an error of less than 0.005 in the value of spectral transmittance being measured.

A5. Sample fluorescence. The contribution of sample fluorescence on spectral transmittance measurements on certain dyes and whitening agents present in fabrics that may fluoresce could result inartificially high values of spectral transmittance.

A5.1 Monochromatic illumination. In spectrophotometers where the monochromator precedes the specimen within the optical path, the artificially high values of transmittance appear at the excitation wavelengths of the fluorescing agent. This includes nearly all wavelengths in the UVR spectral region. The error due to the fluorescence can be removed by placing a UV transmitting, visible blocking filter after the sample. A Schott Glass UG11 filter has been found to be satisfactory. However, the decrease in transmission of the filter with increasing wavelength may reduce the usefulness of the long wavelength UVA measurement.

A5.2 Polychromatic illumination. In spectrophotometers and spectroradiometers where the illumination is polychromatic and the monochromator follows the specimen in the optical path, the artificially high values of transmittance appear at the emission wavelengths of the fluorescing agent. The effects of fluorescence are, therefore, eliminated at most UVR wavelengths. The use of an illuminating light source that conforms to the spectral distribution requirements for solar simulators will most accurately include the contribution of sample fluorescence to the long wavelength UVA measurement. However, because the fluorescent component does not contribute to the UPF, the spectral distribution of the source is irrelevant, so long as it provides sufficient energy to cover the spectral range of interest to acceptable signal to noise ratios in the spectral data.

Appendix B
Summary of Methodology

Manufacturers and producers of UV protective textile materials and/or products must be aware of the proper existing standards and test methods to meet published voluntary standards for labeling UV protective clothing for the United States.

B1. There are three (3) published voluntary standards: ASTM D6544, AATCC TM183, and ASTM D6603.

B2. To summarize at a high level, these standards are used in conjunction with one another to develop the proper UPF values to appropriately label products as "UV Protective."

B2.1 Evaluate UV transmission of original specimen ("unprepared") or ("laundered once") specimen using AATCC TM183. The value reported is the mean TM183 UPF value.

B2.2 Refer to ASTM D6544 to determine the exposure conditions depending on the product end use.

B2.3 After subjecting the specimens to the specified exposure conditions of ASTM D6544, evaluate UV transmission using AATCC TM183. The specimen is referred to as the "prepared-for-testing" specimen.

B2.4 Refer to ASTM D6603 for additional calculations. Classifications for labeling UPF value are also outlined in ASTM D6603.

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Test Method for Dusting Behavior of Dyes

1. Purpose and Scope

1.1 This test method provides a means to assess the dusting behavior of dyes.

2. Principle

2.1 Dust is generated from a dye sample by means of a dust-generating device, extracted from the dust-bearing air by vacuum and conveyed to a detection point, where the amount of dust generated is estimated visually or determined quantitatively by a gravimetric or photometric option.

3. Terminology

3.1 **dust,** n.—fine particles of solid material dispersed in a gas.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Follow good laboratory practices. Wear safety glasses in all laboratory areas.

4.2 An eyewash safety shower should be located nearby and a self-contained breathing apparatus should be readily available for emergency use.

5. Uses and Limitations

5.1 Dyestuff dust is formed during handling operations such as dispensing, transferring, sprinkling, etc.

5.2 The dustiness of dyestuffs is an important characteristic when considering aspects of hygiene, health and safety in the dyestuff consuming industry. It is important, therefore, that a reliable and reproducible method exists to measure this property.

5.3 Although other methods of dust measurements exist, the method given in this test procedure is both more representative of and comparable with actual practice when handling dyestuffs. With respect to a comparison of dyestuffs or the reliability limits, it should be understood that the resulting value is not a specific value like density.

5.4 Solid dyes are sold in different physical forms (powdered, granulated, etc.). The particle-size distribution of commercial products varies considerably. The mean particle diameter may be less than 50 µm or as much as several millimeters. The range of particle-size distribution of a solid dye may also be narrow or wide.

5.5 The particle-size distribution of dyestuff dust is largely independent of the physical form of the dyestuff. Two typical particle-size distributions for dyestuff dust are shown in Fig. 1.

6. Apparatus and Materials (see 13.1)

6.1 Balance, accurate to ± 0.1 g, for weighing the dye.

6.2 Dust-generating device, with filter holder and connecting joints, and incorporating the following additional components (see Figs. 2 and 3, and 13.2).

6.2.1 Filter, white, diameter 50 ± 2 mm, capable of trapping the dust quantitatively (pore size < 5 µm), for gravimetric and photometric options made of cellulose acetate nitrate. For the visual option, suitable glass fiber filters can be used.

6.2.2 Vacuum pump with a suction capacity of at least 20 L/min.

6.2.3 Regulator for adjusting the air flow rate.

6.2.4 Flowmeter capable of monitoring the air flow rate between 10-20 L/min.

6.2.5 Timer for opening the slide valve to start suction, and for timing the period of suction.

6.3 Dust assessment apparatus.

6.3.1 For visual option, Gray Scale for Staining (see 13.3 and 13.4).

6.3.2 For gravimetric option, analytical balance.

6.3.3 For photometric option, photometer.

6.4 Analytical balance, accurate to ± 0.01 mg, for weighing the dust collected on the filter (see 6.2.1) (gravimetric assessment option), and/or photometer, for extinction measurement of the dust collected and dissolved in a suitable solvent (photometric assessment option).

6.5 Cleaning device; e.g., brush or vacuum cleaner, for cleaning the apparatus.

6.6 Tweezers for removing filter paper from the holder after dust generation.

7. Procedure

7.1 Place the filter holder with the filter (see 6.2.1) in the dust generating apparatus (see 6.2) and close it to give an airtight fit. If the gravimetric option of assessment is used, condition and weigh the filter before insertion into the filter holder.

7.2 Using the balance (see 6.1), carefully weigh 10.0 ± 0.1 g of the dye and place it in the hopper at the top of the apparatus. Start the timer (see 6.2.5) to open the slide valve with a sudden action, allowing the dye to drop down through the tube into the dust chamber below.

7.3 Five seconds after the slide has opened, apply a vacuum to extract the dust from the chamber for collection on the filter (see 6.2.1) under the following conditions:

7.3.1 Air flow rate: 15 L/min.

7.3.2 Extraction time: 120 s (starting 5 s after dye has dropped).

7.3.3 Height of fall: 815 ± 5 mm.

7.4 Using tweezers (see 6.6), carefully remove the dust-laden filter from the holder and evaluate it by one of the options given in Section 8.

7.5 Clean the equipment thoroughly (see 6.5) after each test. If the equipment is wet-cleaned, take particular care that it is dried thoroughly.

8. Assessment of Filter

8.1 Visual option. Compare the dust-laden filter visually with the Gray Scale for Staining (see 6.3.1) rate the filter using Table I of AATCC EP2, Gray Scale for Staining. Half-grades may be used.

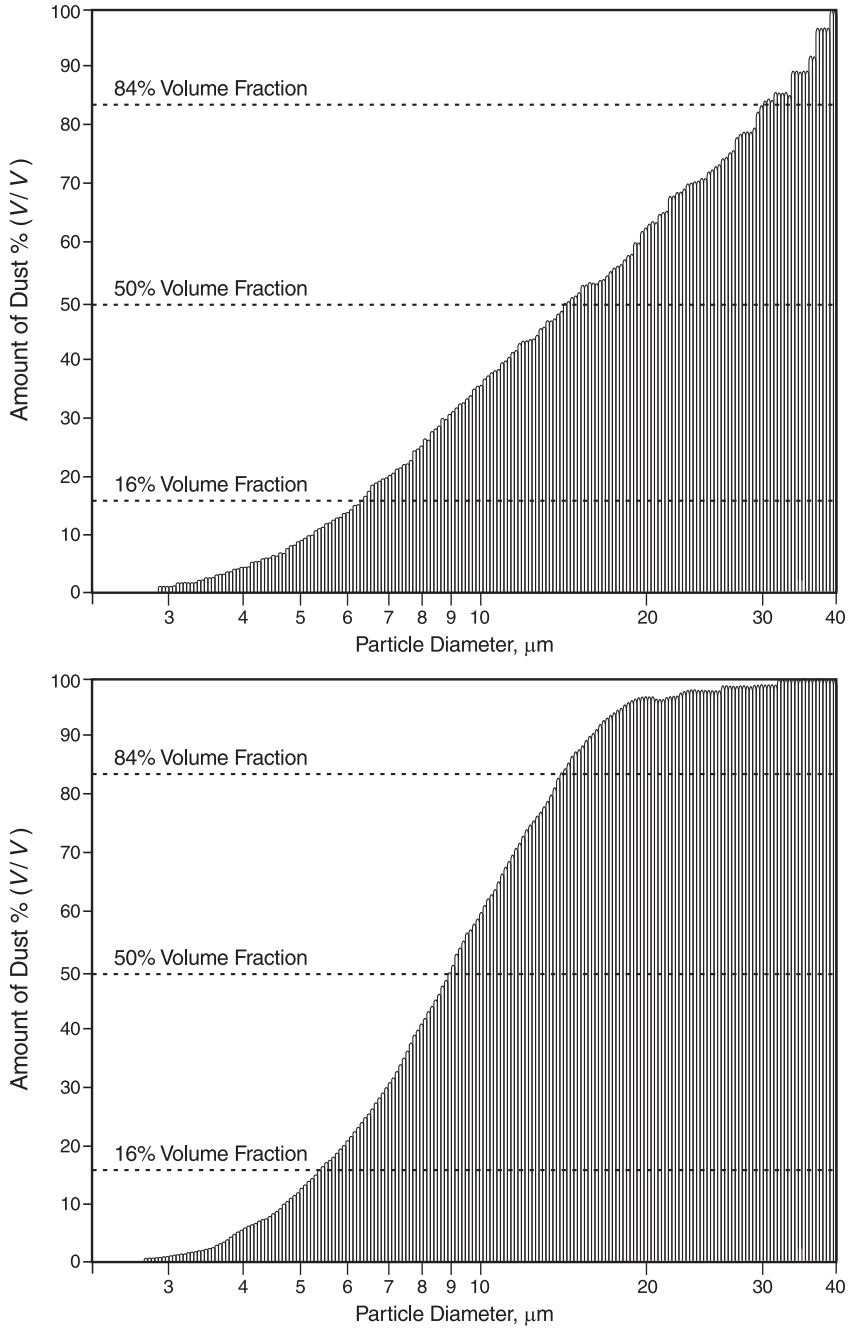
Table I—Dust Assessment Rating Scale

Grade	Description
5	No dust
4	Slightly dusty
3	Moderately dusty
2	Dusty
1	Very dusty

8.2 Gravimetric option. Weigh the dust-laden filter on the analytical balance (see 6.4) to the nearest 0.01 mg. Since with low-dusting products the mass of dust is very small (< 1 mg), considerable errors are to be expected with the gravimetric option. In such cases preference should be given to photometric measurement.

8.3 Photometric option. For photometric measurement of the amount of dust, dissolve the dyestuff by shaking well the

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NOTE — The *x*-axis of both graphs has a logarithmic scale.

Fig. 1—Typical graphical representation of the amount of dust (by volume) as a function of the dust-particle diameter.

dust-laden filter in a suitable solvent at room temperature. When a clear solution has formed, measure the transmission with a photometer and read the corresponding amount of dust off a previously prepared calibration graph (see 12.1).

9. Evaluation

9.1 The generation and detection of dust depend on a great number of parameters. For this reason, the determination of the amount of dust yields results which are valid for the specified test conditions. This means that a direct comparison of either the visual or the quantitative results of dust behavior with results from other test methods is not possible. However, the relative order of results of a set of test specimens from one test method is likely to be comparable with that from other test methods.

9.2 Visual option. Express the results of the visual option as the Gray Scale rating obtained in 8.1.

9.2.1 A quantitative determination of the amount of dust produced by a dye is not possible by the visual option. The main reasons for this are different particle size distributions, particle sizes and shades of the dust in each case.

9.2.2 Visual assessment is subjective and depends on factors such as the experience of the examiner, the hue of the layer of dust and the nature of the filter surface (smooth or rough). Variations of up to half a grading unit are inherent in this system. Empirically, the total error under reproducibility conditions (same apparatus and laboratory) does not exceed this value.

9.3 Gravimetric and Photometric options.

9.3.1 Record the results obtained from 8.2 and 8.3 in milligrams of dyestuff dust collected.

9.3.2 In both of the quantitative options (gravimetric and photometric), the determination is based on the amount of dust trapped on the filter. Because the amount of dust is measured in milligrams, in the gravimetric option, variations in the conditioning of the filter and electrostatic influences can lead to substantial errors. If the amount of dust is determined by photometry, care must be taken that the transmission is measured on clear solutions. According to the experience of various laboratories, a reproducibility of about 10% (coefficient of variation) can be attained under favorable conditions.

9.4 Scatter of results. In some cases, scattering of the results may occur. The main causes of this are:

9.4.1 Factors specific to the apparatus:

9.4.1.1 The air flow rate is incorrectly adjusted.

9.4.1.2 The air flow rate through the apparatus is not constant, or the vacuum is incorrectly applied.

9.4.1.3 The timing control is imprecise.

9.4.2 External factors:

9.4.2.1 Humidity.

9.4.2.2 Electrostatic charges inside the vertical tube and the dust chamber.

9.4.2.3 Inhomogeneous distribution of the dust within the sample.

10. Report

10.1 The report should include the following information:

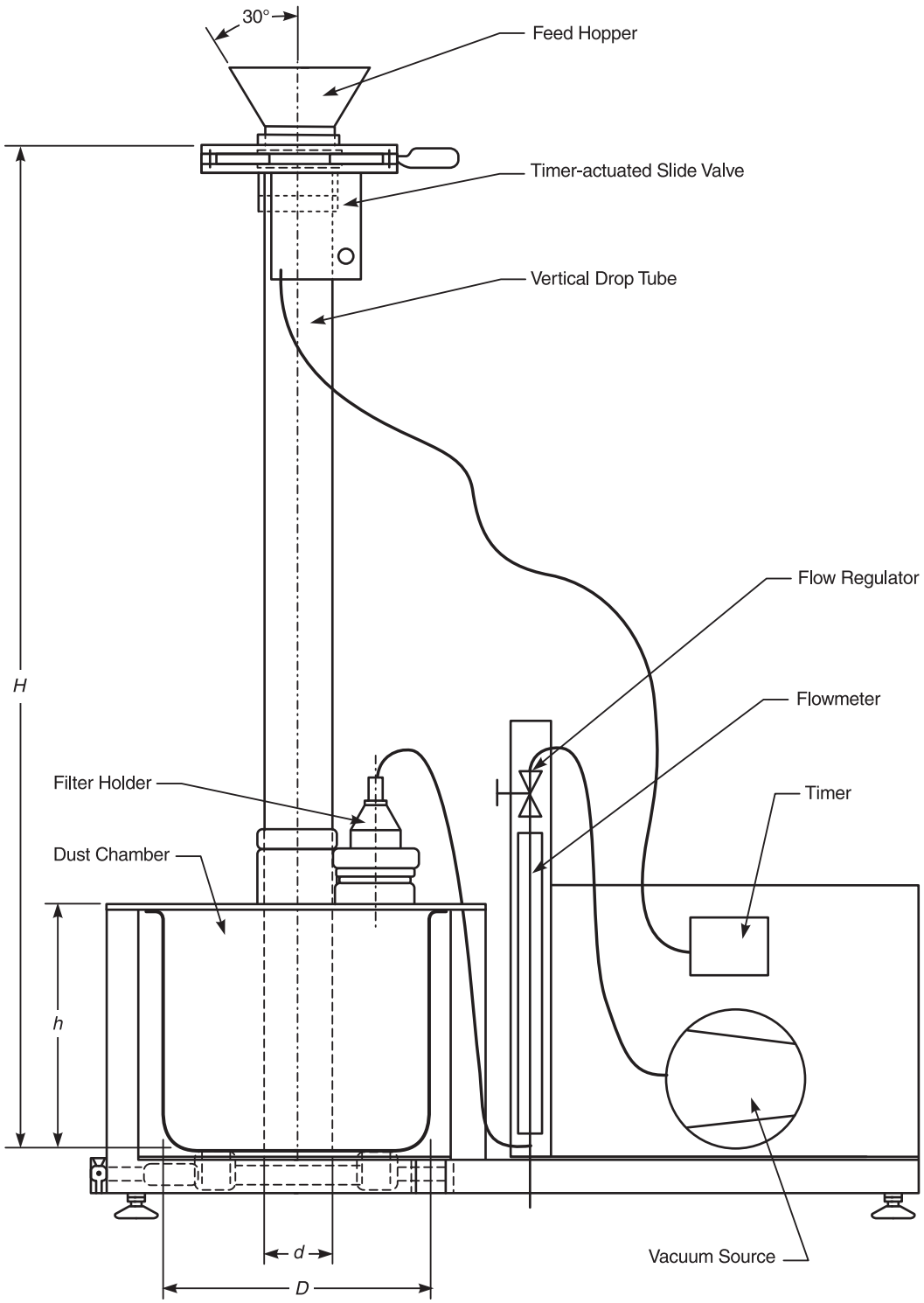
10.1.1 A full description of the sample tested.

10.1.2 The mass of the test specimen.

10.1.3 The assessment option used and the result obtained, expressed as in 9.2 and 9.3.

10.1.4 Details of any deviation from the procedure.

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H	Total fall height[1]	815 mm	± 5 mm
h	Dust chamber height	195 mm	± 5 mm
D	Dust chamber diameter	Ø 210 mm	± 5 mm
d	Drop tube diameter	Ø 47 mm	± 1 mm

[1] Total fall height is measured from the top side of the slide valve plate to the inside face of the dust chamber.

Fig. 2—Apparatus for dust determination.

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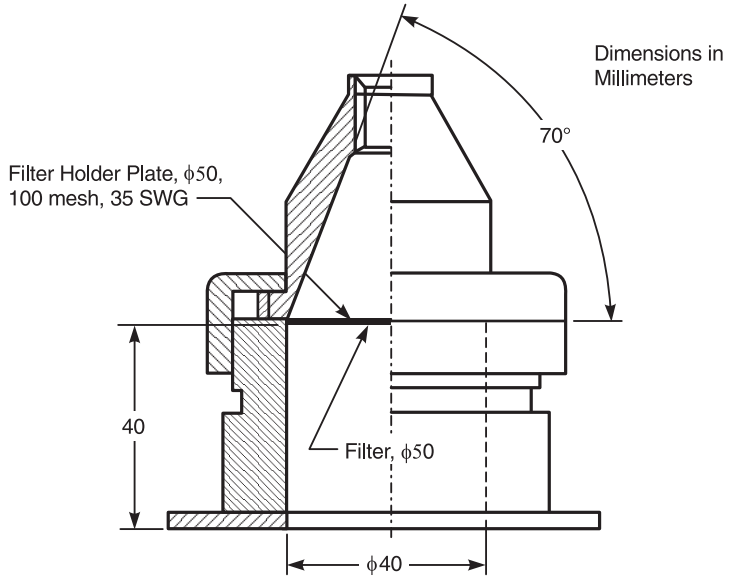


Fig. 3—Filter holder.

11. Precision and Bias

11.1 *Summary.* The precision is based on data for the gravimetric option contained in Annex A of ISO 105-Z05:1996, Determination of the dusting behaviour of dyes. Four laboratories participated in the study, testing three dyes at two time periods at 10 determinations each period. Three laboratories were judged to be at a similar performance level. Testing at the two time periods showed no effect. All of the data from the three laboratories were included in the analysis, using analysis of variance.

11.2 *Precision*. Because of the limited test plan, this is included as an interim precision statement.

11.2.1 Components of variance are given as standard deviations, with units of mg/filter, and as variances, with units of $(mg/filter)^2$, in the following table:

Components of Variance	Std. Deviation	Variance
Between-Laboratory	0	0
Laboratory X Dye Interaction	0.187	0.035
Within-Laboratory	0.276	0.076

Table II

	Single Dye Comparisons		Multiple Dye Comparisons	
N	***Within-Laboratory***	***Between-Laboratory***	***Within-Laboratory***	***Between-Laboratory***
5	0.34	0.34	0.62	0.62
10	0.24	0.24	0.57	0.57
15	0.20	0.20	0.56	0.56

11.2.2 *Critical Differences*. A difference between two averages, under the appropriate precision category, is statistically significant at the 95% confidence level, if the difference is equal to or greater than the table value shown (see Table II).

Most comparisons are likely to be under the single dye category. However, if comparisons are made across dyes, use the multiple dye columns which include the effect of the interaction component of variance.

11.3 *Bias*. This test method has no known bias. There is no referee method for determining a true value for the dusting behavior of dyes by which bias may be established.

12. References

12.1 AATCC TM182, Relative Color Strength of Dyes in Solution (see 13.3).

13. Notes

13.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

13.2 Instead of a filter and filter holder, other dust-detection devices may be fitted to the apparatus, such as an impactor or an optical particle counter.

13.3 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

13.4 It is also possible to carry out a visual assessment of colorless solid materials. However, special care has to be exercised in such cases. Black filters may be helpful, but require separate preliminary trials. It is preferable to use the gravimetric or photometric option.

14. History

14.1 Reaffirmed in 2020. Relocated History to the section 14.

14.2 Reaffirmed 1999, 2000, 2005, 2010, 2014; editorially revised 2008, 2009, 2019. Technically equivalent to ISO 105-Z05.

14.3 Developed in 1998 by AATCC Committee RA87.

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Test Method for Chelating Agent Content in Hydrogen Peroxide Bleach Baths: Copper PAN Indicator

1. Foreword

1.1 While several test methods exist for determining chelation value of chelating agents, this method includes an additional calculation for percent chelating agent in the bleach bath.

2. Purpose and Scope

2.1 This test method measures the chelating agent content in a hydrogen peroxide bleach solution.

2.2 The method is designed to permit routine or periodic titration of the peroxide saturator section of a plant preparation range, or other prepared peroxide bleach solutions, to determine the concentration of chelating agent present.

2.3 The method is restricted to chelating agents based on ethylenediaminetetraacetic acid (EDTA), N-hydroxyethylethylenediaminetriacetic acid (HEDTA), and diethylenetriaminepentaacetic acid (DTPA). This includes any proprietary products (referred to herein as "product") containing possible multiple ingredients, one or more of which may be chelating agents.

3. Principle

3.1 The percent chelating agent or product in the hydrogen peroxide bleach bath is determined in two steps: first, a direct titration of the chelating agent or product using a known concentration of copper sulfate in the presence of an indicator (PAN, 1-(2-pyridylazo)-2-naphthol), and second, a similar titration of the bleach solution containing the chelating agent using a more dilute concentration of the copper sulfate as the titrant.

3.2 Results are reported as percent chelating agent. A higher value indicates more chelating agent or product in the bath.

4. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

4.1 AATCC TM168, Test Method for Chelation Value of Polyaminopolycarboxylic Acids and Their Salts: Copper PAN (see 15.1).

5. Terminology

5.1 **chelating agent,** n.—*in textile chemistry*, a chemical capable of deactivating metal ions by formation of a water soluble complex. SYN. *sequestering agent*.

5.2 **copper chelation value (CuCV),** n.—the milligrams of copper sulfate pentahydrate chelated by one gram of a chelating agent or product containing a chelating agent.

6. Safety Precautions

6.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

6.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

6.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

7. Uses and Limitations

7.1 Results may be affected by any of the following:

7.1.1 Pre-chelation of iron or copper ions contained in the bleach solution will reduce the apparent, calculated concentration of chelating agent originally present in the bath.

7.1.2 Oxidation of uncombined chelating agent by hydrogen peroxide will reduce the apparent concentration in the bleach bath.

7.1.3 Cyanide ions, ammonia, and most amines will form molecular compounds with copper ions without affecting the PAN indicator, resulting in false positive results.

7.2 The percent chelating agent in the bleach bath will not be affected by changes in the active ingredient content of the chelating agent or product.

7.2.1 AATCC TM168 should be conducted on each new lot of the chelating agent or product to determine if the active ingredient content has changed.

7.2.2 The copper chelation value (CuCV) determined in this method is a direct indication of the active ingredient content of the chelating agent or product and can be monitored for this purpose.

8. Reagents

8.1 Use ACS reagent grade quality chemicals. All chemicals should be handled with care. Many of the chemicals used in this procedure are corrosive or a strong irritant.

8.1.1 Acetic acid, glacial (CH_3COOH).

8.1.2 Copper sulfate pentahydrate ($CuSO_4 \cdot 5H_2O$).

8.1.3 Methanol (CH_3OH). Methanol is a flammable liquid and should be stored in the laboratory only in small containers away from heat, open flame and sparks.

8.1.4 PAN, 1-(2-pyridylazo)-2-naphthol ($C_{15}H_{11}ON_3$).

8.1.5 Sodium acetate trihydrate ($NaC_2H_3O_2 \cdot 3H_2O$).

8.1.6 Sodium hydroxide, pellets (NaOH).

8.1.7 Sulfuric acid, 98% (H_2SO_4).

9. Samples

9.1 Test two samples of the chelating agent or product and two samples of the bleach bath.

10. Preparation of Reagents

10.1 Reagent A (12.500 g/L copper sulfate). Dissolve 12.500 g ± 0.002 g of copper sulfate pentahydrate in distilled water and dilute to 1 L in a volumetric flask.

10.2 Reagent B (2.500 g/L copper sulfate). Transfer 200 mL of Reagent A to a 1 L volumetric flask and dilute to volume with distilled water. Retain remainder of Reagent A.

10.3 PAN indicator. Dissolve 0.025 ± 0.001 g of PAN in 50.0 mL of methanol. Store in a stoppered bottle in a refrigerator. Prepare fresh weekly.

10.4 Sodium acetate buffer solution. In preparing, dispensing and handling glacial acetic acid, use chemical goggles, or face shield, impervious gloves and an impervious apron. Concentrated acids should be handled only in an adequately ventilated hood. CAUTION: Always add acid to water. Dissolve 34.0 ± 0.1 g of sodium acetate trihydrate in 500 ± 1 mL of distilled water and add 15.0 ± 0.1 mL of glacial acetic acid. Mix well and store in a closed container.

10.5 Sodium hydroxide (20% by volume). In preparing, dispensing and handling sodium hydroxide, use chemical goggles, or face shield, impervious gloves and an impervious apron. Dis-

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solve 200 ± 1 g of sodium hydroxide in 800 mL of distilled water with stirring. Cool, transfer to a 1 L volumetric flask and adjust to volume with distilled water.

10.6 Sulfuric acid (20% by volume). In preparing, dispensing and handling sulfuric acid (98%), use chemical goggles, or face shield, impervious gloves and an impervious apron. Concentrated acids should be handled only in an adequately ventilated hood. CAUTION: Always add acid to water. Slowly add 200 ± 1 g of sulfuric acid to 500 mL of distilled water. Cool, transfer to a 1 L volumetric flask and adjust to volume with distilled water.

11. Procedure

11.1 Chelation value of chelating agent or product.

11.1.1 Weigh 0.9-1.1 g of the chelating agent or product to nearest 0.01 g. Dilute to 75 mL with distilled water.

11.1.2 Add 25 mL of sodium acetate buffer solution and adjust the pH to 4.5-5.5 with glacial acetic acid (to decrease pH) or 20% sodium hydroxide (to increase pH) as required.

11.1.3 Add 1 mL of PAN indicator and titrate to a permanent purple end point with Reagent A. Record the volume used to the nearest 0.01 mL and use this volume in Eq. 1

11.1.4 A 12.5 g/L solution of copper sulfate (Reagent A) is used to titrate the chelating agent product, whereas a 2.5 g/L solution of copper sulfate (Reagent B) is used for greater accuracy to titrate the more dilute chelating agent contained in the peroxide bleach solution.

11.2 Chelating agent in bleach bath.

11.2 1 Weigh 90-110 g of the hydrogen peroxide bleach solution to the nearest 0.01 g. If the specimen is taken from the saturator, ensure that it is free of debris and foam.

11.2.2 Adjust the pH to 7.0-9.0 with 20% sulfuric acid.

11.2.3 Add 35 mL sodium acetate buffer solution. Adjust the pH to 4.5-5.5 with glacial acetic acid or 20% sodium hydroxide if necessary.

11.2.4 Add 1 mL of PAN indicator and titrate to a permanent purple end point with Reagent B. Record the volume used to nearest 0.01 mL and use this volume in Eq. 2.

12. Calculations

12.1 Calculate the copper chelation value of the chelating agent or product to two decimal places using Eq. 1:

$$CuCV = 12.5\ (V)/W \qquad \text{(Eq. 1)}$$

where:

CuCV = Copper chelation value
V = volume of Reagent A used, mL
W = weight specimen, g

12.2 Calculate the mean copper chelation value of two samples.

12.3 Calculate the chelating agent content in the bleach bath to the nearest 0.01% using Eq. 2:

$$CA = 2.5(100)(V_2)/(W_2)(S) \qquad \text{(Eq. 2)}$$

where:

CA = chelating agent content, %
V_2 = volume Reagent B used, mL
W_2 = weight of specimen, g
S = average CuCV (from 12.2)

12.4 Calculate the average percent chelating agent content in the bleach bath.

13. Report

13.1 Describe or identify the samples/products tested.

13.2 Report that the sample was tested using AATCC TM185-2021.

13.3 Report the test results:

13.4.1 Report the average chelating agent value.

13.4.2 Report the average percent chelating agent content in the bleach bath.

13.5 Describe any modification(s) of the published standard.

14. Precision and Bias

14.1 *Precision*.

14.1.1 In 1990, an interlaboratory study was completed which included seven laboratories, one operator in each, running two determinations per sample on three samples in Step 11.1 of the procedure, and running two determinations per bath on three baths for Step 11.2 of the procedure. No prior assessment was made of the relative level of the participating laboratories in performance of the test method. Results from one lab were not included in the analysis.

14.1.2 Analysis of the data sets yielded components of variance and critical differences as displayed in Tables I, II and III. Differences between two averages of N determinations, for the appropriate precision parameter, should reach or exceed the table value to be statistically significant at the 95% confidence level.

14.2 *Bias*.

14.2.1 The values for CuCV can only be defined in terms of a test method. There is no independent, referee method for determining the true values. The method has no known bias.

Table I—Components of Variance

Component	Chelation Value	% Chelating Agent
Laboratory V(L)	2.61	0.000111
Interaction V(SL)	0.76	0.000157
Within V	0.81	0.000030

Table II—Chelating Agent or Product Critical Differences (95% Confidence)

Det. in Avg (N)	Within Laboratory	Between Laboratory
	(Single Sample Comparisons)	
1	2.49	5.13
2	1.76	4.81
3	1.44	4.71
	(Multiple Sample Comparisons)	
1	2.49	5.67
2	1.76	5.39
3	1.44	5.29

Table III—Bleach Bath Critical Differences (95% Confidence)

Det. in Avg (N)	Within Laboratory	Between Laboratory
	(Single Bath Comparisons)	
1	0.015	0.033
2	0.011	0.031
3	0.009	0.031
	(Multiple Bath Comparisons)	
1	0.015	0.048
2	0.011	0.047
3	0.009	0.046

15. Notes

15.1 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

16. History

16.1 Revised in 2021 to clarify and align with prescribed AATCC style guidelines.

16.2 Editorially revised 2019. Reaffirmed 2016, 2011. Editorially revised 2010. Reaffirmed 2006, 2000, 1999.

16.3 Developed in 1998 by AATCC Committee RR90. Maintained by RA99.

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Test Method for Weather Resistance: UV Light and Moisture Exposure

1. Purpose and Scope

1.1 This test method provides a procedure for the exposure of textile materials of all kinds, including coated fabrics and products made thereof, in a laboratory artificial weathering exposure apparatus employing fluorescent UV lamps as a light source and using condensing humidity and/or water spray for wetting.

2. Principle

2.1 Specimens are exposed to a fluorescent UV light source and to periodic wetting under controlled conditions. Resistance to degradation is rated in terms of a comparison standard and the exposure criteria, percent loss in strength or percent residual strength (breaking or bursting as appropriate) and/or color change of the material when evaluated under standard textile conditions.

3. Referenced Documents

3.1 The following AATCC documents are referenced:

3.1.1 AATCC EP6, Evaluation Procedure for Instrumental Color Measurement (see 17.4).

3.1.2 AATCC TM3.3, Test Method for Colorfastness to Light: Xenon-Arc (see 17.4).

3.2 The following ASTM documents are referenced:

3.2.1 ASTM D123, Standard Terminology Relating to Textiles (see 17.5).

3.2.2 ASTM D3787, Test Method for Bursting Strength of Knitted Goods: Constant Rate of Traverse (CRT) Ball Burst Test (see 17.5).

3.2.3 ASTM D5034, Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test) (see 17.5).

3.2.4 ASTM G151, Standard Practice for Exposing Non-Metallic Materials in Accelerated Test Devices that use Laboratory Light Sources (see 17.5).

3.2.5 ASTM G154, Standard Practice for Operating Fluorescent Light Apparatus for UV Exposure of Non-Metallic Materials (see 17.5).

3.3 The following SAE document is referenced:

3.3.1 SAE J2020, Standard Test Method for Accelerated Exposure of Automotive Exterior Materials Using a Fluorescent UV and Condensation Apparatus (see 17.6).

4. Terminology

4.1 **breaking strength,** n.—the maximum force applied to a specimen in a tensile test carried to rupture.

4.2 **bursting strength,** n.—the force or pressure required to rupture a textile by distending it with a force, applied at right angles to the plane of the fabric, under specified conditions.

4.3 **fluorescent UV lamp,** n.—a lamp in which radiation at 254 nm from a low-pressure mercury arc is transformed to longer wavelength UV by a phosphor.

4.4 **irradiance,** n.—radiant power per unit area as a function of wavelength expressed as watts per square meter, W/m^2.

4.5 **radiant energy,** n.—energy traveling through space in the form of photons or electromagnetic waves of various lengths.

4.6 **spectral irradiance,** n.—the variation of energy due to the source over the wavelength span of the emitted radiation.

4.7 **standard atmosphere for testing textiles,** n.—air maintained at 21 ± 2°C (70 ± 4°F) and 65 ± 5% relative humidity.

4.8 **standard atmosphere for storing textiles,** n.—laboratory conditions maintained as required in ASTM D1776.

4.9 **ultraviolet radiation**, n.—radiant energy for which the wavelengths of the monochromatic components are smaller than those for visible radiation and more than about 100 nm.

NOTE: The limits of the spectral range of ultraviolet radiation are not well defined and may vary according to the user. Committee E-2.1.2 of the CIE distinguishes in the spectral range between 400 and 100 nm:

UV-A	315-400 nm
UV-B	280-315 nm
UV-C	100-280 nm

4.10 **UV-A Type Fluorescent UV lamp,** n.—a fluorescent UV lamp where radiant emission below 300 nm is less than 2% of its total light output.

4.11 **UV-B Type Fluorescent UV lamp,** n.—a fluorescent UV lamp where radiant emission below 300 nm is more than 10% of its total light output.

4.12 **weather,** n.—climatic conditions at a given geographic location, including such factors as sunlight, rain, humidity and temperature.

4.13 **weather resistance,** n.—ability of a material to resist degradation of its properties when exposed to climatic conditions.

5. Safety Precautions

5.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

5.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

5.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

5.4 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5.5 Do not operate the test equipment until the manufacturer's operating instructions have been read and understood. It is the responsibility of whoever operates the test equipment to conform to the manufacturer's directions for safe operation.

5.6 The test equipment emits ultraviolet radiation. Do not look directly at the light source. During normal operation, care must be taken to shield operators from ultraviolet radiation, either by closing the chamber door, using specimen holders that prevent light from escaping, or some other effective method.

5.7 Before servicing the light sources, allow 30 min for cool-down after the lamp operation has been terminated.

5.8 When servicing the test equipment, shut off both the "off" switch on the front panel and the main power disconnect switch. When equipped, ensure that the main power light on the machine front panel goes out.

6. Uses and Limitations

6.1 The use of this procedure is intended to simulate the deterioration caused by the UV energy in sunlight and water. Exposures are not intended to simulate the deterioration caused by localized weather phenomena, such as atmospheric pollution, biological attack and salt water exposure.

6.2 Cautions. Variation in results may be expected when operating conditions are varied within the accepted limits of this procedure. Therefore, no reference shall be made to results from the use of this procedure unless accompanied by a report detailing the specific operating conditions in conformance with the sec-

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tion on Report.

6.3 Results obtained from this procedure can be used to compare the relative durability of materials subjected to the specific test cycle used. Comparison of results from specimens exposed in different types of apparatus should not be made unless correlation has been established among devices for the material to be tested. Variations in results may be expected when operating conditions vary within the limits of this procedure. Because of the variability in results obtained using this practice and the variability in results from exterior exposures, use of a single "acceleration factor" that relates hours of an accelerated exposure to a specific period of outdoor exposure is not recommended. Because of possible variations in results, no reference should be made to results obtained from tests conducted in the apparatus using this procedure unless accompanied by the information required in the section on Report.

6.4 There are a number of factors that may decrease the degree of correlation between accelerated tests using laboratory light sources and actual use exposures.

6.4.1 Differences in the spectral distribution between the laboratory light source and sunlight.

6.4.2 Shorter than normal wavelength exposures are often used to obtain faster failure rates in laboratory accelerated exposure tests. For outdoor exposures, the cut-on for short wavelength UV radiation is generally considered to be 300 nm. Exposures to UV radiation of wavelengths less than 300 nm, may produce degradation reactions, which do not occur when the material is used outdoors. If a laboratory light source used in an accelerated test contains UV radiation of wavelengths shorter than that found in the actual use condition, the mechanism of degradation and stability ranking of materials can be dramatically different in the accelerated test.

6.5 It may not be necessary to simulate daylight over the entire spectrum, if radiation in a specific region is known to produce the type of degradation of interest in the materials being tested and does not alter stability ranking of materials. Laboratory light sources, which have a very strong emission in a narrow band relative to the rest of the ultraviolet or visible spectrum, however, may cause a particular reaction to be favored relative to others which may be very important. This type of light source also may not produce changes caused in exposures to daylight. Exposures to light sources, which only produce ultraviolet radiation may not produce color fade caused by visible radiation and may cause polymer yellowing that is more pronounced than that produced in exposures to daylight.

7. Apparatus (See 17.1, 17.2 and 17.3)

7.1 Test chamber (see 17.3).

7.2 UV-A type fluorescent UV lamp (see 17.7).

7.3 Moisture system.

7.3.1 Condensation. The moisture system may be used to generate either condensation or water spray or both (see 17.8).

7.3.2 Water spray. The test chamber may be equipped with a means to introduce intermittent water spray onto the test specimens under specified conditions. The spray shall be uniformly distributed over the samples. The spray system shall be made from corrosion resistant materials that do not contaminate the water employed.

7.4 Black panel thermometer (see 17.9 and 17.10).

7.5 Specimen holders (see 17.11).

7.6 Test chamber location (see 17.12).

8. Test Samples

8.1 Number of Specimens Replicate specimens of the material to be tested and a comparable number of the standard for comparison should be used as required to ensure accuracy. It is recommended that at least three replicates of each material evaluated be exposed in each test to allow for statistical evaluation of results.

Expose a sufficient number of specimens so that the expected test result is not more than 5% of the average above or below the true average of the lot at the 95% probability level. Determine the number of specimens using standard deviation with one-sided limits as directed in ASTM D2905, Standard Practice for Statements on Number of Specimens for Textiles.

8.2 Specimen Dimensions. Certain materials may exhibit a dimensional change as a result of exposure. The test equipment manufacturer, physical test apparatus and number of replicate specimens required will affect the needed specimen dimension. The test procedures which will be used to assess the change in properties should be reviewed to ensure that specimen size is appropriate for performing the subsequent procedures.

Cut strips of fabric at least 102 mm x 152 mm with the longer dimension parallel to the warp of machine direction, unless otherwise specified, for the following tests:

8.2.1 Bursting Strength (Ball Burst).

8.2.2 Breaking Strength (Grab Tensile).

8.2.3 Color Change.

8.2.4 When required to prevent raveling, the specimens may be edged using a flexible epoxy resin or similar material.

8.2.5 Label each specimen for identification using material resistant to the environment encountered during the test.

9. Test Cycle Determination

9.1 The test cycle is best determined by the influencing factors of the end use, in particular, the climatic conditions. Not all materials are affected equally by the same environment. Results obtained by the use of any one test cycle may not be representative of those of any other test cycles or any outdoor weathering test. Any acceleration factor derived for one geographic location does not necessarily apply to any other geographic location. However, certain test cycles are suggested to group similar climates with respect to the test cycle.

9.2 The nature of the test material contributes to the selection of the appropriate test cycle with respect to UV exposure, wetting, wet time and temperature. The following test cycles options have been used for textile materials.

9.2.1 Option 1, General Applications: 8 h UV at an irradiance of 0.77 W/m^2 @ 340 nm at 60°C (140°F) followed by 4 h condensation at 50°C (122°F). This option is used for general applications such as outdoor furniture fabrics, tent material, etc.

9.2.2 Option 2, Thermal Shock Applications: 8 h UV at an irradiance of 0.77 W/m^2 @ 340 nm at 60°C (140°F); followed by 0.25 h water spray; followed by 3.75 h condensation at 50°C (122°F). This option has been used for architectural and other applications where thermal shock may be an issue.

9.2.3 Option 3, Automotive Exterior: 8 h UV at an irradiance of 0.72 W/m^2 @ 340 nm at 70°C (158°F) followed by 4 h condensation at 50°C (122°F). UV irradiance may be monitored and maintained by the manual method or by the feedback-loop method as described in SAE J2020.

9.3 The use of these cycles does not imply, expressly or otherwise, an accelerated weathering test, nor is this method restricted to the use of these cycles. The degree of correlation to any actual outdoor weathering exposure must be determined by quantitative analysis.

9.4 The test conditions in this method are subject to acceptable test apparatus operational fluctuation of ± 0.02 W/m^2 @ 340 mm for irradiance and ± 2°C for temperature. In addition, a transition time of approximately one hour between test conditions is acceptable.(See ASTM G151-19, Annex A.3)

10. Standards of Comparison

10.1 Standards for comparisons can be made of any suitable textile material where a history of strength degradation or color change rates are known depending upon individual test needs. They must be exposed simultaneously with the test specimen to be investigated. The intent of

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these standards is to demonstrate uniformity between separate machine and test runs. When test data of the exposed standards differ by more than 10% from the known data, the test machine operating conditions must be thoroughly reviewed, and any malfunctions or defective parts corrected. The test is then repeated. If the data still differ by more than 10% from the known data and there is no evidence of machine malfunction, then the test standard should be questioned and re-evaluated. Test data obtained with questionable standards must be treated with caution and resolution provided with quantitative analysis.

11. Procedure

11.1 Maintain and calibrate the apparatus in accordance with the manufacturer's recommendations.

11.2 Before beginning the exposure test, bring all specimens, control and test, to moisture equilibrium in the atmosphere for testing textiles in accordance with ASTM D1776, Standard Practice for Conditioning and Testing Textiles. Equilibrium is considered to have been reached when the increase in mass of the specimen in successive weightings made at intervals of not less than 2 h does not exceed 0.2% of the mass of the specimen. Perform any necessary tests or evaluations necessary to establish a base line for comparing the unexposed specimens to the exposed specimens.

11.3 Specimen Mounting. Mount the specimens in the frames which are supplied with the cabinet with the test surfaces facing the lamp. When the test specimens do not completely fill the racks, the empty spaces must be filled with blank panels to maintain the test conditions within the chamber.

11.3.1 To provide rigidity, flexible specimens may be attached to a backing panel made of aluminum or other noncorrosive heat conductive material.

11.3.2 Holes in specimens and any openings larger than 1 mm around irregularly shaped specimens shall be sealed to prevent loss of water vapor. Porous specimens shall be backed with a vapor barrier such as aluminum or plastic.

11.3.3 Fabrics. Flexible fabric specimens are mounted by simply wrapping them around an aluminum blank and clamping them into place with the spring ring clip. The specimens should present a smooth face to the inside of the chamber (see Fig. 1).

11.3.4 Yarns. Yarns should be wound on frames to a length of 150 mm minimum. Only that portion of the yarns directly facing the radiant energy is tested for breaking (tensile) strength. Single strand or multiple strand testing may be performed. When multiple strand testing

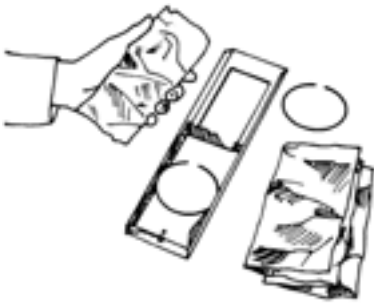


Fig. 1—Typical mounting for flexible fabric specimens.

is to be performed, the yarns are wound on the frame closely packed to 25.4 mm width. The control specimens must contain the same number of strands as the specimen subjected to exposure. After the exposure has been completed and before the yarns are unwound for testing, those yarns facing the light source are bound together by the use of 20 mm masking or other suitable type tape to maintain the closely packed arrangements on the exposure frame.

11.3.5 In the case of woven, knitted and nonwoven fabrics ensure that the side directly exposed to the radiant source is the one normally used as the face.

11.4 Program the device to achieve the required test conditions and operate the apparatus continuously within the limits specified above. Use test conditions specified in 8.2 or agreed to by mutual consent or as required by a product quality specification.

11.5 Operate continuously, repeating the cycle, except for servicing the instrument and inspection of specimens. Inspect specimens daily during the middle of a condensation cycle to make sure all specimens are uniformly wetted.

11.6 To minimize any effects from temperature or UV light variation, it is recommended that specimens be repositioned. Reposition specimens horizontally by (1) moving the two extreme right hand specimen holders to the far left of the exposure area and (2) sliding the remaining specimen holders to the right as shown in Fig. 2. For short duration tests (less than one week), reposition specimens at least once a day. For longer duration tests, reposition specimens at least five times during the duration of the test.

12. Periods of Exposure

12.1 Use one of the following methods to determine the duration of the exposure:

12.1.1 A specified number of total hours.

12.1.2 The number of total hours of exposure required to produce an amount of change in either the test specimen or an agreed upon standard sample.

13. Conditioning

13.1 If the test and control samples are wet upon removal from the tester, dry them at ambient laboratory conditions or at a temperature not exceeding 71°C (160°F).

13.2 Condition the test and control samples in the controlled atmosphere for testing textiles. Bring all specimens to moisture equilibrium. Equilibrium is considered to have been reached when the increase in mass of the specimen in successive weighings made at intervals of not less than 2 h does not exceed 0.2% of the mass of the specimen. In general practice, the industry approaches equilibrium from the "As Received" side.

13.3 For each test to be made on the material and control, exposed and unexposed, prepare test specimens by marking and raveling or cutting the central portion of each exposed specimen to the dimensions specified in the respective test procedure. Marking and raveling or cutting of the test specimens is preferred after the exposure but may be done before exposure. Control specimens not exposed are similarly prepared and are wet-out and allowed to dry without tension before testing.

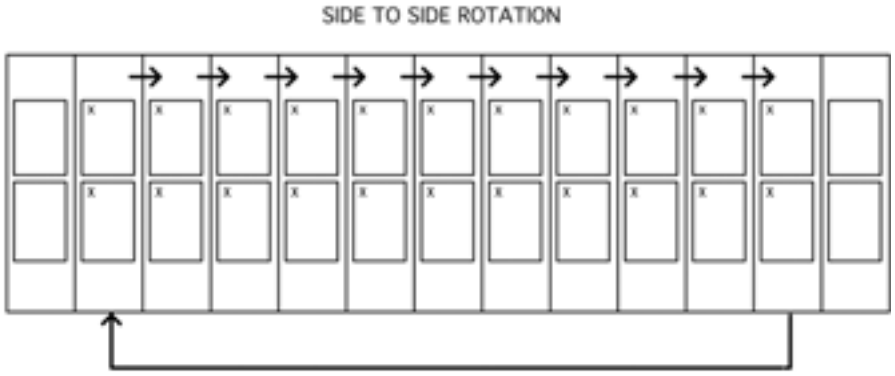


Fig. 2—Specimen rotation.

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14. Evaluation of Results

14.1 Changes in exposed test specimens shall be evaluated or rated by applicable AATCC, ASTM or ISO test methods.

14.2 Physical Properties

14.2.1 Ball Bursting Strength of Fabrics. Determine the ball bursting strength of fabrics as directed in ASTM D3787, Test Method for Bursting Strength of Knitted Goods: Constant Rate of Traverse (CRT) Ball Burst Test.

14.2.2 Grab Tensile Test. Determine the grab tensile strength as directed in ASTM D5034, Test Method for Breaking Strength and elongation of Textile Fabrics (Grab Test).

14.3 Color Change.

14.3.1 Evaluate the color change as directed in AATCC TM16.3, Test Method for Colorfastness to Light: Xenon-Arc.

15. Report

15.1 Report the following information on exposure conditions:

15.1.1 Reference to AATCC TM186-2023.

15.1.2 Manufacturer and model of fluorescent UV/condensation apparatus.

15.1.3 The manufacturer's designation for the fluorescent UV lamp.

15.1.4 Exposure cycle, for example, 4 h UV/60°C (140°F), 4 h CON/50°C (122°F).

15.1.5 Total exposure time.

15.1.6 Total UV light exposure time.

15.1.7 Any deviation from the exposure test method

15.2 Report the following information on the test specimen:

15.2.1 The type of fiber(s) of which the material is composed, which side of the fabric was exposed (in the event that the fibers differ on the face and back of the fabric), the fabric weight in g/m^2 and the nature of the fabric finish, if known.

15.3 Report the following information on the evaluation:

15.3.1 The evaluation method, rating and relative data for each property evaluated.

15.3.2 The standard used for comparative evaluation, if any.

15.3.3 Data. Average the data for the various replicates, or handle statistically as appropriate, and record a significant value of breaking or bursting strength retention and/or color change after exposure in relation to original strength or color as applicable. Report must contain as a minimum:

(a) Arithmetic mean or average,

(b) The number of tests,

(c) Standard deviation or coefficient of variation.

Statement of a mean without the number of tests and precision is essentially useless.

16. Precision and Bias

16.1 *Precision*.

16.1.1 *Laboratory Study*. In early 1999, a small study was done in a single laboratory to get an estimate of intra-laboratory precision. One fabric (greige, #400 style cotton print cloth) was exposed under conditions of this test method and values determined for DE^*_{ab}, grab tensile after exposure, and ball burst after exposure.

16.1.2 *Intra-Laboratory Precision*. Components of variance and *within-laboratory* precision shown as critical differences are given for each of these properties, respectively, in Tables I, II and III.

16.1.3 For each of the properties noted, when differences are due only to chance causes, the differences between test results should not exceed the value shown in 95 of the 100 comparisons.

16.1.4 Analysis of variance or *t*-tests may be used to compare averages. See any standard statistical text for more information.

Table I—ΔE^*_{ab}

Mean	Sample Variance	Standard Deviation
9.7	0.2	0.4
	95% Level	
Values in Average	Standard Error	Critical Difference
1	0.4	1.2
2	0.3	0.8
3	0.2	0.7
4	0.2	0.6
5	0.2	0.5
6	0.2	0.5
7	0.2	0.5
8	0.2	0.4
9	0.1	0.4
10	0.1	0.4

Table II—Grab Tensile after Exposure

Exposed Grab		
Mean	Sample Variance	Standard Deviation
59	29	5.4
	95% Level	
Values in Average	Standard Error	Critical Difference
1	5.4	15.1
2	3.8	10.4
3	3.1	8.7
4	2.7	7.5
5	2.4	6.7
10	1.7	4.8
Control Grab		
Mean	Sample Variance	Standard Deviation
75.2	6.6	2.6

Table III—Ball Burst after Exposure

Exposed Ball Burst		
Mean	Sample Variance	Standard Deviation
83	81	9
	95% Level	
Values in Average	Standard Error	Critical Difference
1	9.0	25.2
2	6.4	17.8
3	5.2	14.6
4	4.5	12.6
5	4.0	11.3
10	2.8	8.0
Control Ball Burst		
Mean	Sample Variance	Standard Deviation
87	55	7.4

16.2 *Bias*.

16.2.1 There is no referee test method for determining definitive values to establish bias in this test method. The test method has no known bias.

17. Notes

17.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

17.2 Refer to ASTM G151 and G154 for guidance on the design and performance requirements for instruments specified in this method (see 3.2).

17.3 The exposure chamber shall be a Fluorescent UV/Condensation Apparatus constructed of corrosion-resistant materials enclosing eight fluorescent UV lamps, a heated water pan, water spray system (optional), test specimen racks and provisions for controlling and indicating operating times and temperatures

17.4 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

17.5 Available from ASTM International 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832.9500; fax: +1.610.832.9555; web site: www. astm.org.

17.6 Available from SAE International, 400 Commonwealth Dr., Warrendale PA 15098-0001; tel: +1.412.776.4841; web site: www.sae.org.

17.7 The limits of the spectral range of ultraviolet radiation are not well defined and may vary according to the user. Committee E2.1.2 of CIE distinguishes in the spectral range between 400 and 100 nm:

UV-A	315-400 nm
UV-B	280-315 nm
UV-C	100-280 nm

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Unless otherwise specified, the lamps shall be UV-A Type Fluorescent UV lamps with a peak emission at 343 nm and a spectral irradiance as shown in Fig. 3.

17.8 Condensation Mechanism. Water vapor shall be generated by heating a water pan extending under the entire specimen mounting area and containing a minimum water depth of 25 mm. Specimen racks and the test specimens themselves shall constitute the sidewalls of the chamber. The backside of the specimens shall be exposed to cooling effects of ambient room air. The resulting heat transfer causes water to condense on the test surface.

17.8.1 The specimens are arranged so that condensate runs off the test surface by gravity and is replaced by fresh condensate in a continuous process. Vents along the bottom of the test chamber shall be provided to permit an exchange of ambient air and water vapor to prevent oxygen depletion of the condensate.

17.8.2 Water supply with an automatic control to regulate the level in the water pan shall be provided. Distilled, deionized or potable tap water are equally acceptable for purposes of the test, because the condensation process itself distills water onto the test surface.

17.9 Specimen temperature shall be measured by a Black Panel Thermometer with a remote sensor attached to a black aluminum panel. The thermometer shall be precise to ± 1°C through a range from 30-80°C and shall be positioned in the center of the exposure area so that the sensor is subject to the same conditions as the specimens.

17.10 During UV exposure, the selected equilibrium temperature shall be maintained within ± 2°C by supplying heated air to the test chamber. During condensation exposure, the selected equilibrium temperature shall be maintained within ± 2°C by heating the water in the water pan.

17.11 The test specimens shall be mounted in a stationary specimen exposure area with the plane of the test surface facing the lamps. It is recommended that specimens are mounted and exposed within an area 210 mm in height × 900 mm wide on each side of the apparatus as shown in Fig. 4. Specimen repositioning is good practice and is always recommended. However, if specimens are mounted above, below, or to the side of the 210mm × 900 mm area, specimen repositioning is required as indicated by the dashed line in Fig. 2.

17.12 Apparatus should be located in an area maintained at a temperature between 20°C and 30°C. The room temperature shall be measured by thermometers mounted on interior walls or columns approximately 1500 mm above the floor level and at least 300 mm from any heated apparatus. Three or more thermometers located at various points will show any temperature variation in the area.

17.12.1 It is recommended that the apparatus be located at least 300 mm from walls or other apparatus. Nearby heat sources, such as ovens or heated test apparatus, should be avoided or shielded, because such heat sources can reduce the cooling required for condensation.

17.12.2 The room where the apparatus is located shall be ventilated to remove the heat and moisture produced and to maintain the temperatures specified above. Two to four air changes per hour will normally provide sufficient ventilation.

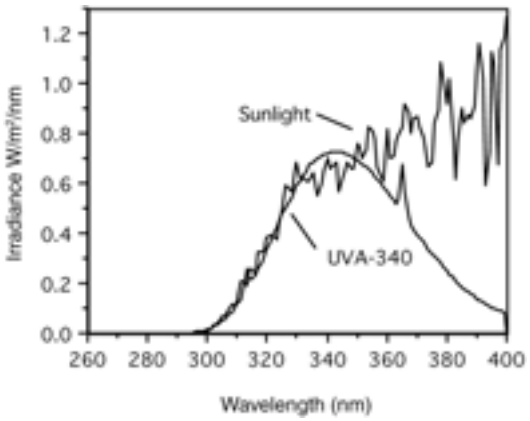


Fig. 3—Representative spectral power distribution, UVA-340 fluorescent lamps

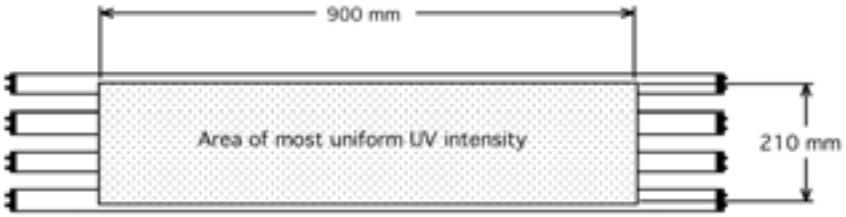


Fig. 4—Area of most uniform light intensity.

18. History

18.1 Revised in 2023 for clarity and to align with style guidelines.

18.2 Revised 2022; editorially revised 2019, 2016; revised 2015; editorially revised and reaffirmed 2013; revised 2009; editorially revised 2008, 2007; reaffirmed 2006, 2001; revised 2000.

18.3 Developed in 1999 by AATCC Committee RR64; jurisdiction transferred in 2007 to Committee RA5.

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Test Method for Dimensional Changes of Fabrics: Accelerated

Developed in 2000 by AATCC Committee RA42; reaffirmed 2001, 2002; revised 2004, 2013; editorially revised 2006, 2008, 2011, 2019; editorially revised and reaffirmed 2009. Related to ISO 23231.

1. Purpose and Scope

1.1 This test method is an accelerated procedure for determination of the dimensional change of fabrics. The procedure uses an apparatus with programmable settings that simulate multiple home or commercial launderings or wet processing.

1.2 This method does not replace dimensional change test methods currently in use.

2. Principle

2.1 Dimensional changes are determined by comparing the distances between length and width direction bench marks before and after a programmed test cycle.

3. Terminology

3.1 **dimensional change,** n.—a generic term for changes in length or width of a fabric specimen subjected to specified conditions. The change is usually expressed as a percentage of the initial dimension of a specimen.

3.2 **growth,** n.—*of textile materials*, a dimensional change resulting in an increase of length or width of a specimen.

3.3 **laundering,** n.—*of textile materials*, a process intended to remove soils and/or stains by treatment (washing) with an aqueous detergent solution and normally including subsequent rinsing, extracting and drying.

3.3 **shrinkage,** n.—*of textile materials*, dimensional change resulting in a decrease in the length or width of a specimen.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all-inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Manufacturer's safety recommendations should be followed when operating accelerated testing equipment.

5. Uses and Limitations

5.1 Although data have been generated comparing the dimensional change of some textile fabrics after home laundering and this accelerated apparatus, the user will find it necessary to determine the correlation between the results from a selected program for this apparatus and the dimensional change results from other chosen test methods or wet processes (see 14.4).

5.2 While the term "laundering" includes the use of an aqueous detergent (during washing) and subsequent drying, this accelerated method does not use detergents.

6. Apparatus and Materials

6.1 Accelerated laundering apparatus (see 14.1).

6.2 Indelible ink marking pen. Select marking pens with the smallest tip (see 14.2).

6.3 Tape or ruler marked in millimeters.

6.4 A tape or ruled template marked with, or from which, percent dimensional change to 0.5% or smaller increment can be directly read (see 14.2).

6.5 Digital Imaging System (see 14.3).

7. Sampling

7.1 Take a sufficient number of samples from a lot to provide an adequate number of data points to generate satisfactory precision.

7.2 It is recommended that a minimum of four specimens with different length and widthwise yarns in each specimen be taken from each sample.

8. Specimens

8.1 Cut specimens parallel to the lengthwise fabric direction as shown in Fig. 1a. Diagonal alignment of specimens may be selected to minimize the need for overedging specimens and to obtain larger bench mark dimensions (see Fig. 1b).

8.2 The size of specimens may depend upon the model of the apparatus and the type, number and configuration of its basket chambers.

8.2.1 Specimens, 190 × 190 mm, aligned with the fabric length and width directions and marked with 125 mm bench mark distances are recommended by the instrument manufacturer. Alternatively, 255 × 255 mm specimens can be cut diagonally to accommodate 255 mm bench marks distances that are aligned with the actual length and width directions of the fabric (see Fig. 1b).

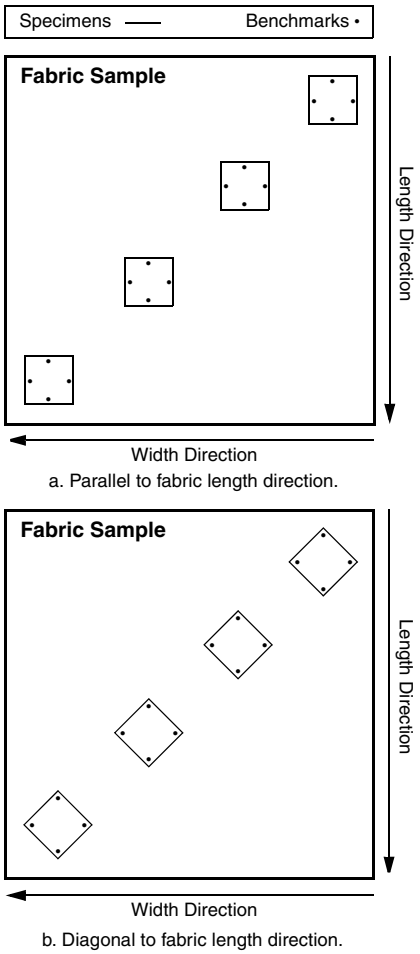


Fig. 1—Specimen and bench mark placement.

8.3 The use of standard textile conditioning prior to or after a test cycle program completion is not required.

8.4 To improve the accuracy and precision of dimensional change calculations based on the bench mark distances applied to the specimens as in 8.2.1, measure and record the distance between each pair of bench marks with suitable tape or ruler to the nearest millimeter before starting the test. This is measurement A. If using a template or ruler from which direct percent dimensional change can be read, verify that the original bench mark distances are exact.

9. Test Procedure

9.1 Using the program in Table I, de-

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termine whether the results obtained from this program correlate with testing the same fabric by selected end-use test procedure. If the program in Table I does not generate satisfactory correlation with dimensional change results from the selected end-use test procedure, alter program elements such as number of rinses and drying time.

9.2 Follow the equipment manufacturer's recommendations for the number of specimens to be tested in each basket chamber.

9.3 Start the machine and allow the program to run through to completion.

9.4 Ensure that specimens are completely dry at the end of the drying program. If specimens hang up in the chamber during drying, they will not be completely dry. If repeated tests with the same program cycles results in specimens hanging up in the chamber, the users should try an alternate program.

9.4.1 If test specimens continue to hang up, contact the equipment manufacturer for other possible alternate programs.

9.5 After completion of the selected program, remove specimens and place them on a flat surface for a minimum of 5 min prior to measurement.

10. Measurement

10.1 Measure and record bench mark distances for each specimen in each fabric direction to the nearest increment of the ruler or device used. This is measurement B (see 11.2).

10.2 Measure and record, to the nearest incremental unit in each fabric direction, the percent dimensional change from the ruler or device used.

11. Calculation

11.1 If measurements were taken in units of percent dimensional change, average the measurements in each direction separately to the nearest 0.1%.

11.2 If bench mark measurements were taken, calculate the individual and average dimensional change results in each direction separately to the nearest 0.1% as follows:

$$\%DC = 100[(B-A)/A]$$

where:

DC = Dimensional change
A = Original bench mark distance
B = Bench mark distance after laundering test cycle

11.3 A final measurement smaller than the original measurement indicates negative dimensional change which is shrinkage (minus sign). A final measurement larger than the original measurement indicates positive dimensional change which is growth (plus sign).

12. Report

12.1 For each sample tested, report:

(a) Individual and average percent length and width dimensional change, separately, to the nearest 0.1% with the appropriate dimensional sign (shrinkage negative/growth positive);

(b) Program used; and

(c) Specimen size, alignment (see Figs. 1a or 1b) and number of specimens in each basket chamber.

13. Precision and Bias

13.1 *Precision*.

13.1.1 In a preliminary multiple lab study, five fabrics were tested in ten labs. One operator tested samples on two different days in each lab. Study samples were: 80 cotton/20 polyester knit fleece, 100% cotton twill, 50 cotton/50 polyester knit jersey, 100% cotton oxford and 100% cotton knit pique. Lab results were reported as the average of ten specimens for each fabric. Lab results were obtained on two different days for each fabric. While the total study included test results from five accelerated cycles, the data used in this precision analysis were the reported dimensional change results obtained from only the first accelerated test cycle.

13.1.2 The reported precision was obtained from a component of variance analysis. Two different standard deviations were calculated: the standard deviation of within lab variation (Se) and the standard deviation of lab-to-lab and within lab variations combined (Sc).

13.1.3 The preliminary precision was obtained from a component of variance analysis. Tables II and III show the interaction of the analysis and standard deviations: the standard deviation of within-lab variation (Se) and the standard deviation of lab-to-lab and within-lab variations combined (Sc) for each fabric direction. Average dimensional change percent and component of variance analysis for each sample is shown in Appendix A.

13.1.4 Tables IV and V show single fabric and multiple fabric *within-* and *between-laboratory* standard error (SE) and critical difference (CD) values.

13.2 *Bias*.

13.2.1 Dimensional change (%) can be defined only in terms of a test method or procedure. This test method has no known bias.

14. Notes

14.1 Quickwash Plus™ System may be obtained from SDL Atlas L.L.C., 3934 Airway Dr., Rock Hill, SC 29732-9200, tel: +1.803.329.2110, fax: +1.803.329.2133; e-mail: info@sdlatlas.com; web site: www.sdlatlas.com.

14.2 Marking pens are available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709, tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org. Measuring ruler available from AATCC.

14.3 A digital imaging system may be used as a measuring device in place of the prescribed manual measurement devices if it is established that its accuracy is equivalent to the manual devices.

14.4 Dimensional changes exhibited by articles produced from textile fabrics are primarily (but not entirely) dependent on dimensional changes exhibited by the fabric.

Table I—Accelerated Machine Program Settings*

Program Operation	Number of Cycles	Time per Cycle, s	Temperature, °C
Wash**	1		60°C
Agitation time		165 s	
Rinse/dry	3		60°C
Agitation time		45 s	
Spin time		35 s	
Dry time		240 s	
Air pressure, 3.8 bars			
Water level, 3 L			

*According to the machine manufacturer, prior tests have established that this program generates 95% of the total dimensional change in 95% of fabrics tested.

**The term "wash" implies the use of detergent, but correlation tests by users have shown that the addition of detergent is not necessary in this accelerated test. Due to the nature of the action of the accelerated test procedure, even a slight amount of detergent foams and impedes the test.

Table II—Component of Variance Analysis

COV	Variance Length	Variance Width
Lab V(L)	0.26	0.11
Fab/Lab V(FL)	0.32	0.23
Days W/I Fabric V(D(F))	0.36	0.00
Samples W/I Days and Fabric V(S(DF))	0.00	0.08
Error V	0.19	0.08

Table III—Within Lab and Combined Lab-to-Lab and Within Lab Percent Dimensional Change at the 95% Probability Level = 2Se and 2Sc

	LENGTH		WIDTH	
FABRIC	Se	Sc	Se	Sc
Knit Fleece	0.49	0.69	0.38	0.70
Twill	0.34	1.01	0.20	0.35
Knit Jersey	0.34	0.95	0.37	0.52
Oxford	0.53	0.88	0.55	1.06
Knit Pique	0.29	0.66	0.17	0.48

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Table IV—Length Precision

	Single Fabric Within Lab		Comparison Between Labs		Multiple Fabric Within Lab		Comparisons Between Labs	
Det/Avg	(Single Op)				(Single Op)			
N	SE	CD	SE	CD	SE	CD	SE	CD
1	0.42	1.17	0.66	1.84	0.71	1.96	0.87	2.41
2	0.30	0.83	0.59	1.64	0.64	1.77	0.82	2.27
3	0.24	0.68	0.57	1.57	0.62	1.71	0.80	2.22
4	0.21	0.59	0.55	1.53	0.60	1.67	0.79	2.19
5	0.19	0.52	0.54	1.51	0.60	1.65	0.79	2.18
7	0.16	0.44	0.53	1.48	0.59	1.63	0.78	2.16

Table V—Width Precision

	Single Fabric Within Lab		Comparison Between Labs		Multiple Fabric Within Lab		Comparisons Between Labs	
Det/Avg	(Single Op)				(Single Op)			
N	SE	CD	SE	CD	SE	CD	SE	CD
1	0.29	0.80	0.44	1.23	0.56	1.54	0.65	1.80
2	0.21	0.57	0.39	1.09	0.52	1.42	0.61	1.71
3	0.17	0.46	0.38	1.04	0.50	1.39	0.61	1.68
4	0.15	0.40	0.37	1.02	0.50	1.38	0.60	1.67
5	0.13	0.40	0.36	1.00	0.49	1.36	0.60	1.65
7	0.11	0.30	0.35	0.98	0.49	1.35	0.59	1.64

Appendix A. Component of Variance Analysis for Individual Fabrics*

Fabric	Direction	DC% AVG	Lab V(L)	Day V(D)	Lab/Day V(L/D)	Fabrics V(F W/I D)	Error V
Fleece	length	11.34	0.24	1.19	0.12	0.00	0.49
80c/20p	width	5.40	0.16	0.15	0.00	0.19	0.14
Twill	length	3.84	0.91	0.17	0.26	0.28	0.34
100c	width	1.21	0.08	0.01	0.01	0.00	0.04
Jersey	length	6.06	0.61	0.10	0.36	0.90	0.34
50c/50p	width	1.11	0.13	0.00	0.04	0.00	0.14
Oxford	length	2.87	0.48	0.05	0.14	0.00	0.54
100c	width	4.72	0.83	0.04	0.00	0.00	0.30
Pique	length	9.05	0.35	0.29	0.11	0.02	0.29
100c	width	6.98	0.20	0.01	0.08	0.00	0.03

*DC = Dimensional change
V = Variance
L = Lab
D = Day
F = Fabric
W/I = Within

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Test Method for Colorfastness to Sodium Hypochlorite Bleach in Home Laundering

Developed in 2000 by AATCC Committee RA60; editorially revised and reaffirmed 2001, 2008; reaffirmed 2002, 2017; revised 2003, 2010; editorially revised 2004, 2012, 2013, 2016, 2019.

1. Purpose and Scope

1.1 This test method is used to determine the colorfastness to sodium hypochlorite bleach, commonly called "chlorine bleach," in home laundering of textiles which are expected to withstand frequent laundering.

1.2 Results of this test method may be used in combination with other testing for establishing care instructions (see 5.1, 9.6 and 12.8).

1.3 If a chlorine bleach contains ingredients other than the sodium hypochlorite, it is the total effect of these chemicals on the color change that is being evaluated.

1.4 This test method is designed for home laundering devices. An accelerated colorfastness test simulating multiple home laundering (including those with sodium hypochlorite) is described in AATCC TM61, Test Method for Colorfastness to Laundering: Accelerated. There is no known correlation between those sections of AATCC TM61 and this method.

2. Principle

2.1 Specimens are laundered under appropriate conditions of temperature, detergent solution, chlorine bleach solution and abrasive action for five home laundering cycles. The laundered specimens are evaluated for color change.

3. Terminology

3.1 **bleach,** n.—*in home laundering*, a product that will clean, whiten, brighten and aid in the removal of soils and stains from textile materials by oxidation that is inclusive of chlorine and non-chlorine products.

3.2 **care instructions,** n.—*in textiles*, a series of directions describing which care practices should refurbish a product without adverse effects and warning of those care practices expected to have a harmful effect (see 12.1).

3.3 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the materials.

3.4 **laundering,** n.—*of textile materials*, a process intended to remove soil and/or stains by treatment (washing) with an aqueous detergent solution and normally including subsequent rinsing, extracting and drying.

3.5 **sodium hypochlorite bleach,** n.—4-6% solution of sodium hypochlorite (NaOCl), pH 9.8-12.8, commonly called "chlorine bleach."

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

4.3 The 1993 AATCC Standard Reference Detergent WOB and the 2003 AATCC Standard Reference Liquid Detergent WOB may cause irritation. Care should be taken to prevent exposure to skin and eyes.

4.4 All chemicals should be handled with care.

4.5 In preparing, dispensing and handling bleach and detergent, use chemical goggles or face shield, impervious gloves and an impervious apron.

4.6 If concentrated sulfuric acid is diluted to prepare the 10% acid (see 14.8.1 and 14.8.3), use chemical goggles or face shield, impervious gloves and an impervious apron. Concentrated acids should be handled only in an adequately ventilated hood. CAUTION: Always add acid to water.

4.7 An eyewash/safety shower should be located nearby and an organic vapor respirator should be available for emergency use.

4.8 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienist (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 14.1).

5. Uses and Limitations

5.1 Specimens may be tested using these procedures, but without sodium hypochlorite bleach, to determine the effect of laundering with water and/or water and detergent alone (see 9.6 and 12.8). To test for colorfastness to non-chlorine bleach, see AATCC TM172, Colorfastness to Powdered Non-Chlorine Bleach in Home Laundering (see 14.2).

6. Apparatus and Materials (see 14.3)

6.1 Automatic washing machine (see 14.4).

6.2 Automatic tumble dryer (see 14.5).

6.3 Conditioning/drying racks with pull-out screens or perforated shelves (see 14.6).

6.4 Ballast of 920 × 920 mm (± 30 mm) hemmed pieces of cloth. Choose one of the following.

6.4.1 Ballast type 1, bleached cotton sheeting.

6.4.2 Wash load ballast type 3, 50/50 polyester/cotton plain weave.

6.5 1993 AATCC Standard Reference Detergent WOB or 2003 AATCC Standard Reference Liquid Detergent WOB (see 14.7).

6.6 Drip and line drying facilities.

6.7 Balance with at least 5 kg minimum capacity and sensitivity to ± 0.1 g.

6.8 Lab timer.

6.9 Gray Scale for Color Change (AATCC EP1) (see 14.7) or suitable colorimeter or spectrophotometer for color change.

7. Reagents

7.1 Sodium hypochlorite (NaOCl), 4-6%.

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7.2 Distilled water.

7.3 Sulfuric acid, H_2SO_4, (10% solution).

7.4 Potassium iodide, KI, (10% solution).

7.5 Sodium thiosulfate, $Na_2S_2O_3$, (0.1*N*).

8. Test Specimens

8.1 Use one specimen weighing 110.0 ± 10.0 g from a sample. If a garment or textile item (towel, bed sheet, etc.) is to be used as a specimen, weigh the entire item to 0.1 g. If the garment weight exceeds the specified load weight of 1.8 kg, the total weight is to be included in the report. See 9.2.3 for exact weight specifications on load.

9. Procedure

9.1 Table I summarizes washing and drying conditions. Information on the machine and laundering conditions may be found in the AATCC LP1 (see 14.10).

9.2 Washing.

9.2.1 Fill the washer with the specified water volume at the selected water temperature using the normal cycle setting. Measure and record the water hardness in any test.

9.2.2 Add 66 ± 1 g of 1993 AATCC Standard Reference Detergent WOB or 100 ± 1 g of 2003 AATCC Standard Reference Liquid Detergent WOB (see 6.5). Add 240 ± 5 mL (1 cup) of sodium hypochlorite bleach or the amount recommended by the manufacturer for a full load of laundry. Run the washer for 2 min to ensure proper mixing of liquor.

9.2.3 Add enough ballast to the specimen to make a 1.8 ± 0.1 kg load. Set the washer on the cycle selected for washing and set the wash timer (see Table I and 9.1). Start the agitation and independently time the wash for reproducible results. When the desired wash time is complete, stop washer and advance the dial to the end of the wash cycle. Restart washer to rinse, drain and spin the load.

9.2.4 For the specimen to be dried by procedures A, B or D, allow washing to proceed through the rinse and final spin cycle. Remove test specimens immediately after the final spin cycle, separate any tangled pieces, taking care to minimize distortion, and dry by procedures A, B or D [see Table I and AATCC LP1 (see 9.1)].

9.2.5 For the specimen to be dried by procedure C, Drip Dry, stop the washer at the end of the final rinse cycle just before the water begins to drain. Remove the specimens soaking wet.

9.3 Drying.

9.3.1 (A) Tumble Dry. Place the washed load (test specimens and ballast) in the tumble dryer, set on the cycle which generates the correct exhaust temperatures as specified in AATCC LP1 (see 9.1). For fabrics that are heat sensitive, lower temperatures consistent with producer's recommendation are required and must be reported. Operate the dryer until the total load is dry. Remove the load immediately after the machine stops.

9.3.2 (B) Line Dry. Hang each specimen by two corners with the fabric length in the vertical direction. Allow the test specimens to hang in still air at room temperature until dry.

9.3.3 (C) Drip Dry. Hang each dripping wet specimen by two corners, with the fabric length in the vertical direction. Allow specimens to hang in still air at room temperature until dry.

9.3.4 (D) Screen Dry. Spread each specimen on a horizontal screen or perforated surface, removing wrinkles but not distorting or stretching and allow to dry in still air at room temperature.

9.4 Repeat 9.2 and 9.3 for a total of five launderings or to an end point of color change (see 14.2).

9.5 On completion of the fifth laundry cycle, (or after each cycle where end point criteria are to be met), condition the specimens for at least 4 h by laying each specimen separately on the screen or perforated shelf of a conditioning rack at an atmosphere of 21 ± 2°C (70 ± 4°F) and 65 ± 5% RH before evaluating color change.

9.6 If this test method is to be used to develop care instructions, specimens may also have to undergo further testing. Nonchlorine bleach sensitivity may need to be tested using AATCC TM172. To separate the effects of components such as hardness, pH or bleach content (see 5.1, 12.8 and ASTM D3938, Standard Guide for Determining or Confirming Care Instructions for Apparel and Other Textile Products), this method with detergent only and/or this method using water only may be used.

Table I—Washing and Drying Conditions (see 14.10)

Machine Cycle	Wash Temperatures	Drying Procedures
(1) Normal or Cotton/Sturdy	(II) 27 ± 3°C (80 ± 5°F)	(A) Tumble:
(2) Delicate	(III) 41 ± 3°C (105 ± 5°F)	i. Cotton/Sturdy
(3) Permanent Press	(IV) 49 ± 3°C (120 ± 5°F)	ii. Delicate
	(V) 60 ± 3°C (140 ± 5°F)	iii. Permanent Press
		(B) Line
		(C) Drip
		(D) Screen

10. Evaluation

10.1 Visual evaluation.

10.1.1 Using an unwashed specimen from the sample as a control, rate the color change of the test specimen in at least three places on the specimen or three times in one place using the Gray Scale for Color Change (AATCC EP1), or using AATCC EP7, Instrumental Assessment of the Change in Color of a Test Specimen, and record the numerical rating that corresponds to the appropriate one on the Gray Scale.

10.1.2 Ratings may be made compared to an unexposed control cut from the test sample or compared to a laundered control specimen of known fastness.

10.2 Instrumental evaluation.

10.2.1 Instrumental measurements comparing unexposed and exposed, (washed), specimen or garments may be used (see AATCC EP6, Instrumental Color Measurement, AATCC EP7, and AATCC TM173, Test Method for Calculation of Small Color Differences for Acceptability).

11. Interpretation of Results

11.1 This test is a satisfactory end use test for demonstrating the effect of the 1993 AATCC Standard Reference Detergent WOB or the 2003 AATCC Standard Reference Liquid Detergent WOB and chlorine bleach on textile materials in home laundering. The color change results will establish a level of performance that can be used to prepare care instructions (see 14.9 and ASTM D3938).

11.2 If a significant color change is observed, an untreated duplicate sample should be retested using this method with water only and/or by following AATCC TM172 (see 9.6).

12. Report

12.1 Report the numerical average color change grade for each specimen tested.

12.2 Report washing conditions (Arabic and Roman Numerals) and drying conditions (capital letter from Table I; [i.e., (1) IIIA (iii)] means NORMAL cycle, wash water temperature of 41 ± 3°C, tumble dry [permanent press]).

12.3 Report the type and amount of detergent used.

12.4 Report the brand, sodium hypochlorite activity, and amount of the chlorine bleach used.

12.5 Report the available chlorine content delivered to the wash by the dosage

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(see 14.8.3).

12.6 Report the weight of fabric specimen or garment tested, if other than found in 8.1.

12.7 Report the number of wash cycles tested, if other than 5.

12.8 Report water hardness.

12.9 If the information gained through this testing is to be used in establishing care instructions, results of washing using detergent and powdered non-chlorine bleach, detergent alone or water alone may have to be included (see 5.1 and 9.6).

13. Precision and Bias

13.1 *Precision*.

13.1.1 In 2001 a single laboratory study was performed. The same operator tested all samples.

13.1.2 Samples tested consisted of three fabrics dyed with three different dyes. Washing conditions tested consisted of 60°C (140°F) and 49°C (120°F) wash temperatures. Both tests used 200 ppm levels of sodium hypochlorite. Each sample was evaluated instrumentally three times and averages were calculated.

13.1.3 *Within-laboratory* Standard Errors and Critical Differences are shown in Table II. Data are on file at the AATCC Technical Center.

13.2 *Bias*. The colorfastness to chlorine bleach in home laundering can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

14. Notes

14.1 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1.513.742.2020; web site: www.acgih.org.

14.2 While the design of this method is to expose specimens to 5 cycles of washing and drying, the number of cycles may be shortened or lengthened if the end point is a specific grade for color change.

14.3 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

14.4 Contact AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; web site: www.aatcc.org, for model number(s) and source of current recommended washer(s). Any other washer which is known to give comparable results may be used. Washing machine conditions given in M6 represent the actual speeds and times available on current specified model(s). Other washers may vary in one or more of these settings.

14.5 Contact AATCC (see 14.4) for model number(s) and source(s) of current recommended dryer(s). Any other dryer which is known to give comparable results can be used. Dryer machine conditions given in M6 represent the actual speeds and times available on current specified model(s). Other dryers may vary in one or more of the settings.

14.6 Screen or perforated conditioning/drying racks available from Somers Sheet Metal Inc., 5590 N. Church St., Greensboro NC 27455; tel: +1.336.643.3477. Drawings of racks are available from AATCC (see 14.4).

14.7 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

14.8 Use sodium hypochlorite bleach purchased within the last six months. Keep the bleach sealed in an airtight container and in a cool, dry place.

14.8.1 To confirm the sodium hypochlorite bleach's activity, weigh 2.00 g sodium hypochlorite into an Erlenmeyer flask and dilute to 50 mL with deionized water. Add 10 mL of 10% sulfuric acid and 10 mL of 10% potassium iodide. Titrate with 0.1*N* sodium thiosulfate until colorless.

Calculation:

% sodium hypochlorite =

$$\frac{\text{mL } Na_2S_2O_3(0.1N)(0.03722)}{2.0 \text{ g NaOCl}} \times 100$$

The factor 0.03722 is derived by multiplying the molecular weight of NaOCl (74.45 g/mol) by 0.001 (mL to L conversion) and dividing by 2 (mols of thiosulfate per hypochlorite).

14.8.2 Oxidizing power of sodium hypochlorite is typically expressed in terms of available chlorine, the equivalent amount of diatomic chlorine present. A 5.25% NaOCl solution contains 50,000 ppm available chlorine.

14.8.3 To determine the available chlorine delivered to the wash solution, weigh a 50 g wash water sample into an Erlenmeyer flask. Add 10 mL of 10% sulfuric acid and 10 mL of 10% potassium iodide. Titrate with 0.01*N* sodium thiosulfate until colorless.

Calculation:

ppm available chlorine =

$$\frac{\text{mL } Na_2S_2O_3(0.01N)(35.45)(1000)}{(50 \text{ g wash water sample})}$$

NOTE: No buffering agent is required since the solution is already buffered by the detergent.

14.9 ASTM methods available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832.9500; fax: +1.610.832.9555; web site: www.astm.org.

14.10 Laundering temperatures and other parameters listed in AATCC LP1 may not be the same as those listed in various test methods. Parameters in AATCC LP1 are updated periodically to reflect consumer experience, taking into account current technology and regulations regarding home water heater temperatures, particularly in the United States.

Typically, test method committees pursue a deliberate effort to maintain the same test method conditions over a long period of time in order to produce comparable results over time. Additionally, significant changes may invalidate precision data developed using the original conditions.

Temperatures listed in AATCC test methods were established at the time the methods were developed or revised, and closely mirror the temperature ranges specified in 16 CFR Part 423 for care labels. It should also be noted that AATCC test methods were developed using traditional top loading washing machines. Specified water level, detergent, and other details are not applicable to high efficiency top loading or front loading machines. New, commercially available home laundering machines may have to be modified by the user to meet the specifications in various test methods. All test reports should indicate the exact testing conditions, and any modification of the standard method, including use of AATCC LP1 parameters. Results of tests performed with different conditions may not be comparable.

Table II—Within-Laboratory Standard Errors and Critical Differences* (95% Probability) n-3)

Wash Temperature	Average DE*	Standard Error	Critical Difference
60°C (140°F) wash	22.19	0.22	0.61
49°C (120°F) wash	19.56	0.31	0.86

*Note: In this single laboratory test, estimates of standard error and critical differences may be either underestimated or overestimated to a considerable extent and should be used with special caution. The values should be viewed as minimal data with regards to precision. Confidence intervals are not well established.

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Test Method for Fluorine Content of Carpet Fibers

Developed in 2000 by AATCC Committee RA57; editorially revised and reaffirmed 2001, 2007; reaffirmed 2002, 2012; editorially revised 2005, 2010, 2019; revised 2017.

1. Purpose and Scope

1.1 This method is for determining the amount of fluorocarbon soil-resist agent present on carpet fibers by measurement of elemental fluorine on the pile fiber. It is useful for measurement of fluorocarbon present on carpet pile fiber as tested within the practical range of 50 to 1000 µg/g (ppm). It may also be used for yarns treated with fluorocarbon anti-soiling agents.

2. Principle

2.1 A weighed fiber specimen is burned in an oxygen combustion flask and the released hydrogen fluoride is absorbed into a sodium hydroxide solution. The fluorine content is measured as dissolved sodium fluoride at a constant pH and ionic strength using a fluoride ion activity electrode and a specific ion meter.

3. Terminology

3.1 **fluorine content,** n.—*in textile floor coverings*, the ratio of the weight of total elemental fluorine to the total weight of carpet fibers.

3.2 **soil resist agent,** n—a material applied to, or incorporated in, carpet face fiber which retards and/or limits the build-up of soil.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care. Use chemical goggles or face shield, impervious gloves and an impervious apron during the dispensing and mixing of the fluoride and sodium hydroxide standards. The standards should be made and dispensed in a hood.

4.3 An eyewash/safety shower should be located nearby and a self-contained breathing apparatus should be readily available for emergency use.

4.4 Exercise care in cutting the specimens with the scalpel or razor blade; protective gloves may provide additional protection.

4.5 Oxygen supply pressure from the pressure regulator should not exceed 10 psi.

4.6 Do not operate the test equipment until the manufacturer's operating instructions have been read and understood. It is the responsibility of whoever operates the test equipment to conform to the manufacturer's directions for safe operation.

4.7 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). Fluoride salts and sodium hydroxides are listed with OSHA permissible exposure limits [PEL]. In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 12.1).

5. Apparatus, Reagents, Materials

5.1 500-mL combustion flask with ball stopper (with platinum wire hook) and clamp (one each per specimen; reusable, see Fig. 1).

5.2 Platinum carrier basket (one per sample; reusable).

5.3 Indirect source of ignition.

5.4 Black paper specimen wrappers (one per sample).

5.5 Specific ion meter.

5.6 Fluoride ion selective electrode.

5.7 Reference electrode.

5.8 0.001*N* Sodium Hydroxide (NaOH).

5.9 TISAB (Total Ionic Strength Adjustment Buffer with CDTA (1,2-cyclohexanediaminetetraacetic acid) or formulated for fluorine analysis to prevent aluminum interference) (see 12.2).

5.10 Re-pipette or bottle-top dispenser (optional).

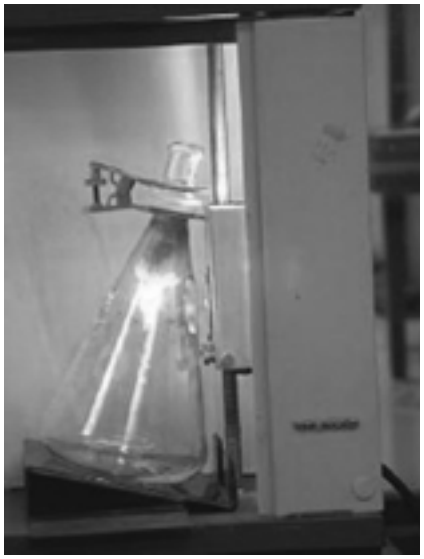


Fig. 1—Ignited specimen suspended from platinum wire attached to ball stopper.

5.11 Sodium Fluoride (NaF) standard (see Appendix A).

5.12 Mechanical shaker (optional).

6. Calibration

6.1 Calibrate the specific ion meter using the procedures described in the operation manuals for the fluoride ion activity electrode and the specific ion meter. Calibration should be done on a daily or every shift basis.

7. Sampling

7.1 Prepare the yarn sample by removing the tufts from the carpet sample with a scalpel or razor blade. Cut the tufts as close to the primary backing as possible having a stubble not greater than 1 mm. Twenty to 40 tufts are required for a single sample, depending on carpet construction. Place the tufts in a paper envelope and identify by test number.

7.1.1 *Sampling Option #1*. To determine an overall average fluorine content of carpet fibers, take samples from across the entire width of the carpet to compensate for possible variability in application and combine the tufts in a single envelope.

7.1.2 *Sampling Option #2*. To determine the variability of fluorine content across a carpet sample, take multiple samples from various areas of the carpet and analyze each set of carpet tufts separately (see 12.3).

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8. Procedure

8.1 Perform single-step (8.4) *or* two-step (8.5) process.

8.2 Weigh samples using tared weighing pans. Weigh between 0.110 g and 0.150 g of randomly selected tufts from each sample and record the weight. Fold all of the weighed tufts into the black paper sample holder and place the paper holder into the platinum basket. Suspend the basket from the platinum wire hook on the flask stopper.

8.3 Number each combustion flask and plastic cup.

8.4 Buffer Addition: Single-Step Method

8.4.1 Dispense 20.0 ± 1 mL of the 0.001*N* sodium hydroxide solution into the oxygen combustion flask (see 12.4).

8.4.2 Dispense 20 mL of TISAB into the oxygen combustion flask.

8.5 Buffer Addition: Two-Step Method

8.5.1 Dispense 20.0 ± 1 mL of the 0.001*N* sodium hydroxide solution into the oxygen combustion flask. TISAB will be added later.

8.6 Attach a flexible tube on the oxygen supply. Turn ON the oxygen supply. Place the flexible plastic tube into the oxygen combustion flask. Turn ON the pressure regulator and allow oxygen to fill the flask for 5-15 s. Turn OFF the oxygen flow (see 12.5).

8.7 Immediately place the flask stopper with the platinum basket suspended from the hook on the flask and clamp in place. Do not allow the sample basket to become wet or the sample will not ignite.

8.8 Place the flask in the infrared ignition cabinet. Any indirect source of ignition is acceptable. The cabinet serves as a contaminant in the event the flask over pressurizes and fails. Individual labs can determine method of ignition. Source of ignition should be turned OFF once the sample ignites. Allow the flame to burn out. Do not open the cabinet until the flaming has completely ceased. Confirm that the test specimen has burned completely. If any portion of the sample remains unburned, abort the test. Remove the flask from the ignition cabinet using a flask grip or gloves, as the flask will be hot.

8.9 Shake the flask.

8.9.1 If a mechanical shaker is used, place the flask into the holder and secure into position. Turn on the mechanical shaker for 1.5-2 min. Remove the flask.

8.9.2 If shaking is done by hand, shake vigorously for a minimum of 30 s.

8.10 *For Two-Step Method Only.* Dispense 20.0 ± 1 mL of TISAB into the empty flask. When dispensing, lift the flask stopper as little as possible to prevent any loss of gases. Replace the stopper and clamp in place. Repeat step 8.9.

8.11 Open the flask and pour all of the buffer solution (sodium hydroxide/TISAB) into a 100-mL plastic cup labeled with the test number.

8.12 Remove the platinum basket from the solution using tweezers. Cover the cup and allow the mixture to cool by sitting at room temperature for a minimum of 30 min. (see 12.6).

8.13 Place the fluoride ion specific electrode into the cup containing the cooled test solution from the flask. Stir slowly and allow the meter to stabilize. Read and record the meter reading alongside the sample weight (see 12.7).

8.14 Decontaminate the electrode. Discard the solutions in the plastic cups and to prevent cross-contamination, do not reuse the cups. Wash the platinum baskets and glassware, rinse with deionized or distilled water, and dry prior to using in a new test.

9. Calculation

9.1 Calculate the results as follows:

a) If the meter was calibrated multiplying the calibration standards by 40 (volume of the solution in mL), then the calculation is:

$$\frac{\text{Meter Reading}}{\text{Sample Weight}}$$

b) If the meter was calibrated without multiplying by the sample volume, then the calculation is:

$$\frac{\text{Meter Reading}}{\text{Sample Weight}} \times \text{Sample Volume (mL)}$$

9.1.1 Quantitation limits should be set by the individual lab based on the application levels of interest. These limits are determined by the calibration standards used and the meter/electrodes' capability. Any result below 25 ppm could be considered "background noise."

10. Precision & Bias

10.1 *Interlaboratory test data*. An interlaboratory study was conducted in 1998 utilizing eight laboratories and two nylon face fiber carpets. Each laboratory made five determinations from each specimen using a single operator.

10.2 *Components of variance*. The components of variance shown as variances and the square root of the variances (to show appropriate units of measure) are listed in Tables I and II.

10.3 *Precision parameters*. For the components of variance reported in Table I, two averages of observed values should be considered significantly different at the 95% probability level if the difference equals or exceeds the critical differences listed in Table II.

10.3.1 Because only two carpets were used in this evaluation, users of the test method are advised to apply statistical caution in comparison of other carpets.

10.4 *Bias*. Fluorine content can be defined only in terms of a test method. While there are other independent means to determine fluorine content, any bias of this method in comparison to them have not been established.

Table I—Components of Variance for Fluorine Content (Expressed as Variances and Standard Deviations[a])

Component	Variance	Standard Deviation μg/g (ppm)
Specimens within materials and laboratories	630.0	25.1
Laboratories	378.0	19.4
Interaction: Materials times laboratories	59.6	7.7

[a]Square roots of the components of variance are reported to express the variability in the appropriate units of measure rather than as the squares of those units of measure.

Table II—Critical Differences for the Conditions Noted 95% Probability Level, Single Material Comparisons μg/g (ppm) Fluorine

Number of Test Results in each Average	Within-Laboratory Precision		Between-Laboratory Precision	
	Standard Error	Critical Difference	Standard Error	Critical Difference
1	25.1	70	31.7	88
2	17.8	49	26.3	73
3	14.5	40	24.2	67
4	12.6	35	23.1	64
5	11.2	31	22.4	62
7	9.5	27	21.6	61
9	8.9	23	21.2	59

NOTE 1—The tabulated values of the critical differences should be considered to be a general statement, particularly with respect to between-laboratory precision.

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11. Report

11.1 Report carpet description and date of testing.

11.2 Report any variation from procedure.

11.3 Report test result from 9.1.

12. Notes

12.1 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

12.2 Total Ionic Adjustment Buffer (TISAB) is available from most laboratory supply companies. The buffer solution contains the following: sodium citrate, sodium hydroxide, acetic acid and water.

12.3 In order to prevent confusion, a sample numbering system should be established wherein all items to be used with a specific test specimen are numbered the same; i.e., paper envelopes, plastic cups, combustion flasks, etc.

12.4 For convenience in processing large numbers of samples, re-pipette dispensers may be used for the stock solutions of sodium hydroxide and TISAB buffer. If so, they should be checked for calibration daily.

12.5 The pressure from the pressure regulator should not exceed 14 kPa (2 psi). Excessive pressure may cause splashing.

12.6 An equilibrating bath can be used to bring the temperature of the solution to the temperature at which the meter was calibrated more rapidly. Use of a temperature correcting electrode can allow for some tolerance around the initial calibration temperature. Reference the manufacturer's guidelines for electrode and meter operation.

12.7 Keep the electrodes immersed in the 2 µg/g (ppm) fluoride standard solution when not in use to maintain electrode conditioning.

Appendix A
Standard Preparation

A1 Examples of fluoride standards:

A1.1 2000 µg/g (ppm) fluoride standard: Weigh 4.4200 ± 0.0001 g of oven dried sodium fluoride (NaF) and transfer to a 1 L volumetric flask. Add distilled water to the 1 L mark and shake until dissolved.

A1.2 20 µg/g (ppm) fluoride standard: Pipette 10.0 mL of the above 2000 µg/g (ppm) solution into a 1 L volumetric flask. Add distilled water to the 1 L mark and shake.

A1.3 2 µg/g (ppm) fluoride standard: Pipette 100.0 mL of the 20 µg/g (ppm) fluoride solution into a 1 L volumetric flask. Add distilled water to the 1 L mark and shake.

A1.4 0.001*N* sodium hydroxide solution: Pipette 2.00 mL of 0.500*N* NaOH into a 1 L volumetric flask. Fill to the 1 L mark with distilled water and shake.

Appendix B
Maintaining Laboratory Level

B1 It is strongly suggested that nylon fiber with known amount of non-volatile fluorocarbon treatment be used to develop and maintain a control chart of testing.

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Test Method for Acid Cellulase Enzyme Treatment of Cellulosic Fabrics

1. Purpose and Scope

1.1 This test method describes a procedure for evaluating the reaction of cellulosic and cellulosic blend fabrics to treatment with acid cellulase enzymes. Such treatments usually result in improvements in the hand, drape, fuzz and pill appearance of cellulosic fabrics together with the removal of linters and immature/dead cotton fibers. The agitation in a home washing machine simulates the action in rotary drum machines, paddle machines and jets.

2. Principle

2.1 This test method utilizes an automatic top-loading washing machine to simulate the action of rotary drum machines, paddle machines, or jets used in production type operations (that usually operate at low liquor ratios). The concentrations of enzyme, buffer, and detergent required in this test method are based on grams/liter (g/L), to compensate for the difference in liquor ratios. Other cellulase enzymes, e.g., modified acid and neutral types, may be evaluated by this test procedure; however, the concentration, pH, and time should be followed as recommended by the enzyme supplier.

2.2 Fabric hand is evaluated using AATCC EP5. Improvement in fabric fuzz, pill appearance, and degree of removal of linters and immature/dead cotton fibers is visually evaluated. Fabric strength, if applicable, is determined by ASTM D3786 or ASTM D5035.

3. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

3.1 AATCC EP5 Evaluation Procedure for Fabric Hand (see 12.1).

3.2 AATCC LP1 Laboratory Procedure for Home Laundering: Machine Washing.

3.3 ASTM D3786 Hydraulic or Pneumatic Bursting Strength of Knitted Goods for Knits.

3.4 ASTM D5035 Breaking Force and Elongation of Textile Fabrics-Strip Method for Woven Fabrics.

4. Terminology

4.1 **cellulase enzyme,** n.—an enzyme that attacks cellulose.

5. Safety Precautions

5.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

5.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

5.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

6. Apparatus (see 12.2)

6.1 Standard washing machine (see Table I, 12.3).

6.2 Standard tumble dryer (see Table III, 12.3).

6.3 Laundering ballast, Type 1 or Type 3 (see Table IV).

6.4 Apparatus and conditioning facilities as needed for AATCC EP5, ASTM D3786 (optional, knitted or nonwoven specimens), and ASTM D5035 (optional, woven specimens). Refer to standards for details.

6.5 Reagents

6.5.1 Nonionic detergent; linear alcohol with 9 moles of ethylene oxide.

6.5.2 Buffer AC, sodium acetate.

6.5.3 Acid cellulase enzyme.

7. Specimens

7.1 Use four to eight garments, or four to eight full-width fabric specimens (0.9 m (1 yd) in length). Seal the cut edges of fabric specimens by sewing or some other means.

7.1.2 Use two to four garment/fabric specimens to run the test cycle with enzyme.

7.1.3 Use the other two to four garment/fabric specimens to run a control cycle without enzyme, for comparison.

8. Procedure

8.1 Enzyme Wash.

8.1.1 Set washing machine controls to obtain the parameters listed in Table I.

8.1.2 Load for laundering shall consist of half the specimens in the sample, plus enough laundering ballast pieces to make a total of 4 pieces. Record initial weight of ballast if used.

8.1.3 Begin selected wash cycle. Allow machine to fill to specified water level.

8.1.4 Add 1 g/L nonionic detergent

8.1.5 Add 6 g/L sodium acetate buffer.

8.1.6 Stir well and verify pH of 4.5-5.0. If too high, add acetic acid. If too low, add caustic, 50%.

8.1.7 Add acid cellulase enzyme as shown in Table II. Stir.

8.1.8 Add wash load (test specimens and ballast, if applicable), distributing evenly around center agitator. Restart wash cycle.

8.1.9 Stop wash cycle after 12 min. agitation and restart the agitation portion. Do not allow washing machine to drain or rinse. Repeat agitation as needed to reach the total time indicated in Table II. Specimens must be exposed to the enzyme bath for sufficient time to ensure treatment.

8.1.10 Allow washing to proceed through the final spin cycle.

8.1.11 Separate tangled specimens and ballast pieces, taking care to minimize distortion.

8.1.12 Place the washed load (test specimens and ballast) in the tumble dryer and set the temperature control to generate a maximum exhaust temperature of 66°C (see Table III). Allow the dryer to run until the load is fully dry. Remove specimens immediately. Avoid over drying.

8.1.13 In addition to acting on the cellulose of the test specimens, the cellulase enzyme used in this test will also act on the cellulose content of the ballast fabric. Hence, ballast used in this method may not be reusable to the same extent as in other methods. If ballast sheets are reused, weigh prior to each use to ensure that they have not incurred weight loss greater than 75% from their original weight. If they have, replace with new ballast.

8.2 Control Wash.

8.2.1 Repeat the wash procedure with the remaining specimens, omitting enzyme. Wash and dry specimens and ballast (if applicable) exactly as for enzyme wash.

9. Evaluation

9.1 Evaluate and compare the hand of tested specimens exposed to acid cellulase enzyme with that of tested specimens not exposed to acid cellulase enzyme using AATCC EP5.

9.2 Visually compare the appearance of tested specimens exposed to acid cellulase enzyme with that of tested specimens not exposed to acid cellulase enzyme to determine the degree of im-

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Table I—Standard Washing Machine Parameters (see 12.3, 12.4)

Cycle	Normal
Water Level, L (gal)[1]	45 ± 4 (12 ± 1)
Agitation Speed (strokes/min.)	86 ± 2
Stroke Length	Up to 220°
Washing Time, min.[2]	12 ± 1 (repeat as needed)
Final Spin Speed, rpm	660 ± 15
Final Spin Time, min.	5 ± 1
Wash Temp., °C (°F)[1]	Very Hot: 60 ± 3 (140 ± 5)

[1] Water Level and Wash Temperature indicated in the table may not correspond with standard machine settings. Verify and adjust as needed to achieve the indicated parameters.
[2] Extended agitation time is required for this test method. Stop and repeat agitation portion as needed before washing machine drains.

Table II—Acid Cellulase Enzyme Portions

	Acid Cellulase Enzyme		
Agitation Time[1]	(g/L)	g[4]	Test Specimens
24 min. (2 cycles) (0.2-0.5)		9.0-22.5	100% linen and its blends
24 min. (2 cycles) (1.0 -2.0)		45-90	100% cotton and its blends[2]
48 min. (4 cycles)3 (4.0)		180	rayon or lyocell

[1] Repeat agitation portion of wash cycle as needed before draining and rinsing.
[2] Use the same enzyme concentration for 100% cotton and cotton/polyester blends to achieve the desired effect on the cotton portion of the fabric.
[3] The acid cellulase enzyme attacks these fibers much more slowly; therefore a longer agitation (48 min) is required.
[4] Quantity based on a 45 liter water volume.

Table III—Standard Tumble Dryer Parameters (see 12.3)

Cycle	Normal
Max. Exhaust Temp., °C (°F)[1]	66 ± 6(155 ± 10)
Cool Down Time, min.	≤ 10

Table IV—Laundering Ballast Parameters

	Type 1	Type 3
Fiber Content	100% cotton	50% cotton/50% polyester ± 3%
Greige Fabric Yarns	16/1 ring spun	16/1 or 30/2 ring spun
Greige Fabric Construction	52 × 48 ± 5 yarns/in., plain weave	52 × 48 ± 5 yarns/in., plain weave
Finished Fabric Weight	155 ± 10 g/m^2	155 ± 10 g/m^2
Edges	All edges hemmed or over-edged	All edges hemmed or over-edged
Finished Piece Size	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)
Finished Piece Weight	130 ± 10 g	130 ± 10 g

provement in fabric fuzz, pill appearance, and degree of removal of linters and immature/dead cotton fibers. Report visual grades for each property as none, slight, distinct, or considerable difference/improvement.

9.3 If desired, determine the strength of tested specimens exposed to acid cellulase enzyme and compare with that of tested specimens not exposed to acid cellulase using ASTM D3786 for knitted or nonwoven specimens or ASTM D5035 for woven specimens.

10. Report

10.1 Describe or identify the sample tested.

10.2 Report that the sample was tested using AATCC TM191-2020.

10.3 Report the testing conditions:

10.3.1 Number of specimens tested.

10.3.2 Cellulase enzyme used and concentration.

10.3.3 Type of ballast (Type 1, Type 3, or none).

10.4 Report the test results:

10.4.1 Average hand value for specimens treated in the enzyme wash (see AATCC EP5).

10.4.2 Average hand value for specimens treated in the control wash (see AATCC EP5).

10.4.3 Average visual fuzz, pill, linter, and immature/dead cotton fiber grades for specimens treated in the enzyme wash.

10.4.4 Average visual fuzz, pill, linter, and immature/dead cotton fiber grades for specimens treated in the control wash.

10.4.5 Average strength of specimens treated in the enzyme wash, if applicable.

10.4.6 Average strength of specimens treated in the enzyme wash, if applicable.

10.4.7 If desired, report difference between corresponding results for specimens treated in enzyme and control washes.

10.4.8 Describe any modification(s) of the published standard.

11. Precision and Bias

11.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages. Analysis of variance or *t*-tests may be used to compare averages. See any standard statistical text for more information.

11.2 *Bias*. The evaluation of acid cellulase enzyme effect on cellulose fabric can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

12. Notes

12.1 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

12.2 For potential apparatus, reagents or materials sources, visit the AATCC Buyer's Guide at www.aatcc.org/bg. AATCC provides its Corporate members the option to list their items and services. AATCC does not qualify, or in any way approve, endorse or certify that any of the listings meet the specifications in its standards.

12.3 For model number(s) of washer(s) and tumble dryers(s) reported to meet the standard parameters, visit www.aatcc.org/testing/laundering or contact AATCC, P.O. Box 12215, Research Triangle Park NC 27709; +1.919.549.8141; ordering@aatcc.org.

12.4 Prior to 2021, water temperature and agitation time were the only washing parameters defined for this test method. The additional parameters listed in Table I are standard conditions for testing purposes, selected to align with machines specified in other AATCC laundering methods. Note that this test method must be performed in a traditional top-loading washing machine.

13. History

13.1 Revised in 2021 to clarify and align with prescribed AATCC style guidelines.

13.2 Editorially revised 2019. Reaffirmed 2013. Editorially revised 2012. Editorially revised and reaffirmed 2009. Editorially revised 2008, 2005. Editorially revised and reaffirmed 2004. Reaffirmed 2003.

13.3 Developed in 2002 by AATCC Committee RR41. Maintained by RA99.

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Test Method for Weather Resistance of Textiles: Sunshine-Arc Lamp Exposure With and Without Wetting

1. Purpose and Scope

1.1 This test method provides a means for determining the weather resistance of textiles.

1.2 The test options described below are applicable to fibers, yarns and fabrics and products made therefrom, including coated fabrics, whether natural, colored, finished, or unfinished. Test options included are:

A—Sunshine-Arc Lamp Exposure with Wetting

B—Sunshine-Arc Lamp Exposure without Wetting

1.3 The use of these options does not imply, expressly or otherwise, an accelerated test for a specific application. The degree of correlation between any weather resistance test and the actual exposure in use must be determined mathematically and agreed upon by the contractual parties.

1.4 Test duration and performance evaluation procedures are covered in material specifications agreed upon by the contractual parties.

2. Principle

2.1 Test specimens and the agreed upon comparison standard(s) are exposed simultaneously to light source under specified conditions of either relative humidity or wetting. The amount of exposure is determined with respect to a specified degree of degradation such as color change, strength loss, etc. The weather resistance of the specimen is evaluated by comparison of a specified degradation of exposed portion of the test specimen to the unexposed original material: using one or more of the procedures recommended in this test method.

3. References

3.1 **ASTM Standards** (see 17. 12).

3.1.1 D5034, Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test).

3.1.2 D5035, Test Method for Breaking Strength and Elongation of Textile Fabrics (Strip Test).

3.1.3 D2256, Test Method for Tensile Properties of Yarns by the Single-Strand Method.

3.1.4 D3787, Test Method for Bursting Strength of Knitted Goods.

3.1.5 D3786, Test Method for Hydraulic Bursting Strength of Knitted Goods and Nonwoven Fabrics.

3.1.6 D1424, Test Method for Tear Resistance of Woven Fabrics by Falling Pendulum (Elmendorf) Apparatus.

3.1.7 D1117, Methods of Testing Nonwoven Fabrics, Trapezoid Tear Procedure.

3.1.8 G24, Standard Practice for Conducting Exposures to Daylight Filtered through Glass

3.1.9 G152, Standard Practice for Operating Open Flame Carbon-Arc Light Apparatus for Exposure of Nonmetallic Materials.

3.1.10 D2905, Number of Specimens Required to Determine the Average Quality of a Textile.

3.1.11 D1776, Standard Practice for Conditioning and Testing Textiles.

3.1.12 G151, Standard Practice for Exposing Non-Metallic Materials in Accelerated Test Devices that use Laboratory Light Sources.

3.2 **AATCC Test Methods** (see 17.4).

3.2.1 TM16.2, Colorfastness to Light: Carbon-Arc.

3.2.2 EP1, Gray Scale for Color Change.

3.3 **SAE Test Methods** (see 17.14).

3.3.1 J1545, Instrumental Color Difference Measurement for Exterior Finishes, Textiles, and Colored Trim.

3.4 **ISO Test Methods** (see 17.13).

4. Terminology

4.1 **AATCC Blue Wool Lightfastness Standard,** n.—one of a group of dyed wool fabrics distributed by AATCC for use in determining the amount of light exposure of specimens during lightfastness testing.

NOTE: Because of the unstable nature of these materials, they are also sensitive to the heat and moisture conditions which exist before exposure testing, after exposure testing, and prior to measurement.

4.2 **Black Panel Thermometer,** n.—a temperature measuring device, the sensing unit of which is coated with black designed to absorb most of the radiant energy encountered in lightfastness testing.

4.2.1 This device provides an estimation of the maximum temperature a specimen may attain during exposure to artificial light (see 17.1). Any deviation from the geometry of this device described in 17.1 may have an influence on the measured temperature.

4.3 **breaking strength,** n.—the maximum force applied to a specimen in a tensile test carried to rupture.

4.4 **bursting strength,** n.—the force or pressure required to rupture a textile by distending it with a force applied at right angles to the plane of the fabric, under specified conditions.

4.5 **color change,** n.—as used in colorfastness testing, a change in color of any kind (whether a change in lightness, hue or chroma or any combination of these, discernible by comparing the test specimen with a corresponding untested specimen.

4.6 **irradiation,** n.—the time integral of irradiance expressed in joules per square meter (J/m^2).

4.7 **laboratory sample,** n.—a portion of material taken to represent the lot sample, or the original material, and used in the laboratory as a source of test specimens.

4.8 **reference fabric,** n.—one or more blue wool lightfastness standards selected for exposure as a check on a test apparatus and operating conditions.

4.9 **reference plastic,** n.—a clear polystyrene plastic standard selected for exposure as a check on a test apparatus and operating conditions.

NOTE: It has not been verified that these reference materials can be used as a check on a test cabinet or operating conditions for outdoor exposure tests due to seasonal variations in ultraviolet, temperature, relative humidity, and time of wetness during the uncontrolled portion of the test.

4.10 **specimen,** n.—a specific portion of a material or a laboratory sample upon which a test is performed or which is selected for that purpose.

4.11 **spectral energy distribution,** n.—the variation of energy due to the source over the wavelength span of the emitted radiation.

4.12 **standard atmosphere for testing textiles,** n.—air maintained at 21 ± 2°C (70 ± 4°F) and 65 ± 5% relative humidity.

4.13 **tearing strength,** n.—the average force required to continue a tear previously started in a fabric.

4.14 **weather,** n.—climatic conditions at a given geographic location, including such factors as sunlight, rain, humidity and temperature.

4.15 **weather resistance,** n.—ability of a material to resist degradation of its properties when exposed to climatic conditions.

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5. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive, It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted on specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

5.1 Do not operate the test equipment until the manufacturer's instructions have been read and understood. It is the operator's responsibility to conform to the manufacturer's directions for safe operation.

5.2 The test equipment contains high intensity light source. The door of the test chamber must be kept closed when the equipment is in operation.

5.3 Before servicing light sources, allow 30 min for cool-down after lamp operation is terminated.

5.4 When servicing the test equipment, shut off both the off switch on the front panel and the main power disconnect switch. When equipped, ensure that the main power indicator light on the machine front panel goes out.

5.5 Daylight exposure of the skin and eyes for prolonged periods may be hazardous and therefore caution should be employed to protect these areas. Do not look directly at the sun under any circumstances.

5.6 Good laboratory practices should be followed. Wear prescribed safety glasses in all laboratory areas.

6. Uses and Limitation

6.1 Not all materials are affected equally by the same light source and environment. Results obtained by the use of any one test option may not be representative of those of any other test option or any end-use application has been established between contractual parties.

6.2 Radiation emitted by the open flame carbon arc contains significant levels of very short wavelength UV (less than 260 nm) and must be filtered. None of the filter types used changes the spectral power distribution of the open flame carbon arc to make it match daylight in the long wavelength UV or the visible light regions of the spectrum. Although these filters are specified in many tests because of historical precedent, they transmit significant energy below 300 nm (the typical cut-on wavelength for terrestrial sunlight) and may result in aging processes not occurring outdoors.

6.3 Option A is intended for use in evaluating the resistance to degradation of a textile and related material, including coated fabrics, when subjected to simulated weathering including wetting, and radiant energy factors, and are further intended to check the level of durability represented in a material specification. Option B is intended for use in evaluating the resistance to degradation due to weathering and sunlight exposure under conditions where wetting is not a factor.

6.4 In interpreting the results of these test options, it should be remembered that resistance to degradation under simulated sunlight exposure is dependent upon such factors as:

6.4.1 The inherent properties of the material, its physical state, mass and compactness.

6.4.2 Spectral energy distribution and density of the radiant flux (from artificial source).

6.4.3 The temperature and relative humidity of the air about the textile specimen during exposure.

6.4.4 The effect of leaching or degradation of additive products, including fiber stabilizers, by rain or water sprays.

6.4.5 Atmospheric contaminants.

6.4.6 The action of additive finishes and colorants including spectral absorption characteristics.

6.4.7 Where applicable, the action of residual laundry or dry-cleaning chemicals.

6.5 The relative rates of degradation of different textile materials do not necessarily change to the same degree as those factors themselves change. Consequently, the relative durability of textile and related materials under the varying conditions of use cannot be predicted with certainty with any one test. Therefore, it is common practice to investigate the durability of materials by exposure to the variety of conditions under which they can be used to arrive at a full understanding of the performance that can be expected from them.

6.6 Having once established the acceptability of a given material, subsequent lots of it can be rated by direct comparison between the test specimen and the initially accepted material using the agreed upon option(s) provided in this test method. Although these options lead to results that are in general agreement for the majority of textiles, they do not lead to the same result for all textiles. When significantly different levels of resistance to degradation are found between the natural sunlight and weather methods and their artificial (laboratory) counterparts, the results obtained from direct sunlight and the elements of the weather will be regarded as the level of durability of the particular material for the purposes of intended end use.

6.7 When using this test method, the test option selected should incorporate light, humidity, wetting and temperature effects based upon historical data and experience. The test option selected should also reflect expected end-use conditions associated with the material to be tested.

6.8 When using this test method, use a standard of comparison. This comparison material should have a known history in terms of a change in property value after a specific exposure.

6.9 Variations in results may be expected when operating conditions are varied within the accepted limits of this method. Therefore, no reference shall be made to results from the use of this method unless accompanied by a report detailing the specific operating conditions and locations in conformance with Section 15.

6.10 In case of a dispute arising from differences in reported test results when using this test method for acceptance testing of commercial shipments, the purchaser and the supplier should conduct comparative tests to determine if there is a statistical bias between their laboratories. Statistical analysis is recommended for the investigation of bias. As a minimum, the two parties should take a group of test specimens that are as homogeneous as possible and that are from a lot of material of the type in question. Test specimens should then be randomly assigned in equal numbers to each laboratory for testing. The average results from the two laboratories should be compared using the appropriate Student's t-test and an acceptable probability level chosen by the two parties before testing is begun. If a bias is found and corrected or the purchaser and the supplier must agree to interpret future test results with consideration for the known bias.

7. Apparatus and Materials (see 17.2)

7.1 If possible, install the test apparatus in a room where temperature and relative humidity are controlled to minimize the effects of air supply variations.

7.2 Sunshine-arc type exposure apparatus with means of providing water sprayed to the specimen (see Appendix A and 17.3).

7.3 Sunshine-arc type exposure apparatus without providing water sprayed to the specimen (see Appendix A and 17.3).

8. Comparison Standards

8.1 Standards for comparison must be determined and agreed upon by the contractual parties. They can be made of any suitable textile material where a history of strength degradation or color change rates are known, depending upon individual test needs. They must be exposed simultaneously with the test specimen to be investigated. The intent of these standards is to demonstrate uniformity between separate machine exposures or outdoor test locations. When test data of the

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exposed standards differ by more than 10% from the known data, the test apparatus machine malfunction, then the test standard should be questioned and re-evaluated. Test data obtained with questionable standards must be treated with caution and resolution provided with quantitative analysis.

8.2 When colorfastness is the sole evaluation criterion, Blue Wool Standards, as defined in AATCC Test Method (TM) 16.2, Colorfastness to Light: Carbon-Arc (see TM192), have been found acceptable for exposures without water spray (see 17.4). However, caution should be employed since the rate of fade of the Blue Wool Standards by any one method may not agree with other methods.

9. Test Specimen Preparation and Mounting

9.1 Specimen Preparation.

9.1.1 Prepare specimens for the specified test procedures of at least the size given in Table I. When applicable, cut specimens with the longer direction parallel to the warp or filling direction as required in a material specification or contract order.

9.1.2 Prepare specimens as required to meet the needs of the applicable test method given in Table I (see 17.5, 17.6, 17.7 and 17.8).

9.1.3 These dimensions are given as a general guide and in most cases will be adequate to perform the required evaluation. Certain materials may exhibit a dimensional change with respect to the exposure. The test equipment manufacturer, physical test apparatus and number of specimens required will affect the needed specimen size. In any event, the test procedures given in Table I should be reviewed to ensure sufficient specimens are exposed for individual needs.

9.1.4 Unless otherwise specified, test specimens shall be a maximum thickness of 25 mm (1 in.). For specimens over 25 mm (1 in.) thick, formed or configured specimens, and large components, specific instructions must be agreed upon between the purchaser and the supplier.

9.1.5 When required to prevent raveling, the specimens may be edged using a flexible epoxy resin or similar material, or they may be edged by sewing, pinking or fusing.

9.1.6 Label each specimen. Replicant specimens are preferred to provide a record of the changes encountered at different radiant exposure levels. Retain an unexposed specimen for comparison to the exposed specimen.

9.2 Specimen Mounting.

9.2.1 Mount the specimens on frames which are installed in the test chamber without backing, unless otherwise specified (see 17.9).

9.2.2 Fabrics. Secure the specimens to the frames to hold them smoothly without curling of the edges. Fabrics may be stitched to a gauze backing. For colorfastness determinations, mount the specimens in frames as directed in TM16.2.

9.2.3 Yarns. Wind yams on frames. Only that portion of the yarns directly facing the radiant energy is tested for breaking strength. Single strand or multiple strand testing may be performed. When multiple strand testing may be performed, the yarns are wound on the frame closely packed to 2.5 cm (1.0 in.) width. The control specimens must contain the same number of strands as the specimen subjected to exposure. After the exposure has been completed and before the yarns are removed from the frame for testing, those yarns facing the light source are bound together using 2.0 cm (0.75 in.) wide masking or other suitable tape to maintain the closely packed arrangements on the exposure frame.

10. Preparation of Test Apparatus

10.1 Prior to running a new set of conditions, operate the machine for 24 h to ensure conformance to the specified operating conditions as described in Table II. Verify the test apparatus temperature and relative humidity monitoring sensors at the start of each test (see 17.7 and Appendix B).

10.2 Check to see that the machine has been calibrated and maintained within the manufacturer's recommended calibration schedule interval.

10.3 Turn off all rack and specimen spray units.

10.4 Set machine operating conditions according to Table II and the specified option.

10.5 Program the test apparatus to achieve the specified cycle. Fill the specimen rack with blank or dummy specimens and the black panel thermometer unit. This simulated airflow in the test chamber during the test exposure and should not include the actual test specimens. Support the black panel thermometer unit in the specimen drum or rack in the same manner as the test specimen frames. When exterior indicators are not available, read the black panel thermometer unit through the window in the test chamber door. Operate and control the test apparatus as specified in Table II and further defined by the manufacturer. Operate the test apparatus in this mode and adjust the chamber dry and wet bulb temperatures, or chamber thermometer and hygrometer, to provide the required black panel temperature and relative humidity.

10.6 Shut off the test apparatus after operating at the controlled conditions for a minimum of 24 h. Remove the blank or dummy specimens from the specimen rack.

11. Calibration and Verification

11.1 Maintain and calibrate the apparatus described in the manufacturer's instructions.

12. Machine Exposure Test Procedures

12.1 Mount the framed test specimens on the specimen rack. Make sure that all materials are adequately supported, both top and bottom, in proper alignment. Any displacement of the material toward or away from the source, even by a small distance, may lead to variation between specimens. The specimen rack must be filled: blank or dummy specimens are used when the number of specimens being tested is insufficient to fill the specimen rack. When alternate light and dark cycles are required, begin exposure at the start of the light cycle.

12.2 In the case of textile fabrics, unless otherwise specified, ensure that the side normally used as the face is directly exposed to the radiant source.

12.3 Check at regular intervals to ensure the apparatus provides the designated temperatures and relative humidity. When available, monitor exposure test

Table I—Specimen Size and Preparation

Property	Test Method	Specimen Size
Breaking Strength:		
Strip Test	ASTM D5034	5 × 20 cm (2 × 8 in.)
Grab Test	ASTM D5035	13 × 28 cm (5 × 7 in.)
Single Strand	ASTM D2256	15 cm (6 in.)
Bursting Strength:		
Woven Fabric	ASTM D3786	15 × 15 cm (6 × 6 in.)
Nonwoven Fabric	ASTM D3786	15 × 15 cm (6 × 6 in.)
Knit Fabric	ASTM D3787	15 × 15 cm (6 × 6 in.)
Test Strength:		
Elmendorf	ASTM D1424	10 × 13 cm (4 × 5 in.)
Trapezoid	ASTM D1117	10 × 18 cm (4 × 7 in.)
Colorfastness to Light: Carbon-Arc	AATCC TM16.2	3 × 6 cm (1.25 × 2.4 in.)

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Table II—Machine Start-Up Conditions by Cycle

Component	Cycle 1	Cycle 2	Cycle 3	Cycle 4
Cycle Duration, min	120	120	Continuous	120
Light Only, min	90	60	Continuous	102
Dark Only, min	0	60	0	0
Light & Water Spray, min	30	0	0	18
Black Panel Temperature	77 ± 3°C (170 ± 5°F)	77 ± 3°C (170 ± 5°F)	77 ± 3°C (170 ± 5°F)	63 ± 3°C (145 ± 5°F)
Relative Humidity, %				
Light Cycle	70 ± 5	70 ± 5	27 ± 3	50 ± 5
Specimen Rack, rpm	1			
Light Source	Sunshine Open-Flame			
Carbon Type	Copper Clad			
Voltage				
Across Arc	50 V			
Power In	200 V			
Ampere	60 A			
Water Requirements (Input)				
Type	Demineralized, Distilled, or Reverse Osmosis			
Solid, ppm	One or Less			
PH	7 ± 1			
Temperature	Ambient 16 ± 5°C (61 ± 9°F)			

chamber conditions with suitable recorders. If necessary, readjust the controls to maintain the specified test conditions and verify calibrations of the test apparatus.

12.4 Install the appropriate cycle for the agreed upon operating cycle using an electronic controller or mechanical device to program periods with or without radiation.

12.5 **For Option A,** ensure that the water from the specimen spray strikes the test specimens in the form of a fine spray equally distributed over the test specimens.

12.6 **For Option B,** ensure that specimen water sprays are turned off.

12.7 Ensure the current, voltage and water supply during the test meet the requirements of Table II. Ensure the test cycle provides the designated black panel temperature and relative humidity.

12.8 To ensure uniform total irradiation over the specimen surface, reposition specimens vertically in a sequence which will provide each specimen equivalent exposure periods in each location. When the exposure interval does not exceed 24 h, each specimen should be located equidistant from the horizontal axis of the arc. Unless otherwise specified, rotate the test specimens after each 250 h of exposure.

12.9 In some laboratories, the test apparatus is operated on a time cycle of continuous five days exposure followed by two days of rest (machine off) until the designated period of exposure is completed. This provides for operation during weekdays followed by non-operation on weekends, which is a normal schedule for many laboratories. In some laboratories, the test apparatus is operated on a continuous schedule. In any event, the operating schedule must be reported.

12.10 Operate the test apparatus continually until the carbons are consumed. Avoid unnecessary delays when carbons are changed and the exposure period continued, as such delays may contribute to variations in results or lead to errors.

12.11 Conduct physical test using the test methods listed in Section 16, as applicable (see 17.10).

12.12 Average the data for the various replicates, or handle statistically as appropriate, and record a significant value of breaking, tearing, or bursting strength retention and/or colorfastness after exposure in relation to original strength or color, as applicable. It is appropriate to record the percent elongation characteristics of the unexposed controls and exposed specimens at rupture or at some prescribed point in the force-elongation curve as this is frequently significant supplementary information.

13. Conditioning After the Exposure

13.1 After the test exposure is completed, remove the test specimens and control specimens from exposure. Condition in a dark room at standard conditions for testing textiles, as directed in ASTM D1776, Standard Practice for Conditioning and Testing Textiles, [65 ± 5% RH and 21 ± 2°C (70 ± 4°F)] for a minimum of 4 h before evaluation.

14. Evaluation and Interpretation

14.1 Classify the durability or resistance to degradation of the material with respect to the standard of comparison by one of the following:

14.1.1 Percent Strength Retained or Percent Strength Loss. Record percent strength loss or percent strength retained (breaking, tearing, or bursting as appropriate) of the material after the prescribed exposure time period.

14.1.2 Residual Strength. Record the initial and final strength values along with all other pertinent data as above.

14.1.3 Colorfastness. Classify the colorfastness of the material as directed in TM16.1, Colorfastness to Light: Outdoor (see 17.11).

14.2 In Terms of an Agreed Upon Comparison Sample or Standard. Satisfactory: As durable as or more durable than the comparison sample at the amount of radiant exposures and/or exposure time period prescribed in the material standard. Unsatisfactory: Less durable than the comparison sample at the amount of radiant exposure and/or exposure time period prescribed in the material standard.

14.3 For the purpose of defining the relative durability of a test material to an agreed upon comparison standard an index, S_nX, can be defined as the ratio of the percent residual strength in the test specimen compared to be as resistant to degradation as the standard. With the index a specimen is considered to be as resistant to degradation as the standard when the S_nX value is 1; more durable when the S_nX index exceeds 1; less durable when the S_nX index is less than 1.

NOTE: An index of this nature has particular value when recording the relative durability of a series of materials to a common standard and can actually be more useful in research than in the routine assessment of the durability of materials for purpose of commerce.

15. Report

15.1 Use the following guide and report all applicable information:

15.2 Report any deviation from this test method or the performance of the comparison standard.

15.3 Report all information in 15.1 for the same conditions that the samples and reference materials are exposed.

15.4 Report all applicable properties evaluated in Section 14.

15.5 Report the direction of the test if not in warp direction.

16. Precision and Bias

16.1 *Interlaboratory Test Data, Breaking Strength Determination*. The sunshine carbon-arc test was convened by Suga Weathering Technology Foundation, and the exposure was carried out at five laboratories in Japan in 2000 and 2001.

The randomly drawn samples of four test fabrics and three specimens of each

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Exposure Option Used (A or B) _____

Operators Name _____ Date _____

Sample Identification _____

Material exposed: Face _____ Back _____

Colorfastness to Light Rating _____

Lightfastness Classification: _____ No _____

Test Specimen Compared To: Masked Portion _____

Unmasked Portion _____

Unexposed Original _____

Colorfastness to Light Rating Determined By:

Gray Scale For Color Change _____

Instrumentally, Name Type _____

Classification Method: _____

Comparison Standard: _____

Temperature Controlled: By Ambient _____ °C

By Black Panel _____ °C

By Black Standard _____ °C

Exposure Controlled: By AATCC Blue Wool Lightfastness Standards _____

By Radiant Energy _____

Other _____

Total Radiant Energy _____

Type of Test Apparatus _____ Model number _____

Specimen Rack: _____ Inclined _____ 1-Tier _____ 3-Tier _____ Horizontal

Serial Number _____ Manufacturer's Name _____

Type of Water Supply _____

Option Employed _____ Elapsed Exposure Time _____

Mounting Procedure: Backed _____ Unbacked _____

Sample Rotation Schedule _____ Relative Humidity % _____

were exposed at the laboratories according to TM111, Weather Resistance of Textiles: Exposure to Daylight and Weather. All the machine start up conditions Cycle 1 to 4 in TM111 were carried out. These conditions by Cycle are as in TM111 and listed in Table III. And the specifications of the four test fabrics are in Table IV.

Each laboratory made the same test three times. The running hours of these tests were 230 h for all four Cycles. The test of the same cycle condition was repeated every 230 h. So, these *within-laboratory* and *between-laboratory* precision programs were performed in short-term repeats.

All the samples exposed were gathered to Suga Weathering Technology Foundation, Technical committee, and measured by the same machine.

The breaking strength of all specimens was determined as directed in ASTM D5035, Test Method for Breaking Strength and Elongation of Textile Fabrics (Strip method), using a constant rate of extension tensile tester in two directions.

Mean value, repeatability standard deviation Sr, 95% repeatability limit (in measurement units) r, repeatability coefficient of variation (in percent) (CV%r), reproducibility standard deviation S_R, 95% reproducibility limit (in measurement units) R, and reproducibility coefficient of variation (in percent) (CV%R) are used as specified in ASTM E177, Standard Practice for Use of the Terms Precision and Bias in ASTM Test Methods, and listed in Table IV.

16.2 Interlaboratory Test Data, Color Change Determinations. For the same Interlaboratory program described above, color measurements were made prior to and after the sunshine carbon arc exposure. Color change determinations were determined as directed in AATCC Evaluation Procedure (EP) 6, Instrument Color Measurement. The specifications of the four test fabrics are in Table V. The running hours for the color measurement tests were 80 h for all four cycles.

Mean value, repeatability standard deviation Sr, 95% repeatability limit (in measurement units) r, repeatability coefficient of variation (in percent) CV%r, reproducibility standard deviation S_R, 95% reproducibility limit (in measurement units) R, and reproducibility coefficient

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Table III—Specification of Test Specimens for the Breaking Strength

	Textile		Dyestuff	Color Index	Color
A	Acrylic	Warp Woof	Aizen Cation Blue K-GLH	Blended	Blue
B	Cotton	Warp Woof	Remazol Brill. Orange 3R	C.I. Reactive Orange 16	Orange
C	Wool	Warp Woof	Kayanol Milling Brown 4GW	C.I. Acid Brown 13	Brown
D	Polyamide (Nylon)	Warp Woof	Kayanol Milling Red BW	C.I. Acid Red 138	Red

Table IV—Precision All Cycles Breaking Strength, *N*

				Within a Lab			Between Labs		
Sample	Orientation	Test	Mean Level (*M*)	Sr	r	CV%r	SR	R	CV%R
Acrylic	warp	cycle 1	337.26	8.43	23.60	7.00	8.43	23.61	7.00
		cycle 2	336.37	8.04	22.50	6.69	8.27	23.16	6.88
		cycle 3	333.58	8.79	24.61	7.38	9.52	26.66	7.99
		cycle 4	342.65	6.14	17.19	5.02	6.14	17.19	5.02
	woof	cycle 1	210.41	9.52	26.67	12.68	11.33	31.72	15.08
		cycle 2	215.06	6.90	19.33	8.99	6.90	19.32	8.98
		cycle 3	212.57	8.75	24.50	11.53	8.75	24.50	11.53
		cycle 4	213.26	8.59	24.05	11.28	8.99	25.17	11.80
Cotton	warp	cycle 1	276.30	16.68	46.72	16.91	16.68	46.70	16.90
		cycle 2	370.62	11.70	32.75	8.84	23.36	65.41	17.65
		cycle 3	307.00	10.52	29.47	9.60	16.65	46.62	15.19
		cycle 4	349.80	12.93	36.20	10.35	20.88	58.46	16.71
	woof	cycle 1	132.86	11.42	31.97	24.06	18.75	52.51	39.52
		cycle 2	162.40	10.30	28.83	17.75	15.55	43.54	26.81
		cycle 3	162.69	10.55	29.54	18.16	15.10	42.28	25.99
		cycle 4	170.81	11.16	31.25	18.30	14.95	41.86	24.51
Wool	warp	cycle 1	370.36	24.31	68.08	18.38	24.31	68.07	18.38
		cycle 2	380.37	17.11	47.91	12.60	25.62	71.74	18.86
		cycle 3	326.63	11.88	33.27	10.19	21.31	59.67	18.27
		cycle 4	410.40	13.31	37.27	9.08	13.31	37.27	9.08
	woof	cycle 1	323.18	19.82	55.50	17.17	26.03	72.88	22.55
		cycle 2	295.08	11.18	31.31	10.61	17.17	48.08	16.29
		cycle 3	244.79	11.23	31.44	12.85	12.73	35.64	14.56
		cycle 4	394.35	22.06	61.76	15.66	27.46	76.89	19.50
Polyamid	warp	cycle 1	59.14	4.40	12.32	20.84	4.40	12.32	20.83
		cycle 2	104.79	5.14	14.39	13.73	7.73	21.64	20.65
		cycle 3	51.24	4.64	13.00	25.37	4.64	12.99	25.36
		cycle 4	99.40	5.18	14.50	14.59	5.18	14.50	14.59
	woof	cycle 1	79.52	5.44	15.22	19.14	6.27	17.56	22.08
		cycle 2	122.46	5.28	14.80	12.08	8.18	22.90	18.70
		cycle 3	73.20	4.15	11.62	15.88	4.15	11.62	15.87
		cycle 4	114.28	7.07	19.80	17.33	8.49	23.77	20.80
Pooled or Average values	warp	cycle 1	260.76	13.46	37.68	14.45	13.46	37.68	14.45
		cycle 2	298.04	10.50	29.39	9.86	16.25	45.49	15.26
		cycle 3	254.61	8.96	25.09	9.85	13.03	36.48	14.33
		cycle 4	300.56	9.39	26.29	8.75	11.38	31.86	10.60
	woof	cycle 1	186.49	11.55	32.34	17.34	15.60	43.67	23.42
		cycle 2	198.75	8.42	23.57	11.86	11.95	33.46	16.84
		cycle 3	173.31	8.67	24.28	14.01	10.18	28.51	16.45
		cycle 4	223.17	12.22	34.22	15.33	14.97	41.92	18.78

of variation (in percent) CV%R are used as specified in ASTM E177 and listed in Table VI.

16.3 Precision

16.3.1 95% repeatability limit r and repeatability coefficient of variation CV%r of the breaking strength and color difference ΔE has been established. For example, the repeatability coefficient of variation CV%r of the color difference ΔE of Acrylic fabric tested by Cycle 1 was 9.77%. Two single measurements performed in the same laboratory that differ by more than 9.77% (expressed as percentage of their color difference, ΔE) must be considered suspect, i.e., having arisen from different sample population.

95% reproducibility limit R and reproducibility coefficient of variation CV%R of the breaking strength and color difference ΔE, has been established. The reproducibility coefficient of variation CV%R of the color difference ΔE of Acrylic fabric tested by Cycle 1, was 10.70%. Two

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Table V—Specification of Test Specimens for the Color Measurement

	Textile	Dyestuff	Color Index	Color
A	Acrylic	Aizen Cation Blue K-GLH	Blended	Blue
B	Polyester	Kayacelon Yellow E-3GL	C.I. Disperse Yellow 64	Yellow
C	Polyamide (Nylon)	Lanyl Red B	C.I. Acid Red 215	Red
D	Wool	Suminol Leveling Sky Blue R extra conc	Acid Blue 62	Blue

Table VI—Precision All Cycles Colorfastness, ΔE

			Within a Lab			Between Labs		
Sample	Test	Mean Level	Sr	r	CV%r	SR	R	CV%R
Acryl	cycle 1	6.12	0.21	0.60	9.77	0.23	0.65	10.70
	cycle 2	2.52	0.11	0.31	12.29	0.11	0.31	12.29
	cycle 3	3.37	0.17	0.48	14.12	0.19	0.54	15.99
	cycle 4	4.97	0.15	0.43	8.72	0.25	0.69	13.87
Polyester	cycle 1	3.90	0.19	0.53	13.55	0.19	0.53	13.72
	cycle 2	2.74	0.10	0.27	9.83	0.18	0.50	18.20
	cycle 3	3.77	0.13	0.37	9.68	0.22	0.61	16.03
	cycle 4	2.55	0.12	0.34	13.25	0.32	0.91	35.47
Polyamid	cycle 1	6.20	0.27	0.75	12.12	1.24	3.48	56.19
	cycle 2	3.12	0.14	0.39	12.57	0.84	2.36	75.48
	cycle 3	5.15	0.09	0.24	4.71	0.72	2.02	39.15
	cycle 4	4.08	0.15	0.43	10.57	0.52	1.45	35.51
Wool	cycle 1	32.73	1.23	3.45	10.53	1.79	5.00	15.28
	cycle 2	15.14	0.57	1.59	10.47	1.04	2.92	19.29
	cycle 3	18.92	0.25	0.70	3.68	0.94	2.62	13.86
	cycle 4	27.06	0.47	1.31	4.82	1.07	3.00	11.08
Pooled or Average values	cycle 1	12.24	0.48	1.33	10.88	0.86	2.42	19.76
	cycle 2	5.88	0.23	0.64	10.87	0.54	1.52	25.88
	cycle 3	7.80	0.16	0.44	5.70	0.52	1.45	18.52
	cycle 4	9.67	0.22	0.63	6.49	0.54	1.51	15.63

single measurements performed in separate laboratories that differ by more than 10.70% (expressed as a percentage of their average value) must be considered as suspect, i.e., having come from different sample population.

16.4 Bias

16.4.1 An accepted reference value is not available; therefore, the bias cannot be established.

17. Notes

17.1 For Black Panel Thermometers. Testing temperature is measured and regulated by a Black Panel Thermometer unit mounted on the specimen rack to permit the face of it to receive the same exposure as the test specimen. Black Panel Thermometers shall consist of a metal panel at least 70 × 150 mm whose temperature is measured with a thermometer or thermocouple whose sensitive portion is located in the center of and in good thermal contact with the panel. The side of the panel facing the light source shall be black with a reflectance of less than 10% throughout the spectrum of sunshine lamp reaching the specimen; the side of the panel not facing the light source shall be open to the atmosphere within the exposure chamber.

17.2 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

17.3 Refer to ASTM G151 and G152 for guidance on the design and performance requirements for instruments specified in this method (see 3.1).

17.4 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

17.5 Unless otherwise agreed upon, as when specified in an applicable material specification, take a number of specimens such that the user may expect at the 95% probability level that the test result is not more than 5.0% of the average above the true average of the lot. Determine the number of specimens using standard deviation with one-sided limits as directed in ASTM D2905 (see 3.1.10).

17.6 Pile fabrics, such as carpets, which have fibers that may shift position, or texture which may make evaluations in small areas difficult should be tested with an exposed area of not less than approximately 40.0 mm (1.6 in.) × 50.0 mm (2.0 in.). Exposed sufficient size or multiple specimens to include all colors in the sample.

17.7 In some cases high humidity, in combination with atmospheric contaminants, has been found to produce property changes as great as those produced by light. When requested, prepare a duplicate set of test specimens and standards and expose simultaneously in another cabinet or rack of the same type used in the light exposures but covered with an opaque material to exclude the light. Because there is a combined effect of light, temperature, humidity and atmospheric contaminants, it cannot be assumed that a comparison between specimens exposed in the covered cabinet and in the uncovered cabinet under glass will permit separating the effects produced by light only. However, a comparison of the two sets of specimens with a piece of the original which has not been an exposure cabinet will indicate whether a material is sensitive to moisture and atmospheric contaminants. This may also help to explain why different results may be obtained with the same amount of radiant energy in daylight exposures made at different times and at different locations.

17.8 In general, the warp direction is used, but the filling direction may be used to complete or in place of the warp direction when specified. Warp yarns sometimes are protected from the radiation due to fabric structure. When the filling direction is used, it must be reported.

17.9 Specimen frames must be made of stainless steel or suitably coated steel to avoid contaminating the specimens with metallic impurities that might catalyze or inhibit the degradation. When specimens are fastened with staples, they should be of the nonferrous type over-coated to avoid contamination of the

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specimen by corrosion products. Metal frames must have a dull finish and be designed to avoid reflectance that could influence the performance of the material. Frames shall conform to the curvature of the specimen rack. The size of the frame is determined by the type specimens required for individual property requirements.

17.10 Under certain conditions, by agreement between principals to a transaction, tearing strength of the material can be used in lieu of, or to complement, the breaking or bursting strength. The material specification may further define the use of wet breaking, tearing, or bursting strength tests in lieu of or to complement those tests done under standard textile testing conditions. Such conditions of test shall be reported along with the data.

17.11 A difference in color change properties between original material and the covered portion of the exposed specimen indicated that the textile has been affected by some agent other than light, such as heat or a reactive has in the atmosphere. Although the exact cause difference in color may not be known, it should be noted in the report when it occurs.

17.12 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832.9500: fax: +1.610.832.9555; web site: www.astm.org.

17.13 Available from American National Standard Institute Inc., 11 West 42nd St., New York 10036; tel: +1.212.642.4900; fax: +1.212.302.1286; web site: www.ansi.org.

17.14 Available from Society of Automotive Engineers International, 400 Commonwealth Dr., Warrendale PA 10596-0001; tel: +1.412.776.4841; web site: www.sae.org.

18. History

18.1 Revised in 2023 to align to the style guide.

18.2 Revised in 20202. Editorially revised 2016; reaffirmed 2013;revised in 2009; editorially revised 2008, 2007; reaffirmed 2005, 2005.

18.3 Developed in 2003 by AATCC Committee RR64; jurisdiction transferred in 2007 to Committee RA50.

Appendix A
Apparatus and Materials—Machine Exposure

A1. Sunshine Carbon Arc Lamp Exposure Apparatus-Option A and B

A1.1 Sunshine arc type artificial, weathering apparatus consisting of a vertical carbon open-flame arc composed of less than four sets of copper clad carbons mounted at the center of a vertical shell. The arc is enclosed by extended UV filters according to ASTM G152. A lamp exhaustion unit provides for flow of air around the arc and for removal of by-products of combustion of the arc. A blower unit provides a flow of air over the specimens and through the test chamber. Instrumentation provides control of irradiation, wetting, relative humidity, air temperature and black panel temperature is provided. A detailed description of the test apparatus is provided in ASTM G152, Standard Practice for Operating Open Flame Carbon-Arc Light Apparatus for Exposure of Nonmetallic Materials.

A1.2 For Option A, ensure that the water spray strikes the test samples as a fine spray equally distributed over the surface exposed to the radiation.

A1.3 Vaporizing units add moisture to the air as it passes through the conditioning chamber prior to its entry into the test chamber. Relative humidity of the air in the test chamber is determined from readings of wet and dry bulb thermometers, either indicating or recording, whose sensing portion is located in the air flow at its point of exit from the test chamber.

A1.4 Testing temperature is measured and regulated on the basis of a black-panel thermometer unit. Control of the black-panel temperature is accomplished preferably by a continuous flow of air over the specimen at a controlled temperature, but an on-off flow of room temperature air is permissible.

A1.5 Water supply, demineralized, distilled or reverse osmosis. Water transport lines must be constructed of stainless steel or other acceptable materials that will not provide contamination to the water. Provision must be made to maintain water temperature entering the test chamber at 16 ± 4.5°C (60 ± 9°F).

Appendix B
Suggested Machine Calibration and Maintenance Schedule

Follow the manufacturer's instructions for suggested calibration and maintenance schedule.

B1. Component Replacement

B1.1 Carbons. Change according to manufacturer's instructions.

B1.2 Lamp Assembly. Replace when worn or excessively pitted.

B1.3 Extended UV Filters. Replace filters after 2000 h of use or when pronounced discoloration or milkiness develops, whichever occurs first. Filters should be replaced on a rotating replacement schedule to provide more uniformity over long periods of exposure according to manufacturer's instructions.

B1.4 Black Panel Sensor. When bare metal is observed or surface luster can no longer be maintained (see 17.1).

B2. Cleaning

B2 Extended UV Filters. Wash filters with detergent and water.

B2.2 Black Panel Sensor. Minimum is weekly.

B2.3 Test chamber. When discoloration or mineral deposits are observed, minimum is once per month. Use stainless steel cleaning agent that contains NO chloride, flush with deionized water or equivalent.

B2.4 Conditioning Chamber. Minimum is once per month. Drain and flush out sediment with deionized water or equivalent.

B2.5 Lamp Assembly. Brush loose carbon residue according to manufacturer's instructions.

B3. Verification of Operation

B3.1 Controls. Inspect daily to ensure proper set points.

B3.2 Calibration. Minimum is once per week. Use procedures detailed in the manufacturers instructions.

B3.4 Records. Maintain weekly records of operating set point and calibrations.

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Test Method for Aqueous Liquid Repellency: Water/Alcohol Solution Resistance

1. Purpose and Scope

1.1 This test method can be used to determine the efficacy of a protective finish that is capable of imparting a low energy surface on all types of fabrics, by evaluating the fabric's resistance to wetting by a selected series of water/alcohol solutions of different surface tensions.

2. Principle

2.1 Drops of standard test liquids, consisting of a selected series of water/alcohol solutions with varying surface tensions, are placed on the fabric surface and observed for wetting, wicking and contact angle. The aqueous repellency grade is the highest numbered test liquid which does not wet the fabric surface. (The scale ranges from zero to eight, with a rating of eight signifying the most repellent surface.)

3. Terminology

3.1 **grade,** n.—*in textile testing*, the symbol for any step of a multi-step standard reference scale for a quality characteristic.

NOTE: The grade is assigned to test specimens exhibiting a degree of quality comparable to that step of the standard.

3.2 **aqueous repellency,** n.—*in textiles*, the characteristic of a fiber, yarn or fabric whereby it resists wetting by aqueous liquids.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses and impervious gloves when handling test liquids in all laboratory areas.

4.2 The alcohol specified in this method is flammable. Keep away from heat, sparks and open flame. Use with adequate ventilation. Avoid prolonged breathing of vapor or contact with skin. Do not ingest.

4.3 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities [e.g., Occupational Safety and Health Administration's (OSHA) permissible exposure limits (PEL) as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version]. In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprising time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 12.1).

5. Uses and Limitations

5.1 This test method is not intended to give an absolute measure of the resistance of the fabric to staining by all aqueous materials. Other factors, such as composition and viscosity of the aqueous substances, fabric construction, fiber type, dyes, other finishing agents, etc., also influence stain resistance. This test can, however, provide a rough index of aqueous stain resistance, in that generally the higher the aqueous solution repellency grade, the better resistance to staining by aqueous materials, especially liquid aqueous substances. This is particularly true when comparing various finishes for a given fabric.

6. Apparatus and Materials (see 12.2)

6.1 Test liquids prepared and numbered according to Table IV.

6.2 Dropping bottles (see 12.3).

6.3 White AATCC Textile Blotting Paper (see 12.4).

6.4 Laboratory gloves (general purpose is sufficient).

7. Test Specimens

7.1 Test two specimens of the same size from each sample. Specimen size should be sufficient to allow for the complete range of test liquids to be evaluated, but shall be no smaller than 20 × 20 cm (8×8 in.) and no larger than 20 × 40 cm (8×16 in.). Specimens from sample to sample should be the same size. Condition the test specimens for a minimum of 4 h at 21 ± 2C (70 ± 4F) and 65 ± 5% RH prior to testing (see 12.5).

8. Procedure

8.1 Place the test specimen flat on the white textile blotting paper on a smooth, horizontal surface.

8.1.1 When evaluating open weave or "thin" fabrics, conduct the test on at least two layers of the fabric; otherwise, the test liquid may wet the underlying surface, not the actual test fabric, and thereby cause confusion in the reading of the results.

8.2 Wearing clean laboratory gloves, brush the pile of napped or pile fabrics lightly with your hand in the direction giving the greatest lay of the surface prior to placing the drops of the test liquid.

8.3 Beginning with the lowest-numbered test liquid (AATCC Aqueous Solution Test Grade Liquid No. 1), *carefully* place small drops [approximately 5 mm (0.19 in.) in diameter or 0.05 mL volume] on the test specimen in three locations along the filling direction. The drops should be approximately 4.0 cm (1.5 in.) apart. The dropper tip should be held at a height of approximately 0.6 cm (0.25 in.) from the fabric surface while placing drops. DO NOT TOUCH THE FABRIC WITH THE DROPPER TIP. Observe the drops for 10 ± 2 s, from approximately a 45° angle.

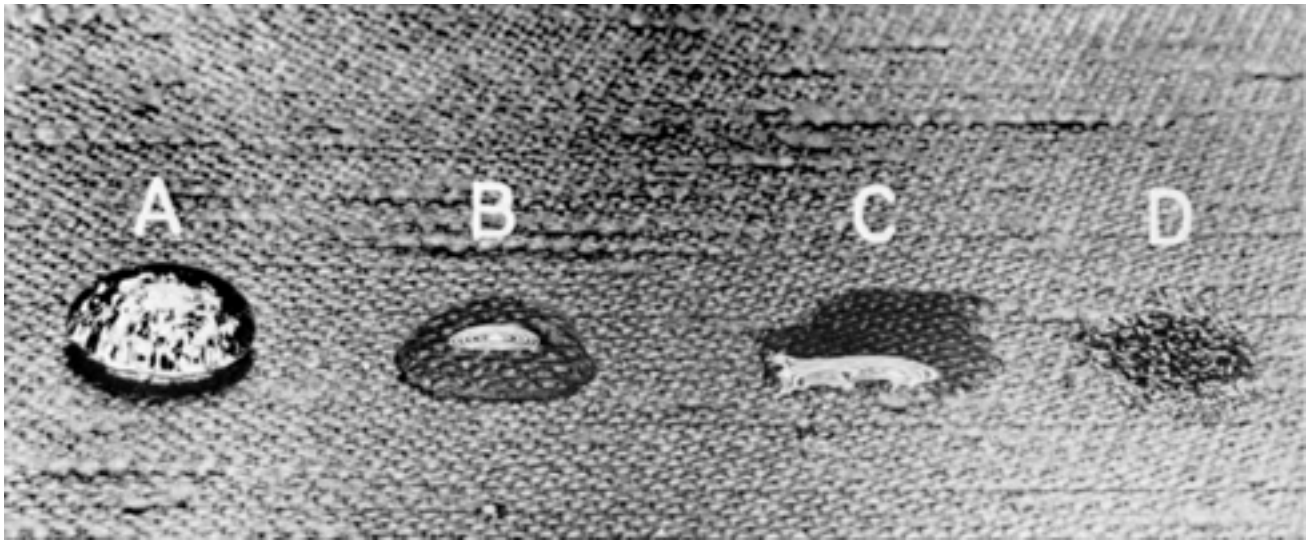
8.4 If no penetration or wetting of the fabric at the liquid-fabric interface and no wicking around the drops occur, place drops of the next higher-numbered test liquid at an adjacent site on the fabric and again observe for 10 ± 2 s.

8.5 Continue this procedure until one of the test liquids shows obvious wetting or wicking of the fabric under or around the drop within 10 ± 2 s.

9. Evaluation

9.1 The AATCC Aqueous Solution Repellency Grade of a fabric is the numerical value of the highest-numbered test liquid which will not wet the fabric within a period of 10 ± 2 s. A grade of zero (0) is assigned when the fabric fails the 98:2 water/isopropyl alcohol solution test liquid. Wetting of the fabric is normally evidenced by a darkening of the fabric at the liquid-fabric interface or wicking and/or loss of contact angle of the drop. On black or dark fabrics, wetting can be detected by loss of "sparkle"

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A = Passes; clear well-rounded drop
B = Borderline pass; rounding drop with partial darkening
C = Fails; wicking apparent and/or complete wetting
D = Fails; complete wetting

Fig. 1—Grading example.

within the drop.

9.2 Different types of wetting may be encountered depending on the finish, fiber, construction, etc.; the determination of the end point can be difficult on certain fabrics. Many fabrics will show complete resistance to wetting by a given test liquid (as indicated by a clear drop with a high contact angle, see Fig. 1, Example A) followed by immediate penetration by the next higher-numbered test liquid. In these instances, the end point and aqueous solution repellency grade is obvious. However, some fabrics will show progressive wetting under several test liquids as evidenced by a partial darkening of the fabric at the liquid-fabric interface (see Fig. 1, Examples B, C and D). For such fabrics, the point of failure is considered to be the test liquid which exhibits complete darkening of the interface or any wicking within 10 ± 2 s.

9.3 A **failure** occurs when two (or more) of the three drops applied from a given test liquid show complete wetting (Fig. 1[D]) or wicking with loss of contact angle (Fig. 1[C]). A **pass** occurs if two (or more) of the three drops applied show clear well-rounded appearance with high contact angle (Fig. 1[A]). The grade is expressed as the integer value of the pass test liquid immediately prior to the fail test liquid. A **borderline pass** occurs if two (or more) of the three drops applied show the rounded drop with partial darkening of the test specimen (Fig. 1[B]). The grade is expressed to the nearest 0.5 value determined by subtracting one-half from the number of the borderline pass test liquid.

10. Report

10.1 The specimen size used for testing should be reported (see 7.1).

10.2 The aqueous solution repellency grade should be measured on two separate specimens. If the two grades agree, report the value. When the two grades are not in agreement, a third determination should be made. Report the grade of the third determination if that value is the same as either of the first two determinations. When the third determination is different from either of the first two, report the median value. Report the aqueous solution repellency grade to the nearest 0.5 value (see Fig. 1 and 9.3).

11. Precision and Bias

11.1 *Summary.* Intralaboratory tests were conducted in November 2002 and January 2003 to establish the precision of this test method. Both intralab testings involved two participants in one laboratory rating two specimens of each of seven fabrics each day for three days. The grades of these intralabs use the scale from 1-10. All materials necessary for the intralabs were provided by customer fabric trials and were processed at two finish load levels. The fabrics used included nylon, polyester, cotton and polyester/cotton materials. The unit of measure was the median of the grades of the two (or three) specimens rated each day.

11.2 The components of variance as standard deviations of the aqueous solution repellency grade were calculated as shown in Table I.

11.3 *Critical differences*. For the components of variance in 11.2, two observations should be considered significantly different at the 95% confidence level if the difference equals or exceeds the critical differences shown in Table II.

Table I—Components of Variance Table for Alcohol/water test

Single operator	0.26
Between operator/within laboratories	0.43

Table II—Critical Differences[a]

No. of Observations[b]	Single Operator	Within Laboratory
1	0.50	0.79
2	0.18	0.59
3	0.15	0.48

[a] The critical differences were calculated using t – 1.950, which is based on infinite degrees of freedom.

[b] An observation is a unit of measure obtained from the median of the grades for 2 (or 3) specimens.

Table III—Mean Values from Testing of Specimens

	Finish load level	
Fabric	**Low**	**High**
Cotton	3.5	5.5
Polyester		7.5
Cotton/Polyester	1.5	2.5
Nylon	6	8

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Table IV—Standard Test Liquids

AATCC Aqueous Solution Repellency Grade Number	Composition	Surface Tensions *N
0	None (fails 98% water)	
1	98:2/Water : isopropyl alcohol (vol:vol)	59.0
2	95:5/Water : isopropyl alcohol (vol:vol)	50.0
3	90:10/Water : isopropyl alcohol (vol:vol)	42.0
4	80:20/Water : isopropyl alcohol (vol:vol)	33.0
5	70:30/Water : isopropyl alcohol (vol:vol)	27.5
6	60:40/Water : isopropyl alcohol (vol:vol)	25.4
7	50:50/Water : isopropyl alcohol (vol:vol)	24.5
8	40:60/Water : isopropyl alcohol (vol:vol)	24.0

*N = dynes/cm at 25°C

11.4 *Bias*. The true value of the aqueous solution repellency grade can only be defined in terms of this test method. Within this limitation, this test method has no known bias.

12. Notes

12.1 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1.513.742.2020; web site: www.acgih.org.

12.2 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

12.3 For convenience, it is desirable to transfer the test liquids from stock solutions to dropping bottles; each marked with the appropriate AATCC Aqueous Solution Repellency Grade number. A typical system found useful consists of 60 mL dropping bottles with ground-in pipettes and Neoprene bulbs. Prior to use the bulbs should be soaked in heptane for several hours and then rinsed in fresh heptane to remove soluble substances. It has been found helpful to place the test liquids in sequential order in a wooden platform on the grading table. NOTE: Ratio of test liquids does affect surface tension of the liquid. Use only analytical grades of test liquids. The surface tension of the liquids should be checked every month or the liquids in the dropper bottles should be replaced every month from a sealed stock solution bottle since the isopropyl alcohol concentration decreases due to evaporation.

12.4 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

12.5 Often AATCC TM118 (Test Method for Oil Repellency: Hydrocarbon Resistance) and AATCC TM193 are done concurrently. It is recommended that the specimen sizes for each test be the same.

13 History

13.1 Revised in 2023 to add History Section and clarity.

13.2 Revised 2019, 2018; reaffirmed 2017;revised 2016; reaffirmed 2012;revised 2011, 2010,2008; revised 2007; editorially revised 2006; reaffirmed and editorially revised 2005.

13.3 Developed in 2004 by AATCC Committee RA56; Technically equivalent to ISO 23232.

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Test Method for Anti-House Dust Mite Properties of Textiles under Long-Term Test Conditions

1. Purpose and Scope

1.1 This test method is for the evaluation of the degree of anti-house dust mite activity in a long-term testing environment for textiles treated at the manufacturing level for this purpose.

2. Principle

2.1 Test and control specimens are inoculated with the test organism and nutrients. After six weeks of incubation, sufficient time for mite colonies to flourish under optimal conditions, the dust mites are recovered from the specimens by heat extraction. Results are expressed as per cent reduction on the treated sample versus the untreated control.

3. Terminology

3.1 **activity,** n.—*of an anti-dust mite agent*, a measure of the effectiveness of the agent.

3.2 **anti-house dust mite agent,** n.—any chemical which kills (acaricide) or repels house dust mites.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedure and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as Safety Data Sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 This test should be carried out by persons with training and experience in the use of acarological techniques.

4.2 CAUTION: Although house dust mites are not considered to be a direct hazard to humans, their fecal pellets have been demonstrated to be a potential potent allergen for those susceptible to asthma. Therefore, every necessary and reasonable precaution must be taken to eliminate this risk to the laboratory personnel and to personnel in the associated environment. Where appropriate, wear protective clothing and respiratory protection that prevents penetration by the matter.

4.3 All work should be conducted using standard safe laboratory practices.

4.4 All chemicals should be handled with care.

4.5 An eyewash/safety shower should be located nearby for emergency use.

4.6 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 12.1).

5. Limitations

5.1 The method cannot be used to determine the specific mode of action of a given acaricide treatment.

5.2 The method should not be used to predict the performance of a finished article if the test specimen will go through additional processing steps or represents only a component of the final article.

5.3 While this test method provides some insight into the effect of different textile treatments in controlling the ability of mites to successfully establish breeding colonies, direct conclusions cannot be made with regard to a treatment removing or reducing allergens.

5.4 This test method does not allow recovery of house dust mite eggs. However, it does provide a good measurement of the effectiveness of anti-dust mite treatment on a breeding mite colony.

6. Materials

6.1 Test Organisms

6.1.1 Test mites: *Dermatophagoides pteronyssinus* or *D. farinae*. Any other species suitable to a given country or region can also be used.

6.2 Maintenance of House Dust Mite Stock Cultures

6.2.1 Dust mite colonies should be maintained at 25 ± 1°C (77 ± 1°F) and 73-76% relative humidity on TetraMin Tropical Flakes (see 12.2).

6.2.2 Before use, the flakes should first be ground with a mortar and pestle, then sieved so that the particle size is between 500–750 µm.

6.2.3 Care must be taken to ensure that stock cultures of mites used for testing have not previously been exposed to chemicals or treatments that might have some interaction with the mites.

7. Specimens

7.1 Test Specimens.

7.1.1 Prepare a minimum of three test specimens by cutting them to fit snugly in the bottom of a 10 cm diameter glass or polystyrene Petri dish. This can accurately be done by tracing around the dish on the sample, or by using a suitably sized die. For loose fibers, sufficient material to cover the bottom of the Petri dish should be used.

7.1.2 Petri dishes or other test chambers of larger or smaller size can be substituted where desired. The size of the test specimen should change accordingly to ensure a snug fit of the sample in the bottom of the dish.

7.2 Control specimens.

7.2.1 A minimum of three specimens of the same fiber type and construction as the test specimen but containing no anti-mite finish (negative control) should also be prepared.

7.2.2 In addition, an internal lab control previously determined to support dust mite colonies should be included for each test. The purpose of this internal control is to provide validation that the mite colony has developed at the expected rate on a known sample over the course of the six weeks test.

7.3 Sterilization of Specimens.

7.3.1 Specimens may be sterilized when there is potential for fungal growth on the specimen over the test period due to the presence of spores. The method of sterilization used will depend on the sample composition and finish, as well as the particular anti-house dust mite treatment. Report method of sterilization, if used.

8. Procedure

8.1 Test Setup.

8.1.1 Distribute 50 mg of the ground/sieved nutrient mixture on each test specimen. The mixture can be applied through a sieve to enable an even distribution of the material.

8.1.2 To prevent mites from escaping, the sides of the container can be coated with petroleum jelly. A heavy layer of

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coating should be avoided; as it can melt and interfere with mite recovery during the subsequent heat step. Tangle Trap has also been found to provide an effective barrier (see 12.3). However, final recovery numbers may be negatively influenced due to the numbers of mites becoming irreversibly trapped on the material. A third option is to place a textile sample that has previously been demonstrated to be effective as a mite barrier fabric tightly across the top of the test chamber and fix in place.

8.1.3 Close each test chamber and place the test units at 25 ± 1°C (77 ± 1°F) and 73-76% relative humidity for approximately 48 h to acclimatize the specimen microenvironment.

8.1.4 Place 25 male and 25 female mites from a robust colony on each acclimatized test and control specimen.

8.1.5 If possible, use mating pairs to ensure that the females are at a similar stage of oviposition.

8.1.6 Close the test chamber and incubate the samples at 25 ± 1°C (77 ± 1°F) and 73-76% relative humidity for six weeks.

8.2 Mite Recovery.

8.2.1 After the six weeks of incubation, remove one test chamber at a time from the incubator. For each plate, pre-cut a sample of nylon mesh material (see 12.4) to fit snugly within the test plate. Cover the mesh with adhesive film so that the sticky side of the tape is in direct contact with the mesh, and the edges of the tape extends beyond the edge of the mesh by approximately 5 mm.

8.2.2 Remove the lid from the test chamber and firmly place the tape/mesh combination directly on top of the test specimen, with the sticky side of the tape facing down. Ensure the outer edges of the tape adhere to the sides of the test plate. The mesh should serve to limit the amount of food particles and dead mites that might adhere to the sticky tape during the subsequent recovery step.

8.2.3 Place the plate, minus the lid, directly on a heating plate (see 12.5) set at 50°C (122°F). Place a weight on top of the tape/mesh combination to ensure intimate contact between the mesh and the test specimen. Use of a pre-cut circle of Styrofoam that will fit in the test plate between the tape/mesh and the weight can aid in evenly distributing the weight, and may also limit moisture buildup on the tape that might otherwise occur.

8.2.4 Leave each test chamber on the hot plate for a minimum of 5 h. This should be sufficient for mite recovery from textiles, as well as thicker samples or those with a heavy backing, such as carpets.

8.2.5 After the heat exposure, remove the weight and recover the tape/mesh. Cover the second side of the mesh with a clear polyethylene film or a second coat of sticky tape, to secure the mites for subsequent counting.

9. Evaluation

9.1 For each specimen, count the total number of mites recovered on the film using a low power stereo-binocular microscope.

9.2 For each set of test and control specimens, calculate the mean and standard deviation.

10. Report

10.1 Express results as percent reduction versus the control, using the following formula:

$$R = \frac{A - B}{A} \times 100$$

where:

R = percent reduction of the test specimen versus the control.
A = the mean number of dust mites found on the control specimen.
B = the mean number of dust mites found on the test specimen.

10.2 Indicate that test was conducted following AATCC TM194-2023.

10.3 For a valid test: If a healthy mite colony has been established on the negative control specimens during the six week test period, mites on the internal lab control do not need to be recovered and counted. However, if the mite numbers on the negative controls are lower than expected for a control in this time period, mites from the internal lab control must be recovered as outlined in 8.2. Recovery numbers outside of the normal range of mites previously established for each of the internal controls, invalidate the test.

10.4 Any deviation from this procedure as written must be reported.

10.5 The criterion for passing the test must be determined by the interested parties.

11. Precision and Bias

11.1 *Precision*. Precision for this test method has not been established. Until a precision statement is generated for this test method, use standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages.

11.2 *Bias*. Anti-house dust mite properties of textiles under long-term test conditions can be defined only in terms of a test method. There is no independent method for determining the true values. As a means of estimating these properties, the method has no known bias.

12. Notes

12.1 Available from Publication Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1.513.742.2020; web site: www.acgih.org.

12.2 TetraMin Tropical Flakes con be obtained from Spectrum Brands Pet LLC 3001 Commerce St., Blacksburg VA 24060-6617. phone 800.526.0650 and via their website at https://www.tetra-fish.com or other online retailers.

12.3 Tangle Trap can be obtained from The Tanglefoot Company, 314 Straight Ave. S.W., Grand Rapids MI 49504-6485; tel: +1.616.459.4139; fax: +1.616.459.4139; e-mail: info@tanglefoot.com; web site: www.tanglefoot.com.

12.4 A mesh consisting of 447 denier nylon filaments, with 40 × 40 threads/inch, thickness of 0.36 mm and with a hole size of 0.63 mm has been found effective. This mesh can be obtained from Industrial Textiles Ltd., 62 Patiki Rd., Avondale, Auckland 1007 NZ; tel: +64.98283188; free fax: +64.98281022; email: info@vakeattack.co.nz; web site: www.indtex.co.nz.

12.5 Other means of heating the samples, such as the use of an incandescent light source, can also be used. However, the process must be optimized to obtain maximum mite recovery for each type of sample before initiation of the test.

13. History

13.1 Revised in 2023 to add dust mite food source and to align to the AATCC Style guidelines.

13.2 Editorially revised 2019; reaffirmed 2013; editorially revised 2010; reaffirmed 2008, 2007.

13.3 Developed 2006 by AATCC Committee RA49.

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Test Method for Liquid Moisture Management Properties of Textile Fabrics

1. Purpose and Scope

1.1 This test method is for the measurement, evaluation and classification of liquid moisture management properties of textile fabrics. The test method produces objective measurements of liquid moisture management properties of knitted, woven and nonwoven textile fabrics.

1.2 The results obtained with this test method are based on water resistance, water repellency and water absorption characteristics of the fabric structure, including the fabric's geometric and internal structure and the wicking characteristics of its fibers and yarns.

2. Principle

2.1 The liquid moisture management properties of a textile are evaluated by placing a fabric specimen between two horizontal (upper and lower) electrical sensors each with seven concentric pins. A predetermined amount of test solution that aids the measurement of electrical conductivity changes are dropped onto the center of the upward-facing test specimen surface. The test solution is free to move in three directions: radial spreading on the top surface, movement through the specimen from top surface to the bottom surface, and radial spreading on the bottom surface of the specimen. During the test, changes in electrical resistance of specimen are measured and recorded.

2.2 The electrical resistance readings are used to calculate fabric liquid moisture content changes that quantify dynamic liquid moisture transport behaviors in multiple directions of the specimen. The summary of the measured results are used to grade the liquid moisture management properties of a fabric by using predetermined indices.

3. Terminology

3.1 **absorption rate (AR_T) (top surface) and (AR_B) (bottom surface),** n.—the average speed of liquid moisture absorption for the top and bottom surfaces of the specimen during the initial change of water content during a test.

3.2 **accumulative one-way transport capability (R),** n.—the difference between the area of the liquid moisture content curves of the top and bottom surfaces of a specimen with respect to time.

3.3 **bottom surface (B),** n.—*for testing purposes*, the side of the specimen placed down against the lower electrical sensor which is the side of the fabric that would be the outer exposed surface of a garment when it is worn or product when it is used.

3.4 **maximum wetted radius (MWR_T) and (MWR_B) (mm),** n.—the greatest ring radius measured on the top and bottom surfaces.

3.5 **moisture management,** n.—*for liquid moisture management testing*, the engineered or inherent transport of aqueous liquids such as perspiration or water (relates to comfort) and includes both liquid and vapor forms of water.

3.6 **overall (liquid) moisture management capability (OMMC),** n.—an index of the overall capability of a fabric to transport liquid moisture as calculated by combining three measured attributes of performance: the liquid moisture absorption rate on the bottom surface (AR_B) the one way liquid transport capability (R), and the maximum liquid moisture spreading speed on the bottom surface (SS_B).

3.7 **spreading speed (SS_i),** n.—the accumulated rate of surface wetting from the center of the specimen where the test solution is dropped to the maximum wetted radius.

3.8 **top surface (T),** n.—*for testing purposes*, the side of a specimen that, when the specimen is placed on the lower electrical sensor, is facing the upper sensor. This is the side of the fabric that would come in contact with the skin when a garment is worn or when a product is used.

3.9 **total water content (U) (%),** n.—the sum of the percent water content of the top and bottom surfaces.

NOTE: Total water content measurements may be more accurately termed, "total surface water content" particularly in the case of fabric with cellulosic content. Total water content implies that all water in the specimen is being measured which may be the case with some manufactured fabrics. However, when testing cellulosic fibers, moisture trapped in the interior of the fiber (for example, in the lumen of cotton fibers) will not be included with a specimen's detected surface liquid moisture.

3.10 **wetting time (WT_T) (top surface) and (WT_B) (bottom surface),** n.—the time in seconds when the top and bottom surfaces of the specimen begin to be wetted after the test is started.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all-inclusive. It is the users' responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturers' recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5. Uses and Limitations

5.1 AATCC test methods are not performance specifications as the methods produce measurements only. It is the responsibility of test method users to choose the minimum (or maximum) measurement value that is acceptable for the product being tested.

5.2 This test method focuses on liquid moisture transport in the flat state. The test method may be applicable to the evaluation of fabrics in garments or textile products as they would be exposed to liquid moisture (e.g., perspiration) present on the surface of human skin. It does not measure gaseous moisture transport properties (e.g., water vapor transmission) or tactile properties that also influence human perceptions of comfort.

5.3 Because human perceptions of comfort are influenced by multiple liquid movement properties, as well as ergonomic (garment fit) factors, the use of a single unit of measurement from this test method or any other test method could be misleading as explained in the AATCC/ASTM International's *Moisture Management Technical Supplement as Related to Textile Apparel, Linens and Soft Goods* (see 13.1). Consequently, this test method alone will not give an overall rating of the comfort of a garment or textile product. Overall performance schemes, such as the Grading Graph in 9.2.1, should be developed when trying to correlate a combination of absorption, wicking, liquid and vapor movement that can be related to the environment and preferences of fit and

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style in which the textile product is to be used and worn.

5.4 This method may not be applicable to coated, laminated, or complex fabric constructions. Caution should be used when analyzing fabrics with repellent surface finishes. This method may not be applicable to specimens exhibiting high overall absorbent capacity such as terry cloth or other thick knitted and woven fabrics. Thicker or highly absorbent fabrics may not allow for proper liquid moisture movement to be analyzed using the test solution volume.

5.5 This test method does not measure drying performance directly. Drying performance is inferred based on the area of liquid moisture spreading.

5.6 The wetting times measured in this method may be related to absorbency as measured by AATCC TM79, Test Method for Absorbency of Textiles (see 13.1).

5.7 The maximum wetted radius defined in 3.4 above should not be used to infer maximum spread areas. As this testing apparatus employs the use of concentric circular rings for measuring wetted radius, for specimens that exhibit noncircular, elliptical or amoeboid spread patterns, the radius may be misrepresented. For example, fabrics with a linear symmetry such as ribbed knits or fabrics with repellent finishes may exhibit irregular spreading patterns.



Fig. 1—Moisture management tester.

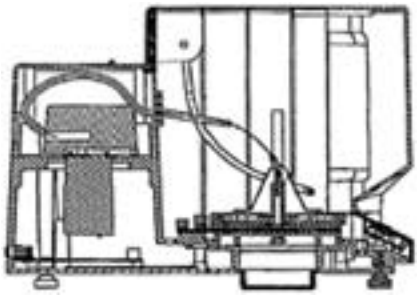


Fig. 2—A cross-sectional side view of the equipment.

6. Apparatus and Materials

6.1 Moisture Management Tester (MMT) (see 13.2, Figs. 1 and 2).

6.2 Computer with MMT software installed.

6.3 Distilled water.

6.4 Sodium chloride solution (0.9% NaCl).

6.5 Conductivity meter.

6.6 White AATCC Textile Blotting Paper (see 13.1) or soft paper towels.

7. Test Specimens

7.1 Before cutting specimens, launder the sample(s) according to the conditions selected from the AATCC LP1 Laboratory Procedure for Home Laundering: Machine Washing (see 13.1) or as agreed between parties. It may be suitable to test a material unlaundered or after repeated launderings. Removal of sizing and/or finishes may affect the liquid moisture management properties of a fabric (see 13.3).

7.2 Cut five 8 × 8 cm specimens, taken diagonally across the width of a sample, to ensure that different sets of length and width yarns are in each specimen or from different sites on a product.

7.3 Place the specimens on a flat smooth, horizontal surface without tension before testing, to condition them to moisture equilibrium in an environment controlled at 21 ± 2°C (70 ± 4°F) and 65 ± 5% RH as recommended by ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 13.5).

8. Procedure

8.1 Prepare the test solution by dissolving 9 g sodium chloride (USP Grade) in 1 L of distilled water and adjust its electrical conductivity to 16 ± 0.2 milli Siemens (mS) at 25°C (77°F) by adding sodium chloride or distilled water as necessary. The test solution is used to provide a conductive medium for the instrument's sensors and does not duplicate perspiration.

8.2 Follow the manufacturer's instructions for starting the instrument, addition of the test solution, and the computer software's set up to collect test data.

8.3 Raise the upper sensor to its locked position and place a paper towel on the lower sensor. Press the "Pump" button for 1-2 min until the amount (0.22 cc) of test solution is drawn from the container and drips onto the paper towel and no air bubbles are present inside the tubing. Remove the paper towel.

8.4 Place the conditioned test specimen on the lower sensor with the specimen's top surface up (see 3.8). Release the upper sensor until it freely rests on the test specimen and shut the door of the tester. Confirm that the "Pump-On Time" is set at 20 s to assure the predetermined amount (0.22 cc) of test solution is dispensed. For each specimen, the percent (%) water content point on the graph should be 0.0 at the start of each test to avoid erroneous test results. Set the "Measure Time" for 120 s and start the test. At the end of the 120 s test time, the software will automatically stop the test and calculate all of the indices.

8.5 Raise the upper sensor and remove the tested specimen.

8.6 Before inserting the next specimen, keep the upper sensor in its locked position. Dry between the rings of pins on both upper and lower sensors using White AATCC Textile Blotting Paper or a soft paper towel cut into narrow (0.5 cm) strips. Wait 1 min, or longer, to ensure there is no residual test solution present on the sensors, otherwise any leftover moisture will cause an erroneous start (see 8.4). If salt deposits are observed on the sensors after drying, use distilled water to remove.

8.7 Load a new specimen on top of the lower sensor with the fabric top surface up and repeat steps 8.4-8.6.

8.8 When testing has been completed for the day, use distilled water to clean and purge the pump and tubing.

9. Evaluation Measurement Units, Grading, and Classification

9.1 Measurement units – For each sample tested, compile the average values for each measurement unit as follows:

Wetting Time – WT_T (top surface) and WT_B (bottom surface),

Absorption Rate – AR_T (top surface) and AR_B (bottom surface),

Maximum Wetted Radius – MWR_T (top surface) and MWR_B (bottom surface),

Spreading Speed – SS_T (top surface) and SS_B (bottom surface),

Accumulative One-way Transport Capability – (R), and

Overall (liquid) Moisture Management Capability (OMMC).

9.1.1 Formulae used to calculate the units of measurement shown in 9.1 are given in Appendix A.

9.2 Grading – Using the average values from 9.1 and Table I, grade the tested sample(s). The grading developed is based on a study referenced in 13.2 which classifies material moving moisture from the back to the face with higher values.

9.2.1 Table II – Grading Summary Table can be used to summarize and illustrate the liquid moisture management properties of the tested sample(s).

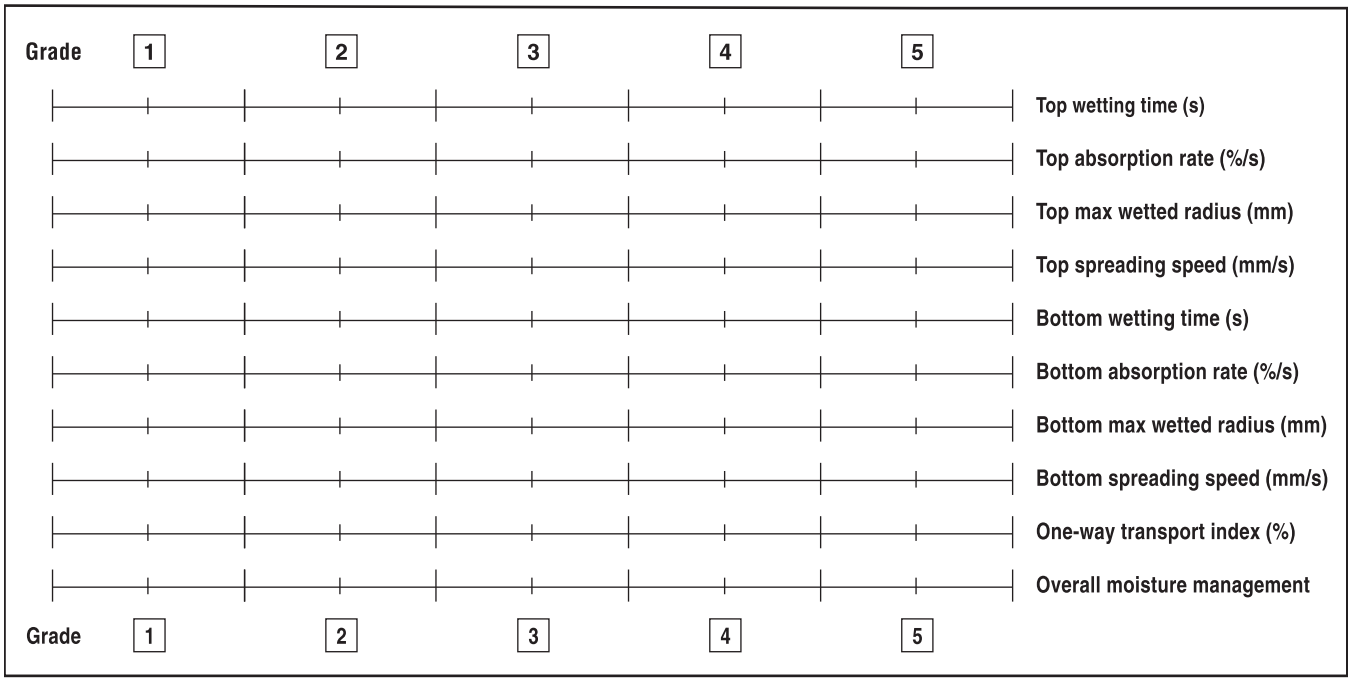
9.2.2 Tables I and II are examples of grading schemes. Other schemes may be developed.

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Table I—Grading Table of all Indices

Index		Grade 1	2	3	4	5
Wetting time (sec	Top	≥ 120	20 – 119	5 – 19	3 – 5	< 3
	Bottom	≥ 120	20 – 119	5 – 19	3 – 5	< 3
Absorption rate (%/sec)	Top	0 – 9	10 – 29	30 – 49	50 – 100	> 100
	Bottom	0 – 9	10 – 29	30 – 49	50 – 100	> 100
Max wetted radius (mm)	Top	0 – 7	8 – 12	13 – 17	18 – 22	> 22
	Bottom	0 – 7	8 – 12	13 – 17	18 – 22	> 22
Spreading speed (mm/sec)	Top	0.0 – 0.9	1.0 – 1.9	2.0 – 2.9	3.0 – 4.0	> 4.0
	Bottom	0.0 – 0.9	1.0 – 1.9	2.0 – 2.9	3.0 – 4.0	> 4.0
One-way transport capability (R)		<-50	-50 – 99	100 – 199	200 – 400	> 400
Overall Moisture Management Capability (OMMC)		0.00 – 0.19	0.20 – 0.39	0.40 – 0.59	0.60 – 0.80	> 0.80

Table II—Grading Summary Table



10. Report

10.1 Record the average and standard deviations of the measurements listed in 9.1.

10.2 Using the averages, grade the sample according to Table I and summarize using Table II.

10.3 Report the average, standard deviation and grade for each sample or the agreed upon measurements.

11. Precision

11.1 A single-lab study was performed in November 2008 using a SDL Atlas Moisture Management Tester, Model 290, Software Version 3.06. The six fabrics tested are identified in Table III.

11.2 The samples were laundered one time in a single 1.8 kg (4 lb) load according to AATCC TM135 (1)(III)(A)i; Normal water level with 2003 AATCC Standard Reference Liquid Laundry Detergent. A normal tumble dry setting was used for a 30 min interval. Ten new specimens from each sample were tested three separate days by one operator. Specimens were conditioned 48 h and tested in prescribed standard atmospheres (see 7.2).

11.3 Table III Summary Data for Each MMT Measurement by Sample shows the average values of each MMT unit of measurement for each sample. The table also shows the calculated standard deviation (SD), coefficient of variation percent (CV%), and 95% confidence interval for each sample. As noted in 5.2-5.5, caution in comparing data between selected samples or among all of the samples in the study cannot be made as the study samples represent different fabric types, yields, fiber contents and technology applications.

11.4 T-tests on all of the reported measurements for the three sets of 10 replicate data within a sample, at the 95% confidence level, do not show any statistically significant differences within a sample. Therefore, the confidence intervals may be used to suggest a range of values that may be obtained with each test measurement for various types of samples.

11.5 *Between-laboratory* precision has not been established for this test method. Until such precision information is available, users of the method should use standard statistical techniques in making any comparison of test results for *between-laboratory* averages.

11.6 Earlier studies regarding the Moisture Management Tester's precision

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Table III—Summary Data for Each MMT Measurement by Sample

	Yield G/M²	Fiber Content		Top Wetting Time (sec)	Bottom Wetting Time (sec)	Top Absorption Rate (%/sec)	Bottom Absorption Rate (%/sec)	Top Max Wetted Radius (mm)	Bottom Max Wetted Radius (mm)	Top Spreading Speed (mm/sec)	Bottom Spreading Speed (mm/sec)	Accumulative One-Way Transport Index (%)	OMMC
Woven	117	100% C	average	2.32	2.37	86.55	71.22	30.00	30.00	8.22	8.07	-120.41	0.42
			min	2.05	2.20	82.44	66.70	30.00	30.00	7.96	7.69	-143.57	0.41
			max	2.59	2.59	90.09	75.10	30.00	30.00	8.50	8.27	-98.18	0.43
			std dev	0.11	0.10	1.71	1.69	0.00	0.00	0.14	0.13	11.11	0.0047
			cv %	0.05	0.04	0.02	0.02	0.00	0.00	0.02	0.02	-0.09	0.01
			95% CI	0.04	0.04	0.61	0.60	N/A	N/A	0.05	0.05	3.98	0.00
Knit	168	100% PET	average	2.93	3.06	57.11	48.52	23.50	22.50	4.53	4.42	-89.25	0.36
			min	2.77	2.77	52.54	44.19	20.00	20.00	4.11	4.08	-110.25	0.35
			max	3.41	3.41	94.06	52.44	25.00	25.00	4.93	4.93	-55.66	0.37
			std dev	0.13	0.14	7.26	2.28	2.33	2.54	0.22	0.21	12.54	0.01
			cv %	0.04	0.05	0.13	0.05	0.10	0.11	0.05	0.05	-0.14	0.03
			95% CI	0.05	0.05	2.60	0.82	0.83	0.91	0.08	0.08	4.49	0.00
Knit	204	100% C	average	5.28	4.33	38.12	56.21	52.50	27.67	4.97	5.73	417.00	0.87
			min	3.72	2.84	29.74	47.24	25.00	25.00	4.30	4.71	342.02	0.81
			max	6.84	5.64	98.17	64.78	30.00	30.00	5.91	6.88	507.58	0.90
			std dev	0.89	0.75	11.93	4.18	1.53	2.54	0.34	0.53	36.56	0.02
			cv %	0.17	0.17	0.31	0.07	0.03	0.09	0.07	0.09	0.09	0.02
			95% CI	0.32	0.27	4.27	1.50	0.55	0.91	0.12	0.19	13.08	0.01
Knit	199	100% C	average	3.65	3.06	39.25	49.63	20.17	20.67	3.74	4.23	296.73	0.74
			min	2.84	2.13	32.64	44.64	15.00	20.00	3.11	3.77	209.59	0.64
			max	4.44	3.64	43.90	54.07	25.00	25.00	4.78	5.41	378.72	0.85
			std dev	0.41	0.33	3.18	2.32	1.60	1.73	0.37	0.35	39.86	0.05
			cv %	0.11	0.11	0.08	0.05	0.08	0.08	0.10	0.08	0.13	0.07
			95% CI	0.15	0.12	1.14	0.83	0.57	0.62	0.13	0.13	14.26	0.02
Knit	648	65N/21PET/14EL	average	6.74	3.55	17.71	68.51	14.17	20.50	2.21	3.55	722.30	0.87
			min	3.80	3.08	11.73	50.45	10.00	20.00	1.10	3.32	649.96	0.81
			max	14.84	3.88	32.40	90.84	15.00	25.00	2.97	3.88	785.22	0.93
			std dev	2.67	0.21	5.11	14.57	1.90	1.53	0.52	0.13	36.53	0.04
			cv %	0.40	0.06	0.29	0.21	0.13	0.07	0.24	0.04	0.05	0.05
			95% CI	0.96	0.08	1.83	5.21	0.68	0.55	0.19	0.05	13.07	0.01
Knit	168	65N/21PET/14EL	average	7.16	6.23	54.94	80.52	10.00	11.00	1.29	1.43	233.47	0.52
			min	5.72	4.92	36.67	42.76	10.00	10.00	0.73	1.30	23.50	0.36
			max	14.67	7.08	106.95	168.13	10.00	15.00	1.51	1.75	574.27	0.77
			std dev	1.60	0.49	19.13	30.62	0.00	2.03	0.13	0.08	112.50	0.11
			cv %	0.22	0.08	0.35	0.38	0.00	0.18	0.10	0.06	0.48	0.21
			95% CI	0.57	0.18	6.85	10.96	N/A	0.73	0.05	0.03	40.26	0.04

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have been performed (see 13.4.)

11.7 Data from the 2008 study are on file at AATCC Headquarters for reference purposes.

12. Bias

12.1 Bias. The true value of liquid moisture management property of textile fabrics can be defined only in terms of a test method. There is no independent method for determining the true value. In estimating this property, the test method has no known bias.

13. Notes

13.1 White AATCC Textile Blotting Paper, the AATCC/ASTM International's *Moisture Management Technical Supplement as Related to Textile Apparel, Linens and Soft Goods*, and other noted documents are available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

13.2 The information about the Moisture Management Tester is available from SDL Atlas L.L.C., 3934 Airway Dr., Rock Hill, SC 29732-9200; tel: +1.803.329.2110; fax: +1.803.329.2133; e-mail: info@sdlatlas.com; web site: www.sdlatlas.com; and may be built according to patent USP 6,499,338 B2 and information in the research paper "Moisture Management Tester: A Method to Characterize Fabric Liquid Moisture Management Properties," *Textile Research Journal*, Vol. 75, No. 1, 2005, pp57-62, and the research paper "An Improved Test Method for Characterizing the Dynamic Liquid Moisture Transfer in Porous Polymeric Materials," *Polymer Testing*, Vol. 25, 2006, pp677-689.

13.3 "Influence of Pretreatments on Moisture Management Test Results," Jane Batcheller, University of Alberta, AATCC International Conference Poster Presentation, May 2010.

13.4 "Precision of New Test Method for Characterizing Dynamic Liquid Moisture Transfer in Textile Fabrics" by Bao-guo Yao, Zhejiang University and The Hong Kong Polytechnic University and Yi Li and Yi-lin Kwok, the Hong Kong University, published in the AATCC *Review*, Vol. 8, No 7, July 2008, pp44-48.

13.5 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832.9555; fax: +1.610.832.9555; web site: www.astm.org.

14. History

14.1 Jurisdiction transferred to Committee RA114 in 2022; Editorially revised to note change.

14.2 Editorially revised in 2020 to add History section and to replace reference to M6 with LP1.

14.3 Editorially revised 2019, 2016. Editorially revised and reaffirmed, 2017, 2012. Reaffirmed 2010. Revised 2011.

14.3 Developed in 2009 by AATCC Committee RA63.

Appendix A

A1. Formulae Used in the Moisture Management Tester Software for the Calculation of Units of Measurement

A1.1 Absorption rate AR_T and AR_B (%/sec) formulae are:

AR_T = Average ($SLOPE_T$) and AR_B = Average ($SLOPE_B$)

A1.2 Spreading speed (S_i) formulae are:

$$S_i = \frac{\Delta r_i}{\Delta t_i} = \frac{\Delta r_i}{t_i - t_{i-1}}$$

where:

concentric ring = i (i = 1, 2, 3, 4, 5, or 6)
wetting time + t_i,I = the liquid moisture spreading speed (S_i) from ring i–1 to ring i
the distance between ring i and ring i–1 = Δr_i

A1.3 Accumulative Spreading Speeds (SS_T and SS_B) formulae are:

$$SS_T = \sum_{i=1}^{N_T} S_i = \sum_{i=1}^{N_T} \frac{\Delta r_i}{t_i - t_{i-1}},$$

$$SS_B = \sum_{i=1}^{N_B} S_i = \sum_{i=1}^{N_B} \frac{\Delta r_i}{t_i - t_{1-1}}$$

where:

N_T and N_B are the numbers of the maximum wetted rings of the top and bottom surfaces

A1.4 Accumulative one-way transport capability (R) formula is:

[Area (U_B) – Area (U_T)]/Total testing time.

A1.5 Overall moisture management capability (OMMC) formula is:

$$\text{OMMC} = C_1 * AR_{B_\text{ndv}} + C_2 * R_{\text{ndv}} + C_3 * SS_{B_\text{ndv}}$$

where:

C_1, C_2, and C_3 are the weighting values * for AR_{B_ndv}, R_{ndv} and SS_{B_ndv}
(AR_B)= absorption rate
(R) = one-way transport capability, and
(SS_B) = spreading speed

$$AR_{B_ndv} = \begin{cases} 1 & , AR_B \geq AR_{B_\max} \\ \dfrac{AR_B - AR_{B_\min}}{AR_{B_\max} - AR_{B_\min}} & , AR_B \in [AR_{B_\min}, AR_{B_\max}] \\ 0 & , AR_B \leq AR_{B_\min} \end{cases}$$

$$R_{ndv} = \begin{cases} 1 & , R \geq R_{\max} \\ \dfrac{R - R_{\min}}{R_{\max} - R_{\min}} & , R \in [R_{\min}, R_{\max}] \\ 0 & , R \leq R_{\min} \end{cases}$$

$$SS_{B_ndv} = \begin{cases} 1 & , SS_B \geq SS_{B_\max} \\ \dfrac{SS_B - SS_{B_\min}}{SS_{B_\max} - SS_{B_\min}} & , SS_B \in [SS_{B_\min}, SS_{B_\max}] \\ 0 & , SS_B \leq SS_{B_\min} \end{cases}$$

$AR_{B_\max}$, $AR_{B_\min}$, $R_{\max}$, $R_{\min}$, $SS_{B_\max}$ and $SS_{B_\min}$ are the maximum and the minimum grading values for each index for summed specimens tested from a sample.

*C_1, C_2, and C_3 can be adjusted according to the relative importance of the three indices based on the type of fabric and the end use of the product. The weighting values used in the development of the MMT software were C_1 = 0.25, C_2 = 0.5 and C_3 = 0.25, based on human perception studies, in which the one-way transport capability was found to be twice as important as absorption rate and spreading speed.

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Test Method for Colorfastness to Sodium Hypochlorite of a Textile Floor Covering

1. Purpose and Scope

1.1 This test method is used to determine the fastness of colored pile yarn floor coverings to the effects of a sodium hypochlorite solution, commonly called chlorine bleach. It is applicable to predyed, postdyed, printed, or other otherwise colored pile yarn floor coverings.

2. Principle

2.1 A specimen of pile yarn floor covering is treated with a small volume of sodium hypochlorite bleach. After allowing the treated specimen to remain at controlled conditions for a specific time, it is rinsed in water to remove all of the bleach. Any color loss is assessed after drying.

3. Terminology

3.1 **color change,** n.—a change in color of any kind whether in lightness, hue or chroma or any combination of these, discernible by comparing the test specimen with a corresponding untested specimen.

3.2 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of material.

3.3 **sodium hypochlorite bleach,** n.—4-6% solution of sodium hypochlorite (NaOCl), pH 9.8-12.8, commonly called "chlorine bleach."

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all-inclusive. It is the user's responsibility to use safe and proper techniques in the handling of materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Caution should be used when cutting specimens. All chemicals should be handled with care.

4.2 In preparing, dispensing and handling aqueous chemical solutions, use chemical goggles or face shield, impervious gloves and an impervious apron.

4.7 An eyewash/safety shower should be located nearby and an organic vapor respirator should be available for emergency use.

4.8 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; (see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienist (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 12.1).

5. Apparatus and Materials

5.1 AATCC Stain Cup and 50 mm (2.0 in.) Diameter Staining Ring (see 12.2).

5.2 Gray Scale for Color Change (AATCC EP1) (see 12.2).

5.3 Water, deionized.

5.4 Sodium Bisulfite ($NaHSO_3$) (see 6.2).

5.5 Balance, capable of weighing to the nearest 0.01 g.

5.6 Sodium Hypochlorite Bleach Standards, (NaOCl).

5.7 Illumination conditions. The source under which the comparison is made shall be determined prior to evaluation. Sources simulating any of the following CIE illuminants may be utilized (see 12.3 and 12.4):

Color Illuminant	Description	Temperature
D_{65}	Daylight	6500K ± 200
D_{75}	Daylight	7500K ± 200
A	Incandescent	2856K ± 200
CWF	Cool White Fluorescent	4150K ± 350

Note: Additional illuminants or changes in color temperature can be negotiated between contractual parties.

6. Preparation of Standard Sodium Hypochlorite Bleach Solution

6.1 Sodium Hypochlorite Bleach standards with traceable certificates shall be used without titration. The solution must be used within 36 h after exposure to atmospheric conditions. Titration is necessary if the purchased standard is diluted to another strength.

6.2 The Sodium Bisulfite Solution used to neutralize the treated specimens should be prepared by diluting 17.25 g in 1000 mL of deionized water.

7. Test Specimens

7.1 Cut two test specimens from the finished pile yarn floor covering. The specimens shall be a minimum of 150 mm (6 in.) square.

7.2 Remove any extraneous surface materials by brushing and/or vacuuming.

8. Test Procedure

8.1 Condition all test specimens, pile side up, on a non-absorbent surface for 24 h at standard conditions of 21 ± 2°C (70 ± 4°F) and 65 ± 5% RH. Avoid contamination or contact with extraneous materials.

8.2 Place the 50 mm (2 in.) diameter-staining ring in the center of one of the test specimens. The other specimen is a control used for rating. While pressing down on the ring, pour 20 mL of the sodium hypochlorite bleach solution into the center of the ring, taking care not to splatter bleach outside of the staining ring. Press the bottom of the stain cup through the ring into the pile, moving it up and down five times to facilitate wetting of the pile fiber from top to bottom. Do not rotate or twist the cup inside the ring as surface abrasion shall affect the results. Carefully remove the staining ring and stain cup.

8.3 Recondition the treated specimen for 24 ± 4 h in a flat position, pile side up, at standard conditions of 21 ± 2°C (70 ± 4°F) and 65 ± 5% RH. Avoid any air draft which could result in an accelerated drying.

8.4 Neutralize the bleached specimens in a bath containing the prepared Sodium Bisulfite solution for a minimum of 5 mins.

8.5 Rinse the bleached specimen with running water at 21 ± 6°C (70 ± 10°F) until rinse water indicates removal of all unfixed dye. Removal of unfixed dye is enhanced by the use of a laboratory wringer.

8.6 Extract (centrifuge or vacuum) the wet specimen to remove excess water.

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8.7 Allow specimens to air dry at room temperature in conditioning racks.

9. Evaluation or Results

9.1 Rate the color loss of the test specimen by using AATCC EP1, Evaluation Procedure for Gray Scale for Color Change, where a grade of 5 represents no color change and 1 represents a severe color change.

9.2 Illuminate the surface with an illumination level at the specimen plane in the range of 1080-1340 lx (100-125 fc) (see 12.5). The light should be incident upon the surfaces at 45 ± 5° and the direction of viewing at 90 ± 5° to the place of the surfaces. Compare the perceived visual differences between the original and tested carpet specimens with the perceived differences represented by the Gray Scale. The colorfastness grade is that number of the Gray Scale pair whose contrast corresponds to the contrast between the perceived difference in color or contrast between the original material and tested specimen. The cleanliness and physical condition of the Gray Scale is extremely important in obtaining consistent results.

9.3 Only observers whose color vision has been tested normal using an appropriate color vision test for color acurity shall perform this procedure.

10. Report

10.1 Report the numerical color change grade for each specimen tested.

10.2 Report the sodium hypochlorite activity and the amount of the solution used to treat the specimen.

10.3 Report any deviation to this test method.

11. Precision and Bias

11.1 *Precision*.

11.1.1 An interlaboratory study involving five laboratories and three tufted carpets was conducted in 2009 using the procedure described in this test method. A total of three replicates of each material were tested on different days by each of the five laboratories. The interlaboratory precision study is characterized by repeatability (Sr and r), reproducibility (SR and R), and the average results shown in Table I.

Table I—Interlaboratory Precision Analysis for Colorfastness to Bleach[1]

Materials Used	Average	Sr	SR	r	R
A-Tufted Nylon Carpet with Bleach Resistance	3.03333	0.54772	0.62138	1.53362	1.73986
B-Tufted Nylon Carpet Acid Dyed	1.63333	0.28868	0.46248	0.80829	1.29495
C-Solution Dyed Carpet	4.83333	0.12910	0.25820	0.36148	0.72296

[1]Laboratories 5; Materials 3; Determinations 3.

11.2 *Bias*.

11.2.1 Colorfastness to Sodium Hypochlorite Bleach can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

12. Notes

12.1 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: +1. 513.742.2020; web site: www.acgih.org.

12.2 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

12.3 For visual assessments to be made, it is recommended that the correlated color temperature be calibrated to 6500 ± 200 K by a service traceable to a national standard.

12.4 Filters and lamps shall be properly maintained and cleaned periodically following manufacturer's recommendations.

12.5 The illumination range of 1080-1340 lx (100-125 fc) is recommended for critical visual assessments of materials of medium lightness and is the same range specified for critical visual assessment in ASTM D1729, Standard Practice for Visual Appraisal of Colors and Color Differences of Diffusely Illuminated Opaque Materials.

13. History

13.1 Reaffirmed 2021.

13.2 Editorially revised 2020 to add Table I which was previously omitted and to align History section to AATCC style guidelines; editorially revised 2019, 2016; reaffirmed, 2013, 2012.

13.3 Developed in 2011 by AATCC Committee RA57.

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Test Method for Vertical Wicking Rate of Textiles: to Specified Distances

Foreword

Various versions of the vertical wicking test have been used to evaluate textiles for many years. In 2004, AATCC and ASTM International jointly published *Technical Supplement*, a collection of unofficial industry procedures. This test method is a more thorough and standardized version of the vertical wicking procedure published in the *Technical Supplement*. When it was first published, this test method included options for measuring wicking rate to specified distances and to specified times. In 2022, the rate to specified times option was transferred to a separate test method, AATCC TM213.

1. Purpose and Scope

1.1 This test method is used to evaluate the ability of fabric specimens to transport liquid vertically when a cut edge is submerged.

1.2 This method is applicable to woven, knitted, or nonwoven fabrics.

2. Principle

2.1 The distance liquid travels along and/or through a fabric specimen from a cut edge is visually observed, manually timed and recorded at specified intervals.

2.2 The movement of liquid along a fabric may be influenced by fiber content, fabric construction, mechanical or chemical processing or a combination of these.

2.3 Results are reported as wicking rates (distance per unit of time) for a short period (to 20 mm) and a long period (to 150 mm).

3. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

3.1 AATCC LP1, Laboratory Procedure for Home Laundering: Machine Washing (see 15.1).

3.2 AATCC TM213, Test Method for Vertical Wicking Rate of Textiles: to Specified Times (see 15.1).

3.3 ASTM D123, Standard Terminology Relating to Textiles.

3.4 ASTM D1331, Standard Test Methods for Surface and Interfacial Tension of Solutions of Paints, Solvents, Solutions of Surface-Active Agents, and Related Materials.

3.5 ASTM E1402, Standard Guide for Sampling Design.

3.6 ASTM D1776, Standard Practice for Conditioning and Testing Textiles.

4. Terminology

4.1 **absorption**, n.—a process in which one material (the absorbent) takes in another (the absorbate); as the absorption of moisture by fibers. (ASTM D123)

4.2 **capillary action**, n.—the process by which a liquid flows through narrow spaces via forces of adhesion, cohesion, and surface tension, and without the use of external forces, such as gravity.

4.3 **fabric,** n.—*in textiles*, a planar structure made from yarns or fibers (ASTM D123).

4.4 **vertical wicking,** n.—in a textile held vertically, the upward movement of liquid via absorption or capillary action.

4.5 **vertical wicking rate,** n.—the speed at which liquid travels along or through a textile

4.6 **wicking,** n.—*in textiles*, the movement of a liquid, by capillary action, along or through a material.

4.7 **wicking distance,** n.—the linear measurement that the liquid travels along or through a textile from a starting point to a stopping point.

4.8 **wicking time,** n.—the measurable period during which liquid travels along or through a textile.

5. Safety Precautions

5.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

5.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

5.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

6. Limitations

6.1 Liquids other than distilled or deionized water (tinted water, dye solutions, etc.) may be used in this test. Solutions of different surface tensions may yield different results.

6.2 The procedure measures movement of a liquid vertically upward from the submerged cut edge of a specimen, but does not duplicate an end-product's exposure during wear.

6.3 The results obtained by this test are not a measure of comfort, which is beyond the scope of this method.

6.4 The relationship between results obtained by vertical and horizontal wicking tests is not known.

7. Apparatus, Reagents, Materials (see 16.2)

7.1 Distilled or deionized water, 21 ± 2°C (70 ± 4°F).

7.2 Marking pens, fine-point, water-soluble ink (see 16.3).

7.3 Stopwatch or digital timer.

7.4 Measuring tape or ruler, mm graduations.

7.5 Surface tensiometer, if using liquid other than water (see 11.1).

7.6 Erlenmeyer flask, 500 mL; or rectangular pan, long enough to accommodate all specimens side-by-side.

7.7 Scissor jack, if using Erlenmeyer flask.

7.8 Means of suspending specimen(s) into a flask or pan (see Figs. 2 and 3).

7.9 Small straight pins (optional, see 11.2).

7.10 Powder-free disposable gloves, e.g., latex or nitrile.

8. Sampling

8.1 Test results are valid only when the samples are statistically representative (see ASTM E1402).

8.2 Sampling must be random. Every unit of product must have a mathematically equal chance of becoming a sample; and every portion of each sample must be equally likely to become a test specimen.

8.3 All specimens must be alike within the variations due to pure chance. There must be no differences within samples assignable to known causes.

9. Specimens

9.1 Liquid may wick up face and back of fabric at different rates. Upon agreement among interested parties, determine whether both fabric sides are to be measured. If only one fabric side is to be measured, mark the selected side. If testing is to be performed after laundering, use a marking pen with permanent ink to denote the fabric side chosen for testing.

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9.2 Upon agreement between interested parties, determine in which direction(s) the fabric will be tested.

9.3 To test the durability of finishes or evaluate wicking properties after laundering, launder fabric swatches or items according to conditions chosen from AATCC LP1 before cutting specimens.

9.4 Wear gloves and minimize handling of the test specimens as oils transferred from the skin may affect liquid movement.

9.5 Cut specimens at least 100 mm from the selvage. If testing sewn products, take specimens from different panels, at least 100 mm away from seams, pockets, plackets and other assembly features. Take each specimen such that different sets of (fabric) length and width yarns are present. For nonwoven fabrics, spread specimens across length and width of sample.

9.6 Cut 3 specimens, each 25 ± 3 mm wide and at least 165 mm long.

9.6.1 Cut the long dimension parallel to the fabric direction(s) being tested. Accurate dimensions and alignment of specimens are critical for repeatable testing.

9.7 Cut an extra specimen to determine if a pin is needed to submerge the specimen end (see 11.2).

9.8 Using a marking pen with soluble ink, mark a line across the width of each specimen, 5 ± 1 mm from the end to be submerged, on the fabric side to be tested. The line denotes the level to which the specimen will be submerged at the start of the test (see Fig. 1).

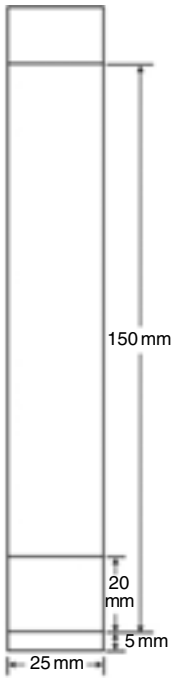


Fig. 1—Vertical wicking specimen and marking.



Fig. 2—Specimen prepared for vertical wicking testing in an Erlenmeyer flask.



Fig. 3—Rectangular pan for testing multiple specimens.

9.9 Using a marking pen with soluble ink, mark lines across the width of each specimen, 20 ± 1 mm and 150 ± 1 mm above the first line (see Fig. 1).

9.9.1 Alternative wicking distances may be marked depending on the desired end use of the fabric. When comparing results, use the same wicking distance for all specimens being compared.

10. Conditioning

10.1 Prior to testing, condition test specimens as directed in ASTM D1776. (Use conditions indicated in Table 1 for Textiles, general. Estimate conditioning time as indicated in Table 2 for the appropriate fiber content.) Lay each specimen, separately, on a screen or perforated shelf of a conditioning/drying rack.10.2 Perform all tests in the standard atmosphere for testing (21 ± 2°C [70 ± 4°F], 65 ± 5% RH).

11. Preparation

11.1. For liquids other than distilled or deionized water (including tinted water or dye solution), measure the surface tension as directed in ASTM D1331.

11.2 Some lightweight or hydrophobic fabrics may float on the water. Before testing, lower one short end of the extra specimen into water.

11.2.1 If the specimen is easily submerged, proceed with marking remaining specimens.

11.2.2 If the end of the specimen floats on the surface of the water, insert a straight pin across the end. Use the minimum weight necessary to allow the specimen to submerge. Use the same size pin for all specimens in the sample. When comparing results for various samples, it is best to use the same size pin for all.

11.2.3 Do not reuse the water or sample. Use a clean container and fresh water for each test.

12. Procedure

12.1 Perform test on a stable surface to minimize vibration.

12.2 To test a single specimen in an Erlenmeyer flask, add approximately 90 mL of water (or alternate liquid) to the flask. Ensure water depth is at least 10 mm. Ensure that the lip and neck of the flask remain dry to prevent specimen from premature contact with water.

12.2.1 Place the flask on a scissor jack and suspend the specimen in the opening, above the water line. The specimen should hang perpendicular to the flask floor, with no folds or waves. Insert a pin across the bottom end as needed (see 11.2).

12.2.2 Raise the jack until the specimen end is submerged to the 5-mm line and start the timer.

12.3 To test multiple specimens in a rectangular pan, suspend specimens with bottom ends just above the floor of the pan. Ensure that no part of any specimens is in contact with the pan. The specimens should hang perpendicular to the pan floor, with no folds or waves. Insert a pin as needed (see 11.2). It is critical to ensure the 5-mm line on each specimen is at the same level before adding water.

12.3.1 Carefully add water to the pan. Avoid pouring or splashing water onto specimens. Start a timer as soon as the specimens are submerged to the 5-mm

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line.

12.4 Monitor the rise of the water, visible as migration of the marking ink and/or darkening of the specimen(s). Record the time, to the nearest second, that it takes for the soluble ink at the 20-mm line to start to migrate. This is the short-period wicking time. Multiple observers or timers may be necessary to accurately record the wicking time of multiple specimens tested simultaneously in a rectangular pan. If wicking is uneven across the width of the specimen, record the highest point and note observation in the test report.

12.5 Continue monitoring the rise of the water and record, to the nearest second, the time that it takes for the soluble ink at the 150-mm line to start to migrate. This is the long-period wicking time.

12.6 Terminate the test when any of the following conditions are met:

12.6.1 Water does not reach the 20-mm line within 300 s (5 min.)

12.6.2 Water does not reach the 150-mm line within 1,800 s (30.0 min.)

12.6.3 Water wicks to the 150-mm line.

12.7 Measure the distance the water has migrated. Record the time (s), distance (mm) and the reason for termination of the test.

12.8 For tests in a flask, remove the specimen from the flask. Repeat the procedure for the remaining specimens.

12.8.1 Use a clean flask with fresh water for each specimen.

13. Calculation

13.1 Calculate the vertical wicking rates using Eq. 1. Two different rates, a short period rate and a long period rate, are obtained for each specimen.

$$W = d/t \quad (1)$$

where:

W = vertical wicking rate, mm/s
d = wicking distance, mm (20 for short period, 150 for long period, or as agreed)
t = wicking time, s

NOTE: The standard wicking distance is 20 mm for the short period and 150 mm for the long period. Alternate distances may be used if the test was terminated before reaching the marks or as agreed by interested parties.

13.2 Calculate the mean short-period wicking rate for each direction tested, to the nearest 0.1 mm/s.

13.2.1 If desired, calculate standard deviation.

13.3 Calculate the mean long-period wicking rate for each direction tested, to the nearest 0.1 mm/s.

13.3.1 If desired, calculate standard deviation.

14. Report

14.1 Describe or identify the sample tested.

14.2 Report that the sample was tested using AATCC TM197-2022.

14.3 Report the testing conditions:

14.3.1 If testing was performed after laundering, report the laundering conditions used and the number of cycles completed.

14.3.2 Report the fabric side(s) and direction(s) tested.

14.3.3 Report the liquid used, and surface tension if other than water.

14.3.4 Report the weight of pin, if used to submerge specimen end.

14.3.5 Report the short- and long-period wicking distances.

14.3.6 Report the time at which test was terminated, if applicable (see 12.6)

14.4 Report the test results:

14.4.1 Report the mean short-period wicking rate for each direction tested.

14.4.2 Report the individual specimen rates or standard deviation for the short period, in each direction tested.

14.4.3 Report the mean long-period wicking rate for each direction tested.

14.4.4 Report the individual specimen rates or standard deviation for the long period, in each direction tested.

14.5 Describe any modification(s) of the published method.

15. Precision and Bias

15.1 *Precision*.

15.1.1 Tests for vertical wicking of textiles were conducted in 2009, with 1 laboratory, 3 operators, and 5 samples. The samples used in the study were (a) 100% cotton jersey knit, (b) 100% cotton interlock knit, (c) 100% polyester woven, (d) 100% cotton twill weave, and (e) 50/50 cotton/polyester blend woven.

15.1.2 Table I shows the means and confidence intervals for each fabric in the length direction. Table II shows the means and confidence intervals for each fabric in the width direction.

15.1.3 *Between-laboratory* precision has not been established for this test method. Until such precision information is available, users of the method should use standard statistical techniques in making any comparison of test results for *between-laboratory* averages.

15.2 *Bias*. The true value of vertical wicking rates of textile fabrics can be defined only in terms of a test method. There is no independent method for determining the true value. In estimating this property, the test method has no known bias.

16. Notes

16.1 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

16.2 For potential apparatus, reagents or materials sources, visit the AATCC Buyer's Guide at www.aatcc.org/products/bg. AATCC provides its Corporate members the option to list their items and services. AATCC does not qualify, or in any way approve, endorse or certify that any of the listings meet the specifications in its standards.

16.3 Dark fabric colors or prints and designs may be difficult to test. Using soluble marking ink of a contrasting color may help make the marks more clearly visible. Fluorescent ink or wicking liquid and observation under ultraviolet light (black light), flashlight or use of thermal Infra-Red meters may also improve visibility.

17. History

17.1 Revised in 2022 (with title change) in 2022 to remove wicking rate to specified times (now TM213) and define other parameters. Jurisdiction transferred to RA114.

17.2 Editorially revised 2019. Reaffirmed 2018. Editorially revised 2016. Reaffirmed 2013. Editorially revised and reaffirmed 2012.

17.3 Developed in 2011 by AATCC Committee RA63.

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Table I—Summary of Statistics—Length Direction (rate in mm/s)

	Jersey	Interlock	Polyester Woven	Cotton Woven	Poly/Cotton Woven
RECTANGULAR PAN METHOD	**SHORT PERIOD**				
Mean	1.6	0.2	2.6	1.4	0.3
Standard Deviation	0.5	0.1	0.9	0.4	0.1
Count	18	18	18	18	18
Confidence Level (95.0%)	0.3	0.0	0.4	0.2	0.1
FLASK METHOD	**SHORT PERIOD**				
Mean	1.4	0.4	2.6	1.3	0.3
Standard Deviation	0.4	0.1	0.7	0.6	0.0
Count	18	18	18	18	18
Confidence Level (95.0%)	0.2	0.1	0.4	0.3	0.0
RECTANGULAR PAN METHOD	**LONG PERIOD**				
Mean	0.1	0.0	0.1	0.1	0.1
Standard Deviation	0.0	0.0	0.0	0.0	0.0
Count	18	18	18	18	18
Confidence Level (95.0%)	0.0	0.0	0.0	0.0	0.0
FLASK METHOD	**LONG PERIOD**				
Mean	0.1	0.0	0.1	0.1	0.1
Standard Deviation	0.0	0.0	0.0	0.0	0.0
Count	18	18	18	18	18
Confidence Level (95.0%)	0.0	0.0	0.0	0.0	0.0

Table II—Summary of Statistics—Width Direction (rate in mm/s)

	Jersey	Interlock	Polyester Woven	Cotton Woven	Poly/Cotton Woven
RECTANGULAR PAN METHOD	**SHORT PERIOD**				
Mean	1.9	0.9	2.3	1.0	0.3
Standard Deviation	0.7	0.2	0.7	0.3	0.1
Count	18	18	18	18	18
Confidence Level (95.0%)	0.3	0.1	0.4	0.1	0.1
FLASK METHOD	**SHORT PERIOD**				
Mean	1.8	1.0	2.2	1.2	0.2
Standard Deviation	0.6	0.4	0.7	0.7	0.1
Count	18	18	18	18	18
Confidence Level (95.0%)	0.3	0.2	0.4	0.3	0.1
RECTANGULAR PAN METHOD	**LONG PERIOD**				
Mean	0.1	0.1	0.1	0.1	0.0
Standard Deviation	0.0	0.0	0.0	0.0	0.0
Count	18	18	18	18	18
Confidence Level(95.0%)	0.0	0.0	0.0	0.0	0.0
FLASK METHOD	**LONG PERIOD**				
Mean	0.1	0.1	0.1	0.1	0.0
Standard Deviation	0.0	0.0	0.0	0.0	0.0
Count	18	18	18	18	18
Confidence Level (95.0%)	0.0	0.0	0.0	0.0	0.0

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Test Method for Horizontal Wicking of Textiles

Foreword

Historically, the textile industry has utilized many different test procedures for determining the wicking characteristics of textile fabrics, i.e., the movement of water or liquid through fabrics. Within the last decade, the industry has developed new technologies that have changed water movement and absorbency responses of textiles leading to the use of the labeling term "moisture management" to describe these phenomena. While, traditionally, wicking tests measured the movement of liquids through the vertical plane of fabrics (commonly referred to as vertical wicking), (see AATCC TM197, Test Method for Vertical Wicking of Textiles), the improvements in the last decade have also led to an interest in measuring the movement of liquids across the horizontal plane of fabric surfaces (horizontal wicking). Many interested groups (textile manufacturers, chemical suppliers, and retailers, as well as independent testing laboratories) participated in the efforts to standardize test methods to measure the vertical and horizontal wicking properties of fabrics with moisture management attributes.

Unofficial techniques for determining the absorbency and wicking properties of textile fabrics were published in the 2004 AATCC/ASTM International's *Technical Supplement: A Compilation of Procedures and Guidelines for Textile Products* and the 2008 AATCC/ASTM International's *Moisture Management Technical Supplement: Applicable to Apparel, Linens and Soft Goods* (see 14.1). This test method is based on the testing procedures for wicking published in those Technical Supplements.

1. Purpose and Scope

1.1 This test method is used to evaluate the ability of horizontally aligned fabric specimens to transport liquid along and/or through them, and is applicable to woven, knitted, or nonwoven fabrics.

2. Principle

2.1 The rate (area per unit of time) liquid travels through a fabric specimen is visually observed, manually timed and recorded at specified intervals.

3. Terminology

3.1 **fabric,** n.—*in textiles*, a planar structure made from yarns or fibers (ASTM D123 and see 14.3).

3.2 **horizontal wicking,** n.—*in textiles*, the movement of liquid across the horizontal plane of a textile from a site where a specific volume has been applied.

3.3 **horizontal wicking rate,** n.—the change in the area of the liquid with respect to time as the liquid travels through a textile.

3.4 **wicking,** n.—*in textiles*, the movement of a liquid, by capillary action, along or through a material.

3.5 **wicking distance,** n.—the linear measurement that the liquid travels along or through a textile from a starting point to a stopping point.

3.6 **wicking time,** n.—the measurable period during which liquid travels along or through a textile.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5. Uses and Limitations

5.1 This method is primarily applicable to fabrics that absorb the full volume of the test liquid, without pooling on the surface or allowing dripping. If the full volume of the test liquid is not available to measure the wicking rate, inconsistent results should be expected. Hence, the results obtained on fabrics that fully absorb the test liquid cannot be compared with those obtained on fabrics that do not. In addition, the movement of liquid within a fabric may be influenced by fiber content, fabric construction, mechanical or chemical processing or a combination of these.

5.2 This method evaluates the wicking ability of horizontally aligned test specimens to distilled or deionized water. In this test, wicking may be influenced by the properties of the test liquid, but not by gravity.

5.3 Liquids other than distilled or deionized water (tinted water, dye solutions, etc.) may be used to evaluate the wicking of textiles. If a liquid other than water is chosen, its surface tension should be measured and reported as solutions of different surface tensions may yield different results.

5.4 Dark fabric colors or prints and designs may be difficult to test. The use of a soluble ink of a contrasting color may help in making the liquid more clearly visible.

5.5 The results obtained by this test are not a measure of comfort which is beyond the scope of this method.

5.6 The relationship between vertical and horizontal wicking test results is not known. Caution should be used when comparing results using different alignments.

6. Apparatus, Reagents, Materials

6.1 Distilled or deionized water at 21 ± 2°C (70 ± 4°F).

6.2 Marking pens, with fine point, permanent and soluble ink (see 14.2).

6.3 Stopwatch or digital timer.

6.4 Tape or ruler, mm graduations.

6.5 Surface tensiometer, if using liquid other than water (see 10.1).

6.6 Beaker or Petri dish.

6.7 Burette, 10 (± 0.05 mL) mL with 0.5 mL numbered graduations that allows a delivery rate of 15-25 drops per milliliter, or handheld electronic pipette.

6.8 Embroidery hoop, 152 ± 5 mm (6.0 ± 0.2 in.) diameter (see 14.1).

6.9 Laboratory sealing film (optional).

6.10 Ring stand and clamp.

6.11 Template, 100 ± 3 mm diameter.

7. Sampling

7.1 Determine whether the fabric has a hydrophobic or hydrophilic side. Upon agreement between interested parties, determine whether both fabric sides are to be tested. If only one fabric side is to be tested, mark the selected side. If testing is to be performed after laundering, use a marking pen with permanent ink to denote the fabric side chosen for testing.

7.2 All specimens should be cut at least 100 ± 5 mm from the selvage. If testing garments, take specimens from different garment panels away from seams, pockets, plackets or other assembly features. Each specimen shall be taken so that different sets of length and width yarns are present.

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8. Specimens

8.1 Cut five specimens, 200 × 200 ± 5 mm, aligned with the fabric selvage or longitudinal alignment of the garment panels. For testing horizontal wicking of garment samples, it may not be necessary to cut out individual specimens.

8.2 Cut an extra specimen from either fabric or garment samples to determine placement of the drop in the center of the marked circle before starting the testing (see 10.2).

8.3 If testing the durability of finishes or evaluating wicking properties after laundering, take specimens as described in 8.1 from fabric swatches that have been laundered according to conditions chosen from the AATCC LP1.

9. Conditioning

9.1 Condition specimens for at least 4 h in an atmosphere of 21 ± 2°C (70 ± 4°F), 65 ± 5% RH prior to testing according to ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 14.3).

9.2 Perform all tests in the standard atmosphere for testing.

10. Procedure

10.1 If necessary, the surface tension of the chosen liquid (including tinted water or dye solution) should be measured as directed in ASTM D1331-89, Standard Test Methods for Surface and Interfacial Tensions of Solutions of Surface-Active Agents (see 14.3) and reported.

10.2 Use a template and a fine-point marking pen with soluble ink to mark a 100 ± 3 mm diameter circle in the middle of each specimen. Mark the circle on the side of the fabric to be tested. The placement of the water volume for the test is to be in the center of the circle (see Fig. 1).

10.3 Fill a 10.0 mL burette with distilled or deionized water and lower the water level to the top graduation level, sighting to the bottom of the meniscus.

10.3.1 Insert a trial specimen into an embroidery hoop, ensuring that its surface is taut, without creases or wrinkles. Place the embroidery hoop horizontally on the top of the beaker or Petri dish. Lower the burette so that the tip is 10 ± 1 mm above the fabric surface. Slowly open the burette stopcock to verify that 1.0 ± 0.1 mL volume of water can be dispensed into the center of the marked circle. Determine (and record) the approximate position of the stopcock opening that ensures a consistent delivery of the test liquid, i.e., the position that permits delivery of the specified number of drops of the test liquid in 10 ± 2 s. Discard the trial specimen.

10.4 Insert test specimen into the embroidery hoop with the chosen fabric side to be tested facing up for exposure to the water. The use of laboratory sealing film on the embroidery hoop ring surfaces may be utilized to prevent slippage of light weight fabric and help maintain a taut fabric surface.

10.5 Open the stopcock, to the position determined in 10.3.1, and dispense 1.0 ± 0.1 mL of water from the 10 ± 1 mm height onto the center of the 100 ± 3 mm marked circle. A handheld electronic pipette may be used instead of a burette but the same height of 10 ± 1 mm from the fabric surface should be maintained. Immediately after opening the stopcock or pressing the dispense button, start the timer. The test should be terminated when the water reaches the 100 ± 3 mm circle. Record the distance the liquid has spread in the length and width direction of the fabric and the time elapsed.

10.6 The test should be terminated if the water stops migrating and does not reach the 100 ± 3 mm circle in 5 ± 0.1 min. Record the time that the test was terminated.

10.7 Repeat steps 10.4-10.6 for each remaining specimen.

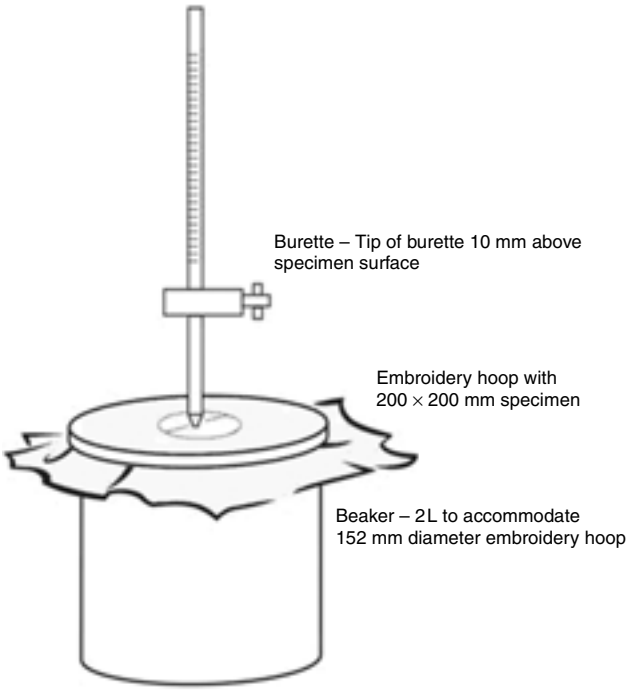


Fig. 1—Horizontal alignment test configuration.

11. Calculation

11.1 Record the time to the nearest second for each specimen.

11.2 Calculate the horizontal wicking rate for each specimen and the average wicking rate for each sample. Use formula 1 below for the calculation of horizontal wicking rate:

$$W = \pi(1/4)(d_1)(d_2)/t \qquad \text{(Eq. 1)}$$

where:

W = wicking rate, mm^2/s
d_1 = wicking distance in length direction, mm and d_2 = wicking distance in width direction, mm
t = wicking time, s

11.3 The wicking rate can be calculated from either the time it takes to reach the edge of the circle or the distance traveled in the 5.0 ± 0.1 min time period.

12. Report and Interpretation

12.1 Report the wicking time, the wicking distance in the length and width direction, and the average wicking rate for each sample. Report any dripping or pooling of test liquid.

12.2 Report if a solution other than distilled or deionized water at 21 ± 2°C (70 ± 4°F) was used and its surface tension and temperature.

12.3 Report if testing was performed after laundering, the laundering conditions used and the number of launderings completed.

12.4 Shorter horizontal wicking times indicate faster wicking (see 5.1 and 5.5).

13. Precision and Bias

13.1 *Precision*.

13.1.1 *Interlaboratory Study*. Tests for wicking of textiles using horizontal orientations were conducted in 2009, with one laboratory, three operators, and five

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samples. The five samples used in the study were: (a) 100% cotton jersey knit, (b) 100% cotton interlock knit, (c) 100% cotton twill weave, (d) 50/50 cotton/polyester blend woven, and (e) 100% polyester woven. However, two of the woven samples (d and e) were found to be not suitable for the measurement of horizontal wicking, as water pooled on the surface and did not spread during the test. Therefore, the single lab precision statement shown is based on only three fabrics: two knits and one woven.

13.1.2 The data is shown in Table I. The analysis of variance technique was applied to the data set. Because the ANOVA indicated that the effect due to the fabric was significant, the results, shown in Table II, for each of the three fabrics are separate. The analysis is being retained for reference at the AATCC Technical Center.

13.1.3 *Between-laboratory* precision has not been established for this test method. Until such precision information is available, users of the method should use standard statistical techniques in making any comparison of test results for *between-laboratory* averages.

13.2 *Bias*. The true value of horizontal wicking rates of textile fabrics can be defined only in terms of a test method. There is no independent method for determining the true value. In estimating this property, the test method has no known bias.

Table I—Data (rate in mm²/s)[1]

	Operator 1 Day 1	Operator 1 Day 2	Operator 2 Day 1	Operator 2 Day 2	Operator 3 Day 1	Operator 3 Day 2
100% Cotton Jersey	40	60	37	39	44	68
	34	57	84	70	56	62
	41	32	51	60	59	61
	64	47	68	48	41	60
	60	58	55	30	62	46
100% Cotton Interlock	29	25	24	40	23	22
	28	22	25	16	19	19
	27	17	35	29	19	22
	16	19	31	20	21	19
	37	14	18	18	27	20
100% Cotton Woven	120	49	33	120	43	31
	47	90	130	110	46	140
	47	45	130	170	38	35
	76	66	160	220	44	29
	98	51	43	71	27	43

[1]All values in this table are expressed with exactly two significant figures.

Table II—Statistical Results (rate in mm²/s)[1]

	100% Cotton Jersey	100% Cotton Interlock	100% Cotton Woven
Average	53	23	78
Std Dev	13	6.0	50
95% CI	5.0	2.0	18

[1]All values in this table are expressed with exactly two significant figures.

14. Notes

14.1 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

14.2 A suitable marking pen contains ink that is water soluble, available commercially such as Paper Mate® Flair® Fiber Tip Pen from Sanford Corporation, 2707 Butterfield Rd., Oak Brook IL 60523; tel: +1.630.481.2200; fax: +1.866.666.8735; web site: www.papermate.com.

14.3 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610. 832.9500; fax: +1.610.832.9555; web site: www.astm.org.

15. History

15.1 Jurisdiction transferred in 2022 to AATCC Committee RA114; Editorially revised to note the change.

15.2 Reaffirmed 2020

15.3 Reaffirmed 2018, 2013. Editorially revised and reaffirmed 2012. Editorially revised 2019, 2016.

15.4 Developed in 2011 by AATCC Committee RA63.

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Test Method for Drying Time of Textiles: Moisture Analyzer

Developed in 2011 by AATCC Committee RA63; jurisdiction transferred in 2022 to AATCC Committee RA114; reaffirmed 2018; revised 2012, 2013; editorially revised 2017, 2019, 2022.

Foreword

Historically, the textile industry has utilized many different test procedures for determining the drying characteristics of textile fabrics, whether those characteristics are drying rate, time or some other drying parameter. Within the last decade, the industry has developed new technologies that have resulted in an improvement in these characteristics, as one of many attributes associated with the use of the term "moisture management." Many interested groups (textile manufacturers, chemical suppliers, retailers, as well as independent testing laboratories) participated in the effort to develop and publish standard test methods to measure the drying characteristics of fabrics. Unofficial techniques for determining the drying properties of textile fabrics were published in the 2008 AATCC/ASTM International's *Moisture Management Technical Supplement: Applicable to Apparel, Linens and Soft Goods*. This test method formalizes an additional technique that has been used by several companies.

1. Purpose and Scope

1.1 This test method is intended to evaluate the drying time of knit, woven or nonwoven fabrics at an elevated temperature using a gravimetric moisture analyzer. By performing the test at non-standard textile testing conditions, it is possible to simulate drying at body temperature or to perform testing at temperatures that simulate conditions of use.

2. Principle

2.1 Water is applied to the test specimen and then dried at a pre-selected temperature (37° ± 2°C [99 ± 4°F]); if an alternative temperature is used it should be reported, in an automated moisture analyzer. The time required for a test specimen to reach a designated endpoint is measured and recorded as the drying time.

3. Terminology

3.1 **drying rate,** n.—the change in volume per unit time of a liquid that evaporates from a textile.

NOTE: Drying rate is dependent upon textile construction, fiber content, apparel construction, finishes, the atmosphere in which the test is performed, and the volume challenge of the liquid.

3.2 **drying time,** n.—the time it takes for a specified amount of liquid to evaporate from a textile under controlled testing conditions.

NOTE: In this test method the amount of water is determined in a preparatory step so that the specimen is wet-out to a given level. The testing conditions are controlled by the analyzer.

3.3 **endpoint,** n.—the point of termination of a drying test that is either to the original dry weight of a specimen or some other agreed value such as dry weight +4.0% moisture content.

3.4 **moisture retention,** adj.—*for this test method*, the percent moisture retention of the specimen after being submerged in deionized water for 1 min, and hanging vertically for 5 min in a controlled environment (see 9.3.2 and formula 1).

3.5 **test side,** n.—the side, either the face or back of a specimen, which is uppermost when placed on the moisture balance analyzer support platform; the side onto which the test volume of water is applied.

3.6 **weight loss,** n.—the difference between the saturated weight of a specimen and the weight after drying.

3.7 **wet pick-up,** n.—*in textile processing*, the amount of liquid, and material carried by the liquid, applied to a textile.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the users' responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturers' recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5. Uses and Limitations

5.1 The drying of a fabric may be influenced by fiber content, fabric construction (e.g., thickness or raised surface, such as fleece), mechanical or chemical processing or a combination of these aspects.

5.2 Other factors that may influence drying include the temperature and humidity conditions of the test and the amount of liquid to be dried. This test method is performed in non-standard textile testing conditions. Since this test can be performed at various temperature settings of the moisture analyzer, test temperatures can be selected to simulate the temperature of a human body at rest, during exercise, or when exposed to outdoor temperatures.

5.3 This method is based on the application of a controlled amount of water to test specimens, measurement of the drying rate of that application, and calculation of the moisture retention based on that application.

5.4 This method is limited to fabrics with at least one side that exhibits an absorbency time within 30 ± 2 s (see AATCC TM79, Test Method for Absorbency of Textiles). It is not applicable to fabrics that exhibit an absorbency of more than 30 s, as in such fabrics water would begin evaporating from the surface of the fabric as the temperature is being raised, thereby skewing the results.

5.5 One possible use of this method is to compare the moisture retention characteristics of untreated versus treated fabrics, or to compare the moisture retention characteristics of a textile material with or without an additive.

5.6 The results obtained by this test are not a measure of comfort which is beyond the scope of this method.

5.7 The relationship between drying time measured by this test and absorbency has not been defined. Although TM79 is used in the preparatory steps to determine which side of a fabric should be used in testing, this test does not measure absorbency.

6. Apparatus and Materials

6.1 Moisture analyzer with a ceramic heating element and accurate to at least 0.001 g (see 14.1).

6.2 Deionized water.

6.3 Vertical specimen stand. The stand should have a horizontal hanger of suitable size to hang the specimen using binder clips. (see Fig. 2).

6.4 Tweezers.

6.5 400 mL beaker.

6.6 Wire screen—standard 0.25 × 0.25 in. mesh (see 14.2).

6.7 Specimen support platform (see

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Fig. 3).

6.8 Electronic motorized pipette with 2.5 mL tip, and a dispensing speed of 4 mL/min (see 14.1).

6.9 Computer with software for data capture (optional).

7. Specimens

7.1 Cut ten 70 ± 1 mm round specimens diagonally across the width of a sample, to ensure that different sets of length and width yarns are in each specimen or from different sites from a product are taken. If the sample being tested is not large enough to cut ten 70 ± 1 mm diameter specimens across the width, it is acceptable to test a smaller specimen size, but must be reported with the test results. Two of the specimens are to be used for the preliminary steps, and eight are for the testing (see Section 9).

8. Conditioning

8.1 Place the specimens on a flat smooth, horizontal surface without tension before testing. Condition them to moisture equilibrium in a standard atmosphere for testing according to ASTM D1776, Standard Practice for Conditioning and Testing Textiles, Table I, Textiles, General (see 14.3).

9. Preparation

9.1 Turn on the moisture analyzer to 37 ± 2°C (99 ± 4°F) and allow it to warm up for at least 30 min.

9.2 Using TM79, place a drop of water on the face of an extra fabric specimen to judge absorbency; on the back of another extra specimen, repeat to determine which is the most absorbent side to use for testing. If there is no difference, either side can be used for testing by this method. If both sides of the specimen have absorbency times greater than 30 s, see 5.4.

9.3 Weigh and record the original weight (W_1) of the specimen. Submerge



Fig. 1—Submersion of specimen in water.

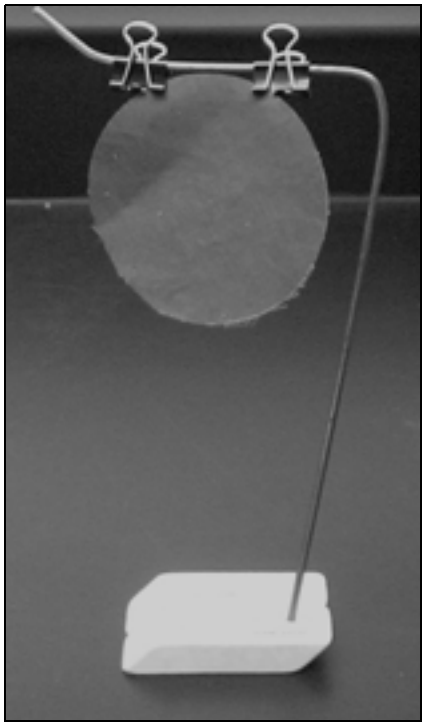


Fig. 2—Vertical specimen stand.

the specimen in water, as shown in Fig. 1.

9.3.1 Using tweezers, remove the specimen after 1 min and hang on the vertical stand, as shown in Fig. 2.

9.3.2 After 5 min, remove the specimen with tweezers. Weigh and record the specimen weight (W_2). Use formula 1 to determine the moisture retention.

$$\text{Moisture Retention (\%)} = \frac{W_2 - W_1}{W_1} * 100 \qquad \text{(Eq. 1)}$$

where:

W_1 = dry weight, in g
W_2 = saturated weight, in g

Use formula 2 to determine the amount of water to add to the fabric:

$$y = \frac{x * W_1}{100} \qquad \text{(Eq. 2)}$$

where:

y = amount of water to add, in mL
x = moisture retention (see result of formula 1)
W_1 = dry weight, in g

NOTE: This equation assumes the density of water is 1 g/mL at 25°C (77°F).

9.4 Set the electronic motorized pipette to pickup and dispense the calculated amount of water determined in 9.3. This amount will be used for all specimens from a sample without regard to variation in weight of individual specimens from the same sample.

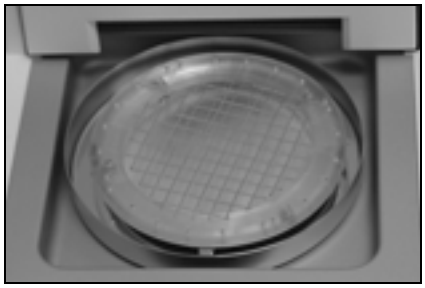


Fig. 3—Specimen support platform.

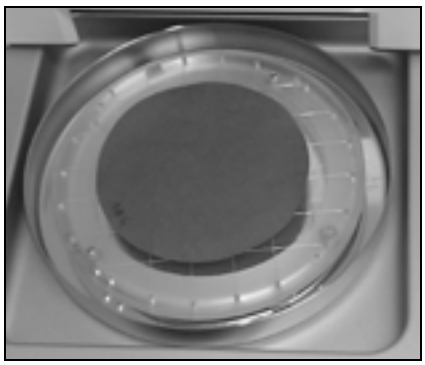


Fig. 4—Placement of specimen on support platform.

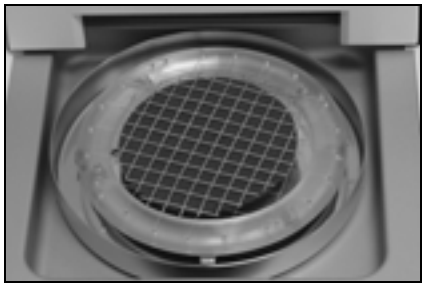


Fig. 5—Placement of wire screen on top of specimen.

9.5 Follow the manufacturer's instruction for setting the program operation. The test endpoint should be agreed upon by interested parties.

10. Procedure

10.1 Open the sample chamber and place the support platform and wire screen in the sample chamber (as shown in Fig. 3).

10.2 Tare the moisture analyzer. Place the specimen with the selected test side up on the support platform (see Fig. 4).

10.3 Using an electronic motorized pipette, apply the specified amount of deionized water in a uniform manner over the surface of the specimen.

10.4 Place the wire screen on top of the specimen (see Fig. 5) and start the ana-

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lyzer and software (if applicable). The moisture analyzer or the software will automatically terminate the test when the selected endpoint is reached.

10.5 Repeat 10.1-10.4 for remaining specimens.

10.6 Record the drying time to the nearest minute for each specimen.

11. Calculations and Evaluation

11.1 Calculate the average drying time and the standard deviation for the sample.

12. Report

12.1 Report moisture retention, average drying time and standard deviation, amount of water applied, temperature, and the end point.

12.2 If the specimens tested were not 70 ± 1 mm in diameter, report the specimen diameter used.

13. Precision and Bias

13.1 *Precision*. Tests for drying time of textiles using a moisture analyzer were conducted in 2010, with one laboratory, two operators, and five samples. The five samples used in this study were (a) 100% cotton interlock knit, (b) 100% polyester twill weave, (c) 100% cotton jersey knit, (d) 65/35 polyester/cotton blend woven, and (e) 100% polyester interlock knit.

13.2 *Table I Statistical Summary.* Data includes mean drying time, standard deviation and the 95% confidence level for both operators.

13.3 *Between-laboratory* precision has not been established for this test method. Until such precision information is available, users of the method should use standard statistical techniques in making any comparison of test results for *between-laboratory* averages.

13.4 *Bias*. The true value of the drying time of textile fabrics can be defined only in terms of a test method. There is no independent method for determining the true value. In estimating this property, the test method has no known bias.

Table I—Statistical Summary Data

	Operator 1			Operator 2		
Drying Time by Fabric	Mean Drying Time, min	Std Dev	95% CI	Mean Drying Time, min	Std Dev	95% CI
A	147	4.7	3	158	5.2	4
B	115	4.2	3	121	4.5	3
C	73	2.5	2	78	2.6	2
D	28	1.7	1	28	1.3	1
E	85	1.8	1	93	3.5	2

14. Notes

14.1 These items may be obtained from any lab equipment supplier.

14.2 This item may be obtained from any hardware or home improvement store.

14.3 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610. 832.9500; fax: +1.610.832.9555; web site: www.astm.org.

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Test Method for Drying Rate of Textiles at their Absorbent Capacity: Air Flow

Developed in 2012 by AATCC Committee RA63; jurisdiction transferred in 2022 to AATCC Committee RA114; reaffirmed 2013, 2014; revised 2015, 2017; editorially revised 2016, 2019, 2022.

Foreword

As moisture management products have entered the marketplace, measurement of the drying characteristics of textiles has gained additional importance. The 2008 AATCC-ASTM International *Moisture Management Technical Supplement: Applicable to Apparel, Linens, and Soft Goods* contains three techniques for the measurement of drying times of textiles. This test method is an additional approach to the drying test methods outlined in the *Technical Supplement*.

1. Purpose and Scope

1.1 This test method determines the drying rate of textiles at their absorbent capacity.

1.2 This method is not for testing the drying rate for textiles taken from socks or hosiery.

2. Principle

2.1 This method determines the drying rate of textiles based on the evaporation rate that occurs at their approximate absorbent capacity.

3. Terminology

3.1 **absorbent capacity,** n.—the maximum amount of liquid a material can hold; dependent on the specific test method used.

3.2 **drying rate,** n.—the change in volume per unit time a liquid evaporates from a textile.

NOTE: Drying rate is dependent upon textile construction, fiber content, apparel construction, finishes, the atmosphere in which the test is performed, and the volume challenge of the liquid.

3.3 **drying time,** n.—the time it takes for a specified amount of liquid to evaporate from a textile under controlled testing conditions.

NOTE: In this test method a series of different amounts of water are used to challenge the specimen. The testing conditions are controlled by the analyzer and room conditions.

3.4 **end time,** n.—the time at which the temperature goes through an inflection between the steepest slope and the flat sections of the temperature versus time plot (see Fig. 2 and 11.2).

3.5 **start time,** n.—the time at which the water challenge is added to the specimen.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as materials safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 The manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

5. Uses and Limitations

5.1 This test is performed under controlled laboratory temperature and relative humidity conditions.

5.2 In order to provide consistent results, consistent air flow is needed. The apparatus used in this method uses 2.5 ± 0.5 m/s air flow through a 13 ± 0.1 cm diameter opening with no specimen present. The results will vary if other rates of air flow are used. While the test may be performed at alternative conditions, for comparison of results, the same conditions must be used.

5.3 This method is limited to textiles—tested on the non-face side—that exhibit a maximum absorbency time of 30 s as measured by AATCC TM79, Test Method for Absorbency of Textiles (see 14.1). It is not applicable to textiles that exhibit absorbency of more than 30 s.

6. Apparatus and Materials

6.1 Temperature recorder, with capabilities to take readings every 1 s, data storage and transmittal to computer data file (see Fig. 1).

6.2 Infrared thermocouple probe—temperature range of 15.0-50.0 ± 0.1°C (59.0-122.0 ± 0.18°F) (see Fig. 1).

6.3 Fan box (16.5 ± 0.1 × 16.5 ± 0.1 × 27.5 ± 0.1 cm) which generates 2.5 ± 0.5 m/s of air flow through a 13.0 ± 0.1 cm diameter opening (see Fig. 1).

6.4 Variable volume micropipette(s) (range from 0.050 to 1.000 ± 0.8% mL) (see Fig. 4) or a high precision pump.

6.5 Paper towels, 5 × 14 cm, a color that will exhibit a color change when wet (light blue preferred) (see Fig. 1).

6.6 Anemometer—hot wire-type, capable of measuring air flow from 0.1 to 5.0 ± 0.1 m/s (see Fig. 1).

6.7 Embroidery hoop (see 14.1).

6.8 Deionized or distilled water.

6.9 Rigid material (e.g., cardstock) for circular mounting template.

6.10 Water drop detection sensor, capable of detecting a water drop after it has passed through a test specimen (optional).

6.11 Commercially available tester (see 14.3).

7. Sampling

7.1 Take rolls representative of a lot sample. If testing a textile from end items, take three items from each lot.

8. Test Specimens

8.1 From each textile sample, cut three specimens (15.0 × 15.0 ± 0.5 cm) from right, center, and left locations across the sample width for each test.

8.2 If testing garments or end-products, take specimens from different sections of the garment; i.e., sleeve, back and front.

8.3 Construct a circular mounting template (see 6.9) the size of the tester opening (13 ± 0.1 cm diam). Trace the template on the back side of the specimen.

9. Conditioning

9.1 Prior to testing, condition test specimens as directed in ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 14.2). Condition the specimens for at least 4 h in an atmosphere of 21 ± 2°C (70 ± 4°F), 65 ± 5% RH, by laying each specimen separately on a screen or perforated shelf of a conditioning rack.

9.2 Perform all tests in the standard atmosphere for testing.

10. Procedure

10.1 Mount the textile face down with the traced circle at the rim of the hoop (see 6.7) to ensure even tension across the specimen. The specimen should not sag in the middle of the hoop before the fan is turned on.

10.2 Using AATCC TM79 (see 14.1), place a drop of water on the specimen

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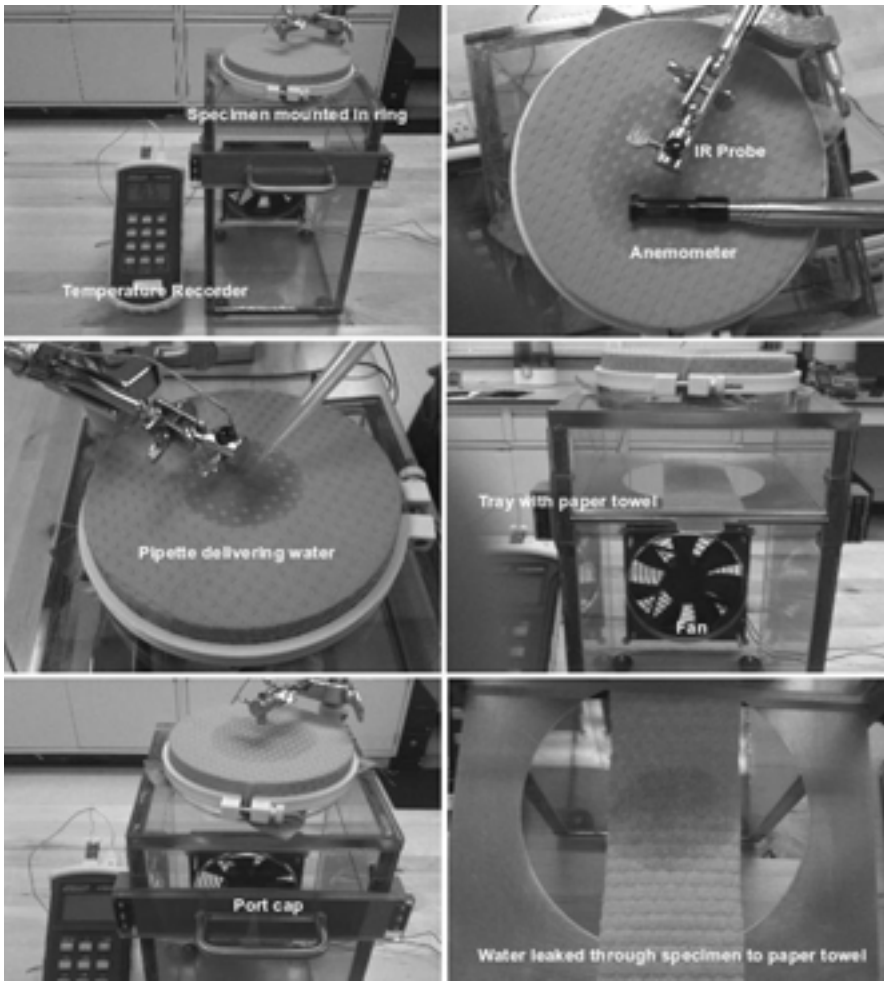


Fig. 1—Different views of the apparatus used to perform test.

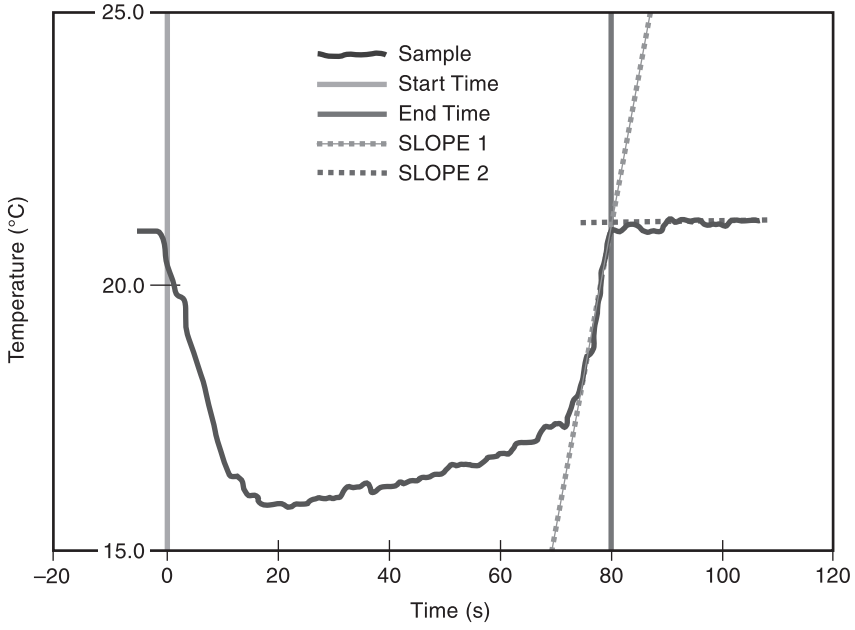


Fig. 2—Plot of temperature versus time.

and determine the absorbency. If the absorbency is less than or equal to 30 s then proceed with the test. If greater than 30 s, then stop test. Textiles with an absorbency time greater than 30 s cannot be tested with this method. Allow the specimen to dry for 5 min with the fan turned on, then turn the fan off.

10.3 Insert a blue paper towel (see 6.5) into the fan assembly if using an apparatus with transparent walls. If the apparatus has opaque walls, the use of a sensor to detect water is required (see 6.10).

10.4 Turn on the fan.

10.5 After 5 min measure the temperature of the textile. This temperature is the dry, equilibrated temperature of the textile.

10.6 Dispense 0.100 ± 0.003 mL of water onto the surface of the specimen with the pipette touching specimens. If delivering water manually, the pipette should be held at a 45° angle. Observe if any water passed through the specimen onto the paper towel.

10.6.1 The delivery rate of the water is critical. To keep a constant delivery rate, deliver that water at a speed of 0.200 ± 0.0016 mL/s. Deliver the water while touching the pipette to the textile. Alternately, a peristaltic pump or a tube with a valve may be used to deliver water at a consistent speed of 0.200 ± 0.0016 mL/s. The tube can be prefilled with the correct amount of water using a pipette.

10.7 If the paper towel is wet, allow the specimen to dry. Replace the wet paper towel in the tray with a new paper towel. Reduce the volume in the pipette by 10%, and repeat 10.6. If water did not leak, let the sample dry, then increase the volume by 0.100 mL and test again. Repeat until the absorbent capacity can be estimated (see 10.8).

10.8 The maximum volume of water that is absorbed by the textile without passing through and wetting the paper towel is equivalent to the absorbent capacity (V_m) for this method.

10.9 Position the IR thermocouple probe in the middle of the specimen 1 cm above the surface. Adjustment of the thermocouple probe may be necessary as test specimens will sag slightly after the fan is turned on.

10.10 Remove the water detection tray. Close port to the shelf.

10.11 Let the specimen dry by waiting for the temperature to return to the value determined in 10.5.

10.12 Place the anemometer probe above and near the center of the specimen. Record air speed through the specimen.

10.13 Select the maximum amount of water to be at least 10% less than V_m. Select four additional volumes of water at approximately 10%, 25%, 50%, and 75% of the maximum amount. For example, a textile with a V_m = 0.470 mL the volumes

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would be 0.050, 0.100, 0.200, 0.300, and 0.400 ± 0.001 mL.

10.14 Start the testing process by recording temperature data and delivering the first of a series of increasing volumes of water on the specimen in the area of the IR thermocouple probe view.

10.15 Record data every second until the temperature returns to the starting temperature.

10.16 Repeat 10.11-10.15 for each of the other four selected volumes of water.

11. Calculation and Evaluation

11.1 View the data on the recorder or plot the temperature versus time data using a spreadsheet program for each of the three specimens.

11.2 Determine the start and end times on the graph. The start time is when the water is added to the specimen (Time 0 in Fig. 2). The end time is the time at the intersection of the section with the steepest slope and the flat section of the temperature plot. To determine this intersection time value two linear fits are required. Generate a linear fit for the steepest section of the plot by choosing seven data points within the steepest portion of the plot (Slope 1 in Fig. 2). Generate a linear fit for the flat section of temperature plot by choosing 25 data points after the inflection in the temperature plot (Slope 2 in Fig. 2). The end time is the intersection of the two linear lines. The drying time is the difference of the end time and start time.

11.3 Calculate the drying rate (***R***), using formula 1:

$$R = V / Drying\ Time \quad \text{(Eq. 1)}$$

where:

R = drying rate, in mL/h

V = volume of water used in the test, in mL

Drying Time = End time – Start Time, convert seconds to hours

Example: ***R*** = 0.100 (mL)/0.022 (h) = 4.5 mL/h

11.4 Plot R_i (mL/h) versus V_i (mL) (see Fig. 3) where ***i*** represents each volume challenge.

11.5 Fit the plot of ***R*** versus ***V*** to Equation 2 using a non-linear least squares fitting algorithm:

$$R = a[1-\exp(-bV)] \quad \text{(Eq. 2)}$$

where:

R = Drying Rate, in mL/h

V = volume of water used, in mL

a and ***b*** = fitting constants

11.6 The fit yields constants ***a*** and ***b***. The fit for Fig. 3 is ***a*** = 11.9 mL/h and ***b*** = 4.7 mL^{-1}.

11.7 The drying rate at the absorbent capacity (R_{max}) is ***a***. In Fig. 3 R_{max} is 12 mL/h.

12. Report

12.1 Report the drying rate at each volume and the absorbent capacity for each textile. Determine an average and standard deviation for R_{max} for each textile along with the linear flow rate of air through the textile.

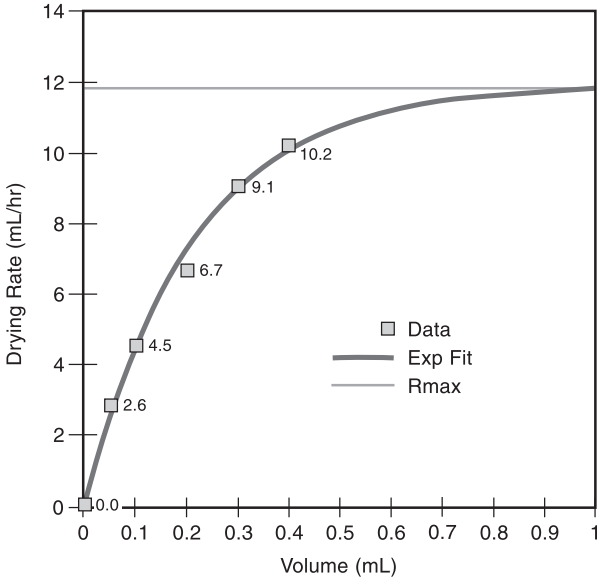


Fig. 3—Plot of drying rate versus challenge volume.



Fig. 4—Example of micropipette.

13. Precision and Bias

13.1 *Precision*.

13.1.1 In 2017, an interlaboratory study was completed, which included three laboratories with single operators, testing three specimens of each of three fabrics. Mean values for drying rate of each fabric are shown in Table I. All tested fabrics were 100% polyester circular knits. Fabric A was a 130-g/m^2 double knit. Fabric B was a 178-g/m^2 tricot knit. Fabric C was a 207-g/m^2 double knit. No prior assessment was made of the participating laboratories on performance of the test method.

13.1.2 Analysis of the data yielded critical differences shown in Table II. Differences between two averages of (***n***) specimens should reach or exceed the table value to be statistically significant at the 95% confidence level. A 2011 single-lab study had similar within-laboratory differences.

Table I—Interlaboratory Precision Study Results

Fabric	*n*	Mean (mL/h)	St Dev
A	9	23.92	3.13
B	9	8.611	1.34
C	9	17.11	1.76

Table II—Critical Differences for Averages, 95% Probability

n	Within-Laboratory	Between-Laboratories
1	2.449	6.927
2	1.734	4.906
3	1.414	3.999
4	1.224	3.464
5	1.095	3.098
6	1.000	2.828

13.2 *Bias*. The drying rate can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

14. Notes

14.1 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

14.2 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832.9500: fax: +1.610.832.9555; web site: www.astm.org.

14.3 For potential equipment information pertaining to this test method, please visit the online AATCC Buyer's Guide at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

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Test Method for Drying Rate of Fabrics: Heated Plate

Developed in 2012 by AATCC Committee RA63; jurisdiction transferred in 2022 to AATCC Committee RA114; reaffirmed 2013, 2014; editorially revised 2016, 2019, 2022.

Foreword

As moisture management products have entered the marketplace, measurement of the drying characteristics of fabrics has gained additional importance. The 2008 AATCC-ASTM International's *Moisture Management Technical Supplement: Applicable to Apparel, Linens, and Soft Goods*, contains three techniques for the measurement of drying times of fabrics. This test method is an additional approach to the drying test methods in the *Technical Supplement*.

1. Purpose and Scope

1.1 This test method determines the drying rate of a fabric, exposed to a prescribed volume of water, while in contact with a heated plate set at 37°C (99°F), the skin surface temperature at which the human body starts to perspire.

1.2 This method is applicable to all types of fabrics, including knits, wovens, and non-wovens, as well as to fabrics taken from end product items.

2. Principle

2.1 This method determines the drying rate of a fabric based on the evaporative rate that occurs when the fabric is placed in contact with a prescribed amount of water at the interface of a heated metal plate, held to a constant temperature.

3. Terminology

3.1 **drying rate,** n.—the change in volume per unit time a liquid evaporates from a textile.

NOTE: Drying rate is dependent upon textile construction, fiber content, apparel construction, finishes, the atmospheres in which the test is performed, and the volume challenge of the liquid.

3.2 **drying time,** n.—the time it takes for a specified amount of liquid to evaporate from a textile under controlled testing conditions.

NOTE: In this test method the amount of water is constant between tests. The testing conditions are controlled by the analyzer and the room.

3.3 **end time,** n.—the time at which the temperature goes through an inflection between the steepest slope and the flat sections of temperature versus time plot (see Fig. 2 and 11.2).

3.4 **start time,** n.—the time at which the water challenge is added to the specimen.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as materials safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 The manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

5. Uses and Limitations

5.1 This test is performed under controlled laboratory temperature and relative humidity conditions. While the test may be performed at alternative conditions, for comparison of results, the same conditions must be used.

5.2 This test is performed with a set amount of water. While different amounts of water may be used, care should be taken to ensure that the water does not wick to the edge of the test specimen. If water migrates to the edge of the sample, then it is necessary to increase the specimen size or reduce the volume and repeat the test with a new specimen. Further, when comparing results, tests must be conducted with the same amount of water.

5.3 The drying rates of fabrics with a membrane in its structure can be tested using this method as long as the moisture vapor transmission rate of the fabric is greater than 5,000 g/m^2/day (see 14.1 of JIS L 1099, Method B1,) to allow for detection of water evaporation. Multi-layered composite fabrics can also be tested using this method.

5.4 When testing fabric in socks, the socks should be cut open and the inner portion of the sock placed in contact with the heated plate, to simulate contact of fabric with human skin.

6. Apparatus and Materials

6.1 Drying rate test apparatus (see Fig. 1).

6.2 Temperature recorder—with capabilities to take readings every 1 s, data storage, and transmittal to computer data file.

6.3 IR thermocouple probe—temperature range of 15.0-50.0 ± 0.1°C (59.0-122.0 ± 0.18°F).

6.4 Fan box – production of 1.5 ± 0.5 m/s air flow across the width of the hot

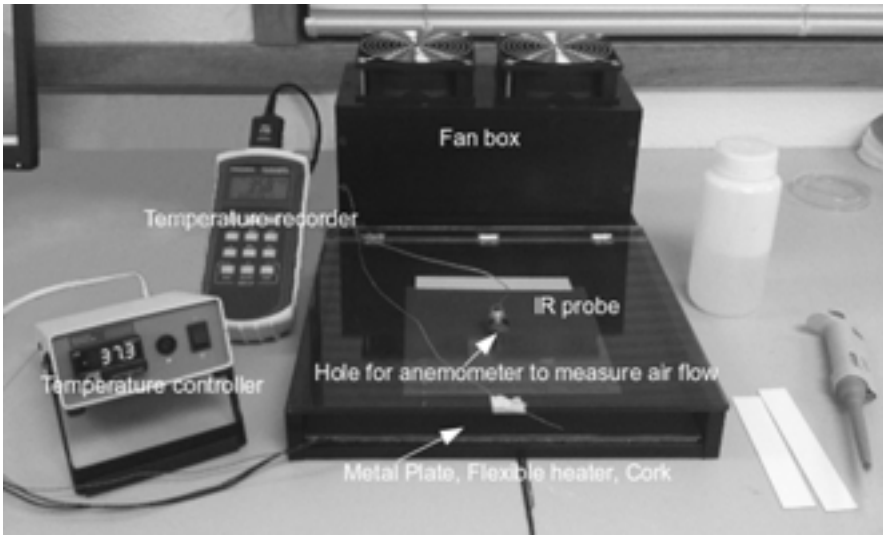


Fig. 1—Test apparatus.

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plate, measured directly behind the IR thermocouple probe.

6.5 Metal plate—30.5 × 30.5 ± 0.5 cm.

6.6 Flexible heater—30.5 × 30.5 ± 0.5 cm with controller to maintain temperature of 37 ± 1°C (98 ± 2°F).

6.7 Cork board for insulation—30.5 × 30.5 ± 0.5 cm.

6.8 Micropipette, adjustable volume, 0.100-1.000 ± 0.003 mL.

6.9 Anemometer—hot wire-type, capable of measuring air flow from 0.5-2.5 ± 0.1 m/s.

6.10 Magnetic, plastic or metal strip, 15.0 cm long, 4.0 ± 2.0 cm wide, 0.2 ± 0.1 cm thick, can be used to hold the specimen in place.

6.11 Deionized or distilled water.

7. Sampling

7.1 Take rolls representative of a lot sample. If testing fabric in end items, take three items from each lot.

8. Test Specimens

8.1 From each fabric sample, cut three specimens (15.0 × 15.0 ± 0.5 cm) from right, center, and left locations across the sample width for each test.

8.2 If testing garments or end-products, take specimens from different sections of the garment; i.e., sleeve, back and front.

8.3 If testing socks or hosiery, slit the specimen lengthwise and test each unique construction area. If the specimen is less than the desired size, then monitor the water migration closely. If water migrates to the edge of the sample, then it is necessary to reduce the volume and repeat the test with a new specimen.

9. Conditioning

9.1 Prior to testing, condition test specimens as directed in ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 14.2). Condition the specimens for at least 4 h in an atmosphere of 21 ± 2°C (70 ± 4°F), 65 ± 5% RH, by laying each specimen separately on a screen or perforated shelf of a conditioning rack.

9.2 Perform all tests in the standard atmosphere for testing.

10. Procedure

10.1 Turn on the temperature controller for the flexible heater and fan to let the metal plate temperature stabilize to 37 ± 1°C (99 ± 2°F).

10.2 Use a hot wire anemometer to verify the air flow across the plate is 1.5 ± 0.5 m/s. Measure the air flow directly behind the IR thermocouple probe.

10.3 Place a test specimen on the metal plate for five minutes to allow the specimen to equilibrate to the metal plate temperature with the side of the specimen intended to be next to the skin placed against the metal plate surface. Use a strip to hold the top edge of the specimen, the one closest to the air fan to the metal plate surface.

10.4 Position the IR thermocouple probe in the middle of the specimen 1.0 ± 0.1 cm above the specimen. It is helpful to have a mark on the metal plate which is centered to the field of view of the IR thermocouple probe.

10.5 Start the recorder, lift the free end of the specimen (the opposite side of the magnetic strip) and apply 0.200 ± 0.003 mL of water on the plate below the specimen directly under the IR thermocouple probe. Reposition the specimen so the specimen covers the water drop. The start time is when the specimen comes in contact with the water.

10.6 View the specimen to determine the wicking profile generated by the water. Verify that the water did not wick to the edge of the specimen. If the water does wick to an edge of the sample then use a larger specimen or reduce the volume.

10.7 Collect and record the temperature every second until temperature returns to the initial temperature. Repeat the test on the additional specimens.

11. Calculation and Evaluation

11.1 View the data on the recorder or plot the temperature versus time data using a spreadsheet program for each of the three specimens.

11.2 Determine the start and end times

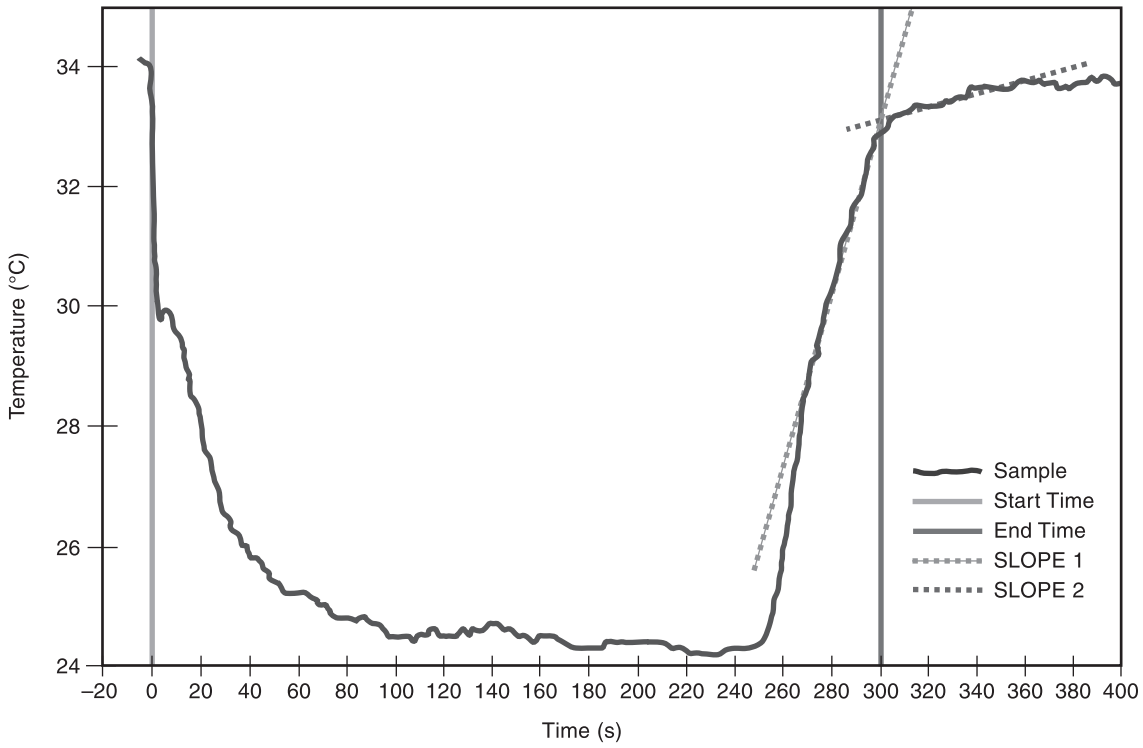


Fig. 2—Plot of temperature versus time to determine dry time.

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on the graph. The start time is when the water is added to the specimen (Time 0 in Fig. 2). The end time is the time at the intersection of the section with the steepest slope and the flat section of the temperature plot. To determine this intersection time value two linear fits are required. Generate a linear fit for the steepest section of the plot by choosing seven data points within the steepest portion of the plot (Slope 1 in Fig. 2). Generate a linear fit for the flat section of temperature plot by choosing 25 data points after the inflection in the temperature plot (Slope 2 in Fig. 2). The end time is the intersection of the two linear lines. The drying time is the difference of the end time and start time.

11.3 Average the drying time of the three specimens.

11.4 Calculate the drying rate (***R***), using the following equation:

$$R = V / Drying\ Time$$

where:

R = drying rate, in mL/h

V = volume of water used in the test, in mL

Drying Time = End time – Start Time, in hours

Example: ***R*** = 0.200 (mL)/0.0836 (h) = 2.39 mL/h

11.5 Average the three values for each sample.

12. Report

12.1 Report the individual and average drying rates for each fabric stating the water volume used. When comparing data, laboratory temperature, laboratory relative humidity, instrument air flow, and the water volume must be the same.

13. Precision and Bias

13.1 *Precision*.

13.1.1 An internal laboratory test was run in 2010 in which five different fabrics were tested in one laboratory over two days using two operators using a 0.200 ± 0.003 mL water challenge. The analysis of variance technique was applied to the data set. The analysis is being retained in the RA63 committee files. The number of tests (n), mean average (Mean), 95% confidence interval (95% CI), standard error (SE), and standard deviation (SD) are shown in Table I.

13.1.2 *Between laboratory* precision has not been established for this test method. Until such precision information is available, users of the method should use standard statistical techniques in making any comparison of test results for *between-laboratory* averages.

13.2 *Bias*. The drying rate can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

14. Notes

14.1 Japanese Industrial Standards (JIS) may be obtained through web site: www.jsa.or.jp.

14.2 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832.9500; fax: +1.610.832.9555; web site: www.astm.org.

Table I—Analysis of Variance

Drying Rate by Fabric (mL/hr @ 0.2 mL challenge)	n	Mean	95% CI	SE	SD
40% polyester, 60% cotton woven (145 g/m^2)	12	1.83	± 0.16	0.07	0.25
100% polyester double knit (150 g/m^2)	12	1.93	± 0.07	0.03	0.11
100% cotton jersey knit (230 g/m^2)	12	0.84	± 0.05	0.02	0.08
100% polyester fleece, (136 g/m^2)	12	1.34	± 0.07	0.03	0.11
Sock—38% polyester, 25% cotton, 25% nylon, 12% spandex	12	0.66	± 0.04	0.02	0.07

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Test Method for Relative Hand Value of Textiles: Instrumental

1. Purpose and Scope

1.1 This method is for the determination of an instrumental relative hand value (RHV) using the mechanical properties measured on a selected reference fabric and a comparable candidate fabric.

1.2 This method does not purport to replace AATCC EP5, Evaluation Procedure for Fabric Hand, nor are the values obtained by this method known to be comparable to actual tactile sensations felt by human evaluators.

2. Principle

2.1 A specimen is placed over a plate with a specified diameter opening through which a plunger forces the fabric specimen resulting in mechanical deformation data. Force displacement curve data from a reference fabric and a test fabric are used in calculations to produce a Relative Hand Value (RHV).

3. Terminology

3.1 **hand,** n.—the tactile sensations or impressions that occur when fabrics are touched, squeezed, rubbed or otherwise manipulated.

3.2 **reference fabric,** n.—a fabric selected to represent a baseline for comparison of tactile properties of hand.

3.3 **relative hand value,** (RHV) n.—an index of properties, relative to a reference fabric, as determined by either an instrumental system or a subjective hand evaluation that can be used to predict tactile sensations a human perceives when handling a fabric.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5. Uses and Limitations

5.1 The results from this method yield a quantitative description of RHV for use in research and production quality control. A publication describing the historical, theoretical and technical background for the instrumentation on which this test method is based is available (see 13.1).

5.2 The test instrument system produces other hand measurement attributes but this method is specific only to the RHV attribute. Refer to the instrument's instruction manual.

5.3 Caution should be used when comparing the RHVs from fabrics with different yields, constructions, fiber contents and finishes to avoid erroneous conclusions. Correlation to hand panel studies has not been broadly determined. It is the responsibility of the user to determine applicability to their product/needs.

6. Apparatus and Materials

6.1 A relative hand value instrument system (RHV-IS), PhabrOmeter™ System and accessories and PhES™ software (see 13.2 and Figs. 1 and 2).

6.2 Laboratory gloves and forceps.

6.3 Reference fabric(s). A reference fabric must be of the same density type and class and may be of the same fiber content, construction, processing stage or a fabric that is chosen for comparison against all fabrics to be tested.

7. Conditioning

7.1 Condition reference and test specimens flat, according to ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 13.3).



Fig. 1—PhabrOmeter™ System

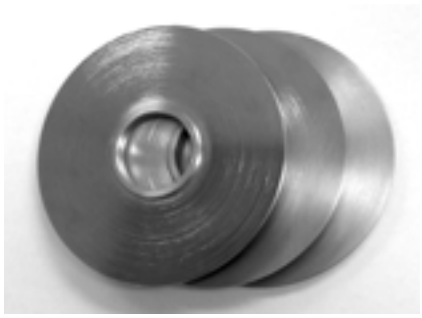


Fig. 2—Test Load Plates

8. Preparation of Reference Fabric

8.1 Multiple reference fabrics should be established and indentified using the New Fabric form in the RHV-IS. Each reference fabric should have a separate New Fabric file.

8.2 Perform tests on multiple sets of specimens (see 9.1-9.3) of a selected reference fabric specimen to provide statistical confidence in the force displacement curves for each reference fabric (see 10.1-10.7) and save test data.

8.2.1 For example, if checking the RHV values of an application and variations of the application on woven, 400 g/m^2 (11.8 oz/yd^2), polyester/cotton specimens, the reference fabric data base would be the same fabric that does not have an application or that has been prepared for finishing; therefore, the experimental applications can be compared to the unfinished reference specimen data set.

8.3 Selected reference fabric should be identified according to its class (or type of material). For instance, if one is testing woven fabric specimens one would not select a knitted reference fabric against which to compare.

9. Test Specimens

9.1 Identify the face and back of the specimen to ensure the same side will be tested.

9.1.1 The technical face of a specimen may not always be the one that will be 'touched' when selecting an item to purchase or the one that will be in contact with the skin when used or worn. Before taking specimens, agreement should be reached between parties as to whether the technical face, back or both sides shall be

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tested.

9.2 Wear gloves during specimen preparation, as excessive handling and distortion of fabric in any manner must be avoided. Take specimens no nearer to the selvage or edge of the specimen than 10.0% of the fabric width.

9.3 Cut three 113 ± 2 mm (4.5 ± 0.1 in.) diameter specimens from a fabric specimen using the specimen cutter accessory. Where possible, each specimen should contain different groups of lengthwise and widthwise yarns or sites (nonwovens) from specimen locations across the specimen's width that are free of wrinkling.

9.3.1 If sampling from products or garments, select specimens from various locations away from seams, trims, hems or buttons.

10. Procedure

10.1 Start the instrument and software system for testing and open New Fabric form.

10.2 Determine mass of one of the three specimens and record in grams (0.00). If a specimen fabric's construction has the propensity for variable weights (i.e., nonwoven, cheesecloth), measure all three specimens and use the average mass.

10.3 Determine the fabric thickness according to ASTM D1777, Standard Test Method for Thickness of Textile Materials, and record to the nearest mm (see 13.3). If a specimen fabric's construction has the propensity for variable thicknesses (i.e., nonwoven, upholstery), measure all three specimens and use the average thickness.

10.4 Enter the mass (see 10.2) and the thickness (see 10.3) into the opened New Fabric form and select the System's Choice check box then save. Entering this information allows the software to calculate linear density (µg/cm) automatically. Select the appropriate test load plate for the calculated linear density of the specimen being tested (see Table I).

10.4.1 If the candidate specimen has a different linear density as the result of exposure to textile preparation, finishing, or laundering processes than the reference fabric's linear density, then the System's Choice check box should not be checked. The appropriate linear density must be selected manually to assure that the candidate fabric specimen is entered with the same linear density as its reference fabric.

10.5 Set the instrument to Loading Test Plate status and place the appropriate number of test load plates onto the load plate platform.

10.6 Activate the Loading Test Specimen status icon. With gloved hands or forceps, mount the specimen on the specimen holder with the side to be tested facing down and start the test.

10.7 Save the data before loading the next specimen.

10.8 Repeat 10.6 and 10.7 for each remaining specimen.

11. Calculation, Report and Interpretation

11.1 From the deformation extraction curves generated from each specimen, the software calculates the averaged fabric attributes and once a reference fabric is selected a RHV can automatically be produced (see 13.4 and see Appendix A for the formula used in the calculation of RHV).

11.2 Report the produced average RHV for each specimen.

11.3 If a common reference fabric has been selected and confidence in its database has been established, a general ranking of the RHVs for all tested fabrics may be undertaken (see 5.3).

11.4 A smaller RHV value indicates a specimen performs closer to the reference fabric. Other analysis tools may be used to interpret the results.

12. Precision and Bias

12.1 *Interlaboratory Study*. Testing using the Relative Hand Value: Instrumental Method was conducted in 2010 by four laboratories with seven instruments. Laboratories tested five processed specimens that were softened or had other after preparation, dyeing or process treatment. The specimens used in the study were: 100% polypropylene, nonwoven, 1.9 oz/yd^2; 100% cotton interlock knit, 6.2 oz/yd^2; 100% cotton woven print cloth, 3.2 oz/yd^2; 76% Nylon/24% Spandex warp knit, 5.6 oz/yd^2; and 100% nylon woven, 3.5 oz/yd^2. In each laboratory, one operator tested three specimens of each fabric on two different days using different specimens.

12.1.1 One laboratory whose instrument's calibration was verified by the instrument manufacturer performed testing on each of the specimen fabrics in their unfinished state. Those data files were sent to each laboratory for use with their instrument's calculation of Relative Hand Value for each specimen in the study set.

12.1.2 The analysis of variance technique was used on the data collected. A full report is on file for reference at the AATCC Technical Center.

12.1.3 *Precision*. The components of variation of the Relative Hand Values by fabric as standard deviations of are given in Table II.

Results showed that the hand values evaluated on Day 1 and Day 2 were not significantly different at the 95% confidence level. Results also showed that the hand values obtained under different standard atmospheres (23°C [73°F], 50% RH and 22°C [72°F], 65% RH) were not significantly different at 95% confidence level. *Within laboratory* and *between laboratories* precision and critical differences are given in Tables III and IV.

12.2 *Bias*. The true value of the relative hand value can be defined only in terms of a test method. There is no independent method for determining the true value. As an estimate of this property, this test method has no known bias.

13. Notes

13.1 Pan, Ning, Quantification and Evaluation of Human Tactile Sense Towards Fabrics, *International Journal of Design & Nature*, Vol. 1, 2007, pp48-60; web site: www.phabrometer.com/FAQ/MajorPaper.pdf.

13.2 A RHV-IS, the PhabrOmeter™ System, accessories and PhES™ software, can be purchased from Nu Cybertek Inc., 2925 Spafford., Ste. D, Davis CA 95618; tel: +1.530.758.3258; toll-free: +1.877.718.1880; fax: +1.888.566.1298; e-mail: info@nucybertek.com; web site: www.nucybertek.com.

13.3 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832.9555; fax: +1.610.832.9555; web site: www.astm.org.

13.4 A fabric's end use may determine which side of a specimen will be tested but if unknown, either side of the fabric may be tested. However, the direction in which specimens are tested needs to be consistent. For

Table I—Test Load Plates by Linear Density

Fabric Description	Linear Density, µg/cm	# of Plates to Use
Super light	<280	None
Light	280 ~ 1200	1
Medium	1200 ~ 3440	2
Heavy	>3440	3

Table II—Summary Data for Day 1 and Day 2 with Standard Deviations

Specimen ID	RHV Minimum	RHV Maximum	RHV Mean	SD
Nonwoven	0.056	0.151	0.086	0.03
Cotton knit	1.493	1.913	1.772	0.15
Cotton woven	0.273	0.541	0.385	0.10
Nylon/spandex warp knit	0.300	0.534	0.399	0.08
Nylon woven	0.201	0.397	0.287	0.06

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Table III—Precision *N*=3 (Number of specimens per sample)

	Pooled Standard Deviation	DF	Precision
Between Laboratory	0.5068	7	0.1916
Within Laboratories	0.5068	10	0.1603
Between Test Intervals (days)	0.4991	35	0.0856

Table IV—Critical Differences for One Operator per Laboratory (Instrument)

Number of Specimens	Between Laboratories	Within Laboratory
1	0.3319	0.2276
2	0.2347	0.1964
3	0.1916	0.1603
4	0.1659	0.1444
5	0.1483	0.1241

example, if you want the specimen to be face up, then all the sample specimens to be tested in your defined group or class should be face up.

14. History

14.1 Reaffirmed 2020.

14.2 Reaffirmed 2014, 2013. Editorially revised 2019.

14.3 Developed in 2012 by AATCC RA89.

Appendix A
Calculation of the Relative Hand Values (RHV)

A1. After the feature selection process from the extraction curve, a fabric tested Y_k will be represented by an array of p=8 data points as

$$Y_k = (y_{k1}, y_{k2}, \ldots, y_{kp})$$

A2. Since the reference fabric Y_s has been selected and tested the same way, there is

$$Y_s = (y_{s1}, y_{s2}, \ldots, y_{sp})$$

with weight pre-determined for each feature contained in a weight array W

$$W = (w_1, w_2, \ldots, w_p)$$

A3. The RHV is actually the weighted Euclidian Distance between the two fabrics

$$\mathrm{RHV} = \sqrt{\sum_{i=1}^{p} w_i (y_{ki} - y_{si})^2}$$

A4. Based on the reference fabric, the RHV value now becomes an indicator of hand preference and provides hand preference ranking. The smaller the RHV value, the closer the fabric tested to the reference fabric in terms of fabric hand.

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Test Method for Light Blocking Effect of Textiles: Spectrophotometric

Foreword

AATCC TM148-1989, Light Blocking Effect of Curtain Materials, was discontinued in 1994. During the time of its discontinuance, it was found that a standard laboratory sphere spectrophotometer used in transmission mode could be used to measure the light blocking effect of textiles used in end uses such as partial and full blackout curtains in both residential and contract applications. This method uses spectrophotometric measuring equipment that provides precise, reproducible and traceable light measurements. These instruments are currently in wide industrial use.

1. Purpose and Scope

1.1 This test method is intended to spectrophotometrically measure the light blocking properties of window covering textiles against light passing through them and being seen by a standard observer on the darkened side of the textile (typically a curtain or drape).

1.2 This method is applicable to all types of fabrics, including knits, wovens, and non-wovens, as well as to fabrics taken from end product items. However, it is not recommended for mesh structures with mesh holes greater than 3 mm across.

2. Principle

2.1 The test method assesses the percentage of light from a light source blocked by a textile material by measuring the original light and subtracting from it the light passing through the materials. The amount of light transmitted through the fabric is compared on a percentage basis, to the amount of light transmitted without any blocking of the light (total transmission). The percent of light that is not transmitted, out of the total transmission, represents the material's light blocking capability.

2.2 The method uses the CIE *Y* tristimulus value, calculated using illuminant D65 and the 10° observer as the measure of total transmission.

3. Terminology

3.1 **diffuse transmission,** n.—process by which incident light, while being transmitted through an object, is redirected or scattered over a range of angles.

3.2 **light blocking,** n.—the ability of a textile to prevent the transmission of light.

3.3 **regular transmission,** n.—process by which incident light is transmitted through an object in a rectilinear, straight-through manner, without diffusion.

3.4 **spectrophotometer,** n.—an instrument used for measuring the transmittance and/or reflectance of specimens as a function of wavelength.

3.5 **total transmission,** n.—the amount of light transmitted comprising diffuse plus non-diffuse transmission.

3.6 **transmission,** n.—process by which incident light is transmitted through an object.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear eye protection in all laboratory areas.

4.2 The manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

5. Uses and Limitations

5.1 This test is performed under controlled laboratory temperature and relative humidity conditions. While the test may be performed at alternative conditions, such as household ambient temperature and relative humidity, for comparison of results, the same conditions must be used.

5.2 The light this test is performed with is from a xenon lamp. Different light sources may be used and results may, therefore, also be different, such as with spectrophotometric instruments with band width infrared (IR) capacity.

6. Apparatus and Materials

6.1 Spectrophotometer with transmission measuring capability with a means of precisely mounting and anchoring a specimen in the transmission holding clamp.

6.2 Means of covering reflectance or external port with a white standard tile to establish the one-hundred percent transmission standard and to complete the sphere for further transmission readings.

6.3 Means of blocking transmission to determine zero percent transmission.

6.4 If the spectrophotometer is equipped with adjustable ultraviolet energy, standardize the amount of ultraviolet energy in the spectrophotometer light source (see 14.1).

7. Sampling

7.1 Take five representative test specimens from the sample. Increase the number of specimens accordingly to include different areas that may have varying light transmission characteristics within the total sample area. Textiles such as long repeat jacquards may require a larger number of specimens in order to obtain a more representative sample. Textiles containing large repeats of alternating diaphanous and opaque areas may have the light blocking effect of each area sampled separately.

8. Test Specimens

8.1 Each specimen should be large enough to fit onto the specimen holding frame and completely cover the transmission port of the spectrophotometer.

8.2 Take multiple specimens to provide an appropriate random sample from different width and length areas of the fabric, not taking any specimens from within 10% of the width from the selvedge or machine edge.

9. Conditioning

9.1 Prior to testing, condition test specimens as directed in ASTM D1776, Standard Practice for Conditioning and Testing Textiles. (Condition the specimens adequately in an atmosphere of 21 ± 2°C (70 ± 4°F), 65 ± 5% RH, by laying each specimen separately on a screen or perforated shelf of a conditioning rack. Conditioning time will be dependent on factors indicated in ASTM D1176.)

9.2 Perform all tests in the standard atmosphere for testing.

10. Procedure (see Fig. 1)

10.1 Operational check of total transmission with no light blocking.

10.1.1 Block the external/reflectance port with a white standard to complete the sphere.

10.1.2 Set the machine to transmission

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Fig. 1—Specimens Mounted in Two Different Instrument Manufacturers' Spectrophotometers

measuring mode without any sample in the transmission holder.

10.1.3 Measure and record total transmission, using the CIE $Y_{D65\text{-}10}$ tristimulus value, which should result in a measurement of 100.

10.2 Operational check of total transmission with opaque light blocking.

10.2.1 Keep the white standard in place at the external/reflectance port to maintain the completed sphere.

10.2.2 Block the transmission port with an opaque, black standard, such as a metal plate.

10.2.3 Measure and record total transmission. The CIE $Y_{D65\text{-}10}$ value of the measurement should be 0.

10.3 Record transmission through the specimen material.

10.3.1 Keep the white standard in place to maintain the completed sphere.

10.3.2 Place the specimen to be measured in the transmission holder clamp.

10.3.3 Measure and record the light transmission using the CIE $Y_{D65\text{-}10}$ tristimulus value.

11. Calculation

11.1 Calculate the light blocking effect (x) in percent (%), using Equation 1:

$$x = 100\text{-}A \quad \text{(Equation 1)}$$

where:

A = CIE $Y_{D65\text{-}10}$ tristimulus value as measured in 10.3

11.2 Average the calculated values from each of the five specimens.

12. Report

12.1 Cite this test method as having been used.

12.2 Report the light source used and whether or not the amount of ultraviolet energy in the spectrophotometer light source has been standardized.

12.3 Report the average light blocking effect for all of the specimens tested.

12.4 Report the number of specimen tested.

12.5 Report the standard deviation for the resultant data.

12.6 Report the temperature and relative humidity during testing.

13. Precision and Bias

13.1 A piece-dyed woven polyester fabric of blue color, designed as a partial light blocking curtain, coated and flocked subsequent to dyeing, was measured (n=5) in single, double and triple layers on two different manufacturers' instruments in one lab and on the same type of one manufacturer's instrument in a second lab (see Table I).

To provide more detailed information on single operator, single instrument repeatability and sample size at a 95% confidence level, 30 specimens of the same fabric were measured at Lab 2, with the instrument from Manufacturer A (see Table II).

13.2 Thirty crock squares of 100% cotton plain weave sheeting from the same lot were measured at Lab 1, with the instrument from Manufacturer A (see Table III).

13.3 *Bias.* Light blocking capacity of fabrics by the spectrophotometric method can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

14. Notes and References

14.1 See AATCC EP11, Evaluation Procedure for Spectrophotometer UV Energy Calibration Procedure for Optically Brightened Textiles, as included in the AATCC *Manual of International Test Methods and Procedures*, for details regarding ultraviolet energy standardization of spectrophotometer light sources. EP11 is available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; email: ordering@aatcc.org; web site: www.aatcc.org.

15. History

15.1 Revised in 2021 for clarity and to add history section per the AATCC style guide.

15.2 Revised in 2016.

15.3 Developed in 2014 by AATCC Committee RA36.

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Table I—Statistics for $n = 5$, Two Labs, Two Instruments

Single Layer

Lab 1, Instrument Manufacturer A		Lab 2, Instrument Manufacturer A		Lab 1, Instrument Manufacturer B	
Average	**St Dev**	**Average**	**St Dev**	**Average**	**St Dev**
96.46	0.03	96.50	0.15	96.58	0.01

Combined Average	Combined Standard Deviation
96.51	0.10

95% Confidence (Single Layer, Combined Data)

n	Standard Error
1	0.20
2	0.14
3	0.11
4	0.10
5	0.09

Double Layer

Lab 1, Instrument Manufacturer A		Lab 2, Instrument Manufacturer A		Lab 1, Instrument Manufacturer B	
Average	**St Dev**	**Average**	**St Dev**	**Average**	**St Dev**
99.83	0.00	99.63	0.01	99.84	0.01

Combined Average	Combined Standard Deviation
99.77	0.10

95% Confidence (Double Layer, Combined Data)

n	Standard Error
1	0.20
2	0.14
3	0.12
4	0.10
5	0.09

Triple Layer

Lab 1, Instrument Manufacturer A		Lab 2, Instrument Manufacturer A		Lab 1, Instrument Manufacturer B	
Average	**St Dev**	**Average**	**St Dev**	**Average**	**St Dev**
99.98	0.00	99.97	0.00	99.99	0.00

Combined Average	Combined Standard Deviation
99.98	0.01

95% Confidence (Triple Layer, Combined Data)

n	Standard Error
1	0.02
2	0.01
3	0.01
4	0.01
5	0.01

Table II—Statistics for Blue Polyester: $n = 30$, Single Lab, Single Instrument

Single Layer		Double Layer		Triple Layer	
Average	**St Dev**	**Average**	**St Dev**	**Average**	**St Dev**
96.45	0.30	99.63	0.01	99.97	0.00
95% Confidence		**95% Confidence**		**95% Confidence**	
n	**Standard Error**	***n***	**Standard Error**	***n***	**Standard Error**
1	0.582	1	0.026	1	0.008
2	0.411	2	0.019	2	0.006
3	0.336	3	0.015	3	0.005
4	0.291	4	0.013	4	0.004
5	0.260	5	0.012	5	0.004
6	0.237	6	0.011	6	0.003
7	0.220	7	0.010	7	0.003
8	0.206	8	0.009	8	0.003
9	0.194	9	0.009	9	0.003
10	0.184	10	0.008	10	0.003
30	0.106	30	0.005	30	0.002

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Table III—Statistics for White Cotton: *n* = 30, Single Lab, Single Instrument

Single Layer	
Average	**St Dev**
61.45	0.23
95% Confidence	
n	**Standard Error**
1	1.374
2	0.971
3	0.793
4	0.687
5	0.614
6	0.561
7	0.519
8	0.486
9	0.458
10	0.434
30	0.251

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Test Method for Water Vapor Transmission of Textiles

Foreword

This test method is similar in concept to other "upright cup" methods; however, it was developed specifically to evaluate comfort of vapor permeable textiles used in outdoor products. The temperature differential between the heated bath and the ambient air creates a driving force to promote water vapor transmission. This driving force is likely greater than in actual use, but it serves as an effective condition to differentiate materials by their water vapor transmission performance. Measuring water vapor transmission through test fabric as a percentage of transmission through control paper improves precision of the results, even with small fluctuations in test time or temperature. The control paper can also serve as a verification tool to ensure consistent results.

1. Purpose and Scope

1.1 This test method is used to determine the relative rate of water vapor transmission of textiles.

2. Principle

2.1 Test and control specimens are attached to water-filled canning jars where the fabric replaces the flat lid. The jars are weighed and suspended in a water bath at 54°C (129°F) for 24 h. The amount of water vapor that passes through the specimens is determined by re-weighing each jar at the end of the 24 h test and subtracting from the original weights. Vapor transmission through test specimens is reported as a percentage of the average loss through control specimens. Vapor transmission may also be reported in units of $g/m^2/24$ h.

3. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

3.1 ASTM C168, Standard Terminology Relating to Thermal Insulation (see 15.1).

3.2 ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 15.1).

4. Terminology

4.1 **control paper,** n.— *for liquid moisture management testing,* a standard filter paper able to transmit a consistent volume of water vapor under controlled conditions.

4.2 **moisture management,** n.—*for liquid moisture management testing*, the engineered or inherent transport of aqueous liquids such as perspiration or water (relates to comfort) and includes both liquid and vapor forms of water.

4.3 **water vapor transmission rate,** n.—the steady water vapor flow in unit time through unit area of a body, normal to specific parallel surfaces, under specific conditions of temperature and humidity at each surface. [ASTM C168].

5. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

5.1 Follow good laboratory practices (GLP). Wear safety glasses in all laboratory areas.

6. Uses and Limitations

6.1 This test method may not be applicable to materials that depend on direct contact with water for vapor transmission.

7. Apparatus (see 15.2)

7.1 Water bath, with a thermostat that can be set to maintain a water bath temperature of 54 ± 2°C (129 ± 5°F) (see 15.3).

7.2 Stainless steel plate, 11 or 12 gauge, with a minimum of seven 88.3-mm (3.5-in.) diam holes. The plate must be stiff enough to suspend jars slightly above the bottom of the bath. The plate will also cover the water bath to minimize evaporation.

7.3 Wide-mouth pint (nominal 16 oz) canning jars, or other glass jars with nominal 83 mm (3.25 in) diam opening and 121 mm (4.75 in.) inside depth, to fill all holes in the stainless steel plate.

7.4 Canning jar bands, or other means of securing specimens over jar mouths.

7.5 Balance, accurate to ± 0.1 g (± 0.001 oz).

7.6 Distilled water.

7.7 Timer.

7.8 Thermometer, suitable for measuring water temperature at 54 ± 2°C (129 ± 5°F).

7.9 AATCC vapor transmission control paper (see 15.4), 3 sheets.

7.10 Anemometer, with precision to measure air flow at 0.1 m/s.

8. Specimens

8.1 Cut all specimens at least 100 ± 5 mm (4 ± 0.2 in.) away from the selvage. If testing garments, cut specimens at least 100 ± 5 mm (4 ± 0.2 in.) from all seams and stitching.

8.2 Take specimens such that each contains different warp yarns or wales and different fill yarns or courses.

8.3 Cut four specimens approximately 108 mm (4.5 in.) in diam. Specimens should be large enough to cover the jar opening and be secured by the band, but exact size and shape are not critical to the test.

8.4 Control specimens are pre-cut filter paper (see 7.9).

9. Conditioning

9.1 Prior to testing, precondition and condition all specimens and control papers as directed in ASTM D1776. (Use conditions indicated in Table 1 for Textiles, general. Estimate conditioning time as indicated in Table 2 for the appropriate fiber content.) Lay each test and control specimen, separately, on a screen or perforated shelf of a conditioning/drying rack.

9.2 Perform all tests in the standard atmosphere for textile testing, 21 ± 2°C (70 ± 4°F) and 65 ± 5% RH.

9.3 Perform tests in still air, with a maximum air flow of 0.1 m/s (0.33 ft/s). Greater air flow will result in higher values for vapor transmission. Use an anemometer, following manufacturer's instructions, to verify maximum air flow in test area does not exceed 0.1 m/s (0.33 ft/s).

10. Preparation of Apparatus and Specimens

10.1 Clean and dry the canning jars.

10.2 Number each jar. The number must remain legible after wetting.

10.3 Fill the water bath approximately three quarters full with distilled water. Cover with the metal plate.

10.4 Heat and maintain the water bath at 54 ± 2°C (129 ± 5°F).

10.5 Place the first jar on the balance and tare. Add 390 ± 1 g distilled water to

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the tared jar.

10.6 Repeat for a total of 7 jars (3 control, 4 test).

10.6.1 If multiple samples are being tested simultaneously, only one set of control jars is required.

10.7 Place one control specimen, face up, over the opening of jar number "1" and secure with a band.

10.8 Repeat for jars numbered "2" and "3."

10.9 Place test specimens from a sample over the openings of jars numbered "4" through "7," face side up, and secure with jar bands (see Fig. 1).

10.9.1 If multiple samples are being tested simultaneously, secure specimens from the second sample on jars numbered "8" through "11." Continue to secure all specimens, keeping specimens from each sample on consecutively numbered jars.

11. Procedure

11.1 Tare empty balance and weigh the first filled and covered jar assembly. Record the jar number and total original weight of the assembly.

11.2 Weigh each remaining jar assembly, recording the jar number and total weight.

11.3 Place the prepared jars through the holes in the metal plate. Handle carefully to avoid wetting the specimens. If control paper gets wet during preparation, replace with a dry specimen and reweigh. Distribute test and control specimens evenly across the length and width of the plate (see Fig. 2).

11.3.1 If there are more holes than specimens, place empty jars in the unused holes. Add water or other ballast to the jars as needed to prevent floating. If water is used, place flat lids and bands on the jars to avoid creating excess humidity in the test environment.

11.3.2 Jars should be approximately three-quarters submerged in the water bath.

11.4 Set the timer for 24 h and start.



Fig. 1—Jar assembly prepared for testing.



Fig. 2—Test apparatus with 8 test specimens, 3 control specimens, and 4 empty jars (see 11.3.1).

11.5 After 24 h, remove the jars from the water bath. Avoid dripping on other specimens.

11.6 Dry the outsides of the jars.

11.7 Weigh each jar assembly, recording the jar number and total final weight.

11.8 Carefully unscrew bands and remove each specimen. Note any condensation on the underside of the specimen.

12. Calculations

12.1 Calculate the water vapor transmitted through each specimen by subtracting the final weight from the corresponding original weight of each jar assembly.

$$T_n = O_n - F_n \quad \text{(Eq. 1)}$$

where:

T = water vapor transmitted, g
n = jar number
O = original weight of jar assembly, g
F = final weight of jar assembly, g

12.2 Calculate the average water vapor transmitted through the control specimens.

$$T_{\text{control avg}} = (T_1 + T_2 + T_3)/3 \quad \text{(Eq. 2)}$$

12.3 Divide each value for water vapor transmission through test fabric by the average water vapor transmitted through control samples and multiply by 100% to get the percent of control paper vapor transmission.

$$T_{4\%} = 100\% * (T_4/T_{\text{control avg}}) \quad \text{(Eq. 3)}$$

12.4 Calculate the average percent of control paper vapor transmission for each sample.

$$T_{\text{avg}\%} = (T_{4\%} + T_{5\%} + T_{6\%} + T_{7\%})/4 \quad \text{(Eq. 4)}$$

12.5 Optional: Calculate the individual and average vapor transmission through test specimens in units of g/m^2/24 h. For canning jar bands, exposed specimen area is 0.004243 m^2 (inside diam of lid ring = 73.5 mm). Test duration is 24 h.

13. Report

13.1 Describe or identify the fabric tested.

13.2 Report that the sample was tested using AATCC TM204-2019.

13.3 Report the testing conditions and preliminary data:

13.3.1 Original and final weights of all the test and control jar assemblies (O_n, F_n).

13.3.2 Weight of water vapor transmitted through each specimen (T_n). Review the range of this data to determine the need for retesting.

13.4 Report the test results:

13.4.1 Report the individual and average percent of control paper vapor transmission for test specimens from each sample ($T_{n\%}$, $T_{\text{avg}\%}$).

13.4.2 Optional: Report the individual and average vapor transmission through test specimens in units of g/m^2/24 h.

13.4.3 Report any condensation observed on the underside of specimens after testing.

13.5 Describe any modification(s) of the published method.

14. Precision and Bias

14.1 *Precision*

14.1.1 In 2014, an interlaboratory study was completed, which included five laboratories with single operators, testing four specimens of each of six fabrics. No prior assessment was made of the participating laboratories on performance of the test method (see Table I).

14.1.2 Analysis of the data yielded critical differences shown in Table II. Differences between two averages of (n) specimens, for the appropriate precision

Table I—Precision Study Results

Fabric	*n*	Mean	St Dev
A	20	79.365	3.065
B	20	88.598	4.819
C	20	74.860	2.100
D	20	81.621	2.103
E	20	89.186	2.403
F	20	71.343	5.531

Table II—Critical Differences for Averages, 95% Probability

n	Within-Laboratory	Between-Laboratories
1	1.366	7.460
2	0.967	5.283
3	0.789	4.307
4	0.683	3.730
5	0.611	3.336
6	0.558	3.046

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parameter, should reach or exceed the table value to be statistically significant at the 95% confidence level.

14.2 *Bias*. The water vapor transmission can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

15. Notes

15.1 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428, USA; +1.610.832.9500; www.astm.org.

15.2 For potential equipment information pertaining to this test method, please visit the online AATCC Buyer's Guide at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

15.3 A commercially available food warmer provides a suitable water bath for this test. A 12 × 20 in. food warmer will fit 15 specimens, including 3 controls; a minimum of 7 specimens (3 controls, 4 test) is required.

15.4 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

16. History

16.1 Jurisdiction transferred in 2022 to AATCC Committee RA114; editorially revised to note change.

16.2 Revised in 2019 to add anemometer to apparatus list and to align format with AATCC style.

16.2 Editorially revised and reaffirmed 2017. Editorially revised 2016.

16.3 Developed in 2015 by AATCC Committee RA63.

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Test Method for Carpet: Liquid Penetration by Spillage

Developed in 2016 by AATCC Committee RA57; reaffirmed and editorially revised 2019.

1. Purpose and Scope

1.1 This test method is intended to simulate the actual use of carpets in the field, and evaluates the propensity of the face and back of carpets to be penetrated by a spilled liquid.

1.2 Unless specifically agreed by interested parties, this method is not intended for the evaluation of seams/joints of the carpet. This test method is used to evaluate whether or not a carpet (face and backing) is penetrated by a spilled liquid through the carpet face and backing. The method does not assess seams/joints unless reported. This method is intended to simulate the actual use of carpet in the field.

2. Principle

2.1 A colored liquid is spilled from a specified height onto the pile side of a carpet. The test liquid is allowed to stand, undisturbed, on the specimen for approximately 24 h.

2.2 The specimen is evaluated for liquid penetration.

3. Terminology

3.1 **blotting paper,** n.—any absorbent cloth-like material that is white in color, and will readily absorb small droplets of liquid.

3.2 **carpet backing,** n.—materials (fabrics, yarns or chemical compounds) in the carpet other than the face yarns.

3.3 **carpet composite,** n.—the finished carpet product, including: face fibers, primary backing, adhesive coating, and secondary backing (if applicable).

3.4 **liquid spillage resistance,** n.—*for carpet*, the ability to prevent the penetration of liquids through the carpet composite.

3.5 **primary backing,** n.—*for tufted carpet*, the substrate into which yarns are inserted.

3.6 **seam,** n.—*for carpet*, a line formed by the junction of any two carpet composite edges.

3.7 **secondary backing,** n.—a substrate fabric attached to the back of a carpet with an adhesive binder.

3.8 **spill,** adj.—The unintended falling of liquid onto a carpet face.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials used in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practice should be followed. Wear safety glasses in all laboratory areas.

4.2 All chemicals should be handled with care.

4.3 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

4.4 Ground all electrical equipment.

5. Uses and Limitations

5.1 This test method is not applicable to liquid penetration due to flooding, sub floor leakage, etc. It is not designed to classify a carpet as liquid impervious or to assess the long term effects of liquids on product deterioration or degradation.

5.2 The test is not intended to evaluate broad loom or modular carpet seams unless the seams are specifically tested as agreed between the buyer and seller.

6. Apparatus and Materials (see 14.1)

6.1 Glass or plastic laboratory funnel with a 35 ± 2 mm (1.43 ± 0.08 in.) length tube. The tube inside diameter should be 10 ± 2 mm (0.40 ± 0.08 in.).

6.2 Methylene Blue dye stuff**.**

6.3 Glassine paper or equivalent (see 14.2)

6.4 Splash ring guard, at least 150 mm (6 in.) in diameter (not to exceed width of the specimen), with a minimum height of 127 mm (5.0 in.), and made from a non-absorbent material.

6.5 White blotting paper or equivalent (see 14.2).

6.6 Funnel and funnel holding device, capable of centering the funnel over the specimen at a height of 100 cm ± 0.3 cm (39.4 ± 0.125 in.).

7. Reagents

7.1 Methylene Blue 0.1% aqueous, diluted with de-ionized water. Prepare enough to apply 100 mL of solution to every specimen to be tested.

8. Preparation of Test Specimens

8.1 Cut three specimens with a minimum dimensions of 300 × 300 mm (12.0 × 12.0 in.) from the test sample, one from each side of the full width of the carpet, at least 300 mm (12.0 in.) from the edge, and one specimen from the approximate center of the carpet width.

8.2 Prior to testing, remove any loose tufts or fibers from the specimens.

9. Conditioning

9.1 Prior to testing, condition test specimens as directed in ASTM D1776, Standard Practice for Conditioning and Testing Textile (see 14.3). After testing, the test specimens shall sit undisturbed for 24 ± 2 h in the standard atmosphere for textiles, prior to evaluation.

9.2 Perform test in standard atmosphere for textiles.

10. Procedure

10.1 Position laboratory funnel with the aperture of funnel 100 cm ± 0.3 cm (39.4 ± 0.125 in.) above a flat, level surface on which the test specimen will be placed (see Fig. 1).

10.2 Place the test specimen face up on the level surface. Locate the funnel aperture above the approximate center of the specimen.

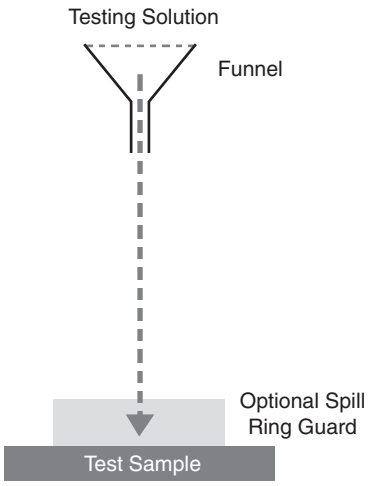


Fig. 1—Relative position of spillage equipment.

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10.2.1 Place splash ring guard (optional) (optional) on the specimen, with center of guard coinciding with center of test specimen. Ensure that the splash ring makes good contact with the test specimen.

10.3 Prepare a separate flat**,** level surface covered with the glassine paper (or other non-absorbent paper). The paper should have sufficient coverage to capture any liquid transfer from the test specimen.

10.4 Pour 100 ± 1 mL of the 0.1% Methylene Blue test solution through the funnel and on to the test specimen. Allow the solution to completely soak into its face or allow a minimum of 30 min. before removing the ring. If the test specimen will be moved to a holding area, carefully remove the test specimen, maintaining the horizontal orientation so no liquid runs over the edge. Transfer the test specimen to a flat, level surface that is covered with glassine paper (see 14.5). The specimen(s) will remain undisturbed in the conditioned atmosphere for 24 ± 2 h.

10.5 Repeat 10.4 until all specimens have been tested.

10.6 After the 24-hour exposure time (10.4), remove any liquid standing from the face of the test specimen by use of blotting paper (see 14.2).

11. Evaluation

11.1 Carefully, raise the test specimen off of the glassine paper, and observe if the test solution has transferred to the glassine paper, or if the back of the test specimen has been stained by the solution. both the paper and the secondary backing for any visible signs of liquid penetration.

11.2 Use blotting paper to assess if any liquid has migrated through to the back. Place the test specimen face-up, for at least 10 seconds, on a piece of blotting paper. Lift specimen and observe blotting paper for visible signs of liquid penetration through the specimen.

11.3 A transfer of liquid to the glassine or blotting paper also constitutes liquid penetration of the test specimen.

12. Report

12.1 Include in the report the identification information of the carpet samples being assessed. Indicate that the evaluation was accomplished using AATCC TM205-2016e.

12.2 For each specimen evaluated, report whether the specimen exhibited liquid penetration, as determined in 11.1, 11.2 or 11.3.

12.3 Report the specimen size used in the testing.

12.4 Report any deviation from the test method.

12.5 Report if a Spill Ring was used during testing.

13. Precision and Bias

13.1 Precision and Bias statements are not applicable because data is not generated by this test method.

14. Notes

14.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

14.2 Available from AATCC, P.O. Box 11215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; ordering@aatcc.org; www.aatcc.org.

14.3 Available from ASTM International, 100 Barr Harbor Dr., West Conshohocken PA 19428; tel: +1.610.832.9500; fax: +1.610.832.9555; www.astm.org.

14.4 When a carpet pile yarn is adequately treated, is properly treated it is not unusual for a liquid to stand on the face of the specimen.

14.5 If the test solution runs over the edge while a test specimen is being moved, the test specimen should be discarded and the specimen retested.

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Test Method for Free and Hydrolyzed Formaldehyde: Water Extraction

1. Purpose and Scope

1.1 This test method is applicable to wettable (non-hydrophobic) textile fabrics that may contain free or hydrolyzed formaldehyde, particularly fabrics finished with chemicals containing formaldehyde. It provides a water extraction process and a subsequent colorimetric analysis of the amount of free and hydrolyzed formaldehyde extracted.

2. Principle

2.1 A weighed fabric specimen is submerged in a flask filled with deionized water. The sealed or capped flask is placed in a water bath at a controlled temperature for a specified length of time. The amount of formaldehyde extracted into the water is then determined colorimetrically.

3. Terminology

3.1 **free formaldehyde,** n.—formaldehyde that is not bonded to a finish or a fabric. In this form, formaldehyde can be readily extracted from the fabric by immersion in water.

3.2 **hydrolyzed formaldehyde,** n.—although originally part of larger chemical structure, such as a crosslinking reactant, the formaldehyde is cleaved from the larger molecule and dissolved into the water.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practice should be followed. Wear safety glasses in all laboratory areas.

4.2 When handling glacial acetic acid to prepare Nash reagent, use chemical goggles or face shield, impervious gloves and an impervious apron during preparation. Concentrated acids should be handled only in an adequately ventilated laboratory hood. CAUTION: Always add acid to water.

4.3 Formaldehyde is a carcinogen. Use in an adequately ventilated laboratory hood. Avoid inhalation or skin contact. Use chemical goggles or face shield, impervious gloves and an impervious apron when working with formaldehyde (see 8.1 and 9.1).

4.4 An eyewash/safety shower should be located nearby and a self-contained breathing apparatus should be readily available for emergency use.

4.5 Exposure to chemicals used in this procedure must be controlled at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 1910.1000; see www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVs) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL) and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 14.3).

5. Uses and Limitations

5.1 The procedure is intended for use in the range of free and hydrolyzed formaldehyde on non-hydrophobic fabric from 20 µg/g up to about 3500 µg/g. This test method approximates hydrolyzed formaldehyde and closely simulates normal user conditions. However, TM206 is an aqueous immersion method and fabrics with a propensity for bleeding could be a concern; if the dye has an absorbance in the range of 412nm, it could interfere with the spectrophotometric measurement. It is not intended to approximate formaldehyde released under hot storage conditions.

5.2 Alternatively, AATCC TM112, Test Method for Formaldehyde Release from Fabric: Sealed Jar (see 14.1), determines free and hydrolyzed formaldehyde and additional formaldehyde as generated under more severe conditions than this test method. The appropriate test should be chosen based on that distinction (see 14.2).

6. Apparatus, Reagents, Materials

6.1 250-mL Erlenmeyer flasks with stoppers.

6.2 Weighing balance, with a minimum sensitivity of 0.01 g.

6.3 Thermostatically controlled water bath: 40 ± 1°C (105 ± 2°F).

6.4 Nash reagent prepared from ammonium acetate, acetic acid and acetylacetone and water (see 7.1).

6.5 Formaldehyde solution, approximately 37% (see 4.3).

6.6 Volumetric flasks, 50, 250, 500 and 1000-mL, Class A.

6.7 Volumetric pipettes, 1, 2, 3, 5, 10, 15, 20, 25, 30 and 100-mL, all calibrated "to deliver" and meeting Class A volume accuracy and flow time requirements, or a digital, variable-volume pipette capable of meeting Class A requirements.

6.8 A 1-L clean brown glass bottle.

6.9 Spectrophotometer, capable of reading absorbance to a minimum of three decimal places, at a wavelength of 412 nm.

6.10 Test tubes (see 14.4).

6.11 Distilled or deionized water. Hereafter, referred to as deionized water.

6.12 pH meter, calibrated and accurate to 0.01

7. Preparation of Nash Reagent

7.1 Nash Reagent: In a 1000-mL volumetric flask, dissolve 150 g of ammonium acetate in 800-mL of deionized water; add 3.0 $\pm$ 0.1 mL of glacial acetic acid and 2.0 $\pm$ 0.1 mL of acetylacetone. Add deionized water up to the 1000-mL level and mix. Store in a capped brown bottle.

7.1.1 Newly made Nash Reagent requires a 12 h wait period before being used as the reagent darkens slightly in color during that time period. The reagent may be used up to 8 weeks after it is made. However, since the sensitivity may change slightly over a long period of time, it is good practice to run a calibration curve weekly or when used to correct for the slight changes in the standard curve.

8. Preparation of Standard Solution and Calibration

8.1 Prepare a 1500 µg/mL stock solution of formaldehyde by diluting 3.8-mL of reagent grade formaldehyde solution (approximately 37%, see 4.3) to 1.0-L with deionized water in a volumetric flask. Equilibrate the stock solution for a minimum of 24 h in a capped/stoppered flask before standardization. Determine the concentration of formaldehyde in the

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stock solution by a standard method (see Section 9. Standardization of Stock Formaldehyde Solution, or any other suitable procedure such as sodium sulfite titration using 0.1 *N* HCl (see 14.5). Record the actual concentration of this standardized stock solution. This stock solution will keep for at least four weeks and is used to prepare standard dilutions.

8.1.1 A 1:10 dilution of the standardized formaldehyde stock solution is prepared by pipetting 25-mL of the standardized stock solution into a 250-mL volumetric flask and diluting to the mark with deionized water. The stock solution is then titrated and its concentration is determined as µg/mL.

8.2 A 5-concentration calibration curve, which may include zero, is suggested. Prepare the calibration curve by pipetting from the following: 0, 1, 3, 5, 10, 15, 20 and 30-mL volumes of the 1:10 dilution into 500-mL volumetric flasks then adding deionized water to the 500-mL level. (If, for example, the standardized stock solution were found to be 1470 µg/mL by titration, calculate new values for the calibration curve abscissa; i.e., 0.0, 0.29, 0.88, 1.47, 2.94, 4.41, 5.88, 8.82 µg/mL using regression analysis.)

8.2.1 When 1, 3, 5, 10, 15, 20 and 30-mL aliquots of the 1:10 dilution of the standardized stock solution from 8.1 are diluted with deionized water in 500-mL volumetric flasks, formaldehyde solutions containing approximately 0.30, 0.90, 1.5, 3.0, 4.5, 6.0 and 9.0 µg/mL formaldehyde respectively will be obtained. Record the calculated concentration of each solution as determined by section 9 Standardization of Stock Formaldehyde. The equivalent concentrations of the formaldehyde in the test fabric based on the weight of accurately known 1 g of the test fabric and 100-mL of water in the Erlenmeyer flasks will be 100 times the accurate concentrations of these standard solutions.

8.3 Use 5 mL aliquots of each of the standard solutions and the procedure described in 11.4-11.7 to prepare a calibration chart in which µg/mL formaldehyde are plotted against absorbance.

9. Standardization of Formaldehyde Stock Solution

9.1 The stock solution containing approximately 1500 µg/mL of formaldehyde must be standardized in order to produce precise calculations from the calibration curve required in colorimetric analysis.

9.2 An aliquot of the stock solution is reacted with an excess of sodium sulfite followed by a back-titration with standard acid solution to a pH of 9.5.

9.2.1 Apparatus: 10 mL-volumetric pipette, 50-mL volumetric pipette, 50-mL burette, 150-mL beaker.

9.2.2 Reagents: 1.0 *M* sodium sulfite (126 g Anhydrous Na_2SO_3/L), 0.02 *N* sulfuric acid (can be purchased in standardized form from chemical supply companies or must be standardized from standard NaOH solution). Do not use commercial standardized sulfuric acid that has been stabilized with formaldehyde. If there is a doubt, check with the chemical supplier.

9.2.3 Procedure:

A. Pipette 50-mL of the 1.0 *M* sodium sulfite (Na_2SO_3) into the beaker.

B. Pipette 10-mL of the stock formaldehyde solution to a beaker.

C. Titrate the solution with the standard 0.02 *N* H_2SO_4 to an endpoint of pH of 9.5.

9.2.4 Record the volume of acid used to the nearest 0.01-mL. (The volume of acid should be approximately 25-mL for 0.02 *N* acid.)

Calculations:

$$C = (30{,}030)\ (A)\ (N)/10$$

where:

C = Wt/Vol concentration of formaldehyde (µg/mL)
A = Vol of acid used (mL)
N = Normality of acid

9.3 Repeat the titration standardization in duplicate or triplicate. Average the results.

9.4 Use the determined concentration in preparing the calibration curve for the colorimetric
analysis.

10. Test Specimens

10.1 Samples should immediately be placed into separate zipper-type closure plastic bags when sampled. As an extra precaution for maintaining their original state the samples may be wrapped in aluminum foil before being placed in the plastic bags.

10.2 Cut approximately 1-g specimens; weigh each one to ± 0.01 g. Perform in duplicate for each sample. If there is a delay in testing the 1-g specimens once they are cut, they should be put back into the zipper-type closure plastic bags (and rewrapped in foil if previously done) until testing commences.

11. Procedures

11.1 Pipette 100.0-mL of deionized water into each Erlenmeyer flask. Place a fabric specimen in each flask. The fabric should be completely submerged in the water and not float on the surface (see 14.6). A stirring rod may be used to push and keep the fabric below the water surface. Cap the flasks and place them in the water bath at 40 ± 1°C (105 ± 2°F) for 60 ± 2 min. Shake the flasks every 5 min by hand or use an ultrasonic water bath for agitation. If using an ultrasonic bath, check the water frequently for temperature changes.

11.2 Remove the flasks from the water bath.

11.3 Remove the fabric from the flasks. Recap the flasks and shake them to mix any condensation formed on the sides. Allow solutions to cool for 30 min.

11.4 Pipette 5-mL of Nash reagent into a suitable number of test tubes, small (50-mL) Erlenmeyer flasks, or other suitable flasks (colorimeter or spectrophotometer tubes can be used directly, see 14.4) and pipette 5-mL of the reagent into at least one additional tube for a reagent blank. Add 5-mL aliquots from each of the sample incubation jars to the tubes. Reagent blanks are prepared by adding 5-mL of the Nash reagent and 5-mL of deionized water to the tubes.

11.5 Mix and place the tubes in a 40 ± 1°C (105 ± 2°F) water bath for 30 ± 1 min. Remove tubes from bath. Cap with stoppers or plastic paraffin film and allow to cool to room temperature away from direct light.

11.6 Read the absorbance in the spectrophotometer against the reagent blank at 412 nm. **Caution:** Exposure of the developed yellow color to direct sunlight for a period of time will cause some fading. If there is appreciable delay in reading the tubes after color development and strong sunlight is present, care should be exercised to protect the tubes from light. Otherwise the color is stable for considerable time (at least overnight) and reading may be delayed (see 14.7 and 14.8).

11.7 Determine the µg/mL formaldehyde (HCHO) in the sample solutions using the prepared calibration curve (see 8.3 and 14.9).

12. Calculations and Report

12.1 Calculate the amount of formaldehyde released for each specimen to the nearest µg/g using the following equation:

$$F = (C)\ (100)/W$$

where:

F = concentration of formaldehyde (µg/g),
C = concentration of formaldehyde in solution as read from the calibration curve
W = weight of the test specimen, g.

12.2 If performing both this test and AATCC TM112, care should be used in reporting calculated results as this test uses 100-mL of water and AATCC TM112 uses 50-mL.

13. Precision and Bias

13.1 *Precision*.

13.1.1 *Interlaboratory tests*. One inter-

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laboratory study (ILS) of AATCC TM206 was conducted in 2015. Some of the three participating laboratories used two operators, each of whom tested multiple specimens of the same material. Single operators in each participating laboratory ran minimum of two determinations on the same fabric having low formaldehyde levels in the range of 50-100 µg/mL. The results from all participants were evaluated using basic descriptive statistics and analyzed by analysis of variance (ANOVA).

13.1.2 Calculation of components of variance (see Table I) was thought to be appropriate and critical differences (see Table II) and confidence limits (see Table III) were calculated for this low level-formaldehyde fabric as shown below.

13.1.3 When two or more laboratories wish to compare test results, it is recommended that laboratory level be established between them prior to beginning test comparisons.

13.1.4 If comparisons are made between laboratories on a single fabric containing a low level of free formaldehyde content, the critical differences in the column, single level, in Table II should be used.

13.1.5 The number of determinations per laboratory average also determines the critical difference.

13.2 *Bias*.

13.2.1 The formaldehyde release of a fabric can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating free formaldehyde content from a fabric under the conditions of AATCC TM206, the method has no known bias.

13.2.2 AATCC TM206 generally is accepted by the textile and apparel industries as a referee method.

Table I—Components of Variance as Standard Deviations, Expressed in Percentage Points

Component	Variance Fabric 1
Laboratory	2.4
Operator	1.1
Specimen	2.2

Table II—Critical Differences—95.0% Confidence, Expressed in Percentage Points

Observations in Avg (*N*)	Single Operator	Within Laboratory	Between Laboratory
1	13.3	6.8	9.6
2	6.7	5.3	8.6
3	4.4	4.7	8.2
4	3.3	4.3	8.0
5	2.7	4.1	7.9
6	2.2	3.9	7.8

Note: Critical differences were calculated using $z = 1.960$.

Table III—Confidence Limits—95.0% Confidence, Expressed in Percentage Points

Observations in Avg (*N*)	Single Operator	Within Laboratory	Between Laboratory
1	9.4	4.8	6.8
2	4.7	3.7	6.1
3	3.1	3.3	5.8
4	2.4	3.0	5.7
5	1.9	2.9	5.6
6	1.6	2.8	5.5

Note: Confidence limits were calculated using $z = 1.960$.

14. Notes

14.1 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; ordering@aatcc.org; www.aatcc.org.

14.2 Formaldehyde is a component of, or precursor to, many fabric finishes. As part of the application the finish is cured, which may result in self-crosslinking of the finish and/or crosslinking of the finish with the fabric. Incomplete crosslinking, or hydrolysis in later storage, can result in the presence of hydrolyzed formaldehyde on the fabric. This test is to determine the amount of free and hydrolyzed formaldehyde. Warm moist conditions can (further) hydrolyze cross-linked finishes and release more formaldehyde. AATCC TM112 determines free and hydrolyzed formaldehyde and additional formaldehyde as generated under more severe conditions than this test method. The appropriate test should be chosen based on that distinction.

14.3 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; tel: 513-742-2020; www.acgih.org.

14.4 The ratio of reagent to sample solutions can be adjusted, within limits, to suit the individual absorbance range and optical path length of the sampling tubes or cuvettes of the particular photometric instrument being used. For example, although 5-mL reagent to 5-mL sample solution has been found convenient for several types of instruments, other 1:1 ratios, such as a ratio of 2-mL reagent to 2-mL sample may be more suitable for others. The same ratio should be used with the standards as with the sample. The use of spectrophotometer tubes directly for color developments avoids the transfer step of test tubes to spectrophotometer cells, and may save considerable time when many determinations are to be made. Repipettes or similar devices can be used for reagent dispensing, and disposable tip automatic pipettes can be used for preparing the sample solutions.

14.5 J. Frederick Walker, *Formaldehyde*, 3rd Ed. Reinhold Publication Co., New York, 1964, p486. An alternate reference is the *Analytical Methods for a Textile Laboratory*, Third Edition, 1984, edited by J. William Weaver (see 14.1).

14.6 It is important that the fabric be totally submerged in the water during the extraction process. If necessary, use a non-formaldehyde containing surfactant. A concentration of 0.2% surfactant is recommended.

14.7 If extracted solutions are colored in flask, refer to publication by Dilip Pasad "Optimization of the AATCC Sealed Jar and HPLC Methods for Measurement of Low Levels of Formaldehyde," *AATCC Textile Chemist and Colorist*, June 1989, Volume 21, No 6 (see 14.1).

14.8 If the extracted solutions contain particulate matter, the solutions should be centrifuged or filtered to remove contaminant.

14.9 The procedure in Section 10 has been set up to cover the range from about 0 µg/g on the weight of the fabric to about 500 µg/g. In fabrics containing hydrolyzed formaldehyde in the range exceeding 1.0 Absorbance unit, it is recommended that a ratio of 10-mL Nash reagent to 1-mL sample be used. If this is done, it is necessary to prepare an additional calibration chart with 10:1 ratios of standard solutions to Nash reagent by diluting 1, 3, 5, 10, 15 and 20 µg/mL, respectively, of the approximately 1500 µg/mL standardized stock solution of formaldehyde to the mark with deionized water in 500-mL volumetric flasks. Formaldehyde solutions containing approximately 3, 9, 15, 30, 45 and 60 µg/mL respectively will be obtained (see 8.3).

15. History

15.1 Revised in 2020 to update multiple sections for clarity.

15.2 Reaffirmed 2017; editorially revised 2018, 2020 (with title change).

15.3 Developed in 2016 by AATCC Committee RA45.

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Test Method for Seam Twist in Garments Before and After Home Laundering

Developed in 2017 by AATCC Committee RA42; revised 2019.

Foreword

A preliminary version of this test method was published as TS-004 in the 2004 AATCC/ASTM International (Committee D13) *Concept 2 Consumer Technical Supplement*. It is now being published as a standard test method under the jurisdiction of AATCC Research Committee RA42, Dimensional Changes Test Methods.

Seam twist in garments may be influenced by the direction of the fabric in garment panels, the manner of garment assembly, as well as skewness change of the fabric.

AATCC TM207 is for evaluation of seam twist *before and after* home laundering. AATCC TM179, Skew Change of Fabrics After Home Laundering, is used to evaluate the *change* in fabric skew *after* home laundering. ASTM D3882, Standard Test Method for Bow and Skew of Woven and Knitted Fabrics is only used to evaluate fabric skew in its original state (*before* laundering). The relationship among results of these methods is unknown; results of one method should not be compared to those of another method.

1. Purpose and Scope

1.1 This test method determines the seam twist of garments before, as well as after, home laundering.

1.2 This test procedure is specifically intended for garments with vertical seams, such as tops and pants, but may be applied to other end products such as curtains, drapes, or pillowcases. The method is applicable to garments constructed from knitted and/or woven textiles suitable for home laundering.

2. Principle

2.1 Displacement of seams from their locations along natural folds is measured before and after laundering. Seam twist is calculated as a percentage of original seam length. A value close to zero indicates little or no twist; a higher value indicates greater twist; the calculations in this method do not produce negative twist values.

3. Referenced Documents

3.1 AATCC LP1, Home Laundering: Machine Washing (see 16.1).

3.2 AATCC LP2, Home Laundering: Hand Washing (see 16.1).

3.3 AATCC TM133, Colorfastness to Heat: Hot Pressing (see 16.1).

3.4 AATCC TM143, Appearance of Apparel and Other Textile End Products after Home Laundering (see 16.1).

3.5 AATCC TM150, Dimensional Changes of Garments after Home Laundering (see 16.1).

3.6 AATCC TM179, Skew Change of Fabrics After Home Laundering (see 16.1).

3.7 ASTM D123, Standard Terminology Relating to Textiles (see 16.2).

3.8 ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 16.2).

3.9 ASTM D3882, Standard Test Method for Bow and Skew of Woven and Knitted Fabrics (see 16.2).

3.10 ISO 6330, Textiles—Domestic washing and drying procedures for textile testing (see 16.3).

4. Terminology

4.1 **laundering,** *n.—of textile materials*, a process intended to remove soils and/or stains by treatment (washing) with an aqueous detergent solution and normally including rinsing, extraction and drying.

4.2 **seam twist,** *n.*—a rotation, usually lateral, between different panels of a garment connected by a seam.

NOTE: Seam twist may also be referred to as twist, torque or spirality.

5. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

5.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

5.2 Detergent may cause irritation. Care should be taken to prevent exposure to skin and eyes.

5.3 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

6. Uses and Limitations

6.1 This method is applicable to vertical seams aligned with the side folds of a garment when laid on a flat surface. The folds are used as a reference for seam displacement measurements. The side seams of most garments, and inseams of pants, are typically evaluated.

6.2 This test method is applicable to seams oriented with the vertical direction, regardless of location (e.g., side seams, center seams, inseams, etc.) It is not applicable to seams not oriented with the vertical direction, such as curved seams.

6.3 Seam length, fabric construction, and garment assembly, individually or in combination, influence seam twist. Results obtained from matching seams, with identical seam constructional details (i.e. stitch type, seam shape, stitches/inch, etc.) as, for example, from the left and right sides of an item, may be averaged. However, results obtained from seams with different seam constructional details, even within the same garment, should not be compared, or averaged.

6.4 Different laundering temperatures, machine specifications or load sizes may result in different seam twist results. Use caution in comparing results obtained using different testing parameters.

7. Apparatus and Materials (see 16.4)

7.1 Conditioning facilities to accommodate hanging garments and/or conditioning/drying racks with pull-out, perforated shelves for laying garments flat (see 16.1).

7.2 Indelible ink marking pen (see 16.1).

7.3 Tape or rule marked in millimeters, eighths, or tenths of an inch.

7.4 Standard washing machine (see Table I and 16.5).

7.5 AATCC 1993 Standard Reference Detergent (powder, with brightener) (see 16.1).

7.6 Balance with at least 5.0 kg (10.0 lb) capacity.

7.7 Laundering ballast, Type 1 or Type 3 (see Table IV).

7.8 Standard tumble dryer (see Table

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III and 16.5), or facilities for drip/line drying.

8. Test Specimens

8.1 Consider individual items as specimens. When available, test three specimens per manufacturing lot.

8.2 The specimens tested by this method may also be tested at the same time for appearance and/or dimensional change in accordance with AATCC TM143, AATCC TM150, and/or AATCC TM179.

9. Conditioning

9.1 Prior to testing, condition test specimens as directed in ASTM D1776. Use conditions indicated in Table 1 for Textiles, general and conditioning time as indicated in Table 2 for the appropriate fiber content.

9.1.1 Lay specimens flat, separately, on a screen or perforated shelf of a conditioning rack. For garments normally stored hung, hang separately on hangers to condition. Allow space for air movement on all sides of the specimens.

9.2 Specimens will be conditioned again after laundering.

10. Measurement before Laundering

10.1 Determine the seam(s) to evaluate for twist (see 6.1). Evaluation of all vertical seams is recommended to get a complete picture of garment appearance.

10.2 After conditioning, lay one test specimen on a flat, smooth, horizontal surface in its natural configuration. Arrange specimen so the first seam to be measured is visible (if the seam has twisted to the back of the specimen, lay the specimen face down). It may be helpful to gently shake the specimen to allow it to hang loosely, but do not stretch or distort the specimen in shaking. Smooth, but do not stretch, all panels, including seams and hems.

10.3 Mark Point *A* at the lower edge of the seam, including hem. Mark Point *B* 25 mm (1 in.) below the point at which the seam intersects another seam. For example, mark Point *B* of pant inseam 25 mm below intersection with rise seam; mark Point *B* of shirt side seam 25 mm below intersection with armhole (see Figs. 1 and 2).

10.4 Measure and record seam length *AB* to the nearest millimeter, or eighth or tenth of an inch.

10.5 Determine if seam twist exists before laundering.

10.5.1 If the seam falls along the natural fold of the garment, there is no seam twist (see Figs. 1 and 2). Record original twist distance as 0 mm or 0.0 in.

10.5.2 If the seam does *not* fall along the natural fold of the garment, mark Point *A′* at the lower edge of the natural fold, including hem (see Figs. 3 and 4).

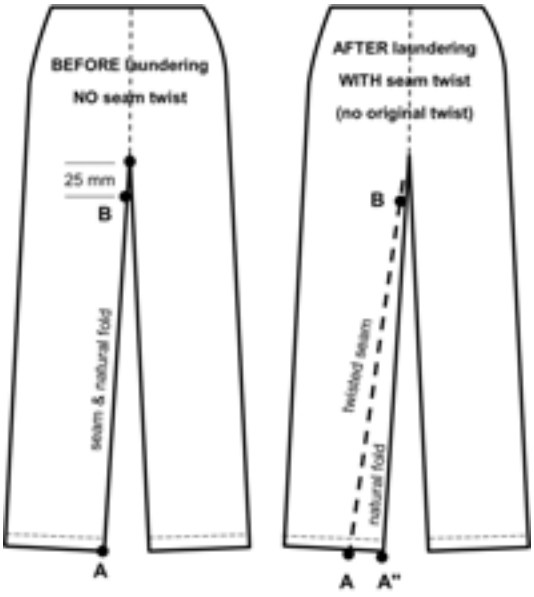


Fig. 1—Pants, before and after laundering, no original seam twist. Garment has no seam twist before laundering; length AB coincides with both seam and natural fold. Garment after laundering has side seam twist (in practice, it is likely that vertical seams in other locations of the same garment will also exhibit twist).

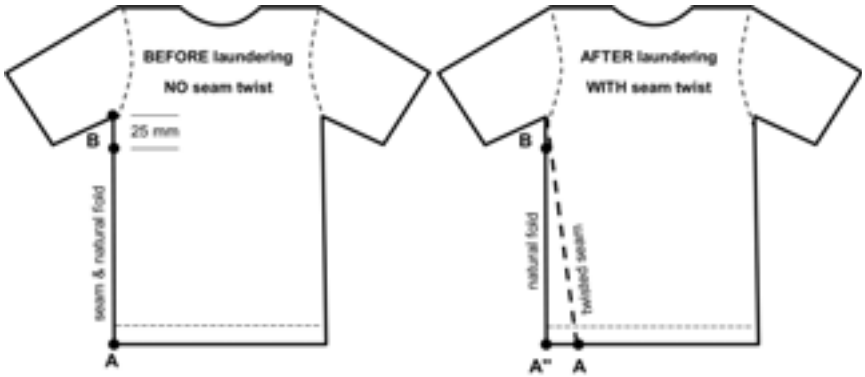


Fig. 2—Shirts, before and after laundering, no original seam twist. Garment has no seam twist before laundering; length AB coincides with both seam and natural fold. Garment after laundering has side seam twist (in practice, it is likely that vertical seams in other locations of the same garment will also exhibit twist).

10.5.3 Measure and record original twist distance *AA′* for each seam to the nearest millimeter, or eighth or tenth of an inch.

10.6 Repeat 10.2 through 10.5 for remaining seams on all specimens.

11. Laundering Procedure

11.1 Washing.

11.1.1 Select washing conditions for testing from Table I. Set washing machine controls to generate the selected cycle parameters.

11.1.2 Load for laundering shall consist of all specimens in the sample, plus enough laundering ballast pieces to make a total load weight of 1.8 ± 0.1 kg (4.0 ± 0.2 lb).

11.1.3 Begin selected wash cycle. Allow machine to fill to specified water level.

11.1.4 Add 66 ± 1 g of AATCC 1993 Standard Reference Detergent to washing machine as directed by washing machine manufacturer's instructions. If detergent

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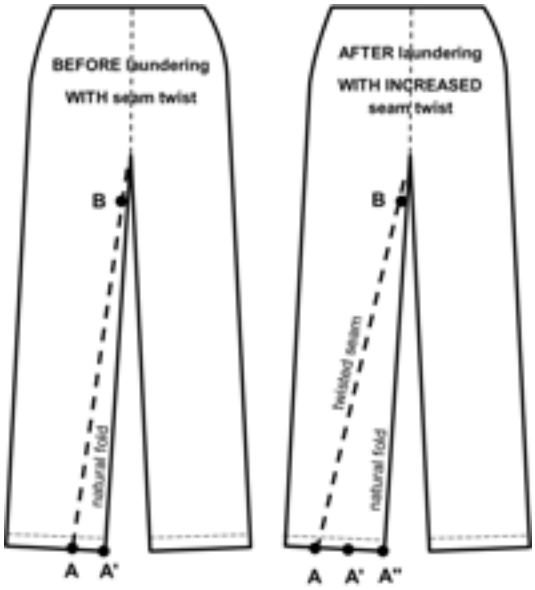


Fig. 3—Pants, before and after laundering, with original seam twist and increased seam twist after laundering (in practice, it is likely that vertical seams in other locations of the same garment will also exhibit twist).

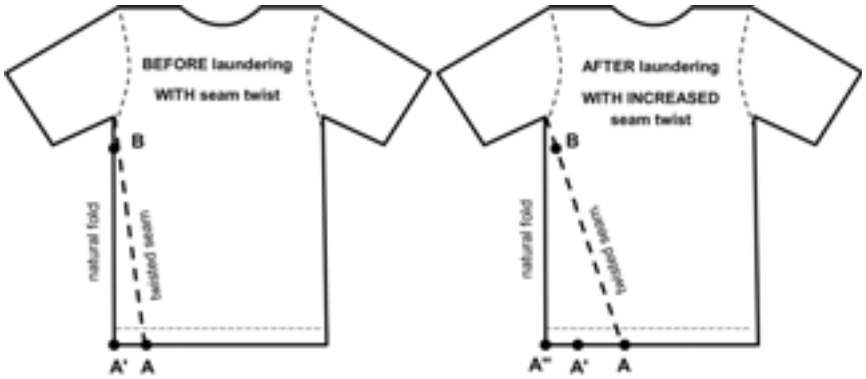


Fig. 4—Shirts, before and after laundering, with original seam twist and increased seam twist after laundering (in practice, it is likely that vertical seams in other locations of the same garment will also exhibit twist).

is added directly to wash water, agitate briefly to dissolve completely. Stop agitation before adding wash load.

11.1.5 Add wash load (test specimens and ballast), distributing evenly around center agitator. Restart wash cycle.

11.1.6 For specimens to be drip dried (drying procedure C), stop wash cycle before the water begins to drain from the final rinse cycle. Remove specimens soaking wet. For specimens to be tumble (A), line (B), or screen/flat (D) dried, allow washing to proceed through the final spin cycle.

11.1.7 After each washing cycle, separate tangled specimens and ballast pieces, taking care to minimize distortion. Proceed to appropriate drying procedure.

11.2 Drying.

11.2.1 Select drying conditions from Table II.

11.2.2 (A) Tumble dry. Place the washed load (test specimens and ballast) in the tumble dryer, and set the temperature control to generate the selected exhaust temperature (see Table III).Allow the dryer to run until the load is fully dry. Remove specimens immediately.

11.2.3 (B) Line/hang dry. Hang each specimen on an appropriate hanger, straightening and smoothing facings, seams, etc. Specimens should generally be oriented in the direction in which they are worn. Do not fold or stretch specimens over the hanger. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

11.2.4 (C) Drip dry. Hang dripping wet specimen on an appropriate hanger, straightening and smoothing facings, seams, etc. Specimens should generally be oriented in the direction in which they are worn. Do not fold or stretch specimens over the hanger. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

11.2.5 (D) Screen/flat dry. Spread each specimen on a horizontal screen or perforated surface, removing wrinkles without distorting or stretching it. Allow specimens to stand in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

11.2.6 For all drying methods, allow specimens to dry completely before washing again.

11.2.7 Repeat the selected washing and drying procedure for a total of 3 complete cycles, or an agreed upon number of cycles.

11.3 Conditioning.

11.3.1 After final drying cycle is complete, condition test specimens as described in section 9.

11.4 Ironing (optional).

11.4.1 The wrinkles in most fabrics flatten sufficiently under pressure of a measuring instrument at the time of measurement, so as not to cause measurement error. If the specimens are extremely wrinkled and the consumer would ALWAYS expect to iron a garment made from the fabric, test specimens may be hand ironed prior to measurement.

11.4.2 Use safe ironing temperatures appropriate to the fibers in the fabric being ironed. See AATCC TM133, Table I, Safe Ironing Temperature Guide (see 16.1). Exert only the minimum pressure necessary to remove wrinkles.

11.4.3 Due to the extremely high variability of hand ironing procedures performed by individual operators (no standard test method exists for hand ironing procedures), the reproducibility of dimensional change results after hand ironing has been found to be extremely poor. Consequently, caution is advised when comparing dimensional change results after laundering and hand ironing, reported by different operators.

11.4.4 After ironing, condition test specimens as described in section 9.

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Table I—Standard Washing Machine Parameters (see 16.5, 16.6)

Cycle	(1) Normal	(2) Delicate	(3) Permanent Press
Water Level, L (gal)	72 ± 4 (19 ± 1)	72 ± 4 (19 ± 1)	72 ± 4 (19 ± 1)
Agitation Speed, strokes/min.	86 ± 2	27 ± 2	86 ± 2
Washing Time, min.	16 ± 1	8.5 ± 1	12 ± 1
Final Spin Speed, rpm	660 ± 15	500 ± 15	500 ± 15
Final Spin Time, min.	5 ± 1	5 ± 1	5 ± 1
Wash Temp, °C (°F)[1]	**(II) Cold:** 27 ± 3 (80 ± 5)	**(II) Cold:** 27 ± 3 (80 ± 5)	**(II) Cold:** 27 ± 3 (80 ± 5)
	(III) Warm: 41 ± 3 (105 ± 5)	**(III) Warm:** 41 ± 3 (105 ± 5)	**(III) Warm:** 41 ± 3 (105 ± 5)
	(IV) Hot: 49 ± 3 (120 ± 5)	**(IV) Hot:** 49 ± 3 (120 ± 5)	**(IV) Hot:** 49 ± 3 (120 ± 5)
	(V) Very Hot: 60 ± 3 (140 ± 5)	**(V) Very Hot:** 60 ± 3 (140 ± 5)	**(V) Very Hot:** 60 ± 3 (140 ± 5)

[1] Due to US Department of Energy requirements, many washing machines use cooler water. An external control box may be used to override the machine set temperatures.

Table II—Standard Drying Conditions

(A)	Tumble Dry
(Ai)	Tumble Dry Normal
(Aii	Tumble Dry Delicate
(Aiii)	Tumble Dry Permanent Press
(B)	Line/Hang Dry
(C)	Drip Dry
(D)	Screen Dry/Dry Flat

Table III—Standard Tumble Dryer Parameters (see 16.5)

Cycle	(Ai) Normal	(Aii) Delicate	(Aiii) Permanent Press
Max. Exhaust Temp, °C (°F)	68 ± 6 (155 ± 10)	60 ± 6 (140 ± 10)	68 ± 6 (155 ± 10)
Cool Down Time, min.	≤ 10	≤ 10	≤ 10

Table IV—Laundering Ballast Parameters

	Type 1	Type 3
Fiber Content	100% cotton	50% cotton/50% polyester ± 3%
Greige Fabric Yarns	16/1 ring spun	16/1 or 30/2 ring spun
Greige Fabric Construction	52 × 48 ± 5 yarns/in., plain weave	52 × 48 ± 5 yarns/in., plain weave
Finished Fabric Weight	155 ± 10 g/m^2	155 ± 10 g/m^2
Edges	All edges hemmed or over-edged	All edges hemmed or over-edged
Finished Piece Size	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)
Finished Piece Weight	130 ± 10 g	130 ± 10 g

12. Measurement after Laundering

12.1 After conditioning, lay one test specimen on the same surface used for marking and measurement in section 10. Use the same measuring device as for initial marking and measurement. Arrange specimen in its natural configuration, so the first seam to be measured is visible (if the seam has twisted to the back of the specimen, lay the specimen face down). It may be helpful to gently shake the specimen to allow it to hang loosely, but do not stretch or distort the specimen in shaking. Smooth, but do not stretch, all panels, including seams and hems.

12.2 If the seam falls along the natural fold of the garment, there is no seam twist. Record after-laundering twist distance as 0 mm or 0.0 in.

12.2.1 If the seam does *not* fall along the natural fold of the garment, mark Point *A″* at the lower edge of the natural fold (see Figs. 3 and 4).

12.3 Measure and record after-laundering twist distance *AA″* for each seam to the nearest millimeter, or eighth or tenth of an inch.

12.4 Repeat 12.1-12.3 for remaining seams on all specimens.

13. Calculations and Interpretation

13.1 For each seam evaluated, calculate percent seam twist before laundering, if present, to the nearest 0.1% using equation 1:

$$X_0 = 100\% \times (AA'/AB) \qquad \text{(Eq. 1)}$$

where:

X_0 = seam twist before laundering (%)
AB = seam length before laundering (mm or in.)
AA' = twist distance before laundering (mm or in.)

13.2 Average the percent seam twist before laundering for all matching seams (e.g., left and right inseams for all specimens in the sample).

13.3 For each seam evaluated, calculate percent seam twist after laundering to the nearest 0.1% using equation 2 (note that seam length is measured *before* laundering):

$$X_n = 100\% \times (AA''/AB) \qquad \text{(Eq. 2)}$$

where:

X_n = seam twist after n laundering cycles (%)
AB = seam length before laundering (mm or in.)
AA'' = twist distance after n laundering cycles (mm or in.)

13.4 Average the percent seam twist after laundering for all like seams (e.g., left and right inseams for all specimens in the sample).

13.5 Calculate change in seam twist after laundering using equation 3:

$$X_c = X_n - X_0 \qquad \text{(Eq. 3)}$$

where:

X_c = change in seam twist after laundering (%)

13.6 It is recommended that equation 2 be used when determining if items being tested meet percent seam twist specified in product performance specifications.

14. Report

14.1 Report for each sample tested:

14.1.1 A description or identification of the sample.

14.1.2 Evaluation was performed using AATCC TM207-2019.

14.1.3 Type of ballast used, i.e., Type 1 or Type 3.

14.1.4 Number of laundering cycles (default is 3 cycles, see 11.2.7).

14.1.5 Laundering conditions, including washing machine cycle, wash temperature, drying procedure and tumble-drying temperature, if applicable. Alphanumeric designations may be used if they are clearly understood by all parties. For example, 1-IV-A(ii) indicates normal wash cycle at 49°C and Delicate tumble dry cycle.

14.1.6 If specimens were distorted or wrinkled in their original state.

14.1.7 If specimens were hand ironed.

14.1.8 Any modifications of the test method.

14.2 Report for each seam type evaluated:

14.2.1 Seam description or location. (Example: Pants may have a double-needle, felled side seam while the inseam may

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Table V—Precision Values

n	Within-Laboratory	Between-Laboratory
1	0.5489	2.5940
2	0.3887	1.8371
3	0.3169	1.4976
4	0.2745	1.2970
5	0.2455	1.1601
6	0.2241	1.0590

Note: Grand mean of seam twist after laundering for all interlab measurements was 1.9%. Mean seam twist for jersey leggings was 0.9%. Mean for jersey T-shirt was 4.3%. Mean for denim jeans was 1.1%.

be a single seam with a safety stitch.)

14.2.2 Number of like seams included in average (3 specimens, each with 2 like seams, for a total of 6 seams, is default).

14.2.3 Average percent seam twist before laundering (X_0).

14.2.4 Average percent seam twist after laundering (X_n).

14.2.5 Percent change in seam twist (X_c) after laundering.

15. Precision and Bias

15.1 *Precision.* Four different laboratories participated in an interlaboratory study to measure seam twist after home laundering (twist was not evaluated before laundering). All 4 labs measured the seam twist of 3 different garments (jersey leggings, jersey T-shirt, denim jeans), following this test method. Each lab used only 1 operator and repeated the test 2 times, on separate days. Table V gives the *within-lab* and *between-lab* precision values for this study. Laundering was performed in traditional top-loading washing machines with 1993 AATCC Standard Reference Detergent (powder), and tumble dryers. Precision of tests performed with other laundering conditions has not been determined.

15.2 *Bias*. Seam twist resulting from home laundering can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

16. Notes

16.1 Available from AATCC, PO Box 12215, Research Triangle Park, NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www. aatcc.org.

16.2 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428, USA; +1.610.832. 9500; www.astm.org.

16.3 Available from ISO, www.iso.org.

16.4 For potential equipment information pertaining to this test method, please visit the online AATCC Buyer's Guide at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

16.5 For model number(s) of washer(s) and tumble dryers(s) reported to meet the standard parameters, visit www.aatcc.org/test/washers or contact AATCC, PO Box 12215, Research Triangle Park, NC 27709, USA; +1.919. 549.8141; www.aatcc.org. An alternate load size was described in previous versions of this method (3.6-kg load, 83-L water level, 80 g AATCC 1993 Standard Reference Detergent), but no machine is reported to meet the standard parameters for this load size. In addition, skew change results obtained with the alternate load size may not be equal to those obtained with a standard load size.

16.6 The laundering temperatures and other parameters listed in this test method are standard conditions for testing purposes. As with most lab procedures, they represent, but may not exactly replicate, current consumer practices. Consumer practices vary over time and among households; lab practices must be consistent to allow valid comparison of results. If laundering equipment or conditions other than those specifically listed in this test method are used, they must be described in detail and noted as a modification of the standard method. Alternate laundering conditions are outlined in AATCC LP1, AATCC LP2, and ISO 6330.

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Test Method for Water Resistance: Hydrostatic Pressure Using a Restraint

Developed in 2017 by AATCC Committee RA63; reaffirmed and editorially revised 2019. Related to ISO 811.

1. Purpose and Scope

1.1 This test method measures the resistance of a fabric to the penetration of water under hydrostatic pressure while restrained by an acrylic plate. It is applicable to all types of fabrics, including those treated with a water resistant or water repellent finish.

1.2 Water resistance depends on the repellency of the fibers and yarns, as well as the fabric construction.

1.3 The results obtained by this method may not be the same as the results obtained by the AATCC methods for resistance to rain or water spray. Results may not be the same as those obtained by hydrostatic water resistance testing *without* a restraint.

2. Principle

2.1 One surface of the test specimen is subjected to a hydrostatic pressure, increasing at a constant rate, until three points of leakage appear on its other surface. The water may be applied from above or below the test specimen.

2.2 This test method is the same as AATCC TM127, Test Method for Water Resistance: Hydrostatic Pressure except a restraint is used to prevent deformation of the fabric.

3. Terminology

3.1 **hydrostatic pressure,** n.—the force distributed over an area exerted by water.

3.2 **water resistance,** n.—*of fabric*, the characteristic to resist wetting and penetration by water.

3.3 **water repellency,** n.—*in textiles*, the characteristic of fiber, yarn, or fabric to resist wetting.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details, such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

5. Apparatus and Materials (see 11.1)

5.1 Hydrostatic Tester.

5.1.1 Option 1, Hydrostatic Pressure Tester (see 11.2).

5.1.2 Option 2, Hydrostatic Head Tester (see 11.3).

5.2 Water, distilled or de-ionized.

5.3 Restraint: 200 × 200 ± 10 mm, made of 6.0 ± 0.5 mm thick clear cast acrylic. Other restraint materials may be used with the agreement of all parties.

5.4 Paper towel: single ply, with no finish, 200 × 200 ± 10 mm, a color that will exhibit a color change when wet.

6. Test Specimens

6.1 A minimum of three fabric specimens should be taken diagonally across the width of the fabric to be representative of the material. Cut specimens at least 200 × 200 mm to allow proper clamping.

6.2 Handle the specimens as little as possible and avoid folding or contaminating the area to be tested.

6.3 Condition the test specimens at 21 ± 2°C (70 ± 4°F) and 65 ± 5% RH for at least 4 h before testing.

6.4 The surface of the fabric to be exposed to water must be specified because different results may be obtained on the face and the back. Identify that surface on a corner of each specimen.

7. Procedure

7.1 Verify the water in contact with the test specimen is regulated at 21 ± 2°C (70 ± 4°F) (see 11.4).

7.2 Dry the clamping surface.

7.3 Position the specimen with the surface to be tested facing the water (see 11.6). Place the paper towel on the side of the fabric not facing the water, then place the restraint so the paper towel is sandwiched between the fabric and restraint.

7.4 Clamp the fabric, paper and restraint assembly in place (see 11.5). Clamp slowly to avoid a sudden increase in pressure that may affect results.

7.5 Operation.

7.5.1 Option 1—Hydrostatic Pressure Tester. Turn on the motor, press the lever to raise the overflow device at the rate of 10 mm/s, and close the air vent as soon as water flows from it.

7.5.2 Option 2—Hydrostatic Head Tester. Select the gradient of 60 mbar/min, press the start button (see 11.6).

7.6 Disregarding water droplets that appear within approximately 3 mm of the specimen clamping ring, record the hydrostatic pressure at the moment water droplets penetrate the fabric in three different places. Droplets will be visible as darkened spots on the paper towel (see 11.7).

8. Calculation

8.1 Calculate the average hydrostatic pressure for each sample.

9. Report

9.1 Results for each specimen and the average for each sample.

9.2 The material and the side tested.

9.3 Water temperature and type.

9.4 Gradient (rate of increasing water pressure).

9.5 Tester option used.

9.6 Any modification to the method.

10. Precision and Bias

10.1 *Precision*.

10.1.1 In 2017, a single-laboratory study was completed with two fabrics (see Table I). Fabric A was a bottom-weight woven twill with a durable water-repellent finish. Fabric D was a 43-g/m^2 nonwoven material constructed of spunbond olefin fiber.

10.1.2 Analysis of the data yielded critical differences shown in Table II. Differences between two averages of (n) specimens, for the appropriate precision parameter, should reach or exceed the table value to be statistically significant at

Table I—Precision Study Results (mm water)

Fabric	*n*	Mean	St Dev
A	7	433.7	14.8
D	8	563.6	57.7

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Table II—Critical Differences for Averages, 95% Probability (mm water)

n	Within- Laboratory
1	29.76
2	21.08
3	17.18
4	14.88
5	13.31

the 95% confidence level (see Table II).

10.2 *Bias*.

10.2.1 Water resistance of fabrics can only be defined in terms of a test method. There is no independent, referee method for determining the true value. This test method has no known bias.

11. Notes

11.1 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

11.2 Hydrostatic Pressure Tester (Suter).

11.2.1 The apparatus consists essentially of an inverted conical well equipped with a coaxial ring clamp to fasten the cloth specimen under the well bottom. The apparatus introduces water from above the specimen over an area 114 mm in diameter and at a rate of 10.0 ± 0.5 mm of hydrostatic head per second. A mirror is affixed below the specimen to enable the operator to ascertain penetration of the specimen by drops of water. A valve is provided for venting the air in the well.

11.2.2 Hydrostatic testing apparatus of the type described is no longer available for sale.

11.3 Hydrostatic Head Tester.

11.3.1 The apparatus applies hydrostatic pressure at 60 mbar/min (selectable) to the bottom side of the fabric. A reservoir with a circular test area of 100 ± 5 cm^2 (15.5 ± 0.8 $in.^2$) contains distilled or deionized water which is applied to the fabric surface. The fabric specimen is secured with a clamp.

11.4 Some laboratories use water at ambient temperature. If water is other than 21 ± 2°C (70 ± 4°F), so state.

11.5 Lateral water leakage can be minimized by sealing the fabric with paraffin at the clamping area.

11.6 1 mbar = 10.2 mm H_2O.

11.7 An automated droplet detection system may be used in place of visual observation of water droplets if it is established that its accuracy is equivalent to visual observation.

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Test Method for pH and Total Alkali in Wet Processed Textiles: Combined

Foreword

This test method was written as an alternative to AATCC TM81 and AATCC TM144 to offer a combined method for performing these tests on a single specimen (see 14.1).

1. Purpose and Scope

1.1 This test method is used to determine the pH and total alkali content of wet processed textiles. pH and total alkali can be used to determine washing and/or neutralizing efficiency after certain wet processing steps, particularly bleaching, and can be used as a measure of the suitability of prepared fabric for subsequent dyeing and finishing operations.

1.2 To make a quantitative determination, the chemicals which influence pH must be removed from the textile specimen, collected as a water extract and then accurately measured by a pH meter.

2. Principle

2.1 Each specimen is immersed in boiling, distilled or deionized water. The specimen is boiled for an additional time, then covered and allowed to cool to room temperature. The pH of the bath is measured, then titrated with an acid to a predetermined endpoint. The percentage of total alkali on the textile can then be calculated from the amount of acid used and the weight of the specimen.

3. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

3.1 AATCC TM81, Test Method for pH of the Water-Extract from Wet Processed Textiles (see 14.2).

3.2 AATCC TM144, Test Method for Alkali in Wet Processed Textiles: Total (see 14.2).

3.3 W. W. Scott, *Standard Methods of Chemical Analysis*, 6th Ed., Van Nostrand, New York, 1962, p1343.

4. Terminology

4.1 **bleaching,** n.—elimination of unwanted coloring matter from a textile substrate by oxidative or reductive chemical treatment.

4.2 **pH,** n.—the negative logarithm of the effective hydrogen ion concentration or hydrogen ion activity in gram equivalents per liter used in expressing both acidity and alkalinity on a scale with values from 0 to 14, with 7 representing neutrality, numbers less than 7 representing increasing acidity and numbers greater than 7 representing increasing alkalinity.

4.3 **total alkali,** n.—*in textile wet processing*, residual alkaline materials in a wet processed textile expressed as percent sodium hydroxide on dry weight of the textile.

4.4 **wet processing,** n.—*in textile manufacturing*, a collective term for processes included in preparation, dyeing, printing, and finishing in which the textile material is treated with a liquid, normally water, or with chemicals in solution or dispersion in a liquid.

5. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Users MUST consult manufacturers for specific details such as safety data sheets and other recommendations. Consult and follow all applicable OSHA standards and rules.

5.1 Follow good laboratory practices. Wear safety glasses in all laboratory areas.

5.2 Handle all chemicals with care.

5.3 If concentrated sulfuric acid is diluted to prepare the 0.02*N* sulfuric acid (see 7.5), use chemical goggles or face shield, impervious gloves and an impervious apron. Handle concentrated acids only in an adequately ventilated laboratory hood. CAUTION: Always add acid to water.

5.4 Ensure an eyewash/safety shower is located nearby and a high-efficiency particulate respirator with a full facepiece is readily available for emergency use.

5.5 Control exposure to chemicals used in this procedure at or below levels set by governmental authorities (e.g., Occupational Safety and Health Administration's [OSHA] permissible exposure limits [PEL] as found in 29 CFR 910.1000; see web site: www.osha.gov for latest version). In addition, the American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Values (TLVS) comprised of time weighted averages (TLV-TWA), short term exposure limits (TLV-STEL), and ceiling limits (TLV-C) are recommended as a general guide for air contaminant exposure which should be met (see 14.3).

6. Limitations

6.1 pH and total alkali can be used to determine the suitability of wet processed textiles for subsequent dyeing and finishing operations or to evaluate the washing and/or neutralizing efficiency following any wet processing operation.

6.2 In this test method all alkalis detected, such as sodium hydroxide, sodium carbonate, sodium bicarbonate, and other alkaline salts, are calculated as sodium hydroxide. The results are reported as total alkali, expressed as NaOH.

7. Apparatus and Reagents (see 15.4)

7.1 pH meter, 0.1-unit graduations.

7.2 Beakers, glass, 600 mL.

7.3 Burette, glass, 10 mL, 0.10 mL graduations.

7.4 Buffer solutions, pH 4.0, 7.0, 10.0, or others as needed.

7.5 Sulfuric acid, 0.02*N* H_2SO_4 (see 3.3 for method of preparation of standard acid).

8. Calibration

8.1 Calibrate the pH meter with pH 4.0 buffer solution in accordance with the manufacturer's instructions. For dual calibration instruments, use pH 4.0 buffer and pH 7.0 or pH 10.0 buffer.

9. Specimens

9.1 Specimens may be taken dry or wet.

9.2 Dry cloth. Select two (2) or more representative dried specimens. Place in oven at 100°C (212°F) for 1 h. Cool in a desiccator and weigh to the nearest 0.1 g. Specimens should weigh 5-10 g.

9.3 Wet cloth. Select two (2) or more representative specimens weighing approximately 10-20 g in the wet state.

9.4 If the mass per unit area of the fabric is excessively low, or difficult to wet, cut the specimen into small pieces to promote better immersion in the distilled water.

10. Procedure

10.1 Since distilled or deionized water can contain dissolved carbon dioxide, take care to remove it by boiling before use. Boil 450-500 mL of distilled water

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in a 600-mL beaker at a moderate rate for 10 min. Place each specimen in a separate beaker, cover with a watch glass, and continue boiling for an additional 10 min.

10.2 Allow the covered beaker and contents to cool to room temperature. Stir the specimen, then insert the pH meter electrode(s), being careful to avoid contact with the specimen. Determine the pH of the bath using a pH meter operated according to manufacturer's instructions.

10.3 Titrate the water and specimen with 0.02*N* sulfuric acid (H_2SO_4) to a steady (10 s) pH 3.9 endpoint. Add titrant dropwise followed by gentle agitation of the specimen without touching the electrode(s). Read burette to the nearest graduation (see 14.5 and 14.6).

10.4 If the specimen was taken from a wet sample, rinse and dry the titrated specimen to a constant weight (as in section 9.2) being careful to collect any large threads that may have unraveled during the titration procedure. Weigh to the nearest 0.1 g

10.5 Titrate a water blank by boiling (10 min.) and cooling 450-500 mL of distilled water, then titrating as in 10.3, without a specimen present.

11. Calculation

11.1 Calculate the percentage of total alkali from the titer using Eq. 1.

$$X = [(A-B)(0.04)(N)\ 100]/W \quad \text{(Eq. 1)}$$

where:

X = % weight total alkali as sodium hydroxide (NaOH);
A = Burette reading of specimen;
B = Burette reading of water blank;
N = Normality of sulfuric acid (0.02*N*);
W = Weight of specimen.

11.2 Calculate the mean percentage of total alkali for all specimens.

12. Evaluation

12.1 pH and total alkali of wet-processed textiles depend on the chemical treatment previously given the textile. For example, the pH and alkali would normally be higher after caustic boiling than after bleaching.

12.2 pH and total alkali content are important because of their effect on the subsequent processing operations to which the textile is subjected. Textiles with high pH and/or alkali values may exhibit yellowing tendencies, exhibit shade changes, or alter the exhaustion and fixation of dyestuffs, and could produce a decrease in the cure of resin finishes or exhaustion of softeners.

12.3 Total alkalinity quantifies the amount of residual alkali in a specimen, per unit of weight, provided the specimen contains alkali.

13. Report

13.1 Describe or identify the sample
13.2 Report that sample was testing using AATCC TM209-2022
13.3 Report the average percentage of total alkalinity for all samples

14. Precision and Bias

14.1 *Precision*. Three (3) operators in a single laboratory participated in an intralaboratory study to measure pH and alkalinity using this method. All operators measured 6 different fabrics and repeated each test 2 times, on separate days. Table I gives the within-lab precision values from this study. Table II provides mean and standard deviation values for each fabric, for reference.

14.2 *Bias*. The true values of pH and alkalinity can only be defined in terms of a test method. Within this limitation, this test method has no known bias.

15. Notes

15.1 A single-laboratory study (3 technicians and 6 fabrics) was conducted in 2017. In this study, all three (3) methods were compared. The data from this study indicated that the pH determined by AATCC TM81 was slightly different from the pH determined by this combined method. Specimen weight and water volumes are different for AATCC TM81 and AATCC TM144. This combined method uses parameters similar to those of AATCC TM144; this may be the source of variations in pH results between AATCC TM81 and this method.

15.2 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

15.3 Available from Publications Office, ACGIH, Kemper Woods Center, 1330 Kemper Meadow Dr., Cincinnati OH 45240; +1.513.742.2020; www.acgih.org.

15.4 For potential apparatus, reagents, or materials sources, visit the AATCC Buyer's Guide at www.aatcc.org/bg. AATCC provides its Corporate members the option to their items and services. AATCC does not qualify, or in any way approve, endorse, or certify that any of the listings meet the specifications in its standards.

15.5 If preferred, titration may be made with a suitable colorimetric indicator such as methyl orange (pH 3.1-4.4). The precision may be less with this method than with a good pH meter.

15.6 An auto-titrator may be used if it has been established by the user that its accuracy is equivalent.

16. History

16.1 Revised in 2022 to align to the AATCC Style Guide, and to correctly reference section 7.5.

16.2 Developed in 2019 by AATCC Committee RA34.

Table I—Precision Values

n	Operators	pH (Within Lab)	Alkalinity (Within Lab)
1	1	0.31044	0.02478
1	2	0.43905	0.03505
1	3	0.62091	0.04956
2	1	0.25401	0.01752
2	2	0.31111	0.02478
2	3	0.43997	0.03504
3	1	0.12701	0.01011
3	2	0.17958	0.01429
3	3	0.25401	0.02021
4	1	0.11173	0.00876
4	2	0.15800	0.01239
4	3	0.22345	0.01752
5	1	0.09838	0.00970
5	2	0.13913	0.01372
5	3	0.19676	0.01567

Table II—Precision Study Values

Fabric	Mean pH	St Dev	Mean Alk	St Dev
Twill, Khaki	7.1132	0.2175	0.02185	0.00883
Twill, Navy	7.3803	0.1314	0.02596	0.01010
Twill, White	5.9948	0.1728	0.01035	0.00992
Denim, Indigo	10.104	0.0497	0.00512	0.01255
Denim, Black Overdye	9.3405	0.0829	0.1873	0.0294
Knit, White	9.2165	0.1691	0.07803	0.01288

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Test Method for Electrical Resistance Before and After Various Exposure Conditions

Foreword

This test method provides guidance for testing durability of e-textile resistance. The method includes a procedure for testing electrical resistance as well as modifications of established industry methods for laundering, drycleaning, water, perspiration, acids and alkalis, ultraviolet (UV) radiation, and microbe exposure. Test exposures relevant to the expected end use of sample. Similar modifications to other established methods may also be possible. Additional modifications may be required to accommodate large or unusually-shaped samples that cannot be cut into standard swatches. If the referenced industry methods are also used for their original purpose (e.g., colorfastness), use a separate set of specimens.

1. Purpose and Scope

1.1 This test method is used to determine the electrical resistance of e-textiles before and after exposure to laundering, drycleaning, water, perspiration, acids and alkalis, ultraviolet (UV) radiation, and/or microbes.

1.2 this test method is applicable to e-textile fabrics or end products with incorporated (woven, knitted, printed, embroidered, etc.) conductive paths/traces. It is not for evaluation of yarns or fibers.

2. Principle

2.1 Electrical resistance is measured along a conductive path or segment of a conductive path before and after various exposure conditions. Any or all of the provided exposure methods may be used: laundering, drycleaning, water, perspiration, acids and alkalis, ultraviolet (UV) radiation, and microbes. Resistance is reported in ohms (Ω), where a higher value indicates more resistance (less conductance). Change in resistance after exposure is also calculated and reported as percent change, where a higher value indicates more change (typically a greater increase in resistance).

3. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

3.1 AATCC EP13, Evaluation Procedure for Electrical Resistance of Electronically-Integrated Textiles (see 16.1).

3.2 AATCC LP1, Laboratory Procedure for Home Laundering: Machine Washing (see 16.1).

3.3 AATCC LP2, Laboratory Procedure for Home Laundering: Hand Washing (see 16.1).

3.4 AATCC TM6, Test Method for Colorfastness to Acids and Alkalis (see 16.1).

3.5 AATCC TM15, Test Method for Colorfastness to Perspiration (see 16.1).

3.6 AATCC TM22, Test Method for Water Repellency-Spray Test (see 16.1).

3.7 AATCC TM100, Test Method for Antibacterial Finishes on Textile Materials: Assessment of (see 16.1).

3.8 AATCC TM106, Test Method for Colorfastness to Water: Sea (see 16.1).

3.9 AATCC TM127, Test Method for Water Resistance: Hydrostatic Pressure Test (see 16.1).

3.10 AATCC TM158, Test Method for Dimensional Changes on Drycleaning in Perchloroethylene: Machine Method (see 16.1).

3.11 AATCC TM186, Test Method for Weather Resistance: UV Light and Moisture Exposure (see 16.1).

3.12 ASTM D1776, Standard Practice for Conditioning and Testing Textiles.

4. Terminology

4.1 **electronically-integrated textile (e-textile)**, n. —a fabric or textile product with permanently-integrated electrical circuits or parts of electrical circuits.

4.2 **electrical resistance**, n. — A specific textile's ability to reduce electric current flowing through it at a given voltage. Resistance, R, is expressed in units of ohms (Ω).

5. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Users MUST consult manufacturers for specific details such as safety data sheets and other manufacturer's recommendations. Consult and follow all applicable OSHA standards and rules.

5.1 Follow good laboratory practices. Wear safety glasses in all laboratory areas.

5.2 Follow manufacturer's safety recommendations when operating laboratory testing equipment.

6. Uses and Limitations

6.1 Each conductive path to be measured must be accessible, at least at the two measurement points. Connectors may be used to provide accessibility. The measured specimen should mimic the intended design of the end product, if known.

6.2 The exposure procedures referenced in this document were not developed to measure changes in electrical resistance. Electrical resistance changes after various exposure conditions may not accurately reflect end use results but provide a relative measure of electrical resistance durability.

7. Apparatus and Materials (see 16.2)

7.1 **Electrical Resistance Measurement**

7.1.1 Conditioning facilities and racks with pull-out, perforated shelves for laying specimens flat (see 16.1).

7.1.2 Flat workbench with grounded electronic static dissipation (ESD) mat. The surface should be at least as large as the specimen.

7.1.3 Indelible ink marking pen (see 16.1) or needle and sewing thread, for making benchmarks.

7.1.4 Digital multimeter or ohmmeter, resolution to at least 0.1 Ω and 0.9% accuracy. For samples with resistance less than 1 Ω, use a four-wire (four-point) system with resolution to at least 0.01 Ω. Select probes appropriate to the test material. Weighted probes of an appropriate size may improve repeatability.

7.1.5 Tape or rule marked in millimeters.

7.2 **Exposure 1: Home Laundering**

7.2.1 For home laundering using a washing machine, refer to AATCC LP1 for apparatus and materials.

7.2.2 For home laundering by hand washing, refer to AATCC LP2 for apparatus and materials.

7.3 **Exposure 2: Drycleaning**

7.3.1 Refer to AATCC TM158 for apparatus and materials related to drycleaning exposure.

7.3.2 The measuring scale referenced in AATCC TM158 is NOT necessary for evaluation of electrical resistance.

7.4 **Exposure 3: Water: Spray**

7.4.1 Refer to AATCC TM22 for apparatus and materials related to water spray exposure.

7.5 **Exposure 4: Water: Hydrostatic Pressure**

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7.5.1 Refer to AATCC TM127 for apparatus and materials related to hydrostatic pressure exposure.

7.6 **Exposure 5: Water: Salt (Sea)**

7.6.1 Refer to AATCC TM106 for apparatus and materials related to salt-water exposure.

7.6.2 Multifiber fabric referenced in AATCC TM106 is NOT necessary for evaluation of electrical resistance.

7.6.3 Scales for chromatic transference, color change and staining referenced in AATCC TM106 are NOT necessary for evaluation of electrical resistance.

7.6.4 Swatches or items larger than 60 × 120 mm require a modified apparatus capable of applying the specified pressure over the entire area and an oven large enough to accommodate the apparatus.

7.7 **Exposure 6: Perspiration**

7.7.1 Refer to AATCC TM15 for apparatus and materials related to perspiration exposure.

7.7.2 Multifiber fabric referenced in AATCC TM15 is NOT necessary for evaluation of electrical resistance.

7.7.3 Scales for chromatic transference, color change and staining referenced in AATCC TM15 are NOT necessary for evaluation of electrical resistance.

7.7.4 Swatches or items larger than 60 × 120 mm require a modified apparatus capable of applying the specified pressure over the entire area and an oven large enough to accommodate the apparatus.

7.8 **Exposure 7: Acids and Alkalis**

7.8.1 Refer to AATCC TM6 for apparatus and materials related to acids and alkalis exposure.

7.8.2 Gray Scale for Color Change referenced in AATCC TM6 is NOT necessary for evaluation of electrical resistance.

7.8.3 Additional reagent volume and vessels may be required to test large or unusually-shaped samples.

7.9 **Exposure 8: Ultraviolet (UV) Radiation and Moisture**

7.9.1 Refer to AATCC TM186 for apparatus and materials related to UV radiation and moisture exposure.

7.9.2 Swatches or items larger than 70 × 120 mm may require a modified sample holder and a test chamber large enough to accommodate the sample.

7.10 **Exposure 9: Microbes**

7.10.1 Refer to AATCC TM100 for apparatus and materials related to microbe exposure.

7.10.2 Swatches or items larger than 48 mm in diameter may require a larger volume of inoculum and a Petri dish or other shallow container large enough to accommodate the sample.

8. Specimens

8.1 Each conductive path (or section of a path) is considered a specimen. Measure at least three identical specimens for each exposure condition.

8.1.1 Additional identical specimens may be used to improve accuracy. Various path types or lengths may be tested as separate samples. Do not compare results for different path types or lengths.

8.1.2 If possible, cut swatches that each contain at least one straight conductive path. Cut swatches such that specimens are centered. Swatch and specimen dimensions are specified for individual exposure conditions.

8.1.3 If end products cannot be cut without interrupting the pathway, specimens will depend upon product design. Test identical specimens from three end-product items. Modification of the exposure conditions may be required to accommodate non-standard sample sizes or forms.

8.2 **Exposure 1: Home Laundering**

8.2.1 If possible, cut three swatches, 400 × 400 mm, each containing at least one straight conductive path (specimen), 150-300 mm in length. Cut each swatch such that the specimen is near the center, and at least 65 mm from all edges.

8.2.2 The procedure for home laundering exposure is easily adapted to most uncut and other non-standard samples.

8.3 **Exposure 2: Drycleaning**

8.3.1 If possible, cut three swatches, 500 × 500 mm, each containing at least one straight conductive path (specimen), 150-300 mm in length. Cut each swatch such that the specimen is near the center, and at least 65 mm from all edges.

8.3.2 The procedure for drycleaning exposure is easily adapted to most uncut and other non-standard samples.

8.4 **Exposure 3: Water: Spray**

8.4.1 If possible, cut three swatches, 180 × 180 mm, each containing at least one straight conductive path (specimen), about 150 mm in length. Cut each swatch such that the specimen is near the center.

8.4.2 The procedure for water spray exposure can be adapted to some uncut and other non-standard samples, but only a circular test area with about 155-mm diameter will be exposed to water. The test area must be flat. Conductive elements at the perimeter of the test area may be damaged by the specimen holder.

8.5 **Exposure 4: Water: Hydrostatic Pressure**

8.5.1 If possible, cut three swatches, 200 × 200 mm, each containing at least one straight conductive path (specimen), about 90 mm in length. Cut each swatch such that the specimen is near the center.

8.5.2 The procedure for hydrostatic pressure exposure can be adapted to some uncut and other non-standard samples, but only a circular test area with about 100-mm diameter will be exposed to water. Conductive elements at the perimeter of the test area may be damaged by the clamping mechanism.

8.6 **Exposure 5: Water: Salt (Sea)**

8.6.1 If possible, cut three swatches, 60 × 120 mm, each containing at least one straight conductive path (specimen), about 125 mm in length, diagonally across the swatch. Align swatch to achieve the longest possible conductive path; it is not necessary to align swatch edges with machine direction of the fabric.

8.6.2 Adaptation of the procedure for salt-water exposure to uncut and other non-standard samples may be difficult, particularly if the sample is not a single flat layer.

8.7 **Exposure 6: Perspiration**

8.7.1 If possible, cut three swatches, 60 × 120 mm, each containing at least one straight conductive path (specimen), about 125 mm in length, diagonally across the swatch. Align swatch to achieve the longest possible conductive path; it is not necessary to align swatch edges with machine direction of the fabric.

8.7.2 Adaptation of the procedure for perspiration exposure to uncut and other non-standard samples may be difficult, particularly if the sample is not a single flat layer.

8.8 **Exposure 7: Acids and Alkalis**

8.8.1 For each reagent [HCl, CH_3COOH, NH_3, Na_2CO_3, NH_4OH, $Ca(OH)_2$], use three swatches or end-product items of any convenient size, each containing at least one conductive path (specimen)—18 specimens in total. If possible, select straight specimens, 150-300 mm in length.

8.8.2 No adaptation of the procedure for acids and alkali exposure is required to test most large or unusually-shaped samples.

8.9 **Exposure 8: UV Radiation and Moisture**

8.9.1 If possible, cut three swatches, 70 × 120 mm, that each contain at least one straight conductive path (specimen), about 130 mm in length, diagonally across the swatch. Align swatch to achieve the longest possible conductive path; it is not necessary to align swatch edges with machine direction of the fabric.

8.9.2 Adaptation of the procedure for UV radiation and moisture exposure to uncut and other non-standard samples may be difficult, particularly if the sample is not a single flat layer.

8.10 **Exposure 9: Microbes**

8.10.1 For each test bacteria (*Staphyloccoccus aureus* and *Klebsiella pneumoniae*), cut three circular swatches with a 48-mm diameter that each contain at least one straight conductive path (specimen),

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about 40 mm in length—6 specimens in total.

8.10.2 Cut several additional 48-mm diameter swatches; inclusion and position of conductive path is not critical for these swatches.

8.10.3 Adaptation of the procedure for microbe exposure to uncut and other non-standard samples may be difficult.

9. Conditioning

9.1 Prior to marking, condition test specimens as directed in ASTM D1776. (Use conditions indicated in Table 1 for Textiles, general. Estimate conditioning time as indicated in Table 2 for the appropriate fiber content.) Lay each swatch or product, separately, on a screen or perforated shelf of a conditioning/drying rack.

10. Preparation of Specimens

10.1 **Exposure 1: Home Laundering**

10.1.1 Mark the ends of three identical specimens (one per swatch or end-product item) (see Fig. 1).

10.2 **Exposure 2: Drycleaning**

10.2.1 Mark the ends of three identical specimens (one per swatch or end-product item) (see Fig. 2).

10.3 **Exposure 3: Water: Spray**

10.3.1 Mark the ends of three identical specimens (one per swatch or end-product item) (see Fig. 3).

10.4 **Exposure 4: Water: Hydrostatic Pressure**

10.4.1 Mark the ends of three identical specimens (one per swatch or end-product item) (see Fig. 4).

10.5 **Exposure 5: Water: Salt (Sea)**

10.5.1 Do NOT attach multifiber fabric or other adjacent fabric.

10.5.2 Mark the ends of three identical specimens (one per swatch or end-product item) (see Fig. 5).

10.6 **Exposure 6: Perspiration**

10.6.1 Do NOT attach multifiber fabric or other adjacent fabric.

10.6.2 Mark the ends of three identical specimens (one per swatch or end-product item) (see Fig. 5).

10.7 **Exposure 7: Acids and Alkalis**

10.7.1 Mark the ends of three identical specimens per reagent (one per swatch or end-product item).

10.8 **Exposure 8: UV Radiation and Moisture**

10.8.1 Mark the ends of three identical specimens (one per swatch or end-product item) (see Fig.6).

10.9 **Exposure 9: Microbes**

10.9.1 Mark the ends of three identical specimens per bacteria (one per swatch) (see Fig. 7).

10.9.2 Stack each marked swatch in a closed container with additional swatches as needed to absorb 1.0 ± 0.1 mL of inoculum and leave no free liquid.

10.9.3 Control swatches are NOT necessary for evaluation of electrical resistance.

10.9.4 Sterilization is NOT necessary for evaluation of electrical resistance.

11. Procedure

11.1 **Initial Resistance Measurement**

11.1.1 Arrange the first specimen on flat, non-conductive surface for measurements.

11.1.2 Set multimeter or ohmmeter to measure resistance in the expected range.

11.1.3 Place the positive and negative probes of the meter at the marked points.

11.1.4 Pressure and contact area between probes and specimen during resistance measurement may greatly affect

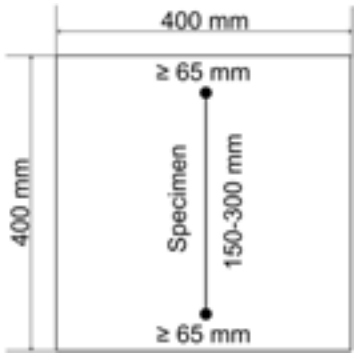


Fig. 1—Swatch marked for Exposure 1: Home Laundering.

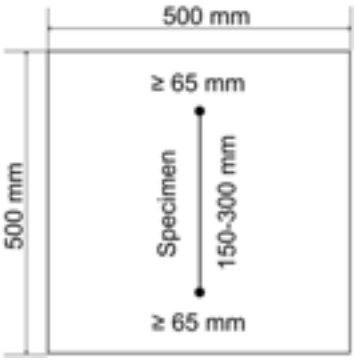


Fig. 2—Swatch marked for Exposure 2: Drycleaning.

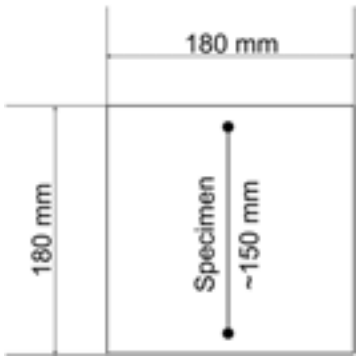


Fig. 3—Swatch marked for Exposure 3: Water: Spray.

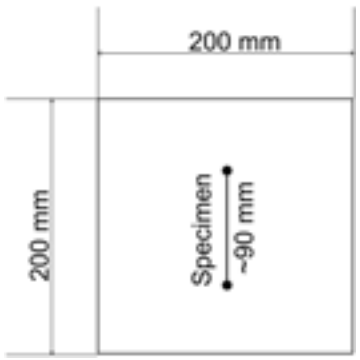


Fig. 4—Swatch marked for Exposure 4: Water: Hydrostatic Pressure.

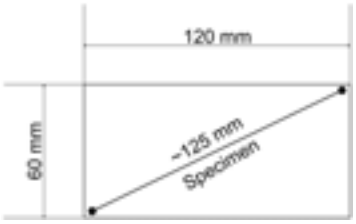


Fig. 5—Swatch marked for Exposure 5: Water: Salt (Sea) or Exposure 6: Perspiration.

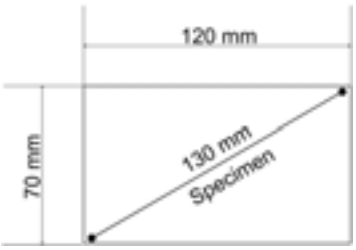


Fig. 6—Swatch marked for Exposure 8: UV Radiation and Moisture

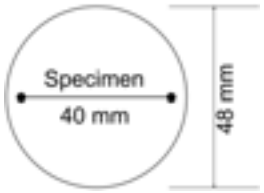


Fig. 7—Swatch marked for Exposure 9: Microbes.

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readings. Maintain consistent force to ensure repeatable results. Limiting measurements to a single operator, while not required, may also improve reliability and consistency of the measurements collected.

11.1.5 Connect the multimeter to the specimen using any suitable probes or connections system. Note the probes or connection system in the report. Do not compare measurements taken using different probes or connection systems.

11.1.6 Record the resistance as soon as the reading stabilizes for a period of 3 seconds. Take one resistance measurement per specimen.

11.1.7 Measure and record distance between marked points, to the nearest mm.

11.1.8 Repeat resistance and distance measurements for the remaining specimens.

11.2 **Exposure 1: Home Laundering**

11.2.1 Select laundering parameters—wash temperature, drying method, etc.—based on expected care instructions for the end product.

11.2.2 Perform 3 laundering cycles (wash and dry) unless otherwise agreed between buyer and seller.

11.2.3 For home laundering using a washing machine, follow the standard washing and drying procedures described in AATCC LP1. Include all three swatches or end products in the same laundering load. Follow ironing procedure if necessary.

11.2.4 For home laundering by hand washing, follow the standard washing and drying procedures described in AATCC LP2. Wash each swatch or end product separately.

11.3 **Exposure 2: Drycleaning**

11.3.1 Perform the drycleaning procedure described in AATCC TM158. Include all three swatches or end products in the same load.

11.3.2 Perform 1 drycleaning cycle unless otherwise agreed between buyer and seller.

11.3.3 Marking and measurement of swatch dimensions are NOT necessary for evaluation of electrical resistance.

11.4 **Exposure 3: Water: Spray**

11.4.1 Perform the spray test procedure described in AATCC TM22.

11.5 **Exposure 4: Water: Hydrostatic Pressure**

11.5.1 Perform the hydrostatic pressure procedure described in AATCC TM127, using a predetermined end point.

11.5.2 Increase pressure at the specified rate until it reaches 2,000 mm H_2O. Maintain this pressure for 2 min, then immediately decrease pressure to 0 and remove specimen.

11.5.3 Do NOT stop test if water drops appear before 2,000 mm or 2 min. Do NOT continue test beyond 2 min. at 2,000 mm.

11.6 **Exposure 5: Water: Salt (Sea)**

11.6.1 Perform the salt-water procedure described in AATCC TM106 for each specimen.

11.6.2 Adjust pressure as needed to avoid damage to the conductive components. If pressure other than that specified in AATCC TM106 is used, note it in the test report.

11.7 **Exposure 6: Perspiration**

11.7.1 Perform the perspiration procedure described in AATCC TM15 for each specimen.

11.7.2 Adjust pressure as needed to avoid damage to the conductive components. If pressure other than that specified in AATCC TM15 is used, note it in the test report.

11.8 **Exposure 7: Acids and Alkalis**

11.8.1 Perform the acids and alkalis procedure described in AATCC TM6, being sure to perform spot tests directly on each conductive path specimen.

11.9 **Exposure 8: UV Radiation and Moisture**

11.9.1 Perform the UV radiation and moisture procedure described in AATCC TM186, Option 1 for 100 total hours.

11.10 **Exposure 9: Microbes**

11.10.1 Perform the microbial exposure procedure described in AATCC TM100 for each specimen, except for dilution and plating.

11.10.2 Remove swatches from jar after neutralization. Perform final resistance measurement on dried and conditioned specimen; discard liquid.

12. Evaluation

12.1 Recondition specimens (see Section 9) before final resistance measurement.

12.2 Measure and record electrical resistances after treatment. Use the same marked points and connection system as for the initial resistance measurement (see 11.1).

12.3 Measure and record distance between marked points to the nearest mm.

13. Calculation

13.1 For each exposure condition, calculate the mean of initial resistance measurements for all specimens. This is R_i.

13.2 For each exposure condition, calculate the mean of final resistance measurements for all specimens. This is R_f.

13.3 For each exposure condition, calculate the mean initial distance for all specimens. This is d_i.

13.4 For each exposure condition, calculate the mean final distance for all specimens. This is d_f.

13.5 For each exposure condition, calculate % resistance change after treatment using Equation 1.

$$RC = (R_f - R_i)/R_i \times 100\% \qquad \text{(Eq. 1)}$$

where:

RC = % resistance change after treatment,
R_f = Mean final resistance, after treatment (Ω),
R_i = Mean initial resistance (Ω).

14. Report

14.1 Describe or identify the sample tested.

14.2 Report that the sample was tested using AATCC TM210-2019. Report each exposure condition tested.

14.3 Report the general testing conditions:

14.3.1 Multimeter or ohmmeter resolution and accuracy.

14.3.2 Probe type and connection system.

14.4 For **Exposure 1: Home Laundering**, report testing conditions and results:

14.4.1 Exposure procedure (AATCC LP1 or AATCC LP2) and version referenced.

14.4.2 Laundering conditions, including number of cycles, wash temperature, and drying procedure.

14.4.3 Water hardness of the wash water (ppm).

14.4.4 Description of specimen (conductive path length, width, shape, material, etc.)

14.4.5 Initial and final distance between marked points for individual specimens (mm).

14.4.6 Mean initial distance, d_i, and final distance, d_f, between marked points (mm).

14.4.7 Initial and final resistance between marked points for individual specimens (Ω).

14.4.8 Mean initial resistance, R_i, and final resistance, R_f, between marked points (Ω).

14.4.9 Resistance change, RC, after laundering (%).

14.4.10 Any modification(s) of the published method.

14.5 For **Exposure 2: Drycleaning**, report testing conditions and results:

14.5.1 Exposure procedure (AATCC TM158) and version referenced.

14.5.2 Procedure used (normal or sensitive material).

14.5.3 Percentage by mass of test swatches in the load.

14.5.4 Ballast type.

14.5.5 Maximum inlet or outlet air temperature during drying.

14.5.6 Details of the finishing treatment used.

14.5.7 Number of drycleaning cycles.

14.5.8 Description of specimen (conductive path length, width, shape, material, etc.)

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14.5.9 Initial and final distance between marked points for individual specimens (mm).

14.5.10 Mean initial distance, d_i, and final distance, d_f, between marked points (mm).

14.5.11 Initial and final resistance between marked points for individual specimens (Ω).

14.5.12 Mean initial resistance, R_i, and final resistance, R_f, between marked points (Ω).

14.5.13 Resistance change, *RC*, after laundering (%).

14.5.14 Any modification(s) of the published method.

14.6 For **Exposure 3: Water: Spray**, report testing conditions and results:

14.6.1 Exposure procedure (AATCC TM22) and version referenced.

14.6.2 Description of specimen (conductive path length, width, shape, material, etc.)

14.5.3 Initial and final distance between marked points for individual specimens (mm).

14.6.4 Mean initial distance, d_i, and final distance, d_f, between marked points (mm).

14.6.5 Initial and final resistance between marked points for individual specimens (Ω).

14.6.6 Mean initial resistance, R_i, and final resistance, R_f, between marked points (Ω).

14.6.7 Resistance change, *RC*, after laundering (%).

14.6.8 Any modification(s) of the published method.

14.7 For **Exposure 5: Water: Salt (Sea)**, report testing conditions and results:

14.7.1 Exposure procedure (AATCC TM106) and version referenced.

14.7.2 Description of specimen (conductive path length, width, shape, material, etc.)

14.7.3 Initial and final distance between marked points for individual specimens (mm).

14.7.4 Mean initial distance, d_i, and final distance, d_f, between marked points (mm).

14.7.5 Initial and final resistance between marked points for individual specimens (Ω).

14.7.6 Mean initial resistance, R_i, and final resistance, R_f, between marked points (Ω).

14.7.7 Resistance change, *RC*, after laundering (%).

14.7.8 Any modification(s) of the published method.

14.8 For **Exposure 6: Perspiration**, report testing conditions and results:

14.8.1 Exposure procedure (AATCC TM15) and version referenced.

14.8.2 Description of specimen (conductive path length, width, shape, material, etc.)

14.8.3 Initial and final distance between marked points for individual specimens (mm).

14.8.4 Mean initial distance, d_i, and final distance, d_f, between marked points (mm).

14.8.5 Initial and final resistance between marked points for individual specimens (Ω).

14.8.6 Mean initial resistance, R_i, and final resistance, R_f, between marked points (Ω).

14.8.7 Resistance change, *RC*, after laundering (%).

14.8.8 Any modification(s) of the published method.

14.9 For **Exposure 7: Acids and Alkalis**, report testing conditions and results:

14.9.1 Exposure procedure (AATCC TM6) and version referenced.

14.9.2 Description of specimen (conductive path length, width, shape, material, etc.)

14.9.3 Initial and final distance between marked points for individual specimens (mm).

14.9.4 Mean initial distance, d_i, and final distance, d_f, between marked points (mm).

14.9.5 Initial and final resistance between marked points for individual specimens (Ω).

14.9.6 Mean initial resistance, R_i, and final resistance, R_f, between marked points (Ω).

14.9.7 Resistance change, *RC*, after laundering (%).

14.9.8 Any modification(s) of the published method.

14.10 For **Exposure 8: UV Radiation and Moisture**, report testing conditions and results:

14.10.1 Exposure procedure (AATCC TM186, Option 1) and version referenced.

14.10.2 Description of specimen (conductive path length, width, shape, material, etc.)

14.10.3 Manufacturer and model of fluorescent UV/condensation apparatus.

14.10.4 Manufacturer's designation for the fluorescent UV lamp.

14.10.5 Exposure cycle (Option 1: 8 h UV/60°C, 4 h con/50°C).

14.10.6 Total exposure time.

14.10.7 Total UV light exposure time.

14.10.8 Initial and final distance between marked points for individual specimens (mm).

14.10.9 Mean initial distance, d_i, and final distance, d_f, between marked points (mm).

14.10.10 Initial and final resistance between marked points for individual specimens (Ω).

14.10.11 Mean initial resistance, R_i, and final resistance, R_f, between marked points (Ω).

14.10.12 Resistance change, *RC*, after laundering (%).

14.10.13 Any modification(s) of the published method.

14.11 For **Exposure 9: Microbes**, report testing conditions and results:

14.11.1 Exposure procedure (AATCC TM100) and version referenced.

14.11.2 Description of specimen (conductive path length, width, shape, material, etc.)

14.11.3 Number of swatches per jar.

14.11.4 Test bacteria used (*Staphyloccoccus aureus* and *Klebsiella pneumoniae)*.

14.11.5 Neutralizing solution used.

14.11.6 Initial and final distance between marked points for individual specimens (mm).

14.11.7 Mean initial distance, d_i, and final distance, d_f, between marked points (mm).

14.11.8 Initial and final resistance between marked points for individual specimens (Ω).

14.11.9 Mean initial resistance, R_i, and final resistance, R_f, between marked points (Ω).

14.11.10 Resistance change, *RC*, after laundering (%).

14.11.11 Any modification(s) of the published method.

15. Precision and Bias

15.1 *Precision.* Precision studies have not been completed on each of the exposure conditions listed in this standard. For precision of resistance measurements, see AATCC EP13.

15.2 *Bias.* Electrical resistance can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

16. Notes

16.1 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1. 919.549.8141; ordering@aatcc.org; www.aatcc.org.

16.2 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

17. History

17.1 Reaffirmed 2020.

17.2 Developed in 2019 by AATCC Committee RA111.

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Test Method for the Reduction of Bacterial Odor on Antibacterial-Treated Textiles

1. Purpose and Scope

1.1 This test method is for the evaluation of the reduction of odor generated from textiles treated with antibacterial finishes.

1.2 This procedure is meant to evaluate single layer of a textile. For multiple layered textiles, such as carpet, the test is to be used only on the-layer of the material that has been treated.

1.3 The reduction of bacterial odor is to be determined by testing a comparable fabric that has not been treated with an antibacterial finish.

1.4 For evaluation of the antibacterial efficacy of a textile, existing antibacterial test methods for the bactericidal/static effect of the product should be used. See Note 13.6.

2. Principle

2.1 This procedure is intended to be used to supplement results from industry standard antibacterial efficacy tests that, do not measure the issue of odor reduction.

2.2 Ammonia-generating organisms are employed as the surrogate test organisms of which their ability to produce ammonia while they metabolize urea or protein residue is utilized. Ammonia is a component of bacterial odor that can be smelled in many problem areas including sink drains, diaper pails, incontinence products, and pet bedding.

2.3 *Proteus vulgaris* is the surrogate gram-negative organism while *Staphylococcus saprophyticus* is a gram-positive surrogate organism.

2.4 The results obtained from this test may be used as a relative comparison of the effects of antibacterial treatments on odor generation over time when compared to each other and to untreated fabrics. The lower limit of detection for ammonia by this method is 2 - 3 ppm, which is at or near the mean concentration for olfactory detection according to at least one study (see 13.1) although another source has reported it at 17 ppm (see 13.2).

3. Terminology

3.1 **activity,** n.—*of an antibacterial agent*, a measure of effectiveness of the agent.

3.2 **antibacterial agent,** n.—*in textiles*, any chemical that kills bacteria (bactericide) or interferes with the multiplication, growth or activity of bacteria (bacteriostatic).

3.3 **odor,** n.—the sensation that results when olfactory receptors in the nose are stimulated by particular chemicals in gaseous form. (Source Princeton Word Net)

4. Safety Precautions

4.1 The safety precautions specified in the method are ancillary to the testing procedures and are not intended to be all inclusive.

4.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

4.3 Users MUST consult manufacturers for specific details such as safety data sheets, equipment operating instructions, and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

4.4 This test should be performed only by trained personnel. Consult the U.S. Department of Health and Human Services publication *Biosafety in Microbiological and Biomedical Laboratories* (see 13.13).

4.5 CAUTION: The bacteria used in this test may be pathogenic; i.e., capable of infecting humans and producing disease. Therefore, every necessary and reasonable precaution must be taken to eliminate this risk to the laboratory personnel and to personnel in the associated environment. Wear protective clothing and respiratory protection that prevents penetration by the bacteria.

5. Limitations

5.1 This method, as written, is limited to bacterial organisms capable of producing ammonia as a metabolic by-product. Other bacteria that can produce a metabolic by-product which can be measured by different gas detector tubes may be used, but these variations must be reported. It is not known whether results from different gas detector tubes are comparable.

6. Apparatus, Reagents, Materials (see 13.4)

6.1 Volumetric flasks, 5 mL, glass, with a #8 stopper.

6.2 Gas detector color diffusion tubes for ammonia, calibrated 2.5 to 1500 ppm-h (Draeger ammonia columns is an example of a suitable gas detector tube).

6.3 Sealing film, thermoplastic, with two-way stretch.

6.4 Fabric scissors.

6.5 Inoculating loop.

6.6 Incubator, 37 ± 2°C, 75% RH equipped with an orbital shaker used to agitate overnight cultures.

6.7 Serological pipettes, 1.0 mL, sterile.

6.8 Brain-Heart Infusion Broth (BHIB)

6.9 Urea, Molecular Biology Certified, >99.9%.

6.10 Filter sterilization apparatus.

6.11 Test Organisms, *Proteus vulgaris* or *Staphylococcus saprophyticus*.

6.11.1 *Proteus vulgaris* ATCC 29905 (or 8427), CIP 104989, DSM 13387, NCIMB 13426, CCUG 35382T, (see 13.5).

6.11.2 *Staphylococcus saprophyticus* ATCC 15305, CIP 76.125, DSM 20229, NCIMB 8711.

6.12 Bacterial agar for organism maintenance slants (BHIA, TSA, or NA).

7. Specimens

7.1 Cut three-1.0 g specimens from each sample fabric of each test. Each specimen is then cut into sufficiently small pieces that can be inserted through the neck of the 5-mL volumetric flasks.

7.2 Prepare one specimen from a control sample as in 7.1.

7.2.1 The control sample should be from a fabric with the same fiber content and fabric construction (weave/knit, fabric count, yield) as the test sample, but with no antibacterial finish.

7.2.2 If a true control sample cannot be generated due to manufacturing constraints, use a textile fabric of similar construction.

7.2.3 A suitable viability control fabric will generate ammonia during the testing period with the specified testing conditions.

8. Preparation of Reagents

8.1 Two media preparations are required for the test.

8.1.1 Growth media: BHIB.

8.1.2 Test media: 5% BHIB with 2% urea.

8.2 Heat BHIB as needed to dissolve completely; dispense into 10 mL volumes in conventional bacteriological culture tubes. Plug and sterilize at 103 kPa (15 psi) for 15 min.

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8.3 Just prior to testing, and after cooling BHIB to room temperature, determine the amount of BHIB needed to conduct the testing described below. Dilute BHIB with sterile deionized water to achieve 5% BHIB. Add 2% w/v of urea to the predetermined amount of BHIB and mix well. Sterilize using a 0.2-uM filter.

8.4 Using a 4-mm inoculating loop, transfer the culture in BHIB from stock culture. Incubate cultures at 37 ± 2°C.

8.5 Maintain stock cultures on Tryptic Soy Agar (TSA) or appropriate agar slants. Store at 5 ± 1°C and transfer once a week to fresh agar.

9. Procedure

9.1 Sterilization is not necessary for the test. But if fabrics are sterilized, include method of sterilization in the final test report.

9.2 Place 1.0 g of the cut pieces of each specimen ~~i~~n a separate volumetric flask so that they rest in a pile within the bulb of the flask.

9.3 Prepare test organism *P. vulgaris* or *S. saprophyticus* using a broth culture incubated for 18 - 24 hour at 37 ± 2°C and shaking at 110-250 rpm.

9.4 Adjust the suspension with 5% BHIB containing 2% w/v urea to obtain the test inoculum containing 1-5 × 10^5 cfu per mL of the test organism. Store this test inoculum at 2-8°C until ready for testing.

9.5 If the evaluation of antibacterial finish(es) is to be undertaken after home laundering, interest parties should agree on the laundering conditions from AATCC LP1 and the number of launderings. The laundering conditions should be stated with the test results.

NOTE: Inoculum must be used within 2 hours following adjustment.

NOTE: Other concentrations of nutrient may be used in testing as long as they support ammonia generation on the untreated control fabric. These changes must be outlined in the report.

9.5 Apply 1.0 mL of the inoculum with 2% urea directly onto each specimen, taking care to ensure total uptake of the inoculum by the fabric.

NOTE: If reduction of the test volume is necessary to prevent oversaturation of a test specimen, do not use less than 0.5 mL of test inoculum. When the volume of test inoculum is decreased, the concentration of the bacterial cells in the inoculum shall be increased to provide the same number of bacterial cells as when the normal volume of test inoculum is applied unless otherwise specified.

9.6 Seal the flask opening with film or a similar substrate.

9.7 Place all the flasks in an incubator operating at 37 ± 2°C and incubate for 20 ± 2 h.

9.8 For each flask, open a gas detector tube by snapping off the neck as directed in the detector's instructions, and insert the open end into the neck of each volumetric flask, sealing off the neck from the atmosphere.

9.9 Secure the gas detector tubes in the necks using strips of sealing film as required to ensure the seal.

9.10 Place all the flasks in an incubator operating at 37 ± 2°C and record the test start time.

9.11 Monitor the flasks once each hour for signs of color change from yellow to blue within the packing of each gas detector tube. For each flask, record the reading associated with the color change on the tube scale and the time elapsed from the start.

NOTE: The gas detector tubes are only certified for an 8-hour exposure timeframe. The exposure time with the column must be stopped after eight hours.

9.12 It is recommended that~~,~~ once a color change is detected, the flasks are to be read and recorded every 30 minutes thereafter.

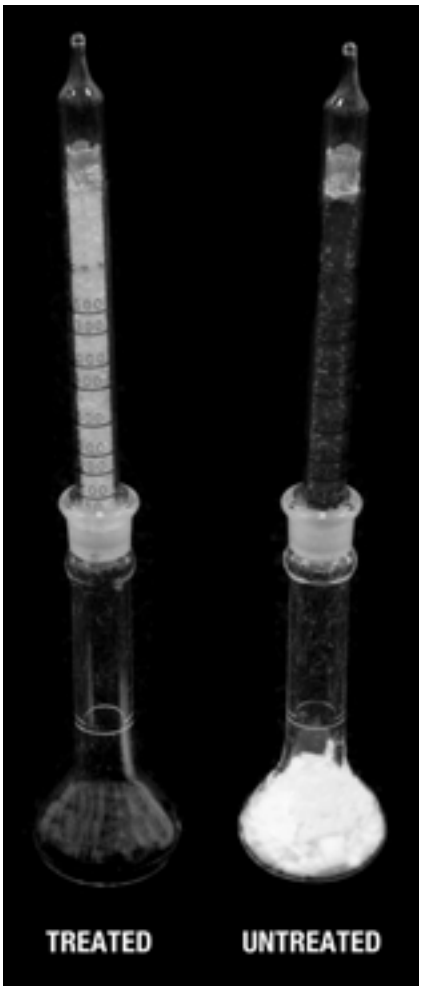


Fig. 1—Test vessels containing adequately treated and untreated fabrics after 6 h of incubation at 37 ± 2°C.

9.13 The test is complete for each specimen when it reaches the top of the scale or the 8-h time frame has elapsed (whichever is first). The end point time will be utilized for ppm and % reduction calculations.

10. Calculations

10.1 For each test and control specimen, divide the endpoint reading by the corresponding elapsed time to obtain the ppm of ammonia generated (Eq. 1).

$$\text{ppm} = \frac{\text{(ppm*h reading)}}{\text{h exposure}} \qquad \text{(Eq. 1)}$$

10.2 If replicates are used, calculate the mean ppm for all test specimens in the sample. Calculate the mean ppm for all control specimens.

10.3 Using the ppm from Eq. 1, calculate the % reduction of ammonia by the treated fabric using Eq. 2.

$$R = 100\frac{(B - A)}{B} \qquad \text{(Eq. 2)}$$

where:

R = % reduction
A = mean ppm of ammonia from the treated sample
B = mean ppm of ammonia from the untreated sample (control)

11. Report

11.1 Describe or identify the sample tested.

11.2 Report that the sample was tested using AATCC TM211-2021

11.3 Report the testing conditions:

11.3.1 Control sample used.

11.3.2 Test organism used, including supply source and strain number.

11.4 Report the test results:

11.4.1 Ammonia concentration (ppm) for each test and control specimen.

11.4.2 End point (hours) for each test and control specimen.

11.4.3 Reduction of ammonia (%) by test sample.

12. Precision and Bias

12.1 *Precision*. In 2020, 3 laboratories completed a study with 2 treated fabrics and an untreated control fabric. Fabric A was the control. Fabric B and C had the same antibacterial treatment, but Fabric C was tested after 25 home laundering cycles. Each lab used 3 specimens and 1 operator. The test organism was *Proteus vulgaris* (two different strains).

12.1.1 Table I gives the repeatability and reproducibility for this test method,

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Table I—Precision for % Ammonia Reduction

Sample	Mean	Repeatability (SD) within labs	Reproducibility (SD) among labs
B	98.9	0	0.12
C	4.5	4.44	5.44

based on the 2020 study.

12.2 *Bias*. Ammonia reduction can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

13. Notes

13.1 Smeets, Bulsing, van Rooden, et. Al., Odor and Irritation Thresholds for Ammonia: A Comparison between Static and Dynamic Olfactometry, Chemical Senses, 2006.

13.2 Aerias, (Air Quality Sciences Indoor Air Quality Resource Center), 2211 Newmarket Pkwy, Suite 106, Atlanta GA 30067, (770) 933-0638.

13.3 Publication available for US Department of Health and Human Services CDC/NIH_HHS Publication No. (CDC) 84-8395.

13.4 For potential apparatus, reagents or materials sources, visit the AATCC Buyer's Guide at www.aatcc.org/bg. AATCC provides its Corporate members the option to list their items and services. AATCC does not qualify, or in any way approve, endorse or certify that any of the listings meet the specifications in its standards.

13.5 ATCC is the American Type Culture Collections (USA), CIP is the Pasteur Institute Collection (France), DSM is the German Collection of Microorganisms and cell cultures (Germany), and CCUG is the Culture Collection University Göteborg (Sweden). Equivalent bacteria Strains obtained for agencies of the World Federation of Culture Collection WFCC) may be used by agreement between the interested parties. The strains used in the test shall be documented with their supply source.

13.6 Test methods suitable for testing bactericidal/static effect of finish would be AATCC TM100 or AATCC TM147.

14. History

14.1 Developed in 2021 by AATCC Committee RA31.

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Test Method for Fiber Fragment Release During Home Laundering

Foreword

The use of the term "microfiber" in the context of environmental pollution is a source of ambiguity, AATCC has undertaken the initiative to develop definitions for fiber fragment and microfiber. Although there are still many questions regarding fiber fragments that occur, specifically in water, in the environment, textile industry stakeholders determined the development of a standard method for quantifying shedding propensity of fabrics was needed. Degradability, chemical toxicity, and physical hazards are other important components of the shedding of textiles and will need additional test method development. As research continues, some testing parameters may be revised to improve reproducibility and relevance of the method. Options are offered where best practices are not yet defined.

1. Purpose and Scope

1.1 This test determines the mass of fiber fragments released in an accelerated laundering testing machine. Accelerated laundering is expected to provide a relative approximation of fiber fragment release in full-scale home laundering, but an exact correlation has not been determined.

1.2 This test is applicable to textiles that are expected to withstand home laundering.

2. Principle

2.1 Specimens are tested under conditions of temperature, detergent (optional), and agitation such that fiber fragments are generated and released in a short time. Agitation of the canister speeds up abrasion of the fabric using stainless steel balls and a low liquor-to-fabric ratio. The wash liquor is filtered, and the fiber fragment mass is measured.

2.2 All parties should agree to Options before testing. Washing Option A uses a detergent solution to allow complete wetting of fabric specimens. Washing Option B uses water only. Drying Option A uses filters conditioned at standard textile testing conditions. Drying Option B uses oven-dried filters. Oven-drying may allow for tighter control of filter mass but is impractical for some labs due to space limitations and moisture regain during measurement.

2.3 Results are reported as the mass of fiber fragments deposited on the filter and as a percentage of the original fabric specimen mass.

3. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

3.1 ASTM D6193, Standard Practice for Stitches and Seams (see 18.1).

3.2 ASTM E1402, Standard Guide for Sampling Design (see 18.1).

4. Terminology

4.1 **fiber,** n.—*in textiles*, a generic term for any one of the various types of matter that form the basic elements of a textile and which are generally characterized by flexibility, fineness and high ratio of length to thickness.

4.2 **fiber fragment,** n.—a short piece (typically $< 5 \times 10^{-3}$ m long) of textile fiber, broken away (or separated) from a textile construction.

NOTE: Fiber fragments are of concern as environmental pollutants; they are often referred to as "microfibers" due to their small size.

4.3 **microfiber,** n.—a fiber with linear density less than 1 denier or 1 dtex.

NOTE: Polyester microfibers typically have diameter of $< 10^{-5}$ m; this is a frequently referenced dimension, but not the formal definition of a microfiber.

NOTE: The term "microfiber" has been widely adopted as synonymous with "fiber fragment." To avoid any ambiguity, AATCC uses "microfiber" ONLY as a category of linear density.

5. Safety Precautions

5.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

5.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

5.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

6. Limitations

6.1 This test method is not intended to provide an exact correlation to the mass of fiber fragments released from a finished product in a full-scale washing machine.

6.2 Results are reported as the mass change of a glass fiber filter. Other materials from the tested specimen, handling, or atmosphere may also deposit on the filter, increasing the apparent mass of fiber fragments. Loss of fiber fragments due to incomplete rinsing may also affect the apparent mass.

6.3 Because the glass fiber filter pore size is 1.6 µm, fibers of a smaller size have the potential of escaping through the filter. Because only one filter of one size is used, there is a potential for fibers bigger than 5 mm to be captured on the filter as well.

7. Apparatus (see 18.2)

7.1 Sewing machine capable of creating a single-needle lockstitch, type 301 (see ASTM D6193).

7.1.1 Sewing machine needle, 0.65 mm (size 9) or a size appropriate for specimen fabric (see 18.3).

7.2 Analytical balance, 0.0001 g readability.

7.2.1 Deionizer, recommended for accurate weighing results

7.2.2 Static free brush

7.3 Accelerated laundering testing machine.

7.3.1 A laundering machine for rotating closed canisters in a thermostatically controlled water bath at 40 ± 2 rpm.

7.3.2 Stainless steel lever lock canisters, Type 2, 1200 mL (2.54 pt), 90 × 200 mm (3.5 × 8.0 in.) with lids and neoprene gaskets.

7.3.3 PTFE canister seals.

7.3.4 Mechanism for holding canisters on the accelerated laundering testing machine shaft.

7.3.5 Stainless steel balls, 6 mm (0.25 in.) in diameter, 50 per canister.

7.4 Conditioning facilities and conditioning/drying racks with perforated shelves (Drying Option A, see 18.5).

7.4.1 Alternatively (Drying Option B), non-circulating air drying oven, capable of maintaining 70 ± 2°C.

7.4.2 For Drying Option B (oven drying), Glass desiccator(s), with sufficient space for at least 20 aluminum weighing trays.

7.4.3 For Drying Option B (oven drying), Desiccant gel, indicating, such as silica gel.

7.5 Vacuum filtration system for 47-mm diameter filter.

7.5.1 Glass filtration funnel and clamp.

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7.5.2 Filter support; fritted-glass, sintered-metal, or stainless-steel screen.
7.5.3 Vacuum flask.
7.5.4 Vacuum source.

8. Reagents (see 18.2)

8.1 Water, distilled or deionized (see 18.4).
8.2 For Washing Option A, detergent, AATCC High Efficiency (HE) Standard Reference Liquid Detergent WOB (without optical brightener) (see 18.5).

9. Materials (see 18.2)

9.1 Wipes, non-shedding, low-lint, or lint-free.
9.2 Nitrile gloves.
9.3 Lab coat.
9.4 Aluminum foil.
9.5 Filament sewing thread, appropriate for the specimen fabric (see 18.6).
9.6 Forceps, non-serrated tips.
9.7 Aluminum weighing dishes.
9.8 Sieve, with aperture size of less than 5 mm (0.2 in).
9.9 Beaker, at least 1500 mL.
9.10 Wash bottle.
9.11 Borosilicate glass fiber filter with no binder, 1.6 µm pore size, 47 mm diameter.
9.12 Medicine dropper/bulb pipette, glass, 2 mL capacity, nominal 20 drops per mL.

10. Sampling

10.1 Test results are valid only when the samples are statistically representative (see ASTM E1402).
10.2 Sampling must be random. Every unit of product must have a mathematically equal chance of becoming a sample; and every portion of each sample must be equally likely to become a test specimen.
10.3 All specimens within the sample must be alike within the variations due to pure chance. There must be no differences within samples assignable to known causes.
10.4 All parties shall agree on the face and back of sample for testing purposes prior to testing.

11. Specimens

11.1 This test method measures very small mass changes. Minimize all sources of contamination.
11.1.1 Wipe all work surfaces with a non-shedding, low-lint, or lint-free wipe.
11.1.2 Wear a laboratory coat over clothing and tie back long hair.
11.1.3 Wear nitrile gloves. Do not handle filters, specimens or weighing dishes with bare hands. Analytical balances are sensitive enough to register touch contamination.
11.1.4 Clean sieve and all glassware, then triple rinse before use.
11.1.5 Avoid contamination of specimens by minimizing contact with other specimens. To remove contamination from common surfaces such as cutting table and sewing machine between samples, wipe with a non-shedding, low-lint, or lint-free wipe.
11.2 Cut 4 specimens, each 200 mm × 340 mm (8 in × 13.5 in), diagonally across the sample with the longest edge parallel to the selvage.
11.3 Finish all specimen edges with a clean-finish hem, seam type EFb-1 (see ASTM D6193).
11.3.1 Fold specimen edges under (to the back) 50 mm (2 in) from each edge. Do not press. Wide hem allows easier folding of bulky specimens. For consistency, use the same hem width for all test specimens.
11.3.2 Turn under raw edge to meet fold, leaving a clean-finish hem. Do not press. Stitch hem in place close to inner fold using lockstitch (stitch type 301), 2.0-2.5 mm per stitch (10-12 stitches per inch). Backstitch to secure stitching. Trim sewing threads.
11.3.3 Finished specimen size is 100 ± 10 mm × 240 ± 10 mm (4 ± 0.4 in × 9.5 ± 0.4 in).
11.3.4 In rare cases where seam type EFb-1 is not feasible with a fabric specimen, an alternate seam and/or edge finish may be used and a detailed description shall be included in the test report.
11.4 Avoid marking specimen identification before laundering. Marking may affect fiber fragment shedding due to damage from marking pens or contamination from pigment flakes.

12. Conditioning

12.1 Drying Option A: Place fabric specimens in a standard atmosphere of 21 ± 2°C (70 ± 4°F) and 65% ± 5% RH for at least 4 h, laying each specimen separately on a screen or perforated shelf of a conditioning rack. Cover specimens with aluminum foil to reduce possible contamination from air deposition.
12.2 Drying Option B: Place fabric specimens in clean, non-air circulating oven at 70°C for at least 4 h. Place oven-dried fabric specimens in desiccator for at least 30 min. to cool. Clean oven and desiccator again after removing specimens to minimize fiber contamination on the filters.
12.3 Filters are conditioned or oven dried and cooled after rinsing (see section 13.3). No additional conditioning is required.

13. Preparation

13.1 Label each aluminum weighing dish BEFORE pre-weighing.
13.2 Rinse each filter and place in a separate labeled aluminum weighing dish. Carefully place one filter in the sieve. Use a wash bottle with distilled/deionized water to rinse the filter to remove soluble compounds. Using forceps, remove the filter from the sieve and place in a dish.
13.3 Condition or oven-dry filters.
13.3.1 Drying Option A: Place filters (in weighing dishes) in a standard atmosphere of 21 ± 2°C (70 ± 4°F) and 65% ± 5% RH for at least 4 h. Keep filters covered with aluminum foil to avoid contamination.
13.3.2 Drying Option B: Dry filters (in weighing dishes) in a non-air-circulating oven at 70 ± 2°C for at least 1 h. Use oven-dried filters for testing. Place oven-dried filters and dishes in desiccator for 30 min. to cool. Maintain filters in the desiccator until immediately before use.
13.4 Washing Option A: Prepare 0.25% detergent solution.
13.4.1 Add 2.5 g detergent per 1000 mL distilled water. Stir until well combined.
13.4.2 Use fresh solution for each test or re-stir to ensure consistency and include solution creation date in test report

14. Procedure

14.1 Place each filter and weighing dish on the balance and record the total pre-test mass to the nearest 0.0001 g (W_1). Keep filters away from possible contamination sources.
14.1.1 Drying Option A: If testing is not performed in a conditioned atmosphere, return filters to conditioned atmosphere immediately after measuring mass.
14.1.2 Drying Option B: Return filters to oven, desiccator, or other enclosed environment immediately after measuring mass.
14.2 Place each fabric specimen on the balance and record the total pre-test mass to the nearest 0.0001 g (S_1). Keep specimens away from possible contamination sources. Remove loose fibers from balance between specimens using a static-free brush. Use a deionizer to minimize static interference with the mass.
14.3 Fill the accelerated laundering testing machine with water to the designated level. Adjust temperature to 40 ± 2°C (104 ± 4°F).
14.4 Ensure all canisters, lids, seals, gaskets, and balls are thoroughly cleaned and rinsed before use.
14.5 Add 50 stainless steel balls and 360 mL of detergent solution (Washing Option A) or 360 mL of water (Washing Option B) to each canister.
14.6 Place one neoprene gasket inside each canister lid. Place a PTFE seal over the gasket to reduce contamination of the

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wash water. Clamp a lid with gasket and seal on each canister.

14.7 Preheat canisters and contents for approximately 10 min. BEFORE adding specimen.

14.7.1 Follow manufacturer's directions for preheating in the main bath or external heater.

14.8 Remove canisters from the accelerated laundering testing machine or external heater and immediately add 1 specimen to each canister. Do not allow solution/water in canisters to cool.

14.9 Add solution/water and stainless steel balls, but no specimen, to a laundering procedure verification canister. Include at least 1 "verification" canister in every laundering cycle. It may be convenient to include 1 verification canister for every 4 test specimen canisters to balance the instrument. The verification canister is a "blank" to check for any contamination or ineffective rinsing during the testing process.

14.10 Re-clamp the lid with gasket and seal on each canister.

14.11 Fasten the canisters horizontally on the rotor shaft of the accelerated laundering testing machine in such a manner that when the canisters rotate, the covers strike the water first. Place an equal number of canisters on each side of the shaft. Distribute canisters with test specimens and verification canisters evenly in the rotating testing machine shafts.

14.12 Start the laundering testing machine and run it at 40 ± 2 rpm for 45 min.

14.13 When laundering cycle ends, remove canisters.

14.14 Pour the contents of the first canister into the sieve held over a beaker. Be careful to avoid splashing. Capture all wash water in the beaker. Capture specimen and steel balls in the sieve.

14.15 Rinse all contents and components of the canister, collecting all rinse water in the same beaker.

14.15.1 Without excessive handling of the specimen, lift it out of the sieve and rinse it using a wash bottle, spraying the specimen on the front and back sides to free loose fiber fragments from the surface. Collect all rinse water in the beaker, through the sieve. Do not squeeze or press the specimen.

14.15.2 Keeping the sieve over the beaker, use the wash bottle to rinse the steel balls with distilled/deionized water. Repeat for a total of 3 rinses. Collect all rinse water in the beaker, through the sieve. Remove balls from the sieve.

14.15.3 Rinse all sides of the empty sieve with the wash bottle, collecting all rinse water in the beaker. Repeat for a total of 3 rinses. Set sieve aside for use with the next specimen.

14.15.4 Using the wash bottle, rinse the gasket, seal, canister and lid, collecting all rinse water in the same beaker. Repeat for a total of 3 rinses.

14.15.5 Rinse gloved fingers that may have collected fibers through contact with specimen or canister components. Collect all rinse water in the same beaker.

14.16 Filter contents of the beaker. Ensure that specimen designation on the canister and filter weighing dish match.

14.16.1 Attach a freshly cleaned and rinsed filter support to a filtration flask and attach the flask to the vacuum pump.

14.16.2 Using forceps, carefully transfer filter from weighing dish to the support.

14.16.3 Place the funnel over the filter and clamp in place. Turn on the vacuum and seat the filter by wetting thoroughly with squirt bottle.

14.16.4 If an alternate filtration system is used, set up according to manufacturer's directions and note as a modification in the test report.

14.16.5 Carefully pour the contents of the collection beaker through the filter. Avoid splashing. Rinse the beaker with water at least 3 times to remove residual fibers from the beaker sides. Pour all rinse water through the filter.

14.16.6 Rinse funnel sides at least 3 times with the wash bottle.

14.16.7 Stop the vacuum, then unclamp and remove the funnel.

14.17 Using forceps, lift the edge of the filter and transfer the filter onto its weighing dish. Do not tip the filter.

14.18 If fibers remain on the underside of the funnel, use a medicine dropper to rinse fibers into the weighing dish. (Do not use a wash bottle as it may splash or spread fibers beyond the dish.) Use as little water as possible to avoid long dry times.

14.19 Cover the filter and dish to avoid contamination.

14.20 Repeat the rinse and filtration processes (14.14-14.19) for all remaining canisters, including verification canisters.

14.21 Condition or oven-dry filters.

14.21.1 Drying Option A: Place filters (in weighing dishes) in a standard atmosphere of 21 ± 2°C (70 ± 4°F) and 65% ± 5% RH for at least 4 h. Keep filters covered with aluminum foil to avoid contamination. Ideally, filters and collected fibers should reach moisture equilibrium at the standard conditions, but it is not always practical to verify this for every specimen; it is assumed that filters and fibers will reach equilibrium within 4 hours. If filters or fibers are still wet after 4 hours, maintain in standard atmosphere for at least 4 more hours before measuring post-test mass.

14.21.2 Drying Option B: Dry filters (in weighing dishes) in a non-air-circulating oven at 70 ± 2°C for at least 1 h. Use oven-dried filters for post-test mass. Place oven-dried filters and dishes in desiccator for 30 min. to cool. Maintain filters in the desiccator until immediately before post-test measurement.

14.22 Place each conditioned or oven-dried filter and weighing dish on the balance and record the total post-test mass to the nearest 0.0001 g (W_2). Keep filters away from possible contamination sources. Remove loose fibers from balance between specimens using a static-free brush. Use a deionizer to minimize static interference with the mass.

15. Calculations

15.1 Calculate the mass of fiber fragment release from each specimen using Eq. 1.

$$F_{m,n} = W_{2,n} - W_{1,n} \quad \text{(Eq. 1)}$$

where:

$F_{m,n}$ = Mass of fiber fragment release (g) for specimen n,
$W_{1,n}$ = Pre-test mass of conditioned or oven-dry filter and weighing dish (g), and
$W_{2,n}$ = Post-test mass of conditioned or oven-dry filter and weighing dish (g).

15.2 Calculate the mean mass of fiber fragment release and the standard deviation for each sample.

15.3 Calculate the percent of fiber fragment release of each specimen using Eq. 2.

$$F_{p,n} = 100\% * (W_{2,n} - W_{1,n})/S_{1,n} \quad \text{(Eq. 2)}$$

where:

$F_{p,n}$ = Fiber fragment release percentage (%) for specimen n,
$W_{1,n}$ = Pre-test mass of conditioned or oven-dry filter and weighing dish (g),
$W_{2,n}$ = Post-test mass of conditioned or oven-dry filter and weighing dish (g), and
$S_{1,n}$ = Pre-test mass of conditioned or oven-dry specimen (g).

15.4 Calculate the mean percent of fiber fragment release and standard deviation for each sample.

15.5 Calculate the accumulated contamination from each verification canister using Eq. 3.

$$F_{m,v} = W_{2,v} - W_{1,v} \quad \text{(Eq. 3)}$$

where:

$F_{m,v}$ = Accumulated contamination (g) from verification canister v,
$W_{1,v}$ = Pre-test mass of conditioned or oven-dry filter and weighing dish (g), and
$W_{2,v}$ = Post-test mass of conditioned or oven-dry filter and weighing dish (g).

16. Report

16.1 Describe or identify the tested sample.

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16.2 Report that the sample was tested using AATCC TM212-2021

16.3 Report the testing conditions:

16.3.1 Report which side of the fabric was used as the face.

16.3.2 Report whether distilled or deionized water was filtered before use.

16.3.3 Report whether Washing Option A (detergent solution) or Washing Option B (water only) was used.

16.3.4 Report detergent formulation and concentration, if applicable (default is AATCC HE detergent, WOB, 0.25%).

16.3.5 Report if Drying Option A (conditioned specimens and filters) or Drying Option B (oven-dried specimens and filters) was used for measurements.

16.4 Report the test results:

16.4.1 Report the average mass of fiber fragment release of each sample to the nearest 0.0001g. Note the number of significant digit(s). For example, if the mass of fiber fragment release is 0.0123 g, the number of significant digits is 3 (the digit "0" here does not count as significant digits).

16.4.2 Report the standard deviation for mass of fiber fragment release of each sample.

16.4.3 Report the average percent fiber fragment release of each sample, to the same number of significant digits as that of the average mass of fiber fragment release (see 16.4.1).

16.4.4 Report the standard deviation for percent fiber fragment release of each sample.

16.4.5 Report the accumulated contamination for each verification canister. Do not subtract from sample fiber fragment release values; it is reported for reference only. If verification canister values are higher than expected, consider retesting all samples, taking additional precautions to avoid contamination.

16.5 Describe any modification(s) of the published method, including alternate edge finish or filtration system.

17. Precision and Bias

17.1 *Precision*. In 2020, 2 laboratories completed a study with 3 fabrics. Fabric A was a cotton woven loop terry. Each lab used only 1 operator. Fabric B was a rayon/cotton woven flannel. Fabric C was a polyester double-sided fleece. All tests were performed with Drying Option A (conditioned specimens and filters). Data was collected for Washing Option A (AATCC HE detergent).

17.1.1 Table I gives the *within-lab* precision values for mass of fiber fragment release.

17.1.2 Table II gives the *within-lab* precision values for percent fiber fragment release. The extent of variation in fiber shedding measured as a percentage of the base fabric mass is much higher than that measured in simple shedding mass terms. This is expected because of the influence of fabric base mass on fiber shedding potential.

17.1.3 The precision tables represent a limited amount of data obtained by two different labs on three different fabrics. While the tables can be used for preliminary guidance/interpretation purposes, they need to be updated on the basis of data obtained for a bigger and more representative population of woven, knitted and nonwoven fabrics.

17.2 *Bias*. Fiber fragment release during home laundering can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

Table I—Precision Values for Mass Release

N	Within-Laboratory Precision (Drying Options A, Washing Option A)*
1	0.0005240
2	0.0003711
3	0.0003025
4	0.0002620
5	0.0002343
6	0.0002139
7	0.0001977
8	0.0001853
9	0.0001747
10	0.0001657

*Grand mean of fiber fragment release for all interlab measurements was 0.0020 g. Mean fiber fragment release for Fabric A was 0.0016 g. Mean release for Fabric B was 0.0034 g. Mean release for Fabric C was 0.0013 g.

Table II—Precision Values for Percent Release

N	Within-Laboratory Precision (Drying Options A, Washing Option A)*
1	0.0004740
2	0.0003357
3	0.0002737
4	0.0002370
5	0.0002120
6	0.0001935
7	0.0001792
8	0.0001676
9	0.0001580
10	0.0001499

*Grand mean of fiber fragment release for all interlab measurements was 1.2753%. Mean fiber fragment release for Fabric A was 0.007197%. Mean release for Fabric B was 0.026414%. Mean release for Fabric C was 0.006355%.

18. Notes

18.1 Available from ASTM International, www.astm.org.

18.2 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse, or certify that any of the listed equipment or materials meets the requirements in its test methods.

18.3 A sewing needle should be selected based on fabric construction. In general, the smallest suitable needle is preferred to limit damage and distortion of the specimen.

18.4 Best practice is to filter water to minimize contamination. This is a time-consuming process and may not be practical in all cases but choosing not to filter water before use must be recognized as a potential source of error.

18.5 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc. org.

18.6 Filament thread is recommended to minimize contamination from sewing thread fiber fragments. All thread ends should be carefully trimmed and discarded. Multifilament thread may be easier to use than monofilament thread, but either is acceptable. Nylon or polyester filament is acceptable.

19. History

19.1 Developed in 2021 by AATCC Research Committee RA100.

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Test Method for Vertical Wicking Rate of Textiles: to Specified Times

Foreword

Various versions of the vertical wicking test have been used to evaluate textiles for many years. In 2004, AATCC and ASTM International jointly published *Technical Supplement*, a collection of unofficial industry procedures. This test method is a more thorough and standardized version of the vertical wicking procedure published in the *Technical Supplement*. When it was first published, this test method included options for measuring wicking rate to specified distances and to specified times. In 2021, the rate to specified distance remained in AATCC TM197 while the rate to specified times option was transferred to this method.

1. Purpose and Scope

1.1 This test method is used to evaluate the ability of fabric specimens to transport liquid vertically when a cut edge is submerged.

1.2 This method is applicable to woven, knitted, or nonwoven fabrics.

2. Principle

2.1 The distance liquid travels vertically up a fabric specimen from a submerged cut edge is visually observed, manually timed and recorded at specified intervals.

2.2 The movement of liquid along a fabric may be influenced by fiber content, fabric construction, mechanical or chemical processing or a combination of these.

2.3 Results are reported as wicking rates (distance per unit of time) for a short period (120 s) and a long period (600 s).

3. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

3.1 AATCC LP1, Laboratory Procedure for Home Laundering: Machine Washing (see 16.1).

3.2 AATCC TM197, Test Method for Vertical Wicking Rate of Textiles: to Specified Distances (see 16.1).

3.3 ASTM D123, Standard Terminology Relating to Textiles.

3.4 ASTM D1331, Standard Test Methods for Surface and Interfacial Tension of Solutions of Paints, Solvents, Solutions of Surface-Active Agents, and Related Materials.

3.5 ASTM E1402, Standard Guide for Sampling Design.

3.6 ASTM D1776, Standard Practice for Conditioning and Testing Textiles.

4. Terminology

4.1 **absorption,** n.—a process in which one material (the absorbent) takes in another (the absorbate); as the absorption of moisture by fibers. (ASTM D123)

4.2 **capillary action,** n—the process by which a liquid flows through narrow spaces via forces of adhesion, cohesion, and surface tension, and without the use of external forces, such as gravity.

4.3 **fabric,** n.—*in textiles*, a planar structure made from yarns or fibers (ASTM D123).

4.4 **vertical wicking,** n.—*in a textile held vertically*, the upward movement of liquid via absorption or capillary action.

4.5 **vertical wicking rate,** n.—the speed at which liquid travels along or through a textile

4.6 **wicking,** n.—*in textiles*, the movement of a liquid, by capillary action, along or through a material.

4.7 **wicking distance,** n.—the linear measurement that the liquid travels along or through a textile from starting point to a stopping point.

4.8 **wicking time,** n.—the measurable period during which liquid travels along or through a textile

5. Safety Precautions

5.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

5.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

5.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

6. Limitations

6.1 Liquids other than distilled or deionized water (tinted water, dye solutions, etc.) may be used in this test. Solutions of different surface tensions may yield different results.

6.2 The procedure measures the time for water to move vertically upward from the submerged cut edge of a specimen, but does not duplicate an end-product's exposure during wear.

6.3 The results obtained by this test are not a measure of comfort which is beyond the scope of this method.

6.4 The relationship between results obtained by vertical and horizontal wicking tests is not known.

7. Apparatus, Reagents, Materials (see 16.2)

7.1 Distilled or deionized water at 21 ± 2°C (70 ± 4°F).

7.2 Marking pens, with fine point, permanent and soluble ink (see 16.3).

7.3 Stopwatch or digital timer.

7.4 Measuring tape or ruler, mm graduations.

7.5 Surface tensiometer, if using liquid other than distilled or deionized water (see 11.1).

7.6 Erlenmeyer flask, 500 mL; or rectangular pan, long enough to accommodate all specimens side-by-side

7.7 Scissor jack, if using Erlenmeyer flask.

7.8 Means of suspending specimen(s) into a flask or pan (see Fig. 2 and Fig. 3).

7.9 Small straight pins (optional, see 11.2).

7.10 Powder-free disposable gloves, e.g., latex or nitrile.

8. Sampling

8.1 Test results are valid only when the samples are statistically representative (see ASTM E1402).

8.2 Sampling must be random. Every unit of product must have a mathematically equal chance of becoming a sample; and every portion of each sample must be equally likely to become a test specimen.

8.3 All specimens must be alike within the variations due to pure chance. There must be no differences within samples assignable to known causes.

9. Specimens

9.1 Liquid may wick up front and back of fabric at different rates. Upon agreement among interested parties, determine whether both fabric sides are to be measured. If only one fabric side is to be

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measured, mark the selected side. If testing is to be performed after laundering, use a marking pen with permanent ink to denote the fabric side chosen for testing.

9.2 Upon agreement between interested parties, determine in which direction(s) the fabric will be tested.

9.3 To test the durability of finishes or evaluate wicking properties after laundering, launder fabric swatches or items according to conditions chosen from AATCC LP1 before cutting specimens.

9.4 Wear gloves and minimize handling of the test specimens as oils transferred from the skin may affect liquid movement.

9.5 Cut specimens at least 100 mm from the selvage. If testing sewn products, take specimens from different panels, at least 100 mm away from seams, pockets, plackets and other assembly features. Take each specimen such that different sets of (fabric) length and width yarns are present. For nonwoven fabrics, spread specimens across length and width of sample.

9.6 Cut 3 specimens, each 25 ± 3 mm wide and at least 165 mm long.

9.6.1 Cut the long dimension parallel to the fabric direction(s) being tested. Accurate dimensions and alignment of specimens are critical for repeatable testing.

9.7 Cut an extra specimen to determine if a pin is needed to submerge the specimen end (see 11.2).

9.8 Using a marking pen with soluble ink, mark a line across the width of each specimen, 5 ± 1 mm from the end to be submerged, on the fabric side to be tested. The line denotes the level to which the specimen will be submerged at the start of the test. This is 0 for measuring wicking distance (see Fig. 1).

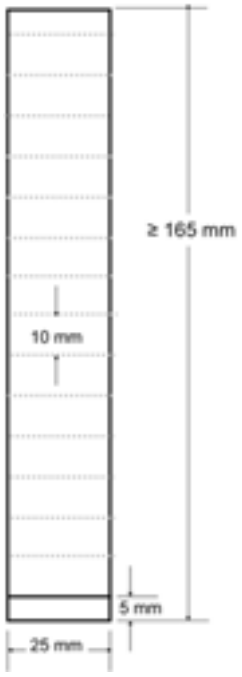


Fig. 1—Vertical wicking specimen and marking.

9.9 To facilitate the measurement of wicking distances, use a marking pen with soluble ink to mark intervals of 10 ± 1 mm above the first line (see Fig. 1).

10. Conditioning

10.1 Prior to testing, condition the specimens as directed in ASTM D1776. (Use conditions indicated in Table 1 for Textiles, general. Estimate conditioning time as indicated in Table 2 for the appropriate fiber content.) Lay each specimen, separately, on a screen or perforated shelf of a conditioning/drying rack.10.2 Perform all tests in the standard atmosphere for testing (21 ± 2°C [70 ± 4°F], 65 ± 5% RH).



Fig. 2—Specimen prepared for vertical wicking testing in an Erlenmeyer flask.

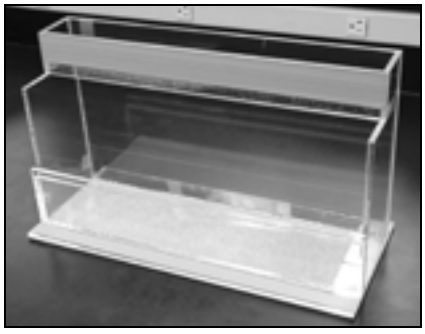


Fig. 3—Rectangular pan for testing multiple specimens.

11. Preparation

11.1 For liquids other than distilled or deionized water (including tinted water or dye solution), measure the surface tension as directed in ASTM D1331.

11.2 Some lightweight or hydrophobic fabrics may float on the water. Before testing, lower one short end of the extra specimen into water.

11.2.1 If the specimen is easily submerged, proceed with marking remaining specimens.

11.2.2 If the end of the specimen floats on the surface of the water, insert a straight pin across the end. Use the minimum weight necessary to allow the specimen to submerge. Use the same size pin for all specimens in the sample. When comparing results for various samples, it is best to use the same size pin for all.

11.2.3 Do not reuse the water or sample. Use a clean container and fresh water for each test.

12. Procedure

12.1 Perform test on a stable surface to minimize vibrations.

12.2 To test a single specimen in an Erlenmeyer flask, add approximately 90 mL of water (or alternate liquid) to the flask. Ensure water depth is at least 10 mm. Ensure that the lip and neck of the flask remain dry to prevent specimen from premature contact with water.

12.2.1 Place the flask on a scissor jack and suspend the specimen in the opening, above the water line. The specimen should hang perpendicular to the flask floor, with no folds or waves. Insert a pin across the bottom end as needed (see 11.2).

12.2.2 Raise the jack until the specimen end is submerged to the 5-mm line and start the timer.

12.3 To test multiple specimens in a rectangular pan, suspend specimens with bottom ends just above the floor of the pan. Ensure that no part of any specimens is in contact with the pan. The specimens should hang perpendicular to the pan floor, with no folds or waves. Insert a pin as needed (see 11.2). It is critical to ensure the 5-mm line on each specimen is at the same level before adding water.

12.3.1 Carefully add water to the pan. Avoid pouring or splashing water onto specimens. Start a timer as soon as the specimens are submerged to the 5-mm line.

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12.4 Monitor the rise of the water, visible as migration of the marking ink and/or darkening of the specimen(s). Record the distance from the start line to the point of ink migration and/or darkening, to the nearest mm, at 120 s (2 min.) This is the short-period wicking distance. Multiple observers or timers may be necessary to accurately record the wicking distance of multiple specimens tested simultaneously in a rectangular pan. If wicking is uneven across the width of the specimen, record the highest point and note observations in the test report.

12.5 Continue monitoring the rise of the water and record, to the nearest mm, the distance at 600 s (10 min.) This is the long-period wicking time. If wicking is uneven across the width of the specimen, record the highest point and note observations in the test report.

12.6 Terminate the test when any of the following conditions are met:

12.6.1 Water does not wick beyond the 5-mm line within 600 s (10 min)

12.6.2 Water wicks beyond the top of the specimen in less than 600 s (10 min)

12.6.3 600 s (10 min) have elapsed

12.7 Record the time, distance (mm) and the reason for termination of the test.

12.8 For tests in a flask, remove the specimen from the flask. Repeat the procedure for the remaining specimens.

12.8.1 Use a clean flask with fresh water for each specimen.

13. Calculation

13.1 Calculate the vertical wicking rates using Eq. 1. Two different rates, a short period rate and a long period rate, are obtained for each specimen.

$$W = d/t \quad (1)$$

where:

W = vertical wicking rate, mm/s
d = wicking distance, mm
t = wicking time, s (120 for short period, 600 for long period, or as agreed)

NOTE: The standard wicking time is 120 s for the short period and 600 s for the long period. Alternate times may be used if the test was terminated or as agreed by interested parties.

13.2 Calculate the mean short-period wicking rate for each direction tested, to the nearest 0.1 mm/s.

13.2.1 If desired, calculate standard deviation.

13.3 Calculate the mean long-period wicking rate for each direction tested, to the nearest 0.1 mm/s.

13.3.1 If desired, calculate standard deviation.

14. Report

14.1 Describe or identify the sample tested.

14.2 Report that the sample was tested using AATCC TM213-2022.

14.3 Report the testing conditions:

14.3.1 If testing was performed after laundering, report the laundering conditions used and the number of cycles completed.

14.3.2 Report the fabric side(s) and direction(s) tested.

14.3.3 Report the liquid used, and surface tension if liquid other than distilled or deionized water was used.

14.3.4 Report the weight of pin, if used to submerge specimen end.

14.3.5 Report the short- and long-period wicking times (usually 120 s and 600 s).

14.3.6 Report the time at which test was terminated, if applicable (see 12.6)

14.4 Report the test results:

14.4.1 Report the mean short-period wicking rate for each direction tested.

14.4.2 Report the individual specimen rates or standard deviation for the short period, in each direction tested.

14.4.3 Report the mean long-period wicking rate for each direction tested.

14.4.4 Report the individual specimen rates or standard deviation for the long period, in each direction tested.

14.5 Describe any modification(s) of the published method.

15. Precision and Bias

15.1 *Precision*. Tests for vertical wicking of textiles to a specified time were conducted in 2021, with 1 laboratory, 1 operator, and 4 samples. The samples used in the study were (a) 65% polyester/35% cotton lightweight woven broadcloth, (b) 100% polyester felt, (c) 50% polyester/25% cotton/25% rayon jersey knit, and (d) cotton blend brushed woven twill (exact fiber composition unknown).

15.1.2 Table I gives the *within-lab* precision values for short-period wicking rate. Table II gives the *within-lab* precision values for long-period wicking rate.

15.1.3 *Between-laboratory* precision has not been established for this test method. Until such precision information is available, users of the method should use standard statistical techniques in making any comparison of test results for *between-laboratory* averages.

15.2 *Bias*. The true value of vertical wicking rates of textile fabrics can be defined only in terms of a test method. There is no independent method for determining the true value. In estimating this property, the test method has no known bias.

Table I—Precision Values for Short-Period Wicking Rate

n	Within-lab Precision
1	0.15
2	0.11
3	0.09
4	0.08
5	0.07

Note: Grand mean of short-period wicking rate for all measurements was 0.25 mm/s. Mean rate for Fabric A was 0.32 mm/s. Mean rate for Fabric B was 0.27 mm/s. Mean rate for Fabric C was 0.003 mm/s. Mean rate for Fabric D was 0.41 mm/s.

Table II—Precision Values for for Long-Period Wicking Rate

n	Within-lab Precision
1	0.06
2	0.04
3	0.03
4	0.03
5	0.03

Note: Grand mean of long-period wicking rate for all measurements was 0.09 mm/s. Mean rate for Fabric A was 0.14 mm/s. Mean rate for Fabric B was 0.06 mm/s. Mean rate for Fabric C was 0.01 mm/s. Mean rate for Fabric D was 0.15 mm/s.

16. Notes

16.1 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

16.2 For potential apparatus, reagents or materials sources, visit the AATCC Buyer's Guide at www.aatcc.org/bg. AATCC provides its Corporate members the option to list their items and services. AATCC does not qualify, or in any way approve, endorse or certify that any of the listings meet the specifications in its standards.

16.3 Dark fabric colors or prints and designs may be difficult to test. Using soluble marking ink of a contrasting color may help make the marks more clearly visible. Fluorescent ink or wicking liquid and observation under ultraviolet light (backlight) may also improve visibility.

17. History

17.1 Developed in 2022 (previously part of AATCC TM197), by AATCC Committee RA114.

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Test Method for Measuring Condensation on Textiles in a Humid Microclimate

1. Foreword

1.1 While several test methods exist to measure the effectiveness of moisture transport mechanisms through fabric, they do not measure the amount of water condensation created within the microclimate of the textile product. This method provides a measure of the fabric contribution to the end-product condensation accumulation.

2. Purpose and Scope

2.1 This test method measures the water condensation accumulation on the innermost (back) surface of fabrics in a humid microclimate, with and without simulated precipitation on the outermost (face) fabric surface.

2.2 The method is applicable to fabrics or multi-layer fabric systems.

3. Principle

3.1 The test method measures the amount of water condensation produced in 15 min. on a specimen placed in a humid microclimate created by controlling temperature and relative humidity of chambers on either side of the specimen. Specimens are tested with a sprinkler to simulate rain, and with no sprinkler.

3.1.1 The amount of water is calculated as the mass gain of a piece of preweighed filter paper after absorbing the condensation from the specimen.

3.2 Results are reported as g/m^2, where a higher value indicates more water condensation. Area and time are constant for all specimens.

4. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

4.1 ASTM D1776, Standard Practice for Conditioning and Testing Textiles.

4.2 ASTM E1402, Standard Guide for Sampling Design.

5. Terminology

5.1 **back,** n—*in textiles*, the side of a fabric that is intended to be the inner surface in an end product.

NOTE: *Back* is the opposite of *face*.

5.2 **face,** n—*in textiles*, the side of a fabric that is intended to be the outer, visible surface in an end product.

NOTE: The technical face of the fabric is defined by production processes but may *not* be the side selected as face in the end product.

5.3 **microclimate,** n— *in textiles*, the air layer immediately adjacent to one side of a textile.

NOTE: In apparel, the microclimate typically refers to the air layer between skin and clothing.

5.4 **water condensation,** n—liquid water droplets formed when airborne water vapor contacts a colder surface.

6. Safety Precautions

6.1 The safety precautions specified in the method are ancillary to the testing procedures and are not intended to be all inclusive.

6.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

6.3 Users MUST consult manufacturers for specific details such as safety data sheets, equipment operating instructions, and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

7. Limitations

7.1 While this method may be used for any textile fabric, results will vary based on absorbency as well as vapor permeability (see Appendix B).

7.2 This test method measures condensation accumulation at a specific timepoint (15 min humidity exposure). Because accumulation is not linear nor infinite, this result cannot be used to predict accumulation at other timepoints (see Appendix B).

7.3 This test method does not address phenomena such as rain penetration, ventilation, wind resistance, air permeability or breathability.

8. Apparatus and Materials (see 17.1)

8.1 Apparatus with inner and outer chambers to simulate the microclimate between human skin and clothing (including headwear and footwear), and sprinklers to simulate precipitation. See Appendix A for apparatus specifications.

8.2 Specimen mounting rings.

8.2.1 Inner mounting rings (4): 165.1 mm inside diameter, 171.5 mm outside diameter, 12.7 mm height.

8.2.2 Outer clamping rings with splash guards (2): 172.7 mm inside diameter, 177.8 mm outside diameter, 12.7 mm height.

8.2.3 Outer clamping rings (2): 172.7 mm inside diameter, 177.8 mm outside diameter, 12.7 mm height.

8.3 Distilled water, for sprinkler system reservoir and water vaporization plate reservoir.

8.4 Filter paper, Grade 1 qualitative, 163-mm circles (see 17.2).

8.5 Balance, minimum resolution of 0.01 g.

8.6 Stopwatch, minimum resolution of 0.1, accuracy of ± 0.1 s.

8.7 Weight, 2 kg, 163-mm diameter.

9. Sampling

9.1 Test results are valid only when the samples are statistically representative (see ASTM E1402).

9.2 Sampling must be random. Every unit of product must have a mathematically equal chance of becoming a sample; and every portion of each sample must be equally likely to become a test specimen.

9.3 All specimens must be alike within the variations due to pure chance. There must be no differences within samples assignable to known causes.

10. Specimens

10.1 Cut 4 specimens of the test fabric per sample, with a minimum dimension of 200 mm in each direction.

10.1.1 Specimens may be round, rectangular, or a non-standard shape, but must be large enough to cover the ports of the test apparatus and be secured by the specimen rings.

10.1.2 Mark pertinent identification of construction (e.g., warp, course, face, water repellent finish side) on each specimen. Mark which 2 specimens will be tested with simulated precipitation from sprinklers and which 2 specimens will be tested dry.

10.1.3 For multi-layer fabric systems, secure all layers together, leaving an open area (no stitches or other means of attachment) of at least 152.4 mm diameter. Use the open area for testing.

10.2 Use 4 pieces of filter paper. Cut to 163-mm circles.

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10.2.1 With indelible ink, mark each piece of filter paper with consecutive numbers or letters for identification purposes.

11. Conditioning

11.1 Bring specimens and filter paper to moisture equilibrium in a standard atmosphere according to ASTM D1776. Use conditions listed for Textiles, General.

12. Preparation of Apparatus

12.1 Secure a cap or other non-porous material over each opening in the test apparatus.

12.2 Fill the sprinkler reservoir. Do not add water above the maximum fill level indicated by the manufacturer.

12.3 Fill the water vaporization plate reservoir to the top with distilled water.

12.4 Set the sprinkler reservoir temperature to 10°C.

12.5 Set the inner chamber air temperature to 35°C.

12.6 Set the inner chamber humidity to 95%.

12.7 Allow the apparatus to reach the prescribed settings.

13. Procedure

13.1 Measure the mass of each conditioned filter paper circle. Record filter paper identification and conditioned mass to the nearest 0.01 g.

13.2 Perform tests in the standard atmosphere for testing textiles.

13.3 When the apparatus inner chamber reaches 35°C and 95% RH, remove the cap from each opening.

13.3.1 If the machine has been idle for more than 30 min., run a full test cycle with the caps or non-porous material in place. Remove caps before mounting fabric specimens.

13.4 Place the specimens over the openings using the mounting and clamping rings. Place the rings with splash guards on the side with sprinkler heads. See Fig. 1. It is important that the specimens are pulled taut such that the material is free from creases or dips that can fill with moisture. First-level saturation of the material will impair transmission and affect the results.

13.5 Begin test within 5 minutes of mounting the test specimen Start sprinklers and timer. Keep inner chamber settings at 35°C and 95% RH throughout the test.

13.6 After 15 min., stop sprinklers and prepare stopwatch.

13.7 Remove one specimen. If additional tests are to be performed. cover the opening with a cap. Immediately place the specimen on a flat surface, face down.

13.8 After 20 s, place a pre-weighed filter paper circle on the back of the specimen, inside the ring. Place the 2-kg weight on the filter paper for 15 s.

13.9 Remove the weight. Measure the mass of the wet filter paper circle. Record filter paper identification and wet mass to the nearest 0.01 g.

13.10 Repeat 13.7-13.9 for remaining specimens. After the final specimen, dry all rings and splash covers.

14. Calculation

14.1 For each specimen, calculate the condensation accumulation to the nearest g/m² using Eq. 1.

$$C = (W_a - W_b)/0.02088 \qquad \text{(Eq. 1)}$$

where:

C = Condensation accumulated,
W_a = Mass of filter paper after blotting specimen (g),
W_b = Mass of filter paper before blotting specimen (g),
0.02088 = Surface area of 163-mm diameter filter paper (m²),

Round *up* to the nearest whole number

15. Report

15.1 Describe or identify the sample tested.

15.2 Report that the sample was tested using AATCC TM214-2022.

15.3 Report the testing conditions:

15.3.1 Specimen side tested (face or back).

15.4 Report the test results:

15.4.1 Quantity of water condensation accumulated for each specimen (g/m²), to the nearest whole number. Note which specimens were tested with and without simulated precipitation.

15.4.2 Mean condensation accumulation of the dry specimens.

15.4.3 Standard deviation of condensation accumulation for the dry specimens.

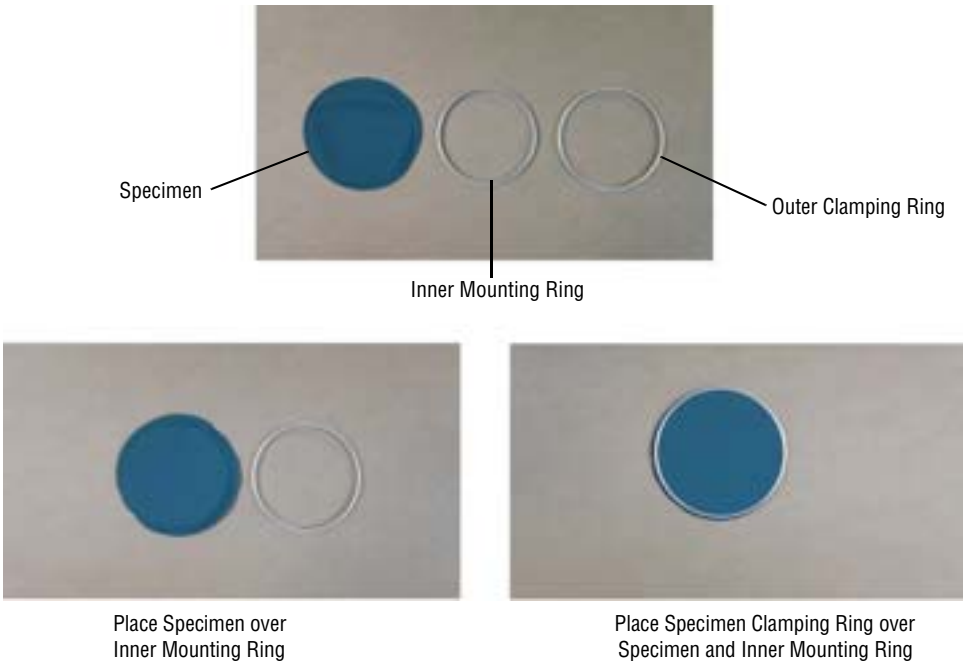


Fig 1a. Mounting procedure for specimens tested dry.

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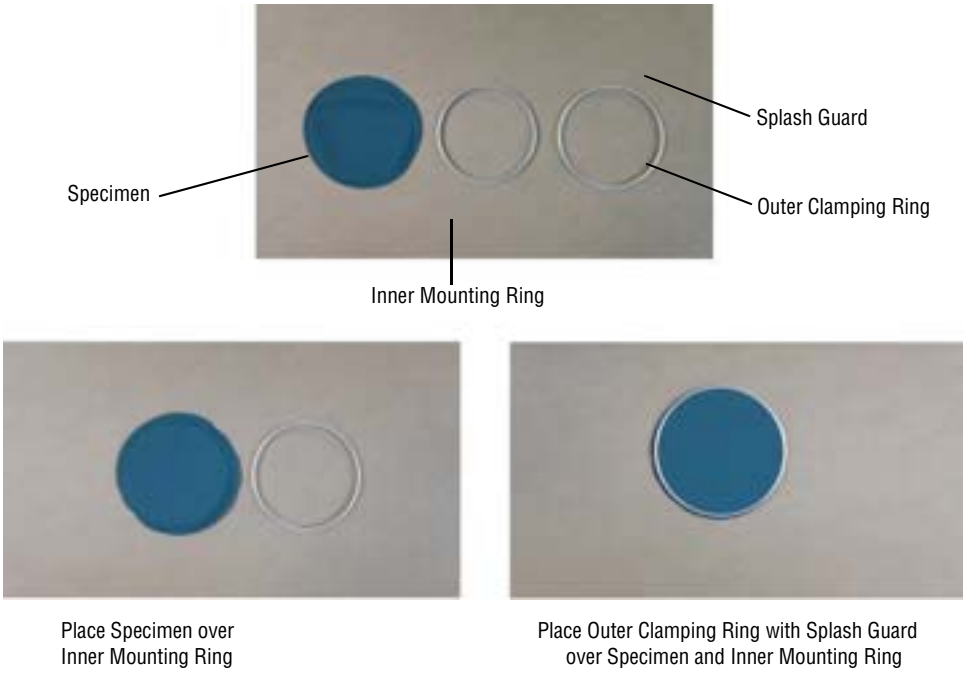


Fig 1b. Mounting procedure for specimens tested with simulated precipitation.

15.4.4 Mean condensation accumulation of the sprayed specimens.

15.4.5 Standard deviation of condensation accumulation for the sprayed specimens.

15.5 Describe any modification(s) of the published method, including alternate test duration.

16. Precision and Bias

16.1 *Precision—Between-laboratory* precision has not been established for this test method. Until such precision information is available, users of the method should use standard statistical techniques in making any comparison of test results for *between-laboratory* averages. A single-laboratory test was run in 2019 in which 10 randomly-drawn specimens of a 3-layer water-resistant fabric were tested by a single operator. Confidence intervals for the single-lab study are found in Table I.

16.2 A single-laboratory test run in 2021 in which 2 operators used 10 samples each from the same roll of water-resistant fabric on a wet test and a further 10 samples each on a dry test (see 3.3 for test options) to define the possible operator variable on achieved results. The results can be seen in Table 2 and Table 3.

16.3 *Bias.* Condensation accumulation can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

17. Notes

17.1 For potential apparatus, reagents or materials sources, visit the AATCC Buyer's Guide at www.aatcc.org/bg. AATCC provides its Corporate members the option to list their items and services. AATCC does not qualify, or in any way approve, endorse or certify that any of the listings meet the specifications in its standards.

17.2 Use Grade 1 qualitative filter paper meeting the following specifications:

Pore size:	11 µm
Thickness:	160-200 µm
Basis Weight:	85-90 g/m²

18. History

18.1 Developed in 2022 by AATCC Committee RA63.

Appendix A
Test Apparatus

A1. The apparatus produces water condensation on test specimens. Condensation is formed by different temperature and humidity conditions in the inner and outer chambers.

A2. The outer chamber contains a sprinkler system to simulate rain. The water temperature is regulated at 10°C ± 1.5°C. The flow rate is regulated to cause falling droplets to land on test specimens at a 30° ± 5° angle. See Fig. A1 and Fig. A2.

A3. The inner chamber contains a variable heat output water vaporization

Table I—Within-Laboratory Confidence Intervals

Sample	Mean (g/m²)	St. Dev.	95% Confidence
A	54	5.8	4.2

Table 2—Operator 1 Comparison Results (g/m²/h)

Option A (Wet Test)		Option B (Dry Test)	
Test 1-5 Average	Test 6-10 Average	Test 1-5 Average	Test 6-10 Average
222.991	224.673	7.850	8.073

Table 3—Operator 2 Comparison Results (g/m²/h)

Option A (Wet Test)		Option B (Dry Test)	
Test 1-5 Average	Test 6-10 Average	Test 1-5 Average	Test 6-10 Average
218.131	219.065	7.290	8.224

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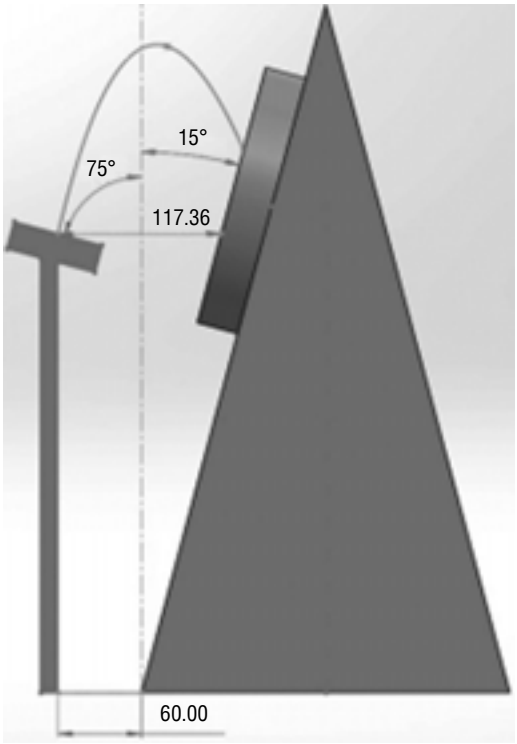
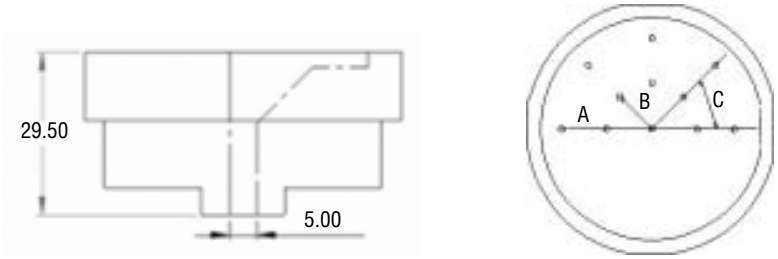


Fig A1. Schematic drawing of a sprinkler head (left) and mounted test specimen (right).



A = 18 mm radius
B = 9 mm radius
C = 45° angle
Flow rate = 60 ± 5 L/h per sprinkler

Fig A2. Detailed drawing of sprinkler head nozzle. Nozzle face has 11 holes of 1-mm diameter arranged in two semi-circles of five and one central.

plate with a 100-mL water reservoir and a supplemental variable heating plate to maintain air temperature. The heat plates can be adjusted to produce humidity levels up to 100% RH and air temperatures between 20°C and 60°C. Actual temperature and humidity are controlled within 5% of set values. Vent holes at the top maintain atmospheric pressure of the inner chamber. See Fig A3.

A4. Openings (at least 2) between the inner and outer chambers are used to mount test specimens. Each opening is 152.4 mm in diameter.

A5. Temperature and humidity sensors are level with the openings to maintain inner chamber temperature and RH parameters within 5% of set values.

A6. Non-porous caps fit securely over each opening for the preparation phase. Sheets of non-porous film (e.g., polyethylene or polythene with a minimum thickness of 40 m) may be used as an alternative to caps.

Appendix B

This test relies upon the initial 15 minutes a fabric is subjected to a higher humidity/water exposure condition to characterize the amount of condensate that is produced. This time period might not be sufficiently long for the fabric under investigation to reach an equilibrium state for the greater moisture levels, particularly for fabrics incorporating hydrophilic components. Consequently, test results obtained for this initial 15-minute exposure period may be low compared to a 15-minute period at equilibrium with the greater moisture level. For example, a cotton-based fabric or a fabric with a hydrophilic water barrier component conditioned to standard laboratory temperature and humidity (per 11.1) will have greater moisture diffusivity than after absorbing moisture to its equilibrium state; consequently, less condensation will form during the initial 15-minute period represented by this test than during subsequent 15-minute periods.

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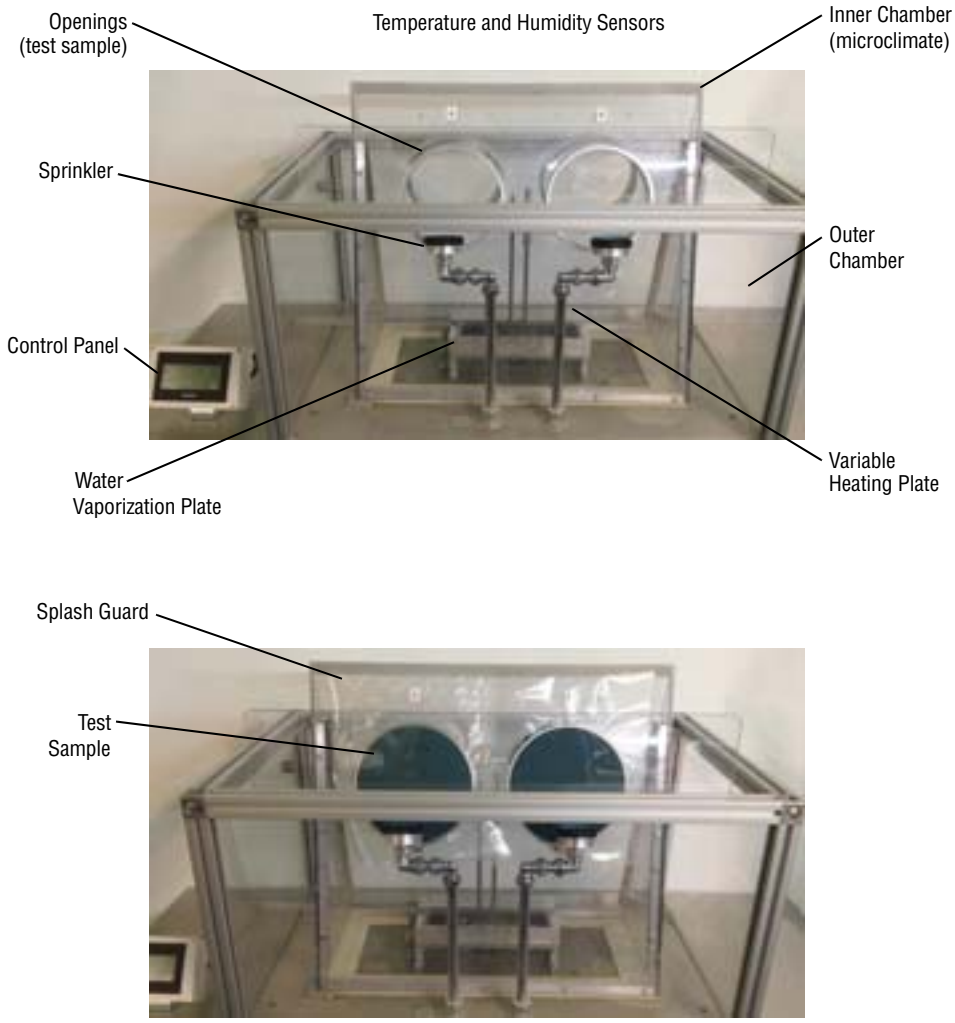


Fig A3. Condensation test apparatus.

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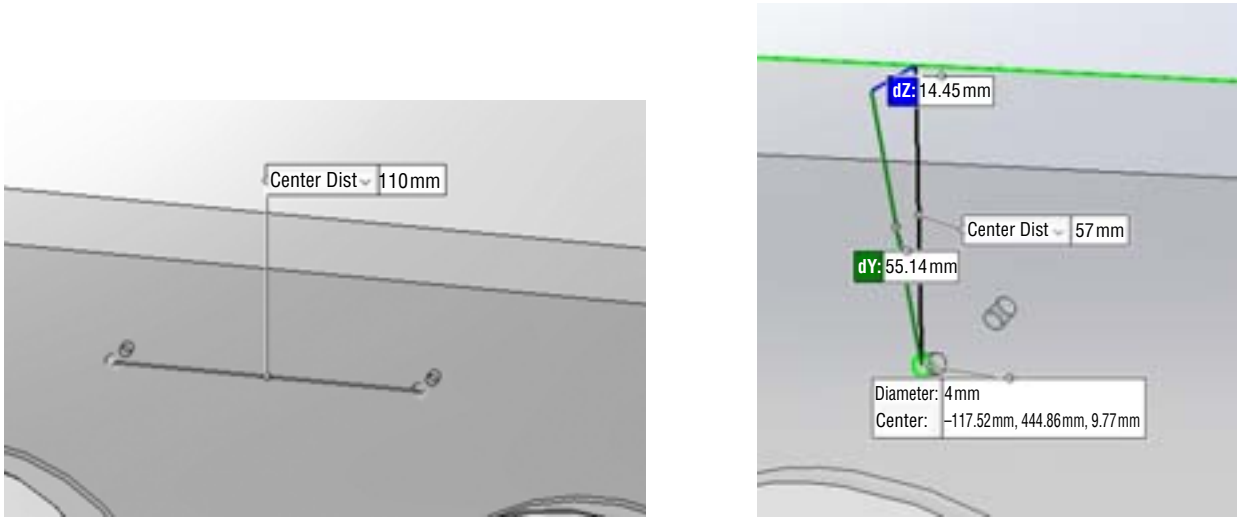


Fig A4. Vent holes—4 holes, Diameter 4 mm, 110 mm pitch.

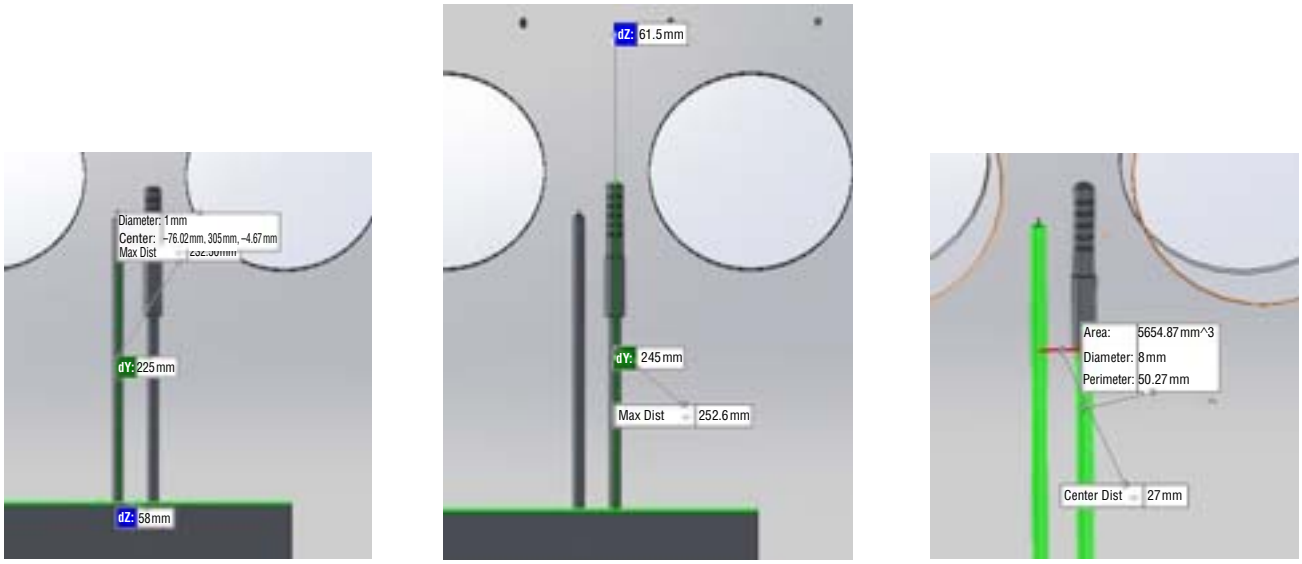


Fig A5. Thermocouple and Hygrometer position 27 mm pitch, 225 mm from heat plate.

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Test Method for Soil Release: Household Soils

1. Foreword

1.1 This test method measures the ability of fabrics to release household soils during home laundering. Standard laundering procedures represent, but may not exactly replicate, current consumer practices, which vary by sample, over time and among households.

Alternate soils or care procedures may be addressed by modifications of this method or other methods, including AATCC TM130, Test Method for Soil Release: Oily Stain Release.

In this method, a soil is forced into the fabric before laundering. To evaluate a fabric's ability to repel oils, use AATCC TM118, Test Method for Oil Repellency: Hydrocarbon Resistance.

2. Purpose and Scope

2.1 This test method is designed to measure and evaluate the ability of fabrics and garments to release typical household soils during home laundering.

2.2 The method is applicable to textile fabrics suitable for home laundering, particularly those with soil-release finishes. It is not applicable to yarns and fibers.

2.3 This method is not intended to evaluate differences or efficacy in commercial detergents or laundering equipment.

3. Principle

3.1 A soil is applied to a test specimen. The soil is allowed to dwell for a specified period of time to set. The soiled specimen is then laundered in a prescribed manner and any residual stain is evaluated by comparison with a stain release scale showing a graduated series of stains.

3.2 Results are reported as a grade from 5 to 1, with 5 representing no residual stain and 1 representing the most residual stain.

4. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

4.1 AATCC LP1, Laboratory Procedure for Home Laundering: Machine Washing (see 17.1).

4.2 AATCC LP4, Laboratory Procedure for Synthetic Soil (see 17.1).

4.3 AATCC TM118, Test Method for Oil Repellency: Hydrocarbon Resistance (see 17.1).

4.4 AATCC TM130, Test Method for Soil Release: Oily Stain Release Method (see 17.1).

4.5 AATCC TM133, Test Method for Colorfastness to Heat: Hot Pressing (see 17.1).

4.6 ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 17.2).

4.7 ASTM E1402, Standard Guide for Sampling Design (see 17.2).

5. Terminology

5.1 **soil,** n.— dirt, oil or other substances not normally intended to be present on a substrate such as a textile material.

5.2 **stain,** n.— a local deposit of soil or discoloration on a substrate that exhibits some degree of resistance to removal, as by laundering or drycleaning.

5.3 **soil release,** n.—the degree to which a soiled substrate approaches its original, unsoiled appearance as a result of a care procedure.

6. Safety Precautions

6.1 The safety precautions specified in the method are ancillary to the testing procedures and are not intended to be all inclusive.

6.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

6.3 Users MUST consult manufacturers for specific details such as safety data sheets, equipment operating instructions, and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

7. Limitations

7.1 Standard laundering conditions are not intended to replicate consumer care for all samples. By laundering all samples under the same conditions, this method provides a relative measure of soil release properties. The method may be modified to use recommended care for a particular sample, but results should not be compared with those for samples laundered under different conditions.

7.2 Specimens prepared for this method should not be used for dimensional change or appearance evaluation. Specimens for this method should be laundered separately from specimens for dimensional change or appearance. Although specimen dimensions are the same for these methods, oily stains may interfere with evaluation of other properties.

8. Apparatus and Materials (see 17.3)

8.1 Conditioning facilities and racks with pull-out, perforated shelves for laying specimens flat (see 17.1).

8.2 AATCC Blotting Paper (see 17.1).

8.3 Glassine paper (see 17.1).

8.4 Timer.

8.5 Cylindrical weight, 6.4 cm (2.5 in.) diam., 2.268 ± 0.045 kg (5.0 ± 0.1 lb) (stainless steel is preferable, see 17.1). Multiple weights are recommended to efficiently process multiple samples.

8.6 Dropper, capable of dispensing 20 drops per milliliter (for liquid soils).

8.7 Syringe, oral type, plastic, 10 mL (for thick/solid soils).

8.8 100% cotton cloth, white, for blotting soils.

8.9 Automatic washing machine (see AATCC LP1-2021 Table I, 17.4).

8.10 Tumble dryer, normal cycle: maximum exhaust temperature 68°C ± 6°C (155°F ± 10°F), cool down time ≤ 10 min. or facilities for drip/line/flat drying (see 17.4).

8.11 1993 AATCC Standard Reference Detergent (powder, with optical brightener) (see 17.1).

8.12 Balance with at least 5.0-kg (10.0-lb) capacity.

8.13 Laundering ballast, Type 1 (see AATCC LP1-2021 Table II, 17.5).

8.14 Water hardness test kit, residential use (see 17.6).

8.15 Evaluation area, as described in Appendix A.

8.16 AATCC Stain Release Scale (see 17.1).

8.17 Typical household soils (see Table I).

9. Sampling

9.1 Test results are valid only when the samples are statistically representative (see ASTM E1402).

9.2 Sampling must be random. Every unit of product must have a mathematically equal chance of becoming a sample; and every portion of each sample must be equally likely to become a test specimen.

9.3 All specimens must be alike within the variations due to pure chance. There must be no differences within samples assignable to known causes.

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Table I—Typical Household Soils[1]

Liquid Soils	Thicker/Solid Soils
Red wine	Ketchup
Grape juice	Mustard
Orange juice	Standard dirt[2]
Sweat (lab grade only)	Grass slurry (local/regional grasses as agreed upon)
Betadine	Butter
Urine (lab grade only)	Mayonnaise
Blood (lab grade only)	
Chocolate syrup	
Coffee	
Cola	
Salad dressing	

[1] This list is not all-inclusive. The brand of soils used may be agreed upon between interested parties and documented within the reporting section of the testing report.
[2] Refer to AATCC LP4 Laboratory Procedure for Synthetic Soil for formula and mixing instructions.

10. Specimens

10.1 Prepare 2 test specimens 38 × 38 ± 1 cm (15.0 × 15.0 ± 0.4 in.). Apply no more than 4 different soils per specimen so that there are 2 replicates of each soil. Use additional pairs of specimens if more than 4 soils are tested.

10.2 Mark 4, 7.6 × 7.6 cm (3.0 × 3.0 in.) squares on each specimen and number them 1 through 4 (see Fig. 1).

10.3 If unable to cut specimens as described, use entire item for testing.

11. Conditioning

11.1 Prior to application of soil, condition test specimens as directed in ASTM D1776. Use conditions indicated in Table 1 for Textiles, general. Estimate conditioning time as indicated in Table 2 for the appropriate fiber content. Lay each specimen or item, separately, on a screen or perforated shelf of a conditioning/drying rack.

12. Soil Application Procedure

12.1 Place a single thickness of blotting paper on a smooth, horizontal surface.

12.2 Position the first unsoiled specimen face up, flat, on the blotting paper. If a multilayered item is being tested, place a single thickness of blotting paper between the item layers, directly under the numbered square.

12.3 Using the dropper, apply 5 drops (approx. 0.2 mL) of one liquid soil in the approximate center of the drawn square labeled number 1 on the test specimen. When applying thick/solid soils, place approximately 0.5 g of one soil in the approximate center of the appropriate drawn square on the test specimen.

12.4 Repeat 12.3 for each soil in squares 2 through 4 with other soils as needed.

12.5 Place a 7.6 × 7.6 cm (3.0 × 3.0 in.) square of glassine paper over each soiled area.

12.6 Place a weight directly over the soiled areas.

12.7 Allow the weight to sit undisturbed for 60 ± 5 s. Then remove the weight and discard the glassine paper.

12.8 Using a clean cloth, remove any excess soil from the test site by blotting, applying minimal pressure.

12.8.1 If excess soil is not sufficiently removed by blotting, use a back and forth rubbing motion, applying minimal pressure.

12.9 Repeat 12.2-12.8 for the remaining specimen(s). If testing multiple soils or samples to be laundered together, multiple weights allow greater efficiency. Do not put unsoiled specimens over soiled areas of blotting paper. Replace blotting paper as needed.

12.10 Wash specimens 20 ± 5 minutes after soiling to allow soils to set in. Do not allow soiled test specimens to contact each other in a manner that would transfer soils. The maximum number of test specimens per washer load should be 30, with no more than 4 soils per specimen.

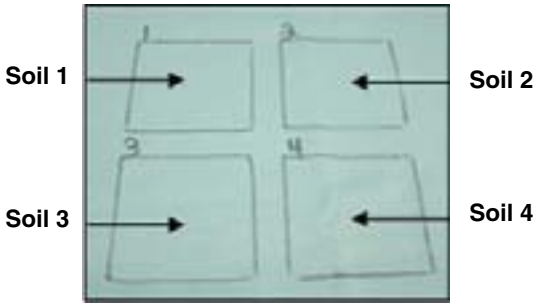


Fig. 1—Specimen marking for soil placement.

13. Laundering Procedure

13.1 Washing.

13.1.1 Set washing machine controls to generate the parameters indicated in AATCC LP1-2021 Table I, standard laundry cycle, normal setting, warm water 41°C (105°F). If fabric or garment being tested includes care instructions, launder accordingly.

13.1.2 Load for laundering shall consist of all specimens in the sample, plus enough laundering ballast pieces to make a total load weight of 1.8 ± 0.1 kg (4.0 ± 0.2 lb).

13.1.3 Begin selected wash cycle. Allow machine to fill to water level specified in AATCC LP1-2021 Table I.

13.1.4 Add 66 ± 1 g of AATCC 1993 Standard Reference Detergent to washing machine as directed by washing machine manufacturer's instructions.

13.1.5 If detergent is added directly to wash water, agitate briefly to dissolve completely. Stop agitation before adding wash load.

13.1.6 Add wash load (test specimens and ballast), distributing evenly around center agitator. Restart wash cycle.

13.1.7 For specimens to be drip dried (drying procedure C), stop wash cycle before the water begins to drain from the final rinse cycle. Remove specimens soaking wet. For specimens to be tumble (A), line (B), or screen/flat (D) dried, allow washing to proceed through the final spin cycle.

(B) Line/Hang Dry. Hang each specimen by two corners, with the fabric length in the vertical direction. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

(D) Screen/Flat Dry. Spread each specimen on a horizontal screen or perforated surface, removing wrinkles without distorting or stretching it. Allow specimens to rest in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

13.1.8 After each washing cycle, separate tangled specimens and ballast pieces, taking care to minimize distortion. Proceed to appropriate drying procedure.

13.2 Tumble Drying (A).

13.2.1 Place the washed load (test

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specimens and ballast) in the tumble dryer and set the temperature control to generate a maximum exhaust temperature of 68°C ± 6°C (155°F ± 10°F).

13.2.2 Allow the dryer to run until the load is fully dry.

13.2.3 Remove specimens from dryer immediately on completion of the cycle and lay flat to prevent formation of wrinkles or creases that can affect the stain release rating.

13.3 Ironing (optional)

13.3.1 If the specimens are extremely wrinkled and the consumer would ALWAYS expect to iron the end product, hand iron swatches or products prior to evaluation. Use caution due to the extremely high variability in hand ironing procedures performed by individual operators (no standard test method exists for hand ironing procedures).

13.3.2 Use safe ironing temperature for the sample. See AATCC TM133, Table I, Safe Ironing Temperature Guide for guidance regarding textile fibers. Exert the minimum pressure necessary to smooth specimen.

13.4 Evaluate the grade of soil release of each specimen within 4 h after drying. Reconditioning is not required.

14. Evaluation

14.1 Mount the Stain Release Scale on the viewing board, with the center of the scale 114 ± 3 cm (45 ± 1 in.) from the floor (see Appendix A). If the AATCC Gray Scale for Color Change or other scale is agreed upon, report the scale used.

14.2 Place the test specimen flat with face up in the center of the table. Rotate the fabric to the angle that produces the lowest rating.

14.3 Stand directly in front of the specimen for evaluation, 76 ± 3 cm (30 ± 1 in.) from the viewing board, with eyes approximately 157 ± 15 cm (62 ± 6 in.) from the floor. Varying the viewing angle either horizontally or vertically can affect grades obtained on some fabrics.

14.4 Two observers shall independently assign a grade to each stained area on each test specimen. The grade is that of the Stain Release Scale that corresponds most closely to the appearance of the stained area. Half-step grades (i.e., 1.5, 2.5, 3.5, 4.5) may be assigned to residual stains that are between grades on the Stain Release Scale.

14.5 Calculate the mean of 4 grades for each soil, (2 observers, 2 specimens), to nearest 0.1. This is the unit of measure for this test method.

15. Report

15.1 Describe or identify the fabric tested.

15.2 Report that the sample was tested using AATCC TM215-2023.

15.3 Report the testing conditions:

15.3.1 Water hardness of the washing procedure in terms of parts per million (ppm).

15.4 Soils applied, including brand or formulation.

15.5 Report the test results:

15.5.1 Mean stain release grade calculated for each soil.

15.5.2 Individual stain release grades for each specimen and observer, or standard deviation of the mean.

15.6 Describe any modification(s) of the published method, including alternate soiling or laundering procedures.

16. Precision and Bias

16.1 *Interlaboratory Test Data*—An interlaboratory evaluation study involving 3 different labs was carried out in 2021 to establish the precision values for this test method. Each lab performed the test as described in the procedure, using 1 operator and 3 independent evaluators who assigned stain ratings according to the AATCC Stain Release Scale. Laundering conditions as per AATCC LP1-2021 Table I (standard laundry cycle, normal setting, warm water at 41°C (105°F), tumble dry) were used.

Four reference soils used in the study were i) Welch's Grape Juice, ii) Kraft Italian Dressing, iii) French's Mustard, and iv) Heinz Ketchup. These were purchased locally and supplied to all participants. Six fabrics, 3 representing lock knit structure and 3 twill weave construction were used. It is worth noting that both lock knit and twill weave constructions represent tightly constructed fabrics.

16.2 *Precision*—Table II lists the within lab and between lab precision values for four different staining agents, Welch's grape juice, Kraft Italian dressing and French mustard. As expected, the between lab precision values are greater than the within lab precision values. Among the four staining agents, French mustard shows the highest precision values while Kraft Italian shows the lowest precision values.

The 6 different fabrics with 6 different constructions and 6 different material compositions DO NOT exhibit the same or identical staining potential, as indicated by major differences in the stain ratings of the 6 fabrics. This means one soil type cannot be interchanged with another.

Each subgroup (soil type) within individual fabrics did not show a significant difference, which means results are considered repeatable and reproducible among the labs that participated in the study.

16.3 *Bias*—The true value of each stain release grade can only be defined in terms of the test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

17. Notes

17.1 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

17.1.1 The AATCC Stain Release Scale should be replaced every 12 months and stored in the dark to prevent fading.

17.2 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428, USA; +1.610.832.9500; www.astm.org.

Table II—Within Lab and Between Lab Precision Values for Selected Soils

n	Welch' s Grape Juice		Kraft Italian Dressing		French's Mustard		Heinz Ketchup	
	Within Lab	Between Labs	Within Lab	Between Labs	Within Lab	Between Labs	Within Lab	Between Labs
1	0.3012	0.852	0.1783	0.5031	0.3707	1.0468	0.2041	0.5768
2	0.213	0.6034	0.126	0.3563	0.263	0.7414	0.145	0.4085
3	0.174	0.4919	0.103	0.2905	0.214	0.6044	0.118	0.3330
4	0.151	0.4260	0.089	0.2516	0.185	0.5234	0.102	0.2884
5	0.135	0.3810	0.080	0.2250	0.166	0.4681	0.091	0.2579
6	0.123	0.3478	0.073	0.2054	0.151	0.4274	0.083	0.2355
7	0.114	0.3221	0.067	0.1902	0.140	0.3958	0.077	0.2181
8	0.107	0.3013	0.063	0.1779	0.131	0.3702	0.072	0.2040
9	0.100	0.2840	0.059	0.1677	0.124	0.3489	0.068	0.1923
10	0.095	0.2694	0.056	0.1591	0.117	0.3311	0.065	0.1824

Note: The standard method calls for n = 2.

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17.3 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

17.4 For model number(s) of washer(s) and tumble dryers(s) reported to meet the standard parameters, visit www.aatcc.org/test/washers or contact AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org.

17.5 There is a possibility of finish and/or stain redeposition on ballast leading to subsequent transfer to test specimens. Fresh ballast should be used whenever it is deemed that buildup has occurred to the point where stain release results would be affected significantly or in cases where results are critical in adopting new finishes. The ballast should be replaced when it becomes obviously worn or frayed.

17.6 Water hardness may be measured with a kit designed for residential use, following manufacturer's instructions. Hardness can make detergent less effective and result in lower stain release grades. Hardness range is not specified for this method but use caution in comparing test results obtained with vastly different water hardness values.

17.7 The laundering temperatures and other parameters listed in this test method are standard conditions for testing purposes. As with most lab procedures, they represent, but may not exactly replicate, current consumer practices. Consumer practices vary over time and among households; lab practices must be consistent to allow valid comparison of results. If laundering equipment or conditions other than those specifically listed in this test method are used, they must be described in detail and noted as a modification of the standard method. Alternate laundering conditions are outlined in AATCC LP1.

17.8 The use of two 96-in. lamps for viewing laundered specimens is specified in this method. It is recognized, however, that physical space limitations in certain laboratories will prevent the use of 96-in. lamps. In those situations, two 48-in. lamps and a narrower viewing board may be used.

18. History

18.1 Developed in 2023 by AATCC Research Committee RA56.

Appendix A
Evaluation Area

A1. Viewing Board

A1.1 Plywood board, 1829 mm (72 in.) high × 6 mm (0.25 in.) thick × wide enough to accommodate scale.

A1.2 Painted to match the Grade 2 chip on the AATCC Gray Scale for Staining. Approximate CIELAB values for the chip are L* = 77, a* = 0, b* = 0. A tolerance of two units for each parameter is acceptable for the board color in this test method.

A2. Lighting

A2.1 Overhead fluorescent tube light fixture (see 17.8).

A2.1.1 Two parallel F96 T12 cool white lamps (without baffle or glass).

A2.1.2 One white enamel reflector (without baffle or glass).

A2.1.3 Mount as shown in Fig. A1.

A2.1.4 Eliminate all light sources except the specified fluorescent lamps.

A3. Walls

A3.1 It has been the experience of many observers that light reflected from the side walls near the viewing board can interfere with the evaluation results. It is recommended that the side walls be painted matte black (85° gloss less than 5 units) or that blackout curtains be mounted on both sides of the viewing board to eliminate the reflective interference.

A4. Specimen Table

A4.1 Table with non-glare black top 610 × 920 mm (24 × 36 in.); 890 ± 30 mm (35 ± 1 in.) high.

A4.2 One long edge of table centered and touching the viewing board (see Fig. A1).

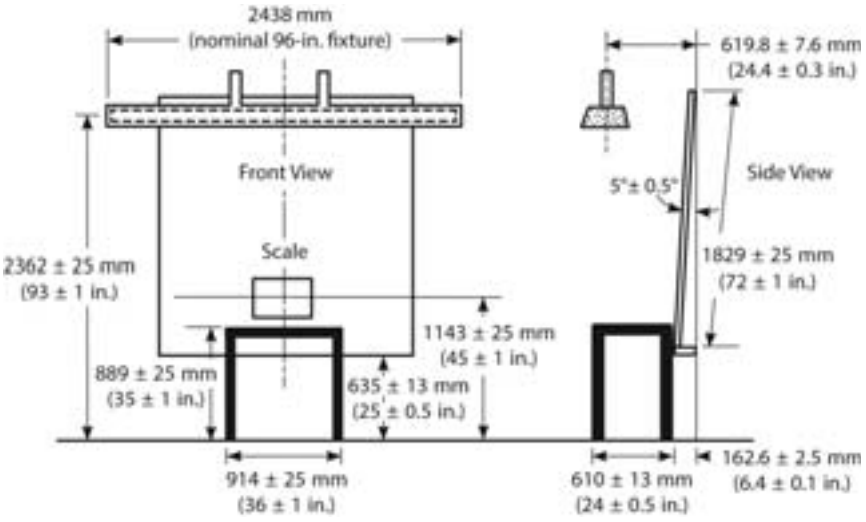


Fig. A1—Evaluation area for stain release evaluation.

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Laboratory Procedure for Home Laundering: Machine Washing

Foreword

This procedure is based on laundering methods and parameters originally developed as part of various AATCC standards. As a stand-alone laundering protocol, it may be combined with other test methods, including those for appearance, care label verification, and flammability. A procedure for hand laundering may be found in AATCC LP2, Laboratory Procedure for Home Laundering: Hand Washing.

Standard laundering procedures remain consistent to allow valid comparison of results. Standard parameters represent, but may not exactly replicate, current consumer practices, which vary over time and among households. *Alternate* laundering parameters (water level, agitation, temperature, etc.) are updated periodically to more closely mirror consumer practices and allow use of available consumer machines, though different parameters may produce different test results.

1. Purpose and Scope

1.1 This procedure provides standard and alternate home laundering conditions using an automatic washing machine. While the procedure includes several options, it is not possible to include every existing combination of laundering parameters.

1.2 This test is applicable to all fabrics and end products suitable for home laundering.

2. Principle

2.1 Home laundering procedures, including washing in an automatic washing machine and several drying methods are described. Parameters for washing machines and tumble dryers are also included. The procedures described herein need to be combined with an appropriate test method to obtain and interpret results.

3. Terminology

3.1 **laundering**, n.—*of textile materials*, a process intended to remove soils and/or stains by treatment (washing) with an aqueous detergent solution and normally including rinsing, extraction and drying.

3.2 stroke, n.—*of washing machines*, a single rotational movement of the washing machine drum.

NOTE: This motion can be in one direction (i.e., clockwise or counter-clockwise), or alternate back and forth. In either case, the motion shall be counted at each pause, or change of direction. For example, one set of back and forth motions shall be considered two strokes. In the case of unidirectional agitation, each pause of the motion shall be counted

3.3 **stroke length,** n.—*of washing machines*, the degree of rotation in a single movement (stroke) during the agitation cycle of the wash.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this procedure. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Detergent may cause irritation. Care should be taken to protect skin and eyes from exposure.

4.3 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

5. Uses and Limitations

5.1 *Standard* laundering parameters represent, but may not exactly replicate, current consumer practices, which vary over time and among households. *Alternate* laundering parameters (water level, agitation, temperature, etc.) are updated periodically to more closely mirror consumer practices and allow use of available consumer machines, though different parameters may produce different test results.

5.2 Equipment meeting all listed parameters may not be readily available.

6. Apparatus and Materials (see 17.1)

6.1 Automatic washing machine (see Tables I, IIA-D, IIIA-C, IVA-B, 17.2).

6.2 Tumble dryer (see Table VI, 17.2) or facilities for drip/line/flat drying.

6.3 1993 AATCC Standard Reference Detergent (powder) or AATCC High Efficiency (HE) Standard Reference Liquid Detergent (see 17.3, 17.4).

6.4 Scale with at least 5.0-kg (10.0-lb) capacity.

6.5 Laundering ballast Type 1, Type 3, or Type 4 (see Table VII).

NOTE: When Type 4 ballast is selected, the ballast shall be agreed upon by both parties, and shall be described in the report.

7. Verification

7.1 Verify top-loading washing machine parameters before running a test, or at minimum, once a year. No laboratory verification methods are currently available for front loading machines.

7.2 Water level verification

7.2.1 Using a graduated pail, fill machine with room-temperature water until it totals the specified volume.

7.2.2 Vertically, (perpendicularly to water surface), submerge an 1-m ruler until it touches the bottom of the machine drum. Using a permanent ink marker, draw a line on the ruler at the point of contact with the surface of the water. Drain water from drum.

7.2.4 Begin cycle, stopping when fill is complete (before agitation begins). Use the marked ruler to verify water level. Ensure ruler is submerged at the same place as when it was marked.

7.3 Agitation rate

7.3.1 To facilitate counting the number of strokes per minute during agitation in the wash cycle, tape one end of a short ruler or rod (approximately 150-mm) to the center point on the top of the agitator or impeller.

7.3.2 Place a small piece of colored tape at the free end of the rod.

7.3.3 Begin wash cycle and count the number of strokes per minute by focusing your eyes on the colored tape.

7.4 Spin speed (rpm)

7.4.1 Use a tachometer and reflective tape to measure the rpm of the machine during the spin cycle. Follow the manufacturer's instructions for the tachometer used.

8. Specimens

8.1 Cut and prepare specimens as directed in the applicable test method. This laundering procedure may be performed on any size or shape specimen(s) with a total dry mass less than or equal to 1.8 kg (4.0 lb) or 3.6 kg (8.0 lb).

8.2 If required, condition and evaluate

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Table I—Standard Washing Machine Parameters

Cycle	(1) Normal	(2) Delicate	(3) Permanent Press
Water Level, L (gal)	72 ± 4 (19 ± 1)	72 ± 4 (19 ± 1)	72 ± 4 (19 ± 1)
Agitation Rate, strokes/min.	86 ± 2	27 ± 2	86 ± 2
Stroke Length	Up to 220°	Up to 220°	Up to 220°
Washing Time, min.	16 ± 1	8.5 ± 1	12 ± 1
Final Spin Speed, rpm	660 ± 15	500 ± 15	500 ± 15
Final Spin Time, min.	5 ± 1	5 ± 1	5 ± 1
Wash Temp, °C (°F)[1]	(II) Cold: 27 ± 3 (80 ± 5) (III) Warm: 41 ± 3 (105 ± 5) (IV) Hot: 49 ± 3 (120 ± 5) (V) Very Hot: 60 ± 3 (140 ± 5)	(II) Cold: 27 ± 3 (80 ± 5) (III) Warm: 41 ± 3 (105 ± 5) (IV) Hot: 49 ± 3 (120 ± 5) (V) Very Hot: 60 ± 3 (140 ± 5)	(II) Cold: 27 ± 3 (80 ± 5) (III) Warm: 41 ± 3 (105 ± 5) (IV) Hot: 49 ± 3 (120 ± 5) (V) Very Hot: 60 ± 3 (140 ± 5)

[1] The temperatures in this table are similar to those specified by the US Federal Trade Commission for care label verification (see Table VIII). Due to US Department of Energy requirements, many consumer washing machines use cooler water. An external control box may be used to override the machine set temperatures.

Table IIA—Alternate Laundering Parameters (Traditional Top-loading Washing Machines 2000-2008)

Cycle	Normal	Delicate	Permanent Press
Water Level, L (gal) 1.8-kg load	68 ± 4 (18 ± 1)	68 ± 4 (18 ± 1)	68 ± 4 (18 ± 1)
3.6-kg load	83 ± 4 (22 ± 1)	83 ± 4 (22 ± 1)	83 ± 4 (22 ± 1)
Agitation Rate, strokes/min.18	179 ± 2	119 ± 2	179 ± 2
Stroke Length	Up to 90°	Up to 90°	Up to 90°
Washing Time, min.	12 ± 1	8 ± 1	10 ± 1
Final Spin Speed, rpm	645 ± 15	430 ± 15	430 ± 15
Final Spin Time, min.	6 ± 1	6 ± 1	4 ± 1
Wash Temp, °C (°F)	Very Cold: 16 ± 3 (60 ± 5) Cold: 27 ± 3 (80 ± 5) Warm: 41 ± 3 (105 ± 5) Hot: 49 ± 3 (120 ± 5) Very Hot: 60 ± 3 (140 ± 5)	Very Cold: 16 ± 3 (60 ± 5) Cold: 27 ± 3 (80 ± 5) Warm: 41 ± 3 (105 ± 5) Hot: 49 ± 3 (120 ± 5) Very Hot: 60 ± 3 (140 ± 5)	Very Cold: 16 ± 3 (60 ± 5) Cold: 27 ± 3 (80 ± 5) Warm: 41 ± 3 (105 ± 5) Hot: 49 ± 3 (120 ± 5) Very Hot: 60 ± 3 (140 ± 5)
Rinse Temp.	Very Cold: < 18°C (< 65°F) Other: < 29°C (< 85°F)	Very Cold: < 18°C (< 65°F) Other: < 29°C (< 85°F)	Very Cold: < 18°C (< 65°F) Other: < 29°C (< 85°F)

Table IIB—Alternate Laundering Parameters (Traditional Top-loading Washing Machines 2009-2010)

Cycle	Normal	Delicate	Permanent Press
Water Level, L (gal)			
1.8-kg load	68 ± 4 (18 ± 1)	68 ± 4 (18 ± 1)	68 ± 4 (18 ± 1)
3.6-kg load	83 ± 4 (22 ± 1)	83 ± 4 (22 ± 1)	83 ± 4 (22 ± 1)
Agitation Rate,[1] strokes/min.	179/119 ± 2	119 ± 2	179/119 ± 2
Stroke Length	Up to 90°	Up to 90°	Up to 90°
Washing Time, min.	12 (6 at stepdown agitation) ± 1	6 ± 1	9 (3 at stepdown agitation) ± 1
Final Spin Speed, rpm	645 ± 15	430 ± 15	430 ± 15
Final Spin Time, min.	6 ± 1	3 ± 1	4 ± 1
Wash Temp, °C (°F)	Very Cold: 16 ± 3 (60 ± 5) Cold: 27 ± 3 (80 ± 5) Warm: 41 ± 3 (105 ± 5) Hot: 49 ± 3 (120 ± 5) Very Hot: 60 ± 3 (140 ± 5)	Very Cold: 16 ± 3 (60 ± 5) Cold: 27 ± 3 (80 ± 5) Warm: 41 ± 3 (105 ± 5) Hot: 49 ± 3 (120 ± 5) Very Hot: 60 ± 3 (140 ± 5)	Very Cold: 16 ± 3 (60 ± 5) Cold: 27 ± 3 (80 ± 5) Warm: 41 ± 3 (105 ± 5) Hot: 49 ± 3 (120 ± 5) Very Hot: 60 ± 3 (140 ± 5)
Rinse Temp.	Very Cold: < 18°C (< 65°F) Other: < 29°C (< 85°F)	Very Cold: < 18°C (< 65°F) Other: < 29°C (< 85°F)	Very Cold: < 18°C (< 65°F) Other: < 29°C (< 85°F)

[1] Around 2009-2010, many of the top selling vertical axis washing machines featured a step-down agitation that started at a higher agitation (e.g., 179 spm) and then changed to a lower agitation (e.g., 119 spm) during the cycle.

Table IIC—Alternate Laundering Parameters (Traditional Top-loading Washing Machines 2011-2012)

Cycle	Normal	Delicate	Permanent Press
Water Level, L (gal)			
1.8-kg load	72 ± 4 (19 ± 1)	72 ± 4 (19 ± 1)	72 ± 4 (19 ± 1)
3.6-kg load	83 ± 4 (22 ± 1)	83 ± 4 (22 ± 1)	83 ± 4 (22 ± 1)
Agitation Rate, strokes/min.	86 ± 2	27 ± 2	86 ± 2
Stroke Length	Up to 220°	Up to 220°	Up to 220°
Washing Time, min.	16 ± 1	8.5 ± 1	12 ± 1
Final Spin Speed, rpm	660 ± 15	500 ± 15	500 ± 15
Final Spin Time, min.	5 ± 1	5 ± 1	5 ± 1
Wash Temp, °C (°F)	Very Cold: 16 ± 3 (60 ± 5) Cold: 27 ± 3 (80 ± 5) Warm: 41 ± 3 (105 ± 5) Hot: 49 ± 3 (120 ± 5) Very Hot: 60 ± 3 (140 ± 5)	Very Cold: 16 ± 3 (60 ± 5) Cold: 27 ± 3 (80 ± 5) Warm: 41 ± 3 (105 ± 5) Hot: 49 ± 3 (120 ± 5) Very Hot: 60 ± 3 (140 ± 5)	Very Cold: 16 ± 3 (60 ± 5) Cold: 27 ± 3 (80 ± 5) Warm: 41 ± 3 (105 ± 5) Hot: 49 ± 3 (120 ± 5) Very Hot: 60 ± 3 (140 ± 5)
Rinse Temp.	Very Cold: < 18°C (< 65°F) Other: < 29°C (< 85°F)	Very Cold: < 18°C (< 65°F) Other: < 29°C (< 85°F)	Very Cold: < 18°C (< 65°F) Other: < 29°C (< 85°F)

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Table IID—Alternate Laundering Parameters (Traditional Top-loading Washing Machines 2013-2017)

Cycle	Normal	Delicate
Water Level, L (gal)		
1.8-kg load	72 ± 8 (19 ± 2)	72 ± 8 (19 ± 2)
3.6-kg load	83 ± 4 (22 ± 1)	83 ± 4 (22 ± 1)
Agitation Rate, strokes/min.	86 ± 5	27 ± 5
Stroke Length	Up to 220°	Up to 220°
Washing Time, min.	16 ± 2	8.5 (5: integrated soak) ± 1
Number of Rinses	1	1
Final Spin Speed, rpm	660 ± 15	500 ± 15
Final Spin Time, min.	5-10	5-10
Wash Temp, °C (°F)	Cold: 16 ± 4.2 (60 ± 7.5)	Cold: 16 ± 4.2 (60 ± 7.5)
	Warm: 30 ± 4.2 (86 ± 7.5)	Warm: 30 ± 4.2 (86 ± 7.5)
	Hot: 44 ± 4.2 (111 ± 7.5)	Hot: 44 ± 4.2 (111 ± 7.5)
	Very Hot: 54 ± 4.2 (130 ± 7.5)	Very Hot: 54 ± 4.2 (130 ± 7.5)
Rinse Temp. (see 17.5)	Tap Cold	Tap Cold

Table IIIA—Alternate Laundering Parameters (High-efficiency Top-loading Washing Machines 2013-2017)

Cycle	Normal	Delicate
Water Level, L (gal)	30 ± 8 (8 ± 2)	57 ± 4 (15 ± 1)
Agitation Rate, strokes/min.	60 ± 5	75 ± 5
Washing Time, min.	11 ± 2	9 ± 2
Number of Rinses	1	1
Final Spin Speed, rpm	770 ± 20	500 ± 20
Final Spin Time, min.	5-18	5-10
Wash Temp, °C (°F)	Cold: 16 ± 4.2 (60 ± 7.5)	Cold: 16 ± 4.2 (60 ± 7.5)
	Warm: 24 ± 4.2 (75 ± 7.5)	Warm: 24 ± 4.2 (75 ± 7.5)
	Hot: 35 ± 4.2 (95 ± 7.5)	Hot: 35 ± 4.2 (95 ± 7.5)
	Very Hot: 54 ± 4.2 (130 ± 7.5)	Very Hot: 54 ± 4.2 (130 ± 7.5)
Rinse Temp. (see 17.5)	Tap Cold	Tap Cold

Table IIIB—Alternate Laundering Parameters (High-efficiency Top-loading Washing Machines, Agitator Style, 2018)

Cycle	Normal	Delicate
Water Level, L (gal)	68 ± 8 (18 ± 2)	68 ± 8 (18 ± 2)
Agitation Rate, strokes/min.	60 ± 5	70 ± 5
Washing Time, min.	14 ± 2	8 ± 2
Number of Rinses	1	1
Final Spin Speed, rpm	660 ± 20	660 ± 20
Final Spin Time, min.	5-10	1-6
Wash Temp, °C (°F)	Tap Cold: 16 ± 4.2 (60 ± 7.5)	Tap Cold: 16 ± 4.2 (60 ± 7.5)
	Cold: 16 ± 4.2 (60 ± 7.5)	Cold: 16 ± 4.2 (60 ± 7.5)
	Warm: 35 ± 4.2 (95 ± 7.5)	Warm: 30 ± 4.2 (86 ± 7.5)
	Hot: 54 ± 4.2 (130 ± 7.5)	Hot: 54 ± 4.2 (130 ± 7.5)

Table IIIC—Alternate Laundering Parameters (High-efficiency Top-loading Washing Machines, Impeller Style, 2018)

Cycle	Normal	Delicate
Water Level, L (gal)	44 ± 8 (11.5 ± 2)	68 ± 8 (18 ± 2)
Agitation Rate, strokes/min.	60 ± 5	70 ± 5
Washing Time, min.	14 ± 2	8 ± 2
Number of Rinses	1	1
Final Spin Speed, rpm	660 ± 20	500 ± 20
Final Spin Time, min.	5-10	1-6
Wash Temp, °C (°F)	Tap Cold: 16 ± 4.2 (60 ± 7.5)	Tap Cold: 16 ± 4.2 (60 ± 7.5)
	Cold: 16 ± 4.2 (60 ± 7.5)	Cold: 16 ± 4.2 (60 ± 7.5)
	Warm: 35 ± 4.2 (95 ± 7.5)	Warm: 30 ± 4.2 (86 ± 7.5)
	Hot: 54 ± 4.2 (130 ± 7.5)	Hot: 44 ± 4.2 (111 ± 7.5)

specimens before laundering, as directed in an applicable test method.

9. Standard Washing Procedure

9.1 Select washing conditions for testing from Table I. Set washing machine controls to generate the selected cycle parameters.

9.2 Load for laundering shall consist of all specimens in the sample, plus enough laundering ballast pieces to make a total load weight of 1.8 ± 0.1 kg (4.0 ± 0.2 lb).

9.3 Begin selected wash cycle. Allow machine to fill to water level specified in Table I.

9.4 Add 66 ± 1 g of 1993 AATCC Standard Reference Detergent to washing machine as directed by washing machine manufacturer's instructions. (Do not use AATCC High Efficiency (HE) Standard Reference Liquid Detergent for standard washing procedure.)

9.4.1 If detergent is added directly to wash water, agitate briefly to dissolve completely. Stop agitation before adding wash load.

9.5 Add wash load (test specimens and ballast), distributing evenly around center agitator. Restart wash cycle.

9.6 For specimens to be drip dried (drying procedure C), stop wash cycle before the water begins to drain from the final rinse cycle. Remove specimens soaking wet. For specimens to be tumble (A), line (B), or screen/flat (D) dried, allow washing to proceed through the final spin cycle.

9.7 After each washing cycle, separate tangled specimens and ballast pieces, taking care to minimize distortion. Proceed to appropriate drying procedure.

10. Alternate Laundering Procedure (Traditional Top-loading Washing Machines)

10.1 Select washing conditions for testing from Tables IIA-D. Set washing machine controls to generate the selected cycle parameters.

10.2 Load for laundering shall consist of all specimens in the sample, plus enough laundering ballast pieces to make a total load weight of 1.8 ± 0.1 kg (4.0 ± 0.2 lb) or 3.6 ± 0.1 kg (8.0 ± 0.2 lb).

10.3 Begin selected wash cycle. Allow machine to fill to specified water level.

10.4 Add detergent to washing machine as directed by washing machine manufacturer's instructions.

10.4.1 If detergent is added directly to wash water, agitate briefly to dissolve completely. Stop agitation before adding wash load.

10.4.2 For 1.8-kg load, use 66 ± 1 g of 1993 AATCC Standard Reference Detergent or 50 ± 1 mL of AATCC High Efficiency (HE) Standard Reference Liquid

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Table IVA—Alternate Laundering Parameters (High-efficiency Front-loading Washing Machines Prior to 2013)

Cycle	Normal	Delicate	Permanent Press
Water Level, L (gal)	22 ± 4 (5.75 ± 1)	22 ± 4 (5.75 ± 1)	22 ± 4 (5.75 ± 1)
Agitation Rate, rpm	40 ± 10	30 ± 10	30 ± 10
Washing Time, min.	18 ± 1	14 ± 1	16 ± 1
Number of Rinses	2	2	2
Final Spin Speed, rpm	1100 ± 100	400 ± 100	800 ± 100
Final Spin Time, min.	9.5 ± 1	3 ± 1	6 ± 1
Wash Temp, °C (°F)	Cold: 16 ± 2.9 (60 ± 5)	Cold: 16 ± 2.9 (60 ± 5)	Cold: 16 ± 2.9 (60 ± 5)
	Warm: 25 ± 2.9 (77 ± 5)	Warm: 25 ± 2.9 (77 ± 5)	Warm: 25 ± 2.9 (77 ± 5)
	Hot: 35 ± 2.9 (95 ± 5)	Hot: 35 ± 2.9 (95 ± 5)	Hot: 35 ± 2.9 (95 ± 5)
	Very Hot: 54 ± 2.9 (130 ± 5)	Very Hot: 54 ± 2.9 (130 ± 5)	Very Hot: 54 ± 2.9 (130 ± 5)
Rinse Temp. (see 17.5)	Tap Cold	Tap Cold	Tap Cold

Table IVB—Alternate Laundering Parameters (High-efficiency Front-loading Washing Machines 2013-2017)

Cycle	Normal	Delicate
Water Level, L (gal)	15 ± 4 (4 ± 1)	17 ± 4 (4.5 ± 1)
Agitation Rate, rpm	45 ± 10	40 ± 10
Washing Time, min.	11 ± 1	11 ± 1
Number of Rinses	2	2
Final Spin Speed, rpm	1300 ± 150	400 ± 150
Final Spin Time, min.	12-18	11-17
Wash Temp, °C (°F)	Cold: 16 ± 2.9 (60 ± 5)	Cold: 16 ± 2.9 (60 ± 5)
	Warm: 25 ± 2.9 (77 ± 5)	Warm: 25 ± 2.9 (77 ± 5)
	Hot: 35 ± 2.9 (95 ± 5)	Hot: 35 ± 2.9 (95 ± 5)
	Very Hot: 54 ± 2.9 (130 ± 5)	Very Hot: 54 ± 2.9 (130 ± 5)
Rinse Temp. (see 17.5)	Tap Cold	Tap Cold

Table V—Standard Drying Conditions

(A)	Tumble Dry
(Ai)	Tumble Dry Normal
(Aii)	Tumble Dry Delicate
(Aiii)	Tumble Dry Permanent Press
(B)	Line/Hang Dry
(C)	Drip Dry
(D)	Screen Dry/Dry Flat

Table VI—Standard Tumble Dryer Parameters

Cycle	(Ai) Normal	(Aii) Delicate	(Aiii) Permanent Press
Max. Exhaust Temp, °C (°F)	68 ± 6 (155 ± 10)	60 ± 6 (140 ± 10)	68 ± 6 (155 ± 10)
Cool Down Time, min.	≤ 10	≤ 10	≤ 10

Table VII—Laundering Ballast Parameters

	Type 1	Type 3	Type 4
Fiber Content	100% cotton	50% cotton/50% polyester ± 3%	Varies (Similar Material to fabric under test, as agreed upon by both parties)
Greige Fabric Yarns	16/1 ring spun	16/1 or 30/2 ring spun	
Greige Fabric Construction	52 × 48 ± 5 yarns/in., plain weave	52 × 48 ± 5 yarns/in., plain weave	
Finished Fabric Weight	155 ± 10 g/m^2	155 ± 10 g/m^2	
Edges	All edges hemmed or over-edged	All edges hemmed or over-edged	
Finished Piece Size	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)	
Finished Piece Weight	130 ± 10 g	130 ± 10 g	

Detergent.

10.4.3 For 3.6-kg load, use 80 ± 1 g of 1993 AATCC Standard Reference Detergent or 50 ± 1 mL of AATCC HE Standard Reference Liquid Detergent. Additional HE detergent is not required for larger load.

10.5 Add wash load (test specimens and ballast), distributing evenly around center agitator. Restart wash cycle.

10.6 For specimens to be drip dried (drying procedure C), stop wash cycle before the water begins to drain from the final rinse cycle. Remove specimens soaking wet. For specimens to be tumble (A), line (B), or screen/flat (D) dried, allow washing to proceed through the final spin cycle.

10.7 After each washing cycle, separate tangled specimens and ballast pieces, taking care to minimize distortion. Proceed to appropriate drying procedure.

11. Alternate Laundering Procedure [High-efficiency (HE) Washing Machines]

11.1 Select washing conditions for testing. Set washing machine controls to generate the selected cycle parameters.

11.1.1 For HE top-loading washing machines, use parameters from Tables IIIA-C.

11.1.2 For HE front-loading washing machines, use parameters from Tables IVA-B.

11.2 Load for laundering shall consist of all specimens in the sample, plus enough laundering ballast pieces to make a total load weight of 1.8 ± 0.1 kg (4.0 ± 0.2 lb) or 3.6 ± 0.1 kg (8.0 ± 0.2 lb).

11.3 Add 50 ± 1 mL of AATCC High Efficiency (HE) Standard Reference Liquid Detergent as directed by washing machine manufacturer's instructions. Use the same amount of detergent and same water level for 1.8-kg load or 3.6-kg load. Use *only* detergent specifically formulated for use in HE machines.

11.4 Add wash load (test specimens, then ballast), distributing evenly around center agitator or impeller for top-loading

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Table VIII—Water Temperatures Published by US Federal Trade Commission[1]

Designation	Initial Wash Temp, °C (°F)
Cold	≤ 30 (≤ 86)
Warm	31-44 (87-111)
Hot	45-63 (112-145)

[1] Refer to 16 CFR, Part 423, Appendix A for current care label temperature ranges.

machines. Begin selected wash cycle.

11.5 For specimens to be drip dried, stop wash cycle before the water begins to drain from the final rinse cycle. Remove specimens soaking wet. For specimens to be tumble, line, or screen (flat) dried, allow washing to proceed through the final spin cycle.

11.6 After each washing cycle, separate tangled specimens and ballast pieces, taking care to minimize distortion. Proceed to appropriate drying procedure.

12. Standard Drying Procedures

12.1 Select drying conditions from Table V.

12.2 (A) Tumble Dry. Place the washed load (test specimens and ballast) in the tumble dryer, and set the temperature control to generate the selected cycle exhaust temperature (see Table VI). Allow the dryer to run until the load is fully dry. Remove specimens immediately.

12.3 (B) Line/Hang Dry. Hang each fabric specimen by two corners, with the fabric length in the vertical direction. Hang product specimens on an appropriate hanger, straightening and smoothing facings, seams, etc. Product specimens should generally be oriented in the direction in which they are used. Do not fold or stretch specimens over the hanger. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

12.4 (C) Drip Dry. Hang dripping wet fabric specimens by two corners, with the fabric length in the vertical direction. Hang dripping wet product specimens on appropriate hangers, straightening and smoothing facings, seams, etc. Specimens should generally be oriented in the direction in which they are worn. Do not fold or stretch specimens over the hanger. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

12.5 (D) Screen/Flat Dry. Spread each specimen on a horizontal screen or perforated surface, removing wrinkles without distorting or stretching it. Allow specimens to dry in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

12.6 Allow specimens to dry completely before washing again.

12.7 Repeat the selected washing and drying procedures for an agreed upon number of cycles.

13. Ironing

13.1 No standard test method exists for hand ironing and procedures performed by individual operators have extremely high variability. Follow guidance of applicable standard regarding hand ironing. If no guidance is provided, hand iron prior to evaluation only if specimens are extremely wrinkled and the consumer would *always* expect to iron the fabric or end product.

13.1.1 If specimens are to be hand ironed, use safe ironing temperatures appropriate to the fibers in the fabric being ironed. See Table I, Safe Ironing Temperature Guide, in AATCC TM133, Test Method for Colorfastness to Heat: Hot Pressing (see 17.3). Exert the minimum pressure necessary to remove wrinkles.

14. Evaluation

14.1 If required, condition and evaluate specimens as directed by the applicable test method.

15. Report

15.1 Report specific laundering conditions as directed by applicable test method. In addition to cycle and temperature designations, it is recommended to include the specific tables referenced for washing and tumble drying parameters.

16. Precision and Bias

16.1 Precision and Bias statements are not applicable because data are not generated by this procedure.

17. Notes

17.1 For potential equipment information pertaining to this procedure, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

17.2 For model number(s) of washer(s) and tumble dryers(s) reported to meet the parameters in Tables I-V, visit www.aatcc.org/test/washers or contact AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933. Cycle and temperature names on machines may not match those in the tables. Where possible, setting information is also provided online.

17.3 Materials or drawings are available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

17.4 Detergent with or without optical brightener may be used for this procedure. Refer to applicable test method for guidance on selecting a detergent. If no guidance is given, use detergent *without* optical brightener for all colorfastness evaluations.

17.5 Tap Cold rinse temperature is not meant to be controlled. Tap cold is equivalent to the temperature of water entering the home, which is dependent upon geography and time of year. In the extreme, water temperatures in consumer homes can vary from 4°C to 49°C (40°F to 120°F); however, on average, these temperatures vary from 13°C to 24°C (55°F to 75°F).

18. History

18.1 Revised in 2021 to update the tables and add Type 4 Ballast.

18.2 Editorially revised 2019.

18.3 Supersedes AATCC M6.

18.4 Developed in 2018 by AATCC Committee RA88.

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Laboratory Procedure for Home Laundering: Hand Washing

Foreword

This procedure is based on laundering methods originally developed as part of various AATCC standards. As a standalone laundering protocol, this procedure may be combined with other test methods, including those for appearance and care label verification. Procedures for home laundering using an automatic washing machine may be found in AATCC LP1, Laboratory Procedure for Home Laundering: Machine Washing.

1. Purpose and Scope

1.1 This procedure provides standard home laundering conditions using hand washing and line or flat drying.

1.2 This test is applicable to all fabrics and end products suitable for hand washing with detergent.

2. Principle

2.1 Home laundering procedures, including hand washing, hang drying, and flat drying are provided. The procedures described herein need to be combined with an appropriate test method to obtain and interpret results.

3. Terminology

3.1 **laundering,** n.—*of textile materials*, a process intended to remove soils and/or stains by treatment (washing) with an aqueous detergent solution and normally including rinsing, extraction and drying.

4. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this procedure. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

4.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

4.2 Detergent may cause irritation. Care should be taken to protect skin and eyes from exposure.

5. Uses and Limitations

5.1 Standard laboratory laundering procedures represent, but may not exactly replicate, current consumer practices, which vary over time and among households.

5.2 Tumble and drip drying procedures are not included herein as they are not commonly used for delicate items that may be hand washed.

6. Apparatus and Materials (see 13.1)

6.1 Wash tub, 9.5-L.

6.2 White towels, large enough to accommodate test specimen, any weight.

6.3 Appropriate hangers, for hang drying.

6.4 Drying racks with pull-out, perforated shelves, for drying flat (see 13.2).

6.5 1993 AATCC Standard Reference Detergent (see 13.2, 13.3).

7. Specimens

7.1 Cut and prepare specimens as directed in the applicable test method. This laundering procedure may be performed on any size or shape specimen(s) with a total dry mass less than or equal to 0.7 kg (1.5 lb).

7.2 If required, condition and evaluate specimens before laundering, as directed in an applicable test method.

8. Standard Washing Procedure

8.1 Select washing temperature from Table I. Add 7.6 ± 1.9 L (2.0 ± 0.5 gal) of water at this temperature to the wash tub.

8.2 Add 20 ± 1 g of 1993 AATCC Standard Reference Detergent to wash tub.

8.3 Agitate by hand to dissolve detergent.

8.4 Add specimen(s) to solution and gently squeeze to distribute detergent solution. Do not twist or wring.

8.5 Let specimen(s) soak for 2 min.

8.6 Gently squeeze specimen(s) in detergent solution for 1 min. Do not twist or wring.

8.7 Repeat 2-min. soak and 1-min squeeze in detergent solution.

8.8 Remove specimen(s) from wash tub and gently squeeze to remove excess detergent solution. Do not twist or wring.

8.9 Place specimen(s) on clean white towel. Empty and rinse wash tub.

8.10 Add 7.6 ± 1.9 L (2.0 ± 0.5 gal) of clean water at the specified rinse temperature (see Table I) to the wash tub.

8.11 Place washed specimen(s) from towel in rinse water and gently squeeze to distribute. Do not twist or wring.

8.12 Let specimen(s) soak for 2 min.

8.13 Gently squeeze specimen(s) in rinse water for 1 min. Do not twist or wring.

8.14 Repeat 2-min. soak and 1-min squeeze in rinse water.

8.15 Remove specimen(s) from wash tub and gently squeeze to remove excess water. Do not twist or wring.

9. Standard Drying Procedure

9.1 Using clean white towels, blot water from washed specimen(s). Do not wring or twist.

9.2 Line/Hang Dry. Hang each fabric specimen by two corners, with the fabric length in the vertical direction. Hang product specimens on an appropriate hanger, straightening and smoothing facings, seams, etc. Product specimens should generally be oriented in the direction in which they are worn. Do not fold or stretch specimens over the hanger. Allow specimens to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion

9.3 Screen/Flat Dry. Spread each specimen on a horizontal screen or perforated surface, removing wrinkles without distorting or stretching it. Allow specimens to lie in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on specimens as it may cause distortion.

9.4 Allow specimens to dry completely before washing again.

9.5 Repeat the selected washing and drying procedures for an agreed upon number of cycles.

Table I—Standard Washing and Rinsing Temperatures[1]

Designation	Wash Temp, °C (°F)	Rinse Temp, °C (°F)
Very Cold	16 ± 3 (60 ± 5)	< 18 (< 65)
Cold	27 ± 3 (80 ± 5)	< 29 (< 85)
Warm	41 ± 3 (105 ± 5)	< 29 (< 85)
Hot[2]	49 ± 3 (120 ± 5)	< 29 (< 85)

[1]The temperatures in this table are similar to those specified by the US Federal Trade Commission for care label verification (see 16 CFR, Part 423, Appendix A).

[2]Hot water may not be a reasonable selection for hand washing or for the delicate items typically requiring hand washing.

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10. Evaluation

10.1 If required, condition and evaluate specimens as directed by the applicable test method.

11. Report

11.1 Report specific laundering conditions as directed by applicable test method. It is recommended to report, at a minimum, wash and rinse temperature, number of complete cycles, and drying procedure (hang or flat).

12. Precision and Bias

12.1 Precision and Bias statements are not applicable because data are not generated by this procedure.

13. Notes

13.1 For potential equipment information pertaining to this procedure, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

13.2 Materials or drawings are available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

13.3 1993 AATCC Standard Reference Detergent with or without optical brightener may be used for this procedure. Refer to applicable test method for guidance on selecting a detergent. If no guidance is given, use detergent *without* optical brightener for all colorfastness evaluations. Consumer detergents specifically formulated for use in hand washing may be used with prior agreement among all interested parties. The use of such detergents may increase *between laboratory* variability. Even if all comparison testing is done in one laboratory, repeatability may be lessened by use of such detergents due to formulation changes to meet market or consumer demands.

14. History

14.1 Reaffirmed 2020.

14.2 Editorially revised 2019.

14.3 Developed in 2018 by AATCC Committee RA88. Supersedes AATCC M5.

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Laboratory Procedure for Transfer of Free Permethrin from Textile Surfaces

Developed in 2018 by AATCC Committee RA49; editorially revised 2019.

Foreword

This laboratory procedure was originally developed as a method for transfer, extraction, and analysis of "free" permethrin insecticide. Because specific analysis methods vary among equipment manufacturers and models, the committee chose to standardize and publish only the transfer procedure. The practice may also be used for substances other than permethrin if paired with suitable extraction and analysis techniques. Although there is no result or data produced directly by the procedure, results reported after final analysis should note the use of this procedure.

1. Purpose and Scope

1.1 This procedure is used to transfer "free" permethrin insecticide from the surface of a textile material to a clean cloth for subsequent extraction and analysis. It may also be used to transfer other treatment substances

2. Principle

2.1 Using a Martindale abrasion tester, specimens of a treated sample fabric (with permethrin) are rubbed with an untreated white test fabric to transfer the permethrin from the specimen to the white test fabric. Trials at various (increasing) numbers of rubs are used to determine the total amount of free permethrin present on the surface of the test specimen.

3. Referenced Documents

3.1 ASTM D4850, Standard Terminology Relating to Fabrics and Fabric Test Methods (see 11.1).

3.2 ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 11.1).

3.3 ASTM D4966, Standard Test Method for Abrasion Resistance of Textile Fabrics (Martindale Abrasion Tester Method) (see 11.1).

4. Terminology

4.1 **abrasion,** n.—the wearing away of any part of a material by rubbing against another surface (ASTM D4850).

5. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this procedure. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

5.1 Good laboratory practices should be followed. Wear safety glasses and gloves when manipulating treated materials.

5.2 All chemicals should be handled with care.

5.3 An eyewash/safety shower should be located nearby for emergency use.

5.4 After testing, and to avoid any risk of contamination by permethrin, eliminate all activity of the treated samples, referring to the manufacturer's recommendation.

6. Uses and Limitations

6.1 The degree of permethrin transfer is affected by the number of rubs. The number should be varied to produce varying degrees of abrasion and to determine the total free permethrin on the surface.

6.2 Extraction and analysis are not included in this method procedure. Final results may vary based on the selection of solvent and analytical technique.

7. Apparatus, Reagents, and Materials

7.1 Martindale Abrasion Tester (see Fig. 1) with the following replaceable items:

7.1.1 Standard Felt, of mass 750 ± 50 g/m^2 and 3 ± 0.3 mm thick, cut into three 140-mm diameter round specimens.

7.1.2 Polyurethane Foam Backing, 3 ± 0.01 mm thick, 29-31 kg/m^3 density, and 170-210 N hardness, cut into three 38-mm diameter round specimens.

7.1.3 Weights, to create 12 kPa pressure for woven fabric and 9 kPa pressure for knitted (see Fig. 2).

7.2 AATCC Crocking Cloth, 100% cotton, plain weave, 32 ± 5 warp ends/cm, 33 ± 5 filling picks/cm, 100 ± 3 g/m^2, cut into three 38-mm diameter round specimens.



Fig. 1—Martindale abrasion tester.

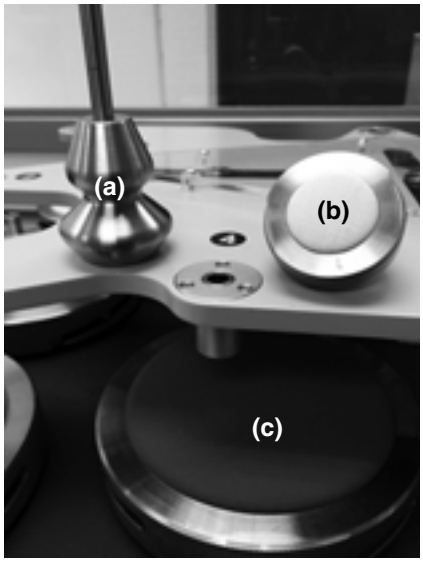


Fig 2—(a) Weight, (b) Sample holder with foam and untreated white specimen, and (c) Testing table with felt and test specimen.

8. Specimens

8.1 Minimize handling of the sample to avoid transfer of permethrin prior to testing.

8.2 Cut three 140-mm diameter round specimens diagonally across the width of the sample.

8.3 All specimens should be cut at least

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100 ± 5 mm from the selvage.

9. Conditioning

9.1 Prior to testing, condition test specimens, as well as white test fabric, standard felt, and foam backing specimens, as directed in ASTM D1776, Standard Practice for Conditioning and Testing textiles (see 11.1). Condition for at least 4 h in an atmosphere of 21 ± 2°C, 65 ± 5% RH, by laying each specimen separately on screen or perforated shelf of a conditioning rack.

9.2 Perform all tests in the standard atmosphere for testing. This will minimize variation in abrasion due to wet friction.

10. Abrasion

10.1 Using the manufacturer's directions, set the abrasion figure to a Lissajous figure with 60-mm stroke length.

NOTE: Check the rubbing area by placing a 140-mm untreated white specimen on the Martindale tester table and a 38-mm colored specimen in the sample holder. Run for several minutes, then measure the area of transferred color. Transferred area should be approximately 75 cm^2.

10.2 Place a felt specimen, followed by a test specimen on a testing table of the Martindale Abrasion Tester (see Fig. 2). Place the mounting weight (supplied with the tester) on the test specimen and secure the fabric/felt to the table with the mounting weight in place. Remove the weight and inspect for tucks, ridges, or distortions. If necessary, loosen and repeat until smooth.

10.3 Repeat 10.2 for each remaining felt specimen and test specimen.

10.4 Assemble a holder by placing one untreated white test fabric specimen face down into the specimen holder. Place a foam specimen between the specimen and the metal insert. Assemble the holder according to the manufacturer's instructions (see Fig. 2).

10.5 Repeat 10.4 for each remaining white test fabric specimen and foam specimen.

10.6 Place each assembled holder on the Martindale Abrasion Tester above a table mounted with fabric/felt specimens. Add the required weight to give a pressure on each specimen of 12 kPa for woven fabric and 9 kPa for knitted fabric (see Fig. 2).

10.7 Using the manufacturer's directions, set the counter system to record the desired rubs and start the abrasion test. (Complete transfer of the free permethrin is usually obtained at greater than 300 rubs).

10.8 After the set number of rubs is complete, carefully remove the untreated white test fabric specimens.

10.9 Repeat the procedure with fresh specimens and increasing numbers of rubs as needed.

10.10 Untreated white test fabric specimens may be subsequently extracted, and extract analyzed, to determine the maximum amount of "free" permethrin on the test fabric surface (see 11.2).

11. Notes

11.1 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; +1.610.832.9500; www.astm.org.

11.2 An ultrasonic method and solvent such as 80% acetonitrile/20% methanol is recommended for extraction of permethrin from the untreated white test fabric specimens after completion of the transfer procedure. The quantity of extracted permethrin may be determined by gas chromatography or other analytical technique. To determine the *maximum* "free" permethrin, the transfer procedure should be repeated with an increasing number of rubs, until the quantity of extracted permethrin ceases to increase. Concentration of free permethrin may be expressed as mg/m^2, using the rubbing area from section 11.1 (75 cm^2 = 0.0075 m^2).

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Laboratory Procedure for Synthetic Soil

1. Foreword

1.1 The AATCC Synthetic Soil Formula is applicable for several methods within AATCC and other organizations. Historically, AATCC listed the formula within select AATCC methods. It was determined that the formula should be listed independently.

1.2 AATCC TM123 Carpet Soiling: Accelerated Soiling Method—was written in 1967 and used throughout the 1970s, 80s, and 90s. Many types of soil application devices were developed during this time. During this time, the soil ingredients were determined, a formula was developed, and recorded in TM 123. During this period, soil resistance technology was improved in the carpet industry. Improving the variation of the test soil application on the carpet specimen became necessary. A workgroup of carpet & fiber manufacturers and chemical suppliers formed an ASTM workgroup to improve the methods' variation. They conducted many trials with different tumblers, mixing media, and other attributes. Finally, a reproducible way to apply the soil mixture, detailed in AATCC TM123, was agreed on.

1.3 A new method was written within ASTM D13.21 to describe the method; it was designated ASTM D6540—Standard Test Method for Accelerated Soiling of Pile Yarn Floor Covering. ASTM D6540 referenced the formula found in AATCC TM123.

1.4 TM123 was dropped as an active soiling test method in subsequent years. It was withdrawn in 2001. The Synthetic Soil formula was transferred to AATCC TM122 Carpet Soiling: Service Soiling Method to preserve the soiling formula. As TM122 may eventually suffer the same faith as TM123, to keep the formula active, it was determined that it should be listed independently.

1.5 To ensure that the soil formula is available regardless of the status of the test standard, the formula is now stated in AATCC LP4 Laboratory Procedure for Synthetic Soil.

1.5.1 Previous references to AATCC Synthetic Soil Formula have been removed from AATCC TM122. ASTM D6540 references to the formula have been updated to align with AATCC LP4.

2. Purpose and Scope

2.1 Synthetic Soil is used as a soiling option for stain and soiling evaluations.

2.2 Synthetic soil originated from carpet-related methods; however, its use is not limited and can be used in other methods requiring a synthetic soil application.

3. Principle

3.1 This lab procedure provides the historical background of the Synthetic Soil Formula (see 1.2 and 1.3) and includes the formula (see Table I) and the mixing guidelines (see 7.2)

4, Referenced Documents

4.1 AATCC TM122-2019 Carpet Soiling: Service Soiling Method (see 8.2).

4.2 AATCC TM123-2000 Carpet Soiling: Accelerated Soiling Method (see 8.2).

4.3 ASTM D6540—Standard Test Method for Accelerated Soiling of Pile Yarn Floor Covering (see 8.1).

5. Terminology

5.1 **soil, n.**—dirt, oil, or other substances not normally intended to be present on a substrate such as a textile material.

5.2 **soiling, n.**—*in textiles*, a process by which a textile substrate becomes more or less uniformly covered with/or impregnated with soil.

6. Safety Precautions

NOTE: These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all-inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test method. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturers' recommendations. All OSHA standards and rules must also be consulted and followed.

6.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas. Wear impervious gloves when handling soil.

6.2 All chemicals should be handled with care.

7. Procedure

7.1 The soil Formula is listed in Table I.

7.2 Synthetic Soil Mixing Guidelines.

7.2.1 Canadian Peat moss should be dry and free of lumps.

7.2.2 Portland cement must be dry. If it contains lumps, discard it and use a fresh supply.

7.2.3 Mix all dry ingredients thoroughly before adding mineral oil.

7.2.4 Dry mix at 50°C (122°F) for 6-8 hours.

7.2.5 Place the dry mixture in a ball mill with ceramic balls. Turn the mill on and run it for approximately 24 hours.

7.2.6 Store mixture in vapor-tight containers with a desiccant.

7.3 The synthetic soil may be purchased pre-mixed (see 8.2).

8. Notes

8.1 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428: +1.610.832.9500; www.astm.org.

8.2 For potential material information about this lab procedure, please visit the online AATCC Buyer's Guide at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate Member, but AATCC does not qualify or in any approve, endorse, or certify that any of the listed equipment or materials meet the requirements of its test methods.

9. History

9.1 Developed in 2023 by AATCC Committee RA57.

Table I—Synthetic Soil Formula

Ingredients	% By Weight
Canadian peat moss	38.0
Portland cement	17.0
Kaolin clay	17.0
Silica Gel Sorbent (230–400 mesh)	17.0
Carbon black (Lamp or furnace black)	1.75
Black iron oxide	0.50
Mineral oil (medical grade)	8.75

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Laboratory Procedure for Home Laundering with Bleach

Foreword

This procedure is based on laundering methods and parameters originally developed as part of various AATCC standards. As a stand-alone laundering protocol, it may be combined with other test methods, including those for appearance and care label verification.

Standard laundering procedures remain consistent to allow valid comparison of results. Standard parameters represent, but may not exactly replicate, current consumer practices, which vary over time and among households. *Alternate* laundering parameters (water level, agitation, temperature, etc.) are listed in AATCC LP1.

1. Purpose and Scope

1.1 This procedure provides standard conditions for home laundering with bleach, using an automatic washing machine. Option 1 is for sodium hypochlorite ("chlorine bleach"). Option 2 is for sodium percarbonate ("non-chlorine bleach").

1.2 This test is applicable to all fabrics and end products suitable for home laundering. It may be used to determine whether a sample is suitable for laundering with chlorine and/or non-chlorine bleach.

2. Principle

2.1 Home laundering procedures with chlorine and non-chlorine bleach are described. Textile samples may be tested for compatibility with one or both types of bleach, but bleaches must not be combined in a single test. The procedures described herein must be combined with an appropriate test method or evaluation procedure to obtain and interpret results.

3. Referenced Documents

3.1 AATCC EP1, Evaluation Procedure for Gray Scale for Color Change (see 19.1).

3.2 AATCC LP1, Laboratory Procedure for Home Laundering: Machine Washing (see 19.1).

3.3 AATCC TM133, Test Method for Colorfastness to Heat: Hot Pressing (see 19.1).

3.4 ASTM D3938, Standard Guide for Determining or Confirming Care Instructions for Apparel and Other Textile Products.

3.5 ASTM E1402, Standard Guide for Sampling Design.

4. Terminology

4.1 **bleach,** n.—*in home laundering*, a product that will clean, whiten, brighten and aid in the removal of soils and stains from textile materials by oxidation that is inclusive of chlorine and non-chlorine products.

4.2 **chlorine bleach,** n.—a solution of sodium hypochlorite (NaOCl) that will clean, whiten, brighten and aid in the removal of soils and stains from textile materials.

4.3 **laundering**, n.—*of textile materials*, a process intended to remove soils and/or stains by treatment (washing) with an aqueous detergent solution and normally including rinsing, extraction and drying.

4.4 **non-chlorine bleach,** n.—a product that will clean, whiten, brighten and aid in the removal of soils and stains from textile materials without the release the hypochlorite ion in solution.

NOTE: *Powdered* non-chlorine bleach typically contains sodium percarbonate ($Na_2CO_3 \cdot 1.5\ H_2O_2$) as its active ingredient. *Liquid* non-chlorine bleach typically contains hydrogen peroxide (H_2O_2) as its active ingredient. When dissolved in water, sodium percarbonate liberates hydrogen peroxide; the chemistry for laundering with both forms of non-chlorine bleach is the same.

5. Safety Precautions

5.1 The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

5.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

5.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

6. Limitations

6.1 *Standard* laundering parameters represent, but may not exactly replicate, current consumer practices, which vary over time and among households. Use of alternate laundering parameters may produce different test outcomes.

6.2 Water quality and hardness are not defined in this procedure. Since most laboratories use local water sources for laundering, it is generally impractical to control parameters. In cases of critical testing or unexpected results, additional testing and comparison may help isolate sources of variability.

7. Apparatus and Materials (see 19.2)

7.1 Automatic washing machine (see Table I, 19.3).

7.1.1 Graduated pail, for washing machine water level verification.

7.1.2 Ruler, 1-m, for washing machine water level verification.

7.1.3 Stopwatch or timer, for washing machine agitation speed verification

7.1.4 Tachometer, for washing machine spin speed verification.

7.2 Tumble dryer (see Table III, 19.3) or facilities for drip/line/flat drying.

7.3 Balance with at least 5.0-kg capacity, for laundering load.

7.4 Laundering ballast Type 3 (see Table IV). Use of alternate ballast material may affect test results.

7.5 Analytical balance, 0.001-g readability, for reagents.

8. Reagents

8.1 1993 AATCC Standard Reference Detergent WITHOUT brightener (WOB, powder) (see 19.1).

8.2 Laundering bleach (see 19.4).

8.2.1 For Option 1 (chlorine bleach): sodium hypochlorite (NaOCl), 5% available chlorine (see 19.5).

8.2.2 For Option 2 (powdered non-chlorine bleach): sodium percarbonate ($Na_2CO_3 \cdot 1.5\ H_2O_2$), >99% (see 19.6).

9. Verification

9.1 Verify washing machine water level before running a test, or at minimum, once a year.

9.1.1 Using a graduated pail, fill machine with room temperature water until it totals the specified volume.

9.1.2 Vertically, (perpendicularly to water surface), submerge a 1-m ruler until it touches the bottom of the machine drum. Using a permanent ink marker, draw a line on the ruler at the point of contact with the surface of the water. Drain water from drum.

9.1.3 Begin cycle, stopping when fill is complete (before agitation begins). Use the marked ruler to verify water level. Ensure ruler is submerged at the same

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Table I—Standard Washing Machine Parameters (see 19.4)

Cycle	(1) Normal		(2) Delicate		(3) Permanent Press	
Water Level, L (gal)	72 ± 4 (19 ± 1)		72 ± 4 (19 ± 1)		72 ± 4 (19 ± 1)	
Agitation Speed, strokes/min.	86 ± 2		27 ± 2		86 ± 2	
Washing Time, min.	16 ± 1		8.5 ± 1		12 ± 1	
Final Spin Speed, rpm	660 ± 15		500 ± 15		500 ± 15	
Final Spin Time, min.	5 ± 1		5 ± 1		5 ± 1	
Wash Temp, °C (°F)[1]	**(II) Cold:**	27 ± 3 (80 ± 5)	**(II) Cold:**	27 ± 3 (80 ± 5)	**(II) Cold:**	27 ± 3 (80 ± 5)
	(III) Warm:	41 ± 3 (105 ± 5)	**(III) Warm:**	41 ± 3 (105 ± 5)	**(III) Warm:**	41 ± 3 (105 ± 5)
	(IV) Hot:	49 ± 3 (120 ± 5)	**(IV) Hot:**	49 ± 3 (120 ± 5)	**(IV) Hot:**	49 ± 3 (120 ± 5)
	(V) Very Hot:	60 ± 3 (140 ± 5)	**(V) Very Hot:**	60 ± 3 (140 ± 5)	**(V) Very Hot:**	60 ± 3 (140 ± 5)

[1] The temperatures in this table are similar to those specified by the US Federal Trade Commission for care label verification (see Table VIII). Due to US Department of Energy requirements, many consumer washing machines use cooler water. An external control box may be used to override the machine set temperatures.

place as when it was marked.

9.2 Verify washing machine agitation before running a test, or at minimum, once a year.

9.2.1 To facilitate counting the number of strokes per minute during agitation in the wash cycle, tape one end of a short ruler or rod (approximately 150-mm) to the center point on the top of the agitator or impeller.

9.2.2 Place a small piece of colored tape at the free end of the rod.

9.2.3 Begin wash cycle and count the number of strokes per minute by focusing your eyes on the colored tape.

9.3 Verify washing machine spin speed (rpm) before running a test, or at minimum, once a year.

9.3.1 Use a tachometer and reflective tape to measure the rpm of the machine during the spin cycle. Follow the manufacturer's instructions for the tachometer used.

9.4 *If desired*, verify concentration of laundering bleach (see 19.4 and 19.5)

10. Sampling

10.1 Test results are valid only when the samples are statistically representative (see ASTM E1402).

10.2 Sampling must be random. Every unit of product must have a mathematically equal chance of becoming a sample; and every portion of each sample must be equally likely to become a test specimen.

10.3 All specimens must be alike within the variations due to pure chance. There must be no differences within samples assignable to known causes.

11. Specimens

11.1 The mass of laundered test sample can influence the outcome of this procedure. Launder specimens or uncut sample with a total mass of 110.0 ± 10.0 g. Cut and prepare specimens as directed in the applicable test method. (Load with ballast will be 1.8 ± 0.1 kg.)

11.1.1 For garment testing, launder garments together or separately as needed to obtain the 110.0 ± 10.0 g specimen mass. If a single garment exceeds 110.0 ± 10.0 g, launder separately and report the specimen mass.

11.2 If applicable, retain an unlaundered specimen for evaluation of changes (color, pilling, etc.) after laundering.

12. Conditioning

12.1 If required, condition and evaluate specimens before laundering, as directed in an applicable test method.

13. Standard Washing Procedure

13.1 Select washing conditions for testing from Table I. Set washing machine controls to generate the selected cycle parameters.

13.2 Load for laundering shall consist of 110.0 g of test sample, plus enough laundering ballast pieces to make a total load weight of 1.8 ± 0.1 kg.

13.3 Begin selected wash cycle. Allow machine to fill to water level specified in Table I.

13.4 Add 66.0 ± 1.0 g of 1993 AATCC Standard Reference Detergent to washing machine as directed by washing machine manufacturer's instructions.

13.5 Add bleach as directed by washing machine manufacturer's instructions.

13.5.1 For Option 1, add 240 ± 5 mL of 5% NaOCl (chlorine bleach).

13.5.2 For Option 2, add 20.0 ± 1.0 g $Na_2CO_3 \cdot 1.5\ H_2O_2$ (powdered non-chlorine bleach).

13.6 If detergent and bleach are added directly to wash water, agitate for 2 min. to dissolve and mix all chemicals. Stop agitation before adding wash load.

13.7 Add wash load (test sample or specimens and ballast), distributing evenly around center agitator. Restart wash cycle.

13.8 For samples to be drip dried (drying procedure C), stop wash cycle before the water begins to drain from the final rinse cycle. Remove test sample or specimens soaking wet. For samples to be tumble (A), line (B), or screen/flat (D) dried, allow washing to proceed through the final spin cycle.

13.9 After each washing cycle, separate tangled pieces, taking care to minimize distortion. Proceed to appropriate drying procedure.

14. Standard Drying Procedures

14.1 Select drying conditions from Table II.

14.2 (A) Tumble Dry. Place the washed load (test sample or specimens and ballast) in the tumble dryer and set the temperature control to generate the selected cycle exhaust temperature (see Table III). Allow the dryer to run until the load is fully dry. Remove test sample or specimens immediately.

14.3 (B) Line/Hang Dry. Hang each test swatch or specimen by two corners, with the fabric length in the vertical di-

Table II—Standard Drying Conditions

(A)	Tumble Dry
(Ai)	Tumble Dry Normal
(Aii)	Tumble Dry Delicate
(Aiii)	Tumble Dry Permanent Press
(B)	Line/Hang Dry
(C)	Drip Dry
(D)	Screen Dry/Dry Flat

Table III—Standard Tumble Dryer Parameters (see 19.4)

Cycle	(Ai) Normal	(Aii) Delicate	(Aiii) Permanent Press
Max. Exhaust Temp, °C (°F)	68 ± 6 (155 ± 10)	60 ± 6 (140 ± 10)	68 ± 6 (155 ± 10)
Cool Down Time, min.	≤ 10	≤ 10	≤ 10

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Table IV—Laundering Ballast Parameters

	Type 3
Fiber Content	50% cotton/50% polyester ± 3%
Greige Fabric Yarns	16/1 or 30/2 ring spun
Greige Fabric Construction	52 × 48 ± 5 yarns/in., plain weave
Finished Fabric Weight	155 ± 10 g/m^2
Edges	All edges hemmed or over-edged
Finished Piece Size	920 × 920 ± 30 mm (36.0 × 36.0 ± 1 in.)
Finished Piece Weight	130 ± 10 g

Table V—Water Temperatures Published by US Federal Trade Commission[1]

Designation	Initial Wash Temp, °C (°F)
Cold	≤ 30 (≤ 86)
Warm	31-44 (87-111)
Hot	45-63 (112-145)

[1] Refer to 16 CFR, Part 423, Appendix A for current care label temperature ranges.

rection. Hang product specimens on an appropriate hanger, straightening and smoothing facings, seams, etc. Product specimens should generally be oriented in the direction in which they are used. Do not fold or stretch specimens over the hanger. Allow samples to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on samples as it may cause distortion.

14.4 (C) Drip Dry. Hang dripping wet test swatch or specimen by two corners, with the fabric length in the vertical direction. Hang dripping wet product specimens on appropriate hangers, straightening and smoothing facings, seams, etc. Specimens should generally be oriented in the direction in which they are worn. Do not fold or stretch specimens over the hanger. Allow samples to hang in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on samples as it may cause distortion.

14.5 (D) Screen/Flat Dry. Spread each test swatch or specimen on a horizontal screen or perforated surface, removing wrinkles without distorting or stretching it. Allow samples to dry in still air at room temperature not greater than 26°C (78°F) until dry. Do not blow air directly on samples as it may cause distortion.

14.6 Allow samples to dry completely before washing again.

14.7 Repeat the selected washing and drying procedures for an agreed upon number of cycles.

15. Ironing

15.1 No standard test method exists for hand ironing and procedures performed by individual operators have extremely high variability. Follow guidance of applicable standard regarding hand ironing. If no guidance is provided, hand iron prior to evaluation only if specimens are extremely wrinkled and the consumer would *always* expect to iron the fabric or end product.

15.1.1 If specimens are to be hand ironed, use safe ironing temperatures appropriate to the fibers in the fabric being ironed (see AATCC TM133). Exert the minimum pressure necessary to remove wrinkles.

16. Evaluation

16.1 Condition and evaluate specimens as directed by the applicable test method. If applicable, condition laundered and retained (unlaundered) specimens in the same way.

16.2 For color change evaluation, refer to AATCC EP1. Compare laundered specimens to an unlaundered swatch from the same sample.

16.2.1 If a significant color change is observed, launder a duplicate sample without bleach (AATCC LP1) to determine whether the color change is due to dye instability or a bleach effect. Use the same laundering cycles and temperatures.

16.3 For care label determination, evaluate color change and dimensional change with chlorine and non-chlorine bleach. If results are unsatisfactory with all bleaches, evaluate color change and dimensional change after laundering without bleach (AATCC LP1). Refer to ASTM D3938 for guidance on determining or confirming care instructions.

17. Report

17.1 Describe or identify the sample laundered.

17.2 Report that the sample was laundered using AATCC LP04-2023.

17.2.1 Report the option used (i.e., Option 1: chlorine bleach or Option 2: non-chlorine bleach).

17.3 Report the laundering conditions:

17.3.1 Washing cycle (i.e., Normal, Delicate, Permanent Press).

17.3.2 Washing temperature (i.e., 27°C, 41°C, 49°C, 60°C).

17.3.3 Drying procedure (i.e., Tumble, Line/Hang, Drip, Screen/Flat).

17.3.4 For tumble drying, cycle (i.e., Normal, Delicate, Permanent Press).

17.4 If applicable, report the designation and parameters of test method(s) or evaluation procedure(s) used in combination with this laboratory procedure.

17.5 If applicable, report the results or other information as directed in test method(s) or evaluation procedure(s) used.

17.5.1 In most cases, results should include, as a minimum: arithmetic mean (X), number of tests (n), and standard deviation (s) or coefficient of variation ($\%CV$).

17.6 Describe any modification(s) of the published standard(s), including alternate ballast.

18. Precision and Bias

18.1 Precision and Bias statements are not applicable because data are not generated by this procedure.

19. Notes

19.1 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

19.2 For potential equipment information pertaining to this procedure, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

19.3 For model number(s) of washer(s) and tumble dryers(s) reported to meet the parameters in Table I and Table III, visit www.aatcc.org/test/washers or contact AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

19.4 It is not necessary to test with both powdered and liquid forms of non-chlorine bleach. When dissolved in water, sodium percarbonate ($Na_2CO_3 \cdot 1.5\ H_2O_2$, powdered non-chlorine bleach) liberates hydrogen peroxide (H_2O_2, liquid non-chlorine bleach); the chemistry for laundering with both forms is the same. The powdered form is more commonly used in home laundering and is more stable than the liquid form.

19.5 Do NOT use commercial laundering bleach for this test because concentration and added ingredients may vary. For unexpired, reagent-grade 5% available chlorine hypochlorite solution, it is not necessary to verify concentration. If you choose to verify the available chlorine, the following titration process may be used:

19.5.1 Pour 2.00 ± 0.01 g NaOCl solution into an Erlenmeyer flask and dilute to 50.0 mL with deionized water.

19.5.2 Add 10 mL of 10% (w/v) H_2SO_4 and

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10 mL of 10% KI (100 g/L).

19.5.3 Back-titrate with 0.1*N* $Na_2S_2O_3$ until colorless.

19.5.4 Calculate % NaOCl using Eq. 1.

% NaOCl = 100% * (mL $Na_2S_2O_3$ * 0.1*N* * 0.03722)/2.00 g NaOCl Eq. 1

Where:

The factor 0.03722 is derived by multiplying the molecular weight of NaOCl (74.45 g/mol) by 0.001 (mL to L conversion) and dividing by 2 (mols of $Na_2S_2O_3$ per NaOCl).

19.5.5 To determine % available chlorine, divide % NaOCl by 1.05 ((74 g/mol NaOCl)/(71 g/mol CL_2)).

19.5.6 Example: 28.0 mL of $Na_2S_2O_3$ used in back titration yields approximately 5.2% sodium hypochlorite and 5% available chlorine.

19.5.7 If available chlorine is 4.5%-5.5%, use as indicated in the Standard Washing Procedure. If available chlorine is outside the specified range, discard and obtain fresh solution.

19.6 Do NOT use commercial laundering bleach for this test because concentration and added ingredients may vary. For unexpired, reagent-grade $Na_2CO_3 \cdot 1.5\ H_2O_2$, it is not necessary to verify concentration. If you choose to verify the available oxygen, the following titration process may be used:

19.6.1 Grind the granules in a blender prior to weighing.

19.6.2 Add 3.000 ± 0.001 g $Na_2CO_3 \cdot 1.5\ H_2O_2$ to a 250-mL Erlenmeyer flask.

19.6.3 Place the flask on a magnetic stirrer. Add a coated stir bar. Use a cardboard box or other device to eliminate light exposure. Put a hole in the top of the box or otherwise allow a funnel to be used for addition of chemicals. Support the funnel with a ring and stand.

19.6.4 Add 100 mL of 1.5*N* H_2SO_4, 15 mL of 15.0% KI (150 g/L) and 10 mL of CH_2Cl_2.

19.6.5 Stir, *in the dark*, for 15-30 min. Do not exceed 30 min.

19.6.6 Provide indirect light as you titrate slowly but steadily with 0.1*N* $Na_2S_2O_3$ with constant stirring, to a pale yellow.

19.6.7 Add about 2.0 mL of starch indicator (5.0% solution) and continue the titration one drop at a time until blue color disappears.

19.6.8 Calculate % available oxygen using Eq. 2.

% available oxygen = 100% * (mL $Na_2S_2O_3$ * 0.1*N* * 0.008)/ 3.000 g $Na_2CO_3 \cdot 1.5\ H_2O_2$ Eq. 2

19.6.9 If available oxygen is 17%-18%, use as indicated in the Standard Washing Procedure. If available oxygen is outside the specified range, discard and obtain fresh solution.

20. History

20.1 Developed in 2023 by AATCC Committee RA88.

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Evaluation Procedure for Gray Scale for Color Change

Foreword

This procedure is used to standardize visual evaluation of color change due to colorfastness tests. It is not intended for use in the color approval (lab dip) process. For instrumental assessment of color change, refer to AATCC EP7, Evaluation Procedure for Instrumental Assessment of the Change in Color of a Test Specimen. The colors of AATCC Gray Scale color chips are the same as those specified in ISO 105-A02.

1. Scope

1.1 This evaluation procedure describes the use of a Gray Scale for visually evaluating overall color change resulting from colorfastness tests. Individual color difference components of lightness, chroma and hue may be noted, but are not quantified by this procedure. To determine if a Gray Scale is still suitable for use, see Appendix A.

1.2 Color change due to any colorfastness test may be evaluated.

2. Principle

2.1 The result of a colorfastness test is evaluated by visually comparing the difference in color between the original (untested) and tested specimens with the differences represented by the Gray Scale for Color Change. The color change grade is equal to that of the Gray Scale pair that is judged to have the same color difference. Results are reported as a Gray Scale for Color Change grade of 5 to 1, with 5 representing no color change and 1 representing the most color change.

3. Terminology

3.1 **color change,** n.—a change in color of any kind whether in lightness, hue or chroma, or any combination of these, discernible by comparing the test specimen with a corresponding untested specimen.

3.2 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

3.3 **grade,** n.—the number assigned to a test specimen resulting from comparison to a scale.

3.4 **Gray Scale,** n.—a scale consisting of pairs of standard gray chips, the pairs representing progressive differences in color corresponding to numerical colorfastness grades.

NOTE: Gray Scale grades are expressed as half-steps in decimal form (e.g., 1 or 1.5). Dashed form is also acceptable (e.g., 1 or 1-2).

4. Materials

4.1 Gray Scale for Color Change, including gray mask with *Y* tristimulus of 63 ± 2 (see 8.1 and 8.2).

4.2 White backing material (cardstock or other), *Y* tristimulus value of at least 85 (see 8.3), no optical brightener. Small pieces (approximately the same size as specimens) may be used for evaluation only, or larger pieces may be used for permanent mounting of specimens.

4.3 Illumination and viewing environment as described in AATCC EP9, Evaluation Procedure for Visual Assessment of Color Difference of Textiles (see 8.1).

5. Description of the Scale

5.1 Color change grade 5 is represented on the scale by two neutral gray reference chips mounted side by side, having a *Y* tristimulus value of 12 ± 1. This pair has no perceptible color difference (see Appendix A).

5.2 Color change grades 4.5 (or 4-5) to 1, inclusive, are represented by reference chips like those used in grade 5, paired with lighter neutral gray chips of similar dimensions and gloss. The visual differences in the whole-grade pairs—grades 4, 3, 2 and 1—are in geometric progression of color difference. The differences in the half-grade pairs—4.5, 3.5, 2.5 and 1.5 (or 4-5, 3-4, 2-3 and 1-2)—are intermediate between the whole-grade pairs (see 8.2 and Fig. 1).

6. Procedure

6.1 Place an original (untested) specimen and its corresponding tested specimen side by side, edges touching, in the same plane and oriented in the same direction. Take special care to create a sharp junction between the two specimens. Trim loose threads as needed.

6.2 Place the specimens, both the original and tested, over a piece of white backing material. Any means of attachment (staples, etc.) must not be in the area viewed. If necessary, place two or more layers of the original textile under both original and tested specimens to create opacity.



Fig. 1—Gray Scale for Color Change.

6.3 Place the Gray Scale along the specimen edges, with the junctions of the specimens and the Gray Scale pair aligned.

6.4 Place the mask provided with the scale over the specimens and the scale to eliminate any influence from the surrounding areas (see Fig. 2).

6.5 Illuminate the surfaces and use viewing conditions as described in AATCC EP9, with preference to Option A (45°/0°).

6.6 Compare the perceived visual difference between original and tested specimens with the differences represented by the Gray Scale pairs. The grade is that of the Gray Scale pair whose color difference corresponds most closely to the color difference between the original and tested specimens.

6.6.1 A grade of 5 is given only when there is no perceived difference in color between the original and tested specimens.

6.6.2 A grade of 0 may be given when the perceived difference in color between the original and tested specimens is decidedly greater than a grade of 1.

6.7 The cleanliness and physical condition of the Gray Scale is extremely important in obtaining consistent results (see 8.4).

6.8 When several assessments have been made, it is very useful to compare all the pairs of original and tested specimens which have been given the same numerical grade. This gives a good indication of the consistency of the grades, since any errors become prominent. Pairs that do not appear to have the same degree of color difference as others with the same grade should be re-checked against the Gray Scale and, if necessary, the grades should be changed.

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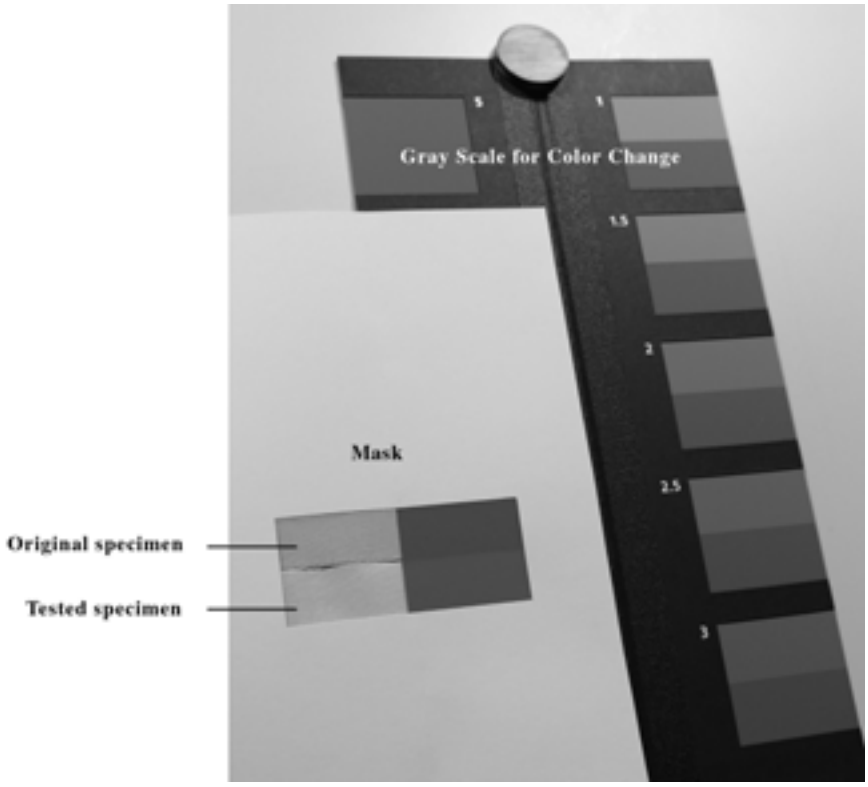


Fig. 2—Illustration of how to use scale.

7. Report

7.1 Report the grade assigned to each specimen pair as directed in the relevant colorfastness test method. The Gray Scale grade is an evaluation of the overall, total color difference between the original and tested specimens.

7.2 This procedure is not used to evaluate the changes in the individual components of lightness, chroma and hue. If a description of these component changes is required, the observer may add appropriate qualitative terms to the numerical grade, as listed in Table I.

8. Notes

8.1 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

8.2 The grades of the Gray Scale pairs, and the corresponding total color differences and tolerances, determined by the CIE 1976 L*a*b* (CIELAB) formula, are given in Table II. Colorimetric data is calculated using the CIE 1964 10° observer data for Illuminant D_{65}.

Table I—Color Difference Direction Descriptors

Lightness	Lighter, Darker
Chroma	More (or higher) chroma and Less (or lower) chroma
Hue	Redder, Greener, Yellower, Bluer

8.3 Colorimetric data is calculated using the CIE 1964 10° observer data for Illuminant D_{65}.

8.4 The Gray Scale should be inspected frequently for fingerprints and any other marks. If the marks interfere with the evaluation process, the scale should be replaced immediately. The scale can also be physically damaged through handling. To minimize damage, do not place the mask in the folded scale. If the physical damage to the Scale interferes with the evaluation process, it should be replaced immediately. The scale should be measured quarterly on a spectrophotometer or colorimeter (see Appendix A) to ensure that the total color differences are within specification as shown in Table II. If not within specification, the scale should be replaced. If the Scale is not measured on a spectrophotometer or colorimeter, replace within 1 year of initial use. Keep the scale in its sleeve when it is not in use.

Table II—Color Differences for Gray Scale for Color Change Chip Pairs

Grade	ΔE_{CIELAB}	Tolerance
5	0.0	+ 0.2
4.5	0.8	± 0.2
4	1.7	± 0.3
3.5	2.5	± 0.3
3	3.4	± 0.4
2.5	4.8	± 0.5
2	6.8	± 0.6
1.5	9.6	± 0.7
1	13.6	± 1.0

9. History

9.1 Revised to include Gray Scale maintenance and replacement instructions approved by the committee in 2018, but omitted from the previous published version.

9.2 Revised 2018, 2012, 2002, 1987, 1979. Editorially revised 2011, 2009, 1991. Editorially revised and reaffirmed 1992. Reaffirmed 2007.

9.3 Technically equivalent to ISO 105-A02. Adopted in 1954 by AATCC; under jurisdiction of AATCC Committee RA36.

Appendix A

A.1 Table II provides the color difference values in CIE 1976 L*a*b* (CIELAB) units for each pair of chips on the Gray Scale for Color Change, calculated using the CIE 1964 10° observer data for Illuminant D_{65}.This table is for instrumental measurement and confirmation that a Gray Scale is within tolerance, only. Table II is NOT to be used for assigning a Gray Scale grade based on instrumental measurement of two specimens (see AATCC EP7).

A.2 The reference for each of the color pairs must be the reference for *that pair* (not the grade 5 reference). Each color difference measurement is the difference between the bottom section of a pair and the top section of the same pair.

To check a Gray Scale for compliance, spectrophotometric measurements may be made under the following conditions:

A.1.1 di:8° (diffuse, specular included/8 degree) sphere instrument.

A.1.2 Small area of view, approximately 10-mm diameter.

A.1.3 Specular component included.

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Evaluation Procedure for Gray Scale for Staining

Foreword

This procedure is used to standardize visual evaluation of staining due to colorfastness tests. It is not intended for evaluating color change of white or light-colored materials. For instrumental assessment of staining, refer to AATCC EP12, Evaluation Procedure for Instrumental Assessment of Degree of Staining. The colors of AATCC Gray Scale color chips are the same as those specified in ISO 105-A03.

1. Scope

1.1 This evaluation procedure describes the use of a Gray Scale for visually evaluating colorant staining on a control material, resulting from colorfastness tests of a colored textile. To determine if a Gray Scale is still suitable for use, see Appendix A.

1.2 Staining due to any colorfastness test may be evaluated.

2. Principle

2.1 The result of a colorfastness test is evaluated by visually comparing the difference in color between the original (unstained) and stained control materials with the differences represented by the Gray Scale for Staining. The staining grade is equal to that of the Gray Scale pair that is judged to have the closest color difference. Results are reported as a Gray Scale for Staining grade of 5 to 1, with 5 representing no staining and 1 representing the most staining.

3. Terminology

3.1 **colorant staining,** n.—the unintended pickup of colorant by a substrate due to a transfer from another material.

3.2 **colorfastness,** n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the materials to any environment that might be encountered during the processing, testing, storage or use of the material.

3.3 **grade,** n.—the number assigned to a test specimen resulting from comparison to a scale.

3.4 **Gray Scale,** n.—a scale consisting of pairs of standard gray chips representing progressive differences in color corresponding to numerical colorfastness grades.

NOTE: Gray Scale grades are expressed as half-steps in decimal form (e.g., 1 or 1.5). Dashed form is also acceptable (e.g., 1 or 1-2).

4. Materials

4.1 Gray Scale for Staining, including gray mask with *Y* tristimulus of 63 ± 2 (see 8.1 and 8.2).

4.2 White backing material (cardstock or other), *Y* tristimulus value of at least 85 (see 8.3), no optical brightener. Small pieces (approximately the same size as specimens) may be used for evaluation only, or larger pieces may be used for permanent mounting of specimens.

4.3 Illumination and viewing environment as described in AATCC EP9, Evaluation Procedure for Visual Assessment of Color Difference of Textiles (see 8.1).

5. Description of the Scale

5.1 Staining grade 5 is represented on the scale by two white reference chips mounted side by side, having a *Y* tristimulus value of 85 ± 2. This pair has no perceptible color difference (see Appendix A).

5.2 Staining grades 4.5 (or 4-5) to 1, inclusive, are represented by reference chips like those used for grade 5, paired with darker neutral gray chips of similar dimensions and gloss. The visual differences in the whole-grade pairs—grades 4, 3, 2 and 1—are in geometric progression of color difference. The differences in the half-grade pairs—4.5, 3.5, 2.5 and 1.5 (or 4-5, 3-4, 2-3 and 1-2)—are intermediate between the whole-grade pairs (see 8.2 and Fig. 1).

6. Procedure

6.1 General stained material.

6.1.1 Place an original (unstained) specimen and its corresponding stained specimen side by side, in the same plane and oriented in the same direction. Take special care to create a sharp junction between the two specimens. Trim loose threads as needed.

6.1.2 Place the specimens, both the unstained and stained, over a piece of white backing material. Any means of attachment (staples, etc.) must not be in the area viewed. If necessary, place two or more layers of the original textile under both original and tested specimens to create opacity.

6.1.3 Place the Gray Scale along the specimen edges, with the junctions of the specimens Gray Scale pair aligned.



Fig. 1—Gray Scale for Staining.

6.1.4 Place the mask provided with the scale, printed side down, over the specimens and the scale. Position the specimen and Gray Scale pair junctions to be visible through the large bottom hole labeled, "For Evaluating General Staining" (see Fig. 2). For small or irregular stains, a smaller mask hole may be used.

6.2 Stained crocking squares.

6.2.1 Back crocking cloth squares as indicated in the relevant test method. Any means of attachment (staples, etc.) must not be in the area viewed.

6.2.2 Place the mask provided with the scale, printed side down, over the crock square. Position the stained area to be visible through the center of the middle hole labeled, "For Evaluating Crock Squares."

6.2.3 Place the Gray Scale over the mask, next to the test specimen.

6.3 Stained multifiber fabric.

6.3.1 Place an original (unstained) multifiber fabric specimen and its corresponding stained multifiber specimens side by side, in the same plane and oriented in the same direction, aligning each pair of fiber strips. Take special care to create a sharp junction between the two multifiber specimens. Trim loose threads as needed.

6.3.2 Place the multifiber fabric specimens, both the original and stained, over a piece of white backing material. Any means of attachment (staples, etc.) must not be in the area viewed.

6.3.3 Place the Gray Scale along the edge of the first pair of fiber strips, with the junctions of the Gray Scale and the strips aligned. Each pair of fiber strips will be evaluated separately.

6.3.4 Place the mask provided with the

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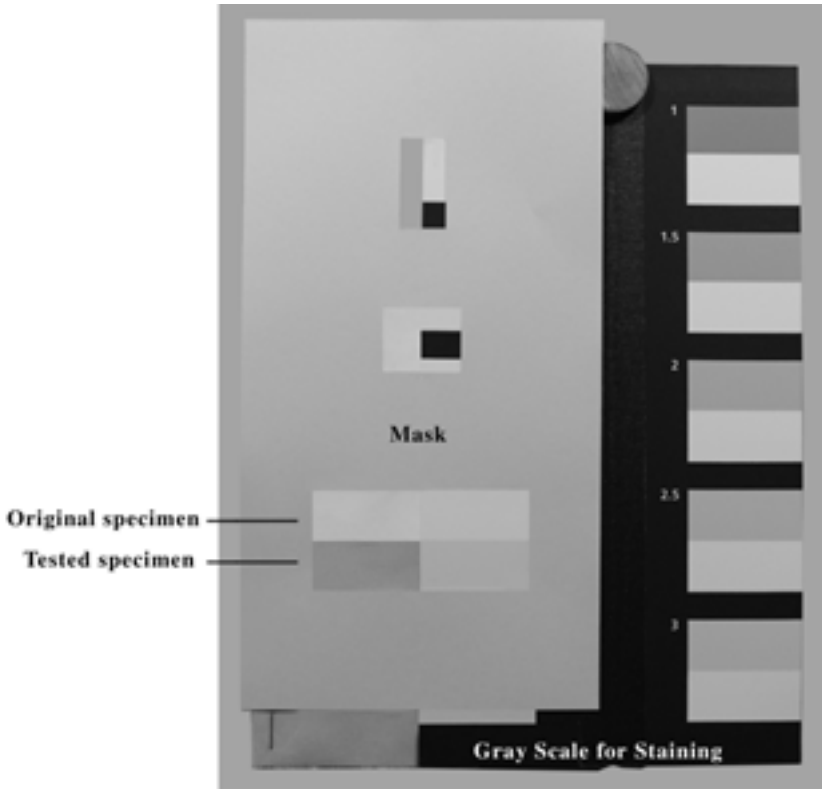


Fig. 2—Specimen, scale and mask arrangement for general colorant staining evaluation.

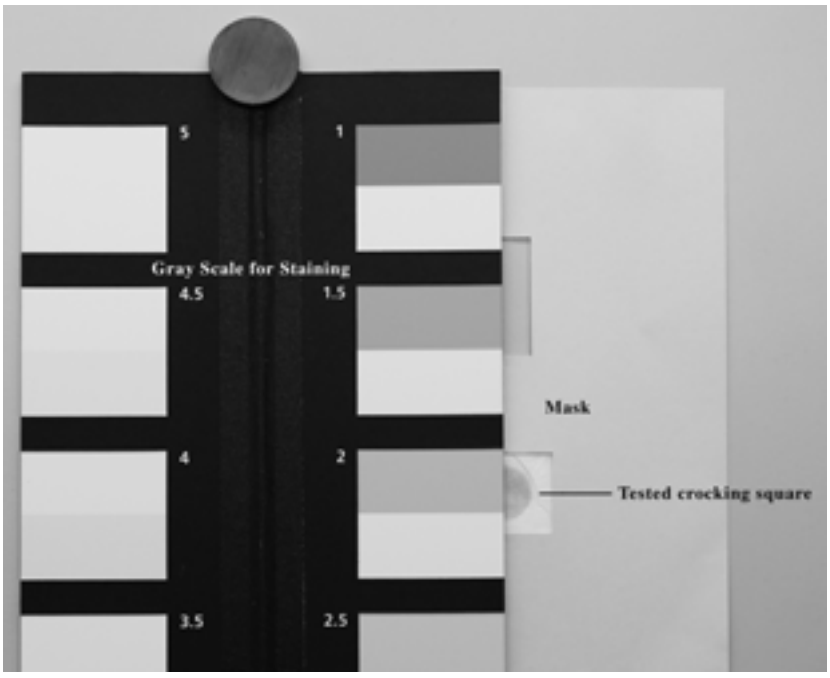


Fig. 3—Specimen, scale and mask arrangement for crocking staining evaluation.

scale, printed side down, over the multifiber specimens and the scale. Position the fiber strip and Gray Scale pair junctions to be visible through the small top hole labeled, "For Evaluating Stained Multifiber Fabric."

6.4 Illuminate the surfaces and use viewing conditions as described in AATCC EP9, with preference to Option A (45°/0°).

6.5 Compare the perceived visual difference between original and stained specimens with the differences represented by the Gray Scale pairs. The grade is that of the Gray Scale pair whose color difference corresponds most closely to the color difference between the original and stained specimens.

6.5.1 A grade of 5 is given only when there is no perceived difference in color between the original and stained specimens.

6.5.2 A grade of 0 may be given when the perceived difference in color between the original and stained specimens is decidedly greater than a grade of 1.

6.6 The cleanliness and physical condition of the Gray Scale is extremely important in obtaining consistent results (see 8.4).

6.7 When several assessments have been made, it is very useful to compare all the pairs of original and stained specimens which have been given the same numerical grade. This gives a good indication of the consistency of the grades, since any errors become prominent. Pairs that do not appear to have the same degree of color difference as others with the same grade should be re-checked against the Gray Scale and, if necessary, the grades should be changed.

7. Report

7.1 Report the grade assigned to each specimen pair as directed in the relevant colorfastness test method.

8. Notes

8.1 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

8.2 The grades of the Gray Scale pairs and the corresponding color differences and tolerances, determined by the CIE 1976 L*a*b* (CIELAB) formula, are given in Table I. Colorimetric data is calculated using the CIE 1964 10° observer data for Illuminant D_{65}.

8.3 Colorimetric data is calculated using the CIE 1964 10° observer data for Illuminant D_{65}.

8.4 The Gray Scale should be inspected frequently for fingerprints and any other marks. If the marks interfere with the evaluation process, the scale should be replaced immediately. The scale can also be physically damaged through handling. To minimize damage, do not place the mask in the folded scale. If the physical damage to the Scale interferes with the evaluation process, it should be replaced immediately. The scale should be measured quarterly on a spectrophotometer or colorimeter (see Appendix A) to ensure that the total color differences are within specification as shown in Table I. If not within specification, the scale should be replaced. If the Scale is not measured on a spectrophotometer or colorimeter, replace within 1 year of initial use. Keep the scale in its sleeve when it is not in use.

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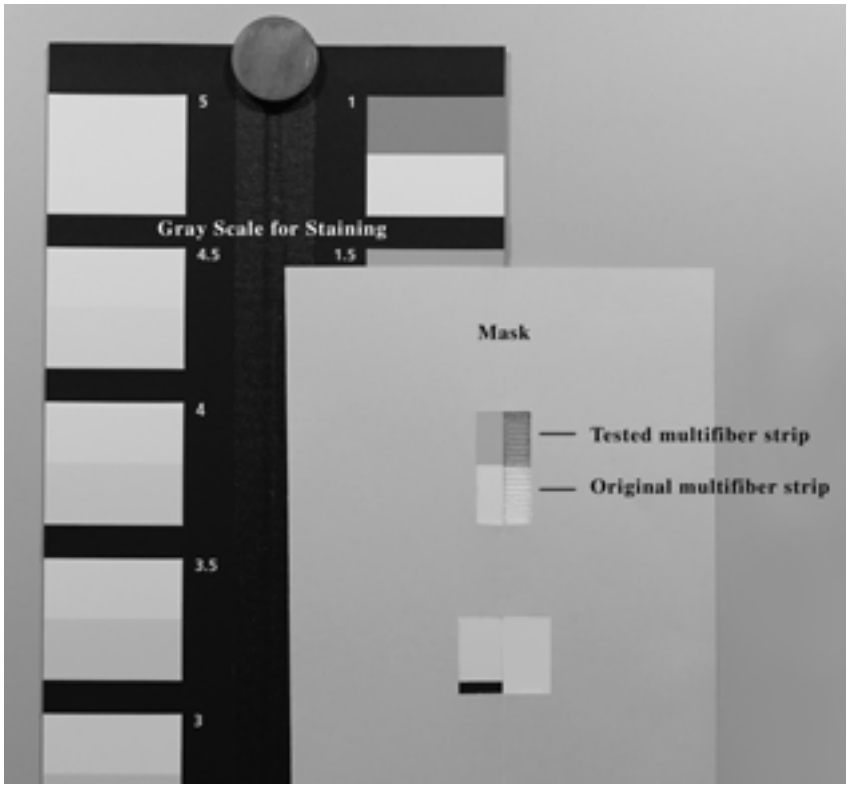


Fig. 4—Specimen, scale and mask arrangement for multifiber staining evaluation.

9. History

9.1 Revised to include Gray Scale maintenance and replacement instructions approved by the committee in 2018 but omitted from the previous published version.

9.2 Revised 2018, 2005, 2002, 1996, 1981, 1979. Editorially revised 2006, 1992. Editorially revised and reaffirmed 2012. Reaffirmed 2007, 1989.

9.3 Related to ISO 105-A03. Adopted in 1954 by AATCC; under jurisdiction of AATCC Committee RA36.

Appendix A

A.1 Table I provides the color difference values in CIE 1976 L*a*b* (CIELAB) units for each pair of chips on the Gray Scale for Staining, calculated using the CIE 1964 10° observer data for Illuminant D_{65}. This table is for instrumental measurement and confirmation that a Gray Scale is within tolerance, only. Table I is NOT to be used for assigning a Gray Scale based on instrumental measurement of two specimens (see AATCC EP12).

Table I—Color Differences for Gray Scale for Staining Chip Pairs

Grade	ΔE_{CIELAB}	Tolerance
5	0.0	+ 0.2
4.5	2.2	± 0.3
4	4.3	± 0.3
3.5	6.0	± 0.4
3	8.5	± 0.5
2.5	12.0	± 0.7
2	16.9	± 1.0
1.5	24.0	± 1.5
1	34.1	± 2.0

Reference chips are white, with a *Y* value of 85 ± 2.

A.2 The reference for each of the color pairs must be the reference for *that pair* (not the grade 5 reference). Each color difference measurement is the difference between the bottom section of a pair and the top section of the same pair.

To check a Gray Scale for compliance, spectrophotometric measurements may be made under the following conditions:

A.1.1 di:8 (diffuse, specular included/8 degree) sphere instrument.

A.1.2 Small area of view, approximately 10-mm diameter.

A.1.3 Specular component included.

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Evaluation Procedure for Fabric Hand

1. Purpose and Scope

1.1 This evaluation procedure describes guidelines for the presentation of fabrics for the evaluation of hand. Its purpose is to standardize the conditions under which a fabric is evaluated for one or more of the constituent elements of hand (for references on constituent elements of hand see Appendix A).

1.2 These guidelines may be used in the following circumstances:

1.2.1 When different people at different times wish to follow the same protocol for examining fabrics.

1.2.2 In training evaluators to detect and distinguish among different constituent elements or components of hand.

1.2.3 When an individual wishes to duplicate the conditions under which a fabric has been previously evaluated.

1.2.4 With a panel of individuals evaluating the same fabric(s).

2. Principle

2.1 An evaluator is presented a specimen in a prescribed manner and is asked to handle the specimen in a prescribed sequence.

3. Terminology

3.1 **hand,** n.—the tactile sensations or impressions that occur when fabrics are touched, squeezed, rubbed or otherwise handled.

3.2 **constituent elements of hand,** n.—those components, qualities, attributes, dimensions, properties or impressions which make the sensation of touching one fabric different from that of touching another.

NOTE: The various terms comprising elements of hand can be categorized by physical attributes of compression, bending, shearing and surface (see Appendix A).

4. Uses and Limitations

4.1 Effective use of these guidelines is limited by the ability of the evaluator to describe the sensations being felt. Care should be taken to determine if any evaluator exhibits impairment in touch and to determine if any variation between evaluators exists.

4.2 The validity of the data collected will be dependent on prior agreement regarding which constituent elements of hand are to be evaluated and mutual acceptance of a scale for evaluation.

5. Specimens

5.1 Test Specimens.

5.1.1 Take large enough test specimens from each sample to allow an evaluator to hold the specimen in both hands. In general, cut all specimens to greater than 200 mm (8 in.) and less than 900 mm (35.4 in.) in either length or width direction. All specimens are to be cut to the same size and shape, even if ratings are done at different times/dates.

5.1.2 Identify the length and width direction of each specimen to provide a means to assess differences that may exist.

5.1.3 Avoid using a specimen more than one time for evaluation because stretching and crushing may change the fabric hand.

5.1.4 The number of specimens of each sample used is to be consistent with accepted statistical analysis and with the number of evaluators.

5.2 Specimen Marking.

5.2.1 Prior to conditioning and evaluation, specimens are to be prepared and marked by someone other than the evaluator.

5.2.2 Mark specimens to denote sample and specimen identification, the surface of the specimen to be evaluated and the length direction of the fabric (see 5.1.2). Use a pen or pencil, not attached labels, to mark the specimens with identification, directional and surface information.

5.2.3 Condition specimens prior to testing according to ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 11.1). Report if other conditions are used (see 9.6).

6. Evaluator Preparation

6.1 Evaluators are to wash their hands 0.5 h prior to the evaluation of specimens using the same washing procedure and soap, preferably a hand soap that does not contain moisturizers.

6.2 Evaluators are to dry their hands with the same toweling; i.e., all use cotton fabric towels or all use paper towels.

6.3 Evaluators are to avoid activities involving extreme exercise and should not expose hands to temperature changes or moisture after washing hands prior to specimen evaluation.

7. Procedure

7.1 Evaluation Set-up.

7.1.1 The evaluator is to be relaxed and comfortable in a room free of distractions. The evaluation may be performed seated or standing.

7.1.2 The evaluator is to be assisted by a facilitator who is to provide instructions about the specific elements of hand to be evaluated, the rating scale to be used, the number of samples and specimens to be rated or compared, the order of presentation, the expected duration of the rating session, and other pertinent information regarding the evaluation protocol.

7.1.3 Evaluators may communicate evaluation ratings, rankings and other tactile responses verbally to the facilitator or to a recording device. They may record hand sensations on a rating form.

7.1.4 The evaluator may or may not view specimens during a judging session. Blocking the view of specimens is generally preferred (see 11.2). This may be accomplished by handling the specimens behind a screen or drape, closing the eyes and/or using a blindfold.

7.2 Handling Sequence.

7.2.1 The facilitator is to place a specimen on a smooth, nonmetallic surface. The specimen is to be placed with the surface to be evaluated uppermost and correctly aligned as indicated by the markings on the specimen.

7.2.2 If the thermal element of hand (warm/cool) is to be evaluated, the evaluator is to make this assessment first with the initial contact of the finger tips to the fabric surface.

7.2.3 While still on a flat surface, the specimen is to be held down with one hand and stroked or touched with the other hand (see 11.3).

7.2.4 The evaluator is to then touch the specimen by lightly pressing it with the fingers and palm of the hand.

7.2.5 The evaluator is then to pick up the specimen and rub it between the thumb and fingertips.

7.2.6 Next, the evaluator is to squeeze the specimen gently between the thumb, fingers and palm by making a fist.

7.2.7 If the ease of stretch is to be judged, the specimen is to be held so there is at least 90 mm (3.5 in.) and no more than 250 mm (10 in.) of fabric between the hands. With elbows close to the body, hands are to be pulled apart noting the ease of extending the specimen. The specimen is to be stressed in the lengthwise, widthwise and diagonal (bias) directions.

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7.2.8 If recovery from squeezing (resiliency) is to be judged, the evaluator must look at the specimen. The specimen is to be clenched tightly in a fist and then released quickly; i.e., in less than 5 s. Such evaluation is to be made after all other handling procedures have been completed.

7.2.9 Repeat steps 7.2.1-7.2.8 for each specimen.

8. Evaluation

8.1 Specimens can be compared in pairs or sets and judged for the direction and magnitude of the differences in the constituent elements of hand to be assessed. One of the following techniques may be used:

8.1.1 Establish a fabric standard and rate each specimen against the reference. Using agreed upon terminology (see Appendices A and B), describe the specimen as (smoother) than the reference specimen or not as (smooth) as the reference specimen. Several specimens can be rated one at a time against the standard.

8.1.2 Establish two extremes for a property of interest described by selected hand terminology and assign arbitrary numerical values to the extremes. For example, the descriptor (limp) can be selected and arbitrarily assigned a value of 1 while the other extreme descriptor (stiff) could be assigned the value of 5. Specimens to be evaluated could then be assigned values within the established numerical scale according to the differences felt.

8.1.3 Specimens may be ranked by comparative assessments such as "most" (rough), "least" (rough) or "moderately" (rough). Ranking specimens may be difficult with increasing numbers of specimens.

8.1.4 Perception scales for the description of change in a constituent element of hand when comparing an original sample specimen against processed, treated, or other finished specimens can be developed. For example, perception scales using numerical values assigned to judgment descriptions such as 1 = no difference perceived; 2 = slight difference perceived; 3 = moderate difference perceived; 4 = extreme difference detected. Numerical values and perception descriptions may be expanded.

8.2 The evaluation should be repeated by the same individual within 1-5 days.

8.2.1 If the individual fails to rate specimens at the same points on the scale or rank them in the same order on the second day; i.e., fails to obtain consistent results on a repeat evaluation, appropriate statistical analysis shall be used as a means to determine the agreement or non-agreement of the two sets of data.

9. Report

9.1 The number of samples and specimens evaluated by each individual.

9.2 The elements of hand evaluated.

9.3 The visual blocking procedure used.

9.4 The order of presentation if different than listed in Section 7.

9.5 The evaluation scale, ranking or other assessment procedure used.

9.6 The conditions under which specimens were rated.

9.7 The numerical or assessment values.

10. Precision and Bias

10.1 *Precision*. Precision of the techniques for subjective evaluation of hand described by this guideline have not been determined. There are no predetermined rating scales for assigning numeric values to constituent elements of hand that evaluators may apply in making material comparisons to facilitate derivation of components of variance needed for precision calculations. Consequently, the user must resort to other statistics than analysis of variance in assessing the utility of the methodology for comparative purposes.

10.1.1 Results from the subjective evaluation of hand usually are in the form of arbitrary rankings, weighted descriptors, or other arbitrary, discrete, discontinuous rating scales agreed upon by those making a comparison. Non-parametric analysis techniques, based usually on the chi-square or "*t*" statistics for determining statistical significance and probability levels of ranking orders, are suggested for data of this type. With this kind of analysis, significance between ratings is determined, but there can be no determination of magnitude associated with variability measurement.

10.2 *Bias*. Within the guideline techniques, bias, if any, cannot be determined, since there are no known procedures for determining the true values for the constituent elements of hand.

11. Notes

11.1 Available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610. 832.9500; fax: +1.910.832. 9555; web site: www.astm.org.

11.2 Visual impressions of a specimen can cause biased tactile judgments. For example, shiny fabrics are not always exclusively slick or smooth; looped, bulky fabrics are not exclusively soft. Color preferences can also influence hand judgments.

11.3 Use of left or right hand, associated with an evaluator's dominant hand, may cause biased tactile judgments. Users may wish to specify evaluators use either their dominant hand, or opposite to dominant hand, for touching and stroking the specimens during the evaluation process.

12. History

12.1 Reaffirmed 2020

12.2 Revised 1996(with title change). Reaffirmed 2006, 1992, 1991. Editorially revised 2019 (with title change), 1997. Editorially revised and reaffirmed 2011. 2001.

12.3 Developed in 1990 by AATCC Committee RA89.

Appendix A

A1. Constituent Elements of Hand: Terminology Categorized by Physical Attribute*

PHYSICAL ATTRIBUTE			
Compression	**Bending**	**Shearing**	**Surface**
hard	stiff	supple	coarse
thin	pliable	clinging	rough
thick	supple	tight	slippery
springy	crisp	loose	harsh
fullness	limp	firm	smooth
bulky	papery	pliable	fuzzy
firm	lively	elastic	soft
soft	springy	stretchy	scratchy
lively	boardy		slick
lofty			waxy
resilient			nappy
			oily
			raspy
			warm
			cool

*Appendix A is intended as a reference listing of hand descriptors and should not be considered an exhaustive compilation of terms.

NOTE: Some terms can be attributed to more than one physical property category; i.e., the term "soft" can be considered as a compression or as a surface attribute.

Appendix B

B1. References Concerning Specific Constituent Elements of Hand

B1.1 AATCC *"Bibliography on Fabric Hand*," compiled by AATCC Committee RA89, Hand Evaluation Test Methods, August 1995.

B1.2 ASTM D123, Terminology Relating to Textiles, Annex 3, Terms Relating to the Hand of Fabrics, *Annual Book of ASTM Standards*, Vol. 07.01, 1986, p92, available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832.9500; fax: +1. 610.832.9555; web site: www.astm.org.

B1.3 ASTM *Manual on Sensory Testing Methods*, Special Technical Publication 434, ASTM, 1968.

B1.4 Brand, R. H., Measurement of Fabric Aesthetics, Analysis of Aesthetic Components, *Textile Research Journal*, Vol. 34, 1964, pp791-804.

B1.5 Civille, G. V., and Dus, C. A., Development of Terminology to Describe the Hand/Feel Properties of Paper and Fabrics, *Journal of Sensory Studies*, Vol.

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5, 1990, pp19-32.

B1.6 Kawabata, Sueo, *The Standardization and Analysis of Hand Evaluation*, 2nd Edition, The Textile Machinery Society of Japan, 1980.

B1.7 Kim, C. J. and Vaughn, E. A., Physical Properties Associated with Fabric Hand, *AATCC Book of Papers*, 1995, pp78-95.

B1.8 Wiczynski, M. E., Psychometric Properties of the Hand of Polyester/Cotton Blend Fabrics, Unpublished doctoral dissertation, University of North Carolina at Greensboro, 1988.

B1.9 Winakor, G., Kim, C. J. and Wolins, L., Fabric Hand: Tactile Sensory Assessment, *Textile Research Journal*, Vol. 50, 1980, pp601-610.

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Evaluation Procedure for Instrumental Color Measurement

1. General Description of Instrumental Color Measurement

1.1 Purpose

1.1.1 This evaluation procedure is a reference document to support the proper measurement of the color stimulus of specimens by instrumental means as required in many of the current AATCC test methods. The document contains three major subsections covering *reflectance measurement*, *transmittance measurement* and *related calculations*. In addition, an appendix is supplied which details specific techniques and specimen handling procedures.

1.2 Terminology

1.2.1 **area-of-view,** n.—*of color measuring instrument*, the dimensions of the surface area that a color measuring instrument is capable of covering in a single color measurement.

1.2.2 **color measurement,** n.—a numerical representation of the color stimulus of an object obtained by use of a color measuring instrument. A single measurement may represent an average of multiple readings of a specimen.

1.2.3 **color measuring instrument,** n.—any device, such as a colorimeter or spectrophotometer, used to measure the relative amounts of energy reflected from (or transmitted through) a specimen in the visible region of the energy spectrum (comprising the wavelengths from 360 to 780 nm, and including as a minimum the region from 400 to 700 nm).

1.2.4 **fluorescence,** n.—a phenomenon in which radiant flux of certain wavelengths is absorbed and re-emitted at other visible, usually longer, wavelengths.

1.2.5 **geometry,** n.—*of a color measuring instrument*, one of the following terms (diffuse/8, 8/diffuse, 0/45 or 45/0) which describe the angle or manner in which a color measuring instrument:

(1) illuminates the specimen (diffuse, 8°, 45°).

(2) views the resulting reflected light (8°, diffuse, 45°, 0°).

Diffuse/8 (written as d:8°) and 8/diffuse (written as 8°:d) geometry instruments contain a sphere used to diffuse the light illuminating [or reflected from] the specimen, while 0/45 (written as 0°:45°) and 45/0 (written as 45°:0°) geometry instruments generally use mirrors or fiber optics to direct the illumination [or viewing] at a 45° angle to/from the specimen.

NOTE: Instruments of different geometrics may produce different colorimetric results on most textile materials.

1.2.6 **reflectance,** n.—the ratio of the reflected radiant or luminous energy (light) to the incident energy in the given conditions.

1.2.7 **reflectance factor,** n.—the ratio of the light reflected from the specimen to the light reflected from the perfect reflecting diffuser under the same geometric and spectral conditions of measurement.

1.2.8 **specular reflection,** n.—the reflection without diffusion, in accordance with the laws of optical reflection, as in a mirror.

1.2.9 **standardization,** n.—*of color measuring instrument*, the act of measuring one or more standard materials with a color measuring instrument for the purpose of calculating a set of correction factors to be applied to all subsequent measurements.

1.2.10 **transmittance,** n.—the ratio of transmitted light to incident light under specified geometric and spectral conditions.

NOTE: Regular transmittance (of transparent materials) is the ratio of undiffused transmitted light to incident light.

1.2.11 **transmittance factor,** n.—the ratio of the light measured after being transmitted through a transparent or translucent specimen to that measured in the absence of the specimen. In case of liquid samples, the test measurement is usually that of the colored solution in a cuvette and the reference measurement is that of the identical cuvette filled with the solvent used for dissolution (usually distilled water).

1.2.12 **verification standard,** n.—*in color measurement*, any stable material which is used for the purpose of confirming (or verifying) the validity of an instrument standardization. Color measurements, which are made immediately following a standardization, are compared to original measurements of the standard to detect improper standardization.

1.3 Safety Precautions

1.3.1 These safety precautions are for information purposes only. The precautions are ancillary to the testing pr0cedures and are not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handling materials in this test procedure. Manufacturers MUST be consulted for specific details such as safety data sheets and other manufacturer's recommendations. All OSHA standards are rules must also be consulted and followed.

1.3.2 Good laboratory practices should be followed. Wear eye protection in all laboratory areas.

1.3.3 The manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

1.4 General Guidelines

1.4.1 In general, instrumental color measurement procedures are dictated by the type of specimen to be measured and the instrument with which it will be measured. Many types of color measuring instruments are available, differing in such features as area-of-view, illumination method and geometry. The user is cautioned that conflicting results may be obtained by comparing data acquired on instruments of different designs.

1.5 Procedure

1.5.1 Operators who utilize AATCC EP6 as a reference for color measurement should generally conduct such measurements as follows:

1.5.2 Refer to the appropriate section (Reflectance or Transmittance) for specimen to be measured.

1.5.3 Standardize instrument (see 2.4). Maintain a record of the procedure and of the results for any verification standards measured.

1.5.4 Obtain and prepare specimen, noting any special sampling and/or conditioning procedures that may be required (see Appendix A.1.2).

1.5.5 Present specimen to color measuring instrument, again following any special techniques required for the type of material being measured (see Appendix A).

1.5.6 Measure the color of the specimen, obtaining the appropriate spectral reflectance or transmittance factors or colorimetric values.

1.5.7 Perform any appropriate numerical calculations as specified in Section 4 or as is required by the specific test method.

2. Measurement of Color by Reflectance Methods

2.1 Principle

2.1.1 Materials of an opaque or nearly opaque nature (but not translucent) are measured by reflectance methods to obtain a numerical representation of the color of the specimen. Proper equipment setup, standardization of the color measuring instrument, and proper presentation of the test specimens to the instrument are required to achieve consistent,

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reliable, and meaningful reflectance measurement results. In addition, calculation of the colorimetric values required for evaluating the results must be performed in the prescribed manner.

2.2 Uses and Limitations

2.2.1 This section is restricted to the measurement of the color of opaque and nearly opaque specimens by reflectance spectrophotometers or reflectance colorimeters. The use of special procedures and/or aids to the presentation of specimens is covered in the appendix.

2.3 Apparatus and Materials

2.3.1 Reflectance color measuring instrumentation illuminates a specimen and measures the amount of light which is reflected from the surface of the specimen. Illumination is normally done in a polychromatic manner (white light), however monochromatic mode is acceptable for nonfluorescent specimens. Reflectance color measuring instruments may be broadly divided between two groups: Spectrophotometers and Colorimeters.

2.3.2 Spectrophotometers (typically d:8°, using polychromatic illumination) separate and measure the spectrum of light reflected from the specimen, relative to a reference, white at regular intervals (wavelength intervals of 5 nm, 10 nm, and 20 nm are most common). This data may be used to calculate the needed tristimulus values (*X,Y,Z*) for the selected illuminant/observer combination. Some spectrophotometers (typically 8°:d) illuminate the sample with monochromatic light, and measure the amount of light reflected from the surface as the sample is illuminated at regular wavelength intervals.

2.3.3 Colorimeters measure the tristimulus values (*X,Y,Z*) directly through broadband filters which are designed to produce colorimetric values for one illuminant and observer (typically C/2°). Measurement of reflectance factors at specific wavelengths is not possible with a colorimeter.

2.3.4 Within these two categories, the instruments are further defined by their geometry. Two types of geometries are prevalent: Sphere (also referred to as d:8° or 8°:d and 45/0 or 0/45. The first term in each geometry type refers to the method (or angle) by which the sample is illuminated (ex: 45° in a 45/0 instrument). The second term refers to the angle at which the instrument views the illuminated specimen (ex: 0° in a 45/0 instrument).

2.3.5 Sphere (d:8°) instruments illuminate the specimen indirectly when the specimen is placed against a port opening into a diffusely illuminated sphere, and view the specimen at an angle between 0° and 10° from the perpendicular. This arrangement is designed to capture all light reflected from the specimen. Some sphere instruments with a viewing angle greater than 0° include a *specular port* which allows for the inclusion or exclusion of the *specular reflectance*. The majority of spectrophotometers in use in the textile industry use d:8°, where the actual viewing angle is 8°. Measurements made on other geometries will likely produce different spectral or colorimetric results.

2.3.6 Sphere (8°:d) instruments are similar, but the path of illumination and viewing are reversed. This method illuminates the sample at an angle between 0° and 10°, and measures the amount of light reflected from the surface into the sphere.

2.3.7 Instruments with 45°:0° or (0°:45°) geometry illuminate the specimen at the first angle and view the specimen at the second. These two geometries can be either circumferential (viewing or illuminating at 45° to the specimen in a complete circle) or directional. For most textile samples, either 45°:0° or 0°:45°) yield equivalent results.

2.3.8 All color measuring instruments require a white calibrated standard with which to standardize the instrument. The colorimetric values for this calibration standard are stored in the instrument or the software and require only that a specific standard be used to standardize the instrument. The correct white standard is usually identified with a serial number.

2.4 Standardization

2.4.1 Proper standardization of any color measuring instrument is absolutely necessary to achieve uniform, accurate results. While different types of instruments require varying methods of standardization, there are common principles which must be observed.

2.4.2 In general, an instrument standardization involves measuring a clean white surface of known reflectance factors (referenced to a perfect reflecting diffuser) and calculating (through software built into the instrument or computer program) a series of correction factors which will be applied to all subsequent measurements. Some instruments also require a black tile (or light trap), and possibly a gray tile. Each of these materials must be maintained in its original clean, unscratched condition. Refer to manufacturer's recommendations for cleaning instructions.

2.4.3 The frequency with which this standardization must be performed depends on many factors including the type of instrument, the environmental conditions in which the instrument operates, and the required accuracy of the results. For most applications, an interval no more than 8 hours is typical.

2.4.4 Once the standardization step has been performed, it is important to verify the success of the procedure by measuring some colored materials (**verification standards**) and comparing the resulting colorimetric values to the original values for these materials. If the measured values do not fall within an acceptable variation from their original values, the standardization is not considered valid. The number of **verification standards** and the acceptability limits depend on user requirements, but are typically 1-3 standards and an acceptance limit of 0.20 ΔE_{CMC} (2:1) (D_{65}/10°) units.

2.5 Sampling

2.5.1 All measurements taken on color measuring instrumentation involve "**sampling**." The area-of-view of the instrument, the number of presentations averaged to produce a single measurement, the difficulty of presenting the specimen to the instrument, and the accuracy with which the sample represents the object of concern (garment, roll, dye lot, etc.) all play important parts in achieving meaningful and reproducible results. Refer to ASTM E1345 (see 7.1) and SAE J1545 (see 7.3) for techniques in establishing sampling procedures.

2.6 Specimen Preparation

2.6.1 The ideal specimen to measure is a rigid, non-textured, inert, opaque specimen of uniform color. Such ideal specimens do not exist in textiles, so it becomes necessary to employ techniques and practices when measuring most textile materials which eliminate or reduce the effect any objectionable characteristics have on the instrumental color measurement. Specific procedures and techniques for handling specimens which meet the following characteristics are presented in Appendix A.

2.6.2 *Fluorescence* of the specimen (from dyes or fluorescent whitening agents [FWAs]) will influence the results depending on the amount of fluorescing material present and the amount and quality of ultraviolet and visible energy in the instrument light source. Results may be particularly hard to duplicate between instruments. Example materials are white or lightly colored materials treated with FWAs (see A1.1).

2.6.3 *Moisture Content* of textile materials can affect their color and appearance characteristics. The amount of conditioning time necessary to achieve a stable moisture state varies with fiber, fabric construction, dyes and surrounding conditions. Examples of materials which are typically affected by moisture content are cotton and rayon fabrics (see A1.2).

2.6.4 *Non-Rigid* specimens tend to protrude (or "pillow") into the viewing port of instruments which contact the specimen at the measurement port. The amount of intrusion may vary depending on number of layers, softness of material and the backing pressure applied to mount the specimen. Variations in the amount of intrusion will result in significant deviations in the resulting color

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measurement which are both unpredictable and non-reproducible. Examples of these materials are fiber, yarn, knits and layers of lightweight fabric (see A1.4.3).

2.6.5 *Non-Opaque* specimens allow some light to pass through the material during measurement. Most textile materials, by nature of their structure, fit this category. During measurement, any light which passes through the material to reach the backing plate (or escape from the instrument) will yield false and unpredictable results. Examples of these materials are knits, lightweight materials, and fiber (see A1.3 and A1.3.1).

2.6.6 *Sensitivity* of the specimen to light (photochromism) and/or heat (thermochromism) will result in unpredictable and nonreproducible results, depending on the degree of sensitivity and the amount of time the specimen is exposed to objectionable conditions (see A2.5).

2.6.7 *Size* of the specimen is important in obtaining a representative measurement by the instrument. When the specimen is too small for normal measurement, special techniques may be required to achieve a proper color measurement (see A1.6).

2.6.8 *Surface Texture* of the specimen (including pile lay, twill, gloss, and luster) will affect the results of the color measurement. The color measurement of specimens with such physical characteristics will be affected in different ways depending on the geometry of the instrument. Results between instruments may be non-reproducible. Examples of these specimens are carpet, corduroy and wound yarn (see A1.7).

2.6.9 *Variation in Color* (non-uniformity) within the specimen, as related to the area-of-view of the instrument can cause inaccurate and non-reproducible results. Examples are denim and heathers (see A1.8).

3. Measurement of Color by Transmittance Methods

3.1 Principle

3.1.1 Materials of a transparent nature are measured by transmittance methods to obtain a numerical representation of their color and/or their relative strength (concentration) compared to that of a standard material. The most common application is the measurement of solutions of dyes in clear solvents contained in a glass cuvette or flow cell for the purposes of determining colorant identity, relative strength or color difference.

3.2 Uses and Limitations

3.2.1 This method is generally suitable for the measurement of true solutions which are transparent and void of particulate matter. Although translucent or turbid liquids are sometimes measured by transmittance methods, such methods are beyond the scope of this evaluation procedure.

3.3 Apparatus, Reagents, and Materials

3.3.1 Transmittance instruments may be dedicated instruments (capable only of measuring transmittance specimens), or may be combined with a sphere for an instrument capable of performing both reflectance and transmittance measurements. Most transmittance instruments are spectrophotometers, although a few are colorimeters.

3.3.2 Transmittance cells or cuvettes (used to hold the solution during measurement) are typically made of glass or quartz and are designed for specific pathlengths (normally 10 mm) suitable for measurement of most textile related solutions (usually requiring some dilution steps to be taken before measurement). Flow-through cells, which allow the solution to be pumped or fed through them, are more efficient for high volume applications and usually result in better reproducibility of results.

3.3.3 When using volumetric procedures, volumetric glassware (pipettes and flasks) is **absolutely required** for proper preparation of specimens, and only glassware which is rated at *class A* should be used.

3.3.4 Balances which are capable of accurate results within 0.1% of the quantity being weighed must be used when weighing specimens used to prepare the solution for measurement.

3.4 Standardization

3.4.1 Before measuring specimen(s), the instrument must be standardized according to the manufacturer's directions. Normally this requires measuring a clear solution of the solvent used for dissolution (usually distilled water) in a clean transmittance cell to produce a 100% reference line. Some instruments require the measurement of a "blocked beam" to set a 0% reference. This procedure produces a set of correction factors which are applied to all subsequent measurements. The use of one or more colored filters as a **verification standard** to check both photometric and wavelength accuracy is recommended.

3.5 Sampling

3.5.1 Sampling procedures for transmission methods depend on the nature of the material and the type of specimen collected. Powder and paste specimens should be taken from enough locations to ensure a reproducible sample within the population being sampled and thoroughly mixed prior to proceeding with weighing and dissolving.

3.6 Specimen Preparation

3.6.1 Specimens must be prepared by prescribed laboratory procedures using analytical balances and volumetric glassware. Solutions should be prepared at a concentration at which the percent transmittance of the solution in the transmittance cell is between 10% and 50% at the wavelength of maximum absorbance (see 7.4). Each specimen must be in true solution, not in suspension, dispersion or association. When preparing specimens where solubility is unknown, it is necessary to let the prepared solution stand for a period of time to observe any settling of contents. If such settling is observed, a true solution does not exist and other solvent systems are required. All specimens should be allowed to reach room temperature prior to presenting the specimen to the instrument.

4. Calculations

Most calculations of colorimetric nature are performed by the software being used to operate the color measuring instrument. In normal cases of reference to this Evaluation Procedure it will not be necessary for the user to perform these calculations; however they are described in this Evaluation Procedure as a means of reference and standardization for those who may need to perform such calculations.

4.1 Tristimulus Values

4.1.1 The tristimulus values ($\boldsymbol{X,Y,Z}$) are derived from spectral data and are the basis for all colorimetric calculations. The exact $\boldsymbol{X,Y,Z}$ values derived from a set of spectral data depend on several factors including the wavelength range (and interval) of measurement and the user's choice of illuminant/observer functions used in the calculation. While most calculations of tristimulus values are performed by computer programs, interested persons should refer to ASTM E308 (see 7.1) for definitive procedures (see 7.2).

4.2 1976 CIE L*,a*,b*, C*, and hue angle (h_{ab})

4.2.1 Calculate the L^*, a^*, b^*, C^*_{ab}, h_{ab} values from the X,Y,Z tristimulus values for both the reference and specimen as follows:

$$L^* = 116(Y/Y_n)^{1/3} - 16$$

if

$$Y/Y_n > 0.008856$$

but

$$L^* = 903.3\ (Y/Y_n)$$

if

$$Y/Y_n \le 0.008856$$

$$a^* = 500[f(X/X_n) - f(Y/Y_n)]$$

$$b^* = 200[f(Y/Y_n) - f(Z/Z_n)]$$

where:

$$f(X/X_n) = (X/X_n)^{1/3}$$

if

$$X/X_n > 0.008856$$

or

$$f(X/X_n) = 7.787(X/X_n) + 16/116$$

if

$$X/X_n \le 0.008856$$

$$f(Y/Y_n) = (Y/Y_n)^{1/3}$$

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if

$Y/Y_n > 0.008856$

or

$f(Y/Y_n) = 7.787(Y/Y_n) + 16/116$

if

$Y/Y_n \leq 0.008856$

$f(Z/Z_n) = (Z/Z_n)^{1/3}$

if

$Z/Z_n > 0.008856$

or

$f(Z/Z_n) = 7.787(Z/Z_n) + 16/116$

if

$Z/Z_n \leq 0.008856$

$C^*_{ab} = (a^{*2} + b^{*2})^{1/2}$

h_{ab} = arctan(b^*/a^*) expressed on a 0° to 360° scale with the a^* positive axis being 0° and the b^* positive axis at 90°.

For these equations, X_n, Y_n and Z_n are the tristimulus values of the illuminant. For daylight the preferred illuminant/observer combination is D_{65}/10°. Table I gives the values for all combinations in ASTM E308.

4.3 Color Strength Value using Reflectance Measurement

4.3.1 Color Strength Value is a single numerical value related to the amount of color absorbing material (colorant) contained in a specimen. It is most often used to calculate the difference in strength (% strength) between two colored specimens. Color Strength Value may be calculated by any one of **four** acceptable methods. The value which results from one method may not agree with any other method. The choice of method is usually dependent on the nature of the specimen and the need for obtaining a color strength value. Colorimeters must calculate this value using a tristimulus function method. The methods used are labeled as SWL, SUM, WSUM and TSVSTR, and are calculated as follows:

4.3.2 Color Strength Value of specimens measured on a spectrophotometer most often involves calculation of a **K/S value** at one or more wavelength intervals. The commonly used equation for calculation of the **K/S value** for opaque specimens (i.e., textiles) at a specified wavelength (λ) is:

$$\mathbf{K/S = ((1.0 - R_\lambda)^2)/(2.0R_\lambda)}$$

where:

$\mathbf{R}_\lambda$ is the reflectance factor of the specimen at wavelength λ (the **%R** value is normally measured by a spectrophotometer and normalized to 1.0 [i.e., 100% = 1.0]).

4.3.3 For applications where the Pineo correction form of the equation is required (generally for deeply dyed textiles), the formula used is:

$$\mathbf{K/S = (1.0 - (R_\lambda - s)^2)/2.0(R_\lambda - s)}$$

where:

s is the minimum reflectance achievable in the deepest dyed shade and is used for all wavelengths.

4.3.4 The four types of Color Strength Value calculations that are most commonly used are:

(SWL) K/S at single wavelength (usually wavelength of maximum absorbance). Use equation 4.3.2 and calculate **K/S** for a single wavelength.

(SUM) K/S summed over wavelength interval within the visible spectrum. Use equation 4.3.2, calculate **K/S** at each desired wavelength interval and sum. The resulting value may be normalized by dividing it by the number of intervals summed.

(WSUM) K/S weighted by visual function (such as the x,y,z functions and the D_{65} illuminant energy function) and summed over a wavelength interval within the visible spectrum, then divided by the number of wavelength intervals summed.

$$WSUM = \Sigma_\lambda [(K/S_\lambda * \bar{x}_\lambda * E_\lambda) + (K/S_\lambda + \bar{y}_\lambda * E_\lambda) + (K/S_\lambda * \bar{z}_\lambda * E_\lambda)]/n$$

where:

K/S = K/S as calculated in equation 4.3.2 above

E = Energy of selected illuminant (normally D_{65})

$\bar{x},\bar{y},\bar{z}$ = tristimulus weighting values for selected observer (normally 10°)

n = number of wavelength intervals used

(TSVSTR) tristimulus color strength value (*X*,*Y*,*Z* function). Normally the *Y* value is used as a gross color strength value related to the visual lightness function, although either the *X* or *Z* may be chosen for measurement of components where the absorption characteristics are known and fall into widely separated regions of the visible spectrum. In most applications, the smallest of the 3 tristimulus values (*X*, *Y* or *Z*) is chosen and is used in place of the %R value in equation in 4.3.2 above. While the TSVSTR equation is used in the textile industry, no specific scientific support of it can be found in normally accepted reference books.

NOTE: See 4.5 below to calculate the relative difference in strength between two specimens using one of these calculation methods.

4.4 Color Strength Value using Transmittance Measurement

4.4.1 Color Strength Value is a single numerical value related to the amount of color absorbing material (colorant) contained in a solution. It is most often used to calculate the difference in strength (% strength) between two colored solutions. Color Strength Value may be calculated by any one of **four** acceptable methods. The value which results from one method may not agree with any other method. The choice of method is usually dependent on the nature of the specimen and the need for obtaining a color strength value. Colorimeters must calculate this value using a tristimulus function method. The methods used are labelled as SWL, SUM, WSUM and TSVSTR, and are calculated as follows:

4.4.2 Color Strength Value of specimens measured on a spectrophotometer most often involves calculating an **absorbance value** at one or more wavelengths. The equation for calculating the **absorbance value** at a specified wavelength (λ)

Table I—Tristimulus Values for Illuminant/Observer Combinations

Illuminant/Observer Combinations	Tristimulus Values		
	X_n	Y_n	Z_n
		(10° observer)	
A/10°	111.146	100.000	35.203
C/10°	97.285	100.000	116.145
D_{50}/10°	96.720	100.000	81.427
D_{55}/10°	95.799	100.000	90.926
D_{65}/10°	94.811	100.000	107.304
D_{75}/10°	94.416	100.000	120.641
F2/10° (Cool White Fluorescent)	103.279	100.000	69.027
F7/10° (Daylight Fluorescent)	95.792	100.000	107.686
F11/10° (Ultralume 4000, TL84)	103.863	100.000	65.607
		(2° observer)	
A/2°	109.850	100.000	35.585
C/2°	98.074	100.000	118.232
D_{50}/2°	96.422	100.000	82.521
D_{55}/2°	95.682	100.000	92.149
D_{65}/2°	95.047	100.000	108.883
D_{75}/2°	94.972	100.000	122.638
F2/2°(Cool White Fluorescent)	99.186	100.000	67.393
F7/2°(Daylight Fluorescent)	95.041	100.000	108.747
F11/2° (Ultralume 4000, TL84)	100.962	100.000	64.350

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is:

$$\mathbf{A_\lambda = \log_{10}\,(1.0/\tau_\lambda)}$$

where:

τ_λ is the internal transmittance of the specimen (the **%T** value normally measured by a spectrophotometer and normalized to 1.0 [i.e., 100% = 1.0]).

4.4.3 The four types of Color Strength Value calculations that are most commonly used are:

(SWL) absorbance at single wavelength (usually wavelength of maximum absorbance). Use equation in 4.4.2 and calculate **A** for a single wavelength.

(SUM) absorbance summed over wavelength interval within the visible spectrum. Use equation in 4.4.2, calculate **A** at each desired wavelength interval and sum. The resulting value may be normalized by dividing it by the number of intervals summed.

(WSUM) absorbance weighted by visual function (such as the x,y,z functions and the D_{65} illuminant energy function) and summed over a wavelength interval within the visible spectrum, then divided by the number of wavelength intervals summed.

$$\text{WSUM} = \Sigma_\lambda\,[(A_\lambda * \bar{x}_\lambda * E_\lambda) + (A_\lambda * \bar{y}_\lambda * E_\lambda) + (A_\lambda * \bar{z}_\lambda * E_\lambda)]/n$$

where:

- A = Absorbance calculated in equation in 4.4.2 above
- E = Energy of selected illuminant (normally D_{65})
- $\bar{x},\bar{y},\bar{z}$ = tristimulus weighting values for selected observer (normally 10°)
- n = number of wavelength intervals used

(TSVSTR) tristimulus color strength value (*X*,*Y*,*Z* function). Normally the *Y* value is used as a gross color strength value related to the visual lightness function, although either the *X* or *Z* may be chosen for measurement of components where the absorption characteristics are known and fall into widely separated regions of the visible spectrum. In most applications, the smallest of the three tristimulus values (*X*, *Y* or *Z*) is chosen and is used in place of the %T value in equation in 4.4.2 above. While the TSVSTR equation is used in the textile industry, no specific scientific support of it can be found in normally accepted reference books.

4.5 Relative Strength

4.5.1 Color Strength Values calculated by the above method(s) may be used to calculate the relative difference in *strength* between two specimens, one of which is considered to be a *standard*. The result of this comparison yields a numerical quantity which is termed **%strength**.

$$\mathbf{\%strength = (ColorStrValue_{specimen}/ColorStrValue_{standard})\,(100.0)}$$

5. Reporting

Reporting of procedures and data accomplished through reference to this evaluation procedure should include, as a minimum, the following information:

5.1 The instrument geometry type that was used to measure the specimen(s).

5.2 The identification of the spectrophotometer or colorimeter used.

5.3 The illuminant\observer used to calculate the colorimetric values.

5.4 Identification of specimen.

5.5 The color strength value calculation (if any) used (**SWL**, **SUM**, **WSUM**, **TSVSTR**).

5.6 The specimen presentation and averaging methods employed, such as area-of-view, number of thicknesses, temperature and% relative humidity if not at standard conditions and number of readings in a measurement.

6. Precision and Bias

6.1 AATCC EP6 is meant to be used as a supportive reference method. Specific precision and bias statements contained within each referring AATCC test method must be used to evaluate the significance of any colorimetric results obtained as part of that test method procedure. The user is cautioned that precision and bias are affected in part by sample presentation, averaging methods, and the repeatability and accuracy of the individual color measuring instrument used.

7. Notes and References

7.1 ASTM standards are available from ASTM, W. Conshohocken, PA; web site: www.astm.org.

7.1.1 ASTM E1345; Standard Practice for Reducing the Effect of Variability of Color Measurement by Use of Multiple Measurements.

7.1.2 ASTM E308; Standard Test Method for Computing the Colors of Objects by Using the CIE System.

7.2 CIE Publication 15.22018, Colorimetry Fourth Edition is available through the US National committee of the CIE or via the CIE Webshop (www.tech-street.com/cie/). One method to verify that the correct calculations are being made is to enter 100% reflectance values into the computer program being used and have the system calculate the tristimulus values. These values should agree with the values in Table I (from ASTM E308) to the second decimal place for the illuminant being checked.

7.3 SAE J1545; Instrumental Color Difference Measurement for Exterior Finishes, Textiles and Colored Trim, available from SAE International, 400 Commonwealth Dr., Warrendale, PA 15096; web site: www.sae.org.

7.4 Stearns, E. I., *The Practice of Absorption Spectrophotometry*, Wiley-Interscience, 1969.

7.5 Wyszecki and Stiles, *Color Science*, 2nd Ed.

7.6 ASTM D1776; Standard Practice for Conditioning of Textiles for Testing.

8. History

8.1 Revised in 2021 for clarity and to add history section to align with AATCC Style Guide.

8.2 Editorially revised 2019; revised 2016; reaffirmed 2008 and 2003; reaffirmed and editorially revised 1998; reaffirmed 1997 and 1996.

8.3 Developed in 1995 by AATCC Committee RA36. Technically equivalent to ISO 105-J01.

Appendix A

A1. Reflectance Measurement Problems and Guidelines

A1.1 Fluorescence—When employing measuring instruments that are equipped with capabilities for calibrating the UV content of the specimen illumination, the evaluations of whiteness can be directly compared to results from similarly calibrated instruments and that are adequate for commercial use. For color measuring instruments which do not have the ability to control accurately the amount of UV energy illuminating the specimen, the operator may insert a UV-absorbing filter between the light source and the sample, effectively eliminating the UV induced fluorescence. Note that this technique may result in disagreement with visual results. Note also that this technique is only applicable in those cases where the fluorescence is caused by absorption of ultraviolet radiation. Instruments which can control the UV energy will produce results more consistent with visual observation. Previously measured data (standards, controls, etc.) should not be used for direct comparison unless the data were collected on UV calibrated instruments. Fluorescent specimens which fluoresce due to the absorption of visible energy cannot be measured consistently except on an instrument where the quantity and quality of incident energy can be controlled. If this cannot be achieved, then both specimens being compared should be measured at nearly the same time and on the same instrument.

A1.2 Moisture Content—If the existence of moisture affects the measurement

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of color it is necessary to "condition" the specimens to allow the moisture content to stabilize. This conditioning should take place in a room or chamber where the temperature and humidity are constant, and enough time should be allowed for all samples to regain moisture to a reproducible amount. This is typically several hours for most specimens containing cotton or hydroscopic fiber, but can vary significantly with different ambient conditions. The condition of the specimen should be maintained as much as possible during measurement. Standard atmosphere for textile testing is defined by AATCC to be a relative humidity of 65% (± 5%) and ambient temperature of 21 ± 2ûC (70 ± 4ûF) (see 7.6).

A1.3 Non-Opaque—Most textile samples are non-opaque to some extent. All specimens **must** be measured using the same procedure. If sufficient quantity of material is available, it is advisable to layer the material until light will no longer penetrate through the layers. If the measurement of the specimen backed with a white tile is not significantly different to the measurement of the same specimen backed with a black tile, then the number of layers is considered sufficient. Note that many layers of soft material can cause other problems (see Non-Rigid below) and may require a compromise. In such cases it may be necessary to use a nominal number of layers (always the same number for each specimen measured) backed by the same material or tile that contains no fluorescent whitening agents.

A1.3.1 When it is not possible to achieve complete opacity a correction equation may be used which yields a corrected %R ($R_¥$) value. $R_¥$ is defined as the opaque reflectance of a layer so thick that no change in reflectance is seen between the over white and over black measurements. This procedure requires measurement of the specimen backed by a white tile, and measurement of the specimen backed by a black tile. The following equations are then used to produce a corrected %R value ($R_¥$) at each wavelength interval (see 7.5):

$$a = 0.5\,[R_W + ((R_B - R_W + R_g)/(R_B R_g))], \text{ then}$$
$$R_\infty = a - (a^2 - 1)^{1/2}$$

where:

R_W = R of specimen backed by white tile
R_B = R of specimen backed by black tile
R_g = R of white tile alone

Where R_W, R_B and R_g are expressed in decimal fraction form; i.e, 0-1.0.

A1.4 Non-Rigid—To avoid intrusion of soft specimens into the measuring port of the color measuring instrument use one of the following procedures:

A1.4.1 Winding, fastening, or mounting to a card or other sufficiently rigid structure. The backing structure must be of a neutral color, reproducible for all specimens to be measured and the opacity rules (see above) must be observed. When yarn specimens are wound onto cards it is necessary to control the tension, orientation, and thickness of the material to achieve reproducible results.

A1.4.2 A few instruments are designed to measure specimens without contacting their surfaces. Specimens to be measured must be flat, backed by a rigid structure and have sufficient thickness to eliminate any effects from non-opacity.

A1.4.3 Measurement of some specimens behind glass increases repeatability of color measurement for spectrophotometers, particularly on fiber and yarn specimens. The resulting reflectance measurements must be corrected by a glass correction equation, or the effects of the glass may bias the results. In addition, the amount of material and the pressure used to press the material against the glass must be controlled. The equation normally used is:

$$R_\lambda = (R_g + T_c - 1)/(R_g + T_c - 1.0 - (T_d * R_g) + T_d)$$

NOTE: All R and T values are expressed in decimal fraction form.

where:

R_g = measured %R behind glass
T_c = transmittance of glass to collimated light (nominally equal to 0.92 for glass with a refractive index of 1.50 and no absorption)
T_d = transmittance of glass to diffuse light (nominally equal to 0.87 for glass as described above)
R_λ = corrected % reflectance with no glass

NOTE: See 7.4. This form of the glass equation is derived from that in the reference, page 307.

A1.5 Sensitivity to Light or Heat—Specimens which are sensitive to light and/or heat are best measured in a color measuring instrument which exposes the specimen to light for only a brief moment during the actual measurement. Flash illumination instruments and instruments with automatic light shutters provide mechanisms for limiting the amount of time that the specimen is exposed to light. Instruments which must scan the visible spectrum (taking several seconds to perform a single measurement) should not be used for these specimens. In all cases, sample preparation must include precautions for limiting the amount of light the specimen is exposed to prior to measurement. Instruments which illuminate the specimen monochromatically are also acceptable for measuring these types of specimens.

A1.6 Small Specimens—Specimens which are small enough to require the use of a SAV (small area view) option on the color measuring instrument must be read multiple times and averaged to improve the precision of the measurement. Specimens smaller than the area-of-view of the instrument cannot be measured reliably.

A1.7 Surface Texture—The difficulty in measuring specimens with pronounced physical surface characteristics first lies in the decision as to what physical attribute interests the user. The ability of an instrument to separate color from *appearance* is an advantage in some situations, and a disadvantage in others. When it is advisable to measure only the *color* of the specimen, the most effective means is to mount the specimen behind glass and apply enough pressure to eliminate any texture differences. The same precautions and requirements apply as in the use of glass for non-rigid samples above. When the texture of the surface results in *directionality variation*, the specimen may be measured in multiples of 4, rotating 90° after each measurement. All measurements are then averaged to produce a single set of colorimetric values. An example of such a procedure developed for automotive fabrics is SAE J1545 (see 7.3).

A1.8 Variation in Color—When the specimen to be measured is not uniform in color, measurement averaging (of spectral data from a spectrophotometer or tristimulus value data from a colorimeter) is necessary to achieve uniform, reproducible measurements. This requires determining the number of readings necessary for the area-of-view employed which, when averaged together to produce a single result, can be reproduced by repeating the procedure at random locations on the specimen (see SAE J1545).

A2. Transmittance Measurement Problems and Guidelines

A2.1 Chemical Instability—Solutions which are chemically unstable should not be measured. The reason for the instability must be determined and an appropriate procedure and/or solvent developed which results in a stable solution.

A2.2 Foaming, Air Bubbles—The addition of a drop or two of isobutyl alcohol (2-methyl propanol) will help to remove any bubbles or foam during the specimen preparation phase. When adding the liquid specimen to the transmission cell, the operator should hold the cell at an angle and pour the solution down the side of the cell to minimize the formation of air bubbles.

A2.3 pH Sensitivity—Many dyes are sensitive to pH changes and will exhibit a marked shift in the color of the prepared specimen if the pH is changed. The use of distilled water (or demineralized water) as well as the addition of a pH buffer during preparation is usually required.

A2.4 Plating—Dye solutions (espe-

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cially *Basic Class* dyes) have a tendency to form a single layer of dye molecules adhering to the walls of the glass transmission cell. The transmission cell should always be rinsed (or flushed) with the solution to be measured, emptied, then refilled with fresh solution before measuring the color. Note that this procedure assumes that the cell was clean (free of plating) prior to testing, and that all specimens are treated identically. Where practical, an instrument standardization measurement should be made immediately prior to the specimen measurement, and after the reference cell has been similarly flushed with the colored solution, then filled with clear solution (usually distilled water). If a dual beam instrument is being used (where both a specimen and reference cell are simultaneously employed in the measurement), the reference cell (containing clear solution) is flushed with the colored solution before being filled with the clear solution. In this case, the normal method of instrument standardization is sufficient.

A2.5 Sensitivity to Light or Heat (photochromism or thermochromism)—When possible transmission measurement should be performed using an instrument which illuminates the specimen monochromatically (at a single wavelength). If this is not possible, an instrument which provides a light blocking shutter, or which illuminates the specimen using a momentary flash of light may be used to reduce the effect.

A2.6 Resolution (wavelength interval)—In general, the transmission curves for dyes in solution have sharper, more pronounced peaks than those obtained from reflectance measurement. Spectrophotometers which measure at wavelength intervals of 10 nm or less are required to obtain accurate, reproducible results. In addition, the bandpass of the instrument being used must be less than or equal to 10 nm.

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Evaluation Procedure for Instrumental Assessment of the Change in Color of a Test Specimen

Foreword

This Evaluation Procedure was developed in 1995 by AATCC Committee RA36. Adopted as ISO 105-A05 in 1996. The metric used in this method was developed in *ISO TC38 SC1 WG-7*.

1. Scope

1.1 This evaluation procedure is intended as an alternative to the visual method of AATCC EP1. It is applicable to any test method referring to gray scales except those which involve treatment with solutions containing fluorescent whitening agents (FWA).

2. Principle

2.1 The color of the specimen which has been submitted to the colorfastness test and the color of an identical untreated specimen are measured instrumentally. The CIELAB coordinates for lightness L^*, chroma C^*_{ab}, and hue h_{ab} are determined for both specimens as defined in CIE 015:2018 (see 3.1.3). The CIELAB differences L^*, C^*_{ab}, and H^*_{ab} are calculated and converted to a gray scale grade by means of a series of equations.

3. Referenced Documents

3.1 NOTE: Use current versions of all publications unless otherwise specified.

3.1.1 AATCC *Manual of International Test Methods and Procedures*, American Association of Textile Chemists and Colorists, AATCC EP1. Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

3.1.2 Jaeckel, S. M., The Variability of Grey-Scale Assessment and its Contribution to the Variability of a Test Method, *Journal of the Society of Dyers and Colourist*, Vol. 96, 1980, pp540-544.

3.1.3 CIE Publication 15:2018 Colorimetry, 4th Edition. Available through the US National Committee of the CIE or via the CIE web shop (www.techstreet.com/cie/).

4. Terminology

4.1 **color change,** n.—a change in color of any kind whether in lightness, hue or chroma or any combination of these, discernible by comparing the test specimen with a corresponding untested specimen.

4.2 **gray scale grade,** n.—*for color change* (*GSc*), the numerical value that is assigned to the change in color of a test specimen as compared to an original or untreated specimen. This shall be reported as a half step number between 1 and 5 (i.e., 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5).

5. Safety Precautions

5.1 Users must consult color measuring equipment manufacturers for specific details such as equipment operating instructions and other recommendations.

6. Apparatus

6.1 Spectrophotometer or colorimeter meeting any one of the geometric definitions described in CIE publication 15-2018, subclause 6.4.

7. Test and Reference Specimens

7.1 Choose a representative sample of material, which has been subjected to a colorfastness test, of adequate size to fit the sample holder of the instrument used. Back this specimen with enough thicknesses of the original material to make the specimen opaque. (Opacity, or at least consistency, is required for accurate measurement.) Backing a single layer of the material with a white material that contains no FWAs is a permitted alternative provided both the reference specimen (original material) and test specimen are backed in an identical manner.

7.2 Form a layer of the reference specimen (original material) consisting of the same number of thicknesses as that prepared in 7.1.

8. Procedure

8.1 Measure the color of the reference specimen and calculate the CIELAB L^*, C^*_{ab}, and h_{ab} values using the data for the 10° observer and for illuminant D_{65} (see 11.1).

8.2 Measure the color of the test specimen which has been subjected to the colorfastness test and make the same colorimetric calculations as done for the reference specimen. Then make the calculations as described below in Section 9.

9. Calculations

9.1 Symbols

9.1.1 The subscript '*ab*' indicates association with the CIELAB color-difference equation.

9.1.2 The subscript '*M*' is used to represent a function of the mean of the test and reference specimen.

9.1.3 The subscript '*K*' is used to represent chroma corrected hue functions.

9.1.4 The subscript '*F*' is used to represent the special gray scale color difference and separate this color difference (ΔE) from others in normal use.

9.1.5 The subscript '*T*' is for the test specimen.

9.1.6 The subscript '*R*' is for the reference specimen.

9.1.7 The 'Round' function, Round (x,n) used below, rounds the value of x to the specified number of decimal places n, i.e., 3 in the usages below.

9.1.8 All hue angles are in degrees and not radians.

9.2 Calculate ΔE_F using the following pseudo code:

If Abs($h_{ab,T} - h_{ab,R}$) ≤ 180 then
  $h_M = (h_{abT} + h_{abR})/2$
else
  If Abs($h_{ab,T} + h_{ab,R}$) < 360 then
    $h_M = (h_{abT} + h_{abR})/2 + 180$
  else
    $h_M = (h_{abT} + h_{abR})/2 - 180$
  endif
endif

if Abs($h_m - 280$) ≤ 180 then
  $x = [(h_m - 280)/30]^2$
else
  $x = [(360 - \text{Abs}(h_m - 280))/30]^2$
endif

$\Delta L^* = L^*_{abT} - L^*_{abR}$
$\Delta C^*_{ab} = C^*_{abT} - C^*_{abR}$
$C_M = (C^*_{abT} + C^*_{abR})/2$
$D = (\Delta C^*_{ab} \cdot C_M \cdot e^{-x})/100$
$\Delta E^*_{ab} = [(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2]^{1/2}$

$\Delta H^*_{ab} = [(\Delta E^*_{ab})^2 - (\Delta L^*)^2 - (\Delta C^*_{ab})^2]^{1/2}$
If ($h_{abT} - h_{abR}$) < 0 then
  $\Delta H^*_{ab} = (-1) \cdot \Delta H^*_{ab}$
$\Delta C_K = \Delta C^*_{ab} - D$
$\Delta H_K = \Delta H^*_{ab} - D$
$\Delta C_F = \Delta C_K/[1 + (20 \cdot C_M/1000)^2]$
$\Delta H_F = \Delta H_K/[1 + (10 \cdot C_M/1000)^2]$
$\Delta E_F = [(\Delta L^*)^2 + (\Delta C_F)^2 + (\Delta H_F)^2]^{1/2}$
$\Delta E_F = \text{Round}\ (\Delta E_F, 3)$

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9.3 Calculate the gray scale grade for color change (GSc) according to the following function:

if $\Delta E_F \leq 3.400$ then
GSc = 5 – ≤ ΔE_F/1.7
else
if $\Delta E_F \leq 13.600$ then
GSc = 5 – Log_{10} (ΔE_F/0.85)/
Log_{10} (2)
else
GSc = 1
endif
endif

At this point, the GSc is a decimal number between 1 and 5.

10.4 To produce half-step grades (i.e., 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5), perform the following function:

GSc = Round (GSc,3)
GSc = [Int (2 · GSc + 0.5)]/2

10. Report

Report at least the following information:

10.1 The gray scale grade(s) [GSc value(s)] for change in color of the specimen(s); follow the directions and report as directed in the appropriate test method.

10.2 The number and date of this procedure, i.e., AATCC EP7-2021.

10.3 All details necessary for the identification of the sample(s) tested.

10.4 CIE geometry type that was used.

10.5 Identification of the spectrophotometer or colorimeter used.

10.6 Whether illuminant D65 or illuminant C was used.

10.7 Whether the 1964 10° or the 1931 2° observer was used.

11. Notes

11.1 NOTE: Permitted alternative is D65/2°.

12. History

12.1 Editorially revised in 2023 to update a reference in Section 2: Principle, and to put the history in the correct order.

12.2 Revised in 2021 to remove table option and restructure calculations for clarity; editorially revised in 2019; revised in 2015; reaffirmed 2009, 2003; revised 1998; reaffirmed 1996.

12.3 Adopted as ISO 105-A05 in 1996.

12.4 Developed in 1995 by AATCC Committee RA36.

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Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale

Developed in 1996 by AATCC Committees RA36/RA38; under jurisdiction of AATCC Committee RA36; reaffirmed and editorially revised 1997, 1998; revised 2002, 2010; reaffirmed 2007, 2017; Technical Correction 2008; editorially revised 2012, 2019.

1. Scope

1.1 This evaluation procedure describes the use of the AATCC 9-Step Chromatic Transference Scale for evaluating staining on undyed textiles in colorfastness tests.

2. Principle

2.1 The staining of undyed cloth in colorfastness tests is rated by visually comparing the color of the stained cloth and unstained cloth to the differences in color represented by the Scale (see 7.1).

3. Terminology

3.1 **colorant staining**, n.— the unintended pickup of colorant by a substrate due to (1) exposure to a colored or contaminated liquid medium or (2) direct contact with dyed or pigmented material, from which colorant transfers by sublimation or mechanical action (as in crocking).

3.2 **colorfastness**, n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material.

3.3 **grade**, n.—*in textile testing*, the symbol for any step of a multistep standard reference scale for a quality characteristic.

NOTE: The grade is assigned to test specimens exhibiting a degree of the quality comparable to that step of the standard.

3.4 **rating**, v.—*in textile testing*, the process for determining or assigning a grade to a material by comparing it to a standard reference scale.

4. Description of Scale

4.1 The Scale makes use of 54 color chips. The five hues (red, yellow, green, blue and purple) were selected from the *Munsell Book of Color*. The neutral chip representing row number 5 as well as the neutral gray chips correspond to the gray chips on the Gray Scale for Staining (see 7.1).

4.2 The chips on the Scale appear on white cardboard in ten rows. The rows are placed and aligned so that every color shows a similar gradation in depth in a vertical line, ranging from the lightest colors on top to the darkest colors on the bottom. For rating purposes, the rows are designated numerically with number 5 being assigned to the top row which depicts no color, number 4.5 to the second row or lightest colors and so on down to number 1 for the bottom row or darkest colors (see 7.2).

4.3 The chips are mounted in rows with sufficient separation between them to show 9.5 mm (3/8 in.) circular holes cut in the card between rows of chips in the ten rows. A white cardboard mask, as shown in Fig. 1, is provided so that when it is held vertically over the scale, it discloses one of the circular openings in the color scale and the two adjacent color chips (see 7.3).

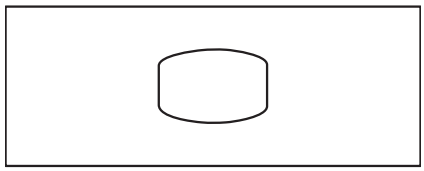


Fig. 1—White cardboard mask.

5. Use of the Scale

5.1 Illuminate the surfaces with a daylight simulator with illumination level at the specimen plane in the range of 1080-1340 lx (100-125 fc) (see 7.4). The light should be incident upon the surfaces at 90 ± 5° and the direction of viewing 45 ± 5° to the plane of the surfaces (see Fig. 2).

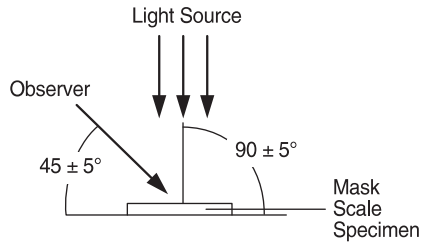


Fig. 2—Illumination and viewing angle for rating specimens.

5.2 The material exhibiting the transferred color to be evaluated is placed behind the card on which the chips are mounted, so that a representative part of the colored portion is visible through one of the circular holes in the column closest to it in hue. Before rating a color transferred to a thin material such as white cotton sheeting, the test specimen should be backed with a number of layers of the clean test cloth. This applies also to samples of thin material showing transferred color mounted on test cards. The color of the cardboard will influence the judgment of the examiner, unless the color of the card is eliminated by adding several layers of clean test material behind the specimen showing color transfer. This can be done without removing the sample from the card. With the mask in place, the specimen and the mask are then moved up or down in this column until the specimen most closely approximates the chroma of one of the chips in the column. In order to eliminate or minimize the shadow on the specimen through the holes on the Scale, the Scale should be held so that the specimen and the Scale are perpendicular to the light source.

5.3 Numerical grades are determined by comparison with the numbered rows on the Scale. The rating method given in the Scale is as follows:

Colorant Staining (Transfer)	Grade
No transfer	5
Transfer equal to Row 4.5	4.5
Transfer equal to Row 4	4
Transfer equal to Row 3.5	3.5
Transfer equal to Row 3	3
Transfer equal to Row 2.5	2.5
Transfer equal to Row 2	2
Transfer equal to Row 1.5	1.5
Transfer equal to Row 1	1

6. Evaluation of Results

6.1 Evaluations made with the AATCC 9-Step Chromatic Transference Scale should give essentially the same grades as evaluations made with the Gray Scale for Staining (AATCC EP2) (see 7.1)

6.2 The AATCC 9-Step Chromatic Transference Scale is believed to fulfill evaluation requirements when the results are analyzed on a statistical basis. It may facilitate evaluations of staining, especially when used by less experienced raters.

6.3 Reports on grades arrived at by the use of the AATCC 9-Step Chromatic

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Transference Scale must clearly state that this scale was used and not the Gray Scale for Staining.

7. Notes

7.1 Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

7.2 Any test specimen which has a change in contrast decidedly greater than a grade of 1 may be rated a 0.

7.3 The cleanliness and physical condition of the Scale is extremely important in obtaining consistent results. Inspect the Scale frequently for fingerprints and any other marks. If the marks are considered to interfere with the ratings, then the Scale should be replaced. The Scale can also be physically damaged through handling. Again, if the physical damage to the chips; e.g., chipped edges, loose or buckled chips; interferes with the rating process, replace the Scale. At least annually, the Scale should be measured on a spectrophotometer or colorimeter to ensure that the total color differences are within specification. Sphere geometry spectrophotometric measurements of the chips shall be taken with the specular component included. A 0°/45° (45°/0°) geometry is an acceptable alternate. The colorimetric data shall be calculated using the CIE 1964 10° observer data for llluminant D_{65}. Measure the colors on the Scale upon receipt and, using the initial color readings, total color difference for each chip should not exceed a ΔE_{CMC} value of 0.3. A CMC ratio of 2:1 should be used. Keep the Scale in its case when it is not in use.

7.4 See AATCC EP9, Evaluation Procedure for Visual Assessment of Color Difference of Textiles, for notes on choice of daylight simulator and illumination level.

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Evaluation Procedure for Visual Assessment of Color Difference of Textiles

1. Purpose and Scope

1.1 This evaluation procedure provides general principles and a procedure for determining and describing the color difference of test specimens by visual comparison with a standard. The scope is to standardize the procedure for visual color difference evaluations of textile materials.

2. Principle

2.1 The color of a specimen or specimens and a reference standard are compared and evaluated under specific lighting and viewing conditions. The observed color difference can also be compared with tolerance range specimens.

3. Terminology

3.1 **chroma,** n.—the proportion of spectrally pure color that expresses the degree of departure from the gray of the same lightness.

3.2 **hue,** n.—the attribute of visual perception according to which a color field is judged similar to one conventionally named yellow, red, blue, or green or a combination of an adjacent pair of these colors, considered to be arranged in a closed ring.

3.3 **lightness,** n.—the attribute of visual perception by which a non-self-luminous body is judged to reflect more or less light.

3.4 **metamerism,** n.—visual phenomenon according to which spectrally different materials have, for a given observer in a given illumination, matching color appearance. The fields may no longer match when viewed in a different light or by a different observer.

3.5 **reference standard,** n.—a material that defines the specific color to be matched, and may also be used to define other appearance properties, such as finish, texture and construction.

3.6 **tolerance range specimens,** n.—selected specimens which deviate from the reference standard in hue, lightness, chroma, or in combinations of the three, and impose a range of observable color differences around the reference standard for evaluation purposes.

4. Uses and Limitations

4.1 This evaluation procedure is used for the determination and description of the magnitude of color difference of textile materials against a standard color specimen.

4.2 When evaluating multicolored patterns, each color is evaluated separately.

4.3 To determine if illuminant metamerism is present, evaluate specimens under more than one illuminant.

4.4 This procedure shall be performed only by observers with normal color vision according to color discrimination test methods such as the Farnsworth Munsell 100 Hue Test (see 10.1) and pseudoisochromatic plate tests such as the Ishihara Color Vision Test (see 10.2).

5. Apparatus

5.1 Illumination conditions. The source under which the comparison is made shall be determined prior to evaluation. Sources simulating any of the following illuminants may be utilized (see 10.3 and 10.4):

Illuminant	Description	Color Temperature
D_{65}	Daylight	6500K ± 200
D_{75}	Daylight	7500K ± 200
A	Incandescent	2856K ± 200
CWF	Cool White Fluorescent	4150K ± 350

Note: Additional illuminants or changes in color temperature can be negotiated between the contractual parties.

5.1.1 Specimens may be viewed with UV illumination to test for the presence of Fluorescent Whitening Agents (FWAs).

5.1.2 The recommended illumination level at the specimen plane is in range of 1080-1340 lx (100-125 fc).

5.2 Viewing environment. The viewing environment consists of an illumination source (such as a lighting cabinet or luminaire) and a viewing surface located within the lighting cabinet or on a table in a designated visual evaluation area of a room and is limited in area to that which corresponds to the normal field of view of an observer performing normal visual evaluations. Within the viewing environment the specimens will be placed on a flat, uniform surface having no distortions or depressions, and having a color matching the range of grays from Munsell N5/ to N7/ (see 10.5). The color of the area surrounding the viewing surface, that is, the interior surfaces of the lighting cabinet or the walls surrounding the designated visual evaluation area of a room, shall be a matte gray matching to within +/1 N value the Munsell gray selected for the viewing surface (see Table I). The gray selected for the viewing surface and surrounding area may be negotiated between the contractual parties. The viewing environment shall contain no other items, and *shall be shielded from extraneous light*. Observers shall wear clothing of neutral color, and there shall be no reflection on the specimens from surrounding surfaces.

5.3 Viewing geometry.

5.3.1 Option A. The illumination source is horizontal. The specimen plane shall be at a 45 ± 5° angle to horizontal, resulting in illumination incident at a 45° angle relative to the specimen plane. The observer shall view at a 90° angle relative to the specimen. This option shall be used for either glossy or matte specimens.

5.3.2 Option B. The illumination source is horizontal. The specimen plane shall be horizontal, resulting in illumination incident at a 90° angle, relative to the specimen plane. The observer shall view at a 45 ± 5° angle relative to the specimen. This option shall be used for either glossy or matte specimens.

5.3.3 Option C. The specimen plane and illumination source shall be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 ± 5° angle relative to the horizontal. The observer shall view the specimens at a 90° angle, relative to the plane of the specimens. This option shall not be used for evaluations of glossy or non-matte specimens. The observer shall take care when evaluating specimens not to block the incident light from the source.

5.3.4 Examples (see Fig. 1). Either Option A or B is the preferred viewing position.

5.4 Instrumentation used for Assessing

Table I—Color of Background and Surround

Evaluation Category	Color of Background	Color of Surround	Maximum Munsell Chroma of Background and Surround
Critical	similar to standard	N5/ to N7/	0.2
General	N5/ to N7/	N5/ to N7/	0.3

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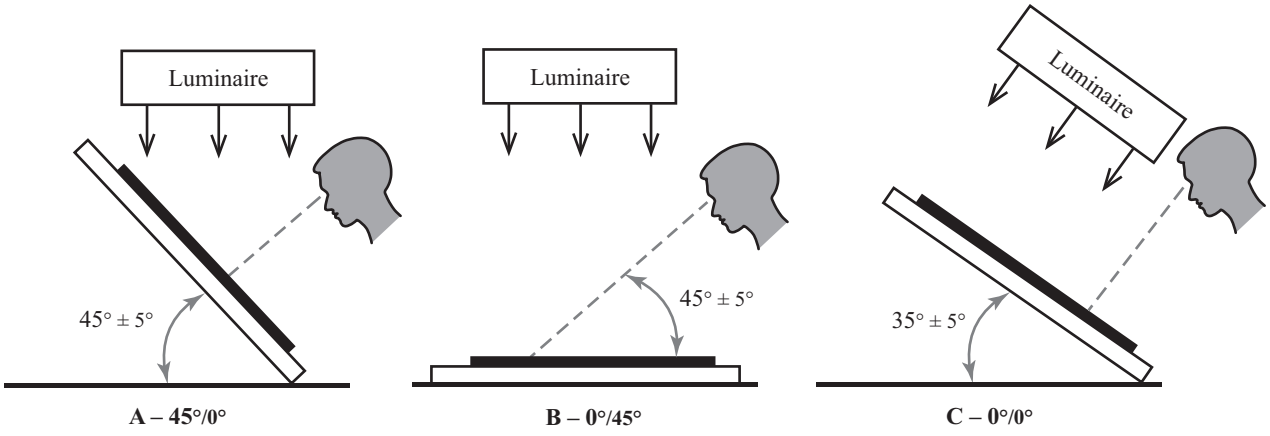


Fig. 1—Viewing positions for Options A (preferred), B (preferred) and C.

Light Intensity and Color Temperature. Equipment must be operated in accordance to manufacturer's specifications. Ensure that the instrumentation used for assessing light intensity and color temperature is within factory calibration specifications. Prior to measurement, the illuminant simulators should be powered on and allowed to stabilize. Generally this may take from 15-30 min. Place the instrument inside the illuminated viewing area. The instrument's sensor should be facing the incident light, placed on the viewing surface and in the center. Without obstructing the path between the incident light and the sensor, measure and record the colorimetric data and/or illuminance data.

6. Test Specimens

6.1 Prior to comparison, all color standards and test specimens shall be conditioned to avoid differences, which may be caused by temperature and humidity (see 10.7).

6.2 Recommended specimen sizes noted below are to be used only if the test method being used does not include specimen size requirements (see 10.6).

6.3 Dyed woven or knitted cloths. The test specimen shall be a representative rectangle of cloth, a minimum dimension of 100 mm in the lengthwise direction by 200 mm wide, whenever possible. Multicolored specimens shall be of size to incorporate all colors in the pattern. All surfaces shall be clean and free from stains or soiling. Remove torn edges, markings, or other marks from the specimen. Specimens shall be cut at least 150 mm from the selvage, and seam areas shall be avoided.

6.4 Narrow cloths, braids, tapes, webbings. The test specimen shall be a minimum of 200 mm in length and full width, as received.

6.5 Loose fiber. The test specimen shall be carded and made into a suitable size lap or pad.

6.6 Threads and yarns. The test specimen shall be in skein form, wound on a card to form a compact area of parallel yarns, or knitted to suitable size. Card windings shall be of sufficient thickness to conceal the card. The card shall not be bowed by tension of the wound yarns.

7. Procedures

7.1 The lighting conditions at the specimen plane shall be tested using instrumentation for assessing light intensity and color temperature to ensure compliance with the specification in 5.1 above. In addition, the user shall test the viewing area for illumination uniformity. The intensity of illumination shall not deviate more than 25% from the center to the viewing edges of the specimen.

7.1.1 The illuminant simulator shall be tested four times per year, or as recommended by the equipment manufacturer (see 10.4).

7.2 Different color difference evaluation procedures, as listed below under 7.2.1 and 7.2.2, may be required for different applications. Each of these procedures may be used with apparatus from Options A, B or C. Observers shall allow their vision to become adapted to the lighting conditions for at least 2 min prior to making an evaluation. This requirement shall apply again if the lighting conditions are changed; i.e., use of another source.

7.2.1 Tolerance range specimen procedure. The reference standard shall be placed on the specimen plane. The tolerance range specimens shall be placed around the edge of the reference standard, contacting it at its edge, with lengthwise directions parallel. The reference standard and specimen shall both face up unless otherwise specified. Any directional specimens shall be oriented in the same direction. Non-directional specimens shall be placed with their lengths in the same or opposite (180° rotation) orientation. The viewing distance shall be 700 ± 150 mm while performing the color difference evaluation.

7.2.2 Test specimen vs. reference standard procedure. The test specimen shall be placed on the specimen plane adjacent to the reference standard with lengthwise directions parallel and adjacent. The reference standard and specimen shall both face up unless otherwise specified. Any directional specimens shall be oriented in the same direction. Non-directional specimens shall be placed with lengthwise directions in the same or opposite (180° rotation) orientation. The observer shall stand in front of, and perpendicular to the specimen plane, taking care to avoid blocking the light. The viewing distance shall be 700 ± 150 mm while performing the evaluation.

7.3 Multiple specimen procedure. Evaluations should be made as quickly as possible, since prolonged viewing affects the observers' ability to evaluate.

8. Evaluation

8.1 Determine the color difference in terms of lightness, chroma and hue relative to the standard.

8.2 Evaluation options:

8.2.1 Tolerance range evaluation. The observer shall determine if the color difference between the reference standard and a test specimen is within the tolerance range, or is greater than the tolerance range.

8.2.1.1 A description of the color difference between the reference standard and a test specimen shall include terms as designated in Table II.

8.2.2 Magnitude evaluation. The observer shall determine the magnitude of color difference between a test specimen and reference standard by assigning val-

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Table II—Color Difference Direction Descriptors

Lightness	Lighter, Darker
Chroma	More (or higher) chroma and Less (or lower) chroma
Hue	Redder, Greener, Yellower, Bluer

Table III—Color Difference Magnitude Descriptors

Verbal Description	Gray Scale for Color Change Corresponding Grade
Equal	5
Slight	4.5
Noticeable	4
Considerable	3
Much	2
Off Shade	1

ues from Table III. For evaluating color changes resulting from colorfastness tests, see AATCC EP1, Evaluation Procedure for Gray Scale for Color Change. See 10.8 for specific application within the dyestuff industry. See 10.9 for additional description of use of magnitude descriptors.

9. Report

9.1 The color difference between each specimen and reference standard shall be reported, according to the test method used. A within-tolerance evaluation which approaches the limits of the tolerance range should be reported.

9.2 Any deviations from the specified apparatus, procedure, and/or evaluation shall be documented.

9.3 As a minimum, the following information should be included when referring to data acquired using this evaluation procedure:

9.3.1 Specimen identification and date of evaluation,

9.3.2 Illuminant simulator under which evaluation was performed,

9.3.3 Geometry option used (A, B or C),

9.3.4 Specimen type (flat goods, wound card, carpet, etc.),

9.3.5 Specimen condition (number of thicknesses, pile lay, etc.).

9.3.6 Evaluation option and result:

9.3.6.1 Color difference direction descriptor,

9.3.6.2 Color difference magnitude descriptor.

9.4 Operator and site where evaluation performed.

10. Notes

10.1 The Farnsworth Munsell 100 Hue Test is available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549.8933; e-mail: ordering@aatcc.org; web site: www.aatcc.org.

10.2 The Ishihara Color Vision Test may be obtained from a variety of sources.

10.3 For visual assessments to be made under daylight illumination, it is recommended that the correlated color temperature be calibrated to 6500 K (± 200 K) and that the quality of daylight simulation be preferably rated as BC (CIELAB) or better, as determined by the method specified in CIE Publication SO12E, Standard method of assessing the specified quality of daylight simulators for visual appraisal and measurement of colour, available from TechStreet, 3916 Ranchero Dr., Ann Arbor MI 48108; tel: +1.800.699.9277; web site: www.techstreet.com. The above rating is the same as that specified in ASTM D1729-96, Standard Practice for Visual Appraisal of Colors and Color Differences of Diffusely-Illuminated Opaque Materials.

10.4 Filters and lamps shall be properly maintained and cleaned periodically following manufacturers' recommendations.

10.5 Samples of Munsell neutral color N5/ to N7/ may be obtained from Munsell Division of X-Rite Inc., 4300 44th St. SE, Grand Rapids MI 49512; tel: +1.800.248.9748; web site: www.xrite.com.

10.6 See AATCC EP6, Evaluation Procedure for Instrumental Color Measurement, Section 2.6 "Specimen Preparation" for additional recommendations for various specimen types, as included in the AATCC *Manual of International Test Methods and Procedures*.

10.7 See AATCC EP6, Appendix A1.2-Moisture Content, for details regarding specimen conditioning.

10.8 Specific application for the dyestuff industry: During formulation, synthesis and standardization, the reference standard and test specimen shall be dyed in steps; i.e., reference standard at 95%, 100% and 105%, and the test specimen at 95% and 100%, to facilitate the determination of the test specimen strength compared to the reference standard. This evaluation is performed by placing a test specimen between two reference standards of varying strengths to determine one reference standard and one test specimen which appear equal or near-equal in tinctorial strength. Once this is done, the selected pair is evaluated as in 8.2 for the color difference direction and magnitude descriptors with respect to chromaticity and hue. Lightness magnitude and direction are often omitted in this case, due to the determination of strength.

10.9 Table III includes whole-step and half-step designations. Some industries require further dissemination, with application-specific language to describe the intermediate steps. For example, the term "trace" is widely used for an imagined-step, between 5 and 4.5, within the automotive industry.

11. History

11.1 Revised in 2021 to update steps in table three and to align with the AATCC style guide.

11.2 Revised 2002, 2010, 2011, 2017; reaffirmed 2007; editorially revised 2008, 2012, 2019.

11.3 Developed in 1999 by AATCC Committee RA36.

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Evaluation Procedure for Multifiber Adjacent Fabrics

1. Purpose and Scope

1.1 This procedure is intended to permit the qualification of candidate multifiber test fabrics, by comparing the staining of their components with that of a control multifiber fabric, under controlled conditions.

2. Principle

2.1 The staining properties of individual components of candidate multifiber test fabrics are evaluated by comparing the shades obtained on them and on the components of a control fabric, using a soap solution and each of four different dye baths. The bath containing the yellow disperse dye, Terasil Yellow 2GE 200%, stains the secondary acetate and polyamide components (of the control fabrics and fabrics under test); the bath containing the orange acid dye, Irgalan Orange RL-KWL 250%, stains the polyamide components; the bath containing the blue direct dye, Solophenyl Blue GL 250%, stains the cotton components; and, the bath containing the navy disperse dye, Terasil Navy Blue BGLN 200%, stains the polyester and polyamide components. Evaluations of the stained components are conducted by colorimetric measurements to eliminate the subjectivity of visual evaluations.

3. Safety Precautions

NOTE: These safety procedures are for information purposes only. The precautions are ancillary to the testing procedures and not intended to be all inclusive. It is the user's responsibility to use safe and proper techniques in handing materials in this procedure. Manufacturers MUST be consulted for specific details such as material safety data sheets and other manufacturer's recommendations. All OSHA standards and rules must also be consulted and followed.

3.1 Good laboratory practices should be followed. Wear safety glasses in all laboratory areas.

3.2 All chemicals should be handled with care.

3.3 Manufacturer's safety recommendations should be followed when operating laboratory testing equipment.

4. Uses and Limitations

4.1 The procedure was tested on multifiber test fabrics containing 15 mm wide fiber stripes. Multifiber test fabrics containing narrower fiber stripes may also be evaluated as long as the aperture on the spectrophotometer used will accommodate the smaller size.

4.2 This procedure was developed using Staining Scale Rating (SSR) values as calculated in ISO 105-A04 (see 15.1). These calculated SSR values are used to assign the reported Gray Scale for Staining (see 15.2). This reported value is based upon a range that is broader than would be found by using the calculated SSR values between 4 and 3 (see 15.2).

5. Apparatus and Materials (see 15.3)

5.1 Accelerated laundering machine.

5.1.1 A laundering machine for rotating closed canisters in a thermostatically controlled water bath at 40 ± 2 rpm.

5.1.2 Stainless steel lever lock canisters Type 1, 500 mL (1 pt), 75 × 125 mm (3.0 × 5.0 in.)

5.1.3 Stainless steel lever lock canisters Type 2, 1200 mL, 90 × 200 mm (3.5 × 8.0 in.).

5.1.4 Adapter plates for holding canisters (see 5.1.3) on laundering machine shaft.

5.1.5 Teflon fluorocarbon gaskets (see 11.5 and 15.4).

5.2 Spectrophotometer geometry d/0° specular included, small area view; calculation

$D_{65}/10°$; tolerance ΔE_{cmc} 0.5

5.3 pH meter accurate to ± 0.01.

5.4 Balance with a weighing accuracy of ± 0.001g.

5.5 Control Multifiber Fabric (see 15.6)

6. Reagents

6.1 Soap, specified in ISO 105-C02 (see 15.1 and 15.3).

6.2 Terasil Yellow 2GE 200% (see 15.5).

6.3 Irgalan Orange RL-KWL 250% (see 15.5).

6.4 Solophenyl Blue GL 250% (see 15.5).

6.5 Terasil Navy Blue BGLN 200% (see 15.5).

6.6 Distilled or Deionized water.

7. Specimens

7.1 Multifiber fabric, to be used as a control (see 15.6); Weight: (3.00 ± 0.1) g.

7.2 Multifiber fabric under test; Weight: (3.00 ± 0.1) g; approximately 100 × 100 mm.

7.3 Mark the specimens to distinguish between the control specimens and the specimens under test.

8. Preparation of Reagents

8.1 Soap Solution.

8.1.1 Dissolve 50 g soap in 1 L water at 45°C.

8.2 Terasil Yellow 2GE 200% Dispersion.

8.2.1 Dispersion: 2.45 g dye into 1000 mL (see 8.6).

8.2.2 Dilution: 20 mL dispersion into 1000 mL

8.2.3 Test Solution for 0.00049 g/L Terasil Yellow 2GE 200%: 10 mL from first dilution and 100 mL soap solution into 1000 mL (see 8.7).

8.3 Irgalan Orange RL-KWL 250% Solution.

8.3.1 Solution: 1.10 g dye into 1000 mL (see 8.6).

8.3.2 Test Solution for 0.011 g/L Irgalan Orange RL-KWL 250%: 10 mL solution and 100 mL soap solution into 1000 mL (see 8.7).

8.4 Solophenyl Blue GL 250% Solution.

8.4.1 Solution: 0.775 g/L dye into 1000 mL (see 8.6).

8.4.2 Dilution: 20 mL into 1000 mL.

8.4.3 Test Solution for 0.000155 g/L Solophenyl Blue GL 250%: 10 mL from first dilution and 100 mL soap solution into 1000 mL (see 8.7).

8.5 Terasil Navy Blue BGLN 200% Dispersion.

8.5.1 Dispersion: 2.4 g dye into 1000 mL (see 8.6).

8.5.2 Test Solution for 0.024 g/L Terasil Navy Blue BGLN 200%: 10 mL dispersion and 100 mL soap solution into 1000 mL (see 8.7).

8.6 Original Dispersion or Solution. Disperse or dissolve a specified quantity of dye in a 200 mL beaker. Using a funnel, transfer it into a 1 L volumetric flask. Using a spray bottle, spray some water around the sides and to the bottom of the beaker and add it to the flask. Repeat until all the dye is transferred. Fill the flask about 3/4 full with water, close it with a stopper, tip it upside down several times to ensure complete dispersion or solution and fill the flask to the meniscus. Transfer the liquid into a 1 L Erlenmeyer flask and close it with a stopper and mark 'Original Dye Solution.'

8.7 Test Solution. To prepare the test solution for the orange and navy dyes, shake the Erlenmeyer flask containing

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the 'Original Dye Solution' to ensure a homogeneous dispersion or solution. Using a volumetric pipette, transfer the specified aliquot into a volumetric flask. To avoid excess foaming slowly add 100 mL soap solution and water to fill the flask to the meniscus. Transfer the liquid to a 1 L Erlenmeyer flask, close and mark the flask with 'Test Solution.'

8.7.1 The yellow and blue dyes require an additional dilution prior to making the 'Test Solution.' Shake the Erlenmeyer flask containing the 'Original Dye Solution' to ensure a homogeneous dispersion or solution and transfer the specified aliquot into a 1 L volumetric flask. Dilute with water to the meniscus. Transfer the liquid to a 1 L Erlenmeyer flask, close and mark the flask 'Dilution.' To prepare the 'Test Solution' transfer the specified aliquot into a 1 L volumetric flask. To avoid excess foaming, slowly add 100 mL soap solution and water to fill the flask to the meniscus. Transfer the liquid to a 1 L Erlenmeyer flask, close and mark the flask with 'Test Solution.'

9. Preparation of Canisters

9.1 Rinse canisters to be used with acetone and then with water.

9.2 Immerse lids and seals in a small beaker with acetone for 2 min, rinse with water.

10. Pre-Wetting Specimens

10.1 Wet out the specimens in 1 L water at 50°C (122°F) for 10 min.

10.2 Rinse the specimens in cold water.

10.3 Squeeze the specimens between blotting paper, or with a wringer to approximately 80% wet pickup.

11. Procedure

11.1 It is essential to attain a level staining of the respective fibers. Evaluations are conducted by colorimetric measurements to eliminate objectivity from visual evaluations. To ensure reproducibility it is necessary to adhere closely to the testing protocol.

11.2 In each of the four canisters, add 300 mL of one of the test solutions.

11.3 Check the pH of the test solution. This should be 9.5-10.

11.4 Enter a control specimen and a specimen under test in a canister and immediately stir for 5-10 s.

11.5 Clamp the covers on the canisters with a Teflon gasket inserted between the neoprene gasket and the top of each canister to prevent contamination.

11.6 Repeat this for the other three canisters

11.7 Load the canisters as rapidly as possible into the accelerated laundering machine.

11.7.1 If an operator is conducting the tests without an assistant, it is recommended that only two treatment baths (canisters) be prepared and run at one time.

11.8 Start the test at 20 ± 2°C (68 ± 4°F), raise the temperature to 50 ± 2°C (122 ± 4°F) in 20 min and continue at this temperature for 45 min.

11.9 Stop the machine, remove the canisters and place the specimens separately by color in 500 mL beakers. Rinse the specimens with cold water for 2 min. Repeat the rinse with fresh water for an additional 2 min.

11.10 Place the specimens flat on blotting paper and dry in an oven at 50-60°C and condition at 65 ± 5% RH and 21 ± 2°C (70 ± 4°F) for a minimum of 4 h prior to evaluation.

12. Evaluation

12.1 Fold the specimens to form two layers of the strip to be measured. Place the specimen on the instrument with the weave at a 45° angle. Take two measurements, each at a different place, rotate the specimen 90° and take two additional measurements, each at a different place.

12.2 Obtain the CIELAB L*, a*, b*, h_{ab} values for the following components:

12.2.1 Secondary acetate for the specimens treated with the yellow dye.

12.2.2 Polyamide for the specimens treated with the orange dye.

12.2.3 Cotton for the specimens treated with the blue dye.

12.2.4 Polyester and polyamide for the specimens treated with the navy dye.

12.2.5 The secondary acetate, polyamide, polyester and cotton for the untreated control multifiber fabric and for the untreated multifiber fabric under test (the values should be practically identical).

13. Calculations

13.1 The ΔE^*_{ab} values are calculated, using the formulae in Clause 4.2.1 of AATCC TM173, Test Method for Calculation of Small Color Differences for Acceptability. The staining scale ratings (SSR values) are calculated (see Table I) using the formulae in Clauses 6.4 and 6.5 in ISO 105-A04.

13.2 The staining scale ratings for the control multifiber fabric and of the multifiber fabric under test are based on the respective CIELAB values from the appropriate fibers of the tested versus the original specimens.

13.3 If the software does not give the SSR values, the calculations may be made with a hand-held calculator. However, it is more efficient to enter the formulas in a spreadsheet. (Table I)

13.4 Enter the formulas into the template in cells L3, M3, N3, and O3, exactly as shown in Table II, starting with the equal sign.

13.5 Copy the CIELAB values from the instrument's software into the appropriate cells of the template.

13.6 The adjacent multifiber fabric under test is acceptable for its staining properties when the SSR values are less than 4.00 and greater than 3.00 (see 15.2).

14. Report

14.1 The report should contain, as a minimum, the following information: The Stain Scale Rating [(SSR) value(s)]

Table I. Spreadsheet Template for the Calculation of Staining Scale Rating

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1			MF Type	Specimen Under Test				Untreated Specimens				Calculated Values			
2	Dyes	Fibers		L^*_S	a^*_S	b^*_S	h^*_S	L^*_R	a^*_R	b^*_R	h^*_R	DL*	DE*	DE_{GS}	SSR
3	Yellow	DiA	Control												
4			Test												
5	Orange	PA	Control												
6			Test												
7	Blue	CO	Control												
8			Test												
9	Navy	PES	Control												
10			Test												
11		PA	Control												
12			Test												

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Table II. Staining Scale Ratings Formulae

Cell L3	DL*	=D3-H3
Cell M3	DE*	=SQRT(L3^2+(E3-I3)^2+(F3-J3)^2)
Cell N3	DE_{GS}	=M3-0.4*SQRT(M3^2-L3^2)
Cell O3	SSR	=IF(6.1-1.45*LN(N3)<4,6.1-1.45*LN(3),5-0.23*N3)

for both the control specimen as well as the specimen under evaluation for each dyestuff.

14.2 The number and date of this procedure; i.e., AATCC EP#.

14.3 All details necessary for the identification of the sample(s) tested.

15. Notes

15.1 ISO 105-C02 Colour fastness to Washing: Test 2 and ISO 105-A04 Method for the Instrumental Assessment of the degree of staining of adjacent fabrics, may be obtained from ANSI, 11 West 42nd St., New York NY 10036; tel: +1.212.302.1286; fax: +1.212.398. 0023; web site: www.ansi.org; or ISO web site: www.iso.org.

15.2 Specifying SSR values allows for a tighter tolerance for acceptability than would be possible with conventional Gray Scales ratings.

15.3 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

15.4 Teflon is a registered trademark of the DuPont Co., Wilmington DE 19898.

15.5 Although the specific dyes listed are not in current commercial production, small quantities may be requested by those laboratories interested in performing EP10. Be aware there are limited quantities of these dyes and once they are depleted no dyes will be available for use. Available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919.549.8141; fax: +1.919.549. 8933; e-mail: ordering@aatcc. org; web site: www.aatcc.org.

15.6 A fabric known to meet the staining specifications.

16. History

16.1 Editorially revised 2023 to clarify the Notes section and to move the History section.

16.2 Editorially revised 2019; revised 2018; reaffirmed 2017; editorially revised 2016; 2013; reaffirmed 2012; editorially revised 2008; reaffirmed 2007, 2006.

16.3 Developed in 2005 by AATCC Committee RA59.

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Evaluation Procedure for UV Energy of Optically Brightened Textiles: Spectrophotometer Calibration

1. Scope

1.1 This evaluation procedure describes the use of the Textile UV Calibration Standard (TUVCS) for inter-instrument calibration of UV content in the light source of spectrophotometers for the purpose of measuring white or light to medium colored textiles that have been optically brightened with a fluorescent whitening agent. UV absorption properties of fluorescent whitening agents (FWAs) used for textiles differ from those of FWAs used for plastics or other non-textile materials, necessitating the use of a different reference standard than those typically used for these other materials. This procedure is based on a semi-annually replaced textile calibration standard that facilitates accurate UV content adjustment of the light source of spectrophotometers for use with textile materials independent of instrument geometry and the methodology used for adjusting the amount of UV energy in the instrument.

2. Principle

2.1 The amount of UV energy in a spectrophotometer light source is standardized by adjusting it (mechanically or by computation) until the calibrated value of the light source is in agreement with the CIE Whiteness Index (CIE WI) of the TUVCS (see 7.1).

3. Terminology

3.1 **calibration procedure,** n.—a method for adjusting certain parameters in an instrument so that different instruments from the same or different manufacturers produce the same results in a test for a given property.

3.2 **CIE Whiteness Index (CIE WI),** n.—an index predicting the apparent degree of whiteness of a material based on tristimulus values determined according to one of the CIE standard methods, proposed by the Commission Internationale de l'Éclairage (CIE).

3.3 **fluorescent whitening agent (FWA),** n.—colorant that absorbs near ultraviolet (UV) radiation and re-emits visible (violet-blue) radiation. This causes a yellowish material to which it has been applied to appear whiter (ASTM E284).

4. Description of the Textile UV Calibration Standard

4.1 The TUVCS consists of a set of 4 layers of combed cotton broadcloth that has been bleached, mercerized and treated with a specific formulation of optical brighteners. It measures 80 × 80 ± 5 mm and is attached to an opaque white, non-optically brightened blotting paper by staples on one edge. This facilitates the handling of the TUVCS without touching the portion that is to be used for measurement. It has been found that temperature and humidity changes of normal instrument operating ranges do not significantly alter the determined WI values. These changes are normally not in excess of 0.1 WI. However, exposure to light can significantly deteriorate the FWA and cause changes in the WI. Therefore, it is important that the TUVCS be kept in its UV protective bag when not being used (see 7.4).

5. Use of the TUVCS Calibration Standard

5.1 The described procedure is applicable to large area view measurements (the recommended condition) for sphere or angular measuring geometry. Measurement on a given instrument in other conditions or configurations requires separate calibrations for the respective condition/configuration.

5.2 Any UV calibration standard must be stored in a container protecting it from UV exposure. The TUVCS should be stored in the UV protective brown plastic bag it is supplied in to keep it from becoming dirty. Also, take care not to touch the TUVCS when handling it to prevent soiling.

5.3 For multiple instruments in one location, a single TUVCS can be used. Each instrument must be separately calibrated with the TUVCS. CAUTION: This should be limited to no more than three instruments, as excessive use of the TUVCS may shorten its useful life to less than the required six months.

6. Procedure

6.1 Set the conditions of the spectrophotometer color measurement software to CIE standard illuminant D_{65} and CIE 10º (1964) standard observer (see 7.2).

6.2 Set the instrument conditions to those shown in Table I. Calibrate the spectrophotometer according to the manufacturer's procedure for the measuring geometry used.

Table I—Instrument Conditions

	Sphere	45/0 (0/45)
Area of view	Large	Large
UV Filter Mode	Calibrated	Calibrated
Specular component	Included	Excluded

6.3 In the spectrophotometer color measurement software, go to the UV calibration procedure. Enter the CIE WI value printed on the label of the TUVCS calibration standard fabric. Ensure that the software is set to accept CIE WI and not another kind of whiteness index.

6.4 Mount the TUVCS on the instrument's sample mount, centered over the instrument's port. Back the TUVCS with its attached white backing card. The white backing card used in the TUVCS is sufficiently opaque that the color of the instrument's sample clamping mechanism is of no consequence (see Fig. 1).



Fig. 1—Position of the Sample at the Instrument.

6.5 Perform the instrument's standard automatic or manual method for calibrating UV energy. The value obtained should be within ± 0.5 CIE WI units of the value entered in 6.3 above.

6.6 Remove the textile standard and perform a normal black and white reference standard instrument calibration in the UV calibrated mode.

6.7 For the next six (6) months use the textile standard for bi-weekly UV calibrations (see 7.3) and in any other optical modes you may choose to use (see 5.1).

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7. Notes

7.1 Textile UV Calibration Standards are obtainable from AATCC at P.O. Box 12215, Research Triangle Park NC 27709; +1.919. 549.8141 or online at www.aatcc.org. Because of the limited stability of textile standard materials to light degradation, new standards must be ordered every six months. When not in use, the standards must be stored in the supplied UV protective bag.

7.2 Colorimetric calculations are performed using ASTM E308 *Standard Practice for Computing the Colors of Objects by Using the CIE System*, illuminant/observer data values. The instruments must be performing to original manufacturer specifications for non-optically brightened materials before this procedure is performed. Any deviation from these specifications will be reflected in the whiteness measurements and will not be corrected by this procedure.

7.3 Illumination changes typically do not occur abruptly in instruments. Unless extenuating circumstances warrant a higher frequency, UV calibrations should be performed once every other week. If any maintenance is performed on an instrument, then a UV calibration should always be performed prior to use.

7.4 The supplied UV protective bag provides the TUVCS protection from environmental or chemical color change.

8. History

8.1 Revised in 2021 to update reference to ASTM E308 table references, to add history section per the AATCC style guide. Editorially revised the AATCC ordering process and clarify stability and storage.

8.2 Editorially revised in 2019 (with title change); revised in 2016; editorially revised in 2012 and 2010; reaffirmed in 2008.

8.3 Developed in 2007 by AATCC Committee RA36.

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Evaluation Procedure for Instrumental Assessment of Degree of Staining

Developed in 2010 by AATCC Committee RA36; editorially revised and reaffirmed 2012, 2017; editorially revised 2019. Technically equivalent to ISO 105-A04.

1. Scope

1.1 This evaluation procedure is intended as an alternative to the visual method of AATCC EP2, Gray Scale for Staining (see 8.1), for assessing the degree of staining on a fabric test specimen used in colorfastness tests (see 8.2). This procedure is applicable to any test method referring to the Gray Scale for Staining.

2. Principle

2.1 An undyed fabric specimen which has been subjected to a fastness test in contact with the dyed fabric being tested and an equivalent undyed reference fabric are measured for comparison. The undyed reference fabric has been independently subjected to the same fastness test in the absence of the dyed fabric. The comparison is assessed in CIELAB units, and numerically converted to a staining scale grade. The numerical conversion uses an equation specifically for the determination of degree of staining.

3. Terminology

3.1 **colorant staining,** n.—the unintended pickup of colorant by a substrate due to (1) exposure to a colored or contaminated liquid medium, or (2) direct contact with dyed or pigmented material, from which colorant transfers by sublimation or mechanical actions (as in crocking).

3.2 **colorfastness, n.**—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the materials to any environment that might be encountered during the processing, testing, storage or use of the material.

3.3 **gray scale for staining,** n.—a scale consisting of pairs of standard gray chips representing progressive differences in color or contrast corresponding to numerical colorfastness grades.

4. Apparatus

4.1 Spectrophotometer or colorimeter meeting any one of the geometric definitions described in CIE publication 15.2, subclause 1.4 (see 8.3).

5. Procedure

5.1 Mount the specimen fabric and the reference fabric, on non-optically-brightened white card stock.

5.2 Measure the reference fabric, according to AATCC EP6, Evaluation Procedure for Instrumental Color Measurement. It is recommended that the arithmetic mean of at least two measurements be utilized. If the instrument permits different viewing conditions to be used, the preferred method is to include the specular component.

5.3 Measure the specimen fabric by the same procedure used in 5.2. If the staining on the specimen fabric is uneven, several measurements shall be made and the arithmetic mean value employed in the calculations (see AATCC EP6, Section A1.8).

6. Calculations

6.1 Calculate (see 8.4) the CIELAB color difference ΔE^*_{ab} and the magnitude of the lightness difference ΔL^* between the adjacent fabrics, as described in 6.1 and 6.2, to two decimal places. Calculations shall be performed using the CIE 1964 10° supplemental standard observer and illuminant D_{65}, with the 1931 2° observer and illuminant C being a permitted alternative.

6.2 Calculate, to two decimal places, the Gray Scale difference ΔE_{GS} from ΔE^*_{ab} and ΔL^* using the following equation:

$$\Delta E_{GS} = \Delta E^*_{ab} - 0.4 * [(\Delta E^*)^2 - (\Delta L^*)^2]^{1/2}$$

6.3 Calculate, to two decimal places, the staining-scale grade (SSG) using the following equation:

$$SSG = 6.1 - 1.45 * \ln(\Delta E_{GS})$$

If SSG is greater than 4, recalculate using the following equation:

$$SSG = 5 - 0.23 * \Delta E_{GS}$$

6.4 Determine from Table I the Gray Scale for Staining Grade (GSs) to be reported.

Table I—Gray Scale for Staining Step Values

Calculated SSG	Reported GSs
5.00 to 4.75	5
4.74 to 4.25	4-5
4.24 to 3.75	4
3.74 to 3.25	3-4
3.24 to 2.75	3
2.74 to 2.25	2-3
2.24 to 1.75	2
1.74 to 1.25	1-2
Less than 1.25	1

7. Report

7.1 The report shall contain, as a minimum, the following information:

7.1.1 The Gray Scale for Staining Grade(s) [GSs value(s) from Table 1]. Follow the directions and report as directed in the appropriate test method.

7.1.2 The number and date of this procedure; i.e., AATCC EP12-20XX;

7.1.3 All details necessary for the identification of the sample(s) tested;

7.1.4 The configuration of the instrument;

7.1.5 Identification of the spectrophotometer or colorimeter used;

7.1.6 Whether illuminant D_{65} or illuminant C was used;

7.1.7 Whether the 1964 10° supplemental standard observer or the 1931 2° observer was used.

8. Notes and References

8.1 AATCC *Manual of International Test Methods and Procedures*, American Association of Textile Chemists and Colorists. AATCC EP2 available from AATCC, P.O. Box 12215, Research Triangle Park NC 27709; tel: +1.919. 549.8141; fax: +1.919.549. 8933; e-mail: ordering@aatcc. org.

8.2 Jaeckel, S. M., The Variability of Grey-Scale Assessment and its Contribution to the Variability of a Test Method, *Journal of the Society of Dyers and Colourists*, Vol. 96, 1980, pp540-544.

8.3 Commission Internationale de l'Éclairage, Publication CIE No. 15:2004 Colorimetry, 3rd ed., Central Bureau of the CIE, Vienna, 2004. To obtain copies, go to www. cie-usnc.org.

8.4 AATCC TM173, Test Method for Calculation of Small Color Differences for Acceptability, and reference 8.3.

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Evaluation Procedure for Electrical Resistance of Electronically Integrated Textiles

Foreword

This evaluation procedure was initially developed as a tool for iterative design and development purposes, using a 2-point probe system. With a 4-point system, it is also suitable for quality control applications. The procedure may be used for resistance measurement of any e-textile material or product. As e-textile technology evolves, more specialized methods, or options within this method, may be developed.

1. Purpose and Scope

1.1 This evaluation procedure determines the electrical resistance of e-textiles. E-textile fabrics or end products with incorporated conductive paths/traces may be evaluated. This method is not for evaluation of yarns or fibers.

1.2 Electrical resistance may be measured as received, or after treatment such as stretch or laundering, etc. The change in resistance due to treatment may also be calculated. For additional guidance in measurement before and after various exposures, refer to AATCC TM210.

2. Principle

2.1 Electrical resistance of specimens is measured using a digital multimeter and surface probes. Option 1 uses a 4-point system. Option 2 uses a 2-point system. Change in resistance is calculated from measurements before and after treatment, if desired.

3. Referenced Documents

3.1 AATCC TM76, Test Method for Electrical Surface Resistivity of Fabrics (see 15.1).

3.2 AATCC TM210, Test Method for Electrical Resistance Before and After Various Exposure Conditions (see 15.1).

3.3 ASTM D1776, Standard Practice for Conditioning and Testing Textiles.

3.4 ASTM E1402, Standard Guide for Sampling Design.

4. Terminology

4.1 **conductive element, n.**—*in e-textiles*, a fiber, fabric, ink, film or other material that is permanently integrated into a textile that will support the flow of electrons, generally with electrical resistance of less than 500 kΩ.

4.2 **electrical resistance, n.**—a conductive element's response to limit the flow of electrical current in response to an applied voltage or electric field.

4.3 **electronically-integrated textile (e-textile),** n.—a fabric or textile product with permanently-integrated electrical or electronic circuits, or parts of electrical or electronic circuits.

5. Safety Precautions

5.1 These safety precautions are for information purposes only. The precautions are ancillary to the testing procedures and are not intended to be all inclusive.

5.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this test method.

5.3 Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

6. Limitations

6.1 Each conductive path to be measured must be accessible, at least at the 2 measurement points. Connectors may be used to provide accessibility. The measured specimen should mimic the intended design of the end product, if known.

6.2 Refer to multimeter or ohmmeter user's manual for applicable range. Test low-resistance samples (< 1 Ω) with a 4-point system to eliminate lead resistance.

6.3 Refer to AATCC TM76 for samples having a resistance higher than 10^5 Ω.

7. Apparatus and Materials (see 15.2)

7.1 Conditioning facilities and racks with pull-out, perforated shelves for laying specimens flat (see 15.2).

7.2 Flat workbench.

7.3 Indelible ink marking pen (see 15.1) or needle and sewing thread, for making benchmarks.

7.4 Multimeter or ohmmeter.

7.4.1 For Option 1 (4-point measurement), digital multimeter or ohmmeter, stable current, 6.5-digit resolution.

7.4.2 For Option 2 (2-point measurement), digital multimeter, also known as DMM or volt-ohm-meter, VOM. Resolution to 0.1 Ω and 0.9% accuracy.

7.5 Probes, suitable for test material (see 18.3 and Fig. 1).

7.5.1 For magnetic probes, metal plate, ferromagnetic, at least as large as specimen.

7.6 Tape or rule marked in millimeters.

8. Verification

8.1 Verify accuracy of the multimeter or ohmmeter according to manufacturer's instructions. Readings may be verified by measuring the resistance of one or more resistors in the range of specimen resistance.

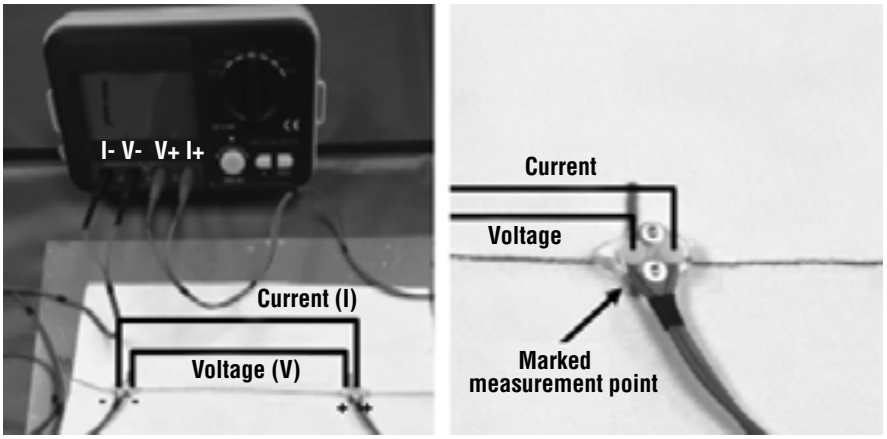


Fig. 1—Ohmmeter and probes for 4-point resistance measurements.

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9. Sampling

9.1 Test results are valid only when the samples are statistically representative (see ASTM E1402).

9.2 Sampling must be random. Every unit of product must have a mathematically equal chance of becoming a sample; and every portion of each sample must be equally likely to become a test specimen.

9.3 All specimens must be alike within the variations due to pure chance. There must be no differences within samples assignable to known causes.

10. Specimens

10.1 Each conductive path (or section of a path) is considered a specimen and should mimic the intended design of the end product. Measure at least 3 specimens.

10.1.1 Select specimens that can be arranged to lie flat on the measurement surface with no tension and no risk of a short circuit.

10.1.2 For convenience, straight specimens, 150-300 mm in length, are recommended for development testing.

10.1.3 Specimens from end products will depend upon product design. Complete conductive pathway should be tested.

10.2 Select identical specimens from at least 3 swatches or end products.

10.2.1 Size and shape of swatches or end products is not critical to this evaluation procedure.

10.2.2 If measuring resistance after treatment (stretch, laundering, etc.), refer to the appropriate treatment standard or AATCC TM210 for swatch dimensions.

10.3 Handle specimens with care to avoid contamination, in particular, salts or moisture from human hands, which may affect resistance measurements.

11. Conditioning

11.1 Prior to marking, condition test specimens as directed in ASTM D1776. (Use conditions indicated in Table 1 for Textiles, general. Estimate conditioning time as indicated in Table 2 for the appropriate fiber content.) Lay each swatch or product, separately, on a screen or perforated shelf of a conditioning/drying rack.

12. Preparation

12.1 Mark the ends of each specimen. If sewing thread is used, ensure that needle holes do not interrupt the circuit.

12.1.1 If the conductive path to be measured connects to an electronic system, the mark should be made at the point where the path in the textile connects to the electronic system.

12.2 If fraying of swatches is expected in treatment, edges may be pinked or serged.

13. Procedure

13.1 For magnetic probes, place the metal plate on a flat work surface.

13.2 Arrange the first specimen on flat, non-conductive surface. For magnetic probes, place specimen directly over the metal plate. Smooth the specimen to eliminate as many folds and wrinkles as possible, but do not stretch.

13.3 Set multimeter to resistance measurement mode. Connect probes to the meter. Use a 4-point system to eliminate lead resistance for all measurements of less than 1 Ω. A 4-point system may also be used for resistance above 1 Ω, but is not required as lead resistance will have less impact on the meter reading. Note the probes or connection system in the report. Do not compare measurements taken using different probes or connection systems.

13.4 Option 1: *4-point measurement*.

13.4.1 Place the positive and negative probes of the meter with voltage leads at the marked points. Current leads should be along the same conductive path, outside the voltage leads, but as close to the marked points as possible (see Fig. 1). Ensure consistent pressure throughout the test; weighted or magnetic probes may be used for this purpose.

13.5 Option 2: *2-point measurement*.

13.5.1 Place the positive and negative probes of the meter at the marked points. Ensure consistent pressure throughout the test; weighted or magnetic probes may be used for this purpose.

13.6 Record the resistance as soon as the reading stabilizes for a period of 3 s. Take 1 resistance measurement per specimen.

13.7 Measure and record distance between marked points, to the nearest mm.

13.8 Repeat resistance and distance measurements for the remaining specimens.

13.8.1 If only a single specimen is available, measure the resistance of the specimen 3 times. Note this modification in the test report.

14. Evaluation after Exposure (optional)

14.1 Subject specimens to exposures as directed in an applicable standard or AATCC TM210. Record exposure details as directed.

14.2 Recondition specimens (Section 11.1) before final measurement.

14.3 Measure and record electrical resistances after treatment. Use the same marked points and connection system as for the initial resistance measurement.

14.4 Measure and record distance between marked points to the nearest mm.

15. Calculation

15.1 Calculate the mean resistance for all measured specimens. This is R.

15.1.1 If testing was performed before treatment, calculate the mean of initial resistance measurements for all specimens. This is R_i.

15.1.2 If testing was performed after treatment, calculate the mean of final resistance measurements for all specimens. This is R_f.

15.1.3 If only a single specimen was used, calculate the mean of 3 resistance measurements for the specimen.

15.2 Calculate the mean distance between marked points for all specimens, to the nearest mm. This is d.

15.2.1 If testing was performed before treatment, calculate the mean initial distance for all specimens. This is d_i.

15.2.2 If testing was performed after treatment, calculate the mean final distance for all specimens. This is d_f.

15.3 If applicable, calculate% resistance change after treatment using Equation 1.

$$RC = (R_f - R_i)/R_i \times 100\% \qquad \text{(Eq. 1)}$$

where:

RC = % resistance change after treatment,
R_f = Mean final resistance, after treatment (Ω),
R_i = Mean initial resistance (Ω).

16. Report

16.1 Describe or identify the tested sample.

16.2 Report that the sample was tested using AATCC EP13-2020.

16.3 Report the testing conditions:

16.3.1 Option used (Option 1: 4-point measurement or Option 2: 2-point measurement)

16.3.2 Treatment description and number of cycles, if applicable.

16.3.3Number of specimens tested.

16.3.4 Description of specimen (conductive path length, width, shape, material, etc.)

16.3.5 Multimeter or ohmmeter resolution and accuracy.

16.3.6 Probe type and connection system.

16.3.7 Distance between marked points for individual specimens.

16.3.8 Mean distance, d, between marked points, to the nearest mm. (Initial distance, d_i, and final distance, d_f, if applicable.)

16.4 Report the test results:

16.4.1 Resistance between marked points for individual specimens.

16.4.2 Mean resistance, R, between marked points. (Initial resistance, R_i, and final resistance, R_f, if applicable.)

16.4.3 Resistance change after treatment, if applicable.

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16.5 Describe any modification(s) of the published standard.

17. Precision and Bias

17.1 *Precision.* Five different laboratories participated in an interlaboratory study to measure resistance of e-textiles according to this method. The experiment was replicated on a second day by each of the participating labs.

17.1.1 Samples were a stainless steel conductive thread embroidered on a woven base fabric and silver-based conductive ink printed on thermoplastic polyurethane (TPU) film and applied to a knitted fabric. Printed lines of 0.5-mm width and 1.0-mm width were evaluated separately. All specimens were 100 cm in length.

17.1.2 ANOVA test indicated that there is no significant difference between the results of day 1 and day 2. Among the 5 different labs, the mean values of 3 labs clustered around a measured average resistance of 5.5 Ω and the mean values of the 2 other labs clustered around a measured average resistance of 4.6 Ω. While the standard deviations of the 5 labs were somewhat similar, they were all high, thus implying that there is an inherently high variation in the measurement and that at least 3 replicate specimens are needed for this test. Table I lists *within-laboratory* and *between-laboratory* precision values for different sample sizes.

17.2 *Bias.* Electrical resistance of e-textiles can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

Table I—Within-Laboratory and Between-Laboratory Precision Values

n	Within-Laboratory	Between-Laboratory
1	0.5278	1.2253
2	0.3738	0.8678
3	0.3047	0.7074
4	0.2639	0.6127
5	0.2360	0.5480
6	0.2155	0.5002

Note: Grand mean of resistance for all interlab measurements was 5.3 Ω. Mean resistance for conductive thread specimens was 5.9 Ω. Mean resistance for 0.5-mm printed specimens was 6.7 Ω. Mean resistance for 1.0-mm printed specimens was 3.5 Ω.

18. Notes

18.1 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1. 919.549.8141; ordering@aatcc.org; www.aatcc.org.

18.2 For potential equipment information pertaining to this test method, please visit the online *AATCC Buyer's Guide* at www.aatcc.org/bg. AATCC provides the possibility of listing equipment and materials sold by its Corporate members, but AATCC does not qualify, or in any way approve, endorse or certify that any of the listed equipment or materials meets the requirements in its test methods.

18.3 Weighted or magnetic probes provide more consistent pressure than hand-held probes. Flat or slightly rounded contact areas are suitable for most textile substrates. For 4-point measurements, probes must be aligned along the same conductive path; do not use clip-style probes. An example of suitable probes for e-textile resistance measurements is shown in Fig. 1.

19. History

19.1 Revised in 2021 to add 4-point measurement.

19.2 Reaffirmed in 2020. Editorially revised in 2019 to correct materials sources.

19.3 Developed in 2018 by AATCC Committee RA111.

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AATCC Evaluation Procedure for Small Color Differences

1. Purpose and Scope

1.1 Ellipsoidal color difference equations are used to calculate a volume of acceptable color difference, based on the color of the Reference and the specified tolerance. C*IEDE2000* and *CMC (l:c)* are both ellipsoidal color difference equations, and they may both be used for color evaluation of textile and other materials in many applications. The current data (see references 6.1 through 6.7) indicate that the *CIEDE2000* metric performs slightly better in terms of agreement with visual evaluations. It is therefore being recommended as the primary color difference method to use for instrumental evaluations.

1.2 The CIEDE2000 formula is denoted as ΔE_{00}. The *CMC (l:c)* formula is generally shown as ΔE_{cmc}. Both *CIEDE2000* and *CMC (l:c)* are extensions of the *CIE 1976 L*a*b** color-difference formula with corrections for variation in color-difference perception dependent on lightness, chroma, and hue. *CIEDE2000* additionally includes corrections for chroma-hue interaction.

1.3 The *CMC (l:c)* modification to *CIELAB* provides a unit of measurement for the acceptance volume about a reference color. This volume takes the shape of an ellipsoid whose base semi-axes are defined by *SL*, *SC*, and *SH* in the directions of lightness, chroma, and hue differences, respectively, in *CIELAB* color space. The actual size and shape of the ellipsoid is further controlled by modifying parametric factors *l* and *c*, see section 3.3. The *CMC (l:c)* formula varies the ratio of the lengths of these three semi-axes systematically throughout *CIELAB* color space according to the equality of lengths with perceptually equal color differences, regardless of both the color of the Reference and the direction of difference of any trial specimen from it. Around any given reference color, the ratio *SL:SC:SH* is fixed regardless of the industrial application. However, it should be noted that color difference evaluations may vary if the reference and trial specimen are switched. Note: The difference in the resulting total color difference computed in the two directions (reference to specimen vs. specimen to reference) is not significant for color differences of the magnitude that is relevant for textile applications (under 3.0 units).

The *CIEDE2000* modification to *CIELAB* also provides an ellipsoidal acceptance volume about a reference color. It is analogous to *CMC (l:c)* in the use of three semi-axes for the lightness, chroma, and hue dimensions. *CIEDE2000* was developed to improve on shortcomings identified in the performance of various color difference equations including *CMC (l:c)*. *CIEDE2000* differs from *CMC (l:c)* in an important application, however. *CIEDE2000* determines the arithmetic mean between the Reference and Specimen and uses this as the point in *CIELAB* color space to determine the shape of the ellipsoidal space. This resolves the associated color differences encountered when the reference and the specimen are reversed. This also results in the boundary for the acceptance volume to be somewhat imprecise for a group of specimens compared to a single reference, since the calculations are based on the relation of each specimen to the reference,). Because of this difference in computing the acceptance boundary, *CIEDE2000* may not be suited for some applications, especially when visual plotting is utilized.

1.4 Both *CIEDE2000* and *CMC(l:c)* support the use of component color difference metrics, that is *DL, DC,* and *DH* values which are weighted for visual acceptance by the *CMC (l:c)* or *CIEDE2000* internal math, respectively. These weighted values of *DL, DC*, and *DH* are correlated more strongly to visual color evaluation than legacy metrics such as *DL*, Da*, Db*, DC*, DH**. For *CMC (l:c),* these component color difference terms are DL_{cmc}, DC_{cmc}, DH_{cmc}. For C*IEDE2000*, these terms are DL_{00}, DC_{00}, and DH_{00}.

1.5 In applications which require rating within or about a critical tolerance, multiple levels consisting of ellipsoidal volumes representing various levels of color difference may be generated in both ΔE_{00} and ΔE_{cmc}. This will result in a set of concentric *volumes/tolerances* which provide a uniform grading system when correlated to and associated with a predefined set of terms. An example of this is the association of Gray Scale Grades with word descriptions as shown in Table III of AATCC EP9-2021.

2. Principle

2.1 These equations offer significantly improved correlation between visual assessments and instrumentally measured color differences compared to *CIELAB DE*ab*. Industrial users of these equations are able to adopt a single-number tolerance for judging the acceptability of a color match in most situations, regardless of both the color of the reference and the direction of the color difference of any specimen from it.

3. Terminology

3.1 **CIE 1976 L*a*b* equation,** n.— a commonly used equation which transforms CIE tristimulus values into a three-dimensional opponent color space. Generally abbreviated as *CIELAB* (see Reference section 6.1).

3.2 **ΔE_{cmc},** n.—*in color difference evaluation*, a single number defining the total color difference in *CMC (l:c)* units of a specimen from a reference.

3.3 **semi-axes (S_L, S_C, and S_H),** n.—*in CMC (l:c) color difference evaluation*, basic individual dimensions of the CMC tolerance volume, related to lightness, chroma and hue respectively, whose ratios are determined by the position of the reference in CIELAB space. The actual semi-axes of the ellipsoid are determined by these values and modified by parametric factors *l* and *c*.

3.4 **parametric factors (*l* and *c*)** n.—*in CMC (l:c) color difference equation*, weighting factors used to adjust the component difference ratios, lightness and chroma respectively, based on the effect of the surface characteristics of the stimuli, such as texture. The values $l = 2$ and $c = 1$ are usually used for commercial pass/fail tolerance predictions of textile samples, although *l* may be varied depending on the texture of the material. The value of *c* is always set to one.

3.5 **ΔE_{00},** n.—*in color difference evaluation*, a single number defining the mean total color difference in CIEDE2000 units of a specimen from a reference and vice versa.

3.6 **parametric factors (k_L, k_C, *and* k_H),** n.—*in CIEDE2000 color difference equation*, weighting factors used to adjust the component difference ratios, lightness, chroma and hue respectively, based on the effect of the surface characteristics of the stimuli, such as texture.

4. Calculations

In all calculations below, the subscript ***R*** refers to the **R**eference material and the subscript ***S*** refers to the **S**pecimen. In some documents this pair is referred to as ***S*** for **S**tandard and ***B*** for **B**atch. In other documents these are also referred to as ***S*** for **S**tandard and ***T*** for Trial or **0** and **1,** respectively. Users should clearly define the method being used.

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4.1 Calculation of CIELAB Values

4.1.1 Calculate the *CIELAB* L^*, a^*, b^*, C^*_{ab} and h_{ab} values from the CIE tristimulus values X, Y, Z of each specimen using Equations (1), (2), (3) and (4):

$$L^* = 116 f(Q_Y) - 16 \quad (1)$$

$$a^* = 500[f(Q_X) - f(Q_Y)] \quad (2)$$

$$b^* = 200[f(Q_Y) - f(Q_Z)] \quad (3)$$

where:

$$Q_X = \left(\frac{X}{X_n}\right);\ Q_Y = \left(\frac{Y}{Y_n}\right);\ Q_Z = \left(\frac{Z}{Z_n}\right)$$

and

$$f(Q_i) = Q^{1/3} \text{ if } Q > \left(\frac{6}{29}\right)^3$$

else

$$f(Q_i) = \left(\frac{841}{108}\right) Q_i + \frac{4}{29} \text{ if } Q_i \le \left(\frac{6}{29}\right)^3$$

Here, i varies as X, Y and Z.

$$C^*_{ab} = \sqrt{(a^*)^2 + (b^*)^2} \quad (4)$$

$$h_{ab} = \begin{cases} \arctan\left(\dfrac{b^*}{a^*}\right) & \text{if } a^* > 0 \text{ and } b^* \ge 0 \\ \arctan\left(\dfrac{b^*}{a^*}\right) + 360^\circ & \text{if } a^* > 0 \text{ and } b^* < 0 \\ \arctan\left(\dfrac{b^*}{a^*}\right) + 180^\circ & \text{if } a^* < 0 \\ 90^\circ & \text{if } a^* = 0 \text{ and } b^* > 0 \\ 270^\circ & \text{if } a^* = 0 \text{ and } b^* < 0 \end{cases} \quad (5)$$

$$h_{ab} = 0 \quad \text{if } a^* = 0 \text{ and } b^* = 0 \quad (6)$$

4.1.2 For these equations, X_n, Y_n, and Z_n are the CIE tristimulus values for the chosen illuminant and observer. The preferred combination, for textile applications, is *CIE standard illuminant D65* and the *CIE 1964 supplementary standard (10°) observer*. Values for this combination and for some others are given in Table I. Values for combinations not in Table I should be obtained by referring to ASTM E308 (see Reference section 6.11) or, only if the required combination is not therein, CIE 015:2018 (see Reference section 6.10).

4.2 Calculation of CIELAB Color Difference Values

Table I—Tristimulus Values for Selected Illuminant-Observer Combinations

Illuminant-Observer Combinations	Tristimulus Values		
	X_n	Y_n	Z_n
D_{65}/10°	94.811	100.000	107.304
D_{65}/2°	95.047	100.000	108.883

4.2.1 Calculate the *CIELAB* ΔL^*, ΔC^*_{ab}, and ΔH^*_{ab} color difference values as follows:

$$\Delta L^* = L^*_S - L^*_R \quad (7)$$

$$\Delta a^* = a^*_S - a^*_R \quad (8)$$

$$\Delta b^* = b^*_S - b^*_R \quad (9)$$

$$\Delta C^*_{ab} = C^*_{ab,S} - C^*_{ab,R} \quad (10)$$

$$\Delta E^*_{ab} = \left[(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2\right]^{1/2} \quad (11)$$

$$\Delta H^*_{ab} = pq\left[(\Delta E^*_{ab})^2 - (\Delta L^*)^2 - (\Delta C^*_{ab})^2\right]^{1/2} \quad (12)$$

where:

$p = 1$ if $m \ge 0$ or $p = -1$ if $m < 0$ and $q = 1$ if $|m| \le 180$ or $q = -1$ if $|m| > 180$

$m = h_{ab,S} - h_{ab,R}$ in which $|\ldots|$ indicates that the positive value is to be used regardless of the sign of the expression between the two lines.

4.2.2 An equivalent method of calculating ΔH^*_{ab} is to use Equation (13):

$$\Delta H^*_{ab} = t\left[2(C^*_{ab,S}\, C^*_{ab,R} - a^*_S\, a^*_R - b^*_S\, b^*_R)\right]^{1/2} \quad (13)$$

where:

$t = 1$ if $a^*_S\, b^*_R \le a^*_R\, b^*_S$

or $t = -1$ if $a^*_S\, b^*_R > a^*_R\, b^*_S$

4.3 Calculation of CIEDE2000

4.3.1 Symbols and Definitions

L^*	*CIELAB* Lightness
a^*, b^*	*CIELAB* a^*, b^* coordinates
C^*_{ab}	*CIELAB* chroma
h_{ab}	*CIELAB* hue angle
L'	*CIEDE2000* lightness
$\overline{L}'$	arithmetic mean of the *CIEDE2000* lightnesses of two color stimuli
a', b'	*CIEDE2000* a', b' coordinates
C'	*CIEDE2000* chroma
$\overline{C}'$	arithmetic mean of the *CIEDE2000* chromas of two color stimuli
h'	*CIEDE2000* hue angle
$\overline{h}'$	arithmetic mean of the *CIEDE2000* hue angles of two color stimuli
G	switching function used in the modification of a^*
$\Delta L'$	*CIEDE2000* lightness difference
$\Delta C'$	*CIEDE2000* chroma difference
$\Delta h'$	*CIEDE2000* hue-angle difference
$\Delta H'$	*CIEDE2000* hue difference
ΔE_{00}	*CIEDE2000* color difference
S_L	lightness weighting function
S_C	chroma weighting function
S_H	hue weighting function
T	T-function for hue weighting
R_T	rotation function
$\Delta\theta$	hue dependence of rotation function
R_C	chroma dependence of rotation function
k_L	lightness parametric factor
k_C	chroma parametric factor
k_H	hue parametric factor

4.3.2 CIEDE2000 Calculations

All angular quantities in this Evaluation Procedure shall be evaluated in degrees. *CIELAB* L^*, a^*, b^* and C^*_{ab} coordinates of the two samples shall be calculated according to ISO 11664-4:2008/CIE S 014-4/E:2008 which are shown above in Equations (1) through (13). Modified *CIELAB* coordinates shall be calculated according to Equations (14) to (20).

$$L' = L^* \quad (14)$$

$$a' = (1 + G)a^* \quad (15)$$

$$b' = b^* \quad (16)$$

$$C' = (a'^2 + b'^2)^{1/2} \quad (17)$$

$$h' = \begin{cases} \arctan\left(\dfrac{b'}{a'}\right) & \text{if } a' > 0 \text{ and } b' \ge 0 \\ \arctan\left(\dfrac{b'}{a'}\right) + 360^\circ & \text{if } a' > 0 \text{ and } b' < 0 \\ \arctan\left(\dfrac{b'}{a'}\right) + 180^\circ & \text{if } a^* < 0 \\ 90^\circ & \text{if } a' = 0 \text{ and } b' > 0 \\ 270^\circ & \text{if } a' = 0 \text{ and } b' < 0 \end{cases} \quad (18)$$

$$h' = 0^\circ \quad \text{if } a' = 0 \text{ and } b' = 0 \quad (19)$$

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where:

$$G = 0.5\left[1 - \sqrt{\frac{(\overline{C^*_{ab}})^7}{(\overline{C^*_{ab}})^7 + 25^7}}\right] \quad (20)$$

$\overline{C^*_{ab}}$ is the arithmetic mean of the C^*_{ab} values for the two samples of the color-difference pair.

Equation (18) ensures that h' is the angular position of the point a', b' in the range from 0 to 360 measured from the positive a' axis in the a', b' plane. In the case where $a' = b' = 0$, h' is indeterminate and shall be assigned a value of zero as indicated in Equation (19).

NOTE 1: The L', a', b', C' and h' values should be used only for the calculation of color difference and should not be used as an alternative uniform color space. When reporting *CILEAB* color space coordinates, L^*, a^*, b^*, C^*_{ab} and h_{ab} values should be used.

Differences between two samples denoted by subscripts R (Reference) and S (Specimen) shall be calculated as follows:

$$\Delta L' = L'_S - L'_R \quad (21)$$

$$\Delta C' = C'_S - C'_R \quad (22)$$

$$\Delta H' = 2(C'_R\, C'_S)^{1/2} \sin\left(\frac{\Delta h'}{2}\right) \quad (23)$$

where:

$$\Delta h' = 0° \quad \text{if } C'_R\, C'_S = 0 \quad (24)$$

$$\Delta h' = h'_S - h'_R$$

$$\text{if } C'_R\, C'_S \neq 0 \text{ and}$$

$$|h'_S - h'_R| \leq 180° \quad (25)$$

$$\Delta h' = h'_S - h'_R - 360°$$

$$\text{if } C'_R\, C'_S \neq 0 \text{ and}$$

$$|h'_S - h'_R| > 180° \quad (26)$$

$$\Delta h' = h'_S - h'_R + 360°$$

$$\text{if } C'_R\, C'_S \neq 0 \text{ and}$$

$$|h'_S - h'_R| < (-180)° \quad (27)$$

NOTE 2: Equations (24) to (27) avoid possible computational difficulties when h'_R and h'_S are in different quadrants or when one of the chromas is zero. They are based on Sharma et al., 2005.

NOTE 3: In information technology and other fields the subscripts r (for reference) and t (for test) are sometimes used instead of 0 and 1 respectively. Similarly, in industrial evaluation of small color differences, s (for Standard) and b (for batch) are sometimes used.

The CIEDE2000 color difference, ΔE_{00} between the two samples shall be calculated by:

$$\Delta E_{00} = \left[\left(\frac{\Delta L'}{k_L S_L}\right)^2 + \left(\frac{\Delta C'}{k_C S_C}\right)^2 + \left(\frac{\Delta H'}{k_H S_H}\right)^2 + R_T\left(\frac{\Delta C'}{k_C S_C}\right)\left(\frac{\Delta H'}{k_H S_H}\right)\right]^{1/2} \quad (28)$$

where:

$$S_L = 1 + \frac{0.015(\bar{L}' - 50)^2}{\sqrt{20 + (\bar{L}' - 50)^2}} \quad (29)$$

$$S_C = 1 + 0.045\ \bar{C}'T \quad (30)$$

$$S_H = 1 + 0.015\ \bar{C}'T \quad (31)$$

$$T = 1 - 0.17\cos(\bar{h}' - 30) + 0.24\cos(2\bar{h}') + 0.32\cos(3h' + 6°) - 0.20\cos(4\bar{h}' - 63°) \quad (32)$$

$$R_T = -\sin(2\Delta\theta)\ R_C \quad (33)$$

$$\Delta\theta = 30°\exp\{-[(\bar{h}' - 275°)/25°]^2\} \quad (34)$$

$$R_C = 2\sqrt{\frac{(\bar{c}')^7}{(\bar{c}')^7 + 25^7}} \quad (35)$$

k_L, k_C and k_H are parametric factors explained in section 3.6.

NOTE 4: The $\bar{L}'$, $\bar{C}'$ and $\bar{h}'$ values used in Equations (29) to (35) are the arithmetic means of the corresponding values of the color-difference pair. A consequence of this is that the total color difference is reversible, that is, the total color difference between a pair is the same whether the first or second sample is used as the reference for calculation of color-difference components.

NOTE 5: The locus of points of equal total color difference from a reference is not an exact ellipsoid and is not exactly centered on the reference. Users should take care in calculating the mean hue angle if the color-difference pair has samples in different quadrants. For example, if a color-difference pair has hue angles of 30° and 300°, the simple mean, 165°, is incorrect, the correct value being 345°. To determine the mean correctly, the following equations (Sharma et al., 2005) shall be used.

$$\bar{h}' = \frac{(h'_0 + h'_1)}{2}$$

$$\text{if } |h'_0 - h'_1| \leq 180° \text{ and } C'_0 C'_1 \neq 0 \quad (36)$$

$$\bar{h}' = \frac{(h'_0 + h'_1 + 360°)}{2}$$

$$\text{if } |h'_0 - h'_1| > 180° \text{ and } (h'_0 + h'_1) < 360° \text{ and } C'_0\, C'_1 \neq 0 \quad (37)$$

$$\bar{h}' = \frac{(h'_0 + h'_1 - 360°)}{2}$$

$$\text{if } |h'_0 - h'_1| > 180° \text{ and } (h'_0 + h'_1) \geq 360° \text{ and } C'_0\, C'_1 \neq 0 \quad (38)$$

$$\bar{h}' - h'_0 + h'_1 \quad \text{if } C'_0\, C'_1 = 0 \quad (39)$$

NOTE 6: Some worked examples of the calculations are given in CIE 142-2001, Luo et al., 2001 and Sharma et al., 2005. Sharma et al. also give some useful implementation notes and mathematical observations.

If required, the components of *CIEDE2000* color difference (ΔL_{00}, ΔC_{00}, and ΔH_{00}) may be calculated as below:

4.3.3 Color Sorting and Statistical Quality Control

For certain applications, such as apportioning a color difference into lightness, chroma and hue components for shade sorting and statistical quality control, it is important to have these three directional attributes that are based on the position of the reference specimen. The best way to accomplish this is to use the *CMC* attributes DL_{cmc}, DC_{cmc} and DH_{cmc} as defined in Equations (41), (42) and (43) below in section 4.4.1.

4.3.4 Reporting of Results

The use of illuminant D65 and 10° observer calculations and ($k_L : k_C : k_H$) ratio of 1:1:1 are recommended as the standard for calculating ΔE_{00} values. If other ΔE_{00} ($k_L : k_C : k_H$) ratios are used, they must be specified as part of the value (i.e., ΔE_{00}(2:1:1), would indicate using a $k_L : k_C : k_H$ ratio of 2:1:1.

4.4 Calculation of ΔE_{cmc}

4.4.1 Calculate the color difference in *CMC (l:c)* units using Equation (40):

$$\Delta E_{CMC} = \left[\left(\frac{\Delta L^*}{lS_L}\right)^2 + \left(\frac{\Delta C^*_{ab}}{cS_C}\right)^2 + \left(\frac{\Delta H^*_{ab}}{S_H}\right)^2\right]^{1/2} \quad (40)$$

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where for the Reference:

$$\text{for } L_R^* > 16 \; S_L = \frac{0.040975 L^*}{1 + 0.01765 L^*}$$

$$\text{for } L_R^* \leq 16 \; S_L = 0.511$$

$$S_C = \frac{0.0638 C_{ab}^*}{1 + 0.0131 C_{ab}^*} + 0.638$$

$$S_H = (FT + 1 - F) S_C$$

where:

$$F = \left[\frac{(C_{ab}^*)^4}{(C_{ab}^*)^4 + 1900} \right]^{1/2}$$

and

$$T = 0.36 + abs\,[0.4 \cos(35 + h_{ab})]$$

unless h_{ab} is between 164° and 345° then

$$T = 0.56 + abs\,[0.2 \cos(168 + h_{ab})]$$

For the last two equations, '*abs*' indicates the absolute—i.e., positive value—of the term inside the square brackets.

NOTE 9: If required, the components of *CMC (l:c)* color difference (L_{cmc}, C_{cmc}, and H_{cmc}) may be calculated using the terms in the round brackets in the first line of the Equation (40) namely:

$$\Delta L_{CMC} = \frac{\Delta L^*}{l S_L} \quad (41)$$

$$\Delta C_{CMC} = \frac{\Delta C_{ab}^*}{c S_C} \quad (42)$$

$$\Delta H_{CMC} = \frac{\Delta H_{ab}^*}{S_H} \quad (43)$$

4.4.1.1 When $l = 2.0$ this equation fixes the ratio of the three terms (*SL:SC:SH*) to correlate with visual assessment of typical textile samples. Other values of *l* may be required in cases where the surface characteristics are significantly different. For example, other values may be required when measuring very dark trials, but the user should assume an *l* value of 2.0 until actual results indicate a need for adjustment. In practice, "*c*" is always set to unity and it may be omitted from the equation.

4.4.2 The equation for *CMC (l:c)* describes an ellipsoidal volume, with axes in the direction of lightness, chroma and hue centered about a reference. Ellipsoid semi-axis lengths of S_L, S_C and S_H, calculated for a given reference, describe a volume of acceptance which contains all samples within the tolerance.

4.4.3 Reporting of results. The use of illuminant D_{65} and 10° standard observer and *(l:c)* ratio of 2:1 are recommended as the standard for calculating ΔE_{cmc} values. If other *CMC (l:c)* ratios are used, they must be specified as part of the value (i.e., $\Delta E_{cmc(1.37:1)}$, would indicate using a *l:c* ratio of 1.37:1

5. Interpretation of Results

5.1 ΔE is a single number which represents the total color difference, either ΔE_{00} for the number of *CIEDE2000* color difference units or ΔE_{cmc} for the number of *CMC* color difference units of a specimen from a reference.

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7. History

7.1 Editorially revised in 2023 to fix a typo.

7.2 Editorially revised 2022. Developed in 2021 by AATCC Committee RA36.

7.3 Supersedes Test Method 1732009.

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AATCC Standard Reference Detergents and Laundry Detergents in General

Developed in 1995 by AATCC Committee RA88; revised 1981/1982, 1991, 1998 (with title change), 2005, 2017 (with title change); numbered in 2011; editorially revised 2019.

1. Background/History

1.1 The AATCC and other testing and development groups have used standard reference detergents since the 1960s. AATCC Standard Reference Detergent 124 and WOB (without brightener) were the first formulations adopted by the AATCC to represent typical home laundering products of that era, primarily phosphate-built powder detergents. The proliferation of detergent formula variations and the elimination of phosphate due to environmental pressures made Standard Reference Detergents 124 and WOB obsolete.

1.2 Since the discontinuation of 124 and WOB, new representative detergents have been available for use. The following paragraphs provide some perspective on laundry detergent trends and the AATCC standard reference detergents.

2. Current and Future Trends in the Detergent Industry

2.1 Consumer laundry detergents today come in a variety of product forms including powders, liquids, and single unit dose packs. All forms are primarily composed of surfactants and builders (except non-built liquid detergents, which contain primarily surfactants). They also contain processing or delivery aids for product stability and can contain many optional ingredients to provide performance or aesthetic benefits, such as fluorescent brighteners, enzymes, bleaches, anti-redeposition agents, fiber and dye protective agents, perfumes, suds control agents, etc.

2.2 A key difference between powder and liquid laundry detergents is their builder system. Powder detergents are typically better built than liquid laundry detergents, have a carbonate-builder system, often contain aluminosilicates (zeolites) for added builder capacity, and perform optimally at higher pHs (pH ≈ 10). Heavy-duty liquid laundry detergents are typically citrate-built and perform optimally at lower pHs (pH ≈ 8.5). Single unit dose packs can contain liquid and/or powder components and the corresponding builder system.

At the time of this writing (2017), the market share for consumer detergents in the United States is approximately 75% for liquids, with the remaining being split between powders and packs.

2.3 The detergent market will continue to evolve as the demands of the consumer and chemical industries change. Detergents of the future will be impacted by many factors, including environmental concerns, conservation efforts, chemical availability, raw material costs, performance benefits, and consumer needs. There is, therefore, a need to ensure that reference detergents are reflective of consumer products and should be updated after major shifts in the detergent industry are noted.

3. Rationale for Using 1993 AATCC Standard Reference Detergent

3.1 With this background and especially the environmental concerns with the use of phosphates, new AATCC Standard Reference Detergents, with and without brightener, were developed in 1993 and are still used today. These concentrated powder formulations use carbonate builder instead of phosphate and were created without enzymes due to possible changes of enzyme strength over the storage life of the product.

3.2 Laboratory comparisons between 1993 AATCC Standard Reference Detergents and consumer products would likely show differences in washing performance. However, comparison among consumer products could also show differences of similar or even greater magnitude.

3.3 One of the factors leading to possible differences between results using the 1993 AATCC Standard Reference Detergents and current consumer detergent products is water hardness. In hard water situations, the 1993 AATCC Standard Reference Detergents may give better soil removal results because they are more effective across a broader range of hardness than most consumer products.

3.4 Other Standard Reference Detergents are used in various ISO test methods. These detergents are specified in methods designed for home laundering equipment sold outside the United states.

3.5 AATCC 1993 formula is suitable for toploading machine and accelerated (Launderometer) testing. It is not suitable for high efficiency top loading or front loading machines.

4. Rationale for using AATCC Liquid Detergents

4.1 Due to the growing market share of liquid detergents, AATCC saw the need and developed a liquid standard reference detergent that was relevant to the current laundry market.

4.2 Although powder and liquid detergents are designed to provide cleaning and stain removal, they achieve these objectives differently. A key difference is the pH regime in which the two product forms operate. Powder detergents perform optimally at higher pHs (pH ≈ 10). This higher alkalinity helps stain removal efficiency; however it also negatively impacts fibers and dyes. Over the years, technologies have been developed to deliver benefits under alkaline conditions and powder detergents have been able to overcome many of their initial disadvantages.

4.3 Liquid detergents perform optimally at lower alkalinity (pH ≈ 8.5). Because this pH is closer to neutrality, liquid laundry detergents tend to be less harsh on fibers and dyes. Over the years, technologies have been developed to help liquid laundry detergents deliver stronger cleaning benefits while remaining less harsh on fibers and dyes. Due to significant formulation differences between powder and liquid laundry detergents, different reference detergents are needed to fairly represent them.

4.4 AATCC's first liquid detergent was the 2003 formula, with and without brightener, and was suitable for traditional toploading machines. Its use was discontinued in 2016 as a newer formula was developed that could be used in both traditional and high efficiency machines.

4.5 The 2015 HE Standard Reference Detergent, available with and without brightener, is the current recommended standard liquid for both traditional toploading and high efficiency machines. Testing of this detergent at the time of its introduction showed that its stain removal performance was comparable to current consumer detergents.

5. Rationale for Use of Reference Detergents in Laboratory Testing

5.1 Many fabric attributes critical to consumer use and acceptability, such as

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dimensional change, surface or smoothness appearance, colorfastness, soil release, and flammability resistance performance are influenced by the manner in which textile products are laundered. The textile industry has adopted standard detergents and laundering conditions to allow for the prediction of the acceptability of textile products to judge the performance of their products. The standard detergents have been designed to represent a cross section of market detergents.

5.2 The use of locally purchased national brands of detergents in testing labs is a fairly common practice. This practice has been driven by several factors: a) inaccurate assumption that the same detergent brand has the same composition from location to location and from one year to the next one, b) the convenience of buying locally, and c) price. The use of "off the shelf" detergents in laboratory testing adds an element of variability that use of standard test methods and detergents intends to control. The percentage of optical brightener or fluorescent brightener agents has a definite effect on colorfastness evaluations. The amount of optical brightener or fluorescent brightening agents is known to vary within a single brand of detergent sold for consumer use.

5.3 Detergent manufacturers have developed detergents with other cleaning components such as non-chlorine color safe bleach systems. AATCC has standard full-scale (washing machine) and accelerated methods for determination of colorfastness using such products. They are AATCC TM172, Test Method for Colorfastness to Powdered Non-Chlorine Bleach in Home Laundering, and AATCC TM190, Test Method for Colorfastness to Home Laundering with Activated Oxygen Bleach Detergent: Accelerated.

5.4 It should be noted that different standard laundering equipment and detergents are used in Europe and Asia.

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Overview of Liquid Fabric Softeners Used in Home Laundering

Developed in 2006 by AATCC Committee RA88; numbered in 2011; revised 2017; editorially revised 2019.

1. Background on Fabric Softeners

1.1 While AATCC and other testing and development groups have used standard reference detergents since the 1960s, there has been no standard reference for fabric softeners. In 2017, about 80% of U.S. households regularly use fabric softeners in their laundering process. The most popular methods of applying softeners to textiles in the home are rinse cycle softeners and dryer sheet softeners. Rinse cycle softeners are used in more than 40% of U.S. households while dryer sheets are used in more than 60% of households, with some households using both forms. This monograph provides information on liquid fabric softener used in home laundering.

1.2 Household liquid fabric softeners were developed in the 1960s to maintain the original pleasant feel of clothes on skin. With repeated laundering, clothes lose some of their original mechanical properties because of the intense stress they experience during laundering, and the textile fibers tend to entangle. Through the drying process, the fibers remain entangled and clothes become stiff. Items made with manufactured fibers tend to get charged with static electricity during tumble drying causing static cling. Industry experts indicate that the use of liquid softeners in home laundering helps minimize these issues.

2. Fabric Softener Attributes

2.1 Softness. Liquid fabric softeners function by depositing cationic active compounds or ingredients onto fabric surfaces during the rinsing cycle. Dialkyl quaternary softening agents consist of a positively charged amine group and fatty chains. Once the quaternary active attaches to the fabric, the fatty chains orient themselves perpendicular to the fabric surface. This helps prevent fibers from becoming entangled creating a thick, full density. Full fiber lubrication creates less friction on the surface and in between fibers providing a soft hand to fabrics. A softener's active ingredient or compound deposition in the rinse cycle can be very high, approaching 90% in some cases. The rinse deposition provided by a liquid fabric softener provides an advantage over dryer sheet softener deposition, as liquid softeners have greater exposure to the fabric surfaces, thus deeper penetration potential to provide a softer hand. Softer garments provide consumers with the perception of increased tactile comfort during wear.

2.2 Static. Static is an imbalance of electrical charge, or electrons, on textile surfaces. Dissimilar materials brought into contact and separated can transfer electrons, leaving a charge imbalance. Friction and low moisture content enhance this process. For example, when clothing is dried in a tumble dryer, items made of fibers that naturally contain some moisture—like cotton—can dissipate charges better than items made of manufactured fibers like polyester. Fabric softeners reduce friction by lubricating fibers, which helps to minimize static buildup in the dryer. In addition, due to the better conductivity properties of fabric softeners' active compounds or ingredients, their use can help neutralize electric charge build-up in a dryer from mixed fiber content loads.

2.3 Fabric Scent. Most fabric softeners (except the unscented or "free" versions) provide freshness to fabrics by fragrances incorporated into the formulations. These generally provide a fresh dry fabric scent that in some cases can last for several days. Many consumers correlate a fresh wet and dry fabric scent with an impression of cleanliness.

2.4 Appearance/Color. The use of liquid fabric softeners usually results in better color appearance of laundered fabrics. The deposition of the softening component lubricates fibers and yarns, which reduces fabric abrasion and helps prevent fibers from becoming entangled. This, in turn, helps maintain fabric appearance and may extend the life of the fabric. Fabric abrasion may cause surface disruptions, such as fuzziness and/or pilling, that can contribute to a faded, worn appearance.

2.5 Wrinkle Reduction. Due to the fiber lubrication and reduction in fiber-to-fiber friction from deposited softener active ingredients, liquid softeners help reduce fabric wrinkling.

2.6 Flammability. Liquid fabric softeners are not recommended for use on children's sleepwear or garments labeled as flame resistant as the softener may reduce flame resistance.

3. Trends in Liquid Fabric Softeners

3.1 All effective home laundry liquid fabric softeners contain a cationic surfactant-a quaternary ammonium compound. Liquid softeners were developed in the early 1960s as simple dispersions of ditallow dimethyl ammonium chloride (DTDMAC), fragrance, electrolyte, colorant, and water. A softener's active compound or ingredient is highly water insoluble and exists as a vesicle, a bilayer structure, in water. DTDMAC was very effective as a softening and antistatic agent and was cost effective. In the mid-1980s, manufacturers converted many liquid softeners to compacted "ultra" formulas by concentrating the active level about three times, resulting in smaller, more convenient bottle sizes. To maintain the desired softness performance, manufacturers used combinations of DTDMAC with amidoamine quaternary active ingredients, imidazolinium quaternary active ingredients, or imidazolines.

3.2 The softener market will continue to evolve as the demands of the consumer constantly change. Softeners of the future will be impacted by many factors, including the desire for rapid biodegradability, water conservation efforts, raw material costs, new performance benefits, improved freshness, and other consumer needs.

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A Summary of ASTM Methods for Interlaboratory Testing

Developed in 1992 by AATCC Committee RA102; numbered in 2011; editorially revised 2018, 2019.

1. Introduction

1.1 ASTM D2904, Standard Practice for Interlaboratory Testing of a Textile Test Method that Produces Normally Distributed Data, and ASTM D2906, Standard Practice for Statements on Precision and Bias for Textiles, are guides for planning interlaboratory tests to evaluate proposed test methods and for using the test results to write precision statements of normally distributed data. It is intended that these methods as well as ASTM D4467, Interlaboratory Testing of a Textile Test Method that Produces Non-Normally Distributed Data, be used as guides in the development of information for precision and bias statements in AATCC test methods.[1] This monograph is a summary of the vital parts of the ASTM standards and are the minimum conditions that should be used in developing precision and bias statements for AATCC test methods. Components of variance for single operator, within laboratory, and between laboratory effects are determined. Critical differences calculated from the component variances indicate, for averages of "n" specimens from different samples, the minimum differences which are statistically significant.

2. Parameters in the Interlaboratory Test

2.1 Materials.

2.1.1 There should be a minimum of two materials representing the range of interest of the property being tested. Subsamples of materials should be as homogeneous as possible. Where possible, values for each material should be established by alternative methods to determine if there is a variable bias between the proposed method and the referee method at different levels of the property.

2.2 Laboratories.

2.2.1 There should be a minimum of five laboratories.

2.3 Operators.

2.3.1 A minimum of two operators within each laboratory is recommended; however, testing by a single operator is acceptable.

2.4 Specimens.

2.4.1 There should be a minimum of two specimens tested by each operator in each laboratory *from each material*. The number of specimens tested per operator should be determined from the established variance for the test (determined from tests on one material by one operator in one laboratory) and the smaller systematic effect it is desired to be able to detect. Procedures for calculation of the required number of specimens are detailed in Section 5.5 of ASTM D2905. It is advisable to test a larger number of materials in more laboratories with the number of operators per laboratory and the number of tests per operator at a minimum of two each. The order of tests should be randomized to eliminate any storage or time effects.

2.5 Instruments.

2.5.1 This should not be an effect included in the statistical analysis. When multiple instruments are used in a laboratory, ascertain if differences among instruments exist, and if they do, use known standard samples to obtain correction factors.

3. Procedure

3.1 A preliminary within-laboratory ruggedness test should be performed prior to interlaboratory testing. It is also advisable to conduct a pilot scale interlaboratory test before full scale testing.

3.2 Obtain an adequate number of samples and code for distribution to laboratories. The materials should be completely randomized for assignment to laboratories. It may be necessary in some cases to employ partial randomization. For example, in testing yarn from different spinning frames, specimens from each frame may be allocated to each laboratory. Determine subsample homogeneity before distribution to laboratories.

3.3 Conduct tests at each laboratory according to the procedure of the proposed test method.

4. Analysis

4.1 An analysis of variance (ANOVA) is used to determine the significance of effects (operators, laboratories) in the interlaboratory test. This procedure assumes uniformity of variances. If variances are not uniform, data transformations such as those suggested in Section 11 of ASTM D2904 may be required.

4.2 ANOVA for single materials.

4.2.1 Prepare a separate ANOVA for each material, using a specially designed statistics package from ASTM or ANOVA procedures available in other statistical software packages (SAS, SPSS). In the latter case, the effects in the model should include laboratory, operator nested within laboratory and specimen nested within operator and laboratory as the sources of variation. The analysis will yield F-values for each effect, and these can be used to determine if significant differences exist between operators and/or between laboratories. Alternatively, the variances can be calculated by hand using the formulas in Annex A2 of ASTM D2904.

4.2.2 Determine the components of variance using either the ASTM or other statistical package. This calculation is part of the ASTM program, but it will require an additional procedure if other standard statistical packages are used (e.g., VARCOMP in SAS). Formulas for calculation of the components of variance are also given in Annex A2 of ASTM D2904.

4.2.3 Calculate the critical differences for each of the effects using the components of variance. The ASTM program provides these critical differences for selected numbers of specimens, or they may be calculated using the equation in Sections 8.2 and 8.4 of ASTM D2906. The critical differences for each material should be compared to determine if they are sufficiently similar to combine the data for all materials into a single ANOVA table. An engineering decision should be made on the practical importance of the observed variation in the critical differences for materials. The auxiliary tests listed in Section 15 of ASTM D2904 are helpful in making this determination.

4.3 ANOVA for all materials.

4.3.1 If the critical differences for all materials are sufficiently similar, prepare an ANOVA table which includes all materials. Determine significant effects with F-tests. The ASTM program will perform these analyses directly. For other programs, the effects included in the model should be: material, material*laboratory interaction, operator within laboratory, material*operator within laboratory interaction and specimen within operator and laboratory.

4.3.2 Calculate the components of variance and critical differences as be-

[1] ASTM D2904, D2906 and D4467 are available from ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832.9500; fax: +1.610.832.9555; web site: www.astm.org.

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fore. The component of variance for materials is usually not calculated as the materials were deliberately chosen to exhibit different levels of the property of interest.

4.3.3 If neither materials*laboratory or material*operator within laboratory is significant, do not include these in the precision statement. However, if either of these effects is significant, different components of variance apply to situations involving one material or more than one material. This means that the test method evaluates materials differently when tested in different laboratories or by operators within laboratories. In these cases *both* single material and multimaterial components of variance should be calculated, with the latter including the components of variance for the materials interactions. (See Section A2.14.2.2 of ASTM D2904 for further explanation.)

5. Report or Precision Statement

5.1 Describe the interlaboratory test in terms of numbers of materials, laboratories, operators within laboratories and specimens tested by each operator.

5.2 Report the components of variance (as variances or as standard deviations) and the critical differences for selected numbers of specimens, for each of the effects determined from the analysis including all materials. If the material interactions are not significant, report critical differences between specimens tested by each operator, between operators and between operators within laboratories.

5.3 If the materials interactions are significant, report both single and multimaterial components of variance and critical differences, for single operator, within laboratory and between laboratory effects.

5.4 In the case of arbitrary grades, or of other grades that are limited and not continuous, or for which meaningful transformations may not be practicable (such as the Gray Scale), Recommended Text 8—Special Cases of Ratings in ASTM D2906 may be used. Most rating scales in AATCC methods are limited and not continuous.

Appendix A
Definitions of Terms

The following definitions are taken from ASTM D123 and E456.

A1 **nested experiment,** n.—an experiment to examine the effect of two or more factors in which the same level (version) of a factor cannot be used with all levels (versions) of other factors.

A2 **ruggedness test,** n.—a planned experiment in which environmental factors of test conditions are deliberately varied in order to evaluate the effects of such variation.

A3 **standard deviation,** n.—the positive square root of the variance.

A4 **specimen,** n.—a specific portion of a material or laboratory sample upon which a test is performed or which is taken for that purpose. (Syn. *test specimen*.)

A5 **variance,** n.—a measure of the squared dispersion of observed values or measurements expressed as a function of the sum of the squared deviations from the population mean or sample average.

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Barré: Visual Assessment Descriptive Terms and Terminology

Developed by AATCC Committee RA99 in 2012; revised 2018; editorially revised 2019.

1. Introduction

Barré is the result of physical or dye differences in the yarns, geometric differences in the fabric structure or any combination of these differences. This monograph contains a list of the descriptive terms that can be used to describe the appearance and/or pattern of any barré in circular knit fabrics.

2. Terminology

2.1 **barré,** n.—an unintentional, usually repetitive, pattern of continuous bars or stripes, parallel to the courses of circular knit fabric.

3. Descriptive Terms for Barré Patterns (in Circular Knit Fabrics)

3.1 **Band type**—an appearance wherein the barré appears as a regularly repeating band of two or more consecutive ends in the knitting machine creel repeat.

3.2 **Darker appearing**—a barré that appears darker than the normal portion of the fabric. This barré can appear as darker in dyed shade when the fabric is viewed with reflected light, or darker in appearance when the fabric is viewed with transmitted light.

3.3 **Lighter appearing**—a barré that appears lighter than the normal portion of the fabric. This barré can appear as lighter in dyed shade when the fabric is viewed with reflected light, or lighter in appearance when the fabric is viewed with transmitted light.

3.4 **Multiple end type**—a regularly repeating barré wherein more than one course from the knitting machine creel repeat exhibits a barré appearance. However, unlike the band type barré, a multiple end type barré does not involve consecutive ends (from the knitting machine creel).

3.5 **Random**—a barré that does not repeat periodically along the length of the fabric. (Syn: *short term*.)

3.6 **Regularly repeating**—a barré that repeats periodically along the length of the fabric.

3.7 **Single end type**—a regularly repeating barré wherein only one course from the knitting machine creel repeat exhibits a barré appearance.

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Glossary of AATCC Standard Terminology

Legend:

EP = Evaluation Procedure
LP = Laboratory Procedure
M = Monograph
RA = Research Committee
TM = Test Method

AATCC Blue Wool Lightfastness Standard, n.—one of a group of dyed wool fabrics distributed by AATCC for use in determining the amount of light exposure of specimens during lightfastness testing. *Source:* TM16.1, 16.2, 16.3, 111, 192.

AATCC fading unit (AFU), n.—a specific amount of exposure made under the conditions specified in various test methods where one AFU is one-twentieth (1/20) of the light-on exposure required to produce a color change equal to Step 4 on the Gray Scale for Color Change or 1.7 ± 0.3 CIELAB units of color difference on AATCC Blue Wool Lightfastness Standard L4. *Source:* TM16.1, 16.2, 16.3.

abrasion, n.—the wearing away of any part of a material by rubbing against another surface. *Source:* LP3, TM93, 119, 120.

absorbance, n.—the logarithm to the base 10 of the reciprocal of transmittance.

NOTE: Absorbance is proportional to the mass of absorbing material in the path length of a spectrophotometer cell (syn: *absorbance value*). *Source:* TM182.

absorbency, n.—the propensity of a material to take in and retain a liquid, usually water, in the pores and interstices of the material. *Source:* TM70, 79.

absorbent capacity, n.—the maximum amount of liquid a material can hold; dependent on the specific test method used. *Source:* TM200.

absorption, n.—a process in which one material (the absorbent) takes in or incorporates another material (the absorbate) within itself; such as the absorption of moisture by fibers. *Source:* RA99.

absorption rate (AR_T) (top surface) and (AR_B) (bottom surface), n.—the average speed of liquid moisture absorption for the top and bottom surfaces of the specimen during the initial change of water content during a test. *Source:* TM195.

accelerated ageing, n.—*in textile processing and testing*, use of controlled environmental conditions to promote rapid physical and/or chemical change in a textile material. *Source:* TM26.

accumulative one-way transport capability (R), n.—the difference between the area of the liquid moisture content curves of the top and bottom surfaces of a specimen with respect to time. *Source:* TM195.

acid dye, n.—an anionic dye having substantivity for fibers which contain cationic groups usually in acidic or neutral aqueous dyebaths. *Source:* TM159.

activated oxygen bleach, n.—a bleaching system comprising an oxygen bleach and a bleach activator. *Source:* TM190.

activity, n.—*of an antibacterial agent*, a measure of effectiveness of the agent. *Source:* TM90, 100, 147, 174.

activity, n.—*of an anti-dust mite agent*, a measure of the effectiveness of the agent. *Source:* TM194.

add-on, n.—*in textile processing*, the amount of any material, chemical finish, coating, sizing, etc. that is applied to a textile (see also *wet pick-up*).

NOTE: Add-on is usually determined as a percentage of either the dry or conditioned weight of the textile prior to processing. *Source:* RA99.

ageing, accelerated, n.—see *accelerated ageing*.

anionic dye, n.—a dye that dissociates in aqueous solution to give a negatively charged colored ion. *Source:* RA87.

antibacterial agent, n.—any chemical material which kills bacteria (bactericide) or interferes with the multiplication, growth or activity of bacteria (bacteriostat). *Source:* TM90, 100, 147, 174.

antifungal agent, n.—any chemical material which kills or inhibits the growth of fungi. *Source:* TM174.

anti-house dust mite agent, n.—any chemical which kills (acaricide) or repels house dust mites. *Source:* TM194.

antimicrobial agent, n.—*in textiles*, any chemical material which kills or inhibits the growth of microorganisms. *Source:* TM174.

appearance of creases, n.—see *crease retention*.

appearance of textile end products, n.—the overall visual impression of a textile end product quantified by comparison of individual components with appropriate reference standards. *Source:* TM143.

aqueous repellency, n.—*in textiles*, the characteristic of a fiber, yarn or fabric whereby it resists wetting by aqueous liquids. *Source:* TM193.

area-of-view, n.—o*f color measuring instrument*, the dimensions of the surface area that a color measuring instrument is capable of covering in a single color measurement. *Source:* EP6.

azoic composition, n.—a physical mixture of an azoic coupling component and a stabilized azoic diazo component which produces, *in situ*, an insoluble azo colorant in both cellulosic and synthetic substrates. *Source:* RA87.

azoic diazo component, n.—a stabilized diazonium salt of a primary acrylamine or derivative thereof which is capable of reacting with an azoic coupling component. *Source:* RA87.

azoic dye, n.—an insoluble azo compound developed, *in situ*, on a substrate by chemically reacting an azoic diazo component (diazotized amine) with an azoic coupling component. *Source:* RA87.

Bacterial Amylase Unit (BAU), n.—a measure of starch degradation as shown by the quantity of an enzyme that will dextrinize one milligram of starch per minute under the specified experimental conditions. *Source:* TM103.

bacterial resistance, n.—*in textiles*, resistance to the development of visible bacterial growth and accompanying odors, resulting from bacterial degradation of fibers or soil on them, as distinguished from musty fungal odors. *Source:* TM174.

ballast, n.—*in procedures for processing or testing of textiles*, material that is used to bring the total weight or volume of the textiles to an amount specified in the procedure. *Source:* TM88B, 88C, 124, 143.

barré, n.—an unintentional, usually repetitive, pattern of continuous bars or stripes parallel to the courses of circular knit fabric. *Source:* M10.

basic dye, n.—a dye that dissociates in an aqueous medium to give a positively charged colored ion (cation) with affinity for fibers containing acidic groups. *Source:* TM141.

black panel thermometer, n.—a temperature measuring device, the sensing unit of which is coated with black designed to absorb most of the radiant energy encountered in lightfastness testing. *Source:* TM16.2, 16.3, 111, 192.

black standard thermometer, n.—a temperature measuring device, the sensing unit of which is coated with a black material designed to absorb most of the radiant energy encountered in lightfastness testing and is thermally insulated by means of a plastic plate. *Source:* TM16.2, 16.3.

bleach, n.—*in home laundering*, a product that will clean, whiten, brighten and aid in the removal of soils and stains from textile materials by oxidation that is inclusive of chlorine and non-chlorine products. *Source:* TM172, 188.

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bleach activator, n.—a bleach agent precursor, which converts a less potent bleaching species into a more powerful one. *Source:* TM190.

bleaching, n.—elimination of unwanted coloring matter from a textile substrate by oxidative or reductive chemical treatment. *Source:* TM81, 98, 102, 144, 209.

blotting paper, n.—any absorbent cloth-like material that is white in color, and will readily absorb small droplets of liquid. *Source:* TM205.

bottom surface (B), n.—*for testing purposes*, the side of the specimen placed down against the lower electrical sensor which is the side of the fabric that would be the outer exposed surface of a garment when it is worn or product when it is used. *Source:* TM195.

breaking strength, n.—the maximum force applied to a specimen in a tensile test carried to rupture. *Source:* TM111, 169, 186, 192.

breathing resistance, n.—the differential pressure required to draw air through a measured surface area at a constant air flow rate. *Source:* EN 14683, an FDA-recognized consensus method for surgical masks. NOTE: The term "breathability" is also associated with this definition, but due to possible confusion with vapor permeability, it is not used in this document. Fabrics designed to specifically allow escape of perspiration in the form of vapor are not typically suitable for face coverings. *Source: M14*.

broad bandpass radiometer, n.—a relative term applied to radiometers that have a bandpass width of more than 20 nm at 50% of maximum transmittance and can be used to measure irradiance at wavelengths such as 300-400 nm or 300-800 nm. *Source:* TM16.1, 16.2, 16.3, 111.

burnt gas fumes, n.—atmospheric oxides of nitrogen as derived from the combustion of illuminating or heating gas. *Source:* TM23.

bursting strength, n.—the force or pressure required to rupture a textile by distending it with a force, applied at right angles to the plane of the fabric, under specified conditions. *Source:* TM111, 169, 186, 192.

calibration procedure, n.—a method for adjusting certain parameters in an instrument so that different instruments from the same or different manufacturers produce the same results in a test for a given property. *Source:* EP11.

carbonizing, n.—a chemical process for eliminating cellulosic matter from a mixture with animal fibers by degrading the cellulosic material to an easily friable condition. *Source:* RA99.

care instructions, n.—*in textiles*, a series of directions describing which care practices should refurbish a product without adverse effects and warning of those care practices expected to have a harmful effect. *Source:* TM188.

carpet, n.—all textile floor coverings not designated as rugs. *Source:* TM121, 122, 138, 165, 171.

carpet backing, n.—materials (fabrics, yarns or chemical compounds) in the carpet other than the face yarns. *Source:* TM205.

carpet composite, n.—the finished carpet product, including: face fibers, primary backing, adhesive coating, and secondary backing (if applicable). *Source:* TM205.

carpet pile brush, n.—a hand operated brush having long, semi-rigid bristles intended specifically for erecting the pile of small areas of carpet (see also *pile lifter*). *Source:* TM171.

cationic dye, n.—a dye that dissociates in an acidified, aqueous solution to give a positively charged colored ion. *Source:* RA87.

cellulase enzyme, n.—an enzyme that attacks cellulose. *Source:* TM191.

center wavelength, n.—the specified wavelength for a bandpass filter; the wavelength midway between the half power points, for example, 340 nm ± 2 nm. *Source:* TM111.

chelating agent, n.—*in textile chemistry*, a chemical capable of deactivating metal ions by formation of a water soluble complex. SYN: *sequestering agent*. *Source:* TM149, 161, 168, 185.

chemical finish, n.—chemical material other than colorants and residual processing chemicals added to textiles to impart desired functional or aesthetic properties to the textile product. *Source:* TM94.

chemical finishing, n.—the process of applying chemicals, other than colorants, to textiles to impart desired functional and/or aesthetic characteristics which may or may not be durable in normal use. *Source:* RA99.

chroma, n.—the proportion of spectrally pure color that expresses the degree of departure from the gray of the same lightness. *Source:* EP9.

chrome dye, n.—a mordant dye capable of forming a chelate complex with a chromium ion. *Source:* RA87.

CIE 1976 L*a*b* equation, n.—a commonly used equation which transforms CIE tristimulus values into a three-dimensional opponent color space. Generally abbreviated as CIELAB. *Source:* TM173.

CIE chromaticity coordinates, n.—the ratio of each of the tristimulus values of a psychophysical color to the sum of the tristimulus values (ASTM E284). *Source:* TM110.

CIE tristimulus values, n.—amounts of three non-real reference color stimuli required to give a color match with the color stimulus considered, and defined by the CIE for the CIE 1931 standard observer and the CIE 1964 standard colorimetric observer and for a particular illumination condition. *Source:* TM110.

CIE Whiteness Index (CIE WI), n.—an index predicting the apparent degree of whiteness of a material based on tristimulus values determined according to one of the CIE standard methods, proposed by the Commission Internationale de l'Éclairage (CIE). *Source:* EP11.

clean-fiber content, n.—the amount of fiber after removal of nonfibrous content. *Source:* TM20A.

cleaning head, n.—a vacuum head modified with spray nozzles for cleaning solution application. Some types include a powered brush unit to facilitate wetting and soil release. *Source:* TM171.

cleanness, n.—*in carpet soiling tests*, the absence of change in appearance due to soil, specifically the degree to which the specimen approaches the original clean, unsoiled condition.

NOTE: Cleanness is independent of changes of physical structure which may be present because of exposure to traffic or action of cleaning procedures. *Source:* TM121.

CMC unit, n.—*in color difference evaluation*, a measure of acceptability expressed in terms of the boundary for the CMC acceptability ellipsoid of $\Delta E_{cmc} = 1.0$.

NOTE: CMC is an acronym for the Colour Measurement Committee (of the Society of Dyers and Colourists) which was largely responsible for the development of the CMC (ℓ:c) equation. *Source:* TM173.

coated fabrics, n.—a flexible material composed of a textile fabric and an adherent polymeric or other material applied to one or both surfaces. *Source:* RA99.

color change, n.—*as used in colorfastness testing*, a change in color of any kind (whether a change in hue, chroma or lightness) *Source:* TM192.

color change, n.—*as used in colorfastness testing*, a change in color of any kind whether in lightness, hue or chroma or any combination of these, discernible by comparing the test specimen with a corresponding untested specimen. *Source:* EP1, 7, TM16.1, 16.2, 16.3, 111, 161, 196.

color measurement, n.—a numerical representation of the color stimulus of an object obtained by use of a color measuring instrument. A single measurement may represent an average of multiple readings of a specimen. *Source:* EP6.

color measuring instrument, n.—any device, such as a colorimeter or spectrophotometer, used to measure the relative amounts of energy reflected from (or transmitted through) a specimen in the visible region of the energy spectrum (comprising the wavelengths from 360 to 780 nm, and including as a minimum the

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region from 400 to 700 nm). *Source:* EP6.

color strength, n.—a measure of the ability of a dye to impart color to other materials.

NOTE: Color strength is evaluated by light absorption in the visible region of the spectrum. *Source:* TM182.

colorant, n.—a material which is applied to a substrate for the express purpose of changing the transmittance or reflectance of visible light.

NOTE: Dyes, pigments, tints and optical brighteners are examples of colorants; soils are not colorants. *Source:* TM140.

colorant bleeding, n.—the loss of colorant from a textile substrate during contact with a liquid medium (such as water, drycleaning solvent or condensed vapors) with consequent coloring of the medium. *Source:* RA93.

colorant staining, n.—the unintended pickup of colorant by a substrate due to a transfer from another material. *Source:* EP2.

colorant staining, n.—the unintended pickup of colorant by a substrate due to (1) exposure to a colored or contaminated liquid medium, or (2) direct contact with dyed or pigmented material from which colorant transfers by sublimation or mechanical action (as in crocking). *Source:* EP8, 12, TM137.

colorfastness, n.—the resistance of a material to change in any of its color characteristics, to transfer of its colorant(s) to adjacent materials or both, as a result of the exposure of the material to any environment that might be encountered during the processing, testing, storage or use of the material. *Source:* EP1, 2, 8, 12, TM6, 8, 15, 16.1, 16.2, 16.3, 23, 61, 101, 104, 106, 107, 109, 116, 117, 125, 129, 131, 132, 133, 157, 162, 163, 164, 165, 169, 172, 188, 190, 196.

colorfastness to light, n.—the resistance of a material to a change in its color characteristics as a result of exposure of the material to sunlight or an artificial light source. *Source:* TM16.1, 16.2, 16.3.

colorist, n.—*in textile coloration*, a person experienced in developing color formulas, evaluating samples for color and producing colored samples to meet standards. *Source:* RA99.

commercial factor (cf), n.—*in color difference evaluation*, a tolerance (specified in terms of ΔE_{cmc} units) which adjusts all axes of the unit CMC volume equally to create a volume of acceptance for commercial use. *Source:* TM173.

commercial laundering, n.—a process by which textile products or specimens may be washed, rinsed, bleached, dried and pressed in commercial laundering equipment, typically at higher temperatures, higher pHs and longer times than used for home laundering. *Source:* TM96.

compatibility, n.—*in textile dyeing*, propensity of individual dye components in a combination shade to exhaust at similar rates resulting in a buildup of shade that is constant, or nearly constant, in hue throughout the dyeing process. *Source:* TM141.

condensed dye, n.—a dye which, during or after application, reacts covalently with itself or other compounds, other than the substrate, to form a molecule of greatly increased size. *Source:* RA87.

constituent elements of hand, n.—those components, qualities, attributes, dimensions, properties or impressions which make the sensation of touching one fabric different from that of touching another. *Source:* EP5.

control paper, n.—*for liquid moisture management testing*, a standard filter paper able to transmit a consistent volume of water vapor under controlled conditions. *Source:* TM204.

copper chelation value (CuCV), n.—the milligrams of copper sulfate pentahydrate chelated by one gram of a chelating agent or product containing a chelating agent. *Source:* TM185.

crease retention, n.—*in fabrics*, the visual impression of an inserted crease quantified by comparison with a set of reference standards. *Source:* TM88C, 143.

crocking, n.—a transfer of colorant from the surface of a colored yarn or fabric to another surface or to an adjacent area of the same fabric principally by rubbing. *Source:* TM8, 116, 165.

cross dyeing, n.—a process of dyeing textiles containing fibers having different dye affinities to achieve a multicolored effect (see also *union dyeing*). *Source:* RA99.

ΔE_{cmc}, n.—*in color difference evaluation*, a single number defining the total color difference in CMC units of a trial from a standard. *Source:* EP14.

ΔE_{00}, n.—*in color difference evaluation*, a single number defining the mean total color difference in CIEDE2000 units of a specimen from a reference and vice versa. *Source:* TM173, EP14.

depth, n.—departure of a colored object from white and frequently associated with either concentration or efficiency of a colorant. *Source:* RA36.

detergent, n.—a cleaning agent containing one or more surfactants as the active ingredient(s). *Source:* RA88.

differential wear—see *frosting*.

diffuse transmission, n.—process by which incident light, while being transmitted through an object, is redirected or scattered over a range of angles. *Source:* TM148, 203.

dimensional change, n.—a generic term for variation in length or width of a garment or fabric specimen subjected to specified conditions. *Source:* TM96, 158.

dimensional change, n.—a generic term for changes in length or width of a fabric specimen subjected to specified conditions. The change is usually expressed as a percentage of the initial dimension of the specimen. *Source:* TM135, 150, 187.

direct dye, n.—an anionic dye having substantivity for cellulosic fibers, normally applied from an aqueous dyebath containing an electrolyte. *Source:* RA87.

disperse dye, n.—an essentially water insoluble dye having affinity, when properly dispersed, for polyester, polyamide and some other manufactured polymeric fibers. *Source:* TM146, 154, 176.

dispersibility, n.—the degree to which particles can be broken down to some minimum size such that they will pass through the interstices of a standard filter paper. *Source:* TM146.

dispersion, n.—*in textile wet processing*, a suspension of very fine particles in a liquid phase. *Source:* TM176.

drycleaning, n.—the cleaning of fabrics with organic solvents such as petroleum solvent, perchloroethylene or fluorocarbon.

NOTE: The process also includes adding detergent and moisture to the solvent, up to 75% relative humidity, and hot tumble drying to 71°C (160°F). *Source:* TM86, 132, 142, 158.

drying rate, n.—the change in volume per unit time a liquid evaporates from a textile.

NOTE: Drying rate is dependent upon textile construction, fiber content, apparel construction, finishes, the atmospheres in which the test is performed, and the volume challenge of the liquid. *Source:* TM199, 200, 201.

drying time, n.—the time it takes for a specified amount of liquid to evaporate from a textile under controlled testing conditions. *Source:* TM199, 200, 201.

dummy load, n.—see *ballast*.

durable press, adj.—having the ability to retain substantially the initial shape, flat seams, pressed-in creases and unwrinkled appearance during use and after laundering or drycleaning. *Source:* TM88B, 88C, 124, 143.

dust, n.—fine particles of solid material dispersed in a gas. *Source:* TM184.

dusting, n.—the tendency of particles of sufficiently low mass to become airborne when a powder material is handled or agitated. *Source:* TM170.

dye, n.—a colorant applied to or formed in a substrate, via the molecularly dispersed state, which exhibits some degree of permanence.

NOTE: Definitions for individual classes of dyes may be found under their particular class name; i.e., reactive dye. *Source:* TM140, 163, 170.

dyeing, n.—*in textile manufacturing*, application of a dye to a substrate nor-

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mally with the intention of obtaining an even distribution throughout the substrate. *Source:* RA99.

electrical resistance, n.—a specific textile's ability to reduce electric current flowing through it at a given voltage. Resistance, *R*, is expressed in units of ohms (Ω). *Source:* EP13, TM210.

electrical resistance, n.—the physical property of a material which is a measure of the ability of electrons to flow through it when a voltage is applied across two points on the material (resistance [in ohms] equal voltage [in volts] divided by current flow [in amperes]). *Source:* TM84.

electrical resistivity, n.—material property of a substance whose numerical value is equal to the ratio of the voltage gradient to the current density. *Source:* TM76.

electronically-integrated textile (e-textile), n.—a fabric or textile product with permanently-integrated electrical circuits or parts of electrical circuits. *Source:* EP13, TM210.

electrostatic clinging, n.—adherence of one substance to another caused by an electrical charge on one or both surfaces. *Source:* TM115.

electrostatic propensity, n.—*of carpets*, the ability to produce and accumulate an electrostatic charge. *Source:* TM134.

endpoint, n.—the point of termination of a drying test that is either to the original dry weight of a specimen or some other agreed value such as dry weight + 4.0% moisture content. *Source:* TM199.

end time, n.—the time at which the temperature goes through an inflection between the steepest slope and the flat sections of the temperature versus time plot. *Source:* TM200, 201.

erythema, n.—abnormal redness of the skin (sunburn) due to capillary congestion (as in inflammation). *Source:* TM183.

extractable matter, n.—nonfibrous material in or on a textile substrate, not including water, which is removable by water, enzyme or a specified solvent or solvents as directed in a specified procedure. *Source:* TM97.

fabric, n.—*in textiles*, a planar structure made from yarns or fibers (ASTM D123). *Source:* TM197, 198.

face, n.—*in textiles*, the side of a fabric that is intended to be the outer visible surface in an end product. *Source:* TM22.

face covering, n.—an item of clothing designed to reduce distribution of the wearer's respiratory droplets (e.g., from breathing, coughing and/or sneezing). (Compare *medical face mask*.) *Source:* M14.

fiber, n.—*in textiles*, a generic term for any one of the various types of matter that form the basic elements of a textile and which are generally characterized by flexibility, fineness and high ratio of length to thickness. *Source:* TM20A.

fiber fragment, n.—a short piece (typically $< 5 \times 10^{-3}$ m long) of textile fiber, broken away (or separated) from a textile construction. NOTE: Fiber fragments are of concern as environmental pollutants; they are often referred to as "microfibers" due to their small size. *Source:* TM212.

fibrillation—see *frosting*.

finish, n.—*other than in preparation and coloration*, the result of application of mechanical energy, thermal energy, or chemical materials, separately, or together, to a textile product to impart durable or nondurable functional or aesthetic properties not otherwise obtainable. *Source:* RA99.

finishing, n.—the process of applying mechanical energy, thermal energy or chemical materials to a textile product (see also *finish*). *Source:* RA99.

fit, for face coverings, n.—the ability to cover the user's nose and mouth while making consistent and snug contact with user's face at the perimeter. *Source:* M14.

flock, n.—*in textiles*, very short fibers intended for application to a substrate as a partial or complete surface covering or filler. *Source:* RA99.

fluidity, n.—*of a cellulose solution*, a measure of the ease of flow or motion of a solution, and therefore an indicator of the molecular weight of the cellulose. *Source:* TM82.

fluorescence, n.—a phenomenon in which radiant flux of certain wavelengths is absorbed and re-emitted at other visible, usually longer, wavelengths. *Source:* EP6.

fluorescent brightener, n.—*see fluorescent whitening agent*.

fluorescent UV lamp, n.—a lamp in which radiation at 254 nm from a low-pressure mercury arc is transformed to longer wavelength UV by a phosphor. *Source:* TM186.

fluorescent whitening agent (FWA), n.—colorant that absorbs near ultraviolet (UV) radiation and re-emits visible longer wavelength radiation. This enhances the whiteness appearance of yellowish materials to which it has been applied. (ASTM E284). *Source:* EP11, TM110.

fluorine content, n.—*in textile floor coverings*, the ratio of the weight of total elemental fluorine to the total weight of carpet fibers. *Source:* TM189.

foam, n.—a dispersion of a gas in a liquid or solid. *Source:* TM167.

formaldehyde release, n.—that formaldehyde exuded from textiles under the accelerated storage conditions of this test including that which is free (unbound or occluded) from unreacted chemicals or from finish degradation as a result of this test. *Source:* TM112.

free formaldehyde, n.—formaldehyde that is not bonded to a finish or a fabric. In this form, formaldehyde can be readily extracted from the fabric by immersion in water. *Source:* TM206.

frosting, n.—*in textiles*, a change of fabric color caused by localized abrasive wear. (Syn: *differential wear*, *fibrillation*.)

NOTE: Frosting may be the result of differential wear, as in multicomponent blends in which the fibers do not match in shade, or of the abrasion of single-fiber constructions in which there is a variation in or incomplete penetration of dye. *Source:* TM119, 120.

fulling, n.—a textile finishing process in which cloth is subjected to moisture, heat, friction and pressure. *Source:* RA99.

fungicide, n.—a compound used to kill fungi. *Source:* TM30.

fungistatic, adj.—Inhibiting the growth of fungi or fungal spores without destroying them. *Source:* TM30.

garment, n.—a shaped article of textile fabric, or other flexible sheet material, intended to cover portions of the human body. *Source:* TM150.

geometry, n.—*of a color measuring instrument*, one of the following terms (diffuse/0, 0/diffuse, 0/45 or 45/0) which describe the angle or manner in which a color measuring instrument:

(1) illuminates the specimen (diffuse, 0°, 45°)

(2) views the resulting reflected light (0 [0°-10°], diffuse, 45°, 0°).

Diffuse/0 and 0/diffuse geometry instruments contain a sphere used to diffuse the light illuminating [or reflected from] the specimen, while 0/45 and 45/0 geometry instruments generally use mirrors or fiber optics to direct the illumination [or viewing] at a 45° to/from the specimen.

NOTE: Instruments of different geometries may produce different colorimetric results on most textile materials. *Source:* EP6.

grade, n.—the number assigned to a test specimen resulting from comparison to a scale. *Source:* EP1, 2, TM88B, 88C, 124, 141, 143, 161.

grade, n.—*in textile testing*, the symbol for any step of a multistep standard reference scale for a quality characteristic.

NOTE: The grade is assigned to test specimens exhibiting a degree of the quality comparable to that step of the standard. *Source:* EP8, TM118, 193.

Gray Scale, n.—a scale consisting of pairs of standard gray chips representing progressive differences in color or contrast corresponding to numerical colorfastness grades. *Source:* EP1, 2, TM161.

NOTE: Gray Scale grades are expressed as half-steps in decimal form (e.g., 1 or 1.5). Dashed form is also acceptable (e.g., 1 or 1-2).

gray scale for staining, n.—a scale consisting of pairs of standard gray chips representing progressive differences in

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color or contrast corresponding to numerical colorfastness grades. *Source:* EP12.

gray scale grade, n.—*for color change, (GSc)*, the numerical value that is assigned to the change in color of a test specimen as compared to an original or untreated specimen. *Source:* EP7.

growth, n.—a dimensional change resulting in an increase in the length or width of a specimen. *Source:* TM96, 135, 150, 187.

half-power bandpass, n.—the interval between wavelengths at which transmittance is 50% of peak transmittance in a bandpass filter.

NOTE: The interval should not exceed 20 nm for a narrow bandpass filter. *Source:* TM111.

hand, n.—the tactile sensations or impressions that occur when fabrics are touched, squeezed, rubbed or otherwise manipulated. *Source:* EP5, TM202.

healthcare personnel (HCP), n—all persons, paid and unpaid, working in healthcare settings whose activities place them at risk for transmission of respiratory infections from patients; *Source:* CDC. *Source:* M14.

healthcare setting, n.—a place where healthcare is delivered. (Compare *public setting*.); *Source:* FDA. *Source:* M14.

horizontal wicking, n.—*in textiles*, the movement of liquid across the horizontal plane of a textile from a site where a specific volume has been applied. *Source:* TM198.

horizontal wicking rate, n.—the change in the area of the liquid with respect to time as the liquid travels through a textile. *Source:* TM198.

hot pressing, n.—a process for smoothing and possibly shaping textile products by applying mechanical pressure with heat, either dry or in the presence of moisture. *Source:* TM133.

hot water extraction, n.—a method of cleaning carpet by injecting a heated cleaning solution into the pile and quickly removing the solution and soil by vacuum (see also *steam cleaning*).

NOTE: Hot water extraction is often erroneously called "steam cleaning." Hot water is used at a temperature of 60 ± 3°C (140 ± 5°F). These temperatures are far below the temperature of steam of 100°C (212°F). *Source:* TM171.

hue, n.—the attribute of visual perception according to which a color field is judged similar to one conventionally named yellow, red, blue, or green or a combination of an adjacent pair of these colors, considered to be arranged in a closed ring. *Source:* EP9.

hydrolyzed formaldehyde, n.—although originally part of larger chemical structure, such as a crosslinking reactant, the compound is cleaved from the larger molecule and dissolved into the aqueous media. *Source:* TM206.

hydrostatic pressure, n.—the force distributed over an area exerted by water. *Source:* TM127, 208.

infrared radiation, n.—radiant energy for which the wavelengths of the monochromatic components are greater than those for visible radiation and less than about 1 mm.

NOTE: The limits of the spectral range of infrared radiation are not well defined and may vary according to the user. Committee E-2.1.2 of the CIE distinguishes in the spectral range between 780 nm and 1 mm:

IR-A	780-1400 nm
IR-B	1.4-3.0 μm
IR-C	3 μm to 1 mm

Source: TM16.1, 16.2, 16.3.

ingrain dye, n.—a colorant which is formed, *in situ*, in the substrate by the development and coupling of one or more intermediate compounds.

NOTE: The term was originally used for colorants obtained from oxidation bases and by azoic techniques, but is now reserved for other types of colorant formed *in situ*. *Source:* RA87.

irradiance, n.—radiant power per unit area as a function of wavelength expressed as watts per square meter, W/m^2. *Source:* TM16.1, 16.2, 16.3, 111, 169, 186.

irradiation, n.—the time integral of irradiance expressed in joules per square meter (J/m^2). *Source:* TM111, 169, 192.

"L" designation, n.—the sequence number given each AATCC Blue Wool Lightfastness Standard according to the number of AATCC Fading Units required to produce a color change equal to Step 4 on the Gray Scale for Color Change. *Source:* TM16.1, 16.2, 16.3, 177.

laboratory sample, n.—a portion of material taken to represent the lot sample, or the original material, and used in the laboratory as a source of test specimens. *Source:* TM111, 192.

langley, n.—a unit of total solar radiation equivalent to one gram calorie per square centimeter of irradiated surface.

NOTE: The internationally recommended units are: Joule (J) for quantity of radiant energy, watt (W) for quantity of radiant power, and meter squared (m^2) for area. The following factors are to be used: 1 langley = 1 cal/cm^2; 1 cal/cm^2 = 4.184 J/cm^2 or 41840 J/m^2. *Source:* TM16.1, 16.2, 16.3.

laundering, n.—*of textile materials*, a process intended to remove soils and/or stains by treatment (washing) with an aqueous detergent solution and normally including subsequent rinsing, extracting and drying. *Source:* LP1, 2, TM61, 88B, 88C, 96, 124, 135, 142, 143, 150, 172, 179, 187, 188, 190, 207.

laundering creases, n.—sharp folds or lines running in any direction in a washed or dried specimen.

NOTE: Laundering creases are an unintended result of restricted movement of specimens in the washer or the dryer. *Source:* TM88B, 88C, 124, 143.

leuco dye, n.—a soluble reduced form of a dye from which the original dye may be regenerated by oxidation (see also *vat dye* and *sulfur dye*). *Source:* RA87.

leuco sulfur dye, n.—a sulfur dye composition consisting of the leuco compound in the presence of a small excess of a reducing agent. *Source:* RA87.

leveling, n.—*in textile dyeing and finishing*, the process leading to more uniform distribution of a chemical or dye within a substrate or between substrates. (compare *transfer*, *migration*.) *Source:* RA99.

light, n.—electromagnetic radiation weighted by the luminosity function of the human eye. *Source:* TM148.

light blocking, n.—the ability of a textile to prevent the transmission of light. *Source:* TM148, 203.

lightfastness, n.—the property of a material, usually an assigned number, depicting a ranked change in its color characteristics as a result of exposure of the material to sunlight or an artificial light source. *Source:* TM16.1, 16.2, 16.3, 125.

lightness, n.—the attribute of visual perception by which a non-self-luminous body is judged to reflect more or less light. *Source:* EP9.

liquid non-chlorine bleach, n.—a product that contains hydrogen peroxide as its active ingredient. *Source:* TM172.

liquid spillage resistance, n.—*for carpet*, the ability to prevent the penetration of liquids through the carpet composite. *Source:* TM205.

mass coloration, n.—a method of coloring manufactured fibers by incorporation of the colorant in the spinning composition before extrusion into filaments. *Source:* RA87.

maximum wetted radius (MWR_T) and (MWR_B) (mm), n.—the greatest ring radius measured on the top and bottom surfaces. *Source:* TM195.

mechanical finishing, n.—the process of applying mechanical energy to textiles to impart functional and/or aesthetic characteristics. *Source:* RA99.

medical device, n.—any item intended for use in the diagnosis of disease or other conditions, or in the cure, mitigation, treatment, or prevention of disease. *Source:* FDA. *Source:* M14.

medical face mask, n.—an item of protective clothing designed to protect portions of the wearer's face, including the mucous membrane areas of the wearer's nose and mouth, from contact with blood and other body fluids during medical procedures. (Compare *face covering*.) *Source:* ASTM F2100. *Source:* M14.

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mercerization, n.—a process for irreversibly altering the physical characteristics and appearance of natural cellulosic fibers by swelling in strong alkali. *Source:* TM43, 89.

metal-complex dye, n.—a dye having a coordinated metal atom in its molecule.

NOTE: Unless the term *metal-complex* dye is used in direct association with a particular application class of dye, e.g., *metal-complex disperse dye* or *metal-complex reactive dye*, its use is inexact and inadvisable. *Source:* RA87.

metal sensitivity, n.—the propensity of a dye to produce an abnormal color on a textile material when in the presence of certain metal ions. *Source:* TM161.

metallizable dye, n.—a dye capable of forming a dye-metal complex either in substance (see *metal-complex dye*) or on the substrate. *Source:* RA87.

metamerism, n.—visual phenomenon according to which spectrally different materials have, for a given observer in a given illumination, matching color appearance. The fields may no longer match when viewed in a different light or by a different observer. *Source:* EP9.

microfiber, n.—a fiber with linear density less than 1 denier or 1 dtex.

NOTE: Polyester microfibers typically have diameter of $< 10^{-5}$ m; this is a frequently referenced dimension, but not the formal definition of a microfiber.

NOTE: The term "microfiber" has been widely adopted as synonymous with "fiber fragment." To avoid any ambiguity, AATCC uses "microfiber" ONLY as a category of linear density. *Source:* TM212.

migration, n.—the nonuniform movement and distribution of dyes, pigments, finishes or other materials from one part of a material to another. *Source:* TM157.

migration, n.—*in textile processing, testing, storage and use*, movement of a chemical, dye or pigment between fibers within a substrate or between substrates due to capillary forces (see also *transfer*). *Source:* TM140.

mildew resistance, n.—*in textiles*, resistance to development of unsightly fungal growths and accompanying unpleasant, musty odors on textile materials exposed to conditions favoring such growths. *Source:* TM30, 174.

moisture content, n.—that part of the total mass of a material that is absorbed or adsorbed water, compared to the total mass. *Source:* TM20A.

moisture management, n.—*for liquid moisture management testing*, the engineered or inherent transport of aqueous liquids such as perspiration or water (relates to comfort) and includes both liquid and vapor forms of water. *Source:* TM195, 204.

moisture retention, adj.—*for this test method*, the percent moisture retention of the specimen after being submerged in deionized water for 1 min, and hanging vertically for 5 min in a controlled environment. *Source:* TM199.

mordant, n.—a substance, usually a metallic compound, applied to a substrate to form with a dye a complex which is retained by the substrate more firmly than the dye itself. *Source:* RA87.

mordant dye, n.—a dye that is fixed with a mordant. *Source:* RA87.

nap, n.—the raised fibrous surface of a fabric obtained by a mechanical brushing or other raising process. (compare *flock*, *pile*.) *Source:* RA99.

narrow bandpass radiometer, n.—a relative term applied to radiometers that have a bandpass width of 20 nm or less at 50% of maximum transmittance and can be used to measure irradiance at wavelengths such as 340 or 420, ± 0.5 nm. *Source:* TM16.1, 16.2, 16.3, 111.

non-chlorine bleach, n.—a product that does not release the hypochlorite ion in solution; i.e., sodium perborate, sodium percarbonate or hydrogen peroxide. *Source:* TM172.

non-diffuse transmission, n.—process by which incident light is transmitted through an object in a rectilinear, straight-through manner, without diffusion. *Source:* TM148.

nonfibrous content, n.—products such as fiber finishes, yarn lubricants, slasher sizing, fabric softeners, starches, china-clay, soaps, waxes, oils and resins which are applied to fiber, yarn, fabric or apparel. *Source:* TM20A.

oil repellency, n.—*in textiles*, the characteristic of a fiber, yarn or fabric whereby it resists wetting by oily liquids. *Source:* TM118.

onium dye, n.—a cationic dye that is solubilized by a labile ammonium, sulfonium, phosphonium or oxonium substituent which splits off during fixation to leave an insoluble colorant in the fiber. *Source:* RA87.

optical brightener, n.—see *fluorescent whitening agent*.

odor, n.—the sensation that results when olfactory receptors in the nose are stimulated by particular chemicals in gaseous form. *Source:* TM211.

overall (liquid) moisture management capability (OMMC), n.—an index of the overall capability of a fabric to transport liquid moisture as calculated by combining three measured attributes of performance: the liquid moisture absorption rate on the bottom surface (AR_B) the one way liquid transport capability (R), and the maximum liquid moisture spreading speed on the bottom surface (SS_B). *Source:* TM195.

oxygen bleach, n.—a bleach agent which can liberate hydrogen peroxide by hydrolysis upon dissolving in water. *Source:* TM190.

parametric factors (l and c), n.—in CMC (l:c) color difference equation, weighting factors used to adjust the component difference ratios, lightness and chroma respectively, based on the effect of the surface characteristics of the stimuli, such as texture. The values l = 2 and c =1 are usually used for commercial pass/fail tolerance predictions of textile samples, although l may be varied depending on the texture of the material. The value of c is always set to one. *Source:* EP14.

parametric factors (kL, kC, and kH), n.—in CIEDE2000 color difference equation, weighting factors used to adjust the component difference ratios, lightness, chroma and hue respectively, based on the effect of the surface characteristics of the stimuli, such as texture. *Source:* EP14.

particle filtration efficiency, n.—the efficiency of the filter material in capturing aerosolized particles, expressed as the percentage of a known number of particles that does not pass the medical face covering material at a given flow rate. Source: ASTM F2299, the FDA recognized consensus standard for surgical masks. *Source:* M14.

percent UV blocking, n.—100 minus the UV transmission. *Source:* TM183.

perfect reflecting diffuser, n.—ideal reflecting surface that neither absorbs nor transmits light, but reflects diffusely, with the radiance of the reflecting surface being the same for all reflecting angles, regardless of the angular distribution of the incident light. (ASTM E284).

NOTE: The perfect reflecting diffuser is the basis of calibration of reflectance measuring instruments. The equations for whiteness and tint are formulated so that the CIE concept of the perfect reflecting diffuser has a whiteness index of 100.0 and a tint value of 0.0. *Source:* TM110.

permanent press, n.—see *durable press*.

personal protective equipment (PPE), medical, n.—clothing, helmets, gloves, face shields, goggles, face coverings and/or respirators or other equipment designed to protect the wearer from injury or the spread of infection or illness. *Source:* FDA. *Source:* M14.

perspiration, n.—a saline fluid secreted by the sweat glands. *Source:* TM15, 125.

pH, n.—the negative logarithm of the effective hydrogen ion concentration or hydrogen ion activity in gram equivalents per liter used in expressing both acidity and alkalinity on a scale whose values run from 0 to 14 with 7 representing neutrality, numbers less than 7 representing increasing acidity, and numbers greater than 7 representing increasing alkalinity. *Source:* TM81, 144, 209.

photochromism, n.—a qualitative designation for a reversible change in

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color of any kind (whether a change in hue or chroma) which is immediately noticeable upon termination of light exposures when the exposed area of a specimen is compared to the unexposed area.

NOTE: The reversion of the color change or instability of the hue or chroma upon standing in the dark distinguishes photochromism from fading. *Source:* TM16.1, 16.2, 16.3.

photometer, n.—an instrument used for measuring light. *Source:* TM148.

pigment, n.—a colorant in particulate form which is insoluble in a substrate, but which can be dispersed in the substrate to modify its color. *Source:* TM140, 176.

pilate, n.—see *pile lifting*. *Source:* TM171.

pile, n.—*in textile fabrics*, yarn loops or tufts protruding from the plane of a fabric, incorporated into the fabric by any means, and which later might be cut, sheared or brushed. *Source:* RA99.

pile lifter, n.—a vacuum cleaning device with motorized rotating brushes designed to erect and agitate carpet pile to facilitate soil removal (see also *carpet pile brush*). *Source:* TM171.

pile lifting, n.—the act of erecting the pile tufts of a carpet with a *carpet pile brush*, *pile lifter* or *pile rake* to allow imbedded soil to be more easily removed and erecting the pile after cleaning to restore a uniform overall appearance. (also called *pilate*.) *Source:* TM171.

pile rake, n.—a hand tool with smooth circular plastic tines used for pile lifting. *Source:* TM171.

pleating, n.—the process of making one or more desirable folds in a cloth by doubling the material over on itself. *Source:* TM131.

pool water, n.—water to which various chemicals are added to maintain purity and clarity, normally used for swimming. *Source:* TM162.

powdered non-chlorine bleach, n.—a product that contains either sodium perborate or sodium percarbonate as its active ingredient. *Source:* TM172.

premetallized acid dye, n.—an acid dye manufactured by reacting an equivalent of a suitable metal ion with one equivalent of a dye (1:1 premetallized acid dye), or with two equivalents of the same or different dyes (1:2 premetallized acid dyes), capable of chelating the metal. *Source:* TM159.

preparation, n.—*in textile manufacturing*, those processing operations performed on greige fabric, colored fabric, textile yarns or fibers to ready them for dyeing, printing or finishing.

NOTE: For example, typical greige cotton fabric preparation includes singeing, desizing, scouring, bleaching and (optionally) mercerizing. *Source:* RA99.

prespotting, n.—*in cleaning*, a pretreatment to remove or enhance removal of soil or stains in a local area (see also *spotting*). *Source:* RA99.

primary backing, n.—*for tufted carpet*, the substrate into which yarns are inserted. *Source:* TM205.

printing, n.—*in textile manufacturing*, a process for applying colorants or other materials to the surface of a substrate, usually to produce a design. *Source:* RA99.

professional wetcleaning, n.—a process for cleaning sensitive textiles (e.g., wool, silk, rayon, linen) in water by professionals using special technology, detergents and additives to minimize the potential for adverse effects. It is followed by appropriate drying and restorative finishing procedures. *Source:* RA43.

public setting, n.—a venue in which people may be in close proximity to others outside their immediate family or household. (Compare *healthcare setting*.) *Source:* M14.

pyranometer, n.—a radiometer used to measure the global solar irradiance or, if inclined, hemispherical solar irradiance. *Source:* TM16.1, 16.2, 16.3, 111.

radiant energy, n.—energy traveling through space in the form of photons or electromagnetic waves of various lengths. *Source:* TM111, 169, 186.

radiant exposure, n.—time integral of irradiance. *Source:* TM111.

radiant flux density, n.—rate of flow of radiant energy past the specimen. *Source:* TM111, 169.

radiant power, n.—energy per unit time emitted, transferred or received as radiation. *Source:* TM16.1, 16.2, 16.3, 111, 169.

radiometer, n.—an instrument used to measure radiant energy. *Source:* TM16.1, 16.2, 16.3, 111.

rating, v.—*in textile testing*, the process for determining or assigning a grade to a material by comparing it to a standard reference scale. *Source:* EP8.

reactive dye, n.—a dye that, under suitable conditions, is capable of reacting chemically with a substrate to form a covalent dye-substrate linkage. *Source:* RA87.

reference fabric, n.—one or more blue wool lightfastness standards selected for exposure as a check on a test apparatus and operating conditions. *Source:* TM111, 192.

reference fabric, n.—a fabric selected to represent a baseline for comparison of tactile properties of hand. *Source:* TM202.

reference plastic, n.—a clear polystyrene plastic standard selected for exposure as a check on a test apparatus and operating conditions. *Source:* TM192.

reference standard, n.—a material that defines the specific color to be matched, and may also be used to define other appearance properties, such as finish, texture and construction. *Source:* EP9.

reflectance, n.—the ratio of the reflected radiant or luminous energy (light) to the incident energy in the given conditions. *Source:* EP6.

reflectance factor, n.—the ratio of the light reflected from the specimen to the light reflected from the perfect reflecting diffuser under the same geometric and spectral conditions of measurement. *Source:* EP6.

regular transmission, n.—process by which incident light is transmitted through an object in a rectilinear, straight-through manner, without diffusion. *Source:* TM203.

relative color strength, n.—*in spectrophotometric testing of dyes*, the percentage of the color strength of a sample relative to that of a reference dye assigned a color strength of 100%. *Source:* TM182.

relative hand value (RHV), n.—an index of properties, relative to a reference fabric, as determined by either an instrumental system or a subjective hand evaluation that can be used to predict tactile sensations a human perceives when handling a fabric. *Source:* TM202.

restoration force, n.—the energy brought to bear on a fabric to accomplish a dimensional restoration. *Source:* TM160.

retained chlorine, n.—*in textiles bleached with chlorine-type bleaches*, available chlorine that remains in the material after washing and drying. *Source:* TM92, 114.

rewetting agent, n.—*in textile preparation, dyeing and finishing*, a surfactant which, after application and drying onto textiles, promotes rapid wetting on subsequent exposure to an aqueous solution. *Source:* TM27.

rhe, n.—the unit of fluidity; the reciprocal of the unit of viscosity (the poise). *Source:* TM82.

rot resistance, n.—*in textiles*, resistance to deterioration of a textile material as a result of fungal growth in or on it.

NOTE: Such deterioration is normally assessed by measuring loss in tensile strength. *Source:* TM30, 174.

rug, n.—a textile floor covering of limited area which is complete in itself and is intended for use as a partial covering of a floor or another floor covering. *Source:* TM137, 165, 171.

rug back, n.—(1) that part of a rug normally in contact with the floor; (2) the underside of a rug as opposed to the use surface. *Source:* TM137.

scouring, n.—*in textile processing*, treatment of textile materials in aqueous or other solutions to remove nature fats, waxes, proteins and other constituents as well as dirt, oil and other impurities. *Source:* RA99.

seam smoothness, n.—*in fabrics*, the

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visual impression of planarity of a seamed specimen quantified by comparison with a set of reference standards. *Source:* TM88B, 143.

seam twist, n.—a rotation, usually lateral, between different panels of a garment connected by a seam.

NOTE: Seam twist may also be referred to as twist, torque, or spirality. *Source:* TM207.

secondary backing, n.—a substrate fabric attached to the back of a carpet with an adhesive binder. *Source:* TM205.

semi-axes (ℓS_L, cS_C, S_H), n.—*in color difference evaluation*, individual dimensions of the CMC volume which are used to calculate a ΔE_{CMC} value. *Source:* EP14.

sequestering agent, n.—see *chelating agent*.

service life, *for textiles*, n.—the maximum number of laundering cycles that can be performed before an item can no longer perform its intended end use. *Source:* M14.

shading, n.—*in colored textile fabrics*, gradual changes in hue, chroma and/or lightness lengthwise or widthwise.

NOTE: When unintended, shading is considered a defect; may be intentional for styling purposes. *Source:* RA99.

shrinkage, n.—a dimensional change resulting in a decrease in the length or width of a specimen. *Source:* TM96, 135, 150, 187.

skew, n.—a fabric condition resulting when filling yarns or knitted courses are angularly displaced from a line perpendicular to the edge or side of the fabric (see ASTM D123). *Source:* TM179.

smoothness appearance, n.—*in fabrics*, the visual impression of planarity of a specimen quantified by comparison with a set of reference standards. *Source:* TM124, 128, 143.

sodium hypochlorite bleach, n.—4-6% solution of sodium hypochlorite (NaOCl), pH 9.8-12.8, commonly called "chlorine bleach." *Source:* TM188, 196.

soil, n.—dirt, oil or other substances not normally intended to be present on a substrate such as a textile material. *Source:* TM121, 122, 130.

soiling, n.—*in textiles*, a process by which a textile substrate becomes more or less uniformly covered with/or impregnated with soil. *Source:* TM121, 122.

soil release, n.—the degree to which a soiled substrate approaches its original, unsoiled appearance as a result of a care procedure. *Source:* TM130.

soil resist agent, n.—a material applied to, or incorporated in, carpet face fiber which retards and/or limits the build-up of soil. *Source:* TM189.

solubilized sulfur dye, n.—a thiosulfuric acid derivative of a sulfur dye which during dyeing is converted to the substantive alkali-soluble thiol form. *Source:* RA87.

solubilized vat dye, n.—a water-soluble salt of the sulfuric ester of a leuco vat dye.

NOTE: After application to the fiber the parent vat dye is regenerated by hydrolysis and oxidation. *Source:* RA87.

solvent dye, n.—a dye which is soluble in organic solvents, but not in water, and is widely used in lacquers, inks, waxes, plastics, soaps, cosmetics, fuels and colored smokes. *Source:* RA87.

specimen, n.—a specific portion of a material or laboratory sample upon which a test is performed or which is selected for that purpose. *Source:* TM111, 192.

speck, n.—a small particle, such as an agglomerate in a liquid dispersion, or a very small spot of dark color on a dyed substrate. *Source:* TM176.

speckiness, n.—*in textile dyeing and printing*, the quality or state of containing specks. *Source:* TM176.

spectral energy distribution, n.— the variation of energy due to the source over the wavelength span of the emitted radiation. *Source:* TM111, 169, 186, 192.

spectral transmittance, n.—the percent of incident radiant energy passing through a given material and not absorbed in the process, as a function of wavelength. *Source:* TM111, 169.

spectrophotometer, n.—an instrument used for measuring the transmittance and/or reflectance of specimens as a function of wavelength. *Source:* TM203.

specular reflection, n.—the reflection without diffusion, in accordance with the laws of optical reflection, as in a mirror. *Source:* EP6.

spill, adj.—The unintended falling of liquid onto a carpet face. *Source:* TM205.

spin finish, n.—a material applied to manufactured fibers by the producer to facilitate subsequent processing.

NOTE: Spin finishes may lubricate yarns, alter hand and bulk, control slippage in wet finishing, maintain package density, etc. *Source:* RA99.

spotting, n.—*in cleaning*, application of solvent or solution to a material prior to or after commercial scouring, laundering, or drycleaning to remove or enhance removal of soil or stains in a local area (see also *prespotting*). *Source:* TM157.

spreading speed, (SS_i), n.—the accumulated rate of surface wetting from the center of the specimen where the test solution is dropped to the maximum wetted radius. *Source:* TM195.

stability, n.—*of dye dispersions*, the resistance to chemical decomposition, physical disintegration, agglomeration or any combination of these. *Source:* RA87.

stain, n.—a local deposit of soil or discoloration on a substrate that exhibits some degree of resistance to removal, as by laundering or drycleaning. *Source:* TM130.

stain, n.—*for pile floor covering*, a discoloration due to a color adding material, such as food or liquid, that exhibits resistance to removal by standard cleaning methods. *Source:* TM175.

stainblocker, n.—a chemical substance which, when applied to a textile substrate, imparts partial or total resistance to staining. *Source:* TM175.

standard atmosphere for storing textiles, n.—laboratory conditions maintained as required in ASTM D1776. *Source:* TM111, 186.

standard atmosphere for testing textiles, n.—air maintained at 21 ± 2°C (70 ± 4°F) and 65 ± 5% relative humidity. *Source:* TM111, 169, 186, 192.

standard depth scale, n.—*in color measurement*, a series of dyed samples of different hue and chroma that have been accepted to have the same depth. *Source:* RA36.

standardization, n.—*of color measuring instrument*, the act of measuring one or more standard materials with a color measuring instrument for the purpose of calculating a set of correction factors to be applied to all subsequent measurements. *Source:* EP6.

start time, n.—the time at which the water challenge is added to the specimen. *Source:* TM200, 201.

steam cleaning, n.—see the correct term, *hot water extraction*. *Source:* TM171.

steaming, n.—*in textile dyeing or printing*, treatment with moist steam to promote penetration and/or fixation of the dye.

NOTE: The steam may be neutral, acid, or air-free depending on the dye class. *Source:* RA99.

stone-washed, adj.—*in garment processing*, a descriptive term denoting alteration of the appearance of a new garment to give it a worn or laundered appearance. SYN: *washdown*. *Source:* RA99.

substrate, n.—*in textiles*, a fiber, fiber assembly, yarn, fabric or film to which another material is applied. *Source:* RA99.

sulfur dye, n.—a dye, containing sulfur both as an integral part of the chromophore and in attached polysulfide chains, normally applied in the alkali-soluble reduced (leuco) form from a sodium sulfide solution and subsequently oxidized to the insoluble form in the fiber. *Source:* TM26.

surface active agent, n.—see *surfactant*. *Source:* RA99.

surfactant, n.—a soluble or dispersible material which reduces the surface tension of a liquid, usually water. *Source:* RA99.

tearing strength, n.—the average force required to continue a tear previously started in a fabric. *Source:* TM111, 169, 192.

test side, n.—the side, either the face or back of a specimen, which is uppermost when placed on the moisture balance analyzer support platform; the side

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onto which the test volume of water is applied. *Source:* TM199.

textile floor covering, n.—an article having a use-surface composed of textile material and generally used for covering floors. *Source:* TM121, 122, 138, 165, 171.

textile product, n.—an article of fabric, or other flexible material, made to protect or decorate the body (garments), for use in the home (bedding, drapery, towels, table linens), or other uses (i.e handkerchiefs, etc.). *Source:* TM79.

thermal finishing, n.—the process of applying heat to textiles to impart desired functional and/or aesthetic characteristics.

NOTE: Thermal finishing includes heat setting, hot embossing, etc. but excludes drying. *Source:* RA99.

thermal fixation, n.—the use of dry heat to achieve a degree of permanence when applying colorants to textile materials. *Source:* TM154.

tint, n.—a colorant, applied to a substrate for purposes of identification, which is easily removed from the substrate by subsequent wet treatment. *Source:* RA99.

tint, n.—*in whiteness measurement*, the hue of a white material as influenced by the wavelength of peak emission or reflectance (CIE 15.2).

NOTE: Tint, if other than zero, is an indication of a reddish or greenish hue having shifted away from a bluish hue with dominant wavelength of 466 nm. *Source:* TM110.

tolerance range specimens, n.—selected specimens which deviate from the reference standard in hue, lightness, chroma, or in combinations of the three, and impose a range of observable color differences around the reference standard for evaluation purposes. *Source:* EP9.

top surface (T), n.—*for testing purposes*, the side of a specimen that, when the specimen is placed on the lower electrical sensor, is facing the upper sensor. This is the side of the fabric that would come in contact with the skin when a garment is worn or when a product is used. *Source:* TM195.

total alkali, n.—*in textile wet processing*, residual alkaline materials in a wet processed textile expressed as percent sodium hydroxide on dry weight of textile. *Source:* TM144, 209.

total irradiance, n.—radiant power integrated over all wavelengths at a point in time expressed in watts per square meter (W/m^2). *Source:* TM16.1, 16.2, 16.3, 169.

total transmission, n.—the amount of light transmitted comprising diffuse plus non-diffuse transmission. *Source:* TM148, 203.

total water content (U) (%), n.—the sum of the percent water content of the top and bottom surfaces.

NOTE: Total water content measurements may be more accurately termed, "total surface water content," particularly in the case of fabric with cellulosic content. Total water content implies that all water in the specimen is being measured which may be the case with some manufactured fabrics. However, when testing cellulosic fibers, moisture trapped in the interior of the fiber (for example, in the lumen of cotton fibers) will not be included with a specimen's detected surface liquid moisture. *Source:* TM195.

transfer, n.—*in textile processing*, *testing, storage and use*, movement of a chemical, dye or pigment between fibers within a substrate or between substrates (see also *migration*, *leveling*). *Source:* TM159, 163.

transmission, n.—process by which incident light is transmitted through an object. *Source:* TM148, 203.

transmittance, n.—*of light*, that fraction of the incident light of a given wavelength which is not reflected or absorbed, but passes through a substance.

transmittance, n.—the ratio of transmitted light to incident light under specified geometric and spectral conditions.

NOTE: Regular transmittance (of transparent materials) is the ratio of undiffused transmitted light to incident light. *Source:* EP6.

transmittance factor, n.—the ratio of the light measured after being transmitted through a transparent or translucent specimen to that measured in the absence of the specimen. In case of liquid samples, the test measurement is usually that of the colored solution in a cuvette and the reference measurement is that of the identical cuvette filled with the solvent used for dissolution (usually distilled water). *Source:* EP6.

ultraviolet protection factor (UPF), n.—the ratio of the average effective ultraviolet radiation (UV-R) irradiance transmitted and calculated through air to the average effective UV-R irradiance transmitted and calculated through fabric. *Source:* TM183.

ultraviolet radiation, n.—radiant energy for which the wavelengths of the monochromatic components are smaller than those for visible radiation and more than about 100 nm.

NOTE: The limits of the spectral range of ultraviolet radiation are not well defined and may vary according to the user. Committee E-2.1.2 of the CIE distinguishes in the spectral range between 400 and 100 nm:

UV-A	315-400 nm
UV-B	280-315 nm
UV-C	100-280 nm

Source: TM16.1, 16.2, 16.3, 111.

ultraviolet radiation, n.—radiant energy for which the wavelengths of the monochromatic components are smaller than those for visible radiation and more than about 100 nm.

NOTE: The limits of the spectral range of ultraviolet radiation are not well defined and may vary according to the user. Committee E-2.1.2 of the CIE distinguishes in the spectral range between 400 and 100 nm:

UV-A	315-400 nm
UV-B	280-315 nm
UV-R	280-400 nm

Source: TM183, 186.

union dyeing, n.—a process of dyeing textiles containing fibers having different dye affinities to achieve the appearance of a uniform, homogenous color (see also *cross dyeing*). *Source:* RA99.

unlevelness, n.—*in textile dyeing and finishing*, nonuniform distribution of a dye or chemical in or on a substrate. (compare *leveling* and *shading*.) *Source:* RA99.

use-surface, n.—*of textile floor covering*, the part of a textile floor covering directly exposed to foot traffic. *Source:* TM121, 122, 138, 165, 171.

vat dye, n.—a water-insoluble dye, usually containing keto groups, which is normally applied to the fiber from an alkaline aqueous solution of the reduced enol (leuco) form which is subsequently oxidized in the fiber to the insoluble form. *Source:* TM176.

verification standard, n.—*in color measurement*, any stable material which is used for the purpose of confirming (or verifying) the validity of an instrument standardization. Color measurements, which are made immediately following standardization, are compared to original measurements of the standard to detect improper standardization. *Source:* EP6.

vertical wicking, n.—*in a textile held vertically*, the upward movement of liquid from a cut edge. *Source:* TM197.

vertical wicking rate, n.—the speed at which liquid travels along or through a textile. *Source:* TM197.

visible radiation, n.—any radiant energy capable of causing a visual sensation.

NOTE: The limits of the spectral range of visible radiation are not well defined and may vary according to the user. The lower limit is generally taken between 380 and 400 nm and the upper limit between 760 and 780 nm (1 nanometer, 1 nm = 10^{-9} m). *Source:* TM16.1, 16.2, 16.3, 111.

volume of acceptability, n.—*in color difference evaluation*, the volume of the ellipsoid obtained when each semi-axis (ℓS_L, cS_C, S_H) is multiplied by cf—creating an agreed volume describing the limits of commercial acceptability for the color difference about a standard. *Source:* TM173.

wand, n.—a tool used to deliver cleaning solution to carpets and apply vacuum to remove the solution.

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NOTE: A *wand* usually consists of an extension handle and a *cleaning head*. Various types are light weight wand, drag wand (heavy duty) and power wand which has a motorized rotating or vibrating part to aid soil release. *Source:* TM171.

washdown, n.—*in processing new fabrics or garments*, a change in appearance to give a worn or laundered look resulting from washing, scouring, chemical, mechanical, or other treatment, including any combination of such treatments (see also *stone-washed*). *Source:* RA99.

washing, n.—*in textile processing*, vigorous treatment with water, usually heated, to remove impurities, chemicals, or dyes in preparation, dyeing, or finishing. *Source:* RA99.

washing, n.—*in testing textile floor coverings*, a specific wet cleaning process involving the use of detergent and scrub brush to remove soil and/or extraneous matter residing in the pile fibers. *Source:* TM138.

water repellency, n.—*in textiles*, the characteristic of a fiber, yarn or fabric to resist wetting (see also *water resistance*). *Source:* TM22, 70, 127, 208.

water resistance, n.—*of fabric*, the characteristic to resist wetting and penetration by water (see also *water repellency*). *Source:* TM35, 42, 127, 208.

water vapor transmission rate, n.—the steady water vapor flow in unit time through unit area of a body, normal to specific parallel surfaces, under specific conditions of temperature and humidity at each surface (ASTM C168). *Source:* TM204.

weather, n.—climatic conditions at a given geographical location including such factors as sunlight, rain, humidity and temperature. *Source:* TM111, 169, 186, 192.

weather resistance, n.—ability of a material to resist degradation of its properties when exposed to climatic conditions. *Source:* TM111, 169, 186, 192.

weight loss, n.—the difference between the saturated weight of a specimen and the weight after drying. *Source:* TM199.

wet pick-up, n.—*in textile processing*, the amount of liquid, and material carried by the liquid, applied to a textile (see also *add-on*). *Source:* TM199.

wet processing, n.—*in textile manufacturing*, a collective term for processes included in preparation, dyeing, printing and finishing in which the textile material is treated with a liquid, normally water, or with chemicals in solution or dispersion in a liquid. *Source:* TM81, 144, 209.

wetting, n.—the time required for a drop of water to lose its light reflection characteristic, i.e., when it changes to a dull, wet spot due to the absorbent propensity of a textile material. *Source:* TM79.

wetting agent, n.—a chemical compound which when added to water lowers both the surface tension of the liquid and its interfacial tension against the solid material. *Source:* TM17, 27, 43.

wetting time (WT_T) (top surface) and (WT_B) (bottom surface), n.—the time in seconds when the top and bottom surfaces of the specimen begin to be wetted after the test is started. *Source:* TM195.

whiteness, n.—the attribute by which an object color is judged to approach a preferred white (ASTM E284).

NOTE: Whiteness as measured by this test method is an indication of how white the textile appears to an average viewer. *Source:* TM110.

wicking, n.—*in textiles*, the movement of a liquid, by capillary action, along or through a material. *Source:* TM197, 198.

wicking distance, n.—the linear measurement that the liquid travels along or through a textile from a starting point to a stopping point. *Source:* TM197, 198.

wicking time, n.—the measurable period during which liquid travels along or through a textile. *Source:* TM197, 198.

wool, n.—*used in the generic sense*, fiber from the fleece of the sheep or lamb, hair from the Angora or Cashmere goat, rabbit hair, and the so-called specialty hair fibers from camel, alpaca, llama and vicuna. *Source:* TM96.

wool oil, n.—a lubricant applied to wool or hair fibers to facilitate subsequent processing. *Source:* RA99.

wrinkle recovery, n.—that property of a fabric which enables it to recover from folding deformations. *Source:* TM66, 128.

yarn lubricant, n.—a material applied to yarn to reduce friction and static generation and thus facilitate subsequent processing. *Source:* RA99.

zone of inhibition, n.—clear area of no growth of a microorganism, cultured onto the surface of agar growth medium, in proximity to the borders of a specimen placed in direct contact with this agar surface.

NOTE: A zone of inhibition occurs as result of the diffusion of an antimicrobial agent from the specimen. *Source:* TM90, 147, 174.

History

Revised in 2021, 2019; numbered and revised 2018; Compiled by AATCC Committee RA99.

Appendix A
References

A.1 *A Dictionary of Textile Terms*, 12th ed., 1976; Dan River Inc., 111 West 40th St., New York NY 10018.

A.2 ASTM D123, Standard Terminology Relating to Textiles; ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832.9500; fax: +1.610.832. 9555; web site: www.astm.org.

A.3 ASTM E284, Standard Definitions of Terms Relating to Appearance of Materials; ASTM International, 100 Barr Harbor Dr., W. Conshohocken PA 19428; tel: +1.610.832. 9500; fax: +1.610.832.9555; web site: www.astm.org.

A.4 *Colour Terms and Definitions*, 2nd ed., 1988; Society of Dyers and Colourists, P.O. Box 244, Perkin House, 82 Grattan Rd., Bradford, West Yorkshire BD1 2JB, ENGLAND.

A.5 *Fairchild's Dictionary of Textiles*, 7th edition, 1995; Phyllis Tortora and Robert S. Merkel, Eds., Fairchild Publications, New York NY.

A.6 *Knitting Dictionary*, 1966; National Knitted Outerwear Association, 51 Madison Ave., New York NY 10010.

A.7 *Dictionary of Fiber and Textile Technology*, 1990; Hoechst Celanese Corp., P.O. Box 32414, Charlotte NC 28231.

A.8 *Textile Terms and Definitions*, 9th edition, 1991; The Textile Institute, 10 Blackfrairs St., Manchester M3 5DR, ENGLAND.

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AATCC Style Guide for Writing Test Methods and Procedures

1. Foreword

1.1 AATCC standards are used in many different ways, and by many different types of users. Hence, it is important that they be written in a precise manner.

1.2 The numerical results obtained by AATCC test methods and procedures may be used by interested parties to establish pass/fail criteria. AATCC policy does not allow the Association to establish such criteria.

2. Purpose and Scope

2.1 This style guide provides instruction for the organization and content of AATCC test methods, evaluation procedures, and laboratory procedures. AATCC monographs are not required to follow the style listed in this document though it may still serve as a useful guide.

2.2 This document includes general guidelines and examples to cover most cases. Committees should consult with AATCC RA99 and/or ECR to address special cases requiring modification of the guidelines.

3. Referenced Documents

3.1 AATCC M9, A Summary of ASTM Methods for Interlaboratory Testing (see 33.1).

3.2 AATCC M11, A Glossary of AATCC Standard Terminology (see 33.1).

3.3 ASTM D1776, Standard Practice for Conditioning and Testing Textiles (see 33.2).

3.4 ASTM D2906, Standard Practice for Statements on Precision and Bias for Textiles (withdrawn, see 33.2)

3.5 ASTM E1402, Standard Guide for Sampling Design (see 33.2).

3.6 *The ISCC-NBS Method of Designating Colors and Dictionary of Color Names*, K. L. Kelly and D. B. Judd, NBS Circular No. 553, 1955, U.S. Government Printing Office, Washington, DC, USA (available at https://archive.org/details/circularofbureau553unse/page/2).

4. Guide for: Sections

4.1 All AATCC test methods, evaluation procedures, and laboratory procedures are required to contain the sections identified in the list below by an asterisk. Standards may also contain the additional sections shown in the list. However, to promote uniform style, the additional sections may only be used in the sequence shown. Titles or headings of the sections may not be replaced with other terms.

4.1.1 The following is the prescribed order for the text of AATCC test methods and procedures:

*Title
Foreword
*Purpose and Scope
*Principle
Referenced Documents
Terminology
*Safety Precautions
Limitations
Apparatus, Reagents, Materials**
Verification, Calibration**
Sampling
*Specimens
Conditioning
Preparation of Apparatus, Reagents**
*Procedure
*Calculations, Evaluation**
*Report (not required for laboratory procedures)
*Precision and Bias (not required for laboratory procedures)
Notes
*History
Appendices
*Required section
**Use appropriate heading

4.2 Using the modified decimal system, number each series of sections and subsections serially and set off these numbers by periods (decimal points) to indicate the particular part of the standard from the largest section, down to the individual paragraphs. For example: "2." indicates the second principal section of a standard; "2.3 and 2.14" designate respectively the third and fourteenth subsections of primary section 2.; similarly, "2.14.10" designates the tenth paragraph of the fourteenth subsection of the second section of the standard.

4.3 Use no longer than three-place numbers. For example: A two-place number means 2.14; a three-place number means 2.14.1 or 2.14.10. This numbering not only shows at a glance the arrangement of the written material, but also permits simple and specific cross referencing. Numbers with more than three places defeat the purpose of simplicity.

5. Guide for: Designation

5.1 The official designation for the standard appears at the top of the first page, above the title.

5.2 All designations begin with "AATCC."

5.3 Designations include a 1- or 2-letter code indicating the type of standard. The words are *not* spelled out in the designation.

TM: Test Method
EP: Evaluation Procedure
LP: Laboratory Procedure
M: Monograph

5.4 Designations include a number (Latin numerals) immediately after the code. There is no space between code and number. Numbers are assigned chronologically for each type of standard. If a standard is withdrawn, that number is not reused unless the standard is reinstated. Numbers may be 1-, 2- or 3-digits. For example, AATCC TM6 or AATCC LP1.

5.5 The code and number are followed by a hyphen and the year the standard was approved or the year of the last technical revision. For example, AATCC TM6-2016.

5.6 If a standard is technically revised a second time in the same year, append the letter "t" to the year. For example, AATCC TM135-2018t was revised in June 2018, following a revision in January 2018.

5.7 If a standard is reviewed and reaffirmed with no changes, append the year of reaffirmation in parentheses. For example, AATCC TM198-2011(2013) was technically revised in 2011 and reaffirmed in 2013. No changes were made in 2013.

5.8 If a standard is editorially revised, append the letter "e." It is not necessary to indicate the year of the editorial revision in the designation. For example, AATCC TM61-2013e was technically revised in 2013 and editorially revised in 2016.

5.9 The same designation format should be used throughout the standard and in any external references to the standard. The portion after the hyphen may be omitted where is it clear that the reference is intended to indicate the most recent version of the standard. For example, AATCC TM6 indicates the most recent version published at the time it is read. AATCC TM6-2016 indicates the version of the standard published in 2016, regardless of when the reader comes across the reference, or how many times it has since been revised.

6. Guide for: Title

6.1 Begin the title with the type of standard. For example, "Test Method

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for…," "Evaluation Procedure for…" or "Laboratory Procedure for…" Monograph titles are not required to begin with the word Monograph.

6.2 Name the property to be measured, not some quality to be inferred. Keep the title explicit and terse. For example: "Strength Loss in Rayon Cloth by Exposure to Sulfurous Acid"; not "Resistance of Fabric to Acid Damage." "Biological Oxygen Demand in Textile Mill Effluents" not "Stream Sanitation."

6.3 To simplify finding standards in alphabetical listing, use key words that describe the general nature of the test or procedure in the beginning of the title, followed by more specific descriptive terms. For example, AATCC TM162, Test Method for Colorfastness to Water: Chlorinated Pool. Review existing standard titles and use similar wording where applicable. For example, "Colorfastness to…" not "Color Change in…"

7. Guide for: Foreword

7.1 A history of the rationale for the development of the standard may be included to help clarify the need for the standard, particularly if similar standards exist. Keep the source of all standards anonymous; individual companies should not be credited for AATCC standards.

8. Guide for: Purpose and Scope

8.1 Name the properties to be addressed (including any characteristics that may be related to those properties.

8.2 Name the material(s) to which the standard is applicable.

9. Guide for: Principle

9.1 Briefly state the technique covered in the standard, outlining the fundamental physical and chemical concepts involved.

9.2 Provide the metric or scale for reporting results.

9.2.1 EXAMPLE: Results are reported as a Gray Scale for Color Change grade of 5 to 1, with 5 representing no color change and 1 representing the most color change.

10. Guide for: Referenced Documents

10.1 Begin the section with instruction to use current publications. Place the statement prior to any numbered subsections.

10.1.1 EXAMPLE: NOTE: Use current versions of all publications unless otherwise specified.

10.2 List all references in alphabetical order.

10.3 List any AATCC, ASTM, ISO or other standard cited, by numerical designation, and title. Include a reference to the note indicating how to obtain the standard. For example, AATCC TM6, Colorfastness to Acids and Alkalis (see 15.1).

10.4 List any articles, chapters, or other publications cited, using the ACS format.

11. Guide for: Terminology

11.1 Define all terms not found in the ordinary desk dictionary, and all terms used in some specialized sense. Define terms that are used only in one restricted branch of the textile industry. If a definition is taken from some other publication, quote it in full and give due credit by complete attribution.

11.2 Define all key terms in titles to ensure that all persons referring to or using the standard understand its intent.

11.3 All terms listed in AATCC standards are compiled in AATCC M11. Review existing terms to avoid duplication or conflicting definitions. Consider if terms will be used beyond the standard. Whenever possible, write definitions that may be used more broadly. If definitions are specific to the standard only, include limitations in italics.

11.3.1 EXAMPLE: **transmittance**, n.—*of light*, that fraction of the incident light of a given wavelength which is not reflected or absorbed but passes through a substance.

12. Guide for: Safety Precautions

12.1 Include a generic caveat on precautions in all test methods and procedures.

12.1.1 EXAMPLE:

X.X The safety precautions specified in the method/procedure are ancillary to the testing procedures and are not intended to be all inclusive.

X.X It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment in handling materials in this standard.

X.X Users MUST consult manufacturers for specific details such as equipment operating instructions and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

12.2 DO NOT include additional guidance regarding OSHA standards, safety glasses, or other precautions covered by the generic caveat.

12.3 If specific precautionary statement(s) are necessary, include them in the body of the test method or procedure. Do not prescribe specific remedial measures and actions. However, the text may reference authoritative sources where reliable information concerning remedial measures can be obtained.

12.4 When a specific precautionary statement(s) exists, reference the appropriate section(s) following the generic caveat.

12.4.1 EXAMPLE: See 6.7 for additional safety precautions related to handling acids.

13. Guide for: Limitations

13.1 State where the results are not useful and where they might be misleading.

14. Guide for: Apparatus, Reagents, Materials

14.1 Separate the subject into two or three separate sections if there will be more than 10 items per section.

14.2 At the beginning of the section(s), reference a text note regarding sources.

14.2.1 EXAMPLE (to appear in Notes section at end of document): For potential apparatus, reagents or materials sources, visit the AATCC Buyer's Guide at www.aatcc.org/bg. AATCC provides its Corporate members the option to list their items and services. AATCC does not qualify, or in any way approve, endorse or certify that any of the listings meet the specifications in its standards.

14.3 For items available directly from AATCC, reference a text note for contact information.

14.3.1 EXAMPLE (to appear in Notes section at end of document): Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

14.4 For items available from *only one known sourc*e, reference a note indicating the source. If multiple sources exist, include only the general reference to the AATCC Buyer's Guide.

14.5 List general name of each apparatus, reagent or material. Include descriptive details following a comma, or in an Appendix.

14.5.1 EXAMPLE: Bell jar, 4 L, with a glass plate base.

14.6 Do not include preparation of reagents or calibration of apparatus in this section.

14.7 Include as apparatus only special equipment that is not in the catalogs of laboratory supply houses, or apparatus that is rarely used. Include spectrophotometers and projection microscopes. Do not include such ordinary items as scissors. Common glassware, such as beakers, burettes, and flasks, need not be listed, but may be listed, if efficacy is enhanced.

14.8 List all reagents—acids, bases, salts, etc.—using chemical names not trade names.

14.8.1 EXAMPLE: sodium hydroxide. (not "caustic soda").

14.9 Include chemical formulae for all reagents. If a complex organic compound is used, the formula may be omitted but the proper name as accepted by the

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Geneva Convention should be used.

14.10 List all dyes using CI numbers.

14.11 Unless otherwise stipulated, assume all chemical reagents are of ACS Reagent quality.

14.12 Include as materials, specialty items like multifiber fabric, standard color swatches, standard photographs of defects and reference spectra.

15. Guide for: Verification, Calibration

15.1 Instruments and equipment must be periodically verified to protect against any drift due to time, wear or accident. Checks may be made by the operator every time a test is performed. Include these checks in the Preparation or Procedure section.

15.2 The check may be rarely performed, as when installing new apparatus or setting up after a major relocation. Infrequent checks belong in a separate Appendix.

15.3 Checks may be performed daily, like checking the zero point of analytical balances. The check may be part of good laboratory management that is performed as weekly or monthly routine. Include in this section verifications that are part of laboratory routine and which should be put on a regular schedule.

15.4 Include only verification steps that are applicable to all suitable apparatus. For machine-specific verification or calibration, reference manufacturer's instructions.

15.5 Besides mechanical adjustments, include calibration curves, standard curves and verification of normality (or molarity) of standard solutions.

16. Guide for: Sampling

16.1 Include a generic caveat on sampling in all test methods and procedures. Include ASTM E1402 in the Referenced Documents section.

16.1.1 EXAMPLE:

X.X Test results are valid only when the samples are statistically representative (see ASTM E1402).

X.X Sampling must be random. Every unit of product must have a mathematically equal chance of becoming a sample; and every portion of each sample must be equally likely to become a test specimen.

X.X All specimens must be alike within the variations due to pure chance. There must be no differences within samples assignable to known causes.

17. Guide for: Specimens

17.1 Describe the size, shape and weight of specimens. Include any limitations on the location or choice of test material.

17.2 Indicate the preferred number of specimens per sample.

17.3 Include a generic statement regarding the number of specimens and a reference to the Precision subsection of the standard if desired.

17.3.1 EXAMPLE: Additional specimens may be used to increase the precision of the average (see Precision).

17.4 Include all preparatory or preliminary steps for specimens such as trimming edges, pulling threads to trace weave patterns or attaching multifiber fabric swatches.

18. Guide for: Conditioning

18.1 If specimens or other materials must be conditioned for testing in the standard atmosphere for textiles (or other atmospheric conditions), indicate the conditions and other details. Refer to ASTM D1776 for complete conditioning information. In most cases, a general summary of conditioning parameters (see 18.1.1) is sufficient. For testing moisture-sensitive properties, additional details or reference to ASTM D1776 may be required.

18.1.1 EXAMPLE: Condition specimens [and/or other materials] for at least 4 h in an atmosphere of 21 ± 2°C (70 ± 4°F) and 65% ± 5% RH by laying each specimen [and/or other materials] separately on a screen or perforated shelf of a conditioning rack.

18.2 If testing should be performed in the standard atmosphere for textiles, note this in the Procedure section.

19. Guide for: Preparation of Apparatus and Reagents

19.1 Include all preparatory steps for reagent preparation and apparatus set-up (e.g., leveling and adjusting). Do not include steps already described in Verification or Calibration sections. Reference, but do not repeat steps included in an Appendix. Include only those steps that are applicable to all suitable apparatus. For machine-specific verification or calibration, reference manufacturer's instructions.

20. Guide for: Procedure

20.1 Procedures are operating instructions for technicians and must permit trained technicians, working in separate laboratories, to obtain results that are comparable within agreed limits, with little or no other guidance.

20.2 State all operating instructions clearly, simply and unequivocally. Leave no room for differing techniques. Recite every needed detail in proper sequence.

20.3 Write procedures in the second person imperative format.

20.3.1 EXAMPLE: Cut specimens… (not "Specimens shall be cut…")

20.4 If the container is a critical item, name it.

20.4.1 EXAMPLE: a 250-mL wide-mouthed Erlenmeyer flask.

20.5 If time is important, state narrow limits.

20.6 Specify temperature (and other measures) objectively.

20.6.1 EXAMPLE: water at 140-160°F (not "hot water.")

20.6.2 EXAMPLE: water at room temperature (not "rinse in cold water.")

20.7 Include tolerances (or minimum/maximum) for all measurements.

20.7.1 EXAMPLE: add 10.00 ± 0.02 mL (not "add exactly 10 mL.")

20.8 Indicate the number of significant figures to record in observations. Be sure the accuracy is mathematically justified.

20.9 When two equally acceptable procedures give statistically interchangeable results, spell out each procedure and state that either may be used.

21. Guide for: Calculation, Evaluation

21.1 Calculation involves straightforward computation without exercise of opinion. Include as calculation all requisite algebra and arithmetic. State exactly what is to be computed and how. State the number of significant figures required in all calculations.

21.2 For subsequent reference in texts, number each equation in a sequence by "Eq." and a numeral in parentheses at the right margin of the line.

21.3 Indicate multiplication by parentheses (( )) and division by the solidus (/).

21.3.1 EXAMPLE: $x = 100\ (A - B)/C$ *(Eq. 1)*

21.3.2 EXAMPLE: $y = 100\ (0.00587)\ (A - B)/C$ *(Eq. 2)*

21.4 Keep formulas on one line of typescript if possible.

21.5 Use symbols in current use.

21.5.1 EXAMPLE: For x is less than or equal to y, write: $x \leq y$

21.6 Restrict equations to symbols and numerals.

21.6.1 EXAMPLE:

$x = 100\ (A - B)/C$ *(Eq. 1)*

where:

x = iron as Fe, percent of oven dry specimen weight.

21.7 Keep all numerals at the left side of expressions that contain both numerals and letters.

21.7.1 EXAMPLE: $x = 100\ (0.00587)\ (A - B)/C$

21.8 Do not condense chemical or physical equations. It makes checking of computations difficult.

21.8.1 EXAMPLE $x = 100\ (0.00587)\ (A - B)/C$ (not $0.587\ (A - B)/C$

21.9 Confirm the position of the decimal point in all decimal fractions by putting a zero in the units column.

21.9.1 EXAMPLE: $a = 0.3010B$

21.10 When a mathematical development requires two or more lines, place

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the statements in column, repeating only the equality sign.

21.10.1 EXAMPLE:

$n = (t^2)(v^2)/(e^2)$ *(Eq. 3)*

$= (1.96^2)(7.5^2)/(5^2)$

$= 8.6$

$= 10$ (to the nearest higher multiple of 5)

21.11 Show the basis of percentage statements.

21.11.1 EXAMPLE: Moisture content, % of conditioned weight.

21.12 Include a sample computation whenever an equation is long, involved or in the least degree difficult to reduce to English.

21.13 Use the heading "Evaluation" in place of or in addition to "Calculation" when the results are expressed in descriptive form, relative terms or as abstract values. Such results may be expressed in terms of a 5 to 1 rating scale with 5 being best and 1 being worst.

22. Guide for: Report

22.1 Require the tested sample to be described or identified.

22.2 Require that the standard be cited, including the option or test used if there is a choice.

22.3 Indicate the testing conditions to be reported.

22.4 Indicate the test results to be reported.

22.4.1 Include a sample work sheet or report form if data collection, reporting or calculation may be misunderstood based on the written text.

22.4.2 Most reports containing numerical data should include, as a minimum:

(a) Arithmetic mean or average (X)

(b) The number of tests (n)

(c) Standard deviation (s) or coefficient of variation ($\%CV$)

Statement of a mean without the number of tests and precision is essentially useless.

22.4.3 AATCC policy prohibits endorsement of specifications or pass/fail criteria. The ratings suggested in some of the standards serve as a guide; but they are not intended for and must not be construed as pass/fail criteria.

22.5 Require any modifications of the standard to be reported.

22.6 EXAMPLE:

X. Report

X.1 Describe or identify the sample tested.

X.2 Report that the sample was tested using AATCC TM15-2018.

X.3 Report the testing conditions:

X.3.1 Wet pickup (%).

X.3.2 Evaluation procedure for color change (AATCC EP1 or AATCC EP7).

X.3.3 Evaluation procedure for staining (AATCC EP2, AATCC EP8, or AATCC EP12).

X.4 Report the test results:

X.4.1 Color change grade for tested specimen.

X.4.2 Staining grades for each fiber type in the multifiber fabric and undyed adjacent fabric, if used.

X.5 Describe any modification(s) of the published standard, including use of an alkaline perspiration solution.

23. Guide for: Precision

23.1 *Precision.* A statement on precision allows potential users of the test method to assess in general terms its usefulness in proposed applications. A statement on precision is not intended to contain values that can be exactly duplicated in every user's laboratory. Instead, the statement provides guidelines as to the kind of variability that can be expected between test results when the method is used in one or more reasonably competent laboratories, and when the test method's use is in statistical control. No valid statement can be made about precision unless the use is in statistical control.

23.2 Refer to AATCC M9 or ASTM D2906 for details on conducting interlaboratory studies and analyzing data for a precision statement. File the data obtained in the interlaboratory study and the detailed analysis of the data at the AATCC Technical Center.

23.3 Every test method should contain a statement (1) about the precision of test results obtained in the same laboratory under specifically defined conditions of *within-laboratory* variability, and (2) about the precision of test results obtained in different laboratories. The specifically defined *within-laboratory conditions* may concern test results obtained on the same material by the same operator using the same equipment within a short period of time, or *within-laboratory* precision may be reported for other specific conditions; for example, between days or between operators. Describe the particular within-laboratory variability for which precision is reported in detail. The statement regarding *between-laboratory* variability must pertain to test results obtained in different laboratories on the same material.

23.3.1 Any new AATCC test method that produces data shall contain, as a minimum, a statement of single operator precision when first submitted to committee and TCR ballots, but full *within-laboratory* and *between-laboratory* precision statements are encouraged.

23.3.2 At its first 5-year review, any AATCC test method that produces data shall be brought into full conformance, including *within-laboratory* and *between-laboratory* precision statements prior to submission to committee and TCR ballots.

23.3.3 It is the policy of AATCC that no longstanding test methods will be dropped due to the lack of a precision statement.

23.3.4 In any AATCC test method that produces data in which more than one testing option is allowed, a precision statement based on the most used option satisfies the requirement. A committee may include precision for the other options, and is encouraged to do so, especially single laboratory precision for each option available.

23.4 If the test result data are continuous variates, give the standard deviation or coefficient of variation, whichever is appropriate, and the applicable components of variance for each type of precision that is reported. In any case, the precision statement shall give the 95% critical difference for *within-laboratory* and for *between-laboratory* test data. If the precision is not the same for all materials, then give the precision for each material used in the interlaboratory test that was used to obtain the precision measurements.

23.5 If the precision varies with the test level, describe this variation.

23.6 Include other related information that may help users assess the degree of applicability of the statement to the materials of interest to them. It may be desirable to note the presence of other types of variability in test results on which information can be derived by supplemental studies.

23.7 Many rating scales in AATCC methods are limited and not continuous. If analysis of variance is used with such data, statistical errors may occur and the possibility should be noted in the precision statement. A subjective basis for evaluation the precision of test results may be offered as an alternative.

23.7.1 EXAMPLE:

X.1 *Interlaboratory Test Data*—An interlaboratory test was run in 20XX in which randomly-drawn samples of 2 materials were tested in each of 5 laboratories. Each laboratory used 2 operators, each of whom tested 4 specimens of each material. Calculation of components of variance was thought to be inappropriate due to the restricted and discontinuous rating scales, the non-linear relationships between the rating scales and color difference units, and the increased variability in color difference units as the true value of the ratings decrease.

X.2 *Precision*—Based on the observations described in X.1 and on general practice in the trade, a lot or consignments is generally considered as having a rating that is significantly worse than a specified value when a specimen from the lot or consignment has a rating for (insert here the name of the property) that is more than one-half step below the

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specified rating on the Gray Scale for Color Change.

23.8 *Precision Statement Format.* The precision statement included in a test method must contain three basic sections: (1) a brief description of the testing plan from which the data set was derived, citing the number of materials tested, the number of laboratories participating, the number of operators per laboratory, the number of tests per operator, and any other pertinent information; (2) a listing of the components of variance derived from the data set; and (3) a listing of the precision parameters calculated from the components of variance, usually in the form of critical differences, or confidence intervals.

23.9 The required statement of precision shall contain the information specified above or an explanation as to why a statement is not practicable. The absence of a statement on precision is not warranted if the reason is that an interlaboratory test has revealed that the precision is poor.

23.9.1 EXAMPLE: A precision statement is not applicable because data are not generated by this procedure.

23.9.2 EXAMPLE: Precision and Bias statements are not applicable because data are not generated by this test method.

23.10 Test methods that do not contain full precision statements should note the status and explanation.

23.10.1 EXAMPLE: Precision for this test method has not been established. Until a precision statement is generated for this test method, caution should be used when testing materials with this method. In most cases the use of standard statistical techniques in making any comparisons of test results for either *within-laboratory* or *between-laboratory* averages have been found to be generally accepted.

23.10.2 EXAMPLE: *Between-laboratory* precision has not been established for this test method. Until such precision information is available, users of the method should use standard statistical techniques in making any comparison of test results for *between-laboratory* averages.

24. Guide for: Bias

24.1 A statement on bias furnishes guidelines as to whether the test method can be used when comparisons with accepted reference values are to be made. If the bias is known, the method can be modified to include a correction for the bias, and thus the corrected method would be without known bias.

24.1.1 EXAMPLE: *Bias.* The <property name> can be defined only in terms of a test method. There is no independent method for determining the true value. As a means of estimating this property, the method has no known bias.

24.2 If bias varies with the test level, describe the bias.

24.3 Any statement on bias shall describe the bias and how the method has been modified to provide corrected test results. If bias cannot be determined, include a statement to this effect.

24.4 File the data and details of the experiment to determine bias at the AATCC Technical Center.

24.5 For test methods that do not generate data, the statement on bias should state this.

24.5.1 EXAMPLE: *Bias.* A bias statement is not applicable because data are not generated by this test method.

25. Guide for: Notes

25.1 Notes should contain only explanatory matter, never any mandatory detail of performing the test or procedure.

25.2 Number notes in sequence as they appear in the text.

25.3 Place notes in a separate section at the end of the method, before the appendices. *Exception:* Notes that are part of tables belong with the tables.

26. Guide for: History

26.1 State the last date revised and a brief description of the revision. Indicate any subsequent reaffirmations or editorial changes in the same subsection.

26.2 List all revisions, editorial, revisions, and reaffirmations, in reverse chronological order. Where applicable, indicate title changes in parentheses.

26.3 Where applicable, list the year in which the AATCC standard was adopted by ISO.

26.4 State the number of the committee that developed the standard and the first year of publication. If jurisdiction was transferred to another committee, include the year and committee in the same section.

27. Guide for: Tables

27.1 Use tables to avoid repetition in the text.

27.2 Number tables in sequence by Roman numerals.

27.3 Place tables in proper sequence in the text, not in the appendices.

27.4 Head each table with a terse explicit title, in title case.

27.5 Head each column of figures with a pertinent legend.

27.6 Below the table, add all necessary notes. Identify the notes by lower case letters. Put the corresponding letter in the body of the table within parentheses.

28. Guide for: Figures

28.1 Title every line drawing or photograph, using sentence case and ending with a period.

28.2 Number figures and photographs in sequence, with Arabic numerals.

28.3 Preferably, use line drawings, with all lettering and the figure drawn to double scale. Drawings usually are clearer than photographs, and have the advantage of showing dimensions and interior surfaces. The printer will size the illustration to fit the printed page.

28.4 Use photographs of 300 dpi or more. All photographs will be printed in grayscale. Ensure that critical features are visible.

28.5 Place figures and photographs in proper sequence in the text, not in an appendix.

29. Guide for: Appendices

29.1 Include in appendices, supplementary information that might be needed but which is too voluminous to put in the text for fear of interrupting the sequence of thought.

29.2 The following are typical information for appendices:

Flow charting of test methods.

Glossaries of special terms.

Lists of chemical or mathematical symbols.

Detailed descriptions of apparatus of specialized nature.

Verification or calibration procedures.

Derivations of mathematical equations.

Charts and monographs.

Report forms.

30. Guide for:Flow Charts (Optional Appendix Item)

30.1 A flow chart is not a replacement for a written test procedure, but is a diagram illustrating the flow and logic of any process, or system (i.e., test procedure, interlaboratory studies, etc.). It is a drawing made up of various, but specific, shapes of boxes connected by directional straight lines. Flow charting (1) gives a bird's-eye view of the flow and logic of a procedure in a pictorial fashion, (2) facilitates communication and understanding among individuals having different areas of expertise, (3) makes more obvious some of the strong and weak points of a system, (4) helps to generate ideas and provides a basis of discussion for improving a system, and (5) aids in the writing of new standards.

30.2 The box shapes of a proper flow chart are simple, but specific. The start and stop of a flow chart are denoted by a horizontal ellipse, with the appropriate word, START or STOP, within the ellipse. A process step, where no decision (yes or no) is required, is denoted by a

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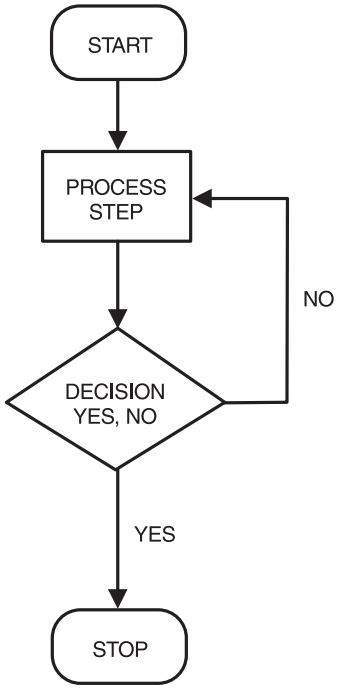


Fig. 1—Sample flow chart.

rectangle. A process box usually has just one output arrow. If a decision is required, a diamond is used and two output arrows are required (yes or no). Fig. 1 is an example of a flow chart.

30.3 Flow charts should be used to illustrate major pieces of equipment, all inputs to processes (i.e., raw materials, utilities, etc.), all points where measurements are taken, and all points where adjustments can be made. The step number within the test method should be in the appropriate box or diamond of the flow chart.

31. Manuscript Format

31.1 Submit all manuscripts to the AATCC Technical Center in electronic format, Word document, doubled spaced.

31.2 Designation of Colors.

31.2.1 In presenting textile color data, use the method of color nomenclature developed by the Inter-Society Color Council and the National Bureau of Standards, commonly referred to as the ISCC-NBS method. See *The ISCC-NBS Method of Designating Colors and Dictionary of Color Names*.

31.2.2 Identify colors by the three attributes of hue, lightness and chroma. Do not use the words shade, tint, depth of color or intensity. For example: rose and vermillion are hues of red, not tints or shades.

31.3 Units of Measurements.

31.3.1 Use metric or SI units wherever feasible. Use English units only where they are the accepted trade practice.

31.3.2 Do not mix systems in one paragraph.

EXAMPLE: DO NOT USE "a specimen 2 × 3 in. and weighing 2.0-2.5 g."

31.3.3 State measurements in both systems when both are in current American usage. Round off conversions to comparable accuracy.

EXAMPLE: 1.0 in. becomes 2.5 cm, not 2.54 cm; but 1.00 in. is converted to 25.4 mm not 25 mm.

31.3.4 For liquid measure and for volumetric glassware, use milliliter (mL), not the obsolete (cc) cubic centimeter. For measures of capacity or volume, use the cubic centimeter, for which the accepted abbreviation is cm^3.

31.3.5 Laboratory procedures require metric measurement and Celsius temperatures. Fahrenheit temperature is permissible for physical testing. Mill and dyehouse practices govern units for process control, even if they are obsolescent or obsolete, such as Baumé, Twaddell, Brix and Rohmer in place of specific gravity hydrometers.

31.3.6 For cases when it is necessary to express measurements when range and accuracy must be included.

EXAMPLE: weigh to within ± 0.001 g a specimen with weight in the range of 4-6 g.

31.4 Numbering.

31.4.1 Use numbers instead of words in every case, unless confusion will ensue.

EXAMPLES: 4 samples; 5 days; 15.43 g.

31.4.2 Designate tables by Roman numerals, and figures or drawings by Arabic numerals.

EXAMPLES: Table IV; Fig. 6.

31.4.3 Do not begin a sentence with a numeral.

31.4.4 Use decimal fractions.

31.4.5 Place a zero before decimal numbers to ensure that no digit has been omitted or misplaced.

EXAMPLE: 0.36 cm (not ".36 cm.")

31.4.6 Point off numbers in excess of 4 digits with commas in the text (1,234,567) but with spaces in tabulated material (1 234 567). Do not point off numbers of 4 digits except when they occur in columns containing numbers of more than 4 digits.

31.5 Spelling

31.5.1 In general, use the preferred spelling in the Merriam-Webster *International Dictionary.*

31.5.2 Use American forms.

EXAMPLES: "color" not "colour," "liter" not "litre."

31.5.3 Hyphenate compound adjectives, particularly such forms as "2-g specimens."

31.6 Punctuation

31.6.1 Use a comma in a series of words but not before a conjunction.

EXAMPLE: wash, dry and condition specimens.

31.6.2 Always place commas and periods *inside* of quotation marks.

31.6.3 Place semicolons and colons *outside* quotation marks.

31.7 Capital Letters

31.7.1 When in doubt, use lower case letters.

31.7.2 Capitalize the principal words in headings and titles of standards, names of books or papers. Use lower case for prepositions and conjunctions in titles.

31.7.3 Use initial capital "C" for "committee" when used as a title, but not when used as a collective noun.

EXAMPLES: "Committee RA60" or "Committee on Industrial Pollution"; but "the committee recommends."

31.7.4 Use capital letters in referring to tables, figures, inserted plates and volumes.

EXAMPLES: Table III, Fig. 2, Plate VI, Vol. 25.

31.8 Abbreviations

31.8.1 Use abbreviations in the singular number only. An exception is abbreviations preceding numerical values.

EXAMPLES: "2 in." not "2 ins; Figs. 1 and 2, Vols. I and II, Nos. 1 and 2.

31.8.2 Use abbreviations only after numbers denoting a definite quantity, except in tabulations.

EXAMPLES: Do not say "mix rinse liquor in a *bbl.*" nor, "wash in H_2O."

31.8.3 Abbreviate "percentum" or "percent" with the symbol (%), which is far easier to read.

31.8.4 Do not use the symbol (#) to mean "pounds" or "number."

31.8.5 Put a period (.) after an abbreviation only when the omission would cause confusion.

EXAMPLES: "in." not "in" for inches, "no." not "no" for number, "Fig." not "Fig" for figure, and "vol." not "vol" for volume.

31.8.6 A list of acceptable conventional abbreviations is shown in Appendix A.

31.8.7 Besides the conventional abbreviations, frequently repeated series of words may be abbreviated if the words are spelled out before the abbreviation at the first use.

EXAMPLES: TNT (trinitrotoluene); BOD (biochemical oxygen demand); TCE (trichloroethylene).

31.8.8 Use chemical symbols only to represent chemical entities, never as abbreviations.

EXAMPLES: "water" not H_2O unless water is used as a reagent. Write "rinse in water," not "rinse in H_2O; say "platinum crucible," not Pt crucible."

31.8.9 Spell out the names of unusually complex or of uncommon organic or inorganic reagents if the chemical for-

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mula will waste space or obscure clarity of text.

EXAMPLES: $CuSO_4 \cdot 5H_2O$ rather than cupric sulfate pentahydrate. "Aspirin" is simpler than "acetylsalicylic acid" and certainly simpler than to write the structural formula.

31.8.10 Do not, as a general rule, abbreviate terms that are infrequently used without spelling out the entire word or phrase the first time the expression appears.

32. History

32.1 Revised in 2020 to include Designation and History sections, along with various updates and clarifications.

32.2 Numbered July 2018. Revised January 2018.

32.3 Developed by AATCC Committee RA99.

33. Notes

33.1 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

33.2 Available from ASTM International, www.astm.org.

Appendix A
Abbreviations

absolute	abs
alternating current (noun)	ac
(adjective)	a-c
ampere	amp
angstrom unit	Å
anhydrous	anhyd
average	avg
barrel	bbl
Birmingham wire gage	Bwg
British thermal unit	Btu
Brown and Sharpe (gage)	B&S
calorie	cal
Celsius	°C (Note 1)
centigram	cg
centimeter	cm
centipoise	cp
centistoke	cs
chemically pure (discontinued)	CP
coefficient of variation	*%CV*
Colour Index	C.I.
concentration	conc.
cubic centimeter (volume)	cm^3 (Note 2)
day	Spell out
degree	deg (Note 3)
diameter	diam
direct current (noun)	dc
(adjective)	d-c
equation	Eq
Fahrenheit	°F (Note 1)
figure	Fig.
foot	ft
foot-pound	ft-lb
formula weight	FW
gallon	gal
grain	gr
gram	g
horsepower	hp
hour	h
inch	in.
inside diameter	ID
Kelvin	K (Note 1)
kilocycle	kc
kilogram	kg
kilojoule	kJ
kilometer	km
kilovolt	kV
kilowatt	kW
kilowatt-hour	kW-h
linear	lin
liter	L
logarithm (common)	log
lux	lx
maximum	max.
meter	m
microamperes	µa
microgram	µg
microliter	µL
micromicron	µµ
micron (micrometer)	µm
microvolt	µV
microwatt	µW
mile	mi
miles per hour	mph
milliampere	ma
milliequivalent	meq
milligram	mg
milliliter	mL (Note 2)
millimeter	mm
millivolt	mV
minimum	min.
minute	min (Note 4)
molal	*m* (Italic)
molar	*M* (Italic)
molecular weight	MW
nanometer	nm
Newton	N
normal	*N* (Italic)
number	No.
ohm	ohm or
on weight of fiber	owf
ounce	oz
ounces per square yard	oz/sq yd
outside diameter	OD
page	p
pages	pp
parts per billion	ppb
parts per million	ppm
per	Spell out or use solidus (/)
percent	%
pint	pt
pound	lb
pounds per square foot	lb/sq ft
pounds per square inch	psi
quart	qt
radian	rad
relative humidity	RH
revolutions per minute	rpm
second	s
solution	soln.
specific gravity	sp gr
square	sq
standard deviation	*s*
ton	T
Twaddell	Tw
United States (wire gage)	US
United States Pharmacopeia	USP
volt	V
volume	vol.
watt	W
watt-hour	W-h
week	wk
yard	yd
year	yr

Notes

Note 1. Always use the abbreviation for the temperature scale °F, °C or K in statements of numerical temperatures, but omit the abbreviation "deg" for "degrees." Write "69°F," not "69 deg F."

Note 2. Use the abbreviations "cm^3" rather than "cc" as the unit of capacity. Use "mL" for milliliter" as the unit of volume.

Note 3. Restrict the use of the degree symbol (°) to °API, °BRIX, °Baumé, °Twaddell, °C, °F, °Rohmer and to degrees of angle or of arc.

Note 4. Spell out "minute" and "minimum" if there is any danger of confusion by use of the abbreviation "min."

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Rules of Procedure for AATCC Test Method and Technology Committees

Developed by AATCC Committee C2; revised 2017; numbered 2018; editorially revised 2019.

Introduction

The Board of Directors is AATCC's governing body which administers its affairs under the provisions of the Constitution and Bylaws. AATCC Committees are responsible for nearly every phase of the Association's activities. Thus, the strength, vigor and authority of the Association are derived from the broad participation of its membership.

These Rules of Procedure are the responsibility of the Executive Committee on Research (ECR) which may amend or change items included after consultation with the Technical Committee on Research (TCR) and subject to the Board of Directors approval as long as they do not conflict with the AATCC Constitution or Bylaws.

A—Executive Committee on Research

A1. Function

A1.1 ECR is a standing committee of the Board of Directors. This Committee shall be responsible for establishing policy for, and programming of, all test method development and technical activities not specifically delegated to other special interest groups, subject to the Board of Directors approval.

A1.2 ECR directives for the execution of policy and program as approved by the Board of Directors are carried out through TCR.

A1.3 ECR has the responsibility to make sure that the test method development procedure follows the consensus principle and that any negative votes are fairly considered and resolved.

A1.4 ECR is the supervisory body within AATCC charged with the responsibility to attest that the necessary procedures have been followed and that the approval requirements set forth in these procedures have been met. The role of ECR in overseeing the development procedures and handling of negative votes at the committee level is covered in G1.1 through G1.4. Letter ballots of TCR are reviewed by a special Reviewing Committee consisting of the Chairs of ECR, TCR and the Test Method Committee concerned (G2).

A2. Organization

A2.1 The Chair of ECR is a member of the Board of Directors of the Association, appointed annually by the Board of Directors. The Chair may not serve more than three years consecutively.

A2.2 The Chair of TCR becomes Vice-Chair of ECR.

A2.3 The President of the Association, immediate past president, president elect, Chairs of interest groups, Chair of the International Test Method Committee, Chair of the Education Advisory Board, Education Director and the Executive Vice President are members ex-officio.

A2.4 In addition, six senior members are appointed by the Board of Directors, two new appointments being made each year for terms of three years each.

A2.5 Members of ECR shall represent a broad spectrum of activities and interests and are appointed in a manner to maintain a balance of interest. No more than 50% of ECR shall represent any one interest area.

A2.6 The Technical Director serves as Secretary of ECR.

A3. Reports

A3.1 ECR reports to the Board of Directors through its Chair.

A3.2 In cooperation with the TCR Chair, the Chair of ECR prepares an annual report for publication in the AATCC *Manual of International Test Methods and Procedures*.

B—Technical Committee on Research

B1. Function

B1.1 TCR is a subcommittee of ECR. It translates into action program and policy developed by ECR. It functions in an administrative and coordinating capacity and is responsible for directing the work of the Test Method, Technology, and Reference Committees.

B1.2 The Chair of TCR, subject to ECR approval, authorizes technical and research assignments to the AATCC Technical Center following requests for assistance by Test Method and Technology Committees.

B2. Organization

B2.1 The Chair of TCR is appointed annually by the Board of Directors after nomination by ECR. The Chair's term is limited to three consecutive years.

B2.1.1 Subject to the approval of ECR and the Board of Directors, the Chair nominates a Vice Chair and the members of TCR.

B2.1.2 Regional Board Members, members of the Executive Committee on Research, and Chairs of the active Research Committees shall be ex officio members of TCR.

B2.1.3 Appointed members of TCR may serve up to five years; this term may be extended by majority vote of ECR.

B2.2 The Chair of TCR appoints, subject to ECR approval, members of Research Committees and Chairs of these committees after consulting with their members.

B2.3 The Technical Director serves as Secretary of TCR.

B3. Meetings and Reports

B3.1 TCR shall hold at least two meetings each year called by the Chair.

B3.2 TCR through its Chair reports its activities to ECR.

B3.3 In cooperation with the ECR Chair, the Chair of TCR prepares an annual report for publication in the AATCC *Manual of International Test Methods and Procedures* by January 31 of each year.

C—Test Method Committees

C1. Function

C1.1 Test Method Committees are appointed to carry out specific technical projects leading to the establishment of AATCC Test Methods, the development of scientific data or the assembly of information for the benefit of the textile industry and in the public interest.

C2. Organization

C2.1 The organization of new committees or the development of proposed new test methods is announced in AATCC's publication, printed or electronic and other media of the press. Non-members as well as AATCC members are invited to participate in these activities. Non-members are given every opportunity to voice their opinion and are invited to become members and can subsequently have a voting interest. Non-members often provide the basis or request for a new test method to be developed.

C2.1.1 Members of a Test Method

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Committee are appointed by the Chair subject to the approval of the Chairs of TCR and ECR.

C2.1.2 Non-members may participate fully at the subcommittee (initial) stage of test method development including interlaboratory tests and voting.

C2.1.3 A tentative completion date for development of any proposed new method must be set by the Test Method Committee and approved by ECR.

C2.2 Test Method Committee Chairs are appointed after the Chair of TCR has consulted with the Test Method Committee members concerned. Consultation may be by voice vote of members present at a regularly scheduled meeting or by letter ballot of all voting members. They must be Senior members of AATCC.

C2.2.1 The Chair of a Test Method Committee appoints the Chairs of its subcommittees.

C2.2.2 A Test Method Committee Chair may appoint a Steering Committee, subject to the approval of the Chair of TCR.

C2.2.3 A Test Method Committee Chair shall appoint a Secretary.

C3. Committee Officers

C3.1 Senior members of AATCC only may hold office in Test Method Committees and Subcommittees; i.e., Chair, Vice-Chair or Secretary.

C3.2 No member may be Chair of more than two active Research Committees at one time.

C3.3 A member may not serve as Chair of a Test Method Committee for more than three years consecutively, but may continue as a member.

C3.4 An ex-Chair may again become Chair after a lapse of one year.

C4. Voting

C4.1 Senior members of AATCC only may exercise a vote in AATCC Test Method Committees.

C4.2 When additional expertise is required, committee chairs may invite an associate or non-member of the Association to serve as members of a Test Method Committee without vote, for a specified term.

C4.3 A Senior member may be a voting member of three active Test Method and three Technology Committees at the same time, but may serve on additional committees as a non-voting member. Exceptions to this may be granted by the Executive Committee on Research upon written petition from the member.

C4.4 Only one individual from any organization may be a voting member of any one test method committee. Other individuals from the same organization may participate in committee activities as non-voting members.

C5. Reports

C5.1 Following each committee meeting a Test Method Committee shall present a written report to TCR. The annual report may serve as a basis for the Committee's report in the AATCC *Manual of International Test Methods and Procedures*. A copy of the annual report should be forwarded to the Technical Director on or before January 31.

C6. Meetings

C6.1 Test Method Committee meetings shall be open to members of AATCC and non-members.

C6.2 The schedule of committee meetings shall be published in AATCC's journal and/or electronically such as on the web site.

C6.3 There shall be no registration fee for attendance to anyone (member or non-member) at these committee meetings.

C7. Publicity

C7.1 Members of the press may attend Test Method Committee meetings. They should be advised by the Chair, during the course of the meeting, that any deliberations or actions of the Committee shall not be published without clearance from the Chair, who should be guided by Association policy regarding publication of AATCC material.

Technical papers, committee reports and other documents and test methods are the property of the Association. They may not be published in full elsewhere unless written permission has been granted by the Association. Publication of Association material in other publications must give appropriate source and credit.

C8. Committee Actions

C8.1 Formal actions of a Test Method Committee shall be affirmed by letter ballot when requested by any voting member of a committee.

C9. Review of Test Methods

C9.1 A Test Method Committee shall review test methods under its jurisdiction every five years for the purpose of revision, reaffirmation or withdrawal. If a method is to be reaffirmed, reaffirmation shall be by majority approval of committee members or by a committee letter ballot followed by a TCR letter ballot. Such action must be initiated in the fourth year after approval so that final action may be concluded by the fifth year.

In cases where the Chair of a Reference Committee is no longer active or resigns from their duties in AATCC, Committee RA99, Technical Manual Editorial Review, in conjunction with the Laboratory Director, may appoint one or two experts who are knowledgeable in the area pertaining to a specific test method as reviewers of the method.

The reviewers shall recommend reaffirmation, revision or withdrawal of the method and shall initiate the appropriate TCR letter ballot.

C9.1.1 A new test method will be reviewed for reaffirmation each year for the first three years immediately following its publication through a TCR ballot.

C9.2 Workshops and seminars concerning AATCC test methods are available to non-members and members equally. At these programs, critical comments on AATCC test methods are brought to the attention of the responsible committees for their consideration and action.

D—Duties of a Test Method Committee's Chair

D1. Responsibility

D1.1 The Chair of a Test Method Committee is responsible for the operation of that Committee.

D2. Meetings

D2.1 The Chair shall call such meetings as are necessary and see that the committee's work proceeds as planned.

D2.2 In general, meetings are scheduled in conjunction with meetings of TCR in order that the latest activities of the Test Method Committee may be reported. However, the Chair may call additional meetings at the Chair's discretion.

D3. Personnel

D3.1 The Chair appoints members of a Test Method Committee, subject to the approval of the Chairs of TCR and ECR.

D3.2 Every effort is made to have a balance of voting interests on each committee in order to develop test methods according to the consensus principle.

D3.3 If a member of a Test Method Committee resigns or is unable to take an active part in the work of the Committee and is dropped the Chair shall notify the Chair of TCR and the Laboratory Director.

D3.4 A voting member who fails to return three consecutive ballots may be dropped from the committee's membership rolls. A terminated member may appeal to ECR for reinstatement if the member demonstrates satisfactory reasons for the inactivity. Committee membership can also be terminated by nonpayment of annual Association dues.

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D4. Test Methods—Changes and Revisions

D4.1 A Chair on behalf of their Committee makes such recommendations for changes and revisions in test methods as are required and reports them to the Chair of TCR, who will in turn, submit these recommendations to ECR. Subject to ECR approval these recommendations for changes or revisions shall be referred to TCR for letter ballot.

D5. Financing

D5.1 Each calendar year Test Method Committees will be contacted regarding their possible needs for funds to be made available at the beginning of the fiscal year, January first.

D5.2 Should a committee require special funds not included in the annual budget for its work, the Chair should make application to the Chair of TCR who will submit the request to ECR with a recommendation. Should there be contingent provision in the existing budget to cover such expenditures, ECR may authorize the funds. Otherwise, ECR, on approval will request an authorization for expenditures from the Appropriations Committee, which, in turn, will submit the request to the Board of Directors.

D6. Requests for Assistance from AATCC Technical Center

D6.1 Technical and administrative assistance from the AATCC Technical Center, required for committee operations, should be requested by the Test Method Committee Chair from the Chair of TCR. The Chair of TCR, in consultation with ECR and Senior Staff, will determine if the work requested can be undertaken within the existing budget and program facilities available.

E—Duties of a Test Method Committee's Secretary

E1. Minutes and Records

E 1.1 The Secretary shall keep accurate minutes of all meetings and records of all test method developments of the Committee.

E1.1.1 Minutes prepared for distribution should contain on the first page the legend "**Committee Document—Not for Publication**."

E1.2 Minutes shall be circulated at an early date after a meeting. Copies shall be sent to the AATCC Technical Center, all committee members, the Chair of TCR, the Technical Director, and any other desired information addresses.

E1.3 The Secretary shall keep records of attendance at meetings.

E2. Correspondence and Reports

E2.1 At the direction of the Chair, the Secretary shall carry on such correspondence and prepare such reports as are required by the Committee.

F—Procedure for Establishing Methods of Test

F1. Need and Feasibility

F1.1 Proposals for the development of new test methods may be made by members or non-members of AATCC. They are brought to the attention of the Chair of TCR who in consultation with members of ECR determines whether the development of a proposed new method is feasible through an existing Test Method Committee. If the organization of a new committee is necessary, ECR may approve an announcement in AATCC's publication inviting members and non-members of AATCC to participate in an exploratory meeting to determine general interest in the subject. If there is sufficient interest, ECR may approve the establishment of a new committee that will function as indicated under Section C.

F2. Development Procedure

F2.1 A subcommittee which may include non-members of AATCC is organized to study available background information and to conduct interlaboratory tests according to a suitable statistical design. When the subcommittee has developed a test method which is considered reproducible and technically valid it is submitted along with all data available to the full Test Method Committee.

F2.2 A letter ballot of the Test Method Committee is conducted on the proposed test method according to G1. Upon receipt of an affirmative vote and evidence of consideration of all negative votes, a letter ballot of TCR is conducted on a proposed test method according to G2. The results are reported to ECR for final action.

F3. Designation and Publication

F3.1 ECR gives final approval or denial on a proposed test method on the basis that all substantially concerned persons have had an opportunity to express their views and that these views shall have been considered; also, that there is evidence of the technical validity of the proposed test method.

F3.2 Following confirmation by ECR a new test method is officially designated and published as soon as practical in the AATCC *Manual of International Test Methods and Procedures* and in other AATCC publications when appropriate. Research committee chairs may request that an approved test method be made available as soon as practical via electronic format prior to publication of the AATCC *Manual of International Test Methods and Procedures* by written request to the ECR secretary.

G—Letter Ballots

G1. Committee Letter Ballots

G1.1 A majority of members present at a meeting of a Test Method Committee may vote to send a new test method or revision of an existing test method to a letter ballot of the entire Committee. Before requesting a letter ballot, however, those committee members present at the meeting should assure themselves that the new method or revision of an existing method is in prescribed form and that sufficient data are in hand to support the method or revision. All members of a Test Method Committee, both voting and non-voting, shall receive copies of all letter ballots issued by the Committee. Nonvoting members as well as voting members of the Committee may submit in writing any comments which they deem appropriate concerning the proposed test method or revision. To be counted, all letter ballots must be received by the committee concerned within 30 days of the mailing date. A ballot is considered valid when, of the voting members sent ballots, the number of ballots returned is equal to, or greater than, 50% of the mailing plus one ballot.

G1.2 If there are no negative votes on the ballot it would progress to a TCR letter ballot. If there are negatives then refer to Section G3.

G2. TCR Letter Ballots

G2.1 Once a committee letter ballot has been approved within the committee then it will progress to a TCR letter ballot. The ballot will be sent to all members of TCR. To be counted, all letter ballots must be received by the committee concerned within 30 days of the mailing date. A ballot is considered valid when, of the voting members receiving ballots, the number of ballots returned is equal to, or greater than, 50% of the mailing plus one ballot.

G2.2 If there are no negative votes on the ballot then it would be reviewed by ECR for publication in the next edition of the AATCC *Manual of International Test Methods and Procedures*. If there are negatives refer to G3.

G3. Negatives on Ballots

G3.1 Within 60 days after the closing date for the Committee or TCR letter ballots the Chair of the Test Method Committee shall attempt to resolve any nega-

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tive votes or comments, either personally or by a subcommittee which the Chair may appoint. If at the end of the 60 day period there are still unresolved negative votes, the Chair of the Committee shall forward to the Chair of TCR the matter on which the ballot was taken, copies of unresolved negative votes with accompanying supporting comments, a summary of attempts to resolve the negative votes, and background data and actions which led to the ballot.

G3.2 The Chair of TCR shall present the material provided by the Committee Chair to ECR for decision at its next regular meeting. Copies of the material shall be distributed to members of ECR prior to the meetings or as early as possible to facilitate thorough study and consideration.

G3.3 After studying the ballot and the background information provided, ECR shall take one of the following actions.

G3.3.1 Return the matter to the Test Method Committee for further work and another ballot if ECR decides that material and data submitted are insufficient to justify the action taken by the Test Method Committee.

G3.3.2 Approve a TCR letter ballot on the basis of the Test Method Committee majority or approve the publication on the basis of TCR majority if ECR reasons that the unresolved negative votes and comments are not persuasive and do not justify further investigation.

G3.4 Should the Test Method Committee recommend that a revision of the questions be put to letter ballot, a new ballot shall be circulated, and again, until no material objections exist.

G3.5 Resolution of all negative ballots must be recorded.

G4. Disqualifications and Challenges

G4.1 Members of ECR and the Board of Directors shall disqualify themselves when appeals are made on test methods if they or a member of their immediate family have financial interest in or other close relationship with any of the parties (or with the subject matter) that would make it inappropriate for them to participate in the consideration and decision of any appeal. All parties to an appeal shall have the right to challenge the qualification of the appeals board members, with justification based on the above stated reasons, at least one week prior to consideration of an appeal by ECR or the Board of Directors who are disqualified from a specific appeal shall absent themselves from the argument, deliberations and decisions.

H—Correspondence and Records

H1. Copies

H1.1 Copies of all correspondence, reports and research records should be forwarded to the Chair of TCR and the Technical Director in addition to any other desired information addresses.

H2. Files

H2.1 Test Method Committee files in the hands of Chairs and Secretaries shall be turned over promptly to their successors. The files that are no longer active but are of historical value and interest should be forwarded to the AATCC Technical Center for permanent record.

I—Reference Committees

I1. Function

I1.1 When a Test Method Committee has completed its assignment, it may assume an inactive status and function as a Reference Committee with approval of TCR and ECR.

I2. Organization

I2.1 The Chair of the Test Method Committee shall continue as Chair of the Reference Committee. The Chair shall continue the responsibilities of the committee in terms of reviewing methods requiring reaffirmation. If changes to the method are necessary the Chair should reactivate the committee.

I2.1.1 The Chair may retain such members of the Test Method Committee as he deems advisable, otherwise, the remainder of the committee is dissolved. Should a Reference Committee be reactivated, the Chair shall act as temporary Chair until the committee is reorganized.

I2.2 After its first meeting, a reactivated Test Method Committee shall submit to the Chair of TCR its recommendations for Chair.

I3. Review of Test Methods

I3.1 Reference Committees shall follow the same review procedure for test methods as active Research Committees (see C9).

I3.2 In cases where the Chair of a Reference Committee is no longer active in AATCC, see C9.1.

J—Technology Committees

J1. Scope

J1.1 To coordinate and direct the planning of technical programs of special interest to individuals and groups concerned with technology subjects; to recommend and encourage presentation of these programs as a part of the regular program of technical papers, and to promote in every way increased interest and attendance at the technical sessions of AATCC.

J1.2 To encourage and promote active participation in the research and test method development programs of the Association by members whose primary interest is in the technology subject in order that problems in that area of technology may be part of the program of research of the Association.

J1.3 To act as a clearinghouse for information on the technology subject for members of AATCC.

J1.4 To develop a consensus by polling AATCC members whose primary interest is in the field of the technology subject.

J1.5 To promote the interests of the Association whenever the opportunity may arise in accordance with the Committee's objectives.

J1.6 To encourage and solicit membership in the Association on the part of individuals interested in the technology subject.

J2. Organization

J2.1 Members of a Technology Committee are appointed by the Chair subject to the approval of the Chair of TCR and ECR. Members of the Committee shall be selected to provide representation from all Sections of AATCC which have sufficient interest in the technology subject.

J2.2 The Committee shall have a Chair, and may have a Vice-Chair. The names of the individuals to be recommended to the Chair of TCR for appointment to these offices, subject to the approval of ECR, shall be determined by vote of the Technology Committee and submitted to the Chair of TCR.

J2.2.1 A Chair of a Technology Committee shall appoint a Secretary from the membership of the Committee.

J2.2.2 A Chair of a Technology Committee shall appoint the Chair of such subcommittees as are necessary to carry out its program.

J2.2.3 A Chair of a Technology Committee may appoint a Steering Committee, subject of the approval of the Chair of TCR.

J2.2.4 A Chair of a Technology Committee may not serve in that capacity for more than three years consecutively, but may continue as a member of the Committee. An ex-Chair may again become Chair after a lapse of one year.

J2.3 Only Senior members of AATCC may hold office in Technology Committees and subcommittees, and Senior members only may exercise a vote. Associate members may serve without vote on a Technology Committee. No more than

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one member of an organization may be a voting member of any one Technology Committee.

J3. Committee Actions

J3.1 Formal actions of a Technology Committee shall be affirmed by letter ballot when requested by any voting member of the committee. Procedures for balloting including resolution of negatives shall be the same as test method committees.

J4. Meetings

J4.1 A Technology Committee shall call such meetings as are necessary and see that the committee's work proceeds as planned.

J5. Reports

J5.1 Following each committee meeting a Technology Committee shall send a written report to the Chair of TCR and the Technical Director. A Technology Committee shall submit an annual report to TCR. A report shall serve as a basis for the Committee's report to be published in the AATCC *Manual of International Test Methods and Procedures*. A copy of the annual report should be forwarded to the AATCC Technical Center for publication purposes.

J6. Publicity

J6.1 A Technology Committee shall publish announcements of its meetings and activities in the Association's publication.

J6.2 The Committee shall maintain an up-to-date mailing list of both members of AATCC and prospective members who are interested in the technology subject, and shall advise such interested individuals regarding AATCC technical sessions where papers on the technology subject will be presented; notify such interested individuals by mail of the Committee members who can be contacted for information or help in problems of the technology subject; and solicit from such individuals recommendations for membership on the Technology Committee.

J6.3 The interested Sections of AATCC will be encouraged to include in their Sectional or Program Committees, a member of that Section who is a member of the Technology Committee, in order that the officers of the Section may be kept informed of the work of the Technology Committee and the availability of technical programs of interest to the members of the Section concerned with the technology subject.

J6.4 Technical papers, committee reports and other committee documents are the property of the Association. They may not be published in full elsewhere until they have appeared in AATCC's publication or unless permission has been granted by the AATCC. Publication of Association material in other publications must give appropriate source and credit. For publicity purposes, when immediate release of news is desirable, abstracts, digests, summaries of reports must be limited to 500 words or less, unless written permission is granted by the Association.

J7. Financing

J7.1 Each calendar year Technology Committees will be contacted regarding their possible needs for funds to be made available at the beginning of the fiscal year, January first.

J7.2 Should a Committee require special funds not included in the annual budget for its work, the Chair should make application to the Chair of TCR who will submit the request to ECR with a recommendation. Should there be contingent provision in the existing budget to cover such expenditures, ECR on approval will request through the Board of Directors an authorization for expenditures from the Appropriations Committee.

K—AATCC Technical Center

K1. Function

K1.1 The AATCC Technical Center furnishes supporting services, through its staff and laboratory facilities, to the Association's committees.

K1.2 The functions of the laboratory at the AATCC Technical Center are as follows:

a. To serve as a bureau of standards for test methods, a place where AATCC methods can be run according to every detail of the test procedure as outlined.

b. To serve as a proving ground for test methods developed by various test method committees of AATCC. In doing so, the test methods will be conducted and the results analyzed statistically to make sure that they are reproducible in every sense.

c. To serve as a focal point in the organization as a site for demonstrating and developing test methods, for conducting workshops and seminars for the education of our members and industry at large.

d. To serve as a place in which test equipment necessary to conduct many AATCC test methods is on exhibit as well as a central location for equipment which might be of interest to AATCC members.

K2. Organization

K2.1 The AATCC Technical Center located in the Research Triangle Park of North Carolina provides a permanent organizational structure for the Association.

K2.2 The Executive Vice President is responsible for the overall management of the AATCC Technical Center.

K2.3 The Technical Director supervises the technical and research assignments authorized by ECR.

K3. Clearance of Assignments

K3.1 No assignment can be accepted by the AATCC Technical Center unless it has been approved by ECR. The AATCC Technical Center operations are intended to supplement and support AATCC Committees through preplanned co-ordination in an effort to provide maximum service within the limitations of the budget.

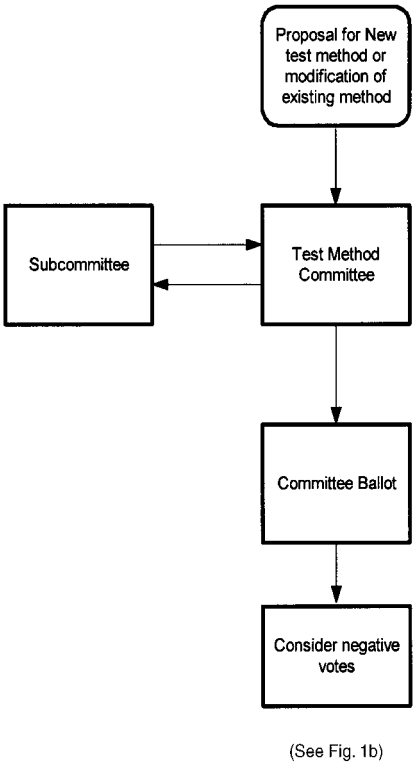


Fig. 1a—AATCC Test Method Development.

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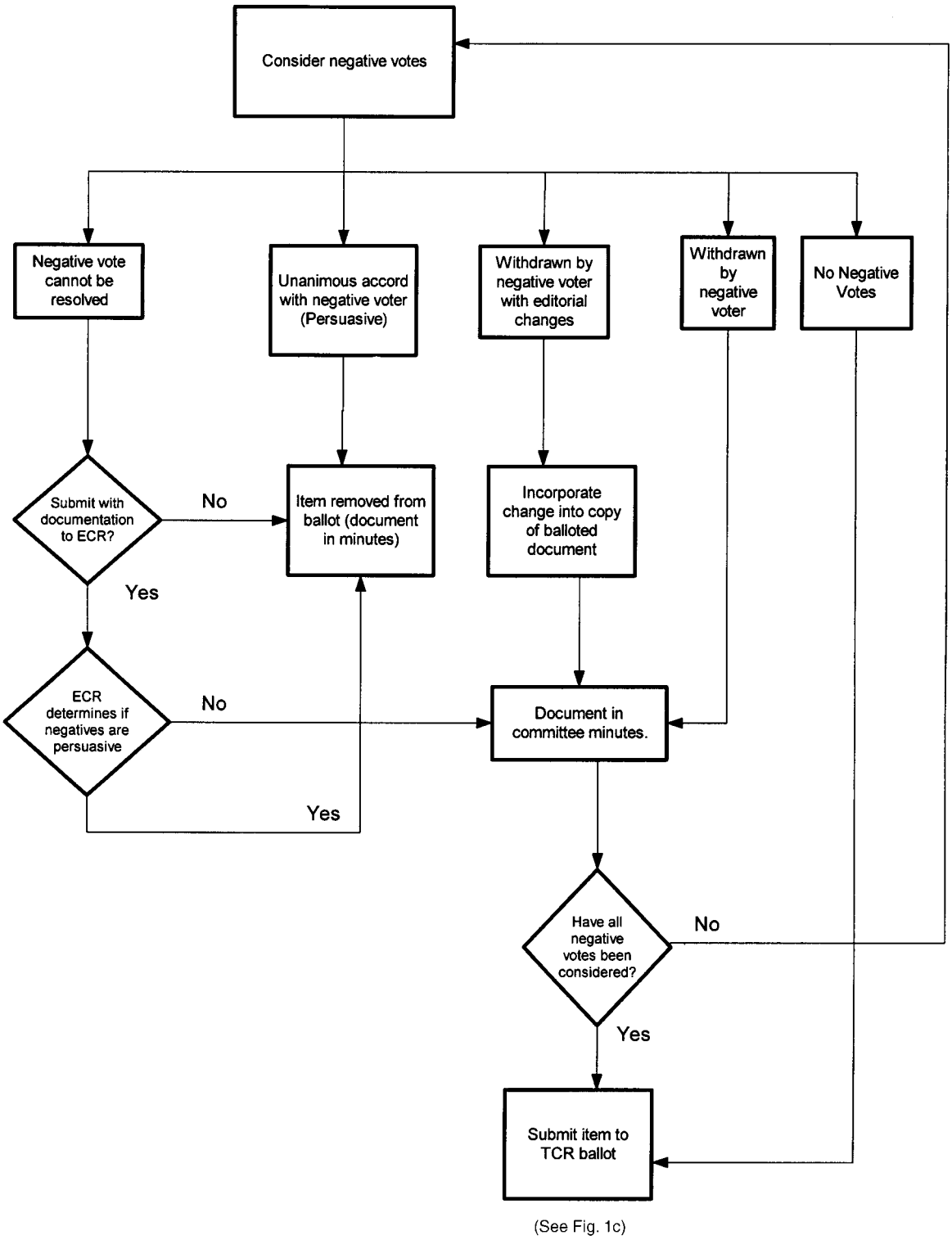


Fig. 1b—AATCC Test Method Development.

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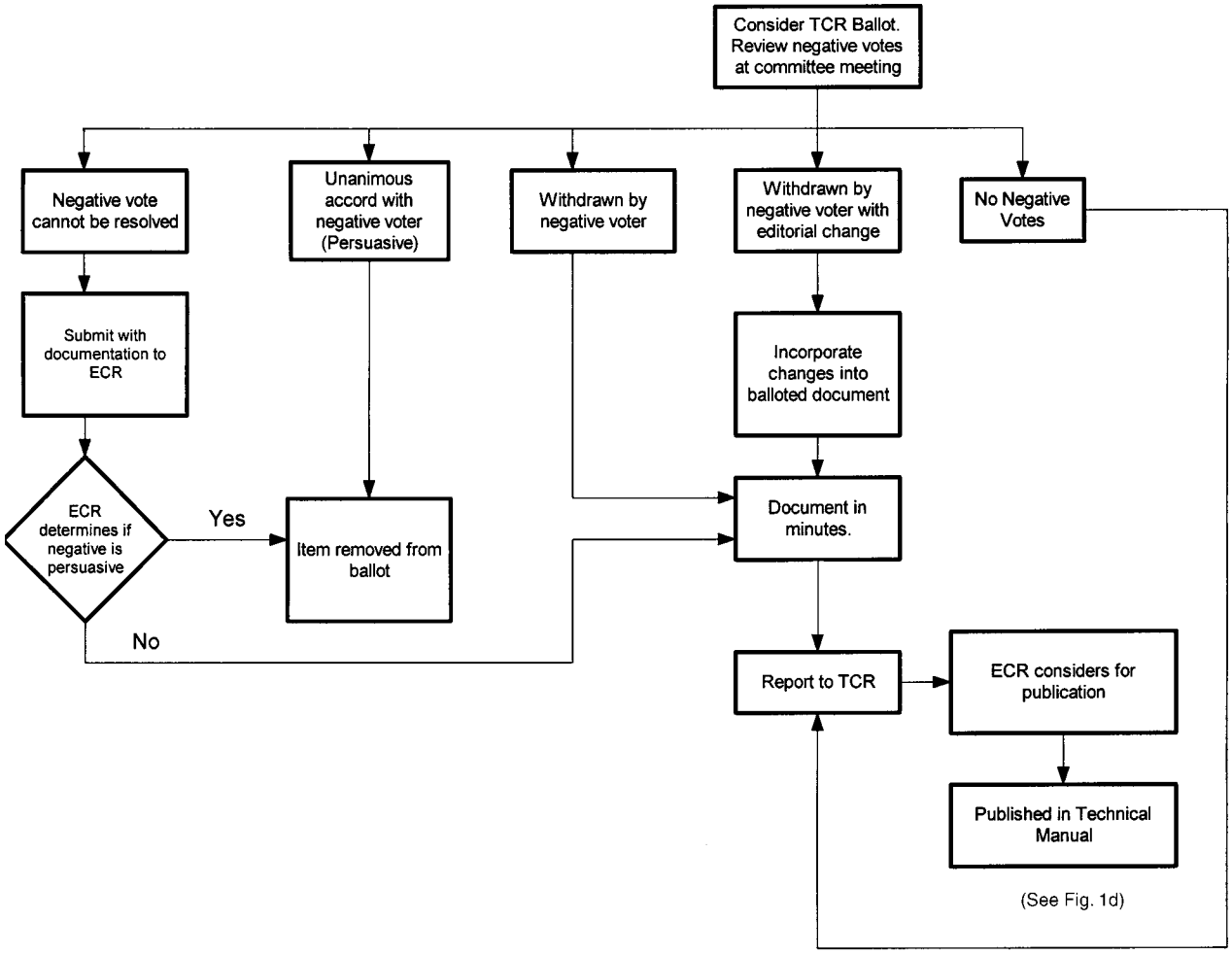


Fig. 1c—AATCC Test Method Development.

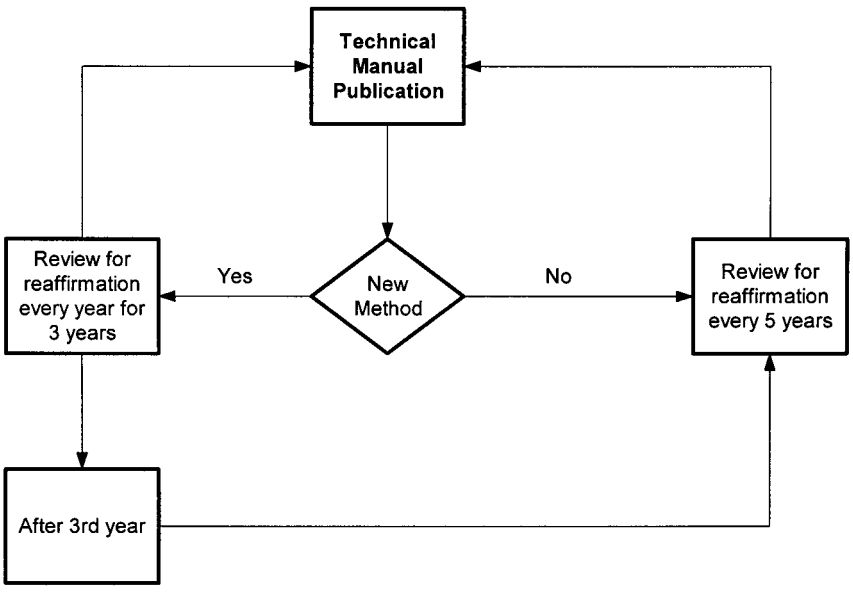


Fig. 1d—AATCC Test Method Development.

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Guidance and Considerations for General Purpose Textile Face Coverings: Adult

1. Foreword

1.1 Given the COVID-19 pandemic and related government recommendations of face covering use in public settings, this guidance was developed by a group of industry stakeholders to provide information to manufacturers. Considerations and recommendations are intended to help manufacturers more efficiently design and produce face coverings for general purpose (non-medical) use. This document contains voluntary guidance and practical considerations for:

- Regulatory Considerations
- Fit and Sizing Considerations
- Material and Construction Considerations
- Particle Filtration Considerations
- Breathing Resistance Considerations
- Laundering and Service Life Considerations
- Tie and Ear Loop Considerations
- Product Labeling and Marking Considerations

2. Purpose and Scope

2.1 This monograph contains voluntary guidance for design, construction and labeling of general-purpose, reusable textile face coverings for adults.

2.2 Guidance in this document applies to general-purpose face coverings intended for daily use by asymptomatic individuals.

2.3 Face coverings described in this document are intended for adult-use only.

2.4 Face coverings described in this document are not medical devices per the US Food and Drug Administration (FDA) or Health Canada; they are not personal protective equipment (PPE) as defined by the National Institute for Occupational Safety and Health (NIOSH).

3. Referenced Documents

NOTE: Use current versions of all publications unless otherwise specified.

3.1 **Industry Methods:**

3.1.1 AATCC LP1 Laboratory Procedure for Home Laundering: Machine Washing (see 19.1).

3.1.2 ASTM D3938 Standard Guide for Determining or Confirming Care Instructions for Apparel and Other Textile Products.

3.1.3 ASTM D4964 Standard Test Method for Tension and Elongation of Elastic Fabrics (Constant Rate-of-Extension Type Tensile Testing Machine).

3.1.4 ASTM D5034 Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test).

3.1.5 ASTM F2100 Standard Specification for Performance of Materials Used in Medical Face Masks.

3.1.6 ASTM F2299 Standard Test Method for Determining the Initial Efficiency of Materials Used in Medical Face Masks to Penetration by Particulates Using Latex Spheres.

3.1.7 ASTM D737 Standard Test Method for Air Permeability of Textile Fabrics.

3.1.8 EN 14683 Medical face masks—Requirements and test methods.

3.1.9 ISO 9237 Textiles—Determination of the permeability of fabrics to air.

3.2 **US Regulations:**

3.2.1 16 CFR 300 Rules and Regulations under the Wool Products Labeling Act of 1939

3.2.2 16 CFR 303 Rules and Regulations under the Textile Fiber Products Identification Act

3.2.3 16 CFR 423 Care Labeling of Textile Wearing Apparel and Certain Piece Goods

3.2.4 16 CFR 1610 Standard for the Flammability of Clothing Textiles

3.3 **Canada Regulations:**

3.3.1 Canada Consumer Product Safety Act (S.C. 2010, c. 21)

3.3.2 Textile Flammability Regulations (SOR/2016-194)

3.3.3 Textile Labelling Act (R.S.C., 1985, c. T-10)

3.3.4 Textile Labelling and Advertising Regulations (C.R.C., c. 1551)

4. Terminology

4.1 **breathing resistance,** n.—the differential pressure required to draw air through a measured surface area at a constant air flow rate. *Source*: EN 14683, an FDA-recognized consensus method for surgical masks.

NOTE: The term "breathability" is also associated with this definition, but due to possible confusion with vapor permeability, it is not used in this document. Fabrics designed to specifically allow escape of perspiration in the form of vapor are *not* typically suitable for face coverings.

4.2 **face covering,** n.—an item of clothing designed to reduce distribution of the wearer's respiratory droplets (e.g., from breathing, coughing and/or sneezing). (Compare *medical face mask*.)

NOTE: For the purpose of this guidance, face coverings specifically:

- Fully cover user's nose and mouth.
- Fit snugly against the sides of the face so there are no gaps.
- Do not cause difficulty breathing while wearing.

For the purpose of this guidance, face coverings specifically are not:

- Medical Devices.
- Personal Protective Equipment (PPE)
- Intended for use by healthcare personnel

4.3 **fit,** for face coverings, n.—the ability to cover the user's nose and mouth while making consistent and snug contact with user's face at the perimeter (see Fig. 1).

4.4 **healthcare personnel (HCP),** n.—all persons, paid and unpaid, working in healthcare settings whose activities place them at risk for transmission of respiratory infections from patients. *Source*: CDC.

NOTE: Examples of such activities include those that require direct contact with patients and/or exposure to the patient care environment, including being in the patient room or in a triage or examination room or other potentially contaminated areas, and handling blood, body fluids, secretions, or excretions (except sweat) or soiled medical supplies, equipment or environmental surfaces.

4.5 **healthcare setting,** n.—a place where healthcare is delivered. (Compare *public setting*.) *Source*: FDA.

NOTE: Healthcare settings include, but are not limited to, acute-care hospitals; long-term care facilities, such as nursing homes and skilled nursing facilities; physicians' offices; urgent-care centers; outpatient clinics; home healthcare (i.e., care provided at home by professional healthcare providers), and emergency medical services. Settings include specific sites within non-healthcare settings where healthcare is routinely delivered (e.g., a medical clinic embedded within a workplace or school).

4.6 **medical device,** n.—any item intended for use in the diagnosis of disease or other conditions, or in the cure, mitigation, treatment, or prevention of disease.

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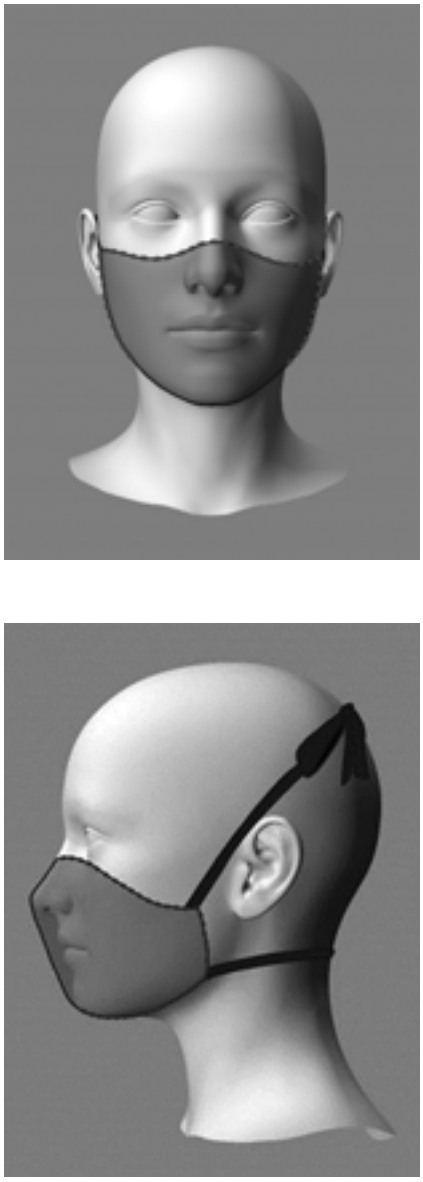


Fig. 1—Face covering covers the user's nose and mouth and makes contact at perimeter (dotted line).

Source: FDA

4.7 **medical face mask,** n.—an item of protective clothing designed to protect portions of the wearer's face, including the mucous membrane areas of the wearer's nose and mouth, from contact with blood and other body fluids during medical procedures. (Compare *face covering*.) *Source*: ASTM F2100.

4.8 **particle filtration efficiency,** n.—the efficiency of the filter material in capturing aerosolized particles, expressed as the percentage of a known number of particles that does not pass the medical face covering material at a given flow rate. *Source*: ASTM F2299, the FDA recognized consensus standard for surgical masks.

4.9 **personal protective equipment (PPE),** medical, n.—clothing, helmets, gloves, face shields, goggles, face coverings and/or respirators or other equipment designed to protect the wearer from injury or the spread of infection or illness. *Source*: FDA.

4.10 **public setting,** n. —a venue in which people may be in close proximity to others outside their immediate family or household. (Compare *healthcare setting*.)

NOTE: Public settings include (but are not limited to) grocery stores, pharmacies, parks, public transportation, and office environments.

4.11 **service life,** *for textiles*, n.—the maximum number of laundering cycles that can be performed before an item can no longer perform its intended end use.

NOTE: The properties defining service life for a face covering are:

- User Fit
- Particle filtration
- Breathing resistance
- Ear loop or tie attachment
- Ear loop or tie elastic recovery (if applicable)

5. Safety Precautions

5.1 The safety precautions specified in the monograph are not intended to be all inclusive.

5.2 It is the user's responsibility to reference applicable safety data sheets, use safe and proper techniques, and wear appropriate personal protective equipment when required.

5.3 Users MUST consult manufacturers for specific details such as safety data sheets, equipment operating instructions, and other recommendations. Consult and follow all applicable health and safety regulations (e.g., OSHA standards and rules).

6. Limitations

6.1 This guidance does not address all aspects of textile face covering design and performance.

6.2 Barrier and breathability properties apply to fabric only and may not be representative of complete face covering properties. This document does not include detailed protocols for evaluating complete face coverings.

6.3 This guidance does not apply to regulated respiratory protection. Refer to applicable regulations and standards for guidance on these items.

7. Regulatory Considerations

7.1 Follow all applicable national and local regulations. The list below is NOT intended to be comprehensive. Additional testing and/or certifications may be necessary.

7.2 **For the United States:**

7.2.1 Review US Consumer Products Safety Commission (CPSC) flammability guidance regarding 16 CFR 1610. Some fabrics may be exempt from testing based on weight/construction and/or fiber content.

7.2.2 Review US Federal Trade Commission (FTC) guidance regarding fiber labeling, country of origin, and manufacturer identification according to 16 CFR 303 and 16 CFR 300. Non-wool face coverings may be exempt from labeling requirements if they do not cover any part of the neck or shoulders.

7.2.3 Review FTC care label guidance regarding 16 CFR 423. Care labels are typically required to be permanently affixed to the apparel item, but it may be possible to apply for an exemption.

7.3 **For Canada:**

7.3.1 Review SOR/2016-194 for flammability guidance.

7.3.2 Review the Textile Labelling Act & the Textile Labelling and Advertising Regulations to ensure product includes all applicable labeling, fiber content, and claims requirements.

7.4 Labels should generally be "conspicuous and accessible" whether they are affixed to the face covering or the packaging. Label information should also be conspicuous and accessible in online, print media or other sales channels.

8. Fit and Sizing Considerations

8.1 Size face coverings to sufficiently cover the user's nose and mouth.

8.1.1 See Appendix 1 for CDC's and Health Canada sizing recommendations for homemade face coverings.

8.2 Design face covering so they do not obscure the wearer's visibility.

8.3 Design perimeter of face covering to make consistent and snug contact with user's face (and under chin), maximize airflow through covering material and minimize side leakage (see Fig. 1).

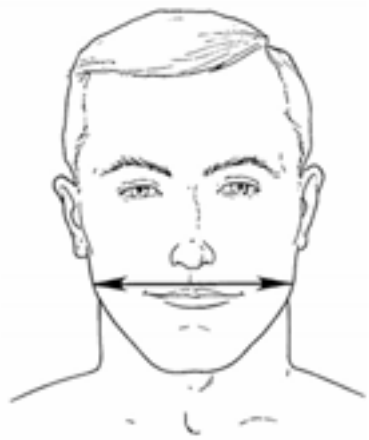
8.4 See Fig. 2 for key facial measurements to consider when sizing covering.

8.4.1 For specific measurements one may consult NIOSH's survey on anthropometric head and face data of respirator users. (www.nap.edu/resource/11815/Anthrotech_report.pdf)

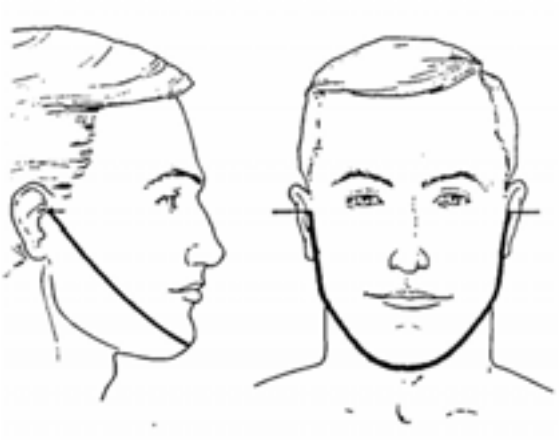
9. Materials and Construction Considerations

9.1 This section is not intended to be design restrictive, but to provide practical options to manufacturers as to which fabrics and processing techniques have demonstrated promise, and those that have not.

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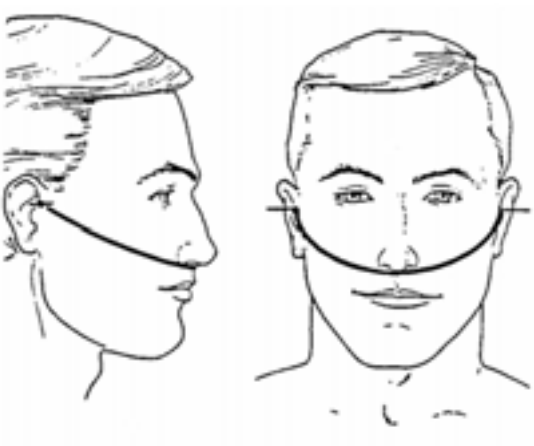
BIGONIAL BREADTH



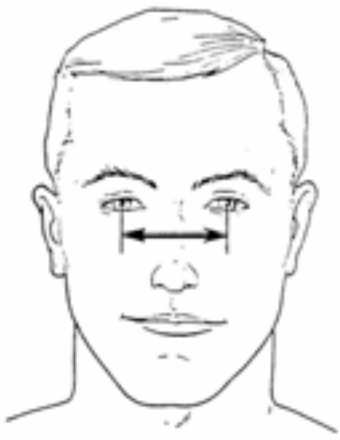
BITRAGION CHIN ARC



BIZYGOMATIC BREADTH



BITRAGION SUBNASALE ARC



INTERPUPILLARY BREADTH



MENTON-SELLION LENGTH

Fig. 2—Facial measurements to consider for sizing.

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9.2 **Fiber and Fabric Selection**

9.2.1 Fiber content, construction, and fabric weight or thread count are good screening tools, but do not guarantee performance. Final material selections should be based on performance properties of particle filtration and breathing resistance.

9.2.2 Any mention of a specific fabric type in this document is an example only. This section is not intended to be an exhaustive list of suitable materials.

9.2.3 Fabrics marketed for apparel use that comply with regulatory requirements and meet minimum industry specifications that may be applicable to face coverings.

9.2.4 Multiple (2 to 3) layers of fabrics may be used to achieve desired performance. Single-layer face coverings may not provide adequate filtration unless they are proved by testing. More than three layers may restrict breathing and may be heavy and bulky for the wearer.

9.2.5 Consider fabric compatibility for multi-layer constructions. Fabrics with significantly different dimensional stability to laundering may cause fit or function issues if used together.

9.2.6 While aesthetic and quality characteristics are beyond the scope of this document, manufacturers may wish to consider evaluation for properties such as wrinkle resistance, pilling, and colorfastness to crocking, laundering, and perspiration.

9.2.7 Denser constructions and heavier fabric weights typically provide better particle filtration due to smaller spaces between the yarns or fibers. Keep in mind that fabrics and multi-layer assemblies providing high levels of particle filtration may also have high breathing resistance. A balance of particle filtration and breathing resistance is required for a functional face covering.

9.2.8 Fabrics with high stretch, including jersey knits, may pose a challenge because stretching increases the spaces between yarns. If used, such fabrics should be combined with a more stable layer in the construction of face coverings.

9.2.9 Fabrics with brushed or napped ("fuzzy") surfaces may trap droplets well but may be irritating in direct contact with a wearer's nose or mouth. Brushed or napped fabrics are best used as an outer or middle layer, with the raised surface towards the face (place at least one additional layer between the raised surface and the wearer's face). Flannel is one example of a brushed surface fabric.

9.2.10 Some researches indicate that fabrics made of staple fibers and yarns show better filtration properties. These yarn types may help decreasing the open space between and within the yarns. Texturized multifilament yarns may also achieve this.

9.2.11 Wearer comfort is an important consideration in selecting fabric for the inner layer of a face covering. Consider fabrics that are soft or smooth for use against the skin.

9.2.12 Consider moisture management of the inner and/or middle layer. Fabrics that move or hold moisture away from the face may be more comfortable for the wearer. Humid environments also promote formation of larger droplets which are easier to retain with a face covering.

9.2.13 Coated or laminated fabrics may have undesirable properties including high breathing resistance and chemicals unsuitable for direct contact with mouth and nose.

9.2.14 Avoid fabrics and findings containing hazardous chemicals. Consult applicable Restricted Substances Lists (RSL, e.g., www.aafaglobal.org/AAFA/Solutions_Pages/Restricted Substance List.aspx) for guidance.

9.2.15 Studies have been performed measuring the filtration performance of various commonly available fabrics. These can be consulted when selecting a fabric. (See Appendix B)

9.2.16 Examples of fabric types and fiber contents can be found in Appendix C.

9.3 **Face Covering Construction**

9.3.1 Avoid unnecessary seams as they may introduce potential areas for leakage. Seams and stitching create holes that may allow passage of more droplets than unstitched areas of fabric.

9.3.2 Avoid seams at nose and mouth area. Bulky seam allowances can cause pressure and discomfort to the wearer.

9.3.3 Use appropriate seam construction to prevent seam allowances from interfering with fit or breathability, particularly for multi-layer assemblies.

9.3.4 Use appropriate needle and thread to make the smallest possible holes in sewing.

9.3.5 Use appropriate stitch length to ensure secure construction without unnecessary holes.

9.3.6 Use pleats, darts, or other construction techniques to shape face coverings for a proper fit.

10. Particle Filtration Efficiency Considerations

10.1 The filtration efficiency is measured on the fabric of the covering only and indicates the potential for the fabric to be effectively used in a covering. It does not provide final (face covering) product effectiveness. The manufacturer should consider that the face covering design and seal to the face are very important.

10.2 Manufacturers can evaluate a covering's fabric particle filtration using the below **recommended** methodology and specifications.

10.3 **Base Particle Filtration Method:** ASTM F2299—Standard Test Method for Determining the Initial Efficiency of Materials Used in Medical Face Masks to Penetration by Particulates Using Latex Spheres (or a technical equivalent method).

10.4 **Particle Size:** The best level of protection will be achieved with small particle sizes, but this will be hardest to achieve. Report efficiency at 3-micron. The user may consider testing at smaller sizes as well, for example, 0.5- and 1-micron particles will be more challenging and representative of actual use conditions.

10.5 **Face Velocity:** 10.4 cm/s

10.6 **Detection:** Optical with range at least 0.5-5 micron and able to report effective filtration at 3-micron. Calibration per ISO 21501-4.

10.7 **Other acceptable methods:** Methods that have been determined a practical equivalence to the one outlined above.

10.8 **Recommendation:** Woven face coverings should not fall below 70% efficiency at 3-micron before and after laundering to specified number of cycles.

10.9 The standard filtration test described will only evaluate the fabric performance. A crucial aspect of the face covering as an effective end product would be to assess the performance of the face covering, not just the fabric. Formal PPE guidelines would require human panel testing or head-form testing (cf. NIOSH), and require fit-testing on workers when using these face coverings for protection. As the latter would not be possible to conduct for general population face coverings, but end product evaluation of fit is still relevant, it is suggested and encouraged to evaluate face coverings for filtration efficiency, ideally maintaining a protection level of 70% at maximum 3-micron particles, and possibly lower sizes.

10.10 Available methodologies may include mannequin head forms, with specified air flow and particle distributions as described above. These tests may not be as accessible as fabric filtration tests, but are crucial to establish complete face coverings performance, whereas the tests above will only establish fabric performance.

11. Breathing Resistance Considerations

11.1 Face coverings must not impede the wearer's breathing. Manufacturers can evaluate a fabric or fabric assembly for breathing resistance using a suitable air permeability test method. When preparing specimens, cut all materials to the specimen size specified in the applicable test method, or slightly larger. Stack layers in the order and orientation in which

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they will be used in the complete face covering. Baste or overlock the perimeter of the assembly, stitching through all layers to create an assembly for testing.

11.2 Perform one of the following test methods on the material assembly. It is not necessary to evaluate the air permeability of individual layers in a multilayer assembly.

11.2.1 EN 14683 (Annex C) using 8 L/min air flow, with standard diameter of 25 mm.

11.2.2 ASTM D737 using a standard 125 Pa pressure drop. Use the standard 38.3 cm^2 test area.

11.2.3 ISO 9237 using a standard 100 Pa pressure differential.

11.3 The following air permeability values are recommended for face coverings:

11.3.1 For EN 14683 (Annex C), a maximum of 36.7 Pa/cm^2.

11.3.2 For ASTM D737, a minimum of 37.5 $ft^3/min/ft^2$ (1.14 $L/min/cm^2$).

11.3.3 For ISO 9237, a minimum of 0.91 $l/min/cm^2$ (or 15 cm/s).

11.4 Test additional specimens after multiple laundering cycles (see Laundering and Care Considerations). Do not retest the same specimens.

11.5 Air permeability is measured on the fabric only. While this test indicates the potential for the fabric to be effectively used in a face covering, construction, printing, and other processes may also impact the breathing resistance of the finished face covering.

12. Laundering and Care Considerations

12.1 Machine laundering instructions are preferred. Avoid hand laundering instructions due to increased potential for contamination. Dry cleaning instructions may also be impractical for face coverings, which should be cleaned after every use.

12.2 Consider the most sensitive element in determining care instructions (e.g., the textile layer most prone to dimensional change.)

12.3 Determine the service life of the face covering.

12.3.1 Launder as directed in AATCC LP1, using temperature and cycle settings indicated by face covering care instructions (see 15.1).

12.3.2 Evaluate fit, particle filtration, air permeability and loop or tie attachment after multiple laundering cycles. Each cycle includes both washing and drying procedures.

12.3.3 To evaluate fit and loop/tie attachment after laundering, launder complete face covering(s).

12.3.4 To evaluate filtration and air permeability after laundering, make and launder one or more material assemblies with all textile layers in the order in which they will be used in the complete face covering.

12.3.5 Multiple face coverings and/or assemblies may be laundered together.

12.3.6 Service life is the maximum number of laundering cycles after which the face covering is still fit for use based on the suggested criteria in this document, regulations and/or internal criteria.

12.3.7 It is not generally practical to evaluate performance properties after every laundering cycle. Select an interval based on the expected or desired service life. For example, for 50 cycles, evaluate after 25, 35, 45 and 50 cycles.

12.3.8 Evaluate a different area of the assembly, or a different assembly, for filtration after each interval.

12.3.9 Evaluate a different area of the assembly, or a different assembly, for breathing resistance after each interval.

12.3.10 Evaluate attachment of loops/ties after each interval.

12.3.11 Evaluate elastic recovery (if applicable) of different loops/ties after each interval.

13. Tie and Ear Loop Considerations

13.1 Provide ties or ear loops to secure face covering to user's face.

13.1.1 Ties typically offer a more customizable fit.

13.2 Ensure ties and ear loops are durable enough to withstand normal use of the face covering.

13.2.1 Verify attachment strength using ASTM D5034.

13.2.2 The recommended attachment strength for face covering ties and ear loops is a minimum of 10 lbf (44.5 N).

13.2.3 Test additional specimens after multiple laundering cycles (see Laundering and Care Considerations). Do not retest the same specimens.

13.3 For elastic ties and ear loops, ensure adequate elastic recovery to retain shape and fit.

13.3.1 Verify elastic recovery using ASTM D4964. Determine the maximum load based on elastic width:

- Width ≥ 1": 10 lbf (44.5 N)
- 1" > width > ½": 5 lbf (22.3 N)
- ≤ ½" width: 2 lbf (8.93 N)

13.3.2 The recommended elastic recovery for face covering ties and ear loops is a minimum of 90% after 60 s.

14. Product Labeling and Marking Considerations

14.1 **Product Description:** Accurately describe the product as a *face covering* on all packaging and promotional material.

14.2 **Adult Use:** Clearly label product as "Adult Use Only", "Not for Children" or practical equivalent.

14.3 **Intended Use:** Do not include any label, image or marking that suggests use beyond that what face covering was intended.

14.3.1 Intended User: Do not suggest use by a healthcare personnel, or user requiring respiratory protection.

14.3.2 Intended Use Setting: Do not suggest use in any setting that requires specific fluid and/or inhalation precautions to be taken, such as a healthcare setting or commercial setting.

14.3.3 Social Distancing: Do not suggest that use of face covering reduces the need for social distancing.

15. User Instruction Labeling

15.1 Follow all applicable national and local regulations.

15.2 **Wearing Instructions:** Include instructions on how a user puts on, wears, and removes face covering.

Recommendation: One example of removal instructions is: "Individuals should be careful not to touch their eyes, nose, and mouth when removing their face covering and wash hands immediately after removing." Refer to CDC, Health Canada, or other official sources for additional recommendations.

15.3 **Washing Instructions:** Include instructions for home laundering (see Laundering and Care Considerations).

Recommendation: May include a recommendation to wash after every use.

15.4 **Service Life (Maximum number of laundering cycles):** Include the maximum number of laundering cycles (per care instructions) before disposal. (see Laundering and Care Considerations).

15.5 **Fit:** Include guidance for assessing fit (see Fit Considerations) and a statement instructing user to dispose of face covering if it no longer covers nose and mouth or makes snug contact with face at perimeter.

15.6 **Disposal Instructions:** Include instructions for proper disposal of face covering. One example of disposal instructions is:

> "At end of service life, dispose of face covering in a plastic bag to avoid cross-contamination."

16. Performance Claims, Advertising and Promotional Literature

16.1 Follow all applicable national and local regulations.

16.2 Particular claims may be allowable but must follow federal guidelines (See Section 7: Regulatory Considerations) for claims substantiation in US and Canada respectively. Given the context of this guidance, manufactures should avoid making the specific claims regarding:

- Use as surgical mask, or liquid barrier protection

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- Anti-microbial or anti-viral properties
- Respiratory protection and particulate filtration efficiency (e.g., 95%)

16.3 Manufacturers should not include claims that would create undue risk to the user and/or categorize product as a "medical device" or "personal protective equipment".

16.3.1 The product should not make any reference to a medical purpose.

16.3.2 The product should not make any reference to "PPE" or personal protective equipment.

16.4 Do not make claims or logos that imply approval by US National Institute for Occupational Safety and Health (NIOSH), US Centers for Disease Control and Prevention (CDC), Health Canada, or another safety-related agency.

16.5 COVID-19 (or other infections disease): No printed matter, including advertising or promotional materials, relating to the use of the authorized face covering may represent or suggest that such product is safe or effective for the prevention or treatment of patients during the COVID-19 pandemic.

17. Summary of Recommendations (Evaluation Protocol)

17.1 Table I is a summary of recommendations in the form of an evaluation protocol. This protocol does not consider applicable regulations.

18. Notes

18.1 Available from AATCC, PO Box 12215, Research Triangle Park NC 27709, USA; +1.919.549.8141; ordering@aatcc.org; www.aatcc.org.

19. History

19.1 Developed in 2020 by AATCC Committee RA113.

Appendix A
Government Positions on Cloth/Textile Coverings

A1. Below are links to government guidance regarding textile coverings worn by general public:

Health Canada:

- www.canada.ca/en/public-health/services/diseases/2019-novel-coronavirus-infection/prevention-risks/about-non-medical-coverings-face-coverings.html
- www.canada.ca/en/public-health/news/2020/04/ccmoh-communication-use-of-non-medical-coverings-or-facial-coverings-by-the-public.html

United States CDC:

- www.cdc.gov/coronavirus/2019-ncov/prevent-getting-sick/cloth-face-cover-faq.html
- www.cdc.gov/coronavirus/2019-ncov/prevent-getting-sick/cloth-face-cover.html

See Fig. A1 for CDC's sizing recommendations for homemade coverings:

See information on cloth face coverings for additional information:

CDC:

- www.cdc.gov/coronavirus/2019-ncov/downloads/cloth-face-coverings-information.pdf
- www.cdc.gov/coronavirus/2019-ncov/prevent-getting-sick/cloth-face-cover-faq.html

Health Canada:

- www.canada.ca/en/public-health/services/diseases/2019-novel-coronavirus-infection/prevention-risks/about-non-medical-masks-face-coverings.html#_Appropriate_non-medical_mask
- www.canada.ca/en/public-health/services/diseases/2019-novel-coronavirus-infection/prevention-risks/sew-no-sew-instructions-non-medical-masks-face-coverings.html#_Sew_method

Appendix B
Studies of Fabric Filtration and Air Permeability

B1. Below are links to government guidance regarding textile coverings worn by general public.

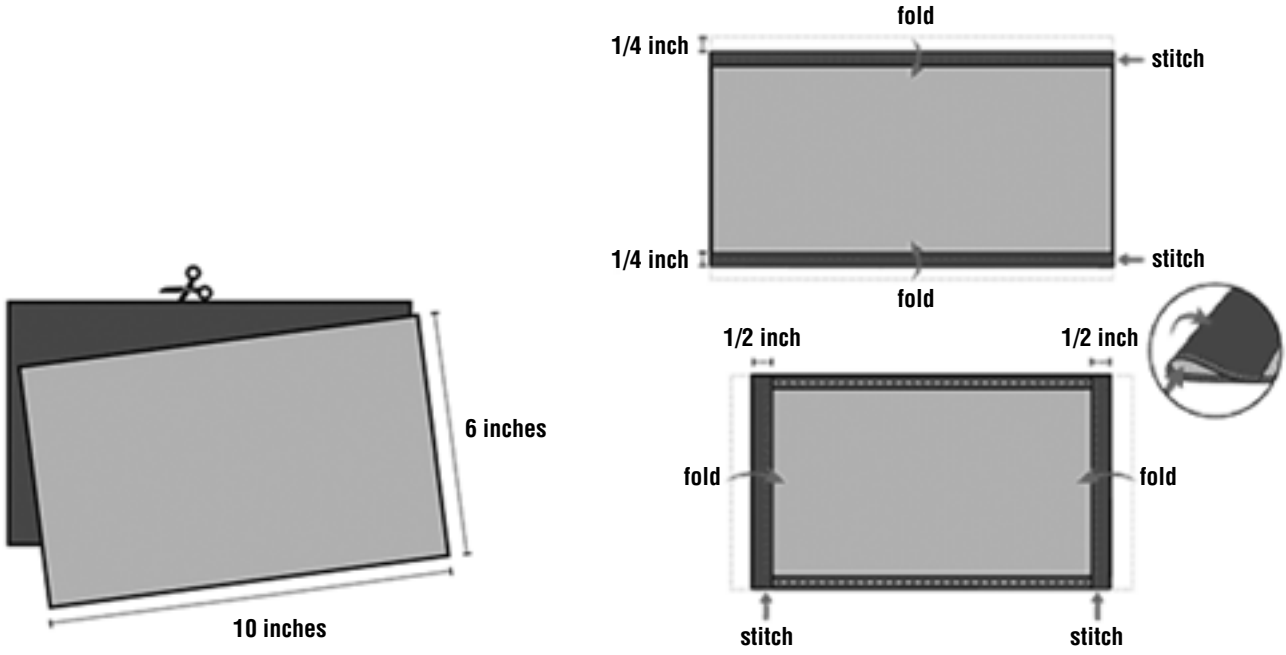


Fig. A1—CDC recommended dimensions and construction for a cloth face covering.

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B1.1 *Testing the Efficacy of Homemade Masks: Would They Protect in an Influenza Pandemic?*

Davies, A., Thompson, K., Giri, K., Kafatos, G., Walker, J., & Bennett, A. (2013). Testing the Efficacy of Homemade Masks: Would They Protect in an Influenza Pandemic? Disaster Medicine and Public Health Preparedness, 7(4), 413-418. doi: 10.1017/dmp.2013.43

www.cambridge.org/core/journals/disaster-medicine-and-public-health-preparedness/article/testing-the-efficacy-of-homemade-masks-would-they-protect-in-an-influenza-pandemic/0921A05A69A9419C862FA2F35F819D55/core-reader

B1.2 *Aerosol Filtration Efficiency of Common Fabrics Used in Respiratory Cloth Masks*

Abhiteja Konda, Abhinav Prakash, Gregory A. Moss, Michael Schmoldt, Gregory D. Grant, and Supratik Guha *ACS Nano* **2020** *14* (5), 6339-6347. doi: 10.1021/acsnano.0c03252

https://pubs.acs.org/action/showCitFormats?doi=10.1021%2Facsnano.0c03252&href=/doi/10.1021%2Facsnano.0c03252

B1.3 *Simple Respiratory Protection—Evaluation of the Filtration Performance of Cloth Masks and Common Fabric Materials Against 20–1000 nm Size Particles*

Samy Rengasamy, Benjamin Eimer, Ronald E. Shaffer, Simple Respiratory Protection—Evaluation of the Filtration Performance of Cloth Masks and Common Fabric Materials Against 20–1000 nm Size Particles, The Annals of Occupational Hygiene, Volume 54, Issue 7, October 2010, Pages 789–798, https://doi.org/10.1093/annhyg/meq044

https://academic.oup.com/annweh/article/54/7/789/202744

B1.4 *Study performed by the Manufacturing Development Center at the Wake Forest Institute for Regenerative Medicine*

https://newsroom.wakehealth.edu/News-Releases/2020/04/Testing-Shows-Type-of-Cloth-Used-in-Homemade-Masks-Makes-a-Difference

Appendix C
Considerations and Examples of Fiber Contents and Fabrics

C1. For your reference, Table A1 shows examples of fiber contents and fabrics used in studies. This is not an exhaustive list of acceptable options.

NOTE: Fiber content, construction, and fabric weight or thread count are good screening tools, but do not guarantee performance. Final material selections should be based on performance properties of particle filtration and breathing resistance.

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Table I—Summary of Recommended Evaluation and Criteria for General Purpose Textile Face Coverings

Evaluation	Method/Citation	Recommendation Criteria
General Labeling		
Product Description	Visual Inspection	Should accurately describing the product as face covering.
Intended Use	Visual Inspection	Should not include any labeling or marking that would convey use beyond that which face covering was intended.
Intended User	Visual Inspection	Should not include any labeling or marking that would convey use by a healthcare personnel, or user requiring respiratory protection
Intended Use Setting	Visual Inspection	Should not include any labeling or marking that would convey use in setting that requires specific fluid and/or inhalation precautions to be taken, such as a healthcare setting or commercial setting
Adult Use Only—Labeling	Visual Inspection	Should be labeled "Adult Use Only", "Not for Children" or practical equivalent
User Instruction Labeling		
Wearing Instructions	Visual Inspection	Should include manufacturer's instructions on how user is put on, wear, and remove face covering. Instructions for removal of covering should include (or practical equivalent) to: **"Individuals should be careful not to touch their eyes, nose, and mouth when removing their face covering and wash hands immediately after removing"**
Washing Instructions	Visual Inspection	Should include instructions for manufacturer's recommended laundering
Service Life Labeling	Visual Inspection	Face coverings should include the maximum number of laundering cycles (per care instructions) before disposal.
Fit	Visual Inspection	Should include a statement instructing user to dispose of face covering if it no longer fits.
Disposal Instructions	Visual Inspection	Should include instructions for manufacturer's recommended procedure for proper disposal of face covering.
Performance Claims, Advertising and Promotional Literature		
Medical Device/ Medical Purpose	Visual Inspection	The product should not make any claims referring to a medical purpose, to use in the diagnosis of disease or other conditions, or in the cure, mitigation, treatment, or prevention of disease
Personal Protective Equipment Use	Visual Inspection	The product should not make any claims referring to "PPE" or personal protective equipment.
Specific Claims	Visual Inspection	Should not make the following claims: • Use as surgical covering, or liquid barrier protection • Anti-microbial or anti-viral properties • Respiratory protection and particulate filtration efficiency (e.g., 95%)
Health Agencies Logos and Approvals	Visual Inspection	Should not make claims or logos that imply approval by US National Institute for Occupational Safety and Health (NIOSH), US Centers for Disease Control and Prevention (CDC), Health Canada, or another safety-related agency.
COVID-19 Related Labeling	Visual Inspection	No printed matter, including advertising or promotional materials, relating to the use of the authorized face covering may represent or suggest that such product is safe or effective for the prevention or treatment of patients during the COVID-19 pandemic.
Attachment Ties and Straps		
# of test specimens shall be per method instruction. Confirm face covering meets below recommendations before and after total claimed washes.		
Elastic Extension Straps	ASTM D4964	Should exhibit a 90% recovery after 60 seconds. Maximum Load determined by elastic width. • Width ≥ 1": 10 lbf (44.5 N) • 1" > width > ½": 5 lbf (22.3 N) • ≤ ½" width: 2 lbf (8.93 N)
Tie and Attachments on Reusable Face Coverings	ASTM D5034	Attachments should withstand a minimum of 10 lbf (44.5 N),
Laundering and Care		
# of test specimens shall be per method instruction.		
Laundering	AATCC LP1	After laundering to specified number of washes, face covering should not exhibit: • Particle filtration and/or breathability falls below recommended 70% PFE or air permeability limits **OR** • Change in dimensions that no longer allows acceptable user "fit" **OR** • Physical change that renders covering no longer useable (e.g., strap failure)
Particle Filtration		
# of test specimens shall be per method instruction. Confirm face covering meets below recommendations before and after total claimed washes.		
Filtration efficiency	ASTM F2299 (or technical equivalent)	Face covering shall demonstrate a particle filtration efficiency of >70% with maximum 3-micron particle size at a face velocity of 10.4 cm/s. Testing shall be performed on "as-received" samples and after claimed number of washes.
Breathing Resistance—Air Permeability		
# of test specimens shall be per method instruction. Confirm face covering meets below recommendations before and after total claimed washes.		
Air permeability	EN 14683 (Annex C)	Using 8 L/min air flow, with standard diameter of 25 mm. Should exhibit a maximum of 36.7 Pa/cm^2.
	ASTM D737	Using a standard 125 Pa pressure drop. Use the standard 38.3 cm^2 test area. Should exhibit a minimum of a minimum of 37.5 ft^3/min/ft^2
	ISO 9237	Using a standard 100 Pa pressure differential. Should exhibit a minimum of 0.91 L/min/cm^2 (or 15 cm/s).

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Table A1—Example Fabrics for General Purpose Textile Face Coverings.

Examples of Fiber Content for Outer Layer
Cotton or cotton rich blends (over 60% are preferable because of their absorbency can draw moisture). **Reference:** https://academic.oup.com/cid/article/41/7/e67/310340. • Polyester may be an option, depending on the fabric density.
Examples of Fabric for Outer Layer
• Woven—plain, satin, sateen, twill 190 g/m^2 or more — 400 thread count or more, sheet/ pillow case (sateen weave) — 180 picks per inch heavy quilting fabric (plain weave) — 120-160 picks per inch plain weave • Knit—interlock, brushed/napped fleece, spacer, rib, pique
Examples of Fiber Content for Inner Layer (against skin)
• Cotton • Nylon • Polyester **Reference:** https://academic.oup.com/annweh/article/54/7/789/202744
Examples of Fabric for Inner Layer (against skin)
• Woven—plain, satin, sateen, terry, — Taffeta 45 g/m^2 — flannel (twill or plain weave) 160 g/m^2 • Knit—interlock, brushed/napped fleece, space, rib, pique — 28 gauge to 36 gauge (finer gauges create thinner fabric) — Stitch counts greater than 20 wales per inch × 28 courses per inch
Examples of Fiber Content for Optional Filter Layer (between inner and outer layer)
• Natural or synthetic fiber spacers • Nylon or polyester (nonwoven) interfacing • Nylon microfiber (woven or knit) • Polypropylene nonwoven (e.g., common in reusable grocery bags) • Wool felt
Examples of Fabric for Optional Filter Layer (between inner and outer layer)
• Woven — satin, plain, twill—120 to 180 g/m^2 — terry—300-400 g/m^2 cotton as a kitchen towel • Knit—interlock, brushed/napped fleece, spacer, rib, pique (28 gauge or finer) • Nonwoven—(must not restrict breathing) Blue shop towels—hydro knit fabric • Composite/laminate—(must not restrict breathing) — Gore-tex (50 D, 101 g/m^2) is not recommended. See www.gore-tex.com/pressroom/press-release/gore-tex-brand/statement-from-wl-gore-associates-regarding-gore-tex-materials-and-covid-19.

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AATCC Administrative Committees for 2023

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Interest groups shall be based on a commonality of interest in a particular subject matter and not on geographic designation. Individual AATCC members can join more than one interest group.

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Mission: To stimulate and expand the knowledge base of applying chemicals (preparation aids, colorants, finishes, polymers, etc.) to substrates; concentrating on the technologies, equipment, systems and processes used, and including the tested physical/mechanical properties of treated materials.

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RA23 Colorfastness to Water Test Methods

Scope: To develop test methods for measuring the resistance of colors in all kinds of textiles to change, transfer or spotting by any type of water such as fresh water, sea water and chlorinated swimming pool water. TM15, 104, 106, 107, 162.

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RA24 Fiber Analysis Test Methods

Scope: To develop test methods for the qualitative identification and the quantitative determination of fiber mixtures. TM20, 20A.

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RA31 Antimicrobial Activity and Odor Control Test Methods

Scope: To develop test methods for detection and measurement of antimicrobial and odor control activity of treated textiles. TM24, 28, 30, 90, 100, 147, 174, 211.

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RA32 Static Electricity Test Methods

Scope: To develop test methods for producing and measuring the electrostatic behavior of static electrical charges. TM 76, 84, 115.

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RA33 Colorfastness to Atmospheric Contaminant Test Methods

Scope: To study the factors in atmospheric exposure, other than photochemical influences, which cause color changes; to develop test methods for measuring color changes that will reliably correlate with end use. Contaminants under study include oxides of nitrogen, ozone, sulfur dioxide and products of hydrocarbon combustion. TM23, 109, 129, 164.

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RA34 Preparation Test Methods

Scope: To develop test methods for (1) measuring the effect of preparation (defined as all wet steps prior to dyeing, printing or finishing of white textile fibers, yarns or fabrics) upon the chemical, physical and colorfastness properties of these textiles; (2) evaluating the effectiveness of preparation chemicals; and (3) controlling preparation processes including methods for obtaining information about materials on the textiles to be prepared. TM81, 82, 89, 97, 98, 101, 102, 144, 209.

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RA36 Color Measurement Test Methods

Scope: To develop test methods relating to color science, to advise other AATCC committees involved with special problems on color, and to provide liaison between AATCC and other groups directly concerned with problems in color science. EP1, 2, 5, 6, 7, 8, 9, 10, 11, 12; TM110, 148, 173, 182, 203.

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RA38 Colorfastness to Crocking Test Methods

Scope: To develop test methods for measuring resistance of colors in textiles to transfer by rubbing; to modify and improve the crockmeter to enlarge its adaptability. TM8, 116.

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RA42 Dimensional Change Test Methods

Scope: To develop test methods for measuring dimensional changes in fabrics or textile articles when subjected to atmospheric change or to home and commercial laundering. TM96, 99, 135, 150, 179, 187.

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RA43 Professional Textile Care Test Methods

Scope: To develop test methods for measuring the effect of professional textile care (including drycleaning, wet cleaning, finishing, and spot removal) upon the properties of textiles. TM86, 132, 158.

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RA45 Finish Analysis Test Methods

Scope: To develop dependable methods on chemical analysis for identifying and quantitatively measuring finishing materials on textiles. TM94, 112, 206.

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RA49 Insect Resistance Test Methods

Scope: To provide quantitative and qualitative evaluations of textiles treated with insecticides. The test methods may be used to a) quantify the amount of active ingredient applied to the material; b) assess the ability of the insecticide treated material to protect the user; or c) assess the ability of the insecticide to protect the material. LP3; TM194.

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RA50 Lightfastness and Weathering Test Methods

Scope: To develop test methods for determining or predicting resistance of textile materials to degradation when exposed to light and moisture alone, in combination with elevated temperatures, and/or other environmental factors. Test method exposure environments are intended to reproduce the degradation effects of outdoor weathering exposure and/or indoor lightfastness. Degradation modes include loss of appearance properties, such as lightfastness, and/or physical properties, such as tensile strength. TM16.1, 16.2, 16.3, 111, 125, 169, 186, 192.

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RA56 Stain Resistance Test Methods

Scope: To develop test methods for evaluating the resistance of apparel fabrics to stain, to study the phenomenon of soil deposition on apparel fabrics of various fibers and finish types during laundering, and to study the soiling of apparel fabrics in service with the intent of developing a test method. TM118, 130, 193.

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RA57 Floor Covering Test Methods

Scope: To develop test methods that will reliably predict the use characteristics of all types of floor coverings. TM121, 122, 134, 137, 138, 165, 171, 175, 189, 196, 205.

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RA59 Fibrous Test Materials

Scope: To prepare detailed specifications for AATCC standard test fibers, yarns and fabrics other than colorfastness fading standards; specifications to be developed with the assistance and concurrence of those test method development committees responsible for the test methods in which the standard fibers, yarns or fabrics are used. EP10.

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RA60 Colorfastness to Washing Test Methods

Scope: To develop test methods that will reliably predict the colorfastness of textiles during laundering or mill scouring. TM61, 172, 188, 190.

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RA61 Appearance Retention Test Methods

Scope: To develop test methods for evaluating the appearance of textiles, component parts, garments, and other end products. TM66, 88B, 88C, 124, 128, 143.

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RA63 Water Resistance, Absorbency & Wetting Agent Evaluation Test Methods

Scope: To develop test methods for evaluating the water resistance, absorbency and effects of wetting agents on textiles. TM17, 22, 27, 35, 42, 43, 70, 79, 127, 195, 197, 198, 199, 200, 201, 204.

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RA75 Evaluation of Materials & Products for End Use Performance

Scope: To coordinate and direct technical inquiries on issues related to end products; to be a forum for examining end-use performance in relation to lab test methods and technique(s); and to communicate with research committees when knowledge learned might be germane to committee activity.

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RA80 Printing Technology

Scope: To promote, coordinate and encourage technical programs of special interest to individuals and groups concerned with printing and to provide increased technical knowledge.

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RA87 Applied Dyeing & Characterization of Dyes Test Methods

Scope: To develop test methods for evaluating the dyeing properties of different dye classes and dyeing systems; to encourage discussions and publications of papers on dyeing theory; and to promote the understanding of modern dyeing theory through symposia, workshops and/or seminars TM140, 141, 146, 154, 157, 159, 163, 167, 170, 176, 184.

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RA88 Home Laundering Technology

Scope: To encourage and promote discussions on current developments and problems associated with the use of standard and new chemical materials, including soaps and detergents, bleaches, water and fabric softeners, enzymes and any other auxiliaries available to the consumer for use in home laundering and coin-operated machines, which may affect color, performance and special functional finishes and the ecology. The information gathered by this committee will be directed to the attention of specific AATCC research committees for their use and orientation. Where appropriate, this committee will plan technical programs and will serve as a clearinghouse for the dissemination of information on this subject area. LP1, 2; M1, 2, 3, 4, 7.

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RA89 Hand Evaluation Test Methods

Scope: To develop specialized test methods and terminology for the assessment and description of differences in the hand of fabrics. EP5; TM202.

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RA99 Technical Manual Editorial Review

Scope: To maintain the test method style manual; to review all new and existing test methods for compliance with the style manual, clarity of thought and technical consistency between methods; to recommend layout and content changes for the TECHNICAL MANUAL that primarily affect its ease of use. M10, 11, 12.

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RA100 Global Sustainability Test Methods

Scope: To develop test methods related to current developments and industry practice in the fields of sustainability and the environment for the textile and its supplier industries; and to provide a forum for discussion of related federal and state legislation and rules and regulations of government agencies as well as to promote the understanding of sustainability through symposia, workshops and/or seminars.

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RA102 Statistics Advisory

Scope: To provide assistance in statistical methodology to Research and Technology Committees, to provide guidelines for the use of correct statistical practices and terminology in the development of new test methods and revisions of existing test methods, and to review precision and bias statements in published test methods. M9.

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RA103 Spectroscopic Technologies

Scope: To promote the advancement of all forms of spectroscopic technologies (NIR, NMR, ultraviolet, visual, IR, Raman, microwave, etc.) by serving as a clearinghouse for information for the benefit of individuals and companies in the textile industry.

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RA104 Garment Wet Processing Technology

Scope: To coordinate and direct technical discussions on current areas of interest in garment dyeing, washing, finishing, and other areas; to provide sources for increased technical knowledge for the benefit of individuals and groups concerned with garment wet processing.

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RA106 UV Protective Textiles Test Methods

Scope: To develop test methods for measurement of ultraviolet protective performance of fabrics and clothing. TM183.

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RA109 Flammability Technology Committee

Scope: To promote and encourage the investigation, study and discussion of problems associated with flammability characteristics of textile fabrics; to promote and coordinate technical programs of special interest to individuals and groups concerned with the application of flame retardants in dyeing or finishing; and to provide sources for increased technical knowledge for the benefit of individuals and groups concerned with flame resistance of textile fabrics.

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RA111 Electronically Integrated Textiles Test Methods

Scope: To develop test methods and terminology for testing electronically integrated textiles. EP13; TM210.

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RA112 Thermal Analysis Test Methods

Scope: To develop test methods for quantitative assessment of thermal regulation properties of textiles.

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RA113 Emerging Issues Test Methods

Scope: To develop test methods, guidelines, and other educational tools for emerging issues related to textiles. M14.

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RA114, Moisture Management Test Methods

Scope: To develop test methods for evaluation of wetting agents and moisture management of textiles. TM17, TM27, TM43, TM79, TM195, TM197, TM198, TM199, TM200, TM201, TM204, TM213. The committee was formed in 2022, when absorbancy and moisture management were removed from the scope of RA63.

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TenCate Protective Fabrics

KYLE WATSON
VF Denver

Reference Committees

Committees are inactive and listed standards are currently managed by RA99, Technical Manual Editorial Review.

RR1 Colorfastness to Acids and Alkalis Test Methods
Test Method: 6

RR9 Ageing of Sulfur-Dyed Textiles Test Methods
Test Method: 26

RR29 Abrasion Resistance Test Methods
Test Methods: 93, 119, 120

RR35 Damage by Retained Chlorine Test Methods
Test Methods: 92, 114

RR41 Enzyme Activity Test Methods
Test Methods: 103, 191

RR53 Colorfastness to Pleating Test Methods
Test Method: 131

RR54 Colorfastness to Heat Test Methods
Test Methods: 117, 133

RR79 Coated, Bonded and Laminated Fabrics Test Methods

RR81 Flock Testing
Test Method: 142

RR90 Chelating Agent Evaluation Test Methods
Test Methods: 149, 161, 168, 185

***RR92 Interaction of Dyes and Finishes Test Methods**
Test Methods: 157, 163

*Currently managed by RA87, Applied Dyeing & Characterization of Dyes Test Methods

* Nonvoting
†Ex-Officio

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AATCC Representatives to Other Organizations for 2022

American National Standards Institute

USA Technical Advisory Group (TAG) for ISO*/TC38—Textiles (2021-2023)

Chair, Primary AATCC Representative
RICHARD SLOMKO
Atlas Material Testing Technology LLC

Secretary, Alternate AATCC Representative
DIANA WYMAN
AATCC

ASTM International

ASTM D13.96.01 ASTM/AATCC Liaison Task Group (2022-2026)

Co-Chair/Secretary
ERIKA SIMMONS
AATCC

Members 2021 (Appointed by ECR Chair)
ANGELA MASSENGILL
Cotton Incorporated

JOHN CROCKER
SDL Atlas

Member 2018-2021 (Appointed by ECR Chair)
RICHARD SLOMKO
Atlas Material Testing Technology LLC

Science History Institute

AATCC Representative
MARTHA CARPER
University of Delaware

Inter-Society Color Council

Liaison (Appointed by RA36 Chair)
ANN C LAIDLAW
ACL Color Consulting LLC

National Fire Protection Association

Liaison (Appointed by RA109 Chair)
BRIAN SHIELS
PBI Performance Products Inc

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Color Solutions International/Dystar

ROBERT F MILLER
Hickory Yarns

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*International Organization for Standardization

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Joint Report for 2022
C2 Executive Committee on Research

CARLA MACCLAMROCK, Cotton Incorporated, Chair

C3 Technical Committee on Research

APURBA BANNERJEE, Vice Chair

The following report summarized activities in which the technical portion of AATCC through the guidance of the Executive Committee on Research and the Technical Committee on Research were involved during 2023. More comprehensive reports of committee activities can be found in the annual reports filed by individual committees. Beginning on page 568.

New Standards

AATCC TMTM214-2022, Test Method for Measuring Condensation on Textiles in a Humid Microclimate. New – Developed in 2022 by AATCC Committee RA63.

AATCC TM215-2023, Test Method for Soil Release: Household Soils. New. Developed in 2023 by AATCC Committee RA56.

AATCC LP4-2023, Laboratory Procedure for Synthetic Soil: New – Developed in 2023 by AATCC Committee RA57.

AATCC LP5-2023, Laboratory Procedure for Home Laundering with Bleach: New – Developed in 2023 by AATCC Committee RA88.

Revised Standards

AATCC TM16.1-2023, Test Method for Colorfastness to Light: Outdoor. Revised in 2023 to align to AATCC style guide.

AATCC TM16.2-2023, Test Method for Colorfastness to Light: Carbon-Arc. Revised in 2023 to align to AATCC style guide.

AATCC TM111-2023, Test Method for Weather Resistance of Textiles: Exposure to Daylight and Weather. Revised in 2023 to align with the AATCC style guide.

AATCC TM149-2023, Test Method for Chelation Value of Aminopolycarboxylic Acids and Their Salts: Calcium Oxalate. Revised in 2023 to clarify and align with prescribed AATCC style guidelines.

AATCC TM161-2023, Test Method for Color Change of Disperse Dyes Caused by Metals, with and without Chelating Agent. Revised in 2023 to clarify and align with prescribed AATCC style guidelines.

AATCC TM168-2023, Test Method for Chelation Value of Polyaminopolycarboxylic Acids and Their Salts: Copper PAN. Revised in 2023 to clarify and align with prescribed AATCC style guidelines.

AATCC TM179-2023, Test Method for Skew Change in Fabrics After Home Laundering.

Revised in 2023 to add that optionally, the direction of skew may be reported.

AATCC TM186-2023, Test Method for Weather Resistance: UV Light and Moisture Exposure: Revised in 2023 for clarity and to align with style guidelines.

AATCC TM192-2023, Test Method for Weather Resistance of Textiles: Sunshine-Arc Lamp Exposure with and Without Wetting: Revised in 2023 to align to the AATCC style guide.

AATCC TM193-2023, Test Method for Aqueous Liquid Repellency: Water/Alcohol Solution Resistance: Revised in 2023 to add History section and clarity.

AATCC TM194-2023, Test Method for Anti-House Dust Mite Properties of Textiles under Long-Term Test Conditions: Revised in 2023 to add dust mite food source and to align with the AATCC Style Guide.

Editorially Revised Standards

AATCC TM20A-2021e, Test Method for Fiber Analysis: Quantitative. Editorially revised in 2023 to update numbering in section 17.

AATCC TM43-1952e6(2018)e3, Test Method for Wetting Agents for Mercerization. Editorially revised in 2023 to move the History section.

AATCC TM61-2013e(2020)e2, Test Method for Colorfastness to Laundering: Accelerated. Editorially revised in 2023 to update designation and History section.

AATCC TM79-2010e2(2018)e3, Test Method for Absorbency of Textiles. Editorially revised in 2023 to move History section.

AATCC TM88B-2018te, Test Method for Seam Smoothness in Fabrics after Home Laundering. Editorially revised in 2023 to add History section.

AATCC TM88C-2018te, Test Method for Crease Retention in Fabrics after Home Laundering. Editorially revised in 2023 to add History section.

AATCC TM107-2022e, Test Method for Colorfastness to Water. Editorially revised in 2023 to correct misspellings.

AATCC TM124-2018te, Test Method for Smoothness Appearance of Fabrics after Home Laundering. Editorially revised in 2023 to add History section.

AATCC TM142-2016e, Test Method for Appearance of Flocked Fabrics after Home Laundering and/or Coin-Op Drycleaning. Withdrawn

AATCC TM163-2013(2020)e4, Test Method for Colorfastness to Storage: Dye Transfer.

Editorially revised in 2023 to update history:

AATCC TM174-2022e, Test Method for Antimicrobial Activity of New Carpets. Editorially revised in 2023 to correct section formatting.

AACC EP7-2021e, Evaluation Procedure for Instrumental Assessment of the Change in Color of a Test Specimen: Editorially revised to update a reference in the section 2.Principle, and to place the history in correct order.

AATCC EP10-2018e2, Evaluation Procedure for Multifiber Adjacent Fabrics: Editorially revised 2023 to clarify the Notes section and to move the History section.

AATCC EP14-2021e2, AATCC Evaluation Procedure for Small Color Differences: Editorially revised in 2023 to correct a typo.

DIGITAL LABS

Fluorescence and High-Visibility for Textiles was held February 23-24, 2022. Presentations included The History of Safety Colors, Roland Connelly, RoLyn Group Color Consultants; The Life of a High Visibility Shirt: A Manufacturer's View, Denise Statham, Workwear Outfitters; ANSI 107 Testing Requirements – Challenges in Meeting Requirements, Sravanth Kanukuntla, SGS North America Inc.; Detergent Use in the Lab: Beyond Just Coming Clean, Erika Sim-

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mons, AATCC; Detergent Technologies with Focus on Optical Brighteners, Rachel Applegate, Procter & Gamble; Effectively Measuring Fluorescent Textiles (or Not!), Ken Butts, Datacolor; High-Vis Apparel and Sustainability in Polyester Exhaust Dyeing, Werner Tissen, DyStar Colours Distribution GmbH; Optical Brightening Agents Fundamentals, Randy Mumford, Archroma; Weathering Test: AATCC 169 Option 4,Michael Murphy, e-Dye; Science of Visibility, Michael Hayes and Katie Aune, 3M Personal Safety Division; A Brand Perspective on Managing Hi-Vis Product Claims and Controlling Production; and Best Practices, Gregg Woodcock, Hanesbrands Inc.

Dyestuff Chemistry Fundamentals Digital Lab, hosted by Bryan Dill, Archroma, US Inc on March 30, 2022, outlined a digital lab as a primer in basic dyestuff chemistry and application for the beginner to intermediate level. The presentation begam with an overview of how we see color and proceeded to an overview of key dye classes. Dye classes and their reaction mechanisms were shown for natural and synthetic substrates. Optical Brightening Agents were also covered.

On June 8, 2022, **Textile Labeling Essentials: United States Softlines Labeling Requirements Digital Lab,** was hosted by Jennifer Hargrave, Bureau Veritas Consumer Products Services**.** This Digital Lab covered United States softlines labeling requirements: including Fiber Content Labeling, New York State Faux Fur Labeling, Care Labeling, Manufacturer's Identification, Country of Origin, Law Labeling, and Tracking Labels for Children's Products.

Sustainability in Textiles, **August 17-18, 2022**

Lewis Shuler, Paradise Textiles, moderated the first day of meetings that included the following presenters: Dimitri Deheyn, Scripps Institution of Oceanography, Comparative Analysis of Biodegradability of Textile Materials in the Marine Environment; Jan Beringer, Hohenstein, Useful Data from Microfiber Analysis; Heather Elliot, Smartwool, Test method for Fiber Fragment Release During Home Laundering; Jeremy Stangeland, Under Armour, Turn Off the Tap; Siva Pariti, BluWin, Tackling Challenges beyond sustainability – Reshaping Tomorrow; Sharon Perez, Lenzing, Lenzing fiber innovation: Providing solutions for the creation of sustainable materials; Mary Ann Ankeny, Cotton Incorporated, From Dirt to Shirt and Beyond: Cotton's Circularity Story; and a hosted panel, Fiber Fragmentation Panel hosted by Heather Eliot, Jeremy Strangeland, and Lewis Shuler.

The second day notes Muhammad Zubair, Organic Dyes & Pigments, as moderator of the following presentations: Kasper Nossent, Dyecoo, Water Free and Chemical Free Dyeing of Polyester on Industrial Scale in Super Critical CO; Karin Fleck, Vienna Textile Lab, White and Green Biotech Creating New and Old Dyes and Pigments; Randy Mumford, Archroma, Sustainability in Textile Finishing; Dee Roche, Alchemie, Dyeing for Change: How Waterless, Low Energy Dyeing Technology is Set to Kickstart the Revolution; Akhil Sivanandan, Greenstory, Understanding Life Cycle Assessment (LCA) with Pioneering Sustainability Platform Green Story; Seshadri Ramkumar, Texas Tech University, Sustainable Oil Absorbent Textiles; and Scott Echols, ZDHC, A Journey not a Destination: How to engage with the ZDHC Roadmap to Zero Programme.

SHORT-COURSES & WORKSHOPS

The Color Acuity Short-Course, A Deeper Dive into Color Vision in Screening, was held on both August 23, 2022, and October 25, 2022 at the AATCC Technical Center and hosted by Erika Simmons, AATCC. Agenda topics included**:** Color Discernment the Business Case; Color Vision Defect the Quality Management Case; Applicable Industry Standards; Tools and Test Kits: AATCC Color Quality Tools; and Lighting Considerations.

Hosted August 24-25, 2022, at the AATCC Technical Center, **AATCC Color Management Workshop** outlined presentations: Color Basics: Balancing Theory and Application, Jean Hoskin, SJH Designs; Color Communication, Tim Williams, Color Solutions International; Training a Color Approval Team, Jean Hoskin, SJH Designs; Understanding Whites and other Fluorescent Materials, Chris Hipps; Archroma; and Fundamentals of Dyeing and Challenges to Meet Performance Standards; Fred Gliddon, Archroma.

Varied outlined Breakouts included: Illumination and Observer Issues (Color Lab) with lead facilitator, Jean Hoskin and Tim Williams; Sample Analysis and Measurement Technique (General Lab) with lead facilitator, Chris Hipps and Carol Revels; and Implementing Virtual Development in a Digital Workflow (Council Room) with lead facilitator, Ken Butts, and Andrew Fraser.

The second day of the workshop hosted the following presentations: Color and Lighting: Control, Efficiency, and Compliance, Andrew Fraser, LKH Sourcing; Supply Chain Conformance – Why Don't my Numbers Match Yours?, Ken Butts, Datacolor; The Cost of Color Quality, Andrew Fraser, LKH Sourcing; The Evolution and Revolution of Global Color Management, Carol Revels, Lands' End Inc.; and Ultra-Portable Color Measurement: A Spectro in Your Pocket?, Ken Butts, Datacolor.

Breakout sessions included: Processes and Technologies for Better Color Control in the Supply Chain with lead facilitator, Tim Williams, and Chris Hipps; and Realities of Color Management in Retail (Council Room) with lead facilitator, Ken Butts, and Carol Revels.

The Textile Discovery Summit was held, October 4-6, 2022 at the Hilton University Place located in Charlotte, NC.

Highlights included the Fiber Innovation session with moderator, Nelson Houser, Retired Archroma, introducing the following presentations: Super Sorptive and Super Active Cotton Fabrics for Biological and Chemical Protections, Gang Sun, University of California, Davis; Squitex: All-Natural Performance Fabric, Melik Demirel, Pennsylvania State University; A Spandex Fiber with Hybrid Properties for Customized Fitting, Guan Wang, The LYCRA Company; Fibers and Textiles from Chicken Feathers, Yiqi Yang, University of Nebraska-Lincoln; Kyorene Graphene Functional Fibers Properties, Applications, and Proper Test Methodology, David Vanek, Graphene One, LLC.

Moderated by Heather Johnson, American & Eifrid LLC, the Innovations in Textiles and Testing session included presentations: Recycled Polyester Testing and Verification, Sandeep Khatua, TUV SUD America; Wicking Tests Revisited – Overview of Relevance and Discrepancies Between Methods; Emiel DenHartog, North Carolina State University; Antimicrobial Technology Using Sequestered Hydrogen Peroxide in Performance Finish Applications Suitable for Protective Fabrics, Bernd Liesenfield, QuickMed Technologies.

The Smart Textiles session moderated by Robina Hogan, Frankl & Thomas Inc included A Colorimetric Biosensor Made by Polydiacetylene Nanofibers for a Rapid Detection of SARS-CoV-2, Abhishek Bhattacharjee, Colorado State University; Gel-Free Velcro Hook Textile Electroencephalography Electrode for Brain Activity Monitoring, Granch Berhe Tseghai, Ghent University; and SmartTex: Health Monitoring Research for Aerospace and Earth, Samantha Shintay, Hohenstein Institute America.

Carol Revels, Lands' End Inc. moderated the Sustainability session hosting presentations: Recycling in the Global Textile Complex: The Future and Technology That Will Take Us There, Karen K. Leonas, North Carolina State University; Recycled Content Verification Methods from Chain of Custody to Testing to Ensure Claim Validity & Regulatory Compliance; Min Zhu, SGS North

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America; Upcycling Textile Wasters into Bio-based Building Blocks: Challenges and Opportunities Toward a Novel High Yield, and Low Carbon Footprint Conversion Process; Ramon E. Vera, North Carolina State University.

The Sustainable Dyeing & Finishing Session, moderated by Nelson Houser, included Dyeing Innovations Improve Sustainability and Cost, Hardy Sullivan, Thies Corp.; and Sustainable Finishing Technologies, Michael Schuhmann, Archroma.

Andrew Fraser moderated Electronic Color Management Presentations and Roundtable, with Doug Bynum, Natific USA, and Tim Williams, Colors Solutions International.

Break-out sessions included PFAS Tutorial, moderated by Dennis Scheer, SST Technology Solutions, Inc; PFAS-Regulations and Understanding the Implications for your Products, Pankajkumar Sarda, Intertek; Odor Control session moderated by Bob Monticello, International Antimicrobial Council; The AATCC TM30-2017e Test Method and ASTM E3227-20 Standard Test Practice are Complementary Approaches for Evaluating the Performance of Antifungal Technologies, Alicia McClain, International Antimicrobial Council; and Assessment of Odor Control Technologies Using Chemical Analysis by Headspace Chromatography, Peter Nastasi, Microban Products Company.

The Break-out session, Recycled Fibers Roundtable, hosted by Carol Revels, included facilitators, Min Zhu, SGS North America Inc; Sandeep Khatu, TUV SUD America, Inc., and Karen K. Leonas, North Carolina State University.

The **AATCC Textile Testing Workshop** was held October 26 – 27, 2022 at AATCC Technical Center. Hosted by the staff members, Diana Wyman, Erika Simmons, Angela Jabara, and Carrie Gray of AATCC, workshop presentations on Wednesday, October 26th included Introduction to Standardized Testing and Fabric Preparation; An Overview of Color Basics; Visual Color and Instrumental Evaluation, Ken Butts, Datacolor; AATCC Grading Scales; and An Overview of Standard Laundering Conditions.

Break-out sessions on this day included Color Evaluation, Color Laboratory Session, Dimensional Changes Test Methods; Appearance Retention Test Methods and Oil Resistance Test Methods.

On the second day, Thursday, October 27th, workshop presentations included Colorfastness Crocking Methods; Colorfastness Water and Perspiration Methods; Colorfastness Light Method; Colorfastness Laundry Methods; and Water Resistance Test Methods, led by Kiarash Arangdad, ITG Burlington.

Breakout sessions included presentations of Colorfastness Test Methods; ASTM Test Methods; and Water Resistance Test Methods.

A roundtable discussion followed by certificate presentations closed out the workshop session.

AATCC Printing United Alliance Digital Textile Printing: 2022 held at Sheraton Imperial I Durham, on December 7–8, 2022.

On Wednesday, December 7th, Ray Weiss, PRINTING United Alliance and Diana Wyman, AATCC opened the session before the presentation of the following:

Business Growth and TDG, Victor Pena, Omniprint International; Textile Industry Update: 2022 and Beyondm Johnny Shell, Keypoint Intelligence; Show Me The Color! New Spot Color Library Optimization, Jim Rael, ColorCasters, LLC and Shelby Sapusek, ColorCasters, LLC; Décor-Print story – a conversation with Kerry King from Spoonflower, Dee Dee Davis, Décor Print; Empowering the Print Community to Better Understand the Interior Design Industry Panel Discussion Adrienne Palmer, Big Picture magazine (moderator); Alyssa McNamarra, Spoonflower; Michael Sanders, Digital Bias Consulting; and Kathryn Sanders, Western Sensibility.

Other presentations included: On Demand Direct Print, Steven Amitai, Greentex America; Digital Product Passports; Lily Hogan, Toxnot.com; How to Formulate Pigment Ink Jet inks for Fashion and Apparel Applications, Scott Donovan, DuPont Industrial Solutions; and Computer to Screen Application for Garment Printing, Carl Blue, Clemson University.

The session held Thursday, December 8th included: Digital Textile Printing as an Artistic Medium, Kathryn Sanders, Western Sensibility; Performing Color Process Control in Digital Textile Printing, Viktor Lazzeri, Barbieri Electronic; Textile RGB (Fogra58): The Overarching Digital Textile Working Space, Joe Tschudi, Tschudi Software Technology GmbH; LED Lighting & Color Matching – How Textile Companies are Dealing with the Revolution.

James Summers, JUST Normlicht, Inc. & GL Optic; Advancements in Digital Textile Manufacturing, Frank Henderson, Henderson Sewing; and How to Get the Best Fitting Apparel Using Today/Tomorrow Technologies; Daniel Watts, Pattern Room Sales Pty Ltd.

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Reports of Research Committees 2022

RA23, Colorfastness to Water

Chair: Jodi Geis
Manufacturing Solutions Center

AATCC TM15, Test Method for Colorfastness to Perspiration is current.

AATCC TM107, Test Method for Colorfastness to Water passed and was submitted to TCR Ballot in August.

AATCC TM106, Test Method for Colorfastness to Water: Sea has been aligned with TM15 and now needs a P&B statement.

A task group was created to develop a new method around colorfastness to Water: Bromine (Spa).

Following ASTM D1776 meeting in June 2022, AATCC TM15 targeted to have changes to the reference conditioning of specimens.

RA24, Fiber Analysis

Chair: Joseph Lin
Vartest Laboratories, Inc.

Committee to vote on new PT fabrics for February and August 2023.

The committee will review the AATCC 1992 Cross-Sectioning Kit to validate tools for current practice.

RA31, Antimicrobial Activity

Chair: Matthew Hardwick
Resinnova Laboratories

Bob Monticello, International Antimicrobial Council, was elected committee chair.

Yihong Li, Microban International, was elected as committee secretary.

AATCC TM90, Test Method for Antibacterial Activity of Textile Materials: Agar Plate and AATCC TM174, Test Method for Antimicrobial Activity Assessment of New Carpets passed TCR ballot and will be in 2023 manual.

AATCC TM 147, Test Method for Antibacterial Activity of Textile Materials: Parallel Streak is working on stats for a Precision & Bias statement.

The committee is planning to write a monograph on scouring methods to remove microbial contaminants on materials.

The committee will review the scope and expand it to include odor control methods.

RA32, Static Electricity

Chair: Victor Khlolodkov
Ascend Performance Materials, LLC

A subcommittee was formed to draft a new electrostatic decay method. Susan Gassett, US Army Soldier Center, provided a Static Decay procedure and Victor Kholodkov, Ascend Performance Materials, is writing first draft of the procedure into AATCC version.

Electro-Tech Systems of Pennsylvania will supply the testing apparatus.

RA33, Colorfastness to Atmospheric Contaminants

Chair: John Crocker
Cotton Incorporated

AATCC TM109, Test Method for Colorfastness to Ozone in the Atmosphere under Low Humidities, and AATCC TM129, Test Method for Colorfastness to Ozone in the Atmosphere under High Humidities, were approved by the committee.

AATCC TM109 Temperature Set Point Study has paused until committee is able to meet onsite.

RA34, Preparation

Chair: William Garrett
Cotton Incorporated

Several test methods due for 5-year review. They include AATCC TM81, Test Method for pH of the Water-Extract from Wet Processed Textiles, AATCC TM82, Test Method for Fluidity of Dispersions of Cellulose from Bleached Cotton Cloth, AATCCTM98, Test Method for Alkali in Bleach Baths Containing Hydrogen Peroxide, AATCCTM102, Test Method for Determination of Hydrogen Peroxide by Potassium Permanganate Titration; AATCC TM144, Test Method for Alkali in Wet Processed Textiles: Total, and AATCC TM209, Test Method for ph and Total Alkali in Wet Processed Textiles: Combined. These methods were reviewed by a subcommittee and are scheduled for Summer 2022 ballot cycle.

Ken Greeson, Cotton Incorporated, gave an update on subcommittee's efforts to investigate use of alternative solvents; such as ionic liquids as compared to hexanes in AATCCTM97, Test Method for Extractable Content of Textiles; concluding subcommittee's recommendation to consider an exploratory search of solvents to replace hexanes as new solvents become available.

RA36, Color Measurement

Chair: Andrew Fraser (Acting)
InMocean Group, LLC

The committee agreed to hold a digital lab for DE1000 v. DECMC and to add is to the AATCC Color Management workshop.

Liaison roles for ASTM/ISO/CIE discussed with Roland Connelly noting that the liaison jobs are primarily communication driven. ASTM, ISO and CIE have standards that AATCC references or contra-wise.

AATCC EP8, Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale; and AATCC EP12, Evaluation Procedure for Instrumental Assessment of Degree of Staining; will change to an annual replacement to verify scales within Tolerances and will change from CMC to CIELAB to align with other EPs. The committee does not see the need to change the current verification method nor the need to switch to an annual replacement unless the scale is out of tolerance of a CMC 0.30.

A subcommittee will be formed to address calculation of methods by CIELAB or CMC.

RA36SC will reactivate to focus on changes LED brings to Color Management and evaluate use of best practices by organization.

RA38, Colorfastness to Crocking

Chair: John Crocker
Cotton Incorporated

AATCC TM8, Test Method for Colorfastness to Crocking: Crockmeter, and AATCC TM116, Test Method for Colorfastness to Crocking: Rotary Vertical Crockmeter, were reaffirmed in February 2021.

Committee only hosting meetings as needed; noting that all methods are well established in the industry.

RA42, Dimensional Changes

Chair: Seemanta Mitra
UL, LLC

Paul Pinkston volunteered to serve as the

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committee's chairperson.

Diana Wyman, AATCC, and John Crocker, SDL Atlas, will research options for developing a monograph on commercial options. Currently noted as only exploratory and not considered as committee project.

AATCC TM96, Test Method for Dimensional Changes in Commercial Laundering of Woven and Knitted Fabrics Except Wool, calls for a wash wheel that is no longer available. With no new wash wheels being manufactured, new users are unable to follow the method as written. The committee agreed to ballot discontinuation of method.

AATCC TM179, Test Method for Skew Change in Fabrics After Home Laundering, approved to add optional reporting for the direction of skew in November.

RA45, Finish Analysis

Chair: Carla MacClamrock
Cotton Incorporated

An interlaboratory study will be conducted to establish Precision and Bias for AATCC TM206, Test Method for Free and Hydrolyzed Formaldehyde: Water Extraction.

New fabric samples with varying levels of formaldehyde that are suitable for interlaboratory study have been created. Volunteer labs will receive the samples along with testing instructions.

AATCC TM112, Test Method for Formaldehyde Release from Fabric: Sealed Jar and AATCC TM206 now have an option of using HPLC to determine concentration. HPLC data will be provided by participating interlab study labs with available equipment.

RA49, Insect Resistance

Chair: Brent Smith
Burlington Fabrics

AATCC TM194, Test Method for the Assessment of the Anti-House Dust Mite Properties of Textiles under Long-Term Test Conditions will move to TCR ballot.

The committee discussed CLB #491217, TCR ballot T0518 (chemical transfer via crocking), was rejected at TCR due to lack of an extraction method. Volunteers are being invited to review. If there are no volunteers, the method will be retired from the agenda.

Volunteers were solicited to work on the conversion of Natick test method for insect repellent content into an AATCC method.

Committee discussed AATCC Test Method 194, Test Method for Assessment of the Anti-House Dust Mite Properties of Textiles under Long-Term Test Conditions. The committee recommended food substitute (Tetramin Fish Flakes) as the new nutrient within the method.

RA50, Lightfastness & Weathering

Chair: Smrithi Kumar
Q-Lab Corp.

The AATCC 2023 manual will include technical changes made to AATCC TM111, Test Method for Weather Resistance of Textiles: Exposure to Daylight and Weather; AATCC TM186, Test Method for Weather Resistance: UV Light Moisture Exposure; AATCC TM192, Test Method for Weather Resistance of Textiles: Sunshine-Arc Lamp Exposure with and Without Wetting, AATCC TM16.1, Test Method for Colorfastness to Light: Outdoor; and AATCC TM16.2, Test Method for Colorfastness to Light: Carbon-Arc

Each method will be updated into the new style guide format for availability midyear.

Andy Francis, Q-Lab, reported on the US response to the DIS ballot on IS0 105-B04.

ISO Standards Update – ISO 105 BO4 passed PWI and will move on to next stage.

RA56, Stain Resistance

Chair: Suzanne Holmes
Cotton Incorporated

Interlabs will develop P&B for draft methods on Soil Release: Household Soils Method and Stain Release: Spot Cleaning Method.

ANOVA statistics was done to develop a P&B statement for Soil Release draft that is scheduled to be balloted in December 2022.

Based on received comments, Interlab for Stain Release: Spot Cleaning method draft require repeating, with goal of completion by May 2023 meeting.

AATCC TM193, Test Method for Aqueous Liquid Repellency: Water/Alcohol Solution Resistance was reaffirmed in July 2022.

AATCC staff to forward video of remade video of AATCCTM 118, Test Method for Oil Repellency-Hydrocarbon Resistance Method. Committee members are invited to submit comments upon review.

RA57, Floor Covering

Chair: Alan Buttenhoff
Shaw Industries Group

The committee conducted a study of the synthetic soil formulas from SDL Atlas and decided on the appropriate version. The formula will be removed from AATCC TM122, Test Method for Carpet Soiling: Service Soiling, and a separate Monograph will be created that lists the formula.

RA59, Fibrous Test Materials

Chair: Miranda Klaas
TestFabrics Inc.

Multifiber monograph is ready for committee review and will be distributed to members.

Polystyrene chip for SAE/ASTM Committee and a 5-years supply of Gas Fading Control fabric is available from Test Fabrics

Lynee Willis, Association for Contract Textiles, addressed bristle impact on test results for pilling – an ongoing study using a verification cloth from Testfabrics; and John Crocker, SDL Atlas, shared results of bristles impact.

RA60, Colorfastness to Washing

Chair: Suzanne Gassett
US Army Natick Soldier Systems

AATCC TM190, Test Method for Colorfastness to Home Laundering with Activated Oxygen Bleach Detergent: Accelerated will be balloted to discontinue due to lack of NOBS.

Committee to review AATCC TM172, Test Method for Colorfastness to Powdered Non-Chlorine Bleach in Home Laundering and AATCC TM188, Test Method for Colorfastness to Sodium Hypochloride Bleach in Home Laundering.

RA61, Appearance Retention

Chair: Renuka Dhandapani
Cotton Incorporated

AATCC TM88B, Test Method for Seam Smoothness in Fabrics after Home Laundering; AATCC TM 88C, Test Method for Crease Retention in Fabrics after Home Laundering; AATCC TM124, Test Method for Smoothness Appearance of Fabrics after Home Laundering; and AATCC TM143, Test Method for Appearance of Apparel and Other Textile End Products after Home Laundering are due for 5-year review. Only editorial changes are needed to satisfy the Style Guide recommendations.

Study to be conducted at AATCC using a portable viewing board from Verivide, to compare the regular fluorescent light set up. Ken Greeson and Angela Massengill, Cotton Incorporated, will work on reducing the number of gradings to be collected to a level that will be easy on the rater. For TM88C, traditional 8-ft fluorescent overheat lights and traditional flood lamp set up will also be compared to LED overhead light with LED floodlight.

AATCC TM66, Test Method for Wrin-

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kle Recovery of Woven Fabrics: Recovery Angle and AATCC TM128, Wrinkle Recovery of Fabrics: Appearance are due for 5-year review and have been submitted for Summer 2022 balloting.

Chair Renuka Dhandapani, Cotton Incorporated, presented, 'Things to Know before Replacing Fluorescent Lights with LED'. Several test methods will be affected by the change from fluorescent to LED lighting. A committee task force will be formed to address the effects.

RA63, Water Resistance

Chair: Kiarash Arangdad
ITG Burlington

Stephen Payne, Cotton Incorporated, presented recent test data regarding Water Absorbency Spray Rating (WASR) and the impact of residual detergents after 30 successive launderings on water repellency. A task group will be created to discuss potential options for the shade discoloration on the exterior fabric surface after spray testing.

Two proposed new test methods discussed: Proposed new test method for Measuring Condensation on Textiles within Microclimates and proposed new test method for Measuring Water Absorbency with Spray Test.

AATCC TM22, Test Method for Water Repellency: Spray, is targeted for review.

RA75, Evaluation for End Use Performance

Chair: Tyler Pickett
Adidas

A committee brainstorming session was conducted to identify topics of interest. Topics included: wear testing, sharing presentations on ISO/ASTM trial wear tests; developing monographs for best practices; using drycleaning institute complaints about field failures, colorfastness issues to develop monographs; and contrast in lab results for comfort of wool versus actual wearer comfort of wool.

Julian Rojas, Henkel, volunteered as Interim Committee Chair.

Steve Arcidiacono, US Army, discussed potential publication of findings from survey reviewed at the 2019 AATCC Antimicrobial Conference. Volunteers interested in assisting with outline and write-up are asked to contact Garry Atkinson, AATCC at atkinsonga@aatcc.org.

RA80, Printing Technology

Chair: Alyssa McNamara
Spoonflower

Victoria Nelson Harris, Mimaki, presented, "*An Overview of Color Management for Digital Textile Printing*". Her presentation included an introduction to Mimaki textile product line and new innovations.

The committee reviewed technology planned for demonstration in Berlin, Germany at FESPA 2022 scheduled May 31 – June 3, 2022.

RA87, Applied Dyeing and Characterization of Dyes

Chair: Barry Brady
Organic Dyes and Pigments, LLC

Randy Mumford, Archroma, presented, "*Sustainable Dyeing of Polyamide*".

The committee discussed AATCC TM157, Test Method for Colorfastness to Solvent Spotting: Perchloroethylene, noting that while awaiting a collective P&B with same solvents and fabrics, they will move forward to approve the modification of an alternative solvent and will update the method once the P&B is complete.

The Committee is targeting presentation on non-halogen and nonmetal navy and black dyes from Archroma for next meetings.

RA88, Home Laundering

Chair: Gregg Woodcock
Hanesbrands Inc.

The committee motioned to propose by ballot to deactivate Monograph M3,High Efficiency Washers in North America.

A motion was made to add an optional requirement to document water hardness (as specified in ISO 6059) for AATCC LP1, Laboratory Procedure for Home Laundering: Machine Washing and AATCC LP2, Laboratory Procedure for Home Laundering: Hand Washing.

The committee discussed AATCC's detergent delivery methods with consideration of individual dose packages (pods/pucks).

Discussions addressed developing a standard related to Chlorine and Non-Chlorine bleach laundering procedures.

RA89, Hand Evaluation

Chair: Seshadri Ramkumar
TIEH Texas Tech Univ

The Committee decided to add verbiage to the Appendix of AATCC EP5, Evaluation Procedure for Fabric Hand; to inform users that instrumentation is available for measuring/evaluating fabric hand and to include a reference to the AATCC Buyer's Guide.

Emtec made a recommendation to create a terminology draft on haptics. It was suggested that AATCC M11, Glossary of AATCC Standard Terminology and ASTMD123 be consulted to confirm existing definitions to prevent overlap.

RA99, Technical Manual Editorial Review

Chair: Norma Keyes

In efforts to consistently format method's history and insert placement, the committee will be accessible to chairs for review of standard's drafts before submission to ballot. Invitations to committee chairs will be distributed.

The committee discussed past due and upcoming methods including AATCC TM158, Test Method for Dimensional Changes on Dry cleaning in Perchloroethylene: Machine; noting that to include new solvents, the definition of drycleaning may need to be redefined.

Other methods due for review include: AATCC TM149, Test Method for Chelation Value of Aminopolycarboxylic Acids and Their Salts: Calcium Oxalate; AATCC TM161, Test Method for Color Change of Disperse Dyes Caused by Metals, with and without Chelating Agent and AATCC TM168, Test Method for Chelation Value of Polyaminopolycarboxylic Acids and Their Salts: Copper PAN.

Discussion included creating a lab procedure to evaluate appearance characteristics that are not currently covered in AATCC TM142, Test Method for Appearance of Flocked Fabrics after Home Laundering and/or Coin-Op Drycleaning.

RA100, Global Sustainability

Chair: Shulong Li
Milliken

Jeff Strahan, Milliken & Company, presented "*History of PFAS/Overview of Existing and Upcoming Legislation*".

Rock J. Vitale, Environmental Standards, & Jeffrey Stull, International Personnel Protection, Inc. presented, "*Topics in PFAS Testing/ PFAS Testing & ASTM*".

A motion was made to form a subcommittee to explore future methods of AATCC TM212, Test Method for Fiber Fragment Release During Home Laundering to include dryer shedding methods and identify/quantify methods of fiber fragment types. RA24 will be referenced.

RA102, Statistics

Chair: Radhakrishnaiah Parachuru
Georgia Inst of Technology

AATCC Monograph 9, A Summary of ASTM Methods for Interlaboratory Testing, references discontinued ASTM standards and will be reviewed for update.

Krishnaiah Parachuru., Georgia Tech, will be available to offer statistical guidance for AATCC ISO 17043 development: Erika Simmons, AATCC, will send a copy of the standard for reference.

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RA103, Spectroscopic Technologies

Chair: Nelson Vinueza
NC State University

Januka Budhathoki-Uprety, NCSU, presented, "Using Spectroscopic Techniques to Access Textile Dyes Removal from Aqueous Solution by Polymer".

Dr. Ericka Ford, Associate Professor, NC State University, presented 'Raman and Infrared Analysis of Blended and Nanocomposite Fibers'.

RA104, Garment Wet Processing

Chair: R Michael Tyndall
Tyndall Textile Consulting Ltd

Dennis Scheer and Dallas Crotts agreed to become members of a speaker task force to assist with the task of selecting and securing speakers for RA104 committee meetings.

Committee discussion hosted several member-interest topics including offsite education visits, forming a speaker task force, identifying garment processing techniques, challenges from garment dyeing, new technologies, and more.

RA111, Electronically Integrated Textiles

Chair: Bethany Pollack
Draper Knitting Co

Several proposed work items for the committee includes Future IEC 63203-20X-X: Wearable Electronic Devices and Technologies – Part 20X-X: Test method for Measuring Triboelectric Nanogenerator Performance of Fabric under Sliding Contact Separation Mode; Future IEC 63203-20X-X: Wearable Electronic Devices and Technologies – Part 20X-X: Test method for Measuring Performance of Fabric Based Piezoelectric Nanogenerator on Stretching Mode; Future IEC 63203-402-X: Wearable electronic devices and technologies – Part 402-X: Performance of Stress Measurements in Wearables; Future IEC 63203-20X-X: Wearable Electronic Devices and Technologies – Part 20X-X: Standard Test Method for Measuring Resistance-based Textile Tensile Strain Sensors (RTTSS) in dry and wetted conditions; and Future IEC 63203-20X-X: Wearable Electronic Devices and Technologies – Part 20X-X: Test Methods for Performance of Wearable Heating Products.

Committee will meet with ASTMD 13.50 in Spring/Summer 2023

RA112, Thermal Analysis

Chair: Apurba Banerjee

Reviewed and discussed new testing measurements to identify the standard of cool to touch testing.

The committee chair shared an overview presentation of the University of Oregon team research paper. Members to receive a copy.

RA113, Emerging Issues

Chair: Min Zhu
SGS North America, Inc

AATCC Monograph 14, Guidance and Considerations for General Purpose Textile Face Coverings: Adult, is approved to be reaffirmed as is. If updates on requirements or language for children's mask are needed, they will be added in the following version.

ISOTC 338 Menstrual products was established in early 2022 with US Technical Advisory Group (TAG) available later in March 2022. ANSI is an observer member and only the scope of this ISO standard has been drafted. With the newly established task force, Diana Wyman, AATCC will contact ISO for status report of this project.

RA114, Moisture Management

AATCC M14, Guidance and Considerations for General Purpose Textile Face Coverings: Adult is approved to be reaffirmed as is. If the updates on the requirements or language for children's mask are needed, they may be added to a subsequent version.

ISO/TC 338 Menstrual products was established early 2022 with a US Technical Advisory Group (TAG) establishing later in March 2022. ANSI is an observer member. The scope of this ISO standard is currently being drafted.

The committee formed its Task Force with Diana Wyman, AATCC, tasked to contact ISO to determine their work status.

Committee outlined methods under its jurisdiction as AATCC TM017, Test Method for Wetting Agents, Evaluation of; AATCC TM27, Test Method for Wetting Agents: Evaluation of Rewetting Agents; AATCCTM43, Test Method for Wetting Agents for Mercerization; AATCC TM79, Test Method for Absorbency of Textiles; AATCC TM195, Test Method for Liquid Moisture Management Properties of Textile Fabrics; AATCC TM197, Test Method for Vertical Wicking of Textiles; AATCC TM198, Test Method for Horizontal Wicking of Textiles; AATCC TM199, Test Method for Drying Time of Textiles: Moisture Analyzer; AATCC TM200, Test Method for Drying Rate of Textiles at their Absorbent Capacity: Air Flow; AATCC TM201, Test Method for Drying Rate of Fabrics: Heated Plate; and AATCC TM204, Test Method for Water Vapor Transmission of Textiles.

Trial samples were received with results pending from Thermetrics and PPT Group. The material will be tested against AATCC TM201.

Committee discussion noted that AATCCTM199 will be reviewed to confirm whether there are ceramic options still available for analyzer.

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